Ordered Chiral Geometry

Volume I of the integrated monograph: two associative directions, aligned locality, three bars, and bar–cobar reconstruction

Raeez Lorgat

26 July 2026

Abstract

Holomorphic collision and composition along an oriented interval impose different operations on local observables. This volume studies the bar constructions attached to those operations and the comparisons required for reconstruction. Transverse associative composition, associative chiral composition, and the chiral Lie bracket retain their own coefficient categories, units, and differentials. The ordered bar–cobar resolution supplies the algebraic starting point. A geometric identification requires a map that also preserves collision coefficients and their compatibility with ordered composition.

The first finite quantization comparison illustrates why the operation and its coefficient parameter must be retained together. For a finite-dimensional complex Lie algebra, antisymmetrization identifies the relative Hochschild complex of its Rees enveloping algebra with the adjoint Chevalley complex whose differential is multiplied by the quantization parameter $\hbar$. This quasi-isomorphism holds over every truncated coefficient ring $\mathbb{C}[\hbar]/(\hbar^{N+1})$ and passes to the $\hbar$-adic limit.

The quadratic calculation determines an associative relation that the Lie brackets alone do not detect. For the Moyal product $\star$ with $\{q, p\} = 1$, set $e = q^2/2$, $f = -p^2/2$, and $h = -qp$. Then

$$e \star f + f \star e + \frac{1}{2} h \star h = -\frac{3}{8}\hbar^2.$$

The corresponding Casimir quotient of the Rees enveloping algebra of $\mathfrak{sl}_2$ is the even Weyl algebra. Scalar reduction and an augmentation give continuous stable trace maps of commutative algebras; monoidal module fusion uses a separately specified completed bialgebra. Extending these constructions to Hamiltonian jets and mixed open–closed operations requires compatible cochain transition maps, continuous limit identifications, and operadic refinements. Geometric and physical realizations additionally require their comparison maps and orientation, descent, and quantization data. The finite calculations thus determine concrete algebraic constraints on the proposed comparisons.

Operations and reconstruction

A boundary algebra records compositions of boundary operators. Its derived centre records actions compatible with these compositions, including their higher homotopies. A selected boundary module instead has its own derived endomorphism algebra. These two constructions answer different questions: coherent action on all boundaries and reconstruction from a chosen boundary.

The field-theoretic problem starts with independently defined bulk observables and boundary conditions. It asks for an action map into the appropriate geometric centre. Its domain includes the support, position dependence, and allowed collisions of the insertions. Quantization must deform this map together with its source and target. An elementary example shows why separate quantizations are insufficient. Let k be a field. Over the dual numbers $k[\varepsilon]/(\varepsilon^2)$, the free rank-two algebra with relation $x^2 = \varepsilon$ deforms $k[x]/(x^2)$. The augmentation $x \mapsto 0$ cannot lift to the undeformed coefficient algebra: a lift would send x to εc, whose square is zero. Thus the action itself has a deformation obstruction.

For a finite-dimensional complex Lie algebra, the relative Rees comparison gives a quasi-isomorphism from relative Hochschild cochains to the Chevalley complex whose differential is scaled by the Rees parameter. Quadratic Hamiltonians generate the even Weyl algebra through the oscillator Casimir quotient. Scalar reduction and an augmentation give continuous stable trace maps of commutative algebras. Monoidal module fusion uses a separately specified coassociative, counital bialgebra in the chosen completed coefficient category.

The Hamiltonian jet limit requires compatible cochain transition maps and limit identifications. Higher mixed open–closed operations require the stated operadic refinements. A Hochschild centre and the Drinfeld centre of a monoidal module category are separate comparison targets. Hall, quantum-double, and physical bulk identifications require their actual comparison morphisms and the relevant orientation, descent, pairing, and analytic hypotheses. A proper filtration specifies an inverse-limit carrier; compatibility of the operations with that limit remains part of each comparison theorem.

Notation and normalization

The ground field has characteristic zero. Stable categories are presentable and idempotent-complete. Tensor products preserve colimits separately. Completions are separated inverse limits of finite weight or proper-height quotients. The Igusa normalization is

$$\Delta_5^{\theta} = 64 D_5, \qquad \chi_{10} = D_5^{\,2},$$

with D_5 monic. Physical charge is additive; Gram degree is carried through its polarization cocycle. Bar, factorization, and Hopf coproducts retain distinct notation and distinct geometric meanings.

Contents

Abstract iii
Operations and reconstruction v
Notation and normalization vii
Contents vii

I Ordered collisions and reconstruction **1**

1 Ordered words and collision coefficients 3
1.1 The coefficient category 3
1.2 Consecutive cuts 4
1.3 Finite normalized bar and cobar 5
1.4 The support left by all cuts 6
1.5 The empty unit and the local vacuum 7
1.6 Finite summands and geometric extensions 7
2 Multiplicities and the local vacuum 9
2.1 Additive labels on collision diagrams 9
2.2 Reassembling the full coefficient 10
2.3 Derived specialization of the vacuum generator 12
2.4 Extending a comparison across the collision support 13
2.5 Higher comparison equations 14
3 Primitive Heisenberg coefficients 17
3.1 Principal parts and differentiation 17
3.2 The differential module on the collision diagonals 19
3.3 Collision and cut maps 22
3.4 Free comparisons and completed mapping complexes 22
3.5 A telescope with all spatial derivatives 24

3.6 The polynomial Heisenberg complex . . . 27
3.7 The binary holonomic dual . . . 27
4 Residues, intersections and cut transfer . . . 29
4.1 Principal parts with spectators . . . 29
4.2 The finite first-cut contraction . . . 31
4.3 Transferred products . . . 32
4.4 Wick comparisons across a cut . . . 35
4.5 Derived intersections of diagonal coefficients . . . 37
4.6 Exterior powers of the primitive coefficient . . . 38
4.7 The support left by every consecutive cut . . . 41
5 Covariant fields and Gaussian elimination . . . 45
5.1 Charges and spectral translation . . . 45
5.2 The completed line connection . . . 47
5.3 A Gaussian complex with retained coordinates . . . 49
5.3.1 The finite heat contraction . . . 49
5.4 Curvature and products of antifields . . . 51
5.5 The inclusion in every antifield degree . . . 52
5.5.1 A homotopy and the flatness criterion . . . 53
5.6 Frames and Gaussian normalization . . . 54
5.7 Higher comparison maps . . . 54
5.8 Formal potential interactions . . . 56
5.9 Composition of normalized eliminations . . . 58
6 Vacuum support and derived actions . . . 59
6.1 The four-position vacuum extension . . . 59
6.2 Horizontal symmetries of the primitive coefficient . . . 60
6.3 Derived scalar maps and the residue simplex . . . 61
6.4 Pointed actions and their comparison paths . . . 63
6.4.1 Free fields and the diagonal tensor . . . 65
6.4.2 A cochain action . . . 66
6.5 A vacuum probe against finality . . . 67
6.5.1 Extracting the separated vacuum coefficient . . . 69
6.5.2 The coherent loop and its consequence . . . 69
6.6 Derivation profiles . . . 70
6.6.1 The fixed-point construction . . . 71
6.6.2 An inner gauge and its domain . . . 75
6.7 Multiplicative recipients for the profiles . . . 75
6.7.1 Tensoring the entire coefficient construction . . . 76
6.7.2 A bar lift of the derivation action . . . 76
6.7.3 The complex of natural higher comparisons . . . 78
6.8 Relations between separated inputs . . . 80
6.8.1 The subset resolution . . . 81
6.8.2 Preserving the mixed generators . . . 82
6.8.3 Grouping and its higher comparisons . . . 83

6.9 Extension spaces . . . 84
6.9.1 Choices that survive and choices that disappear . . . 85
6.9.2 A triple obstruction after all pair choices . . . 85
6.10 The bar of the full action diagram . . . 86
7 Analytic coefficients and collision monodromy . . . 89
7.1 Convergent jets and numerical translation . . . 89
7.2 The separated gauge and its collision boundary . . . 92
7.3 Punctures with their regular coefficient . . . 94
7.3.1 The marked pair and its ternary products . . . 95
7.3.2 The primitive three-point chart . . . 97
7.4 Two punctures through a collision . . . 99
7.4.1 Finite traces and the inverse telescope . . . 100
7.4.2 Restoring the parameter connection . . . 101
7.5 Spatial Hom and intersections of supports . . . 102
7.5.1 The global regular and localized pair . . . 104
7.6 Higher braid transport . . . 105
7.7 Logarithmic families and their curvature . . . 108
8 Fields, centers and reconstruction maps . . . 111
8.1 The regular and punctured normal line . . . 111
8.2 The completed center of the normal pair . . . 112
8.3 Marked deformations and the Kummer residue . . . 114
8.4 Independent normal directions . . . 116
8.5 The shape of a triple collision . . . 117
8.6 Three-point resonance . . . 118
8.7 The center of the puncture category . . . 120
8.8 A scale and two punctures . . . 122
8.8.1 A product deformation . . . 122
8.9 Transport of Hochschild operations . . . 123
8.10 An analytic coefficient dual for the regular Heisenberg bar . . . 123
8.10.1 Growth in the charge direction . . . 124
8.10.2 The completed bar comparison . . . 125
8.10.3 Spectral operations and charge transport . . . 128
8.11 Multiplication from commuting resolutions . . . 130
8.12 The endomorphisms of a moving vacuum . . . 134
8.13 The collision class of the product . . . 136
8.13.1 A constant coefficient model . . . 136
8.13.2 The normal relative complex . . . 137
8.13.3 Computing the class on a resolved map . . . 138
8.13.4 The value for every input order . . . 139
8.14 A map invisible to natural actions . . . 140
8.14.1 Derivations of exterior coefficients . . . 141
8.14.2 Marked coefficients and composition . . . 141
8.14.3 Vanishing under natural actions . . . 143
8.15 An obstruction for every pointed recipient . . . 144

8.15.1 The two ordered products . . . 144
8.15.2 The singleton class cannot vanish . . . 145
8.15.3 The parallel pair survives . . . 147
8.16 Field reconstruction and vacuum normalization . . . 148
The five coupled constructions . . . 153

Ordered and Associative Chiral Geometry **155**
Bar constructions and their comparisons . . . 155

II The invariant object **157**
9 Two directions of locality . . . 159
9.1 The local experiment . . . 159
9.2 Carrier and operation . . . 159
9.3 Principal symbols and residues . . . 160
10 The Ran coefficient category . . . 161
10.1 Decompositions and alignment . . . 161
10.2 Coefficients and chiral convolution . . . 161
10.3 The geometric aligned projector . . . 163
10.4 Factorization coalgebras and pointwise products . . . 165
10.5 The empty configuration and the two units . . . 166
11 The two noncommutative chiral objects . . . 169
11.1 The two products and their units . . . 169
11.2 A factorizing family supported at a marked point . . . 170
11.3 An independently variable chiral product . . . 171
11.4 Unitization and factorization . . . 172
11.5 The mixed compatibility equation . . . 173
11.6 Formal fields and their comparison problem . . . 173
11.7 The geometry supplying each operation . . . 176
12 Transversely ordered chiral algebras . . . 177
12.1 Locality on the coefficient family . . . 177
12.2 Fusion along an interval . . . 178
12.3 Algebraization and product reconstruction . . . 179
13 Block-ordered operations . . . 181
13.1 Chiral operations and ordered profiles . . . 181
13.2 The operadic exponential law . . . 181
13.3 The sign of a mixed contraction . . . 182
13.4 Resolution of an operadic model . . . 184

III Bar, cobar, and the boundary state **187**
14 Three bar constructions . . . 189
14.1 The products and their units . . . 189
14.2 Normalization and the differential . . . 190
14.3 The bar of an associative envelope . . . 191

14.4 Realization and factorization . . . 193
14.5 Two orders of realization . . . 194
15 The internal ordered bar . . . 197
15.1 Normalized words and their cuts . . . 197
15.2 The shuffle comparison . . . 199
15.3 Three coproducts . . . 200
15.4 Interval factorization homology . . . 201
16 Cobar and twisting cochains . . . 203
16.1 The tensor cobar differential . . . 203
16.2 The twisting equation . . . 204
16.3 The two universal extensions . . . 204
17 Complete bar–cobar reconstruction . . . 207
17.1 The counit and the gap contraction . . . 207
17.2 Positive-weight completion . . . 208
17.3 The quadratic coalgebra and its resolutions . . . 209
17.4 A torsion boundary for the Tor comparison . . . 211
17.5 Coefficient duality . . . 211
18 Modules, comodules, and Morita theory . . . 213
18.1 The weighted representation categories . . . 213
18.2 The twisting adjunction . . . 214
18.3 The two bar contractions . . . 214
18.4 Other completions and two-sided comparisons . . . 216
18.5 Reconstruction from an endomorphism algebra . . . 216
18.6 Factorization descent . . . 217
19 Derived center and bulk action . . . 219
19.1 The action represented by the center . . . 219
19.2 Hochschild cochains and the first operation homotopy . . . 220
19.3 The polynomial example . . . 221
19.4 The bulk map and a change of sector . . . 222
19.5 Cyclic orientation and the circle action . . . 223
19.6 The circle operator for the polynomial algebra . . . 224
20 Cyclic closure and modular sewing . . . 227
20.1 The cyclic pairing . . . 227
20.2 Surfaces and the determinant twist . . . 228
20.3 The open theory and its circle states . . . 229
20.4 A finite open algebra with unbounded circle homology . . . 229
20.5 Stable curves and modular operations . . . 230
20.6 The mixed sewing equation . . . 231

IV Exact noncommutative models 233

21 Free and square-zero pairs . . . 235
21.1 The tensor algebra and its primitive summand . . . 235
21.2 The square-zero algebra . . . 236

21.3 Dual multiplication and completion . . . 237
21.4 Realization in the chiral coefficient category . . . 238
22 The quantum plane . . . 241
22.1 Normal forms and multiplication . . . 241
22.2 The linear resolution . . . 241
22.3 The finite bar coalgebra . . . 242
22.4 The Yoneda product . . . 243
22.5 Geometric realization . . . 244
23 Quiver path algebras . . . 245
23.1 Vertices, paths and left modules . . . 245
23.2 The relative free bar . . . 246
23.3 Relations and the quadratic complex . . . 247
23.4 Geometric realization . . . 248
24 Weyl and Clifford deformations . . . 249
24.1 Polynomial families and ordered bases . . . 249
24.2 Specialization, Rees construction and completion . . . 250
24.3 The curved quadratic dual . . . 251
24.4 A cobar resolution with a scalar term . . . 252
25 Associative chiral algebras on the formal disc . . . 255
25.1 Regular fields and differential algebras . . . 255
25.2 A braiding defined on every state . . . 256
25.3 From generating fields to an exchange operator . . . 257
25.4 The geometric realization problem . . . 259
26 Zamolodchikov–Faddeev exchange . . . 261
26.1 Ordered colors and normal forms . . . 261
26.2 Spectral matrices and the rational solution . . . 262
26.3 Scalar normalization and the product . . . 263
26.4 The obstruction and the realization problem . . . 264
27 The RTT Yangian of $\mathfrak{gl}_2$ . . . 265
27.1 Coefficients and evaluation . . . 265
27.2 The coefficient Hopf algebra . . . 266
27.3 Two filtrations . . . 267
27.4 The determinant in rank two . . . 267
27.5 Exchange on evaluation modules . . . 268
28 The Hall algebra of the A_2 quiver . . . 271
28.1 Representations and multiplication . . . 271
28.2 The root vector and the Serre relation . . . 272
28.3 A basis and the positive quantum group . . . 272
29 Matrix traces and the necklace bracket . . . 275
29.1 Cyclic words and matrix Hamiltonians . . . 275
29.2 Hamiltonian derivations and the moment element . . . 276
29.3 Power sums and the matrix rank . . . 277

29.4 The relative two-particle surface . . . 278

V The nonlocal ladder 279

30 Spectral exchange over configuration spaces . . . 281
30.1 The coefficient space and its algebra . . . 281
30.2 Changing the order of tensor factors . . . 282
30.3 Multiplication between disjoint groups of generators . . . 283
30.4 Diagonal exchange and its three-particle coefficient . . . 284
30.5 A coefficient connection is additional structure . . . 284
31 RTT and reflection algebras . . . 287
31.1 Matrix composition and the two coproduct conventions . . . 287
31.2 The reflection equation and boundary braids . . . 288
31.3 Covariance under an RTT Hopf algebra . . . 289
31.4 Ordered bases for reflection entries . . . 290
32 Quantum vertex realization . . . 293
32.1 Two expansion regions . . . 293
32.2 The Yangian field . . . 293
33 Shuffle realization and the positive current algebra . . . 295
33.1 The product and its normalization . . . 295
33.2 The cubic kernel and two currents . . . 296
33.3 Polynomial coefficients and the generated algebra . . . 297
33.4 The generic positive affine Yangian . . . 298
33.5 The tensor required by cooperation . . . 298
33.6 Rational cooperation with exchange . . . 299
33.7 Geometric realization and the double . . . 300
34 The ladder in one diagram . . . 301

VI Field theory and master equations 303

35 A finite cotangent BF model . . . 305
35.1 Coordinates and the cotangent action . . . 305
35.2 The divergence class . . . 306
35.3 Finite dg Lie algebras . . . 308
35.4 A correction that requires completion . . . 309
35.5 Local fields and interval gluing . . . 309
36 BRST and ordered reduction . . . 311
36.1 The signed bar complex . . . 311
36.2 Length and cohomology . . . 312
36.3 Comparison before and after reduction . . . 313
36.4 The vertex-algebra application . . . 314
37 Stokes and algebraic identities . . . 315
37.1 Cellular coefficients and the full square . . . 315
37.2 Choosing higher homotopies . . . 316
37.3 Integration constructs an operation homotopy . . . 317

37.4 The associahedral equations . . . 318
37.5 A nonassociative product with a coherent extension . . . 319
38 Two edge colors and two loop operators . . . 321
38.1 Weights compatible with contraction . . . 321
38.2 A domain with two invariant genera . . . 322
38.3 The dual graph construction . . . 323
38.4 Generator amplitudes and the master equation . . . 324
38.5 The crossed derivation identity . . . 326
38.6 An explicit polynomial model . . . 326
39 Typed anomalies . . . 329
39.1 The differential at the classical solution . . . 329
39.2 Central extensions and nilpotency . . . 330
39.3 A master anomaly invisible to operator nilpotency . . . 331
39.4 Curvature as algebraic structure . . . 333
39.5 The Atiyah extension, curvature and holonomy . . . 334
39.6 Comparison maps . . . 335

VII Calabi–Yau and automorphic interfaces 337
40 Calabi–Yau categories . . . 339
40.1 The two Calabi–Yau variances . . . 339
40.2 The moduli functor and its tangent complex . . . 340
40.3 From a cyclic class to a closed form . . . 341
40.4 A smooth category with nonproper total space . . . 343
40.5 Cyclicity and closedness . . . 344
40.6 The ordered chiral comparison . . . 344
41 Hall and BPS algebras . . . 347
41.1 The extension correspondence . . . 347
41.2 The BPS algebra and its Lie generators . . . 348
41.3 Point sheaves on a K3 surface . . . 350
41.4 Quiver and surface Yangian comparisons . . . 351
41.5 Cooperation and geometric reconstruction . . . 353
42 The Igusa denominator as a bar character . . . 355
43 Finite-height currents and an exact physical model . . . 357
44 Character, bracket, parity, and orientation . . . 359
45 Exact dependency theorem . . . 361
46 Synthesis . . . 363

VIII Technical appendices to the preceding book 365
47 Sign calculation for the length-three bar . . . 367
48 Rational R-matrix verification . . . 369
49 Carriers and comparison maps . . . 371
50 Enveloping bars and planar path sequences . . . 373

50.1 Chevalley comparison . . . 373
50.2 The Virasoro row . . . 375
50.3 Planar path sequences . . . 376
51 The obstruction constitution . . . 379

Mixed Holomorphic–Topological Deligne Theory **381**
Book entrance: Mixed Holomorphic–Topological Deligne Theory . . . 381

IX Finite Hamiltonian geometry **383**
52 The formal symplectic plane . . . 385
52.1 Hamiltonians and the Poisson bracket . . . 385
52.2 Unimodularity . . . 386
53 The shifted cotangent theory . . . 387
53.1 The Lie algebra and its functions . . . 387
53.2 The BV action . . . 387
54 The universal boundary brane . . . 389
54.1 Polarization . . . 389

X Relative Hochschild comparison **391**
55 Hochschild cochains of the universal brane . . . 393
55.1 The coefficient ring and the Chevalley differential . . . 393
55.2 The formal coefficient limit . . . 394
56 The Kontsevich–Duflo map . . . 397
56.1 Linear Poisson formality . . . 397
56.2 The boundary action . . . 397
57 The formal Hamiltonian limit . . . 399
57.1 Weight completion and comparison data . . . 399

XI The finite-jet laboratory **401**
58 The first three Hamiltonian jets . . . 403
58.1 Quadratic Hamiltonians . . . 403
58.2 Cubic Hamiltonians . . . 403
58.3 The first nonlinear layer . . . 403
59 Moyal quantization in coordinates . . . 405
59.1 The bidifferential formula . . . 405
59.2 Weyl generators . . . 406
60 Chevalley coordinates and the Poisson differential . . . 407
60.1 The $\mathfrak{sl}_2$ invariant . . . 407
61 Residue duality and the formal current algebra . . . 409
61.1 The continuous dual . . . 409
62 All-arity open–closed operations . . . 411
62.1 Hochschild braces . . . 411
62.2 The first mixed identities . . . 411

63 Finite Chan–Paton ranks . . . 413
63.1 Rank one . . . 413
63.2 Rank two and Cayley–Hamilton reduction . . . 413
63.3 Stable passage . . . 413

XII Global holomorphic symplectic descent 415
64 Fedosov descent . . . 417
64.1 The Weyl algebroid . . . 417
65 Factorization and descent . . . 419
65.1 The algebraic mixed-HT realization . . . 419
65.2 BV realization . . . 419

XIII Matrix branes and trace geometry 421
66 The derived commuting stack . . . 423
67 Necklace Hamiltonians . . . 425
68 Stable traces . . . 427
69 The two-brane cone . . . 429

XIV Wheel corrections and exact quantum structure 431
70 The Duflo wheel . . . 433
71 The local centre with comparison data . . . 435
72 Cyclic classes, stable traces, and Lie homology . . . 437
73 Synthesis . . . 439

XV Technical appendices to the preceding book 441
74 Signs in the shifted cotangent complex . . . 443
75 A direct Jacobi check for Hamiltonian jets . . . 445
76 Bibliographic note . . . 447

The Igusa Square Root 449
Book entrance: The Igusa Square Root . . . 449

XVI The automorphic determinant 451
77 The protected Jacobi input . . . 453
77.1 The K3 elliptic genus . . . 453
77.2 The virtual half-index . . . 453
78 The Borcherds product . . . 455
78.1 The effective chamber . . . 455
78.2 The theta constant normalization . . . 455
79 Divisor, character, and the meaning of the square root . . . 457
79.1 Automorphic section, orientation line, and Pfaffian operator . . . 457

XVII The denominator algebra 459
80 The Lorentzian root lattice . . . 461
80.1 Real simple roots and Weyl vector . . . 461
81 The denominator algebra . . . 463
82 The ordered bar as the denominator complex . . . 465
83 Finite-height currents . . . 467
83.1 Classical local theory and exact quantum zero modes . . . 467

XVIII The compact scalar character 469
84 Reduced curve counting on $K3 \times E$ . . . 471
85 Extensions over the elliptic classifying stack . . . 473
86 Additive physical charge and quadratic automorphic degree . . . 475
86.1 The polarized central extension . . . 475
87 Point-local and wrapped sectors . . . 477

XIX The Borcherds algebra and its quantum group 479
88 The local–wrapped root algebra . . . 481
89 The formal Igusa quantum group . . . 483
90 A holomorphic–topological BF realization . . . 485
91 The determinant and Fredholm square-root operators . . . 487
91.1 Trace-class realization on the root Hilbert superspace . . . 487
92 What the automorphic character determines . . . 489

XX The universal Borcherds Hall and three-Calabi–Yau models 491
93 Finite simple-root windows . . . 493
94 The real-root seed window . . . 495
94.1 The first Ginzburg differential . . . 495
95 Hall realization of the positive algebra . . . 497
95.1 Parity and the quantized superdatum . . . 497
96 The three-Calabi–Yau completion . . . 499
97 Hall states and protected superstates . . . 501
98 Extension determinants and critical orientations . . . 503
99 The colored elliptic Ran space . . . 505
100 Physical charge and Gram degree . . . 507
101 The universal protected field theory . . . 509
102 Absolute convergence of the virtual root operator . . . 511
103 The virtual root Fredholm operator . . . 513
103.1 Automorphic orientation . . . 513
104 The universal Igusa protected effective theory . . . 515
105 The primitive comparison and its extensions . . . 517
106 The eight diagonal-divisor forms . . . 519

107 Algebraic and physical closure 521

XXI Technical appendices to the preceding book 523
108 Exact theta-series calculation 525
109 The root reflection matrices 527
110 Automorphic, homological, and Hall comparisons 529

Calabi–Yau Quantum Groups 531
Book entrance: Calabi–Yau Quantum Groups 531

XXII Categorical and field-theoretic foundations 533
111 Calabi–Yau categories 535
111.1 Smoothness, properness, and the Calabi–Yau class 535
112 Local BV theories and factorization observables 537
112.1 Cyclic elliptic local Lie algebras 537
112.2 Boundary and defect restriction 537
113 Three curve-level algebraic outputs 539
114 Holomorphic Chern–Simons theory 541
115 Pushforward to a curve 543
116 The derived centre and the physical bulk 545

XXIII Hall and BPS algebras 547
117 Extension correspondences 549
118 Orientation data and the threefold Hall product 551
119 The geometric E-equivariant Hall source 553
120 Local and wrapped geometry inside the Hall algebra 555
121 Cohomological refinement and the Gram character 557
122 BPS primitives 559
123 The integral Hall–Borcherds amalgam 561
124 Coproducts, pairings, and doubles 563
125 Ordered chiral bialgebras 565

XXIV Exact model cases 569
126 The A_2 Hall algebra 571
127 The tripled Jordan quiver and $\mathbb{C}^3$ 573
128 Surface CoHAs and ADE Yangians 575
129 Coulomb branches and shifted Yangians 577

XXV K3 symmetries 579
130 The Mukai lattice 581
131 The Nakajima Heisenberg algebra 583
132 The orthogonal LLV algebra 585

133 Local ADE Yangians 587
134 The K3 symmetry trichotomy 589
135 K3 correspondences on fixed coefficients 591
136 The algebra generated by geometric actions 593
137 Tensor reconstruction and geometric kernels 595

XXVI The exact $K3 \times E$ protected completion 597
138 The protected input 599
139 Cyclic open-string operations 601
140 An algebraic open–closed comparison 603
141 The exact scalar and first-order observables 605
142 Physical meaning 607
143 Primitive comparisons and their algebraic extensions 609
144 Finite quivers and Calabi–Yau completions 611

XXVII Quantum groups, exchange, and representation theory 613
145 From positive half to full quantum group 615
146 Tannaka reconstruction 617
147 KZ and rational exchange 619
148 The centre and the double 621

XXVIII A compact six-dimensional protected source 623
149 Holomorphic cotangent BF theory on $K3 \times E$ 625
150 Tree exactness and the quantum master equation 627
151 Classical reduction along K3 629
152 Protected states and the Igusa trace 631
153 The compact-source comparison 633

XXIX Physics 635
154 Holomorphic–topological boundary theories 637
155 Six-dimensional hCS on $K3 \times E$ 639
156 Omega backgrounds 641
157 Anomalies as typed classes 643

XXX Automorphic forms 645
158 Borcherds products as characters 647
159 The universal Borcherds weight formula 649
160 The Gritsenko–Cléry atlas 651
161 Scalarization hierarchy 653

XXXI The unified theorem system 655
162 The foundational theorem 657

163 Landscape of established realizations . . . 659
164 Constructed comparisons and their geometric extensions . . . 661
165 Actions and comparison maps . . . 663
166 Comparison data for Calabi–Yau constructions . . . 665

XXXII Technical appendices to the preceding book 667
167 Typed data table . . . 669
168 The three K3 brackets . . . 671
169 Exact arithmetic checks . . . 673

Universal Chiral BV and Einstein Completion 675
Coherent boundary actions and BV fields . . . 675

XXXIII The full mixed boundary datum 677
170 Three products and one interface . . . 679
171 Compatible modules and the mixed centre . . . 681
172 Relative induction from a chiral algebra . . . 683

XXXIV The deformation algebra and its BV envelope 685
173 The centre-extended deformation algebra . . . 687
173.1 Field equations and Bianchi identities . . . 687
174 Shifted cotangent transgression . . . 689
175 Intrinsic algebraic quantization . . . 691
176 Perturbative factorization realization . . . 693

XXXV Boundary duality, exchange, and reduction 695
177 The Koszul–Morita square . . . 697
178 Planar exchange and Tannaka reconstruction . . . 699
179 BRST reduction . . . 701

XXXVI Conformal covariance and Einstein completion 703
180 Projective connections and the Virasoro tangent complex . . . 705
181 Three-dimensional gravity as a flat Cartan theory . . . 707
182 Four equivalent linear complexes . . . 709
183 Brown–Henneaux and the projective boundary . . . 711
184 Einstein completion of a conformal pair . . . 713
185 Higher-spin completion . . . 715

XXXVII Exact gravitational complexes and quantum tests 717
186 The four complexes in coordinates . . . 719
186.1 Flat adjoint complex . . . 719
186.2 Einstein–Cartan complex . . . 719

186.3 Metric complex . . . 719
186.4 Calabi complex . . . 720
187 Boundary BFV theory and the Virasoro moment map . . . 721
188 Matter stress tensor and the Einstein moment map . . . 723
189 One-loop gravity and analytic torsion . . . 725
190 BTZ, Cardy, and the area law . . . 727
191 Higher-spin holonomy and boundary charges . . . 729

XXXVIII Anomalies, Hall sources, and automorphic completion 731
192 Four native anomaly classes . . . 733
193 Hall and Calabi–Yau sources . . . 735
193.1 Mixed realization of the amalgam . . . 735
194 Igusa root states and the mixed-boundary comparison . . . 737
195 The carrier lattice . . . 739
196 Sixteen structural laws . . . 741
197 Coherent actions and gravitational reconstruction . . . 743

XXXIX Technical appendices to the preceding book 745
198 The Chern–Simons normalization . . . 747
199 The principal embedding count . . . 749
200 Low-arity mixed signs . . . 751

XL Detailed treatment of Volume I 753
201 Five carriers, three noncommutativities, one type discipline . . . 755
201.1 The local experiment . . . 755
201.2 The Ran coefficient and its two products . . . 756
201.3 Three forms of noncommutativity . . . 757
201.4 Boundary and defect origin . . . 758
201.5 The complete diagonal coefficient . . . 758
202 Aligned decompositions and the exact locality law . . . 761
202.1 Linear species . . . 761
202.2 Sheaf-level locality . . . 762
202.3 The two coproducts on the transverse bar . . . 762
202.4 Aligned projection and mixed locality . . . 763
203 Rectangular recognition and its exact hypotheses . . . 765
203.1 Product-disk factorisation . . . 765
203.2 Block-Ran recognition . . . 765
203.3 Opposites and three reversals . . . 766
204 Associative chiral algebras and nonlocal vertex fields . . . 767
204.1 Why the second E_1 is necessary . . . 767
204.2 Formal-disc realization . . . 767
204.3 The genuinely doubly noncommutative object . . . 769

204.4 Elementary free construction . . . 769
205 Homotopy-coherent generators and relations . . . 771
205.1 Why ordinary quotients are insufficient . . . 771
205.2 Two-coloured levelled forests . . . 771
205.3 Rectification and the ∞-category . . . 772
205.4 Low arity . . . 772
205.5 Homotopy transfer . . . 772
206 The transverse interval bar . . . 775
206.1 The simplicial object . . . 775
206.2 Normalized words and signs . . . 775
206.3 Interval factorisation homology . . . 776
206.4 Associahedra and the monadic bar . . . 777
206.5 Completion . . . 777
207 The associative chiral bar and the theorem of two bars . . . 779
207.1 Operadic chiral bar . . . 779
207.2 Chiral Chevalley chains are a third construction . . . 779
207.3 Universal enveloping comparison . . . 780
208 Bar–cobar reconstruction and completion . . . 783
208.1 Ordinary tensor cobar . . . 783
208.2 A finite-word contraction . . . 783
208.3 Positive-weight completion . . . 784
208.4 Deformation complexes . . . 784
209 Koszul–Morita theory, relative bases, and the centre . . . 787
209.1 The augmentation boundary . . . 787
209.2 One-sided and two-sided twisting . . . 787
209.3 Relative augmentation bases . . . 788
209.4 The internal derived center . . . 788
210 Quadratic, PBW, curved, and Verdier comparisons . . . 791
210.1 Quadratic recognition . . . 791
210.2 PBW deformations and curvature . . . 791
210.3 Verdier duality . . . 792
211 Free and square-zero ordered chiral algebras . . . 793
211.1 Algebraic coefficients and factorization . . . 793
211.2 The free algebra . . . 793
211.3 The square-zero algebra . . . 794
211.4 A geometric realization in both directions . . . 794
212 The quantum and Jordan planes . . . 795
212.1 The quantum plane . . . 795
212.2 The Jordan plane . . . 796
212.3 Products beyond the Betti numbers . . . 796
212.4 Finite bar dimensions and ranks . . . 797
212.5 Internal chiral realization . . . 798

213 Augmentation obstructions, Weyl curvature, and matrix sectors 799
213.1 The Weyl obstruction . . . 799
213.2 The matrix obstruction . . . 800
213.3 An augmented algebra with a matrix sector . . . 800
214 Relative quiver bars and boundary conditions . . . 803
214.1 The natural base . . . 803
214.2 A relation and its coproduct . . . 803
214.3 Simple modules and the Yoneda product . . . 804
214.4 Scalar boundary objects and geometric families . . . 805
215 Enveloping algebras and planar-sequence obstructions . . . 807
215.1 Cartan–Eilenberg comparison . . . 807
215.2 Three chain complexes . . . 808
215.3 The planar sequences . . . 808
215.4 The Virasoro row computed . . . 809
216 Finite-label exchange and its bar coalgebra . . . 813
216.1 The exchange algebra and its ordered basis . . . 813
216.2 The scalar algebra and its linear resolution . . . 813
216.3 The actual bar coalgebra . . . 814
216.4 The Yoneda product and its variance . . . 815
217 RTT Yangians and the bar filtration . . . 817
217.1 The augmented coefficient algebra . . . 817
217.2 The current bar differential . . . 817
217.3 The algebraic spectral sequence . . . 818
217.4 Completion and the inverse limit . . . 819
217.5 The abelian calculation and its geometric scope . . . 820
218 The A_2 Hall algebra in complete detail . . . 821
218.1 Untwisted products and the two flag tables . . . 821
218.2 The twist and divided powers . . . 822
218.3 Flags as correspondences . . . 822
218.4 Coefficients over a space of supports . . . 823
219 Necklace Hamiltonians and the mixed trace algebra . . . 825
219.1 The derived commuting equation . . . 825
219.2 An infinitesimal BRST complex . . . 825
219.3 The necklace action on the complexes . . . 826
219.4 The ordinary commuting quotient . . . 826
219.5 The stable commutative algebra and its rank-dependent bracket . . . 827
219.6 Two particles and the nilpotent cone . . . 828
219.7 The formal closed comparison . . . 828
220 Modules, orbifolds, and higher E_n directions . . . 831
220.1 Pointed bars and interfaces . . . 831
220.2 Crossed products . . . 831
220.3 Symmetric products and DMVV . . . 831
220.4 Higher transverse dimensions . . . 831

221 Perturbative BV realization of the ordered chiral object 833
221.1 Classical data 833
221.2 Renormalised quantum observables 833
221.3 Free holomorphic–topological BF theory 834
222 BRST descent and Drinfeld–Sokolov reduction 835
222.1 The comparison and its hypotheses 835
222.2 A comparison map from tilings 835
222.3 A nonzero higher differential 836
222.4 The universal affine reduction 837
222.5 The $\mathfrak{sl}_2$ stress tensor 837
222.6 Associative coefficients and the vacuum 839
223 From order to exchange: KZ, Tannaka, and four-dimensional Chern–Simons 841
223.1 Infinitesimal braid relations 841
223.2 Meromorphic Tannaka reconstruction 841
223.3 Four-dimensional Chern–Simons theory 842
224 Cyclic traces, modular sewing, and the two-edge master equation 843
224.1 Closing the transverse interval 843
224.2 Chiral sewing over stable curves 843
224.3 Two kinds of contraction 844
225 Prime forms, plumbing, and the analytic chiral genus 847
225.1 The normalized bidifferential 847
225.2 Plumbing a node 847
225.3 Genus one 848
225.4 Higher genus and Siegel theta series 848
226 A concrete two-genus model 849
226.1 The transverse Frobenius algebra 849
226.2 A rank-one Deligne–Mumford theory 849
226.3 Strict mixed interchange 849
226.4 What the model proves and what it does not 850
227 Typed anomalies and invertible theories 851
227.1 Classes, curved data and descent 851
227.2 Cancellation requires a relative theory or a trivialization 851
227.3 A projective representation and its actual compensator 852
227.4 Scalar evaluations and central charge 852
228 Bar Euler products and Borcherds denominators 855
228.1 Root-graded Lie superalgebras 855
228.2 The Igusa product 855
228.3 Finite positive windows 856
228.4 Polarized Gram transform 856
229 Higher-dimensional sources and factorisation pushforward . . . 857
229.1 Pushforward along an actual map 857

229.2 Calabi–Yau categories and Hall products . . . 857
229.3 Coulomb branches and shifted Yangians . . . 857
229.4 Source-to-chiral comparison . . . 858
230 Obstructions to comparison . . . 859
230.1 Carrier prohibitions . . . 859
230.2 Monoidal and operadic prohibitions . . . 859
230.3 Algebra and calculation prohibitions . . . 860
230.4 Physics and sewing prohibitions . . . 860
230.5 Positive replacements . . . 860
231 Obstructions and finite calculations . . . 863
231.1 Carrier-to-construction atlas . . . 863
231.2 Finite calculations and their boundaries . . . 863
231.3 Data in the worked examples . . . 864
232 Structure theorem for the complete datum . . . 865
232.1 The primitive datum . . . 865
232.2 Two associative structures . . . 866
232.3 Mixed locality and derived presentations . . . 866
232.4 Geometric sources of the operations . . . 866
233 Suspensions, orientations, and low-arity signs . . . 867
233.1 Cohomological suspension . . . 867
233.2 Four-letter bar word . . . 867
233.3 Two edge colours . . . 867
233.4 Arnold relation and KZ curvature . . . 868
233.5 Rational Yang–Baxter expansion . . . 868
234 Exact finite computations . . . 869
234.1 Finite chain-matrix construction . . . 869
234.2 Quantum-plane and Jordan-plane windows . . . 869
234.3 The $\mathfrak{sl}_2$ complex . . . 870
234.4 The A_2 Hall relation at $q = 2$ . . . 870
234.5 Finite stable-trace Jacobi calculation . . . 870
235 Generators-and-relations templates . . . 871
235.1 Quadratic transverse presentation . . . 871
235.2 Associative chiral presentation . . . 871
235.3 Doubly associative presentation . . . 871
235.4 Relative presentation . . . 872
236 Logical status of comparison statements . . . 873
237 Extended comparison table . . . 875

Book I

Ordered collisions and reconstruction

Chapter 1

Ordered words and collision coefficients

Three ordered insertions can have the same first and third position while the second remains separate. A construction that remembers only adjacent collisions loses this configuration. A construction that remembers only the three singleton state spaces loses the normal derivatives along its collision diagonal. Both kinds of information enter reconstruction.

Let k be a field of characteristic zero and let X be a smooth separated curve over k, and write $[n] = \{1, \dots, n\}$. The positions of a word of length n form X^n, including its diagonals. The order belongs to the labels in $[n]$. It does not order the points of X. Cutting the labels between p and $p+1$ is permitted where every point of the first block is distinct from every point of the second. Collisions inside either block remain permitted.

1.1 The coefficient category

We use cochain complexes: the differential has degree 1, and $M[r]^i = M^{i+r}$. Thus an element sa of $M[1]$ has degree $|a| - 1$. Tensor products of maps and differentials use the Koszul sign rule. All tensor products of complexes are derived unless a flat model has been specified.

Fix dg-enhanced stable coefficient categories $\mathscr{D}(X^n)$, with external tensor products, exceptional inverse images $f^!$, and direct images j_* for open embeddings. These functors are exact, and external tensor products preserve colimits separately. We require the following six-functor identities on the diagrams used below: open base change, composition of open direct images, compatibility of external products with those operations, and localization for a closed complement. For the diagonals that identify consecutive labels we also require the base-change identifications in (1.2). These identities are part of the coefficient setting, with their coherent enhancements. They are the hypotheses on a differential-module or sheaf realization of the constructions.

A monotone surjection $\alpha : [n] \twoheadrightarrow [r]$ has interval fibres. Its diagonal map is

$$d_\alpha : X^r \longrightarrow X^n, \qquad (y_1, \dots, y_r) \longmapsto (y_{\alpha(1)}, \dots, y_{\alpha(n)}).$$

An *ordered word coefficient* is a family $F_n \in \mathscr{D}(X^n)$, $n \geq 1$, with coherent equivalences $d_\alpha^! F_n \simeq F_r$. We adjoin a separate coefficient $F_0 \in D(k)$ for the empty word, where $D(k)$ is the derived category of k-complexes. There is no collision map from a nonempty word to that empty word. Write $\mathscr{W}_X$ for this stable category of diagrams.

Stability concerns the ambient diagram category. A factorization condition selects objects whose specified structure maps are equivalences, and need not define a stable subcategory. For example, suppose a coefficient has rank one on every separated configuration and is strongly factorizing. Its direct sum with itself has rank two on a separated pair. The external square of its singleton coefficient has rank four. The componentwise direct sum therefore cannot retain the same strong factorization condition.

An arbitrary partition of $[n]$ still defines a diagonal in X^n. The restriction of F_n to it, its supported sections, and its normal derivatives remain meaningful. The definition of $\mathscr{W}_X$ does not identify such

a restriction with a shorter word unless the partition is consecutive. Comparisons on the full collision diagram must supply their own maps and coherences.

1.2 Consecutive cuts

For $0 \le p \le n$, set

$$U_p^{(n)} = \{(x_1, \dots, x_n) : x_i \neq x_j \text{ whenever } i \le p < j\}, \qquad j_p : U_p^{(n)} \hookrightarrow X^n.$$

The two end cuts have $U_0^{(n)} = U_n^{(n)} = X^n$. Define

$$(F \star_c G)_n = \bigoplus_{p=0}^{n} j_{p,*} j_p^{!}(F_p \boxtimes G_{n-p}). \tag{1.1}$$

The coefficient K, equal to k at the empty word and zero at every nonempty word, will be its unit.

Here is the diagonal compatibility required of the coefficient setting. If p lies strictly inside a fibre of α, the image of d_α misses $U_p^{(n)}$. Exceptional pullback of the corresponding localized term is zero. Otherwise $p = |\alpha^{-1}([q])|$ for a unique cut q of $[r]$, and the surviving base-change map gives

$$d_\alpha^{!} j_{p,*} j_p^{!}(F_p \boxtimes G_{n-p}) \simeq j_{q,*} j_q^{!}(F_q \boxtimes G_{r-q}). \tag{1.2}$$

The indices on the right use the word-coefficient identifications. We require these maps to commute with composition of monotone surjections. Summing them makes (1.1) an object of $\mathscr{W}_X$.

Proposition 1.1 (The cut product)**.** *Under the stated coefficient identities, $(\mathscr{W}_X, \star_c, K)$ is a monoidal stable category. The product is exact separately in each variable. No interchange of the two ordered blocks is part of this monoidal structure.*

Proof. An iterated product of r factors is indexed by a consecutive decomposition of the word into r blocks, with empty blocks allowed. For any parenthesization, its coefficient is the external product of the block coefficients, localized to the open where distinct blocks have disjoint positions. Open base change identifies these coefficients. For four or more factors, every composite associator is the same identification with the coefficient of the common block decomposition. This proves the pentagon and, in the enhanced categories, the higher coherence identities. An empty factor contributes only the corresponding end cut, giving both unit maps. Exactness follows from the assumed exact operations and the finite sum in each arity. □

Call F *positive* if $F_0 = 0$. Positivity bounds the number of factors that can contribute at a fixed word length.

Lemma 1.2 (Arity bound)**.** *If $F^1, \dots, F^r$ are positive, then*

$$(F^1 \star_c \cdots \star_c F^r)_n = 0 \qquad \text{for } r > n.$$

Proof. A nonzero summand requires a nonempty block for every factor. A word of length n has at most n such blocks. □

This bound does not assert a bound on the pole order or on normal derivatives. Each surviving coefficient may contain all of them. It bounds the number of positive factors, which is the number needed in the bar–cobar argument.

1.3 Finite normalized bar and cobar

An augmented associative algebra in $\mathscr{W}_X$ has unit and augmentation $K \to A \to K$. Assume that $A_0 \to k$ is an equivalence. Its augmentation ideal $\overline{A}$, the fibre of $A \to K$, is then positive. For a coaugmented counital coassociative coalgebra C with $C_0 = k$, write $\overline{C} = \operatorname{cofib}(K \to C)$. The counit splits the coaugmentation, so this is the reduced coefficient.

The normalized constructions have underlying coefficients

$$B_c A = K \oplus \bigoplus_{r \geq 1} (\overline{A}[1])^{\star_c r}, \tag{1.3}$$

$$\Omega_c C = K \oplus \bigoplus_{r \geq 1} (\overline{C}[-1])^{\star_c r}. \tag{1.4}$$

Both sums are finite in each arity. The first has deconcatenation coproduct. The second has concatenation product. A flat differential graded model makes their signs explicit. Writing s for shift by $[1]$, the bar differential is the coderivation with terms

$$b_1(sa) = -s(da), \qquad b_2(sa \otimes sb) = (-1)^{|a|} s(ab). \tag{1.5}$$

Its extension past earlier shifted factors uses their total degree. If t denotes shift by $[-1]$ and $\overline{\Delta} c = \sum c' \otimes c''$, the cobar differential is the derivation determined by

$$d(tc) = -t(dc) + \sum (-1)^{|c'|+1} (tc')(tc''). \tag{1.6}$$

The derivation and coderivation identities, together with associativity and coassociativity, give square-zero differentials. For instance, the two contractions of three suspended factors in the bar have signs $(-1)^{|b|}$ and $(-1)^{|b|-1}$.

Theorem 1.3 (Bar–cobar on positive word coefficients)**.** *In the enhanced coefficient setting above, normalized bar and cobar give inverse equivalences between augmented associative algebras with empty coefficient k and coaugmented counital coassociative coalgebras with empty coefficient k. The unit and counit are equivalences in every arity. No extra completion in bar length is required for this statement.*

Proof. Consider $\Omega_c B_c A$ in arity n. Flatten each tensor of bar words into its original positive factors from $\overline{A}$. There are at most n factors. Filter by their number. The associated graded differential retains the internal differential and the part that splits a bar word into two cobar factors. It omits the multiplication in A, which reduces the number of original factors.

For r original factors, each of the $r - 1$ gaps either lies inside a bar word or separates two cobar factors. Changing the first state to the second is the identity of the same localized coefficient, with the suspension sign. Hence the gap complex is the tensor product of $r - 1$ two-term identity complexes. It is contractible for $r > 1$: contract one chosen gap and leave the others unchanged, using the tensor sign rule. For $r = 1$ it is $\overline{A}$. The counit sends $t(sa)$ to a, sends longer single bar words to zero, and extends multiplicatively. Formulae (1.5)–(1.6) make it a chain map. On the associated graded it is the identity on the only noncontractible term. The filtration is finite in arity n, so the counit is an equivalence there.

For $B_c \Omega_c C$, use the number of original reduced coalgebra factors with the descending filtration. The reduced coproduct increases this number, and concatenation preserves it. The associated graded again consists of the same gap complexes, with the direction of the gap differential reversed. The unit on a reduced coefficient has the formula

$$\eta(c) = \sum_{r \geq 1} \sum [tc_{(1)}| \cdots |tc_{(r)}], \qquad \overline{\Delta}^{(r)} c = \sum c_{(1)} \otimes \cdots \otimes c_{(r)}.$$

The brackets suspend each cobar generator once, so the composite map $c \mapsto s(tc)$ has degree zero. The chain-map identity follows from coassociativity and (1.6). Reduced iterates with more than n positive factors vanish by Lemma 1.2. The one-factor term of the unit is the identity on $\overline{C}$. The same finite filtration argument proves that the unit is an equivalence. These are the natural unit and counit of the bar–cobar adjunction, so their triangle identities give the asserted inverse equivalences. □

An ordinary augmented algebra concentrated at a single position does not satisfy this arity argument merely because it is augmented. Its powers can survive in every tensor length. Positive weight, conilpotence or an appropriate completeness hypothesis must then replace Lemma 1.2.

1.4 The support left by all cuts

A coalgebra C is *strong* if, for every cut, its specified coproduct restricts to an equivalence

$$C_n|_{U_p^{(n)}} \longrightarrow (C_p \boxtimes C_{n-p})|_{U_p^{(n)}}. \tag{1.7}$$

These maps retain the coalgebra coherences. A collection of unrelated equivalences is insufficient. Define the closed subset

$$Z_n = X^n \setminus \bigcup_{p=1}^{n-1} U_p^{(n)} = \bigcap_{p=1}^{n-1} \bigcup_{i \le p < j} \Delta_{ij}, \qquad Z_1 = X. \tag{1.8}$$

Thus a configuration lies in Z_n precisely when every consecutive cut is crossed by a coincident pair.

For a closed subset $Z \subseteq Y$ with open complement $j : U \hookrightarrow Y$, the supported coefficient $R\Gamma_Z M$ is the fibre of $M \to j_* j^! M$.

Theorem 1.4 (Cut-supported reconstruction)**.** *Let C be a strong coalgebra with empty coefficient k. Then there is a natural equivalence of coefficients*

$$(\overline{\Omega_c C})_n \simeq R\Gamma_{Z_n}(C_n)[-1]. \tag{1.9}$$

Conversely, if the augmentation ideal of A is supported on Z_n in every arity, then $B_c A$ is strong. Consequently bar–cobar restricts to an equivalence between strong coalgebras and augmented algebras with these support conditions.

Proof. For a set of cuts $S \subseteq \{1, \dots, n-1\}$, put $U_S = \bigcap_{p \in S} U_p^{(n)}$, with $U_\varnothing = X^n$, and write $L_S M = j_{S,*} j_S^! M$. An r-factor term in $\overline{\Omega_c C}$ corresponds to $r-1$ cuts. Iterated strong coproducts identify its coefficient with $L_S C_n$. Coassociativity identifies the splitting differential with the maps that insert one more cut. The resulting complex is

$$\operatorname{Tot}\left(\bigoplus_{S \subseteq [n-1]} L_S C_n[-|S|-1] \right),$$

with the tensor suspension signs. Apart from the common shift $[-1]$, it is the total fibre of the localization cube for the opens $U_p^{(n)}$. Finite open descent identifies that fibre with $R\Gamma_{Z_n} C_n$, proving (1.9).

For the converse, restrict a normalized bar word to $U_p^{(n)}$. If one of its consecutive factors crosses the cut p, the positions in that factor lie in its own cut open. The support hypothesis makes that factor vanish. Every surviving word has a separator at p. Cutting there is a bijection between surviving bar words and pairs of bar words on the two blocks. The bar differential has no remaining term that multiplies across this separator, again by the support hypothesis. Thus deconcatenation is the equivalence (1.7), including its differential and iterated-cut coherences. The restricted equivalence follows from Theorem 1.3. □

The right side of (1.9) describes the coefficient. Its multiplication still uses the inverses of the strong coproducts and their coherence. Local cohomology alone does not specify that multiplication.

Example 1.5 (Three and four positions)**.** For three labels,

$$Z_3 = (\Delta_{12} \cup \Delta_{13}) \cap (\Delta_{13} \cup \Delta_{23}) = \Delta_{13}.$$

The locus with $x_1 = x_3 \neq x_2$ is therefore part of the reconstructed support. For four labels,

$$Z_4 = \Delta_{14} \ \cup \ (\Delta_{13} \cap \Delta_{24}).$$

To check this equality, suppose $x_1 \neq x_4$. The first cut requires $x_1 = x_2$ or $x_1 = x_3$. If $x_1 = x_2$, the last cut forces $x_3 = x_4$, leaving the middle cut uncrossed. Hence $x_1 = x_3$, and the last cut then forces $x_2 = x_4$. On the affine line, take the independent normal coordinates $a = x_1 - x_4$, $b = x_1 - x_3$, $c = x_2 - x_4$. The reduced ideal of this union is

$$(a) \cap (b, c) = (ab, ac).$$

Its codimension-one and codimension-two parts explain why a single total-diagonal coefficient cannot replace the full support.

1.5 The empty unit and the local vacuum

Suppose the coefficient setting has dualizing objects $J_n = p_n^! k$, where $p_n : X^n \to \operatorname{Spec} k$, with the coherent product identifications $J_p \boxtimes J_q \simeq J_{p+q}$. Set $J_0 = k$. Localization gives cut coproducts on J. They are equivalences on their cut opens. Thus J is a strong coalgebra. Its cobar algebra $\mathcal{U}_X = \Omega_c J$ carries the local vacuum coefficient.

A *vacuum-marked coalgebra* has a coalgebra map $J \to C$. Applying bar–cobar gives the equivalence of undercategories

$$(J \downarrow \mathrm{Coalg}^{\mathrm{strong}}) \simeq (\mathcal{U}_X \downarrow \mathrm{Alg}^{\mathrm{aug},Z}).$$

Here the algebra on the right is augmented over K and its augmentation ideal satisfies the support condition of Theorem 1.4. The two maps

$$\mathcal{U}_X \longrightarrow A, \qquad A \longrightarrow K$$

have different meanings. The first is a local vacuum marking; the second is an empty-word augmentation. Neither is a unital scalar augmentation of a vertex algebra.

For example, the Heisenberg field with generator a, vacuum $\mathbf{1}$, and nonzero level κ satisfies $a_{(1)}a = \kappa\mathbf{1}$. A unital map to the scalar vertex algebra would send the left side to zero and the right side to κ. Such a map cannot exist. The contradiction uses preservation of the vertex product $a_{(1)}a$, a different condition from either geometric map.

1.6 Finite summands and geometric extensions

Suppose a coefficient is explicitly a finite direct sum indexed by two decompositions of the same finite set. The summands for which those decompositions agree admit an inclusion and a projection. Both maps come from the given direct-sum presentation.

A collision stratum need not be a direct summand of a geometric coefficient. Already over $R = k[u]$, the sequence

$$0 \longrightarrow R \longrightarrow R[u^{-1}] \longrightarrow R[u^{-1}]/R \longrightarrow 0$$

does not split as R-modules. If a retraction ρ existed, then $1 = \rho(1) = u\rho(u^{-1})$, impossible in $k[u]$. Thus a finite indexing argument does not produce a retraction across a collision divisor. A geometric interchange requires its actual maps, including any extension class supported on that divisor. Failure of this particular projection does not rule out a different distributive law.

The cut product avoids choosing such a projection: its open localizations and their base-change maps are specified in (1.1)–(1.2). Extending reconstruction from this fixed-curve word category to arbitrary collision presentations still requires the comparisons on those presentations. The support calculation identifies what they must retain, including crossed collisions and the local vacuum.

Chapter 2

Multiplicities and the local vacuum

When collision adds generator degrees, two degree-one states combine to degree two. Replacing their product by a degree-one singleton changes the coefficient. The free commutative example already detects the change.

Let

$$F = k[x_0, x_1, \dots], \qquad T x_i = x_{i+1},$$

where T acts as a derivation and every x_i has generator degree one. The degree-one subspace is the free $k[T]$-module generated by x_0. In degree two, the map

$$x_i x_j \longmapsto \tfrac{1}{2}(t_1^i t_2^j + t_1^j t_2^i)$$

identifies the state space with $k[t_1, t_2]^{\mathfrak{S}_2}$. Translation acts there by multiplication by $t_1 + t_2$. Consequently

$$F^{(2)} \simeq k[T, P], \qquad T = t_1 + t_2, \quad P = t_1 t_2, \tag{2.1}$$

as $k[T]$-modules. This module has the infinite basis $1, P, P^2, \dots$. Its rank differs from that of $F^{(1)}$. The comparison is a comparison of state modules, not of algebras: the product of two degree-two states lies in degree four.

Multiplicity labels retain this degree change. They also distinguish a position that carries the vacuum, of weight zero, from the empty word, which has no position.

2.1 Additive labels on collision diagrams

Retain the coefficient hypotheses and cochain conventions of Chapter 1. For the realizations below, assume also that the exceptional pullbacks commute with the direct-sum simplicial totalizations being used. A multiplicity word is a pair $([n], \mathbf{m})$, where $\mathbf{m} = (m_1, \dots, m_n)$ and $m_i \in \mathbb{N}$, including zero. A monotone surjection $\alpha : [n] \twoheadrightarrow [r]$ gives the target label

$$(\alpha_* \mathbf{m})_j = \sum_{\alpha(i)=j} m_i.$$

A coefficient system on these labelled words has coherent equivalences

$$d_\alpha^! F_{n,\mathbf{m}} \simeq F_{r,\alpha_*\mathbf{m}}.$$

Its positions still lie in X^n. Only the coefficient labels have changed. The empty word carries its own coefficient at $n = 0$.

The cut product keeps the labels on their respective blocks:

$$(F \star_c G)_{n,\mathbf{m}} = \bigoplus_{p=0}^{n} j_{p,*} j_p^! \big(F_{p,(m_1,\dots,m_p)} \boxtimes G_{n-p,(m_{p+1},\dots,m_n)}\big).$$

The common-block proof of associativity still applies. Positivity still means vanishing at the empty word. Thus the number of positive factors in arity n remains at most n, even when some nonempty positions have weight zero.

To apply these labels to residue coefficients, fix a nonnegatively graded state algebra $A = \bigoplus_{d\geq 0} A^{(d)}$. Assume that translation preserves degree and that every residue operation adds state degrees. In particular, this is an assumption on the chosen generator grading, not on conformal weight in an arbitrary vertex algebra.

Suppose a full residue complex has coefficients

$$C_n(A) = \bigoplus_{\pi,\mathbf{d}} V_{\pi,\mathbf{d}}, \qquad \mathbf{d} : \pi \longrightarrow \mathbb{N}. \tag{2.2}$$

Here π is a partition of $[n]$, and its blocks are ordered by their largest labels. The coefficient $V_{\pi,\mathbf{d}}$ contains the states $A^{(d_B)}$ at its blocks, with shift $[|\pi|]$. It includes the complete differential-module coefficient along the partition diagonal, including normal derivatives and localization where distinct blocks are separate. It is not a tensor product of the underlying state spaces alone.

The residue differential merges two blocks and adds their degrees. Assume its square is zero and that its cut and exceptional-collision maps are specified coherently. In particular, under a monotone collision, the localized partition terms killed by pullback vanish. The remaining terms identify with the quotient partitions and their summed block degrees. These are hypotheses on the full residue construction in (2.2).

For a label $\mathbf{m}$, define

$$C_{n,\mathbf{m}}(A) = \bigoplus_{\pi} V_{\pi,\,(\sum_{i\in B} m_i)_{B\in\pi}}. \tag{2.3}$$

Proposition 2.1 (Multiplicity and collision)**.** *The coefficients* (2.3) *inherit the residue differential and form a coefficient system on additive multiplicity words. If the full residue coalgebra is strong, its labelled version is strong. These statements preserve the specified normal derivatives and collision maps.*

Proof. Merging blocks B and B' replaces their degrees by $\sum_{i\in B\cup B'} m_i$. Hence the differential stays in the labelled complex. For a surviving block D of a quotient partition, its degree is

$$\sum_{j\in D} (\alpha_*\mathbf{m})_j = \sum_{i\in\alpha^{-1}(D)} m_i.$$

This is exactly the degree in its source coefficient. The given exceptional-collision comparison therefore restricts to the labelled summands. Composition of collisions adds the same integers in either order. On a cut open, surviving partition blocks lie on one side or the other. Their degree assignments and residue differentials split between the two sides, so the strong cut maps restrict as well. All operations on the differential-module coefficients remain the original ones. □

The label $(1,1)$ collides to (2), not to (1). Equation (2.1) shows why these targets cannot be interchanged even in the free commutative example.

2.2 Reassembling the full coefficient

Summing labelled complexes over all $\mathbf{m}$ duplicates merged states. At two positions, a merged state of weight two appears in the labels $(2,0)$, $(1,1)$ and $(0,2)$. The full residue coefficient contains that state once. A simplicial resolution identifies these copies while preserving their collision maps.

For a fixed summand of (2.2), put

$$M(\pi, \mathbf{d}) = \{\mathbf{m} \in \mathbb{N}^n : \sum_{i \in B} m_i = d_B \text{ for every } B \in \pi\}.$$

This is a nonempty finite set. It has $\prod_{B \in \pi} \binom{d_B + |B| - 1}{|B| - 1}$ elements: each block requires a weak composition of its weight into $|B|$ parts. Give it the groupoid structure with exactly one morphism between any two objects. Its nerve in degree q is $M(\pi, \mathbf{d})^{q+1}$.

Replace each coefficient by

$$R_q V_{\pi,\mathbf{d}} = \bigoplus_{(\mathbf{m}^0, \ldots, \mathbf{m}^q) \in M(\pi,\mathbf{d})^{q+1}} V_{\pi,\mathbf{d}}. \tag{2.4}$$

Faces delete an assignment and degeneracies repeat one. The augmentation sends every copy to $V_{\pi,\mathbf{d}}$ by the identity. A residue merger keeps each assignment tuple and adds the weights of its two blocks. Every source assignment satisfies the merged weight condition. Thus the original residue map extends to (2.4) and commutes with all faces and degeneracies.

In degree zero this construction is

$$R_0 C_n(A) = \bigoplus_{\mathbf{m} \in \mathbb{N}^n} C_{n,\mathbf{m}}(A).$$

The higher degrees impose the identifications absent from this direct sum. We realize the simplicial complex by direct-sum totalization. A coefficient of internal cochain degree p in simplicial degree q has total degree $p - q$, and its total differential is $d_{\mathrm{coef}} + (-1)^p \partial$, where ∂ is the alternating sum of faces.

Theorem 2.2 (Reassembly of multiplicities)**.** *The augmentation induces an equivalence*

$$|R_\bullet C(A)| \longrightarrow C(A).$$

It preserves the given cut coproducts and exceptional-collision maps. The realized coalgebra is strong when $C(A)$ is strong. A specified local-vacuum marking through the weight-zero subcomplex lifts unchanged, and the empty coefficient is preserved. These assertions use the direct-sum coefficient category above.

Proof. For each $(\pi, \mathbf{d})$, choose an assignment $\mathbf{m}_*$ that places d_B at the largest label of B and zero at its other labels. Prepending $\mathbf{m}_*$ defines an extra degeneracy of the augmented nerve. If ∂ is its chain boundary and h this extra degeneracy, cancellation of the alternating faces gives

$$\partial h + h \partial = 1$$

on the augmented chain complex. On the unaugmented complex the right side is $1 - i\varepsilon$, where i selects $\mathbf{m}_*$ and ε is the augmentation. After tensoring with a coefficient complex, the total homotopy on internal degree p is $(-1)^p h$.

The chosen assignments need not commute with residue mergers. Filter instead by the number of partition blocks. This filtration is finite in arity n, and a merger lowers it. On its associated graded, the preceding contraction proves the augmentation is an equivalence. Induction on the finite filtration proves the same assertion before passing to the associated graded. Equivalently, correcting the contraction by the merger differential takes at most $n - 1$ successive corrections. The unbounded simplicial direction therefore causes no infinite sum on an element.

On a cut open, a surviving partition separates into partitions of the two blocks. Its assignment set is the product of their assignment sets. This identifies the corresponding simplicial coefficient with the degreewise external product. Geometric realization preserves that product because tensor products preserve colimits separately and simplicial colimits are sifted. The resulting cut map is an equivalence whenever the original cut map is an equivalence. The common assignment tuples also give the iterated-cut coherences.

Under a monotone collision, use the prescribed exceptional comparison on the surviving partition coefficient and send each assignment to $\alpha_* \mathbf{m}$. This is a functor between nonempty groupoids with exactly one morphism between each pair of objects. It is therefore an equivalence of groupoids. Its realization supplies the collision equivalence. Associativity of addition gives compatibility with composite collisions. The degreewise diagrams $R_q C$ need not themselves have exceptional pullback equivalences. Those equivalences hold after realization.

At weight zero the only assignment is the zero assignment, so the simplicial resolution is constant on the weight-zero subcomplex. The prescribed vacuum marking therefore lifts through it. At the empty word the only assignment is the empty tuple. Its constant resolution preserves the empty coefficient. □

The positive-multiplicity sectors omit positions carrying the vacuum. The all-ones sector also omits the simplices that identify different assignments to a merged block. Both omissions change the full coefficient. If a completion in generator weight is introduced, its interchange with realization requires a separate limit argument.

2.3 Derived specialization of the vacuum generator

A central field that represents the vacuum after specialization must satisfy every translated vacuum relation. Let A° be a vertex algebra over k with a central regular differential subalgebra

$$E = k[e_0, e_1, \dots], \qquad T e_i = e_{i+1}.$$

Regular means that these fields have no singular products with any state. Their normally ordered products give the displayed commutative algebra. Assume A° is flat over E for this central action. The specialization is

$$A = A^\circ / (e_0 - 1, e_1, e_2, \dots).$$

Translation forces the relations after the first one because $T1 = 0$.

Adjoin odd central regular generators η_i of cochain degree -1. They form an exterior algebra. Set

$$Q(A^\circ) = A^\circ \otimes_k \Lambda(\eta_0, \eta_1, \dots), \quad T\eta_i = \eta_{i+1}, \quad d\eta_i = e_i - \delta_{i0}, \quad dA^\circ = 0. \tag{2.5}$$

All expressions are finite polynomials in these generators.

Theorem 2.3 (Translation-compatible specialization)**.** *Formula* (2.5) *defines a differential graded vertex algebra. The map*

$$Q(A^\circ) \longrightarrow A, \qquad \eta_i \longmapsto 0,$$

is a quasi-isomorphism preserving the unit and translation. Without the flatness assumption, the same complex computes $A^\circ \otimes^L_E k$, and its higher Tor classes remain.

Proof. The images $e_i - \delta_{i0}$ are central regular fields. Thus the displayed differential extends as a degree-one derivation of every vertex product. It squares to zero on generators and commutes with translation because

$$dT\eta_i = e_{i+1} = T d\eta_i.$$

These identities extend to every state by the derivation rule.

Put $x_i = e_i - \delta_{i0}$. The complex

$$K_E = (E \otimes \Lambda(\eta_i), d = \textstyle\sum_i x_i \partial_{\eta_i})$$

is a free resolution of the augmentation $E \to k$ that sends all x_i to zero. To verify exactness, let N be total polynomial degree, counting each x_i and each η_i once. For a monomial with $N > 0$, define

$$h = \frac{1}{N} \sum_i \eta_i \partial_{x_i}, \qquad h(1) = 0. \tag{2.6}$$

The graded anticommutator of $\sum_i x_i \partial_{\eta_i}$ and $\sum_i \eta_i \partial_{x_i}$ is the total Euler operator. Hence $dh + hd = 1 - i\varepsilon$, where $i : k \to K_E$ includes constants and ε evaluates all generators at zero. Each sum is finite on a polynomial. Characteristic zero makes division by N valid.

Translation acts by

$$T = \sum_i x_{i+1} \partial_{x_i} + \sum_i \eta_{i+1} \partial_{\eta_i}.$$

It preserves N. In its commutator with the numerator of (2.6), the terms $\eta_{i+1} \partial_{x_i}$ cancel after reindexing. Thus $Th = hT$. This is a translation-compatible contraction of the scalar complex.

The homotopy h is not E-linear. Exactness after tensoring with A° follows instead from flatness of A° over E. It gives $H^\circ Q(A^\circ) = A$ and zero cohomology in other degrees. In general, tensoring the free resolution K_E computes the derived tensor product and its Tor groups. The vertex map to the ordinary specialization preserves the unit and translation by its defining formulas. □

The relation $e_0 = 1$ generally destroys generator grading. Multiplicity reassembly applies to the graded object before specialization. The differential graded specialization then acts on its full state algebra, without assuming that the old weight decomposition survives.

Suppose the full residue functor on differential graded state algebras uses derived tensor products, localization and diagonal direct images on each partition coefficient. Suppose also that its differential and vacuum maps are functorial for vertex maps. The quasi-isomorphism of Theorem 2.3 then induces an equivalence

$$C(Q(A^\circ)) \longrightarrow C(A)$$

of vacuum-marked collision coalgebras. Indeed, it is an equivalence on each partition coefficient, and the finite filtration by partition blocks proves the assertion with the residue differential included. Theorem 1.3 gives an equivalence of their cut cobars. This argument does not choose a module-theoretic splitting of any vacuum extension.

2.4 Extending a comparison across the collision support

A comparison defined on the cut opens must extend across their closed complement. A prescribed vacuum map adds a relative condition to that extension problem.

Let $j : O \hookrightarrow Y$ be an open embedding with closed complement Z. Work in a stable coefficient category with the localization triangle

$$S = R\Gamma_Z Q \longrightarrow Q \longrightarrow j_* j^! Q.$$

Fix a map $u : V \to P$. Write $R\,\mathrm{Hom}$ for the derived cochain complex of maps, and define

$$\begin{aligned} M &= R\,\mathrm{Hom}(P, Q), \\ B &= R\,\mathrm{Hom}(V, Q) \underset{R\,\mathrm{Hom}(j^! V, j^! Q)}{\overset{h}{\times}} R\,\mathrm{Hom}(j^! P, j^! Q), \\ L &= R\,\mathrm{Hom}(\mathrm{cofib}(u), S). \end{aligned}$$

The superscript h denotes a homotopy pullback. A point of B contains a map on V, a map on O, and a specified homotopy between their restrictions to $j^! V$.

Theorem 2.4 (Relative supported extension)**.** *There is a natural fibre sequence*

$$L \longrightarrow M \longrightarrow B \longrightarrow L[1].$$

For compatible degree-zero data b, the obstruction to a lift with the prescribed homotopy is $\partial[b] \in H^1 L$. It vanishes exactly when the space of such lifts is nonempty. After choosing one lift, the connected components of that space form a torsor for $H^0 L$, and its higher homotopy groups are $H^{-i} L$, $i > 0$.

Proof. Form the square of mapping complexes obtained by restricting $P \to Q$ along u and along j. Localization and adjunction identify its two vertical fibres with $R\,\mathrm{Hom}(P, S)$ and $R\,\mathrm{Hom}(V, S)$. Its total fibre is therefore $R\,\mathrm{Hom}(\mathrm{cofib}(u), S) = L$. Taking the other order of fibres identifies the same complex with the fibre of $M \to B$. The long exact cohomology sequence gives the obstruction. A lift identifies its fibre with the additive mapping space of L. The cochain convention then gives the stated homotopy groups. □

For an explicit lift, replace $M \to B$ by a degreewise surjective mapping-path model with kernel L. Choose a graded lift m_0 of a cocycle representing b. Its differential dm_0 is a closed element of L^1. If h contracts L, the corrected lift is

$$m = m_0 - h(dm_0).$$

It is closed and has the same prescribed data. Vanishing of $H^1 L$ supplies existence. Contractibility of the space of lifts also requires $H^i L = 0$ for $i \le 0$.

For strong coalgebras F, C with vacuum $J \to F$, suppose a transverse tensor product $\otimes^!$ is defined with unit J. At configuration size n, take

$$P_n = F_n \otimes^! C_n, \qquad V_n = C_n, \qquad u_n = (J_n \to F_n) \otimes^! 1, \qquad Q_n = C_n.$$

If an action has already been specified on $O_n = \bigcup_{p=1}^{n-1} U_p^{(n)}$, compatibly with the vacuum, its relative extension complex is

$$L_n = R\,\mathrm{Hom}\Big(\mathrm{cofib}(C_n \xrightarrow{u_n} F_n \otimes^! C_n),\ R\Gamma_{Z_n} C_n\Big). \tag{2.7}$$

This complex controls the specified open-and-vacuum extension. To obtain a coalgebra action on the full collision diagram, one must also impose its prescribed exceptional restrictions and all iterated-cut homotopies. Their relative complex is the homotopy fibre of the restriction map to that matching diagram. It cannot be replaced by a product of independent equations.

2.5 Higher comparison equations

A linear comparison must be a map at every configuration size. An isomorphism of singleton state spaces provides only one of these components. Input arity gives a second index: an A_∞ Taylor component with r inputs is a map of whole coefficient systems, not a component at configuration size r.

Let C, D be augmented uncurved A_∞ algebras in the coefficient category. Their structures are degree-one square-zero coderivations b^C, b^D on the corresponding tensor coalgebras of suspended reduced coefficients. Write b^C_r, b^D_r for their components with r inputs. Uncurved means $b^C_0 = b^D_0 = 0$. A degree-zero coaugmentation-preserving coalgebra map F is determined by its components f_r with r inputs. Assume f_1 is a chain map. The equation $b^D F = F b^C$ in input arity $r \ge 2$ reads

$$\begin{aligned} \partial f_r = &\sum_{a+s+t=r,\ s\ge 2} f_{a+1+t}(1^{\otimes a} \otimes b^C_s \otimes 1^{\otimes t}) \\ &- \sum_{\substack{k\ge 2,\ i_1+\cdots+i_k=r \\ i_j\ge 1}} b^D_k(f_{i_1} \otimes \cdots \otimes f_{i_k}) =: O_r. \end{aligned} \tag{2.8}$$

Here $a, t \geq 0$ in the first sum. The unary mapping differential ∂ is the graded commutator with the source and target unary differentials. Every tensor composition uses the Koszul signs. Each comparison component on the right has smaller input arity. If the lower equations hold, then $\partial O_r = 0$: expand the arity-r part of

$$b^D(b^D F - F b^C) + (b^D F - F b^C) b^C = 0.$$

Thus $[O_r]$ lies in degree-one cohomology of the appropriate relative operation complex, including its collision conditions. A contraction h_r supplies $f_r = h_r O_r$ when the prescribed boundary component is zero. For a nonzero prescribed component, first choose a lift and correct its relative defect as in Theorem 2.4.

By contrast, an unknown structure operation b_r has degree one, and the known defect in b^2 has degree two. Constructing that operation requires a degree-two obstruction to vanish. The two extension problems use different cohomology groups.

On positive word coefficients, input arities greater than configuration size vanish. The recursion is therefore finite at each configuration, provided all the relative contractions and coefficient-domain compatibilities have been supplied. A nonzero arity-zero operation removes this triangular form: the arity-r equation can then contain f_{r+1}. Such a comparison needs a separate square-zero realization or a complete positive filtration controlling these insertions.

A surviving obstruction class refutes the comparison whose domain and prescribed data define that class. To exclude all field realizations or all pointed actions, one must prove that every permitted realization preserves the obstructed datum.

Chapter 3

Primitive Heisenberg coefficients

Modulo derivatives, a rational differential with poles at one fixed point is determined by its residue. Moving the points changes this calculation: the coefficients of partial fractions can acquire poles on the collision diagonals. The Heisenberg coefficient must retain these terms before the points are identified. Its primitive cohomology is flat over the position ring, but its diagonal constituents form nonsplit extensions. Those extensions determine both the collision maps and their duals.

Throughout this chapter, k is a field of characteristic zero. Complexes have cohomological grading, and differential modules are left modules. All rational functions have finite pole order. No completion in pole order is taken.

3.1 Principal parts and differentiation

Let $n \geq 1$, let $z_1, \ldots, z_n$ be the position coordinates, and let w be an auxiliary coordinate. Set

$$R_n = k[z_1, \ldots, z_n], \qquad P(w) = \prod_{i=1}^{n} (w - z_i), \qquad M_n = R_n[w, P^{-1}]/R_n[w]. \tag{3.1}$$

Division by the monic denominator gives each element of M_n a unique proper rational representative. Differentiation in w and in each position preserves the polynomial submodule and therefore acts on M_n. These operators commute. The primitive coefficient is

$$H_n = \operatorname{coker}(\partial_w : M_n \longrightarrow M_n). \tag{3.2}$$

The differential $f(w)\,dw$ records the residue convention for a representative $f(w)$ of this quotient.

For a nonzero level $\kappa \in k$, let xM_n and γM_n be copies of M_n in degrees zero and one. The linear Heisenberg coefficient complex is

$$K_{\kappa,n} = [xM_n \xrightarrow{-\kappa\partial_w} \gamma M_n]. \tag{3.3}$$

Its degree-one cohomology is H_n. The polynomial coefficient is its graded symmetric algebra, treated in Section 3.6. The primitive calculation determines its cohomology.

At one point every class of pole order greater than one is a w-derivative, and H_1 is free on the class of $(w - z_1)^{-1}$. At two points, writing

$$g_i = [(w - z_i)^{-1}], \qquad g_{12} = [((w - z_1)(w - z_2))^{-1}],$$

gives

$$(z_1 - z_2)\, g_{12} = g_1 - g_2. \tag{3.4}$$

This identity holds over the polynomial position ring. Dividing by $z_1 - z_2$ describes the separated residues, but loses the question of how they extend over the diagonal. Finite denominator stages compute the quotient over that ring.

Put $M^{(0)} = 0$ and $M^{(m)} = P^{-m}R_n[w]/R_n[w]$ for $m \geq 1$, with arity n fixed. The elements w^e/P^m, $0 \leq e < mn$, form an R_n-basis of $M^{(m)}$. Repeated monic division also gives a basis of the whole module:

$$M_n = \bigoplus_{m\geq 1} \bigoplus_{e=0}^{n-1} R_n \frac{w^e}{P^m}. \tag{3.5}$$

Indeed, multiplying a finite relation by its largest denominator and reducing modulo P first kills the numerator at that denominator. Induction kills every numerator.

Theorem 3.1 (Finite pole stages)**.** *For $m \geq 0$, the module $F_m = M^{(m+1)}/\partial_w M^{(m)}$ is free of rank n, with ordered representatives*

$$c_{m,0} = \frac{1}{P^{m+1}}, \quad \ldots, \quad c_{m,n-2} = \frac{w^{n-2}}{P^{m+1}}, \qquad c_{m,n-1} = \frac{w^{n(m+1)-1}}{P^{m+1}}. \tag{3.6}$$

For $n = 1$ only the last representative occurs. The inclusions of principal parts induce transitions $T_m : F_m \to F_{m+1}$, and $H_n = \operatorname{colim}_m F_m$. In these bases,

$$\det T_m = (-1)^{n(n-1)/2} \frac{(m+1)^{n-1}}{n \prod_{j=1}^{n-1}(n(m+1)-j)} \prod_{i<j}(z_i - z_j)^2. \tag{3.7}$$

At a geometric position with exactly r distinct points, the specialized transition has rank r.

Proof. For $0 \leq e < mn$, differentiation has numerator

$$\partial_w \frac{w^e}{P^m} = \frac{d_{m,e}}{P^{m+1}}, \qquad d_{m,e} = ew^{e-1}P - mw^e P'.$$

Its leading term is $(e - mn)w^{e+n-1}$. These nonzero integer pivots occupy the distinct degrees $n-1, \ldots, n(m+1)-2$. Descending elimination leaves precisely (3.6). It also proves that $\partial_w : M_n \to M_n$ is injective. Every fraction and every primitive belong to finite stages, so the colimit of these cokernels is H_n.

The transition is computed by multiplying a basis numerator by P and eliminating the pivots for $\partial_w M^{(m+1)}$. Only nonzero integers are divided out. Its entries are therefore polynomials in the coefficients of P.

After specialization to an algebraically closed field, a proper rational differential has a proper rational primitive if and only if its residues vanish. At a root of multiplicity d in P, a zero-residue differential with denominator P^{m+1} has a primitive of pole order at most $(m+1)d - 1$. That primitive lies in $M^{(m+1)}$. Consequently the kernel of the specialized T_m is exactly the zero-residue subspace of F_m. The simple poles at the r remaining points give independent residue vectors and belong to $M^{(m+1)}$. This proves the rank assertion.

The determinant vanishes on every pair diagonal. It is symmetric in the roots, so each difference $z_i - z_j$ divides it to even order. The last representative in (3.6) has total finite residue one, and all the others have total residue zero. The last row of T_m is thus $(0, \ldots, 0, 1)$. Simultaneously scaling w and the roots shows that the determinant of the remaining $(n-1)$-square block has degree $n(n-1)$. It must be a constant multiple of the discriminant.

Evaluate that constant at $P = w^n - a$, with a indeterminate. The block is diagonal, with entries

$$-\frac{an(m+1)}{n(m+1)-j}, \qquad 1 \leq j \leq n-1.$$

They follow by eliminating w^{n+j-1} from $w^{n+j-1} - aw^{j-1}$. The discriminant in the convention of (3.7) is

$$\prod_{i<j}(z_i - z_j)^2 = (-1)^{n(n-1)/2+n-1} n^n a^{n-1}.$$

Taking the product of the diagonal entries gives the formula. For $n = 1$ the transition is the identity. □

Corollary 3.2 (Flatness and collision fibers)**.** *The R_n-module H_n is flat. For every commutative R_n-algebra C, its base change is the derivative quotient formed from $C[w, \overline{P}^{-1}]/C[w]$. Its geometric fiber is the residue space of the distinct points remaining after collision.*

Proof. The stage description expresses H_n as a filtered colimit of finite free modules. Each stage basis and its elimination remain valid over C, because their pivots are nonzero integers in k. Taking the colimit proves the base-change assertion. Over a geometric point, powers of the specialized P and of its reduced polynomial give cofinal denominator systems. A zero-residue stage class dies at the next transition. Every simple residue survives. □

Example 3.3 (Two colliding points)**.** Suppress the common translation coordinate and put $R = k[u]$, $P = w(w-u)$. In the ordered basis $a_m = P^{-m-1}$, $b_m = w^{2m+1}P^{-m-1}$, the residues at $w = u$ are

$$\operatorname{Res}_{w=u} a_m\, dw = (-1)^m \binom{2m}{m} u^{-2m-1}, \qquad \operatorname{Res}_{w=u} b_m\, dw = 1.$$

The total residues are respectively zero and one. Ratios of successive first residues determine the transition on the distinct locus. Since its entries are polynomials, this gives the transition over R:

$$T_m = \begin{pmatrix} -\dfrac{m+1}{2(2m+1)}u^2 & 0 \\ 0 & 1 \end{pmatrix}, \qquad H_2 \simeq R[u^{-1}] \oplus R. \tag{3.8}$$

The coordinates on the right are the residue at $w = u$ and the total finite residue. For $m \geq 1$, the differential $dw/[w^m(w-u)]$ has coordinates $(u^{-m}, 0)$, while $dw/(w-u)$ has coordinates $(1, 1)$. Together with their R-multiples these give the displayed module. The transition formula proves injectivity of this residue description. In residue coordinates the position connection is ordinary differentiation in u: differentiating a moving simple pole adds a double pole, whose residue is zero.

At $u = 0$ the localized summand has zero fiber and the regular summand remains. Thus flatness here does not imply finite projectivity or constancy of fiber dimension.

Proposition 3.4. *In Example 3.3, the quotient $M_2 \to H_2$ has no R-linear section. Hence the complex $[M_2 \xrightarrow{\partial_w} M_2]$ has no R-linear strong deformation retract onto its cohomology on these modules.*

Proof. The free module (3.5) satisfies $\bigcap_{N\geq 0} u^N M_2 = 0$. Every element in the image of an R-linear map $R[u^{-1}] \to M_2$ lies in that intersection, so the map is zero. A section would be injective on the localized summand of (3.8). A strong deformation retract would in particular supply such a section. □

3.2 The differential module on the collision diagonals

Position derivatives retain information beyond a fiber's residue space. Let

$$D_n = R_n\langle \partial_1, \dots, \partial_n \rangle, \qquad [\partial_i, z_j] = \delta_{ij}, \quad [\partial_i, \partial_j] = 0,$$

where ∂_i differentiates a rational representative at fixed w. For every nonempty $S \subseteq [n]$, define

$$g_S = \left[\prod_{i\in S} (w - z_i)^{-1} \right] \in H_n. \tag{3.9}$$

For a singleton write $g_i = g_{\{i\}}$. Divided position derivatives recover arbitrary pole orders:

$$\left[\prod_{i\in S} (w - z_i)^{-m_i} \right] = \prod_{i\in S} \frac{\partial_i^{m_i-1}}{(m_i-1)!}\, g_S, \qquad m_i \geq 1. \tag{3.10}$$

The finite list of g_S can therefore describe an unbounded range of pole orders.

Let $L_s\mathcal{M}_n$ be the sum of the principal-part submodules whose poles use at most s labels, and let F_sH_n be its image in H_n. Set $L_0\mathcal{M}_n = F_0H_n = 0$. These filtrations are preserved by position derivatives, and ∂_w preserves $L_s\mathcal{M}_n$.

For nonempty S, the diagonal Δ_S is given by $z_i = z_j$ for $i, j \in S$. Denote by δ_S the cyclic D_n-module generated by η_S, subject to

$$(z_i - z_j)\eta_S = 0 \ (i, j \in S), \qquad \partial_k \eta_S = 0 \ (k \notin S), \qquad \sum_{i \in S} \partial_i \eta_S = 0. \tag{3.11}$$

It is the direct-image differential module of the trivial connection on Δ_S.

To see its normal directions explicitly, choose $r = \max S$ and put $u_i = z_i - z_r$ for $i \in S \setminus \{r\}$. Over the tangential polynomial ring it has the principal-part basis $\prod_{i \neq r} u_i^{-b_i}$, $b_i \geq 1$. Multiplication by u_i kills a term when $b_i = 1$, and otherwise lowers b_i. Differentiation sends $u_i^{-b_i}$ to $-b_i u_i^{-b_i - 1}$. For $|S| = 1$, $\delta_S = R_n$.

Filter D_n by differential order and write $\xi_i = \mathrm{gr}(\partial_i)$. A compatible exhaustive filtration on a module is good if it is bounded below and its associated graded is finitely generated over $R_n[\xi_1, \dots, \xi_n]$. The characteristic variety is the support of that associated graded module in the cotangent bundle $T^*\mathbb{A}^n$. A nonzero module is holonomic when that variety has dimension n. Its holonomic length is its composition length.

Theorem 3.5 (Diagonal constituents)**.** *The pole-label filtration has differential-module quotients*

$$F_sH_n / F_{s-1}H_n \simeq \bigoplus_{|S| = s} \delta_S, \qquad 1 \leq s \leq n, \tag{3.12}$$

with the image of g_S as cyclic generator of its summand. For $2 \leq s \leq n$, the resulting extension of $\bigoplus_{|S|=s} \delta_S$ by $F_{s-1}H_n$ does not split even as a sequence of R_n-modules. The holonomic length of H_n is $2^n - 1$. Over $k = \mathbb{C}$ it is regular holonomic.

Proof. The variables w and $t_i = w - z_i$, $1 \leq i \leq n$, are independent coordinates on the auxiliary affine space. A Laurent monomial determines its set of negative t_i exponents. Modulo monomials with fewer negative labels, this gives

$$\mathrm{gr}_s^L M_n = \bigoplus_{|S| = s} H^s_{(t_i : i \in S)}(R_n[w]).$$

Here the local cohomology summand consists of negative powers in all its indicated variables, with polynomial coefficients in the remaining variables. This description is the top term of the coordinate Čech complex and fixes its differential module structure.

On the summand for S, use $t = w - z_r$, the normal coordinates u_i, and the tangential coordinates z_r, z_k for $k \notin S$. The support ideal becomes $(t, u_i : i \neq r)$. Differentiation in w at fixed positions becomes ∂_t. Thus the local cohomology module has basis

$$t^{-a} \prod_{i \in S \setminus \{r\}} u_i^{-b_i}, \qquad a, b_i \geq 1,$$

over the tangential polynomial ring. The derivative in t is injective and its cokernel retains precisely $a = 1$.

The induced position actions are (3.11). The linear change of normal coordinates has determinant a unit. Its fundamental fraction sends g_S to a nonzero multiple of the fundamental normal class. Normalize η_S by this image.

Apply the kernel–cokernel sequence for ∂_w to $0 \to L_{s-1}\mathcal{M}_n \to L_s\mathcal{M}_n \to \mathrm{gr}_s^L M_n \to 0$. The injectivity on every graded summand proves, by induction, that the cokernels embed in H_n and have the quotients (3.12).

By Corollary 3.2, H_n is torsion-free over the domain R_n. For $s \geq 2$, each element in a summand of (3.12) is killed by a power of a normal equation. A product of these equations kills an element of the finite direct sum. Every R_n-linear map from that quotient into $F_s H_n$ is therefore zero. It cannot be a section.

Each δ_S is simple. In the displayed basis, multiplication by normal coordinates lowers the normal derivative degree, and tangential differentiation lowers polynomial degree. Choose a term of maximal total normal derivative degree and multiply by its normal coordinate monomial. This kills all other normal monomials of that degree and every lower one, leaving a nonzero tangential polynomial times the fundamental class. A derivative of maximal total tangential degree extracts a nonzero scalar. Normal derivatives and tangential multiplication then generate the whole module.

Its characteristic variety is the conormal bundle of Δ_S, of dimension n, so it is holonomic. The finite filtration has one such constituent for each nonempty subset, proving its length. Over $\mathbb{C}$ the trivial diagonal connections are regular, and regular holonomic modules are closed under extensions [59, Definition 6.1.1]. □

The diagonal quotients are not flat over R_n. Consequently their ordinary exterior powers cannot compute the associated graded of exterior powers of H_n without resolving the intersections of their supports.

Theorem 3.6 (Finite differential-module presentation)**.** *The generators* (3.9) *present H_n over D_n with the relations*

$$\begin{aligned} \partial_k g_S &= 0 && (k \notin S), \\ \sum_{i \in S} \partial_i g_S &= 0, \\ (z_i - z_j) g_S &= g_{S\setminus\{j\}} - g_{S\setminus\{i\}} && (i, j \in S,\ i \neq j). \end{aligned} \tag{3.13}$$

No empty-subset generator is required.

Proof. The first and third equations hold on rational representatives. The second represents the w-derivative of the negative of their product. Formula (3.10) generates every pole multiplicity. Repeatedly writing $w = (w - z_i) + z_i$ reduces any numerator to those fractions and to polynomial terms, which vanish in M_n. The presented module Q therefore surjects onto H_n.

Filter Q by the cardinalities of its generators. On its level-s quotient the right side of the last relation vanishes. The remaining relations give surjections

$$\bigoplus_{|S|=s} \delta_S \longrightarrow \operatorname{gr}_s Q \longrightarrow \operatorname{gr}_s^F H_n.$$

Their composite is the isomorphism of Theorem 3.5 with its chosen generators. Both arrows are thus isomorphisms. Induction on the finite filtration proves $Q = H_n$. □

Example 3.7 (The triple attachment)**.** For $a = z_1 - z_3$ and $b = z_2 - z_3$, the binary and ternary generators satisfy

$$(a-b)\,g_{12} = g_1 - g_2, \qquad a\,g_{13} = g_1 - g_3, \qquad b\,g_{23} = g_2 - g_3,$$
$$a\,g_{123} = g_{12} - g_{23}, \qquad b\,g_{123} = g_{12} - g_{13}.$$

On the distinct locus the residue vector of g_{123} is

$$\left(\frac{1}{a(a-b)}, \frac{1}{b(b-a)}, \frac{1}{ab} \right).$$

Its total residue is zero, but its image in $H_3 / F_2 H_3 = \delta_{123}$ is nonzero. The two attaching equations specify how that constituent meets the three binary diagonals.

3.3 Collision and cut maps

Let $\alpha : [n] \twoheadrightarrow [m]$ identify the positions by $z_i = y_{\alpha(i)}$. The specialized coefficient has its induced tangential connection: differentiation in y_t acts by the sum of position derivatives in the fiber $\alpha^{-1}(t)$. For nonempty S, set $T = \alpha(S)$ and $\nu_t = |S \cap \alpha^{-1}(t)|$.

Proposition 3.8. *The coefficient base-change isomorphism $H_n \otimes_{R_n} k[y_1, \dots, y_m] \simeq H_m$ sends*

$$g_S \longmapsto \left(\prod_{t \in T} \frac{\partial_{y_t}^{\nu_t - 1}}{(\nu_t - 1)!} \right) g_T. \tag{3.14}$$

These maps are horizontal and compose under successive surjections.

Proof. Both sides of (3.14) are represented by $\prod_{t \in T} (w - y_t)^{-\nu_t}$. Powers of the repeated and reduced denominator products are cofinal, so substitution identifies their localizations and principal parts. Corollary 3.2 identifies their derivative cokernels. Successive substitutions agree as rational functions, and differentiation obeys the chain rule. □

If all labels of S collapse to one point and $|S| > 1$, its image is zero, since

$$(w - y)^{-|S|} = \partial_w \left(-\frac{(w - y)^{1-|S|}}{|S| - 1} \right).$$

The proposition concerns coefficient base change with the specified tangential action. An extraordinary inverse image of differential modules also involves its density and shift conventions.

A consecutive cut separates groups of labels while allowing collisions inside each group. More generally, let $\mathcal{B} = (B_1 | \cdots | B_r)$ be an ordered partition of $[n]$ into nonempty blocks. Let $U_{\mathcal{B}}$ be the open where $z_i - z_j$ is invertible whenever i and j lie in different blocks. Write $R_{\mathcal{B}}$ for its coordinate ring and M_{B_a}, H_{B_a} for the coefficients using just the labels in B_a, extended to $R_{\mathcal{B}}$.

Proposition 3.9 (Block decomposition)**.** *There are canonical horizontal isomorphisms*

$$M_n|_{U_{\mathcal{B}}} \simeq \bigoplus_a M_{B_a}, \qquad H_n|_{U_{\mathcal{B}}} \simeq \bigoplus_a H_{B_a}. \tag{3.15}$$

They commute with ordered refinements and with collisions inside the blocks.

Proof. Put $P_a = \prod_{i \in B_a} (w - z_i)$. The resultants of different P_a are units on $U_{\mathcal{B}}$, so their powers are pairwise coprime. The polynomial Chinese remainder theorem gives a unique proper partial-fraction decomposition by these denominators. This proves the first isomorphism.

Differentiation in w preserves every summand, giving the second. Position differentiation also preserves each group's allowed poles, so the maps are horizontal. Uniqueness makes successive refinements agree. After an internal collision the resultants between groups remain units. The same decomposition commutes with substitution. □

3.4 Free comparisons and completed mapping complexes

The absence of a section in Proposition 3.4 does not prevent comparison by free resolutions. The finite stages supply both the resolution and its chain map. Let $E = \bigoplus_{m \geq 0} F_m$, and let $\sigma_T : E \to E$ send the m-th summand to the next by T_m.

Proposition 3.10. *The sequence*

$$0 \longrightarrow E \xrightarrow{1-\sigma_T} E \longrightarrow H_n \longrightarrow 0 \tag{3.16}$$

is a free R_n-resolution. For an R_n-module N, it computes derived Hom as

$$\begin{aligned} R\operatorname{Hom}_{R_n}(H_n, N) &= \left[\prod_m \operatorname{Hom}_{R_n}(F_m, N) \xrightarrow{\partial_T} \prod_m \operatorname{Hom}_{R_n}(F_m, N)\right], \\ (\partial_T f)_m &= f_m - f_{m+1} T_m. \end{aligned} \tag{3.17}$$

in degrees zero and one.

Proof. The first nonzero stage of a finite-support vector survives under $1 - \sigma_T$, proving injectivity. Its cokernel is the defining colimit. Both copies of E are free. Applying Hom to these direct sums gives the products in (3.17) and the stated difference map. □

The elimination in Theorem 3.1 records, along with each matrix column, a primitive $p_{m,j} \in M^{(m+1)}$:

$$c_{m,j} - \sum_i (T_m)_{ij} c_{m+1,i} = \partial_w p_{m,j}. \tag{3.18}$$

Each eliminated numerator contributes its coefficient times the rational function whose derivative was eliminated.

Place the two copies of E in degrees zero and one, with corresponding generators $u_{m,j}$ and $v_{m,j}$. The differential is $du_{m,j} = v_{m,j} - \sum_i (T_m)_{ij} v_{m+1,i}$.

Proposition 3.11. *The maps*

$$u_{m,j} \longmapsto -\kappa^{-1} x(p_{m,j}), \qquad v_{m,j} \longmapsto \gamma(c_{m,j}) \tag{3.19}$$

give an R_n-linear homotopy equivalence from this telescope to $K_{\kappa,n}$.

Proof. Equation (3.18) is exactly the chain-map identity. Both degree-zero cohomologies vanish, and the map is the identity on degree-one cohomology H_n. Both complexes are bounded and free over R_n.

An inverse can be constructed on their free bases. Reduce each degree-one basis vector of M_n at a finite stage and lift its class to the corresponding $F_m \subset E$.

The image of a boundary under this lift lies in $\ker(E \to H_n) = \operatorname{im}(1 - \sigma_T)$. Its unique preimage defines the degree-zero map. To find the preimage, advance its finitely many components to a common stage and telescope the intervening transitions. Differences between either composite and the identity induce zero on H_n. Lift those differences through the injective boundary maps in the same way. These lifts are the chain homotopies. □

For $P = w(w-u)$ the first two primitives are

$$p_{0,\mathrm{low}} = -\frac{w - u/2}{P}, \qquad p_{0,\mathrm{top}} = \frac{u}{w-u}. \tag{3.20}$$

Their derivatives are $P^{-1} + u^2/(2P^2)$ and $(w-u)^{-1} - w/(w-u)^2$. Even the identity transition on the top cohomology coordinate therefore has a nonzero chain correction. The displayed homotopy equivalence is R_n-linear. A connection on a chosen resolution requires its own lifts of position differentiation.

Proposition 3.12 (The effect of a connection)**.** *For $R = k[u]$ and $L = R[u^{-1}]$,*

$$\operatorname{Hom}_R(L,R) = 0, \qquad \operatorname{Ext}^1_R(L,R) = k[[u]]/R, \qquad \operatorname{Ext}^q_R(L,R) = 0 \quad (q > 1).$$

Give L and R their ordinary derivative connections, and put $D = k\langle u, \partial\rangle/([\partial, u] - 1)$. Then $R\operatorname{Hom}_D(L,R) = 0$. With target L, the groups $\operatorname{Hom}_D(L,L)$ and $\operatorname{Ext}^1_D(L,L)$ are each k, and the higher groups vanish.

Proof. Present L by the telescope with generators u^{-2m} and transition u^2. Its Hom differential is $(f_m) \mapsto (f_m - u^2 f_{m+1})$ on $\prod_{m\geq 0} R$. The following series converges in the u-adic topology. The map

$$(a_m) \longmapsto \sum_{m\geq 0} u^{2m} a_m \pmod{R}$$

takes boundaries to polynomials. If the sum is the polynomial p, then

$$f_m = u^{-2m}\left(p - \sum_{j<m} u^{2j} a_j\right) \in R$$

satisfies $a_m = f_m - u^2 f_{m+1}$.

Grouping the even and odd digits of any power series proves surjectivity. Infinite u-divisibility makes the degree-zero kernel zero, and the resolution has length one.

For the differential-module calculation, the assignment $e \mapsto u^{-1}$ gives

$$L \simeq D/D(u\partial + 1). \tag{3.21}$$

Indeed, commuting u past powers of ∂ reduces the quotient to the spanning set $u^j e$ for $j \geq 0$ and $\partial^j e$ for $j \geq 1$. Their images are respectively u^{j-1} and $(-1)^j j! u^{-j-1}$, distinct Laurent monomials, so they are independent.

The free resolution given by right multiplication by $u\partial + 1$ is injective, by the order filtration on the Weyl algebra. Applying Hom gives $u\partial + 1$ on the target. On R its eigenvalue at u^j is $j + 1$, always invertible. On L its kernel and cokernel are both the line represented by u^{-1}. □

For example, the constant sequence $a_m = 1$ represents the nonzero ordinary extension class of $(1 - u^2)^{-1}$ modulo R. That extension has generators $e, f_0, f_1, \ldots$ and relations $f_m - u^2 f_{m+1} = e$. It cannot carry the prescribed connections as an extension of L by R, since the corresponding differential-module extension group vanishes.

3.5 A telescope with all spatial derivatives

Position differentiation can raise the denominator stage. To keep it on a free resolution, the relation generators must move as well. This can be done before taking the auxiliary derivative quotient.

Let $d \geq 0$, let $B = k[s_1, \ldots, s_d]$, and let $f \in B[w]$ be monic of positive w-degree. Define the relative Weyl algebra

$$D_B = B\langle w, \partial_w\rangle/([\partial_w, w] - 1),$$

in which the parameters are central. Write $f_a = \partial_{s_a} f$, $f_w = \partial_w f$, and $f_{wa} = \partial_w \partial_{s_a} f$. Put

$$A_m = f\,\partial_w + m f_w, \qquad A_{m,a} = f_a\,\partial_w + m f_{wa}, \qquad m \geq 1. \tag{3.22}$$

These are operators: $\partial_w g = g\partial_w + g_w$. The identity $A_m f = f A_{m+1}$ gives transition maps, by right multiplication by f, between the free complexes $[D_B \xrightarrow{r_{A_m}} D_B]$ in degrees $-1, 0$.

Their direct-sum telescope $\mathcal{T}_f$ has generators e_m in degree zero, r_m, q_m in degree -1, and h_m in degree -2. Define its D_B-linear differential by

$$\begin{aligned} de_m &= 0, & dr_m &= A_m e_m, \\ dq_m &= e_m - f e_{m+1}, & dh_m &= r_m - f r_{m+1} - A_m q_m. \end{aligned} \tag{3.23}$$

Every vector has finite stage support. The only potentially nonzero square is $d^2 h_m = (A_m f - f A_{m+1}) e_{m+1} = 0$.

Lemma 3.13. *The augmentation $e_m \mapsto f^{-m}$, with negative-degree generators sent to zero, makes $\mathcal{T}_f$ a free D_B-resolution of $B[w, f^{-1}]$. After every scalar extension $B \to C$ to a commutative k-algebra it remains a resolution of $C[w, f^{-1}]$ and computes derived scalar extension.*

Proof. Right multiplication by A_m is injective: its leading coefficient in differential order is multiplication by the monic polynomial f. This remains a non-zero-divisor over C. Thus each two-term complex resolves $P_m = D_B / D_B A_m$.

In their colimit the transition and the next relation give

$$e_m = f e_{m+1}, \qquad \partial_w e_m = -m f_w e_{m+1}.$$

Repeatedly applying the second equation reduces every operator applied to e_m to a polynomial times a later generator. The fraction map has inverse $p/f^m \mapsto p e_m$. The first equation makes it well-defined and the second makes it horizontal. Hence colim $P_m = B[w, f^{-1}]$.

The map $1 -$ shift on the direct sum of the stage complexes is injective at the first nonzero stage. Its cone therefore resolves the filtered colimit, and its differential is (3.23) with the displayed choice of sign on h_m.

Its terms are free over D_B and over B. The same arguments after scalar extension prove the last assertion. □

Example 3.14 (A finite stage at collision)**.** Take $f = w(w - s)$ over $k[s]$ and specialize to $s = 0$. Write $D = k\langle w, \partial_w \rangle / ([\partial_w, w] - 1)$. Then $A_m = w B_m$, where $B_m = w\partial_w + 2m$. In the specialized P_m, put $g_m = B_m e_m$. The identities

$$w g_m = 0, \qquad (B_m - (2m-1)) B_m = \partial_w A_m$$

give $B_m g_m = (2m-1) g_m$.

Consequently $\ell_m = e_m - g_m/(2m-1)$ is killed by B_m, and

$$P_m|_{s=0} \simeq k[w, w^{-1}] \oplus D/Dw. \tag{3.24}$$

For an explicit inverse, send e_m to $(w^{-2m}, \eta/(2m-1))$, where $w\eta = 0$. The operator A_m kills this pair. The element g_m maps to $(0, \eta)$ and ℓ_m to $(w^{-2m}, 0)$. These maps verify both summands of (3.24).

The extra delta is removed by the next denominator transition:

$$g_m \longmapsto B_m w^2 e_{m+1} = w^2 B_{m+1} e_{m+1} = w^2 g_{m+1} = 0.$$

The whole telescope therefore has the correct collision fiber, although a single cyclic stage does not.

Extend ∂_{s_a} to D_B by fixing w, ∂_w. Define $\nabla_a(Pv) = (\partial_{s_a} P)v + P\nabla_a v$ and

$$\begin{aligned} \nabla_a e_m &= -m f_a e_{m+1}, \\ \nabla_a r_m &= -(m-1) f_a r_{m+1} + A_{m,a} q_m, \\ \nabla_a q_m &= -(m+1) f_a q_{m+1}, \\ \nabla_a h_m &= -m f_a h_{m+1}. \end{aligned} \tag{3.25}$$

The term $A_{m,a} q_m$ compares the derivative of a relation with its image at the next stage.

Theorem 3.15 (Flat spatial connection)**.** *The operators* (3.25) *satisfy*

$$[d, \nabla_a] = 0, \qquad [\nabla_a, \nabla_b] = 0, \qquad [\nabla_a, s_b] = \delta_{ab}, \qquad [\nabla_a, w] = [\nabla_a, \partial_w] = 0.$$

They make $\mathcal{T}_f$ a complex over the full Weyl algebra $\mathscr{D} = D_B\langle \partial_{s_1}, \dots, \partial_{s_d}\rangle$, with horizontal augmentation to $B[w, f^{-1}]$ with its ordinary derivatives. No discriminant is inverted.

Proof. The scalar commutators follow from the coefficient Leibniz rule. For the chain equation on r_m, normal ordering gives

$$A_{m,a} f + (m-1) f_a A_{m+1} = m A_m f_a, \tag{3.26}$$

both sides equal $m f f_a \partial_w + m f f_{wa} + m^2 f_w f_a$. On q_m, both orders of differentiation give $-(m+1) f_a e_{m+1} + (m+1) f f_a e_{m+2}$. On h_m, both give

$$-m f_a r_{m+1} + m f f_a r_{m+2} + m f_a A_{m+1} q_{m+1}.$$

The last equality follows by differentiating $A_m f = f A_{m+1}$ and using (3.26). The chain equation on e_m is immediate.

For curvature, mixed derivatives and scalar products cancel on e_m, q_m, h_m. On r_m, the q_m coefficient cancels by equality of mixed derivatives of A_m. The remaining q_{m+1} coefficient is

$$\begin{aligned}(m-1)(f_a A_{m+1,b} - f_b A_{m+1,a}) \\ + (m+1)(A_{m,a} f_b - A_{m,b} f_a).\end{aligned}$$

Writing $W = f_a f_{wb} - f_b f_{wa}$, the first line is $(m^2-1)W$ and the second is $(m+1)(1-m)W$. Their sum is zero. Finally, $\partial_{s_a} f^{-m} = -m f_a f^{-m-1}$ proves horizontality of the augmentation. □

For $f = w^3 + aw + b$, the correction operators are $A_{m,a} = w\partial_w + m$ and $A_{m,b} = \partial_w$. These formulas remain defined on $4a^3 + 27b^2 = 0$. Deleting the mixed term from $\nabla_b r_1$ would give the nonzero chain defect $-\partial_w e_1 + \partial_w f\, e_2$, where $\partial_w f$ denotes operator composition. The correction $d(\partial_w q_1)$ cancels it.

To obtain free terms over the full Weyl algebra, let $v_1, \dots, v_d$ be the standard basis of k^d, and form

$$\mathcal{P}_f^j = \bigoplus_{q-p=j} \mathscr{D} \otimes_{D_B} \mathcal{T}_f^q \otimes_k \Lambda^p k^d, \qquad 0 \le p \le d.$$

On the induced module set $L_a(P \otimes u) = P\partial_{s_a} \otimes u - P \otimes \nabla_a u$. The connection's Leibniz rule makes L_a balanced over D_B. Let δ be their alternating sum deleting the corresponding wedge factor, and define

$$d_{\mathcal{P}}(P \otimes u \otimes \omega) = P \otimes du \otimes \omega + (-1)^q \delta(P \otimes u \otimes \omega), \qquad u \in \mathcal{T}_f^q. \tag{3.27}$$

Flatness makes the L_a commute, and the chain equation makes them commute with the internal differential. The total sign in (3.27) therefore gives $d_{\mathcal{P}}^2 = 0$.

Proposition 3.16. *The action augmentation $\mathcal{P}_f \to \mathcal{T}_f \to B[w, f^{-1}]$ is a free $\mathscr{D}$-resolution, bounded in degrees $-d-2, \dots, 0$.*

Proof. Fix q and normally order the parameter derivatives. The induced module becomes their polynomial module with coefficients in $\mathcal{T}_f^q$. Filter the exterior-degree p term by parameter-derivative order plus p. The leading part of L_a is multiplication by its independent polynomial variable. The associated augmented row is thus the Koszul resolution of $\mathcal{T}_f^q$ for those variables. It is exact.

Lifting a leading boundary lowers the filtration. Every vector has finite order, so this procedure terminates. The augmented row resolves $\mathcal{T}_f^q$.

The internal degrees are bounded, and totalization gives a quasi-isomorphism to $\mathcal{T}_f$. Lemma 3.13 identifies its cohomology. Each induced term is free over $\mathscr{D}$ because $\mathcal{T}_f^q$ is free over D_B. □

For $B = R_n$ and $f = P$, this resolves the auxiliary localization in (3.1). The primitive coefficient H_n is obtained by first quotienting by $R_n[w]$ and then taking the ∂_w-cokernel. It is a different module. Applying Hom to the free spatial resolution takes products over all its stage generators. Those products are part of the mapping complex and cannot be replaced by a finite denominator cutoff.

3.6 The polynomial Heisenberg complex

The primitive differential (3.3) extends as a derivation to

$$Z_{\kappa,n} = \operatorname{Sym}_{R_n}(xM_n) \otimes_{R_n} \Lambda_{R_n}(\gamma M_n), \qquad dx(f) = -\kappa\gamma(\partial_w f), \quad d\gamma(f) = 0. \tag{3.28}$$

Here $x(f)$ has degree zero and $\gamma(f)$ degree one. The product is the auxiliary graded-commutative polynomial product, and each element has finite polynomial degree.

Theorem 3.17. *The primitive projection extends to a quasi-isomorphism of differential graded R_n-algebras*

$$\rho_n : Z_{\kappa,n} \longrightarrow \Lambda_{R_n}(H_n), \qquad x(f) \longmapsto 0, \quad \gamma(f) \longmapsto [f], \tag{3.29}$$

where H_n has degree one and zero differential. It preserves the unit, the position connection, coefficient collision maps, and the block decompositions of Proposition 3.9.

Proof. The primitive map is a quasi-isomorphism between bounded flat complexes by Theorem 3.1 and Corollary 3.2. Tensor powers preserve this quasi-isomorphism. In characteristic zero, each graded symmetric power is the image of its averaging idempotent on a tensor power. It therefore preserves the map as well. Their direct sum is (3.29).

The generator formulas commute with differentiation, substitution, and the direct-sum decomposition of principal parts. They also preserve the multiplicative unit. □

Let V be finite-dimensional with a nondegenerate level form, and identify V with its dual by that form. Apply the same argument to the corresponding copies of $V \otimes M_n$. The resulting cohomology is $\Lambda_{R_n}(V \otimes H_n)$.

Over the normal-coordinate ring $R = k[u]$ of Example 3.3, the rank-one polynomial complex has cohomology

$$H^0(Z_{\kappa,2}) = R, \qquad H^1(Z_{\kappa,2}) = R[u^{-1}] \oplus R, \qquad H^2(Z_{\kappa,2}) = R[u^{-1}],$$

with no higher cohomology. In particular $\Lambda^2_R R[u^{-1}] = 0$, since every pair of elements lies in a common rank-one free submodule. Under a cut, exterior algebra converts (3.15) into a tensor product. Comparison with the transverse product requires its contraction maps. The polynomial projection alone does not specify those operations.

3.7 The binary holonomic dual

Duality must transpose the attachment, including its normal derivative. In one variable let D and L be as in Proposition 3.12. Trivialize the canonical line by du and use the adjoint anti-involution $u^t = u$, $\partial^t = -\partial$. For a holonomic left module N, the convention is

$$\mathbb{D}N = {}^t R \operatorname{Hom}_D(N, D)[1],$$

where a right module becomes a left module by the adjoint. This includes the one-dimensional duality shift [59, Definition 2.6.1 and Corollary 2.6.8].

Proposition 3.18. *There is an isomorphism*

$$\mathbb{D}L \simeq D/D(u\partial). \tag{3.30}$$

If e is its cyclic generator, then $p_1 = \partial e \neq 0$, and there is a nonsplit exact sequence of differential modules

$$0 \longrightarrow D/Du \xrightarrow{1 \mapsto p_1} D/D(u\partial) \xrightarrow{e \mapsto 1} k[u] \longrightarrow 0. \tag{3.31}$$

This sequence splits as a sequence of $k[u]$-modules.

Proof. Apply $\mathrm{Hom}_D(-, D)$ to the free resolution of (3.21). Its only cohomology is in degree one. The adjoint of the relation is

$$(u\partial + 1)^t = -\partial u + 1 = -u\partial.$$

The duality shift puts this cokernel in degree zero, giving (3.30).

A normal spanning set for the quotient is $u^m e$, $m \geq 0$, and $p_j = \partial^j e$, $j \geq 1$. Their actions are

$$\begin{aligned} u p_1 &= 0, & u p_j &= -(j-1) p_{j-1} \quad (j > 1), & \partial p_j &= p_{j+1}, \\ \partial e &= p_1, & \partial(u^m e) &= m u^{m-1} e \quad (m \geq 1). \end{aligned}$$

These rules satisfy $[\partial, u] = 1$ on every indicated vector and $u\partial e = 0$. Conversely they define a module on the free vector space with that basis, generated by e. The normal spanning set is therefore independent.

The span of the p_j is D/Du, and the quotient on the $u^m e$ is $k[u]$, proving exactness.

Every lift of $1 \in k[u]$ is $e + \sum_{j=1}^{N} a_j p_j$. Its derivative has nonzero p_1 coefficient, so no lift is horizontal and no D-linear section exists. The assignment $u^m \mapsto u^m e$ is, however, a $k[u]$-linear section. □

The original localization sequence is $0 \to k[u] \to L \to L/k[u] \to 0$. The adjoints of their relations ∂ and u show that the regular and delta constituents are self-dual. Sequence (3.31) reverses their order and retains the nonzero derivative $\partial e = p_1$. Tensoring with the tangential regular connection gives the corresponding extension with the common translation coordinate. By (3.8), the complete primitive dual over the normal-coordinate ring is

$$\mathbb{D}H_2 \simeq k[u] \oplus D/D(u\partial).$$

This calculation fixes the extension in the primitive holonomic category. A duality for the full field coefficient must additionally specify its completion and its operations.

Chapter 4

Residues, intersections and cut transfer

A collision of the first and last positions can cross every cut while leaving all intermediate positions distinct. Such a coefficient survives the cut cobar. At three positions it supplies the homotopy between the two consecutive products. At four positions the crossed partition 13|24 supplies a further summand. We construct these coefficients and their maps before computing their cohomology.

The Heisenberg example then connects two calculations. Wick contractions compare the coefficients on separated groups, with an explicit correction to their cut maps. Derived intersections compute the exterior powers of the primitive module across the diagonals. The first calculation retains operations. The second retains the extensions of their coefficient modules.

Throughout, k has characteristic zero and complexes have cohomological grading. We first use right differential modules on $X = \mathbb{A}^1_k$. In the sections on primitive coefficients we use left modules, as in Chapter 3. Tensoring with the canonical bundle converts between the two conventions. States and normal derivatives are algebraic. All pole orders on an individual input are finite.

4.1 Principal parts with spectators

Let V be a k-vector space with translation T. Its right differential module is $\mathscr{V} = \omega_X \otimes_k V$, with

$$(f\,dz \otimes a)\partial_z = -f'\,dz \otimes a - f\,dz \otimes Ta.$$

For a partition π of $[n]$, order the blocks $B_1, \dots, B_r$ by their largest labels, called their roots. Let $i_\pi : X^r \hookrightarrow X^n$ be its diagonal, and let j_r include the locus of distinct block positions. Put

$$\mathscr{V}_\pi = i_{\pi,*} j_{r,*} j_r^!(\mathscr{V}^{\boxtimes r}). \tag{4.1}$$

The closed direct image includes every finite normal derivative. The root order orients its density and the list of block states. Merging blocks $i < j$ removes root i and retains root j.

In a normal coordinate u, write $\delta_u = [u^{-1}]$ for the simple principal part. In the right normal transfer convention,

$$\mathrm{pp}_u(u^{-m-1}a) = (a\delta_u)\partial_u^m/m!, \qquad m \geq 0. \tag{4.2}$$

Regular series have zero principal part. A regular multiplier acts by its finite Taylor polynomial on any fixed normal derivative.

For example, let $u = z_i - z_j$, let $t = z_e - z_j \neq 0$ be a spectator separation, and let $r, \ell \geq 1$. Then

$$(u-t)^{-\ell}[u^{-r}] = (-1)^\ell \sum_{a=0}^{r-1} \binom{\ell + a - 1}{a} t^{-\ell-a}[u^{a-r}]. \tag{4.3}$$

This follows by expanding $(1 - u/t)^{-\ell}$. Terms with $a \geq r$ vanish in the principal-part module. Thus taking a residue retains the spectator poles and only the required normal jets.

Suppose V is a vertex algebra with vacuum $\mathbf{1}$, in degree zero. We use $Y(Ta,u) = \partial_u Y(a,u)$ and $Y(a,u)\mathbf{1} = e^{uT}a$. For a multiplier f in (4.1), define

$$r_{ij}(f;a_1,\dots,a_r) = \mathrm{pp}_{z_i-z_j}\big(f\,Y(a_i,z_i-z_j)a_j\big), \tag{4.4}$$

keeping every other state in root order. Extend this formula over the existing normal transfers. Lower truncation of the field and the finite pole order of f make its value finite.

The two translation identities are

$$\begin{aligned} r(f;Ta,b) &= \partial_u r(f;a,b) - r(\partial_u f;a,b),\\ r(f;a,Tb) &= Tr(f;a,b) - \partial_u r(f;a,b) + r(\partial_u f;a,b). \end{aligned}$$

Here ∂_u differentiates the rational representative. After tensoring with its density it is minus the right action. Together with spectator differentiation, they prove that (4.4) is a map of right differential modules.

Proposition 4.1 (The residue relations)**.** *Disjoint mergers commute. For three blocks $i<j<\ell$,*

$$r_{(ij),\ell}r_{ij} - r_{i,(j\ell)}r_{j\ell} + r_{j,(i\ell)}r_{i\ell} = 0. \tag{4.5}$$

The residues commute with restrictions to cut opens and with surviving exceptional collision restrictions. A vacuum in the first merged block gives the principal part of the multiplier. A vacuum in the second gives its principal part multiplied by the creation series $e^{uT}a$.

Proof. Disjoint mergers use disjoint state factors and independent normal variables. Their finite principal parts commute. Cuts and surviving collision restrictions use the same block variables and normal transfer. The vacuum assertions are the two vacuum field identities.

For three blocks put $s = z_i - z_\ell$ and $t = z_j - z_\ell$. Mutual locality and lower truncation give a common germ

$$m \in V[[s,t]][s^{-1},t^{-1},(s-t)^{-1}]$$

whose two separated expansions are $Y(a,s)Y(b,t)c$ and $Y(b,t)Y(a,s)c$. Indeed a sufficiently large power of $s-t$ makes these equal, and their common expansion has bounded negative powers of both s and t. Weak associativity identifies its expansion near $s=t$ with $Y(Y(a,s-t)b,t)c$.

Write $P_{a,b}$ for the oriented iterated principal part along $a=0$, then $b=0$. The three-channel identity for such a germ is

$$P_{s-t,t} - P_{t,s} + P_{s,t} = 0.$$

It is the boundary squared in the localization complex of the three lines. Partial fractions in s over $k((t))$ give a second verification. The poles at 0 and t give the two terms in the expansion at infinity. The polynomial part has zero second principal part.

Multiplication by powers of s, t, $s-t$ and differentiation preserve the identity, so it holds with every finite normal derivative. Transport the density once in each change of normal coordinates. Applying this identity to fm proves (4.5) with its displayed signs. □

For the construction below, a *residue system* means maps on (4.1) satisfying Proposition 4.1. This permits the same construction whenever these coefficient identities have been proved, independently of a presentation by fields.

4.2 The finite first-cut contraction

A cut $c \in \{1, \dots, n-1\}$ is *admissible* for π if each block lies on one side of c. Write $J(\pi)$ for the set of admissible cuts. On the cut open U_c of (1.1), a partition crossing c has zero restriction. An admissible partition splits into its left and right coefficients. Since its distinct-block locus already lies in U_c, its coefficient is the direct image from that open.

Give the partition sum the shifts and differential

$$C_n = \bigoplus_{\pi} \mathcal{V}_\pi[|\pi|], \qquad d_C|_\pi = \sum_{i<j} (-1)^{i-1} r_{ij}, \qquad C_0 = k. \tag{4.6}$$

The two orders of a disjoint merger have opposite signs. The three orders reaching one triple block give (4.5). Thus $d_C^2 = 0$.

Restriction and splitting on every U_c give a strong cut coproduct. The differential on the two sides is the tensor differential. Successive cuts restrict to the same open set.

The positive cut cobar has one cell (π, Q) for each $Q \subseteq J(\pi)$:

$$K_n = \bigoplus_{\pi, Q \subseteq J(\pi)} \mathcal{V}_\pi[|\pi| - |Q| - 1]. \tag{4.7}$$

For states in degree zero its cell degree is $|Q| + 1 - |\pi|$.

Write $D = v + \delta$, where v inserts a cut and δ merges two blocks in one of the segments of Q. If $r_{\le c}$ counts blocks before c, and $i < j$ are global block indices in segment ℓ, the signs are

$$v_c = (-1)^{\#\{q \in Q : q < c\} + 1 - r_{\le c}} \,\mathrm{id}, \qquad c \in J(\pi) \setminus Q, \tag{4.8}$$

$$\delta_{ij} = (-1)^{\ell - 1 + i} r_{ij}. \tag{4.9}$$

These follow from the cobar rule $d(tc) = -t(d_C c) + \sum (-1)^{|c'|+1} (tc')(tc'')$, where t has degree one and r_{before} counts the blocks in the segments preceding ℓ. Moving past the preceding segments contributes $\ell - 1 - r_{\mathrm{before}}$. The internal merger contributes $i - r_{\mathrm{before}}$. The split contributes $1 - r_{\le c} + r_{\mathrm{before}}$. This gives both signs. The residue relations and cut compatibility give $v^2 = \delta^2 = v\delta + \delta v = 0$.

A partition is *indecomposable* when $J(\pi)$ is empty. A merger cannot create an admissible cut, so the surviving cells form the subcomplex

$$G_n = \bigoplus_{J(\pi) = \varnothing} \mathcal{V}_\pi[|\pi| - 1], \qquad d_G|_\pi = \sum_{i<j} (-1)^i r_{ij}. \tag{4.10}$$

Let $i : G_n \to K_n$ be the inclusion and p_0 the summand projection. If $c_0 = \min J(\pi)$ exists, set

$$h_0(\pi, Q) = \begin{cases} (-1)^{1 - r_{\le c_0}} (\pi, Q \setminus \{c_0\}), & c_0 \in Q, \\ 0, & c_0 \notin Q. \end{cases} \tag{4.11}$$

On indecomposable partitions set $h_0 = 0$.

Theorem 4.2 (Finite residue retract)**.** *The finite formulas*

$$\Theta = \sum_{a=0}^{n-1} (-\delta h_0)^a, \qquad \rho = p_0 \Theta, \qquad H = h_0 \Theta \tag{4.12}$$

give maps of differential-module complexes and a homotopy

$$\rho D = d_G \rho, \quad Di = i\, d_G, \quad \rho i = 1, \quad DH + HD = 1 - i\rho, \quad H^2 = Hi = \rho H = 0.$$

They commute with monotone collision restrictions under the coefficient identifications of Chapter 1.

Proof. Adding and removing c_0 in either order gives the identity. The terms at any other cut cancel, since its position in the ordered cut list changes by one when c_0 is removed. Hence

$$vh_0 + h_0v = 1 - ip_0, \qquad h_0^2 = h_0i = p_0h_0 = 0, \qquad p_0i = 1.$$

Each δ lowers the number of blocks, so $(\delta h_0)^n = 0$ and $\Theta = (1 + \delta h_0)^{-1}$.

Put $E = 1 + \delta h_0$. Using $\delta i = i\, d_G$ gives

$$DE - Ev = \delta - \delta(vh_0 + h_0v) = \delta i p_0.$$

Since $h_0\delta i = 0$, one has $\Theta\delta i = \delta i$. Multiplication on both sides by Θ therefore gives $\Theta D = v\Theta + i\, d_G p_0\Theta$. Projecting proves the chain-map identity for ρ. It also gives

$$DH + HD = (vh_0 + h_0v + \delta h_0)\Theta = 1 - i\rho.$$

Finally $\Theta i = i$ and $\Theta h_0 = h_0$ give all side conditions. Every term is a finite composition of coefficient maps and keeps its full normal transfer.

Under a monotone collision only partitions whose blocks are unions of collision fibers survive. No admissible cut of such a partition lies inside a fiber. Its cuts therefore identify with those of the quotient partition, with the same order and block counts. This preserves h_0, its signs and the residues. The identifications compose because they preserve the same lists. □

At two positions,

$$K_2 = [\mathcal{V}_{1|2} \xrightarrow{(-r_{12},1)} \mathcal{V}_{12} \oplus \mathcal{V}_{1|2}], \qquad \rho(p,q) = p + r_{12}q,$$

in degrees $-1, 0$. The residue term in ρ is necessary.

At three positions,

$$G_3 = [\mathcal{V}_{2|13} \xrightarrow{-r_{2,(13)}} \mathcal{V}_{123}].$$

The first term has degree -1.

At four positions the terms, in increasing degree, are

$$\mathcal{V}_{2|3|14} \xrightarrow{\begin{pmatrix} -r_{2,14} \\ r_{3,14} \\ -r_{2,3} \\ 0 \end{pmatrix}} \begin{matrix} \mathcal{V}_{3|124} \oplus \mathcal{V}_{2|134} \\ \oplus\, \mathcal{V}_{23|14} \oplus \mathcal{V}_{13|24} \end{matrix} \xrightarrow{(-r_{3,124}, -r_{2,134}, -r_{23,14}, -r_{13,24})} \mathcal{V}_{1234}. \tag{4.13}$$

Their degrees are $-2, -1, 0$. The composite is $r_{3,124}r_{2,14} - r_{2,134}r_{3,14} + r_{23,14}r_{2,3} = 0$. The crossed summand $13|24$ has its own outgoing residue even though the degree--2 term does not map to it.

4.3 Transferred products

Put $K = (K_n)_{n\geq 1}$ and $G = (G_n)_{n\geq 1}$. Concatenation of cobar words inserts the cut between them. It gives the associative product μ on K, with the empty unit adjoined in arity zero. Let $s : K \to K[1]$ have degree -1. Set

$$b_1(sx) = -sDx, \qquad b_2(sx, sy) = (-1)^{|x|} s\mu(x,y), \qquad B_1(sx) = -sd_Gx.$$

Write $\bar{i} = sis^{-1}$, $\bar{\rho} = s\rho s^{-1}$ and $\bar{H} = -sHs^{-1}$. Define

$$F_1 = \bar{i}, \qquad L_r = \sum_{a+b=r} b_2(F_a \otimes F_b), \qquad F_r = -\bar{H}L_r, \qquad B_r = \bar{\rho}L_r \quad (r \geq 2). \tag{4.14}$$

Tensor composition includes the Koszul signs. Each F_r has degree zero and each B_r degree one.

Theorem 4.3. *The maps B_r define an A_∞-algebra on G for the cut product. The maps F_r define an A_∞ quasi-isomorphism $G \to K$. All sums are finite in each arity. On singleton inputs,*

$$B_n(sx_1, \dots, sx_n) = s\, r_{1n}(x_1, \dots, x_n), \qquad n \geq 2. \tag{4.15}$$

Its target is the coefficient of $2|\cdots|n-1|\{1,n\}$ in degree $2-n$ of G_n.

Proof. Let T^c_{cut} be the reduced tensor coalgebra for the cut product, with deconcatenation. Extend b_1 to $T^c_{\mathrm{cut}}(sK)$, writing the resulting differential as d, and extend b_2 as a coderivation β. The dg associative identities say $d\beta + \beta d = \beta^2 = 0$. The tensor extension of the retract has maps $I_0 = T^c_{\mathrm{cut}}(\bar{\imath})$, $P_0 = T^c_{\mathrm{cut}}(\bar{\rho})$ and homotopy, on length a,

$$H_0 = \sum_{j=0}^{a-1} (\bar{\imath}\bar{\rho})^{\otimes j} \otimes \bar{H} \otimes 1^{\otimes(a-j-1)}.$$

The terms telescope to $dH_0 + H_0 d = 1 - I_0 P_0$, with the three zero side conditions.

Define

$$Q = \beta(1 + H_0\beta)^{-1}, \quad I = (1 + H_0\beta)^{-1} I_0, \quad P = P_0(1 + \beta H_0)^{-1}.$$

Every inverse is a finite geometric series, since β lowers tensor length. Substituting $Q = \beta - \beta H_0 Q$ into the contraction identity gives

$$dQ + Qd = -QI_0P_0Q.$$

It follows by multiplication that $B = B_1 + P_0 Q I_0$ squares to zero, $(d+\beta)I = IB$, $P(d+\beta) = BP$ and $PI = 1$.

For deconcatenation Δ one has $\Delta H_0 = (H_0 \otimes 1 + I_0 P_0 \otimes H_0)\Delta$. Apply this identity to $I = I_0 - H_0\beta I$ and $P = P_0 - P\beta H_0$. Induction on tensor length, using $H_0 I = P H_0 = 0$, shows that I, P are coalgebra maps. Consequently $B = P(d+\beta)I$ is a coderivation. Its length-one components are exactly (4.14). These facts prove the A_∞ identities and the quasi-isomorphism assertion.

For singleton inputs let e_n denote the uncut cell and w_c the cell with sole cut c. The product and suspension signs give, inductively,

$$L_n = \sum_{c=1}^{n-1} (-1)^{1-c} s w_c, \qquad H w_1 = e_n, \qquad H w_c = 0 \ (c > 1).$$

After removing the first cut, there is no cut left for another h_0. Thus $F_n = s e_n$.

Also $\rho w_c = 0$ for $c > 1$, whereas $\rho w_1 = -p_0 \delta e_n$. Of the single mergers of the singleton partition, only $1 \sim n$ leaves no admissible cut. Its sign in δ is negative. This proves (4.15). □

Corollary 4.4 (The coefficients of higher mergers)**.** *Let $m \geq 2$, and take inputs x_a with indecomposable partitions $\pi_1, \dots, \pi_m$ on consecutive groups of positions. Put $r_a = |\pi_a|$ and $r = \sum_a r_a$. Number the blocks of their union in root order. For state coefficients in degree zero,*

$$B_m(sx_1, \dots, sx_m) = s \sum_{i=1}^{r_1} \sum_{j=r-r_m+1}^{r} (-1)^{i+1} r_{ij}(x_1, \dots, x_m). \tag{4.16}$$

Here r_{ij} acts on the union coefficient.

Define the desuspended map $m_m = s^{-1} B_m s^{\otimes m}$. Since $|x_a| = 1 - r_a$, its formula is

$$m_m(x_1, \dots, x_m) = \sum_{i=1}^{r_1} \sum_{j=r-r_m+1}^{r} (-1)^{i+1+\sum_{a=1}^{m}(m-a)(1-r_a)} r_{ij}(x_1, \dots, x_m). \tag{4.17}$$

Every formal merger coefficient for this contraction is 1 or -1.

Proof. The union partition has admissible cuts exactly at the input boundaries. We prove by induction that F_a sends its suspended inputs to the suspended uncut union with coefficient one. This holds for F_1.

A split $p + q = a$ in (4.14) creates a cell with one cut, after input p. On such a cell $H = h_0$. Indeed, removing its cut leaves no cut for the next h_0, and a residue does not insert one. Only $p = 1$ survives, because h_0 removes the first admissible cut. The signs of b_2 and h_0 are both $(-1)^{1-r_1}$. They cancel, proving the induction.

At the root, the only part of ρ that survives on a one-cut cell is $-p_0 \delta h_0$. Again only $p = 1$ contributes. The product and homotopy signs cancel as above. The merger r_{ij} then has coefficient $(-1)^{i+1}$. A single merger removes every input boundary exactly when it joins the first input to the last. Each such merger survives p_0 once, proving (4.16).

Evaluating $s^{\otimes m}$ on homogeneous inputs contributes the sign $(-1)^{\sum_a (m-a)|x_a|}$ by the tensor Koszul rule. This proves (4.17). □

For example, on inputs with partitions $2|13$, 4 and 5, the desuspended ternary map is $r_{2,5} - r_{13,5}$. The formula fixes the first-cut contraction. It makes no claim about coefficients obtained from a different contraction.

On three singleton inputs, Theorem 4.3 gives the homotopy $h = r_{13}$ and

$$d_G h = r_{(12),3} r_{12} - r_{1,(23)} r_{23}.$$

In arity four the expression $L_4 = b_2(F_1, F_3) + b_2(F_2, F_2) + b_2(F_3, F_1)$ contains the five planar binary trees. The square-zero identity for B includes all their boundary terms with the Koszul signs. Its maps are defined on the common localized coefficient. They use no identification of incompatible iterated Laurent series.

The construction also applies to dg state coefficients when residues are cochain maps and all permutations use the Koszul rule. On a cell shifted by $r - |Q| - 1$, the internal differential is multiplied by $(-1)^{r-|Q|-1}$. Tensoring the first-cut contraction with the state complexes gives the same proof.

For a concrete three-position value, take the Heisenberg current with $[J_m, J_n] = \kappa m \delta_{m+n,0}$, so $J = J_{-1}\mathbf{1}$ and $TJ = J_{-2}\mathbf{1}$. Set $s = z_1 - z_3$, $t = z_2 - z_3$ and use the multiplier t^{-2} on three current inputs. Let L, R, N first merge 12, 23, 13, respectively, and then merge the remaining pair. In the full normal module at $s = t = 0$ their values are

$$\begin{aligned} L &= \kappa(s^{-2}t^{-2} + 2s^{-3}t^{-1})J, \\ R &= \kappa(2s^{-2}t^{-2} + 2s^{-3}t^{-1})J + \kappa s^{-2}t^{-1}TJ, \\ N &= \kappa s^{-2}t^{-2}J + \kappa s^{-2}t^{-1}TJ. \end{aligned}$$

These are principal-part classes, with their density understood. Indeed

$$\mathrm{pp}_t(t^{-2}Y(J,t)J) = \kappa t^{-4}\mathbf{1} + t^{-2}J^2 + t^{-1}(TJ)J.$$

Contraction of the remaining current with J^2 gives $2\kappa s^{-2}J$. Contraction with $(TJ)J$ gives $2\kappa s^{-3}J + \kappa s^{-2}TJ$. This proves the value of R. The creation series gives N.

For L, use the normal transfer

$$[(s-t)^{-2}t^{-2}] = [s^{-2}t^{-2}] + 2[s^{-3}t^{-1}],$$

obtained by differentiating the simple double principal part. Thus $L - R + N = 0$ and $N \neq 0$. Removing the nonconsecutive channel would remove the required homotopy while leaving a nonzero associativity defect.

4.4 Wick comparisons across a cut

Two different comparisons arise for polynomial Heisenberg coefficients. The first changes the field kernel. The second contracts a Wick operator by a homotopy of the primitive complex. We specify their domains separately.

Use polynomial states $V = k[x_1, x_2, \dots]$ with $Tx_p = px_{p+1}$, vacuum 1 and current $J = x_1$. For an even Laurent polynomial $P(u)$, put

$$K^P_{pq}(u) = \frac{(-1)^{q-1}P^{(p+q-2)}(u)}{(p-1)!(q-1)!}, \qquad C_P(u) = \sum_{p,q\geq 1} K^P_{pq}(u)\,\partial_{x_p} \otimes \partial_{x_q}. \tag{4.18}$$

With m the ordinary polynomial multiplication, the fields are $Y_P(u) = m(e^{uT} \otimes 1)e^{C_P(u)}$.

Only finitely many derivatives act on each polynomial, and each contraction lowers polynomial degree by two. The exponential therefore terminates. The recurrences $pK^P_{p+1,q} = (K^P_{pq})'$ and $qK^P_{p,q+1} = -(K^P_{pq})'$ give translation covariance. Commuting contractions give the common three-field expansion with exponent $C_{P,12} + C_{P,13} + C_{P,23}$. Evenness gives mutual locality after clearing its finitely many poles. Thus these fields give the residue system just constructed.

For two such kernels put $H = P - Q$. On a partition coefficient with distinct block roots define

$$\Gamma^H_\pi = \exp\left(\sum_{a<b} C_H(z_a - z_b)_{ab}\right).$$

Extend this horizontal operator to the closed direct image in (4.1). For two separated groups A, B in their word order, let $C_{H;A,B}$ be the sum of contractions between the groups. Write $\Delta^P_{A,B}$ for the cut restriction and splitting in the coefficient system of Y_P.

Proposition 4.5. *The operators Γ^H are horizontal isomorphisms of the collision complexes, preserve vacuum coefficients, and satisfy*

$$\Gamma^H_{\pi/(i\sim j)} r^P_{ij} = r^Q_{ij}\Gamma^H_\pi, \tag{4.19}$$

$$(\Gamma^H_A \boxtimes \Gamma^H_B)\Delta^P_{A,B}(\Gamma^H_{A\cup B})^{-1} = e^{-C_{H;A,B}}\Delta^Q_{A,B}. \tag{4.20}$$

The cross factors obey the three-group cocycle identity and commute with mergers inside either group.

Proof. The recurrences in (4.18) give $[\partial_{z_a} + T_a, C_{H,ab}] = 0$ and the corresponding identity at b, so Γ^H is horizontal. At a merger, a derivative of the merged polynomial splits into its derivatives on the two inputs. Taylor translation changes each spectator separation by the collision coordinate. The internal contraction satisfies $Y_P = Y_Q e^{C_H}$. These identities, followed by the finite Taylor rule (4.3), prove (4.19).

Evenness makes the same argument valid after permuting blocks into adjacent positions. It also preserves the root-change transfers. Vacuum derivatives vanish.

Split the exponent of $\Gamma^H_{A\cup B}$ into its two internal sums and $C_{H;A,B}$. All contractions commute. Cancelling the internal sums gives (4.20). For three groups both iterated cross factors have exponent $-C_{H;A,B} - C_{H;A,C} - C_{H;B,C}$. The merger rule already proved gives compatibility within each group. □

For $Q = 0$, applying $(\Gamma^P)^{-1}$ to regular polynomial tensors gives residue-free sections, including

$$J_1J_2 - P_{12}, \qquad J_1J_2J_3 - P_{12}J_3 - P_{13}J_2 - P_{23}J_1, \qquad P_{ij} = P(z_i - z_j).$$

The zero-kernel residue-free lattice itself contains polar sections: $u^{-1}(J \otimes 1 - 1 \otimes J)$ fuses to $(e^{uT}J - J)/u$, which is regular. It is therefore larger than the unlocalized external product. Also $\Gamma^H_1 = 1$ while $\Gamma^H_2(J \otimes J) = J \otimes J + H(u)1 \otimes 1$. The coefficient comparison is generally not the tensor power of a state map.

The primitive complex permits a homotopy of the cross contraction itself. Fix $\kappa \in k^\times$ and use $K_{\kappa,n}$ of (3.3) and its polynomial algebra $Z_{\kappa,n}$, with $dx(f) = -\kappa\gamma(f')$ and $d\gamma(f) = 0$. On an open separating groups s, t, the principal parts split by groups by Proposition 3.9.

For proper rational f in group s and g in group t, let Res_{D_t} sum residues at the whole divisor of group t. Define a degree-zero contraction Δ_{st} by its pair value

$$\Delta_{st}(x(f)x(g)) = \kappa \operatorname{Res}_{D_t} g\, df, \tag{4.21}$$

and zero on pairs involving a ghost. It acts by summing over unordered pairs of factors in a polynomial, with the Koszul sign. Define degree--1 pair contractions by

$$K^L_{st}(\gamma(f)x(g)) = -\operatorname{Res}_{D_t} f\, g\, dw, \qquad K^R_{st}(x(f)\gamma(g)) = -\operatorname{Res}_{D_s} f\, g\, dw, \tag{4.22}$$

and zero on the other oriented ghost pair.

Theorem 4.6 (Polynomial homotopies for Wick contraction)**.** *On this separated-group open, $[d, K^L_{st}] = [d, K^R_{st}] = \Delta_{st}$. The contractions commute with the position connections and internal collisions. For finitely many group pairs e, choose K_e to be either primitive or their average. Put $\Omega = k[t_e, dt_e]_e$, with $|t_e| = 0$, $|dt_e| = 1$ and $d_\Omega t_e = dt_e$. With these polynomial forms written on the left, the map*

$$\mathcal{F} = \exp\left(\sum_e t_e\Delta_e + dt_e K_e\right) \tag{4.23}$$

from $Z_{\kappa,n}$ to $\Omega \otimes_k Z_{\kappa,n}$ is a cochain map. Its faces and their integrals give compatible Wick comparisons and homotopies for refinements of the groups.

Proof. On $x(f)x(g)$, the left anticommutator is $\kappa \operatorname{Res}_{D_t} gf'\, dw$. The right one is $\kappa \operatorname{Res}_{D_s} f\, g'\, dw$. Integration by parts identifies them: the proper rational differential $f\, g'\, dw$ has no residue at infinity, so the sum over the two groups is zero. On pairs with a ghost both anticommutators vanish. Pair values determine these constant contraction operators.

Group residue is computed by monic division. It commutes with parameter differentiation and with internal collision substitution, without inverting differences inside the group. Thus the operators preserve these maps. Constant graded derivatives supercommute.

Since forms stand on the left,

$$[d + d_{\mathbf{t}}, t_e\Delta_e + dt_e K_e] = dt_e\Delta_e - dt_e[d, K_e] = 0.$$

The degree-zero exponent terms commute and lower polynomial degree. Their finite exponential is a cochain map. Restriction to faces gives the endpoint maps. Polynomial Stokes gives their homotopy identities. Refinement partitions the same set of cross pairs, so all its comparisons are faces of this formula. □

For $\Delta = \sum_e \Delta_e$ and $K = \sum_e K_e$, the one-parameter integral is

$$S = K \sum_{j \geq 0} \frac{\Delta^j}{(j+1)!}, \qquad [d, S] = e^\Delta - 1.$$

This sum terminates on every polynomial. For $f = (w - z_1)^{-m}$, $g = (w - z_2)^{-n}$ and $u = z_1 - z_2$, the pair values are

$$\beta_{mn} = \kappa(-1)^m \frac{(m+n-1)!}{(m-1)!(n-1)!} u^{-m-n}, \tag{4.24}$$
$$K^L_{mn} = (-1)^{m+1}\binom{m+n-2}{n-1} u^{1-m-n}, \qquad K^R_{mn} = (-1)^m \binom{m+n-2}{m-1} u^{1-m-n}.$$

These follow by Taylor expansion at the indicated group's pole. At $m = n = 1$ the primitives are u^{-1} and $-u^{-1}$, and the contraction is $-\kappa u^{-2}$. The sign uses the rational-generator convention $dx(f) = -\kappa\gamma(f')$. Equation (4.18) uses the current convention with $Y_P(J,u)J = P(u) + (e^{uT}J)J$.

The polynomial cohomology map of Theorem 3.17 can be composed with these homotopies. Their domain allows every internal collision but keeps different groups disjoint. Extension through an external collision with both vacuum insertions remains an additional comparison problem.

4.5 Derived intersections of diagonal coefficients

The diagonal filtration of H_n in Theorem 3.5 has nonflat quotients. Already for $R = k[u]$ and $P = R[u^{-1}]/R$, every tensor $p \otimes q$ vanishes: some u^m kills p, while $q = u^m q'$ for a principal part q'. Hence $P \otimes_R P = 0$. The flat resolution $[R \to R[u^{-1}]]$ in degrees $-1, 0$ instead gives

$$P \otimes^{\mathrm{L}}_R P \simeq P[1].$$

This degree is needed to pass from the diagonal filtration to exterior powers.

Continue over $R_n = k[z_1, \dots, z_n]$ with its left Weyl algebra D_n. For a partition π, let Δ_π be its diagonal and put $c_\pi = n - |\pi|$. Write

$$A_\pi = R\Gamma_{\Delta_\pi} R_n, \qquad \delta_\pi = H^{c_\pi}_{\Delta_\pi}(R_n).$$

Choose an ordered list of independent linear normal equations $f_1, \dots, f_{c_\pi}$. Then

$$A_\pi \simeq \bigotimes_{a=1}^{c_\pi} [R_n \longrightarrow R_n[f_a^{-1}]] \simeq \delta_\pi[-c_\pi], \tag{4.25}$$

where each bracket has degrees zero and one.

The localizations are flat and carry their derivative connections. The cohomology is concentrated in degree c_π: after a linear coordinate change this is the tensor product of the one-variable principal-part calculation. Its Weyl module is the trivial module on the diagonal, pushed forward with all normal derivatives.

Proposition 4.7. *Let $\pi \vee \sigma$ impose both sets of equalities. There is an equivalence of differential-module complexes*

$$\delta_\pi \otimes^{\mathrm{L}}_{R_n} \delta_\sigma \simeq \delta_{\pi\vee\sigma}[c_\pi + c_\sigma - c_{\pi\vee\sigma}]. \tag{4.26}$$

The localization complexes retain the symmetric tensor coherence. Ordered normal coordinates specify the signs in their top-principal-part identifications.

Proof. Tensoring the localization complexes concatenates the lists of normal equations. This computes local cohomology on the intersection of their supports, so $A_\pi \otimes^{\mathrm{L}} A_\sigma \simeq A_{\pi\vee\sigma}$. Restoring the shifts in (4.25) gives (4.26). Localization, its counit and their tensor maps commute with differentiation. Permutations of equations act by the Koszul signs. A change of independent normal basis uses the corresponding determinant in the principal-part identification. □

For a graph G on $[n]$, allow repeated edges but no loops, and let π_G be its connected-component partition. Repeated application of (4.26) gives

$$\bigotimes_{e\in E(G)}^{\mathrm{L}} \delta_e \simeq \delta_{\pi_G}[\beta_1(G)], \qquad \beta_1(G) = |E(G)| - n + |\pi_G|. \tag{4.27}$$

A spanning forest supplies independent normals. Each remaining edge supplies one shift. For example, with $a = z_1 - z_2$ and $b = z_2 - z_3$ and $p, q \geq 1$, every $[a^{-p}b^{-q}]$ is killed by $(a+b)^{p+q-1}$. Localizing at the third edge therefore gives zero, and the triangle tensor is $\delta_{123}[1]$. The coherent structure sheaf R_n/I_π, where I_π defines the diagonal, is a different module and has different derived intersections.

In characteristic zero, the derived exterior power $\mathrm{L}\Lambda^r M$ is the image of $r!^{-1}\sum_{\sigma\in S_r}\operatorname{sgn}(\sigma)\sigma$ on $M^{\otimes^{\mathrm{L}} r}$. Permutations here include the Koszul signs of the complex.

Lemma 4.8. *If $\delta = H^c_Z(R_n)$ for a linear diagonal Z and $r > 1$, then*

$$\mathrm{L}\Lambda^r \delta \simeq \begin{cases} 0, & c \text{ even}, \\ \delta[(r-1)c], & c \text{ odd}. \end{cases}$$

For $r = 1$ the exterior power is δ.

Proof. Put $A = R\Gamma_Z R_n$. Its counit $A \to R_n$ gives a symmetric equivalence $A^{\otimes r} \simeq A$. Indeed the tensor is supported on Z. Its map to R_n lifts uniquely, in the derived mapping space, through the right adjoint $R\Gamma_Z$. Applying the counit to all but one factor gives equivalences with this same lift. This supplies their symmetric coherence.

Since $\delta = A[c]$, a permutation of the shifted factors introduces the character sgn^c. The alternating idempotent vanishes for even c and is the identity for odd c. In the latter case the shift is $rc - c$. In particular the flip on $P \otimes^{\mathrm{L}} P \simeq P[1]$ is minus the identity. □

4.6 Exterior powers of the primitive coefficient

Let $S(b, j)$ be the number of partitions of a b-element set into j nonempty blocks. For $b \geq 2$ define

$$Q_b(t) = \sum_{j=1}^{b} S(b,j)(j-1)!\,t^j, \qquad s_1(\pi) = \#\{B \in \pi : |B| = 1\}.$$

For example, $Q_2 = t + t^2$ and $Q_3 = t + 3t^2 + 2t^3$.

Theorem 4.9 (Diagonal multiplicities)**.** *For $q \geq 0$, the exterior power $\Lambda^q_{R_n} H_n$ is flat over R_n and has finite length over D_n. It vanishes for $q > n$. The composition multiplicity of δ_π is*

$$m_{\pi,q} = [t^q](1+t)^{s_1(\pi)} \prod_{\substack{B\in\pi \\ |B|\geq 2}} Q_{|B|}(t). \tag{4.28}$$

Consequently

$$\operatorname{length}_{D_n} \Lambda^q H_n = q!\,S(n+1, q+1). \tag{4.29}$$

These are composition multiplicities. The resulting modules need not split into their simple constituents.

Proof. An exterior power is a direct summand of the corresponding tensor power, hence is flat by Corollary 3.2. It is torsion-free over the domain R_n. Its generic fiber is the exterior power of the n-dimensional residue space, which proves vanishing above n.

Consider the Grothendieck group of bounded differential-module complexes whose cohomology has finite filtrations by the δ_π. These diagonal modules are simple. On a finite sum, polynomials in the coordinate Euler operators isolate a nonzero monomial. Tangential differentiation and normal multiplication then reduce it to the simple normal principal part. The reverse operations generate every monomial.

Modules with different supports are nonisomorphic. Thus their classes are independent. Derived tensor remains in this category by Proposition 4.7.

Set $e_\pi = (-1)^{c_\pi}[\delta_\pi]$. Their multiplication is $e_\pi e_\sigma = e_{\pi\vee\sigma}$. Write $\lambda_t(M) = \sum_{r\ge 0}[\mathrm{L}\Lambda^r M]t^r$. For an exact triangle $A \to B \to C$, the finite tensor filtration, followed by the alternating idempotent, gives $\lambda_t(B) = \lambda_t(A)\lambda_t(C)$. The preceding lemma therefore gives

$$\lambda_t(\delta_\pi) = \begin{cases} 1 + e_\pi t, & c_\pi \text{ even}, \\ (1 + e_\pi t)^{-1}, & c_\pi \text{ odd}. \end{cases}$$

Each coefficient of this formal series involves only a finite derived calculation.

For a partition τ, evaluate e_π as one when π refines τ and as zero otherwise. These are ring homomorphisms, and the refinement matrix is triangular with diagonal entries one. The subset filtration of H_n gives

$$\mathrm{ev}_\tau\, \lambda_t(H_n) = (1+t)^{\sum_{B\in\tau}\sum_{s=1}^{|B|}(-1)^{s-1}\binom{|B|}{s}} = (1+t)^{|\tau|}.$$

The n singleton constituents are separate copies of the same regular module. The exponent counts all of them. These evaluations are operations on classes, not fibers of modules.

Write $\lambda_t(H_n) = \sum_\pi a_\pi(t) e_\pi$. Möbius inversion in the refinement order gives

$$a_\pi(t) = \prod_{B\in\pi} p_{|B|}(1+t), \qquad p_b(x) = \sum_{j=1}^{b} S(b,j)(-1)^{j-1}(j-1)!\,x^j.$$

To see the coefficient, a partition of one block into j pieces has Möbius value $(-1)^{j-1}(j-1)!$ when those pieces are joined. Values multiply on different blocks. Their generating series is

$$\sum_{b\ge 1} p_b(x)\frac{z^b}{b!} = \log\bigl(1 + x(e^z - 1)\bigr).$$

At $x = 1 + t$ this becomes $z + \log(1 + t(1 - e^{-z}))$. Hence $p_1(1+t) = 1 + t$ and $(-1)^{b-1}p_b(1+t) = Q_b(t)$ for $b \ge 2$. The sign changing e_π back to $[\delta_\pi]$ cancels these block signs and proves (4.28).

The finite derived tensor filtrations have finitely many diagonal simple constituents in each exterior degree. Flatness identifies the derived exterior power with the ordinary one in degree zero. The computed coefficients are therefore actual composition multiplicities. Summing over partitions, with the empty-set term included, gives

$$\sum_{n\ge 0}\sum_{q} \operatorname{length}\Lambda^q H_n\, t^q \frac{z^n}{n!} = \frac{e^z}{1 - t(e^z - 1)}.$$

The coefficient of t^q is $e^z(e^z-1)^q = (q+1)^{-1}\partial_z(e^z-1)^{q+1}$. Its coefficient of $z^n/n!$ is $q!S(n+1, q+1)$. □

At three positions the exterior lengths are 1, 7, 12, 6. In degree two there are three regular constituents, two on each pair diagonal and three on the total diagonal. The generic rank is three, while the differential-module length is twelve.

For a constant d-dimensional space V_0, with $d \geq 1$, write $m^{(d)}_{\pi,q}$ for the composition multiplicity of δ_π in $\Lambda^q(V_0 \otimes H_n)$ and put $y = (1+t)^d - 1$. The same evaluation and inversion give

$$\sum_q m^{(d)}_{\pi,q} t^q = (1+t)^{d s_1(\pi)} \prod_{\substack{B \in \pi \\ |B| \geq 2}} Q_{|B|}(y), \tag{4.30}$$

$$\sum_q \operatorname{length} \Lambda^q(V_0 \otimes H_n) t^q = \sum_{r=0}^{n} r! S(n+1, r+1)\, y^r.$$

Indeed its exterior series evaluates at τ to $(1+t)^{d|\tau|}$. Flatness and generic rank give vanishing above dn. These are also the polynomial Heisenberg cohomology modules when the constant level form is invertible. The form is an invertible change of variables in the primitive differential of Theorem 3.17.

Proposition 4.10 (Resolutions and the determinant)**.** *A flat differential-module resolution of $\Lambda^q H_n$ has terms*

$$E_q^{-a} = \operatorname{Sym}^a(M_n) \otimes_{R_n} \Lambda^{q-a}(M_n), \qquad 0 \leq a \leq q,$$

with differential, for $x_i, y_j \in M_n$ and $b = q - a$,

$$d(x_1 \cdots x_a \otimes y_1 \wedge \cdots \wedge y_b) = \sum_{j=1}^{a} x_1 \cdots \widehat{x_j} \cdots x_a \otimes \partial_w x_j \wedge y_1 \wedge \cdots \wedge y_b.$$

Its augmentation is the quotient in degree zero. There is also an ordinary free resolution

$$0 \longrightarrow \bigoplus_m \Lambda^q F_m \xrightarrow{1-\mathrm{shift}(\Lambda^q T_m)} \bigoplus_m \Lambda^q F_m \longrightarrow \Lambda^q H_n \longrightarrow 0, \tag{4.31}$$

using the free stages of Theorem 3.1. In residue coordinates, with $\Delta_n = \prod_{i<j}(z_i - z_j)$,

$$\Lambda^n H_n \simeq R_n[\Delta_n^{-1}], \qquad \Lambda^{dn}(V_0 \otimes H_n) \simeq R_n[\Delta_n^{-1}] \otimes \det(V_0)^{\otimes n}. \tag{4.32}$$

These last maps are differential-module isomorphisms. Each module has length $n!$, with multiplicity $\prod_{B \in \pi}(|B|-1)!$ on Δ_π.

Proof. The flat complex $[M_n \xrightarrow{\partial_w} M_n]$ in degrees $-1, 0$ resolves H_n. Alternating its q-fold tensor gives E_q. The negative-degree variables become symmetric, and the differential is the displayed sum. Two replacements cancel by their exterior signs. Since H_n is flat, the alternating idempotent preserves the tensor-power quasi-isomorphism, proving exactness.

Exterior powers commute with filtered colimits, so the cokernel in (4.31) is the required module. The least occupied input stage of a nonzero finite sum survives under $1 - \mathrm{shift}$, proving injectivity. Its matrix entries are the corresponding minors of T_m. This is an ordinary-module resolution: a position derivative can increase the pole stage. The first resolution itself retains the derivative connections, coefficient base change and the block decompositions.

For the top exterior power, every transition determinant is a nonzero scalar times Δ_n^2 by (3.7). After scalar rescaling, its colimit is the localization at Δ_n. In generic residue coordinates the connection is ordinary differentiation. This proves the first isomorphism in (4.32). The determinant identity for $V_0 \otimes H_n$ proves the second.

Permuting labels acts on the residue determinant with its permutation sign, and on the second determinant with that sign to the power d. The top coefficients of (4.28) and (4.30) give the asserted multiplicities. Their sum counts permutations by their cycle partitions, and is $n!$. □

4.7 The support left by every consecutive cut

The same partition geometry determines the range of the local vacuum coefficient. For this section X may be any smooth geometrically connected curve over k. Recall

$$Z_n = \bigcap_{c=1}^{n-1} \bigcup_{i \le c < j} \Delta_{ij}$$

from (1.8). We use its reduced closed structure. Local cohomology depends only on its support. An equality partition lies in Z_n precisely when each gap is crossed by one of its blocks.

A *crossing chain* of length r consists of pair blocks $\{i_a, j_a\}$ and singleton blocks at the other positions, where

$$1 = i_1 < i_2 < j_1 < i_3 < j_2 < \cdots < i_r < j_{r-1} < j_r = n. \tag{4.33}$$

For $r = 1$ this means just the pair $\{1, n\}$.

Proposition 4.11. *The irreducible components of Z_n are exactly the diagonals of crossing chains. There are $\binom{n-2}{2r-2}$ components of codimension r, for $1 \le r \le \lfloor n/2 \rfloor$. For $n \ge 3$ their total number is 2^{n-3}.*

Proof. A component is a diagonal of a refinement-minimal indecomposable partition. A block crosses exactly the gaps between its least and greatest labels. Replacing that block by the pair of these endpoints and making its other labels singletons preserves the crossed gaps. Minimality therefore forces pair and singleton blocks.

The intervals $[i_a, j_a - 1]$ of its pair blocks cover all gaps, irredundantly. In order of increasing i_a, the j_a also increase, since otherwise an interval is contained in another. The next interval must start before the previous pair ends: $i_{a+1} < j_a$. Equality is impossible because pair blocks have disjoint endpoints. Irredundancy of the middle interval then gives $j_a < i_{a+2}$.

The endpoints of the cover are $1, n$. These are precisely (4.33), and the same inequalities prove conversely that every chain is an irredundant cover.

Each pair contributes one independent normal equation. A chain is determined by its $2r - 2$ internal endpoints, whose order fixes their roles. This proves the codimension and count. Summing the even binomial coefficients gives 2^{n-3} for $n \ge 3$. □

On an affine-line chart one can compute the full supported object without splitting it into components. Put $f_c = \prod_{i \le c < j} (z_i - z_j)$. For any complex M of coefficients,

$$R\Gamma_{Z_n} M \simeq M \otimes^{\mathrm{L}}_{R_n} \bigotimes_{c=1}^{n-1} [R_n \longrightarrow R_n[f_c^{-1}]]. \tag{4.34}$$

Each bracket has degrees zero and one. The tensor computes successive supports on $V(f_c)$, hence their intersection. Its signs use the displayed gap order. It is a flat model and commutes with ordinary derived coefficient base change.

Let

$$\mathcal{U}_n = R\Gamma_{Z_n}(\omega_{X^n})[n-1]$$

be the positive coefficient of the cut cobar of the local vacuum. This is Theorem 1.4 applied to $\mathcal{J}_n = \omega_{X^n}[n]$. For a nonzero bounded complex M, define its width to be the largest nonvanishing cohomological degree minus the smallest. Write $i_{1n} : X^{n-1} \hookrightarrow X^n$ for the diagonal identifying the first and last positions.

Theorem 4.12 (Vacuum amplitude). *For $n \geq 2$, the lowest cohomology of $\mathcal{U}_n$ is*

$$H^{2-n}(\mathcal{U}_n) = i_{1n,*}\omega_{X^{n-1}}.$$

For every $1 \leq c \leq \lfloor n/2 \rfloor$, $H^{c+1-n}(\mathcal{U}_n) \neq 0$. Thus

$$\operatorname{width}(\mathcal{U}_n) \geq \lfloor n/2 \rfloor - 1.$$

For $2 \leq n \leq 9$ equality holds, and $H^j(\mathcal{U}_n) = 0$ for $j > \lfloor n/2 \rfloor + 1 - n$.

Proof. Take $V = k$ and $T = 0$ in the residue construction. Its maps are scalar principal parts. The resulting partition complex resolves $\omega_{X^n}[n]$. To see this, filter X^n by the closed unions of collision strata of increasing codimension. The successive localization triangles have, on each stratum of codimension c, only its degree-c normal principal-part module.

Its extension from the distinct-root locus is exactly (4.1). Locally this open inclusion is a principal localization and has no higher direct images. The boundary between successive strata is the oriented binary principal part.

After the common shift $[n]$, totalizing this finite filtration gives (4.6) and resolves $\omega_{X^n}[n]$. Applying the supported-cobar theorem and the finite contraction gives G_n for $\mathcal{U}_n$.

The terms whose partition joins 1 to n form a subcomplex. They are the complete scalar Cousin resolution on Δ_{1n}, shifted by $n-2$, since every further coarsening remains indecomposable. There is only one indecomposable partition with $n-1$ blocks, namely $\{1, n\}$ and all other labels single. Thus the quotient begins in degree $3-n$. The long exact sequence proves the lowest-degree formula and vanishing below it.

At the generic point of a crossing-chain component of codimension c, no other component remains. Its local cohomology is the nonzero top principal-part module in degree c. Chains exist for every stated c. After the shift this proves the nonvanishing and the width bound.

For the upper bound in the finite range, form the poset $\mathcal{P}_n$ of intersections of nonempty families of components, ordered by inclusion of their diagonals. For $L \in \mathcal{P}_n$, let $c(L) = \operatorname{codim} L$ and let Δ_L be the order complex of the diagonals strictly containing L. The local-cohomology filtration for a linear arrangement has graded pieces

$$H^{c(L)}_L(R_n) \otimes_k \widetilde{H}_{c(L)-r-1}(\Delta_L; k) \quad \text{in } H^r_{Z_n}(R_n). \tag{4.35}$$

This is [3, Corollary 1.3]. Tensoring with the canonical bundle gives the right-module convention.

The following finite calculation determines the needed vanishing. Enumerate the chains of each strict interval. If N_q is the number of its q-simplices and ρ_q the rank of the boundary in characteristic two, put $b_q = N_q - \rho_q - \rho_{q+1}$. Use $N_{-1} = 1$, $\rho_{-1} = 0$ and the reduced augmentation: ρ_0 is one for a nonempty complex and zero for the empty one. Thus the empty complex has $b_{-1} = 1$.

Let $B_{n,r} = \sum_L b_{c(L)-r-1}(L)$. The boundary which deletes one vertex gives

n	$\lvert\mathcal{P}_n\rvert$	$B_{n,1}$	$B_{n,2}$	$B_{n,3}$	$B_{n,4}$
2	1	1	0	0	0
3	1	1	0	0	0
4	3	1	2	0	0
5	10	1	10	0	0
6	34	1	34	6	0
7	126	1	98	78	0
8	523	1	258	660	24
9	2430	1	642	4764	600

All other $B_{n,r}$ are zero. Boundary matrices have integer entries. Their ranks over characteristic zero are at least their ranks modulo two. Hence each reduced homology dimension in characteristic zero is at most the corresponding b_q. The table and (4.35) give $H^r_{Z_n} = 0$ for $r > \lfloor n/2 \rfloor$ in this range.

Étale coordinates at the collision strata identify the diagonal arrangement locally with the linear one. Local cohomology commutes with this flat pullback, giving the same vanishing on X^{n}. Shifting by $[n-1]$ and using the nonvanishing already proved completes the argument. □

The table gives dimensions in characteristic two, used only as bounds in characteristic zero. It does not split the filtration. The lower bound holds in every arity, so no choice of shifts makes all vacuum coefficients have a uniformly bounded width. The corresponding upper bound for $n \geq 10$ remains open.

Chapter 5

Covariant fields and Gaussian elimination

A contraction of complexes need not commute with moving the insertion points. The abelian line algebra gives a simple example: its cohomology consists of charges, but translating a positive mode can produce a charge. Differential forms on the space of positions record the missing homotopy. We construct that homotopy and retain the logarithmic connection that survives the contraction.

In Gaussian elimination, the hidden quadratic form can vary with the fields that remain. Its normalization then contributes to their effective differential. The lifts of their antifields also change. We compute both effects in a finite polynomial Batalin–Vilkovisky complex. Products of lifts require curvature corrections, whose connected contractions determine higher comparison maps.

Throughout, k is a field of characteristic zero and differentials have cohomological degree one. Odd differentiation is on the left. Tensor products and operator extensions use the Koszul rule. The line model and the finite Gaussian model have the separate coefficient rings and completions specified below.

5.1 Charges and spectral translation

Fix $\kappa \in k^\times$. Consider the graded commutative algebra

$$Y_\kappa = k[B_0, B_1, \ldots] \otimes \Lambda(C_0, C_1, \ldots), \qquad |B_n| = 0, \quad |C_n| = 1.$$

Define derivations Q, T, L of degrees 1, 0, -1 by

$$\begin{aligned} QB_n &= \kappa n C_{n-1}, & QC_n &= 0, \\ TB_n &= nB_{n-1}, & TC_n &= nC_{n-1}, \\ LC_n &= \kappa^{-1} B_n, & LB_n &= 0. \end{aligned} \tag{5.1}$$

Terms with index -1 are zero. The relations

$$Q^2 = L^2 = 0, \qquad QL + LQ = T, \qquad [T, L] = [T, Q] = 0$$

follow on generators and hence on the algebra.

Give $k[J]$ degree zero and zero differential. Let $\pi : Y_\kappa \to k[J]$ send B_0 to J and every other generator to zero. Let $\iota(J) = B_0$.

Let N count the C_n and the B_n with $n > 0$, with multiplicity. Define

$$h = N^{-1} \sum_{n \geq 0} \frac{B_{n+1}}{\kappa(n+1)} \partial_{C_n} \quad \text{on } N > 0, \qquad h = 0 \quad \text{on } N = 0. \tag{5.2}$$

The sum is finite on each polynomial. If its numerator is H, then $QH + HQ = N$: each pair (B_{n+1}, C_n) contributes its Euler operator. Also $H^2 = 0$, and H preserves N. It follows that

$$Qh + hQ = 1 - \iota\pi, \qquad h^2 = h\iota = \pi h = 0, \qquad \pi\iota = 1. \tag{5.3}$$

Thus (π, ι, h) is a strong deformation retract, linear over $k[B_0]$. Its projection fails to preserve translation already on B_1, since $\pi(TB_1) = J$.

Proposition 5.1 (The de Rham correction)**.** *Let ϵ have degree one, with $\epsilon^2 = d\epsilon = 0$. Place forms on the left and put $W = 1 + \epsilon L$. Then*

$$(Q + \epsilon T)W = WQ.$$

The dg algebra map

$$\widetilde{\pi} = \pi(1 - \epsilon L) : \big(\Lambda(\epsilon) \otimes Y_\kappa, Q + \epsilon T\big) \longrightarrow \big(\Lambda(\epsilon) \otimes k[J], 0\big)$$

is a quasi-isomorphism. Together with ι and WhW^{-1} it is a strong deformation retract.

Proof. The operator ϵL is a degree-zero derivation with square zero. Its exponential is W, whose inverse is $1 - \epsilon L$. Extending Q over forms gives $Q(\epsilon a) = -\epsilon Q a$. Consequently the coefficient of ϵ in the intertwining equation is $T - QL = LQ$. Conjugating (5.3) proves the retract identities. Since $L\iota = 0$, the conjugated inclusion is still ι. □

For example, $\widetilde{\pi}(C_0) = -\epsilon J/\kappa$. The one-form term records the translation homotopy of a state whose strict charge projection vanishes.

Proposition 5.2 (Polynomial spectral transport)**.** *On polynomial inputs, the maps*

$$E_z = (1 + dz\, L)e^{zT} : Y_\kappa \longrightarrow \Omega^\bullet(\mathbb{A}^1_z) \otimes Y_\kappa$$

are dg algebra maps and satisfy $E_{z+w} = E_z E_w$ after pullback to two parameters. Let Δ_0 make every B_n, C_n primitive, and set $\widetilde{\Delta}_z = (E_z \otimes 1)\Delta_0$. The spectral coproducts satisfy

$$(\widetilde{\Delta}_u \otimes 1)\widetilde{\Delta}_v = (1 \otimes \widetilde{\Delta}_v)\widetilde{\Delta}_{u+v}.$$

On a generator a their common value is

$$E_{u+v}a \otimes 1 \otimes 1 + 1 \otimes E_v a \otimes 1 + 1 \otimes 1 \otimes a.$$

Proof. The derivation T lowers the mode index, so e^{zT} is finite on each polynomial. The identity $QL + LQ = T$, with forms on the left, gives $(Q + d_z)E_z = E_z Q$. Both factors of E_z are algebra automorphisms over forms. They commute under composition because $[T, L] = 0$. The possible product of two one-form terms vanishes by $L^2 = 0$. This proves the group law.

Each generator is sent to a linear combination of generators. The primitive coproduct therefore gives the displayed equality, which extends to all inputs by multiplicativity. □

The charge projection of this family is

$$\begin{aligned}
&(\pi \otimes \pi)\widetilde{\Delta}_z(B_0) = J_1 + J_2, \\
&(\pi \otimes \pi)\widetilde{\Delta}_z(B_n) = z^n J_1 && (n > 0), \\
&(\pi \otimes \pi)\widetilde{\Delta}_z(C_n) = \kappa^{-1} z^n J_1\, dz && (n \geq 0).
\end{aligned}$$

Pullback along $z \mapsto tz$, including $dz \mapsto t\, dz + z\, dt$, gives a polynomial homotopy to the primitive charge map at $t = 0$. Its composition law is the group law just proved.

5.2 The completed line connection

To sum all modes, fix a finite ordered set I of positions and write

$$R_I = k[z_i\ (i \in I), z_{ij}^{-1}\ (i < j)], \qquad z_{ij} = z_i - z_j, \qquad Y_I = \bigotimes_{i \in I} Y_\kappa.$$

Give B_n weight n and C_n weight $n+1$. We use the product completion

$$\widehat{Y}_{I,\mathrm{jet}} = \prod_{w \geq 0} (R_I \otimes Y_I)_w. \tag{5.4}$$

At each weight the coefficients are polynomials in the finitely many zero modes $B_0^{(i)}$. Localization in position differences precedes completion, so the finite pole order may vary with w. Multiplication involves only finitely many summands at any fixed weight. Differential forms have weight zero.

For two positions put $z = z_1 - z_2$ and define

$$r(z) = \sum_{n,m \geq 0} (-1)^n \binom{n+m}{n} \frac{C_n \otimes B_m - B_n \otimes C_m}{z^{n+m+1}}, \tag{5.5}$$

$$\mathcal{A} = r(z) + dz\, L_1 r(z), \qquad \rho = \kappa^{-1} J_1 J_2 \frac{dz}{z}, \tag{5.6}$$

where $L_1 = L \otimes 1$. Let $D_0 = Q + d_{\mathrm{dR}}$. The sums define continuous multiplication operators on the form-valued version of (5.4).

The positive-weight element

$$b(z) = -\frac{1}{\kappa} \sum_{N \geq 1} \frac{z^{-N}}{N} \sum_{a+c=N} (-1)^a \binom{N}{a} B_a \otimes B_c \tag{5.7}$$

is a primitive for all but the charge residue.

Lemma 5.3. *In the completed coefficient algebra,*

$$Qb = r, \qquad d_{\mathrm{dR}} b = dz\, L_1 r - \iota\rho, \qquad D_0 b = \mathcal{A} - \iota\rho.$$

In particular $D_0\mathcal{A} = 0$ and $(D_0 + \mathcal{A}\cdot)^2 = 0$ on coefficient modules.

Proof. Applying Q to B_a in (5.7) gives, for $a \geq 1$, the coefficient

$$-\frac{(-1)^a a}{N}\binom{N}{a} = (-1)^{a-1}\binom{N-1}{a-1}$$

of $C_{a-1} \otimes B_c$. Applying Q to B_c gives $-(-1)^a \binom{N-1}{a}$ on $B_a \otimes C_{c-1}$. These are exactly the terms of (5.5).

Differentiating $-z^{-N}/N$ gives $z^{-N-1}dz$. The resulting sum is $dz\, L_1 r$ with its $N = 0$ term $\kappa^{-1} B_0 \otimes B_0\, dz/z$ removed. The first two identities follow weight by weight. The form ρ is closed, so the last identity implies $D_0\mathcal{A} = 0$. Since $\mathcal{A}$ has degree one in a graded commutative algebra, $\mathcal{A}^2 = 0$. □

For instance the weight-one term is

$$b(z) = \frac{B_1 \otimes B_0 - B_0 \otimes B_1}{\kappa z} + \text{terms of weight at least two.}$$

It contains the first-mode information that appeared in the spectral projection.

On several sites use the sum of the numerators in (5.2), divided by the total number of ghosts and positive-index B's. This homotopy and Q preserve weight, so their contraction extends continuously to (5.4).

Theorem 5.4 (The line module contraction)**.** *Let $G = e^{-b}$, which is defined weight by weight. The maps*

$$I_{\mathcal{A}} = G\iota, \qquad P_{\mathcal{A}} = \pi, \qquad H_{\mathcal{A}} = GhG^{-1}$$

give a continuous strong deformation retract

$$\left(\prod_{w\geq 0} \Omega^\bullet(R_{\{1,2\}}) \otimes (Y_{\{1,2\}})_w, D_0 + \mathcal{A}\cdot\right) \rightleftarrows \left(\Omega^\bullet(R_{\{1,2\}}) \otimes k[J_1, J_2], d_{\mathrm{dR}} + \rho\cdot\right).$$

These are maps of module complexes.

Proof. Positive weight makes both G and G^{-1} well defined. Lemma 5.3 gives

$$(D_0 + \mathcal{A})G = G(D_0 + \iota\rho).$$

The original homotopy is linear over the zero-mode polynomial ring. Its extension over forms anticommutes with both d_{dR} and multiplication by the odd form ρ. Thus (5.3) also contracts the ρ-twisted complexes. Conjugation gives the asserted homotopy and its side conditions. Every term of b contains a positive-index B, so $\pi G = \pi G^{-1} = \pi$ and $P_{\mathcal{A}} I_{\mathcal{A}} = 1$. All identities are finite at each fixed weight. □

The endpoint vector is $I_{\mathcal{A}}(1) = G$. On the charge complex the differential sends 1 to ρ. A further scalar gauge with constant term one in the charges would require, at the coefficient of J_1J_2, a rational primitive of $-\kappa^{-1}dz/z$. Every rational derivative has zero residue, so such a gauge cannot remove this connection over $R_{\{1,2\}}$. Analytic choices of a logarithm would change the coefficient domain.

Corollary 5.5. *For any finite I, set*

$$b_I = \sum_{i<j} b_{ij}, \qquad \mathcal{A}_I = \sum_{i<j} (r_{ij} + dz_{ij} L_i r_{ij}), \qquad \rho_I = \kappa^{-1} \sum_{i<j} J_i J_j \, d\log z_{ij}.$$

Theorem 5.4 holds with $G_I = e^{-b_I}$ and the total homotopy. Conjugating specified continuous coefficient maps by these gauges preserves their composition identities.

Proof. The primitive identities add over pairs and all exponent terms commute. For a map f from a source model to a target model, the conjugate is $G_t^{-1} f G_s$. The adjacent gauge factors cancel in every composite. Splitting a set of positions into groups splits b_I into internal and cross terms, whose exponentials multiply. □

At three positions the pair sum includes 13 with its spectator. At four positions it includes the crossed pairs. The corollary uses the distinct-position ring R_I and presupposes the coefficient maps being conjugated. Extension through a collision requires further maps.

The completion also restricts spectral transport. The sequence B_n tends to zero in jet weight, but for fixed $z \in k^\times$

$$\pi(e^{zT} B_n) = z^n J$$

does not tend to zero in the discrete weight-zero coefficient ring. Numerical translation therefore has no continuous extension to (5.4). Proposition 5.2 has polynomial inputs. Giving a formal translation parameter weight one defines a different, combined completion.

5.3 A Gaussian complex with retained coordinates

The line contraction leaves a connection on its charge module. We now compute the corresponding effect of eliminating finitely many hidden fields. The hidden quadratic form may depend on the retained coordinates, so its normalization contributes to their effective differential.

Let R be a commutative k-algebra with commuting derivations ∂_a, $1 \le a \le r$. We write them as derivatives in retained coordinates y_a. Let A be an invertible symmetric $d \times d$ matrix over R, put $C = A^{-1}$, and choose $U \in R$. The ring contains $(\det A)^{-1}$ and is preserved by the derivations.

Introduce hidden even coordinates $x_1, \dots, x_d$, a degree-zero parameter $\hbar$, and odd antifields η_{x_i}, η_{y_a} of degree -1. The algebras are

$$\mathcal{B} = R[x_1, \dots, x_d, \hbar] \otimes \Lambda(\eta_x, \eta_y), \qquad \mathcal{B}_{\mathrm{ret}} = R[\hbar] \otimes \Lambda(\eta_y).$$

All hidden-coordinate dependence is polynomial.

For $S = U + x^{\mathrm{t}} A x/2$, the quantum differential is

$$Q_S = \sum_i \big((Ax)_i - \hbar\partial_{x_i}\big)\partial_{\eta_{x_i}} + \sum_a \big(\partial_a S - \hbar\partial_a\big)\partial_{\eta_{y_a}}. \tag{5.8}$$

Let q range over the hidden and retained coordinates. The even operators $\partial_q S - \hbar\partial_q$ commute, and the odd derivatives anticommute. Hence $Q_S^2 = 0$. This is a complex over $k[\hbar]$, since the retained derivatives need not be R-linear.

For example, take $R = k[y, (1+y^2)^{-1}]$ and $S = y^2/2 + (1+y^2)x^2/2$. Then

$$Q_S \eta_y = y + yx^2.$$

Eliminating x must replace the hidden quadratic term and lift η_y to an element whose differential has no hidden variable.

The differential has order two for the polynomial product. For homogeneous f, g, denote its product deviation by

$$B_S(f,g) = Q_S(fg) - (Q_S f)g - (-1)^{|f|} f(Q_S g). \tag{5.9}$$

It is divisible by $\hbar$. This failure of the Leibniz identity will determine the corrections to products of lifts.

5.3.1 *The finite heat contraction*

Let N_x count hidden polynomial degree and hidden antifield number. With repeated hidden indices summed, put

$$d_{\mathrm{cl}} = (Ax)_i \partial_{\eta_{x_i}}, \qquad h_{\mathrm{cl}} = N_x^{-1} \eta_x^{\mathrm{t}} C \partial_x \quad (N_x > 0),$$

and set $h_{\mathrm{cl}} = 0$ on hidden constants. Let ϵ_x set $x = \eta_x = 0$, and let i include $\mathcal{B}_{\mathrm{ret}}$. The Euler identity gives $d_{\mathrm{cl}} h_{\mathrm{cl}} + h_{\mathrm{cl}} d_{\mathrm{cl}} = 1 - i\epsilon_x$. The other contraction identities follow as in (5.3).

Define

$$W_x = \exp\!\left(-\frac{\hbar}{2}\partial_x^{\mathrm{t}} C \partial_x\right), \qquad P = \epsilon_x W_x^{-1}, \qquad h_x = W_x h_{\mathrm{cl}} W_x^{-1}. \tag{5.10}$$

The exponentials terminate on each polynomial. Their commutator with multiplication by Ax gives $Q_x = W_x d_{\mathrm{cl}} W_x^{-1}$, where Q_x is the hidden part of (5.8). Thus (P, i, h_x) is a strong deformation retract for Q_x. The projection P is Wick evaluation with covariance $\hbar C$ and kills every hidden antifield.

Theorem 5.6 (Variable Gaussian contraction). *Let $\delta_y = Q_S - Q_x$. The finite formulas*

$$I_A = (1 + h_x\delta_y)^{-1} i, \qquad H_A = h_x(1 + \delta_y h_x)^{-1}, \qquad P_A = P \tag{5.11}$$

give a strong deformation retract of $(\mathcal{B}, Q_S)$ onto $\mathcal{B}_{\mathrm{ret}}$ with differential

$$Q_A = \sum_a (E_a - \hbar\partial_a)\partial_{\eta_{ya}}, \qquad E_a = \partial_a U + \frac{\hbar}{2}\operatorname{tr}(C\,\partial_a A). \tag{5.12}$$

Every map has polynomial $\hbar$-coefficients.

Proof. The operator δ_y lowers retained antifield number by one, whereas h_x preserves it. The inverse series in (5.11) terminate after at most r perturbations on each input.

To verify the contraction, set $t = (1 + \delta_y h_x)^{-1}\delta_y = \delta_y(1 + h_x\delta_y)^{-1}$. The identity

$$Q_x t + t Q_x = -t\, i P\, t$$

follows from $(Q_x + \delta_y)^2 = 0$ and $Q_x h_x + h_x Q_x = 1 - iP$. Indeed multiply on the left by $1 + \delta_y h_x$ and on the right by $1 + h_x\delta_y$. The difference becomes

$$Q_x\delta_y + \delta_y Q_x + \delta_y(h_x Q_x + Q_x h_x + iP)\delta_y = 0.$$

Substitution in

$$I_A = i - h_x t i, \qquad P_A = P - P t h_x, \qquad H_A = h_x - h_x t h_x, \qquad Q_A = P t i$$

gives the chain equations and

$$Q_S H_A + H_A Q_S = 1 - I_A P_A, \qquad P_A I_A = 1, \qquad H_A^2 = H_A I_A = P_A H_A = 0.$$

Every value of h_x has a hidden antifield, which δ_y cannot remove. Hence $P\delta_y h_x = 0$, so $P_A = P$. This also kills every higher term of the transferred differential. Its only hidden quadratic expectation is

$$P\Big(\frac{1}{2}x^{\mathrm{t}}(\partial_a A)x\Big) = \frac{\hbar}{2}\operatorname{tr}(C\,\partial_a A).$$

This proves (5.12) on every retained antifield degree. The finite heat and perturbation formulas give the stated coefficient ring. □

In the one-variable example, the formulas give

$$E_y = y + \frac{\hbar y}{1 + y^2}, \qquad I_A(\eta_y) = \eta_y - \frac{x\,y}{1 + y^2}\eta_x.$$

The extra antifield term cancels the hidden part of the differential and leaves the Gaussian variance in the force.

The effective force is the closed algebraic one-form

$$dU + \frac{\hbar}{2}\operatorname{tr}(A^{-1}dA).$$

Here $d\log\det A$ abbreviates $\operatorname{tr}(A^{-1}dA)$. No logarithm need belong to R. The Gaussian normalization can be retained as the rank-one module with connection $d + \frac{1}{2}\operatorname{tr}(A^{-1}dA)$. For $A = y$ on $\mathbb{G}_m$, a local analytic horizontal section is $y^{-1/2}$, with sign monodromy around zero. A positive real quadratic form with a positive branch supplies an additional scalar state.

5.4 Curvature and products of antifields

The finite inclusion first sends a retained antifield to

$$I_A(\eta_{y_a}) = \eta_{y_a} - \eta_x^{\mathrm{t}} M_a x, \qquad M_a = \frac{1}{2} C\,\partial_a A.$$

The relation $\partial_a A = M_a^{\mathrm{t}} A + A M_a$ cancels the hidden quadratic force. Products of these unary lifts have a further defect, measured by the curvature of M. We allow any connection satisfying the same metric relation.

Choose matrices M_a over R with

$$\partial_a A = M_a^{\mathrm{t}} A + A M_a, \qquad F_{ab} = \partial_a M_b - \partial_b M_a + [M_a, M_b]. \tag{5.13}$$

The canonical choice $C\,\partial_a A/2$ always exists. Metric compatibility implies $F_{ab}^{\mathrm{t}} A + A F_{ab} = 0$, so $F_{ab} C$ is antisymmetric.

Define

$$\widehat{\eta}_a = \eta_{y_a} - \eta_x^{\mathrm{t}} M_a x, \qquad R_{ab} = \frac{\hbar}{2} \eta_x^{\mathrm{t}} F_{ab} C \eta_x. \tag{5.14}$$

The element $R_{ab} = -R_{ba}$ has degree -2. Taking the trace of (5.13) gives $E_a = \partial_a U + \hbar \operatorname{tr} M_a$. Thus the effective differential is independent of this choice of metric connection.

Lemma 5.7 (The pair correction)**.** *The elements in* (5.14) *satisfy*

$$\begin{aligned} Q_S \widehat{\eta}_a &= E_a, & B_S(\widehat{\eta}_a, \widehat{\eta}_b) &= \hbar \eta_x^{\mathrm{t}} F_{ab} x, \\ Q_S R_{ab} &= -\hbar \eta_x^{\mathrm{t}} F_{ab} x, & B_S(R_{ab}, R_{cd}) &= 0. \end{aligned} \tag{5.15}$$

Hence $\widehat{\eta}_a \widehat{\eta}_b + R_{ab}$ is a chain lift of $\eta_{y_a}\eta_{y_b}$ compatible with the unary lifts. For the canonical connection, $F_{ab} = -[C\,\partial_a A, C\,\partial_b A]/4$.

Proof. In $Q_S \widehat{\eta}_a$, metric compatibility cancels the quadratic terms. The quantum derivative of $-\eta_x^{\mathrm{t}} M_a x$ contributes $\hbar \operatorname{tr} M_a$. The horizontal vector fields associated to $\widehat{\eta}_a$ are $\partial_a - (M_a x)\cdot\partial_x$. Their commutator is $-(F_{ab}x)\cdot\partial_x$. Expansion of (5.9) gives the second identity.

For an antisymmetric matrix K, the classical hidden differential sends $\eta_x^{\mathrm{t}} K \eta_x/2$ to $-\eta_x^{\mathrm{t}} K A x$. Apply this with $K = F_{ab} C$. Its quantum hidden derivative vanishes because K has no x-dependence, and its retained differential vanishes because it has no retained antifield. This proves the third identity. Two curvature pairings have neither hidden-coordinate dependence nor retained antifields, so their product deviation is zero.

Finally $\theta = C\,dA$ satisfies $d\theta + \theta \wedge \theta = 0$. The curvature of $\theta/2$ is $-\theta\wedge\theta/4$, which proves the canonical formula. □

For $f \in R$, the additional term of the differential on a product by f is $-\hbar \sum_a (\partial_a f)\partial_{\eta_{ya}}$. The pair correction has no retained antifield, so the lift also obeys the chain equation on $f\,\eta_{y_a}\eta_{y_b}$.

Example 5.8. For retained coordinates y, z, take

$$A = \begin{pmatrix} 1+y^2 & yz \\ yz & 1+z^2 \end{pmatrix}, \qquad U = \frac{1}{2}(y^2+z^2), \qquad R = k[y, z, (1+y^2+z^2)^{-1}].$$

The canonical correction and effective coefficients are

$$R_{yz} = -\frac{\hbar(y^2+z^2)}{2(1+y^2+z^2)^2}\eta_{x_1}\eta_{x_2}, \qquad E_y = y + \frac{\hbar y}{1+y^2+z^2}, \quad E_z = z + \frac{\hbar z}{1+y^2+z^2}.$$

At $(y, z) = (1, 0)$ the curvature is

$$F_{yz} = \begin{pmatrix} 0 & -1/8 \\ 1/4 & 0 \end{pmatrix}, \qquad R_{yz} = -\frac{\hbar}{8}\eta_{x_1}\eta_{x_2}.$$

Thus the product of the two specified unary lifts needs a nonzero correction.

5.5 The inclusion in every antifield degree

Four retained antifields can require two pair corrections at once. A sum over partial matchings gives the full inclusion and explains the role of the Bianchi identity.

For an increasing list $L = (a_1, \dots, a_m)$, set $J_M(\varnothing) = 1$ and define

$$J_M(L) = \widehat{\eta}_{a_1} J_M(a_2, \dots, a_m) + \sum_{j=2}^{m} (-1)^j R_{a_1 a_j} J_M(a_2, \dots, \widehat{a_j}, \dots, a_m). \tag{5.16}$$

Let $I_M(\eta_{y_{a_1}} \cdots \eta_{y_{a_m}}) = J_M(L)$ and extend by $R[\hbar]$-linearity. Each unmatched label contributes $\widehat{\eta}_a$, each pair contributes R_{ab}, and collecting the labels gives the exterior sign. In degree four the term with two pairs is

$$R_{ab}R_{cd} - R_{ac}R_{bd} + R_{ad}R_{bc}, \qquad a < b < c < d.$$

Theorem 5.9 (The matching inclusion)**.** *The map $I_M : (\mathcal{B}_{\mathrm{ret}}, Q_A) \to (\mathcal{B}, Q_S)$ is unital and commutes with the differentials. The Wick projection P of (5.10) satisfies $PI_M = 1$. All coefficients lie in $R[\hbar]$.*

Proof. Put $\nabla_a F_{bc} = \partial_a F_{bc} + [M_a, F_{bc}]$. Besides (5.15), direct differentiation gives

$$B_S(\widehat{\eta}_a, R_{bc}) = -\frac{\hbar^2}{2}\eta_x^{\mathrm{t}}(\nabla_a F_{bc})C\eta_x. \tag{5.17}$$

Indeed, differentiate the pairing $\eta_x^{\mathrm{t}} F_{bc} C \eta_x$ and use $\partial_a C = -M_a C - C M_a^{\mathrm{t}}$. The derivative of F and the two actions on its normal factors combine into $\nabla_a F$. The cyclic sum of (5.17) is zero by the Bianchi identity $\nabla_a F_{bc} + \nabla_b F_{ca} + \nabla_c F_{ab} = 0$.

Apply the second-order product formula to each matching. Differentiating an unmatched factor gives its E_a term. The deviation of two unmatched factors cancels the differential of the matching in which they are paired, by (5.15). Deviations between an unmatched factor and a pair cancel in triples by (5.17). Pair–pair deviations vanish.

These are all the terms because Q_S has order two. The sign $(-1)^j$ in (5.16) moves label a_j past the $j - 2$ intermediate odd labels to form the pair. Thus each cancellation has the exterior sign of the differential on the retained monomial.

For coefficients $f \in R$, the additional derivative is $-\hbar \sum_a (\partial_a f)\partial_{\eta_{ya}}$. In the matching formula, retained odd differentiation removes the unmatched factor with that label. It therefore commutes with I_M. This proves the complete chain equation over $k[\hbar]$.

The map P is a chain map by Theorem 5.6. Every term in I_M other than the original retained monomial contains a hidden antifield, so P kills it. Hence $PI_M = 1$. Each matched edge contributes one factor of $\hbar$, and there are finitely many matchings. □

Example 5.10 (Several simultaneous pairs)**.** Take $A = I$ and hidden and retained ranks 2ℓ. Let J_a have entries $1, -1$ at $(2a - 1, 2a)$, $(2a, 2a - 1)$ and zero elsewhere. Choose

$$M_{2a-1} = 0, \qquad M_{2a} = y_{2a-1} J_a, \qquad 1 \le a \le \ell.$$

Distinct rotations commute. The only nonzero pairings are $R_{2a-1,2a} = \hbar\eta_{x_{2a-1}}\eta_{x_{2a}}$. The all-paired term in $I_M(\eta_{y_1}\cdots\eta_{y_{2\ell}})$ is

$$\hbar^\ell \eta_{x_1}\cdots\eta_{x_{2\ell}} \neq 0.$$

Thus these chosen connections require arbitrarily many pairs as the rank increases. At fixed hidden rank, the exterior algebra bounds the number of pairs. The same constant metric also admits $M = 0$, whose inclusion is the ordinary one.

5.5.1 A homotopy and the flatness criterion

The matching formula need not coincide with the particular inclusion I_A chosen by finite perturbation. The same projection permits a direct comparison of their homotopies.

Proposition 5.11. *Set $H_M = H_A(1 - I_M P)$. Then*

$$Q_S H_M + H_M Q_S = 1 - I_M P, \qquad H_M^2 = H_M I_M = P H_M = 0.$$

For two metric connections M, M', the degree-minus-one map $K_{M,M'} = H_A(I_M - I_{M'})$ satisfies

$$Q_S K_{M,M'} + K_{M,M'} Q_A = I_M - I_{M'}.$$

Proof. Since $1 - I_M P$ is a chain map, its commutator with Q_S after multiplication by H_A is

$$(1 - I_A P)(1 - I_M P) = 1 - I_M P,$$

using $P I_M = 1$. The equations $P H_M = H_M I_M = 0$ follow from the same identity and $P H_A = 0$. In the expansion of H_M^2, every term contains H_A^2 or $P H_A$ and vanishes. Finally apply $Q_S H_A + H_A Q_S = 1 - I_A P$ to $I_M - I_{M'}$. Its projection is zero, giving the comparison homotopy. □

Proposition 5.12. *The matching inclusion I_M is a homomorphism of graded commutative algebras if and only if $F_{ab} = 0$ for all a, b. A flat metric connection exists over R whenever there is $G \in \mathrm{GL}_d(R)$ for which $G^{\mathrm{t}} A G$ is diagonal and invertible.*

Proof. Multiplicativity on two retained antifields requires $R_{ab} = 0$. The matrix $F_{ab}C$ is antisymmetric. Independence of the exterior monomials and the absence of $\hbar$-torsion make this equivalent to $F_{ab}C = 0$, hence to $F_{ab} = 0$. Conversely, zero curvature removes every matched term, leaving the algebra map $\eta_{y_a} \mapsto \widehat{\eta}_a$.

Write $D = G^{\mathrm{t}} A G = \mathrm{diag}(d_1, \dots, d_d)$. In this frame the matrices

$$N_a = \frac{1}{2}\,\mathrm{diag}(d_1^{-1}\partial_a d_1, \dots, d_d^{-1}\partial_a d_d)$$

are metric compatible. Their commutators vanish and their logarithmic one-forms are closed, so they are flat. Differentiating $D = G^{\mathrm{t}} A G$ shows that $M_a = G N_a G^{-1} - (\partial_a G) G^{-1}$ is compatible with A. Its curvature is the conjugate of that of N, hence zero. □

For the metric of Example 5.8,

$$G = \begin{pmatrix} 1 & -yz/(1+y^2) \\ 0 & 1 \end{pmatrix} \quad \text{gives} \quad G^{\mathrm{t}} A G = \mathrm{diag}\left(1 + y^2, \frac{1 + y^2 + z^2}{1 + y^2}\right).$$

This gives a multiplicative inclusion over $k[y, z, (1+y^2)^{-1}, (1+y^2+z^2)^{-1}]$. It uses an additional inverse beyond the original ring.

Problem 5.13 (A flat metric connection)**.** Does the metric $\left(\begin{smallmatrix} 1+y^2 & yz \\ yz & 1+z^2 \end{smallmatrix}\right)$ admit a flat metric connection over $k[y, z, (1 + y^2 + z^2)^{-1}]$?

The nonzero canonical curvature only obstructs multiplicativity of its specified lifts. Changing a connection can change its curvature. Changing its frame conjugates curvature and cannot make a nonzero curvature vanish.

5.6 Frames and Gaussian normalization

A change of hidden coordinates also changes the retained antifields. The density determines the corresponding quantum correction. For $x = G(y)x'$ with $G \in \mathrm{GL}_d(R)$, use

$$\begin{aligned} A' &= G^{\mathrm{t}} A G, & M'_a &= G^{-1} M_a G + G^{-1}\partial_a G, \\ \eta_x &= G^{-\mathrm{t}} \eta'_x, & \eta_{y_a} &= \eta'_{y_a} - \eta'^{\mathrm{t}}_x G^{-1}(\partial_a G)x'. \end{aligned} \tag{5.18}$$

These are the cotangent transformation rules. For a density factor $\nu \in R^\times$, define

$$Q_\nu = Q_S - \hbar \sum_a \nu^{-1}(\partial_a \nu)\partial_{\eta_{y_a}}.$$

In a retained coordinate chart this is the differential for the volume density $\nu(y)\, dx_1 \cdots dx_d\, dy_1 \cdots dy_r$.

Proposition 5.14. *The matching inclusion is preserved by* (5.18)*. For Q_ν, the retained force is*

$$dU + \frac{\hbar}{2}\operatorname{tr}(A^{-1}dA) - \hbar\nu^{-1}d\nu. \tag{5.19}$$

It is invariant under $\nu' = (\det G)\nu$.

Proof. Substitution gives $\widehat{\eta}_a = \widehat{\eta}'_a$. Also $F'_{ab} = G^{-1}F_{ab}G$, so $\eta_x^{\mathrm{t}} F_{ab} C \eta_x$ is unchanged. Every matching is therefore preserved.

Both I_M and P commute with retained odd differentiation. The density operator subtracts the last term of (5.19) on the retained complex. Finally $\det A' = (\det G)^2 \det A$. Its logarithmic derivative change cancels that of the density. These are algebraic one-forms, independent of logarithm choices. □

Selecting $C\partial_a A/2$ afresh after a frame change need not give the transported connection in (5.18). The matching theorem uses the actual connection chosen. Where the scalar normalization has no global square root, its rank-one module and connection retain the normalization data.

5.7 Higher comparison maps

The inclusion and projection are chain maps, but their products can differ from the products of their values. Joint cumulants organize these differences. The powers of $\hbar$ carried by connected contractions make the resulting higher maps integral.

For a unital degree-zero linear map F between graded commutative algebras, define

$$\kappa_m(F)(a_1, \dots, a_m) = \sum_{\pi \in \mathrm{Part}([m])} \epsilon(\pi)(-1)^{|\pi|-1}(|\pi| - 1)! \prod_{B \in \pi} F\left(\prod_{i \in B} a_i\right). \tag{5.20}$$

Within a block labels increase. Blocks are ordered by their least label. The Koszul sign $\epsilon(\pi)$ collects the inputs in that order. In particular $\kappa_1(F) = F$ and $\kappa_2(F)(a, b) = F(ab) - F(a)F(b)$.

For a graded $k[\hbar]$-module V, let $S^c(V) = \bigoplus_{m\geq 1} \mathrm{Sym}^m(V)$ be the conilpotent symmetric coalgebra, with its unshuffle coproduct. An $L_\infty[1]$ structure on V is a degree-one square-zero coderivation on $S^c(V)$. A morphism is a coalgebra map intertwining these coderivations. Its Taylor components are its compositions with the projection to the cogenerators. It is a quasi-isomorphism when its unary component is one.

Suppose V is also a graded commutative algebra. Let Q be a degree-one square-zero operator of order at most two, with $Q1 = 0$. Its unary part and symmetric product deviation B_Q define an $L_\infty[1]$ structure. The second-order condition eliminates all higher operations. In the present complexes B_Q is divisible by $\hbar$. The normalized structure has operations $(Q, \hbar^{-1}B_Q)$.

Theorem 5.15. *For $m \geq 1$,*

$$\kappa_m(I_M) \in \hbar^{m-1}\mathcal{B}, \qquad \kappa_m(P) \in \hbar^{m-1}\mathcal{B}_{\mathrm{ret}}.$$

The exact quotients

$$\Psi_m = \hbar^{1-m}\kappa_m(I_M), \qquad \Phi_m = \hbar^{1-m}\kappa_m(P)$$

define $L_\infty[1]$ quasi-isomorphisms over $k[\hbar]$ between the normalized structures on $\mathcal{B}_{\mathrm{ret}}$ and $\mathcal{B}$. They satisfy $\Phi \circ \Psi = 1$. Their components of arity greater than one vanish on a unit input.

Proof. In the matching formula, mark the m input groups. The partition sum (5.20) cancels every graph disconnected on those groups. For a graph with c connected components, its coefficient is

$$\sum_{\sigma \in \mathrm{Part}([c])} (-1)^{|\sigma|-1}(|\sigma|-1)! = \delta_{c1}.$$

This is the coefficient identity for $\log(e^t) = t$. A connected graph on m groups has at least $m-1$ edges, each of which contributes $\hbar$ through R_{ab}. This proves divisibility for I_M.

The same argument for Wick contractions, with edge covariance $\hbar C$, proves it for P. Internal edges within one marked group are allowed. On polynomial inputs there are only finitely many contractions.

Let $\tau_V : S^c(V) \to S^c(V)$ be the triangular coalgebra automorphism whose Taylor components are multiplication. Its unary component is the identity, so conilpotence gives an inverse. The Taylor components of

$$\tau_{\mathrm{out}}^{-1} S^c(F) \tau_{\mathrm{in}}$$

are the cumulants (5.20): inversion of the partition sum gives the coefficient $(-1)^{|\pi|-1}(|\pi|-1)!$. Conjugating the unary coderivation Q by τ gives its unary term and product deviation. Since Q kills the unit and has order two, all higher terms vanish. The chain equations for I_M and P therefore give the raw higher identities.

In arity m each term in these identities has the known factor $\hbar^{m-1}$. Division gives the normalized binary deviation and the displayed higher maps. Equivalently the identities hold after adjoining $\hbar^{-1}$ and hence in the original polynomial modules, which have no $\hbar$-torsion. The unary maps are quasi-isomorphisms by Proposition 5.11.

Raw cumulants compose by cancellation of the middle τ's. For an outer component with q inputs and inner block sizes $m_1, \dots, m_q$, the normalization powers satisfy

$$(1-q) + \sum_{i=1}^{q}(1-m_i) = 1 - \sum_i m_i.$$

Thus normalized cumulants also compose. The equality $PI_M = 1$ gives $\Phi\Psi = 1$. Inserting a unit in the partition sum cancels all higher components. □

Example 5.16. Use Example 5.10 and abbreviate $a_i = \eta_{y_i}$. For $\ell = 2$, the two-edge matching connects the three groups (a_1), $(a_2 a_3)$, (a_4), giving

$$\kappa_3(I_M)(a_1, a_2 a_3, a_4) = \hbar^2 \eta_{x_1}\eta_{x_2}\eta_{x_3}\eta_{x_4}.$$

For $\ell = 3$, the connected three-edge matching gives

$$\Upsilon_4(a_1, a_2 a_3, a_4 a_5, a_6) = \eta_{x_1} \cdots \eta_{x_6} \neq 0.$$

These normalized components survive at $\hbar = 0$. For a flat metric connection the inclusion is multiplicative and all its higher cumulants vanish. Its Wick projection may still have higher cumulants. Even for a single Gaussian variable of variance $\hbar/a$, $\kappa_3(P)(x, x, x^2) = 2(\hbar/a)^2$. The third input is composite.

5.8 Formal potential interactions

Let $V \in R[x_1, \dots, x_d]$ and introduce a formal parameter g. Work in $\mathcal{B}[[g]]$: each coefficient is polynomial in x, $\hbar$ and exterior in the antifields. Define

$$\delta_V = g \sum_i (\partial_{x_i} V)\partial_{\eta_{x_i}} + g \sum_a (\partial_a V)\partial_{\eta_{y_a}}.$$

Then $Q_{S+gV} = Q_S + \delta_V$ squares to zero. Every perturbation raises g-order, so the geometric series below are defined coefficientwise.

Theorem 5.17 (Integral interacting contraction)**.** *For any chosen metric connection, the maps*

$$\begin{aligned} I_{M,g} &= (1 + H_M \delta_V)^{-1} I_M, & P_g &= P(1 + \delta_V H_M)^{-1}, \\ H_{M,g} &= H_M (1 + \delta_V H_M)^{-1}, & Q_g &= Q_A + P\delta_V (1 + H_M \delta_V)^{-1} I_M \end{aligned} \tag{5.21}$$

give a continuous strong deformation retract over $k[\hbar][[g]]$. The projection is independent of M. After embedding in the ring with $\hbar^{-1}$ adjoined, it is

$$\begin{aligned} Z_x &= P(e^{-gV/\hbar}), \\ P_g(F) &= Z_x^{-1} P(e^{-gV/\hbar} F|_{\eta_x = 0}). \end{aligned} \tag{5.22}$$

The effective differential and force are

$$\begin{aligned} Q_g &= \sum_a (E_a(g) - \hbar\partial_a)\partial_{\eta_{y_a}}, \\ \sum_a E_a(g)\, dy_a &= dU + \frac{\hbar}{2} d \log\det A - \hbar\, d \log Z_x. \end{aligned} \tag{5.23}$$

The formal interaction potential $-\hbar \log Z_x$ belongs to $R[\hbar][[g]]$.

Proof. Apply the geometric-series contraction identity to (P, I_M, H_M). Every coefficient in g uses finitely many terms, all polynomial in $\hbar$. It gives

$$(Q_S + \delta_V) H_{M,g} + H_{M,g}(Q_S + \delta_V) = 1 - I_{M,g} P_g$$

and the chain-map and side identities.

To identify P_g, count hidden antifields. Each term of H_A adds at least one. Its terms beyond h_x add at least two, because each further $\delta_y h_x$ adds another hidden antifield. The correction $-H_A I_M P$ in H_M also adds at least two whenever it is nonzero. Indeed the purely retained term of I_M is killed by the rightmost hidden homotopy, and every other term already has a hidden antifield.

Each δ_V removes at most one. Thus only successive leading hidden homotopies and hidden potential derivatives can survive the final P. The projection kills inputs with hidden antifields and acts coefficientwise on retained ones, independently of M.

Its functional on hidden polynomials is determined by normalization and the hidden Ward identities. Write $\delta_{V,x} = g\sum_i V_{x_i}\partial_{\eta_{x_i}}$. Since $f - iPf = Q_x h_x f$, any normalized perturbed functional $\mathcal{L}$ satisfies

$$\mathcal{L}(f) = Pf - \mathcal{L}(\delta_{V,x} h_x f).$$

This recursion determines each successive coefficient of g. The expression (5.22) satisfies it by Gaussian integration by parts, coefficientwise after adjoining $\hbar^{-1}$. It therefore equals the integral perturbation formula.

For a function f, differentiating normalized Wick evaluation gives

$$\partial_a P_g(f) = P_g(\partial_a f) - \hbar^{-1}\Big(P_g((\partial_a S + g\partial_a V)f) - P_g(\partial_a S + g\partial_a V)P_g(f)\Big).$$

Together with the hidden Ward identity this proves the chain equation on all antifield monomials. Since $P_g i = 1$, the retained differential has the displayed potential form with $E_a(g) = P_g(\partial_a S + g\partial_a V)$. Differentiating Z_x, including the varying Gaussian normalization, gives (5.23).

Finally, a Wick graph with v interaction vertices and e edges carries $\hbar^{e-v}$. The logarithm selects connected graphs, for which $e \geq v-1$. The outside factor $\hbar$ then gives a nonnegative power. A hidden-independent interaction is its single classical vertex. At fixed g-order there are finitely many contractions. This proves integrality of $-\hbar\log Z_x$. □

Corollary 5.18. *For polynomial observables, including antifields,*

$$\kappa_m(P_g) \in \hbar^{m-1}\mathcal{B}_{\mathrm{ret}}[[g]].$$

The normalized cumulants give an $L_\infty[1]$ quasi-isomorphism for the normalized BV deviations and obey the composition law of Theorem 5.15.

Proof. Normalization cancels unmarked vacuum components. The cumulant cancels decompositions of the marked inputs. A surviving connected graph with m marked observables and v interaction vertices has $e \geq m + v - 1$, so its power $\hbar^{e-v}$ is at least $m-1$. Retained antifields supply their exterior coefficients and Koszul signs. Moments with a hidden antifield vanish.

The cumulant coalgebra argument then gives the higher identities and composition before inverting $\hbar$. The unary map is a quasi-isomorphism by Theorem 5.17. □

Example 5.19 (A cubic hidden interaction)**.** For one hidden coordinate, let its quadratic coefficient be $a \in k^\times$ and let $V = yx^3/6$. Expanding (5.22) with $P(x^{2m}) = (2m-1)!!(\hbar/a)^m$ gives

$$\begin{aligned}
-\hbar\log Z_x &= -\frac{5g^2\hbar^2 y^2}{24a^3} - \frac{5g^4\hbar^3 y^4}{16a^6} + O(g^6),\\
P_g(x) &= -\frac{g\hbar y}{2a^2} - \frac{5g^3\hbar^2 y^3}{8a^5} + O(g^5),\\
\hbar^{-1}\kappa_2(P_g)(x,x) &= \frac{1}{a} + \frac{g^2\hbar y^2}{a^4} + \frac{25g^4\hbar^2 y^4}{8a^7} + O(g^6),\\
\hbar^{-2}\kappa_3(P_g)(x,x,x^2) &= \frac{2}{a^2} + \frac{8g^2\hbar y^2}{a^5} + \frac{175g^4\hbar^2 y^4}{4a^8} + O(g^6).
\end{aligned}$$

The third normalized comparison is already nonzero at $g = 0$ because one observable is x^2.

5.9 Composition of normalized eliminations

Successive eliminations must retain the intermediate dependence on the remaining coordinates. Let P_x°, P_y° be independent reference Wick evaluations, and let the interaction be $V(x, y)$. Coefficientwise after adjoining $\hbar^{-1}$, put

$$Z_x = P_x^{\circ}(e^{-gV/\hbar}), \qquad P_x(F) = Z_x^{-1} P_x^{\circ}(e^{-gV/\hbar} F), \qquad P_y^{\mathrm{eff}}(G) = \frac{P_y^{\circ}(Z_x G)}{P_y^{\circ}(Z_x)}.$$

The factor Z_x cancels in the composite, so

$$P_y^{\mathrm{eff}} P_x(F) = \frac{P_y^{\circ} P_x^{\circ}(e^{-gV/\hbar} F)}{P_y^{\circ} P_x^{\circ}(e^{-gV/\hbar})}.$$

This is joint normalized evaluation. The equality descends to the integral formal maps above wherever the successive nondegenerate Gaussian contractions are defined. Their normalized cumulants compose by Theorem 5.15.

For a nondegenerate Gaussian block matrix, the same identity is expressed by completing the square. If

$$K = \begin{pmatrix} A & B \\ B^{\mathrm{t}} & D \end{pmatrix}, \qquad S_D = D - B^{\mathrm{t}} A^{-1} B,$$

with A and S_D invertible in the specified coefficient ring, elimination of x leaves the quadratic form S_D. The triangular change $x \mapsto x + A^{-1} B y$ has determinant one, and

$$\det K = (\det A)(\det S_D).$$

It therefore composes both the Wick contractions and their normalization lines.

For an interacting example, take reference variances $P_x^{\circ}(x^2) = P_y^{\circ}(y^2) = \hbar$ and $V = yx^3/6$. Then

$$Z_x = 1 + \frac{5}{24} g^2 \hbar y^2 + O(g^4), \qquad P_y^{\mathrm{eff}}(y^2) = \hbar + \frac{5}{12} g^2 \hbar^3 + O(g^4).$$

The correction uses $P_y^{\circ}(y^4) - P_y^{\circ}(y^2)^2 = 2\hbar^2$. Replacing the intermediate factor by a scalar would remove it.

These constructions determine maps on their polynomial or formal coefficient modules. The line gauge additionally retains all position pairs and their logarithmic residue. To compare them with the ordered collision coefficients of Chapters 3 and 4, one must construct the maps on those coefficient systems.

Problem 5.20 (The chiral comparison)**.** Construct a comparison from the transverse collision coefficients and their varying endpoint modules to the all-mode line connection. Require it to preserve the polynomial spectral maps. Specify the continuous coefficient dual, the maps through collision strata and coherent gauges on nested collision domains. For a BV realization, also construct the supported operation maps, state domains and renormalized contractions whose finite reductions are the Gaussian maps above. Require compatibility with both vacuum insertions and with composition of normalized eliminations.

Chapter 6

Vacuum support and derived actions

The scalar vacuum has a nontrivial extension at four positions. The two components of its support, Δ_{14} and $\Delta_{13} \cap \Delta_{24}$, meet on the total diagonal. A normal coordinate acts invertibly on the coefficient along the crossed component. It acts nilpotently on each element of the coefficient at the intersection. This prevents a splitting of their extension, even before differential operators are imposed.

Maps of the Heisenberg primitive coefficient must respect such normal attachments. We first determine its ordinary horizontal endomorphisms and then its derived maps to the regular coefficient. The difference between these two calculations leads to the extension equations for actions. Those equations retain their prescribed vacuum maps and the paths comparing compositions.

Throughout, k has characteristic zero. Complexes are cohomological, with $M[r]^i = M^{i+r}$, and tensor products of maps use the Koszul rule. All completions will be specified.

6.1 The four-position vacuum extension

Take $X = \mathbb{A}^1_k$ and use the coordinates

$$t = z_4, \qquad a = z_1 - z_4, \qquad b = z_1 - z_3, \qquad c = z_2 - z_4.$$

They are independent. Put $A = k[t, a, b, c]$. The support of Example 1.5 is

$$Z_4 = V(a) \cup V(b, c), \qquad I(Z_4) = (ab, ac).$$

For a coordinate u, write $P_u = k[u, u^{-1}]/k[u]$. Its basis consists of u^{-m}, $m \geq 1$. Multiplication by u kills u^{-1}, and differentiation sends u^{-m} to $-mu^{-m-1}$.

Proposition 6.1. *The only nonzero local cohomology modules of A on Z_4 are*

$$\begin{aligned} H^1_{Z_4}(A) &= k[t, b, c] \otimes_k P_a, \\ H^2_{Z_4}(A) &= k[t, a, a^{-1}] \otimes_k P_b \otimes_k P_c. \end{aligned} \tag{6.1}$$

There is an exact sequence of differential modules

$$0 \longrightarrow k[t, a] \otimes P_b \otimes P_c \longrightarrow H^2_{Z_4}(A) \longrightarrow k[t] \otimes P_a \otimes P_b \otimes P_c \longrightarrow 0. \tag{6.2}$$

It does not split as a sequence of A-modules.

Proof. The localization complex for the generators ab, ac is

$$A \longrightarrow A[(ab)^{-1}] \oplus A[(ac)^{-1}] \longrightarrow A[(abc)^{-1}],$$

in degrees zero, one and two. Its maps are the diagonal and the difference of the two inclusions. The first map is injective. Inside the last localization, the two middle terms intersect in $A[a^{-1}]$. This gives the first module of (6.1).

Their sum contains exactly the Laurent monomials with nonnegative b exponent or nonnegative c exponent. The remaining monomials have both exponents negative and arbitrary integral a exponent. They give the second module. Its nonnegative a powers form the first term of (6.2). Quotienting leaves the negative powers of all three normal coordinates. Multiplication and differentiation preserve this submodule.

Multiplication by a is invertible on the middle term, whereas each element of the last term is killed by a power of a. Every A-linear map from the last term to the middle term is therefore zero. A section cannot exist. □

The attachment is visible on one class. The indicated lift of the total-collision class satisfies

$$a[a^{-1}b^{-1}c^{-1}] = [b^{-1}c^{-1}] \neq 0 \quad \text{in } H^2_{Z_4}(A). \tag{6.3}$$

The right side belongs to the crossed component. On $a \neq 0$, the crossed coefficient remains nonzero although the outer diagonal is absent.

For the scalar vacuum $J_n = \omega_{X^n}[n]$, Theorem 1.4 gives

$$(\overline{\Omega_c J})_4 \simeq R\Gamma_{Z_4}(\omega_{X^4})[3].$$

After trivializing the density, its cohomology is (6.1) in degrees -2 and -1. Thus the local vacuum is already a complex with two cohomological degrees at four positions. The localizations and the inverse strong coproducts specify its multiplication. The two cohomology modules alone do not.

6.2 Horizontal symmetries of the primitive coefficient

Return to the left differential modules of Chapter 3. Retain $R_n = k[z_1, \dots, z_n]$, its Weyl algebra D_n, the primitive module H_n, and its generators g_S. The coefficient of w^{-1} in the expansion at infinity of a proper rational representative defines total finite residue:

$$\operatorname{res} : H_n \longrightarrow R_n, \qquad \operatorname{res}(g_i) = 1, \qquad \operatorname{res}(g_S) = 0 \quad (|S| > 1).$$

This coefficient lies in R_n, and a w-derivative has zero residue. Hence the map is well defined and horizontal.

For $v = (v_1, \dots, v_n) \in k^n$, put

$$s_v : R_n \longrightarrow H_n, \qquad s_v(1) = \sum_i v_i g_i.$$

Each g_i is horizontal, so s_v is a map of D_n-modules.

Theorem 6.2. *For $n \geq 2$, every D_n-linear endomorphism of H_n has a unique expression*

$$T = \lambda \operatorname{id} + s_v \operatorname{res}, \qquad \lambda \in k, \quad v \in k^n. \tag{6.4}$$

In residue coordinates its matrix is $\lambda I + v\mathbf{1}^{\mathrm{t}}$. Composition is

$$(\lambda, v)(\mu, w) = (\lambda\mu, \lambda w + \mu v + (\mathbf{1}^{\mathrm{t}} w)v).$$

The scalar target and the residue kernel satisfy

$$\operatorname{Hom}_{D_n}(H_n, R_n) = k \operatorname{res}, \qquad \operatorname{End}_{D_n}(\ker \operatorname{res}) = k.$$

In particular, the residue kernel is indecomposable.

Proof. Flatness embeds H_n in its generic residue space $k(z_1, \dots, z_n)^n$. The connection there is ordinary position differentiation. A horizontal matrix therefore has entries in k.

Near the generic point of $z_i = z_j$, all other differences are invertible. The binary calculation (3.8) gives one localized direction $e_i - e_j$ and regular complementary directions. An element in the image of a linear map from the localized direction to a regular direction would be divisible by every power of $z_i - z_j$. The regular module has no such nonzero element. The matrix must therefore preserve $k(e_i - e_j)$.

For $n \geq 3$, the relation $(e_i - e_j) + (e_j - e_k) = e_i - e_k$ forces the eigenvalues on these lines to agree. For $n = 2$ there is one line. Thus, in either case, $T - \lambda I$ kills the sum-zero hyperplane and has the form $v\mathbf{1}^{\mathsf{t}}$. The maps in (6.4) realize all these matrices on H_n, proving the assertion and its composition law.

A horizontal map $H_n \to R_n$ is generically a constant row vector. Regularity at every binary diagonal forces it to kill every $e_i - e_j$. It is a multiple of residue.

The map s_{e_1} splits residue as a differential-module map, giving $H_n = R_n \oplus \ker \operatorname{res}$. Extend an endomorphism of the kernel by zero on this regular summand. Formula (6.4) restricts to λI on the kernel. Its endomorphism ring is k, which has no nontrivial idempotents. □

The splitting by s_{e_1} singles out a position. Compatibility with the whole family of coefficients removes this choice.

Proposition 6.3. *A family of horizontal endomorphisms T_n of H_n is multiplication by one scalar independent of n under either of the following hypotheses:*

(i) *it commutes with every permutation and every coefficient collision map of Proposition 3.8.*

(ii) *it commutes with the block decompositions of Proposition 3.9. One nontrivial consecutive cut in each arity $n \geq 2$ suffices.*

Proof. Under the first hypothesis, permutation compatibility gives $T_n = \lambda_n I + c_n \mathbf{1}\mathbf{1}^{\mathsf{t}}$ for $n \geq 2$. On the simple-pole classes, a collision α sends g_i to $g_{\alpha(i)}$. A three-to-two collision of fiber sizes $(2, 1)$ therefore gives, for both possible output labels,

$$\lambda_3 e_j + c_3(2e_1 + e_2) = \lambda_2 e_j + c_2(e_1 + e_2).$$

Subtract the equations for $j = 1, 2$ and then compare their coordinates. One obtains $\lambda_3 = \lambda_2$ and $2c_3 = c_2 = c_3$, hence $c_2 = c_3 = 0$.

For $n > 3$, a collision with fiber sizes $(n - 1, 1)$ similarly gives $c_n = 0$ and $\lambda_n = \lambda_2$. Collision to a singleton fixes the last scalar.

For the second hypothesis, write $T_n = \lambda_n I + v_n \mathbf{1}^{\mathsf{t}}$. A cut into nonempty blocks of sizes p, q forces its off-diagonal matrix blocks to vanish. They are

$$v_{n,[1,p]} \mathbf{1}_q^{\mathsf{t}}, \qquad v_{n,[p+1,n]} \mathbf{1}_p^{\mathsf{t}}.$$

Thus $v_n = 0$, and the two diagonal blocks give $\lambda_n = \lambda_p = \lambda_q$. Induction from $\operatorname{End}_{D_1}(H_1) = k$ proves the result. □

On the diagram with monotone collisions alone, the maps g_1 res and g_n res are natural and distinct for $n \geq 2$. A monotone surjection preserves the first and last labels, and total residue commutes with substitution. They fail the cut condition: on the cut $[1, 2] \mid [3, 4]$, the first map sends the residue vector e_3 to e_1. Thus the cut maps supply an essential part of the rigidity assertion.

6.3 Derived scalar maps and the residue simplex

Ordinary scalarity does not determine derived endomorphisms. The simplest comparison uses the regular module as target. We compute it from the normal directions of the diagonal constituents, retaining their nonsplit attachments.

For an affine linear subspace $L \subseteq \mathbb{A}^n$, let δ_L denote the trivial left differential module on L, pushed forward with all normal derivatives.

Lemma 6.4. *Let affine linear subspaces $L, M \subseteq \mathbb{A}^n$ have nonempty intersection. If their dimensions are l, m, and $d = \dim(L \cap M)$, then choices of linear normal coordinates give*

$$R\operatorname{Hom}_{D_n}(\delta_L, \delta_M) \simeq k[-(l + m - 2d)].$$

Proof. Translate a common point to zero and split the ambient space into common tangents, tangents of L alone, tangents of M alone, and common normals. In one variable put $O = k[u]$ and retain P_u above. Their left Weyl-module resolutions use right multiplication by ∂_u and by u, respectively. Applying Hom gives the two-term complexes

(source, target)	d	cohomology representative
(O, O)	$\partial_u : O \to O$	1 in degree 0
(P_u, P_u)	$u : P_u \to P_u$	u^{-1} in degree 0
(O, P_u)	$\partial_u : P_u \to P_u$	u^{-1} in degree 1
(P_u, O)	$u : O \to O$	1 in degree 1.

Both terms have degrees zero and one. Right inverses on the boundaries are, in the displayed order,

$$u^j \mapsto \frac{u^{j+1}}{j+1}, \quad u^{-j} \mapsto u^{-j-1}, \quad u^{-j} \mapsto -\frac{u^{-j+1}}{j-1}\ (j \geq 2), \quad u^j \mapsto u^{j-1}\ (j \geq 1).$$

Set these homotopies to zero on the degree-one cohomology representatives. They contract the complementary monomials. Tensor their contractions with the Koszul signs. Exactly the $l + m - 2d$ directions tangent to only one subspace contribute degree one. □

Theorem 6.5. *The derived scalar mapping complex has a model*

$$R\operatorname{Hom}_{D_n}(H_n, R_n) \simeq \left[k^{\binom{n}{1}} \longrightarrow k^{\binom{n}{2}} \longrightarrow \cdots \longrightarrow k^{\binom{n}{n}}\right] \simeq k. \tag{6.5}$$

Its terms have degrees zero through $n - 1$. With subsets in increasing order, the differential is

$$(dc)_{i_0 \cdots i_p} = \sum_{j=0}^{p} (-1)^j c_{i_0 \cdots \widehat{i_j} \cdots i_p}.$$

The surviving map is total finite residue.

Proof. Apply derived Hom into R_n to the finite filtration of Theorem 3.5. A constituent indexed by a subset of size s has mapping cohomology k in degree $s - 1$, by Lemma 6.4. Choose the normal contractions in its proof and transfer along this finite filtration. A transferred differential has degree one, so it joins only adjacent subset sizes.

Let $T = S \setminus \{i\}$, and choose $r \in T$. The attaching relation

$$(z_i - z_r) g_S = g_{S \setminus \{r\}} - g_T$$

has coefficient -1 on g_T. In that component its new normal direction is the localization extension $0 \to O \to O[u^{-1}] \to P_u \to 0$. The resulting incidence entry is therefore a unit.

For $|S| \geq 3$, a subset of size $|S| - 1$ not contained in S has no degree-one extension with its diagonal, by Lemma 6.4. At size two all singleton constituents have the same support. The actual relation $(z_i - z_j) g_{ij} = g_i - g_j$ fixes the two opposite endpoint entries and makes every spectator entry zero.

Order the normal coordinates to fix their orientations. The edge entries are the displayed differences. Square-zero on each two-step face fixes the higher incidence signs after one choice of orientation per subset. This gives the complex (6.5).

Choose vertex 1. In degree $p \geq 1$ define

$$(hc)_S = \begin{cases} c_{\{1\}\cup S}, & 1 \notin S, \\ 0, & 1 \in S, \end{cases}$$

with 1 first in the orientation. In degree zero put $h = 0$.

Let $\pi(c) = c_{\{1\}}$ in degree zero and let i include constant vertex cochains. Cancellation of incidences gives $dh + hd = 1 - i\pi$. Thus only the common value on all g_i survives, which is total residue. The finite transfer lifts this contraction to the derived mapping complex. □

For $n = 3$ the two differentials are

$$(c_1, c_2, c_3) \longmapsto (c_2 - c_1, c_3 - c_1, c_3 - c_2), \qquad (x_{12}, x_{13}, x_{23}) \longmapsto x_{23} - x_{13} + x_{12}.$$

The choice of vertex specifies the contraction, not the residue map, which is independent of that choice.

The endomorphism target gives a different answer already at two positions. Suppress common translation, put $R = k[u]$, $L = R[u^{-1}]$, and use $H_2 \simeq R \oplus L$ from (3.8). Proposition 3.12 gives $R\,\mathrm{Hom}_D(L, R) = 0$ and $\mathrm{Ext}^0_D(L, L) = \mathrm{Ext}^1_D(L, L) = k$.

The other two corners are $[R \xrightarrow{\partial_u} R]$ and $[L \xrightarrow{\partial_u} L]$. The first has only its constant class. The second has a constant class in degree zero and the class of u^{-1} in degree one.

Consequently

$$\dim_k \mathrm{End}_D(H_2) = 3, \qquad \mathrm{Ext}^1_D(H_2, H_2) = k^2, \qquad \mathrm{Ext}^q_D(H_2, H_2) = 0 \quad (q > 1).$$

Tensoring back the common tangential regular module does not change these groups. To compute derived natural endomorphisms of the whole coefficient system, one must impose the collision and cut comparisons with their homotopies. Neither the single-arity endomorphism calculation nor (6.5) performs that step.

6.4 Pointed actions and their comparison paths

A vacuum-preserving map of coefficients is only the linear part of an action. To compare actions, one must also retain the path that identifies the restricted action with the given one.

The action calculations use a symmetric convolution on the full ordered collision atlas. We specify this carrier before forming its mapping spaces. On the affine line, its underlying scheme is the ravioli configuration space of Alfonsi–Kim–Young [2, §4.1].

For a nonempty finite set I, take one copy X^I_σ of X^I for each total order σ of I. For two orders σ, ρ, remove the diagonal $z_i = z_j$ whenever they disagree on the order of i, j, and identify the resulting opens by their position coordinates. Denote the glued scheme by $\mathscr{R}_I$ and its position map by $q_I : \mathscr{R}_I \to X^I$.

These are valid gluing data. If two orders disagree on a pair, at least one disagrees with any third order on that pair. Thus $V_{\sigma\rho} \cap V_{\sigma\eta} = V_{\sigma\rho} \cap V_{\rho\eta}$ for the prescribed overlaps. The finite smooth chart cover makes $\mathscr{R}_I$ smooth, quasi-compact and quasi-separated, and q_I is locally an isomorphism. At a position tuple with equality blocks B, its fiber has $\prod_B |B|!$ points, one for each choice of order inside every equality block. In particular the total triple collision has six branches.

A surjection $\alpha : I \twoheadrightarrow H$ with ordered fibers expands an order on H by concatenating those fibers. The diagonal substitution then defines $\delta_\alpha : \mathscr{R}_H \to \mathscr{R}_I$. Indeed the preimage of an overlap between expanded orders is exactly the overlap of the original orders. Concatenation of fiber orders

proves $\delta_\alpha \delta_\beta = \delta_{\beta\alpha}$. For example, a collision identifying 1, 3 on the chart 123 reaches that chart only where $z_1 = z_3 \neq z_2$. Its selected total-collision branches occur on charts where 1, 3 are consecutive.

Let $\mathscr{D}$ be the enhanced category of right differential-module families on these schemes with coherent equivalences $\delta_\alpha^! M_I \simeq M_H$, together with the separate empty coefficient in $D(k)$. Differential modules on $\mathscr{R}_I$ are defined by descent on its finite smooth chart cover. Use the derived six-functor operations with the cochain and density conventions of Chapter 1.

For $I = A \amalg B$, let $U_{A,B}$ separate all cross pairs, and let $j_{A,B}$ be its open inclusion. Restriction of orders identifies this open with $(\mathscr{R}_A \times \mathscr{R}_B)_{\mathrm{disj}}$: all shuffles of the two block orders agree there. Define

$$(M \star N)_I = \bigoplus_{I=A\amalg B} j_{A,B,*}(M_A \boxtimes N_B)|_{U_{A,B}}. \tag{6.6}$$

The pairs of subsets are ordered. Either may be empty.

As in Proposition 1.1, all iterated products have the common coefficient on the open separating their blocks. This proves associativity and its higher coherences. Permuting blocks gives the symmetry.

Extraordinary pullback kills a summand whose two groups meet inside one collision fiber. The other summands are exactly the decompositions of the quotient. Base change therefore gives the required collision maps. The unit is K, supported at the empty configuration. Every product of more than $|I|$ positive factors vanishes on $\mathscr{R}_I$.

Restricting to the increasing chart and projecting onto consecutive decompositions gives the word-cut comparison. No equivalence of the two diagram categories is asserted by this projection. In particular their derived mapping spaces must be distinguished.

Now let $\mathscr{C}$ be the full subcategory of coaugmented counital coassociative coalgebras for $\star$ whose coproducts are equivalences on every block-disjoint open. It carries the diagonal tensor $\otimes^!$, with unit the local vacuum J. Its coproduct uses the pairs of equal decompositions in the two finite sums (6.6). For those summands the two localizations have the same open and combine by base change. Iterated equal decompositions verify coassociativity and the tensor compatibilities.

For a coalgebra F with a pointing $u : J \to F$, and $C \in \mathscr{C}$, define its space of pointed actions by

$$\Phi_C(F,u) = \mathrm{fib}_{\mathrm{id}_C}\Big(\mathrm{Map}_{\mathscr{C}}(F \otimes^! C, C) \longrightarrow \mathrm{Map}_{\mathscr{C}}(C,C)\Big). \tag{6.7}$$

The arrow restricts along $u \otimes^! \mathrm{id}_C$. An element therefore includes both a map α and a specified path from its restriction to id_C. All mapping spaces are derived.

A morphism from (F,u,α) to (G,v,β) consists of a pointed map $f : F \to G$ and a path $\alpha \simeq \beta(f \otimes^! \mathrm{id}_C)$ compatible with the two pointings. Its higher simplices retain the corresponding higher paths. Denote this category by $\mathscr{A}_C$. A representing pointed coalgebra Z would give natural equivalences

$$\mathrm{Map}_{J/}(F,Z) \simeq \Phi_C(F,u).$$

The equivalences must hold for every permitted actor F.

There is a second question when F is a transverse monoid, meaning an associative algebra for $\otimes^!$. An action of this monoid includes its multiplication and unit coherences. Let $\mathrm{Act}_F(C)$ be the space of such coherent actions.

A *local center* is a final object among transverse monoids acting on C, with C fixed. For a candidate (Z,β) and a test action (F,α), its mapping space is

$$\mathrm{fib}_\alpha\Big(\mathrm{Map}_{\mathrm{Alg}(\mathscr{C})}(F,Z) \longrightarrow \mathrm{Act}_F(C)\Big). \tag{6.8}$$

The map composes with β. Thus a loop of test actions outside the image of the source fundamental group makes this fiber disconnected. It disproves finality of that candidate.

6.4.1 Free fields and the diagonal tensor

We will use the field coalgebras of Chapter 4 on the affine line. A free-field vertex algebra here has finitely many homogeneous generators, scalar polynomial λ-brackets, and a basis of ordered monomials in their translates. Its differential is linear in those translates. The examples below are tensor products of Heisenberg fields, regular commutative fields and Weyl–Clifford fields.

Let L_B be the positive diagonal family of the state differential module of B. For $|I| \geq 2$ its coefficient is the sum of the full normal direct images along the small diagonal in every order chart. These diagonals have disjoint images: two different order charts overlap only away from at least one of their pair diagonals. The singleton coefficient is the state differential module itself. The empty coefficient is zero. The operator-product residues define a Lie bracket $L_B \star L_B \to L_B$.

Its Chevalley coalgebra is

$$\mathrm{CE}_\star(L_B) = K \oplus \bigoplus_{r \geq 1} \mathrm{Sym}^r_\star(L_B[1]).$$

It has the unshuffle coproduct, and its coderivation is determined by

$$Q_1(sx) = -s(dx), \qquad Q_2(sx\, sy) = (-1)^{|x|} s[x, y].$$

The square-zero identities are respectively the internal chain equation, the derivation rule and Jacobi. Positivity makes the displayed sum finite at each configuration.

For these algebras B, denote this full residue coalgebra by C_B. In its partition description, each summand is the normal direct image from the diagonals of its blocks, localized away from collisions between blocks. Q_2 is their residue boundary. Thus it is also the partition Cousin model of the factorization lattice on the order charts. Its coproduct keeps all block decompositions.

The consecutive components give the cut structure used earlier. The vacuum map $k \to B$ gives $J = C_k \to C_B$.

Lemma 6.6. *For a positive dg Lie object L, the twisting cochain $sx \mapsto x$ induces an equivalence*

$$\mathrm{CE}_\star(L) \xrightarrow{\sim} B_\star U_\star L.$$

Its length-r component is the unnormalized suspended symmetrization

$$sx_1 \cdots sx_r \longmapsto \sum_{\sigma \in S_r} \epsilon(\sigma; sx_1, \dots, sx_r)[x_{\sigma(1)}| \cdots |x_{\sigma(r)}],$$

where ϵ is the Koszul permutation sign.

Proof. Deconcatenating the permutation sum chooses a subset and independently orders it and its complement. This is its unshuffle coproduct. The binary chain equation is the envelope relation $xy - (-1)^{|x||y|} yx = [x, y]$. Grouping adjacent pairs proves it in every length.

Filter the envelope by the number of original L inputs. The multilinear PBW identity gives $\mathrm{gr}\, U_\star L = \mathrm{Sym}_\star L$: ordering a triple has the same result precisely by Jacobi, and averaging is exact in characteristic zero. Apply this identity to a free simplicial Lie resolution for derived inputs. The associated graded comparison is the abelian Koszul comparison.

Its free resolution is $S \star \mathrm{Sym}_\star(L[1])$, where $S = \mathrm{Sym}_\star L$ and $d_K(sx) = -x$. The derivation $h(x) = -sx$, $h(sx) = 0$ satisfies $d_K h + h d_K = N$, with N the total polynomial weight. It anticommutes with the internal differential. Thus h/N contracts each positive weight. The permutation sum compares this resolution with the free bar resolution and induces the identity on their common augmentation. Tensoring with the augmentation module proves the associated graded equivalence.

Both filtrations stop at $|I|$ on a configuration I, so filtered comparison proves the lemma. The coalgebra map has no factor $1/r!$. That factor belongs to PBW averaging. □

Lemma 6.7. *For free fields A, B, the residue construction has natural equivalences*

$$C_A \otimes^! C_B \simeq C_{A\otimes B}.$$

They preserve units, collisions and iterated factorization. The construction preserves quasi-isomorphisms of dg vertex algebras in this class and polynomial homotopies between their maps.

Proof. The free-field factorization lattice on a position chart is the vacuum module for polar sections at the marked points [9, §§3.7.7, 3.7.20]. Filter this module by the number of generating fields. Its associated graded is the graded symmetric algebra of their principal parts. Those principal parts are free over the position ring by (3.5).

At a fixed polynomial degree the complex is bounded and flat, because there are finitely many generating cohomological degrees. The differential preserves the filtration. Successive extensions and filtered unions therefore give a K-flat lattice. Here K-flat means that tensoring with it preserves acyclic complexes.

The fields of A have zero bracket with those of B. Their Lie algebras of polar sections consequently have commuting noncentral factors. Their joint vacuum representation imposes each scalar central relation at its stated value and is the tensor product of their vacuum modules. This identifies the lattice of $A \otimes B$ with the tensor product of the lattices, compatibly with collisions and separation of groups.

The finite Cousin resolution of a lattice is its partition residue complex: the normal boundary maps are the oriented residues of Section 4.1. Tensoring K-flat lattices computes the derived tensor. Passing to right differential modules contracts one copy of the density unit J, giving the asserted $\otimes^!$ comparison. The maps come from polar sections, so their associativity, symmetry, unit and collision diagrams commute before passing to the derived category.

A quasi-isomorphism gives a quasi-isomorphism on each derived partition coefficient. The finite partition filtration then proves the assertion with its residue differential. For a polynomial homotopy use the fixed target tensored with polynomial forms. Evaluation and constant inclusion are inverse quasi-isomorphisms of the parameter complex. Their maps of residue coalgebras retain all faces. □

The same argument for a free regular commutative field uses its multijet lattice [9, §§3.4.20–3.4.22]. If $P_I(w) = \prod_{i\in I}(w - z_i)$, each jet stage $R_I[w]/P_I^m$ is finite free. Chinese remainders give separation of groups, and cofinal powers give collision. The positive cohomological degree of a generating field changes its graded symmetric powers but not this construction.

6.4.2 A cochain action

Fix $\kappa \in k^\times$ and let V_κ be the Heisenberg vertex algebra, with even current J and $[J_\lambda J] = \kappa\lambda$. Adjoin a regular odd field c of degree one and put

$$\mathcal{Z}_\kappa = V_\kappa \otimes \Lambda[c, Tc, T^2c, \dots], \qquad dJ = \kappa Tc, \quad dc = 0.$$

The differential is the zero mode $(:cJ:)_{(0)}$. It is a derivation of every vertex product and squares to zero.

Introduce even fields t, p of degree zero and odd fields η, ι of degrees 1, −1, with brackets

$$[p_\lambda t] = 1, \qquad [\iota_\lambda \eta] = 1.$$

All other generating brackets vanish, apart from graded skewsymmetry. Their quartet algebra E has differential

$$Qt = \eta, \qquad Q\iota = p, \qquad Q\eta = Qp = 0.$$

On $W = V_\kappa \otimes E$, let Q act trivially on the first factor. The assignments

$$i(J) = J + p, \qquad j(J) = J - \kappa T t, \qquad j(c) = -\eta \tag{6.9}$$

define dg vertex maps $i : V_\kappa \to W$ and $j : \mathcal{Z}_\kappa \to W$. Their images commute: for example, $[J + p_\lambda J - \kappa T t] = \kappa\lambda - \kappa\lambda = 0$. Thus

$$\rho : \mathcal{Z}_\kappa \otimes V_\kappa \longrightarrow W, \qquad \rho(a \otimes v) = j(a)_{(-1)} i(v)$$

is a dg vertex map and restricts to i on the vacuum axis.

The map i is a quasi-isomorphism. In the ordered creation-mode basis use $J + p$ in place of J. The remaining variables form the contractible pairs $(T^n t, T^n \eta)$ and $(T^n \iota, T^n p)$. Their Euler contraction sends η to t and p to ι, with the corresponding formulas for translates, and divides by positive total quartet degree. Each state is a finite polynomial, so the contraction is finite. It commutes with translation.

Give J, p, ι conformal weight one and t, η weight zero, and let T raise weight by one. Both the differential and this contraction preserve conformal weight. In particular, they preserve the subspace of positive conformal weight.

Lemma 6.7 now defines a pointed action $\alpha_{\mathcal{Z}}$ by the roof

$$C_{\mathcal{Z}_\kappa} \otimes^! C_{V_\kappa} \simeq C_{\mathcal{Z}_\kappa \otimes V_\kappa} \xrightarrow{C_\rho} C_W \xleftarrow[\sim]{C_i} C_{V_\kappa}. \tag{6.10}$$

The inverse of C_i is taken in the derived category. The vacuum axis supplies its specified pointing. The obstruction below depends on this action. It applies to every coherent transverse monoid structure on $C_{\mathcal{Z}_\kappa}$ with this underlying action.

6.5 A vacuum probe against finality

Let $C = C_{V_\kappa}$ and let

$$N = k[t_0, t_1, \ldots], \qquad |t_n| = 2, \qquad T t_n = t_{n+1}, \qquad dN = 0.$$

This is the free regular commutative vertex algebra on t_0. Write $\epsilon : N \to k$ for its augmentation and $F = C_N$. Multiplication in N makes F a commutative transverse monoid. Its augmentation defines a coherent trivial action α_0 on C. The composite $g_0 : F \to J \to C_{\mathcal{Z}_\kappa}$ is a pointed map inducing that action.

We use total polynomial degree in the t_n, denoted by G, to separate the deformation equations. This is independent of their cohomological degree $2G$. Put $\mathfrak{g} = L_{V_\kappa}$ and $\mathfrak{h} = L_{N \otimes V_\kappa}$. The symmetric convolution envelope is $U_\star \mathfrak{g}$. Its PBW component with a output blocks is $\mathrm{Sym}^a_\star \mathfrak{g}$.

The relative deformation complex at α_0 is

$$\mathcal{T} = R\,\mathrm{Hom}(\mathrm{CE}_\star(\mathfrak{h})/\mathrm{CE}_\star(\mathfrak{g}), U_\star \mathfrak{g}), \qquad d_{\mathcal{T}} = d + [\tau_0, -]. \tag{6.11}$$

Here $\mathrm{CE}_\star$ is the Chevalley coalgebra on suspended inputs, and τ_0 is the twisting cochain of $\epsilon \otimes \mathrm{id}$. Resolve coefficient Homs before taking this convolution and the limit over configurations. Take products over $G > 0$. Convolution adds G. The differential preserves G and the PBW output number.

The bar–cobar comparison and Lemma 6.6 identify this complex with the relative deformation complex of (6.7). In the Chevalley comparison, the leading PBW filtration is the ordinary symmetric–exterior Koszul complex. Its positive factor count is bounded at each configuration, as in Lemma 1.2. The comparison therefore has no infinite length sum there. The positive G filtration makes the relative convolution pronilpotent.

Lemma 6.8. *In total t-degree G, the component of $\mathcal{T}$ with a output blocks has a model in degrees at least $a - 2G$. For $G = 1$, $a = 2$, a degree-zero cocycle is an ordinary binary coefficient, and it has no degree-zero boundary.*

Proof. With n suspended inputs the source degree is $2G - n$. A coefficient extension of degree e into a degree-zero target contributes degree $e + n - 2G$. Derived Hom between coefficient modules has a model in extension degrees $e \geq 0$.

Support forces $n \geq a$: restriction of an input supported on at most n blocks to the locus of a distinct output blocks is zero when $n < a$. Adjunction gives the same vanishing for derived Hom. These statements give the lower bound. For $G = 1$, $a = 2$, degree zero forces $n = 2$, $e = 0$. The internal cocycle equation gives ordinary Hom, and degree minus one is absent. Products and the configuration limit preserve the lower bound. □

Let $\ell : N \to k$ be the augmentation derivation given by $\ell(t_n) = \delta_{n0}$ and zero on all other polynomial degrees. It satisfies

$$\ell(Tb) = 0, \qquad \ell(ab) = \ell(a)\epsilon(b) + \epsilon(a)\ell(b).$$

On singleton suspended inputs define

$$\kappa_1\big(s(t_n \otimes v)\big) = \delta_{n0} v,$$

and let it vanish elsewhere. This is translation compatible: applying total translation on the source gives the translation of the displayed output. Its degree is minus one. The augmentation-derivation identity makes it a cocycle in $\mathcal{T}$. Equivalently, the infinitesimal vertex map $t_0 \mapsto u$, $t_n \mapsto 0$ for $n > 0$, over $|u| = 2$, $u^2 = 0$, has twisting cochain $\tau_0 + u\kappa_1$.

There are no cochains of degree below minus one at $G = 1$, $a = 1$, by Lemma 6.8. Since $\kappa_1(s(t_0 \otimes \mathbf{1})) = \mathbf{1}$, its class is nonzero.

On the Heisenberg Fock space $k[x_1, x_2, \ldots]$, let $J_1 = \kappa\partial_{x_1}$. The current mode identity and $J_0 = 0$ give

$$[J_1, Y(v, z)] = Y(J_1 v, z), \qquad [T, J_1] = -J_0 = 0.$$

Thus J_1 is a translation-compatible vertex derivation, kills the vacuum and satisfies $J_1 J = \kappa\mathbf{1}$.

At two separated singleton blocks define

$$h\big(s(t_n \otimes v), s(\mathbf{1} \otimes w)\big) = \delta_{n0}\, v \boxtimes J_1 w. \tag{6.12}$$

Extend by suspended symmetry, with all other components zero. This has degree zero and is relative to the boundary copy of C.

If $\widehat{J}_1 = J_1\tau_0$, then $h = -(\kappa_1 * \widehat{J}_1)^{(2)}$ in convolution. The minus sign cancels the sign from a degree-one map crossing the suspended t input of degree one. Both factors are closed and the differential obeys the Leibniz rule. Hence h is closed.

6.5.1 *Extracting the separated vacuum coefficient*

For a component K with a output blocks, add a separated boundary singleton at z carrying J. Apply the vacuum coefficient at that block and take the constant coefficient of its Laurent expansion at $z = \infty$. Use test function 1 in z. Put the suspended current first in the input order, including that permutation's Koszul sign. For a cochain of degree q, multiply the result by $(-1)^{q+1}$ and denote it by $\mathsf{P}(K)$.

This extraction is defined on a coefficient resolution. The input J is a free translation generator: the positive-weight Heisenberg states are free over $k[T]$, and no translate has weight one. To see freeness, translation is injective on nonconstant differential polynomials by the largest-mode argument. Homogeneous complements to its image then give a free $k[T]$-basis. The same argument applies to N_+ and $N_+ \otimes V_\kappa$. On the latter, first use the highest actor weight.

Lemma 6.9. *The extraction has degree minus one and satisfies*

$$d_{\mathcal{T}}\mathsf{P} + \mathsf{P}d_{\mathcal{T}} = 0, \qquad \mathsf{P}(h) = \kappa\,\kappa_1.$$

Proof. Extraction at the separated block commutes with coefficient differentials, residues and derivatives in every remaining position. In the component of a fixed output partition, the differential acts within one block. A bracket in the added singleton would require a second input there. Removing its sole input gives the zero component of that output block. The two convolution terms multiplying by the added J cancel, because different output blocks commute on their disjoint locus.

The remaining differential is the one on the other blocks. Inserting the suspended current has degree minus one. The stated factor $(-1)^{q+1}$ gives the displayed anticommutation in the Hom differential. Extraction preserves G and finite normal order, so it passes to their completed convolution. On (6.12) its value is the vacuum coefficient of $J_1 J = \kappa\mathbf{1}$. The permutation sign cancels its prefactor, giving $\kappa\kappa_1$. □

Consider the pointed comparison induced by (6.10):

$$\Theta_F : \mathrm{Map}_{J/}(F, C_{\mathcal{Z}_\kappa}) \longrightarrow \Phi_C(F, u).$$

At g_0, its derivative sends source cochains to $\mathcal{T}$. Every class in its $G = 1$, two-output, degree-zero image has zero vacuum probe.

Indeed, transfer the roof through the quartet contraction. Its twisting corestriction is Lie-valued. One can construct it by the finite rooted-tree formulas for the bracket and the contraction. Each corestriction has one output block. In the derivative of composition, a source cochain acts on one block and the base point acts on the others.

At a separated boundary singleton carrying J, the resulting state is either J or the contraction of $\zeta_{(-1)}(J + p)$. Here ζ lies in the resolving coefficient. All states have nonnegative conformal weight. The normal product adds weights, and the contraction preserves them. Both possible values have positive weight and hence zero vacuum coefficient.

This factorization through positive conformal weights persists on coefficient resolutions. Lemma 6.8 places the indicated degree-zero component in ordinary binary Hom, so there is no hidden lower-degree correction to this vanishing. It holds before extraction at infinity. But $\mathsf{P}(h) = \kappa\kappa_1$ represents a nonzero class. Thus $[h]$ is outside the image of the derivative of Θ_F.

6.5.2 *The coherent loop and its consequence*

Over polynomial forms $k[s, ds]$, the twisting cochain $\tau_0 + ds\, h$ is a loop at τ_0. It also defines a loop of coherent monoid actions. For p actor factors, set

$$\epsilon_p = \epsilon^{\otimes p}, \qquad \ell_p = \sum_{i=1}^{p} \epsilon^{\otimes(i-1)} \otimes \ell \otimes \epsilon^{\otimes(p-i)}.$$

Let κ_p be its singleton cochain and put $h_p = -(\kappa_p * (J_1 \tau_0^{(p)}))^{(2)}$.

Each is closed. Adjacent actor multiplication preserves ℓ_p by its derivation identity, and actor-unit insertion preserves it because $\ell(1) = 0$. For composition of two actor words, the two linear terms agree with

$$\ell_{p+q} = \ell_p \otimes \epsilon_q + \epsilon_p \otimes \ell_q.$$

Every term containing two copies of ds is zero.

Use these formulas on the free associative bar resolution of an actor word followed by its module input. Each parenthesization of a word of length p gives the same constant term and the same linear term h_p. Thus every face and degeneracy agrees with its prescribed multiplication, action or unit map. Constant fillers on these bar simplices give all higher coherences. The extra unit degeneracy contracts the augmented bar resolution. The formulas therefore descend to the required coherent action, retaining their collision maps by Lemma 6.7.

Theorem 6.10. *The comparison Θ_F is not surjective on fundamental groups at g_0. For any coherent transverse monoid structure on $C_{\mathcal{Z}_\kappa}$ with action $\alpha_{\mathcal{Z}}$, the mapping space from (C_N, α_0) to that action is disconnected. In particular this cochain action is not a local center.*

Proof. For a nilpotent relative convolution Lie algebra, based loops of its Maurer–Cartan space are $\exp H^0$ with the Campbell–Hausdorff law. The class of a constant loop form $ds\, v$ is $\exp[v]$. See [42, §5]. Here one applies this statement at each finite G quotient. The differential preserves G, so the quotients on cohomology are surjective. Taking their inverse limit gives the completed Campbell–Hausdorff group.

In composition of loops, terms involving two deformation forms vanish. Hence Θ_F sends $\exp[v]$ to the exponential of its derivative on $[v]$. The nonzero probe of $[h]$ and the vanishing probe on the image show that $\exp[h]$ is not in that image.

The constructed loop belongs to $\mathrm{Act}_{C_N}(C)$, with all its monoid coherences. If it lifted to a loop of monoid maps into $C_{\mathcal{Z}_\kappa}$, forgetting the monoid structures would contradict the preceding pointed comparison. The homotopy exact sequence of (6.8) therefore gives at least two connected components in that fiber. A final action would make it contractible. □

The probe uses a derivation that sends a positive weight state to the vacuum. It detects this specified action and its weight-preserving resolutions. It does not exclude a final transverse monoid with different coefficients or a different action. That question requires recipients for arbitrary monoid actions with their prescribed paths.

6.6 Derivation profiles

A derivation supplies an infinitesimal action. Extending it to an action requires products of its values and derivatives of its actor inputs. We construct those terms together. Their order matters even when the actor inputs themselves are symmetrized.

Let $\mathfrak{g}$ be a positive Lie object in the symmetric coefficient category above, concentrated in degree zero, and put $A = U_\star \mathfrak{g}$. Let $D : \mathfrak{g} \to \mathfrak{g}$ be a degree-zero Lie derivation that commutes with the full coefficient maps. It extends to an associative derivation of A. In particular, it commutes with collision and normal transfer. A derivation on one separated fiber alone does not meet this hypothesis.

Give $\varepsilon\mathfrak{g}$ degree one and form the semidirect product with brackets

$$[x, \varepsilon m] = \varepsilon [x, m], \qquad [\varepsilon m, \varepsilon n] = 0.$$

Its suspended actor inputs $s\varepsilon m$ have degree zero. Their number is the *actor weight*. The Chevalley coalgebra has underlying comodule

$$\mathrm{CE}_\star(\mathfrak{g}) \star \mathrm{Sym}^c_\star(\varepsilon\mathfrak{g}[1]),$$

where the superscript c specifies the unshuffle coproduct. We identify the second factor's generators with $m \in \mathfrak{g}$ in degree zero.

Write $P_w = \mathrm{Sym}^w_\star \mathfrak{g}$. For $u = m_1 \circ \cdots \circ m_w$, set

$$[x, u] = \sum_i m_1 \circ \cdots \circ [x, m_i] \circ \cdots \circ m_w.$$

Symmetric powers are coinvariants, identified with invariants by averaging.

Lemma 6.11. *Coefficient maps $\theta_w : \mathfrak{g} \star P_w \to A$, $w \geq 1$, are the additional components of a twisting cochain under $\mathrm{CE}_\star(\mathfrak{g})$ if and only if*

$$\begin{aligned} 0 = {} & [x, \theta_w(y;u)] - \theta_w(y;[x,u]) - [y, \theta_w(x;u)] + \theta_w(x;[y,u]) - \theta_w([x,y];u) \\ & + \sum_{\substack{S\sqcup T=[w] \\ S,T\neq\varnothing}} \big(\theta_{|S|}(x;m_S)\theta_{|T|}(y;m_T) - \theta_{|S|}(y;m_S)\theta_{|T|}(x;m_T)\big). \end{aligned} \tag{6.13}$$

The splitting (S, T) is ordered, and m_S denotes the symmetric product of the inputs with labels in S.

Proof. The value of a degree-one cochain on n suspended ordinary Lie inputs has degree $1 - n$ in A. Only $n = 1$ contributes, and its value is $\theta_w(x; u)$. The twisting equation has degree $2 - n$ and therefore only tests $n = 2$.

Use $Q_2(sx\, sy) = s[x, y]$. Its three bracket terms are $\theta_w([x, y]; u) + \theta_w(y; [x, u]) - \theta_w(x; [y, u])$.

In a nonzero convolution product, each tensor factor has one ordinary suspended input. Passing the second degree-one cochain across the first factor contributes a minus sign. The unshuffles with an empty actor part give the two commutators, and those with both parts nonempty give the sum in (6.13). Multiplying the twisting equation by minus one proves the formula. Weight zero is the envelope relation. □

6.6.1 The fixed-point construction

We first prove a universal identity in an associative algebra with derivation. Let z be a central formal parameter, and for $R \in zA[[z]]$ set

$$f\langle R\rangle = \sum_{j\geq 0} \frac{D^j f}{j!} R^j.$$

Every power of R stands on the right. Choose central scalar parameters $t_1, t_2, \ldots$ and put

$$U = \sum_{r\geq 1} \frac{t_r z^r}{r!} m^r, \qquad R = U\langle R\rangle, \qquad G_x = x\langle R\rangle. \tag{6.14}$$

Solve for R by increasing powers of z. Each unknown coefficient on the right is multiplied by a coefficient of positive z-order, so the solution is unique.

Let δ_x vary the actor variable m in the direction $[x, m]$. It replaces $D^j m$ by $D^j[x, m]$ and fixes all external arguments.

Lemma 6.12. *The series in (6.14) satisfy*

$$\delta_x R = [G_x, R], \tag{6.15}$$

$$[G_x, G_y] - G_{[x,y]} - \delta_x G_y + \delta_y G_x = 0. \tag{6.16}$$

Proof. Put $U_j = D^j U/j!$ and $X_i = D^i x/i!$. Since $\partial_x U = [x, U]$, differentiation of the fixed-point equation gives a linear equation for $Z = \partial_x R$:

$$Z = [x, U]\langle R\rangle + \sum_j U_j \sum_{p+q=j-1} R^p Z R^q.$$

Its linear part raises z-order, so its solution is unique.

Taking the commutator of G_x with the fixed-point equation gives the same equation for $[G_x, R]$. To check the constant term in that equation, expand Leibniz for D. The difference is

$$\sum_{i,j} X_i [R^i, U_j] R^j = \sum_i X_i \sum_{p+q=i-1} R^p W R^q, \qquad W = \sum_j [R, U_j] R^j = [R, R] = 0.$$

This proves (6.15).

Write $Y_j = D^j y/j!$. Subtracting $G_{[x,y]} = \sum_{i,j} [X_i, Y_j] R^{i+j}$ from $[G_x, G_y]$ leaves

$$\sum_{i,j} X_i [R^i, Y_j] R^j - \sum_{i,j} Y_j [R^j, X_i] R^i.$$

Expand each commutator with a power of R. The first sum is $-\partial_y G_x$ and the second, with its displayed minus sign, is $\partial_x G_y$, by (6.15). This gives (6.16). □

For a nonempty set B of actor labels, define

$$u_B = \frac{1}{|B|!} \sum_{\sigma \in S_B} m_{\sigma(1)} \cdots m_{\sigma(|B|)}.$$

Let $\mathcal{T}_j$ be the nonnegative tuples $(a, b_1, \ldots, b_j)$ such that

$$a + \sum_{i=1}^{j} b_i = j, \qquad a + \sum_{i=1}^{\ell} b_i \geq \ell + 1 \quad (0 \leq \ell < j).$$

These are the child numbers of a plane rooted tree with j nonroot vertices, in preorder. For arbitrary envelope elements v_i, put

$$T_j^D(x; v_1, \ldots, v_j) = \sum_{(a,b) \in \mathcal{T}_j} \frac{D^a x}{a!} \sum_{\sigma \in S_j} \frac{D^{b_1} v_{\sigma(1)} \cdots D^{b_j} v_{\sigma(j)}}{b_1! \cdots b_j!}, \qquad T_0^D(x) = x.$$

Theorem 6.13. *For every scalar sequence* $\mathbf{t} = (t_1, t_2, \ldots)$*, the operations*

$$\theta_w^{\mathbf{t}}(x; m_1, \ldots, m_w) = \sum_{\pi \in \mathrm{Part}([w])} \left(\prod_{B \in \pi} t_{|B|} \right) T_{|\pi|}^D(x; u_B : B \in \pi) \tag{6.17}$$

satisfy (6.13)*. The partitions here are unordered and have nonempty blocks. They define a map of derived convolution coalgebras*

$$\mathrm{CE}_\star(\mathfrak{g} \ltimes \varepsilon\mathfrak{g}) \longrightarrow \mathrm{CE}_\star(\mathfrak{g})$$

under $\mathrm{CE}_\star(\mathfrak{g})$*. On a configuration* I*, its nonconstant operations use only* t_r *with* $r < |I|$*.*

Proof. Expanding the fixed point orders the subtrees below each vertex. A vertex with j children contributes $D^j U/j!$, and the root contributes $D^a x/a!$. Polarizing a term of U groups the actor labels into the blocks B, with product u_B and coefficient $t_{|B|}$. Ordering these blocks among the tree's vertices and summing their orders gives $T^D_{|\pi|}$.

Consequently

$$G_x = x + \sum_{w\geq 1} \frac{z^w}{w!}\theta^{\mathbf{t}}_w(x; m, \dots, m).$$

The coefficient of $z^w/w!$ in (6.16) is (6.13) on equal inputs. The two variations give its actor substitutions. Multiplication of exponential generating series gives the ordered unshuffles. Polarization proves it for all inputs.

The twisting cochain gives the map to $B_\star A$ by its iterated reduced coproduct. The comparison of Lemma 6.6 identifies this target with $\mathrm{CE}_\star(\mathfrak{g})$. On weight zero this is the canonical comparison, which gives the specified map under that coalgebra. Each positive input occupies a nonempty subset of I. A term of actor weight w also consumes one ordinary input, so $w + 1 \leq |I|$. All coproduct and tree sums are therefore finite on each configuration. □

The first two operations display both sources of higher terms:

$$\begin{aligned}\theta^{\mathbf{t}}_1(x; m) &= t_1 Dx\, m,\\ \theta^{\mathbf{t}}_2(x; m, n) &= t_2 Dx\, \frac{mn + nm}{2}\\ &\quad + t_1^2\Big(D^2x\, \frac{mn + nm}{2} + Dx(Dm\, n + Dn\, m)\Big).\end{aligned}$$

At weight three, the mixed coefficient of t_1t_2 is $\sum_{\{i\}\sqcup B=[3]} T^D_2(x; m_i, u_B)$. Thus independently chosen leading weights also have specified mixed primitives.

To interpret these maps as actions of a field coalgebra, one needs a coefficient comparison. Precisely, let $E = \Lambda(\varepsilon)$, with $|\varepsilon| = 1$ and $d\varepsilon = T\varepsilon = 0$. Suppose the residue construction identifies $C_{E\otimes V}$ and C_V with the two Chevalley coalgebras in Theorem 6.13, respectively. Require these equivalences to preserve the vacuum axis, actor weights, collisions and normal transfer, and to identify their derived mapping spaces. Together with Lemma 6.7, they turn the constructed map into a pointed action of C_E on C_V. Under these hypotheses, no completion of normal derivatives or of state spaces is needed: the formal parameter has proved finite identities at every configuration.

Lemma 6.14. *For $V = V_\kappa$, $\kappa \neq 0$, these comparisons exist. They are natural under rescaling $\varepsilon \mapsto t\varepsilon$, tensor powers of E, and finite scalar extension.*

Proof. Resolve E by the regular commutative field algebra

$$\mathcal{K} = k[u, Tu, \dots] \otimes \Lambda[c, Tc, \dots], \qquad |u| = 0, \quad |c| = 1, \quad du = Tc, \quad dc = 0.$$

The map $q : \mathcal{K} \to E$ sends u to zero, c to ε, and their positive translates to zero. The pairs $(T^n u, T^{n+1}c)$ contract by their Euler homotopy, leaving only $\Lambda(c)$. Thus q is a quasi-isomorphism.

Apply the lattice comparison of Lemma 6.7 to $\mathcal{K}$ and V_κ. The finite partition filtration preserves the quasi-isomorphism induced by q, so it gives $C_E \otimes^! C_{V_\kappa} \simeq C_{E\otimes V_\kappa}$. The diagonal Lie object of $E \otimes V_\kappa$ is $\mathfrak{g} \ltimes \varepsilon\mathfrak{g}$: the exterior coefficient multiplies the chiral bracket, and $\varepsilon^2 = 0$. This identifies the target with its Chevalley model, including the vacuum axis.

The maps $u \mapsto tu$, $c \mapsto tc$ commute with q and with the lattice construction. They give weight t^w on w suspended actor inputs. Tensor powers use the same independent contractible pairs. The constructions preserve finite scalar extension and all collision maps, since these maps already act on the lattices and their residue resolutions. □

Henceforth in the Heisenberg case take $D = \kappa^{-1} J_1 = \partial_{x_1}$, so $D(J) = \mathbf{1}$. It is the derivative of the vertex automorphism $J \mapsto J + t\mathbf{1}$ and is therefore a map of the full coefficient family. Let $\alpha_{\mathbf{t}}$ denote the pointed actions just constructed. Write α_D for the sequence $t_1 = 1$, $t_r = 0$ for $r > 1$.

Proposition 6.15. *For every $r \geq 1$, the first variation of $\alpha_{\mathbf{t}}$ in t_r at the trivial action is a nonzero class. The classes for different r have distinct actor weights. For every $N \geq 1$ there is a nontrivial pointed action equal to the trivial action on all configurations with at most N positions.*

Proof. The first variation is $u_r(x; m_1, \dots, m_r) = Dx\, u_{[r]}$. In its PBW component with $r+1$ output roots, it is $\gamma_r(x; v) = (Dx) \circ v$, where $v \in P_r$ and $\circ$ is symmetric multiplication. Its Chevalley cocycle equation is

$$[x, Dy] + [Dx, y] - D[x, y] = 0.$$

Thus it defines the triangular module extension $P_{r+1} \oplus P_r$ with off-diagonal action γ_r.

There is no coefficient map $P_r \to P_{r+1}$. Indeed P_{r+1} is open direct image from its $(r+1)$-root locus, where the source restricts to zero. The underlying extension therefore has only its displayed splitting. It is not equivariant: on $r+1$ distinct roots, put J at the acting root and the vacuum at the others. Then γ_r gives the nonzero vacuum tensor with coefficient one.

Degree-zero derived module maps here are ordinary equivariant maps: their resolved coefficient Hom and Chevalley complexes have no negative degrees. The extension class is therefore nonzero. PBW projection preserves the adjoint differential, so $[u_r]$ is nonzero as well.

For the last assertion, use $t_N = 1$ and all other parameters zero. Its positive terms require at least $N+1$ supports.

It remains to exclude a homotopy to the trivial action. More generally, if two normalized relative actions vanish below weight r, write a gauge between them as $G = 1 + \sum_{j>0} g_j$. The equation $d_0 G = G\theta - \theta' G$ gives $d_0 g_j = 0$ below weight r and $d_0 g_r = \theta_r - \theta'_r$ in weight r. Their first classes must agree. Here one of them is $[u_N] \neq 0$ and the other is zero. □

This also obstructs the ordinary exterior product for profiles starting above weight one. Let β have first nonzero class $[\beta_r]$ in weight $r > 1$. If the standard product μ of C_E made it multiplicative, pull its action path back along the primitive map $\varepsilon \mapsto \varepsilon_1 + \varepsilon_2$. On the product side this scales the first class by 2^r. On the iterated-action side the first jets add, giving $2[\beta_r]$: all products of positive lower jets vanish. The preceding gauge calculation would give $(2^r - 2)[\beta_r] = 0$, impossible in characteristic zero.

This concerns the standard multiplication on this actor. It does not exclude a map of the action to a different multiplicative action.

Example 6.16. In $k[q]$, take $D = \partial_q$, $x = m = q$ and $t_2 = 1$, with all other parameters zero. Then $G_q = q + R$ and

$$G_q = q + \frac{z^2}{2} G_q^2.$$

The solution with constant term q is

$$G_q = \sum_{n \geq 0} \frac{1}{n+1} \binom{2n}{n} \left(\frac{z^2}{2}\right)^n q^{n+1}.$$

Indeed, $G_q = qC(z^2 q/2)$ reduces the equation to $C = 1 + uC^2$, whose solution at $C(0) = 1$ is $(1 - \sqrt{1-4u})/(2u)$.

6.6.2 *An inner gauge and its domain*

The same fixed point determines when an inner derivation gives a trivial profile. We first work in an ordinary unital associative algebra A. Fix $a \in A$ and take $D = [a, -]$.

The variations δ_x fix a. They satisfy $[\delta_x, \delta_y] = -\delta_{[x,y]}$, so $\rho_x F = [x, F] - \delta_x F$ defines a Lie action on functions of m. Let d_0 be its Chevalley differential. In particular, $d_0 F(x) = \rho_x F$ on zero-cochains.

The positive one-cochain $\theta(x) = G_x - x$ satisfies $d_0\theta + \theta \smile \theta = 0$ by (6.16). A normalized gauge from θ to zero is an invertible zero-cochain $B = 1 \bmod z$ with $d_0 B = B\theta$.

Proposition 6.17. *For the fixed point* (6.14) *with* $D = [a, -]$*, such a gauge is*

$$B = \sum_{n \geq 0} \frac{(-1)^n}{n!} a^n R^n. \tag{6.18}$$

Its powers of a stand to the left of its powers of R.

Proof. The constant term is one, so its geometric inverse converges z-adically. Since $\delta_x a = 0$, transport gives

$$\delta_x B = \sum_{n \geq 0} \frac{(-a)^n G_x R^n}{n!} - B G_x.$$

The coefficient on the left of R^k in the sum is

$$\sum_{n+j=k} \frac{(-a)^n (\operatorname{ad} a)^j x}{n!\, j!} = \frac{(-1)^k}{k!} x a^k.$$

For this equality, left and right multiplication L_a, R_a commute and $\operatorname{ad} a = L_a - R_a$. The coefficient identity is exactly $e^{-tL_a} e^{t(L_a - R_a)} = e^{-tR_a}$. Thus $\delta_x B = xB - BG_x$, and $[x, B] - \delta_x B = B(G_x - x)$, as required. □

For $U = zm$ the first terms are

$$R = zm + z^2 [a, m]m + O(z^3), \qquad B = 1 - zam + z^2 \Big(-a[a, m]m + \frac{1}{2} a^2 m^2 \Big) + O(z^3).$$

The coefficient of z^2 in $B - e^{-aR}$ is $\frac{1}{2} a[a, m]m$, nonzero in the free algebra on a, m. An ordinary exponential does not give this gauge.

The polarized identities also hold in a resolved coefficient convolution under two hypotheses. All operations must preserve that coefficient family, and sufficiently many positive factors must vanish at each configuration. The inverse and the gauge equation are then finite there. These hypotheses include the element that implements the inner derivation.

For the positive convolution envelope $U_\star \mathfrak{g}$, that last hypothesis has only the zero-derivation instance. Its convolution unit K is supported at the empty configuration, whereas $\mathfrak{g}$ is positive. A global element $a : K \to U_\star \mathfrak{g}$ is therefore scalar, and $[a, -] = 0$. A nonzero state mode can still define a derivation of the full coefficient family. It need not arise from a global element of this envelope. In particular, the gauge above does not trivialize the Heisenberg shift used in the vacuum probe.

6.7 Multiplicative recipients for the profiles

The failure of the standard product on a higher profile does not prevent that action from mapping to a multiplicative one. We construct such maps by changing the actor parameter. To compare their actions, the change of parameter must extend to an actual bar map.

6.7.1 Tensoring the entire coefficient construction

Lemma 6.18. *Let C be a cocommutative factorizing $\star$-coalgebra with empty coefficient k. Then $\mathcal{T}_C(F) = F \otimes^! C$ is a colimit-preserving strong symmetric monoidal functor for $\star$. For $C = C_V$, it takes L_k to L_V and L_E to $L_{E\otimes V}$, with their brackets.*

Proof. Use the coproduct of C to compare $(F \star G) \otimes^! C$ with $(F \otimes^! C) \star (G \otimes^! C)$. On the summand for $I = A \amalg B$, this is open direct image of

$$(F_A \boxtimes G_B) \otimes^! C_I|_{U_{A,B}} \longrightarrow ((F_A \otimes^! C_A) \boxtimes (G_B \otimes^! C_B))|_{U_{A,B}}.$$

It is an equivalence by factorization and the open projection formula. That formula follows here from localization: extraordinary restriction to the complement vanishes, and restriction to the open is the displayed tensor. At the empty configuration the unit comparison is $K \otimes^! C = K$.

Every iterated comparison uses the same final partition and the same iterated coproduct. Coassociativity and cocommutativity give all monoidal coherences. Collision kills exactly the summands whose blocks split a collision fiber. On the others the comparison is the same base-change map. Tensor product and convolution preserve colimits.

On a small-diagonal branch, projection for its full normal direct image identifies $L_k \otimes^! C_V$ with the state coefficient L_V. The bracket on L_k is the vacuum principal-part boundary. Tensoring it with the factorization lattice of V gives its two-position residue boundary, hence the bracket of L_V. The same computation on the even and odd summands of L_E gives $L_{E\otimes V} = \mathfrak{g} \ltimes \varepsilon\mathfrak{g}$. Thus the functor transports the Chevalley, envelope and bar constructions, including their coefficient maps. □

For any degree-zero positive Lie object $\mathfrak{l}$, work in $U_\star(\mathfrak{l} \ltimes \varepsilon\mathfrak{l})$ and set

$$\rho_r(m_1, \dots, m_r) = \frac{1}{r!} \sum_{\sigma \in S_r} m_{\sigma(1)} \cdots m_{\sigma(r-1)}(\varepsilon m_{\sigma(r)}).$$

Define a cochain by $\tau_{\mathbf{t}}(sx) = x$ on ordinary singleton inputs and

$$\tau_{\mathbf{t}}(s\varepsilon m_1 \cdots s\varepsilon m_r) = t_r \rho_r(m_1, \dots, m_r)$$

on pure actor inputs, with all other reduced components zero.

It is a twisting cochain. With one ordinary input its equation is adjoint equivariance. With two ordinary inputs it is the envelope relation. The remaining equation polarizes from equal actor inputs, where $\rho_r(m, \dots, m) = m^{r-1}\varepsilon m$. All products of two such values vanish because $[m, \varepsilon m] = 0$ and $(\varepsilon m)^2 = 0$.

Thus $\tau_{\mathbf{t}}$ gives the strict bar map

$$F_{\mathbf{t}} = \sum_{j \geq 0} (s\tau_{\mathbf{t}})^{\star j} \overline{\Delta}^{(j)}. \tag{6.19}$$

After the inverse of Lemma 6.6, it defines a pointed self-map. For $\mathfrak{l} = L_k$ denote it by $\Sigma_{\mathbf{t}} : C_E \to C_E$. Augmentation kills its positive odd outputs, so it also preserves the augmentation to J. Every sum is finite on each configuration.

6.7.2 A bar lift of the derivation action

The substitution identity has a differential version. Let $(\mathcal{G}, d)$ be a complete filtered dg associative algebra, with $F^p\mathcal{G}\, F^q\mathcal{G} \subseteq F^{p+q}\mathcal{G}$. Let D preserve this filtration and commute with d. Suppose a has degree one, $da + a^2 = 0$, and $p \in F^1\mathcal{G}$ has degree zero with $dp + [a, p] = 0$. Put

$$a_j = D^j a/j!, \qquad p_j = D^j p/j!, \qquad R = \sum_{j \geq 0} p_j R^j, \qquad A_R = \sum_{j \geq 0} a_j R^j.$$

Lemma 6.19. *These series satisfy* $dR + [A_R, R] = 0$ *and* $dA_R + A_R^2 = 0$.

Proof. The differentiated equations are $da_j = -\sum_{u+v=j} a_u a_v$ and $dp_j = -\sum_{u+v=j} [a_u, p_v]$. Expanding the fixed point gives

$$\sum_{u,v} [a_u, p_v] R^{u+v} = [A_R, R] - \sum_v p_v [A_R, R^v].$$

The difference between the two sides is $\sum_u a_u [R^u, \sum_v p_v R^v] = 0$.

Consequently $-[A_R, R]$ solves the linear fixed-point equation obtained by differentiating R. Its positive filtration gives uniqueness. Since the a_j are odd, substitution yields

$$dA_R = -\sum_{u,v} a_u a_v R^{u+v} + \sum_u a_u [A_R, R^u].$$

Subtracting the first sum from A_R^2 gives $\sum_u a_u [R^u, A_R]$, which cancels the last sum. This proves both equations. □

Apply the lemma with $B = U_\star(\mathfrak{g} \ltimes \varepsilon\mathfrak{g})$, $A = U_\star\mathfrak{g}$ and the bar convolution $\mathcal{G} = \mathrm{Hom}^\bullet(B_\star B, A)$. Let $\pi : B \to A$ kill the odd ideal. The degree-minus-one projection derivation $q : B \to A$ is determined by

$$q(x) = 0, \quad q(\varepsilon m) = m, \quad q(uv) = q(u)\pi(v) + (-1)^{|u|}\pi(u)q(v).$$

It respects the envelope relations. For example, applying q to $[x, \varepsilon m] = \varepsilon[x, m]$ gives the same $[x, m]$ on both sides. Two-odd-input relations give zero.

Take the length-one cochains induced by π and q for a and p. Their degrees are one and zero, and the projection derivation identity is $dp + [a, p] = 0$. The lemma gives a twisting cochain and a coalgebra map $\mathcal{F}_D : B_\star B \to B_\star A$. On a word $[v_0|\cdots|v_w]$, its corestriction is

$$\sum_{(a,b)\in\mathcal{T}_w} \frac{D^a \pi(v_0)}{a!} \frac{D^{b_1} q(v_1) \cdots D^{b_w} q(v_w)}{b_1! \cdots b_w!}. \tag{6.20}$$

This follows by the same ordered-subtree expansion as the profile fixed point. It vanishes unless v_0 has odd-generator degree zero and each later v_i has odd-generator degree one.

Restriction along the Chevalley comparison puts the ordinary input first and sums the orders of the actor inputs, giving α_D. The bar-length filtration is finite at each configuration, so this is an actual coefficient map there.

Proposition 6.20. *For the Heisenberg actions, there is a natural pointed path*

$$\alpha_D(\Sigma_{\mathbf{t}} \otimes^! \mathrm{id}_C) \simeq \alpha_{\mathbf{t}}.$$

It preserves all collisions, normal transfers and the specified vacuum path.

Proof. Lemma 6.18 transports $\Sigma_{\mathbf{t}}$ from L_k to $\mathfrak{g}$. Represent the composite by $\mathcal{F}_D F_{\mathbf{t}}$. A nonzero term puts the ordinary input in its first singleton slot and partitions the actor inputs into ρ-blocks. The projection derivation satisfies $q\rho_r(m_1, \ldots, m_r) = u_{[r]}$. For each unordered partition, summing its block orders in (6.20) gives exactly (6.17), with coefficient $\prod_B t_{|B|}$.

Thus the corestrictions agree in every weight. The pointed bar comparisons transport this equality to the stated path and its coherences. All operations are maps of the full coefficient families. □

6.7.3 The complex of natural higher comparisons

It remains to construct multiplication coherences for α_D itself. The relevant primitives can be chosen as universal differential polynomials. No contraction of an unspecified geometric mapping complex is assumed.

Let P, Z be finite-dimensional degree-zero spaces. Write $E(P) = \Lambda(P^*[-1])$ and choose commuting derivations $D_1, \dots, D_{\dim Z}$ of a degree-zero Lie algebra $\mathfrak{g}$. A natural cochain on $\mathrm{CE}(\mathfrak{g} \otimes E(P))$ is a universal multilinear differential polynomial with values in $U(\mathfrak{g})$. It applies the chosen derivations and multiplies the resulting inputs, using each input once. Its evaluation includes suspended symmetrization. Brackets are expanded as commutators.

Let $\mathcal{N}(P; Z)$ be their convolution algebra, with the Hom differential twisted by the scalar augmentation action τ. Take the product over actor weights, with finitely many derivative orders at each cochain degree and actor weight. Each specified input is evaluated by a finite polynomial.

For an exterior subset S and derivative multi-index a, let $x_{S,a}$ extract that exterior coefficient and apply $D^a/a!$. It has degree $1 - |S|$, actor weight $|S|$ and derivative weight $|a|$. These unary cochains freely generate $\mathcal{N}(P; Z)$ under convolution. Indeed their products are signed orbit sums of ordered differential monomials. Evaluation in the free differential Lie algebra on distinct named inputs recovers each ordered jet word in its free tensor envelope, proving independence. Polarization handles repeated exterior types.

Put $t = x_{\varnothing,0} = \tau$ and $y_{S,a} = (-1)^{\binom{|S|}{2}} x_{S,a}$. For disjoint subsets let $\epsilon(B, C)$ be the sign sorting their exterior product. The differential is

$$\partial t = t^2, \qquad \partial y_{S,a} = -\sum_{\substack{B \sqcup C = S \\ b+c=a}}{}' (-1)^{|B|} \epsilon(B, C)\, y_{B,b}\, y_{C,c}. \tag{6.21}$$

The prime omits $(\varnothing, 0)$ on either side, and the second formula excludes t. To obtain it, use $Q_2(s(e_B x)s(e_C y)) = (-1)^{|B|} s(e_B e_C [x, y])$, the divided Leibniz rule, and the convolution sign $(-1)^{|g||c'|}$ in $(f * g)(c)$. The terms containing t cancel the commutator with the scalar twisting cochain.

Lemma 6.21. *Weight by weight,*

$$H^\bullet \mathcal{N}(P; Z) = \widehat{\mathrm{Sym}}(P) \otimes \Lambda(Z[-1]).$$

The P generators have degree zero and actor weight one. The $Z[-1]$ generators have degree one and derivative weight one. Thus actor weight q and derivative order q have cohomology $\mathrm{Sym}^q(P) \otimes \Lambda^q(Z)$ *in degree q.*

Proof. Let $\Gamma(Z)$ be the divided-power coalgebra, with $\Delta z^{[a]} = \sum_{b+c=a} z^{[b]} \otimes z^{[c]}$. Formula (6.21) identifies $\mathcal{N}(P; Z)$ with

$$(k\langle t\rangle, \partial t = t^2) * \Omega\big(\Lambda(P[1]) \otimes \Gamma(Z)\big).$$

The augmentation ideal of the first factor contracts by $t^{2j} \mapsto t^{2j-1}$ and $t^{2j-1} \mapsto 0$. On a free-product word, contract its first such factor with the sign of the prefix. Projection killing t is therefore a quasi-isomorphism.

Map the remaining cobar to $\mathrm{Sym}(P) \otimes \Lambda(Z[-1])$ by preserving its primitive generators and killing the others. The differential becomes a graded commutator or an odd square, hence zero. The map is surjective on cohomology.

To prove injectivity, dualize each auxiliary weight. The dual coalgebra algebra is $B = \Lambda(P^*[-1]) \otimes \mathrm{Sym}(Z^*)$. With generators α_i, z_j, its free resolution of k is

$$B \otimes k[v_i] \otimes \Lambda(\eta_j), \qquad |v_i| = 0, \quad |\eta_j| = -1, \quad dv_i = \alpha_i, \quad d\eta_j = z_j.$$

For an odd pair, $d(v_i^{m+1}) = (m+1)\alpha_i v_i^m$, leaving only the scalar class. An even pair is the two-term resolution for multiplication by z_j. Their tensor product resolves k. Its Hom into k has zero differential and the stated symmetric–exterior dimensions.

The weightwise dual bar computes the same Ext as the cobar above, so the surjection has equal finite dimensions in every weight. It is an isomorphism. □

For one derivation, every equal-weight sector with $q \geq 2$ is acyclic. For two derivations, the degree-two sector at $q = 2$ need not vanish. The number of derivations matters to this construction.

We also need the natural associative-bar cochains. At each primitive PBW degree, allow PBW projections followed by the same ordered differential polynomials. Take their product over primitive degrees. This class is closed under composition and extension of the exterior coefficients. An actor face sets one exterior generator to zero. Cochains vanishing on every actor face have positive weight in every actor coordinate.

Restriction along the Chevalley–bar map is a surjective quasi-isomorphism of these natural convolution complexes, also relative to any union of actor faces. Here is a construction of its kernel contraction. For an abelian Lie algebra and fixed multilinear primitive degree, the bar is the finite complex of ordered nonempty partitions, with adjacent-block merging. The Koszul resolution in the preceding proof identifies its homology with the suspended symmetrization image. Choose rational complements to cycles and boundaries and average their projections over permutations. This gives natural maps p, h with $pj = 1$, $hj = ph = h^2 = 0$ and $dh + hd = 1 - jp$.

For a nonabelian Lie algebra, the difference of the PBW product from the symmetric product lowers primitive degree. If b is the resulting bar perturbation, set $p' = p(1+bh)^{-1}$ and $h' = h(1+bh)^{-1}$. The series are finite on every input. The inclusion remains j, since $bj = jQ_2$ and $hj = 0$. Its transferred differential is precisely the Chevalley differential.

Precomposition with p' splits restriction on untwisted Hom complexes, and $f \mapsto (-1)^{|f|} f h'$ contracts its kernel. Twisting by the scalar action preserves that kernel and raises minimum primitive input degree. A second finite perturbation therefore contracts the twisted kernel. All maps preserve actor and derivative weights and commute with the actor faces.

Lemma 6.22. *Fix one derivation and at least two actor coordinates. Prescribe compatible natural action data on all proper actor faces and the boundary of a parameter simplex or cube. These data extend to the interior, and their extension space is contractible. The cochains here have derivative order equal to actor weight.*

Proof. Commuting actor-face projections give a surjective restriction by inclusion-exclusion. Its kernel consists of the cochains using every actor coordinate. Each of its weights is at least two, so it is acyclic by Lemma 6.21 and the relative bar comparison just constructed.

Polynomial forms extend compatible boundary values on a simplex or cube. Choose a linear lift of the boundary and actor-face data, and solve Maurer–Cartan successively in actor weight. At the next weight its curvature is a closed degree-two element of the fully mixed kernel tensored with relative polynomial forms. Closedness is the Bianchi identity. The relative condition is the already satisfied boundary equation.

This complex is acyclic over k, so a degree-one correction removes the curvature without changing the prescribed data. The weight filtration is complete.

The same argument with extra parameter simplices solves every relative lifting problem. Thus the restriction of Maurer–Cartan spaces is a trivial Kan fibration and its fibers are contractible. □

Theorem 6.23. *The Heisenberg action α_D admits a unital homotopy-coherent action of the standard monoid (C_E, μ). Every profile action $\alpha_{\mathbf{t}}$ maps to this multiplicative action by $\Sigma_{\mathbf{t}}$. Its two prescribed unit insertions have the common factorizing recipient*

$$\mu(\Sigma_{\mathbf{t}} \otimes^! \Sigma_{\mathbf{t}}) : C_E \otimes^! C_E \longrightarrow C_E$$

with target action α_D.

Proof. Use the bar lift $\mathcal{F}_D$ for the unary action. Its terms have derivative order equal to actor weight.

For n actors, a coherent monoid action has a family on the cube I^{n-1}. The face $t_i = 0$ multiplies the adjacent actors before applying the shorter family. The face $t_i = 1$ composes the two shorter bar actions. The unit at an end deletes an adjacent parameter. At an interior position it combines the two parameters by $a \vee b = a + b - ab$. This polynomial connection is associative, has identity zero and absorbing element one, so repeated unit deletions agree.

For $n = 2$, these rules prescribe the endpoints and the two unit faces. They coincide where they meet. Lemma 6.22 supplies the binary path. Inductively, associativity of actor multiplication and of bar-map composition makes all prescribed boundary intersections agree. The same lemma fills each higher cube with these values fixed. It also proves contractibility of the natural fillings after the unary action and a binary path are fixed.

Lemma 6.18 identifies extension by another exterior actor with geometric tensoring of the action. It carries D to $D \otimes 1$ by naturality in $J \mapsto J + h\mathbf{1}$ over $k[h]/(h^2)$. Thus the bar endpoints are the required geometric endpoints, including their unit, collision and normal-transfer data. Every universal formula has finitely many primitive inputs at a fixed configuration.

Proposition 6.20 supplies the profile maps. Compose two of them with the constructed multiplicative action. On either unit axis, its unit path reduces to the profile comparison. At the intersection both restrict to the same pointing. The binary coherence supplies the relative two-simplex between these paths. Higher coherences give the same assertion for finite ordered products. □

Contractibility above concerns the specified class of natural differential-polynomial fillings. It is not a classification of all geometric multiplicative structures or of all pointed actions. The general monoid recipient problem still requires their full mapping spaces.

6.8 Relations between separated inputs

Separation imposes two different equations on an action. Operators from different blocks commute, and a prescribed mixed input becomes their product. Their primitives must in turn satisfy higher equations. We construct a resolution that keeps both kinds of relation.

Let $V_1, \dots, V_r$ be vector spaces in degree zero. All tensor products in this section are algebraic over k. For a nonempty subset $S \subseteq [r]$, put

$$V_S = \bigotimes_{i \in S}^{\text{increasing}} V_i, \qquad F_r = T\left(\bigoplus_{\varnothing \neq S \subseteq [r]} V_S \right), \qquad P_r = \bigotimes_{i=1}^{r} T(V_i).$$

Here T is the unital tensor algebra.

Write x_i for its singleton generating maps and c_S for the other generating maps of F_r. Every expression below is multilinear in its indicated inputs. The quotient $\vartheta_r : F_r \to P_r$ sends

$$c_S \longmapsto p_S = x_{i_1} \cdots x_{i_s}, \qquad i_1 < \cdots < i_s,$$

and sends x_i to the ith tensor factor.

For two one-dimensional spaces, the two relations are already independent. Starting with $k\langle x_1, x_2, c\rangle$, adjoin h, e in degree minus one and set

$$dh = x_1x_2 - x_2x_1, \qquad de = c - x_1x_2.$$

In the bases (h, e) and (x_1x_2, x_2x_1, c) the differential is $\left(\begin{smallmatrix} 1 & -1 \\ -1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$. One primitive cannot impose both equations.

6.8.1 The subset resolution

Let $C_i = k \oplus V_i[1]$ be the coaugmented coalgebra with primitive reduced part, and form $C = \bigotimes_i C_i$. Its cobar algebra $R_r = \Omega C$ is free on maps

$$q_S : V_S \longrightarrow R_r^{1-|S|}, \qquad q_{\{i\}} = x_i.$$

For disjoint subsets define $\mathrm{inv}(L, R) = \#\{(\ell, j) \in L \times R : \ell > j\}$. Use the cobar convention $d(tc) = -t(dc) + \sum(-1)^{|c'|+1}(tc')(tc'')$, where t has degree one.

Theorem 6.24. *The differential of R_r is*

$$dq_S = \sum_{\substack{L \amalg R = S \\ L, R \neq \varnothing}} (-1)^{|L|+1+\mathrm{inv}(L,R)} q_L q_R. \tag{6.22}$$

It squares to zero. The map $\epsilon : R_r \to P_r$ that preserves the singleton generators and kills q_S for $|S| > 1$ is a quasi-isomorphism.

Proof. Each factor of $V_i[1]$ is odd. Unshuffling a subset into (L, R) contributes $(-1)^{\mathrm{inv}(L,R)}$. Desuspension contributes $(-1)^{|L|+1}$. These are the two signs in (6.22).

In $d^2 q_S$, each ordered three-part decomposition of S occurs twice, by splitting the first or the second factor. The derivation sign makes their coefficients opposite. Thus $d^2 = 0$.

Under ϵ, the pair differential becomes a commutator of different tensor factors. Every larger split contains a higher generator, so ϵ is a chain map.

Put $A_i = T(V_i)$. The sequence

$$0 \longrightarrow A_i \otimes V_i \xrightarrow{a \otimes v \mapsto av} A_i \longrightarrow k \longrightarrow 0$$

resolves the augmentation module: every nonempty word has a unique last factor. Tensoring these resolutions gives $P_r \otimes_\tau C$, where τ sends suspended singletons to their generators and all larger subsets to zero.

Map C to BP_r by the signed shuffle of its singleton inputs into bar slots. Interior faces cancel in pairs because distinct tensor factors commute. Splitting a shuffle at a bar cut proves the coalgebra identity. Its P_r-linear extension is a map of free resolutions of k inducing the identity on k, hence a chain homotopy equivalence. Tensoring over P_r with k shows that $C \to BP_r$ is a quasi-isomorphism.

Assign weight one to each singleton input. At a fixed weight, reduced bar and cobar words have bounded length. The cobar-length filtration now shows that $\Omega C \to \Omega BP_r$ is a quasi-isomorphism. The bar–cobar contraction of $\Omega BP_r \to P_r$ is finite in that weight. Their composite is ϵ. Taking the direct sum over weights proves the assertion. □

The first higher equation is the six-term relation

$$dq_{123} = x_1h_{23} - x_2h_{13} + x_3h_{12} - h_{12}x_3 + h_{13}x_2 - h_{23}x_1, \qquad h_{ij} = q_{\{i,j\}}. \tag{6.23}$$

Write $k_{ijk} = q_{\{i,j,k\}}$. At four inputs,

$$\begin{aligned} dq_{1234} = {} & x_1 k_{234} - x_2 k_{134} + x_3 k_{124} - x_4 k_{123} \\ & + k_{123} x_4 - k_{124} x_3 + k_{134} x_2 - k_{234} x_1 \\ & - h_{12} h_{34} + h_{13} h_{24} - h_{23} h_{14} - h_{14} h_{23} + h_{24} h_{13} - h_{34} h_{12}. \end{aligned} \tag{6.24}$$

The pair–pair terms are necessary for closedness.

In multidegree $(1, \dots, 1)$, a word of generators is an ordered partition of $[r]$. A partition into a blocks has degree $a - r$. These are the faces of the permutohedron with their incidence differential. For $r = 3$ their dimensions are $(1, 6, 6)$ in degrees $(-2, -1, 0)$. For $r = 4$ they are $(1, 14, 36, 24)$. The algebra also includes repeated inputs of each color, retaining their order within that color.

There is a finite contraction in every multiweight. Fix $\mathbf{n} = (n_1, \dots, n_r)$. After grouping coefficient factors by color, each word with these counts has coefficient $\bigotimes_i V_i^{\otimes n_i}$. The differential is an integer matrix on a finite set of words and preserves within-color order.

Choose the color-increasing word as the degree-zero representative. In each degree, take in lexicographic order the differential columns that increase rank. Their source vectors complement the cycles. Their images form a basis of boundaries. Together with the chosen degree-zero representative, these give bases of boundaries, representatives and source lifts, by Theorem 6.24.

Let h send a boundary basis vector to its chosen source lift and vanish on the other basis vectors. If i includes the color-increasing word, finite matrix inversion gives

$$dh + hd = 1 - i\epsilon, \qquad h^2 = hi = \epsilon h = 0.$$

The scalar matrices depend only on $\mathbf{n}$, so this construction is natural in V_i. It is a contraction of complexes, without a multiplicativity assertion.

For complex inputs one includes their internal differentials and the full Koszul signs in $\Omega \bigotimes_i (k \oplus V_i[1])$. The tensor-resolution argument still applies over k. The scalar signs in (6.22) are the degree-zero specialization.

6.8.2 *Preserving the mixed generators*

The map ϑ_r also has prescribed values on c_S. To retain them, adjoin to F_r the generators q_S of degree $1 - |S|$ and e_S of degree minus one for $|S| \geq 2$. Call the resulting free graded algebra Q_r and put

$$dc_S = 0, \qquad de_S = c_S - p_S,$$

with dq_S as in (6.22).

Theorem 6.25. *The factorization $F_r \longrightarrow Q_r \xrightarrow{\sim} P_r$ of ϑ_r is relative semifree. The second arrow sends c_S to p_S and kills all negative-degree generators.*

Proof. Adjoin the q_S by increasing cardinality. Their differentials use earlier generators. Each de_S lies in F_r. This proves relative semifreeness and $d^2 = 0$.

Change generators from c_S to $y_S = c_S - p_S$. Then $de_S = y_S$ and $dy_S = 0$, so Q_r is the free product of R_r with a tensor algebra on contractible pairs.

Project to R_r by killing the pairs. A word in the kernel has a first pair generator, so it is uniquely $a\, y_S b$ or $a e_S b$, with a a word of R_r. Set

$$h(a\, y_S b) = (-1)^{|a|} a e_S b, \qquad h(a e_S b) = 0.$$

In $dh + hd$, the terms differentiating a or b cancel, and differentiating this first pair returns the original word. This contracts the kernel. Compose the resulting quasi-isomorphism $Q_r \to R_r$ with ϵ. □

For one-dimensional V_i, the squarefree dimensions make the extra mixed-input data visible:

r	$\dim Q_r^{-3}$	$\dim Q_r^{-2}$	$\dim Q_r^{-1}$	$\dim Q_r^{0}$
2	0	0	2	3
3	0	1	13	13
4	1	32	105	75.

Each row has one degree-zero cohomology class. The additional generators remember the chosen paths from mixed inputs to singleton products.

6.8.3 Grouping and its higher comparisons

Let $\mathcal{B} = (B_1|\cdots|B_m)$ be an ordered partition of $[r]$, with increasing order inside each block. Define

$$\gamma_{\mathcal{B}} : R_r \longrightarrow \bigotimes_j R_{B_j}$$

by sending q_S to the same generator in its factor if $S \subseteq B_j$, and to zero otherwise.

Proposition 6.26. *The maps $\gamma_{\mathcal{B}}$ are dg algebra quasi-isomorphisms and commute with ordered refinements of the partition.*

Proof. For S contained in a block, the chain equation is unchanged. Otherwise a differential term can survive only if S meets exactly two blocks and its two parts are those intersections. The two orders of that split cancel: $\mathrm{inv}(L,R) + \mathrm{inv}(R,L) = |L||R|$, while exchanging their generators contributes $(-1)^{(1-|L|)(1-|R|)}$. Their total coefficients are opposite. This proves the chain equation.

Composing with the tensor of the augmentations gives ϵ followed by regrouping the factors of P_r. These augmentations are quasi-isomorphisms, hence so is $\gamma_{\mathcal{B}}$. Under iterated grouping a generator survives exactly when its colors lie in one final block. This is also the direct grouping criterion. □

For the relative map, put $T_j = S \cap B_j$. When $|T_j| \geq 2$, let $C_j = c_{T_j}$ and $E_j = e_{T_j}$. For a singleton use $C_j = x_i$, $E_j = 0$, and for the empty set use $C_j = 1$, $E_j = 0$. Let P_j be the increasing singleton product on T_j. Thus $dE_j = C_j - P_j$.

The extension of $\gamma_{\mathcal{B}}$ to Q_r is

$$c_S \longmapsto \prod_j C_j, \qquad e_S \longmapsto \sum_j C_1 \cdots C_{j-1} E_j P_{j+1} \cdots P_m. \tag{6.25}$$

Its differential telescopes to $\prod_j C_j - \prod_j P_j$.

Ordered refinement expands to the same sum with one E factor, so these relative maps also commute with refinement. Their composition with the augmentations proves they are quasi-isomorphisms.

Changing the order of two groups changes the primitive by the boundary

$$E_1P_2 + C_1E_2 - P_1E_2 - E_1C_2 = d(E_1E_2). \tag{6.26}$$

All higher comparisons follow from one closed form. Let t_j, dt_j have degrees zero and one, and use the total differential on polynomial forms tensored with the coefficient algebra. Then

$$\Psi = \prod_{j=1}^{m} \big(P_j + t_j(C_j - P_j) + dt_j E_j\big) \tag{6.27}$$

is closed. Indeed the differential of $t_j(C_j - P_j)$ cancels that of $dt_j E_j$. The factors lie in separate tensor slots and have total degree zero.

A face sets its corresponding factor to P_j or C_j. Integrating along the edges in their specified order gives (6.25). Square faces give (6.26).

With parameter forms placed first, the top coefficient is $(-1)^{m(m-1)/2} E_1 \cdots E_m$. If I_m integrates the top form over the oriented cube, Stokes' formula gives

$$dI_m \Psi = (-1)^{m+1} I_\partial \Psi.$$

All integrals are polynomial over the characteristic-zero field. The higher comparisons are therefore finite and retain their specified faces.

6.9 Extension spaces

The resolution turns extension of an action into equations in its intended target. Their solutions form a space. Solubility of its first equations, existence of a full solution, and contractibility of this space are distinct questions.

Let E be a unital dg algebra and fix $\phi : F_r \to E$. Write $X_i : V_i \to E^0$ and $C_S : V_S \to E^0$ for its closed singleton and mixed images. A map $Q_r \to E$ under F_r consists of $H_S : V_S \to E^{1-|S|}$ and $T_S : V_S \to E^{-1}$ for $|S| \geq 2$, with $H_{\{i\}} = X_i$, satisfying

$$d_E H_S = \sum_{\substack{L \amalg R = S \\ L, R \neq \varnothing}} (-1)^{|L|+1+\mathrm{inv}(L,R)} H_L H_R, \tag{6.28}$$

$$d_E T_S = C_S - X_{i_1} \cdots X_{i_s}, \qquad i_1 < \cdots < i_s. \tag{6.29}$$

These are exactly the chain equations on the free generators q_S, e_S.

For complexes U, E, use the Hom differential $\partial f = d_E f - (-1)^{|f|} f d_U$. The spaces V_S still have degree zero.

Proposition 6.27. *Fix all H_T for proper nonempty subsets $T \subset S$ with their equations satisfied. The right side O_S of (6.28) is closed in $\mathrm{Hom}(V_S, E)^{2-|S|}$. An extension H_S exists precisely when $[O_S] = 0$. If it exists, its components form a torsor for $H^{1-|S|}\,\mathrm{Hom}(V_S, E)$, and at a chosen solution its higher homotopy groups are*

$$\pi_j = H^{1-|S|-j}\,\mathrm{Hom}(V_S, E), \qquad j \geq 1.$$

All lower-stage maps are fixed in this assertion. For a mixed cell, the obstruction lies in $H^0\,\mathrm{Hom}(V_S, E)$ and its choice groups are $H^{-1-j}\,\mathrm{Hom}(V_S, E)$ for $j \geq 0$.

Proof. Differentiating O_S produces the two opposite terms for each ordered three-part decomposition of S, as in $d^2 q_S = 0$. Thus it is closed. The next equation is $\partial H_S = O_S$, which has a solution exactly when this class vanishes.

Subtract a chosen solution. The fiber becomes the simplicial abelian group of closed degree $1 - |S|$ maps from V_S into $E \otimes \Omega^\bullet(\Delta^\bullet)$. Integration of polynomial forms identifies its normalized homotopy groups with the Hom cohomology in degrees $1 - |S| - j$. In particular, an interval changes two solutions by a boundary, giving the stated torsor. The same argument applies to a generator of degree minus one. □

A contraction h of the relevant Hom complex gives $H_S = hO_S$ when its cohomology projection vanishes. If units or collision values are fixed, this must be a contraction of the corresponding relative Hom complex. A contraction that moves those values does not solve the extension problem.

There is also a model of the complete solution space:

$$\mathrm{Map}_{F_r/}(Q_r, E) \simeq \mathrm{fib}_{\phi}\Big(\mathrm{Map}_{\mathrm{dgAlg}}(P_r, E) \longrightarrow \mathrm{Map}_{\mathrm{dgAlg}}(F_r, E)\Big). \tag{6.30}$$

Indeed F_r is free, $F_r \to Q_r$ is a cofibration, and $Q_r \to P_r$ is a quasi-isomorphism. Polynomial forms compute mapping spaces from these cofibrant sources. Restriction is a Kan fibration, so its strict fiber computes its homotopy fiber. These are the unital associative instances of [56, Theorem 4.1.1 and §§4.8.1–4.8.8].

If E is zero below degree $-N$, with $N \geq 0$, only $|S| \leq N + 2$ can give nonzero higher-cell obstructions. At the last of these sizes the obstruction can survive although its primitive would have to lie in a zero degree. Larger right sides vanish as elements of E. A cohomological bound alone does not give this strict bound on the equations.

6.9.1 Choices that survive and choices that disappear

Take $r = 2$, $V_1 = V_2 = k$, and $E = k \oplus k\eta$ with $|\eta| = -1$, $d = 0$, $\eta^2 = 0$. Prescribe $X_1 = X_2 = C_{12} = 0$. Then

$$H_{12} = \lambda\eta, \qquad T_{12} = \mu\eta \quad (\lambda, \mu \in k)$$

are all the solutions.

A degree-minus-one element of $E \otimes \Omega^{\bullet}(\Delta^n)$ is ηf, with f a polynomial function. Closedness forces $df = 0$, hence f is constant. Thus (6.30) is the discrete space k^2. Existence does not imply contractibility.

An intermediate choice need not survive a later cell. With the same E, maps from $k\langle x\rangle$, $|x| = -1$, $dx = 0$, have components $x \mapsto \lambda\eta$. Adjoin y in degree minus two with $dy = x$. In every polynomial-form simplex, the image of y is zero for degree reasons. Its differential then forces the image of x to be zero.

The complete extension space is a point. Consequently a stage choice group alone cannot disprove contractibility of the full space.

6.9.2 A triple obstruction after all pair choices

Start with the free graded algebra on x_1, x_2, x_3 in degree zero and h_{12}, h_{13}, h_{23} in degree minus one, with $dh_{ij} = x_i x_j - x_j x_i$. Give x_i color weight e_i and h_{ij} color weight $e_i + e_j$. Quotient by every word in which a color occurs twice, and call the result E. The differential preserves color counts, so the quotient is a dg algebra.

Its degree-zero basis consists of the unit, three singletons, six ordered pairs and six ordered triples. Its degree-minus-one basis consists of the three h_{ij} and the six products of an h_{ij} with its complementary singleton, on either side. Thus $\dim E = 25$. Define

$$\omega = x_1 h_{23} - x_2 h_{13} + x_3 h_{12} - h_{12} x_3 + h_{13} x_2 - h_{23} x_1. \tag{6.31}$$

Theorem 6.28. *Every pair of the marked maps $k[x_i] \to E$ extends to the corresponding resolution R_2. No choice of pair primitives extends them to R_3. More precisely,*

$$H^0(E) = k[x_1, x_2, x_3]/(x_1^2, x_2^2, x_3^2), \qquad H^{-1}(E) = k\omega,$$

and every choice gives the triple obstruction ω.

Proof. The displayed h_{ij} solve the pair equations. In each pair weight, their differentials are injective. In weight $(1,1,1)$, the six degree-minus-one basis vectors map to the six ordered triple words by the incidence matrix of a hexagon. Its rank is five and its kernel is its oriented cycle (6.31).

The total differential therefore has rank eight. Its degree-zero quotient identifies the orderings of each set of distinct colors and kills every repeated-color product. This gives the two cohomology formulas.

Every other pair primitive is $h_{ij} + \lambda_{ij}\omega$. The augmentation ideal annihilates ω on both sides, since its terms already use all three colors. Substituting any such primitives in (6.23) therefore leaves ω unchanged. Its six summands are distinct basis vectors, so it is nonzero. Since $E^{-2} = 0$, it cannot be a boundary. □

Prescribe c_S to be the increasing singleton product. Then $T_S = 0$ solves all mixed equations, and the same obstruction excludes a map $Q_3 \to E$ under these data. The regular action of E detects its class: multiplication by ω sends $[1]$ to $[\omega] \neq 0$. This detects the specified lifts in $\mathrm{End}_k(E)$. It does not classify the additional pair primitives in that larger algebra.

The obstruction also gives an integral family. Over $k[s]$, adjoin a vector K of degree minus two with $dK = s\omega$. Set $K^2 = 0$ and let the augmentation ideal annihilate K on both sides. These relations are preserved by the differential because the same ideal annihilates ω. The resulting E_s is graded-free of rank 26.

Proposition 6.29. *One has*

$$H^0(E_s) = k[s] \otimes H^0(E), \qquad H^{-1}(E_s) = k[s]/(s)\cdot\omega, \qquad H^{-2}(E_s) = 0.$$

Let R be a commutative $k[s]$-algebra. The prescribed singleton maps extend to R_3 after base change to R exactly when s is a unit there. This criterion allows every choice of pair primitives. It also applies to Q_3 when the mixed inputs are their prescribed singleton products.

Proof. The differential in degrees minus one and zero is split over k, with kernel $k\omega$. After base change its kernel is $R\omega$, so changes of pair primitives still leave the triple expression unchanged. An image of q_{123} must be bK and solve $bs\omega = \omega$. As ω is part of a free R-basis, this is $bs = 1$.

Over $k[s]$, the remaining negative complex is $k[s]K \xrightarrow{s} k[s]\omega$. Its kernel is zero and its cokernel is $k[s]/(s)\cdot\omega$. The degree-zero differential is unchanged. □

The primitive over $k[s, s^{-1}]$ is K/s. There is none over $k[s]$ or $k[[s]]$. At $s = 0$, both K and ω survive. At a nonzero field value, rescaling K gives the quotient of R_3 by repeated-color words. Graded flatness of the family therefore does not imply flatness of its cohomology.

More generally, over a commutative k-algebra R, adjoin vectors K_j with $dK_j = f_j\omega$, zero mutual products and zero products with the augmentation ideal. The same calculation gives

$$H^{-1} = R/(f_1, \dots, f_m)\cdot\omega, \qquad H^{-2} = \ker\bigl(R^m \xrightarrow{(f_1,\dots,f_m)} R\bigr).$$

A triple extension exists exactly when $1 \in (f_1, \dots, f_m)$. The degree-minus-two classes are the syzygies among these coefficients.

6.10 The bar of the full action diagram

An extension can become possible after mapping to another acting object. To decide a universal property, one must therefore retain the arrows between all permitted actions and their comparison paths. The bar of that diagram gives an explicit chain model for the resulting coefficient map.

First let $\mathcal{A}$ be a small ordinary category and $X : \mathcal{A} \to \mathrm{Ch}(k)$ a covariant diagram. Its simplicial replacement has the length-n term

$$\bigoplus_{a_0 \xrightarrow{f_1} \cdots \xrightarrow{f_n} a_n} X(a_0)[n].$$

Use the direct-sum total complex $\mathcal{B}(X)$. An element $[f_1|\cdots|f_n]x$, with $|x| = q$ internally, has total degree $q - n$. The first face applies $X(f_1)$, the interior faces compose adjacent arrows, and the last face deletes f_n. Degeneracies insert identities. If ∂ is the alternating face differential, put $d = \partial + (-1)^n d_X$ on length n.

For a second diagram Y, define the Alexander–Whitney map by

$$\mathrm{AW}([f_1|\cdots|f_n](x \otimes y)) = \sum_{p=0}^{n} (-1)^{(n-p)|x|}[f_1|\cdots|f_p]x \otimes [f_{p+1}|\cdots|f_n]Y(f_p \cdots f_1)\, y. \tag{6.32}$$

The empty composite for $p = 0$ is the identity. The second string begins at a_p. Its coefficient must be transported to $Y(a_p)$. The sign moves x past its suffix, of degree $-(n - p)$.

Proposition 6.30. *Formula* (6.32) *defines a natural coassociative chain map*

$$\mathcal{B}(X \otimes Y) \longrightarrow \mathcal{B}(X) \otimes \mathcal{B}(Y).$$

It descends to normalization and represents the map of homotopy colimits induced by $\mathcal{A} \to \mathcal{A}^2$, $a \mapsto (a, a)$.

Proof. Before moving coefficients through strings, the formula splits a simplex into a prefix and a suffix. Deleting the prefix vertices from the second factor applies its first face repeatedly, giving $Y(f_p \cdots f_1)\, y$.

A face strictly inside either interval agrees with that face taken before the split. At the splitting vertex, the last face of one prefix cancels the first face of the next suffix. At the initial vertex, functoriality identifies the composite coefficient transport with its first face. The last vertex requires no transport. These account for the face differential.

Restoring the displayed coefficient order gives $(-1)^{(n-p)|x|}$. The same permutation carries the internal differential: its source simplex sign is $(-1)^n$, and the first output has total degree $|x| - p$. Thus it also satisfies the internal chain equation. An identity arrow remains in a prefix or a suffix in every summand, which proves descent to normalization.

For three diagrams, either iterated diagonal sums over $0 \le p \le q \le n$. Both transport the second and third coefficients to a_p, a_q and have sign $(-1)^{(n-p)|x|+(n-q)|y|}$. This proves coassociativity.

The diagonal category functor first sends an arrow string to two identical strings. The prefix–suffix map compares the diagonal of their bisimplicial replacement with its total complex. The shuffle map gives the reverse comparison, with the degeneracy homotopies. The latter total complex is $\mathcal{B}(X) \otimes \mathcal{B}(Y)$. Hence the map has the stated derived meaning. □

For one arrow $f : a \to b$, the formula reads

$$\mathrm{AW}([f](x \otimes y)) = (-1)^{|x|}[\,]_a x \otimes [f]\, y + [f]x \otimes [\,]_b Y(f)\, y.$$

Both the transport and the sign are necessary.

Put $Q(a, b) = X(a) \otimes Y(b)$. Over the field k, tensor product preserves colimits and computes derived tensor product. The fiber of the diagonal-induced comparison therefore has model

$$\begin{aligned} \mathcal{K}^j &= \mathcal{B}(X \otimes Y)^j \oplus (\mathcal{B}(X) \otimes \mathcal{B}(Y))^{j-1}, \\ d(u, v) &= (du, \mathrm{AW}(u) - dv). \end{aligned} \tag{6.33}$$

This describes its differential, without asserting that the fiber is acyclic. Over a general coefficient ring one first takes derived-flat replacements.

For an action category with simplicial mapping spaces, use its coherent simplicial replacement and normalize every simplicial direction. The diagonal and shuffle maps in those directions give the same derived comparison. Passing to the homotopy category would omit the action paths that enter this replacement.

On a binary block-disjoint open, suppose each factorizing actor's coefficient is the external product of its two block coefficients. Require coherence over the full action index. Their diagram comparison then has the form (6.33), with external products. Universal factorization requires its acyclicity on that full index. A class on a restricted index may disappear after adding another recipient and its comparison simplices.

Even bar length matters. For the one-object identity category with constant coefficient k, the unnormalized bar has one copy of k in every length and boundary $\partial_n = \sum_{i=0}^{n} (-1)^i$. Truncating after length one creates $H^{-1} = k$ because $\partial_1 = 0$. In the full bar, $\partial_2 = 1$ kills this class. Normalization removes all positive lengths. A finite unnormalized truncation can therefore create an obstruction absent from the full diagram.

The remaining geometric comparison must join these separated relations to the relations on singleton collisions. It must also retain the positive local vacuum and the specified action paths on the full ordered atlas. For the dual-number pointed actor on C_{V_κ}, the contractibility question requires these cells in its actual relative endomorphism complex. A nonzero obstruction must survive every permitted lower-cell choice there, and a nontrivial choice must survive the rest of the extension tower. The ordinary coefficient algebras above specify these tests. They do not themselves construct the required supported comparison.

Chapter 7

Analytic coefficients and collision monodromy

Formal translation is finite on each mode polynomial. Numerical translation acts on an infinite series of modes and can destroy its convergence. The first question is therefore quantitative: how much jet radius does moving a point consume? For the Heisenberg coefficients, a single series of bosons determines the exact loss. The same seminorms then determine where the full line gauge exists.

A collision also changes the attachment of a coefficient to the regular module. We retain this attachment in finite models of punctured lines. Their higher products distinguish a marked localization from its primitive quotient. Moving two punctures together leads to an inverse telescope, while differentiation in all spatial variables gives a different Hom complex. These calculations provide explicit coefficient categories on which collision transport can be studied.

We first work over $\mathbb{C}$, with $\kappa \neq 0$. Differentials have cohomological degree one, odd derivatives act on the left, and exterior forms stand on the left of coefficient expressions. The algebraic sections below work over a field k of characteristic zero. Every completion and every change of coefficient category will be specified.

7.1 Convergent jets and numerical translation

For a finite set I, let

$$Y_I = \mathbb{C}[B_{i,n} : i \in I,\ n \geq 0] \otimes \Lambda(C_{i,n} : i \in I,\ n \geq 0), \qquad |B_{i,n}| = 0, \quad |C_{i,n}| = 1.$$

Fix an order for the odd generators. For a monomial m, let $d_i(m)$ be its number of factors at site i, counted with multiplicity. Give $B_{i,n}$ weight n and $C_{i,n}$ weight $n+1$, and let $w_i(m)$ be the total weight at that site. Field degree and jet weight will be controlled independently.

Definition 7.1. For positive real numbers A_i, r_i, set

$$\left\| \sum_m f_m m \right\|_{\mathbf{A},\mathbf{r}} = \sum_m |f_m| \prod_{i \in I} A_i^{d_i(m)} r_i^{w_i(m)}.$$

Given positive radii $\mathbf{R} = (R_i)_{i \in I}$, define $\mathcal{Y}_I(\mathbf{R})$ as the completion of Y_I for all these seminorms with $0 < r_i < R_i$ and $A_i > 0$. Its cochain complex uses the homogeneous cohomological-degree subspaces.

Signed multiplication is submultiplicative for each seminorm. A repeated odd generator gives zero. Taking rational amplitudes cofinal toward infinity and rational radii cofinal below R_i gives a countable defining family. Thus $\mathcal{Y}_I(\mathbf{R})$ is a Fréchet algebra. It consists exactly of the coefficient series for which every displayed absolute sum converges. Polynomial truncations converge in this topology.

The subalgebra containing only $J_i = B_{i,0}$ is $\mathcal{O}(\mathbb{C}_J^I)$, the entire functions of the charges. Indeed its seminorms are the absolute power-series sums at all positive amplitudes. The projection and inclusion are

$$\pi : \mathcal{Y}_I(\mathbf{R}) \longrightarrow \mathcal{O}(\mathbb{C}_J^I) \quad \text{and} \quad \iota : \mathcal{O}(\mathbb{C}_J^I) \longrightarrow \mathcal{Y}_I(\mathbf{R}).$$

For an open position domain U, a holomorphic coefficient series $\sum_m f_m(z)m$ is admitted when

$$\sum_m \sup_{z\in K} |f_m(z)| \prod_i A_i^{d_i(m)} r_i^{w_i(m)} < \infty$$

for every compact $K \subset U$ and every defining seminorm. Use these series as coefficients of holomorphic differential forms, denoted by $\Omega^\bullet(U; \mathcal{Y}_I(\mathbf{R}))$. Position derivatives are continuous by Cauchy estimates on slightly larger compact subsets. Entire charge coefficients use the same convention.

Recall the derivations of (5.1):

$$QB_n = \kappa n C_{n-1}, \quad QC_n = 0, \qquad TB_n = nB_{n-1}, \quad TC_n = nC_{n-1}, \qquad LC_n = \kappa^{-1}B_n, \quad LB_n = 0.$$

An index -1 gives zero. At several sites take their sums. If N counts the ghosts and the positive-index bosons, define

$$h = N^{-1} \sum_{i,n\geq 0} \frac{B_{i,n+1}}{\kappa(n+1)} \partial_{C_{i,n}} \quad (N > 0), \qquad h = 0 \quad (N = 0). \tag{7.1}$$

Proposition 7.2. *The operators Q, T, L, h are continuous on $\mathcal{Y}_I(\mathbf{R})$. Their polynomial identities extend to this completion. In particular,*

$$Qh + hQ = 1 - \iota\pi, \qquad h^2 = h\iota = \pi h = 0, \qquad \|hF\|_{\mathbf{A},\mathbf{r}} \leq |\kappa|^{-1}\|F\|_{\mathbf{A},\mathbf{r}}.$$

This is an $\mathcal{O}(\mathbb{C}_J^I)$-linear strong deformation retract onto the entire charges in degree zero.

Proof. We give the one-site estimates. For $r < r' < R$, put $c(r/r') = \sup_{w\geq 0} w(r/r')^w$, which is finite. The operator Q preserves field degree and jet weight. On a monomial of weight w, the sum of its absolute coefficients is at most $|\kappa|w$. The operator T lowers weight by one, with total coefficient at most w. Consequently

$$\|QF\|_{A,r} \leq |\kappa| c(r/r')\|F\|_{A,r'}, \qquad \|TF\|_{A,r} \leq r^{-1}c(r/r')\|F\|_{A,r'}.$$

For $A < A'$, the operator L has the bound

$$\|LF\|_{A,r} \leq \frac{\sup_{d\geq 0} d(A/A')^d}{|\kappa| r} \|F\|_{A',r}.$$

These estimates prove continuity for the full seminorm family. At finitely many sites the estimates add.

Every replacement in (7.1) preserves field degree and jet weight. Its absolute coefficient is $1/(|\kappa|(n+1)N)$. The number of ghosts is at most N, which proves the bound for h. The numerator of h anticommutes with Q to give N, squares to zero, and commutes with N. The resulting polynomial identities therefore extend by density and continuity.

The operator never differentiates a charge. Its charge linearity extends from polynomials to entire functions. □

In particular, the inclusion from polynomial to analytic coefficients changes the cohomology from $\mathbb{C}[J_i : i \in I]$ to $\mathcal{O}(\mathbb{C}_J^I)$. It is not a quasi-isomorphism. The larger coefficient algebra is the cost of allowing arbitrary numerical charges and convergent mode series.

Theorem 7.3 (Translation and its optimal radius)**.** *For $|t| < R$, translation extends continuously to*

$$\tau_t = e^{tT} : \mathcal{Y}(R) \longrightarrow \mathcal{Y}(R - |t|).$$

If $0 < |t| < R$ *and* $S > R - |t|$, *no continuous linear map* $\mathcal{Y}(R) \to \mathcal{Y}(S)$ *agrees with* τ_t *on all mode polynomials.*

For $0 < a < R$ *and* $D_a = \{t : |t| < a\}$, *the formula*

$$E_t = (1 + dt\, L)\tau_t$$

defines a continuous dg algebra map

$$E : \mathcal{Y}(R) \longrightarrow \Omega^\bullet(D_a; \mathcal{Y}(R - a)).$$

It satisfies $E_{t+s} = E_t E_s$ *wherever both sides have a common radius domain.*

Proof. For each generator,

$$\tau_t B_n = \sum_{j=0}^{n} \binom{n}{j} t^{n-j} B_j, \qquad \|\tau_t B_n\|_{A,r} \le A(r + |t|)^n.$$

For C_n the corresponding bound is $Ar(r + |t|)^n \le A(r + |t|)^{n+1}$. Multiplicativity and submultiplicativity give

$$\|\tau_t F\|_{A,r} \le \|F\|_{A,r+|t|}. \tag{7.2}$$

Every target radius satisfies $r + |t| < R$, so the polynomial map extends continuously.

To prove optimality, write $t = se^{i\alpha}$ with $0 < s < R$ and take

$$F = \sum_{n \ge 0} R^{-n} e^{-in\alpha} B_n.$$

Its seminorm at A, r is $A/(1 - r/R)$, so it belongs to $\mathcal{Y}(R)$. In the translated polynomial truncations, the coefficient of B_j tends to

$$\sum_{n \ge j} R^{-n} e^{-in\alpha} \binom{n}{j} t^{n-j} = e^{-ij\alpha} \frac{R}{(R - s)^{j+1}}.$$

Coefficient evaluation is continuous. Any continuous extension must therefore have these coefficients on F. Their absolute sum diverges at every radius $r > R - s$. Choosing $R - s < r < S$ proves the assertion.

For a compact $K \subset D_a$ and $r < R - a$, one has $r + \sup_K |t| < R$. Thus (7.2) is uniform on K. The estimate for L controls the one-form component. On polynomials, the identities $QL + LQ = T$, $[T, L] = 0$, and $L^2 = 0$ give

$$(Q + d_t)E_t = E_t Q, \qquad E_{t+s} = E_t E_s.$$

The operator $dt\, L$ is a degree-zero derivation whose square is zero, so $1 + dt\, L$ is multiplicative. Continuity extends these identities to the stated spaces. For two numerical translations a common target radius is any positive radius below $R - |t| - |s|$. □

The primitive coproduct Δ_0 is continuous for the independently completed sites. For a target seminorm at $(A_1, A_2; r_1, r_2)$, choose a source amplitude at least $A_1 + A_2$ and a source radius at least $\max(r_1, r_2)$, still below R. These choices bound the images of every generator. A translation in the first factor replaces r_1 by $r_1 + |t|$. The spectral coproduct identities of Proposition 5.2 consequently extend on the common domains given by these inequalities.

7.2 The separated gauge and its collision boundary

For two sites with separation z, the full primitive of (5.7) and its differential are

$$b(z) = -\frac{1}{\kappa}\sum_{N\ge 1}\frac{z^{-N}}{N}\sum_{a+b=N}(-1)^a\binom{N}{a}B_{1,a}B_{2,b}, \tag{7.3}$$

$$r(z) = \sum_{a,b\ge 0}(-1)^a\binom{a+b}{a}\frac{C_{1,a}B_{2,b}-B_{1,a}C_{2,b}}{z^{a+b+1}}. \tag{7.4}$$

Let L_1 act at the first site, and put

$$D_0 = Q + d_z, \qquad \mathcal{A} = r + dz\,L_1 r, \qquad \rho = \kappa^{-1}J_1J_2\,\frac{dz}{z}.$$

The formulas $Qb = r$ and $d_z b = dz\,L_1 r - \iota\rho$ hold coefficientwise. We now determine their domain of convergence.

Theorem 7.4 (Exact gauge domain). *Fix $R_1, R_2 > 0$ and $z \neq 0$. Each of $b(z)$, $e^{-b(z)}$, and $e^{b(z)}$ belongs to $\mathcal{Y}_{\{1,2\}}(R_1, R_2)$ exactly when*

$$|z| \ge R_1 + R_2.$$

Their largest open domain of holomorphic dependence in this coefficient space is $|z| > R_1 + R_2$. For $q = (r_1 + r_2)/|z| < 1$,

$$\|b(z)\|_{\mathbf{A},\mathbf{r}} = \frac{A_1A_2}{|\kappa|}\sum_{N\ge 1}\frac{q^N}{N} = \frac{A_1A_2}{|\kappa|}\log\frac{1}{1-q}. \tag{7.5}$$

The tail after weight N_0 is bounded by

$$\frac{A_1A_2}{|\kappa|}\,\frac{q^{N_0+1}}{(N_0+1)(1-q)}.$$

Proof. The monomials $B_{1,a}B_{2,b}$ are distinct. Summing their absolute coefficients at $a + b = N$ gives $A_1A_2q^N/(|\kappa|N)$ by the binomial theorem. The harmonic power series converges precisely for $q < 1$. Requiring this for every $r_i < R_i$ is equivalent to $|z| \ge R_1 + R_2$. The tail estimate follows by replacing $1/N$ with $1/(N_0 + 1)$ in its geometric majorant.

Submultiplicativity gives $\|e^{\pm b}\| \le e^{\|b\|}$ wherever b belongs to the coefficient space. Conversely, projection to total field degree two is continuous and sends $e^{\pm b}$ to $\pm b$. The exponentials are defined coefficientwise, since b has positive jet weight. Their membership therefore implies that of b.

All bounds are uniform on compact subsets of the strict-inequality domain. Differentiated series converge there by the same geometric estimates, which prove holomorphic dependence. Every neighborhood of a boundary point contains positions where membership fails. Boundary membership supplies no larger open domain. □

Theorem 7.5 (Analytic module contraction). *Let $U \subset \{z : |z| > R_1 + R_2\}$ be open and put $G = e^{-b}$. There is a continuous strong deformation retract of module complexes*

$$\big(\Omega^\bullet(U;\mathcal{Y}_{\{1,2\}}(R_1,R_2)),\,D_0 + \mathcal{A}\cdot\big) \;\rightleftarrows\; \big(\Omega^\bullet(U;\mathcal{O}(\mathbb{C}^2_J)),\,d_z + \rho\cdot\big)$$

with inclusion $G\iota$, projection π, and homotopy GhG^{-1}. At any defining seminorm,

$$\|GhG^{-1}\| \le |\kappa|^{-1}e^{2\|b\|}.$$

Proof. In addition to (7.5), the binomial sums in (7.4) give

$$\|r(z)\| \le \frac{A_1A_2(r_1+r_2)}{|z|-r_1-r_2}, \qquad \|L_1r(z)\| \le \frac{A_1A_2}{|\kappa|(|z|-r_1-r_2)}.$$

They make the connection form and the coefficient identities convergent on U.

To check the first identity directly, differentiating a first-site boson in (7.3) uses

$$-\frac{(-1)^a a}{N}\binom{N}{a} = (-1)^{a-1}\binom{N-1}{a-1}.$$

Differentiating a second-site boson supplies the other term of r. Position differentiation of $-z^{-N}/N$ gives $z^{-N-1}dz$. Its missing weight-zero term is $\iota\rho$. Hence

$$D_0 b = \mathcal{A} - \iota\rho, \qquad (D_0 + \mathcal{A}\cdot)G = G(D_0 + \iota\rho\cdot).$$

Extend h over forms with the Koszul sign. Since it is charge-linear, it anticommutes with both d_z and multiplication by ρ. Proposition 7.2 thus contracts the ρ-twisted complex onto the charge connection. Conjugation by G gives the asserted retract.

Every monomial of b contains a positive-index boson, so $\pi G = \pi G^{-1} = \pi$. This proves that the projection remains π and that $\pi G\iota = 1$. The homotopy bound follows by multiplying the three operator bounds. □

For a finite ordered set of positions, require $|z_i - z_j| > R_i + R_j$ for every pair and define

$$b_I = \sum_{i<j} b_{ij}(z_i - z_j), \qquad \mathcal{A}_I = \sum_{i<j}(r_{ij} + dz_{ij}L_i r_{ij}), \qquad \rho_I = \kappa^{-1}\sum_{i<j} J_iJ_j\, d\log z_{ij}.$$

All even primitives commute. The estimates add over finitely many pairs, so the same retract holds with $G_I = e^{-b_I}$. Every pair is retained, including the nonconsecutive channel at three positions and the crossed channels at four.

If a specified continuous coefficient map f relates two untwisted models on a common radius domain, its conjugate is $G_t^{-1} f G_s$. For multiplicative f its extra factor is $\exp(b_t - f(b_s))$. In compositions the intermediate factors cancel. Thus conjugation preserves the identities of the supplied maps. It does not supply a missing map from a native collision coefficient. Also $G\iota(1) = e^{-b}$: the endpoint of this module contraction is not the constant unit.

Proposition 7.6. *The full gauge need not extend through $z = 0$ as a finite-pole meromorphic coefficient map on the same analytic carrier.*

Proof. Set all modes equal to zero except $B_{1,1} = a$ and $B_{2,0} = q$, with $aq \neq 0$. This coefficient evaluation is continuous. For any fixed $r_1 > 0$, choose $A_1r_1 > |a|$ and $A_2 > |q|$. It sends

$$b(z) \longmapsto \frac{aq}{\kappa z}, \qquad G(z) \longmapsto \exp\Big(-\frac{aq}{\kappa z}\Big).$$

The latter has nonzero coefficients in every negative power of z. No power of z makes it holomorphic. A finite-pole extension would remain finite-pole under the continuous evaluation, which is impossible. □

The logarithmic connection has a different obstruction: its cohomology changes at resonant charges.

Proposition 7.7. *At numerical charges q_1, q_2, put $\nu = q_1q_2/\kappa$. On a holomorphic annulus, the complex $d_z + \nu\, dz/z$ is acyclic if $\nu \notin \mathbb{Z}$. If $\nu \in \mathbb{Z}$, its cohomology is one-dimensional in each of degrees zero and one, represented by $z^{-\nu}$ and $z^{-\nu}dz/z$. Its horizontal monodromy is $e^{-2\pi i\nu}$. There is no contraction over the entire charge algebra that eliminates every nonzero Laurent exponent uniformly in the charges.*

Proof. On the Laurent mode z^m, the differential is multiplication by $m + \nu$, with target $z^m\,dz/z$. Except at the possible resonant exponent, division by $m + \nu$ has at most polynomial growth in m. It preserves Laurent convergence on compact subannuli and contracts exactly those modes. The horizontal solution is $z^{-\nu}$ on a simply connected cover, which gives the monodromy.

For any nonzero integer m, the entire-charge inverse $(m + J_1J_2/\kappa)^{-1}$ has a pole along the nonempty hypersurface $J_1J_2 = -\kappa m$. At a charge on this hypersurface the mode has nonzero cohomology. Uniform elimination would contradict that specialization. An inverse exists as a formal charge series near zero, but it does not extend to an entire charge function. □

The essential singularity concerns the full-mode gauge. Resonance concerns the residual charge connection. A collision comparison must account for both on its declared coefficient category. Finite algebraic models retain the regular attachment through a collision.

7.3 Punctures with their regular coefficient

Let k be a field of characteristic zero, and choose distinct $a_1, \dots, a_r \in k$, with $r \geq 1$. Put

$$R = k[t], \qquad f = \prod_{i=1}^{r}(t - a_i), \qquad L_f = R[f^{-1}], \qquad D = k\langle t, \partial\rangle/(\partial t - t\partial - 1).$$

All differential modules in this section are left D-modules. The regular and localized modules have their derivative connections. For each puncture, the delta module is

$$\delta_i = D/D(t - a_i) \simeq R[(t - a_i)^{-1}]/R.$$

Partial fractions give the exact sequence

$$0 \longrightarrow R \longrightarrow L_f \longrightarrow \bigoplus_{i=1}^{r} \delta_i \longrightarrow 0. \tag{7.6}$$

We will compare two choices of objects in the dg category generated by this sequence. Keeping objects *marked* means requiring equivalences to preserve the specified objects individually.

For $q \in D$, write r_q for right multiplication by q. The free resolutions

$$P_0 = [D \xrightarrow{r_\partial} D] \longrightarrow R, \qquad P_i = [D \xrightarrow{r_{t-a_i}} D] \longrightarrow \delta_i \tag{7.7}$$

have terms in degrees $-1, 0$. Their differentials are injective: the order filtration identifies the associated graded of D with the polynomial domain $k[t, \xi]$. The degree-zero augmentations send 1 to the cyclic generator of the target. Products of maps below are compositions, with the right map applied first.

Lemma 7.8. *The mapping complexes of the regular and delta modules have cohomology*

$\mathrm{RHom}_D(-,-)$	R	δ_j
R	k	$k[-1]$
δ_i	$k[-1]$	$\begin{cases} k, & i = j, \\ 0, & i \neq j \end{cases}$

Here $k[-1]$ is in cohomological degree one.

Proof. Mapping the two resolutions into R gives the operators ∂ and $t - a_i$ on polynomials. Integration contracts the first to constants. The second is injective with one-dimensional cokernel.

On δ_j, differentiation is injective with cokernel the simple pole $(t - a_j)^{-1}$. Multiplication by $t - a_j$ is surjective, with kernel that same pole. If $i \neq j$, then $t - a_i = (t - a_j) + (a_j - a_i)$ is invertible on δ_j. Its inverse on each vector is a finite geometric sum because $t - a_j$ acts locally nilpotently. These calculations give all four kinds of corners. □

Define a finite graded algebra A_r with identities $e_0, \dots, e_r$ and degree-one arrows

$$\alpha_i : 0 \longrightarrow i, \qquad \beta_i : i \longrightarrow 0 \quad (1 \le i \le r).$$

Every product of two arrows is zero, and the differential is zero. Its dimension is $3r + 1$. The central vertex represents the regular module and the leaves represent the individual closed coefficients.

Write $\mathrm{Perf}(A_r)$ for perfect right dg A_r-modules. Write thick_D for closure under finite cones, shifts, and direct summands in the derived category of left D-modules.

Theorem 7.9 (The formal star category)**.** *There is a strict dg-algebra quasi-isomorphism*

$$A_r \longrightarrow \mathrm{End}_D(P_0 \oplus P_1 \oplus \cdots \oplus P_r).$$

The dg Yoneda functor on these resolutions gives

$$\mathrm{Perf}(A_r) \simeq \mathrm{thick}_D(R, \delta_1, \dots, \delta_r) = \mathrm{thick}_D(R \oplus L_f).$$

Proof. Send e_i to the projection onto P_i. Send α_i to the degree-one identity $P_0^{-1} \to P_i^0$ and β_i to the degree-one identity $P_i^{-1} \to P_0^0$. These maps are closed. Their pairwise products vanish strictly, since their targets have degree zero, on which the next degree-one map vanishes. They represent the nonzero cokernel generators in Lemma 7.8. The map is therefore a quasi-isomorphism on every corner.

The dg Yoneda functor identifies finite cones and summands of these resolutions with perfect A_r-modules. Sequence (7.6) puts L_f in the category generated by the regular and delta modules. Conversely, its cone recovers the direct sum of the deltas, and direct summands recover each delta. □

Each component of the extension (7.6) is nonsplit. A splitting would embed a nonzero vector annihilated by $t - a_i$ into the torsion-free module L_f. Normalize its nonzero extension class to one. Applying derived Hom into R then gives

$$\mathrm{RHom}_D(L_f, R) \simeq [k \xrightarrow{(1,\dots,1)} k^r] \simeq k^{r-1}[-1]. \tag{7.8}$$

The two-term complex has degrees zero and one. The reverse Hom vanishes for one puncture and acquires a degree-one class at the second puncture. Its compositions will detect the localization attachment.

7.3.1 The marked pair and its ternary products

The twisted complexes needed here are finite sums of objects of A_r, with a strictly triangular degree-one matrix d satisfying $d^2 = 0$. Define

$$O = e_0, \qquad \mathcal{L} = e_0 \oplus e_1 \oplus \cdots \oplus e_r, \qquad d_{\mathcal{L}} = \sum_i \beta_i.$$

With e_0 first, the twisting matrix is triangular.

The corresponding free resolution has generators $v_0, \dots, v_r$ in degree zero and relations

$$\partial v_0 = 0, \qquad v_0 + (t - a_i)v_i = 0.$$

Its augmentation sends v_0 to 1 and v_i to $-(t - a_i)^{-1}$. This identifies its cokernel with L_f: the regular submodule and each quotient delta have the extension components just normalized.

The differential is injective. A vector in its kernel first satisfies $x_i(t - a_i) = 0$ at every leaf, hence $x_i = 0$. Its remaining equation is $x_0\partial = 0$, hence $x_0 = 0$.

Set $B_r = \mathrm{End}(O \oplus \mathcal{L})$ in the category of twisted complexes. Before twisting, write $\mathcal{L}_j$ for its j-th summand. A basis is

element	matrix entry	degree
e	$O \to O$	0
a	$O \to \mathcal{L}_0$	0
p	$\mathcal{L}_0 \to O$	0
u_j	$\mathcal{L}_j \to \mathcal{L}_j$ $(0 \le j \le r)$	0
b_i	$O \to \mathcal{L}_i$ (α_i)	1
c_i	$\mathcal{L}_i \to O$ (β_i)	1
θ_i	$\mathcal{L}_0 \to \mathcal{L}_i$ (α_i)	1
z_i	$\mathcal{L}_i \to \mathcal{L}_0$ (β_i)	1.

Multiply matching entries by matrix composition. Every product of two degree-one entries is zero. The commutator differential with the twisting matrix is

$$dp = -\sum_i c_i, \qquad du_0 = -\sum_i z_i, \qquad du_i = z_i \quad (i > 0), \tag{7.9}$$

and is zero on the remaining basis. In particular,

$$pa = e, \qquad ap = u_0, \qquad ac_i = z_i, \qquad \theta_i a = b_i, \qquad b_i p = \theta_i.$$

This specifies a dg algebra of dimension $5r + 4$.

Its cohomology H has degree-zero representatives $E = e$, $F = \sum_{j=0}^{r} u_j$, and a. In degree one, choose b_i, θ_i for $i \le r$ and c_j for $j < r$. The remaining class is $[c_r] = -\sum_{j<r} c_j$. Thus $\dim H = 3r + 2$. Apart from the two identities, the only nonzero binary products are $\theta_i a = b_i$.

Let $i : H \to B_r$ be the chosen inclusion. Define $\pi : B_r \to H$ by

$$\pi(u_0) = F, \quad \pi(u_j) = 0 \ (j > 0), \quad \pi(p) = \pi(z_i) = 0, \quad \pi(c_r) = -\sum_{j<r} c_j,$$

and by the identity on the other chosen representatives. Define a degree-minus-one map h by

$$h(z_i) = u_i, \qquad h(c_r) = -p,$$

and set it to zero on the other basis vectors. Equation (7.9) gives

$$\pi i = 1, \qquad dh + hd = 1 - i\pi, \qquad h^2 = hi = \pi h = 0.$$

These formulas retain the homotopies of the products ac_j, which vanish in cohomology.

We use the suspension $|sx| = |x| - 1$ and the conventions

$$b_2(sx, sy) = (-1)^{|x|} s(xy), \qquad b_3(sx, sy, sz) = (-1)^{|y|} s\, m_3(x, y, z).$$

Thus an unsuspended m_n has degree $2 - n$.

Theorem 7.10. *The algebra B_r models $\mathrm{REnd}_D(R \oplus L_f)$ with its two objects marked. Its strictly unital minimal A_∞-model on H has the binary product above and the following nonzero ternary products on nonidentity basis elements:*

$$m_3(a, c_j, \theta_i) = -\delta_{ij}\theta_i, \qquad m_3(a, c_j, b_i) = -\delta_{ij} b_i,$$
$$m_3(c_\ell, a, c_j) = -\delta_{\ell j} c_\ell \qquad (j, \ell < r,\ i \le r). \tag{7.10}$$

All m_n with $n \ge 4$ vanish. A suspended comparison to B_r has $F_1 = i$ and $F_2(sa, sc_j) = su_j$ for $j < r$. Its other higher components vanish.

Proof. The strict quasi-isomorphism of Theorem 7.9 extends to finite triangular twisted complexes and their Hom complexes. The displayed resolution identifies O, $\mathcal{L}$ with R, L_f, including their compositions.

Transfer through (i, π, h) gives

$$m_3(x, y, z) = (-1)^{|x|}\pi\big(i(x)h(i(y)i(z))\big) - \pi\big(h(i(x)i(y))i(z)\big).$$

The only product of two representatives on which h is nonzero is $ac_j = z_j$, with $j < r$, and its homotopy is u_j. The products $u_j\theta_i$, $u_j b_i$, and $c_\ell u_j$ give exactly (7.10).

Multiplying u_j by a representative gives zero, u_j, or a representative killed by h. A transfer tree with another homotopy above this multiplication therefore vanishes. The remaining four-input tree has root product $u_i u_j$, whose projection is zero. Every larger tree contains one of these configurations. The same reasoning leaves only the stated F_2. Since $hi = 0$, the two identities remain strict units.

The bar equation $b_{B_r}F = Fb_H$ can also be checked without a tree expansion. Its nonzero terms reduce to $du_j = z_j$ when one pair (a, c_j) occurs, and to $u_i u_j = \delta_{ij}u_j$ when two such pairs occur. The suspended signs are those fixed above. All other terms vanish by the matrix products. This verifies the comparison in every arity. □

Theorem 7.11. *For $r \ge 2$, the marked dg category (R, L_f) is not equivalent, by an object-preserving strictly unital A_∞-equivalence, to its binary cohomology category.*

Proof. Normalize the linear term of a hypothetical equivalence by the inverse of its cohomology isomorphism. Its linear term is then the identity. The arity-three equation would express m_3 as the Hochschild boundary of a binary map G_2 of degree minus one.

Evaluate that boundary on (a, c_j, θ_i). Both adjacent binary products are zero. The term $aG_2(c_j, \theta_i)$ also vanishes: its second factor is a reverse degree-one map, and $ac_\ell = 0$ in H. Only $G_2(a, c_j)\theta_i$ can remain, up to its fixed sign. But $G_2(a, c_j) \in H^0\,\mathrm{End}(\mathcal{L}) = kF$, so it acts by the same scalar on every θ_i.

For fixed $j < r$, formula (7.10) acts by -1 on θ_j and by zero on θ_r. No scalar has both values. The resulting rank-one projector survives modulo scalar endomorphisms, a contradiction. □

For two punctures, the obstruction is already

$$m_3(a, c_1, \theta_1) = -\theta_1, \qquad m_3(a, c_1, \theta_2) = 0.$$

For one puncture, there is no reverse class and the five-dimensional minimal model is formal. The formal star generator and the nonformal marked pair can therefore generate the same dg category. The marked objects determine which formality question is being asked.

7.3.2 The primitive three-point chart

The primitive module H_3 of (3.2) supplies a geometric instance. On $z_1 - z_3 \ne 0$, write

$$z_3 = s_0, \qquad z_1 = s_0 + u, \qquad z_2 = s_0 + ut.$$

Here u is invertible. Let e_i denote the generic residue vectors, and set $v_0 = e_2 - e_3$ and $v_1 = e_1 - e_2$.

Proposition 7.12. *There is a horizontal isomorphism*

$$H_3|_{u\neq 0} \simeq k[s_0, u, u^{-1}] \otimes_k \left(k[t]e_3 \oplus k[t, t^{-1}]v_0 \oplus k[t, (t-1)^{-1}]v_1\right).$$

Its generated dg category has model $\operatorname{Perf}(E \otimes_k A_2)$*, where* $E = \Lambda(\rho)$*,* $|\rho| = 1$*, and* $d\rho = 0$*. Adding the auxiliary exterior coefficients of* H_3 *does not enlarge this generated category.*

Proof. The finite presentation of Theorem 3.6 gives

$$g_{23} = \frac{v_0}{ut}, \qquad g_{12} = \frac{v_1}{u(1-t)}, \qquad g_{13} = \frac{v_0 + v_1}{u}, \qquad g_{123} = -\frac{v_0}{u^2 t} + \frac{v_1}{u^2(1-t)}.$$

The singleton generators are regular combinations of e_3, v_0, v_1. These formulas and their position derivatives place every generator in the proposed module. Conversely, g_{23}, g_{12} and their derivatives generate all finite poles at $t = 0, 1$ in the two stated lines.

Flatness from Corollary 3.2 implies torsion-freeness. The generic residue map is thus injective, and the two submodules with these generators coincide. Differentiating the rational representatives proves horizontality.

The three shape summands generate $k[t]$, δ_0, δ_1 by their localization triangles. Theorem 7.9 identifies their category with $\operatorname{Perf}(A_2)$. It also contains the double localization. Exterior coefficients are sums of these localizations. A rank-one localized summand has zero second exterior power, as follows by flatness and passage to the generic point. No additional generator is needed.

The scale coefficient $k[u, u^{-1}]$ has de Rham cohomology represented by 1, du/u. Their span is the strict algebra E. Polynomial common translation contracts to k. Tensoring the bounded free resolutions in the three independent coordinates gives the asserted dg model. □

The three summands of the primitive are a different marking from the pair consisting of a regular and a doubly localized module.

Proposition 7.13. *Put* $L_1 = k[t, t^{-1}]$*,* $L_2 = k[t, (t-1)^{-1}]$*, and* $K = k[t] \oplus L_1 \oplus L_2$*, with the three summands marked. Its derived endomorphism algebra has a strict formal model* C *of dimension eleven. The degree-zero basis consists of the three identities and* $a_i : O \to L_i$ *for* $i = 1, 2$*. The degree-one basis consists of*

$$b_i : O \to L_i, \qquad \theta_{ji} : L_i \to L_j \quad (i, j = 1, 2).$$

Apart from identity actions, the products are $\theta_{ji}a_i = b_j$*. The marked primitive on the chart consequently has formal model* $E \otimes C$*.*

Proof. Over A_2, use $O = e_0$ and the twisted complexes $\mathcal{L}_i = (e_0 \oplus e_i, \beta_i)$. Represent a_i by the identity into the central summand and b_i by α_i into the leaf. Represent θ_{ji} by α_j from the central summand of $\mathcal{L}_i$ to the leaf of $\mathcal{L}_j$. All these maps are closed.

Each reverse block $\operatorname{Hom}(\mathcal{L}_i, O)$ is a two-term isomorphism and is acyclic. For $i \neq j$, the block $\operatorname{Hom}(\mathcal{L}_i, \mathcal{L}_j)$ has one central map in degree zero and two maps in degree one. Its differential spans the map from the source leaf to the target center, leaving θ_{ji}. On a diagonal block, the sum of its two identities and θ_{ii} represent the two classes. The remaining blocks are represented by 1_O, a_i, b_i.

These eleven maps therefore represent every class and no boundary. They form a subalgebra with the stated products, so their inclusion is a strict quasi-isomorphism. Tensoring with the scale model and the contracted translation factor proves the last claim. □

The exterior coefficients $\Lambda^0 H_3$ and $\Lambda^3 H_3$ restrict instead to the scale factor tensored with $k[t]$ and $k[t, t^{-1}, (t-1)^{-1}]$. Marking this pair retains the ternary projector of Theorem 7.11. Tensoring with E does not remove it: the degree-zero endomorphisms remain scalars, and the reverse degree-one maps are still the c_j. Thus primitive formality does not determine the marked exterior system.

7.4 Two punctures through a collision

We now allow the punctures to meet. Work over $B = k[s]$ with the relative Weyl algebra

$$D_B = B\langle t, \partial_t\rangle/(\partial_t t - t\partial_t - 1).$$

The parameter s is central. At this stage there is no derivative in s. Put

$$f = t(t-s), \qquad L_B = B[t, f^{-1}], \qquad A_m = f\partial_t + mf_t, \qquad P_m = D_B/D_B A_m \quad (m \geq 1).$$

The cyclic generator e_m maps to f^{-m}. The relation $A_m(f^{-m}) = 0$ makes this a differential-module map.

The finite-stage phenomenon of Example 3.14 is explicit here. For a nonzero scalar a, the specialization $s = a$ has two simple roots, and P_m presents their localization.

Near a root, write $f = uv$ with u its linear coordinate and v invertible. Replacing the generator by $v^m e_m$ gives the relation $(u\partial_u + m)v^m e_m = 0$. The Weyl normal form identifies this module with the localization generated by u^{-m}. Away from both roots it is the regular rank-one connection. These local presentations establish the claim on the fiber.

At $s = 0$, put $C_m = t\partial_t + 2m$. Then $A_m = tC_m$, and $g_m = C_m e_m$ satisfies

$$t\,g_m = 0, \qquad C_m\,g_m = (2m-1)\,g_m.$$

There is an exact sequence

$$0 \longrightarrow D/Dt \xrightarrow{r_{C_m}} D/D(tC_m) \longrightarrow D/DC_m \longrightarrow 0.$$

Right cancellation in D proves injectivity of the first map. The last quotient is $k[t, t^{-1}]$, generated by t^{-2m}. The vector $\ell_m = e_m - g_m/(2m-1)$ satisfies $C_m\ell_m = 0$ and splits this quotient. Hence

$$P_m \otimes_B k_0 \simeq k[t, t^{-1}] \oplus \delta_0, \qquad g_m \longmapsto 0 \text{ in } L_B \otimes_B k_0. \tag{7.11}$$

The finite presentation acquires an extra closed coefficient at the collision.

The transition between denominator orders removes it. The identity

$$A_m f = f\,A_{m+1}$$

gives $P_m \to P_{m+1}$ by $e_m \mapsto f\,e_{m+1}$. It lifts to right multiplication by f in both degrees of $[D_B \xrightarrow{r_{A_m}} D_B]$. Lemma 3.13 identifies the direct-sum telescope of these resolutions with L_B.

Explicitly, the colimit relations give

$$\partial_t e_m = -mf_t e_{m+1}, \qquad \frac{p}{f^m} \longleftrightarrow pe_m.$$

These maps are inverse and horizontal. They remain so after every scalar extension $B \to C$: the leading coefficient f is monic in t, hence a non-zero-divisor over C.

At $s = 0$, the transition sends

$$g_m \longmapsto C_m t^2 e_{m+1} = t^2 C_{m+1} e_{m+1} = t^2 g_{m+1} = 0.$$

Thus the full telescope retains exactly the localization at the collision.

7.4.1 *Finite traces and the inverse telescope*

Apply derived Hom into $B[t]$ with its derivative connection. The stage complex is $[B[t] \xrightarrow{A_m} B[t]]$ in degrees zero and one. On monomials,

$$A_m(t^n) = (n+2m)t^{n+1} - (n+m)st^n. \tag{7.12}$$

Write $(a)_n = a(a+1)\cdots(a+n-1)$, with $(a)_0 = 1$.

Lemma 7.14. *The stage complex has zero kernel and a free rank-one cokernel over B. Its quotient map, normalized by $\pi_m(1) = 1$, is*

$$\pi_m(t^n) = \frac{(m)_n}{(2m)_n} s^n.$$

The inverse transition on cokernels satisfies

$$\pi_m(fp) = -\frac{m}{2(2m+1)} s^2 \pi_{m+1}(p). \tag{7.13}$$

Proof. The leading coefficient in (7.12) is an invertible positive integer. The operator raises the degree of every nonzero polynomial by one, so it is injective. Descending elimination reduces each polynomial uniquely to a constant modulo its image. The successive relations are

$$\pi_m(t^{n+1}) = \frac{n+m}{n+2m} s\, \pi_m(t^n).$$

They give the stated moments and a polynomial primitive for every vector with zero moment. Substituting $f = t^2 - st$ and these moments proves (7.13) on each monomial. □

Rescale by constants $c_1 = 1$ and $c_{m+1} = -mc_m/(2(2m+1))$. In the coordinates $a_m = c_m \pi_m(p_m)$, the inverse system becomes

$$B \xleftarrow{s^2} B \xleftarrow{s^2} B \xleftarrow{s^2} \cdots$$

in degree one. Dualizing the full direct-sum telescope requires products and the derived inverse limit of this system.

Theorem 7.15. *With $Q = k[[s]]/k[s]$ taken as an algebraic module quotient, there is an equivalence of B-complexes*

$$\mathrm{RHom}_{D_B}(L_B, B[t]) \simeq Q[-2].$$

In the normalized coordinates, the quotient is realized by

$$(a_m)_{m\geq 1} \longmapsto \left[\sum_{m\geq 1} s^{2m-2} a_m\right].$$

Proof. Hom from the free telescope is the product difference complex of the stage mapping complexes. Lemma 7.14 reduces it to

$$\prod_{j\geq 0} B \xrightarrow{\Delta} \prod_{j\geq 0} B, \qquad \Delta(b)_j = b_j - s^2 b_{j+1},$$

in degrees one and two. Its kernel is zero: the relation forces each b_j to be divisible by every power of s^2.

The sum $\sum_{j\geq 0} s^{2j} a_j$ is well-defined as a formal series. On a boundary it telescopes to the polynomial b_0. Conversely, suppose the sum is a polynomial p. Set

$$b_j = s^{-2j}\left(p - \sum_{i<j} s^{2i} a_i\right).$$

The numerator is a polynomial. Its formal-series expression is a tail of order at least $2j$, so it is divisible by s^{2j} in B. Thus $b_j \in B$ and $\Delta(b) = a$. Finally, any $\sum c_n s^n$ is obtained by taking $a_j = c_{2j} + c_{2j+1} s$. This proves the asserted cokernel and all its degrees. □

The constant sequence $a_m = 1$ gives the nonzero class of $(1 - s^2)^{-1}$ modulo polynomials. The formal-series quotient therefore represents an actual relative mapping class.

Proposition 7.16. *The preceding derived Hom does not commute with the flat extension $B \to B[s^{-1}]$. More precisely,*

$$\mathrm{RHom}_{D_B}(L_B, B[t]) \otimes_B B[s^{-1}] \simeq Q[-2],$$
$$\mathrm{RHom}_{D_{B[s^{-1}]}}(B[s^{-1}][t, f^{-1}], B[s^{-1}][t]) \simeq B[s^{-1}][-1].$$

For a closed fiber $s = a$, however,

$$Q[-2] \otimes^{\mathbf{L}}_B k_a \simeq \begin{cases} k[-1], & a \neq 0, \\ 0, & a = 0 \end{cases}.$$

These are the derived Hom groups of the corresponding closed-fiber localizations.

Proof. Multiplication by s is bijective on Q. If sF is polynomial, then F is polynomial. Removing a constant term and dividing by s gives surjectivity. Thus localizing Q leaves it unchanged. In contrast, after inverting s in the stage system, all transitions are invertible. Its limit is $B[s^{-1}]$, its higher derived limit vanishes, and its surviving degree is one.

For the closed fibers use $0 \to B \to k[[s]] \to Q \to 0$. The formal-series module is torsion-free over the principal ideal domain B, hence flat. For $a \neq 0$, the element $s - a$ is invertible on $k[[s]]$. The exact triangle therefore gives $Q \otimes^{\mathbf{L}}_B k_a \simeq k[1]$. For $a = 0$, bijectivity of s on Q makes the tensor complex acyclic. Formula (7.8) gives the same answers for two distinct punctures and one puncture, respectively. □

In particular, L_B is not perfect over D_B. A bounded resolution by finite projectives would make Hom into $B[t]$ commute with this flat extension. The bounded free telescope has infinitely generated terms, and its Hom involves the products just computed.

7.4.2 Restoring the parameter connection

The quotient Q was computed without differentiating in s. The flat telescope connection of (3.25) restores that derivative before taking Hom. On its lowest generator it is $\nabla_s h_m = -m f_s h_{m+1}$. For the corresponding degree-two Hom coordinate p_m, the induced action is therefore

$$p_m \longmapsto \partial_s p_m + m f_s p_{m+1}.$$

This determines the connection on the quotient.

Theorem 7.17. *On $Q = k[[s]]/k[s]$, the induced parameter derivative is*

$$D_Q[F] = \left[F' + \frac{F - F(0)}{s}\right].$$

It satisfies $[D_Q, s] = 1$ and is invertible as a k-linear map. Its inverse is

$$K_Q[F] = \left[\frac{1}{s}\int_0^s \tau F(\tau)\, d\tau\right].$$

Consequently, for the full Weyl algebra $D_{s,t}$,

$$\mathrm{RHom}_{D_{s,t}}(L_B, k[s,t]) \simeq 0.$$

Proof. Choose constant-in-t representatives $p_m = b_m(s)$ of the stage cokernels. Since $f_s = -t$ and $\pi_m(t) = s/2$, their derivatives have moments $b'_m - (m/2)sb_{m+1}$. Multiplication by c_m converts this to

$$a'_m + (2m+1)sa_{m+1}.$$

For $F = \sum_{m\geq 1} s^{2m-2}a_m$, summing gives $F' + (F - a_1)/s$. Its difference from the stated formula is the polynomial $(F(0) - a_1)/s$: a_1 is a polynomial with constant term $F(0)$. Thus both define the same map on Q.

Both D_Q and K_Q preserve polynomial representatives. On formal monomials, $K_Q(s^n) = s^{n+1}/(n+2)$. Direct calculation gives $D_Q K_Q F = F$ and $K_Q D_Q F = F - F(0)$. The latter constant disappears in Q. Also $D_Q(sF) - sD_Q(F) = F + F(0)$ on representatives, which proves the Weyl relation on the quotient.

The Spencer resolution of Proposition 3.16 computes full spatial Hom from the relative complex with its parameter derivative. The relative cohomology is concentrated in degree two, so truncation leaves $[Q \xrightarrow{D_Q} Q][-2]$. Its inverse K_Q contracts the complex. □

For example, the relative class of $(1 - s^2)^{-1}$ has the spatial primitive

$$-\frac{\log(1 - s^2)}{2s} = \sum_{n\geq 0} \frac{s^{2n+1}}{2n+2}.$$

This is a formal series regular at zero. No analytic convergence is needed in this computation.

There is also a finite verification of the full spatial vanishing. Put $x = t$, $y = t - s$. Then $L_B = k[x^{\pm 1}, y^{\pm 1}]$. The two commuting cyclic operators are $E_x = x\partial_x + 1$ and $E_y = y\partial_y + 1$. Their tensor resolution gives the Hom complex

$$k[x,y] \xrightarrow{p\mapsto(E_x p, E_y p)} k[x,y]^2 \xrightarrow{(a,b)\mapsto E_x b - E_y a} k[x,y].$$

Since $E_x^{-1}(x^i y^j) = x^i y^j/(i+1)$, its contraction is $h_1(a,b) = E_x^{-1}a$ and $h_2(c) = (0, E_x^{-1}c)$. Commutation with E_y proves $dh + hd = 1$ and $h^2 = 0$. The relative nonzero class and the spatial vanishing thus have explicit representatives and comparison differentials.

7.5 Spatial Hom and intersections of supports

Full spatial Hom need not vanish for every localization. Its answer depends on which support intersections exist. Let $X = \mathbb{A}^d_k$, $R = k[x_1, \ldots, x_d]$, and let $H_i = \{\ell_i = 0\}$ be a nonempty finite list of affine hyperplanes defined over k. Use the full Weyl algebra D_X and the ordinary connection on

$$L = R[(\ell_1 \cdots \ell_r)^{-1}].$$

The *intersection nerve* has a face $S \subseteq \{1, \dots, r\}$ when $H_S = \bigcap_{i \in S} H_i$ is nonempty. Put its augmentation in degree zero:

$$B^0 = k, \qquad B^q = \bigoplus_{\substack{|S|=q \\ H_S \neq \varnothing}} k \quad (q \geq 1). \tag{7.14}$$

The differential is alternating incidence in increasing index order. In particular, $B^0 \to B^1$ is the diagonal map.

Theorem 7.18. *There is a natural equivalence*

$$\mathrm{RHom}_{D_X}(L, R) \simeq B^\bullet$$

that retains the support-inclusion maps of the finite intersection diagram.

Proof. First let $W \subseteq X$ be a nonempty affine linear subspace of codimension c. Choose independent normal coordinates $u_1, \dots, u_c$. The local-cohomology complex $R\Gamma_W R$ is the tensor of $[R \to R[u_i^{-1}]]$. Thus it is $\delta_W[-c]$, where δ_W is the tensor of the normal principal-part modules and the regular tangential module.

In one normal coordinate the free resolution of the principal-part module is $[D \xrightarrow{r_u} D]$. Hom into $k[u]$ has cohomology in degree one, and evaluation at zero identifies its cokernel with k. Tangential polynomial de Rham complexes contract to constants. Tensoring gives

$$\mathrm{RHom}_{D_X}(R\Gamma_W R, R) \simeq k.$$

The canonical map to this complex from $\mathrm{RHom}_{D_X}(R, R) \simeq k$ is a quasi-isomorphism. In one normal variable the localization triangle proves this because the reverse Hom from $k[u, u^{-1}]$ is contracted by $(u\partial_u + 1)^{-1}$. Tensoring proves the assertion in all normal directions.

For every inclusion of supports, these canonical maps from the same complex $\mathrm{RHom}_{D_X}(R, R)$ commute with the induced trace map. They therefore give an objectwise equivalence of diagrams, not only separate identifications of their cohomology. Every nonempty intersection contributes k, each incidence coefficient is one, and an empty intersection contributes zero.

Iterating the finite localization complexes gives the Mayer–Vietoris complex for support in $\bigcup_i H_i$. Applying derived Hom and the preceding diagram equivalence gives the ordinary cochains of the intersection nerve. The triangle $R\Gamma_{\cup_i H_i} R \to R \to L$ identifies reverse Hom with the fiber of the augmentation from k to those cochains. This is precisely (7.14). □

When all hyperplanes meet, the nerve is a simplex. Inserting a fixed vertex contracts its augmented cochains. In particular, any nonempty arrangement of equality hyperplanes has zero reverse Hom of this localization into the regular target.

Example 7.19 (An affine triple)**.** For $x = 0$, $y = 0$, and $x + y = 1$, every pair intersects and the triple does not. The complex is

$$k \xrightarrow{(1,1,1)} k^3 \xrightarrow{(a,b,c) \mapsto (b-a, c-a, c-b)} k^3.$$

It retracts onto $k[-2]$. With coordinates (w_{12}, w_{13}, w_{23}) on the last term, take

$$\pi(w) = w_{12} - w_{13} + w_{23}, \qquad i(\lambda) = (0, 0, \lambda),$$

and $h_1(a, b, c) = a$, $h_2(w) = (0, w_{12}, w_{13})$. Substitution gives $dh + hd = 1 - i\pi$ and $h^2 = hi = \pi h = 0$. Thus a full spatial reverse class survives when the incidence diagram has this missing intersection.

Corollary 7.20. *Suppose $r > d \geq 1$ hyperplanes are in affine general position. This means that every subset of at most d intersects in the expected codimension and every $(d+1)$-subset has empty intersection. Then*

$$\mathrm{RHom}_{D_X}(L, R) \simeq k^{\binom{r-1}{d}}[-d].$$

Proof. The nerve is the $(d-1)$-skeleton of the simplex. For a degree-q cochain c, set

$$(h_q c)_T = \begin{cases} c_{\{1\}\cup T}, & 1 \notin T, \\ 0, & 1 \in T, \end{cases} \qquad |T| = q - 1.$$

Choose top representatives supported on the d-subsets avoiding vertex one. For $S = \{s_1 < \cdots < s_d\}$ of this kind, put

$$(\pi_d c)_S = c_S - \sum_{j=1}^{d} (-1)^{j-1} c_{\{1\}\cup(S\setminus\{s_j\})}.$$

Alternating incidence gives $dh + hd = 1 - i_d\pi_d$, with $h^2 = hi_d = \pi_d h = 0$. The number of these representatives is $\binom{r-1}{d}$. □

7.5.1 *The global regular and localized pair*

For $R_n = k[z_1, \ldots, z_n]$ and $\Delta_n = \prod_{i<j}(z_i - z_j)$, all equality hyperplanes meet. Hence, for $n \geq 2$,

$$\mathrm{RHom}_{D_n}(R_n[\Delta_n^{-1}], R_n) = 0.$$

At three points this vanishing determines the marked pair consisting of the regular and fully localized coefficient on the whole position space.

Put $R = R_3$ and $L = R[\Delta_3^{-1}]$. Define

$$E_3 = \Lambda(\rho) \otimes_k (k \oplus ka \oplus kb), \qquad |\rho| = |a| = |b| = 1,$$

with zero differential and every product of two elements of $ka \oplus kb$ equal to zero. The tensor product uses the graded sign.

Proposition 7.21. *The derived endomorphism algebra of the marked pair (R, L) is formal. A thirteen-dimensional model retaining both idempotents and the localization map is*

$$T(E_3) = \begin{pmatrix} k & 0 \\ E_3 & E_3 \end{pmatrix}, \qquad d = 0.$$

The lower-left copy models $\mathrm{RHom}(R, L)$, its element 1 is localization, and composition is graded matrix multiplication.

Proof. On the fully localized ring use $c = z_3$, $u = z_1 - z_3$, and $t = (z_2 - z_3)/(z_1 - z_3)$. Then

$$L = k[c] \otimes k[u, u^{-1}] \otimes k[t, t^{-1}, (t-1)^{-1}].$$

There is a dg-algebra map from E_3 to the algebraic de Rham complex of L given by

$$\rho \longmapsto du/u, \qquad a \longmapsto dt/t, \qquad b \longmapsto dt/(t-1).$$

It is a quasi-isomorphism. Polynomial integration contracts the c factor. In the u factor, only 1 and du/u survive. Partial fractions in t leave the two simple-pole forms. Polynomial forms and higher poles have primitives

in the same ring, while residues prove independence of the two remaining classes. The two shape forms have zero product, so this also verifies the multiplicative assertion.

Let D_L be the localized differential-operator algebra. Exact localization and its restriction adjunction identify $\mathrm{REnd}_{D_3}(L)$ with $\mathrm{REnd}_{D_L}(L)$. To compute its product, use the Spencer resolution

$$P = D_L \otimes_k \Lambda(e_1, e_2, e_3), \qquad |e_i| = -1, \qquad d_P = \sum_i r_{\partial_i}\, \iota_i$$

in coordinates (c, u, t), where ι_i contracts with e_i. Its order filtration reduces exactness to the Koszul complex of the derivative symbols.

There is a unital dg-algebra map

$$\Omega^\bullet_L \longrightarrow \mathrm{End}_{D_L}(P), \qquad f \longmapsto r_f, \qquad dx_i \longmapsto -\iota_i.$$

Indeed $[d_P, r_f] = -\sum_i r_{\partial_i f}\, \iota_i$, and the contractions anticommute and commute with multiplication by functions. Composition with augmentation to $\mathrm{Hom}_{D_L}(P, L)$ is the de Rham complex with these signs. The augmentation is a quasi-isomorphism because P is bounded and projective. Thus the displayed algebra map is a quasi-isomorphism.

The coordinate Spencer resolution of R identifies $\mathrm{RHom}_{D_3}(R, L)$ with the same de Rham complex. Precomposition with localization is the identity in these models. Polynomial de Rham contraction gives $\mathrm{REnd}_{D_3}(R) \simeq k$, and the reverse corner vanishes by Theorem 7.18.

To retain composition, choose a dg enhancement of the pair and a closed localization map $\lambda : R \to L$. Write $A = \mathrm{End}(L)$ there. The strict dg functor from $T(A)$ to this pair uses the scalar unit on $\mathrm{End}(R)$, the identity on A, and $a_0 \mapsto a_0\lambda$ on the forward corner. Its reverse corner is zero. Every corner map is a quasi-isomorphism by the calculations above. The multiplicative zigzag $A \simeq \Omega^\bullet_L \leftarrow E_3$ therefore gives $T(A) \simeq T(E_3)$, preserving the idempotents and localization map. □

On the chart $u \neq 0$, the regular object becomes $R[u^{-1}]$. The chart pair, viewed on the whole space by restriction and direct image, is consequently $(R[u^{-1}], L)$. It has the nonzero reverse corner and ternary obstruction of Theorem 7.11. The whole-space pair in Proposition 7.21 has regular object R. These different objects account for their different formality statements. Neither statement by itself determines the full exterior system with its primitive, second exterior coefficient, diagonal attachments, and vacuum maps.

7.6 Higher braid transport

The formal star algebra A_r has no products between its nonidentity arrows. It can nevertheless carry transport with nonzero higher components. We construct such transport on its bar, keeping the leaf generators as part of the data.

Fix $r \geq 2$ and put $S = \bigoplus_{i=0}^{r} k e_i$. The augmentation $A_r \to S$ kills every arrow. Its reduced relative bar coalgebra is

$$BA_r = T^c_S((A_r/S)[1]),$$

where s lowers degree by one and T^c is the direct sum over tensor lengths, with deconcatenation coproduct. Every suspended arrow has degree zero. Thus BA_r is concentrated in degree zero and its bar differential is zero.

The full dual is the path-length completion $\widehat{P}$ of the reverse quiver. Denote its arrows by

$$x_i : 0 \longrightarrow i, \qquad y_i : i \longrightarrow 0,$$

where x_i pairs with $s\beta_i$ and y_i with $s\alpha_i$. Pairing reverses the order of the tensors. There is no Koszul sign because the suspended arrows have degree zero.

Products are composition with the rightmost map applied first. The central loops and units are

$$z_i = y_i x_i, \qquad g_i = e_0 + z_i, \qquad g_i^{-1} = \sum_{m \geq 0} (-z_i)^m.$$

Every finite path quotient is finite-dimensional. Infinite expressions below mean their compatible values in these quotients.

Theorem 7.22. *For $j = i + 1$, exchange e_i, e_j, fix the other idempotents, and define*

$$\begin{aligned} \sigma_i(x_i) &= x_j g_i^{-1}, & \sigma_i(y_i) &= g_i y_j, \\ \sigma_i(x_j) &= x_i, & \sigma_i(y_j) &= y_i. \end{aligned} \tag{7.15}$$

Fix all other arrows. These substitutions are continuous automorphisms of $\widehat{P}$. They satisfy the braid relations and preserve the ordered product $g_1 \cdots g_r$. Their inverses are

$$\begin{aligned} \sigma_i^{-1}(x_i) &= x_j, & \sigma_i^{-1}(y_i) &= y_j, \\ \sigma_i^{-1}(x_j) &= x_i g_j, & \sigma_i^{-1}(y_j) &= g_j^{-1} y_i. \end{aligned}$$

Proof. Every substituted arrow has positive length, so the formulas define continuous endomorphisms. Substitution verifies both inverse identities. On the central units they give

$$\sigma_i(g_i) = g_i g_j g_i^{-1}, \qquad \sigma_i(g_j) = g_i. \tag{7.16}$$

Thus $g_i g_j$ and the full ordered product are preserved.

For three consecutive leaves, write the central units as g, h, k. Both sides of the adjacent braid relation send the three arrow pairs to

$$(x_3 h^{-1} g^{-1}, g h y_3), \qquad (x_2 g^{-1}, g y_2), \qquad (x_1, y_1).$$

They have the same permutation of vertices. The relation therefore holds on all generators. Distant substitutions act on disjoint pairs and fix the central units used by the other substitution, so they commute. These identities hold in words in invertible units before their series expansions. □

The central formulas (7.16) are the usual Artin substitutions. Formula (7.15) lifts them to the arrows themselves. This lift retains a chosen generator at each leaf, which is the framing of the path action.

Transpose σ_i against the finite path spaces. This gives a coaugmented coalgebra automorphism F_i of BA_r, with its stated permutation of objects. Its corestrictions are specified in every arity by

$$\langle q, F_{i,n}(sa_1| \cdots |sa_n) \rangle = \langle \sigma_i(q), sa_1| \cdots |sa_n \rangle \tag{7.17}$$

for each dual arrow q.

Proposition 7.23. *The components (7.17), extended by strict unitality, define augmented A_∞-automorphisms of A_r. They satisfy the braid relations. Every even component vanishes, while the expansion*

$$\sigma_i(x_i) = \sum_{m \geq 0} (-1)^m x_j z_i^m, \qquad \sigma_i(y_i) = y_j + z_i y_j$$

gives nonzero components in every odd arity in the first family.

Proof. Transposing multiplication gives the coalgebra identity. The zero bar differential gives the chain identity. On an input of fixed length, only finitely many path coefficients can contribute, since substitution never decreases path length. Thus the transpose preserves the direct-sum bar.

Inverses transpose to inverses. Transposition reverses composition, but the adjacent braid words are palindromes and distant commutation is symmetric. The braid relations therefore remain valid. Every path parallel to an arrow has odd length. The displayed series then determines the parity and nonvanishing of the components. Adjoining the identity operations gives strict unitality. □

For the pure braid $\Phi_i = \sigma_i^2$, the vertices and linear terms are fixed. In central loop length,

$$\begin{aligned} \Phi_i(z_i) &= z_i - [z_i, z_j] + O(z^3), \\ \Phi_i(z_j) &= z_j + [z_i, z_j] + O(z^3), \\ \Phi_i(z_k) &= z_k \quad (k \notin \{i, j\}). \end{aligned} \tag{7.18}$$

Here $O(z^3)$ means terms of loop length at least three. On arrows the first corrections include

$$\Phi_i(x_i) = x_i - x_i z_j + O(\text{length } 5), \qquad \Phi_i(y_i) = y_i + z_j y_i + O(\text{length } 5).$$

Their transposes have identity linear term and nonzero ternary terms.

Proposition 7.24. *For $r \geq 3$, Φ_i is not inner on the central completed free algebra. For $r = 2$, the full path action is conjugation by*

$$u = g_1 g_2 + (e_1 + x_1 y_1) + (e_2 + x_2 y_2).$$

In either case, its transpose is not homotopic to the identity through augmented, strictly unital bar maps with the objects fixed.

Proof. An inner automorphism of the central corner is conjugation by a unit with nonzero scalar constant term. Its degree-two correction is the commutator with a linear term $h = \sum_k a_k z_k$. Formula (7.18) forces $a_i = a_j = 1$ and every other $a_k = 0$. For a third generator, however, $[h, z_k] = [z_i + z_j, z_k] \neq 0$, while Φ_i fixes z_k. This proves the first assertion.

For two leaves, set $u_0 = g_1 g_2$ and $u_j = e_j + x_j y_j$. These are units in their corners and satisfy $u_j x_j = x_j g_j$, $y_j u_j = g_j y_j$. Conjugation by their sum sends

$$\begin{aligned} (x_1, y_1) &\longmapsto (x_1 g_1 g_2^{-1} g_1^{-1},\ g_1 g_2 g_1^{-1} y_1), \\ (x_2, y_2) &\longmapsto (x_2 g_1^{-1},\ g_1 y_2). \end{aligned}$$

These are the twofold substitutions of (7.15).

The reduced bars are concentrated in degree zero. A degree-minus-one bar homotopy must therefore vanish. Equivalently, a chain map to the bar tensored with polynomial interval forms has no degree-minus-one coefficient for a dt term. Its degree-zero component has derivative zero and is constant. The nonzero ternary term rules out such a homotopy to the identity. □

The two-leaf action is inner, but that inner conjugation is not an augmented bar homotopy. The equivalence relation is part of the transport statement.

7.7 Logarithmic families and their curvature

Let Φ be a product of the pure-braid substitutions. The operator $\Phi - 1$ raises arrow length by at least two. Its logarithm therefore exists on every finite path quotient:

$$\mathcal{L} = \log \Phi = \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} (\Phi - 1)^n.$$

The resulting operators are compatible under quotient maps.

Lemma 7.25. *The operator $\mathcal{L}$ is a continuous derivation. It raises arrow length by at least two and annihilates every vertex idempotent and $g_1 \cdots g_r$.*

Proof. At any finite path precision, the expression

$$\Phi^v = \sum_{n \geq 0} \binom{v}{n} (\Phi - 1)^n$$

is a polynomial in v. For every nonnegative integer v it is multiplicative. In characteristic zero this is a polynomial identity, so it holds for arbitrary v. Differentiating at zero proves the Leibniz rule for $\log \Phi$. The length assertion follows termwise. Every element fixed by Φ is killed by each term of its logarithm, which proves the remaining assertions. □

Take $k = \mathbb{C}$ and write $D = \mathcal{L}/(2\pi i)$. On a punctured disc with coordinate q, define

$$\nabla = d - D \, \frac{dq}{q}. \tag{7.19}$$

At a finite path quotient D is nilpotent. Horizontal transport is $\exp(D \log q)$, a finite polynomial in $\log q$, with counterclockwise monodromy Φ.

Each quotient has the corresponding holomorphic lattice with a logarithmic connection at $q = 0$. These lattices form a compatible inverse system. The same formula defines a formal logarithmic connection over $\mathbb{C}[[q]]$.

Theorem 7.26. *Connection (7.19) is flat and multiplicative. The paired bar coalgebra has connection*

$$\nabla^\vee = d + D^\vee \, \frac{dq}{q}$$

and monodromy $\Phi^{-\vee}$. It defines a strictly unital augmented A_∞-algebra over logarithmic de Rham coefficients. Its new reduced bar components are

$$b_n^{\mathrm{new}} = \frac{dq}{q} (D^\vee)_n, \qquad n \geq 3.$$

Only odd arities occur. Every point evaluation recovers A_r.

Proof. The derivation property of D proves multiplicativity. The form dq/q is closed and has square zero, so the connection is flat. The plus sign on the dual makes evaluation horizontal and gives inverse-transpose monodromy.

The transpose of a derivation is a coderivation. Here it lowers length by at least two, so it acts finitely on each bar input and has no unary component. Path parity excludes even arities. On the bar tensored with logarithmic de Rham coefficients, $d + (dq/q)D^\vee$ is a degree-one coderivation of square zero. This is the A_∞ bar equation.

After desuspension, the arity-n operation has degree $2 - n$. The original unit and binary product remain, and all new operations vanish when an input is a unit. Point evaluation kills dq/q. □

The pointwise algebra is constant while its higher transport is nontrivial. The boundary product is horizontal by Lemma 7.25. These statements concern the completed path algebra and its paired bar.

Two collision parameters impose an additional equation. For three leaves put $\mathcal{L}_{12} = \log(\sigma_1^2)$ and $\mathcal{L}_{23} = \log(\sigma_2^2)$.

Proposition 7.27. *The two derivations have nonzero commutator:*

$$[\mathcal{L}_{12}, \mathcal{L}_{23}](x_1) = x_1(z_3 z_2 - z_2 z_3) + O(\mathit{length}\, 7).$$

Consequently, on a product of two punctured discs the connection

$$d - \frac{\mathcal{L}_{12}}{2\pi i}\, d\log q_1 - \frac{\mathcal{L}_{23}}{2\pi i}\, d\log q_2$$

is not flat.

Proof. The first nonzero terms of a logarithm agree with those of its pure map minus the identity. Thus

$$\mathcal{L}_{12}(x_1) = -x_1 z_2 + O(\text{length } 5), \qquad \mathcal{L}_{23}(x_1) = 0, \qquad \mathcal{L}_{23}(z_2) = [z_3, z_2] + O(z^3).$$

The derivation rule gives the stated commutator. Its leading paths are distinct, so their difference is nonzero. The curvature is

$$(2\pi i)^{-2}[\mathcal{L}_{12}, \mathcal{L}_{23}]\, d\log q_1 \wedge d\log q_2.$$

□

For commuting pure monodromies the logarithms commute, and the sum of their one-variable connections is flat. The displayed adjacent pure monodromies instead require further terms satisfying the curvature equation if they are to belong to a simultaneous family. One-variable transport alone does not construct those terms.

The analytic line gauge and the framed path action were constructed on specified coefficient spaces. To identify either with native collision transport requires maps from the supported collision and endpoint modules. Such maps must preserve the normal derivatives, the regular attachments, both vacuum insertions, and nested compositions on a common continuous domain. The separated estimates supply that domain for the line calculation. The finite marked models specify which objects and higher operations a collision comparison must retain. An identification with the full ordered collision atlas requires this further construction.

Chapter 8

Fields, centers and reconstruction maps

A collision coefficient has endomorphisms before it has a universal interpretation. The preceding chapters constructed coefficients, comparison maps and actions on specified carriers. We now compute which operations commute with all morphisms in the categories those coefficients generate. The first example has only two objects: the regular normal line and its punctured localization. Their derived center is a formal power-series algebra with a logarithmic odd operation. The arrow from the regular line to the punctured line forces that logarithmic condition.

This calculation leads to two different extensions. Several punctures produce completed path algebras and transport on their centers. The Heisenberg field algebra instead gives a tower of cochain actions whose products retain a relative collision residue. A universal action must also account for maps and homotopies between actors. Those maps will supply the final obstruction: two parallel maps of pointed actions remain nonhomotopic after every receiving pointed factorizing action.

8.1 The regular and punctured normal line

Let k be a field of characteristic zero. Put

$$R = k[u], \qquad L = k[u, u^{-1}], \qquad D = k\langle u, \partial\rangle/(\partial u - u\partial - 1).$$

We regard R, L as left D-modules with their derivative connections. Example 3.3 identifies the binary primitive coefficient with $R \oplus L$. Its normal derivative is part of the coefficient. We therefore take $C_2 = \mathrm{thick}_D(R \oplus L)$, the full dg category generated by these modules under shifts, finite cones and direct summands.

For $v \in D$, write r_v for right multiplication by v, a map of free left D-modules. Use the resolutions

$$P_0 = [D \xrightarrow{r_\partial} D] \longrightarrow R, \qquad P_1 = [D \xrightarrow{r_{u\partial+1}} D] \longrightarrow L \tag{8.1}$$

in degrees -1, 0, augmented by 1 and u^{-1}, respectively. The Weyl algebra is a domain, so both differentials are injective.

The first cokernel is R. In the second, the relation $u\partial = -1$ on the cyclic generator reduces mixed normally ordered monomials to pure powers of u or ∂. Their images are the distinct Laurent monomials u^{m-1}, $m \geq 0$, and $(-1)^n n! u^{-n-1}$, $n \geq 1$. This proves the second cokernel identification, including injectivity of its map to L.

Applying $\mathrm{Hom}_D(P_i, -)$, and changing the sign in cochain degree one if necessary, gives

	R	L
R	$[R \xrightarrow{\partial} R]$	$[L \xrightarrow{\partial} L]$
L	$[R \xrightarrow{u\partial+1} R]$	$[L \xrightarrow{u\partial+1} L]$.

Polynomial integration, Laurent integration away from the residue, and the formula $(u\partial + 1)u^m = (m+1)u^m$ therefore yield

$$\begin{array}{c|cc} \mathrm{RHom}_D(-,-) & R & L \\ \hline R & k & k \oplus k[-1] \\ L & 0 & k \oplus k[-1]. \end{array} \tag{8.2}$$

Here $k[-1]$ lies in cohomological degree one.

To retain composition, define a graded path algebra T on objects $0, 1$. Its generators are $a : 0 \to 1$ of degree zero and $\theta : 1 \to 1$ of degree one, subject to $\theta^2 = 0$. Products are written in composition order. Put $b = \theta a$. Thus

$$T^0 = ke_0 \oplus ke_1 \oplus ka, \qquad T^1 = k\theta \oplus kb, \qquad d_T = 0.$$

Apart from identities, the only nonzero product of basis elements is $\theta a = b$. We use perfect right dg modules in $\mathrm{Perf}(T)$.

Theorem 8.1. *There is a strict quasi-isomorphism of dg algebras*

$$T \longrightarrow \mathrm{End}_D(P_0 \oplus P_1).$$

Consequently $C_2 \simeq \mathrm{Perf}(T)$*, as dg categories.*

Proof. Map e_i to the projections. Map a to $\alpha : P_0 \to P_1$, with $\alpha^{-1} = 1$, $\alpha^0 = r_u$. Its chain equation is $\partial u = u\partial + 1$. Map θ to the degree-one identity $\Theta : P_1^{-1} \to P_1^0$, and b to $\Theta\alpha$. These maps are closed and satisfy $\Theta^2 = 0$ strictly.

After augmentation, α represents the inclusion $R \to L$. Θ and $\Theta\alpha$ represent u^{-1} in the Euler and derivative cokernels. They give bases of all the corners in (8.2). The resulting algebra map is a quasi-isomorphism. The functor $\mathrm{Hom}_D(P_0 \oplus P_1, -)$ and its tensor adjoint identify the generators and their Hom complexes. Exactness and passage to direct summands extend this identification to their thick closures, proving the last assertion. □

Restrict the auxiliary binary Heisenberg complex to $(z_1, z_2) = (0, u)$, with its induced normal connection, and denote it by $Z^{\mathrm{nor}}_{\kappa,2}$. At nonzero level, Theorem 3.17 gives the coefficient decomposition

$$Z^{\mathrm{nor}}_{\kappa,2} \simeq R \oplus (R \oplus L)[-1] \oplus L[-2].$$

It generates the same dg category. This statement concerns the coefficient complex. The native transverse multiplication supplies additional structure.

8.2 The completed center of the normal pair

For a dg category generated by T, its categorical center is the derived endomorphism complex of the diagonal T-bimodule, with its Hochschild operations. We compute it using the normalized relative two-sided bar over $S = ke_0 \oplus ke_1$. The algebra S is separable. Removing the unit degeneracies from the two-sided bar therefore leaves a resolution

$$T \otimes_S T^c_S((T/S)[1]) \otimes_S T \longrightarrow T.$$

Here $T^c_S(W) = \bigoplus_{n \geq 0} W^{\otimes_S n}$, with the length-zero term S, is the tensor coalgebra over S.

Applying $\mathrm{Hom}_{T^e}(-, T)$ gives the *product* over arities. Denote this cochain algebra by $C(T)$. This construction and dg Morita invariance identify it with the center of C_2. See [67, §5.4].

Fix the following signs. An n-cochain of internal degree r has total degree $n + r$. Its differential is $D = (-1)^r b_H$, where

$$\begin{aligned}(b_H f)(v_1, \dots, v_{n+1}) = {} & (-1)^{r|v_1|} v_1 f(v_2, \dots, v_{n+1}) \\ & + \sum_{i=1}^{n} (-1)^i f(v_1, \dots, v_i v_{i+1}, \dots, v_{n+1}) \\ & + (-1)^{n+1} f(v_1, \dots, v_n) v_{n+1}.\end{aligned}$$

For an m-cochain g of internal degree s, use

$$(f \smile g)(v_1, \dots, v_{n+m}) = (-1)^{s(\sum_{i=1}^{n} |v_i| - n)} f(v_1, \dots, v_n) g(v_{n+1}, \dots, v_{n+m}). \tag{8.3}$$

These are the suspended-bar differential and product expressed on unsuspended inputs. In particular D satisfies the total-degree Leibniz rule.

Theorem 8.2. *There is a continuous strict quasi-isomorphism of associative dg algebras*

$$k[[x]] \otimes \Lambda(\eta) \longrightarrow C(T), \qquad |x| = 0, \quad |\eta| = 1, \tag{8.4}$$

with zero differential on the source. The Gerstenhaber bracket on cohomology satisfies

$$[\eta, x] = -x, \qquad [x, x] = [\eta, \eta] = 0.$$

Proof. Assign weight one to θ, b and weight zero to e_i, a. For a cochain, let d be input weight minus output weight. The differential preserves d.

Every nonempty composable relative word is $\theta^{\otimes n}$, $\theta^{\otimes(n-1)} \otimes a$, or $\theta^{\otimes(n-1)} \otimes b$. The output must have the same endpoints as its word. In weight $d \geq 1$, this gives exactly the following ordered bases:

$$\begin{array}{c|l} 0 & \theta^{\otimes d} \mapsto e_1, \quad \theta^{\otimes(d-1)} \otimes b \mapsto a \\ 1 & \theta^{\otimes(d+1)} \mapsto \theta, \quad \theta^{\otimes d} \otimes a \mapsto a, \quad \theta^{\otimes d} \otimes b \mapsto b \\ 2 & \theta^{\otimes(d+1)} \otimes a \mapsto b. \end{array}$$

The differential matrices are

$$k^2 \xrightarrow{M} k^3 \xrightarrow{N} k, \qquad M = \begin{pmatrix} 0 & 0 \\ -1 & 1 \\ -1 & 1 \end{pmatrix}, \qquad N = \begin{pmatrix} 1 & 1 & -1 \end{pmatrix}. \tag{8.5}$$

Indeed $\theta a = b$ is the only possible nonzero interior face. The two endpoint faces give the other entries.

In weight zero the degree-zero basis is e_0, e_1, and the first matrix is $-M$. Weight -1 consists of the identity map $k \to k$ in degrees one and two. Its source is the nullary θ, and its target sends a to b. No lower weight occurs.

For each $d \geq 0$, define representatives and projections by

$$i_0(c) = (c, c), \quad p_0(p, q) = p, \qquad i_1(c) = (c, 0, c), \quad p_1(r, s, t) = t - s.$$

A contracting homotopy off these representatives is

$$h_1(r, s, t) = \begin{cases} (0, s) & d > 0, \\ (0, -s) & d = 0, \end{cases} \qquad h_2(z) = (z, 0, 0).$$

Matrix multiplication gives $Dh + hD = 1 - ip$ and $h^2 = hi = ph = 0$. Use the inverse identity in weight -1. Every block is finite, and these maps preserve weight, so their product contracts the full cochain complex continuously.

Define derivations X, Y by

	e_0	e_1	θ	a	b
X	0	0	e_1	0	a
Y	0	0	θ	0	b.

Their internal degrees are -1, 0, and their total degrees are 0, 1. The displayed product table proves the graded derivation identities, hence $DX = DY = 0$. Formula (8.3) gives $X \smile Y = Y \smile X$ and $Y \smile Y = 0$.

The powers $X^{\smile d}$ and $Y \smile X^{\smile d}$ are the representatives $(1, 1)$ and $(1, 0, 1)$ above. Thus $x \mapsto X, \eta \mapsto Y$ defines the asserted quasi-isomorphism. Each formal coefficient has a specified arity, so evaluation on a fixed word involves a finite sum. The unary Gerstenhaber bracket is the graded commutator of derivations. Here $YX - XY = -X$. The biderivation rule on cohomology gives the bracket on all formal series. □

Thus η acts as the logarithmic vector field $-x\partial_x$. In particular

$$[\eta f, g] = -xfg', \qquad [\eta f, \eta g] = \eta x(gf' - fg').$$

The full bar, with its braces, retains the higher center operations. The associative quasi-isomorphism and the cohomology bracket above do not assert formality as an E_2-algebra. Using direct sums of cochains instead would replace $k[[x]]$ by $k[x]$. That convention omits endomorphisms of the direct-sum bar.

The restriction maps explain the logarithmic condition. The full subcategory on L has endomorphism algebra $E = \Lambda(\theta)$, and its normalized cochains are $C(E) = E[[x]]$, with zero differential. Restricting relative words gives

$$k[[x]] \otimes \Lambda(\eta) \longrightarrow E[[x]], \qquad x \longmapsto x, \quad \eta \longmapsto \theta x. \tag{8.6}$$

Restriction to R sends x, η to zero and preserves constants. These are restrictions along full dg subcategories. The extra regular object forces the odd coefficient in $E[[x]]$ to vanish at the formal origin.

One can also see this from the relative bar triangle

$$C(T) \longrightarrow C(k) \oplus C(E) \longrightarrow \mathrm{RHom}_E(Ea, Ea).$$

Split words into those in a diagonal corner and those crossing from 0 to 1. Crossing occurs once. The latter words are the shifted module bar of Ea. Since Ea is free over E, that bar contracts to E.

On the diagonal models the last arrow is $(c, f) \mapsto c - f(0)$, a surjective map of complexes. Its kernel is $k \oplus xE[[x]]$. The identification $\eta = \theta x$ recovers (8.4).

8.3 Marked deformations and the Kummer residue

First fix the two objects, or equivalently the inclusion $S = ke_0 \oplus ke_1 \subset T$. For uncurved S-linear deformations with zero nullary comparison term, retain only positive arities. Let $C_{\geq 1}(T)$ denote this subcomplex of $C(T)$, with the same product totalization. Write $C^k_{\geq 1}(T)$ for the positive-arity normalized Hochschild complex relative to k instead of S. It fixes the unit but permits the object decomposition to move.

Proposition 8.3. *The S-relative complex $C_{\geq 1}(T)$ has cohomology*

$$H^0 = xk[[x]], \qquad H^1 = k[[x]]\eta \oplus k\sigma, \qquad H^2 = kq,$$

and vanishes otherwise. Here $\sigma(a) = a$, $\sigma(b) = b$, and $q(a) = b$. Their other generator values vanish. For the complex relative to k, the answer is

$$H^0 = xk[[x]], \qquad H^1 = k[[x]]\eta \oplus k\sigma \oplus kDa, \qquad H^2 = kD\theta \oplus kDb,$$

with no other cohomology. The last two generators have total degree two, and $D\theta = q$.

Proof. The positive-weight blocks in (8.5) do not change. Removing the two nullary identities in weight zero removes their degree-zero cohomology and leaves their former boundary $(0,1,1) = \sigma$ in degree one. Removing the nullary θ in weight -1 leaves q in degree two. The enumeration in the preceding proof exhausts the cochains, and gives the first formula.

For the second assertion, restore arity zero. Its quotient from the full k-relative complex is T, with zero differential, giving

$$0 \longrightarrow C^k_{\geq 1}(T) \longrightarrow C^k(T) \longrightarrow T \longrightarrow 0.$$

Separability of S identifies the cohomology of $C^k(T)$ with that computed in Theorem 8.2. Projection to arity zero sends $f(x)$ to $f(0)1_T$ and sends $\eta k[[x]]$ to zero: all these latter representatives have positive arity.

The cohomology exact sequence therefore contributes $T^0/k1_T$, with representatives e_0, a, to degree one, and all of $T^1 = k\theta \oplus kb$ to degree two. The connecting maps are the Hochschild differentials of these nullary elements. In particular $(Db)(e_0) = b$, whereas $(D\theta)(e_0) = 0$ and $(D\theta)(a) = b$, so the two degree-two classes are distinct. The positive-arity representatives $\eta k[[x]]$ split the degree-one exact sequence. □

Put $A = k[[s]]$. Let T_s have the graded multiplication of $T \otimes_k A$, but differential

$$d_s a = sb, \qquad d_s\theta = d_s b = d_s e_i = 0. \tag{8.7}$$

It is a dg algebra: $d_s = s[\theta, -]$ is a derivation of square zero. Its graded A-module is free, whereas

$$H^0(T_s) = Ae_0 \oplus Ae_1, \qquad H^1(T_s) = A\theta \oplus (A/(s))b.$$

Even modulo s^2, there is no uncurved A_∞-map $T_s \to T \otimes A$ with $F_0 = 0$ and $F_1 = 1 \pmod{s}$. Its unary equation would be $F_1 d_s = 0$, but on a the left side is $sb \neq 0 \pmod{s^2}$. Higher components do not enter this equation. The full Hochschild complex has $D\theta = q$ and $H^2C(T) = 0$, but its primitive is excluded from this comparison problem. The example concerns graded flatness. Its cohomology is not flat over A.

The family has a realization by differential modules. Define $D_A = A\langle u, \partial\rangle/(\partial u - u\partial - 1)$, $R_A = A[u]$ with its derivative connection, and

$$L_s = A[u, u^{-1}], \qquad \partial f = f' - \frac{s}{u} f.$$

These are polynomial and Laurent polynomial modules over A. No completion in u is taken.

Theorem 8.4. *There is a strict quasi-isomorphism*

$$T_s \longrightarrow \mathrm{End}_{D_A}(P_{0,A} \oplus P_{1,s}), \qquad P_{1,s} = [D_A \xrightarrow{r_u\partial + 1 + s} D_A],$$

where $P_{0,A} = P_0 \otimes_k A$, and $P_{1,s} \to L_s$ is augmented by u^{-1}. It induces

$$\mathrm{Perf}(T_s) \simeq \mathrm{thick}_{D_A}(R_A \oplus L_s).$$

For $K = k((s))$, the generic fiber satisfies

$$\mathrm{Perf}(T_s \otimes_A K) \simeq \mathrm{Perf}(K) \oplus \mathrm{Perf}(\Lambda_K(\theta)).$$

Consequently T_s is not Morita equivalent over A to the constant family $T \otimes_k A$.

Proof. The ordered Weyl filtration proves that D_A is a domain. In the cyclic quotient by $u\partial + 1 + s$, mixed monomials reduce to pure powers: commuting one ∂ past u lowers both exponents, using the cyclic Euler relation. The pure powers act on u^{-1} as

$$u^m u^{-1} = u^{m-1}, \qquad \partial^n u^{-1} = (-1)^n \prod_{j=1}^{n} (j+s) u^{-n-1}.$$

The factors $j + s$ are units of A. These images form a Laurent basis. This proves exactness of the displayed resolution.

Keep the maps $\alpha^{-1} = 1$, $\alpha^0 = r_u$, Θ, and $\Theta\alpha$ from Theorem 8.1. Now

$$d_{\mathrm{Hom}}\alpha = r_{u\partial+1+s} - r_{\partial u} = s\Theta\alpha.$$

All other differentials and all multiplicative relations agree with (8.7), giving a strict dg algebra map.

Compute its four corners after augmenting the target resolutions. For $R_A \to L_s$, the derivative maps u^m to $(m-s)u^{m-1}$. All modes with $m \neq 0$ contract, since $m - s$ is a unit. With the Hom differential sign, the remaining complex is $[A \xrightarrow{s} A]$, represented by $\alpha, \Theta\alpha$. For $L_s \to R_A$, the Euler differential acts by $m + 1 + s$ on u^m, $m \geq 0$. Every coefficient is a unit, so this corner is acyclic.

For $L_s \to L_s$, it acts by $m + 1$, leaving the $m = -1$ mode in both degrees. The remaining regular corner is A in degree zero. Thus the map is a quasi-isomorphism on every corner. The same generator argument as before proves the dg equivalence.

Over K, the two-term ideal spanned by a, b is contractible, because its differential is multiplication by the unit s. The quotient map $T_s \otimes_A K \to K \times \Lambda_K(\theta)$ is therefore a dg algebra quasi-isomorphism, proving the generic-fiber assertion. Its degree-zero categorical center is $K \times K[[x]]$. Indeed the center of a product category is the product of its centers, and the exterior corner has the cochain model $\Lambda_K(\theta)[[x]]$ already computed above.

Applying Theorem 8.2 directly over the field K gives $HH^0(T \otimes_k K) = K[[x]]$. The first ring has a nontrivial idempotent, whereas the second has none. Morita equivalence would preserve these center rings. An A-linear Morita equivalence would remain one after extension to K. Neither equivalence exists. This argument computes both generic centers directly and uses no base-change assertion for completed Hochschild cochains. □

The special fiber has a horizontal inclusion $R \to L$. In the family, the ordinary inclusion of coefficient modules fails to commute with the connections by $-s/u$. The equation $d_s a = sb$ records this failure. Thus the generic splitting is caused by the loss of that horizontal arrow. It makes no assertion that completed Hochschild cochains commute with specialization in s.

8.4 Independent normal directions

Let $R_d = k[u_1, \ldots, u_d]$, and let D_d be its Weyl algebra. The normal-crossing coefficient generator is

$$G_d = \bigoplus_{I \subseteq \{1,\ldots,d\}} R_d \left[\left(\prod_{i \in I} u_i \right)^{-1} \right] = \bigotimes_{i=1}^{d} (k[u_i] \oplus k[u_i, u_i^{-1}]),$$

where the tensor products are external over k and the empty product is one. Each summand has its derivative connection.

Theorem 8.5. *The categorical center of* $\mathrm{thick}_{D_d}(G_d)$*, with full product-total Hochschild cochains, has the associative dg model*

$$k[[x_1, \ldots, x_d]] \otimes \Lambda(\eta_1, \ldots, \eta_d), \qquad |x_i| = 0, \quad |\eta_i| = 1,$$

with zero differential. Its cohomology bracket satisfies

$$[\eta_i, x_j] = -\delta_{ij} x_j, \qquad [\eta_i, \eta_j] = [x_i, x_j] = 0.$$

Proof. Tensor the bounded free resolutions (8.1). Their endomorphism algebra identifies with the tensor product of the individual endomorphism algebras, with the Koszul signs. The tensor of the strict maps in Theorem 8.1 is a quasi-isomorphism from $T^{\otimes d}$. It gives the required dg Morita comparison.

Use the tensor product of the relative two-sided bars as a resolution of its diagonal bimodule. The tensor of their bar diagonals gives the cup product. The shuffle and Alexander–Whitney comparisons with the bar of $T^{\otimes d}$ identify the resulting derived endomorphism algebras. They also identify insertions up to coherent homotopy. This is a comparison of resolutions, not an equality of their terms.

At each multiweight, the cochain spaces are finite-dimensional and bounded. Hence tensor Hom identifies this model with the tensor of the corresponding binary cochain blocks. Their contractions tensor: for two factors the homotopy is $h_1 \otimes 1 + i_1 p_1 \otimes h_2$, with the graded tensor sign. Iteration gives all d factors.

Taking products over multiweights gives precisely the formal power series in the statement. The multiplicative maps of Theorem 8.2 tensor to an associative quasi-isomorphism to this diagonal model. The insertion comparison makes the bracket the sum of the factor brackets. Operations in distinct factors have zero mixed bracket. □

The theorem applies to a normal slice of two independent pair collisions, such as $12|34$, with constant tangential coefficients. The exterior coefficient complex contains all four normal localizations as shifted direct summands and generates this category. A global cut open can carry additional tangential cohomology. At a triple collision the three normal equations $u, v, u - v$ are dependent, so the preceding independent-direction hypothesis fails. Its shape provides a different algebra.

8.5 The shape of a triple collision

On the distinct three-point configuration, use coordinates

$$z_3 = z, \qquad z_1 = z + u, \qquad z_2 = z + ut.$$

They identify it with $\mathbb{A}^1_z \times \mathbb{G}_{m,u} \times V$, where

$$V = \operatorname{Spec} k[t, t^{-1}, (t-1)^{-1}].$$

The scale is u. The two punctures of V retain the two finite pair-collision channels within that scale.

Let

$$B = k \oplus k\epsilon_0 \oplus k\epsilon_1, \qquad |\epsilon_i| = 1, \qquad \epsilon_i \epsilon_j = 0,$$

with zero differential. The map $\epsilon_0 \mapsto dt/t$, $\epsilon_1 \mapsto dt/(t-1)$ is a strict multiplicative quasi-isomorphism $B \to \Omega^\bullet(V)$. Indeed partial fractions integrate every polynomial term and every pole of order at least two. The two simple residues are independent, and the horizontal functions are constants. Products of the one-forms vanish on the curve.

Thus B models the endomorphisms of the trivial differential module on V, and $\Lambda(\rho) \otimes B$, with $\rho = du/u$, models its product with the scale direction. The ordinary de Rham dimensions of the latter are 1, 3, 2.

Theorem 8.6. *Let $F = k\langle\langle x, y\rangle\rangle$ be the completed free associative algebra, with its word-length topology. The normalized full Hochschild cochains of B identify with the dg algebra*

$$F \xrightarrow{d} F \oplus F, \qquad df = ([x, f], [y, f]), \tag{8.8}$$

in degrees zero and one. The degree-one ideal has square zero, and its left and right actions are componentwise multiplication by F. In particular

$$H^0C(B) = k, \qquad H^1C(B) = \frac{F \oplus F}{\{([x,f],[y,f]) : f \in F\}}, \qquad H^jC(B) = 0 \quad (j \neq 0, 1).$$

The word-length m component of H^1 has dimension 2 for $m = 0$, 4 for $m = 1$, and $3 \cdot 2^{m-1}$ for $m \geq 2$.

Proof. Both suspended augmentation generators have degree zero. A cochain with scalar output is a formal series in their words, in total degree zero. Outputs in $k\epsilon_0 \oplus k\epsilon_1$ give two such series, in total degree one. All interior faces vanish, because the augmentation ideal is square zero. With the conventions of (8.3), the endpoint faces give the two commutators in (8.8). The cup product concatenates words and multiplies outputs, giving exactly the asserted algebra.

A homogeneous word polynomial commuting with x is a polynomial in x. To see this, the relation $xf = fx$ first excludes every word ending in y. Cycling a final x to the beginning then excludes every word containing y. Commutation with y leaves only constants. Apply this argument in each length to obtain $Z(F) = k$.

At output length m, the differential comes from scalar words of length $m - 1$. Its rank is zero for $m = 1$, and 2^{m-1} for $m \geq 2$. There is no source for $m = 0$. Subtracting these ranks from $2 \cdot 2^m$ gives the dimensions. The image is the product of its finite-dimensional length components, so it is closed. The quotient and cohomology are therefore compatible with the stated completion. □

This computes the separated shape corner. It does not identify the full triple coefficient category with the category generated by that one corner.

8.6 Three-point resonance

For $n \geq 2$, $c \in k$, and numerical charges $q_1, \dots, q_n \in k$, consider the rank-one connection on the distinct configuration

$$\nabla_n = d + c \sum_{i<j} q_i q_j \, d\log(z_i - z_j).$$

Normalize two positions and write every difference as u times a shape function. Its radial residue is

$$\lambda_n = c \sum_{i<j} q_i q_j.$$

The radial differential on u^m is multiplication by $m + \lambda_n$, followed by du/u. If $\lambda_n \notin \mathbb{Z} \subset k$, division by these scalars contracts every Laurent mode. The global algebraic de Rham complex is then acyclic.

If $\lambda_n \in \mathbb{Z}$, its radial cohomology has representatives $u^{-\lambda_n}$ and $u^{-\lambda_n} du/u$. Tensoring with the shape complex computes the remaining cohomology. The translation coordinate contracts to constants by polynomial integration.

Proposition 8.7. *Take $q_1 = q_2 = q_3 = 1$ and $c = 1/3$. Every two-point global algebraic de Rham complex is acyclic, but the three-point complex has*

$$H^0 = 0, \qquad H^1 = k\left[u^{-1}\frac{dt}{t}\right], \qquad H^2 = k\left[u^{-1}\frac{du}{u} \wedge \frac{dt}{t}\right],$$

and no other cohomology.

Proof. Each pair has radial residue $1/3$, while the triple has residue one. In the scale and shape coordinates its normal connection is

$$d + \frac{du}{u} + \frac{1}{3}\left(\frac{dt}{t} + \frac{dt}{t-1}\right).$$

It remains to compute the shape operator

$$D_{\mathrm{sh}} f = f' + \left(\frac{1}{3t} + \frac{1}{3(t-1)}\right) f$$

on $k[t, t^{-1}, (t-1)^{-1}]$. Every r has a decomposition

$$r = D_{\mathrm{sh}} f + \frac{c(r)}{t}. \tag{8.9}$$

For a pole at $p = 0, 1$ of order $j \geq 2$ and leading coefficient a, subtract

$$D_{\mathrm{sh}}\left(\frac{a(t-p)^{1-j}}{1/3-(j-1)}\right).$$

This lowers the pole order at p and introduces at most a simple pole at the other point.

After eliminating higher poles, remove a polynomial leading term at^j by subtracting $D_{\mathrm{sh}}(at^{j+1}/(j+1+2/3))$. The remainder is $A_0/t + A_1/(t-1)$. Subtracting $D_{\mathrm{sh}}(3A_1)$ leaves $(A_0 - A_1)/t$. All denominators used are nonzero, and every reduction terminates. For instance,

$$\frac{1}{t^2(t-1)^2} = D_{\mathrm{sh}}\left(\frac{3-6t^2}{2t(t-1)}\right) + \frac{2}{t}.$$

There is no nonzero rational horizontal function: its order at zero would have to be $-1/3$. Also $1/t$ is not in the image of D_{sh}. A pole of order $m \geq 1$ of a proposed primitive at either puncture would produce a higher pole with coefficient $-m + 1/3 \neq 0$. The primitive must therefore be polynomial. Degree $j \geq 1$ produces a nonzero polynomial part with coefficient $j + 2/3$, so the primitive must be constant. A constant produces equal coefficients at the two simple poles, and cannot produce $1/t$.

The decomposition (8.9) thus defines a contraction of the shape complex onto $k\,dt/t$ in degree one: its homotopy sends $r\,dt$ to the unique f. The radial homotopy sends $u^m\,du/u$ to $u^m/(m+1)$ for $m \neq -1$, and sends the remaining mode to zero. Tensor these two contractions with the graded tensor signs. The retained radial modes u^{-1}, $u^{-1}du/u$ give exactly the two classes in the statement. □

The dependence on the three residues can be determined without restricting them to products of charges. Let k be a field of characteristic zero, let $a_{12}, a_{13}, a_{23} \in k$, and set $\lambda = a_{12} + a_{13} + a_{23}$. On the ordered configuration of three points of $\mathbb{A}^1_k$, take the rank-one connection

$$\nabla = d + \sum_{i<j} a_{ij}\, d\log(z_i - z_j).$$

All functions and differential forms below are algebraic, with finite Laurent support.

Proposition 8.8. *The dimensions of the global de Rham cohomology are*

	$\dim_k H^0$	$\dim_k H^1$	$\dim_k H^2$
$\lambda \notin \mathbb{Z}$	0	0	0
$\lambda \in \mathbb{Z},\ (a_{12}, a_{13}, a_{23}) \notin \mathbb{Z}^3$	0	1	1
$(a_{12}, a_{13}, a_{23}) \in \mathbb{Z}^3$	1	3	2

Here $\mathbb{Z}$ denotes its image in k. All other cohomology groups vanish.

Proof. Put $z_1 = z + u$, $z_2 = z + ut$, $z_3 = z$. The coordinate ring is $k[z, u, u^{-1}, t, t^{-1}, (t-1)^{-1}]$, and the connection becomes

$$d + \lambda\frac{du}{u} + a_{23}\frac{dt}{t} + a_{12}\frac{dt}{t-1}.$$

Polynomial integration contracts the translation variable. The scale differential on u^m is multiplication by $m + \lambda$. It is contractible unless λ is integral, in which case it has one class in each of degrees zero and one.

Write $a = a_{23}$, $b = a_{12}$. The shape differential is $Df = (f' + (a/t + b/(t-1))f)\,dt$ on $B = k[t, t^{-1}, (t-1)^{-1}]$. Choose an integer $M \geq 1$ such that none of $a - j$, $b - j$, or $j + a + b$ vanishes for an integer $j > M$. Let $V_M \subset B$ consist of functions with poles of order at most M at zero and one and polynomial part of degree at most M. Let $W_M \subset B\,dt$ consist of forms with pole orders at most $M + 1$ and polynomial part of degree at most $M - 1$. Then $D(V_M) \subset W_M$, and $\dim V_M = 3M + 1$, $\dim W_M = 3M + 2$.

This finite subcomplex computes the shape cohomology. Indeed, a pole of order $j \geq M + 2$ is removed by the derivative of a function of order $j - 1$, since its leading coefficient is $a - (j-1)$ or $b - (j-1)$. Such a subtraction introduces at most a simple pole at the other puncture. A polynomial leading term of degree $j \geq M$ is removed using t^{j+1}, whose leading coefficient is $j + 1 + a + b$. These reductions terminate.

Conversely, if $Df \in W_M$, an excessive pole or polynomial degree of f would give an excessive order in Df with one of these nonzero coefficients. Thus $f \in V_M$, which also proves injectivity on cohomology.

A nonzero rational horizontal function has orders $-a$ and $-b$ at zero and one, so both residues must be integral. When they are, $t^{-a}(t-1)^{-b}$ is a horizontal function and spans the kernel. The ratio of any two horizontal functions is constant. The dimension difference of the finite complex therefore gives shape dimensions $(1, 2)$ when $a, b \in \mathbb{Z}$, and $(0, 1)$ otherwise. Tensoring with the two retained scale classes proves the table. For integral λ, integrality of a_{12} and a_{23} is equivalent to integrality of all three residues. □

Thus pairwise global de Rham cohomology cannot replace the full coefficient diagram: it would discard this ternary class. The example concerns global cohomology of the displayed connection. It does not contradict coefficient-level factorization with its actual maps.

8.7 The center of the puncture category

Return to the regular module and the r distinct closed coefficients of Theorem 7.9. Their algebra A_r has identities $e_0, \dots, e_r$ and degree-one arrows $\alpha_i : 0 \to i$, $\beta_i : i \to 0$. Every product of two arrows is zero. Write $O = e_0$ for the regular object and $\mathcal{L}$ for the finite twisted complex representing localization away from all r punctures: $\mathcal{L} = (\bigoplus_{i=0}^{r} e_i, d_{\mathcal{L}} = \sum_i \beta_i)$, as in Theorem 7.10.

Put $S = \bigoplus_{i=0}^{r} k e_i$. The normalized S-relative two-sided bar of A_r is a resolution of its diagonal bimodule. Indeed the unnormalized bar contracts by unit insertion, and separability of S permits normalization. Write $C(A_r) = \mathrm{Hom}_{A_r^e}(\mathrm{Bar}_S(A_r), A_r)$ with the full product totalization. It carries its cup product and bar insertion operations.

Let $Q_r^\vee$ have the same vertices and one reversed dual arrow for each α_i, β_i. Its path-length completion is $\widehat{kQ_r^\vee}$. Define

$$\mathcal{P} = \prod_{i=0}^{r} e_i \widehat{kQ_r^\vee} e_i, \qquad \mathcal{V} = \mathrm{Der}_S^{\mathrm{cont}}(\widehat{kQ_r^\vee}).$$

A derivation in $\mathcal{V}$ is specified by a formal path series with the same endpoints as each arrow. These paths have positive odd length, so every such assignment is continuous.

Theorem 8.9. *The full cochain center of* $\mathrm{Perf}(A_r)$ *is the two-term complex*

$$\mathcal{P} \xrightarrow{\mathrm{Inn}} \mathcal{V}, \qquad \mathrm{Inn}(p)(x) = xp - px, \tag{8.10}$$

in degrees zero and one. For $r \geq 2$, its cohomology is

$$\mathrm{HH}^0(A_r) = k, \qquad \mathrm{HH}^1(A_r) = \prod_{m \geq 0} V_{r,m}, \qquad \mathrm{HH}^q(A_r) = 0 \quad (q \geq 2),$$

where $V_{r,m}$ is the quotient of the block of arrow-output cochains on words of length $2m + 1$, and

$$\dim V_{r,0} = r, \qquad \dim V_{r,m} = 2(r-1)r^m \quad (m \geq 1). \tag{8.11}$$

The cup product of two degree-one classes is zero. With the negative-transpose identification of arrow-output cochains with suspended derivations, the Gerstenhaber bracket is the commutator of derivations modulo inner derivations.

Proof. Every radical input has degree one, hence every suspended input has degree zero. Scalar outputs occur on closed even words and have total degree zero. Arrow outputs occur on odd words with matching endpoints and have total degree one. There are no other outputs.

The radical products vanish, so every interior face of the Hochschild differential vanishes. Its endpoint faces are precisely $xp - px$, with the totalization convention of (8.3). This proves (8.10). Cup products concatenate words and multiply outputs. Two arrow outputs multiply to zero. Inserting cochains into cochains gives the stated derivation commutator with the chosen transpose sign.

At positive even input length $2m$, the closed-path space has dimension $2r^m$. There are r^m paths from the central vertex and r^m in total from the leaves. The next odd block has dimension $2r^{m+1}$. At length zero the dimensions are $r + 1$ and $2r$.

The block differential is explicit. For an odd path, deleting the first edge gives the closed-path column with coefficient $+1$. Deleting the last gives the column with coefficient -1. A nonclosed remainder contributes zero.

A kernel element is central in the completed path algebra. Its central-vertex component commutes with the r free return loops. For $r \geq 2$, a homogeneous word polynomial commuting with two distinct free generators is scalar. Commutation with the first restricts it to powers of that generator. Commutation with the second removes every positive power.

Apply this at each word length. The arrow equations force the same scalar at every leaf. Thus the kernel has dimension one at length zero and vanishes at every positive length. Subtracting the ranks gives (8.11). The differential acts independently on each finite block, so its image is the product of their images and is closed. Its cokernel is exactly the displayed product. □

Order the paths and perform finite elimination in each block of (8.10). Choose pivot columns for the image and extend them to a basis of the target. Inverting the differential on its pivot complement constructs a continuous contraction. Split off the common identity in length zero. These choices give $dh + hd = 1 - i\pi$ with the usual side conditions, coefficientwise in path length.

Choosing the quotient representatives accordingly gives an associative dg model

$$k \oplus \left(\prod_{m \geq 0} V_{r,m} \right)[-1]$$

with zero differential and square-zero positive ideal. Its full higher center operations remain on the bimodule bar. For $r = 1$, the two cyclic placements of the unique return loop instead give the logarithmic center $k[[z]] \otimes \Lambda(\eta)$ of Theorem 8.2, with $[\eta, z] = -z$.

Restriction to the full subcategory on e_0, e_i gives an actual map $C(A_r) \to C(A_1)$. For $r \geq 2$, its degree-zero map is the inclusion $k \to k[[z]]$, and its degree-one map is onto. To see the latter, extend each arrow-output cochain on those two vertices by zero on all other words. All degree-one cochains are closed, so it remains a closed lift before taking the quotient.

The marked objects also receive this action. Extend a cochain to finite twisted complexes by inserting their twisting differentials between the matrix entries of its inputs, including the two ends. For $\mathcal{L}$, the twisting arrows go from a leaf to the central vertex, so two consecutive insertions cannot compose. There is at most one insertion in each gap. The extension is therefore finite at every arity and defines the action on $O, \mathcal{L}$.

The bar identities give its differential and higher composition identities. This construction uses the specified full dg subcategory and its twisted complexes, including the ternary operation in (7.10).

8.8 A scale and two punctures

On the chart $z_1 - z_3 = u \neq 0$, Theorem 7.12 identifies the category generated by the primitive coefficient with $\mathrm{Perf}(E \otimes_k A_2)$, where $E = \Lambda(\rho)$, $|\rho| = 1$, and $d\rho = 0$. Here ρ is represented by du/u. The shape coordinate $t = (z_2 - z_3)/u$ retains its collisions at 0 and 1. We now compute this category's center and an actual deformation in its degree-two direction.

Put $V_2 = \prod_{m \geq 0} V_{2,m}$. Its block dimensions are $2, 4, 8, 16, \ldots$. The scale Hochschild complex is $E[[x]]$, with $|x| = 0$. Tensor the two bimodule bars at finite multiweight and then take the product over both weights. Their blockwise contractions give the center cohomology on this chart:

$$\begin{aligned} H^0 &= k[[x]], \\ H^1 &= k[[x]]\rho \,\oplus\, k[[x]]\widehat{\otimes}V_2, \\ H^2 &= k[[x]]\rho\widehat{\otimes}V_2. \end{aligned} \tag{8.12}$$

Here the completed tensor product is the product of the finite multiweight tensor factors.

The tensor bars retain the products and higher center operations. The chart includes the two shape collisions $t = 0, 1$. On its overlap with the second scale chart, the coordinates are $t' = 1/t$, $u' = ut$. Extending the category through $u = 0$ requires the remaining attachment and comparison maps on that chart.

8.8.1 A product deformation

Define even derivations D_u of E and D_t of A_2 by

$$D_u(\rho) = \rho, \qquad D_t(\alpha_1) = \alpha_1,$$

with D_t zero on the other arrows. Both kill the identities. Extend them to their separate factors of $E \otimes A_2$, whose product μ uses the Koszul sign. Their commuting additive weights determine the associativity equation.

Proposition 8.10. *Over $k[[\varepsilon]]$, the product*

$$\mu_\varepsilon = \mu \exp(\varepsilon D_u \otimes D_t)$$

is associative and unital, with zero differential and the same idempotents. Its first-order Hochschild class is nonzero in $\mathrm{HH}^2(E \otimes A_2)$.

Proof. Every formal coefficient is a defined bilinear map. On weight eigenvectors the product acquires the factor $\exp(\varepsilon w_u(a) w_t(b))$. Additivity of the two weights gives

$$w_u(a)w_t(b) + (w_u(a) + w_u(b))w_t(c) = w_u(b)w_t(c) + w_u(a)(w_t(b) + w_t(c)),$$

which proves associativity, including the original Koszul signs. The identities have weight zero.

The first-order cocycle is the cup product of the two derivation classes. The scale class is $\rho x \neq 0$ in $\mathrm{HH}^1(E)$. The shape class is nonzero in $V_{2,0}$: an inner diagonal derivation scales the two directions of an edge oppositely, whereas D_t scales only α_1. The finite-multiweight Künneth calculation used in (8.12) makes their tensor cup nonzero. □

For the odd elements $X = \rho \otimes 1$ and $Y = 1 \otimes \alpha_1$, the change is visible as

$$X *_\varepsilon Y + Y *_\varepsilon X = (e^\varepsilon - 1)\rho \otimes \alpha_1.$$

A realization by native chiral fields must in addition preserve their collision products, vacuum maps and higher substitutions. The categorical center computed here represents the bimodule endomorphism problem on its stated coefficient category. Its comparison with the full pointed transverse-action problem requires those additional maps.

8.9 Transport of Hochschild operations

The path substitutions of Theorem 7.22 act on the same completed reverse path algebra $\widehat{P}$. Retain $\mathcal{P} = \prod_{i=0}^{r} e_i \widehat{P} e_i$, and let $\mathrm{Der}_S^{\mathrm{cont}}(\widehat{P})$ denote continuous derivations fixing S. Full normalized Hochschild cochains have only total degrees zero and one: closed even paths have identity outputs, and odd paths have arrow outputs. The only differential terms are the two endpoint actions, since products of radical arrows vanish. Their model is

$$C^{\bullet}(A_r, A_r) = [\mathcal{P} \longrightarrow \mathrm{Der}_S^{\mathrm{cont}}(\widehat{P})], \qquad f \longmapsto (a \longmapsto af - fa),$$

with product totalization in path length.

For its Lie structure one shifts Hochschild cochains by one. Changing the sign on the resulting degree -1 term gives the convention

$$\mathfrak{g}_r^{-1} = \mathcal{P}, \qquad \mathfrak{g}_r^{0} = \mathrm{Der}_S^{\mathrm{cont}}(\widehat{P}), \qquad df = -\operatorname{ad}_f .$$

The bracket of two derivations is their commutator, $[\partial, f] = \partial(f)$, and the bracket of two degree -1 elements is zero.

Proposition 8.11. *Let $\Phi = \sigma_i^2$ be a pure substitution and $\mathcal{L} = \log \Phi$, as in Lemma 7.25. Then Φ acts on this dg Lie complex by*

$$f \longmapsto \Phi(f), \qquad \partial \longmapsto \Phi \partial \Phi^{-1}.$$

Its infinitesimal action is $\mathcal{L}$ on $\mathcal{P}$ and $[\mathcal{L}, -]$ on derivations. The corresponding logarithmic connection is defined on the same completion.

Proof. Conjugation takes ad_f to $\operatorname{ad}_{\Phi(f)}$, preserves commutators, and intertwines the evaluation action on $\mathcal{P}$. It therefore preserves the differential and all brackets. Differentiating Φ^u gives the infinitesimal formulas. A relative derivation sends an arrow to positive odd lengths, whereas $\mathcal{L}$ raises length by at least two. Its commutator therefore raises the length filtration on derivations, proving continuity and convergence. □

The full unit-retaining bar carries the Hochschild cup and brace operations. Its transport is obtained from the actual strictly unital higher automorphism and its associated bimodule bar. The two-term model above specifies the dg Lie action. Strict cup invariance of an arbitrary smaller presentation is an additional condition.

This transport acts on the coefficient category and its Hochschild operations. Its identification with a field-theoretic transport would require maps from the ordered collision coefficients which preserve both vacuum insertions and every collision substitution. The relative collision calculation below will supply a normalization that such maps must preserve.

8.10 An analytic coefficient dual for the regular Heisenberg bar

The two-sided bar above computes diagonal-bimodule endomorphisms. We next use an augmented bar to construct a spectral coefficient dual. A nonzero Heisenberg charge requires a coefficient space on

which translation and the collision connection still act. Completing only the charge coalgebra does not address the positive modes. We construct a coefficient dual which retains these modes and admits every complex charge. At fixed positive energy, the bar homotopy uses finitely many mergers, with a bound independent of the charge degree.

Throughout this section the ground field is $\mathbb{C}$, the level κ is nonzero, and degrees are cohomological. Consider the augmented commutative dg algebra

$$A_\kappa = \mathbb{C}[x_n \mid n \geq 0] \otimes \Lambda(\xi_n \mid n \geq 0), \qquad |x_n| = 0, \quad |\xi_n| = 1, \quad dx_n = \kappa(n+1)\xi_{n+1}, \quad d\xi_n = 0. \tag{8.13}$$

Its augmentation kills the generators. The multiplication is the displayed graded-commutative polynomial multiplication. Assign energy $n+1$ to x_n, and energy n to ξ_n. The differential preserves energy, and ξ_0 is the only generator of energy zero.

8.10.1 Growth in the charge direction

Let $\Gamma(f_0)$ have divided-power basis $\gamma_n(f_0)$, $n \geq 0$, with $\Delta\gamma_n(f_0) = \sum_{i+j=n} \gamma_i(f_0) \otimes \gamma_j(f_0)$. For an integer $R \geq 1$, set

$$G_R = \left\{ a = (a_n)_{n\geq 0} : |a|_R := \sup_n |a_n| R^{-n} < \infty \right\}, \qquad G = \varinjlim_R G_R.$$

The limit has the locally convex inductive-limit topology. A sequence represents $\sum_n a_n \gamma_n(f_0)$. We call G the analytic moment space. For several factors we use arrays bounded by $R^{n_1+\cdots+n_r}$ for some R, with the same inductive-limit convention.

Lemma 8.12. *The continuous linear dual of G is the space $\mathcal{E} = \mathcal{O}(\mathbb{C})$ of entire functions. The pairing and its defining seminorms are*

$$\left\langle \sum_n b_n t^n, a \right\rangle = \sum_n b_n a_n, \qquad \|b\|_R = \sum_n |b_n| R^n \quad (R \geq 1). \tag{8.14}$$

Under this pairing, the coproduct $(\Delta a)_{ij} = a_{i+j}$ is dual to multiplication of entire functions. The sequence $a_n = q^n$, for any $q \in \mathbb{C}$, represents evaluation at q.

Proof. The series in (8.14) is absolutely convergent on G_R, with absolute value at most $\|b\|_R |a|_R$. Conversely, let λ be continuous on G, and let $b_n = \lambda(\gamma_n(f_0))$. Continuity on G_R, applied to finite sequences with independently chosen phases and absolute values R^n, gives $\sum_n |b_n| R^n < \infty$. For $a \in G_R$, its finite truncations converge to a in G_S whenever $S > R$.

Consequently $\lambda(a) = \sum_n b_n a_n$. No additional functional on the tails occurs. The coproduct assertion is the Cauchy product, and $|q^n| \leq R^n$ for $R \geq \max(1, |q|)$ proves the last assertion. □

Let $\overline{A}_\kappa$ be the augmentation ideal. In the normalized bar $BA_\kappa = \bigoplus_{p\geq 0} (s\overline{A}_\kappa)^{\otimes p}$, suspension has degree -1: $|[a]| = |a| - 1$. We use the differential $D = d_0 + b$, where

$$d_0[a] = -[da], \qquad b[a|a'] = (-1)^{|a|}[aa'].$$

Extension to a word uses the Koszul sign of the suspended prefix. The coproduct is deconcatenation, including the two empty prefixes.

Fix the usual monomial basis of A_κ, with the odd factors in increasing index order. Write $\mathcal{W}_{E,n}$ for the bar words of energy E in which ξ_0 occurs a total of n times. Both differentials preserve E and n.

Definition 8.13. For each $E \geq 0$, let

$$B_E^{\mathrm{an}} = \varinjlim_{R \geq 1} \left\{ (a_w)_{w \in \coprod_n \mathcal{W}_{E,n}} : \sup_{n, w \in \mathcal{W}_{E,n}} |a_w| R^{-n} < \infty \right\}, \qquad B^{\mathrm{an}} = \prod_{E \geq 0} B_E^{\mathrm{an}}.$$

Each step carries the displayed supremum norm. The product is taken in locally convex spaces, degree by degree. For r coefficient factors, the corresponding target is the product over $(E_1, \dots, E_r)$ of arrays indexed by $\mathcal{W}_{E_1,n_1} \times \cdots \times \mathcal{W}_{E_r,n_r}$ and bounded by $R^{n_1+\cdots+n_r}$ for some R.

At energy zero this definition gives $B_0^{\mathrm{an}} = G$, since the only word is $[\xi_0 | \cdots | \xi_0]$. At positive energy the number of nonzero-energy inputs is at most E. If a word has length p, then

$$n \leq p \leq n + E, \qquad |\mathcal{W}_{E,n}| \leq C_E (n + E + 1)^E \tag{8.15}$$

for a constant C_E. Indeed, each positive-energy input can contain ξ_0 at most once. There are finitely many possible positive monomials at energy at most E, and at most E places among $n + E$ positions to insert them. These observations also bound the cohomological degrees at each fixed E.

8.10.2 The completed bar comparison

Define the dg coalgebra

$$C_\kappa = \Lambda(e_n \mid n \geq 0) \otimes \Gamma(f_n \mid n \geq 0), \qquad |e_n| = -1, \quad |f_n| = 0, \quad de_n = -\kappa(n+1) f_{n+1}, \quad df_n = 0.$$

The generators are primitive. The differential is a derivation for the divided-power product. Give e_n, f_n energies $n + 1$, n, respectively.

Let C_E^+ be the energy-E part generated by e_n and f_n, $n \geq 1$ for the latter. It is finite dimensional. Put

$$C_\kappa^{\mathrm{an}} = \prod_{E \geq 0} (G \otimes C_E^+).$$

Use the moment-array coefficient powers defined above for its coproduct. Finite dimensionality makes the tensor topology in each factor unambiguous.

On a fixed-energy dual we use the seminorms $\sum_{n,w} |b_w| R^n$, where w ranges over the chosen basis at charge degree n. These seminorms specify its Fréchet topology.

Theorem 8.14. *The differentials and coproducts extend continuously to B^{an} and C_κ^{an}. The signed shuffle map extends to a continuous dg coalgebra map*

$$\iota : C_\kappa^{\mathrm{an}} \longrightarrow B^{\mathrm{an}}, \qquad e_n \longmapsto [x_n], \quad f_n \longmapsto [\xi_n]. \tag{8.16}$$

It is a quasi-isomorphism. Both complexes admit continuous contractions onto their common energy-zero subcomplex G.

Topologize their continuous duals by the locally convex direct sum of the fixed-energy duals. Then ι induces a continuous quasi-isomorphism of dg algebras

$$D_\kappa^{\mathrm{an}} := (B^{\mathrm{an}})'_{\mathrm{cont}} \longrightarrow Y_\kappa^{\mathrm{an}} := \boldsymbol{O}(\mathbb{C}_{B_0}) \otimes \mathbb{C}[B_n \mid n \geq 1] \otimes \Lambda(C_n \mid n \geq 0), \tag{8.17}$$

where $|B_n| = 0$, $|C_n| = 1$, and

$$dB_n = \kappa n C_{n-1} \quad (n \geq 1), \qquad dB_0 = dC_n = 0. \tag{8.18}$$

These are the generators of Section 5.1, with $d = Q$. The energy of B_n is n, and that of C_n is $n + 1$. Multiplication is continuous on each pair of finite-energy stages.

Proof. The matrices of d_0 and b on $\mathcal{W}_{E,n}$ have polynomial growth in n. Their entries are bounded in terms of E and κ. The number of entries is polynomially bounded by (8.15). A polynomial times R^n is bounded by a constant times S^n for every $S > R$. Thus both operators extend continuously at fixed energy, and hence to the product.

A coordinate of the deconcatenation coproduct is the coordinate of the concatenated word. The moment growth bound is unchanged, proving continuity into the stated coefficient powers. The same argument applies to C_κ^{an}.

We construct the bar contraction directly. Let $\pi : A_\kappa \to \Lambda(\xi_0)$ kill the positive-energy generators, and let i be the inclusion. On a monomial, let N count all factors among x_n and ξ_{n+1}, with multiplicity. The derivation

$$h_A = \sum_{n\ge 0} \frac{x_n}{\kappa(n+1)} \frac{\partial}{\partial \xi_{n+1}}$$

has degree -1 and satisfies $dh_A + h_A d = N$. Define $H_A = h_A/N$ on $N > 0$, and zero on $\Lambda(\xi_0)$. Then $dH_A + H_A d = 1 - i\pi$, $H_A^2 = 0$, and H_A preserves energy and the number of occurrences of ξ_0.

On the tensor bar with differential d_0, the tensor contraction H applies $-sH_A s^{-1}$ to the first positive-energy input. It is zero if there is no such input. All preceding inputs are $[\xi_0]$, of degree zero, so this description includes the prefix sign. If I, P are the induced inclusion and projection of $B\Lambda(\xi_0)$, then

$$d_0 H + H d_0 = 1 - IP, \quad H^2 = 0, \quad HI = PH = 0, \qquad bI = Pb = 0.$$

The operator bH lowers bar length by one and preserves E, n. Equation (8.15) therefore gives $(bH)^{E+1} = 0$ on the entire energy-E part, independently of n. Consequently the finite formula

$$H'_E = H \sum_{j=0}^{E} (-bH)^j \tag{8.19}$$

satisfies $DH'_E + H'_E D = 1 - IP$.

To check the identity at $E > 0$, put $U = 1 + bH$. The relations above give $d_0 U = U d_0 - b$. Hence

$$DHU^{-1} + HU^{-1}D = 1 + HU^{-1}b(1 - U^{-1}) = 1,$$

since $1 - U^{-1} = bHU^{-1}$ and $b^2 = 0$. At $E = 0$, both sides of the contraction identity vanish.

For fixed E, the matrices in (8.19) are bounded by $K_{E,\kappa}(n + E + 1)^{s_E}$, for some integer s_E. This follows from the finite sum, the bounded entries of H_A, and (8.15). They therefore carry every growth step R continuously into every larger step S. The contractions extend to B^{an}.

On C_κ^{an}, the contracting pairs (e_n, f_{n+1}) give a contraction on each finite space C_E^+. At positive energy it is obtained by dividing the usual Koszul homotopy by the number of positive generators. Tensoring with G and taking the product preserves these identities.

In (8.16), $\gamma_n(f_0)$ maps to the word of n copies of ξ_0, with no factorial. More generally, a basis monomial maps to the signed sum of its distinct shuffles. Exchanging two e's reverses the sign. The adjacent multiplication terms cancel by graded commutativity, and the internal differential has the sign $de_n = -\kappa(n+1)f_{n+1}$. Thus this is a dg coalgebra map.

At fixed positive energy the shuffle coefficients have polynomial growth in n, so the map extends continuously. On energy zero it is the identity, and both positive-energy complements contract. This proves the quasi-isomorphism.

A continuous functional on a product of locally convex spaces depends on finitely many coordinates: a neighborhood controlling it leaves all remaining coordinates unrestricted. At each energy the proof of Lemma 8.12 identifies the dual with coefficient families whose weighted absolute sums converge for every R. The polynomial bound in (8.15) does not affect this condition. Transposing the contractions gives continuous contractions of these Fréchet spaces, and then of their direct sums. In particular, transposition preserves the quasi-isomorphism here by an explicit homotopy, without an exactness assumption on continuous duality.

For C_κ^{an}, its dual is exactly the right side of (8.17). The convolution product is polynomial in the positive modes and the ordinary entire-function product in B_0. The dual differential gives (8.18). The estimate $\|f\,g\|_R \le$

$\|f\|_R\|g\|_R$, together with the finite number of positive monomials at fixed energy, proves the last continuity assertion. The identical weighted-sum estimate proves it for the bar dual. □

The bar contraction satisfies a sharper continuity statement. At fixed energy it preserves each charge-growth radius separately. Write $\mathcal{B}_{E,R}$ for the full space of coefficient arrays on $\coprod_n \mathcal{W}_{E,n}$ with norm

$$\|v\|_{E,R} = \sup_{n,w\in\mathcal{W}_{E,n}} |v_w|R^{-n}, \qquad R \geq 1.$$

This is the bounded-array space in Definition 8.13, not the norm closure of the finite arrays.

Theorem 8.15. *For $E \geq 1$, the operator H'_E of (8.19) extends to a bounded degree-minus-one operator on $\mathcal{B}_{E,R}$ with*

$$DH'_E + H'_E D = 1, \qquad \|H'_E\| \leq \frac{1}{|\kappa|}\sum_{j=0}^{E}\left(\frac{E2^{E+1}}{|\kappa|}\right)^j.$$

The radius R is unchanged. At energy zero the differential vanishes on all arrays of words in $[\xi_0]$, including the empty word. No bound uniform in E is asserted.

Proof. We bound incoming row sums in the bar basis: fix an output word and sum absolute coefficients over its possible inputs. For one output monomial $x^a\xi_S$, an incoming internal differential replaces x_j by ξ_{j+1} and requires $j+1 \in S$. Its coefficient has magnitude $|\kappa|(j+1)(a_j+1)$. Their sum is at most

$$|\kappa|\left(\sum_j (j+1)a_j + \sum_{s\in S\setminus\{0\}} s\right).$$

Summing over the word gives $\|d_0\| \leq |\kappa|E$.

For H, fix the first positive-energy output input, with positive factor count N. An incoming term replaces one of its x_j by ξ_{j+1}. It requires $a_j > 0$ and $j+1 \notin S$. Its coefficient has magnitude $1/(|\kappa|(j+1)N)$. There are at most N eligible indices, so $\|H\| \leq 1/|\kappa|$.

For b, an incoming term splits one output monomial into an ordered pair of nonconstant monomials. The output must have positive energy, because ξ_0 has no such nonzero factorization. There are at most E eligible output inputs. If one has polynomial exponents a_j, r positive-energy odd factors, and possibly ξ_0, the number of ordered factorizations is at most

$$\prod_j (a_j+1)\, 2^{r+\mathbf{1}_{0\in S}} \leq 2^{\sum_j a_j + r + \mathbf{1}_{0\in S}} \leq 2^{E+1}.$$

This count includes forbidden constant factors and therefore remains an upper bound after normalization. Each contribution has magnitude one, giving $\|b\| \leq E2^{E+1}$.

All these matrices preserve n and have finite rows. The weights R^{-n} therefore cancel, so the bounds hold on the full bounded-array space without a density argument. The algebraic contraction identities extend coefficientwise. Since bH lowers length and $n \leq p \leq n+E$, $(bH)^{E+1} = 0$ on this entire space. The finite formula (8.19) gives the asserted identity and norm bound by submultiplicativity. □

For multilinear operations on these duals, use the bornology defined as follows. A bounded set lies in finitely many energy degrees and is bounded for every entire-function seminorm there. Multiplication is bounded for this bornology.

Define the r-factor target $(Y_\kappa^{\mathrm{an}})^{(r)}$ as a direct sum over nonnegative energy indices. Each summand is $O(\mathbb{C}^r)$ tensored with the corresponding finite space of positive monomials. Its seminorms on the entire coefficients are the weighted absolute sums with radii $(R_1, \dots, R_r)$. This specifies the coefficient tensor powers used below.

Corollary 8.16. *One has*

$$H^*(Y_\kappa^{\mathrm{an}}) = \mathcal{O}(\mathbb{C}_{B_0}) \quad \text{in degree zero.}$$

The inclusion of the polynomial algebra Y_κ from Section 5.1 into Y_κ^{an} is not a quasi-isomorphism.

Proof. The degree -1 derivation sending C_n to $B_{n+1}/(\kappa(n+1))$ and killing all B_n has commutator with d equal to the number of positive generators. Divide by this number off the charge subalgebra. This is a continuous contraction at each energy. On cohomology the inclusion is $\mathbb{C}[B_0] \hookrightarrow \mathcal{O}(\mathbb{C}_{B_0})$, which is not surjective because e^{B_0} is not polynomial. □

8.10.3 Spectral operations and charge transport

Define derivations T, L of degrees $0, -1$ by

$$T g_n = n g_{n-1} \quad (g_n = B_n \text{ or } C_n), \qquad LC_n = B_n/\kappa, \qquad LB_n = 0, \tag{8.20}$$

with $T g_0 = 0$. These agree with (5.1). Both kill entire functions of B_0. They lower energy by one whenever they are nonzero. They are continuous and satisfy $dL + Ld = T$. Thus $\tau_z = \exp(zT)$ is a finite sum on every element, for every $z \in \mathbb{C}$.

Define a continuous algebra map $\Delta_z : Y_\kappa^{\mathrm{an}} \to (Y_\kappa^{\mathrm{an}})^{(2)}$ by

$$\Delta_z(g_n) = \tau_z(g_n) \otimes 1 + 1 \otimes g_n, \qquad \Delta_z(f(B_0)) = f(B_0 \otimes 1 + 1 \otimes B_0). \tag{8.21}$$

The counit evaluates B_0 at zero and kills the positive modes. Continuity of the second formula follows from $\|f(t_1 + t_2)\|_{R_1,R_2} \le \|f\|_{R_1+R_2}$.

Let

$$c_{nm}(z) = (-1)^n \binom{n+m}{n} z^{-n-m-1}, \qquad r(z) = \sum_{n,m \ge 0} c_{nm}(z)(C_n \otimes B_m - B_n \otimes C_m).$$

This degree-one tensor belongs to $(Y_\kappa^{\mathrm{an}})^{(2)}[[z^{-1}]]$, with finitely many terms in each coefficient. Translations of this series are formal Taylor translations.

Interpret identities involving several positions as rational jet identities. The coefficients $c_{nm}(z)$ are the Taylor coefficients in u, v of $(z + u - v)^{-1}$. On three positions the coefficient ring is

$$\mathbb{C}[z_{12}, z_{23}, z_{12}^{-1}, z_{23}^{-1}, (z_{12} + z_{23})^{-1}][[u_1, u_2, u_3]].$$

Expansion in a chosen position domain is made only after this rational identity. Numerical convergence requires a further positive-mode completion, such as the one in Theorem 7.4.

Proposition 8.17. *The maps Δ_z, τ_z, the counit and $r(z)$ satisfy the following spectral identities on the specified coefficient powers:*

$$dr = 0, \qquad (T \otimes 1)r = \partial_z r, \qquad (1 \otimes T)r = -\partial_z r,$$
$$(\Delta_w \otimes 1)\Delta_z = (1 \otimes \Delta_z)\Delta_{z+w},$$
$$(\Delta_w \otimes 1)r(z) + r_{12}(w) = (1 \otimes \Delta_z)r(z+w) + r_{23}(z).$$

Moreover, the degree-zero tensor

$$X(z) = (L \otimes 1)r(z) = \frac{1}{\kappa} \sum_{n,m \ge 0} c_{nm}(z) B_n \otimes B_m \tag{8.22}$$

satisfies $dX = \partial_z r$. With exterior forms written on the left, $r + dz\,X$ satisfies the Maurer–Cartan equation for $d + d_z$. On three distinct positions, so does $\sum_{i<j}(r_{ij}(z_{ij}) + dz_{ij}X_{ij}(z_{ij}))$, where $z_{ij} = z_i - z_j$.

Proof. The derivations in (8.20) are checked on generators. Their extensions to entire coefficients follow from their vanishing on B_0. The same argument proves that Δ_z is a cochain map. Both coproduct iterates send g_n to

$$\tau_{z+w}(g_n) \otimes 1 \otimes 1 + 1 \otimes \tau_z(g_n) \otimes 1 + 1 \otimes 1 \otimes g_n,$$

and send $f(B_0)$ to $f(B_0^{(1)} + B_0^{(2)} + B_0^{(3)})$. This proves coassociativity on the whole carrier.

The coefficient identities

$$(n+1)c_{n+1,m} = \partial_z c_{nm}, \qquad (m+1)c_{n,m+1} = -\partial_z c_{nm}$$

prove the translation formulas. The coefficient of $C_n \otimes C_m$ in dr is $-\kappa((n+1)c_{n+1,m} + (m+1)c_{n,m+1}) = 0$. Primitivity gives

$$(\Delta_w \otimes 1)r(z) = r_{13}(z+w) + r_{23}(z), \qquad (1 \otimes \Delta_z)r(z+w) = r_{12}(w) + r_{13}(z+w),$$

in the rational jet domains. The remaining counit and graded-flip identities follow on generators from (8.21) and $c_{nm}(-z) = -c_{mn}(z)$. These arguments give the same identities on entire coefficients by the displayed seminorm estimate.

Finally $[d, L] = T$ and $dr = 0$ give $dX = (T \otimes 1)r = \partial_z r$. In the tensor product with exterior forms, $d(dz\,X) = -dz\,dX$. Thus $(d + d_z)(r + dz\,X) = 0$. The coefficient algebra is graded commutative, so the square of this odd element vanishes. The same calculation applies to the sum on three positions: each differential vanishes and all graded commutators vanish. Every calculation is coefficientwise in the specified formal domains. □

With the sign convention of (5.5), the charge residue is positive. The positive-mode terms of this superconnection can be removed over the formal punctured neighborhood of infinity. The remaining charge connection is retained by the following explicit comparison.

Proposition 8.18. *Put $\mathscr{Y} = (Y_\kappa^{\mathrm{an}})^{(2)}((z^{-1}))$, meaning Laurent series in z^{-1} with coefficients in the indicated coefficient space and a finite upper bound on powers of z. Each fixed-bound space has the product topology. Their union has the inductive-limit topology. Define*

$$b(z) = -\frac{1}{\kappa} \sum_{\substack{n,m \geq 0 \\ n+m>0}} \frac{(-1)^n}{n+m} \binom{n+m}{n} B_n \otimes B_m\, z^{-n-m}, \qquad g(z) = e^{b(z)}. \tag{8.23}$$

These are well-defined formal series, with $g = 1 + O(z^{-1})$.

Writing $D = d + d_z$, multiplication by g is an isomorphism of complexes from

$$\bigl(\mathscr{Y} \otimes \Lambda(dz),\ D + r + dz\,X\bigr) \quad \text{to} \quad \left(\mathscr{Y} \otimes \Lambda(dz),\ D + \frac{B_0 \otimes B_0}{\kappa}\frac{dz}{z}\right).$$

It is continuous in the coefficientwise formal topology. Projection p to energy zero is a quasi-isomorphism from the first complex to the following complex, where $t_i = B_0^{(i)}$ are the charge coordinates:

$$\left(O(\mathbb{C}^2)((z^{-1})) \otimes \Lambda(dz),\ d_z + \frac{t_1 t_2}{\kappa}\frac{dz}{z}\right). \tag{8.24}$$

A homotopy inverse is $f \mapsto e^{-b(z)}f$.

Proof. Write $r = \sum_{j\geq 1} r_j z^{-j}$, $X = \sum_{j\geq 1} X_j z^{-j}$, and $b = \sum_{j\geq 1} b_j z^{-j}$. Proposition 8.17 gives $dX_{j+1} = -jr_j$. Formula (8.23) says $b_j = -X_{j+1}/j$, and therefore

$$db = r, \qquad \partial_z b = X - X_1 z^{-1}, \qquad X_1 = \kappa^{-1} B_0 \otimes B_0.$$

The exponential and its inverse exist because b has strictly positive z^{-1}-order. Each coefficient uses finitely many products. Those products are continuous on fixed-energy stages.

In fact each coefficient of g is a polynomial in finitely many positive modes and the two charge variables. Multiplication by it is continuous on the whole locally convex direct sum. This proves coefficientwise continuity of multiplication by g and g^{-1}. Graded commutativity now gives

$$g^{-1}(D + dz\, X_1 z^{-1})\, g = D + dz\, X_1 z^{-1} + Db = D + r + dz\, X.$$

Sum the two Koszul homotopies of Corollary 8.16 and divide by the total number of positive generators away from energy zero. This contraction commutes with multiplication by $B_0 \otimes B_0$. Extend it to forms with the Koszul sign and to Laurent coefficients term by term. It then contracts the second complex onto (8.24). Conjugating this contraction by g gives the asserted contraction of the first complex. Since $p(b) = 0$, one has $pg = p$, and its inclusion is precisely multiplication by e^{-b} on energy zero. □

Example 8.19 (Nonzero charges)**.** For every $q \in \mathbb{C}$, the formula

$$\chi_q(f(B_0)) = f(q), \qquad \chi_q(B_n) = 0\ (n \geq 1), \qquad \chi_q(C_n) = 0\ (n \geq 0)$$

defines a continuous dg algebra character. On a pair of charges q, q', the superconnection of Proposition 8.17 becomes

$$\nabla_{q,q'} = d_z + \frac{qq'}{\kappa}\frac{dz}{z}.$$

Its horizontal sections on a logarithm branch are scalar multiples of $z^{-qq'/\kappa}$. Counterclockwise full transport has scalar $e^{-2\pi i qq'/\kappa}$, and the half-turn along $z(t) = z_0 e^{\pi i t}$, with the logarithm continued along this path, has scalar $e^{-\pi i qq'/\kappa}$. On three charges the horizontal factor is $\prod_{i<j} z_{ij}^{-q_i q_j/\kappa}$, with a consistent choice of logarithms on the universal cover of the configuration space.

These characters do not commute with translation when $q \neq 0$: $\chi_q(TB_1) = q$, whereas $T\chi_q(B_1) = 0$. Consequently a fixed nonzero charge cannot replace the full algebra in the spectral coproduct identities. The positive modes in (8.17) are necessary even though their cohomology vanishes.

The construction above supplies a continuous coefficient dual of the specified completion of the regular augmented algebra (8.13). A comparison with the state-dependent bar of the nonaugmented Heisenberg transverse monoid must preserve the chosen state and the two-position collision. It must also preserve all three partitions of three positions. In particular, the ordinary augmentation of A_κ is not an augmentation of that transverse monoid. Theorem 8.14 constructs the coefficient dual on the regular polynomial carrier. This comparison with the transverse monoid remains a separate mathematical question.

8.11 Multiplication from commuting resolutions

The action in (6.10) resolves the boundary by one quartet. To compose two actions, use two quartets. This gives strict products with different targets. The maps that merge those targets are quasi-isomorphisms, and their compatibility supplies the coherent product.

Work over a field k of characteristic zero, on $X = \mathbb{A}^1_k$, with $\kappa \in k^\times$. All state spaces in this section are algebraic. Retain the full ordered collision atlas and its category $\mathrm{Coalg}^{\mathrm{f}}$ from Section 6.4. Its diagonal

tensor is $\otimes^!$, with vacuum unit J. The symbol J also denotes the Heisenberg current when it occurs inside a vertex algebra.

Write $\mathcal{Z} = \mathcal{Z}_\kappa$ for the cochain vertex algebra of Section 6.4.2, and write E_{q} for its quartet algebra. Thus

$$[J_\lambda J] = \kappa\lambda, \qquad dJ = \kappa T c, \quad dc = 0, \qquad [p_\lambda t] = [\iota_\lambda \eta] = 1, \quad Qt = \eta, \quad Q\iota = p.$$

The degrees of t, p, η, ι are 0, 0, 1, -1.

For $b \geq 1$ put

$$W_b = V_\kappa \otimes E_{\mathrm{q}}^{\otimes b}, \qquad K_b = J + \sum_{j=1}^{b} p_j, \qquad i_b(J) = K_b.$$

The differential Q fixes J and acts on each quartet as above. Replacing J by K_b in the creation-mode basis leaves b contractible quartets. Their Euler contraction, as in Section 6.4.2, proves that each i_b is a quasi-isomorphism. Let $\mathcal{Z}^{(b)}$ be the commutant of $i_b(V_\kappa)$ in W_b, namely the common kernel of the nonnegative modes of K_b.

If $\phi : [b'] \twoheadrightarrow [b]$ is an order-preserving surjection, let $\phi_*(j) = \min \phi^{-1}(j)$. Define a merge $m_\phi : W_b \to W_{b'}$ by fixing J and putting

$$p_j \longmapsto \sum_{\phi(l)=j} p_l, \qquad \iota_j \longmapsto \sum_{\phi(l)=j} \iota_l, \qquad t_j \longmapsto t_{\phi_*(j)}, \qquad \eta_j \longmapsto \eta_{\phi_*(j)}. \tag{8.25}$$

For $a \geq 1$ the shift $r_a : W_b \to W_{a+b}$ sends J to K_a and sends a quartet with index j to the quartet with index $a + j$. Include W_a by its first a quartets. The candidate product is

$$\mu_{a,b} : \mathcal{Z}^{(a)} \otimes W_b \longrightarrow W_{a+b}, \qquad z \otimes w \longmapsto z_{(-1)} r_a(w). \tag{8.26}$$

Proposition 8.20. *The merges are quasi-isomorphisms of dg vertex algebras, satisfy $m_\phi i_b = i_{b'}$ and compose contravariantly in ϕ. They restrict to quasi-isomorphisms $\mathcal{Z}^{(b)} \to \mathcal{Z}^{(b')}$. The maps (8.26) are dg vertex maps, preserve the commutants, and are strictly associative and natural under concatenation of merges.*

The algebra $\mathcal{Z}^{(b)}$ is freely generated by

$$U_1 = J - \kappa T t_1, \qquad t_j - t_1 \ (2 \leq j \leq b), \qquad p_j, \eta_j, \iota_j \ (1 \leq j \leq b).$$

The map $j_b : \mathcal{Z} \to \mathcal{Z}^{(b)}$ with $j_b(J) = U_1$ and $j_b(c) = -\eta_1$ is a quasi-isomorphism. The iterated product of b cochain factors, together with its boundary action, sends their generators to

$$J^{(i)} \longmapsto U_i = J + \sum_{l<i} p_l - \kappa T t_i, \qquad c^{(i)} \longmapsto -\eta_i, \qquad J_{\mathrm{boundary}} \longmapsto K_b. \tag{8.27}$$

Proof. In (8.25), the sum of the p_l in a fiber pairs with its selected $t_{\phi_*(j)}$ through exactly one term. The same holds for ι and η. All other brackets are preserved, as are Q and T. The fibers partition $[b']$, so $m_\phi(K_b) = K_{b'}$.

The least point over the least point is the least point of a composite fiber. This proves composition. The identity $m_\phi i_b = i_{b'}$ then proves that m_ϕ is a quasi-isomorphism.

For the commutant, write $x_n = J_{-n}\mathbf{1}$ for $n \geq 1$ and $\tau_n^{(j)} = T^n t_j / n!$ for $n \geq 0$. On the creation-mode basis the relevant modes act by

$$(K_b)_0 = \sum_j \partial_{\tau_0^{(j)}}, \qquad (K_b)_n = \kappa n \partial_{x_n} + \sum_j \partial_{\tau_n^{(j)}} \quad (n \geq 1).$$

Replace x_n by $x_n - \kappa n \tau_n^{(1)}$ and each $\tau_n^{(j)}$, $j > 1$, by $\tau_n^{(j)} - \tau_n^{(1)}$. The displayed modes become the derivatives in the remaining $\tau_n^{(1)}$. Their common kernel has precisely the generators in the statement. The change is invertible over $k[T]$, so there are no additional relations.

On these generators,

$$QU_1 = -\kappa T \eta_1, \quad Q(t_j - t_1) = \eta_j - \eta_1, \quad Q\iota_j = p_j.$$

Thus its complex of creation states is that of $\mathcal{Z}$, with $b-1$ contractible pairs of differences and b pairs (ι_j, p_j) adjoined. All states are finite polynomials. Dividing the Euler contraction by positive pair degree contracts the added pairs. This proves the assertion about j_b.

Since every monotone surjection has $\phi_*(1) = 1$, one has $m_\phi j_b = j_{b'}$. Its restriction to the commutants is therefore a quasi-isomorphism.

The current K_a has level κ, is closed, and commutes with the later quartets. Hence r_a is a dg vertex map. The images of $\mathcal{Z}^{(a)}$ and $r_a(W_b)$ commute. Their normally ordered product consequently defines a map of the tensor product vertex algebra. Both images of the commutants commute with K_{a+b}.

The identities $r_a r_b = r_{a+b}$ and $m_{\phi\sqcup\psi} r_a = r_{a'} m_\psi$ prove associativity and naturality. Applying the successive shifts to j_1 gives (8.27). □

The $b+1$ currents $U_1, \dots, U_b, K_b$ commute with one another and each has level κ. The space spanned by J, p_j, Tt_j has the form with J of norm κ and each pair p_j, Tt_j hyperbolic. Thus each additional commuting current uses one additional quartet.

Lemma 8.21. *Let $\mathcal{P}$ have the nonempty finite ordered sets $[b]$ as objects and monotone surjections as morphisms. Let $\mathcal{D}$ be a symmetric monoidal infinity category. Suppose $Z : \mathcal{P}^{\mathrm{op}} \to \mathcal{D}$ is lax monoidal for concatenation, and $M : \mathcal{P}^{\mathrm{op}} \to \mathcal{D}$ has natural associative maps $Z(a) \otimes M(b) \to M(a+b)$. If both functors carry every arrow to an equivalence, then $Z(1)$ is a nonunital associative algebra and $M(1)$ its nonunital left module. Their operations are the roofs*

$$Z(1)^{\otimes n} \longrightarrow Z(n) \xleftarrow{\sim} Z(1), \qquad Z(1) \otimes M(1) \longrightarrow M(2) \xleftarrow{\sim} M(1).$$

Proof. Encode the pair of functors by the two-color operad for a nonunital associative algebra and a left module. Each color carries an index in $\mathcal{P}^{\mathrm{op}}$. Substitution concatenates these indices. Passing to symmetric monoidal envelopes makes the given data a monoidal functor [80, Proposition 2.2.4.9].

The map of envelopes that forgets the indices is a cocartesian fibration. Its fibers are finite products of $\mathcal{P}^{\mathrm{op}}$. Mark every fiber arrow, and every arrow over a base equivalence. Concatenation preserves these markings. Localization over the base is therefore computed by localizing the fibers [57, Proposition 2.1.4].

The nerve of $\mathcal{P}^{\mathrm{op}}$ is contractible because $[1]$ is initial. Each localized fiber is consequently a point. The localized envelope is the envelope of the two-color operad. The original functor inverts the marked arrows, so it factors through this monoidal localization [80, Proposition 4.1.7.4]. This gives the algebra, module, and all their coherences. The unique surjection $[n] \twoheadrightarrow [1]$ gives the displayed roofs. □

Theorem 8.22. *The coalgebra $C_{\mathcal{Z}}$ has a unital associative multiplication for $\otimes^!$, with its vacuum as unit. Its product of b factors is induced by the roof*

$$\mathcal{Z}^{\otimes b} \longrightarrow \mathcal{Z}^{(b)} \xleftarrow{\sim} \mathcal{Z}^{(1)} \xleftarrow{j_1} \mathcal{Z},$$

whose first map is (8.27) and whose middle map merges all quartets. It has a unital coherent action on C_{V_κ} with underlying map (6.10).

Proof. Apply Lemma 6.7 to $Z(b) = C_{\mathcal{Z}^{(b)}}$ and $M(b) = C_{W_b}$. Proposition 8.20 supplies the hypotheses of Lemma 8.21. This first gives a nonunital algebra and module.

Write $m : W_1 \to W_2$ for the merge. Right multiplication by the vacuum on $\mathcal{Z}^{(1)}$ is the inclusion into $\mathcal{Z}^{(2)}$, which agrees with m after j_1. Left multiplication is r_1. Use polynomial forms on the interval, standing to the left of fields. The following two families join $m\,j_1$ to $r_1\,j_1$:

$$J \longmapsto J - \kappa T((1-s)t_1 + st_2), \qquad c \longmapsto -(1-s)\eta_1 - s\eta_2 - ds(t_2 - t_1), \tag{8.28}$$

$$J \longmapsto J + s\,p_1 + ds\,\iota_1 - \kappa T t_2, \qquad c \longmapsto -\eta_2. \tag{8.29}$$

Their images commute with K_2, have the brackets of $\mathcal{Z}$, and satisfy $dJ = \kappa T c$, $dc = 0$ for the total differential. For example, $d(s\,p_1 + ds\,\iota_1) = ds\,p_1 - ds\,p_1 = 0$. The first endpoint of the second family is the last endpoint of the first. Lemma 6.7 carries these polynomial homotopies to the coalgebra category. Since C_{j_1} is an equivalence, both multiplications by the vacuum are homotopic to the identity.

On the module the vacuum acts by $r_1 : W_1 \to W_2$. The equation $r_1 i_1 = i_2$ makes this a quasi-isomorphism. The algebra and module are therefore quasi-unital. The quasi-unit theorems give their unital structures, with a contractible space of choices [80, Theorem 5.4.3.5, Proposition 5.4.3.14]. Transport them along C_{j_1} and C_{i_1}.

For the underlying action, the identity

$$\mu_{1,1}(j_1(z) \otimes i_1(v)) = j_1(z)_{(-1)} i_2(v) = m\big(\rho(z \otimes v)\big)$$

identifies its roof with (6.10). Here m fixes $j_1(z)$, since it changes only p and ι in the first quartet. □

The tensor comparison used here is over the ground field, with derived tensor on the coefficient schemes. Ordinary tensor over a differential parameter algebra has a different behavior.

Example 8.23. Put $\Omega = k[s, ds]$ and consider regular commutative vertex algebras

$$A_+ = \Omega[a_n : n \geq 0], \qquad A_- = \Omega[b_n : n \geq 0], \qquad Da_n = ds\,a_n, \quad Db_n = -ds\,b_n.$$

All a_n, b_n have degree zero, their brackets vanish, $Ta_n = a_{n+1}$, $Tb_n = b_{n+1}$, and $T\Omega = 0$. Both augmentations to Ω are quasi-isomorphisms. On a monomial of positive degree m the coefficient differential is $\partial_s \pm m$, whose inverse on a polynomial is the terminating series

$$(\partial_s + c)^{-1} f = \sum_{n \geq 0} \frac{(-1)^n}{c^{n+1}} \partial_s^n f \quad (c \neq 0).$$

In $A_+ \otimes_\Omega A_-$, however, $a_0 b_0$ is closed and cannot be a boundary.

More precisely, give $B = k[a_i, b_j]$ charge 1 on the a_i and -1 on the b_j, and write B_0 for charge zero. The inclusion $\Omega \otimes_k B_0 \to A_+ \otimes_\Omega A_-$ is a quasi-isomorphism. Indeed the coefficient complex of a charge-c monomial has differential $\partial_s + c$. It is acyclic for $c \neq 0$ and has the constants as its degree-zero cohomology for $c = 0$.

The surviving algebra has the presentation

$$B_0 = k[z_{ij} : i, j \geq 0]/(z_{ij}z_{kl} - z_{il}z_{kj}), \qquad z_{ij} = a_i b_j, \qquad Tz_{ij} = z_{i+1,j} + z_{i,j+1}.$$

Every balanced monomial is a product of the z_{ij}. The displayed relations exchange column partners, so every monomial reduces to the pairing of its sorted row and column multisets. Distinct pairs of multisets give distinct monomials in the independent a_i, b_j. This proves the presentation, including for countably many variables, since every relation uses only finitely many. With r row indices and s column indices, the degree-q dimension is $\binom{r+q-1}{q}\binom{s+q-1}{q}$.

The inclusions $\Omega \to A_\pm$ are quasi-isomorphisms from a K-flat module. Hence the comparison from derived to ordinary tensor has degree-zero cohomology map $k \to B_0$, $1 \mapsto 1$. Neither A_+ nor A_- is K-flat over Ω: tensoring the other's augmentation with it would otherwise make the augmentation $B_0 \to k$ an isomorphism.

8.12 The endomorphisms of a moving vacuum

The cochain action has a universal property at one position. Its tests there are modules for the boundary current. These tests retain the zero mode and the connection as the position moves.

Let $\mathfrak{h}_\kappa$ have generators J_n, $n \in \mathbb{Z}$, and a central generator $\mathbf{c}$, with

$$[J_m, J_n] = \kappa m \partial_{m+n,0} \mathbf{c}.$$

A smooth module has $\mathbf{c} = 1$, and each of its vectors is killed by all sufficiently large J_n. The vacuum is $V_\kappa = k[x_1, x_2, \dots]$, where J_{-n} multiplies by x_n, $J_n = \kappa n \partial_{x_n}$ for $n > 0$, and $J_0 = 0$. Write $\mathfrak{b} = \bigoplus_{n \geq 0} k J_n$ and $\mathfrak{a}_p = \bigoplus_{n \geq p} k J_n$.

For a smooth module M, its continuous cochains are

$$C^\bullet(M) = M \otimes \Lambda(\xi_0, \xi_1, \dots), \qquad d(m\omega) = \sum_{n \geq 0} J_n m \, \xi_n \wedge \omega. \tag{8.30}$$

The sum is finite on each element. These cochains are the filtered union of the finite-dimensional Lie algebra cochains of $\mathfrak{b}/\mathfrak{a}_p$ on $M^{\mathfrak{a}_p}$.

Lemma 8.24. *The complex* (8.30) *computes* $R\,\mathrm{Hom}(V_\kappa, M)$ *in the category of smooth modules. In particular*

$$\mathrm{Ext}^*(V_\kappa, V_\kappa) = \Lambda(\xi_0).$$

The class ξ_0 *is represented by the nonsplit extension* $V_\kappa \otimes k[\epsilon]/(\epsilon^2)$ *on which* J_0 *acts by* ϵ.

Proof. The vacuum is induced from the trivial $\mathfrak{b}$-module, so its homomorphism functor is $M \mapsto M^{\mathfrak{b}}$. The cochain functors are exact. For $S_p = k[J_0, \dots, J_{p-1}]$, induction to smooth Heisenberg modules is exact by the creation-mode basis and is left adjoint to $M \mapsto M^{\mathfrak{a}_p}$.

Thus $Y^{\mathfrak{a}_p}$ is injective over S_p when Y is injective among smooth modules. Its finite Koszul cochains are acyclic in positive degrees. Filtered unions are exact, so the continuous cochains have the same property on Y. They therefore compute the derived invariant functor.

For the vacuum the differential is $\sum_{n \geq 1} \kappa n \xi_n \partial_{x_n}$. The degree-minus-one derivation $h(\xi_n) = x_n/(\kappa n)$ for $n \geq 1$, with $h(\xi_0) = h(x_n) = 0$, satisfies $dh + hd = N$, where N counts the positive-index generators. Dividing by N contracts positive degree and leaves $\Lambda(\xi_0)$. The proposed extension is nonsplit because J_0 vanishes on a split extension and does not vanish on this one. It spans the one-dimensional first extension group. Its square is zero because the second group vanishes. □

Allow the root y to move on X. A *module with a moving root* is a left $\mathcal{D}_X$-module W with a smooth, $\mathcal{O}_X$-linear Heisenberg action satisfying

$$[\nabla_{\partial_y}, J_n] = -n J_{n-1}. \tag{8.31}$$

For $\mathcal{V} = \mathcal{O}_X \otimes V_\kappa$ the connection is $\partial_y + T$. For any left $\mathcal{D}_X$-module N, the tensor $N \otimes_{\mathcal{O}_X} \mathcal{V}$ has its Leibniz connection and the modes acting on the second factor.

Define

$$\mathfrak{z}^\ell = \mathcal{O}_X \otimes C^\bullet(V_\kappa), \qquad T x_n = n x_{n+1}, \quad T\xi_n = (n+1)\xi_{n+1},$$

with connection $\partial_y + T$. The differential commutes with this connection. As a complex this is $\mathcal{Z}^\ell = \mathcal{O}_X \otimes \mathcal{Z}$: $J = x_1$, $c = \xi_0$, and $T^n c/n! = \xi_n$.

Theorem 8.25. *For every complex N of left $\mathcal{D}_X$-modules, the singleton action of Theorem 8.22 induces a natural equivalence*

$$R\operatorname{Hom}_{\mathcal{D}_X}(N, \mathcal{Z}^\ell) \simeq R\operatorname{Hom}_{\mathrm{mov}}(N \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{V}, \mathcal{V}). \tag{8.32}$$

The right-hand mapping complex is formed among modules with a moving root. The tensor with $\mathcal{V}$ is already exact. The derivation $K\xi_n = \kappa^{-1}x_{n+1}$, $Kx_n = 0$, satisfies $dK + Kd = T$. Hence the cohomology of $\mathcal{Z}^\ell$ is $\mathcal{O}_X \otimes \Lambda(\xi_0)$ with trivial connection.

Proof. The invariant module $W^{\mathfrak{b}}$ is preserved by the connection, by (8.31). Induction from the vacuum gives an adjunction

$$\operatorname{Hom}_{\mathrm{mov}}(N \otimes_{\mathcal{O}_X} \mathcal{V}, W) = \operatorname{Hom}_{\mathcal{D}_X}(N, W^{\mathfrak{b}}).$$

Its left adjoint is exact, so the right adjoint preserves injectives. The continuous cochains compute its derived functor.

To verify this last assertion with the connection present, put $A_\kappa = U(\mathfrak{h}_\kappa)/(\mathbf{c} - 1)$. The algebra obtained by adjoining y, D with

$$[D, y] = 1, \quad [y, J_n] = 0, \quad [D, J_n] = -nJ_{n-1}$$

is free as a right A_κ-module on $y^a D^r$. The derivation of the modes preserves their relations, since its value on the relation for J_m, J_n is $-\kappa m(m+n-1)\delta_{m+n-1,0} = 0$. Induction to this algebra is exact and preserves smoothness:

$$J_n D^r = \sum_{j=0}^{r} \binom{r}{j} n(n-1)\cdots(n-j+1) D^{r-j} J_{n-j}.$$

If all J_n with $n \geq p$ kill a vector, all J_n with $n \geq p + r$ kill its induced D^r-multiple.

Thus forgetting the moving connection preserves injectives. Lemma 8.24 proves cochain acyclicity on them, also as differential modules. The first-quadrant double complex of an injective resolution now identifies $R\Gamma_{\mathfrak{b}}\mathcal{V}$ with $\mathfrak{z}^\ell$. Derived adjunction gives (8.32) for all N, with derived mapping complexes understood via injective resolutions and their product totalizations.

It remains to identify this adjunction with the actual singleton action. Put $\mathcal{W} = \mathcal{O}_X \otimes W_1$ and let the current act by the modes of K_1. The action is the roof of i_1 and $z \otimes v \mapsto j_1(z)_{(-1)} i_1(v)$. Under the underived adjunction this map is $z \mapsto j_1(z)$.

The cochains of W_1 have differential $Q + (:cK_1:)_{(0)}$. Its two maps from $\mathcal{Z}$ are

$$f_0 : J \mapsto K_1, c \mapsto c, \qquad f_1 : J \mapsto J - \kappa T t, c \mapsto -\eta.$$

The map fixing J, c is joined to them by

$$\begin{aligned} J &\longmapsto J + sp + ds\,\iota, & c &\longmapsto c, \\ J &\longmapsto J - s\kappa T t, & c &\longmapsto (1-s)c - s\eta - ds\,t. \end{aligned}$$

Both are dg vertex maps.

Here the cochain differential sends t to $\eta + c$, sends J to $\kappa T c$, and kills p, η, c. Its value on ι is p. Integrating the coefficient of ds gives a chain homotopy commuting with T.

The inclusion of the commutant into these cochains is a quasi-isomorphism. In the coordinates of Proposition 8.20, the modes of K_1 differentiate the variables $\tau_n^{(1)}$. Their continuous cochains contract the polynomial de Rham pairs. This contraction is valid for the total complex: give J, p, ι weight one, t, η weight zero, and let T raise weight.

At each weight only finitely many positive-index ghosts can occur, with at most one zero-index ghost. The cochain-degree filtration is therefore finite there. Taking the union of weights preserves the equivalence. The maps f_0, f_1 now show that the singleton roof corresponds to the identity of $\mathfrak{z}^\ell$ under derived adjunction.

Finally $dK + Kd = T$ follows on x_n and ξ_n from $dx_n = \kappa n \xi_n$ and $d\xi_n = 0$. Both sides are derivations, so the equality holds on all states. □

The zero-mode class has disappeared if the allowed modules are required to have $J_0 = 0$. At a fixed nonzero charge μ, the cochain differential instead contains $\mu\xi_0\wedge$, contracted by $\mu^{-1}\partial_{\xi_0}$. Thus the extension class belongs to the stated category of smooth modules with unrestricted zero mode. Its singleton universal property does not assert a universal property on the full collision atlas.

8.13 The collision class of the product

The complex of cochains has only two cohomology classes, but its product distinguishes the input orders. To detect that order after applying the coalgebra functor, retain the homotopy between the binary map and its restriction to a cut. It has a scalar component in the relative de Rham cohomology of the normal punctured line.

8.13.1 A constant coefficient model

Let $\mathcal{K}$ be the regular commutative dg vertex algebra freely generated by h of degree zero and c of degree one, with $dh = Tc$, $dc = 0$. All its brackets vanish. Put $E = \Lambda(\varepsilon)$, where $|\varepsilon| = 1$, $T\varepsilon = 0$, and the differential is zero. The map $q : \mathcal{K} \to E$ sends c to ε and h to zero.

Proposition 8.26. *There is a zigzag of quasi-isomorphisms of dg vertex algebras*

$$\mathcal{Z} \xrightarrow{j_1} \mathcal{Z}^{(1)} \xleftarrow{g} \mathcal{K} \xrightarrow{q} E, \qquad g(h) = \kappa^{-1}(J - \kappa T t + \tfrac{1}{2}p), \quad g(c) = -\eta.$$

It induces equivalences of pointed factorizing coalgebras. No strict dg vertex map $\mathcal{K} \to \mathcal{Z}$ induces an isomorphism on cohomology.

Proof. Write $h_n = T^n h/n!$ and $c_n = T^n c/n!$. Then $dh_n = (n+1)c_{n+1}$. The Euler contraction on the pairs (h_n, c_{n+1}) leaves exactly $\Lambda(c_0)$, so q is a quasi-isomorphism.

Put $U = J - \kappa T t$. The brackets

$$[U_\lambda U] = \kappa\lambda, \qquad [p_\lambda U] = [U_\lambda p] = -\kappa\lambda$$

give $[(U + \frac{1}{2}p)_\lambda (U + \frac{1}{2}p)] = 0$. The image of c commutes with this field, and $Qg(h) = -T\eta = Tg(c)$. Both images commute with K_1. Thus g is a dg vertex map. It carries the surviving class $[c_0]$ to $[-\eta_1]$, so Proposition 8.20 makes it a quasi-isomorphism.

The residue construction preserves these equivalences. For j_1 and g this is Lemma 6.7. For the regular quotient q, use the finite partition filtration of the Cousin complex.

Each derived tensor coefficient preserves its quasi-isomorphism, and the residue maps are natural. Filtered comparison proves the result at every finite configuration, including collisions and all cuts. The same argument applies to the algebraic polynomial path resolutions used below. All maps preserve the vacuum.

Finally suppose $f : \mathcal{K} \to \mathcal{Z}$ is strict. Lemma 8.24 gives $f(c) = ac + dP$ with $P \in V_\kappa$ and $a \in k$. Since the closed degree-zero states of $\mathcal{Z}$ are constants,

$$f(h) = a\kappa^{-1}J + TP + n\mathbf{1} \qquad (n \in k).$$

The coefficient of $\lambda\mathbf{1}$ in $[f(h)_\lambda f(h)]$ is $a^2\kappa^{-1}$. The cross terms have zero such coefficient because $J_{(0)}P = 0$ and translation has no vacuum component. The term $[TP_\lambda TP] = -\lambda(\lambda + T)[P_\lambda P]$ also has zero coefficient there. But $[h_\lambda h] = 0$, so $a = 0$. Thus f kills the surviving ghost class. □

The exact correction $p/2 = Q(\iota/2)$ is essential to this comparison: it cancels the bracket of U with itself. The zigzag identifies the pointed object with C_E. Transport the tower product along this zigzag and denote it by $P_{\mathrm{t}} : C_E \otimes^! C_E \to C_E$. Also let P_{c} be the product induced by the ordinary exterior multiplication. These will have different relative classes.

8.13.2 The normal relative complex

Set $B = E \otimes E = \Lambda(a, b)$, with $a = \varepsilon \otimes 1$ and $b = 1 \otimes \varepsilon$. The tensor comparison identifies C_B with $C_E \otimes^! C_E$. Write $\epsilon_E : E \to k$ and $\epsilon_B : B \to k$ for their augmentations, and $\sigma : B \to B$ for exchange, so $\sigma(ab) = -ab$.

Choose the ordered chart $Y = X^2$ with coordinates (z, w), $u = z - w$, and $j : U = \{u \neq 0\} \hookrightarrow Y$. Put $J_Y = \omega_Y[2]$ with density $dz \wedge dw$ and $J_U = j^! J_Y$. Consider

$$\mathcal{R} = \mathrm{Fib}\big(R\,\mathrm{Hom}_{\mathcal{D}_Y}(J_Y, J_Y) \longrightarrow R\,\mathrm{Hom}_{\mathcal{D}_U}(J_U, J_U)\big). \tag{8.33}$$

All differential modules here are right modules. In global algebraic de Rham complexes its convention is

$$\mathcal{R}^n = \Omega_Y^n \oplus \Omega_U^{n-1}, \qquad D(\alpha, \beta) = (d\alpha,\, j^*\alpha - d\beta).$$

Lemma 8.27. *The only nonzero cohomology of $\mathcal{R}$ is*

$$H^2(\mathcal{R}) = k\gamma, \qquad \gamma = [(0, du/u)].$$

Thus a map $J_Y[-2] \to J_Y$ with a specified nullhomotopy on U has a scalar relative class. Its underlying map on Y has zero class.

Proof. Polynomial integration contracts the de Rham complex of Y to k. On $U = \mathbb{A}^1_w \times \mathbb{G}_{m,u}$, integration in w leaves $k[u, u^{-1}] \xrightarrow{\partial_u} k[u, u^{-1}]du$. Every monomial one-form except $u^{-1}du$ is a derivative. The two spaces of constants agree under restriction. The long exact sequence of the fiber gives the result. The absolute class lies in $H^2_{\mathrm{dR}}(Y) = 0$. □

In $B \otimes B$, with factors at positions z, w, define

$$v_1 = a \otimes b - 1 \otimes ab, \qquad v_2 = b \otimes a + 1 \otimes ab. \tag{8.34}$$

They have degree two and zero product. The constant vertex field is multiplication, so every Laurent multiple of v_i has zero collision residue. Hence they define maps $s_i : J_Y[-2] \to C_{B,2}|_Y$ in the Cousin complex.

For a map $P : C_B \to C_E$ under the vacuum, put $q_P = C_{\epsilon_E} P$. Its cut compatibility supplies a homotopy on U from $(q_P)_2$ to the external square of $(q_P)_1$. Choose a pointed singleton homotopy from $(q_P)_1$ to ϵ_B. Its external square, followed by the cut homotopy, gives a nullhomotopy of $(q_P)_2 s_i$ on U. Let $\lambda_i(P)\gamma$ be the resulting relative class.

Proposition 8.28. *The construction gives well-defined functions*

$$(\lambda_1, \lambda_2) : \pi_0\,\mathrm{Map}_{J/}(C_B, C_E) \longrightarrow k^2.$$

They are independent of the chosen singleton homotopy, and satisfy

$$\lambda(P_c) = (0, 0), \qquad \lambda(PC_\sigma) = (\lambda_2(P), \lambda_1(P)).$$

The domain retains both ordered cuts.

Proof. Spencer resolution and polynomial integration identify

$$R\operatorname{Hom}_{\mathcal{D}_X}(\mathcal{O}_X \otimes B, \mathcal{O}_X) \simeq \operatorname{Hom}_k(B, k).$$

Fixing the value on 1 leaves a connected mapping space, so the required singleton homotopy exists. Its loops have classes $\operatorname{Hom}_k(B^1, k) = k^2$. A representative h has degree minus one and can be nonzero only on a, b.

The first variation of its external square at ϵ_B is $h \boxtimes \epsilon_B + \epsilon_B \boxtimes h$, with tensor Koszul signs. Both terms vanish on v_i: the other odd input is killed by ϵ_B, and $h(ab) = 0$. Equivalently, represent the loop by sending a, b to multiples of ds. Its square vanishes because $ds^2 = 0$. The relative class therefore does not depend on this choice.

Evaluation on a chart and its cut preserves homotopies of full coalgebra maps. The relative class consequently depends only on the component of P. For P_c, q_P is the strict augmentation, which kills v_i with constant cut homotopy. Exchange sends v_1 to v_2 and v_2 to v_1, proving the last formula. □

8.13.3 Computing the class on a resolved map

We compute the invariant through an algebraic path resolution. Suppose a derived character $f : B \to k$ is presented by a roof $B \xleftarrow{\pi} S \xrightarrow{\varphi} k$, where π is a surjective quasi-isomorphism of dg vertex algebras. Put $I = \ker \pi$, an acyclic ideal. Choose closed lifts $\tilde{a}, \tilde{b} \in S^1$. Since $Ta = Tb = 0$, choose $y_a, y_b \in I^0$ with $Qy_a = T\tilde{a}$, $Qy_b = T\tilde{b}$.

The zero mode of y_a commutes with Q, because $(T\tilde{a})_{(0)} = 0$. Thus one can choose $x_1, x_2 \in I^0$ with

$$Qx_1 = (y_a)_{(0)}\tilde{b}, \qquad Qx_2 = (y_b)_{(0)}\tilde{a}. \tag{8.35}$$

The values $\varphi(x_i)$ do not depend on these choices. Changing a primitive by Qe adds a bracket whose image under φ is zero. Changing a closed lift by Qw changes its translation primitive by Tw, whose zero mode vanishes. Two remaining choices of x_i differ by a boundary in I.

Lemma 8.29. *If $C(f) = C_{\epsilon_E} P$, then*

$$\lambda(P) = (\varphi(x_1), \varphi(x_2)).$$

Proof. Write states to the left of differential forms. On two positions the Spencer differential is

$$\begin{aligned} \mathsf{D}(x \otimes y\,\zeta) &= Q(x \otimes y)\,\zeta \\ &\quad + (-1)^{|x|+|y|}\big(Tx \otimes y\,dz \wedge \zeta + x \otimes Ty\,dw \wedge \zeta + x \otimes y\,d\zeta\big). \end{aligned}$$

The two-position Cousin complex has open and diagonal summands with differential $(\sigma, \delta) \mapsto (Q\sigma, r_{12}\sigma - Q\delta)$. A closed Spencer element is therefore a pair (Σ, Γ) with $\mathsf{D}\Sigma = 0$ and $r_{12}\Sigma = \mathsf{D}\Gamma$. On constant coefficients its residue is

$$r_{12}(x \otimes y) = \sum_{n \geq 0} x_{(n)}y\,[u^{-n-1}], \qquad dz = du + dw.$$

Choose $y_{ab} \in I^1$ with $Qy_{ab} = -T(:\tilde{a}\tilde{b}:)$, and put $\Xi_{ab} =: \tilde{a}\tilde{b} : + y_{ab}\,dw$. The element

$$\begin{aligned} \Sigma_1 &= \tilde{a} \otimes \tilde{b} + \tilde{a} \otimes y_b\,dw - y_a \otimes \tilde{b}\,dz \\ &\quad + y_a \otimes y_b\,dz \wedge dw - 1 \otimes \Xi_{ab} \end{aligned}$$

is closed and lifts v_1. Its residue has coefficients in I, whose Spencer complex is acyclic. Choose Γ there with $\mathsf{D}\Gamma = r_{12}\Sigma_1$. Then (Σ_1, Γ) lifts s_1.

A singleton homotopy from φ to $\epsilon_B\pi$ gives, by its external square, a regular form β on Y with $\mathsf{D}\beta = \varphi\Sigma_1$. Its restriction to U is precisely the nullhomotopy in Proposition 8.28. Subtract the relative boundary of $((\beta, 0), 0)$. Since a regular scalar form has zero collision residue, the class becomes $((0, \varphi\Gamma), 0)$. Under this Cousin resolution, $\gamma = (0, du/u)$ is represented by $((0, -[u^{-1}]du), 0)$.

The principal-part de Rham complex is

$$[u^{-m}] \longmapsto -m[u^{-m-1}]\,du \qquad (m \geq 1),$$

tensored with the polynomial de Rham complex in w. Its first cohomology is one-dimensional, detected by the coefficient of $[u^{-1}]\,du$.

If $g \in I^{\circ}$ is that coefficient in Γ, the same coefficient in $\mathrm{D}\Gamma$ is Qg: differentiating a principal part cannot produce a simple pole. In $r_{12}\Sigma_1$ it is $-(y_a)_{(0)}\tilde{b}$. Consequently $g + x_1$ is closed in I and hence exact. The scalar residue of $\varphi\Gamma$ is $\varphi(g) = -\varphi(x_1)$.

The minus sign in the representative of γ gives $\lambda_1(P) = \varphi(x_1)$. Exchange a, b for the second assertion. The added vacuum term in v_2 has no residue. □

8.13.4 The value for every input order

For $1 \leq \alpha \leq n$ put

$$h_\alpha = \kappa^{-1}(U_\alpha + \tfrac{1}{2}p_\alpha), \qquad c_\alpha = -\eta_\alpha, \qquad g_{n,\alpha} : \mathcal{K} \to \mathcal{Z}^{(n)}, \quad (h, c) \longmapsto (h_\alpha, c_\alpha).$$

Each is a quasi-isomorphism by the same half-correction calculation as Proposition 8.26. For different indices their images commute. Hence their joint map is a dg vertex map $A_n : \mathcal{K}^{\otimes n} \to \mathcal{Z}^{(n)}$.

The maps $g_{n,\alpha}$ are polynomially homotopic. For $\alpha > 1$ put

$$\theta_\alpha = t_1 - t_\alpha, \qquad w_\alpha = \kappa^{-1}(\tfrac{1}{2}\iota_1 + \sum_{1<l<\alpha} \iota_l + \tfrac{1}{2}\iota_\alpha).$$

With parameter forms on the left, the homotopy is

$$h \longmapsto (1-s)h_1 + sh_\alpha + ds\, w_\alpha,$$
$$c \longmapsto (1-s)c_1 + sc_\alpha + ds\, \theta_\alpha.$$

Indeed $Q\theta_\alpha = c_\alpha - c_1$ and $Qw_\alpha = h_\alpha - h_1 - T\theta_\alpha$. The only potentially nonzero bracket between these images has ds coefficient $(1-s)/(2\kappa) + s/(2\kappa) - 1/(2\kappa) = 0$.

Transport the n-fold tower product to C_E and call it $P_{\mathrm{t},n}$. Its composite with augmentation is induced by the derived character

$$\epsilon_{\mathcal{K}} \circ g_{n,1}^{-1} A_n (q^{-1})^{\otimes n}. \tag{8.36}$$

To see this, include the first quartet by $\mathrm{in}_n : \mathcal{Z}^{(1)} \to \mathcal{Z}^{(n)}$. This inclusion and the merge agree on $j_1(\mathcal{Z})$, and $\mathrm{in}_n\, g = g_{n,1}$. The iterated product carries $g^{\otimes n}$ to A_n. Substituting these identities in the transported roof proves (8.36).

Theorem 8.30. *The tower product, its opposite, and the ordinary exterior product have respective relative classes*

$$\lambda(P_{\mathrm{t}}) = \Big(-\frac{1}{2\kappa}, \frac{1}{2\kappa}\Big), \qquad \lambda(P_{\mathrm{t}}C_\sigma) = \Big(\frac{1}{2\kappa}, -\frac{1}{2\kappa}\Big), \qquad \lambda(P_{\mathrm{c}}) = (0, 0).$$

They belong to three distinct components of $\mathrm{Map}_{J/}(C_E \otimes^{!} C_E, C_E)$. *More generally, the* $n!$ *permutations of the inputs of* $P_{\mathrm{t},n}$ *are pairwise nonhomotopic.*

Proof. Choose distinct α, β and restrict (8.36) to these two inputs. The preceding homotopy allows $g_{n,1}^{-1}$ to be replaced by $g_{n,\alpha}^{-1}$. We construct the roof needed in Lemma 8.29.

Let P consist of pairs (k_0, γ), where $k_0 \in \mathcal{K}$ and $\gamma \in \mathcal{Z}^{(n)} \otimes k[s, ds]$ satisfy $\gamma(0) = g_{n,\alpha}(k_0)$. Projection $P \to \mathcal{K}$ is a quasi-isomorphism, since the kernel of evaluation at 0 on polynomial forms is contracted by integration. Evaluation at 1 is surjective: $(0, sz)$ lifts every homogeneous z. It is a quasi-isomorphism because its homotopy to evaluation at 0 compares it with $g_{n,\alpha}$.

Take the pullback S with $\mathcal{K} \otimes \mathcal{K}$ along $g_{n,\alpha} \otimes g_{n,\beta}$. Then its projection, followed by $q \otimes q$, is a surjective quasi-isomorphism $\pi : S \to B$. The character is $\varphi(x, k_0, \gamma) = \epsilon_{\mathcal{K}}(k_0)$. The tautological path verifies that this roof presents the specified derived character.

Closed lifts and a translation primitive are

$$\tilde{a} = (c^{(1)}, c, c_\alpha), \qquad y_a = (h^{(1)}, h, h_\alpha), \qquad \tilde{b} = (c^{(2)}, c, \gamma_b),$$

where $\gamma_b = (1 - s)c_\alpha + sc_\beta + ds(t_\alpha - t_\beta)$. The element y_a lies in $\ker \pi$ and satisfies $Q\, y_a = T\, \tilde{a}$.

The first two components of $(y_a)_{(0)}\tilde{b}$ vanish. Its third component is $ds\, \mu$, where

$$\mu = (h_\alpha)_{(0)}(t_\alpha - t_\beta) = \kappa^{-1}\big(\tfrac{1}{2} - [\beta < \alpha]\big).$$

Here $[\beta < \alpha]$ equals one or zero according to the condition, and $(p_l)_{(0)} t_m = \delta_{lm}$. The element $x_1 = (0, -\mu, (s - 1)\mu)$ belongs to $\ker \pi$ and satisfies $Q x_1 = (y_a)_{(0)}\tilde{b}$. Thus $\varphi(x_1) = -\mu$.

Using the roof based on $g_{n,\beta}$ and exchanging the inputs gives the other coordinate. The pair is

$$\kappa^{-1}\big([\beta < \alpha] - \tfrac{1}{2}, [\alpha < \beta] - \tfrac{1}{2}\big). \tag{8.37}$$

Lemma 8.29 identifies it with the relative class.

For $n = 2$, the ordered positions are $\alpha = 1$, $\beta = 2$. Formula (8.37) and the exchange law prove the three values. They are distinct because the characteristic is zero and $\kappa \neq 0$. In any arity, insert these two inputs and vacua in all other factors. The resulting pair tells which of the two selected inputs comes first. The relative orders of all pairs determine the permutation, proving the last assertion. □

Proposition 8.31. *After restriction to pairwise distinct configurations, P_{t}, $P_{\mathrm{t}}C_\sigma$ and P_{c} are homotopic with all cuts and vacua.*

Proof. On the configuration open only the partition into singletons survives in the Cousin complex. For a pointed singleton left differential module M, this restriction is $J_I \otimes M^{\boxtimes I}|_{U_I}$, with the cut maps given by regrouping factors. This is a functor of the pointed derived singleton, since external tensor over k and open restriction preserve quasi-isomorphisms.

Polynomial de Rham contraction identifies the degree-zero singleton maps from B to E with $\mathrm{Hom}^0_k(B, E)$. All three products send a, b to ε, ab to zero, and 1 to 1. Their singleton maps are homotopic. Applying the external-power functor gives the asserted homotopy on every configuration open, with all its cuts and vacua. □

The half in (8.37) comes from the exact correction that made the constant model commute. The integer records the input order. Both are detected by the compatibility at a collision, although the products agree on the configuration opens. A further comparison with a universal action space must retain this compatibility.

8.14 A map invisible to natural actions

The next distinction concerns maps between actors. Their formal coordinate substitution does not record which root carries an odd output. We construct a map with zero substitution in every weight and a nonzero antisymmetric coefficient on two roots.

For a finite-dimensional vector space W put

$$E_W = \Lambda(W[-1]), \qquad F_W = \mathrm{CE}_\star(\mathfrak{l} \otimes E_W), \qquad A_W = U_\star(\mathfrak{l} \otimes E_W), \qquad \mathfrak{l} = L_k.$$

The exterior generators have degree one and zero translation. Write F_r when a basis identifies W with k^r. These are the constant exterior field coalgebras, with pointing from and augmentation to $J = F_0$. All actor maps constructed in this section preserve both. The suspension s has degree minus one, so a pure input $s(em)$, with $e \in W[-1]$ and $m \in \mathfrak{l}$, has degree zero.

8.14.1 Derivations of exterior coefficients

Let D be an endomorphism of a finite-dimensional space V, extended as an even derivation of E_V, of $\mathfrak{g}_V = \mathfrak{l} \otimes E_V$, and of A_V. It fixes $\mathfrak{l}$. Introduce another exterior generator ε, placed on the left. For homogeneous x, m,

$$[x, \varepsilon m] = (-1)^{|x|}\varepsilon[x, m], \qquad [\varepsilon m, \varepsilon n] = 0.$$

The tree set $\mathcal{T}_w$ used in (6.20) consists of nonnegative sequences $(a, b_1, \dots, b_w)$ with

$$a + \sum_{i=1}^{w} b_i = w, \qquad a + \sum_{i=1}^{j} b_i \geq j + 1 \quad (0 \leq j < w).$$

For homogeneous actor coefficients m_i, let $\chi(\pi; m)$ be their Koszul permutation sign and set

$$\theta_w^D(x; m_1, \dots, m_w) = \sum_{(a,b)\in\mathcal{T}_w} \sum_{\pi\in S_w} \frac{\chi(\pi; m)}{a! b_1! \cdots b_w!} D^a x\, D^{b_1} m_{\pi(1)} \cdots D^{b_w} m_{\pi(w)}. \tag{8.38}$$

Put $\theta_0^D(x) = x$. Define a twisting cochain to vanish except on words with one input from $s\mathfrak{g}_V$. Move that input first, with the sign of the suspended permutation, and apply θ_w^D to the remaining actor inputs. The two signs use different degrees: $|sx| = |x| - 1$, whereas $|s(\varepsilon m_i)| = |m_i|$.

Lemma 8.32. *This cochain defines a map $\alpha_D : F_{k\oplus V} \to F_V$ whose restriction to F_V is the identity. On pure inputs, for $z \in V[-1]$ and $n, m_i \in \mathfrak{l}$, its value is*

$$(D^w z)n \ \mathrm{pbw}(m_1 \cdots m_w), \qquad \mathrm{pbw}(m_1 \cdots m_w) = \frac{1}{w!} \sum_{\pi\in S_w} m_{\pi(1)} \cdots m_{\pi(w)}. \tag{8.39}$$

It vanishes on pure inputs with any other number of factors from V.

Proof. In the twisting equation a nonzero term has exactly two inputs from $s\mathfrak{g}_V$. A source bracket either joins these or joins one to an actor input. A product of two twisting values also needs exactly two such inputs. Every other component vanishes.

On the remaining components use the degree-zero tree identity proved in Theorem 6.13. For an input of coefficient degree $d \geq 0$, adjoin d distinct auxiliary exterior variables of degree minus one and multiply the input by their ordered product. Use disjoint variables for different inputs. The new coefficient inputs have degree zero, and D fixes the auxiliary algebra.

Apply the tree identity there and extract the product of all auxiliary variables. Permuting the actor inputs gives precisely $\chi(\pi; m)$, and moving the module input gives the suspended sign. The extracted monomial is nonzero. Thus the identity holds with $Q_2(sx\, sy) = (-1)^{|x|}s[x, y]$ and the convolution tensor sign $(-1)^{|c'|}$ in every degree.

Lemma 6.6 converts the twisting cochain to a coalgebra map. Every positive-weight tree has $a \geq 1$. Its first factor therefore lies in the positive exterior ideal, proving compatibility with augmentation. On the vacuum D vanishes, proving compatibility with the pointing.

Positivity makes the bar sum finite at each configuration. For (8.39), D kills all m_i, so only $(a, b) = (w, 0, \dots, 0)$ survives. Its permutation average is the stated PBW product. □

8.14.2 Marked coefficients and composition

Here natural cochains are universal multilinear expressions in products, brackets, exterior-coefficient operators and the chosen derivations, with finite values at each configuration. The bar comparisons of Section 6.7 retain this class. For a natural map $f : F_W \to F_V$, write $f_0 : W \to V$ for its linear term.

At two distinct labeled roots, project the quadratic term to PBW output with one odd factor. Encode it by a V-valued one-form ω_f on W with homogeneous linear coefficients. The term $t_i\, dt_j\, v$ means

that the input of color j carries the odd output v and the input of color i is unmarked. If $i = j$, sum the two marked-root possibilities. Thus $t\,dt$ encodes $(\varepsilon m)n + (\varepsilon n)m$.

These marked coefficients cannot be changed by a homotopy under and over J. Indeed a pure actor input has degree zero. An output with $b \geq 1$ odd factors, together with n additional ordinary Lie inputs and coefficient resolution degree $q \geq 0$, gives cochain degree at least $b + n + q$. There is no degree-zero homotopy coefficient on a pure actor input.

Its remaining variation would come from a source bracket joining roots. Open adjunction kills every map from that joined support to two distinct output roots. The two choices of marked output are independent PBW summands on the open.

Lemma 8.33. *For composable natural maps,*

$$(gf)_{\circ} = g_{\circ} f_{\circ}, \qquad \omega_{gf} = g_{\circ}\omega_f + f_{\circ}^{*}\omega_g, \qquad (f_{\circ}^{*}\omega)(t) = \omega(f_{\circ}t) \circ f_{\circ}.$$

The first nonlinear form of a diagonal tensor product is the block sum of its forms.

Proof. Lift the inner twisting cochain to the intermediate cobar, using the relative bar contraction following Lemma 6.21. Write $a_{v,m} = t(s(vm))$ and $M_m = t(sm)$ for its singleton generators, where $t = s^{-1}$. On two roots the lift consists of the marked primitive products prescribed by ω_f and the binary generator $t(sx\,sy)$, where $x = f_{\circ}(e_i)m$, $y = f_{\circ}(e_j)n$. On the open its differential is $-t(sx)t(sy) - t(sy)t(sx)$, exactly the source coproduct term. Joined-root corrections vanish there.

Applying g to the primitive products gives $g_{\circ}\omega_f$, and applying it to the binary generator gives $f_{\circ}^{*}\omega_g$. The marked coefficients are homotopy invariant by the preceding degree and support argument.

For the tensor assertion, independently grade the two target exterior factors. The strong tensor comparison preserves these gradings. Any term using both factors has at least two odd outputs, so the one-odd-output component is the block sum. □

The one-variable maps (6.19) realize any polynomial with zero constant term by choosing their coefficients. A pointed polynomial $\Phi : W \to V$ is a finite sum of powers of linear forms, component by component. The corresponding parallel one-variable maps, followed by exterior linear addition, therefore give an actual map $f_{\Phi} : F_W \to F_V$. For a homogeneous component Φ_r of degree r, the symmetric marked form is $r^{-1}d\Phi_r$. Its r input roots contribute the same coefficient, and their sum is the coordinate polynomial. The independent exterior gradings exclude cross terms from parallel factors.

Take $V = \langle z_1, z_2\rangle$ with $Dz_1 = z_2$, $Dz_2 = 0$. Realize

$$\Phi(t_1, t_2) = (t_1, t_2, -t_1t_2), \qquad -t_1t_2 = \frac{(t_1 - t_2)^2 - (t_1 + t_2)^2}{4}.$$

Let e_1, e_2 be the source exterior generators dual to t_1, t_2. Relabel its outputs by $(y_1, y_2, y_3) \mapsto (z_1, \varepsilon, z_2)$, apply α_D, and project $z_1 \mapsto 0$, $z_2 \mapsto \varepsilon'$ to one target generator. Call the resulting composite $f : F_2 \to F_1$.

Proposition 8.34. *The map f has zero linear term. It admits a twisting representative with zero pure binary values on equal colors and mixed value*

$$\tau_f(s(e_1m)s(e_2n)) = \tfrac{1}{2}\big((\varepsilon' m)n - (\varepsilon' n)m\big). \tag{8.40}$$

Its marked form is $\frac{1}{2}(t_2\,dt_1 - t_1\,dt_2)\varepsilon'$, so it is not homotopic to the map through J.

Proof. The last composite from three variables has linear part y_3 and marked form $y_2\,dy_1$. The direct polynomial realization has linear part $(t_1, t_2, 0)$ and form $(0, 0, -\frac{1}{2}d(t_1t_2))$. Lemma 8.33 gives the stated linear term and antisymmetric form.

For the exact mixed representative, lift its binary generator in the intermediate cobar by

$$L_{12}(m, n) = t(s(z_1m)s(\varepsilon n)) - \tfrac{1}{2}\big(a_{z_2,m}M_n + a_{z_2,n}M_m\big).$$

The joined source $s(e_1e_2p)$ lifts to $-t(s(\varepsilon z_1p))$, because $z_1\varepsilon = -\varepsilon z_1$. The cobar differential gives

$$dL_{12} = -t(s(\varepsilon z_1[m, n])) - a_{z_1,m}a_{\varepsilon,n} - a_{\varepsilon,n}a_{z_1,m},$$

which is the required source differential. The primitive correction is closed.

Applying the last map to the first term gives $(\varepsilon' m)n$. Its value on the correction is $-((\varepsilon' m)n + (\varepsilon' n)m)/2$. Their sum is (8.40). On either equal-color pair, the inner quadratic term is zero and the last twisting cochain has either two module inputs or none, so it vanishes.

The relative bar contraction extends the compatible lift without changing these binary values. All constituent maps already exist in every weight. The marked-form invariant proves the last assertion. □

8.14.3 Vanishing under natural actions

The natural cohomology in Lemma 6.21 records actor coordinates and derivation classes. Precomposition by the maps just constructed preserves this natural complex. The bar contraction acts on the Lie and bar factors and retains each exterior coefficient operator. Thus it lifts the graded tree cochains and their parameter homotopies as well. The corrections increase primitive input degree and are finite at each configuration. They preserve the pointing, augmentation and actor faces.

On the coordinate algebra the map of Lemma 8.32 is

$$(u, z) \longmapsto e^{uD}z. \tag{8.41}$$

To verify the normalization, let $\ell \in V^*$ extract one exterior coefficient on bar length one. On (8.39) it gives $\ell(D^w z)n\,\mathrm{pbw}(m_1 \cdots m_w)$. The w identical actor extractors sum $w!$ permutations, while PBW averages them. Its coordinate is therefore $u^w\ell(D^w z)/w!$.

Projection to natural cohomology kills all other nonprimitive exterior labels and fixes the derivation classes. Composition is ordinary substitution, and a diagonal tensor product acts independently on its coordinate blocks.

Proposition 8.35. *The map f of Proposition 8.34 has zero coordinate substitution in every weight. Consider the universal class of Lemma 6.21 for the specified commuting derivations. Every positive-weight natural action in this class pulls back along f to an action naturally homotopic to the trivial one. In particular, this holds for the Heisenberg derivation action α_D with $D = \partial_{x_1}$ from Theorem 6.13.*

Proof. For the last three-variable map, $D^2 = 0$ on $\langle z_1, z_2\rangle$, so (8.41) and the projection give $y_3 + y_1y_2$. Its composite with Φ is identically $-t_1t_2 + t_1t_2 = 0$ in the completed coordinate algebra, not merely to second order.

A positive-weight natural action has class $F \in \prod_{q\geq 1}\mathrm{Sym}^q(V^*) \otimes \mathfrak{d}$, with the chosen derivation space $\mathfrak{d}$. The pullback class is $F(0) = 0$. The natural bar comparison and weightwise cohomology computation following Lemma 6.21 give the equivalence of the corresponding complete Maurer–Cartan spaces. Equivalently, solve the relative path equation successively in positive actor weight, using the kernel contraction there. This gives an all-weight pointed path to the trivial action. Evaluation on each coefficient family preserves the path, and all its sums are finite at a fixed configuration. □

The map has retained information even though all these natural action classes vanish after pullback. The next argument shows that its distinction from the vacuum map survives every receiving pointed factorizing action.

8.15 An obstruction for every pointed recipient

The vacuum forces a degree-zero one-root operation of a factorizing action to be a module map. This prevents a nonzero derivation action from being carried entirely by mixed components after its singleton actor class has vanished.

Retain $C = C_{V_\kappa}$, $\mathfrak{g} = L_{V_\kappa}$, $\mathfrak{l} = L_k$, and set $A = U_\star\mathfrak{g}$. Both diagonal families $\mathfrak{g}, \mathfrak{l}$ are in degree zero, whereas $C_1 = \mathfrak{g}_1[1]$ and $J_1 = \mathfrak{l}_1[1]$. Write $\mu_\mathfrak{g} : \mathfrak{g} \star \mathfrak{g} \to \mathfrak{g}$ for the coefficient bracket. Let $\mathscr{A}_C^{\mathrm{f}}$ be the full category of pointed actions of Section 6.4 whose actor coalgebra is factorizing. Its morphisms retain the specified path between the actions. No lower cohomological bound is imposed on an actor.

For an object (Z, u, γ) put

$$B = \Omega Z, \qquad \mathcal{T}_C = (-) \otimes^! C, \qquad \Phi = \Omega\gamma : \mathcal{T}_C(B) \longrightarrow A.$$

Lemma 6.18 makes $\mathcal{T}_C$ exact and strong symmetric monoidal for $\star$. The pointing gives $\iota : \mathfrak{l} \to B$, and its specified path identifies $\Phi\mathcal{T}_C(\iota)$ with the generating inclusion $\mathfrak{g} \to A$.

For a degree-zero class $a : \mathfrak{l}|_Y \to B|_Y$ on an open $Y \subset X$, define its one-root action

$$\Upsilon(a) = \Phi\mathcal{T}_C(a) \in \mathrm{End}(\mathfrak{g}|_Y).$$

Here $A_1 = \mathfrak{g}_1$. Closed adjunction on each small-diagonal branch identifies a map out of $\mathfrak{l}$ with its singleton map. This also determines the map on the full diagonal family.

8.15.1 The two ordered products

On the two-root collision scheme $\mathscr{R}_2$, let $j : U \hookrightarrow \mathscr{R}_2$ be the disjoint open. Its complement i is the union of the two disjoint diagonal branches. Write μ_{12}, μ_{21} for the two ordered products of B on U.

Lemma 8.36. *Both maps $\mu_{12}, \mu_{21} : (B_1 \boxtimes B_1)|_U \to B_2|_U$ are equivalences. If σ exchanges the external factors, the exchange $R_B = \mu_{12}^{-1}\mu_{21}\sigma$ is natural under algebra maps between such cobars. For $A = U_\star\mathfrak{g}$, R_A is the identity.*

Proof. On two roots the reduced cobar has one generator summand $t\overline{Z}_2$ and two product summands M_{12}, M_{21}, each an external product of $t\overline{Z}_1$ with itself. Its differential from the generator to a product is the corresponding coproduct, with the cobar sign. Factorization makes both coproducts equivalences on U. The quotient by M_{12} is the cone of the other equivalence, hence acyclic. Interchange the two orders for the second assertion. No connectivity is used.

For an algebra map $F : B \to B'$, compatibility with both products gives

$$(F_1 \boxtimes F_1)R_B = R_{B'}(F_1 \boxtimes F_1)$$

after composing with μ'_{12}. In the envelope A, the difference of the two products is the chiral bracket. It is supported on the diagonal, so its open exchange is the identity. □

Lemma 8.37. *For every degree-zero one-root class a, the endomorphism $M = \Upsilon(a)$ satisfies on both ordered collision branches*

$$M\mu_\mathfrak{g} = \mu_\mathfrak{g}(\mathrm{id} \star M) = \mu_\mathfrak{g}(M \star \mathrm{id}). \tag{8.42}$$

If M is also a coefficient derivation, then $M = 0$. On a nonempty open of X, every such one-root endomorphism is a constant scalar.

Proof. Work over Y and put $R = \mathbb{1} \boxtimes \mathbb{1}$ on its two-root scheme. Each product summand of $\mathbb{1} \star \mathbb{1}$ is $j_* j^* R$. Let $\kappa_a = [\iota, a]$ be the difference of the two ordered products on that source. The envelope relation and the vacuum path give

$$\Phi \mathcal{T}_C(\kappa_a) = \mu_{\mathfrak{g}}(\mathrm{id} \star M). \tag{8.43}$$

On U, put $w = (R_B - \mathrm{id})(\iota \boxtimes a)$ and let $P(w) = \mu_{12} j_*(w)$, using the global product from the first convolution summand. The map $D_a = \kappa_a + P(w)$ restricts to zero on U. Moreover $\Phi \mathcal{T}_C(P(w)) = 0$. Indeed $Z \otimes^! C$ is factorizing, and the strong monoidal comparison identifies its cobar exchange with that transported by $\mathcal{T}_C$. Naturality in Lemma 8.36 then kills the image of w, since $R_A = \mathrm{id}$. The open projection formula carries this zero map and a representing nullhomotopy through j_* and multiplication.

Set $Q = \mathrm{cofib}(j_! j^* R \to j_* j^* R)$. Open adjunction lifts D_a to $Q \to B_2$. Only the regular holonomic source R requires $j_!$ here, so the construction allows unbounded B_2. The localization octahedron gives

$$i_* i^* R \longrightarrow Q \longrightarrow i_* i^! R[1]. \tag{8.44}$$

On each branch, the right-module shifts are $i^! R = \mathbb{1}[-1]$ and $i^* R = \mathbb{1}[1]$. The normal-coordinate sequence $\omega \to \omega(*0) \to \delta_0$ fixes the first shift, and duality fixes the second. The composite $j_* j^* R \to Q \to i_* \mathbb{1}$ is precisely the vacuum principal-part bracket.

Closed projection and $i^! C_2 = C_1$ identify the outer terms after $\mathcal{T}_C$ with $i_* \mathfrak{g}[1]$ and $i_* \mathfrak{g}$. The envelope A_2 lies in the right-module heart. Its finite PBW filtration has quotients formed by open and closed direct images of external degree-zero modules. These direct images are exact on the affine collision charts. Also $i^! A_2 = \mathfrak{g}$. Consequently

$$\mathrm{Hom}(i_* \mathfrak{g}[1], A_2) = \mathrm{Ext}^{-1}(\mathfrak{g}, \mathfrak{g}) = 0.$$

After applying $\mathcal{T}_C$ and Φ, the map from Q therefore factors on each branch through an endomorphism N of $\mathfrak{g}$. Equation (8.43) becomes

$$\mu_{\mathfrak{g}}(\mathrm{id} \star M) = N \mu_{\mathfrak{g}}.$$

Insert the vacuum in the first slot. Its bracket is the principal-part map $\mathrm{can} : j_* j^*(\mathbb{1} \boxtimes \mathfrak{g}) \to i_* \mathfrak{g}$, so $(N - M)\mathrm{can} = 0$.

Precomposition by can is injective on these degree-zero maps. Its kernel is an image of $\mathrm{Hom}((\mathbb{1} \boxtimes \mathfrak{g})[1], A_2)$. This group vanishes because both unshifted objects are in the heart. Hence $N = M$. Skew symmetry gives the other slot identity.

If M is a derivation, subtract its module identity from its derivation identity. The remaining slot is zero. Insert the vacuum there and use the same injectivity to obtain $M = 0$.

For the scalar assertion, the full Laurent coefficient source in (8.42) gives commutation with every current mode, including creation modes. To extract them, use the first coordinate as normal derivative, so no derivative falls on the second-coordinate matrix of M. The zeroth normal coefficient of the bracket on $(z - w)^{-n} J \boxtimes v$ is $J_{-n} v$ for $n > 0$.

Put $c = M\mathbf{1}$. Creation-mode commutation gives $M(f) = fc$ on polynomial states. Positive-mode commutation gives $\kappa n \partial_{x_n} c = 0$ for every $n > 0$. Thus c is a scalar function of the position. Horizontalness makes its derivative zero, so it is constant. □

The derivation $D = \partial_{x_1}$ fails the module identity already on the vacuum:

$$(D J_{-1} - J_{-1} D)\mathbf{1} = \mathbf{1}.$$

Commutation with annihilation modes alone would not detect this defect.

8.15.2 The singleton class cannot vanish

We need two facts about derived coefficient maps. The first controls first actor weight. Write $P_2 = \mathrm{Sym}^2_\star \mathfrak{g}$, with its adjoint $\mathfrak{g}$-module structure. The Chevalley filtration for $\mathrm{Ext}^1_{\mathfrak{g}}(\mathfrak{g}, P_2)$ has entries

$$E_1^{n,q} = \mathrm{Ext}^q_{\mathscr{D}}(\Lambda^n_\star \mathfrak{g} \star \mathfrak{g}, P_2), \qquad n, q \geq 0.$$

Open adjunction kills column $n = 0$ in every degree: its source has one-root support, whereas P_2 is open direct image from two distinct roots.

In total degree one, only $(n, q) = (1, 0)$ remains. Its differential is the module cocycle equation. There are no boundaries, and no higher differential can enter or leave it. Thus equality of these extension classes is equality of their actual two-output cocycles. This argument also holds over every open $Y \subset X$.

For a coefficient derivation D, the cocycle is $\gamma_D(x; v) = m(Dx, v)$, where m is symmetric multiplication into P_2. Its module defect is $m([x, Dy] + [Dx, y] - D[x, y], v) = 0$. Its open operation is $D \otimes \mathrm{id}$, so its class is nonzero if $D \neq 0$. This is the first-weight class of the action α_D: its first tree operation is exactly $Dx\, v$.

The second fact is finite tensor expansion. For an affine open $Y \subset X$, an arbitrary complex N of differential modules, and an arbitrary complex W over k,

$$R\,\mathrm{Hom}_Y(\mathfrak{I}, N \otimes_k W) \simeq R\,\mathrm{Hom}_Y(\mathfrak{I}, N) \otimes_k W. \tag{8.45}$$

Under unshifted left/right conversion, $\mathfrak{I}$ becomes $\mathcal{O}_Y$.

Its Spencer resolution on the affine coordinate is

$$0 \longrightarrow D_Y \xrightarrow{P \mapsto P\partial_y} D_Y \longrightarrow \mathcal{O}_Y \longrightarrow 0.$$

It is finite free, so its Hom complex commutes with algebraic tensor by any W, including an unbounded complex. Over k, Künneth is an algebraic direct sum. Every class consequently has a finite expression in its bidegrees.

For a point $q \in X(k)$ normalize a right differential-module fiber by $N_q = i_q^! N[1]$. Then $\mathfrak{I}_q = k$ and $\mathfrak{g}_q = V_\kappa$ are in degree zero. Let Υ_q denote the action on these normalized fibers. Given a morphism $h : (F_1, \alpha_D) \to (Z, \gamma)$ in $\mathscr{A}_C^{\mathrm{f}}$, let h_0 be its twisting component on one pure actor input. Its plain singleton class is

$$[h_0] \in H^1 R\,\mathrm{Hom}(\mathfrak{I}, B).$$

This class forgets equivariance by $\mathfrak{I}$.

Theorem 8.38. *For the action of a nonzero coefficient derivation D, every receiving morphism h has $[h_0]|_Y \neq 0$ on every nonempty affine open $Y \subset X$.*

Proof. Suppose this plain class vanishes. Resolve the pointed arrow before forming its relative convolution complex. Its bar filtration gives a cofibrant graded quotient, so restriction of the resolved Hom complexes is surjective. Fixing the twisting cochain on the resolved vacuum therefore computes the homotopy fiber of restriction with its specified path.

A primitive of h_0 extends to a degree-zero relative cochain, zero on the vacuum. Gauge by its exponential in positive actor weight. At one pure input this changes h_0 by its differential and sets it to zero exactly. Higher terms affect only higher input number. All inverse and exponential series are finite at each configuration, since every positive factor consumes a nonempty set of labels. This gives $h' \simeq h$ with $h'_0 = 0$, while retaining the action-compatibility path.

The first-weight term of h' begins with a closed degree-zero component c_1 on one ordinary input and one actor input. There is no support restriction on c_1. In the normalized bar map of $\mathcal{T}_C(h')$, its binary component is $\mathcal{T}_C(c_1)$ plus products involving $\mathcal{T}_C(h'_0)$. The products vanish. After the action the component is therefore $\Phi\mathcal{T}_C(c_1)$.

Choose $q \in Y(k)$ and put $Y' = Y \setminus \{q\}$. Such a point exists because k is infinite. Place the ordinary input at the varying root in Y' and the actor input at q. Factorization in the first product order identifies the target with $B|_{Y'} \otimes B_q$.

Formula (8.45) expresses $[c_1]$ as a finite sum of bidegrees $(r, -r)$. After applying the action, the second factor maps to $H^{-r} R\,\mathrm{Hom}_k(V_\kappa, V_\kappa)$, which vanishes for $r \neq 0$. Thus its open two-output operation has the form

$$\sum_{i=1}^{m} M_i \otimes N_i, \qquad M_i = \Psi(a_i), \quad N_i = \Psi_q(b_i),$$

with a_i, b_i degree-zero classes.

The specified action path identifies its first-weight extension class with that of α_D. The preceding Chevalley filtration has no boundaries in its P_2 component. Hence on this slice the equality is an equality of coefficient operations:

$$D \otimes \mathrm{id}_{V_\kappa} = \sum_{i=1}^{m} M_i \otimes N_i.$$

Algebraic tensors of these endomorphisms inject into external endomorphisms. For a finite independent list in the second factor, finitely many vector evaluations and linear functionals separate its members. Choose a linear functional λ on the span of $\mathrm{id}, N_1, \dots, N_m$ with $\lambda(\mathrm{id}) = 1$. It gives $D|_{Y'} = \sum_i \lambda(N_i) M_i$.

By Lemma 8.37, every M_i is a module map. Thus $D|_{Y'}$ is both a derivation and a module map, and is zero. The state differential module is torsion-free, and the image of each state is algebraic. Localization is faithful on that image, so $D = 0$ on X, a contradiction. □

8.15.3 The parallel pair survives

Let $f : F_2 \to F_1$ be the map of Proposition 8.34, and let f_0 factor through J. For $D = \partial_{x_1}$, Proposition 8.35 supplies an all-weight pointed path p_f from the pullback of α_D along f to the trivial action. Use the constant path p_0 for f_0. They give two points

$$(f, p_f), (f_0, p_0) \in \mathrm{Map}_{\mathscr{A}_C^{\mathrm{f}}}((F_2, \mathrm{triv}), (F_1, \alpha_D)). \tag{8.46}$$

We first specify the invariant used to separate their images in a general recipient. Suppose two maps $F_2 \to Z$ under J have zero pure single-actor components. A pointed homotopy between them preserves the class of their pure binary component in

$$H^1 R\,\mathrm{Hom}(j^*(\mathbb{I} \star \mathbb{I}), j^* B), \tag{8.47}$$

with both input labels retained.

To prove this, use the full resolved coefficient differential ∂ and write the interval homotopy as $\tau(u) + c(u)\,du$. Its two singleton components have the form $\partial E(u)$, $\partial F(u)$, with $E(0) = F(0) = 0$ and $\partial E(1) = \partial F(1) = 0$. They are obtained by integrating the singleton components of c. On the disjoint open the source bracket vanishes, and the binary interval equation reads

$$\tau'_{1,1} = \partial c_{1,1} + [\partial E, F'] + [\partial F, E'].$$

The full Leibniz rule, with $|E| = |F| = 0$, gives

$$[\partial E, F'] + [\partial F, E'] = \partial [E, F'] - \frac{d}{du}[E, \partial F].$$

The integrated endpoint term vanishes. Thus the change of binary components is a boundary in (8.47). The commutator cochains need not vanish.

Theorem 8.39. *For every morphism $h : (F_1, \alpha_D) \to (Z, \gamma)$ in $\mathscr{A}^{\mathrm{f}}_C$, the maps hf and hf_0 are not homotopic under J. Consequently the category of pointed factorizing actions is not filtered, and*

$$\Phi_C(F, u) = \mathrm{Map}_{C/}(F \otimes^{!} C, C)$$

is not representable on pointed factorizing coalgebras.

Proof. Both composites have zero pure singleton components. In the source cobar, on distinct roots, the signs give

$$d\, t(s(\varepsilon n)s(m)) = -t(s(\varepsilon n))t(s(m)) + t(s(m))t(s(\varepsilon n)).$$

After Ωh, this replaces the reversed product $h_0(n)\iota(m)$ by $\iota(m)h_0(n)$ modulo a boundary. This identity comes from the cocommutative source and imposes no centrality on the recipient's vacuum.

Fix $q \in X(k)$ and put $Y = X \setminus \{q\}$. Let the root of m vary in Y and the root of n be q. In the first product order, the binary class of hf is

$$\tfrac{1}{2}\big([h_0]|_Y \otimes [\iota_q] - [\iota]|_Y \otimes [h_0]_q\big).$$

By (8.45), its two terms lie in distinct bidegrees $(1, 0)$ and $(0, 1)$. The first is nonzero: Theorem 8.38 gives $[h_0]|_Y \neq 0$, and $\Upsilon_q(\iota_q) = \mathrm{id}_{V_\kappa}$ gives $[\iota_q] \neq 0$. No claim about the second term is needed. The binary class of hf_0 is zero.

Normalized restriction to $Y \times \{q\}$ is a map of derived coefficient complexes. It preserves binary components and boundaries, so these distinct slice classes come from distinct classes in (8.47). The homotopy invariant just proved excludes a homotopy under J, and hence excludes one in the action category with its additional action two-cell.

A filtered infinity category equalizes every parallel pair after some receiving map. The pair (8.46) violates this requirement. If Φ_C were represented, its universal action would be a final object of $\mathscr{A}^{\mathrm{f}}_C$. Its mapping space from (F_2, triv) would be contractible and would identify the two composites, again a contradiction. □

The two actor coalgebras admit additive exterior monoid structures. The map f is not a map of those monoids: its first form $\frac{1}{2}(t_2\, dt_1 - t_1\, dt_2)$ has a nonzero cross term under addition. Therefore Theorem 8.39 does not rule out a final transverse monoid action. The specific cochain monoid is already nonfinal by Theorem 6.10. Constructing another final monoid, or proving an obstruction within the category of monoid actions and monoid morphisms, remains a different universal-property problem.

8.16 Field reconstruction and vacuum normalization

A reconstruction from collision coefficients must recover the fields, their exchange, and their vacuum normalization together. Binary field evaluation can lose an exchange operator. An outer collision with a vacuum partner recovers it while retaining the intervening state.

Let R be a commutative algebra over a characteristic-zero field. Let V be a flat R-module, concentrated in degree zero, with R-linear translation T and a vacuum $\mathbf{1}$ spanning a free direct summand. Suppose R-bilinear fields $Y(a, u)b \in V((u))$ satisfy creation and translation,

$$Y(\mathbf{1}, u) = 1, \quad Y(a, u)\mathbf{1} = e^{uT} a, \quad [T, Y(a, u)] = \partial_u Y(a, u) = Y(Ta, u).$$

Also assume weak associativity: for each triple, after multiplication by a sufficiently high power of $u + w$, the expansions of $Y(a, u + w)Y(b, w)c$ and $Y(Y(a, u)b, w)c$ agree.

Specify an invertible $\mathsf{R}[u, u^{-1}]$-linear exchange $\mathcal{S}(u)$ on $(V \otimes_{\mathsf{R}} V)[u, u^{-1}]$, Laurent polynomial on each tensor input. It fixes both vacuum factors.

Assume both shift equations

$$[T \otimes 1, \mathcal{S}(u)] = -\partial_u \mathcal{S}(u), \qquad [1 \otimes T, \mathcal{S}(u)] = \partial_u \mathcal{S}(u),$$

the Yang–Baxter identity, and the following field-exchange identity. For each a, b, require an integer $N \geq 0$ such that $(z_1 - z_2)^N \mathcal{S}(z_1 - z_2)(a \otimes b)$ has no pole and, for every c,

$$(z_1 - z_2)^N Y(b, z_2) Y(a, z_1) c = (z_1 - z_2)^N Y(z_1)(1 \otimes Y(z_2))\big(\mathcal{S}(z_1 - z_2)(a \otimes b) \otimes c\big).$$

The two sides in this equality use their respective ordered formal expansions. Also assume the hexagon

$$\mathcal{S}(z)(Y(w) \otimes 1) = (Y(w) \otimes 1)\mathcal{S}_{23}(z)\mathcal{S}_{13}(z + w), \tag{8.48}$$

where the last factor is expanded in nonnegative powers of w. We leave unitarity as an explicit condition to be tested below.

Use the partition coefficient modules of (4.1), with all normal derivatives retained, external tensors over R, and blocks ordered by their largest labels. Write r°_{ij} for the principal part of the field of block i at block j, leaving every spectator state in the output. For ordered distinct blocks define

$$X_{ab} = \mathcal{S}_{ba}(z_b - z_a), \qquad r_{ij} = r^{\circ}_{ij} X_{i,j-1} \cdots X_{i,i+1} \quad (i < j). \tag{8.49}$$

The shift equations make X_{ab} horizontal. The exchange factors are regular in a collision coordinate whenever their spectator separation is invertible. Their Taylor expansions therefore enter the finite principal-part rule (4.3).

Proposition 8.40. *Put $u = z_1 - z_3$ and $\delta_u = [u^{-1}]$ in the normal principal-part module. On the open where the second position is distinct from the first and third,*

$$r_{13}(u^{-1}; a, b, \mathbf{1}) = \delta_u \, \mathcal{S}_{21}(z_2 - z_3)(a \otimes b). \tag{8.50}$$

The output is written with the merged 13 state first and the spectator second. The canonical flip restores the root order 2|13. The full system of residues determines both $\mathcal{S}$ and the fields.

Proof. By definition $r_{13} = r^{\circ}_{13} X_{12}$. Creation at the last input gives $e^{uT} a$. The exchange is regular as a series in u near zero on this spectator open. The principal part of u^{-1} times that regular series is its constant coefficient times δ_u. This is (8.50).

The map placing a tensor in normal degree zero is injective, so it recovers $\mathcal{S}$. For every integer m, the coefficient of δ_u in $r_{12}(u^{-m-1}; a, b)$ is the coefficient of u^m in $Y(a, u)b$. Thus the binary principal parts recover the entire Laurent field, including its regular coefficients. □

This use of the vacuum is stronger than remembering an empty-configuration coaugmentation. It fixes a state and its creation map in a nonempty coefficient, with the spectator still present. Applying a scalar functional to that spectator would change the reconstruction data.

Example 8.41. Let $V = \mathsf{R}[x_1, x_2, \ldots]$, $J = x_1$, $T x_n = n x_{n+1}$, and use the Wick fields of (4.18) with $\kappa, \lambda \in \mathsf{R}$ and $P(u) = \kappa u^{-2} + \lambda u^{-1}$. The same finite contraction formula defines these fields even when P is not even. Its exchange is $\mathcal{S}(u) = \exp(C_{21}(-u) - C(u))$. Each exponential terminates on polynomial inputs. We compute the defect of omitting this exchange from the outer residue.

Take states $(J, J, \mathbf{1})$ and multiplier s^{-1}, where $s = z_1 - z_3$, $t = z_2 - z_3$. Set $u = s - t$. The first consecutive principal part is that of $P(u)/(u + t)$:

$$\kappa u^{-2} t^{-1} + \lambda u^{-1} t^{-1} - \kappa u^{-1} t^{-2}.$$

The full normal transfer to the common small diagonal is

$$u^{-a}t^{-b} \longmapsto \sum_{m=0}^{b-1} \binom{a+m-1}{m} s^{-a-m}t^{-b+m}.$$

It follows by differentiating the simple double principal part, since differentiation in t at fixed u becomes $\partial_s + \partial_t$.

The two consecutive channels and the outer one without exchange are consequently

$$L = -\frac{\kappa}{st^2} + \frac{\lambda}{st}, \qquad R_{\mathrm{ct}} = 0, \qquad N^\circ = \frac{\kappa}{st^2} + \frac{\lambda}{st}.$$

Each multiplies the vacuum. The second channel vanishes because creation is regular in t. Thus $L - R_{\mathrm{ct}} + N^\circ = 2\lambda \mathbf{1}/(st)$.

The exchange X_{12} contributes $2\lambda(s-t)^{-1}\mathbf{1}\otimes\mathbf{1}$. With the multiplier s^{-1}, its two principal parts contribute $-2\lambda\mathbf{1}/(st)$, which cancels the defect. The uncorrected defect is nonzero for every $\lambda \neq 0$, also over nonreduced R. Indeed 2 is invertible, and the vacuum is a free direct summand.

Theorem 8.42 (Two vacuum partners)**.** *For the field and exchange data above, the commutator of the two crossed residues is*

$$r_{24}r_{13} - r_{13}r_{24} = r^\circ_{24}r^\circ_{13}X_{23}(X_{21}X_{12} - 1). \tag{8.51}$$

Put $p = z_1 - z_3$, $q = z_2 - z_4$, and $d = z_3 - z_4$. On $d \neq 0$, its value on two vacuum partners is

$$\begin{aligned}&(r_{24}r_{13} - r_{13}r_{24})(p^{-1}q^{-1}; a, b, \mathbf{1}, \mathbf{1})\\ &\qquad = [p^{-1}][q^{-1}](\mathcal{S}_{12}(d)\mathcal{S}_{21}(-d) - 1)(a\otimes b).\end{aligned} \tag{8.52}$$

The following conditions are equivalent: unitarity $\mathcal{S}_{21}(-z)\mathcal{S}(z) = 1$, commutation of the crossed residues on the full four-input coefficient, and vanishing of (8.52) for every a, b. The equivalence holds over nonreduced R.

Proof. The hexagon (8.48), followed by principal part in the collision coordinate, gives

$$X_{2,(13)}r^\circ_{13} = r^\circ_{13}X_{23}X_{21}.$$

The coefficients are expanded before taking the principal part, as in (4.3). Thus $r_{24}r_{13} = r^\circ_{24}r^\circ_{13}X_{23}X_{21}X_{12}$. For the opposite order, blocks 1, 3 are adjacent after merging 2, 4, so $r_{13}r_{24} = r^\circ_{13}r^\circ_{24}X_{23}$. The two plain residues act on disjoint state pairs and commute. Subtraction proves (8.51) without unitarity.

After $X_{21}X_{12} - 1$ acts on $a\otimes b\otimes\mathbf{1}\otimes\mathbf{1}$, the third factor is still the vacuum, so X_{23} acts identically. The remaining operator is

$$\mathcal{S}_{12}(d+p-q)\mathcal{S}_{21}(-d-p+q) - 1.$$

It is regular in p, q near zero when d is invertible. Creation at the two partners adds e^{pT} and e^{qT}. The double principal part of $p^{-1}q^{-1}$ times this regular series is its constant coefficient, giving (8.52).

Unitarity implies commutation by (8.51), and commutation implies the stated vacuum vanishing. Conversely the normal-degree-zero inclusion

$$(V\otimes_{\mathsf{R}} V)[d, d^{-1}] \hookrightarrow (V\otimes_{\mathsf{R}} V)[d, d^{-1}][\partial_p, \partial_q][p^{-1}][q^{-1}]$$

is injective. Vacuum vanishing therefore gives $\mathcal{S}_{12}(d)\mathcal{S}_{21}(-d) = 1$. Invertibility reverses the factors to give unitarity. The argument neither specializes to residue fields nor cancels a field evaluation map. □

When the exchanged residues satisfy the relations of Proposition 4.1, the finite residue transfer in Chapter 4 applies to them. Its higher products use the collision of the first and last blocks, after the exchanges through all intervening blocks. The vacuum detectors show why those blocks and the chosen exchange cannot be discarded during a comparison.

Problem 8.43. Construct the global coefficient category and comparison functors relating the ordered word bar–cobar construction to the full collision atlas and to covariant field systems. Specify the exact collision functors, the coefficient dual and its topology, and the vacuum maps on nonempty configurations. A field realization must recover all Laurent coefficients through Proposition 8.40. It must also recover the exchanged outer residues, both vacuum insertions and the compatible maps for nested collisions.

For the Heisenberg case, compare the native transverse cochain monoid and its marked endomorphism system with the completed coefficient centers of this chapter. Construct the comparison on full mapping spaces, specifying the admitted actor category. The nonrepresentability theorem excludes all pointed factorizing coalgebras as a representing domain for Φ_C. Finality among transverse monoid actions, with their multiplication coherences and monoid morphisms, remains to be decided.

The five coupled constructions

Ordered composition and holomorphic collision ask different questions of a field algebra. Composition follows the orientation of a transverse interval. Collision on a smooth curve involves singular coefficients along diagonals and their normal derivatives. Reconstruction must relate these operations while preserving both the empty augmentation and the nonempty local vacuum.

The cut product of Chapter 1 gives a concrete model. It splits an ordered word into consecutive blocks where the blocks are disjoint. Positive word length bounds the number of nonempty blocks, so normalized bar and cobar are finite in each arity. Their inverse equivalence holds in the enhanced coefficient setting of Theorem 1.3. The passage from these coefficients to fields is the comparison problem of Chapter 8.

Five constructions extend this question beyond the word model. Ordered chiral geometry distinguishes transverse associative composition, associative composition for chiral convolution, and the chiral Lie bracket. Each operation has its own bar construction. A mixed distributive law can relate the two associative directions. Comparing the two orders of their iterated bars requires that law on the full bisimplicial bar and preservation of its realizations by the tensor products.

In the finite species model, a mixed coefficient is a direct sum indexed by pairs of ordered decompositions of a finite set. Projection onto equal decompositions is an idempotent, the aligned projector. Transporting this projector to geometric coefficients requires a comparison preserving the splitting and the transition maps under collision. The finite projection alone does not construct these data.

Mixed holomorphic–topological Deligne theory begins with the relative Hochschild complex of a Rees enveloping algebra. Its finite comparison retains the Rees parameter in the Chevalley differential. Compatible jets and operadic refinements are additional hypotheses for the complete open–closed operations.

Igusa–Borcherds theory supplies the root-graded denominator, with its normalization fixed by the product formula. Calabi–Yau Hall geometry starts from extension correspondences. Their pullback and pushforward maps define convolution when the coefficient theory satisfies the required base change. A quantum double additionally uses a continuous Hopf pairing and the stated graded completion.

A mixed boundary datum specifies the algebras, their interface, and coherent actions. Its deformation complex records their simultaneous variation. A local Batalin–Vilkovisky field theory also requires a local realization of this datum. An Einstein comparison must identify the physical action and pairing.

These constructions meet through bar–cobar, open–closed, Hall, current, trace, and determinant maps. Each comparison must specify its coefficients, completion, parity, descent, and anomaly hypotheses. The parts that follow develop the five constructions and the maps between their carriers.

Bar constructions and their comparisons

The three bars use transverse associative composition, associative chiral composition, and the chiral Lie bracket. Mixed distributive laws compare iterated bars when they extend to the full bisimplicial object and tensor products preserve its realizations. Enveloping comparisons use a Poincaré–Birkhoff–Witt isomorphism and strongly convergent bar filtrations.

Book II

The invariant object

Chapter 9

Two directions of locality

9.1 The local experiment

Let X be a smooth complex algebraic curve and let $J \subset \mathbb{R}$ be an oriented interval. Consider a sector of observables that is holomorphic along X and locally constant along J. Two insertions at (z_1, t_1) and (z_2, t_2) can approach one another in either direction. The limit $z_1 \to z_2$ tests the singular coefficients along the diagonal of X^2. Composition as $t_1 \to t_2$ uses the order selected by the orientation of J. A theory carrying both operations must specify their compatibility.

Principle 9.1 (Two-direction principle). Holomorphic collision uses the coefficient system on powers of X and its chiral operation maps. Transverse composition uses an internal E_1 multiplication. Exchange compares orders through a specified transport on the configuration space.

The geometry determines which operation must be supplied:

carrier	operation
diagonal in X^2	singular chiral operation
oriented interval	ordered composition and associative bar
oriented plane	transport along braids
circle	Hochschild trace
stable curve	modular sewing.

The trace, transport and sewing maps belong to the chosen coefficient theory. For ordered word coefficients, the convolution (1.1) implements composition by consecutive cuts. Its realization by observables requires the corresponding operation maps.

9.2 Carrier and operation

Write D_X for the sheaf of differential operators on X. Let $\mathfrak{g}$ be a right D_X-module, let $\Delta : X \hookrightarrow X^2$ be the diagonal, and let $j : X^2 \setminus \Delta(X) \hookrightarrow X^2$ be its open complement. A binary chiral operation on $\mathfrak{g}$ is a morphism of right D_{X^2}-modules

$$\mu : j_* j^* (\mathfrak{g} \boxtimes \mathfrak{g}) \longrightarrow \Delta_* \mathfrak{g}.$$

Here the direct images are those of D-modules. A chiral Lie algebra requires this operation to be skew-symmetric and to satisfy Jacobi, with compositions taken in the chiral pseudo-tensor category [9, §§0.8, 3.1.1–3.1.2]. Thus $\mathfrak{g}$ specifies the carrier, while μ and its identities specify the Lie operation. A unit is additional data.

A factorization family records complexes of D-modules on the powers X^S for finite sets S. It also supplies coherent collision identifications and structure maps on disjoint loci. The chosen algebra or coalgebra structure determines the direction of these maps. The full operator product retains both the family and its operation maps.

9.3 Principal symbols and residues

The simplest collision coefficient already shows why the carrier matters. Regard $\Delta(X)$ as a Cartier divisor, and set

$$F_m = \mathcal{O}_{X^2}(m\Delta) \subset \mathcal{O}_{X^2}(*\Delta), \qquad m \geq 0.$$

These are sheaves of meromorphic functions with pole order at most m. Let $\mathcal{I}_\Delta$ be the ideal of the diagonal. Its normal line is $N_{\Delta/X^2} = (\mathcal{I}_\Delta/\mathcal{I}_\Delta^2)^\vee$.

Proposition 9.2 (Principal symbols along the diagonal)**.** *For $m \geq 1$, the pole filtration has the exact sequence of $\mathcal{O}_{X^2}$-modules*

$$0 \longrightarrow F_{m-1} \longrightarrow F_m \longrightarrow \Delta_* N_{\Delta/X^2}^{\otimes m} \longrightarrow 0.$$

In this sequence, Δ_* denotes the direct image of a sheaf on the closed divisor. The leading symbol is therefore a section of a normal line power on Δ.

Proof. Tensor the divisor sequence $0 \to \mathcal{O}_{X^2}(-\Delta) \to \mathcal{O}_{X^2} \to \Delta_*\mathcal{O}_\Delta \to 0$ with the line bundle $\mathcal{O}_{X^2}(m\Delta)$. The restriction of $\mathcal{O}_{X^2}(\Delta)$ to Δ is the normal line. This gives the asserted quotient and its ambient sheaf. □

Locally choose formal coordinates (z, u) with $z = z_2$ and $u = z_1 - z_2$. The class of $a(z)u^{-m}$ is its leading normal symbol. If $v = \alpha(z)u + O(u^2)$ with $\alpha(z) \neq 0$, then $u^{-m} = \alpha(z)^m v^{-m} + O(v^{-m+1})$. This is the transition rule for $N_{\Delta/X^2}^{\otimes m}$. Coordinate changes can mix the lower pole terms, so the associated graded sheaf alone does not split the pole filtration.

A scalar residue uses a relative one-form. For the projection to the second factor, write such a form as $f(u, z)\,du$. Its residue along Δ is the coefficient of $u^{-1}du$. For a simple pole this pairs the normal symbol with the relative cotangent line. The coefficient is unchanged under a change of the transverse parameter, including its Jacobian in the differential.

Indeed, hold z fixed and expand in the new parameter v. For $m \neq 1$, the form $v^{-m}dv = d(v^{1-m})/(1-m)$ is a Laurent derivative, whose coefficient of $u^{-1}du$ vanishes. For a simple pole,

$$\frac{dv}{v} = \frac{du}{u} + \frac{d(v/u)}{v/u}.$$

The last term is regular because v/u is a unit. Linearity on the finite polar part proves the claimed invariance.

Example 9.3 (A double pole)**.** Set $v = u + u^2$ near $u = 0$. Then

$$v^{-2} = u^{-2} - 2u^{-1} + O(1), \qquad v^{-2}dv = \big(u^{-2} + O(1)\big)du.$$

The function acquires a simple-pole term, while the one-form has zero residue in either coordinate. For $m \geq 1$, normal derivatives detect higher pole orders in the coordinate u:

$$\partial_u^{m-1}(u^{-1}) = (-1)^{m-1}(m-1)!\,u^{-m}.$$

Thus a collision coefficient retains its filtered principal parts and their coordinate transformations as well as its leading symbols.

Collisions of arbitrary subsets require these carriers and operation maps on every finite power of X. Their compatible organization is the purpose of the Ran coefficient category.

Chapter 10

The Ran coefficient category

10.1 Decompositions and alignment

Let X be a nonempty smooth separated algebraic curve over $\mathbb{C}$. Two chiral products can divide the same configuration into different blocks. Their comparison must retain the common divisions and commute with collisions. For two labeled points there are two ordered divisions into nonempty blocks. Two independent divisions give four pairs, of which two agree. This finite projection has a geometric realization on collision coefficients.

For a nonempty finite set I and $r \geq 1$, put $[r] = \{1, \dots, r\}$ and

$$\mathrm{Dec}_r(I) = \{d : I \twoheadrightarrow [r]\}, \qquad I_a = d^{-1}(a).$$

Thus d is an ordered decomposition into r nonempty blocks. The block labels are ordered, while I has no chosen order.

For example, $\{1, 3\} \sqcup \{2\}$ is a division of $\{1, 2, 3\}$. It is not a consecutive cut in the order 1, 2, 3. The consecutive-cut product (1.1) uses its own carrier. A comparison with the present product must account for this difference.

Write $\mathrm{Dec}(I) = \mathrm{Dec}_2(I)$. Let C be a stable ∞-category and let $M_{d,e} \in C$ be objects indexed by $d, e \in \mathrm{Dec}(I)$. Finite sums in C are also products, so they have canonical inclusions and projections.

Proposition 10.1 (Aligned sector in the finite decomposition model)**.** *The object $M = \bigoplus_{d,e} M_{d,e}$ has the split idempotent*

$$e_{\mathrm{al}}\big|_{M_{d,e}} = \begin{cases} \mathrm{id}, & d = e, \\ 0, & d \neq e. \end{cases}$$

Its image is $\bigoplus_d M_{d,d}$. The projection is natural under bijections of I. Repeated alignment, in any parenthesization, selects the summands whose decomposition labels all agree.

Proof. Let q be projection onto the indicated summands and s their inclusion. Then $qs = \mathrm{id}$ and $e_{\mathrm{al}} = sq$, which proves idempotence and the image statement. A bijection preserves equality of decomposition labels.

For three labels, the two composites impose $d = e$ and $e = f$. Both therefore select exactly $d = e = f$. For any binary tree of alignments, equality at each vertex forces equality of all leaf labels. The canonical finite-sum maps give the same splitting. □

To apply this projection to collision coefficients, the summands must form a family compatible with diagonal pullback.

10.2 Coefficients and chiral convolution

Coefficient categories below are $\mathbb{C}$-linear stable ∞-categories. Tensor products are derived. Scalar coefficient complexes use the usual cohomological Koszul signs. For a separated scheme Y of finite type,

let $\mathcal{D}(Y)$ be the presentable stable ∞-category of complexes of right D-modules localized at quasi-isomorphisms.

The correspondence tensor uses de Rham direct image, as in [40, §1.4]. A correspondence $Y \xleftarrow{a} Z \xrightarrow{b} W$ acts by $b_* a^!$. Cartesian composition supplies coherent base-change equivalences, compatible with external products. For an open embedding j, restriction $j^! = j^*$ is left adjoint to j_*.

Let $\mathsf{Fin}_{\mathrm{surj}}$ be the category of nonempty finite sets and surjections. A surjection $p : I \twoheadrightarrow J$ gives the diagonal

$$\Delta_p : X^J \longrightarrow X^I, \qquad (x_j)_{j\in J} \longmapsto (x_{p(i)})_{i\in I}.$$

The positive Ran coefficient category is

$$\mathcal{D}_X = \lim_{I\in \mathsf{Fin}_{\mathrm{surj}}} \mathcal{D}(X^I),$$

where the transition functor for p is $\Delta_p^!$. An object F consists of complexes F_I and coherent equivalences $\Delta_p^! F_I \simeq F_J$ [40, Definition 2.1.2]. The evaluations $F \mapsto F_I$ jointly detect equivalences.

Diagonal pullback defines a pointwise symmetric monoidal product:

$$(F \otimes^! G)_I = F_I \otimes^! G_I.$$

Its unit ω_{Ran} has components ω_{X^I}, the dualizing D-modules. Extraordinary pullback preserves these units and products.

A decomposition $d \in \mathrm{Dec}_r(I)$ determines the disjointness locus

$$U_d = \{(x_i) \in X^I : x_i \neq x_j \text{ whenever } d(i) \neq d(j)\}, \qquad j_d : U_d \hookrightarrow X^I.$$

Points in different blocks must be distinct. Points within one block may coincide.

For $F_1, \dots, F_r \in \mathcal{D}_X$, define the chart expression

$$\left(\bigotimes_{a=1}^{r,\mathrm{ch}} F_a\right)_I = \bigoplus_{d\in \mathrm{Dec}_r(I)} j_{d,*} j_d^! \left(\boxtimes_{a=1}^{r} (F_a)_{I_a}\right). \tag{10.1}$$

The external product uses $X^I = \prod_a X^{I_a}$. The sum retains every ordered decomposition, without dividing by permutations of the blocks. These charts describe the disjoint-union correspondence

$$\mathrm{Ran}(X) \times \mathrm{Ran}(X) \xleftarrow{j} (\mathrm{Ran}(X) \times \mathrm{Ran}(X))_{\mathrm{disj}} \xrightarrow{u} \mathrm{Ran}(X).$$

For $F, G \in \mathcal{D}_X$, the resulting chiral convolution is

$$F \otimes^{\mathrm{ch}} G = u_* j^!(F \boxtimes G), \tag{10.2}$$

with finite-chart expression (10.1) [40, §2.3.3 and Lemma 2.3.4].

The symbol u_* denotes de Rham direct image. An operation defined using extraordinary direct image $u_!$ would require its own domain and a comparison with this tensor. No such comparison is included in the definition. All aligned-projector and factorization statements below use (10.2).

These finite-chart constructions require finite sums and the indicated base changes. For a completed construction, compatibility with its inverse limits is an additional hypothesis. Preservation of arbitrary totalizations is not assumed.

Lemma 10.2 (Collision pullback). *Let $p : I \twoheadrightarrow J$ and $d \in \mathrm{Dec}_r(I)$. If d is nonconstant on a fiber of p, then $\Delta_p^{-1}(U_d)$ is empty. Otherwise there is a unique $\bar{d} \in \mathrm{Dec}_r(J)$ such that $d = \bar{d} \circ p$, and*

$$\Delta_p^{-1}(U_d) = U_{\bar{d}}.$$

Under base change, the summand of (10.1) *indexed by d therefore vanishes or becomes the summand indexed by $\bar{d}$ on X^J. These identifications are coherent under composition of surjections.*

Proof. If $p(i) = p(j)$ but $d(i) \neq d(j)$, the diagonal requires $x_i = x_j$, while U_d requires $x_i \neq x_j$. Their intersection is empty. If each fiber has one color, assign that color to its point in J. This defines $\bar{d}$, uniquely because p is surjective. Its fibers are nonempty because those of d are nonempty. The defining inequalities then give the asserted equality of opens.

For a surviving block, the restriction $p : I_a \twoheadrightarrow \bar{d}^{-1}(a)$ is surjective. The Ran compatibility of F_a identifies its diagonal pullback with $(F_a)_{\bar{d}^{-1}(a)}$. Base change for $j_{d,*}$ gives the claimed summand identification. Descent of a color through two successive quotients equals descent through their composite. Coherence of the correspondence functor supplies the same compatibility for these pullback equivalences. □

The lemma makes (10.1) an object of $\mathcal{D}_X$. Its associativity can also be seen on the charts. Expanding either $(F \otimes^{\mathrm{ch}} G) \otimes^{\mathrm{ch}} H$ or $F \otimes^{\mathrm{ch}} (G \otimes^{\mathrm{ch}} H)$ gives ordered divisions $I = I_1 \sqcup I_2 \sqcup I_3$ into nonempty blocks. In both expansions the open locus separates all three blocks, and the coefficient is the restriction of $F_{I_1} \boxtimes G_{I_2} \boxtimes H_{I_3}$.

Base change identifies the iterated direct images with the direct image from this common open locus. Every longer parenthesization uses the same ordered division into leaf blocks. The coherent correspondence functor therefore gives the associativity coherences in every arity. Permuting the block labels supplies the symmetry. Thus $\otimes^{\mathrm{ch}}$ is a symmetric monoidal product without a unit on the positive Ran category.

10.3 The geometric aligned projector

Fix $F, G, H, K \in \mathcal{D}_X$ and put

$$M = (F \otimes^{\mathrm{ch}} G) \otimes^{!} (H \otimes^{\mathrm{ch}} K), \qquad N = (F \otimes^{!} H) \otimes^{\mathrm{ch}} (G \otimes^{!} K).$$

For $d = (A, B) \in \mathrm{Dec}(I)$, write

$$E_d(F, G) = j_d^{!}(F_A \boxtimes G_B) \in \mathcal{D}(U_d).$$

For $d, e \in \mathrm{Dec}(I)$, set $W_{d,e} = U_d \cap U_e$. Let $\ell_d : W_{d,e} \hookrightarrow U_d$ and $\ell_e : W_{d,e} \hookrightarrow U_e$ be the inclusions, and let $k_{d,e} : W_{d,e} \hookrightarrow X^I$ be their common composite. Define

$$\mathcal{K}_{d,e} = \ell_d^{!} E_d(F, G) \otimes^{!} \ell_e^{!} E_e(H, K) \quad \text{in } \mathcal{D}(W_{d,e}).$$

Lemma 10.3 (The mixed correspondence). *There is a canonical equivalence*

$$M_I \simeq \bigoplus_{d,e \in \mathrm{Dec}(I)} k_{d,e,*} \mathcal{K}_{d,e}. \tag{10.3}$$

For $d = e = (A, B)$, its summand identifies with

$$j_{d,*} j_d^{!}\big((F_A \otimes^{!} H_A) \boxtimes (G_B \otimes^{!} K_B)\big).$$

Consequently the sum of the aligned summands is N_I.

Proof. Expand both chiral factors and distribute the pointwise tensor over their finite sums. Let $\delta_I : X^I \to X^I \times X^I$ be the diagonal. The summand indexed by (d, e) is

$$\delta_I^!(j_d \times j_e)_*\big(E_d(F, G) \boxtimes E_e(H, K)\big).$$

The fiber product of $j_d \times j_e$ with δ_I is $W_{d,e}$. Its base-change equivalence is exactly $k_{d,e,*}\mathcal{K}_{d,e}$. This proves (10.3) on the actual coefficient objects.

When $d = e$, the intersection is U_d. Compatibility of external and pointwise products rearranges the factors as $(F_A \otimes^! H_A) \boxtimes (G_B \otimes^! K_B)$ on that open locus. The middle exchange uses the symmetry of the fixed correspondence theory. Summing these identifications over d gives (10.1) for N_I. □

Definition 10.4 (Aligned correspondence)**.** The labeled mixed correspondence and its aligned subspace are

$$Y_I = \coprod_{d,e \in \mathrm{Dec}(I)} W_{d,e}, \qquad Y_I^{\mathrm{al}} = \coprod_{d \in \mathrm{Dec}(I)} U_d.$$

Let $a_I : Y_I^{\mathrm{al}} \hookrightarrow Y_I$ select the components with equal labels. Let $t_I : Y_I \to X^I$ restrict to $k_{d,e}$ on each component.

The inclusion a_I is finite and clopen because it selects components of a finite disjoint union. The functor $a_{I,*}a_I^!$ selects the aligned part of the object with components $\mathcal{K}_{d,e}$ on Y_I. Pushing its projection and inclusion along t_I gives maps on M_I. The clopen decomposition belongs to the labeled correspondence Y_I. Different labeled components can have the same open image in X^I.

Theorem 10.5 (Global aligned projector)**.** *There are natural maps in $\mathcal{D}_X$*

$$M \xrightarrow{q_{\mathrm{al}}} N \xrightarrow{s_{\mathrm{al}}} M, \qquad q_{\mathrm{al}} s_{\mathrm{al}} = \mathrm{id}_N .$$

The composite $e_{\mathrm{al}} = s_{\mathrm{al}} q_{\mathrm{al}}$ is the idempotent selecting the aligned summands of (10.3)*. These maps commute with collision pullback, permutations and ordered refinement. Repeated alignments give the same common-decomposition map under the canonical associativity identifications. The map q_{al} is an equivalence exactly when every off-aligned summand $k_{d,e,*}\mathcal{K}_{d,e}$ vanishes, for every I and $d \neq e$.*

Proof. On X^I, project to the aligned summands in (10.3), then use Lemma 10.3 to identify their sum with N_I. The inverse identification followed by inclusion defines $s_{\mathrm{al},I}$. The finite-sum splitting gives $q_{\mathrm{al},I} s_{\mathrm{al},I} = \mathrm{id}$.

Let $p : I \twoheadrightarrow J$ be a collision map. Lemma 10.2 describes every component of $Y_I \times_{X^I} X^J$. A component vanishes if either decomposition separates a fiber of p. Every surviving pair is uniquely $(\bar{d} \circ p, \bar{e} \circ p)$ with $\bar{d}, \bar{e} \in \mathrm{Dec}(J)$. Its pullback is $U_{\bar{d}} \cap U_{\bar{e}}$.

This gives a canonical isomorphism

$$Y_I \times_{X^I} X^J \simeq Y_J.$$

Surjectivity of p gives $\bar{d} \circ p = \bar{e} \circ p$ exactly when $\bar{d} = \bar{e}$. The same isomorphism therefore identifies the pullback of Y_I^{al} with Y_J^{al}.

These are Cartesian comparisons of the labeled correspondences, including their clopen inclusions. Base change for the coefficient objects makes their projections and inclusions commute with $\Delta_p^!$. For composed collisions, the unique descended labels and the Cartesian comparisons compose. Coherence of the correspondence functor makes these maps a transformation of the full Ran diagram. They consequently define the asserted maps in its limit $\mathcal{D}_X$.

For repeated pointwise products, each summand has one decomposition label for each factor. Every parenthesized alignment keeps precisely the tuples with all labels equal, as in Proposition 10.1.

For chiral refinement, expand each binary tree into its ordered leaf blocks. Equality at every vertex is equivalent to equality of the resulting leaf decompositions. Both routes thus use the same open locus, coefficient rearrangement

and finite-sum projection. Their graded permutations agree, and their pull–push identifications are coherent by Cartesian composition. Permutations simply relabel these constructions.

Finally, the fiber of $q_{\mathrm{al},I}$ is the sum of its off-aligned summands. It vanishes exactly when each summand vanishes, since each is a retract of that finite sum. The chart evaluations jointly detect equivalences, which proves the last assertion. □

Example 10.6 (Two disjoint points)**.** Take $I = \{1, 2\}$ and let $U = X^2 \setminus \Delta(X)$. The decompositions $d_+ = (\{1\}, \{2\})$ and $d_- = (\{2\}, \{1\})$ have the same disjointness locus U. For $F = G = H = K = \omega_{\mathrm{Ran}}$, restriction to U gives

$$M_I|_U \simeq \omega_U^{\oplus 4}, \qquad N_I|_U \simeq \omega_U^{\oplus 2}.$$

In the order $(+,+)$, $(+,-)$, $(-,+)$, $(-,-)$, the projection is

$$q_{\mathrm{al}}|_U = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad e_{\mathrm{al}}|_U = \mathrm{diag}(1, 0, 0, 1).$$

The two off-aligned copies survive on this open locus. Thus the aligned comparison is generally a split projection, even when its four inputs are the pointwise unit.

The same construction applies to r ordered blocks. Write $q_{\mathrm{al}}^{(r)}$ for the projection that pairs equal elements of $\mathrm{Dec}_r(I)$. Its target pairs the coefficient factors within each block.

To display its scalar signs, let V_a, W_a be coefficient complexes. Take block factors $V_a \otimes \omega_{X^{I_a}}$ and $W_a \otimes \omega_{X^{I_a}}$. For homogeneous coefficients of degrees p_a and q_a, the rearrangement contributes the sign

$$(-1)^{\sum_{a<b} q_a p_b}.$$

This is the sign for interleaving the two ordered lists of factors. The refinement argument in Theorem 10.5 applies in every arity.

10.4 Factorization coalgebras and pointwise products

A chiral commutative coalgebra is a noncounital commutative coalgebra object in $(\mathcal{D}_X, \otimes^{\mathrm{ch}})$. It has coherent coproducts $\delta_r : F \to \otimes^{\mathrm{ch}\,r}_{a=1} F$ for $r \geq 1$. For $d \in \mathrm{Dec}_r(I)$, projection to the d-summand and open adjunction give the map

$$j_d^! F_I \longrightarrow j_d^!\left(\boxtimes_{a=1}^{r} F_{I_a}\right). \tag{10.4}$$

The coalgebra is *factorizing* when these maps are equivalences for every I, r and d. Write $\mathrm{Fact}_D(X)$ for the full subcategory of such coalgebras. This is the positive convention of [40, §2.4].

Proposition 10.7 (Pointwise closure)**.** *The pointwise tensor product defines a symmetric monoidal structure on* $\mathrm{Fact}_D(X)$*. Its unit is* ω_{Ran} *with its canonical factorization coproducts.*

Proof. For chiral commutative coalgebras F and G, define the r-fold coproduct on $F \otimes^! G$ by

$$F \otimes^! G \xrightarrow{\delta_r^F \otimes^! \delta_r^G} \left(\otimes^{\mathrm{ch}}_{a=1}{}^{r} F\right) \otimes^! \left(\otimes^{\mathrm{ch}}_{a=1}{}^{r} G\right) \xrightarrow{q_{\mathrm{al}}^{(r)}} \otimes^{\mathrm{ch}}_{a=1}{}^{r} (F \otimes^! G).$$

The common-decomposition maps commute with refinements and permutations. Hence these maps satisfy the coherent commutative-coalgebra identities. Repeated pointwise products use the same common-decomposition projections, so their associativity and symmetry constraints are coalgebra maps.

On U_d, the d-component of this coproduct is the pointwise tensor of the two maps (10.4), followed by the graded rearrangement of external factors. If F and G factorize, every one of these maps is an equivalence. Their product therefore factorizes.

For the unit, the identifications $j_d^! \omega_{X^I} = \omega_{U_d} \simeq j_d^!(\boxtimes_a \omega_{X^{I_a}})$ give the component maps by adjunction. Their finite sum defines each coproduct on ω_{Ran}. The unit identifications are compatible with all correspondences, so these coproducts satisfy the same refinement identities. Under $F \otimes^! \omega_{\mathrm{Ran}} \simeq F$, the common-decomposition formula recovers each coproduct of F. Thus ω_{Ran} is the unit in the coalgebra category as well. □

The chiral product also has a finite-arity filtration. For $n \geq 1$, let $\mathcal{D}_{X,\leq n}$ be the same compatible-family category restricted to sets I with $1 \leq |I| \leq n$. Restriction gives

$$\mathcal{D}_X \simeq \lim_n \mathcal{D}_{X,\leq n}.$$

The chart formula (10.1) uses only smaller blocks, so the chiral products descend to these categories and restriction preserves them. Colimits of compatible families are computed chartwise: each diagonal pullback preserves colimits. Thus restriction preserves colimits as well.

Suppose $F \in \mathcal{D}_{X,\leq n}$ vanishes on sets of size less than n. For any $G \in \mathcal{D}_{X,\leq n}$, every binary decomposition of a set of size at most n has both blocks smaller than n. The chart formula therefore gives $F \otimes^{\mathrm{ch}} G = 0$. For $n = 1$ the product is zero because no binary decomposition exists.

A limit of stable presentable tensor categories without units, with these restriction and vanishing properties, is called a *pronilpotent tensor category* [55, Definition 14.10]. This arity filtration supplies the setting for the operadic Koszul equivalence of [55, Theorem 14.13]. It concerns the positive chiral product. The pointwise product has the nonzero unit ω_{Ran}.

Let $\Delta_{\mathrm{main},*} : \mathcal{D}(X) \to \mathcal{D}_X$ be the left adjoint to singleton evaluation, the direct image from the main diagonal $X \to \mathrm{Ran}(X)$ [40, §2.1.5]. An object is supported on this diagonal when it belongs to the essential image of that functor.

The chiral Chevalley functor computes Lie homology in $(\mathcal{D}_X, \otimes^{\mathrm{ch}})$. Francis–Gaitsgory's theorem identifies chiral Lie objects in this category with chiral commutative coalgebras [40, Theorems 1.2.4 and 5.1.1]. It restricts to factorization coalgebras exactly on Lie objects supported on the main diagonal $X \to \mathrm{Ran}(X)$ [40, Theorem 5.2.1]. The pointwise product constructed above permits an additional transverse associative multiplication on a factorization coalgebra.

10.5 The empty configuration and the two units

On the singleton chart, $F \otimes^{\mathrm{ch}} G$ is zero because $\mathrm{Dec}_2(\{1\})$ is empty. The positive chiral category therefore has no tensor unit: the singleton component of ω_{Ran} is nonzero. An empty-configuration component supplies a unit.

Adjoin this component by setting

$$\mathcal{D}_X^+ = \mathrm{Vect}_{\mathbb{C}} \times \mathcal{D}_X, \qquad \widehat{F} = (V, F),$$

where $\mathrm{Vect}_{\mathbb{C}}$ denotes complexes of complex vector spaces. The complex V is the coefficient of the empty configuration.

The unital chiral product is

$$(V, F) \otimes^{\mathrm{ch}} (W, G) = \big(V \otimes W, (V \otimes G) \oplus (W \otimes F) \oplus (F \otimes^{\mathrm{ch}} G)\big).$$

Here tensoring a complex with a Ran object uses its $\mathbb{C}$-linear structure. The first two nonempty components correspond to the two possible empty blocks. On each finite set, the expanded chart formula

sums over all maps $I \to [r]$, allowing empty fibers with their specified coefficients. Ordered refinements prove its associativity exactly as before.

The unit is $\mathbf{1}_{\mathrm{ch}} = (\mathbb{C}, \mathrm{o})$, as direct substitution shows. The pointwise product on $\mathcal{D}_X^+$ is instead

$$(V, F) \otimes^! (W, G) = (V \otimes W, F \otimes^! G), \qquad \mathbf{1}_! = (\mathbb{C}, \omega_{\mathrm{Ran}}).$$

The aligned construction extends to empty blocks: a surjection of finite sets still reflects equality of their block-label maps. The two units specify distinct targets for augmentations of the two associative structures.

On $\mathcal{D}_X^+$, both products preserve colimits separately in each variable. For an inverse system (F_n), passage to a completed tensor product requires the comparison

$$(\lim_n F_n) \otimes^? G \longrightarrow \lim_n (F_n \otimes^? G), \qquad ? \in \{!, \mathrm{ch}\},$$

to be an equivalence for the system in question. This limit compatibility is an additional hypothesis of the corresponding completed bar construction.

Chapter II

The two noncommutative chiral objects

Two operators can be composed at the same chiral support or on disjoint chiral configurations. The first composition can record their order along a transverse line. The second uses chiral convolution. Their domains differ even before associativity is imposed. We shall construct both products on one factorizing coefficient family and vary them independently.

II.1 The two products and their units

Retain the smooth complex curve X and the positive Ran category $\mathcal{D}_X$. Its empty-configuration extension is $\mathcal{D}_X^+$. Write $\star^{\mathrm{ch}}$ for the chiral product $\otimes^{\mathrm{ch}}$. The notation $\mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}(C)$ means associative unital algebra objects of a monoidal ∞-category C, with an algebra map to its tensor unit. Associativity includes its coherent higher homotopies.

Definition II.1 (Transversely ordered chiral algebra)**.** A transversely ordered chiral algebra is an object

$$A^{\perp} \in \mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}(\mathrm{Fact}_D(X), \otimes^{!}).$$

Its multiplication $m^{\perp} : A^{\perp} \otimes^{!} A^{\perp} \to A^{\perp}$ and its unit $\omega_{\mathrm{Ran}} \to A^{\perp}$ are maps of factorization coalgebras. We also call it an *ordered chiral algebra*.

The superscript $\perp$ records an independent oriented direction. When a field theory supplies locally constant fusion along that line, the order of transverse intervals determines the order of multiplication. The definition itself specifies the coefficient object and its maps.

Definition II.2 (Associative chiral algebra)**.** An associative chiral algebra is an object

$$\widehat{A}^{\mathrm{ch}}_{\mathsf{Ass}} \in \mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}(\mathcal{D}_X^+, \star^{\mathrm{ch}}).$$

For an augmentation whose empty component is the identity of $\mathbb{C}$, write $\widehat{A} = (\mathbb{C}, A)$. Its augmentation ideal is the positive object A, with an associative product $m^{\mathrm{ch}} : A \star^{\mathrm{ch}} A \to A$. The chiral unit is $\mathbf{1}_{\mathrm{ch}} = (\mathbb{C}, 0)$.

The commutator $m^{\mathrm{ch}} - m^{\mathrm{ch}} \circ \tau$, where τ is the graded symmetry, satisfies the chiral Jacobi identity. It gives a Lie object in $(\mathcal{D}_X, \star^{\mathrm{ch}})$ on the positive ideal. For strict differential graded products this follows by expanding Jacobi and cancelling the associative triple products with their Koszul signs. For coherent products it is the induced map from the Lie operad to the associative operad.

The pointwise unit in $\mathcal{D}_X^+$ is instead $\mathbf{1}_! = (\mathbb{C}, \omega_{\mathrm{Ran}})$. Thus a common object can have two augmentations with different targets and different ideals. No identification of those ideals is implicit.

11.2 A factorizing family supported at a marked point

Fix $x \in X(\mathbb{C})$ and a finite-dimensional complex vector space V in degree zero. Let $i_I : \{x^I\} \hookrightarrow X^I$ be the closed embedding of the point whose coordinates all equal x. Define

$$M(V)_I = i_{I,*}V, \qquad L(V) = M(V) \oplus \big(M(V) \star^{\mathrm{ch}} \omega_{\mathrm{Ran}}\big), \qquad N(V) = \omega_{\mathrm{Ran}} \oplus L(V).$$

Closed direct image is taken in the category of D-modules. In particular, $i_{I,*}V$ retains the normal derivatives at x^I. Proper base change equips $M(V)$ with its Ran collision equivalences.

The other two objects use finite sums on each chart.

On two distinct points, at most one point can equal x. Consequently two marked contributions cannot occur in two separated groups of points. This observation makes $N(V)$ factorize despite its description as a sum. General componentwise sums have no such property.

For a nonempty subset $S \subseteq I$, let $j_S : U_S \hookrightarrow X^I$ be the locus separating S from $I \setminus S$. If $S = I$, take $U_S = X^I$ and $j_S = \mathrm{id}$. Put

$$E_{I,S}(V) = j_{S,*}\, j_S^{!}\big(i_{S,*}V \boxtimes \omega_{X^{I\setminus S}}\big).$$

For $S = I$ this expression means $i_{I,*}V$. The chiral chart formula gives the decomposition

$$L(V)_I = \bigoplus_{\varnothing \neq S \subseteq I} E_{I,S}(V). \tag{11.1}$$

Before open direct image, its S-coefficient sets every S coordinate equal to x and requires every complementary coordinate to differ from x.

Lemma 11.3 (Marked-subset comparisons)**.** *Let $p : I \twoheadrightarrow J$ be a collision map. The pullback $\Delta_p^{!} E_{I,S}(V)$ vanishes unless $S = p^{-1}(S')$ for a nonempty subset $S' \subseteq J$. In that case it is canonically $E_{J,S'}(V)$. For finite-dimensional complex vector spaces V, W, there are natural equivalences*

$$L(V) \otimes^{!} L(W) \simeq L(V \otimes W). \tag{11.2}$$

Both comparisons commute with collisions, relabeling and repeated products.

Proof. If a fiber of p meets both S and its complement, the collision equates two coordinates that U_S separates. Its inverse image is empty, so open base change gives zero. Otherwise membership in S descends uniquely to J. Closed and open base change identify the surviving coefficient with $E_{J,S'}(V)$. Descent of a subset through successive surjections agrees with descent through their composite.

Expand the left side of (11.2) using (11.1) twice. The mixed-correspondence calculation of Lemma 10.3 places the (S, T) coefficient on $U_S \cap U_T$. If $S \neq T$, choose an index in one subset outside the other. The two closed coefficients force it and an index of the other subset to equal x. The latter subset's open condition separates them. Thus this coefficient is zero.

When $S = T$, the closed embedding i_S is the same in both factors. Apply diagonal base change to $i_S \times i_S$. It identifies their extraordinary tensor with $i_{S,*}(V \otimes W)$. The remaining factors are dualizing units. Open base change therefore gives $E_{I,S}(V \otimes W)$. Summing over S proves (11.2).

For any number of factors, the identical argument keeps precisely the tuples with all marked subsets equal. The coefficient there is the tensor product of all the vector spaces. The coherent correspondence functor identifies these maps for every parenthesization. Surjections reflect equality of their inverse-image subsets, which proves compatibility with collisions as well. □

Proposition 11.4 (The common factorizing carrier)**.** *The object $N(V)$ has a canonical structure in* $\mathrm{Fact}_D(X)$*. Its inclusion $\omega_{\mathrm{Ran}} \to N(V)$ and projection $N(V) \to \omega_{\mathrm{Ran}}$ are maps of factorization coalgebras. Suppose V has an associative multiplication $\mu : V \otimes V \to V$. Every $t \in \mathbb{C}$ then gives an augmented pointwise algebra on $N(V)$. Its ideal product on $L(V)$ is $tL(\mu)$ under* (11.2)*.*

Proof. Fix a separated decomposition $I = \bigsqcup_{a=1}^{r} I_a$. An S-coefficient of $L(V)_I$ survives restriction only if S lies in a single part I_a. Two different parts cannot both contain a coordinate equal to x. For $S \subseteq I_a$, base change identifies this restriction with the corresponding coefficient of

$$L(V)_{I_a} \boxtimes \boxtimes_{b \neq a} \omega_{X^{I_b}}.$$

All complementary coordinates in other parts differ from x already by separation from S. Hence these identifications exhaust the restricted sum (11.1).

Now expand $\boxtimes_a N(V)_{I_a}$. Every term containing two $L(V)$ factors vanishes on the separated open by the same argument. Its surviving terms are the dualizing unit and the single marked-part terms just identified. This gives the required factorization equivalence. Open adjunction supplies the coproduct components on X^I.

Under refinement, a surviving marked subset has a unique containing final part. Each iterated comparison uses this part and the same closed and open base-change maps. They therefore give coherent coproducts in all arities. Relabeling merely permutes the parts. The scalar summand is preserved by these maps, which proves the unit and augmentation assertions.

For the multiplication, use the ordinary unital extension of the possibly nonunital product $t\mu$ on the ideal $L(V)$. The comparisons (11.2) respect all iterated tensors, so associativity reduces to associativity of $t\mu$. On a separated open at most one part carries an ideal contribution. The same multiplication acts in that part, while the others have their scalar products. Thus the multiplication respects every factorization map and its refinements. □

For $V = M_2(\mathbb{C})$ with matrix multiplication, extraordinary pullback to the singleton x identifies this algebra with $\mathbb{C} \oplus M_2(\mathbb{C})$ and product

$$(a, v)(b, w) = (ab, aw + bv + t\,vw). \tag{11.3}$$

The unit is $(1, 0)$ and the augmentation is $(a, v) \mapsto a$. At $t = 0$ the ideal has square zero. For matrix units, the commutator of $(0, E_{12})$ and $(0, E_{21})$ is $(0, t(E_{11} - E_{22}))$, so the product is noncommutative for $t \neq 0$.

11.3 An independently variable chiral product

Write

$$P(V) = M(V) \star^{\mathrm{ch}} \omega_{\mathrm{Ran}}, \qquad N(V) = \omega_{\mathrm{Ran}} \oplus M(V) \oplus P(V).$$

The summands and their projections are Ran morphisms by Lemma 11.3. Let $q : N(V) \to M(V)$ and $\epsilon : N(V) \to \omega_{\mathrm{Ran}}$ be the indicated projections, and let $i : P(V) \to N(V)$ be inclusion. For $\lambda \in \mathbb{C}$, define

$$\mu_\lambda = \lambda\, i \circ (q \star^{\mathrm{ch}} \epsilon) : N(V) \star^{\mathrm{ch}} N(V) \longrightarrow N(V). \tag{11.4}$$

Theorem 11.5 (Independent products on one factorizing family)**.** *For every $t, \lambda \in \mathbb{C}$, the object $\widehat{N} = (\mathbb{C}, N(M_2(\mathbb{C})))$ carries both the pointwise algebra of Proposition 11.4 and an augmented chiral algebra whose positive product is* (11.4)*. Its factorization coalgebra is fixed as t and λ vary. Fixing either multiplication leaves the other parameter free.*

Proof. Since $qi = 0$ and $\epsilon i = 0$, both nested positive triple products of μ_λ are zero. In the left-associated product the outer q kills the inner image. In the right-associated product the outer ϵ kills it. This proves associativity. Adjoining the empty component gives its unital extension with unit $(\mathbb{C}, 0)$ and augmentation to that unit. The split projections make every nonunit product of length at least three vanish, giving the higher associative coherences as well.

The positive product is nonzero for $\lambda \neq 0$. On a two-point chart (x, y) with $y \neq x$, take the first input in $M(V)$ and the second in the positive dualizing summand. The corresponding ordered decomposition maps identically, multiplied by λ, to its summand of $P(V)$. Reversing these two inputs gives zero because q kills the dualizing summand. Thus this chiral product is also noncommutative when $\lambda \neq 0$ and $V \neq 0$.

The pointwise multiplication extends to the empty component by scalar multiplication. Its augmentation has target $(\mathbb{C}, \omega_{\mathrm{Ran}})$ and ideal $(0, L(V))$, whereas the chiral augmentation has ideal $(0, N(V))$. Neither formula for a multiplication uses the other parameter. The factorization maps in Proposition II.4 use only the marked subsets and V, so they remain fixed.

For each product, the zero parameter makes the augmentation ideal's multiplication zero, while a nonzero parameter gives a nonzero map. These properties are invariant under equivalence of augmented algebras. Thus the zero and nonzero choices are inequivalent in the respective algebra categories. □

II.4 Unitization and factorization

The marked-point construction has a general algebraic part: adjoining a unit. Its geometric factorization uses the additional support calculation.

Proposition II.6 (Split unitization)**.** *Let $(C, \otimes, \mathbf{1})$ be a stable unital monoidal category whose tensor preserves finite sums separately. Every nonunital associative algebra object M has an augmented unitization $\mathbf{1} \oplus M$. On the four summands of its tensor square, multiplication is the unit multiplication, the two unit actions, and the product of M. The unit is the inclusion of $\mathbf{1}$, and the augmentation is its projection. For any object M, choosing the zero nonunital algebra structure gives the square-zero unitization.*

Proof. For each arity r, expand $(\mathbf{1} \oplus M)^{\otimes r}$ over subsets of the r input positions. Remove unit factors by the monoidal unit constraints. On the remaining ordered M factors apply the given associative operation, and include its value in $M \subset \mathbf{1} \oplus M$. When the subset is empty, use the unit summand.

For a composite operation, removing the unit positions in stages gives the same ordered M positions as removing them at once. The associativity coherences of M and the monoidal unit coherences therefore supply all substitution and unit identities. Projection to $\mathbf{1}$ respects these operations, since every term with an M input has zero scalar component. □

This construction applies in the ambient coefficient categories. To obtain an object of $\mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}(\mathrm{Fact}_D(X), \otimes^!)$, one must also supply a factorization coalgebra on $\omega_{\mathrm{Ran}} \oplus M$. Its multiplication, unit and augmentation must be coalgebra maps. Componentwise direct sums alone need not admit this structure.

For example, $\omega_{\mathrm{Ran}} \oplus \omega_{\mathrm{Ran}}$ has two dualizing summands on both the singleton chart and every separated configuration chart. Factorization over r separated singleton blocks would require 2^r summands on the latter chart. For $r = 2$, the required rank is four, whereas the given rank is two. Thus the factorization equivalence already fails on two distinct points.

The marked-point family supplies the extra factorization maps on an actual common carrier and proves independence of the two products. A mixed distributive law additionally requires its full coherent data. The family uses a fixed point of X, so translation-equivariant field realizations require further structure.

11.5 The mixed compatibility equation

Independence leaves a separate question: when do the two products commute with one another? For a common object $\widehat{A} \in \mathcal{D}_X^+$, set

$$Q = (\widehat{A} \otimes^! \widehat{A}) \star^{\mathrm{ch}} (\widehat{A} \otimes^! \widehat{A}).$$

The aligned section constructed in Theorem 10.5, extended to empty blocks, is a map

$$s_{\mathrm{al}} : Q \longrightarrow (\widehat{A} \star^{\mathrm{ch}} \widehat{A}) \otimes^! (\widehat{A} \star^{\mathrm{ch}} \widehat{A}).$$

It allows two composites with the same source and target:

$$m^{\perp} \circ (m^{\mathrm{ch}} \otimes^! m^{\mathrm{ch}}) \circ s_{\mathrm{al}} \quad \text{and} \quad m^{\mathrm{ch}} \circ (m^{\perp} \star^{\mathrm{ch}} m^{\perp}) : Q \longrightarrow \widehat{A}. \tag{11.5}$$

Definition 11.7 (Mixed associative chiral datum)**.** A mixed associative chiral datum consists of a common object with the two unital associative products and a factorization coalgebra structure on its positive part. The pointwise multiplication and unit are coalgebra maps. It also includes a coherent aligned distributive law. The latter includes a homotopy between the maps (11.5), compatible with both units, every iterated product, relabeling and collision. Its higher homotopies are part of the datum.

A binary equation and the two triple-product faces check only the first coherence conditions. A construction by a mixed operad or by iterated bars must supply its remaining diagrams as well.

The family above detects a failure already on a unit slice of (11.5). Let u be the pointwise unit and choose a in a marked $M(V)$ coefficient. Let b lie in the positive dualizing coefficient at a separated point. Evaluate the two composites on $(a \otimes^! u) \star^{\mathrm{ch}} (u \otimes^! b)$. The first is zero because $m^{\mathrm{ch}}(u, b) = 0$. The second is $m^{\mathrm{ch}}(a, b)$, which is nonzero for $\lambda \neq 0$.

Extraordinary pullback to this configuration gives zero for the first map and $\lambda \, \mathrm{id}_V$ for the second, on V in degree zero. A homotopy therefore cannot equate them. This particular family with $\lambda \neq 0$ therefore fails the aligned mixed law. The calculation imposes no obstruction on other choices of carrier and operations.

11.6 Formal fields and their comparison problem

Choose a formal coordinate z. For a complex vector space V, a field on V is an element of $\mathrm{Hom}_{\mathbb{C}}(V, V((z)))$. Its pole bound may depend on the vector to which it is applied. It need not belong to $(\mathrm{End}_{\mathbb{C}} V)((z))$, where one bound would work on every vector.

Definition 11.8 (Nonlocal vertex algebra)**.** A nonlocal vertex algebra consists of a vector space V, a vector $\mathbf{1} \in V$, and a linear map

$$Y : V \longrightarrow \mathrm{Hom}_{\mathbb{C}}(V, V((z))), \qquad a \longmapsto Y(a, z),$$

with the vacuum and creation properties

$$Y(\mathbf{1}, z)a = a, \qquad Y(a, z)\mathbf{1} \in V[[z]], \qquad Y(a, z)\mathbf{1}\big|_{z=0} = a.$$

It also satisfies weak associativity: for every $a, b, c \in V$ there is $N \geq 0$ such that

$$(z + w)^N \iota_{z,w} Y(a, z + w)Y(b, w)c = (z + w)^N Y(Y(a, z)b, w)c. \tag{11.6}$$

Here $\iota_{z,w}$ expands in nonnegative powers of w relative to z. The factor is chosen so that the indicated substitution and equality of formal distributions are defined.

These are the axioms of [74, Definition 2.3]. Creation already makes $a \mapsto Y(a,z)$ injective. The translation operator $Ta = \partial_z Y(a,z)\mathbf{1}|_{z=0}$ satisfies

$$Y(a,z)\mathbf{1} = e^{zT}a, \qquad [T, Y(a,z)] = Y(Ta,z) = \partial_z Y(a,z)$$

by [74, Lemma 2.5]. Thus weak associativity is an axiom here. A reconstruction theorem starts with fields on a different space and proves these axioms for the space they generate.

For example, let B be a unital associative algebra with derivation T. The formula

$$Y(a,z)b = (e^{zT}a)b$$

defines a nonlocal vertex algebra with no poles. Indeed, e^{wT} preserves multiplication, and both sides of (11.6) with $N = 0$ equal $(e^{(z+w)T}a)(e^{wT}b)c$. The vacuum and creation properties follow from $T1 = 0$. Taking $B = M_2(\mathbb{C})$ and $T = 0$ gives noncommuting constant fields.

S-locality over $\mathbb{C}$ adds a finite exchange relation. For each a, b it requires vectors a_i, b_i, series $f_i(z) \in \mathbb{C}((z))$ and an integer $k \geq 0$ such that

$$(z-w)^k Y(a,z)Y(b,w) = (z-w)^k \sum_i \iota_{w,z} f_i(w-z) Y(b_i,w)Y(a_i,z). \tag{11.7}$$

A nonlocal vertex algebra with this property is a weak quantum vertex algebra [74, Remark 5.6], [75, §3]. The expansion convention in (11.7) is part of the identity. A chosen exchange operator must reproduce it.

There is also a formal deformation convention. Let $\hbar$ be a formal parameter and let $V = V_0[[\hbar]]$ be a topologically free $\mathbb{C}[[\hbar]]$-module. Here V_0 is a complex vector space. Write $V\widehat{\otimes}V = (V_0 \otimes V_0)[[\hbar]]$ and define

$$V_\hbar((z)) = \varprojlim_m (V/\hbar^m V)((z)).$$

Thus a Laurent series has a finite pole bound modulo each power of $\hbar$, with no common bound required as m grows.

A topological state–field correspondence consists of a continuous $\mathbb{C}[[\hbar]]$-linear map $Y : V\widehat{\otimes}V \to V_\hbar((z))$ and a $\mathbb{C}[[\hbar]]$-linear operator T. The map Y has the vacuum and creation properties above, and T satisfies

$$T\mathbf{1} = 0, \qquad \partial_z Y(z) = TY(z) - Y(z)(1 \otimes T) = Y(z)(T \otimes 1).$$

A braiding is a continuous $\mathbb{C}[[\hbar]]$-linear map

$$S(z) : V\widehat{\otimes}V \longrightarrow (V_0^{\otimes 2} \otimes_{\mathbb{C}} \mathbb{C}((z)))[[\hbar]], \qquad S(z) = 1 + O(\hbar).$$

Its topological locality condition is that, for every a, b and m, there is $k \geq 0$ such that for all c

$$\begin{aligned}(z-w)^k Y(z)(1 \otimes Y(w))\big(S(z-w)(a \otimes b) \otimes c\big) \\ = (z-w)^k Y(w)(1 \otimes Y(z))(b \otimes a \otimes c) \pmod{\hbar^m}.\end{aligned} \tag{11.8}$$

The Laurent coefficients are expanded in the order used by each field product, with the factor clearing the required poles.

A *topological quantum vertex algebra* in the convention of [33, Definitions 3.5 and 3.12] is this data satisfying (11.8) and weak associativity modulo every $\hbar^m$. The exponent in weak associativity may depend on a, b, c, m. This convention states associativity explicitly.

There is a useful sufficient exchange condition. Let S^{ij} act on tensor factors i, j, leaving the third fixed. If the topological state–field correspondence satisfies (11.8) and the exact hexagon identity

$$S(w)(Y(z)\otimes 1) = (Y(z)\otimes 1)S^{23}(w)\iota_{w,z}S^{13}(z+w), \tag{11.9}$$

then it is a topological quantum vertex algebra [33, Propositions 3.8 and 3.13].

Indeed, locality gives the quasi-associative identity with the extra factor $S^{23}(w)S^{13}(z+w)$. Substituting (11.9) puts that factor on both sides. Modulo each $\hbar^m$, one clears its poles and cancels its inverse, which exists because $S = 1 + O(\hbar)$. The resulting identity is weak associativity modulo $\hbar^m$.

The converse hexagon follows only after composition with the field map unless an additional injectivity statement is proved [33, Proposition 3.14]. Likewise, operator unitarity and the Yang–Baxter equation must be distinguished from the identities after evaluation by fields. Injectivity of $a \mapsto Y(a,z)$ alone does not give the required injectivity on tensor products of states.

These field axioms specify possible recipients of a formal-disc realization. The geometric-to-field map must supply the state space, vacuum and state–field operation, with the following compatibility.

Proposition 11.9 (Conditional formal-disc field realization)**.** *Suppose an associative chiral algebra has a specified formal-disc realization with state space V, vacuum $\mathbf{1}$ and field map Y. Assume this map satisfies the vacuum, creation and lower-truncation conditions of Definition 11.8. Suppose the comparison with geometric iterated products proves (11.6) for every triple of states. Then $(V, Y, \mathbf{1})$ is a nonlocal vertex algebra.*

In the topological convention above, a topologically free state–field system satisfying (11.8) and (11.9), with $S = 1 + O(\hbar)$, is a topological quantum vertex algebra. The stated translation, vacuum and completion conventions are part of this hypothesis.

Proof. The first assertion follows from the defining nonlocal vertex axioms. Creation also makes $a \mapsto Y(a,z)$ injective by evaluation on the vacuum at $z = 0$. The second assertion is the locality–hexagon implication proved above, using the inverse of S modulo each power of $\hbar$. These implications use the specified realization or state–field system as input. □

The condition $S = 1 + O(\hbar)$ and the $\hbar$-adic completions describe this formal deformation class. An arbitrary unitary rational Yang–Baxter operator does not include those data. Its comparison with this class must specify the deformation parameter, completion and exchange normalization.

Problem 11.10 (The reverse geometric construction)**.** For a given nonlocal or quantum vertex algebra, construct a corresponding formal-disc coefficient system. Fix the category of field systems and the geometric coefficient category, with their morphisms, supports and completions. Construct the coefficients on every finite power of the disc, their collision transitions, relabeling maps, factorization comparisons and associative chiral operations. Supply every required coherence, and identify the recovered state space, vacuum and field operations with the original field system. The empty chiral unit and the chosen vacuum require their own comparison.

An equivalence of categories further requires both functors and coherent natural equivalences from both composites to the respective identity functors. Determine the admissible field class on which these constructions exist, retaining the question for the full stated class. Weak associativity and injectivity of the single-state field map do not alone construct these geometric objects or inverse comparisons.

11.7 The geometry supplying each operation

A pole has more data than its residue. For a normal coordinate z, the successive coefficients of $\sum_{j\geq 1} a_j z^{-j}$ are its principal parts of successive orders. Keeping all these coefficients preserves every normal pole symbol. It does not specify a transverse ordering.

Ordered disjoint intervals in a line have contractible configuration chambers, one for each order. Their composition supplies the coherent associative operation when the theory is locally constant in that direction. Moving points in a plane also supplies braid paths. An exchange operation requires an assignment to those paths and its compatibility with fusion. Principal parts, interval composition and braided exchange therefore come from distinct supplied geometric data.

Chapter 12

Transversely ordered chiral algebras

Fuse two operators, then separate their chiral supports. Alternatively, separate first and fuse in each cluster. Transverse locality requires these operations to agree, including their associativity homotopies. We first express this condition on the coefficient family. We then construct its interval origin and specify the comparison needed to obtain algebraic coefficients from a field theory.

12.1 Locality on the coefficient family

By Definition 11.1, an ordered chiral algebra is an augmented E_1-algebra in $(\mathrm{Fact}_D(X), \otimes^!)$. Write A_I for its coefficient on X^I.

An operation of arity r is parametrized by the little-interval space $\mathsf{E}_1(r)$. Its component will be denoted by

$$\mu_{r,I}(\alpha) : A_I^{\otimes^! r} \longrightarrow A_I, \qquad \alpha \in \mathsf{E}_1(r).$$

The maps vary coherently with α and with operadic substitution. At arity zero, the source is the pointwise unit ω_{X^I}. Write $u_I : \omega_{X^I} \to A_I$ for this unit and $\epsilon_I : A_I \to \omega_{X^I}$ for the augmentation.

Let $d = (I_1, \dots, I_k)$ be an ordered decomposition of I. The open inclusion $j_d : U_d \hookrightarrow X^I$ separates its parts. The factorization structure supplies

$$\delta_d : j_d^! A_I \xrightarrow{\sim} j_d^!\left(\boxtimes_{a=1}^{k} A_{I_a}\right).$$

For $r = 2$ and $k = 2$, compatibility of the binary multiplication $m_I = \mu_{2,I}(\alpha)$ with separation is

$$\delta_d \circ j_d^! m_I = j_d^!(m_{I_1} \boxtimes m_{I_2}) \circ \sigma_d \circ (\delta_d \otimes^! \delta_d). \tag{12.1}$$

Here the tensor comparisons for $j_d^!$ are understood. The map σ_d groups the first cluster's two factors together, then the second cluster's factors.

For homogeneous coefficients, this regrouping reads

$$(a \otimes b) \otimes (c \otimes d) \longmapsto (-1)^{|b||c|}(a \otimes c) \otimes (b \otimes d).$$

Thus the product on two separated clusters is $(-1)^{|b||c|} ac \otimes bd$. If b and c are odd, omitting the middle crossing reverses its sign. The sign records the actual permutation of coefficients.

For general r, k, let $\sigma_{d,r}$ transpose the array of r inputs and k clusters, with its Koszul signs. The full locality equation is

$$\delta_d \circ j_d^! \mu_{r,I}(\alpha) = j_d^!\left(\boxtimes_{a=1}^{k} \mu_{r,I_a}(\alpha)\right) \circ \sigma_{d,r} \circ \delta_d^{\otimes^! r}. \tag{12.2}$$

The same parameter α acts in every cluster. For $r = 0$, this is the unit equation. There is also the corresponding equation for ϵ_I. These equations include homotopies compatible with every substitution, collision, permutation and refinement of d.

Theorem 12.1 (Coherent locality recognition)**.** *An augmented ordered chiral algebra is equivalent to a factorization family A with coherent augmented E_1-algebra structures on its coefficients. The collision equivalences and every δ_d must be equivalences of augmented E_1-algebras, coherently under composition and refinement. On the separated locus, this condition is precisely* (12.2) *with its unit, augmentation and higher homotopies.*

Proof. Evaluation on each finite power preserves the pointwise tensor and its unit. It therefore carries an internal E_1-algebra to the stated component algebras. Its operations are morphisms of factorization coalgebras, so they intertwine every collision and separation map. The tensor product of the component algebras uses $\sigma_{d,r}$, which gives (12.2).

Conversely, fix an arity and a parameter in $\mathsf{E}_1(r)$. The component operations assemble to a Ran morphism because they respect all collision equivalences. Equation (12.2) makes this morphism a map of factorization coalgebras. The prescribed coherent compatibilities perform the same assembly for the entire parameter space and every operadic substitution. They therefore give an E_1-action on A in $\mathrm{Fact}_D(X)$. The arity-zero operations and the maps ϵ_I assemble to its unit and augmentation.

A morphism between either kind of data is the same compatible family of component algebra maps, with its homotopies. Both constructions preserve these mapping spaces and undo each other. Hence they give an equivalence of the stated categories. □

In a strict differential graded model, strictly associative binary products and strict unit equations determine the iterated products. If all transition and factorization maps preserve them, induction proves every higher locality equation. For a general E_1-algebra, the higher homotopies are additional data. The binary square alone does not construct them. The marked family of Proposition 11.4 illustrates (12.2). Only one separated cluster can contain its marked ideal, and the same matrix product acts in that cluster before or after separation.

12.2 Fusion along an interval

Fix a symmetric monoidal coefficient ∞-category $\mathcal{V}$. Let $\mathcal{H}$ be a specified symmetric monoidal ∞-category of holomorphic factorization objects on X^{an}. Write $\odot$ for its tensor product and $\mathbf{1}_{\mathcal{H}}$ for its unit. Assume it is a full subcategory of $\mathcal{V}$-valued prefactorization objects with the pointwise tensor product. Thus $(B \odot C)(U) = B(U) \otimes C(U)$ on each chiral open U. The chosen holomorphic structure and this closure are hypotheses on $\mathcal{H}$.

Let $\mathbb{F}$ be a unital $\mathcal{V}$-valued factorization algebra on $X^{\mathrm{an}} \times \mathbb{R}$. For an open interval J, define its interval coefficient by

$$B_J(U) = \mathbb{F}(U \times J).$$

Require $B_J \in \mathcal{H}$ and require every inclusion of intervals to induce an equivalence in $\mathcal{H}$. For disjoint intervals $J_1 < \cdots < J_r$ inside J, factorization gives

$$B_{J_1} \odot \cdots \odot B_{J_r} \longrightarrow B_J. \tag{12.3}$$

The empty input supplies the unit map. A compatible augmentation of this interval family is a further map to $\mathbf{1}_{\mathcal{H}}$ that preserves every operation.

The rectangular maps in (12.3) respect chiral factorization. To see this, choose disjoint chiral opens $U_1, \dots, U_k \subset U$. The rectangles $U_a \times J_i$ form one disjoint family. Fusing first across the chiral opens and then across intervals, or in the other order, gives its same factorization map. The coefficient permutation is exactly $\sigma_{d,r}$. The field theory's composition homotopies give all refinements of this identity, so these are morphisms in $\mathcal{H}$.

The ordered configurations of r affine subintervals of $(0,1)$ form the convex region

$$0 < a_1 < b_1 < a_2 < b_2 < \cdots < a_r < b_r < 1.$$

Each other component is obtained by permuting the input labels. The contractibility of these components explains associativity: different nested fusions lie in the same ordered component. Their coherent paths, together with substitution, give the little-interval operations. They do not exchange the order of labels.

Proposition 12.2 (The transverse interval action)**.** *Under the preceding hypotheses, restriction to any fixed interval J_0 gives B_{J_0} a coherent unital E_1-algebra structure in $\mathcal{H}$. A compatible augmentation of the interval family makes this an augmented algebra. The construction is natural in the field theory and in its compatible maps.*

Proof. Consider the colored operad whose colors are open intervals in $\mathbb{R}$. Its operations are inclusions of disjoint intervals into one interval. The rectangular maps define an algebra over this operad in $\mathcal{H}$, by their compatibility with chiral factorization. Its unary interval inclusions act by equivalences.

The operadic localization theorem [80, Theorem 5.4.5.9] therefore identifies it with an E_1-algebra, using [80, Example 5.4.5.3] for $M = \mathbb{R}$. The underlying object is its value on J_0. This theorem retains all substitution homotopies. Choosing separate inverses to the unary maps would not by itself specify them. It also identifies the spaces of compatible maps. Consequently it transports an augmentation to the tensor unit and proves the asserted naturality. □

In degree-zero vector spaces, the construction reduces to ordered interval multiplication in [26, 2014 preprint, §3.2]. The unit comes from the empty input. An augmentation instead maps the resulting algebra to its scalar coefficient object and must be supplied separately.

12.3 Algebraization and product reconstruction

The interval action lives in $\mathcal{H}$. Fix a full subcategory $\mathcal{C}_D \subseteq \mathrm{Fact}_D(X)$ that contains ω_{Ran} and is closed under $\otimes^!$. It specifies the admitted algebraic coefficients.

To obtain such coefficients from the field theory, suppose there is a strong symmetric monoidal functor

$$\Phi : (\mathcal{H}, \odot, \mathbf{1}_{\mathcal{H}}) \longrightarrow (\mathcal{C}_D, \otimes^!, \omega_{\mathrm{Ran}}). \tag{12.4}$$

Strong monoidality includes coherent equivalences

$$\phi_r : \Phi(B)^{\otimes^! r} \xrightarrow{\sim} \Phi(B^{\odot r}), \qquad \phi_0 : \omega_{\mathrm{Ran}} \xrightarrow{\sim} \Phi(\mathbf{1}_{\mathcal{H}}).$$

Landing in $\mathrm{Fact}_D(X)$ requires actual collision and separated factorization maps on every finite power of X. The functor must preserve their coherent compositions.

Theorem 12.3 (Holomorphic–topological recognition under coefficient comparison)**.** *Let $\mathbb{F}$ satisfy the hypotheses of Proposition 12.2, with a compatible augmentation. Given the comparison (12.4), the coefficient $A = \Phi(B_{J_0})$ is an augmented ordered chiral algebra. Its multiplication is transverse interval fusion transported by Φ.*

Proof. Write μ_r^B for the interval operations on $B = B_{J_0}$. The transported operations, unit and augmentation are

$$\mu_r^A = \Phi(\mu_r^B) \circ \phi_r, \qquad u^A = \Phi(u^B) \circ \phi_0, \qquad \epsilon^A = \phi_0^{-1} \circ \Phi(\epsilon^B).$$

The monoidal coherence of Φ carries every substitution and unit diagram for B to the corresponding diagram for A. The same holds for the augmentation. All maps lie in $\mathrm{Fact}_D(X)$, so the locality equations and their higher homotopies follow from Theorem 12.1. □

A regular Riemann–Hilbert correspondence can enter the construction of Φ after the chosen analytic coefficients have been identified with the appropriate constructible complexes. One must then verify the tensor normalization, units, diagonal restrictions, external products and separated direct images. An algebraic target also requires an algebraic realization and the relevant regularity conditions. Holomorphic dependence of a field theory alone supplies none of these comparison maps. The distinction between analytic factorization data and algebraic Ran coefficients is already explicit in [26, 2014 preprint, §3.3.1].

The converse recognition question concerns reconstruction on the whole product space. Let $\mathcal{R}(W)$ be the category of finite disjoint unions of rectangles $U_a \times J_a$ contained in an open $W \subset X^{\mathrm{an}} \times \mathbb{R}$. Here each U_a is an analytic disc, each J_a an open interval, and morphisms are inclusions of unions. The empty union is allowed.

Assume the following colimits exist in $\mathcal{V}$. Call a theory *product-generated* when its canonical maps

$$\operatorname*{colim}_{\bigsqcup_a (U_a \times J_a) \in \mathcal{R}(W)} \bigotimes_a \mathbb{F}(U_a \times J_a) \longrightarrow \mathbb{F}(W) \tag{12.5}$$

are equivalences for every W. The colimit is derived.

Starting from an internal E_1-algebra, an inverse must first construct the diagram on the left of (12.5). Its maps must handle every rectangular inclusion, including families whose chiral projections overlap. One must prove their coherent compatibility and factorization descent, then identify both composites with the original objects.

Write $\mathrm{Fact}^{\mathrm{prod,aug}}_{\mathrm{hol,lc}}$ for the category of admitted augmented product-generated theories with interval coefficients in $\mathcal{H}$. If this product reconstruction is inverse to interval restriction, and Φ is a monoidal equivalence, the corresponding categories are equivalent:

$$\mathrm{Fact}^{\mathrm{prod,aug}}_{\mathrm{hol,lc}}(X^{\mathrm{an}} \times \mathbb{R}) \simeq \mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}(C_D, \otimes^!).$$

Without these comparisons, the proved construction is the forward interval action and its transport through (12.4). The general inverse remains a separate reconstruction problem.

Chapter 13

Block-ordered operations

Rectangular locality compares two compositions: row multiplication followed by column factorization, and factorization followed by multiplication. An operadic presentation must encode these comparisons together with their higher homotopies. A forest calculation has a further requirement: every edge contraction must act on the actual collision coefficients. We describe the operadic comparison first and then calculate its possible contraction differential.

13.1 Chiral operations and ordered profiles

An *ordered profile* is a list $(I_1, \dots, I_r)$ of disjoint nonempty finite label sets. Its coefficient space is

$$X^{I_1} \times \cdots \times X^{I_r}.$$

The list records the transverse order of the blocks. Diagonals inside a factor record collisions of its labels. These are different data: a profile specifies neither the coefficient maps for composing adjacent blocks nor the maps on the collision strata.

The chiral operations on a smooth curve already have a precise pseudotensor description. For a nonempty finite set I, let $j_I : U_I \hookrightarrow X^I$ be the inclusion of the locus of pairwise distinct points, and let $\Delta_I : X \hookrightarrow X^I$ be the small diagonal. For right D_X-modules M_i and N, the space of chiral operations is

$$P_I^{\mathrm{ch}}(\{M_i\}, N) = \mathrm{Hom}_{D_{X^I}}\left(j_{I*}j_I^* \boxtimes_{i\in I} M_i, \Delta_{I*}N\right). \tag{13.1}$$

Here the operations are those of the abelian category of right D-modules. Substitution along $J \twoheadrightarrow I$ first applies the operations on its fibers, on the open locus where different fibers are separated. The I-operation then composes their outputs. Diagonal base change identifies the intermediate targets, and composition of these maps is associative.

This is the chiral pseudotensor structure of Beilinson–Drinfeld [9, §§3.1.1–3.1.2]. Its nonempty operation spaces do not by themselves specify a nullary operation.

An operad acting in this pseudotensor category specifies additional algebraic identities. A presentation of the entire derived factorization category must also account for its collision transitions, separated-factorization equivalences and mapping spaces. In particular, the existence of (13.1) is distinct from a symmetric monoidal equivalence between that factorization category and algebras over a chosen infinity-operad.

13.2 The operadic exponential law

Let P and Q be small infinity-operads, and let $\mathcal{V}$ be a symmetric monoidal infinity-category. The category $\mathrm{Alg}_Q(\mathcal{V})$ has the *pointwise tensor*. It tensors underlying objects and applies each operation to both factors, using the symmetry to group the inputs. The tensor unit has its canonical Q-algebra

structure. This is the symmetric monoidal structure of [80, Construction 3.2.4.1, Proposition 3.2.4.3 and Example 3.2.4.4].

An operadic bifunctor is a functor

$$P^{\otimes} \times Q^{\otimes} \longrightarrow \mathcal{V}^{\otimes}$$

over the smash product of pointed finite sets, preserving inert morphisms in both variables. Its two families of operations satisfy interchange, including all coherent substitution relations. The Boardman–Vogt tensor product $P \otimes_{\mathrm{BV}} Q$ represents these bifunctors.

Theorem 13.1 (Operadic exponential law)**.** *Let P and Q be small infinity-operads and $\mathcal{V}$ a symmetric monoidal infinity-category. Equip $\mathrm{Alg}_Q(\mathcal{V})$ with its pointwise tensor. There is a natural equivalence*

$$\mathrm{Alg}_{P \otimes_{\mathrm{BV}} Q}(\mathcal{V}) \simeq \mathrm{Alg}_P\big(\mathrm{Alg}_Q(\mathcal{V})\big).$$

The equivalence identifies operation maps, their coherent substitutions and the full mapping spaces of algebras.

Proof. By the defining universal property of the operadic tensor product, the left side is the infinity-category of the displayed bifunctors [80, Definition 2.2.5.3]. Fixing the P-variable gives a Q-algebra. An operation in P becomes a map from the pointwise tensor of these Q-algebras to another such algebra. Compatibility with the Q-operations is exactly the bifunctor's interchange. Currying therefore gives a P-algebra in $\mathrm{Alg}_Q(\mathcal{V})$.

Conversely, evaluation of a P-algebra in this pointwise algebra category gives the bifunctor. The algebra-category construction in [80, Construction 3.2.4.1] characterizes these two operations on every simplicial parameter space. Currying and evaluation are inverse there and preserve inert maps by Proposition 3.2.4.3. They therefore identify transformations and all their higher homotopies as well as objects, proving the asserted equivalence of infinity-categories. □

To apply this law to the preceding chapter, one needs a specified symmetric monoidal equivalence

$$\Phi : (\mathrm{Fact}_D(X), \otimes^{!}) \xrightarrow{\sim} \big(\mathrm{Alg}_{Q_X}(\mathcal{V}), \otimes_{\mathrm{pt}}\big). \tag{13.2}$$

Here Q_X is the chosen infinity-operad, and $\otimes_{\mathrm{pt}}$ denotes the pointwise tensor just defined. The equivalence must identify the tensor units and the specified augmentation targets. Applying Φ to every ordered operation and then using Theorem 13.1 gives

$$\mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}(\mathrm{Fact}_D(X), \otimes^{!}) \simeq \mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1 \otimes_{\mathrm{BV}} Q_X}(\mathcal{V}). \tag{13.3}$$

On the right, augmentation means a map to the algebra corresponding to the tensor unit on the left. Applying a monoidal inverse to Φ gives the inverse equivalence. Thus (13.3) is conditional on the full comparison (13.2), not merely on the agreement of its objects or its binary operation spaces. Under that hypothesis one may write $\mathsf{OrdCh}_X = \mathsf{E}_1 \otimes_{\mathrm{BV}} Q_X$. The intrinsic ordered algebra does not require this presentation.

13.3 The sign of a mixed contraction

Consider a forest F with one transverse edge e and one chiral edge f, oriented by $e \wedge f$. Write F/e for the forest obtained by contracting e. The degree-zero map c_e^F takes coefficients of F to those of F/e.

Use the same notation for the other contractions. For a coefficient x of degree zero, the contraction boundary is

$$\partial(x \otimes e \wedge f) = c_e^F x \otimes f - c_f^F x \otimes e.$$

The second boundary is

$$\partial^2(x \otimes e \wedge f) = \big(c_f^{F/e} c_e^F - c_e^{F/f} c_f^F\big)x. \tag{13.4}$$

The square vanishes when its two coefficient composites agree. Its sign comes from the two edge orientations. Its commutativity is a separate property of the coefficients.

We now give the general construction with the grading included. Work over a field k of characteristic zero and use cohomological differentials of degree one. Let $\mathcal{F}$ be a collection of forests closed under the chosen edge contractions. For each $F \in \mathcal{F}$, its finite internal-edge set is a disjoint union

$$E(F) = E_\perp(F) \sqcup E_{\mathrm{ch}}(F).$$

The two names distinguish directions of contraction, rather than the object colors of a colored operad. Contracting e gives F/e with the remaining named edges and their directions unchanged. Require the two targets $F/e/f$ and $F/f/e$ to be identified for every pair of distinct edges.

Definition 13.2 (Coefficient contraction system)**.** A coefficient contraction system assigns a cochain complex $C(F)$ to each forest. Write d_C for its differential. Each edge contraction has a degree-zero cochain map

$$c_e^F : C(F) \longrightarrow C(F/e).$$

For distinct edges these maps satisfy, under the specified target identifications,

$$c_f^{F/e} c_e^F = c_e^{F/f} c_f^F \qquad (e \neq f). \tag{13.5}$$

The identifications are required to agree for longer contraction sequences.

Put the orientation line

$$\mathrm{or}(F) = \det(kE_\perp(F)) \otimes \det(kE_{\mathrm{ch}}(F))$$

in degree $-|E(F)|$. Its product orientation places all transverse edges before all chiral edges. Form the total graded space

$$K^n = \bigoplus_{F \in \mathcal{F}} C^{n+|E(F)|}(F) \otimes \mathrm{or}(F). \tag{13.6}$$

The direct sum uses finite support, with no implicit completion. For an ordered wedge $e_1 \wedge \cdots \wedge e_N$, define exterior contraction by

$$\iota_{e_i}(e_1 \wedge \cdots \wedge e_N) = (-1)^{i-1} e_1 \wedge \cdots \widehat{e_i} \cdots \wedge e_N.$$

This determines a map of determinant lines independently of the displayed basis order.

For homogeneous $x \in C^q(F)$ and $\omega \in \mathrm{or}(F)$, define

$$d_{\mathrm{int}}(x \otimes \omega) = d_C x \otimes \omega, \tag{13.7}$$

$$d_\perp(x \otimes \omega) = (-1)^q \sum_{e \in E_\perp(F)} c_e^F x \otimes \iota_e \omega, \tag{13.8}$$

$$d_{\mathrm{ch}}(x \otimes \omega) = (-1)^q \sum_{f \in E_{\mathrm{ch}}(F)} c_f^F x \otimes \iota_f \omega. \tag{13.9}$$

All three maps have total degree one. In the product orientation the chiral-edge sign includes $(-1)^{|E_\perp(F)|}$, because its contraction passes the transverse determinant factor.

Theorem 13.3 (Two-color forest differential)**.** *A coefficient contraction system gives a differential*

$$d = d_{\mathrm{int}} + d_\perp + d_{\mathrm{ch}}$$

on (13.6)*. Its summands satisfy*

$$d_\perp^2 = d_{\mathrm{ch}}^2 = 0, \qquad d_\perp d_{\mathrm{ch}} + d_{\mathrm{ch}} d_\perp = 0, \qquad d_{\mathrm{int}} d_\alpha + d_\alpha d_{\mathrm{int}} = 0 \quad (\alpha = \perp, \mathrm{ch}).$$

Proof. For distinct edges, exterior contractions anticommute:

$$\iota_f \iota_e = -\iota_e \iota_f.$$

The two coefficient composites agree by (13.5). Contracting a fixed pair in the two orders therefore gives opposite terms. Pairing all such terms proves the same-direction squares and the mixed anticommutator. Contracting an edge twice is impossible.

For the internal differential and contraction of e, the two coefficient terms are $d_C c_e x$ and $c_e d_C x$. They agree because c_e is a cochain map. Their totalization signs are $(-1)^q$ and $(-1)^{q+1}$, so they cancel. Finally $d_{\mathrm{int}}^2 = 0$ follows from the coefficient differential. Each forest has finitely many edges, so these calculations define maps on the direct sum and prove $d^2 = 0$ there. □

For the constant coefficient system $C(F) = k$, take $c_e = c_f = \mathrm{id}$. Formula (13.4) is $1 - 1 = 0$. Giving the chiral contraction its sign within E_{ch} alone would instead give $1 + 1 = 2$. Conversely, with the correct orientation but noncommuting coefficient maps, (13.4) is their nonzero commutator. Thus the orientation calculation neither supplies the coefficient square nor replaces its required homotopy in a coherent model.

13.4 Resolution of an operadic model

A Boardman–Vogt resolution carries more data than the contraction complex above. Let $\mathcal{M}$ be a cofibrantly generated symmetric monoidal model category with cofibrant tensor unit $\mathbf{1}_{\mathcal{M}}$. Fix a set of object colors C.

An *interval object* H has endpoints $0, 1 : \mathbf{1}_{\mathcal{M}} \to H$, a weak equivalence $\epsilon : H \to \mathbf{1}_{\mathcal{M}}$, and an associative operation $\vee : H \otimes H \to H$. The element 0 is a unit for $\vee$, and 1 is absorbing. The map ϵ preserves this operation, and both endpoint composites with ϵ are the identity. The endpoint map $(0, 1) : \mathbf{1}_{\mathcal{M}} \sqcup \mathbf{1}_{\mathcal{M}} \to H$ is required to be a cofibration. The usual interval $[0, 1]$ with maximum is the topological example.

Let R be a C-colored operad in $\mathcal{M}$. It is *well pointed* when each operadic unit is a cofibration. It is Σ*-cofibrant* when its underlying collection is cofibrant with its input-permutation actions.

The construction $W(H, R)$ uses trees with R-operations at vertices and lengths in H on internal edges. It identifies a zero-length edge with its operadic contraction. Removing a vertex labeled by an operadic unit combines two adjacent internal lengths by $\vee$, in their order along the tree. If one adjacent edge is external, it removes the other length by ϵ. Tree symmetries give further identifications, and grafting introduces edges of length 1.

The full interval object and these identifications are part of the construction. In a cellular interval model, the boundary of an interval edge has both endpoint faces.

For well-pointed Σ-cofibrant R, the operadic augmentation

$$\epsilon_R : W(H, R) \longrightarrow R \tag{13.10}$$

is a weak equivalence from a cofibrant operad. This is Berger–Moerdijk's resolution theorem [13, Theorem 3.5]. Its uncolored construction and proof are given in [12, §§4–5]. Here weak equivalence is tested on every operation object. Cofibrancy means the lifting property against maps that are trivial fibrations on those objects. The resolution theorem does not require a transferred model structure on all colored operads.

An operad is *admissible* if its algebras have the transferred model structure. Its weak equivalences and fibrations are detected on the underlying objects. Suppose in addition that $\mathcal{M}$ is left proper and that both operads in (13.10) are admissible, well pointed and Σ-cofibrant. The comparison theorem of [13, Theorem 4.1] then makes extension and restriction along ϵ_R a Quillen equivalence. After inverting weak equivalences, it identifies algebras over the resolution with algebras over R.

To model ordered chiral algebras by this route, one must supply (13.2). A second comparison must identify R-algebras with $(\mathsf{E}_1 \otimes_{\mathrm{BV}} Q_X)$-algebras and preserve the specified augmentation. The derived change-of-operad functor, that comparison, and (13.3) then compose to the required equivalence. A concrete forest description must further identify its collision coefficients, substitution maps, interval faces and unit relations with this resolution. Theorem 13.3 proves the contraction identity once its coefficient maps are given. Constructing those maps and the comparison is the geometric problem that remains.

Book III

Bar, cobar, and the boundary state

Chapter 14

Three bar constructions

A bar construction records a multiplication through its strings of composable inputs. Changing the multiplication changes the bar, even when the underlying coefficient object stays the same. Transverse composition uses the pointwise tensor, associative chirality uses chiral convolution, and a chiral Lie bracket gives Chevalley chains. We specify these three constructions and then construct the comparison that exists on associative envelopes.

14.1 The products and their units

Recall the positive Ran coefficient category $\mathcal{D}_X$ and its extension $\mathcal{D}_X^+ = \mathrm{Vect}_{\mathbb{C}} \times \mathcal{D}_X$ by an empty-configuration coefficient. The pointwise tensor on $\mathcal{D}_X$ has unit

$$\mathbf{1}_\perp = \omega_{\mathrm{Ran}}.$$

The unital chiral convolution $\star^{\mathrm{ch}} = \otimes^{\mathrm{ch}}$ on $\mathcal{D}_X^+$ has the different unit

$$K = \mathbf{1}_{\mathrm{ch}} = (\mathbb{C}, \mathrm{o}).$$

The positive chiral product on $\mathcal{D}_X$ has no unit. Its r-fold product vanishes on the chart X^I when $r > |I|$, because I has no decomposition into r nonempty blocks.

Let $A^\perp$ be an associative algebra for $\otimes^!$, with augmentation to $\mathbf{1}_\perp$. An ordered chiral algebra supplies such an object after forgetting its factorization structure. Let A^{ch} be an associative algebra for $\star^{\mathrm{ch}}$, with augmentation to K. Finally, let L be a Lie object for the positive chiral product. These are three specified inputs, with no identification between their operations assumed.

For an augmented associative algebra $A \to \mathbf{1}$ in a monoidal category $(C, \odot, \mathbf{1})$, its two-sided simplicial bar has terms

$$B_n(\mathbf{1}, A, \mathbf{1}) = \mathbf{1} \odot A^{\odot n} \odot \mathbf{1}.$$

Its interior faces multiply adjacent entries. Its two end faces apply the augmentation, and its degeneracies insert the algebra unit. The coherent associative and augmentation identities make these maps a simplicial object. When its realization exists, write

$$B_\odot A = |B_\bullet(\mathbf{1}, A, \mathbf{1})| = \mathbf{1} \odot^{\mathbb{L}}_A \mathbf{1}.$$

Use cohomological grading and let $sV = V[1]$, so $|sa| = |a| - 1$. Choose a differential graded presentation in which the tensors compute the indicated derived tensors and direct-sum normalization computes realization. Write d_L for the differential of the chosen presentation of L.

For a positive object V, $\mathrm{Sym}^r_{\mathrm{ch}}(V)$ means the coinvariants of $V^{\star^{\mathrm{ch}} r}$ under the signed permutation action. Characteristic zero makes these finite-group coinvariants exact. Write $\mathrm{Sym}_{\mathrm{ch}}(V) = K \oplus \bigoplus_{r \geq 1} \mathrm{Sym}^r_{\mathrm{ch}}(V)$.

On $\mathrm{Sym}_{\mathrm{ch}}(sL)$, the coproduct is the unshuffle. Choose a subset of inputs for the first factor and place its complement in the second, with the suspended Koszul sign. Let Q be the coderivation with $Q|_K = 0$ and components to the cogenerators

$$Q_1(sx) = -s(d_L x), \qquad Q_2(sx\, sy) = (-1)^{|x|} s[x, y]. \tag{14.1}$$

All components with more than two inputs are zero. The second formula is symmetric on the suspended inputs by graded antisymmetry of the bracket.

Definition 14.1 (Transverse, associative-chiral, and Lie-chiral bars)**.** The two associative bars are

$$\mathrm{Bar}^{\perp}(A^{\perp}) = \mathbf{1}_{\perp} \otimes^{\mathbb{L},!}_{A^{\perp}} \mathbf{1}_{\perp},$$
$$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A^{\mathrm{ch}}) = K \star^{\mathrm{ch}\,\mathbb{L}}_{A^{\mathrm{ch}}} K.$$

Their ambient coefficient categories are respectively $\mathcal{D}_X$ with $\otimes^!$ and $\mathcal{D}^+_X$ with $\star^{\mathrm{ch}}$. The Lie-chiral bar is the noncounital chiral Chevalley coalgebra

$$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}(L) = \mathrm{CE}_{\mathrm{ch}}(L) = \left(\bigoplus_{r \geq 1} \mathrm{Sym}^r_{\mathrm{ch}}(L[1]), Q \right).$$

For comparison with a counital associative bar, put

$$\mathrm{CE}^+_{\mathrm{ch}}(L) = K \oplus \mathrm{CE}_{\mathrm{ch}}(L).$$

All these sums are direct sums. On each finite configuration the positive chiral sums have only finitely many nonzero terms. Pointwise tensor powers have no corresponding configuration bound. Any product completion of a transverse bar therefore requires its own convergence and continuity hypotheses.

14.2 Normalization and the differential

The augmentation is split by the unit. Write d_A for the coefficient differential and $\bar{A}$ for the augmentation kernel in this presentation. Write

$$[a_1 | \cdots | a_n] = s a_1 \odot \cdots \odot s a_n.$$

The empty word denotes $\mathbf{1}$.

On the tensor coalgebra

$$T^c_{\odot}(s\bar{A}) = \mathbf{1} \oplus \bigoplus_{n \geq 1} (s\bar{A})^{\odot n},$$

the coproduct cuts a word at every position. The differential is the coderivation whose components to words of length one are

$$b_1(sa) = -s(d_A a), \qquad b_2(sa \odot sb) = (-1)^{|a|} s(ab). \tag{14.2}$$

Inserting either component after a prefix gives the Koszul sign of that prefix of suspended factors. Thus multiplication in slots $i, i+1$ has sign

$$(-1)^{\sum_{j<i}(|a_j|-1)+|a_i|}.$$

Theorem 14.2 (Typed bar theorem)**.** *Under the stated differential graded presentation hypotheses, the two associative bars are represented in their respective ambient coefficient categories by*

$$\mathrm{Bar}^{\perp}(A^{\perp}) \simeq \big(T^{c}_{\otimes^!}(s\bar{A}^{\perp}), b_1 + b_2\big), \qquad \mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A^{\mathrm{ch}}) \simeq \big(T^{c}_{\star^{\mathrm{ch}}}(s\bar{A}^{\mathrm{ch}}), b_1 + b_2\big).$$

They are coaugmented counital coassociative coalgebras with deconcatenation. The Lie-chiral bar is the noncounital cocommutative coalgebra with differential (14.1)*. Each differential squares to zero.*

Proof. In the normalized associative bar, a word containing a unit is degenerate. Splitting the augmentation therefore leaves exactly $\bar{A}^{\odot n}$ in simplicial degree n. Totalization suspends every factor. With the convention (14.2), the two end faces vanish on these reduced words and the interior faces give b_2.

The square of b_1 is induced by d^2_A. The relation $b_1b_2 + b_2b_1 = 0$ is the graded Leibniz identity. In b_2^2, contractions of disjoint adjacent pairs cancel with their Koszul signs. The remaining terms contract three consecutive factors in the two orders, and cancel by associativity. For degree-zero inputs this last identity is already visible in

$$b_2[a|b|c] = [ab|c] - [a|bc], \qquad b_2^2[a|b|c] = [(ab)c] - [a(bc)] = 0.$$

Deconcatenation is coassociative because successive cuts enumerate the same three subwords. A contraction and a surviving cut act on one side of the cut, so the differential is a coderivation. The empty word supplies coaugmentation and counit.

For the Chevalley coalgebra, the unary, binary and ternary components of Q^2 are respectively d^2_L, the derivation identity and graded Jacobi. They vanish. All longer components are determined by the coderivation identity. This proves the Lie assertion with its unshuffle coproduct. □

Normalization here takes place in the stable ambient coefficients. Its individual suspended summands need not be factorization objects. The realization criterion below addresses that additional structure on the whole bar.

The positive Chevalley construction is the functor in Francis–Gaitsgory's Lie–commutative chiral duality [40, §4.3 and Theorem 5.1.1]. It factorizes exactly when L is supported on the main diagonal $X \to \mathrm{Ran}(X)$ [40, Theorem 5.2.1]. The equivalence thus identifies a diagonally supported chiral Lie object with its factorization coalgebra. An additional associative multiplication or a second associative algebra is separate data.

14.3 The bar of an associative envelope

Let $U^{\mathsf{Ass}}_{\mathrm{ch}}(L)$ denote the derived associative envelope of L in $(\mathcal{D}^{+}_X, \star^{\mathrm{ch}}, K)$. Among augmented algebras with positive augmentation ideal, it is left adjoint to taking that ideal's commutator Lie object. Its augmentation kills L, and its augmentation ideal has zero empty-configuration component. In a cofibrant strict presentation, its defining relation is

$$xy - (-1)^{|x||y|}yx = [x, y].$$

This is an associative envelope for the chiral tensor. The chiral envelope from star-Lie objects to chiral Lie objects in [40, §6] is a different functor.

Define a map $\kappa : \mathrm{CE}^{+}_{\mathrm{ch}}(L) \to \mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(U^{\mathsf{Ass}}_{\mathrm{ch}}(L))$ by the suspended permutation sum

$$\kappa(sx_1 \cdots sx_r) = \sum_{\sigma \in S_r} \varepsilon(\sigma; sx_1, \dots, sx_r)[x_{\sigma(1)}| \cdots |x_{\sigma(r)}], \tag{14.3}$$

where ε is the Koszul sign of the displayed permutation. Set $\kappa|_K = \mathrm{id}_K$. The sum has no factorial denominator. For degree-zero x, y, its first nontrivial component and its reduced coproduct are

$$\kappa(sx\,sy) = [x|y] - [y|x],$$
$$\overline{\Delta}\kappa(sx\,sy) = [x] \star^{\mathrm{ch}} [y] - [y] \star^{\mathrm{ch}} [x].$$

The same expression is obtained by applying $\kappa \star^{\mathrm{ch}} \kappa$ to the unshuffle coproduct. Dividing only the length-two component by 2 halves $\overline{\Delta}\kappa$ but leaves $(\kappa \star^{\mathrm{ch}} \kappa)\overline{\Delta}$ unchanged.

Theorem 14.3 (Filtered chiral PBW comparison)**.** *For a positive chiral Lie object L, the map* (14.3) *gives a natural equivalence of coaugmented counital coassociative coalgebras*

$$\kappa : \mathrm{CE}^{+}_{\mathrm{ch}}(L) \xrightarrow{\sim} \mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}\big(U^{\mathsf{Ass}}_{\mathrm{ch}}(L)\big).$$

The source has its unshuffle coproduct, and the target has deconcatenation. The comparison preserves the filtration by the total number of original L-inputs. This filtration is finite on every configuration chart.

Proof. Cutting a permutation in (14.3) amounts to choosing a subset of its inputs, ordering that subset and ordering its complement. These are exactly the terms of the unshuffle coproduct followed by $\kappa \star^{\mathrm{ch}} \kappa$, with the same signs. Hence κ preserves coproducts, coaugmentation and counit.

For two homogeneous inputs, the differential identity is

$$b_2\kappa(sx\,sy) = (-1)^{|x|}s\big(xy - (-1)^{|x||y|}yx\big) = (-1)^{|x|}s[x, y].$$

In any length, pair the permutations in which the two factors being multiplied occur in opposite adjacent orders. Their difference is the displayed bracket relation. Summing over the pairs gives the bracket part of the chain-map equation. The unary parts commute by naturality of the internal differential. Thus κ is a chain coalgebra map.

Filter the envelope by the number of original L-factors. The operadic PBW identity identifies its associated graded with $\mathrm{Sym}_{\mathrm{ch}}(L)$. The required identity is the freeness of the associative operad as a right Lie-module, with symmetric operations as generators [36, Theorem 4.2 and Proposition 4.4]. It is an identity of multilinear operations over a characteristic-zero field, so it can be evaluated using the exact, colimit-preserving chiral tensor. A free simplicial Lie resolution computes the derived envelope. Realization preserves this identity because finite tensor powers and finite symmetric-group coinvariants preserve geometric realizations.

Give the Chevalley source its symmetric-degree filtration and the bar target the sum of the PBW degrees of its factors. The associated graded comparison is

$$\mathrm{Sym}_{\mathrm{ch}}(sL) \longrightarrow B_{\star^{\mathrm{ch}}}(\mathrm{Sym}_{\mathrm{ch}} L), \tag{14.4}$$

with only the internal differential on the left.

To check this comparison, put $S = \mathrm{Sym}_{\mathrm{ch}} L$ and consider the free Koszul resolution

$$S \star^{\mathrm{ch}} \mathrm{Sym}_{\mathrm{ch}}(sL).$$

In addition to its internal differential, give it the derivation $d_K(sx) = -x$, $d_K(x) = 0$. The degree-minus-one derivation $h(x) = -sx$, $h(sx) = 0$ satisfies

$$d_K h + h d_K = N,$$

where N counts total polynomial weight. The internal differential anticommutes with h. Thus h/N contracts every positive-weight part, and the augmentation has homology K.

The filtration by suspended degree has subquotients induced from coefficient objects by the free S-module functor. On each configuration this filtration is finite. The chosen presentation computes the tensors in these

subquotients, so this resolution computes the derived tensor with K. The normalized free bar resolution does so as well, by its extra-degeneracy contraction.

Use the sign convention in which the bar resolution's action term sends $1 \otimes [x]$ to $-x$. Extending (14.3) S-linearly then gives the augmentation-preserving comparison between the resolutions. Its action terms match $d_K(sx) = -x$, and its interior terms cancel in adjacent pairs by graded commutativity. Tensoring this comparison with K proves (14.4).

On the chart X^I, a term of PBW weight r contains r positive original inputs and vanishes for $r > |I|$. Every filtration just used is therefore finite on that chart. The cone of κ has acyclic associated graded. Starting at weight zero and successively extending by its graded pieces proves that the cone is acyclic. The chart evaluations are jointly conservative, which proves the claimed equivalence. □

The factorial in ordinary PBW symmetrization and its absence in (14.3) have different purposes. Averaging identifies symmetric powers with the associated graded envelope. The unnormalized sum intertwines unshuffle with deconcatenation.

For comparison, let k be a field of characteristic zero and let $A = k[x]$ with augmentation $x \mapsto 0$. Its augmentation ideal, regarded only as a Lie algebra, is abelian. The Chevalley class $s(x^2)$ is nonzero. Its image in the associative bar is $[x^2] = b_2[x|x]$, a boundary. Thus replacing the envelope hypothesis by the commutator Lie algebra of an arbitrary associative algebra changes the comparison.

Definition 14.4 (PBW-admissible completion)**.** Consider a chiral Lie object and its envelope in a specified completed tensor category. Call this datum PBW-admissible when the following conditions hold. The envelope and the continuous comparison (14.3) exist. The PBW morphism identifies the associated graded envelope with the symmetric algebra. The induced associated graded comparison is (14.4).

The filtrations on both complexes and their comparison cone have strongly convergent spectral sequences. Their limiting pages identify with the associated graded objects of complete, separated, exhaustive filtrations on cohomology.

For a PBW-admissible completion the same comparison is a quasi-isomorphism. Indeed, its cone has zero associated graded cohomology by (14.4). Strong convergence makes the associated graded of its cohomology zero, and separatedness and exhaustiveness make that cohomology zero. The finite-configuration proof above supplies these convergence requirements on positive Ran coefficients. It does not supply continuity for a different infinite product completion.

14.4 Realization and factorization

Factorization compares the coefficient at a separated configuration with the external tensor product of its coefficients at the separate configurations. This condition is nonlinear. In a rank-one example, the singleton and separated two-point coefficients are both k. The componentwise sum of two copies has coefficient k^2 at both configurations, whereas factorization would require $k^2 \otimes_k k^2 \cong k^4$ at two points.

A componentwise shift has the same difficulty. Its two-point coefficient is $k[1]$, whereas the tensor product of its singleton coefficients is $k[2]$. Thus stability of the ambient coefficients does not make factorization objects closed under componentwise sums or shifts.

Geometric realization has a different colimit property. An indexing infinity-category J has a cofinal diagonal $J \to J \times J$ when restriction along it preserves the colimit of every diagram on $J \times J$. Call J *sifted* when it is nonempty and its diagonal is cofinal. Iteration gives the same statement for $J \to J^m$, for every finite $m \geq 1$. The category Δ^{op} indexing simplicial objects is sifted [79, Definition 5.5.8.1 and Lemma 5.5.8.4].

Proposition 14.5 (Sifted realization of factorization objects)**.** *Fix coefficient categories on configuration charts, with specified restriction and collision functors and their coherent identifications. Suppose these categories admit J-indexed colimits, and these functors preserve them. Suppose external tensor products preserve those colimits separately in each variable. If J is sifted, the componentwise colimit of a J-diagram of factorization objects and factorization morphisms is a factorization object.*

Proof. Write $F_I = \operatorname{colim}_{j\in J}(F_j)_I$. Collision maps and their coherences pass to the colimit by the stated preservation hypothesis. For a decomposition $I = I_1 \sqcup \cdots \sqcup I_m$, let u denote restriction to its disjointness locus. Restriction, factorization, cofinality and separate preservation give

$$\begin{aligned} uF_I &\simeq \operatorname*{colim}_{j\in J} u(F_j)_I \\ &\simeq \operatorname*{colim}_{j\in J} \boxtimes_{a=1}^{m} (F_j)_{I_a} \\ &\simeq \operatorname*{colim}_{(j_1,\dots,j_m)\in J^m} \boxtimes_{a=1}^{m} (F_{j_a})_{I_a} \simeq \boxtimes_{a=1}^{m} F_{I_a}. \end{aligned}$$

All comparisons are natural in the diagram and the decomposition, so they preserve refinement and permutation coherences. If an empty-configuration unit is specified, its constant diagram has the same unit as colimit: a sifted category has contractible groupoid completion [79, Proposition 5.5.8.7]. □

For a simplicial bar, every term, face and degeneracy must first be a morphism in the specified factorization category. Restriction and collision functors must also preserve realization. Under these hypotheses the proposition constructs the whole realized factorization object. Its normalized summands remain ambient coefficients, and any coproduct must separately be compatible with the factorization equivalences.

14.5 Two orders of realization

The two associative bars can be iterated only when their remaining structures and augmentations are preserved. A mixed distributive law must therefore extend to the complete simplicial diagrams, including end faces, units and all substitution homotopies.

Definition 14.6 (Bar-admissible mixed datum)**.** Fix a coefficient infinity-category C. A mixed associative datum is bar-admissible when it carries a bisimplicial object

$$B_{\bullet,\bullet}(A) : \Delta^{\mathrm{op}} \times \Delta^{\mathrm{op}} \longrightarrow C$$

whose two directions are identified with the specified two-sided bar constructions. The identifications include their augmentation and unit maps and every coherence. Both monoidal products are required to preserve the realizations used to form the structured iterated bars.

This condition includes the compatibility of the two distinct units. A comparison of binary multiplications alone does not specify the displayed bisimplicial object.

Theorem 14.7 (Mixed Fubini theorem)**.** *Let A be bar-admissible, and suppose its coefficient category C admits the displayed realizations. Then the two iterated realizations and the diagonal realization are naturally equivalent:*

$$\begin{aligned} \big|p \longmapsto \big|q \longmapsto B_{p,q}(A)\big|\big| &\simeq \big|n \longmapsto B_{n,n}(A)\big|, \\ \big|q \longmapsto \big|p \longmapsto B_{p,q}(A)\big|\big| &\simeq \big|n \longmapsto B_{n,n}(A)\big|. \end{aligned}$$

Proof. Mapping either iterated colimit to a test object gives the limit, in the opposite order, of the same diagram of mapping spaces. Limits over the two indexing categories commute. The universal property of colimits therefore identifies both iterated realizations with

$$\operatorname*{colim}_{(p,q)\in\Delta^{\mathrm{op}}\times\Delta^{\mathrm{op}}} B_{p,q}(A).$$

The diagonal $\Delta^{\mathrm{op}} \to \Delta^{\mathrm{op}} \times \Delta^{\mathrm{op}}$ is cofinal. Apply the cofinality criterion of [79, Theorem 4.1.3.1]. For $([p],[q])$, the relevant comma category is opposite to the category of monotone strings in $[p]\times[q]$.

Append the largest vertex (p,q) to each string. Including the old string gives a natural transformation from the identity to this extension functor. Selecting the appended vertex gives another from the constant largest-vertex string. The nerve is therefore contractible. Cofinality identifies the displayed colimit with the diagonal realization, proving both equivalences.

The comparisons are natural in the entire bisimplicial diagram. Any additional operations retained through realization are consequently preserved. The monoidal preservation assumptions ensure that the iterated colimits are the intended structured bars. No tensor assumption is needed for the colimit identity once the bisimplicial object itself is given. □

In the remainder of this part,

$$\mathrm{Bar}_X^{\mathrm{ord}} = \mathrm{Bar}^{\perp}$$

always denotes the transverse bar. The envelope comparison relates an associative chiral bar to a Lie-chiral coalgebra in the same chiral tensor. Fubini compares two orders of an already specified mixed construction. Neither comparison identifies the transverse and chiral products or constructs an additional associative algebra by itself.

Chapter 15

The internal ordered bar

The transverse bar records the associative multiplication while retaining chiral factorization. Its normalized complex uses ordered words. The factorization maps compare entire bar complexes on separated configurations. The suspended shuffle connects these descriptions.

Work in the positive Ran coefficient category $\mathcal{D}_X$ with its pointwise tensor $\otimes^!$ and unit $\mathbf{1} = \omega_{\mathrm{Ran}}$. By Proposition 10.7, the same product and unit belong to $\mathrm{Fact}_D(X)$. If an empty-configuration coefficient is adjoined, the pointwise unit throughout is instead $(\mathbb{C}, \omega_{\mathrm{Ran}})$.

Let A be an associative algebra in $(\mathrm{Fact}_D(X), \otimes^!)$, with an algebra augmentation $\epsilon : A \to \mathbf{1}$. Assume the restriction and collision functors preserve geometric realizations, as do the external and pointwise tensor products separately. The two-sided simplicial bar has terms

$$B_r(\mathbf{1}, A, \mathbf{1}) = \mathbf{1} \otimes^! A^{\otimes^! r} \otimes^! \mathbf{1}.$$

Its inner faces multiply adjacent factors, its end faces use ϵ, and its degeneracies insert the algebra unit. These are factorization morphisms, including their simplicial coherences. Proposition 14.5 therefore constructs the realization

$$BA = \mathrm{Bar}^{\mathrm{ord}}_X(A) := |B_\bullet(\mathbf{1}, A, \mathbf{1})|$$

as a factorization object.

15.1 Normalized words and their cuts

Choose a symmetric monoidal dg presentation of the ambient coefficients. Require the displayed tensors to compute derived tensors and direct-sum normalization to compute realization. The following chain formulas refer to this presentation. Write $\bar{A} = \ker(\epsilon)$. The unit splits $A = \mathbf{1} \oplus \bar{A}$. Let $s = [1]$ lower cohomological degree by one, so $|sa| = |a| - 1$.

The tensor coalgebra on $s\bar{A}$ is the ambient graded object

$$T^c(s\bar{A}) = \mathbf{1} \oplus \bigoplus_{n \geq 1} (s\bar{A})^{\otimes^! n}.$$

Write $[a_1|\cdots|a_n]$ for a tensor of suspended homogeneous inputs, and $[\,]$ for the length-zero component $\mathbf{1}$. These expressions specify coefficient maps in the dg presentation. No choice of global pure-tensor generators is needed. Its deconcatenation is

$$\Delta_{\mathrm{dec}}[a_1|\cdots|a_n] = \sum_{i=0}^{n} [a_1|\cdots|a_i] \otimes^! [a_{i+1}|\cdots|a_n]. \tag{15.1}$$

The counit projects to length zero, and the coaugmentation includes that component.

Define the degree-one maps

$$b_1(sa) = -s(da), \qquad b_2(sa \otimes^! sb) = (-1)^{|a|} s(ab).$$

Extending them as a coderivation means applying each map at every consecutive position, with the sign obtained by passing the preceding suspended factors. Explicitly, if $e_i = \sum_{j<i}(|a_j| - 1)$, then

$$b[a_1|\cdots|a_n] = \sum_{i=1}^{n}(-1)^{e_i+1}[a_1|\cdots|da_i|\cdots|a_n] + \sum_{i=1}^{n-1}(-1)^{e_i+|a_i|}[a_1|\cdots|a_ia_{i+1}|\cdots|a_n].$$

For closed degree-zero inputs, this gives

$$b[a|c] = [ac], \qquad b[a|c|e] = [ac|e] - [a|ce].$$

The second formula displays the associativity relation in b^2.

Theorem 15.1 (Normalized bar)**.** *Under the stated dg and realization hypotheses, the underlying transverse bar is naturally represented by*

$$BA \simeq \big(T^c(s\bar{A}), b\big).$$

Deconcatenation makes it a coaugmented counital coassociative dg coalgebra in the pointwise coefficient tensor. Its word-length filtration

$$F_nBA = \bigoplus_{0\le r\le n}(s\bar{A})^{\otimes^! r}$$

is exhaustive, and the reduced coproduct into $n+1$ positive factors vanishes on F_nBA. This is the conilpotence asserted here.

Proof. In each simplicial degree, split every factor as $\mathbf{1} \oplus \bar{A}$. The degeneracy subobject consists of summands with a unit factor. Its quotient is the tensor power of $\bar{A}$, shifted by the simplicial degree. The end faces vanish on this quotient, and the remaining faces and internal differential give the displayed b with the stated suspension convention.

The unary part squares to zero because $d^2 = 0$. Its commutator with the binary part is zero by the graded Leibniz rule for d. Two disjoint multiplications cancel in opposite orders, whereas the two overlapping multiplications cancel by associativity. Thus $b^2 = 0$.

Two successive cuts enumerate the same three consecutive subwords in either order, proving coassociativity. In $\Delta_{\mathrm{dec}}b$, each cut surviving a multiplication lies wholly before or after the merged pair. These are exactly the terms of $(b \otimes^! 1 + 1 \otimes^! b)\Delta_{\mathrm{dec}}$, with the tensor sign supplied by the degree of the prefix. The unary terms satisfy the same identity. Hence deconcatenation is a chain map.

The differential preserves F_n, since its components preserve or lower length. Direct-sum normalization gives $BA = \operatorname{colim}_n F_nBA$. A reduced coproduct into $n+1$ factors would divide a word of length at most n into $n+1$ nonempty subwords. No such division exists. This proves the filtration form of conilpotence in the coefficient category. □

The pointwise tensor permits arbitrarily many factors on a fixed configuration. Conilpotence follows from the exhaustive word filtration, with no uniform configuration bound and no word-length completion. The filtration terms and their shifted summands are ambient coefficients. The whole realization carries the factorization structure.

15.2 The shuffle comparison

Write $\otimes$ for the tensor of a chosen symmetric monoidal dg coefficient model. Let P, Q be augmented dg algebras in it, satisfying the derived tensor and realization hypotheses above. The multiplication on their tensor product is

$$(p \otimes q)(p' \otimes q') = (-1)^{|q||p'|} pp' \otimes qq'.$$

Embed P and Q by $p \mapsto p \otimes 1$ and $q \mapsto 1 \otimes q$. A shuffle of r inputs from P and t from Q is an interleaving preserving the order within each group. Define

$$\mathrm{sh} : BP \otimes BQ \longrightarrow B(P \otimes Q) \tag{15.2}$$

by summing over these shuffles. Moving $1 \otimes q_j$ before $p_i \otimes 1$ contributes $(-1)^{(|p_i|-1)(|q_j|-1)}$, the sign for the suspended inputs. Empty words act as units for this formula. For degree-zero p, q,

$$\mathrm{sh}([p] \otimes [q]) = [p \otimes 1 | 1 \otimes q] - [1 \otimes q | p \otimes 1].$$

The tensor-product coalgebra also uses a sign. If $\Delta u = \sum u' \otimes u''$ and $\Delta v = \sum v' \otimes v''$, with homogeneous word factors, then

$$\Delta(u \otimes v) = \sum (-1)^{|u''||v'|} (u' \otimes v') \otimes (u'' \otimes v''). \tag{15.3}$$

The degrees here are the total suspended word degrees. In the binary example, the term $[1 \otimes q] \otimes [p \otimes 1]$ in the reduced coproduct has coefficient -1. This is precisely the crossing sign in (15.3).

Proposition 15.2 (Shuffle and factorization)**.** *The shuffle* (15.2) *is a natural quasi-isomorphism of coaugmented counital dg coalgebras. The iterated shuffles satisfy the unit, associativity and symmetry identities.*

For geometric coefficients, assume the chosen restriction, collision and external-product comparisons commute coherently with the normalized bars and shuffles. Then the shuffles give the factorization equivalences of BA, and deconcatenation is a morphism of factorization objects after passage to the derived category.

Proof. Consider two neighboring suspended inputs from different factors. The two shuffles that interchange them contribute mixed multiplication terms with opposite signs, because

$$(p \otimes 1)(1 \otimes q) = (-1)^{|p||q|} (1 \otimes q)(p \otimes 1).$$

Their cancellation leaves only adjacent multiplications within the same factor. These give the multiplication part of the differential of $BP \otimes BQ$. The internal differentials commute with the graded permutations, with the tensor differential sign. Thus $b\,\mathrm{sh} = \mathrm{sh}\, b$.

A cut in a shuffled word determines cuts in both original words and shuffles of the two prefixes and the two suffixes. Conversely, those four choices determine the cut shuffled word. The resulting permutation first moves the Q prefix across the P suffix, giving $(-1)^{|u''||v'|}$, and then shuffles each pair. This proves

$$\Delta_{\mathrm{dec}}\,\mathrm{sh} = (\mathrm{sh} \otimes \mathrm{sh})\Delta_{BP \otimes BQ}.$$

The empty-word terms also give compatibility with counit and coaugmentation.

For the quasi-isomorphism, form the bisimplicial coefficient object

$$(r, t) \longmapsto B_r(\mathbf{1}, P, \mathbf{1}) \otimes B_t(\mathbf{1}, Q, \mathbf{1}).$$

Its iterated realization is $BP \otimes BQ$, because the tensor preserves realization separately. Its diagonal is the simplicial bar of $P \otimes Q$: moving the P and Q factors together gives exactly the displayed graded multiplication.

The diagonal and iterated realizations agree by Theorem 14.7. On normalized chains the comparison is the shuffle.

Indeed, the monotone paths across an r by t rectangle triangulate the product of the two simplices. Each path inserts the degeneracies of its interleaving. Its orientation, together with the internal graded rearrangement, is the suspended shuffle sign. Shared faces cancel, leaving the product boundary. This identifies the written chain map with the realization equivalence.

For several factors, either parenthesization enumerates each interleaving preserving all internal orders exactly once. Its sign is the Koszul sign of the same permutation. The same observation after permuting whole groups proves symmetry, and the empty group proves the unit identity.

Now let $I = I_1 \sqcup I_2$, and let j include the locus where the two groups of points are disjoint. The algebra factorization identifies $j^! A_I$ with $j^!(A_{I_1} \boxtimes A_{I_2})$. Applying the shuffle gives

$$j^!(BA_{I_1} \boxtimes BA_{I_2}) \xrightarrow{\mathrm{sh}} j^! BA_I.$$

Its inverse equivalence is the factorization map. The coalgebra identity gives compatibility with the pointwise deconcatenation, the several-factor identities give refinement and permutation coherence, and the assumed functor comparisons give collision coherence. These are the required compatibilities for a factorization morphism. □

The geometric hypothesis concerns coherent comparisons of functors and their operations. A stable ambient category alone does not supply them. Under that hypothesis, the equivalence concerns the whole bar, even though individual normalized summands need not factorize.

15.3 Three coproducts

There are two coalgebra structures on the transverse bar, with different tensor targets. Factorization separates the support:

$$j^! BA_I \xrightarrow{\sim} j^!(BA_{I_1} \boxtimes BA_{I_2}).$$

These are the disjointness components of its chiral commutative coproduct. Deconcatenation instead repeats the same configuration in a pointwise tensor:

$$BA_I \xrightarrow{\Delta_{\mathrm{dec}}} BA_I \otimes^! BA_I.$$

The shuffle proposition proves compatibility between these structures.

A third coproduct belongs to an acting bialgebra H. It is an algebra map $\Delta_H : H \to H \otimes H$, with an algebra counit, satisfying coassociativity and the counit identities. An antipode makes H a Hopf algebra.

Let M, N be H-representations. To act on $M \otimes N$, first apply Δ_H to the acting factor. Move the second copy of H past M by symmetry, then apply the two representation actions. A completed tensor product requires a specified topology and continuous maps throughout.

Proposition 15.3 (Coproduct separation)**.** *Factorization, deconcatenation and an acting Hopf coproduct are related by comparisons in their specified tensor recipients. In strict dg presentations, let a strong monoidal dg functor Φ carry H to the pointwise coefficient category. Let $f : \Phi H \to BA$ be a chain map preserving counit and coaugmentation. Then f is a coalgebra comparison exactly when*

$$\Delta_{\mathrm{dec}} f = (f \otimes^! f)\Phi(\Delta_H),$$

using the monoidal identification of the target of $\Phi(\Delta_H)$.

For coherent coalgebras, this condition includes the compatible homotopies for every iterated coproduct. A comparison with the chiral coproduct requires transport to the chiral tensor recipient and compatibility on every disjointness locus. Retaining a counit requires the unitalized recipient.

Proof. The displayed equation is the coproduct condition for a strict dg coalgebra morphism in the pointwise tensor. Iterating it and using coassociativity gives every higher coproduct identity. With the stipulated unit maps, it therefore preserves the coaugmented counital coalgebra structure. For coherent coalgebras these iterations are part of the morphism datum. In the chiral recipient, restricting the corresponding coproduct components gives the disjointness equations.

For three representations M, N, P, applying Δ_H twice gives the two candidate actions on their tensor product. Coassociativity identifies them, and the counit gives the unit representation. This explains the representation tensor product supplied by H without identifying its carrier with a Ran coefficient or a bar word. □

15.4 Interval factorization homology

Let $(C, \otimes, \mathbf{1})$ be a symmetric monoidal infinity-category admitting sifted colimits, with tensor preserving them separately. An oriented interval coefficient system assigns an associative algebra A to each interior interval. A left endpoint neighborhood receives a right A-module R, and a right endpoint neighborhood receives a left A-module L. Ordered embeddings act by multiplication and the module actions, with their coherences.

In the unital disk convention, nullary endpoint operations also specify maps $\mathbf{1} \to R$ and $\mathbf{1} \to L$. These are part of the coefficient system [6, Definition 2.30]. Factorization homology is the colimit of its tensor values over disjoint unions of such basic intervals embedded in $[0, 1]$.

Proposition 15.4 (The endpoint bar)**.** *For this coefficient system there is a natural equivalence*

$$\int_{([0,1],\{0,1\})} (A; R, L) \simeq \left| r \longmapsto R \otimes A^{\otimes r} \otimes L \right| \simeq R \otimes^{\mathbb{L}}_{A} L.$$

The superscript records the derived relative tensor product. In an infinity-category this is the relative tensor product itself.

Proof. Replace the interval disk category by its equivalent category of finite ordered configurations with optional endpoint components. The full subcategory containing both endpoint components is the simplex category in the opposite variance. Adjoining a missing first or last endpoint component is left adjoint to its inclusion, so the inclusion is final. This is the interval finality argument of [6, Observations 2.31–2.32 and Proposition 2.34].

The restricted diagram has value $R \otimes A^{\otimes r} \otimes L$ in degree r. Its first and last faces apply the module actions, its inner faces multiply in A, and its degeneracies insert the unit. Finality proves the first equivalence. The realization of this two-sided bar computes the relative tensor product by [80, Theorem 4.4.2.8 and Example 4.4.2.11]. Its tensor-preservation hypotheses are exactly those imposed above. □

For modules without endpoint point maps, the diagram with both endpoints present still defines the relative interval bar and gives $R \otimes^{\mathbb{L}}_{A} L$. To use the full unital disk category, the endpoint point maps must be included in the datum.

Apply this construction internally to the pointwise coefficient tensor. The augmentation $A \to \mathbf{1}$ makes $R = L = \mathbf{1}$ into the two endpoint modules, with identity point maps. The endpoint bar is then $BA = \mathrm{Bar}^{\mathrm{ord}}_{X}(A)$. The earlier functor hypotheses retain its chiral factorization. This identifies the interval coefficient object. A numerical amplitude or prepared boundary state requires its own evaluation or state map in addition to the augmentation.

Chapter 16

Cobar and twisting cochains

A coalgebra map to a tensor bar is determined by its projection to bar length one. The chain-map condition on that projection is a quadratic equation. Tensor cobar represents the same equation by an algebra map in the other direction.

16.1 The tensor cobar differential

Fix a commutative ring k and use cohomologically graded k-modules, with tensor product over k. Let A be a unital augmented dg associative algebra, with augmentation ideal $\bar{A}$. Let C be a counital coaugmented dg coassociative coalgebra. Write I for its coaugmentation and put $\bar{C} = \ker(\epsilon_C : C \to k)$. On $\bar{C}$, the reduced coproduct is

$$\bar{\Delta}(c) = \Delta(c) - \mathrm{I} \otimes c - c \otimes \mathrm{I}.$$

Let $\bar{\Delta}_r : \bar{C} \to \bar{C}^{\otimes r}$ be its iterates, with $\bar{\Delta}_1 = \mathrm{id}$. Coassociativity makes the choice of parenthesization immaterial. Assume C is conilpotent: every $c \in \bar{C}$ is killed by $\bar{\Delta}_r$ for all sufficiently large r.

Retain the bar suspension $|sa| = |a| - 1$ of Theorem 15.1. Write $t = s^{-1}$, so $|tc| = |c| + 1$. Both tensor constructions below use direct sums, with the empty tensor equal to k. In a dg coefficient presentation of the transverse theory, the tensor being modeled is the pointwise product $\otimes^!$.

Definition 16.1 (Tensor cobar)**.** The ordinary tensor cobar is the augmented algebra

$$\Omega C = T(t\bar{C}) = \bigoplus_{r \geq 0} (t\bar{C})^{\otimes r}.$$

Write $\bar{\Delta} c = \sum c' \otimes c''$ for a homogeneous expansion of the reduced coproduct. Its differential is the degree-one derivation determined by

$$d_\Omega(tc) = -t(d_C c) - \sum (-1)^{|c'|} (tc')(tc''). \tag{16.1}$$

The augmentation projects to tensor length zero.

A primitive element p has $\bar{\Delta} p = 0$, so $d_\Omega(tp) = -t(d_C p)$. If p, q, c are closed and $\bar{\Delta} c = p \otimes q$, then

$$d_\Omega(tc) = -(-1)^{|p|} (tp)(tq).$$

The two factors retain their order.

Proposition 16.2. *The derivation* (16.1) *squares to zero and makes* ΩC *an augmented dg algebra.*

Proof. Write $d_\Omega = d_1 + d_2$ for its internal and quadratic terms. The square of an odd derivation is a derivation, so it suffices to check the generators. The identity $d_1^2 = 0$ follows from $d_C^2 = 0$.

For a term $c_1 \otimes c_2 \otimes c_3$ of the iterated reduced coproduct, put $p = |c_1|$ and $q = |c_2|$. Splitting the first factor after the first cut contributes the sign $(-1)^{p+q+1}(-1)^{p+1} = (-1)^{q}$. Splitting the second contributes $(-1)^{p+1}(-1)^{p+1}(-1)^{q+1} = (-1)^{q+1}$. Coassociativity identifies their coefficients, so they cancel. Thus $d_2^2 = 0$.

The chain-map identity for the reduced coproduct is

$$\bar{\Delta}(d_C c) = \sum \big(d_C c' \otimes c'' + (-1)^{|c'|} c' \otimes d_C c'' \big).$$

Substitution in (16.1) gives

$$d_1 d_2(tc) = \sum \big((-1)^{|c'|} t(d_C c')(tc'') - (tc')t(d_C c'') \big) = -d_2 d_1(tc).$$

This proves the square-zero identity. The differential has no scalar term, so the augmentation is a chain map. □

16.2 The twisting equation

For homogeneous graded maps $f, g : \bar{C} \to \bar{A}$, define the convolution product and differential by

$$(f \star g)(c) = \sum (-1)^{|g||c'|} f(c')\, g(c''), \qquad \partial f = d_A f - (-1)^{|f|} f\, d_C.$$

Associativity and coassociativity make $\star$ associative. The chain-map identities for multiplication and coproduct give $\partial(f \star g) = (\partial f) \star g + (-1)^{|f|} f \star (\partial g)$.

Definition 16.3 (Twisting cochain)**.** A twisting cochain is a degree-one map $\tau : \bar{C} \to \bar{A}$ satisfying

$$\partial\tau + \tau \star \tau = 0, \qquad d_A \tau(c) + \tau(d_C c) + \sum (-1)^{|c'|} \tau(c')\tau(c'') = 0. \tag{16.2}$$

Write $\mathrm{Tw}(C, A)$ for this set. Equivalently, extend τ to $C \to A$ by zero on the coaugmentation and require its augmentation component to vanish.

This is the associative Maurer–Cartan equation. It involves no division by two and is defined over the chosen ring k.

16.3 The two universal extensions

The tensor bar $BA = T^c(s\bar{A})$ has the deconcatenation coproduct and the differential of Theorem 15.1. In particular, its components with output of length one are

$$d_B(sa) = -s(d_A a), \qquad d_{B,2}(sa \otimes sb) = (-1)^{|a|} s(ab).$$

Let $p_A : BA \to s\bar{A}$ be the length-one projection.

Theorem 16.4 (Bar–cobar adjunction)**.** *For the augmented dg algebras and coaugmented conilpotent dg coalgebras specified above, there are natural bijections*

$$\mathrm{Hom}_{\mathrm{dgAlg}^{\mathrm{aug}}}(\Omega C, A) \cong \mathrm{Tw}(C, A) \cong \mathrm{Hom}_{\mathrm{dgCoalg}^{\mathrm{coaug}}}(C, BA). \tag{16.3}$$

Thus ordinary tensor cobar is left adjoint to tensor bar.

Proof. Given τ, send tc to $\tau(c)$ and extend multiplicatively, sending the empty tensor to 1_A. This defines the unique augmented graded algebra map $g_\tau : \Omega C \to A$ with the stated generator values. Its chain-map condition on a generator is precisely (16.2). Since both differentials are derivations, that condition suffices on every tensor. Restriction to the generators gives the inverse bijection.

For the coalgebra extension, set $\pi_C(c) = c - \epsilon_C(c)1$ and define

$$f_\tau(c) = \epsilon_C(c)1 + \sum_{r\geq 1}(s\tau)^{\otimes r}\bar{\Delta}_r(\pi_C(c)). \tag{16.4}$$

The map $s\tau$ has degree zero. Conilpotence makes this a finite sum on every input, hence an element of the direct-sum bar. Cutting its length-r term after position i gives the $i, r-i$ component of $(f_\tau \otimes f_\tau)\Delta_C$. This proves the coalgebra identity, including the two counital cuts.

Conversely, any coaugmented coalgebra map to $T^c(s\bar{A})$ has these components, determined by its length-one projection. Thus (16.4) is the unique extension of $s\tau$.

To check its differential, put $D = d_B f_\tau - f_\tau d_C$. The coderivation identities imply

$$\Delta_B D = (D \otimes f_\tau + f_\tau \otimes D)\Delta_C.$$

Iterating this identity expresses the length-r component of D as the sum of tensors with one factor $p_A D$ and all other factors $p_A f_\tau$. Consequently $D = 0$ if and only if $p_A D = 0$.

The two displayed bar components give

$$p_A D = -s\big(d_A\tau + \tau d_C + \tau \star \tau\big).$$

This is zero exactly when τ is a twisting cochain. The inverse map sends a dg coalgebra map f to $t\,p_A f$. Both extension formulas commute with maps of C and A, proving naturality. □

The usual field version appears in [76, Theorem 2.2.9]. The proof above uses the fixed cohomological signs and works over the chosen commutative ring, without flatness or a completed tensor algebra.

The universal twisting maps are

$$\tau_C : C \longrightarrow \Omega C, \qquad \tau_C(c) = t\,\pi_C(c), \qquad \tau_A : BA \longrightarrow A, \qquad \tau_A = t\,p_A.$$

They correspond to the identity maps of ΩC and BA. The unit is $\eta_C = f_{\tau_C} : C \to B\Omega C$. The counit is $\epsilon_A = g_{\tau_A} : \Omega BA \to A$. It sends $t(sa)$ to a and generators of larger bar length to zero.

In $\epsilon_{\Omega C} \circ \Omega\eta_C$, the length-one term of (16.4) returns each generator tc. In $B\epsilon_A \circ \eta_{BA}$, only the decomposition into singleton bar blocks survives, returning the original bar tensor. These are the two triangular identities.

For an internal adjunction on factorization objects, both extensions must preserve the chosen geometric data. One must construct the factorization structure on cobar and verify compatibility with collisions, separated products, permutations and all their coherences. The twisting maps and both inverse correspondences must preserve those structures. The ordinary mapping-set adjunction above then restricts wherever these conditions hold. In that recipient, write $\mathrm{Cobar}^{\mathrm{ord}}_X(C) = \Omega C$ and $\mathrm{Bar}^{\mathrm{ord}}_X(A) = BA$.

An adjunction on derived mapping spaces requires a compatible derived enhancement. A completed cobar also requires its own topology, convergence and comparison maps. These are additional constructions, not consequences of the ordinary mapping-set bijections.

Chapter 17

Complete bar–cobar reconstruction

The tensor counit reconstructs an algebra from its bar coalgebra. Positive weights permit a completion of this reconstruction. The quadratic criterion asks when diagonal cycles contain the homology of the full bar.

17.1 The counit and the gap contraction

Let k be a commutative ring and let $A = k1 \oplus I$ be an augmented associative algebra in cohomological degree zero. All tensors are over k. Retain the suspensions $|sa| = |a| - 1$ and $t = s^{-1}$ from the preceding chapter. Write

$$D = \Omega B A, \qquad c(a_1, \dots, a_n) = t(sa_1 \otimes \cdots \otimes sa_n), \qquad a_i \in I.$$

The generator $c(a_1, \dots, a_n)$ has degree $1 - n$.

The cobar differential becomes

$$dc(a_1, \dots, a_n) = \sum_{i=1}^{n-1} (-1)^{i-1} \big(c(a_1, \dots, a_i) c(a_{i+1}, \dots, a_n) - c(a_1, \dots, a_i a_{i+1}, \dots, a_n) \big). \tag{17.1}$$

Extend it by the graded Leibniz rule. The counit $\epsilon : D \to A$ sends $c(a)$ to a and longer generators to zero. In particular, $dc(a, b) = c(a)c(b) - c(ab)$, whose image under ϵ is zero.

A product of generators is a *block word*. Flatten it to its N original entries. Each gap between consecutive entries is either internal to a block or an outer cut between two blocks. If $S \subseteq \{1, \dots, N-1\}$ is the set of outer cuts, its degree is $|S| - (N - 1)$.

For an internal gap i, put

$$u_i(S) = \#\{ j < i : j \notin S \}.$$

The splitting term inserts the cut i with sign $(-1)^{u_i(S)}$. The multiplication term has the opposite sign and lowers the number of entries. These signs follow from (17.1), including the degrees of all preceding blocks.

Theorem 17.1 (Ordinary reconstruction)**.** *The counit $\epsilon : \Omega B A \to A$ is a quasi-isomorphism of augmented dg algebras. No flatness assumption on I is required.*

Proof. Filter D increasingly by the number of original entries. Every element has finite support in this filtration. On its Nth associated graded piece only splitting remains. For $N \geq 1$, this complex is

$$I^{\otimes N} \otimes E^{\otimes (N-1)}, \qquad E^{-1} = k e_-, \quad E^0 = k e_+, \quad d e_- = e_+.$$

The states e_- and e_+ mean an internal gap and an outer cut, respectively. The tensor differential counts the preceding e_- factors, giving exactly $u_i(S)$.

For $N > 1$, define h by changing the first gap from e_+ to e_- and by sending a word with first gap e_- to zero. On that first factor, $dh + hd = \mathrm{id}$. The differentials on the remaining factors cancel between the two

composites because h has degree -1. Thus this identity holds after tensoring with $I^{\otimes N}$, without any tensor-exactness hypothesis. The pieces $N = 0, 1$ are k, I, both in degree zero.

For a negative-degree cycle, apply h to its highest-entry component and lift the resulting primitive to D. Subtract its differential. The largest entry number strictly decreases, so repetition terminates and proves negative acyclicity. In degree zero, the boundaries impose exactly $c(a)c(b) = c(ab)$. Every element reduces to a scalar plus $c(a)$, with $a \in I$, and its image in $k1 \oplus I$ makes this expression unique. Hence $H^0(D) = A$, through the counit. □

For example, take $A = k[x]/(x^3)$ and give x weight one. The bar complex of weight three is

$$k[x|x|x] \xrightarrow{d} k[x^2|x] \oplus k[x|x^2], \qquad d[x|x|x] = [x^2|x] - [x|x^2],$$

in degrees -3, -2. It has one copy of k in degree -2. Nevertheless the full tensor counit reconstructs A by the theorem.

17.2 Positive-weight completion

Suppose now that $A_{\mathrm{alg}} = \bigoplus_{w \geq 0} A_w$, with $A_0 = k$, multiplication adding weights and augmentation projecting onto A_0. Let D_w be the weight-w subcomplex of $\Omega B(A_{\mathrm{alg}})$. Define

$$\widehat{A} = \prod_{w \geq 0} A_w, \qquad \widehat{D} = \prod_{w \geq 0} D_w.$$

Products of complexes are degreewise. The weight topology has neighborhoods $F^p = \prod_{w \geq p}(-)_w$ of zero. Multiplication is continuous: each weight receives only finitely many products of nonnegative weights.

Theorem 17.2 (Complete reconstruction)**.** *The componentwise counit $\widehat{D} \to \widehat{A}$ is a quasi-isomorphism of complete dg algebras.*

Proof. Every reduced entry has positive weight, so an entry list of total weight w has length at most w. The preceding filtration and contraction therefore apply within each weight and give $H^j(D_w) = 0$ for $j < 0$ and $H^0(D_w) = A_w$. The product differential has the product of the component kernels as its kernel and the product of the component images as its image. Consequently

$$H^j\left(\prod_w D_w\right) = \prod_w H^j(D_w).$$

This proves the assertion. Continuity follows from preservation of weight. □

For $k[x]$ with weight one on x, the ordinary counit recovers $k[x]$, whereas this weight completion recovers $k[[x]]$. A completed geometric bar–cobar equivalence needs a specified completed tensor product, coalgebra category, and comparison maps compatible with collision and inverse limits. An operadic nilcompletion or a boundary double centralizer is not identified with the ordinary tensor counit by this weight calculation. The coalgebra-unit and geometric duality statements require their own convergence and comparison arguments.

17.3 The quadratic coalgebra and its resolutions

Let V be a k-module in cohomological degree zero and weight one, let $R \subseteq V \otimes V$, and set

$$A = T_k(V)/(R), \qquad A_0 = k.$$

Assume each A_w is flat over k. Put $K_0 = k$, $K_1 = V$, and, for $n \geq 2$, define

$$K_n = \bigcap_{i+j=n-2} V^{\otimes i} \otimes R \otimes V^{\otimes j} \subseteq V^{\otimes n}. \tag{17.2}$$

These are embedded submodules: the surrounding tensor factors are flat. Assume also that every K_n is flat. Both flatness conditions hold automatically over a field.

The quadratic dual coalgebra is

$$C = A^{\mathrm{q!}} = \bigoplus_{n \geq 0} s^n K_n,$$

with zero differential and deconcatenation coproduct. Its nth summand has weight n and degree $-n$.

To verify the coproduct target, express each K_n as the common kernel of the adjacent quotient maps with target containing $A_2 = (V \otimes V)/R$. Tensoring these kernel sequences with the flat surrounding factors preserves their kernels. For a cut after i entries, the constraints on the first i and last $n - i$ entries then place the cut tensor in $K_i \otimes K_{n-i}$. Coassociativity and the counit restrict from the tensor coalgebra. Positive weight implies conilpotence.

The weight-n, degree-$-n$ term of BA is $(sV)^{\otimes n}$. Its different adjacent multiplications land in distinct weight-composition summands. Their common kernel is exactly $s^n K_n$, and there is no incoming differential in that degree. Thus inclusion defines a dg coalgebra map

$$i : C \longrightarrow BA,$$

identifying C_n with the diagonal bar cycles.

Define $\kappa : C \to A$ by $\kappa(sv) = v$ and by zero on every other weight. It has degree one. The only possible quadratic term in its twisting equation is multiplication of R, which is zero in A. Hence it is a twisting cochain and induces $q : \Omega C \to A$. The corresponding coalgebra extension is i.

The left and right Koszul complexes are

$$K_L = A \otimes C, \qquad K_R = C \otimes A.$$

For a homogeneous coproduct $\Delta c = \sum c' \otimes c''$, their differentials are

$$d_L(a \otimes c) = -\sum a\kappa(c') \otimes c'', \qquad d_R(c \otimes a) = \sum (-1)^{|c'|} c' \otimes \kappa(c'')a. \tag{17.3}$$

The twisting equation makes both differentials square to zero. Their augmentations to k use the augmentations of both factors. When their augmentations are quasi-isomorphisms, these are Priddy's Koszul resolutions [90, Introduction].

Theorem 17.3 (Quadratic Koszul criterion)**.** *Under the stated flatness hypotheses, the following conditions are equivalent:*

(i) $i : C \to BA$ *is a quasi-isomorphism.*

(ii) $q : \Omega C \to A$ *is a quasi-isomorphism.*

(iii) *both augmented complexes $K_L \to k$ and $K_R \to k$ are quasi-isomorphisms.*

(iv) $\operatorname{Tor}_i^A(k,k)_w = 0$ *whenever* $w \neq i$.

Either one of the two assertions in (iii) suffices. No finiteness assumption on duals is needed for these equivalences.

Proof. First compare the twisted complexes with the augmented bar resolutions. For $P_L = A \otimes BA$, the differential is the bar differential plus $-a_0 a_1 \otimes [a_2|\cdots|a_n]$. Writing $\bar{a}_0 = a_0 - \epsilon(a_0)1$, its augmented contraction is

$$h_L\big(a_0 \otimes [a_1|\cdots|a_n]\big) = -1 \otimes [\bar{a}_0|a_1|\cdots|a_n].$$

The adjacent multiplication involving $\bar{a}_0$ supplies the identity term. The remaining multiplications cancel in $dh_L + h_L d$. Thus $dh_L + h_L d = \mathrm{id} - \iota\epsilon$, where $\iota : k \to P_L$ is the unit.

For $P_R = BA \otimes A$, the last action has sign $(-1)^{n-1}$ and the corresponding contraction is

$$h_R\big([a_1|\cdots|a_n] \otimes a_0\big) = (-1)^n [a_1|\cdots|a_n|\bar{a}_0] \otimes 1.$$

The same cancellation proves its contraction identity.

The maps $\mathrm{id}_A \otimes i : K_L \to P_L$ and $i \otimes \mathrm{id}_A : K_R \to P_R$ respect differentials and augmentations. At total weight w, filter either complex decreasingly by the weight p in its A-factor. The twisting term increases p, and the associated graded comparison is $A_p \otimes i_{w-p}$, with the factors reversed on the right. Flatness of A_p proves that (i) implies either assertion in (iii).

Conversely, suppose one twisted complex resolves k. Its comparison with the corresponding P is then a quasi-isomorphism. Induct on w. The assertion for $w = 0$ is the identity of k.

In the cone at weight w, the subcomplex with positive A-weight has a finite filtration with acyclic quotients $A_p \otimes \operatorname{cone}(i_{w-p})$, by induction. Its quotient is $\operatorname{cone}(i_w)$. The whole cone and that subcomplex are acyclic, so the quotient is acyclic. This proves (i), from either side of (iii).

For comparison with (ii), every weight component of C, BA, ΩC and A is a bounded complex of flat k-modules. At a fixed weight, filter bar increasingly and cobar decreasingly by tensor length. These finite filtrations have tensor products of the input complexes as their associated graded pieces. Tensor products of bounded flat complexes preserve quasi-isomorphisms: apply the finite degree filtration to the cone of each tensor-factor map. Thus B and Ω preserve the quasi-isomorphisms used here. Since $q = \epsilon \circ \Omega i$, the ordinary counit theorem gives (i)⇒(ii).

For the converse, consider the unit $\eta_C : C \to B\Omega C$. Flatten a term to N original reduced C-entries and filter decreasingly by N. Cobar splitting increases N. On the associated graded, only multiplication of neighboring cobar blocks remains. It changes an outer bar cut into an inner product gap.

For homogeneous entries $c_1, \ldots, c_N$ and a cut set S, its coefficient at $i \in S$ is

$$(-1)^{\sum_{j \le i} |c_j| + u_i(S) + 1}.$$

Contract this reversed cube by inserting the first cut, with coefficient $(-1)^{|c_1|+1}$, when it is absent, and by zero when it is present. The first differential gives the identity. The other terms cancel with their graded signs.

All $N > 1$ pieces are therefore contractible. The unit is the identity on the surviving empty and singleton pieces. At weight w one has $N \le w$, so the finite filtration proves that η_C is a quasi-isomorphism. If (ii) holds, Bq is a quasi-isomorphism, and $Bq \circ \eta_C = i$ proves (i).

Finally, P_L is a flat A-module resolution of k, since I and all its tensor powers are flat over k. Tensoring with k over A gives BA, so $H^{-i}(BA)_w = \operatorname{Tor}_i^A(k,k)_w$. The diagonal homology is already C_w by the kernel calculation, and terms with bar length greater than weight vanish. Hence (i) is equivalent to (iv). □

Over a field this is the usual quadratic criterion [76, Theorem 3.4.6]. The proof above isolates the flatness needed over a general ring.

17.4 A torsion boundary for the Tor comparison

Flatness cannot simply be omitted when ordinary bar homology is used for absolute Tor. Take $k = \mathbb{Z}$ and

$$A = \mathbb{Z}[x]/(2x, x^2), \qquad |x| = 0, \quad \mathrm{wt}(x) = 1.$$

Its augmentation ideal is $(\mathbb{Z}/2)x$ with zero product. Thus its ordinary bar has zero differential and one copy of $\mathbb{Z}/2$ in each positive bidegree $(-n, n)$.

The following free A-resolution begins differently. Let $F_0 = A$, $F_1 = Ae$, $F_2 = Af \oplus Ag$, and $F_3 = Ah \oplus Ai \oplus Aj$, with

$$de = x, \qquad df = 2e, \quad dg = xe, \qquad dh = xf, \quad di = 2g, \quad dj = xg.$$

The weights of e, f, g, h, i, j are 1, 1, 2, 2, 2, 3.

The kernel in F_3 is generated by

$$2h, \quad xh, \quad xi, \quad 2j, \quad xj.$$

Take F_4 free on generators mapping to these five elements.

To check exactness through F_3, write each coefficient uniquely as $a + bx$, with $a \in \mathbb{Z}$ and $b \in \mathbb{Z}/2$. The successive kernels are $(2e, xe)$, $(xf, 2g, xg)$ and the displayed five-generator module. This also verifies all differential composites. Extend by free covers of the later kernels.

After tensoring with $\mathbb{Z}$ over A, the weight-two part around homological degree three is

$$\mathbb{Z} \xrightarrow{(2,0)} \mathbb{Z}h \oplus \mathbb{Z}i \xrightarrow{(0,2)} \mathbb{Z}g.$$

Therefore

$$\mathrm{Tor}_3^A(\mathbb{Z}, \mathbb{Z})_2 = \mathbb{Z}/2.$$

The ordinary bar has no such off-diagonal class. Its terms provide a relative resolution, but are not flat A-modules and do not compute this absolute Tor.

Here the quadratic coalgebra is the ordinary bar itself, so its inclusion is the identity and both twisted complexes contract. The tensor counit is also a quasi-isomorphism, while the Tor-purity condition fails. A flat or derived replacement supplies the missing comparison. Finiteness of ordinary coefficient duals is insufficient.

17.5 Coefficient duality

The coalgebra comparison and its dual are separate constructions. Return to $A = T_k(V)/(R)$ and its quadratic coalgebra C. Assume the equivalent quadratic conditions and suppose, in addition, that every A_w and K_w is finitely generated projective over k. Define the degreewise products

$$(BA)^\vee = \prod_{w \geq 0} \mathrm{Hom}_k((BA)_w, k), \qquad C^\vee = \prod_{w \geq 0} \mathrm{Hom}_k(C_w, k).$$

For homogeneous cochains, the convolution product is

$$(f \star g)(c) = \sum (-1)^{|g||c'|} f(c')\, g(c''), \qquad \Delta c = \sum c' \otimes c''.$$

Each output weight involves finitely many weight decompositions, so the product is continuous.

Proposition 17.4. *The dual of i is a quasi-isomorphism of complete augmented dg algebras $i^\vee : (BA)^\vee \to C^\vee$. Moreover $(BA)^\vee$ is a dg algebra model for* $\mathrm{RHom}_A(k,k)$.

Proof. At each weight, the cone of i is an acyclic bounded complex of finitely generated projectives. Such a complex is contractible: split off its highest nonzero degree and proceed downwards. Dualizing preserves that contraction. Taking products therefore proves the quasi-isomorphism, and the coalgebra identity for i gives compatibility with convolution.

For the derived endomorphisms, use $P = A \otimes BA$ with the left twisted differential above. It is a bounded-above complex of projective A-modules resolving k, hence computes derived Hom. Its right BA-coaction is $\rho = \mathrm{id}_A \otimes \Delta$. Lift a homogeneous $f \in (BA)^\vee$ to the A-linear endomorphism

$$L_f = (\mathrm{id}_P \otimes f)\rho, \qquad L_f(a \otimes c) = \sum (-1)^{|f||c'|} a \otimes c' f(c'').$$

Coassociativity gives $L_f L_g = L_{f \star g}$, and the chain-coaction identity gives $[d_P, L_f] = L_{\partial f}$.

Postcomposition with $P \to k$ is a quasi-isomorphism $\mathrm{End}_A(P) \to \mathrm{Hom}_A(P,k)$ because P is bounded above with projective terms. Its composite with L is the identity under $\mathrm{Hom}_A(P,k) = \mathrm{Hom}_k(BA,k)$. Thus L is a quasi-isomorphism of dg algebras, proving the asserted model. □

If k is a field and V is finite-dimensional, the graded pairing identifies

$$C^\vee \cong \prod_{n \geq 0} \big(T(V^*[-1])/(R^\perp)\big)_n.$$

In each finite-dimensional weight, the annihilator of the intersection in (17.2) is the sum of the corresponding orthogonals. That sum is the degree-n part of the ideal generated by $R^\perp$. This proves the quadratic presentation of the dual, including its multiplication.

Over a general ring the preceding proposition retains the actual coalgebra dual. An orthogonal-relations presentation requires its own annihilator comparison. Neither coefficient duality nor the ordinary tensor reconstruction constructs the geometric chiral comparison or a boundary-junction reconstruction.

Chapter 18

Modules, comodules, and Morita theory

The tensor counit reconstructs an algebra. Reconstruction of its representations also requires comparison maps for modules and comodules. Positive weights make the two bar contractions finite in each component. Morita reconstruction asks a different question: when does a chosen object's endomorphism algebra recover its entire category?

18.1 The weighted representation categories

Let k be a field and let $A_{\mathrm{alg}} = \bigoplus_{w \geq 0} A_w$ be an augmented associative algebra in degree zero. Assume $A_0 = k$, multiplication adds weights, and augmentation projects onto A_0. Put $C_{\mathrm{alg}} = B(A_{\mathrm{alg}})$ and

$$A = \prod_{w \geq 0} A_w, \qquad C = \prod_{w \geq 0} (C_{\mathrm{alg}})_w.$$

The bar has degree $-r$ on a word $[a_1 | \cdots | a_r]$. Its differential has the convention $d_C[a|b] = [ab]$. All products of complexes below are degreewise.

A complete weight-graded left A-module is a product

$$M = \prod_{w \geq w_0} M_w$$

for an integer w_0, with differential preserving weight and unital associative actions $A_u \otimes M_v \to M_{u+v}$. The weight topology has neighborhoods $\prod_{w \geq p} M_w$ of zero. Morphisms preserve weight and commute with the action and differential. Such maps are continuous.

For $D = A$ or C, define the completed tensor by

$$(D \widehat{\otimes} M)_w = \bigoplus_{u+v=w} D_u \otimes_k M_v.$$

The sum is finite, since $u \geq 0$ and $v \geq w_0$. A complete weight-graded left C-comodule is a similarly bounded product N with a continuous counital coassociative coaction

$$\rho : N \longrightarrow C \widehat{\otimes} N$$

commuting with its differential. Its morphisms preserve weight and coaction. No coaugmentation of N is chosen.

Each reduced coaction lowers the weight of the final N-factor by at least one. It therefore vanishes after more than $w - w_0$ iterations on N_w. Write $D_{\mathrm{wt}}(A)$ and $D_{\mathrm{wt}}(C)$ for the localizations at maps that are quasi-isomorphisms in every weight. These weak equivalences are specified on the underlying complexes.

18.2 The twisting adjunction

The universal twisting map $\tau : C \to A$ has degree one. It sends $[a]$ to a and vanishes on the empty word and on words of length greater than one. Define

$$F(M) = C\widehat{\otimes}_\tau M, \qquad G(N) = A\widehat{\otimes}_\tau N.$$

The underlying tensors are the ones just defined. Their coaction and action come from Δ_C and left multiplication in A, respectively.

For $\Delta(c) = \sum c_{(1)} \otimes c_{(2)}$ and $\rho(n) = \sum n_{(-1)} \otimes n_{(0)}$, set

$$\begin{aligned} d_F(c \otimes m) &= d_C c \otimes m + (-1)^{|c|} c \otimes d_M m + \sum (-1)^{|c_{(1)}|} c_{(1)} \otimes \tau(c_{(2)}) m, \\ d_G(a \otimes n) &= a \otimes d_N n - \sum a\tau(n_{(-1)}) \otimes n_{(0)}. \end{aligned} \tag{18.1}$$

Every sum is finite in each weight. Expanding either square leaves the twisting equation for τ, followed by the action or coaction. Thus both differentials square to zero. The same identities make the coaction of $F(M)$ and the action of $G(N)$ chain maps.

For instance,

$$d_F([a] \otimes m) = -[a] \otimes d_M m + [\,] \otimes am.$$

The minus sign makes the two resulting copies of $[\,] \otimes a\, d_M m$ cancel in d_F^2. The coefficient a has degree zero, while its bar suspension has degree -1.

Proposition 18.1 (Twisting adjunction)**.** *There is an adjunction $G \dashv F$. Its counit and unit are*

$$\epsilon_M(a \otimes c \otimes m) = a\epsilon_C(c)m, \qquad \eta_N(n) = \sum n_{(-1)} \otimes 1 \otimes n_{(0)}. \tag{18.2}$$

Proof. A degree-zero graded A-linear map $G(N) \to M$ is determined by a weight-preserving map $f : N \to M$. Its value on $a \otimes n$ is $af(n)$. It is a chain map precisely when

$$d_M f(n) - f(d_N n) = -\sum \tau(n_{(-1)}) f(n_{(0)}). \tag{18.3}$$

A graded C-colinear map $N \to F(M)$ is determined by its composite with $\epsilon_C \otimes \mathrm{id}_M$. For the resulting map f, its value is

$$n \longmapsto \sum n_{(-1)} \otimes f(n_{(0)}).$$

Coassociativity and (18.1) show that this is a chain map exactly when (18.3) holds. The free and cofree extension formulas are inverse and natural, proving the adjunction.

Applying them to identity maps gives (18.2). The counit is A-linear, and the unit is C-colinear. In the two composites

$$\epsilon_{G(N)} G(\eta_N), \qquad F(\epsilon_M)\eta_{F(M)},$$

the intervening coalgebra factor is removed by ϵ_C. The coalgebra and comodule counit identities reduce these composites to the respective identity maps. □

18.3 The two bar contractions

The composite $GF(M)$ has underlying tensor $A\widehat{\otimes}C\widehat{\otimes}M$. Write its elements as $a_0[a_1|\cdots|a_r]m$ and put $\bar{a} = a - \epsilon_A(a)1$. Its differential consists of the bar differential, the signed internal differential of M, and the two endpoint terms

$$-a_0a_1[a_2|\cdots|a_r]m, \qquad (-1)^{r-1}a_0[a_1|\cdots|a_{r-1}](a_r m).$$

Endpoint terms are absent when $r = 0$.

Define a degree--1 map

$$h_L\big(a_0[a_1|\cdots|a_r]m\big) = -1[\bar{a}_0|a_1|\cdots|a_r]m. \tag{18.4}$$

Let $i_M(m) = 1[\,]m$. Then

$$dh_L + h_L d = \mathrm{id} - i_M\epsilon_M.$$

Indeed, the new first endpoint returns the reduced part of the original leading coefficient. The first endpoint followed by h_L supplies its scalar part when $r > 0$. All other adjacent multiplications cancel in pairs, including the last module action. At $r = 0$ the remaining term is $a_0[\,]m - 1[\,](a_0m)$.

The internal differential of M cancels because insertion changes the bar-length sign. This proves the identity in every degree. The contraction is k-linear and preserves weight, though i_M need not be A-linear.

The second contraction is on

$$Q = C\widehat{\otimes}_\tau A.$$

Its last endpoint is $(-1)^{r-1}[a_1|\cdots|a_{r-1}]a_ra$. Define

$$h_R\big([a_1|\cdots|a_r]a\big) = (-1)^r[a_1|\cdots|a_r|\bar{a}]1. \tag{18.5}$$

With $j(1) = [\,]1$ and $p(c\otimes a) = \epsilon_C(c)\epsilon_A(a)$, one has

$$dh_R + h_R d = \mathrm{id} - jp.$$

The new last endpoint gives the identity term, its scalar remainder gives jp, and the preceding multiplications cancel. This contraction also preserves weight. Consequently $Q_0 = k$ and every positive-weight component of Q is contractible.

Theorem 18.2 (Koszul module equivalence in positive weights)**.** *The functors F, G preserve weightwise quasi-isomorphisms and induce inverse equivalences*

$$D_{\mathrm{wt}}(A) \simeq D_{\mathrm{wt}}(C).$$

Their unit and counit are (18.2).

Proof. The left contraction proves that ϵ_M is a quasi-isomorphism in each weight. For the unit, the underlying tensor of $FG(N)$ is $C\widehat{\otimes}A\widehat{\otimes}N$. At total weight w, filter it increasingly by the weight v of its final N-factor. There are only the values $w_0 \le v \le w$.

The reduced-coaction term lowers v, while every other differential term preserves it. The associated graded component at v is $Q_{w-v}\otimes_k N_v$ with its tensor differential. Give N_w the filtration concentrated at its largest value $v = w$. The unit is filtered: its leading term is $[\,]\otimes 1\otimes n$, and its other terms have lower final weight.

For $v < w$, the right contraction makes the target component contractible even after tensoring with N_v. For $v = w$, the associated graded unit is the identity of N_w. The cone therefore has a finite filtration with acyclic successive quotients. It is acyclic, so η_N is a quasi-isomorphism in weight w.

To check preservation of weak equivalences by F, filter by the final module weight decreasingly. The endpoint action increases that weight and disappears in the associated graded. For G, use the increasing filtration by final comodule weight, since its reduced coaction lowers that weight. The associated graded maps tensor the given quasi-isomorphisms with C_u or A_u over the field k. Such tensors preserve quasi-isomorphisms, even for unbounded complexes of vector spaces. Both filtrations are finite at each total weight, so the functors preserve the specified weak equivalences.

They consequently descend to the localizations. There their unit and counit are isomorphisms, and the triangle identities prove that the descended functors are inverse. □

The regular module and the augmentation module have different images. The right contraction identifies $F(A)$ with the trivial comodule k through the coaugmentation $k \to F(A)$. In contrast, $F(k) = C$ with its regular coaction, since the reduced algebra acts trivially on k. Thus the equivalence does not identify the regular representation with the augmentation representation.

18.4 Other completions and two-sided comparisons

The lower weight bound is part of the theorem. It bounds both the bar length and the successive reduced coactions in a fixed component. Without that bound the displayed finite-filtration argument does not apply.

Derived I-adic completion, where $I = \ker(A \to k)$, uses the different comparison

$$M \longrightarrow \varprojlim_{p} \big((A/I^p) \otimes^{\mathbb{L}}_{A} M\big).$$

Identifying it with the weight product requires a comparison of the filtrations and their derived limits. A coderived category has its own class of coacyclic complexes. It is not the localization specified here by definition.

For a right module R and a left module M with lower weight bounds, the normalized two-sided bar has terms

$$R\widehat{\otimes}(s\bar{A})^{\widehat{\otimes} r}\widehat{\otimes}M, \qquad \bar{A} = \prod_{u>0} A_u.$$

Only finitely many lengths and weight decompositions occur at any total weight. It computes their relative derived tensor product in these weighted categories. Indeed, the preceding left contraction resolves M by induced A-modules. Their coefficient complexes are complexes of vector spaces, and tensoring with an acyclic right module gives acyclic associated graded components. The finite weightwise filtration therefore justifies the derived tensor computation.

A geometric identification with a derived cotensor product needs a further two-sided twisting comparison. That comparison must preserve the chosen completion and the collision maps. The one-sided algebraic equivalence alone does not supply those operations.

18.5 Reconstruction from an endomorphism algebra

We now work in presentable categories, separately from the lower-bounded weighted categories above. Let $\mathcal{V}$ be a presentable symmetric monoidal ∞-category whose tensor product preserves colimits separately. Let $\mathcal{C}$ be a presentable left $\mathcal{V}$-module category, with the action preserving colimits separately, and choose $b \in \mathcal{C}$. The functor $v \mapsto v \otimes b$ has a right adjoint

$$H_{\mathrm{o}}(x) = \underline{\mathrm{Hom}}_{\mathcal{C}}(b, x) \in \mathcal{V}.$$

This is an internal mapping object, not an external complex chosen without a coefficient category.

Composition makes

$$E = H_{\mathrm{o}}(b)$$

an associative unital algebra in $\mathcal{V}$. Precomposition makes $H_{\mathrm{o}}(x)$ a right E-module: $f \cdot e = f \circ e$. Write $H : \mathcal{C} \to \mathrm{RMod}_E(\mathcal{V})$ for this lift. Evaluation makes b a left E-module object of $\mathcal{C}$. The opposite algebra is required if right modules are written as left modules.

For a right E-module M, define

$$P(M) = M \otimes_E b = \big|[r] \longmapsto M \otimes E^{\otimes r} \otimes b\big|.$$

The bars denote geometric realization of the two-sided simplicial bar. Its faces use the two actions and multiplication, and its degeneracies insert the unit of E. The tensor–internal-Hom adjunction in every simplicial degree identifies compatible maps from this realization with right E-module maps into $H(x)$. Thus $P \dashv H$.

Proposition 18.3 (Relative Morita criterion)**.** *Suppose H_0 is conservative and preserves geometric realizations. Suppose also that the natural map*

$$\theta_v : v \otimes E \longrightarrow H_0(v \otimes b)$$

is an equivalence for every $v \in \mathcal{V}$. Then P, H are inverse equivalences

$$\mathrm{RMod}_E(\mathcal{V}) \simeq \mathcal{C}.$$

Proof. On the free right module $v \otimes E$, the unit $M \to HP(M)$ is θ_v, with its right E-action. It is an equivalence by hypothesis. Every right module is the realization of its simplicial free-module resolution. After forgetting its action, the augmented resolution is split by insertion of the algebra unit.

The functor P preserves realizations because it is a left adjoint. The functor H preserves them because H_0 does and the forgetful functor from right modules creates them. Realizing the free-module unit equivalences proves the unit equivalence for every module.

For $x \in \mathcal{C}$, the triangle identity says that H of the counit $PH(x) \to x$ is inverse to the unit at $H(x)$. It is therefore an equivalence. Conservativity of H_0, and hence of H, proves that the counit itself is an equivalence. □

This is the free-module form of relative Morita reconstruction [80, Proposition 4.8.5.8]. It records the projection comparison θ_v as well as generation. No stability of a category of factorization objects is inferred merely from stability of its ambient coefficient category.

For $\mathcal{V} = D(k)$, let $\mathcal{C}$ be stable and k-linear, and let b be a compact generator. Here generation means that $\mathrm{RHom}_{\mathcal{C}}(b, x) = 0$ implies $x = 0$. Then H_0 is conservative. It is exact, and compactness makes it commute with filtered colimits and therefore with arbitrary coproducts. Exactness and these colimit properties imply preservation of geometric realizations, by their skeletal filtrations.

Both sides of θ_v are exact colimit-preserving functors of v. The map is the identity for $v = k$, and shifts and colimits of k generate $D(k)$. Hence θ_v is an equivalence for every v. The proposition recovers compact-generator Morita equivalence [80, Theorem 7.1.2.1], with the dg version in [67, Theorem 3.8(c)].

For example, in $D(k)$ take $b = k^n$ with $n > 0$. Then $E = \mathrm{Mat}_n(k)$, $H(x) = \mathrm{Hom}_k(k^n, x)$ has the right matrix action, and $P(M) = M \otimes_E k^n$. Evaluation and the matrix units give the inverse maps of classical Morita equivalence.

Compactness alone does not suffice. In $D(k) \times D(k)$, the compact object $b = (k, 0)$ has endomorphism algebra k. Its Hom functor sends (M, N) to M and annihilates the nonzero object $(0, k)$. The missing condition is generation.

18.6 Factorization descent

For a factorization category, the local reconstruction maps must also respect the maps used to assemble the category. Choose a chart diagram with coefficient categories $\mathcal{V}_U$, module categories $\mathcal{C}_U$, and compatible

objects b_U. Put $E_U = \underline{\mathrm{Hom}}_{C_U}(b_U, b_U)$. Assume these algebras are the chart values of an algebra A_b in the chosen category $\mathrm{Fact}_D(X)$. Use specified descent presentations

$$C \simeq \varprojlim_U C_U, \qquad \mathrm{RMod}_{A_b}(\mathrm{Fact}_D(X)) \simeq \varprojlim_U \mathrm{RMod}_{E_U}(\mathcal{V}_U).$$

The limits include the full descent coherence data.

Theorem 18.4 (Factorization Morita theorem)**.** *Suppose the relative Morita criterion holds on every chart. Suppose the local internal-Hom, evaluation and tensor comparison maps are compatible with the chart transitions and all factorization operations, including their higher coherences. Then the descent presentations above give an equivalence of factorization categories*

$$C \simeq \mathrm{RMod}_{A_b}(\mathrm{Fact}_D(X)).$$

Proof. On each chart, the proposition gives mutually inverse functors P_U, H_U. The stated compatibilities make them natural transformations between the two descent diagrams. Their units and counits are equivalences on every chart and commute with the same transition and factorization maps. Taking limits gives inverse equivalences of the displayed categories. The preserved operation maps and their coherences give the factorization structure on this equivalence. □

Thus a compact generator supplies the local theorem in stable k-linear charts, while factorization descent supplies the assembly. For a particular boundary object, both generation and the operation-compatible comparisons must be proved. The tensor counit and the one-sided module equivalence do not establish either condition.

Chapter 19

Derived center and bulk action

A central action must commute with multiplication together with all its homotopies. The derived center represents these actions. Its two compatible compositions give the little-disks structure, while a cyclic orientation supplies the additional circle action. To recognize a physical bulk as this center, one must construct its central action and prove that the representing map is an equivalence.

19.1 The action represented by the center

Let $\mathcal{V}$ be a presentable stable symmetric monoidal ∞-category whose tensor product preserves small colimits separately. All tensors in this section use this specified product. Let A be a unital associative algebra in $\mathcal{V}$, and write

$$A^e = A \otimes A^{\mathrm{op}}.$$

Its left modules are A-bimodules.

The internal derived mapping object

$$Z_{\mathrm{ord}}(A) = \underline{\mathrm{RHom}}_{A^e}(A, A) \in \mathcal{V} \tag{19.1}$$

is characterized by

$$\mathrm{Map}_{\mathcal{V}}(v, Z_{\mathrm{ord}}(A)) \simeq \mathrm{Map}_{A\text{-}A}(v \otimes A, A). \tag{19.2}$$

Here v ranges over $\mathcal{V}$, and the two A-actions on $v \otimes A$ act on its A-factor. Presentability and the colimit-preserving action give this internal Hom by the adjoint functor theorem. Composition makes it an associative algebra.

An E_2-algebra is an algebra over the operad of little two-disks. Write $\mathrm{Alg}_{E_1}(\mathcal{V})$ for the category of unital associative algebras in $\mathcal{V}$. Dunn additivity identifies an E_2-algebra with an associative algebra object in $\mathrm{Alg}_{E_1}(\mathcal{V})$. The latter category has the tensor product induced from $\mathcal{V}$.

For an E_2-algebra R, a coherent central action on A means a left R-module structure on A in $\mathrm{Alg}_{E_1}(\mathcal{V})$. Thus the action

$$\rho : R \otimes A \longrightarrow A$$

preserves the inner algebra multiplications, while its module identities use the outer multiplication on R. Units, associativity and interchange include their full higher coherences. Write $\mathrm{Act}^{\mathrm{cen}}_R(A)$ for the space of these structures on the fixed algebra A.

Theorem 19.1 (Higher Deligne center)**.** *The object* (19.1) *has a canonical E_2-algebra structure and a coherent central action on A. For every E_2-algebra R, this action induces a natural equivalence*

$$\mathrm{Map}_{\mathrm{Alg}_{E_2}(\mathcal{V})}(R, Z_{\mathrm{ord}}(A)) \simeq \mathrm{Act}^{\mathrm{cen}}_R(A). \tag{19.3}$$

Its underlying associative multiplication is composition.

Proof. Apply the center theorem for associative algebras in a presentably symmetric monoidal category [80, Corollary 5.3.1.15]. It gives a final algebra acting on A in $\mathrm{Alg}_{E_1}(\mathcal{V})$. The acting algebra is associative in that category, hence is an E_2-algebra by Dunn additivity [80, Theorem 5.1.2.2].

The underlying object of this center represents morphisms from the regular A-module to itself in the operadic module category [80, Theorem 5.3.1.30]. For the associative operad these modules are A-bimodules [80, Theorem 4.4.1.28]. Its representing property is therefore exactly (19.2). Uniqueness of representing objects identifies it with (19.1), including the evaluation map.

Restriction along a map of acting algebras pulls back the action on A. Finality says that each such action has a contractible space of compatible maps to the center. This identifies its space of actions with the left side of (19.3), naturally in R. The associative multiplication on the represented endomorphism object is composition. □

The two multiplications can also be seen in the monoidal category of A-bimodules. Its tensor is the relative tensor product $\otimes_A$, and its unit is the diagonal bimodule A. Endomorphisms of this unit can be composed or tensored over A. The interchange coherences give the two associative structures. Their E_2-structure is homotopy coherent, so a cochain model need not have a strictly commutative product.

The word *canonical* refers to the center with its universal action. It does not assert uniqueness of every possible E_2-structure on an unmarked complex. The tensor in (19.2) is equally essential: changing it changes the action problem.

19.2 Hochschild cochains and the first operation homotopy

For an ordinary unital algebra A over a field k, the two-sided bar resolution has

$$P_n = A \otimes_k A^{\otimes n} \otimes_k A, \qquad n \geq 0,$$

in cohomological degree $-n$. Its boundary is the alternating sum of adjacent multiplications, starting with sign $+1$, and its augmentation is multiplication $P_0 \to A$. The map

$$a_0 \otimes \cdots \otimes a_{n+1} \longmapsto 1 \otimes a_0 \otimes \cdots \otimes a_{n+1}$$

contracts the augmented complex as a complex of k-vector spaces. On the augmented term A, use $a \mapsto 1 \otimes a$. The first face returns the original tensor and every later face cancels with insertion after the preceding face. Each P_n is free over A^e, so this bounded-above resolution computes (19.1).

Thus the Hochschild cochain model is

$$C^p(A, A) = \mathrm{Hom}_k(A^{\otimes p}, A), \qquad p \geq 0,$$

using the identification $\phi \mapsto (-1)^{p(p+1)/2}\phi(1 \otimes - \otimes 1)$ from the ordinary Hom complex in degree p. Its differential is

$$\begin{aligned}(\delta f)(a_1, \ldots, a_{p+1}) = {} & a_1 f(a_2, \ldots, a_{p+1}) \\ & + \sum_{i=1}^{p} (-1)^i f(a_1, \ldots, a_i a_{i+1}, \ldots, a_{p+1}) \\ & + (-1)^{p+1} f(a_1, \ldots, a_p) a_{p+1}.\end{aligned} \tag{19.4}$$

Its cohomology is denoted $HH^p(A)$. Set $C^r(A, A) = 0$ for $r < 0$. In degree zero, $\delta a(b) = ba - ab$, so $HH^0(A)$ is the ordinary center of A.

For $f \in C^p(A, A)$ and $g \in C^q(A, A)$, the cup product is

$$(f \smile g)(a_1, \ldots, a_{p+q}) = f(a_1, \ldots, a_p)\, g(a_{p+1}, \ldots, a_{p+q}).$$

It represents the product of the derived center. The bar diagonal splits the inputs at p. Its Hom-tensor sign $(-1)^{pq}$ cancels the ratio of the cochain-identification signs, giving the displayed cup formula.

For $0 \le i < p$, let $f \circ_i g$ replace the block of inputs $a_{i+1}, \dots, a_{i+q}$ by $g(a_{i+1}, \dots, a_{i+q})$ and leave the other inputs in order. The signed insertion operation [81, §1] is

$$f \circ g = \sum_{i=0}^{p-1} (-1)^{i(q-1)} f \circ_i g \in C^{p+q-1}(A, A). \tag{19.5}$$

For $q = 0$, the value of g is inserted as a constant, and an empty sum is zero.

The bracket

$$[f, g] = f \circ g - (-1)^{(p-1)(q-1)} g \circ f$$

induces the degree--1 Gerstenhaber bracket on Hochschild cohomology. Together with the higher operations, these give a cochain realization of the little-disks action [71, §7, Corollary 1].

The first homotopy is visible without the full operad. If $D, E : A \to A$ are derivations, then

$$\delta(D \circ E)(a, b) = -D(a)E(b) - E(a)D(b).$$

Indeed, expand $D(E(ab))$ by the two Leibniz rules in (19.4). The remaining terms are the negative of $D \smile E + E \smile D$. Hence the cup products of these degree-one cocycles anticommute up to the displayed cochain homotopy.

This identity illustrates one operation of the E_2-structure. The full action requires all arities and their coherences, as in the theorem.

19.3 The polynomial example

Let k have characteristic zero and take $A = k[x]$. Write $x_\ell = x \otimes 1$ and $x_r = 1 \otimes x$ in $A^e = k[x_\ell, x_r]$. The sequence

$$0 \longrightarrow A^e e \xrightarrow{e \mapsto x_\ell - x_r} A^e \longrightarrow A \longrightarrow 0 \tag{19.6}$$

is a free bimodule resolution, with e in degree -1. The last map sets $x_\ell = x_r = x$. Its kernel is generated by $x_\ell - x_r$, by polynomial division, and multiplication by that element is injective.

Applying $\mathrm{Hom}_{A^e}(-, A)$ makes the differential zero. The chain map to the bar resolution sends e to $1 \otimes x \otimes 1$. Consequently the derivation $D = d/dx$, which takes x to 1, represents the degree-one generator. There are no higher cohomology groups, and therefore

$$HH^*(k[x]) \cong k[x] \otimes_k \Lambda(\xi), \qquad |\xi| = 1, \qquad \xi = [D].$$

The insertion formula gives

$$[a(x)D, b(x)] = a(x)b'(x), \qquad [a(x)D, b(x)D] = (ab' - ba')D.$$

In particular, $[\xi, x] = 1$. The ordinary commutative algebra $HH^0(A) = A$ alone does not contain this operation.

The relation $\xi^2 = 0$ has an explicit representative:

$$\delta\Big(-\frac{1}{2}D^2\Big)(a, b) = D(a)D(b).$$

Both endpoint terms of the Hochschild differential enter this identity. For $a = x$ and $b = x^2$, they give $\delta(D^2)(x, x^2) = -4x$, which cancels $2D(x)D(x^2) = 4x$. The cup square is thus zero in cohomology through a specified homotopy.

19.4 The bulk map and a change of sector

Let B be an E_2-algebra in $\mathcal{V}$, and equip a boundary algebra A with a coherent central B-action. The universal property gives its classifying map

$$\beta : B \longrightarrow Z_{\mathrm{ord}}(A). \tag{19.7}$$

The evaluation of $\beta(b)$ on A is the prescribed action of b. The contractibility in (19.3) specifies the map together with all action coherences.

For a field theory, this construction starts after the bulk and boundary operations have been realized in the common coefficient category with this central-action structure. In a mixed holomorphic–topological theory, that realization must also retain the collision coefficients and their supported maps. Constructing those operations is part of constructing the physical bulk-to-center map. A pointwise action on states supplies only the corresponding pointwise datum.

Centrality does not require (19.7) to be an equivalence. For example, the scalar E_2-algebra $B = k$ acts centrally on $k[x]$. Its map to the derived center includes the constants and misses both x and ξ. This is an example of a central action, with no assertion that it describes a complete field theory.

The failure also occurs in theories satisfying surface sewing. A finite-dimensional Frobenius algebra is a unital associative algebra over a field, with a linear functional ε for which $(a, b) \mapsto \varepsilon(ab)$ is nondegenerate. Its coproduct Δ is adjoint to multiplication. The Frobenius algebra is symmetric when $\varepsilon(ab) = \varepsilon(ba)$.

An oriented two-dimensional open–closed topological quantum field theory assigns vector spaces to intervals and circles, and linear maps to bordisms with corners. These assignments respect gluing and disjoint union. Its data are a symmetric Frobenius algebra A for the interval and a commutative Frobenius algebra B for the circle. They include a unital central algebra map $\iota : B \to A$. Its adjoint $\iota^* : A \to B$ must obey

$$\iota\iota^* = \mu_A \circ \sigma \circ \Delta_A, \tag{19.8}$$

where μ_A is multiplication and σ interchanges the two factors. These conditions are sufficient by the open–closed classification [73, Definition 2.3 and Corollary 4.3].

Example 19.2 (Two closed theories with one boundary algebra)**.** Take $A = k$ with $\varepsilon_A(1) = 1$. One choice of closed data is $B_1 = k$, with the same trace and with $\iota = \iota^* = \mathrm{id}$. A second choice is

$$B_2 = ke_1 \oplus ke_2, \qquad e_i e_j = \delta_{ij} e_i, \qquad 1_{B_2} = e_1 + e_2, \qquad \varepsilon_{B_2}(e_i) = 1.$$

Set $\iota(e_1) = 1$, $\iota(e_2) = 0$ and $\iota^*(1) = e_1$. Both Frobenius pairings are nondegenerate. The closed coproduct is $\Delta_{B_2}(e_i) = e_i \otimes e_i$, while $\Delta_A(1) = 1 \otimes 1$.

The map ι is unital and multiplicative, and its image is central. For $b = \lambda e_1 + \nu e_2$ and $a \in k$,

$$\varepsilon_{B_2}(b\, \iota^*(a)) = \lambda a = \varepsilon_A(\iota(b)a),$$

so the stated maps are adjoint. Both sides of (19.8) are id_k. Thus both choices define oriented open–closed theories.

They have the same open Frobenius algebra and the same image acting on it. Their values on a circle have dimensions one and two, so the theories are not equivalent. Since $Z_{\mathrm{ord}}(k) = k$, the second bulk-to-center map is the projection with kernel ke_2. A single boundary algebra therefore does not determine an arbitrary open–closed surface theory.

Now let $L : \mathcal{V} \to \mathcal{W}$ be a specified strong symmetric monoidal functor, where $\mathcal{W}$ satisfies the same presentability assumptions. Transporting the universal action gives a central action of $LZ^{\mathcal{V}}_{\mathrm{ord}}(A)$ on LA. Its classifying map is

$$c_{L,A} : LZ^{\mathcal{V}}_{\mathrm{ord}}(A) \longrightarrow Z^{\mathcal{W}}_{\mathrm{ord}}(LA). \tag{19.9}$$

The transported bulk action is classified by

$$\beta_L = c_{L,A} \circ L\beta : LB \longrightarrow Z^{\mathcal{W}}_{\mathrm{ord}}(LA).$$

Both composites have the same evaluation, so this factorization follows from the universal property.

No equivalence of $c_{L,A}$ is assumed. If L specifies a protected sector, the boundary is *open–closed complete in that sector* when β_L is an equivalence. This definition requires the sector functor and the central action, rather than only an equality of graded dimensions.

19.5 Cyclic orientation and the circle action

The framed little-disks operad is $E_2 \rtimes SO(2)$. Its circle rotates the disk configurations, so its action must be compatible with the E_2-operations. We now state the extra datum that produces this compatibility.

Let k be a field and let C be a dualizable presentable k-linear category. Dualizability is taken in the symmetric monoidal category of presentable k-linear categories and colimit-preserving k-linear functors. Write

$$R = HH^{\bullet}(C) = \underline{\mathrm{RHom}}_{\mathrm{Fun}^L_k(C,C)}(\mathrm{id}_C, \mathrm{id}_C).$$

Composition of endofunctors makes the functor category monoidal with unit id_C. The two unit-endomorphism multiplications give R its E_2-structure [17, Proposition 2.1.5].

For $C = \mathrm{LMod}_A(D(k))$, this recovers $Z_{\mathrm{ord}}(A)$. Indeed, an A-bimodule P gives the colimit-preserving functor $P \otimes_A -$. A k-linear colimit-preserving functor is determined by its value on the free module A and its right A-action. The assertion holds first on free modules $v \otimes A$ and then on every module by its simplicial free resolution. The same argument identifies natural transformations with bimodule maps. Under this equivalence the diagonal bimodule corresponds to the identity functor, and relative tensor corresponds to composition.

The same free-resolution argument works for $\mathcal{V}$-linear colimit-preserving functors on $\mathrm{LMod}_A(\mathcal{V})$ under the hypotheses of the first section.

Let

$$T = HH_{\bullet}(C)$$

be the categorical trace of id_C, an object of $D(k)$ with its canonical circle action. Explicitly, the trace is the composite of coevaluation, interchange and evaluation for the dual pair $C, C^{\vee}$. For $\mathrm{LMod}_A(D(k))$, it is the Hochschild chain object $A \otimes^{\mathbb{L}}_{A^e} A$. Its homological degree n is placed in cohomological degree $-n$.

Put $H_R = \int_{S^1} R$, the factorization homology of the E_2-algebra R around a circle. It is an associative algebra using the remaining E_1-direction, and it carries the circle action. Noncommutative calculus makes T a circle-equivariant H_R-module [17, Theorem 2.3.7]. An embedded interval gives $R \to H_R$. Restricting the action along this map gives the cap action $R \otimes T \to T$.

For an integer d, a noncommutative orientation of dimension d is a class

$$\theta : k \longrightarrow T^{hS^1}[-d]$$

whose cap map is an equivalence

$$D_\theta : R \xrightarrow{\sim} T[-d]. \tag{19.10}$$

Here hS^1 denotes homotopy fixed points, so the datum includes coherent circle invariance.

There is also a version using a trace. Put $T^\vee = \underline{\mathrm{RHom}}_k(T, k)$. A noncommutative coorientation is a circle-invariant trace

$$\tau : T_{hS^1} \longrightarrow k[-d]$$

for which the pairing with the cap action gives an equivalence

$$D_\tau : R \xrightarrow{\sim} T^\vee[-d].$$

The subscript hS^1 denotes homotopy orbits. In particular, τ includes more data than a functional on $HH_0(C)$.

These conditions are supplied by Calabi–Yau structures as follows. Call C smooth, respectively proper, when its evaluation has a left, respectively right, adjoint in the category of presentable k-linear categories. Put $I = \mathrm{id}_C$. Those adjoints evaluated on k define the inverse dualizing functor $I^!$, respectively the Serre functor S, in $\mathrm{Fun}^L_k(C, C)$. The trace identifications are

$$T \simeq \underline{\mathrm{RHom}}(I^!, I) \quad (C \text{ smooth}),$$
$$T^\vee \simeq \underline{\mathrm{RHom}}(I, S) \quad (C \text{ proper}).$$

A smooth Calabi–Yau structure of dimension d is a circle-invariant class θ as above whose underlying map $I^![d] \to I$ is an equivalence. Precomposition with that map makes D_θ an equivalence. A proper Calabi–Yau structure is a circle-invariant trace τ whose underlying map $I \to S[-d]$ is an equivalence. Postcomposition makes D_τ an equivalence. These are the absolute smooth and proper constructions of Brav–Rozenblyum [17, §§4.1.9 and 4.3.5].

Theorem 19.3 (Cyclic refinement)**.** *An orientation or coorientation as above induces a framed E_2-algebra structure on $HH^\bullet(C)$ refining its existing E_2-structure. Thus a dualizable linear category with a smooth or proper Calabi–Yau structure has the stated cyclic refinement.*

Proof. The noncommutative calculus makes T a circle-equivariant H_R-module. The oriented case supplies the shifted module $T[-d]$ and its circle-invariant vector θ. Its cap map from R is an equivalence by (19.10). The framed-algebra criterion of Brav–Rozenblyum identifies precisely these data with a framed E_2-algebra [17, Theorem 3.3.9]. This proves the absolute case of their cyclic Deligne theorem [17, Theorem 4.1.7].

For a coorientation, use the dual calculus on $T^\vee[-d]$ and the invariant vector corresponding to τ. The assumed cap equivalence is the same nondegeneracy condition in that module. The cooriented cyclic Deligne theorem supplies the framed refinement [17, Theorem 4.3.4]. The original E_2-operations are retained in both constructions. □

19.6 The circle operator for the polynomial algebra

Return to $A = k[x]$ in characteristic zero. Put $\bar{A} = A/k1$ and $C_n(A) = A \otimes_k \bar{A}^{\otimes n}$, placed in degree $-n$. In this normalized Hochschild chain complex, write $a_0[a_1|\cdots|a_n]$ for $a_0 \otimes \bar{a}_1 \otimes \cdots \otimes \bar{a}_n$. The Hochschild boundary is the alternating sum of adjacent products followed by the cyclic last product. In particular, $b(a_0[a_1]) = a_0a_1 - a_1a_0$.

Fix Connes' operator by the convention

$$B(a_0) = 1[a_0].$$

On a chain of length one it is

$$B(a_0[a_1]) = 1[a_0|a_1] - 1[a_1|a_0].$$

An inner occurrence of 1 vanishes in the normalized complex.

The chain $\theta = 1[x]$ has both $b\theta = 0$ and $B\theta = 0$. The negative cyclic complex is the degreewise product $C_\bullet(A)[[u]]$ with differential $b + uB$ and $|u| = 2$. Thus the same constant series θ defines a negative cyclic class, or equivalently a coherent circle-invariant class, of dimension one. Tensoring (19.6) over A^e with A gives

$$HH_0(A) = A, \qquad HH_1(A) = A\,dx, \qquad HH_n(A) = 0 \quad (n > 1).$$

The class of $a_0[a_1]$ maps to $a_0 a_1'(x)\,dx$. The Leibniz rule makes this map vanish on boundaries, and it sends $1[x]$ to the generator dx.

Cap product with θ sends $f(x) \in HH^0(A)$ to $f(x)\,dx$ and sends $a(x)D \in HH^1(A)$ to $a(x)$. Both maps are isomorphisms. The category $\mathrm{LMod}_A(D(k))$ is compactly generated by A and hence is dualizable. Hence θ is an orientation in the sense of (19.10), and the cyclic refinement applies.

With these cap and Connes conventions, the transported degree--1 circle operator is

$$\Delta(f) = 0, \qquad \Delta(aD) = a'.$$

Indeed, $B(a) = 1[a]$ represents $a'(x)\,dx$, while the next homology group is zero. For homogeneous classes u, v, the bracket convention used above is

$$[u, v] = (-1)^{|u|-1}\big(\Delta(u \smile v) - \Delta(u) \smile v - (-1)^{|u|} u \smile \Delta(v)\big).$$

It gives $[aD, f] = af'$ and $[aD, bD] = (ab' - ba')D$, as before. In particular, the circle operation recovers $[\xi, x] = 1$.

The cyclic theorem supplies this framed structure from the orientation data. It does not construct a mixed collision action or identify a specified physical bulk with the center. Those comparisons must preserve their coefficient tensor, support, completion and operation coherences.

Chapter 20

Cyclic closure and modular sewing

Closing an ordered interval requires a pairing that can be used in both directions. The pairing joins two ends, and its inverse cuts an internal edge. Independence of the cuts requires cyclic symmetry of every higher multiplication. We first construct the resulting surface theory over a field, then state the additional maps needed for chiral sewing.

20.1 The cyclic pairing

Let k be a field of characteristic zero. Use cohomological grading and the shift $(V[r])^i = V^{i+r}$. For a complex V, write $V^\vee = \mathrm{RHom}_k(V, k)$. A complex is perfect over k when it is quasi-isomorphic to a bounded complex of finite-dimensional vector spaces. An algebra is called proper when it is perfect over k, and smooth when its diagonal bimodule is perfect over its derived enveloping algebra A^e. Fix an integer d.

Definition 20.1 (Finite Calabi–Yau ordered algebra)**.** A finite Calabi–Yau ordered algebra of dimension d consists of a finite-dimensional, strictly unital A_∞-algebra A and a pairing

$$B : A \otimes_k A \longrightarrow k[-d].$$

The pairing is closed and graded-symmetric, and induces an isomorphism of complexes $A \cong A^\vee[-d]$. Its higher products $m_n : A^{\otimes n} \to A$ have degree $2 - n$, with $m_1 = d_A$ and $m_0 = 0$. They satisfy the A_∞ identities and, for homogeneous inputs, the cyclic identity

$$B(m_n(a_0, \dots, a_{n-1}), a_n) = (-1)^{n+|a_0|\sum_{i=1}^{n}|a_i|} B(m_n(a_1, \dots, a_n), a_0). \tag{20.1}$$

The unit $1 \in A^0$ is closed, is a two-sided unit for m_2, and makes m_n vanish whenever it is an input and $n \neq 2$.

Here graded symmetry means $B(a, b) = (-1)^{|a||b|} B(b, a)$. Closedness means $B(d_A a, b) + (-1)^{|a|} B(a, d_A b) = 0$. For $n = 1$, this is precisely (20.1) after graded symmetry. For $n = 2$, the identity and the unit give

$$\tau(a) = B(1, a), \qquad B(a, b) = \tau(m_2(a, b)).$$

Thus the pairing determines a trace, while the identities for all n retain the compatibility with higher products.

The isomorphism to the shifted dual supplies an inverse pairing. It is a degree-d coevaluation $k \to A \otimes A$ whose two contractions with B are the identity, with the graded tensor signs. Finite dimensionality makes every contraction along a fixed finite graph a finite sum. There is no sum over all genera in this assertion.

For a perfect derived algebra, a proper Calabi–Yau structure uses a coherent circle-invariant functional

$$CC_*(A)_{hS^1} \longrightarrow k[-d].$$

Its underlying bimodule map $A \to A^{\vee}[-d]$ must be an equivalence. Here $CC_*(A)$ is the Hochschild chain object, and hS^1 denotes homotopy orbits, as in the preceding chapter. A functional on $HH_0(A)$ alone does not specify this coherence or a finite cyclic model. The surface construction below uses the model of Definition 20.1.

20.2 Surfaces and the determinant twist

Fix one boundary label. An open boundary is a parametrized interval in the boundary of an oriented bordered Riemann surface. A closed boundary is a parametrized boundary circle. Each is designated incoming or outgoing, and the remaining boundary is called free. The objects of the surface category are finite collections of open intervals and closed circles. Its morphism spaces are moduli spaces of these surfaces, with composition given by gluing matching boundaries.

We use Costello's admissibility convention: every connected component has an incoming closed boundary or a free boundary. In particular, a surface with only closed boundaries must have an incoming one on every component. Disks and annuli with only free boundary are represented by points. The one-input, one-output open disk is an identity, and closed cylinders are replaced by their reparametrization model, including the identity. These conventions define the surface category used in Costello's Theorem A [25, §3].

The degree of the pairing requires a local coefficient line on each moduli space. For a surface Σ with o_{out} outgoing open intervals, put

$$\mathcal{D}(\Sigma) = \det\big(H^0(\Sigma;k) \oplus k^{o_{\mathrm{out}}}\big) \otimes \det\big(H^1(\Sigma;k)\big)^{-1}, \qquad |\mathcal{D}(\Sigma)| = o_{\mathrm{out}} - \chi(\Sigma). \tag{20.2}$$

The last equality places this line in the stated cohomological degree. The inverse on the second determinant denotes the dual line. This is Costello's determinant local system after changing homological to cohomological grading.

Mayer–Vietoris for a gluing identifies the determinant line of the glued surface with the tensor product of those of its two pieces. The outgoing-open factor in (20.2) accounts for the intervals that disappear in the gluing. These identifications are associative and compatible with disjoint union [25, §3, equation defining $\det(\Sigma)$]. For example, a disk with two incoming open intervals gives a line of degree -1. Its dth tensor power therefore permits the pairing of degree $-d$. The disk with two outgoing intervals gives the inverse pairing of degree d.

Write $\mathcal{OC}^d$ for the dg symmetric monoidal category of chains on these moduli spaces with coefficients in $\mathcal{D}^{\otimes d}$. A singular r-chain is placed in cohomological degree $-r$, in addition to the degree of its coefficient line. Write $\mathcal{O}^d$ for its full subcategory on purely open objects. Write $\mathcal{C}^d$ for the subcategory with purely closed objects and no free boundaries in its morphisms. Negative powers of $\mathcal{D}$ mean powers of its dual.

An open topological conformal field theory is a dg symmetric monoidal functor $\Phi : \mathcal{O}^d \to \mathrm{Ch}(k)$ whose tensor comparison maps are quasi-isomorphisms. This last condition is called homological splitting. An open–closed theory is such a functor on $\mathcal{OC}^d$. A chain in a morphism space acts with its chain degree, its boundary acts by the Hom differential, and sewing acts by composition. These theories retain the chains of moduli spaces, rather than just the diffeomorphism classes of surfaces.

20.3 The open theory and its circle states

Theorem 20.2 (Ribbon-surface realization)**.** *A finite Calabi–Yau ordered algebra A determines an open theory Φ_A on O^d, up to quasi-isomorphism, with interval value quasi-isomorphic to A. Let $i : O^d \hookrightarrow OC^d$ be the inclusion. Its derived extension*

$$\Psi_A = OC^d \otimes^{\mathbb{L}}_{O^d} \Phi_A \tag{20.3}$$

is homologically split and is the homotopy-universal open–closed extension of Φ_A. Its restriction to C^d has circle value quasi-isomorphic to $CC_*(A)$. Consequently, the homology of the circle value is $HH_*(A)$. The theorem applies in particular to a smooth proper algebra supplied with such a finite cyclic model.

Proof. We apply Costello's Theorem A and describe the maps that enter it. In the cellular model, disks with one outgoing interval give the higher products, and the two-input and two-output disks give the pairing and its inverse. The inverse relations are the two contraction identities for B. Rotating a disk gives (20.1), with the cell orientation and determinant signs. The boundary of a disk cell is a sum of two-disk gluings, and its image is the corresponding A_∞ identity. Thus the algebra defines a split functor on the cellular open category [25, Theorem 6.2.3 and Lemma 7.3.2].

The cellular category is quasi-isomorphic to O^d. Derived extension along this quasi-isomorphism transports the functor to Φ_A without changing its interval value [25, Lemmas 6.1.7 and 7.3.4]. Formula (20.3) is the derived left adjoint to restriction along i. This gives its universal property and identifies its restriction to open objects with Φ_A.

To compute a circle, use the cellular right module generated by annuli with ordered marked intervals on the outer boundary and one parametrized inner circle. The unit relations remove markings carrying the unit, except at the distinguished starting point. After tensoring with A, these are exactly the normalized Hochschild chains. The annular boundary joins consecutive inputs, including the last with the first. Its differential is therefore the Hochschild differential.

For a dg model this is the explicit identification in Costello's Proposition 7.4.3. The same proposition and its quasi-isomorphism invariance give the statement for A_∞ models.

Filter the annular module by its number of markings. Its differential lowers this filtration, and the unit relations split on the associated graded. This proves the flatness needed for the derived tensor and shows that disjoint unions give quasi-isomorphic tensor comparison maps [25, Lemma 6.3.1]. It proves homological splitting of Ψ_A and completes the construction. □

The open category has no circle object. The circle assertion concerns Ψ_A, and an arbitrary open–closed theory receiving the same open theory receives a comparison from Ψ_A. That comparison need not be an equivalence. The two surface theories in the preceding chapter illustrate why the open states alone do not identify every possible closed sector.

In a ribbon graph, a cyclic order is chosen for the flags at each vertex. The cyclic higher product paired with the remaining input labels a vertex, and the inverse pairing contracts an internal edge. Costello's cellular model uses nodal disks and annuli, whose boundary creates nodes. The dual ribbon description uses edge contractions. The comparison of these models, including their orientations, is part of the geometric theorem, not a consequence of the A_∞ identities alone.

20.4 A finite open algebra with unbounded circle homology

Take $A = k[t]/(t^2)$ in degree zero, with $\tau(1) = 0$ and $\tau(t) = 1$. Multiplication and $B(a, b) = \tau(ab)$ make it a finite cyclic algebra of dimension zero. In the basis $1, t$, the pairing and inverse pairing are

$$[B] = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \operatorname{coev}(1) = 1 \otimes t + t \otimes 1.$$

For $a = \alpha + \beta t$, contraction on either side gives

$$1\,B(t,a) + t\,B(1,a) = \alpha + \beta t, \qquad B(a,1)t + B(a,t)1 = \alpha + \beta t.$$

Both terms of the inverse pairing are necessary. Joining them by multiplication gives the handle element $2t$, whose square is zero.

The circle complex retains more information than this two-dimensional open space. Put $R = A^e = k[x,y]/(x^2, y^2)$ and let $R \to A$ send x, y to t. There is a free bimodule resolution

$$\cdots \xrightarrow{x-y} R \xrightarrow{x+y} R \xrightarrow{x-y} R \longrightarrow A \longrightarrow 0. \tag{20.4}$$

Indeed, multiplication by $x - y$ has kernel spanned by $x + y$, xy, which is the image of multiplication by $x + y$. The reverse kernel is spanned by $x - y$, xy, and the augmentation kernel is also this latter span. This proves exactness in every degree.

Tensoring the resolution over R with A gives alternating differentials 0 and $2t$. Hence

$$HH_0(A) = A, \qquad HH_{2r+1}(A) = A/(t), \qquad HH_{2r+2}(A) = kt \quad (r \geq 0). \tag{20.5}$$

The circle complex is therefore not perfect over k. This algebra is proper and not smooth: a perfect bimodule resolution would make $A \otimes^{\mathbb{L}}_{A^e} A$ bounded, contrary to (20.5).

Smoothness is thus unnecessary for Theorem 20.2, but it changes the finiteness of the closed states. If A is both smooth and proper, a finite construction of the diagonal from perfect A^e-modules makes its Hochschild chains perfect over k. Neither conclusion identifies the bordered-surface category with the compactification by stable algebraic curves.

20.5 Stable curves and modular operations

For $2g - 2 + n > 0$, let $\overline{\mathcal{M}}_{g,n}$ be the moduli stack of stable complex curves of arithmetic genus g with n labeled smooth marked points. Stability means that each rational component has at least three special points and each genus-one component at least one, with nodes counted as special points. Identifying two marked points gives the clutching maps

$$\overline{\mathcal{M}}_{g_1,n_1+1} \times \overline{\mathcal{M}}_{g_2,n_2+1} \longrightarrow \overline{\mathcal{M}}_{g_1+g_2,n_1+n_2}, \tag{20.6}$$

$$\overline{\mathcal{M}}_{g-1,n+2} \longrightarrow \overline{\mathcal{M}}_{g,n}. \tag{20.7}$$

Their images are nodal curves, so these are maps to the compactified moduli stacks.

A stable graph is a finite connected graph with labeled external legs and a nonnegative integer $g(v)$ at each vertex, such that $2g(v) - 2 + n(v) > 0$. Here $n(v)$ counts all incident flags, including both flags of a loop. Its genus is

$$g(G) = \sum_v g(v) + b_1(G).$$

Contracting an edge between distinct vertices adds their vertex genera. Contracting a loop raises its vertex genus by one. Both operations preserve $g(G)$.

A modular operad P assigns complexes $P(g,n)$ with symmetric-group actions and equivariant composition maps

$$\mu_G : \bigotimes_v P(g(v), n(v)) \longrightarrow P(g(G), n(G))$$

for stable graphs. Here $n(G)$ is the number of external legs. The corolla map is the identity, and composition is compatible with every graph contraction and substitution. In particular, the operad includes a contraction of two flags at one vertex. The genus-zero cyclic operations alone do not supply this map.

Clutching gives the stable-curve modular operad, and compatible chains or homology give its linear versions [43, §6.2]. For a finite complex V with a degree-zero symmetric perfect pairing, the collection $V^{\otimes n}$ is an endomorphism modular operad. Its compositions contract the two factors at each internal edge with the pairing. A modular action is a morphism to this endomorphism operad [43, §2.25]. In the dual convention of multilinear forms, the same contraction uses the inverse pairing. Nonzero-degree pairings require the corresponding graded determinant twists.

A chiral coefficient over pointed curves must therefore provide the appropriate pullback and pushforward maps along (20.6) and (20.7), with their support and completion conditions. Its operadic compositions must agree under successive clutching. The bordered-surface theorem constructs neither these coefficient maps nor their comparison with its ribbon operations.

20.6 The mixed sewing equation

Suppose a common collection of decorated graphs has edges of two kinds, transverse and chiral, and is closed under their contractions. Assume all iterated contraction targets are coherently identified, including those involving loops. To a graph G assign a cochain complex $C(G)$ and to an edge e a degree-zero cochain map $c_e^G : C(G) \to C(G/e)$. Require equivariance under graph isomorphisms.

The coefficient equation for distinct edges is

$$c_f^{G/e} c_e^G = c_e^{G/f} c_f^G. \tag{20.8}$$

For different edge kinds this is an additional mixed sewing condition. For one kind it is part of the corresponding modular or ribbon composition law.

Proposition 20.3 (Strict mixed contraction)**.** *Under these assumptions, contraction of any finite set of edges is independent of its ordering. Put* $\mathrm{or}(G) = \det(kE(G))$ *in degree* $-|E(G)|$ *and form the direct sum*

$$K^n = \bigoplus_{[G]} \big(C^{n+|E(G)|}(G) \otimes \mathrm{or}(G)\big)_{\mathrm{Aut}(G)}.$$

For homogeneous $x \in C^q(G)$*, let*

$$D(x \otimes \omega) = d_C x \otimes \omega + (-1)^q \sum_{e \in E(G)} c_e^G x \otimes \iota_e \omega,$$

where ι_e *removes* e *from the oriented wedge with its exterior sign. Then* $D^2 = 0$*. The transverse and chiral edge parts anticommute.*

Proof. Any two orders of a finite edge set differ by adjacent transpositions. Equation (20.8) identifies the composites for each such transposition at the intermediate graph. The coherent target identifications then give the same total contraction. This is the graph-contraction argument of Getzler–Kapranov's Theorem 3.7, with the mixed squares included.

For the differential, contracting two distinct edges in opposite orders gives equal coefficient maps and opposite exterior signs, since $\iota_f \iota_e = -\iota_e \iota_f$. The internal differential anticommutes with each edge term because c_e is a cochain map and the sign $(-1)^q$ changes when d_C is applied. An edge cannot be contracted twice. These facts prove the square-zero and anticommutation identities, as in Theorem 13.3. The argument uses no acyclicity of G, so it includes loops. Equivariance descends the differential to coinvariants, and each graph contributes finitely many terms to the direct sum. □

The determinant sign and the coefficient equation have distinct roles. For two edges e, f, the two-edge term in D^2 is

$$(c_f^{G/e} c_e^G - c_e^{G/f} c_f^G)x.$$

For example, take $C(G) = k^2$ at every contraction target and take the two maps to be

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad Q = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

Then $QP - PQ = -Q \neq 0$. An orientation does not remove this defect. If the coefficient equation holds only up to homotopy, the action also requires compatible higher homotopies for longer contraction sequences.

Nor does an edge coloring alone define two independent genera. Take two genus-zero vertices joined by one transverse and one chiral edge, and give each vertex an external leg so that the graph is stable. Contracting the transverse edge first leaves a chiral loop. Reversing the order leaves a transverse loop. Counting the genus increment by the loop color gives $(g_\perp, g_X) = (0, 1)$ in one order and $(1, 0)$ in the other. The total genus is one in both.

Separate geometric genera must be carried by a more structured sewing object if both contraction orders are allowed.

For an ordered chiral algebra, the remaining construction is consequently a common coefficient system realizing the two geometric sewings, with (20.8) or its coherent replacement. It must preserve the coefficient tensor, support, completion, units and the chosen cyclic pairing. Theorem 20.2 supplies the transverse surface operations when a finite cyclic model is given. An extension to stable curves, and a comparison of the two geometric genera, require their own maps and identities.

Book IV

Exact noncommutative models

Chapter 21

Free and square-zero pairs

The bar differential joins neighboring inputs by multiplication. For a tensor algebra these joins contract every word involving more than one generator. For a square-zero algebra all such joins vanish, so every tensor length remains. The two calculations exhibit the dependence of the bar on multiplication and determine the resulting dual products.

21.1 The tensor algebra and its primitive summand

Let k be a commutative ring and M a cochain complex of k-modules. All tensor products in this section are over k. Put $s = [1]$, so $|sm| = |m| - 1$. The tensor algebra and its augmentation are

$$T(M) = \bigoplus_{n \geq 0} M^{\otimes n}, \qquad \epsilon : T(M) \longrightarrow k,$$

where $M^{\otimes 0} = k$ and ϵ projects to this summand. Multiplication concatenates tensors, and the differential extends that of M by the graded Leibniz rule. Give each generator weight one.

For an augmented dg algebra $A = k \oplus \bar{A}$, use the normalized bar

$$BA = \bigoplus_{r \geq 0} (s\bar{A})^{\otimes r}$$

with the differential and deconcatenation of Theorem 15.1. In particular, $b_1(sa) = -s(da)$ and $b_2(sa \otimes sb) = (-1)^{|a|} s(ab)$. This is the direct-sum bar, with no tensor-length completion.

Let $K(M) = k \oplus sM$ have its suspended internal differential and primitive reduced summand. Thus

$$\Delta(sm) = 1 \otimes sm + sm \otimes 1, \qquad \Delta(1) = 1 \otimes 1.$$

Including a generator as a one-input bar defines a dg coalgebra map

$$i : K(M) \longrightarrow B(T(M)), \qquad i(sm) = [m]. \tag{21.1}$$

Theorem 21.1 (Bar of the tensor algebra)**.** *The map i is a quasi-isomorphism of coaugmented dg coalgebras. There are natural chain maps $p : B(T(M)) \to K(M)$ and a degree--1 map h satisfying*

$$pi = \mathrm{id}, \qquad bh + hb = \mathrm{id} - ip, \qquad h^2 = hi = ph = 0.$$

No flatness hypothesis on M is needed for these chain identities.

Proof. Expand each bar input into its finite sum of generator tensors. A summand of positive generator weight n is specified by a subset $S \subseteq \{1, \dots, n-1\}$ of cuts between adjacent generators. For homogeneous $m_1, \dots, m_n$, denote its tensor by w_S. Its degree is

$$|w_S| = \sum_{a=1}^{n} |m_a| - |S| - 1.$$

This notation describes multilinear tensor maps and does not require a basis of M.

For $j \in S$, let $r_S(j) = \#\{a \in S : a \le j\}$. The multiplication part b_μ removes the cut j with coefficient

$$(-1)^{\sum_{a \le j} |m_a| - r_S(j) + 1}. \tag{21.2}$$

Indeed, $r_S(j) - 1$ suspended blocks precede the block ending at j. Passing those blocks and applying b_2 gives exactly this sign.

For $n \ge 2$, define

$$h(w_S) = \begin{cases} (-1)^{|m_1|} w_{S \cup \{1\}}, & 1 \notin S, \\ 0, & 1 \in S. \end{cases} \tag{21.3}$$

Set $h = 0$ in weights zero and one. Let p be the identity on those two weights and zero on every larger weight. Both b_μ and the internal differential preserve generator weight, so p and i are chain maps.

If $1 \notin S$, removing the inserted first cut contributes w_S to $b_\mu h(w_S)$. Every other cut has its rank increased by one after insertion, so the two orders of removing that cut and inserting the first cancel. If $1 \in S$, only removal of that first cut can contribute to $h b_\mu(w_S)$, and its coefficient is again one. This proves $b_\mu h + h b_\mu = \mathrm{id}$ in weights at least two.

The internal differential replaces m_j by $d_M m_j$ with coefficient

$$(-1)^{\sum_{a<j} |m_a| - \#\{a \in S : a < j\} + 1}.$$

For $j > 1$, inserting the first cut reverses this sign. For $j = 1$, applying d_M reverses the sign $(-1)^{|m_1|}$ in h. Thus the internal differential anticommutes with h in every case. Combining the two parts proves $bh + hb = \mathrm{id} - ip$.

Inserting the first cut twice is impossible, giving $h^2 = 0$. The remaining side identities follow from the weight projections. Finally, a one-input bar is primitive, so i preserves deconcatenation, the counit and the coaugmentation. The chain contraction makes it a quasi-isomorphism. □

The projection p need not preserve the coproduct. Over a field, take $M = kx$ in degree zero and consider $[x|x]$. Its image under p is zero, while its reduced deconcatenation gives

$$(p \otimes p)\overline{\Delta}[x|x] = sx \otimes sx \neq 0.$$

Thus the displayed contraction is a contraction of complexes, and i is the specified coalgebra quasi-isomorphism. The distinction matters when dualizing multiplication.

21.2 The square-zero algebra

The square-zero extension of M is the dg algebra

$$S(M) = k \oplus M, \qquad (a, m)(a', m') = (aa', am' + a'm).$$

Its unit is $(1, 0)$, its augmentation projects to k, and its differential restricts to d_M.

Proposition 21.2 (Square-zero bar)**.** *There is an equality of dg coalgebras*

$$B(S(M)) = T^c(sM) = \bigoplus_{r \ge 0} (sM)^{\otimes r}$$

with deconcatenation and the internal tensor differential. The multiplication part of the bar differential is zero.

Proof. Every adjacent multiplication in the augmentation ideal is a product of two elements of M. Such products vanish by the definition of $S(M)$. The unary bar differential remains $b_1(sm) = -s(d_M m)$, and its tensor extensions give precisely the stated differential. The coproduct is unchanged. □

If M is concentrated in degree zero with zero differential, every reduced bar tensor is a cycle. For a d-dimensional vector space over a field, the homological bar degrees therefore give

$$\dim_k \operatorname{Tor}_r^{S(M)}(k,k) = d^r, \qquad \sum_{r\geq 0} \dim_k \operatorname{Tor}_r^{S(M)}(k,k)\, z^r = \frac{1}{1-dz}.$$

The tensor algebra instead has $\operatorname{Tor}_0 = k$, $\operatorname{Tor}_1 = M$ and zero higher groups, by Theorem 21.1. These are statements about the two specified differentials, not a comparison of their chain dimensions.

21.3 Dual multiplication and completion

Now assume that M is a bounded complex of finitely generated projective k-modules. It represents a dualizable object of $D(k)$. Write $M^\vee = \operatorname{Hom}_k(M,k)$, with its Hom differential. For a dg coalgebra C, its full linear dual is $C^\vee = \operatorname{Hom}_k(C,k)$. Its convolution product is

$$(f \star g)(c) = \sum_{(c)} (-1)^{|g||c_{(1)}|} f(c_{(1)})\, g(c_{(2)}).$$

Every deconcatenation here is finite on an individual input.

For an augmented algebra A, write $A^! = \operatorname{RHom}_A(k,k)$ with its derived composition product. The completion of the tensor algebra on a complex N means

$$\widehat{T}(N) = \prod_{r\geq 0} N^{\otimes r}.$$

Products are degreewise, and the length filtration has neighborhoods $\prod_{r\geq p} N^{\otimes r}$ of zero. Concatenation is continuous, because an output of length r receives only the finitely many products with input lengths summing to r.

Theorem 21.3 (Free–square-zero duality)**.** *Under these hypotheses there are dg algebra quasi-isomorphisms representing*

$$T(M)^! \simeq k \oplus M^\vee[-1], \qquad S(M)^! \simeq \widehat{T}\big(M^\vee[-1]\big).$$

The first algebra has square-zero reduced summand and its internal differential. The second has the completed concatenation product and its internal tensor differential. Its completion comes from dualizing a direct sum of bar tensors.

Proof. For either algebra, the left bar resolution $P = A \otimes BA \to k$ computes derived Hom. For homogeneous inputs its differential is

$$\begin{aligned} d_P(a \otimes [a_1|\cdots|a_r]) = {} & d_A a \otimes [a_1|\cdots|a_r] + (-1)^{|a|} a \otimes b[a_1|\cdots|a_r] \\ & + (-1)^{|a|+1} a a_1 \otimes [a_2|\cdots|a_r], \end{aligned}$$

with the last term omitted when $r = 0$.

Writing $\bar a = a - \epsilon(a)1$, its augmented contraction inserts $\bar a$ as a first bar input, with coefficient -1. The left multiplication term recovers $\bar a$. In positive bar length, first multiplication followed by insertion also recovers the scalar part of a. The other internal and merging terms cancel in pairs. This gives $d_P h_P + h_P d_P = \mathrm{id} - \eta\epsilon$, where $\eta : k \to P$ inserts the algebra unit in bar length zero.

Here is the projectivity assertion needed for that statement. Filter P by bar length. Its associated graded pieces are $A \otimes (s\bar A)^{\otimes r}$ with their internal differential. The coefficient complexes are direct sums of bounded complexes of projective k-modules, since M has this property and tensor powers preserve it. Consequently their Hom complexes into any acyclic complex are acyclic.

Let Q be an acyclic dg A-module and let $f : P \to Q$ be a closed homogeneous map. Construct a primitive for its restriction to each finite bar filtration, extending the previous primitive. The filtration is split as graded A-modules, so a graded extension of that primitive exists. Its differential defect vanishes on the preceding filtration and is a closed map from the associated graded piece to Q. The Hom acyclicity just proved supplies a correction.

Induction gives compatible primitives, and their union is a primitive for f on P. Thus $\mathrm{Hom}_A(P, Q)$ is acyclic, which is the required homotopical projectivity. The augmented bar contraction proves $P \to k$ is a quasi-isomorphism.

The right BA-coaction on P lifts convolution to composition:

$$L_f = (\mathrm{id}_P \otimes f)(\mathrm{id}_A \otimes \Delta) : P \longrightarrow P.$$

Coassociativity gives $L_f L_g = L_{f \star g}$, with the graded tensor sign displayed above. The chain-coaction identity gives $[d_P, L_f] = L_{\partial f}$. Postcomposition with $P \to k$ is a quasi-isomorphism from $\mathrm{End}_A(P)$ to $\mathrm{Hom}_A(P, k)$ by homotopical projectivity. Its composite with L is the identity under $\mathrm{Hom}_A(P, k) = \mathrm{Hom}_k(BA, k)$. Therefore $(BA)^\vee$ is a dg algebra model for $A^!$.

For $A = T(M)$, dualize the coalgebra map (21.1). It gives a dg algebra map

$$i^\vee : (B(T(M)))^\vee \longrightarrow K(M)^\vee = k \oplus M^\vee[-1].$$

Dualizing the explicit chain contraction preserves its homotopy identities, so this map is a quasi-isomorphism. The reduced coproduct of $K(M)$ is zero, hence the reduced convolution product of its dual is zero. This proves the first formula without treating p as a coalgebra map.

For $A = S(M)$, the preceding proposition gives

$$(BA)^\vee = \prod_{r \geq 0} \mathrm{Hom}_k((sM)^{\otimes r}, k) \cong \prod_{r \geq 0} (M^\vee[-1])^{\otimes r}.$$

The last comparison uses finite projectivity and the graded tensor evaluation in each length. Under it, deconcatenation dualizes to concatenation with exactly the convolution sign. The differentials agree by the Hom and tensor conventions. This proves the second formula, including its multiplication and completion. □

Over a field, take $M = kx$ in degree zero, so its dual generator has degree one. The square-zero algebra $S(M)$ has a nonzero dual class in every length. In particular, the square of that odd generator is nonzero. The completed tensor algebra is associative and has no imposed graded-commutativity relation.

The full dual in the proof is a product because the coalgebra is a direct sum. It is not the continuous dual of an unspecified completed coalgebra. Changing either topology requires a separate comparison.

Finite projectivity cannot simply be omitted from the tensor-dual identification. Over a field, let $M = \bigoplus_{j \geq 1} k e_j$. The bilinear form $\beta(e_i, e_j) = \delta_{ij}$ belongs to $(M \otimes M)^\vee$. It does not belong to the image of $M^\vee \otimes M^\vee$. Indeed, a sum of r elementary tensor functionals restricts to a matrix of rank at most r on any finite coordinate subspace. The restriction of β to the first $r + 1$ coordinates has rank $r + 1$.

For a general complex, the actual dual $\prod_r \mathrm{Hom}_k((sM)^{\otimes r}, k)$ must therefore be retained unless its tensor comparison is proved.

21.4 Realization in the chiral coefficient category

The preceding computations take place in the ordinary coefficient tensor. Their formal tensor identities also apply in a specified symmetric monoidal dg coefficient model with the sums, suspensions and normalization used above. For example, an exact strong monoidal dg functor preserving those sums carries the free-bar inclusion and its contraction to the corresponding ambient coefficients. Preserving the dual algebra formulas additionally requires the tensor-dual and product comparisons occurring in their proof.

These requirements do not make a componentwise tensor algebra a factorization object. The normalized summands in Theorem 15.1 are ambient coefficients, whereas factorization compares the entire object on separated configurations. Already $k^2 \otimes \omega_{\mathrm{Ran}}$ has rank two at a singleton and at a separated pair, while factorization would require rank four at the pair. Thus tensoring a factorizing unit by an arbitrary algebra does not supply the missing factorization equivalences.

To interpret $T_X(M)$ or $\mathbf{1} \oplus M$ as ordered chiral algebras, construct their factorization structures and compatible multiplication and augmentation maps. Apply the realization hypotheses of Theorem 15.1 to their full simplicial bars. Then specify the comparison with the free or square-zero coefficient model and prove compatibility of (21.1) with the retained structures. The actual factorizing independence examples of Theorem 11.5 remain available, without this generic tensoring assertion.

A nonlocal vertex algebra supplies neither that factorization structure nor a free geometric associative chiral algebra by itself. The conditional recognition of Theorem 204.4 starts from a specified geometric realization. Constructing its reverse, and comparing the two associative bars on the resulting object, remain separate mathematical problems.

Chapter 22

The quantum plane

The quantum plane has the same graded dimensions as a polynomial algebra in two variables. Its augmentation module also has the same Betti numbers. The parameter remains in the bar coproduct and the Yoneda multiplication, which we compute explicitly.

22.1 Normal forms and multiplication

Let k be a field, choose $q \in k^\times$, and set

$$A_q = k\langle x, y\rangle/(yx - qxy).$$

The augmentation $\epsilon : A_q \to k$ sends x and y to zero. Give both generators polynomial weight one and cohomological degree zero. The differential is zero, and all tensor products below are over k.

Theorem 22.1 (PBW basis of the quantum plane)**.** *The monomials $x^a y^b$, for $a, b \geq 0$, form a basis of A_q. Their product and the Hilbert series are*

$$(x^a y^b)(x^c y^d) = q^{bc} x^{a+c} y^{b+d}, \qquad H_{A_q}(z) := \sum_{w \geq 0} \dim_k (A_q)_w z^w = \frac{1}{(1-z)^2}.$$

Proof. Moving each y past a later x by $yx = qxy$ expresses every word in the stated form. To prove independence, take a vector space with basis $e_{a,b}$ and multiplication

$$e_{a,b} e_{c,d} = q^{bc} e_{a+c,b+d}.$$

This multiplication is associative because

$$bc + (b+d)e = de + b(c+e).$$

Its unit is $e_{0,0}$, and $e_{0,1}e_{1,0} = q e_{1,0}e_{0,1}$. The resulting homomorphism from A_q sends $x^a y^b$ to $e_{a,b}$, proving independence. Commuting y^b past x^c gives the product formula. There are $w+1$ basis monomials of weight w, giving the Hilbert series. □

For $q \neq 1$, the relation has a nonzero commutator $(q-1)xy$. Thus the common Hilbert series does not identify A_q with the commutative plane.

22.2 The linear resolution

Consider a nonnegatively graded algebra with weight-zero part k. It is *Koszul* if its augmentation module has a graded free resolution whose term in homological degree r is generated in polynomial weight r. Write e_x, e_y for free module generators of weight one and e_r for one of weight two. Place them in cohomological degrees $-1, -1, -2$, respectively.

Theorem 22.2 (Koszulness). *The following complex of free left A_q-modules is an augmented resolution:*

$$0 \longrightarrow A_q e_r \xrightarrow{d_2} A_q e_x \oplus A_q e_y \xrightarrow{d_1} A_q \xrightarrow{\epsilon} k \longrightarrow 0,$$

where

$$d_1(u e_x + v e_y) = ux + vy, \qquad d_2(c e_r) = cy e_x - qcx e_y.$$

In particular, A_q is Koszul.

Proof. The composite is $d_1 d_2(c e_r) = c(yx - qxy) = 0$. The image of d_1 is the augmentation ideal, so exactness remains to be proved at the two left terms.

Write the finite sums

$$u = \sum_{i,j \geq 0} u_{ij} x^i y^j, \qquad v = \sum_{i,j \geq 0} v_{ij} x^i y^j.$$

The coefficient of $x^a y^b$ in $ux + vy$ is

$$q^b u_{a-1,b} + v_{a,b-1},$$

with negative indices interpreted as zero.

If $ux + vy = 0$, then $u_{i,0} = 0$, $v_{0,j} = 0$, and

$$v_{a,b} = -q^{b+1} u_{a-1,b+1}.$$

Set $c = \sum_{i,j \geq 0} u_{i,j+1} x^i y^j$. These identities give $u = cy$ and $v = -qcx$, proving $\ker d_1 = \operatorname{im} d_2$. Finally, $cy = 0$ forces $c = 0$ by the ordered basis, so d_2 is injective. The stated generator weights make this a linear resolution. □

22.3 The finite bar coalgebra

Use the normalized bar BA_q with $s = [1]$, so $|sx| = |sy| = -1$. Its differential merges adjacent inputs with alternating signs, beginning with a plus sign. The element

$$\omega_q = [y|x] - q[x|y]$$

is a cycle because its boundary is $yx - qxy$.

Let $C_q \subset BA_q$ be the subspace spanned by 1, $sx = [x]$, $sy = [y]$ and ω_q. Its differential is zero. It is a coaugmented subcoalgebra: sx, sy are primitive and

$$\overline{\Delta}(\omega_q) = sy \otimes sx - q\, sx \otimes sy. \tag{22.1}$$

Here ω_q has cohomological degree -2 and polynomial weight two.

Proposition 22.3. *The inclusion $j : C_q \longrightarrow BA_q$ is a dg coalgebra quasi-isomorphism. Consequently the augmentation Tor groups have dimensions $1, 2, 1$ in homological degrees $0, 1, 2$ and vanish in higher degrees. Their coalgebra structure is the one displayed above.*

Proof. Let $P = A_q \otimes BA_q \to k$ be the left bar resolution, with the differential used in Theorem 21.3. The linear resolution K of Theorem 22.2 maps to P by

$$1 \longmapsto 1 \otimes 1, \qquad e_x \longmapsto -1 \otimes [x], \qquad e_y \longmapsto -1 \otimes [y], \qquad e_r \longmapsto 1 \otimes \omega_q.$$

The minus signs compensate for the left action term in that bar differential. On e_r the chain-map equation is

$$d_P(1 \otimes \omega_q) = -y \otimes [x] + qx \otimes [y].$$

Both complexes are free resolutions, and the map lifts the identity of k. Lift an inverse successively through the exact kernels in each degree, using freeness. The same lifting applied to the differences from the identities constructs chain homotopies. Thus this comparison is a homotopy equivalence.

Tensoring these homotopies with the augmentation right module preserves their identities. The resulting map $k \otimes_{A_q} K \to BA_q$ identifies its source with C_q, with a minus sign on each degree-one generator. It therefore proves that j is a quasi-isomorphism. The coproduct formula proves the claimed coalgebra structure, and the generator degrees give the Tor groups. □

22.4 The Yoneda product

Let $V = kx \oplus ky$ and $R = k(y \otimes x - qx \otimes y) \subset V^{\otimes 2}$. The *quadratic dual* is the algebra $Q_q = T(V^\vee)/(R^\perp)$, where the annihilator uses the ordinary tensor evaluation. For the basis ξ, η dual to x, y, this gives

$$Q_q = k\langle \xi, \eta \rangle / (\xi^2, \eta^2, \xi\eta + q\eta\xi).$$

Source weight records the polynomial weight on which a cochain evaluates. Assign these dual generators cohomological degree one and source weight one. Its graded Hom degree is the negative of that weight. Use the Yoneda convention that $\alpha\beta$ means first β, then α.

Theorem 22.4 (Yoneda algebra of the quantum plane)**.** *There is an isomorphism of bigraded algebras*

$$\operatorname{Ext}^*_{A_q}(k,k) \cong Q_q.$$

The full linear dual of BA_q is a dg algebra model for $\mathrm{R}\operatorname{Hom}_{A_q}(k,k)$*. Restriction along j gives a dg algebra quasi-isomorphism from that model to $C_q^\vee \cong Q_q$, with zero differential on the latter.*

Proof. Applying $\operatorname{Hom}_{A_q}(-,k)$ to K kills its differential, because all matrix entries lie in the augmentation ideal. Let ξ, η, ζ be the resulting cochains dual to e_x, e_y, e_r. Together with the unit they give the full Ext vector space.

Define degree-one A_q-linear endomorphisms F_ξ, F_η of K by

	e_x	e_y	e_r
F_ξ	1	0	$-e_y$
F_η	0	1	qe_x

and set them to zero on $K^0 = A_q$. On e_r, the equations $dF_\xi + F_\xi d = 0$ and $dF_\eta + F_\eta d = 0$ reduce to $-y + y = 0$ and $qx - qx = 0$. On the other generators both equations vanish termwise. These are closed lifts of the two degree-one cochains. Their compositions give

$$\xi^2 = \eta^2 = 0, \qquad \xi\eta = q\zeta, \qquad \eta\xi = -\zeta. \tag{22.2}$$

Postcomposition with $K \to k$ computes the derived composition product, since the finite free resolution is homotopically projective.

The relations of Q_q hold, and $\eta\xi = -\zeta$ shows that the two degree-one classes generate Ext. Those relations reduce every word of length at least three to zero and leave at most four basis elements. Ext has dimension four, so this surjection from Q_q is an isomorphism. All maps preserve the source weights.

The left bar resolution also computes derived Hom: its bar-length filtration has free coefficient modules, and the successive lifting argument proves homotopical projectivity. Its right bar coaction identifies dual convolution with derived composition, as in the proof of Theorem 21.3. The homotopy equivalence proving the preceding proposition dualizes to a homotopy equivalence of complexes. Because j is a coalgebra map, its dual is also an algebra map.

Finally, the convolution sign for two degree-one functionals on a two-input bar is minus one. Dualizing (22.1) therefore gives the same product table (22.2). The two primitive duals pull back to $-\xi, -\eta$ under $K \to P$. Their simultaneous sign change leaves that table unchanged. This identifies $C_q^\vee$ with Q_q as asserted, including the multiplication. □

When $q = 1$, this Yoneda algebra is the exterior algebra on two generators. For $q \neq 1$, its odd anticommutator is

$$\xi\eta + \eta\xi = (q-1)\zeta \neq 0.$$

Thus it is not graded commutative, although its Betti numbers are unchanged. Interchanging x and y replaces the presentation parameter by q^{-1}. The displayed formula keeps their order fixed.

22.5 Geometric realization

These computations concern the ordinary augmented algebra A_q and its coefficient tensor. A geometric realization must supply a factorization object with a compatible transverse multiplication and augmentation, together with the relation $yx = qxy$ in that tensor. It must also compare the full geometric bar with the coefficient bar and preserve (22.1). Tensoring a constant algebra by a factorizing coefficient does not supply those maps, as the rank example in Chapter 21 shows.

The marked-point examples of Theorem 11.5 retain their actual factorization structures. To equip a quantum-plane realization with the second associative chiral operation, construct that operation and its mixed coherences as well. The coefficient presentation and its Yoneda algebra do not by themselves construct this geometric extension.

Chapter 23

Quiver path algebras

A quiver records several objects and the maps between them. Its bar construction must retain the source and target of every map. The vertex idempotents provide the base over which tensors compose, and a quadratic relation becomes a two-input cycle in the relative bar.

23.1 Vertices, paths and left modules

Let k be a field and Q a finite quiver, with vertex set Q_0 and arrow set Q_1. Put

$$S = \bigoplus_{i \in Q_0} k e_i, \qquad e_i e_j = \delta_{ij} e_i, \qquad \sum_i e_i = 1.$$

An arrow $a : i \to j$ belongs to $e_j M e_i$, where M is the S-bimodule spanned by the arrows. Give each arrow weight one and each vertex idempotent weight zero.

The tensor algebra

$$A = T_S(M) = \bigoplus_{n \geq 0} M^{\otimes_S n}, \qquad M^{\otimes_S 0} = S,$$

has multiplication by concatenation and augmentation $\epsilon : A \to S$ given by projection to weight zero. Thus paths multiply from right to left: for $a : i \to j$ and $b : j \to \ell$, the composite is ba. The balancing over S makes a tensor of noncomposable arrows zero.

Write $L_i = S e_i$ for the left simple A-module obtained through ϵ, and $P_i = A e_i$ for the corresponding left projective module. For $Q = (1 \xrightarrow{a} 2)$, the algebra has basis e_1, e_2, a and $a = e_2 a e_1$. The sequence

$$0 \longrightarrow P_2 \xrightarrow{u \mapsto ua} P_1 \longrightarrow L_1 \longrightarrow 0$$

is exact: $P_2 = k e_2$, and the kernel of $P_1 \to L_1$ is ka. Also $P_2 = L_2$. Applying $\operatorname{Hom}_A(-, L_j)$, and using $\operatorname{Hom}_A(P_i, L_j) = e_i L_j$, gives

$$\operatorname{Ext}^1_A(L_1, L_2) = k, \qquad \operatorname{Ext}^1_A(L_i, L_i) = 0, \qquad \operatorname{Ext}^1_A(L_2, L_1) = 0.$$

There are no higher Ext groups.

These endpoints also fix the dual-arrow convention. Let $E = \operatorname{Ext}^*_A(S, S)$, with product given by composition, the right-hand factor acting first. Its vertex corner is

$$e_j E^r e_i = \operatorname{Ext}^r_A(L_i, L_j).$$

Hence the degree-one Ext arrow in this example runs from 1 to 2. The linear dual $M^\vee = \operatorname{Hom}_k(M, k)$ has bimodule action

$$(u f v)(m) = f(vmu), \qquad u, v \in S,$$

so the functional dual to a lies in $e_1 M^\vee e_2$. This bimodule dual reverses endpoints.

For any finite quiver with no relations, multiplication identifies $A \otimes_S M$ with the positive-weight ideal of A. Thus $0 \to A \otimes_S M \to A \to S \to 0$ is a projective resolution. Applying Hom to the simples shows that

$$e_j E^1 e_i \cong (e_j M e_i)^\vee, \qquad E^{\geq 2} = 0.$$

With $u \cdot_{\mathrm{op}} v = vu$ defining the opposite product, this yields

$$E^{\mathrm{op}} \cong S \oplus M^\vee[-1],$$

where the reduced product is zero and $[-1]$ places the dual arrows in cohomological degree one. The ordinary and graded opposite products agree here because every product of two positive-degree elements vanishes.

23.2 The relative free bar

Allow now M to be a cochain complex of S-bimodules, with differential preserving its vertex corners. For an augmented dg S-algebra $A = S \oplus \bar{A}$, define

$$B_S A = \bigoplus_{r \geq 0} (s\bar{A})^{\otimes_S r}, \qquad s = [1].$$

Thus $|sm| = |m| - 1$. Use $b_1(sa) = -s(da)$ and $b_2(sa \otimes sb) = (-1)^{|a|} s(ab)$, extended with the graded tensor signs.

The coproduct is deconcatenation over S. This makes $B_S A$ an S-coring: an S-bimodule with coassociative coproduct to $B_S A \otimes_S B_S A$ and counit to S. The length-zero summand supplies its coaugmentation. All sums are direct sums, without a completion.

Give $S \oplus sM$ its suspended internal differential and zero reduced coproduct. For $m \in e_j M e_i$, its full coproduct is

$$\Delta(sm) = e_j \otimes sm + sm \otimes e_i, \qquad \Delta(e_i) = e_i \otimes e_i.$$

Theorem 23.1 (Relative path-algebra bar)**.** *The map*

$$\iota : S \oplus sM \longrightarrow B_S(T_S(M)), \qquad \iota(sm) = [m], \quad \iota(e_i) = e_i,$$

is a quasi-isomorphism of coaugmented dg S-corings. The underlying complex of the target represents $S \otimes^{\mathbb{L}}_{T_S(M)} S$, with the first S a right module and the second a left module.

Proof. Expand a bar input into generator tensors. A tensor of n homogeneous generators $m_1, \dots, m_n$, with a set $C \subseteq \{1, \dots, n-1\}$ of cuts, defines a bar tensor w_C. Every cut and concatenation is balanced over S, so this notation retains the two endpoint idempotents. Let p project to generator weights zero and one. For $n \geq 2$, set

$$h(w_C) = \begin{cases} (-1)^{|m_1|} w_{C \cup \{1\}}, & 1 \notin C, \\ 0, & 1 \in C, \end{cases}$$

and put $h = 0$ in weights zero and one. These are S-bimodule maps.

Removing the inserted first cut contributes w_C with coefficient one. Insertion increases the rank of every other cut by one, so its two orders with any other cut removal have opposite signs. For an internal differential on m_j with $j > 1$, the inserted suspension reverses the tensor sign. For $j = 1$, the change in $|m_1|$ reverses the coefficient of h instead. These are the cancellations in (21.2), now in each vertex corner.

They give

$$bh + hb = \mathrm{id} - \iota p, \qquad p\iota = \mathrm{id}\,.$$

The inclusion ι preserves deconcatenation, since each $[m]$ is primitive relative to its endpoints. It also preserves the internal differential, counit and coaugmentation. The contraction proves the asserted quasi-isomorphism.

For the derived interpretation, use the left relative bar resolution $A \otimes_S B_S A \to S$. Its augmented contraction inserts the reduced part of the leading coefficient, as in the proof of Theorem 21.3. The same formula is S-balanced and proves exactness.

Its filtration by bar length has quotients induced from complexes of S-modules. Since S is semisimple, acyclic complexes of S-modules contract, so these coefficient complexes are homotopically projective. The filtration argument of that proof therefore makes the relative bar a homotopically projective resolution as well. Tensoring it with the right augmentation module S gives $B_S A$. □

The projection p is generally not a coring map. For composable arrows $a : 1 \to 2$ and $b : 2 \to 3$,

$$p[b|a] = 0, \qquad (p \otimes_S p)\overline{\Delta}[b|a] = sb \otimes_S sa \neq 0.$$

The coring quasi-isomorphism in the theorem is the inclusion ι.

23.3 Relations and the quadratic complex

Return to an ordinary finite-dimensional arrow bimodule M, placed in cohomological degree zero and weight one. Let $R \subseteq M \otimes_S M$ be an S-subbimodule and put $A = T_S(M)/(R)$. For $n \geq 2$, define

$$W_n = \bigcap_{j=0}^{n-2} M^{\otimes_S j} \otimes_S R \otimes_S M^{\otimes_S (n-j-2)}, \qquad W_0 = S, \quad W_1 = M,$$

where the intersection is inside $M^{\otimes_S n}$.

The relative quadratic complex has $K_n = A \otimes_S W_n$ in homological degree n, augmented to S, with differential

$$d(a \otimes m_1 \otimes \cdots \otimes m_n) = am_1 \otimes m_2 \otimes \cdots \otimes m_n.$$

The containment $W_n \subseteq M \otimes_S W_{n-1}$ makes this map well-defined. For $n \geq 2$, the containment $W_n \subseteq R \otimes_S M^{\otimes_S (n-2)}$ gives $d^2 = 0$. Each K_n is projective and generated in weight n. Call A Koszul if S admits a graded projective resolution whose term in homological degree n is generated in weight n.

Theorem 23.2 (Relative quadratic criterion). *The algebra A is Koszul if and only if $K_\bullet \to S$ is exact. When it is exact, the subspace*

$$C_R = \bigoplus_{n \geq 0} s^n W_n \subseteq B_S A$$

is a dg S-coring with zero differential, and its inclusion is a quasi-isomorphism. Here $s^n W_n$ denotes the image of W_n in $(sM)^{\otimes_S n}$, in cohomological degree $-n$.

Proof. Exactness of $K_\bullet \to S$ gives the required linear resolution. For the converse, write A_+ for the positive-weight ideal. A bounded-below graded syzygy Z has a projective surjection obtained by choosing an S-linear section of Z/A_+Z and mapping $A \otimes_S (Z/A_+Z)$ onto Z. Surjectivity follows by induction on weight. Choosing sections successively constructs a minimal graded resolution, whose differentials have image in A_+ times the next term. The choices can preserve the right vertex action, since $S \otimes_k S^{\mathrm{op}}$ is also semisimple.

After tensoring with the right augmentation module S, all minimal differentials vanish. Its generator spaces are therefore $\operatorname{Tor}_n^A(S, S)$. A linear resolution has the same property and puts this Tor group entirely in weight n. Consequently, if A is Koszul, the nth minimal generator space is concentrated in weight n.

Identify the minimal resolution inductively with $K_\bullet$. In degree one, its generators are M, and its differential is multiplication. If the identification holds through degree $n-1$, its next syzygy is generated in weight n. That weight lies in $M \otimes_S W_{n-1}$, and multiplication of the first two factors has kernel

$$(M \otimes_S W_{n-1}) \cap (R \otimes_S M^{\otimes_S (n-2)}) = W_n.$$

Thus $A \otimes_S W_n$ maps onto the entire syzygy, with precisely the displayed differential.

This proves exactness in every homological degree. It is the relative quadratic criterion of Beilinson–Ginzburg–Soergel [10, Theorem 2.6.1].

Each adjacent pair in W_n belongs to the corresponding copy of R, so every bar merger vanishes on $s^n W_n$. Deconcatenation at a cut $n = p + q$ lands in $s^p W_p \otimes_S s^q W_q$. Indeed, the defining intersection imposes all relations internal to either block, and tensoring over the semisimple algebra S preserves these intersections. Hence C_R is the stated subcoring.

Finally, bar length n and weight n force every input to lie in M. In that bidegree the cycle space is exactly $s^n W_n$, because the different possible mergers have distinct input-weight patterns. There are no incoming boundaries of that weight. The linear resolution proves that all Tor groups off this diagonal vanish. The inclusion of C_R therefore induces an isomorphism on every homology group. □

The relation quiver $1 \xrightarrow{a} 2 \xrightarrow{b} 3$, with $ba = 0$, illustrates the last assertion in Chapter 214. Its class $[b|a]$ has nonzero reduced coproduct $sb \otimes_S sa$. That coproduct retains the composable pair even though its product in A is zero.

23.4 Geometric realization

The vertex algebra and arrow bimodules above belong to ordinary coefficient complexes. For a boundary realization, the vertex labels must act on boundary objects, and the arrows must represent fields changing one boundary condition into another. Their composition must realize the path product.

A realization over X requires factorization coefficients carrying these vertex actions, a compatible transverse product and augmentation, and the full relative bar comparison. The signed contraction applies in a coefficient model admitting the displayed balanced tensors and direct sums. Its geometric use requires those constructions and their comparison maps.

A second associative chiral operation requires further maps on composable arrow coefficients and their mixed coherences with transverse composition. An operation on each arrow coefficient alone does not provide the maps between different vertex corners. These are the additional structures needed to turn the relative algebra calculation into a doubly associative quiver construction.

Chapter 24

Weyl and Clifford deformations

A constant in a quadratic relation can preserve the ordered monomial basis while preventing the generators from acting by zero on a one-dimensional module. The relation $yx - xy = \hbar$ exhibits both phenomena. Its constant term survives on the dual side as curvature. We first prove the basis theorem, then construct the curved coalgebra and the cobar map that recover the deformed algebra.

24.1 Polynomial families and ordered bases

Fix a field k, a finite-dimensional vector space V, and the polynomial ring $R_\hbar = k[\hbar]$. All generators have cohomological degree zero. For an alternating bilinear form ω on V, define

$$W_\hbar(V, \omega) = T_{R_\hbar}(V \otimes_k R_\hbar)/(uv - vu - \hbar\omega(u, v)).$$

Alternating means $\omega(v, v) = 0$, including in characteristic two. No nondegeneracy assumption is imposed.

For the Clifford family, choose a quadratic form $q : V \to k$. Thus $q(av) = a^2 q(v)$ and $\beta(u, v) = q(u + v) - q(u) - q(v)$ is bilinear. Define

$$\mathrm{Cl}_\hbar(V, q) = T_{R_\hbar}(V \otimes_k R_\hbar)/(v^2 - \hbar q(v) : v \in V).$$

Its relations imply $uv + vu = \hbar\beta(u, v)$. If 2 is invertible, the latter relations recover the former with $q(v) = \beta(v, v)/2$.

The Clifford generators also have odd internal parity. That parity is distinct from the cohomological grading used in the bar below.

Choose an ordered basis $x_1, \dots, x_d$ of V. Write $q_i = q(x_i)$, $\beta_{ji} = \beta(x_j, x_i)$ and $\omega_{ji} = \omega(x_j, x_i)$. Filter each algebra increasingly by the number of generators, with $\hbar$ in filtration zero.

Theorem 24.1 (Filtered PBW theorem)**.** *The ordered monomials*

$$x_1^{a_1} \cdots x_d^{a_d}, \qquad \begin{cases} a_i \geq 0 & \text{for } W_\hbar(V, \omega), \\ a_i \in \{0, 1\} & \text{for } \mathrm{Cl}_\hbar(V, q), \end{cases}$$

are $R_\hbar$-bases. Consequently their associated graded algebras for the word filtration are $\mathrm{Sym}_k(V) \otimes_k R_\hbar$ *and* $\Lambda_k(V) \otimes_k R_\hbar$*, respectively. Both families are flat over $R_\hbar$ in every characteristic.*

Proof. Move a larger index to the right of a smaller one, using the defining relation. Each correction has two fewer generators. The reordered term has fewer inversions. In the Clifford case, replace x_i^2 by $\hbar q_i$. Induction on length and then inversions proves spanning.

For independence in the Weyl case, act on $R_\hbar[t_1, \dots, t_d]$ by

$$X_i = t_i + \hbar \sum_{r<i} \omega_{ir} \frac{\partial}{\partial t_r}.$$

The multiplication and differentiation operators satisfy $[X_i, X_j] = \hbar\omega(x_i, x_j)$. An ordered product applied to 1 gives

$$X_1^{a_1} \cdots X_d^{a_d}(1) = t_1^{a_1} \cdots t_d^{a_d}.$$

Indeed, when X_i acts, only variables with index at least i are present. Its derivative terms therefore vanish. The resulting polynomial monomials are independent over $R_\hbar$.

For the Clifford case, let B be the bilinear form defined by

$$B(x_i, x_i) = q_i, \qquad B(x_j, x_i) = \beta_{ji}\ (j > i), \qquad B(x_i, x_j) = 0\ (i < j).$$

Then $B(v, v) = q(v)$.

On $\Lambda_k(V) \otimes_k R_\hbar$, let L_v denote exterior multiplication by v and let ι_v be contraction by $B(v, -)$. The operators $T_v = L_v + \hbar\iota_v$ satisfy

$$T_v^2 = \hbar q(v), \qquad T_u T_v + T_v T_u = \hbar\beta(u, v).$$

These identities follow from $L_v^2 = \iota_v^2 = 0$ and $\iota_u L_v + L_v \iota_u = B(u, v)\,\mathrm{id}$.

For $i_1 < \cdots < i_r$, the triangular choice of B gives

$$T_{x_{i_1}} \cdots T_{x_{i_r}}(1) = x_{i_1} \wedge \cdots \wedge x_{i_r}.$$

These vectors form the exterior basis, proving independence. The same ordered bases identify every quotient of the word filtration and prove the associated graded assertions. □

In characteristic two, the square relation is indispensable. For one generator, $2x^2 = 0$ imposes no relation, whereas $x^2 = \hbar$ gives a free rank-two Clifford family. If one instead prescribes $2x^2 = \hbar$, the resulting algebra has $\hbar = 0$ and is not flat over $k[\hbar]$. Thus a symmetric anticommutator without the quadratic data does not state the characteristic-two PBW theorem.

24.2 Specialization, Rees construction and completion

Write $A_\hbar$ for either polynomial family and $A_a = A_\hbar/(\hbar - a)$ for $a \in k$. The fiber A_0 is the symmetric or exterior algebra, and the same ordered monomials form a basis of every A_a. For the word filtration F on A_1, the Rees algebra is

$$\mathcal{R}_F(A_1) = \bigoplus_{m \geq 0} F_m A_1\, z^m \subseteq A_1[z].$$

It is generated by z and $X_i = z x_i$. Its constant terms are quadratic in z: in the first Weyl example, $YX - XY = z^2$. The ordered bases prove the isomorphism

$$A_\hbar \otimes_{k[\hbar],\, \hbar \mapsto z^2} k[z] \xrightarrow{\sim} \mathcal{R}_F(A_1), \qquad x_i \longmapsto z x_i.$$

The formal family is instead the inverse limit

$$\widehat{A_\hbar} = \varprojlim_N A_\hbar/\hbar^N A_\hbar.$$

The ordered bases identify its underlying module with $\mathrm{Sym}(V)[[\hbar]]$ in the Weyl case and $\Lambda(V)[[\hbar]]$ in the Clifford case. In the first expression, every coefficient of $\hbar^n$ is a polynomial, with no uniform bound on its degree.

For example, $\sum_{n \geq 0} \hbar^n x_1^n$ belongs to the completion but to no bounded word-filtration piece. Thus that bounded-word filtration is not exhaustive on a nontrivial completed Weyl family.

Nor is there a unital specialization $k[[\hbar]] \to k$ sending $\hbar$ to 1, since $1 - \hbar$ is invertible. The fiber at one belongs to the polynomial family.

24.3 The curved quadratic dual

Let $A_0 = T_k(V)/(\mathcal{R})$ denote the homogeneous fiber. Its quadratic coalgebra has

$$C_0 = k, \quad C_1 = sV, \quad C_n = s^n \bigcap_{i+j=n-2} V^{\otimes i} \otimes \mathcal{R} \otimes V^{\otimes j} \quad (n \geq 2).$$

Here $|sv| = -1$. The notation sends a tensor $v_1 \otimes \cdots \otimes v_n$ to $[v_1|\cdots|v_n]$ without an additional sign.

The coalgebra $C = \bigoplus_n C_n$ has degree $-n$ and weight n on C_n, zero differential and deconcatenation coproduct. The intersection argument preceding Theorem 17.3 proves that deconcatenation restricts to C. Extend scalars to $R_\hbar$ for the remainder of the construction.

Write a deformed relation as $r - \hbar\kappa(r)$ for $r \in \mathcal{R}$. Define a degree-two functional $\theta : C \otimes_k R_\hbar \to R_\hbar$ by

$$\theta(s^2 r) = \hbar\kappa(r), \qquad \theta(C_n) = 0 \quad (n \neq 2).$$

Let $E = \bigoplus_n \operatorname{Hom}_k(C_n, k) \otimes_k R_\hbar$, graded by n. Its multiplication is the signed convolution

$$(f\,g)(u) = \sum_{\Delta u = \sum u' \otimes u''} (-1)^{|g||u'|} f(u')\, g(u'').$$

The functional θ corresponds to an element $c \in E^2$.

Equivalently, it determines the zero-input operation $m_0 : R_\hbar \to E$, $m_0(1) = c$. This operation and the functional $\theta : C \otimes R_\hbar \to R_\hbar$ have different sources and targets.

Let ξ_i be dual to $[x_i]$. The two curved duals are

$$E_W = \Lambda_{R_\hbar}(\xi_1, \ldots, \xi_d), \qquad c_W = \hbar \sum_{i<j} \omega_{ji}\xi_i\xi_j, \tag{24.1}$$

$$E_{\mathrm{Cl}} = R_\hbar[\xi_1, \ldots, \xi_d], \qquad c_{\mathrm{Cl}} = -\hbar\left(\sum_i q_i \xi_i^2 + \sum_{i<j} \beta_{ji}\xi_i\xi_j\right). \tag{24.2}$$

Every ξ_i has cohomological degree one. The polynomial algebra in the second line has ordinary commuting generators. It is not asserted to be graded commutative. Both dual differentials are zero.

To check the signs, set $\rho_{ji} = [x_j|x_i] - [x_i|x_j]$ in the Weyl case. The convolution gives $(\xi_i\xi_j)(\rho_{ji}) = 1$ for $i < j$. For Clifford relations, use $\rho_{ii} = [x_i|x_i]$ and $\rho_{ji} = [x_j|x_i] + [x_i|x_j]$. Then $(\xi_i^2)(\rho_{ii}) = -1$ and $(\xi_i\xi_j)(\rho_{ji}) = -1$. These evaluations give exactly θ.

The algebra presentations follow by orthogonal quadratic relations and finite-dimensional duality in each weight. The coalgebra C_W consists of alternating tensors. For C_{Cl}, sum the distinct permutations of each multiset to obtain a basis. The latter sums carry no factorial denominators.

In each case c is central, so $(E, 0, c)$ satisfies the curved dg identities

$$d^2 = [c, -] = 0, \qquad dc = 0.$$

This is nonhomogeneous quadratic duality in the sense of [89, §§2.1–2.2], with the suspension signs fixed above. The underlying complex $(E, 0)$ has ordinary cohomology E. Forgetting c discards the constant terms of the deformation.

24.4 A cobar resolution with a scalar term

Put $t = s^{-1}$, so $|tu| = |u| + 1$, and $\bar{C} = \bigoplus_{n>0} C_n$. On the unital tensor algebra $T_{R_\hbar}(t\bar{C} \otimes_k R_\hbar)$, define the degree-one derivation

$$D(tu) = -\sum_{\bar{\Delta}u = \sum u' \otimes u''} (-1)^{|u'|}(tu')(tu'') - \theta(u). \tag{24.3}$$

Denote this dg algebra, once the square-zero identity is proved, by $\Omega_\theta C$. It has no augmentation sending all $t\bar{C}$ to zero when $\theta \neq 0$.

Theorem 24.2. *The derivation in* (24.3) *satisfies* $D^2 = 0$. *The assignments*

$$p : \Omega_\theta C \longrightarrow A_\hbar, \qquad p(t[v]) = v, \qquad p(tC_n) = 0 \quad (n > 1),$$

define a quasi-isomorphism of unital dg $R_\hbar$*-algebras. The degreewise* $\hbar$*-adic completion of* p *is also a quasi-isomorphism.*

Proof. The quadratic part of D squares to zero by coassociativity, as in Proposition 16.2. The remaining terms on a generator are

$$D^2(tu) = t\big((\theta \otimes \mathrm{id} - \mathrm{id} \otimes \theta)\Delta u\big).$$

Centrality of c makes the expression in parentheses zero. Indeed, pairing it with any homogeneous element of E gives the commutator with c. The pairings in each weight are perfect. Thus $D^2 = 0$ on generators and hence everywhere.

For $u = s^2 r$, the image of $D(tu)$ is $r - \hbar\kappa(r)$, which vanishes in $A_\hbar$. For $u \in C_n$ with $n > 2$, every reduced cut has a factor of weight greater than one, and $\theta(u) = 0$. Its image also vanishes. The weight-one generators are closed, so p is a dg algebra map.

We next verify the homogeneous Koszul comparison. For the symmetric algebra, tensor the one-variable resolutions $0 \to k[x_i] \xrightarrow{x_i} k[x_i] \to k \to 0$ over k. They give a linear free resolution, and hence the diagonal criterion applies.

For the exterior algebra, use the complex

$$K_n = \Lambda(V) \otimes_k \Gamma^n(V), \qquad d(a \otimes u^{[\alpha]}) = \sum_{i:\alpha_i>0} a x_i \otimes u^{[\alpha - e_i]}.$$

Here $u^{[\alpha]}$ are divided-power basis symbols with $|\alpha| = n$. No division is used in the differential. The exterior relations give $d^2 = 0$.

For an ordered exterior monomial x_I, fix the total multidegree $\nu = \mathbf{1}_I + \alpha$. If $\nu \neq 0$, choose j with $\nu_j > 0$.

The map

$$H(x_I \otimes u^{[\alpha]}) = \begin{cases} (-1)^{\#\{i \in I : i > j\}} x_{I \setminus \{j\}} \otimes u^{[\alpha + e_j]}, & j \in I, \\ 0, & j \notin I \end{cases}$$

satisfies $dH + Hd = \mathrm{id}$ on that multidegree. The j-term gives the identity. Every other term cancels because the two exterior insertions have opposite signs. Multidegree zero supplies the augmentation to k. This proves a linear free resolution in every characteristic.

Theorem 17.3 now gives $\Omega C \simeq A_0$ for both homogeneous algebras. Tensoring over the field k extends this comparison to $R_\hbar$. Filter $\Omega_\theta C$ by total C-weight, with $\hbar$ in filtration zero, and filter $A_\hbar$ by word length. The quadratic term preserves this filtration and the scalar term lowers it by two. By the PBW theorem, the associated graded map is precisely $\Omega C \otimes_k R_\hbar \to A_0 \otimes_k R_\hbar$.

Its cone is acyclic in every weight. For a cycle in the filtered cone, remove its highest-weight term by a boundary and continue in lower weights. Every element has finite weight and the filtration starts at zero, so this procedure terminates. The cone of p is therefore acyclic.

The same argument works modulo $\hbar^N$ for every N. The transition kernels of these cones are copies of the cone modulo $\hbar$, and are acyclic. Compatible primitives can thus be lifted recursively from level N to level $N+1$. Their inverse limit proves acyclicity of the completed cone. □

For $V = kx \oplus k\,y$ with $\omega(y,x) = 1$, put $\rho = [y|x] - [x|y]$. The formulas give

$$c = \hbar\xi\eta, \qquad D(t\,\rho) = yx - x\,y - \hbar.$$

The scalar term is required for p to respect the differential. Curvature in this calculation belongs to a specified ordinary quadratic dual. A geometric anomaly curvature belongs to its own coefficient complex. Comparing them requires a map carrying the curvature, differential and products, together with the relevant factorization and collision operations.

Chapter 25

Associative chiral algebras on the formal disc

The value of a field at the origin recovers a multiplication when that field has no pole. Translation then recovers the entire field from this multiplication. We prove this algebraic correspondence, construct a complete braided example, and state the geometric realization problem that remains for the resulting fields.

25.1 Regular fields and differential algebras

Work over $\mathbb{C}$ in a fixed formal coordinate z. A nonlocal vertex algebra $(V, Y, \mathbf{1})$, in the sense of Definition 11.8, is *regular* if

$$Y(a,z)b \in V[[z]] \qquad (a, b \in V).$$

Define its product and translation operator by

$$ab = Y(a,z)b\big|_{z=0}, \qquad Ta = \partial_z Y(a,z)\mathbf{1}\big|_{z=0}.$$

A morphism of nonlocal vertex algebras is a linear map preserving the vacuum and satisfying $f(Y(a,z)b) = Y'(f(a),z)f(b)$.

Theorem 25.1 (Reconstruction of regular fields)**.** *The product above makes V a unital associative algebra, T is a derivation, and*

$$Y(a,z)b = (e^{zT}a)b. \tag{25.1}$$

Conversely, this formula gives a regular nonlocal vertex algebra for every unital associative $\mathbb{C}$-algebra with derivation T. These assignments are inverse functors between regular nonlocal vertex algebras and unital associative differential algebras. Morphisms on the latter side preserve multiplication and commute with T.

Proof. Regularity puts both sides of weak associativity in $V[[z,w]]$. Multiplication by $(z+w)^N$ is injective there: the change of variables $u = z + w$ reduces it to multiplication by u^N. Thus weak associativity becomes the exact identity

$$Y(a, z+w)Y(b,w)c = Y(Y(a,z)b, w)c. \tag{25.2}$$

No choice of Laurent expansion is needed for this power-series identity. Setting $z = w = 0$ proves $a(bc) = (ab)c$. Vacuum and creation give $\mathbf{1}a = a\mathbf{1} = a$.

Put $F_z(a) = Y(a,z)\mathbf{1}$, so $F_0 = \mathrm{id}$. In (25.2), first set $w = 0$ and $b = \mathbf{1}$. Then set $b = c = \mathbf{1}$ without specializing the variables. These substitutions give

$$Y(a,z)c = F_z(a)c, \qquad F_{z+w}(a) = F_w(F_z(a)).$$

Differentiating the second identity in w at zero gives $\partial_z F_z = TF_z$. The initial value $F_0 = \mathrm{id}$ determines every coefficient, hence $F_z = e^{zT}$.

Finally set $z = 0$ and $c = \mathbf{1}$ in (25.2). It gives $F_w(ab) = F_w(a)F_w(b)$. The coefficient of w is the derivation identity $T(ab) = (Ta)b + a(Tb)$, while $F_w(\mathbf{1}) = \mathbf{1}$ gives $T\mathbf{1} = 0$. This proves (25.1).

Conversely, a derivation satisfies the iterated Leibniz rule, so e^{wT} preserves products. For the field map (25.1), both sides of (25.2) equal

$$(e^{(z+w)T}a)(e^{wT}b)c.$$

Vacuum and creation follow from $T\mathbf{1} = 0$. The field is regular, and its constant term and vacuum derivative recover the given product and T.

A field morphism preserves those two coefficients. A unital algebra map commuting with T preserves the exponential formula. Thus the assignments are inverse on objects and morphisms. □

For $V = \mathcal{M}_2(\mathbb{C}[t])$ and $T = \partial_t$, the fields are

$$Y(a,z)b = a(t+z)b(t).$$

Let E_{ij} be the matrix units and let L_a denote left multiplication by a. The constant matrix fields need not commute:

$$[Y(E_{12},z), Y(E_{21},w)] = L_{E_{11}-E_{22}},$$

Regularity therefore retains ordinary associative algebras with noncommuting multiplication.

The same correspondence has a complete form. Let $R = \mathbb{C}[[\hbar]]$ and $V = V_0[[\hbar]]$, where V_0 is a complex vector space. Use the completed tensor product $V\widehat{\otimes}_R V = (V_0 \otimes_{\mathbb{C}} V_0)[[\hbar]]$. A regular topological field map is a continuous R-linear map $Y : V\widehat{\otimes}_R V \to V[[z]]$. Require vacuum, creation and weak associativity modulo every $\hbar^m$.

Corollary 25.2. *These field systems are equivalent, by* (25.1)*, to continuous unital associative products on V with continuous R-linear derivation T. The morphisms are continuous and R-linear on both sides.*

Proof. Apply the preceding coefficient argument to $V/\hbar^m V$ for every m. These quotients are complex vector spaces, so cancellation and division by factorials remain valid. The recovered products, operators and identities commute with reduction. Their inverse limits give the claimed correspondence, and the exponential is defined coefficientwise in z. □

25.2 A braiding defined on every state

The complete quantum torus gives an example where the multiplication and the exchange operator can both be written explicitly. Keep $R = \mathbb{C}[[\hbar]]$ and put $q = e^{\hbar}$ as a formal power series. Let $\Gamma = \mathbb{Z}^2$ and let e_g, for $g = (g_1, g_2) \in \Gamma$, be the basis of $\mathbb{C}[\Gamma]$. On $V = \mathbb{C}[\Gamma][[\hbar]]$, define

$$e_g e_j = q^{g_2 j_1} e_{g+j}, \qquad \mathbf{1} = e_0, \qquad T = 0, \qquad Y(a,z)b = ab. \tag{25.3}$$

Every coefficient of $\hbar$ has finite support in Γ. The formula therefore extends continuously to $V\widehat{\otimes}_R V$.

Define the alternating integer form $\omega(g,j) = g_1 j_2 - g_2 j_1$ and the operator

$$S(e_g \otimes e_j) = q^{\omega(g,j)} e_g \otimes e_j. \tag{25.4}$$

It is continuous on the completed tensor product and independent of the spectral variable. In particular, $S = 1 + O(\hbar)$.

Proposition 25.3. *The data* (25.3)–(25.4) *form a topological quantum vertex algebra in the convention of* [33, Definitions 3.5 and 3.12]. *The braiding satisfies exact unitarity, the quantum Yang–Baxter equation and both hexagon identities with the multiplication. The algebra $(V,\cdot)$ has no unital R-algebra homomorphism to R.*

Proof. Associativity of the product is the integer identity

$$g_2 j_1 + (g_2 + j_2)k_1 = j_2 k_1 + g_2(j_1 + k_1).$$

The vector e_0 is a two-sided unit. Corollary 25.2 gives the vacuum, translation and weak associativity axioms for Y.

The locality condition (11.8) holds with exponent zero because

$$q^{\omega(g,j)}(e_g e_j)e_k = (e_j e_g)e_k.$$

Indeed, $\omega(g, j) + g_2 j_1 = j_2 g_1$. This is an identity on every triple of basis states. Linearity modulo $\hbar^m$ and passage to the inverse limit prove it for all states. These checks give the asserted quantum vertex algebra.

Write S^{ij} for the operator on tensor factors i, j and S^{21} for its conjugate by interchange. Alternation gives $S^{21}S = \mathrm{id}$. The operators S^{12}, S^{13}, S^{23} are diagonal on the same basis, so

$$S^{12}S^{13}S^{23} = S^{23}S^{13}S^{12}.$$

Let m denote the multiplication (25.3). The two bicharacter identities

$$\omega(g + j, k) = \omega(g, k) + \omega(j, k), \qquad \omega(g, j + k) = \omega(g, j) + \omega(g, k)$$

give the exact hexagons

$$S(m \otimes 1) = (m \otimes 1)S^{23}S^{13}, \qquad S(1 \otimes m) = (1 \otimes m)S^{12}S^{13}.$$

The unit normalization follows from $\omega(0, g) = \omega(g, 0) = 0$. All translation identities are zero since $T = 0$ and S is constant.

Put $X = e_{(1,0)}$ and $Y_0 = e_{(0,1)}$. They are invertible, with inverses $e_{(-1,0)}$ and $e_{(0,-1)}$, and $Y_0 X = qXY_0$. A unital R-algebra map $\varepsilon : V \to R$ would therefore satisfy

$$(1 - q)\varepsilon(X)\varepsilon(Y_0) = 0.$$

The two images are units, forcing $q = 1$, contrary to $q = e^{\hbar}$ in R. □

The sign in the braiding is tied to the fixed field multiplication. Replacing S by S^{-1} preserves its unit normalization, unitarity, Yang–Baxter equation and both hexagons. On the same states $X, Y_0, \mathbf{1}$, the difference in the locality equation becomes

$$(q^{-1} - q)e_{(1,1)} = -2\hbar e_{(1,1)} \pmod{\hbar^2}.$$

Multiplication by any power of $z - w$ cannot kill this constant nonzero distribution modulo $\hbar^2$. Thus S^{-1} is incompatible with locality for these fields. The vacuum e_0 coexists with the absence of a scalar algebra augmentation.

25.3 From generating fields to an exchange operator

For singular fields, the Laurent expansions and their substitution maps are part of the data. The field space is $\mathrm{Hom}(V, V((z)))$ in the ordinary convention of Definition 11.8, with a pole bound for each input vector. The completed convention instead uses $V_\hbar((z))$ as defined before (11.8). The regular correspondence proved above uses the smaller space $V[[z]]$.

Return to an ordinary nonlocal vertex algebra V over $\mathbb{C}$. Let $U \subseteq V$ be a finite-dimensional subspace generating V from the vacuum by coefficients of its fields, and choose a basis a_i of U. A spectral operator $R(u) \in \mathrm{End}(U \otimes U) \otimes \mathbb{C}((u))$ may prescribe an exchange relation

$$Y(a_i, z_1)Y(a_j, z_2) = \sum_{k,l} R_{ij}^{kl}(z_1 - z_2)Y(a_l, z_2)Y(a_k, z_1).$$

Expand the Laurent coefficients in the specified regions and clear the required poles by a power of $z_1 - z_2$. This is S-locality as in (11.7), with the coefficient functions evaluated at the negative of its difference variable.

To state the hypothesis that recovers an operator on all states, put $K_3 = \mathbb{C}((z_1))((z_2))((z_3))$ and define

$$\pi_3 : V^{\otimes 3} \otimes_{\mathbb{C}} K_3 \longrightarrow \operatorname{Hom}_{\mathbb{C}}\big(V, V((z_1))((z_2))((z_3))\big),$$
$$\pi_3(a \otimes b \otimes c \otimes f)(d) = f Y(a, z_1) Y(b, z_2) Y(c, z_3) d.$$

Proposition 25.4 (Generating exchange relations)**.** *If the fields of U are S-local, then V is a weak quantum vertex algebra. If, in addition, π_3 is injective, the exchange relations determine a unique operator*

$$\mathcal{S}(u) : V \otimes V \longrightarrow V \otimes V \otimes \mathbb{C}((u)).$$

It satisfies unitarity $\mathcal{S}^{21}(-u)\mathcal{S}(u) = \mathrm{id}$ and

$$\mathcal{S}^{12}(u)\mathcal{S}^{13}(u+v)\mathcal{S}^{23}(v) = \mathcal{S}^{23}(v)\mathcal{S}^{13}(u+v)\mathcal{S}^{12}(u),$$

with the Laurent expansions in the indicated order. With that operator, V is a quantum vertex algebra in the convention of [74, Definition 5.10 and Theorem 5.11].

Proof. Write $\mathcal{E}(V) = \operatorname{Hom}_{\mathbb{C}}(V, V((z)))$. Call fields $A(z)$, $B(z)$ compatible if some $N \geq 0$ satisfies

$$(z_1 - z)^N A(z_1) B(z) \in \operatorname{Hom}_{\mathbb{C}}(V, V[[z_1, z]][z_1^{-1}, z^{-1}]).$$

For such an N, their field products are the coefficients of

$$Y_{\mathcal{E}}(A(z), x) B(z) = x^{-N} \Big((z_1 - z)^N A(z_1) B(z) \Big)\Big|_{z_1 = z + x},$$

where substitution expands in nonnegative powers of x relative to z. The expression is independent of a larger clearing exponent. Li's construction identifies the subspace $H \subseteq \mathcal{E}(V)$ generated by id_V and the fields of U under these coefficients as a weak quantum vertex algebra [74, Definition 2.11 and Theorem 5.8].

The state–field map $\Phi : V \to \mathcal{E}(V)$, $\Phi(a) = Y(a, z)$, is injective by creation. The self-module identity gives

$$\Phi(Y(a, x)b) = Y_{\mathcal{E}}(\Phi(a), x)\Phi(b)$$

by [74, Proposition 2.22]. Thus Φ preserves field products and identifies V with H, because U generates V. This proves the first assertion.

For the second assertion, apply [74, Theorems 4.8 and 5.11] to the faithful self-module V. Injectivity of π_3 also gives injectivity for two inputs by inserting the vacuum in the third slot. Consequently the coefficients of an exchange are unique. Reversing twice gives unitarity, and comparing the two reversals of three fields gives the Yang–Baxter equation. The cited theorem controls the Laurent expansions in these comparisons and gives the stated operator. □

The tensor injectivity hypothesis is stronger than injectivity of $a \mapsto Y(a, z)$. For the constant fields of $M_2(\mathbb{C})$, the nonzero tensor $E_{11} \otimes E_{22}$ acts by the zero product, whereas the state–field map is injective. Injectivity of π_3 is a sufficient reconstruction hypothesis. An explicit operator can also be supplied directly, as for the quantum torus above. The ordinary convention in this proposition does not impose a formal deformation or the normalization $S = 1 + O(\hbar)$.

25.4 The geometric realization problem

Problem 25.5 (Formal-disc realization of fields)**.** Fix a nonlocal vertex algebra with covariance, topology and completion. Construct coefficient objects on every finite power of the formal disc. Construct their collision maps, relabeling maps, factorization comparisons and associative chiral operations with all substitution coherences. Give a comparison whose field expansions recover Y, its vacuum and its weak associativity identity (11.6). For a quantum vertex algebra, also realize the full exchange operator and its compatibility with these operations.

Determine the coefficient, support and continuity hypotheses under which these constructions are functorial, and prove both inverse comparisons between geometric objects and fields. Specify any scalar augmentation separately from the vacuum. The full field class remains to be treated, including singular fields outside the regular correspondence.

For a given geometric realization satisfying the stated comparison hypotheses, Proposition 11.9 proves the forward field implication. The converse requires the coefficient objects and collision-compatible maps in the problem. Neither the regular algebraic correspondence nor the explicit quantum-torus braiding supplies those geometric constructions.

Chapter 26

Zamolodchikov–Faddeev exchange

An exchange relation prescribes how two generators change when their order is reversed. For three generators there are two ways to reverse the order. Their agreement is the Yang–Baxter equation. To deduce a basis from this agreement, one must also specify which exchanges are allowed and what happens when a spectral label is repeated.

26.1 Ordered colors and normal forms

Let B be a commutative ring and $1 < \cdots < m$ a finite set of colors. For each color a, let V_a be a finite free B-module. All tensor products in this section are over B.

For each $a > b$, choose a B-linear map

$$c_{ab} : V_a \otimes V_b \longrightarrow V_b \otimes V_a.$$

Write $T_B(V)$ for the unital tensor algebra of a B-module V. Define the *oriented exchange algebra*

$$A(c) = T_B\Big(\bigoplus_{a=1}^{m} V_a\Big) \Big/ \big\langle x \otimes y - c_{ab}(x \otimes y) : x \in V_a,\ y \in V_b,\ a > b \big\rangle. \tag{26.1}$$

Thus an adjacent descending pair may be exchanged. No relation is imposed between two generators of the same color. Choose a basis of each V_a. A word is *ordered* if its colors are nondecreasing. Its generators within any one color retain their order.

Theorem 26.1 (The ordered-word criterion)**.** *Multiplication induces an isomorphism of B-modules*

$$T_B(V_1) \otimes \cdots \otimes T_B(V_m) \xrightarrow{\sim} A(c) \tag{26.2}$$

if and only if the following identity holds for every $a > b > c$:

$$(c_{bc} \otimes 1)(1 \otimes c_{ac})(c_{ab} \otimes 1) = (1 \otimes c_{ab})(c_{ac} \otimes 1)(1 \otimes c_{bc}). \tag{26.3}$$

Both sides map $V_a \otimes V_b \otimes V_c$ to $V_c \otimes V_b \otimes V_a$. Under this condition every sequence of descending adjacent exchanges terminates at the same linear combination of ordered words. The maps c_{ab} need not be invertible.

Proof. For a word with color sequence $(a_1, \dots, a_n)$, set $\operatorname{inv}(a_1, \dots, a_n) = \#\{(i, j) : i < j,\ a_i > a_j\}$. Every term produced by a descending adjacent exchange has inversion number one smaller. Consequently every reduction terminates.

We prove uniqueness of its result by induction on the inversion number. An ordered word has itself as its normal form. Otherwise choose a first exchange and apply the already defined normal form linearly to its output. Two choices of first exchange either coincide, occupy disjoint pairs, or meet in one position.

Disjoint exchanges commute as maps on the tensor word, so their two outputs have a common next reduction. Overlapping exchanges require three strictly descending colors. Performing the next two exchanges on each side

gives the two maps in (26.3). They have equal outputs. In both cases every subsequent word has smaller inversion number, so induction identifies its normal form. This defines a B-linear map N from the free tensor algebra to the free module on ordered words, independent of all choices.

The two sides of each defining relation, in every word context, have the same image under N. Hence N descends to $A(c)$ and is a left inverse to ordered multiplication. Conversely, reducing a word changes it only by a defining relation. Its normal form therefore multiplies back to the original class. These maps are inverse, proving (26.2).

For necessity, both reductions of a descending three-generator tensor represent the same element of $A(c)$. Their outputs already have increasing colors. Injectivity in (26.2) identifies them as tensors, which is (26.3). □

Repeated colors cause no additional overlap: two adjacent reducible pairs can meet only in a strictly decreasing triple. In particular, the theorem concerns all words, including repeated labels.

If reverse maps $c_{ba} : V_b \otimes V_a \to V_a \otimes V_b$ are also prescribed, preserving the ordered basis forces $c_{ab}c_{ba} = 1$. Indeed, reversing an ordered pair and then sorting it must return that pair. Both modules are finite free of the same rank, so a right inverse is also a left inverse. Equivalently,

$$c_{ba}c_{ab} = 1_{V_a \otimes V_b} \qquad (a > b). \tag{26.4}$$

Conversely, taking $c_{ba} = c_{ab}^{-1}$ makes the reverse relations consequences of the oriented ones. The oriented presentation does not require reversal.

26.2 Spectral matrices and the rational solution

Suppose now that V_a is a labeled copy of one finite free module V. Let $t_1, \dots, t_m$ be central spectral parameters and suppose the coefficients of $R(t_a - t_b) \in \mathrm{End}_B(V \otimes V)$ lie in B. If $P(x \otimes y) = y \otimes x$, put $c_{ab} = PR(t_a - t_b)$, with the target labeled $V_b \otimes V_a$. In a basis $e_i^{(a)}$ this means

$$e_i^{(a)} e_j^{(b)} = \sum_{k,l} R_{ij}^{kl}(t_a - t_b) e_l^{(b)} e_k^{(a)} \qquad (a > b).$$

For operators on $V^{\otimes 3}$, subscripts specify the original tensor legs. For example $R_{13} = P_{23}(R \otimes 1)P_{23}$. Removing the common reversal of the three tensor factors from (26.3) gives

$$\begin{aligned} &R_{12}(t_a - t_b)R_{13}(t_a - t_c)R_{23}(t_b - t_c) \\ &= R_{23}(t_b - t_c)R_{13}(t_a - t_c)R_{12}(t_a - t_b). \end{aligned} \tag{26.5}$$

This is the spectral Yang–Baxter equation on the chosen labels. It is a rational identity when the labels are independent variables. A single numerical specialization proves only that specialization.

Writing $R_{21}(u) = PR(u)P$, the reverse condition becomes $R_{21}(-u)R(u) = 1$.

Take a field k of characteristic zero, $V = k^d$, and

$$B_0 = k[\hbar, t_1, \dots, t_m, \Delta^{-1}], \qquad \Delta = \prod_{a>b}(t_a - t_b).$$

Extend V to B_0 and define

$$R(u) = 1 - \frac{\hbar P}{u}. \tag{26.6}$$

Proposition 26.2. *The matrix* (26.6) *satisfies* (26.5)*. Its oriented exchange algebra has the ordered basis of Theorem 26.1. Its reverse product is*

$$R_{21}(-u)R(u) = 1 - \frac{\hbar^2}{u^2}. \tag{26.7}$$

After adjoining the inverses of $t_a - t_b \pm \hbar$, the matrix

$$\widehat{R}(u) = \frac{u - \hbar P}{u - \hbar} \tag{26.8}$$

satisfies both Yang–Baxter and the reverse condition.

Proof. Put $u = t_a - t_b$ and $v = t_b - t_c$, so $t_a - t_c = u + v$. Expand the difference between the two sides of Yang–Baxter in $\hbar$. The constant and linear coefficients cancel. If $C = P_{12}P_{13}$ and $D = P_{13}P_{12}$, the permutation identities

$$P_{13}P_{23} = P_{23}P_{12} = C, \qquad P_{12}P_{23} = P_{23}P_{13} = D$$

make the quadratic coefficient equal to

$$(C - D)\left(\frac{1}{u(u+v)} - \frac{1}{uv} + \frac{1}{v(u+v)}\right) = 0.$$

The cubic coefficient vanishes because $P_{12}P_{13}P_{23} = P_{23}P_{13}P_{12}$. This proves the identity over B_0 in every dimension.

Since $P^2 = 1$, direct multiplication gives (26.7). Multiplying $R(u)$ by $f(u) = u/(u - \hbar)$ preserves Yang–Baxter: both sides acquire the same product of three central factors. Finally, $f(u)f(-u) = u^2/(u^2 - \hbar^2)$ cancels the reverse product. □

The unnormalized matrix can therefore be used for oriented exchange, but not for a reversible presentation with the same free basis. Indeed, imposing both unnormalized orientations forces $\hbar^2 e_i^{(a)} e_j^{(b)} = 0$ for $a \neq b$. Over the generic rational field these mixed quadratic products vanish. Over $k[t_1, \dots, t_m, \Delta^{-1}][[\hbar]]$, with every word-degree component completed in $\hbar$, their reductions modulo $\hbar$ are nonzero ordered words. They are thus nonzero $\hbar^2$-torsion classes.

The normalized construction is also available over this complete ring, where $u \pm \hbar$ are units.

26.3 Scalar normalization and the product

There is an explicit comparison between oriented normalizations, but it changes the multiplication. Suppose the maps c_{ab} satisfy (26.3), and choose central units $f_{ab} \in B$ for $a > b$. The maps $c'_{ab} = f_{ab}c_{ab}$ satisfy the same equation. Both algebras are graded by the color count $\alpha \in \mathbb{N}^m$.

For two color counts define

$$\sigma(\alpha, \beta) = \prod_{a>b} f_{ab}^{\alpha_a \beta_b}.$$

It satisfies

$$\sigma(\alpha, \beta)\sigma(\alpha + \beta, \gamma) = \sigma(\beta, \gamma)\sigma(\alpha, \beta + \gamma), \qquad \sigma(0, \alpha) = \sigma(\alpha, 0) = 1.$$

Consequently $x \star y = \sigma(\alpha, \beta)xy$ for homogeneous x, y of degrees α, β defines an associative unital algebra on the module $A(c)$. Denote it by $A(c)^\sigma$. Sending each generator to itself gives an algebra isomorphism

$$A(c') \xrightarrow{\sim} A(c)^\sigma. \tag{26.9}$$

To verify this, a descending pair acquires the factor f_{ab}, whereas its increasing counterpart acquires none. Thus the new relations hold. An ordered product acquires no factor at any stage, so the map identifies the two ordered bases and is invertible.

More generally, for a word with colors $(a_1, \dots, a_n)$ its normal form for c' is its normal form for c multiplied by

$$\prod_{i<j,\ a_i>a_j} f_{a_i a_j}.$$

Each inverted pair crosses exactly once during sorting. In particular, (26.8) is a scalar twist of the oriented rational algebra. This comparison does not identify their untwisted products.

26.4 The obstruction and the realization problem

Failure of Yang–Baxter obstructs the stated free ordered basis. For example, on k^2 let E_{ij} be the matrix units and take the invertible constant matrix $R = 1 + E_{12} \otimes E_{21}$, whose inverse is $1 - E_{12} \otimes E_{21}$. Direct multiplication gives

$$R_{12}R_{13}R_{23} - R_{23}R_{13}R_{12} = E_{12} \otimes (E_{22} - E_{11}) \otimes E_{21}.$$

For $x = e_1$ and $y = e_2$ this sends $y \otimes x \otimes x$ to $-x \otimes x \otimes y$. Thus three colors with these exchange maps cannot have the ordered basis of Theorem 26.1.

Their associative quotient nevertheless exists and has a nonzero unit: its relations are homogeneous of degree two and leave the scalar summand unchanged. With constant fields $Y(a, z)b = ab$, it is also a regular nonlocal vertex algebra as constructed in the preceding chapter. The obstruction concerns the specified independent ordered products, not the existence of every associative algebra or field system.

The finite-color algebra retains spectral coefficients and word order. An associative chiral realization requires coefficient objects over configurations and their extensions across collisions. One must also construct the substitution maps with all required coherences. A realization by quantum fields must additionally give the state space, vacuum, translation, Laurent expansions and full-state exchange operator. Neither construction follows from an ordered basis alone.

Chapter 27

The RTT Yangian of $\mathfrak{gl}_2$

The rational exchange matrix also defines an algebra whose representations can be tensored. Its generators are the coefficients of a matrix series. Matrix multiplication supplies the coproduct. Keeping the deformation parameter in the coefficient ring reveals two different associated graded algebras and determines the differential in the bar construction.

27.1 Coefficients and evaluation

Work over $B = \mathbb{C}[\hbar]$. The presentation and Hopf formulas hold for every finite $n \geq 1$. The determinant calculation below concerns $n = 2$. Let $V = B^n$, let E_{ij} be its matrix units, and let $P(x \otimes y) = y \otimes x$ on $V \otimes_B V$.

Write

$$T^{(r)} = (t_{ij}^{(r)})_{i,j=1}^{n}, \qquad T(u) = I + \sum_{r \geq 1} T^{(r)} u^{-r}, \qquad t_{ij}^{(0)} = \delta_{ij}.$$

Here u is central, $T_1(u) = T(u) \otimes I$ and $T_2(v) = I \otimes T(v)$ refer to the two auxiliary copies of V. Their matrix entries belong to the same coefficient algebra.

Define $Y_\hbar(\mathfrak{gl}_n)$ to be the unital associative B-algebra on the $t_{ij}^{(r)}$, $r \geq 1$, with relations

$$\big((u-v)I - \hbar P\big) T_1(u) T_2(v) = T_2(v) T_1(u) \big((u-v)I - \hbar P\big). \tag{27.1}$$

The equality is coefficientwise in simultaneous formal Laurent series with bounded positive powers of u and v. Thus the presentation requires no choice of expansion for $(u-v)^{-1}$. The customary rational notation uses $R(w) = I - \hbar P / w$.

Proposition 27.1 (Coefficient relations)**.** *The defining relations are equivalent to*

$$[t_{ij}^{(r+1)}, t_{kl}^{(s)}] - [t_{ij}^{(r)}, t_{kl}^{(s+1)}] = \hbar \big(t_{kj}^{(r)} t_{il}^{(s)} - t_{kj}^{(s)} t_{il}^{(r)} \big), \qquad r, s \geq 0. \tag{27.2}$$

For $r, s \geq 1$ this gives

$$[t_{ij}^{(r)}, t_{kl}^{(s)}] = \hbar \sum_{a=0}^{\min(r,s)-1} \big(t_{kj}^{(a)} t_{il}^{(r+s-1-a)} - t_{kj}^{(r+s-1-a)} t_{il}^{(a)} \big). \tag{27.3}$$

Proof. The (ij, kl) matrix entry of (27.1) is

$$(u-v)[t_{ij}(u), t_{kl}(v)] = \hbar \big(t_{kj}(u) t_{il}(v) - t_{kj}(v) t_{il}(u) \big).$$

Its coefficient of $u^{-r} v^{-s}$ is (27.2), including the boundary cases $r = 0$ or $s = 0$. All entries and coefficients have been retained, so the implication reverses. Iterating the recurrence down to first index zero gives the sum in (27.3) with $a = 0, \dots, r-1$. If $r > s$, its terms with $s \leq a \leq r-1$ cancel in pairs under $a \mapsto r+s-1-a$. This gives the displayed upper limit. $\square$

To identify the first modes, define $U_\hbar(\mathfrak{gl}_n)$ as the B-algebra generated by X_{ij} with relations

$$[X_{ij}, X_{kl}] = \hbar(\delta_{jk}X_{il} - \delta_{il}X_{kj}).$$

For each $a \in \mathbb{C}$ there is an evaluation homomorphism

$$\mathrm{ev}_a : Y_\hbar(\mathfrak{gl}_n) \longrightarrow U_\hbar(\mathfrak{gl}_n), \qquad t_{ij}^{(r)} \longmapsto a^{r-1}X_{ij}. \tag{27.4}$$

For positive r, s, substituting these coefficients in (27.2) makes both sides zero. When $r = 0$ or $s = 0$, the same substitution gives precisely the displayed relation for the X_{ij}. This proves that evaluation is well defined.

The relation with $r = 0$, $s = 1$ also defines a homomorphism $X_{ij} \mapsto t_{ij}^{(1)}$, and ev_0 is its left inverse. The first modes therefore contain an actual copy of $U_\hbar(\mathfrak{gl}_n)$.

The map $X_{ij} \mapsto \hbar e_{ij}$, where e_{ij} are the standard generators of $U(\mathfrak{gl}_n)$, gives the familiar expression

$$T(u) \longmapsto I + \frac{\hbar(e_{ij})}{u - a}.$$

This is a generating series for an algebra map into $U(\mathfrak{gl}_n) \otimes_{\mathbb{C}} B$. Its expansion is at $u = \infty$. The original generators $t_{ij}^{(r)}$ have not been divided by $\hbar$.

27.2 The coefficient Hopf algebra

Theorem 27.2 (Hopf structure)**.** *The Yangian is a Hopf algebra over B, with*

$$\Delta(t_{ij}^{(r)}) = \sum_{k=1}^{n}\sum_{p=0}^{r} t_{ik}^{(p)} \otimes t_{kj}^{(r-p)}, \qquad \varepsilon(t_{ij}^{(r)}) = 0 \quad (r \geq 1),$$

$$S(T^{(r)}) = \sum_{m=1}^{r}(-1)^m \sum_{\substack{r_1+\cdots+r_m=r \\ r_\nu \geq 1}} T^{(r_1)} \cdots T^{(r_m)}.$$

The coproduct takes values in the ordinary algebraic tensor product $Y_\hbar \otimes_B Y_\hbar$. The antipode formula specifies the matrix of its values on the generators.

Proof. Let T' and T'' be RTT matrices whose coefficient algebras commute, and put $\mathcal{R} = (u - v)I - \hbar P$. Their product satisfies RTT:

$$\begin{aligned}\mathcal{R}T_1'T_1''T_2'T_2'' &= \mathcal{R}T_1'T_2'T_1''T_2'' = T_2'T_1'\mathcal{R}T_1''T_2'' \\ &= T_2'T_1'T_2''T_1''\mathcal{R} = T_2'T_2''T_1'T_1''\mathcal{R}.\end{aligned}$$

In the first and last equalities the crossed factors have different auxiliary spaces and commuting coefficients. Taking T' and T'' in the two tensor factors defines Δ. Its coefficients are the finite sums above. Threefold matrix multiplication proves coassociativity, and substitution $T = I$ gives the counit identities.

Give $t_{ij}^{(r)}$ weight r and B weight zero. The resulting filtration has degree-zero part B, and the reduced coproduct of a positive-weight element has smaller positive weights in both factors. For $J = \mathrm{id} - \eta\varepsilon$, the convolution series $\sum_{m\geq 0}(-J)^{*m}$ is consequently finite on each filtered piece. Multiplying this series by $\eta\varepsilon + J$ on either side gives $\eta\varepsilon$. It is the antipode.

On the generating matrix the two identities say $S(T(u))T(u) = T(u)S(T(u)) = I$. Expanding the inverse series gives the formula in the theorem. □

Formal series in u^{-1} thus package finite algebraic formulas. Completion in $\hbar$ is a further operation, discussed after the coefficient basis has been established.

27.3 Two filtrations

Fix a total order on the triples (r, i, j). An ordered monomial means a finite product of generators in that order, including the empty product. Let G_d denote the filtration by total weight $\sum r$, and let F_p denote the filtration by total loop degree $\sum(r-1)$. Both give $\hbar$ degree zero. For the current Lie algebra $L = \mathfrak{gl}_n(\mathbb{C})[z]$, write $L_B = L \otimes_{\mathbb{C}} B$ and equip it with bracket $[x, y]_\hbar = \hbar[x, y]$. Its enveloping algebra over B is denoted $U_B(L_\hbar)$.

Theorem 27.3 (PBW and the current algebra)**.** *The ordered monomials form a B-basis of $Y_\hbar(\mathfrak{gl}_n)$. Moreover,*

$$\operatorname{gr}_G Y_\hbar = B[\,\bar{t}_{ij}^{(r)} : r \geq 1\,], \qquad \operatorname{gr}_F Y_\hbar \cong U_B(L_\hbar), \qquad e_{ij}z^{r-1} \longmapsto \bar{t}_{ij}^{(r)}.$$

If Y_1 is the specialization at $\hbar = 1$ and $G_d Y_1$ has weights $\deg t_{ij}^{(r)} = r$, then

$$Y_\hbar \cong \bigoplus_{d \geq 0} \hbar^d G_d Y_1 \;\subset\; Y_1[\hbar].$$

In this last expression $\hbar$ is the Rees variable.

Proof. At $\hbar = 1$, the ordered-monomial theorem is Molev–Nazarov–Olshanskii's PBW theorem [84, Theorem 1.22]. Denote its generators by $\tau_{ij}^{(r)}$. The assignment $t_{ij}^{(r)} \mapsto \hbar^r \tau_{ij}^{(r)}$ respects (27.2) and hence defines a homomorphism to $Y_1[\hbar]$. Equation (27.3) lowers total weight by at least one when two generators are commuted. Induction on weight, followed by sorting at fixed weight, therefore expresses every word as a B-linear combination of ordered monomials.

The image of an ordered monomial of weight d is $\hbar^d$ times its corresponding PBW monomial in Y_1. These images are B-linearly independent, since the PBW monomials are independent and B has no $\hbar$-torsion. This proves the basis claim and identifies the image with the displayed Rees algebra. It also proves the polynomial description of gr_G.

For the loop filtration, the $a = 0$ term of (27.3) has degree $r + s - 2$. Every other term has degree at most $r + s - 3$. The leading relation is consequently

$$[\bar{t}_{ij}^{(r)}, \bar{t}_{kl}^{(s)}] = \hbar\big(\delta_{kj}\bar{t}_{il}^{(r+s-1)} - \delta_{il}\bar{t}_{kj}^{(r+s-1)}\big).$$

It defines the stated surjection from $U_B(L_\hbar)$. PBW for the free B-module Lie algebra $L_\hbar$ gives an ordered basis of its enveloping algebra. Its image is the basis just proved, so the map is injective. □

In particular, the loop-associated graded retains its scaled bracket. Specializing $\hbar = 1$ gives $U(L)$, whereas specializing $\hbar = 0$ gives the symmetric algebra on L. If W is the complex vector space with basis the Yangian's ordered monomials, its $\hbar$-adic completion has underlying module $W[[\hbar]]$. A coefficient at each power of $\hbar$ is a finite linear combination of PBW monomials. A single finite set of monomials need not suffice for the whole series.

27.4 The determinant in rank two

For $n = 2$ the antisymmetric square of the auxiliary space has rank one. This produces a central scalar series. Put $A_{12} = (I - P)/2$ and define

$$D(u) = t_{11}(u)t_{22}(u - \hbar) - t_{21}(u)t_{12}(u - \hbar). \tag{27.5}$$

Each shifted series is expanded at infinity.

Proposition 27.4 (Quantum determinant). *The coefficients of $D(u) = 1 + \sum_{r\ge 1} d_r u^{-r}$ are central, and*

$$A_{12}T_1(u)T_2(u-\hbar) = D(u)A_{12}, \qquad \Delta D(u) = D(u)\otimes D(u), \qquad \varepsilon(D(u)) = 1.$$

There is a unique central series $c(u) = 1 + \sum_{r\ge 1} c_r u^{-r}$ such that $c(u)c(u-\hbar) = D(u)$. Writing $Y_\hbar^{\circ} = Y_\hbar(\mathfrak{gl}_2)/(d_1, d_2, \dots)$, one has an algebra isomorphism

$$Y_\hbar(\mathfrak{gl}_2) \cong B[c_1, c_2, \dots] \otimes_B Y_\hbar^{\circ}.$$

The determinant-one quotient is a Hopf algebra. At $\hbar = 1$ it is the usual $Y(\mathfrak{sl}_2)$.

Proof. Let $\mathcal{T}(v)$ be the standard Yangian matrix at $\hbar = 1$. After inverting $\hbar$, the preceding PBW proof identifies $T(u)$ with $\mathcal{T}(u/\hbar)$. The antisymmetrizer identity and centrality are then the standard quantum-determinant identity and theorem [84, §2.1 and Theorem 2.10]. Their coefficient identities descend to B because $Y_\hbar$ is B-free. In particular, the order of the cross term in (27.5) is fixed by the column formula.

For two RTT matrices T', T'' with commuting coefficient algebras, the antisymmetrizer identity gives

$$\begin{aligned} A_{12}(T'T'')_1(u)(T'T'')_2(u-\hbar) &= A_{12}T_1'(u)T_2'(u-\hbar)T_1''(u)T_2''(u-\hbar) \\ &= D'(u)D''(u)A_{12}. \end{aligned}$$

This proves the coproduct formula. The counit follows by setting $T = I$. The coefficient of u^{-r} in $c(u)c(u-\hbar)$ is $2c_r$ plus a polynomial in earlier c_s and $\hbar$. Recursion therefore constructs the unique c_r as central polynomials in the d_s. The same uniqueness applied in $Y_\hbar \otimes_B Y_\hbar$ proves $\Delta c(u) = c(u)\otimes c(u)$.

The normalized matrix $\widetilde{T}(u) = c(u)^{-1}T(u)$ satisfies RTT and has determinant one. Conversely, multiply a determinant-one RTT matrix by an independent central series $q(u) = 1 + \sum_{r\ge 1} q_r u^{-r}$. Its determinant is $q(u)q(u-\hbar)$, so the recursion recovers q and normalization recovers the original matrix. These two coefficientwise algebra maps are inverse. This proves the tensor factorization and the algebraic independence of the c_r.

Group-likeness of D makes $(d_1, d_2, \dots)$ a Hopf ideal, since $S(D) = D^{-1}$. The identification at $\hbar = 1$ is the determinant-one presentation of the special-linear Yangian [83, Theorem 2.21]. □

The cross-term order can already be distinguished on a two-dimensional evaluation module. At $\hbar = 1$, $a = 0$ and $u = 3$, let $t_{ij}(u) = \delta_{ij}I + E_{ij}/u$. Then $D(3) = \frac{4}{3}I$, whereas

$$t_{11}(3)t_{22}(2) - t_{12}(3)t_{21}(2) = \begin{pmatrix} 7/6 & 0 \\ 0 & 3/2 \end{pmatrix}.$$

The commutator of this last matrix with E_{12} is $-\frac{1}{3}E_{12}$. Thus the cross-term indices cannot be interchanged.

27.5 Exchange on evaluation modules

An exchange operator must specify the representations it exchanges. Let V_a be the rank-n evaluation module obtained from (27.4) and the dual fundamental action $X_{ij} \mapsto -\hbar E_{ji}$. Its auxiliary matrix is $R_{0V}(u-a)$, where 0 denotes the auxiliary space.

For central parameters a, b, extend scalars to

$$B_{a,b} = \mathbb{C}[\hbar, a, b, (a-b)^{-1}, (a-b-\hbar)^{-1}, (a-b+\hbar)^{-1}].$$

With the coproduct action, there is an invertible intertwiner

$$C_{a,b} = P_{12}R_{12}(b-a) : V_a \otimes_{B_{a,b}} V_b \longrightarrow V_b \otimes_{B_{a,b}} V_a. \tag{27.6}$$

Indeed, put $M = R_{01}(u-a)R_{02}(u-b)$ and $M^{\mathrm{op}} = R_{02}(u-b)R_{01}(u-a)$. Yang–Baxter gives

$$M R_{12}(a-b) = R_{12}(a-b) M^{\mathrm{op}}.$$

Since $R(w)R(-w) = (1-\hbar^2/w^2)I$, this implies $R_{12}(b-a)M = M^{\mathrm{op}} R_{12}(b-a)$. Conjugating the right-hand auxiliary product by P_{12} puts the parameter b in the first module and a in the second, proving (27.6). The same scalar identity proves invertibility over $B_{a,b}$.

We have obtained an ordinary coefficient algebra, a coproduct and exchange maps on specified modules. The series $T(u)$ by itself does not supply a state–field map or coefficient objects on configuration spaces. An associative chiral realization must construct those objects, their collision and substitution maps, and the comparison preserving these operations. A realization by line operators must also identify the tensor product and exchange above with the corresponding physical operations. These are additional reconstruction problems.

Chapter 28

The Hall algebra of the A_2 quiver

Counting extensions gives an algebra in which the quantum Serre relation is a cancellation among flags. For the quiver with one arrow, rank classifies every representation. This makes it possible to prove both the relation and the basis theorem that identifies the entire Hall algebra. The calculation is an elementary instance of the Hall-algebra approach to quantum groups [49, 92].

28.1 Representations and multiplication

Fix a finite field $k = \mathbb{F}_q$, put $v = \sqrt{q} > 0$, and use the coefficient field $K_q = \mathbb{Q}(v)$. A representation of $Q : 1 \to 2$ is a linear map $f : V_1 \to V_2$ between finite-dimensional k-spaces. Its dimension vector is $\dim f = (\dim V_1, \dim V_2)$.

The three indecomposable representations are

$$S_1 = (k \to 0), \qquad S_2 = (0 \to k), \qquad M = (k \xrightarrow{1} k).$$

Indeed, bases adapted to the kernel and image of f give

$$f \simeq S_2^{a} \oplus M^{b} \oplus S_1^{c}, \qquad b = \operatorname{rank} f, \quad a = \dim V_2 - b, \quad c = \dim V_1 - b. \tag{28.1}$$

These integers determine the isomorphism class. The arrow in M also gives the nonsplit exact sequence $0 \to S_2 \to M \to S_1 \to 0$.

For representations A, B, C, let $g^{C}_{A,B}$ be the number of subrepresentations $W \subset C$ with $W \simeq B$ and $C/W \simeq A$. Thus the right factor is the subobject and the left factor is the quotient. The Hall space and its untwisted multiplication are

$$H_q = \bigoplus_{[C]} K_q[C], \qquad [A] * [B] = \sum_{[C]} g^{C}_{A,B}\,[C].$$

The sum is finite: its dimension vector is $\dim A + \dim B$, and (28.1) leaves only finitely many ranks. The zero representation gives the unit $[0]$.

Both bracketings of $[A] * [B] * [D]$ count the same flags

$$0 \subset W \subset U \subset C, \qquad W \simeq D, \quad U/W \simeq B, \quad C/U \simeq A.$$

This proves associativity, with the dimension vector as an additive grading.

The Euler form for this orientation is

$$\langle (a,b), (c,d) \rangle = ac + bd - ad.$$

Define the twisted multiplication by

$$[A] \diamond [B] = v^{\langle \dim A, \dim B \rangle}\,[A] * [B]. \tag{28.2}$$

It is associative because bilinearity gives

$$\langle \alpha, \beta\rangle + \langle \alpha+\beta, \gamma\rangle = \langle \beta, \gamma\rangle + \langle \alpha, \beta+\gamma\rangle.$$

The unit remains $[0]$. Write $e_i = [S_i]$ and, unless a star is displayed, use $\diamond$ for products in H_q.

28.2 The root vector and the Serre relation

A representation of dimension $(1,1)$ has rank zero or one. Each has exactly one subobject S_2 with quotient S_1. Only the rank-zero representation has a subobject S_1. Consequently

$$e_1e_2 = v^{-1}\big([S_1\oplus S_2] + [M]\big), \qquad e_2e_1 = [S_1\oplus S_2].$$

The root vector has the precise normalization

$$e_{12} := e_1e_2 - v^{-1}e_2e_1 = v^{-1}[M]. \tag{28.3}$$

Theorem 28.1 (Quantum Serre relations). *The generators e_1, e_2 satisfy*

$$e_1^2e_2 - (v+v^{-1})e_1e_2e_1 + e_2e_1^2 = 0,$$
$$e_2^2e_1 - (v+v^{-1})e_2e_1e_2 + e_1e_2^2 = 0.$$

Proof. Read a word from its top quotient to its bottom quotient. For a map $f : k^2 \to k$, the three untwisted flag counts are

$\operatorname{rank} f$	112	121	211
0	$q+1$	$q+1$	$q+1$
1	$q+1$	1	0.

If $f = 0$, every line in k^2 is allowed. For rank one, 112 allows every line and 121 forces the kernel line. The word 211 would require the whole space k^2 to map to zero.

The total Euler exponents are $-1, 0, 1$. The coefficients of the first Serre expression on the two rank classes are therefore

$$(q+1)\big(v^{-1} - (v+v^{-1}) + v\big) = 0, \qquad v^{-1}(q+1) - (v+v^{-1}) = 0.$$

Here the second equality uses $q = v^2$.

Let D be the contravariant involution obtained by dualizing a map and exchanging the two vertices. It exchanges S_1, S_2, fixes M, and reverses a flag. Moreover $\langle \dim DB, \dim DA\rangle = \langle \dim A, \dim B\rangle$. It therefore induces an anti-automorphism of the twisted Hall algebra. Applying it to the first relation proves the second. □

28.3 A basis and the positive quantum group

To distinguish a parameter from its numerical specialization, let $\mathcal{A} = \mathbb{Z}[v, v^{-1}]$ in this section, with v indeterminate. Define $U^+_{\mathcal{A}}$ to be the associative $\mathcal{A}$-algebra generated by E_1, E_2 subject to the two relations of Theorem 28.1, with E_i in place of e_i. Its extension of scalars to $\mathbb{Q}(v)$ is the usual positive quantum group $U_v^+(\mathfrak{sl}_3)$. Put

$$X = E_2, \qquad Z = E_1, \qquad Y = ZX - v^{-1}XZ.$$

The degrees of E_1, E_2 are $(1,0), (0,1)$, so Y has degree $(1,1)$.

Proposition 28.2 (Normal forms). *The algebra $U^+_{\mathcal{A}}$ is free over $\mathcal{A}$ with basis $X^aY^bZ^c$, for $a, b, c \geq 0$. Equivalently, it has the presentation*

$$YX = vXY, \qquad ZY = vYZ, \qquad ZX = v^{-1}XZ + Y. \tag{28.4}$$

Proof. In the three-generator presentation, reduce each left side of (28.4) to its right side. Order words first by length and then by their number of inversions for $X < Y < Z$. Each replacement strictly decreases this pair. The only overlapping pair of reductions is ZYX, and both choices give

$$ZYX \longmapsto vXYZ + vY^2.$$

Reductions at disjoint positions commute.

Induction on the decreasing pair now shows that every word has a unique linear combination of ordered words as its normal form. The resulting normal-form map kills every defining relation in every word context and fixes the ordered words. Thus these words are both spanning and linearly independent in the quotient, over $\mathcal{A}$ itself.

Eliminating Y from $YX = vXY$ gives $ZX^2 - (v + v^{-1})XZX + X^2Z = 0$. Eliminating it from $ZY = vYZ$ gives $Z^2X - (v + v^{-1})ZXZ + XZ^2 = 0$. These are exactly the two defining relations of $U^+_{\mathcal{A}}$. Conversely, those relations imply the first two identities in (28.4) after setting $Y = ZX - v^{-1}XZ$. The two substitutions are inverse algebra homomorphisms. □

Specialize along $\mathcal{A} \to K_q, v \mapsto \sqrt{q}$. In the following formula v again denotes this positive square root. Define

$$[m]_v = \sum_{j=0}^{m-1} v^{m-1-2j}, \qquad [m]_v! = \prod_{j=1}^{m} [j]_v, \qquad [0]_v! = 1.$$

All these factorials are nonzero in K_q.

Theorem 28.3 (Hall realization in type A_2)**.** *The map*

$$K_q \otimes_{\mathcal{A}} U^+_{\mathcal{A}} \longrightarrow H_q, \qquad E_i \longmapsto e_i,$$

is an isomorphism of dimension-graded associative algebras. More precisely,

$$\begin{aligned} e_2^a e_{12}^b e_1^c &= v^{a(a-1)+b(b-1)+c(c-1)-b+ab+bc} \\ &\quad \cdot [a]_v![b]_v![c]_v! \, [S_2^a \oplus M^b \oplus S_1^c]. \end{aligned} \tag{28.5}$$

Proof. For $S = S_1, S_2$ or M, an iterated extension with all factors S is isomorphic to S^m. For M this follows because the arrow is invertible on every successive quotient, hence invertible on the whole representation. Its flags are complete flags in k^m. The same count applies to the two simples.

Their number is

$$[m]_q! := \prod_{j=1}^{m} (1 + q + \cdots + q^{j-1}) = v^{m(m-1)/2}[m]_v!.$$

Since $\langle \dim S, \dim S \rangle = 1$, the twist contributes another $v^{m(m-1)/2}$. Hence $[S]^m = v^{m(m-1)}[m]_v![S^m]$. The power of $e_{12} = v^{-1}[M]$ contributes the additional factor v^{-b}.

In $[S_2^a] * [M^b]$, the subobject must have the full source space and its image at the target. Thus the middle representation is $S_2^a \oplus M^b$, and that subobject is unique. In $[S_2^a \oplus M^b] * [S_1^c]$, the subobject must be the full kernel, again unique. The corresponding Euler exponents are ab and bc. Multiplying these factors proves (28.5).

The Serre relations define the displayed homomorphism. By Proposition 28.2, its source has the ordered monomial basis. Equation (28.5) sends that basis bijectively to nonzero multiples of all rank classes (28.1). It proves both injectivity and surjectivity. □

The specialization uses the Laurent-polynomial form $U^+_{\mathcal{A}}$. It does not require evaluating arbitrary elements of $\mathbb{Q}(v)$ at an algebraic number. A geometric Hall realization requires, in addition, a coefficient category and extension correspondences compatible with its factorization operations. Chapter 218 describes these data and the flag correspondences that the finite-field calculation supplies.

Chapter 29

Matrix traces and the necklace bracket

A trace remembers a word only up to cyclic rotation. Differentiating that trace cuts the word at every occurrence of the varied matrix. These cuts give a Lie bracket on cyclic words and Hamiltonian vector fields on every matrix space. Passing to commuting matrices leads to multisymmetric functions, where the trace of the identity retains the matrix rank.

29.1 Cyclic words and matrix Hamiltonians

Work over $\mathbb{C}$ and let $F = \mathbb{C}\langle x, y\rangle$ be the free associative algebra. Write $[F, F]$ for the linear span of commutators and put $F_{\mathrm{cyc}} = F/[F, F]$. This is a vector-space quotient, with a basis consisting of cyclic rotation classes of words, including the empty word. Indeed, a commutator of words is a difference of cyclic rotations. Moving the initial generator of a word to the end changes it by a commutator.

For a word $w = a_1 \cdots a_n$, where each a_i is x or y, define

$$D_x[w]_{\mathrm{cyc}} = \sum_{a_i = x} a_{i+1} \cdots a_n a_1 \cdots a_{i-1} \in F, \tag{29.1}$$

and define D_y in the same way. Both derivatives annihilate the empty word. Rotating w only permutes its cuts, so these maps are well-defined on F_{cyc}. Define the bilinear operation

$$\{f, g\}_{\mathrm{neck}} = [D_x f \, D_y g - D_y f \, D_x g]_{\mathrm{cyc}}. \tag{29.2}$$

We will prove its Jacobi identity by evaluating actual matrix traces. These constructions are the two-generator case of noncommutative symplectic geometry [29, §§6.2,8.4].

Fix $N \geq 1$. The coordinate algebra of a pair of $N \times N$ matrices is

$$O_N = \mathbb{C}[X_{ij}, Y_{ij} : 1 \leq i, j \leq N].$$

Equip it with the Poisson bracket determined by

$$\{X_{ij}, Y_{kl}\} = \delta_{il}\delta_{jk}, \qquad \{X_{ij}, X_{kl}\} = \{Y_{ij}, Y_{kl}\} = 0.$$

This convention fixes the sign of (29.2). Cyclicity of trace defines a linear map

$$J_N : F_{\mathrm{cyc}} \longrightarrow O_N^{\mathrm{GL}_N}, \qquad J_N(f) = \operatorname{Tr} f(X, Y),$$

where GL_N acts by simultaneous conjugation. In particular, $J_N([1]_{\mathrm{cyc}}) = N$.

Theorem 29.1 (Trace realization). *For $f, g \in F_{\mathrm{cyc}}$,*

$$\{J_N(f), J_N(g)\} = J_N(\{f, g\}_{\mathrm{neck}}).$$

Proof. Moving a varied matrix entry to the front of the trace gives

$$\frac{\partial J_N(f)}{\partial X_{ij}} = (D_x f(X,Y))_{ji}, \qquad \frac{\partial J_N(f)}{\partial Y_{ij}} = (D_y f(X,Y))_{ji}.$$

The entrywise Poisson contraction is therefore

$$\sum_{i,j} (D_x f)_{ji}(D_y g)_{ij} - (D_y f)_{ji}(D_x g)_{ij} = \operatorname{Tr}(D_x f\, D_y g - D_y f\, D_x g).$$

This is the required identity. □

For example, $\{[x],[y]\}_{\text{neck}} = [1]$ gives $\{\operatorname{Tr} X, \operatorname{Tr} Y\} = N$. Cutting the two occurrences of x in x^2y gives $D_x[x^2y] = xy + yx$, so

$$\{\operatorname{Tr}(X^2Y), \operatorname{Tr} Y\} = 2\operatorname{Tr}(XY).$$

Lemma 29.2 (Separation by traces)**.** *If $J_N(f) = 0$ for every $N \geq 1$, then $f = 0$ in F_{cyc}.*

Proof. Fix a nonempty word $w = a_1 \cdots a_n$. On an oriented cycle with n vertices, assign the entry $t_i E_{i,i+1}$ to X or Y according as $a_i = x$ or $a_i = y$, with indices read modulo n. The variables t_i are independent. The coefficient of $t_1 \cdots t_n$ in the trace of a word is zero unless that word belongs to the cyclic class of w.

For that class, the coefficient is its positive number of rotation symmetries. Words of other lengths have a different total degree in the t_i. Since the characteristic is zero, this coefficient detects the coefficient of $[w]_{\text{cyc}}$ in f. Evaluating at $N = 1$, $X = Y = 0$ detects the constant class. These tests detect every basis coefficient. □

Proposition 29.3 (The necklace Lie algebra)**.** *Equation* (29.2) *makes F_{cyc} a Lie algebra. The constant class is central, so $F_{\text{cyc}}/\mathbb{C}[1]$ is also a Lie algebra.*

Proof. Skew symmetry follows by moving the two factors cyclically. By Theorem 29.1, every matrix trace sends the Jacobi expression to the Jacobi expression of an ordinary Poisson bracket, hence to zero. Lemma 29.2 proves the claimed identity in F_{cyc}. The constant class has zero cyclic derivatives. □

29.2 Hamiltonian derivations and the moment element

For $f \in F_{\text{cyc}}$, define a derivation δ_f of F by

$$\delta_f(x) = -D_y f, \qquad \delta_f(y) = D_x f. \tag{29.3}$$

Its matrix evaluation is the Hamiltonian vector field $\{J_N(f), -\}$, as the entrywise derivative formula shows. Consequently

$$[\delta_f, \delta_g] = \delta_{\{f,g\}_{\text{neck}}}. \tag{29.4}$$

To justify the implication back in F, evaluate both sides on x and y. Their difference vanishes on matrices of every size. A nonzero polynomial of degree at most d acts nontrivially on 1 by left multiplication in $F/(\text{words of length} > d)$. This finite-dimensional representation separates free polynomials and proves (29.4).

The moment element is $\mu = [x, y]$. Telescoping the cyclic cuts gives

$$[x, D_x f] + [y, D_y f] = 0, \qquad [xD_x f + yD_y f]_{\text{cyc}} = nf \tag{29.5}$$

for f homogeneous of word length n. The first equality implies $\delta_f(\mu) = 0$. The second identifies the kernel of $f \mapsto \delta_f$ with the constant classes.

Proposition 29.4 (Recovering the cyclic Hamiltonian). *Let* $\mathrm{Der}_\mu(F)$ *be the Lie algebra of derivations fixing* μ*. Then* $f \mapsto \delta_f$ *is an isomorphism*

$$F_{\mathrm{cyc}}/\mathbb{C}[1] \xrightarrow{\sim} \mathrm{Der}_\mu(F).$$

If $D(x) = a$ *and* $D(y) = b$ *are homogeneous of degree* m*, its inverse is*

$$f = \frac{[xb - ya]_{\mathrm{cyc}}}{m+1} \pmod{\mathbb{C}[1]}.$$

Proof. Put $n = m + 1$. In word length n, the map

$$\rho : F_m \oplus F_m \longrightarrow [F, F]_n, \qquad (u, v) \longmapsto [x, u] + [y, v]$$

is onto, because successive rotations by one position generate every commutator of words. Since $2 \dim F_m = \dim F_n$, its kernel has dimension $\dim(F_{\mathrm{cyc}})_n$. By (29.5), the cyclic gradient (D_x, D_y) embeds that space in $\ker \rho$, so it is an isomorphism onto the kernel.

The condition $D\mu = 0$ says $\rho(b, -a) = 0$. Its unique cyclic potential satisfies $D_x f = b$, $D_y f = -a$, and the second Euler identity gives the stated inverse. Apply this argument to every homogeneous component of a derivation. Equation (29.4) proves compatibility with the Lie brackets. □

29.3 Power sums and the matrix rank

On N particle pairs (x_i, y_i), set $\{x_i, y_j\} = \delta_{ij}$ and let the symmetric group S_N permute the pairs. Define

$$R_N = \mathbb{C}[x_1, y_1, \ldots, x_N, y_N]^{S_N}, \qquad p_{ab}^{(N)} = \sum_{i=1}^{N} x_i^a y_i^b \quad (a, b \geq 0).$$

Thus $p_{00}^{(N)} = N$. Direct differentiation gives

$$\{p_{ab}^{(N)}, p_{cd}^{(N)}\} = (ad - bc)\, p_{a+c-1,b+d-1}^{(N)}. \tag{29.6}$$

When the coefficient is zero, the right side is zero without requiring a power sum with a negative exponent. In particular, $\{p_{10}^{(N)}, p_{01}^{(N)}\} = N$.

A uniform Poisson algebra retaining this constant is

$$\mathcal{P} = \mathbb{C}[\nu, p_{ab} : a, b \geq 0,\ a + b > 0], \qquad p_{00} := \nu, \tag{29.7}$$

with ν central and bracket given by (29.6). To construct it, regard $\mathbb{C}[x, y]$ as a Lie algebra under its usual Poisson bracket and take its symmetric algebra. The generator ν is the constant polynomial viewed as a Lie-algebra element. The multiplicative unit of the symmetric algebra is a separate element.

The Leibniz extension of this Lie bracket makes $\mathcal{P}$ a Poisson algebra. For each N, evaluation gives a Poisson map

$$\pi_N : \mathcal{P} \longrightarrow R_N, \qquad \nu \longmapsto N, \quad p_{ab} \longmapsto p_{ab}^{(N)}. \tag{29.8}$$

Chapter 219 proves its surjectivity and describes the stable commutative algebra by an explicit orbit basis.

The substitution setting an additional particle pair to zero preserves all positive-degree power sums. It is not a Poisson map for these particle brackets: the bracket of the two linear power sums changes from $N + 1$ to N. The central parameter in (29.7) records the rank needed for each Poisson specialization.

29.4 The relative two-particle surface

For $N = 2$, put $r = x_1 - x_2$, $s = y_1 - y_2$. Then $\{r, s\} = 2$, and the transposition negates both coordinates. The invariant functions

$$A = \frac{r^2}{2}, \qquad B = \frac{rs}{2}, \qquad C = \frac{s^2}{2}$$

satisfy

$$B^2 = AC, \qquad \{A, B\} = 2A, \qquad \{B, C\} = 2C, \qquad \{A, C\} = 4B.$$

The change of variables $e = A/2$, $f = -C/2$, $h = -B$ gives

$$\{h, e\} = 2e, \qquad \{h, f\} = -2f, \qquad \{e, f\} = h, \qquad h^2 + 4ef = 0.$$

This is the Kirillov–Kostant bracket on the nilpotent cone of $\mathfrak{sl}_2$. The full invariant presentation and the Poisson splitting of the center-of-mass plane are proved in Chapter 219. The derived matrix action constructed there retains more information than this degree-zero surface. Comparing that action with a full closed observable algebra requires a further chain-level construction.

Book V

The nonlocal ladder

Chapter 30

Spectral exchange over configuration spaces

An exchange matrix varies with the positions of its inputs. When Yang–Baxter holds, the ordered-word theorem gives a holomorphic family of algebras away from the poles. Reversible exchange compares different orders and determines the product between two groups of generators. We construct these maps before asking for differential structure or extension across collisions.

30.1 The coefficient space and its algebra

Let V be a finite-dimensional complex vector space and let $R(u) \in \mathrm{End}(V \otimes V)$ be a generically invertible meromorphic matrix. Choose a negation-invariant open set $\Omega \subset \mathbb{C} \setminus \{0\}$ on which R and R^{-1} are holomorphic. For a finite set I, define

$$X_I = \{(t_a)_{a\in I} \in \mathbb{C}^I : t_a - t_b \in \Omega \text{ whenever } a \neq b\}.$$

This is an open subset of the configuration space of distinct labeled points. A total order $\prec$ on I orders the labels, independently of their complex coordinates. All tensor products below are ordinary tensor products over $\mathcal{O}_{X_I}$, unless another base is displayed.

Put $E_a = \mathcal{O}_{X_I} \otimes_{\mathbb{C}} V$, with its label a, and let $P(v \otimes w) = w \otimes v$. The exchange map is

$$c_{ab} = PR(t_a - t_b) : E_a \otimes E_b \longrightarrow E_b \otimes E_a.$$

Define $\mathcal{A}_I^{\prec}$ as the unital tensor algebra on $\bigoplus_{a\in I} E_a$, modulo the relations $x\,y = c_{ab}(x \otimes y)$ for $a \succ b$. There is no relation within a single E_a. We use the direct sum of word-degree components, with no completion.

Choose a basis $v_1, \ldots, v_d$ of V, and write $a_i^{(a)}$ for its corresponding generator in E_a. With $R(u)(v_i \otimes v_j) = \sum_{k,l} R_{ij}^{kl}(u) v_k \otimes v_l$, the relation is

$$a_i^{(a)} a_j^{(b)} = \sum_{k,l} R_{ij}^{kl}(t_a - t_b)\, a_l^{(b)} a_k^{(a)} \qquad (a \succ b). \tag{30.1}$$

An ordered word has nondecreasing colors and retains the order of its generators within each color.

For operators on $V^{\otimes 3}$, subscripts refer to the original tensor factors. The spectral Yang–Baxter equation is

$$R_{12}(u) R_{13}(u+v) R_{23}(v) = R_{23}(v) R_{13}(u+v) R_{12}(u). \tag{30.2}$$

It is evaluated where $u, v, u+v \in \Omega$.

Proposition 30.1 (The holomorphic ordered-word family)**.** *Multiplication defines the map of $\mathcal{O}_{X_I}$-modules*

$$\bigotimes_{a\in I}^{\prec} T_{\mathcal{O}_{X_I}}(E_a) \longrightarrow \mathcal{A}_I^{\prec}. \tag{30.3}$$

It is an isomorphism if and only if (30.2) *holds on every triple $a \succ b \succ c$, with $u = t_a - t_b$ and $v = t_b - t_c$. Under this condition every word-degree component is finite free. The presentation and its normal-form inverse commute with base change.*

Proof. At every stalk, apply Theorem 26.1 to the free modules E_a and the descending maps c_{ab}. Its three-color equation becomes (30.2) after removing the common reversal permutation. This proves both directions of the criterion.

The normal-form map consists of finitely many adjacent exchanges on each word. Its coefficients are holomorphic, so the stalkwise formulas define a sheaf map inverse to multiplication. In any fixed degree there are only finitely many ordered words in the chosen bases. They give the asserted free basis, including words with repeated colors.

Both the defining relations and the normal-form formulas can be extended to any new coefficient algebra. Their two inverse identities remain identities after this extension. This proves the base-change assertion, without requiring the new coefficient algebra to be flat. □

Assume henceforth that (30.2) is a meromorphic identity. For $J \subseteq I$, forgetting coordinates gives $p_J : X_I \to X_J$. The generator inclusions define

$$p_J^* \mathcal{A}_J^{\prec|_J} \longrightarrow \mathcal{A}_I^{\prec}.$$

The ordered bases show that this map is injective. These inclusions commute with further forgetting and with base change.

30.2 Changing the order of tensor factors

Write $R_{21}(u) = PR(u)P$. A reversible exchange must also satisfy

$$R_{21}(-u)R(u) = 1. \tag{30.4}$$

Indeed, $c_{ba}c_{ab}$ is the left side of this equation on $E_a \otimes E_b$. Under this hypothesis the reverse relations are consequences of the descending ones. Imposing both orientations therefore defines one algebra $\mathcal{A}_I$, independent of the chosen total order.

For an ordering $\sigma = (a_1, \dots, a_m)$ of I, put $W_\sigma = E_{a_1} \otimes \cdots \otimes E_{a_m}$. Fix a reference order σ_0. Let $N_\sigma : W_\sigma \to W_{\sigma_0}$ be its normal-form map. It is invertible because each elementary exchange is invertible.

Proposition 30.2 (Transport between orders)**.** *Assume* (30.4)*. Every sequence of adjacent exchanges from W_σ to W_τ has the same composite,*

$$C_{\sigma,\tau} = N_\tau^{-1} N_\sigma.$$

Consequently $C_{\tau,\upsilon} C_{\sigma,\tau} = C_{\sigma,\upsilon}$ and $C_{\sigma,\sigma} = 1$.

Proof. For an adjacent exchange $c : W_\sigma \to W_\tau$ descending relative to σ_0, confluence gives $N_\tau c = N_\sigma$. For an ascending exchange, the same equality follows by applying this argument to c^{-1}. Multiplying these identities along any path gives the displayed formula. Its composition and identity laws follow directly. □

These maps compare orders of tensor factors. They do not specify parallel transport along paths in X_I: that would require a connection.

For the raw rational matrix $R(u) = 1 - \hbar P/u$, Proposition 26.2 gives $R_{21}(-u)R(u) = 1 - \hbar^2/u^2$. It supplies the oriented family, but fails (30.4). The normalized matrix

$$\widehat{R}(u) = \frac{u - \hbar P}{u - \hbar}$$

is reversible over

$$\mathbb{C}[\hbar, (t_a)_{a\in I}, (t_a - t_b)^{-1}, (t_a - t_b - \hbar)^{-1}, (t_a - t_b + \hbar)^{-1} : a \neq b].$$

Its algebra is the explicit scalar twist of the oriented raw algebra in (26.9). This comparison changes the product.

30.3 Multiplication between disjoint groups of generators

Suppose exchange is reversible, and write $I = J \sqcup K$. Pull the two smaller algebras to X_I and abbreviate

$$B = p_J^* \mathcal{A}_J, \qquad C = p_K^* \mathcal{A}_K.$$

Multiplication in $\mathcal{A}_I$ gives a map

$$\Phi_{J,K} : B \otimes C \longrightarrow \mathcal{A}_I, \qquad b \otimes c \longmapsto bc.$$

It is an isomorphism of modules: choose the total order with all labels of J before those of K and apply (30.3). Its inverse is the actual normal-form map for this order.

Define

$$\tau_{K,J} : C \otimes B \longrightarrow B \otimes C, \qquad \tau_{K,J}(c \otimes b) = \Phi_{J,K}^{-1}(cb).$$

Thus $\tau_{K,J}$ sorts the K-word past the J-word. If m_B, m_C are the two internal multiplications, put

$$(b \otimes c) \star (b' \otimes c') = (m_B \otimes m_C)\big(b \otimes \tau_{K,J}(c \otimes b') \otimes c'\big). \tag{30.5}$$

Proposition 30.3 (The crossed tensor product)**.** *The product* (30.5) *is associative with unit* $1 \otimes 1$*, and* $\Phi_{J,K}$ *is an algebra isomorphism for this product. The multiplication maps for any finite partition of* I *agree under regrouping and insertion of empty subsets.*

Proof. Write $\tau_{K,J}(c \otimes b') = \sum b_s \otimes c_s$. Its definition gives $cb' = \sum b_s c_s$ in $\mathcal{A}_I$. Hence

$$\Phi_{J,K}\big((b \otimes c) \star (b' \otimes c')\big) = \sum bb_s c_s c' = bcb'c'.$$

The displayed product is therefore the multiplication of $\mathcal{A}_I$ transported by the module isomorphism $\Phi_{J,K}$. It inherits associativity and the stated unit.

For three subsets, both regroupings send $b \otimes c \otimes d$ to bcd. For any number of subsets, every regrouping gives the same product in $\mathcal{A}_I$. Their normal-form inverses agree as well. The empty subset contributes the scalar algebra and its unit. These equalities prove the claimed coherence. □

The ordinary tensor-product multiplication generally differs from (30.5). For $B = \mathbb{C}[x]$, $C = \mathbb{C}[y]$ and $yx = qxy$, the crossing sends $y \otimes x$ to $qx \otimes y$. Ordinary tensor multiplication sends $(1 \otimes y)(x \otimes 1)$ to $x \otimes y$, whose image under Φ is xy. The product of their images is instead $yx = qxy$. For $q \neq 1$, independence of the ordered monomials prevents these maps from being algebra homomorphisms for the ordinary tensor product.

30.4 Diagonal exchange and its three-particle coefficient

Choose meromorphic units $q_{ij}(u)$ satisfying

$$q_{ij}(u)q_{ji}(-u) = 1, \qquad q_{ii}(u) = 1,$$

and set $R(u)(v_i \otimes v_j) = q_{ij}(u)v_i \otimes v_j$. The three fixed-leg operators in (30.2) are diagonal and commute. The family is therefore reversible and has the ordered basis. Its relations are

$$a_i^{(a)} a_j^{(b)} = q_{ij}(t_a - t_b) a_j^{(b)} a_i^{(a)} \qquad (a \succ b).$$

For $a \succ b \succ c$, both complete reductions give

$$\begin{aligned} a_i^{(a)} a_j^{(b)} a_k^{(c)} = {} & q_{ij}(t_a - t_b) q_{ik}(t_a - t_c) q_{jk}(t_b - t_c) \\ & \cdot a_k^{(c)} a_j^{(b)} a_i^{(a)}. \end{aligned} \tag{30.6}$$

Each inverted pair is exchanged exactly once. Scalar commutativity makes the product independent of the chosen sequence. For a general word, the coefficient is the product over all initially inverted color pairs, with the component indices attached to their respective occurrences. Repeated colors retain their internal order.

For constant $q_{12} = q$, $q_{21} = q^{-1}$, select two distinct colors $a \succ b$ and put $x = a_1^{(a)}$, $y = a_2^{(b)}$. Their subalgebra is the quantum plane $\mathbb{C}\langle x, y\rangle/(xy - qyx)$: the independent monomials $y^r x^s$ prove that there are no further relations. This specialization uses distinct colors and imposes no same-color relation.

30.5 A coefficient connection is additional structure

A varying exchange coefficient need not admit a connection preserving its generator lines. Set $K = \mathbb{C}(u)$ and consider

$$A = K\langle x, y\rangle/(yx - q(u)xy), \qquad q(u) = \frac{u+1}{u-1}.$$

Here $q(u)q(-u) = 1$. This scalar example occurs in the diagonal family with $q_{12} = q_{21} = q$ and $q_{11} = q_{22} = 1$. Take $x = a_1^{(b)}$, $y = a_2^{(a)}$ with $a \succ b$, and put $u = t_a - t_b$.

A connection over ∂_u is a $\mathbb{C}$-linear operator ∇ with $\nabla(fa) = f'a + f\nabla(a)$ for $f \in K$. Suppose it is also an algebra derivation and preserves Kx and Ky. Then $\nabla x = \alpha x$, $\nabla y = \beta y$ for some $\alpha, \beta \in K$.

Applying ∇ to the defining relation gives

$$0 = (\alpha + \beta)(yx - qxy) - q'(u)xy = \frac{2}{(u-1)^2} xy.$$

The last element is nonzero by the ordered basis. No connection with these stated properties exists. This includes the prescription making both generators horizontal.

For constant q, coefficientwise differentiation with horizontal generators does preserve the relation and defines such a connection. The obstruction concerns this varying algebra and the prescribed generator lines. A different carrier or differential structure requires its own analysis.

The family $\mathcal{A}_I$ and its crossed products are defined on the distinct-point opens X_I. An associative chiral realization must specify the coefficient category and tensor, differential structure, and extensions at collision diagonals and the excluded poles. It must construct collision and substitution maps, compatible relabelings, factorization comparisons and all their required coherences.

The comparison must identify its chiral operation with the exchange maps constructed here. A field realization additionally needs the state space, vacuum, translation, Laurent expansions and an exchange operator on all states. Constructing these structures and their comparison maps remains the realization problem.

Chapter 31

RTT and reflection algebras

Matrix multiplication and reflection impose different compatibility laws. The product of two independent RTT matrices is again an RTT matrix, which constructs the coproduct. The reflection equation instead gives operators for braids with a boundary. We construct their covariance and then ask when the matrix entries admit an ordered basis.

31.1 Matrix composition and the two coproduct conventions

Let B be a commutative ring and $V = B^n$ with a chosen basis. All algebras are unital B-algebras, and all tensor products are ordinary tensor products over B. For $R \in \mathrm{End}_B(V \otimes V)$, let $\mathcal{T}_R$ be the free algebra on t_{ij}, modulo the entries of

$$RT_1T_2 = T_2T_1R, \qquad T = (t_{ij}), \quad T_1 = T \otimes I_V, \quad T_2 = I_V \otimes T. \tag{31.1}$$

The subscripts here denote auxiliary copies of V. The matrix entries multiply in the coefficient algebra.

There is also a coefficientwise current version. Fix a polynomial matrix $Q(u,v) \in \mathrm{End}_B(V \otimes V)[u,v]$ and put

$$T(u) = I_V + \sum_{r \geq 1} t^{(r)} u^{-r}.$$

Define $\mathcal{T}_Q^{\mathrm{cur}}$ by the entries of every Laurent coefficient in

$$Q(u,v)T_1(u)T_2(v) = T_2(v)T_1(u)Q(u,v). \tag{31.2}$$

Each coefficient is a finite expression because Q is polynomial. The algebra is generated by the $t_{ij}^{(r)}$ without completion. The displayed current is a formal series with coefficients in that algebra. The choice of the cleared kernel Q is part of this presentation.

Lemma 31.1 (Multiplication of independent matrices)**.** *Suppose X and Y satisfy* (31.1) *with coefficients in B-algebras D and E, respectively. Their product XY, with coefficients in $D \otimes E$, satisfies the same equation. The analogous assertion holds for* (31.2)*.*

Proof. Entries from the two coefficient factors commute. Operators on different auxiliary factors also commute, so

$$\begin{aligned} RX_1Y_1X_2Y_2 &= RX_1X_2Y_1Y_2 = X_2X_1RY_1Y_2 \\ &= X_2X_1Y_2Y_1R = X_2Y_2X_1Y_1R. \end{aligned}$$

This is the required equation for XY. Replacing R by $Q(u,v)$ gives the current calculation coefficientwise. No Yang–Baxter identity is needed for this argument. □

Use superscripts [1] and [2] for the two coefficient factors. Thus $(T^{[1]})_{ij} = t_{ij} \otimes 1$ and $(T^{[2]})_{ij} = 1 \otimes t_{ij}$.

Theorem 31.2 (RTT coproduct). *The constant RTT algebra has the bialgebra structure*

$$\Delta_+(t_{ij}) = \sum_k t_{ik} \otimes t_{kj}, \qquad \epsilon(t_{ij}) = \delta_{ij}.$$

Its opposite coproduct is another bialgebra structure:

$$\Delta_-(t_{ij}) = \sum_k t_{kj} \otimes t_{ik}.$$

In matrix notation these are $\Delta_+(T) = T^{[1]}T^{[2]}$ *and* $\Delta_-(T) = T^{[2]}T^{[1]}$. *For currents, set* $t_{ij}^{(0)} = \delta_{ij}$. *Then*

$$\Delta_+(t_{ij}^{(r)}) = \sum_{a+b=r} \sum_k t_{ik}^{(a)} \otimes t_{kj}^{(b)}, \qquad \epsilon(t_{ij}^{(r)}) = 0 \quad (r \geq 1), \tag{31.3}$$

and Δ_- *is the flip of this formula. Both coproducts take values in the ordinary algebraic tensor product.*

Proof. Lemma 31.1 shows that both matrix products preserve the defining relations. Evaluation at the identity matrix defines the counit. In three coefficient factors, the two iterates of Δ_+ are $T^{[1]}T^{[2]}T^{[3]}$, while those of Δ_- are $T^{[3]}T^{[2]}T^{[1]}$. Matrix associativity proves coassociativity. Replacing either coefficient factor by the identity proves the counit identities. The same calculation applies to currents, and every mode sum in (31.3) is finite. □

If the auxiliary factor is numbered first, the two matrix formulas read $\Delta_+(T) = T_{12}T_{13}$ and $\Delta_-(T) = T_{13}T_{12}$. These conventions describe opposite coproducts on the same algebra.

For the Yangian, take $B = \mathbb{C}[\hbar]$ and $Q(u,v) = (u-v)I - \hbar P$, where $P(x \otimes y) = y \otimes x$. The coefficient calculation in Proposition 27.1 gives

$$[t_{ij}^{(1)}, t_{kl}^{(1)}] = \hbar\big(\delta_{kj}t_{il}^{(1)} - \delta_{il}t_{kj}^{(1)}\big).$$

The map $X_{ij} \mapsto t_{ij}^{(1)}$ identifies the first-mode subalgebra with $U_\hbar(\mathfrak{gl}_n)$. Its inverse on that subalgebra is the restriction of the evaluation map (27.4) at zero. It sends $t^{(1)}$ to X and all higher modes to zero. This uses the unscaled current convention $T(u) = I + \sum t^{(r)}u^{-r}$ over $\mathbb{C}[\hbar]$.

31.2 The reflection equation and boundary braids

Assume now that R is invertible and satisfies

$$R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12} \quad \text{on } V^{\otimes 3}.$$

Write $R_{21} = PRP$. The reflection algebra $\mathcal{K}_R$ is the free algebra on entries k_{ij} of a matrix K, modulo the entries of

$$R_{21}K_1R_{12}K_2 = K_2R_{21}K_1R_{12}. \tag{31.4}$$

The matrix K is not assumed invertible. Set $C = PR$. Since $PK_1 = K_2P$, the reflection equation is exactly

$$CK_2CK_2 = K_2CK_2C. \tag{31.5}$$

For $m \geq 2$, define the positive Artin monoid of type B_m by generators $\sigma_1, \dots, \sigma_{m-1}, \kappa$ and relations

$$\begin{aligned} \sigma_i\sigma_{i+1}\sigma_i &= \sigma_{i+1}\sigma_i\sigma_{i+1}, && 1 \leq i < m-1, \\ \sigma_i\sigma_j &= \sigma_j\sigma_i, && |i-j| > 1, \\ \kappa\sigma_i &= \sigma_i\kappa, && i < m-1, \\ \sigma_{m-1}\kappa\sigma_{m-1}\kappa &= \kappa\sigma_{m-1}\kappa\sigma_{m-1}. \end{aligned} \tag{31.6}$$

Adjoining inverses to all generators gives the Artin group of type B_m. The generator κ represents a boundary operation on the last factor.

Theorem 31.3 (Boundary braid coherence)**.** *On the free right $\mathcal{K}_R$-module $M_m = V^{\otimes m} \otimes \mathcal{K}_R$, the operators*

$$\sigma_i \longmapsto C_{i,i+1}, \qquad \kappa \longmapsto K_m$$

define a representation of the monoid (31.6). Here K_m acts on the last auxiliary factor, with its entries acting by left multiplication on $\mathcal{K}_R$. After an algebra map $\mathcal{K}_R \to D$, these operators give a group representation if the resulting matrix $K \in \mathrm{Mat}_n(D)$ is invertible.

Proof. Moving the flips to the left gives

$$C_{12}C_{23}C_{12} = P_{12}P_{23}P_{12}R_{23}R_{13}R_{12},$$
$$C_{23}C_{12}C_{23} = P_{23}P_{12}P_{23}R_{12}R_{13}R_{23}.$$

The permutation products agree, and Yang–Baxter identifies the remaining factors. Operators on disjoint auxiliary factors commute. This proves the first three families of relations in (31.6). The final relation is (31.5) on the last two factors. Thus every defining relation is respected.

When K is invertible, all operators are invertible and the representation extends to the group. □

This theorem compares boundary braid words acting on M_m. A normal form for words in the entries k_{ij} is a separate algebraic question.

31.3 Covariance under an RTT Hopf algebra

Suppose H is a Hopf B-algebra with a matrix $T = (t_{ij})$ satisfying (31.1), $\Delta(T) = T^{[1]}T^{[2]}$ and $\epsilon(T) = I$. If S_H is its antipode, the matrix $U = (S_H(t_{ij}))$ is the inverse of T. Place the entries of K and T in the separate factors of $\mathcal{K}_R \otimes H$.

Proposition 31.4 (Right covariance)**.** *The formula*

$$\delta : \mathcal{K}_R \longrightarrow \mathcal{K}_R \otimes H, \qquad \delta(k_{ij}) = \sum_{a,b} k_{ab} \otimes u_{ia}t_{bj}, \tag{31.7}$$

defines a right H-comodule algebra. In matrix notation, $\delta(K) = UKT$.

Proof. The RTT equation, its conjugate by P, and the inverse identities give

$$T_1RU_2 = U_2RT_1, \qquad R_{21}U_1U_2 = U_2U_1R_{21}, \qquad U_1R_{21}T_2 = T_2R_{21}U_1.$$

For example, multiplying $RT_1T_2 = T_2T_1R$ on the left and right by U_2 gives the first identity. Put $K' = UKT$. Using these identities in order, and commuting only entries from separate coefficient and auxiliary factors, gives

$$\begin{aligned} R_{21}K_1'RK_2' &= R_{21}U_1U_2K_1RK_2T_1T_2 \\ &= U_2U_1R_{21}K_1RK_2T_1T_2 \\ &= U_2U_1K_2R_{21}K_1RT_1T_2 \\ &= U_2K_2U_1R_{21}T_2K_1T_1R \\ &= K_2'R_{21}K_1'R. \end{aligned}$$

Thus (31.7) preserves every defining relation and extends to an algebra homomorphism.

In $\mathcal{K}_R \otimes H \otimes H$, both $(\delta \otimes 1)\delta(K)$ and $(1 \otimes \Delta)\delta(K)$ equal

$$(T^{[2]})^{-1}(T^{[1]})^{-1}KT^{[1]}T^{[2]}.$$

The counit reduces this matrix to K. These are the right coaction identities. □

The side of the coaction and the inverse placement are fixed by the chosen reflection equation. Other transpose and inverse placements have different covariance formulas [72, §6.2].

31.4 Ordered bases for reflection entries

Two invertible Yang–Baxter matrices already distinguish braid coherence from the usual polynomial-sized basis. For $R = I$, equation (31.4) is $K_1K_2 = K_2K_1$. Its entries are exactly the pairwise commutators of the n^2 generators, so $\mathcal{K}_I$ is a polynomial algebra.

For $R = P$, the identity $PK_1P = K_2$ makes both sides equal to K_2^2 before any quotient. Thus $\mathcal{K}_P$ is the free algebra on the same n^2 generators. Over a field, their degree-two dimensions are

$$\dim(\mathcal{K}_I)_2 = \frac{n^2(n^2+1)}{2}, \qquad \dim(\mathcal{K}_P)_2 = n^4.$$

For $n \geq 2$ they differ. Invertibility and Yang–Baxter therefore do not force a basis indexed by nondecreasing words in the matrix entries. Theorem 31.3 still applies in both cases.

A nontrivial positive example is the standard two-dimensional matrix [72, (5)]. Work over $A = \mathbb{C}[q, q^{-1}]$, with $V = A^2$ and basis v_0, v_1. Write $e_{i_1 \cdots i_m} = v_{i_1} \otimes \cdots \otimes v_{i_m}$ and put $\omega = q - q^{-1}$. In the ordered basis $(e_{00}, e_{01}, e_{10}, e_{11})$, take

$$R = \begin{pmatrix} q & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & \omega & 1 & 0 \\ 0 & 0 & 0 & q \end{pmatrix}, \qquad K = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

The checked matrix acts by

$$C(e_{00}) = qe_{00}, \quad C(e_{01}) = \omega e_{01} + e_{10}, \quad C(e_{10}) = e_{01}, \quad C(e_{11}) = qe_{11}.$$

Both three-factor braid products act by q^3 on e_{000} and e_{111}. In each ordered basis $(e_{001}, e_{010}, e_{100})$ and $(e_{011}, e_{101}, e_{110})$ their common matrix is

$$\begin{pmatrix} q^2\omega & q\omega & q \\ q\omega & q & 0 \\ q & 0 & 0 \end{pmatrix}.$$

These eight basis vectors prove the braid identity, using $q\omega + 1 = q^2$. The preceding flip calculation gives quantum Yang–Baxter for R, which is invertible because its determinant is q^2.

Put $\rho = q^{-2}$ and $\lambda = 1 - \rho$. Multiplying the matrices in (31.4) gives the six relations

$$\begin{aligned} bd &= \rho db, & cd &= q^2 dc, & ad &= da, \\ cb &= bc + \lambda da - \lambda d^2, & ab &= ba - \lambda\rho db, & ac &= ca + \lambda dc. \end{aligned} \tag{31.8}$$

For completeness, the intermediate matrix is

$$R_{21}K_1R = \begin{pmatrix} q^2a & q\omega b & qb & 0 \\ q\omega c & a + \omega^2 d & \omega d & qb \\ qc & \omega d & d & 0 \\ 0 & qc & 0 & q^2 d \end{pmatrix}.$$

Let $F = R_{21}K_1RK_2 - K_2R_{21}K_1R$, with rows and columns numbered from zero. The first five relations in the order bd, cd, ad, cb, ac follow respectively from F_{23}, F_{31}, F_{22}, F_{12}, F_{10} by multiplying by units in A. The remaining relation follows from F_{02} and the first one. Conversely, substituting (31.8) makes every entry of F vanish. Thus these are equivalent presentations.

Theorem 31.5 (A complete reflection basis)**.** *For this matrix R, the monomials*

$$d^i b^j c^k a^\ell, \qquad i, j, k, \ell \geq 0,$$

form an A-basis of $\mathcal{K}_R$. They remain a basis after every specialization $q \mapsto q_0 \neq 0$.

Proof. Order the generators by $d < b < c < a$ and orient (31.8) from left to right. Each replacement strictly decreases the length-lexicographic order, so reductions terminate. There are no inclusions between leading words.

The only overlapping pairs of reductions occur in cbd, abd, acd and acb. Reducing first either the left or the right pair gives the same normal expression:

word	normal expression from either first reduction
cbd	$dbc + \lambda d^2 a - \lambda d^3$
abd	$\rho dba - \lambda \rho^2 d^2 b$
acd	$q^2 dca + \lambda q^2 d^2 c$
acb	$bca + \lambda da^2 - \lambda d^2 a$.

Reductions on disjoint pairs commute.

Induct on the word order. Two distinct first reductions either act on disjoint pairs or form one of the four overlaps. In either case they reduce to the same expression. Every remaining word is smaller, so the induction hypothesis gives a unique complete reduction.

Extend the normal map A-linearly to the free algebra. It annihilates every defining relation in every word context, because one may perform that particular replacement first. It therefore descends to $\mathcal{K}_R$ and is the identity on ordered monomials. Multiplication of ordered monomials into the quotient is its inverse, proving both spanning and independence. All rules and inverse identities are defined over A, so base change to any nonzero q_0 preserves the basis. □

The RTT coproduct, reflection coaction and boundary braid action are explicit algebraic operations on different carriers. A physical line realization must specify its bulk line objects, boundary modules and their compositions, and construct maps comparing those operations with the ones above. An interpretation in terms of charged fields additionally requires the state space and field maps of the exchange family. These representation and comparison constructions are not consequences of imposing the RTT or reflection equations.

Chapter 32

Quantum vertex realization

32.1 Two expansion regions

Let V be a complete topological vector space with vacuum $\mathbf{1}$, translation T, state-field map Y, and a unitary rational operator

$$S(z) : V \otimes V \longrightarrow V \otimes V((z)).$$

The two formal regions $|z_1| > |z_2|$ and $|z_2| > |z_1|$ are related by $S(z_1 - z_2)$.

Definition 32.1 (Rational quantum vertex datum)**.** A rational quantum vertex datum satisfies vacuum and translation covariance together with

$$(z_1 - z_2)^N Y(a, z_1)Y(b, z_2) = (z_1 - z_2)^N Y^{21}(S(z_1 - z_2)(a \otimes b); z_2, z_1) \tag{32.1}$$

for a sufficiently large N, and weak associativity in every triple of states.

Theorem 32.2 (Exchange coherence)**.** *Suppose S is unitary,*

$$S^{21}(-z)S(z) = 1,$$

and satisfies the spectral Yang–Baxter equation. Then the pairwise continuation maps obtained from (32.1) *define a braid-group action on every n-point field product. Weak associativity makes the action compatible with iterated operator products.*

Proof. Unitarity identifies a crossing followed by its reverse with the identity. Yang–Baxter identifies the two continuations around every codimension-two triple-collision stratum. These are the defining relations of the braid group. Weak associativity identifies the collision limit in each chamber with the corresponding iterated field. □

An exchange datum describes relations among field products. Its realization by an associative chiral algebra requires the geometric construction and operation comparison of Problem 11.10. The exchange normalization must specify how S combines with the reversal denoted by Y^{21} in (32.1). It must recover ordinary locality when their combined action is the ordinary interchange of the two fields. A comparison with the $\hbar$-adic convention also requires a formal deformation with $S = 1 + O(\hbar)$ and the stated completed tensor products. The field theories of [74, 75] do not by themselves supply this geometric comparison.

32.2 The Yangian field

Let $T(u)$ act on a family of evaluation modules. The RTT relation becomes S-locality for the matrix coefficients of $T(u)$. The normalized Yang operator

$$S(u) = f(u)\left(1 - \frac{\hbar P}{u}\right), \qquad f(u)f(-u) = \left(1 - \frac{\hbar^2}{u^2}\right)^{-1},$$

is unitary. Its expansion at infinity gives the quantum vertex exchange of Yangian currents. The scalar factor changes normalization and preserves the Yang–Baxter equation.

Chapter 33

Shuffle realization and the positive current algebra

A shuffle product combines spectral variables by assigning a scalar factor to each cross-block pair. Associativity follows from the combinatorics of these pairs. The coefficient lattice and the tensor used by a coproduct require further choices. We shall make those choices explicit, beginning with the product of two currents.

33.1 The product and its normalization

Let k be a field of characteristic zero and let $K(u) \in k(u)$. Set

$$\mathrm{Sh}_n = k(z_1, \dots, z_n)^{\mathfrak{S}_n}, \qquad \mathrm{Sh} = \bigoplus_{n \geq 0} \mathrm{Sh}_n .$$

The integer n is the *charge*. All cochain degrees and parities are zero. The direct sum is algebraic, so each element has finite charge support.

For $F \in \mathrm{Sh}_m$ and $G \in \mathrm{Sh}_n$, define

$$(F *_K G)(Z) = \sum_{\substack{I \sqcup J = [m+n] \\ |I| = m}} F(Z_I) G(Z_J) \prod_{i \in I,\, j \in J} K(z_i - z_j). \tag{33.1}$$

Here $[r] = \{1, \dots, r\}$, and Z_I is the subtuple indexed by I. The inputs are symmetric, so their internal order is immaterial. There is one summand for each subset I and no factorial divisor.

The algebra unit is $1_0 = 1 \in \mathrm{Sh}_0 = k$. The constant 1_n in positive charge is a different element.

Theorem 33.1 (Associativity of the shuffle product)**.** *Formula* (33.1) *defines an associative unital charge-graded k-algebra. Every graded subspace containing 1_0 and closed under this product is a unital subalgebra.*

Proof. For inputs of charges m, n, p, both parenthesizations sum over ordered partitions $I \sqcup J \sqcup L = [m + n + p]$ of those sizes. The summand for this partition is

$$F(Z_I) G(Z_J) H(Z_L) \prod_{i \in I,\, j \in J} K(z_i - z_j) \prod_{i \in I,\, l \in L} K(z_i - z_l) \prod_{j \in J,\, l \in L} K(z_j - z_l).$$

Each factor occurs once in either parenthesization. Permuting the variables permutes these summands, proving symmetry of the output. An empty block has one choice and an empty product, which proves both unit equations. The same argument for any number of ordered blocks identifies every iterated product. □

Averaging the displayed summand over all $(m + n)!$ permutations gives $m!n!/(m + n)!$ times (33.1). Multiplication by $n!$ in charge n is an algebra isomorphism from that averaged convention to ours. Summing over all permutations without either normalization is different: for $K = 1$ and constant inputs of charges 1, 1, 2, the two bracketings give 48 and 144.

33.2 The cubic kernel and two currents

To retain parameter specialization, work over

$$R = \mathbb{Q}[h_1, h_2], \qquad h_3 = -h_1 - h_2, \qquad a = \sum_{i<j} h_i h_j, \qquad b = h_1 h_2 h_3.$$

Thus $a = -(h_1^2 + h_1 h_2 + h_2^2)$ and $b = -h_1 h_2 (h_1 + h_2)$. Write

$$N(u) = \prod_{i=1}^{3} (u + h_i) = u^3 + au + b, \qquad K_j(u) = u^{2j} \frac{N(u)}{u} \quad (j = -1, 0, 1).$$

In particular, $K_{-1}(u) = \prod_i (u + h_i)/u^3$ is the cubic rational kernel. Its coefficients lie in R. No parameter has been inverted.

For this kernel take the carrier

$$\mathcal{A}_n^{(-1)} = R[z_1, \dots, z_n, D_n^{-1}]^{\mathfrak{S}_n}, \qquad D_n = \prod_{i<l} (z_i - z_l)^2, \quad D_0 = D_1 = 1.$$

Set $\mathcal{A}^{(-1)} = \bigoplus_n \mathcal{A}_n^{(-1)}$. Subset shuffle preserves these modules. We use $*_j$ for the product with kernel K_j.

For $f, g \in R[z]$ and $u = x - y$, direct multiplication gives

$$f *_j g = f(x) g(y) K_j(u) + f(y) g(x) K_j(-u),$$
$$[f, g]_{*_j} = 2b\, u^{2j-1} \big(f(x) g(y) - g(x) f(y) \big).$$

The bracket is the associative commutator $F *_j G - G *_j F$. If $e_r(z) = z^r$ is the charge-one current, then

$$[e_0, e_1]_{*_{-1}} = -\frac{2b}{u^2}, \qquad e_0 *_{-1} e_0 = 2\Big(1 + \frac{a}{u^2}\Big). \tag{33.2}$$

The alternating numerator of the commutator cancels one diagonal factor. It does not create poles at shifted differences.

At generic parameters the negative shifts $u = -h_i$ are zeros of K_j, and the only other zero or pole is at $u = 0$. In divisor notation, with $D_\lambda = \{u = \lambda\}$,

$$\operatorname{div}(K_j) = \sum_i D_{-h_i} + (2j - 1) D_0.$$

The reverse-to-forward exchange ratio is

$$r(u) = \frac{K_j(-u)}{K_j(u)} = \frac{u^3 + au - b}{u^3 + au + b} = \frac{\prod_i (u - h_i)}{\prod_i (u + h_i)}. \tag{33.3}$$

It is independent of j. The negative shifts are generic poles of this ratio, although they are zeros of the product kernel. When parameters specialize, divisors can coincide and factors can cancel. The rational formulas determine the resulting orders.

33.3 Polynomial coefficients and the generated algebra

For $j = 0, 1$ define the full polynomial carrier

$$\mathcal{A}^{(j)} = \bigoplus_{n \geq 0} R[z_1, \dots, z_n]^{\mathfrak{S}_n}.$$

Both carriers are closed under their respective shuffle products. For K_1 this follows because the kernel is polynomial.

For K_0, every possible denominator in a summand is a simple cross-block difference. Pair that summand with the one obtained by exchanging the two colliding labels. On their diagonal all other factors agree, and the residues of $K_0(u)$ and $K_0(-u)$ cancel. Cancellation at every possible prime denominator in the polynomial unique factorization domain proves polynomiality. The ordered-partition proof gives associativity over R.

The *current-generated algebra* $C^{(j)}$ is the unital R-subalgebra generated by the whole charge-one module $R[z] \subset \mathcal{A}^{(j)}$, for $j = -1, 0, 1$. Its charge-two part can already be smaller than the ambient carrier.

Put

$$c = (x + y)/2, \qquad u = x - y, \qquad t = u^2, \qquad B = R[c, t].$$

Then $R[x, y] = B \oplus uB$, and $B = R[x, y]^{\mathfrak{S}_2}$. For an arbitrary input polynomial $P = A + uC$, its linearized shuffle product is

$$\mu_j(P) = K_j(u)(A + uC) + K_j(-u)(A - uC) = 2t^j\big((t + a)A + bC\big). \tag{33.4}$$

This computes the entire binary map, since pure tensors span $R[x, y]$. Consequently

$$C_2^{(0)} = (t + a, b)B, \qquad C_2^{(1)} = t(t + a, b)B.$$

The first cokernel is $R[c]/(b)$, with R-annihilator (b). The second cokernel maps onto $R[c]$ by setting $t = 0$, so its R-annihilator is zero. A charge-preserving R-linear algebra isomorphism of the two full polynomial carriers would induce an isomorphism of these multiplication cokernels. Thus no such ambient isomorphism exists.

There is nevertheless an actual comparison on the generated algebras. For every split into subsets I, J,

$$D_{m+n}(Z) = D_m(Z_I)D_n(Z_J) \prod_{i \in I, l \in J} (z_i - z_l)^2. \tag{33.5}$$

Multiplying each shuffle summand by this identity proves

$$D_{m+n}(F *_j G) = (D_m F) *_{j+1} (D_n G), \qquad j = -1, 0.$$

Starting with charge-one polynomials, the right side is polynomial for $j = -1$, by the polynomiality just proved for K_0. Since multiplication by D_n is injective and fixes charge one, its image on the generated algebra is precisely the algebra generated by those same currents. We obtain algebra isomorphisms

$$C^{(-1)} \xrightarrow{D} C^{(0)} \xrightarrow{D} C^{(1)}. \tag{33.6}$$

The inverses divide by D_n on the specified images. The map on the full polynomial carriers is not surjective in charge $n \geq 2$, because its image vanishes at $z_1 = z_2$. After diagonal localization it becomes

an isomorphism of the full localized carriers. Factorization (33.5) for any ordered partition proves compatibility with all iterated products. These maps also commute with simultaneous translation of the spectral variables, since D_n depends only on differences.

A wheel condition is a vanishing condition on a triple of shifted roots. Every polynomial in $C_n^{(0)}$ vanishes when three distinct labels satisfy

$$z_i - z_l = h_\alpha, \qquad z_l - z_m = h_\beta, \qquad z_m - z_i = h_\gamma, \qquad \{\alpha, \beta, \gamma\} = \{1, 2, 3\}.$$

Indeed, express a product of currents as a sum over their total orders. No total order respects all three edges of this cycle. A reversed edge supplies a zero factor $K_0(-h_\alpha)$. Over the fraction field of R, the three roots are distinct and this evaluation avoids their diagonal poles. The resulting polynomial vanishing identity holds over R as well.

Such conditions do not determine the integral generated algebra: in charge two there is no triple wheel, whereas $(t + a, b)B$ is a proper submodule of B.

33.4 The generic positive affine Yangian

Now choose complex parameters h_1, h_2, h_3 with sum zero such that

$$m_1h_1 + m_2h_2 + m_3h_3 = 0, \quad m_i \in \mathbb{Z} \quad \Longrightarrow \quad m_1 = m_2 = m_3.$$

Form the shuffle algebra at these numerical parameters, and let $C_h^{(-1)}$ be the subalgebra generated by $\mathbb{C}[z]$. This means specialization of the kernel followed by generation inside the rational carrier.

The positive affine Yangian Y_h^+ is the unital complex algebra on E_r, $r \geq 0$, with relations

$$\begin{aligned} &[E_{r+3}, E_s] - 3[E_{r+2}, E_{s+1}] + 3[E_{r+1}, E_{s+2}] - [E_r, E_{s+3}] \\ &\qquad + a\big([E_{r+1}, E_s] - [E_r, E_{s+1}]\big) = b(E_rE_s + E_sE_r), \end{aligned} \tag{33.7}$$

$$\sum_{\sigma \in \mathfrak{S}_3} [E_{i_{\sigma(1)}}, [E_{i_{\sigma(2)}}, E_{i_{\sigma(3)}+1}]] = 0, \qquad i_1, i_2, i_3 \geq 0. \tag{33.8}$$

Both relations hold for all nonnegative indices. This presentation of the positive half and its shuffle realization are the results of [95, Proposition 1.4(b), Theorem 7.6]:

$$Y_h^+ \xrightarrow{\sim} C_h^{(-1)}, \qquad E_r \longmapsto z^r. \tag{33.9}$$

The kernel there is written $\omega(x, y) = \prod_i (x - y - h_i)/(x - y)^3$ and inserted as $\omega(z_j, z_i)$ for a cross-block pair. Thus it equals our $K_{-1}(z_i - z_j)$. The chargewise factorial conversion above accounts for the averaged symmetrization convention. Composition with (33.6) gives the same generic positive algebra in each of the two polynomial realizations. This conclusion concerns the generated subalgebras. Their ambient coefficient lattices retain the distinction detected by (33.4).

33.5 The tensor required by cooperation

The product alone does not specify a multiplicative coproduct. Here a relation among four currents decides whether the ordinary tensor product can serve as its recipient.

Theorem 33.2 (Obstruction to ordinary counital cooperation). *Fix $j \in \{-1, 0, 1\}$. There is no R-linear, charge-preserving, unital algebra map*

$$\Delta : C^{(j)} \longrightarrow C^{(j)} \otimes_R C^{(j)}$$

with ordinary tensor multiplication and two-sided counit the projection onto charge zero. The same assertion holds for each full ambient carrier $\mathcal{A}^{(j)}$. Coassociativity is not needed for this obstruction.

Proof. The binary formula gives

$$[e_0, e_3]_{*_j} = -2bt^j(x^2 + xy + y^2), \quad [e_1, e_2]_{*_j} = -2bt^j xy, \quad [e_0, e_1]_{*_j} = -2bt^j.$$

Together with $e_0 *_j e_0 = 2t^j(t + a)$, these give

$$[e_0, e_3]_{*_j} - 3[e_1, e_2]_{*_j} + a[e_0, e_1]_{*_j} = -b(e_0 *_j e_0). \tag{33.10}$$

Charge preservation and the two counits force $\Delta(e_r) = e_r \otimes 1_0 + 1_0 \otimes e_r$. Commutators of these even primitive elements have no component of bidegree $(1, 1)$ in the ordinary tensor algebra. The left side of (33.10) therefore has zero such component, whereas the right side has

$$-2b\, e_0 \otimes e_0 \neq 0 \quad \text{in } R[x] \otimes_R R[y] = R[x, y].$$

This is a contradiction over the original parameter ring. □

At $(h_1, h_2, h_3) = (1, 1, -2)$ the detector is $4e_0 \otimes e_0$. The proof also applies to a derived tensor recipient with ordinary Koszul multiplication: the detecting $(1, 1)$ component is the ordinary tensor of two free modules. It makes no claim of a nonzero detector on the divisor $b = 0$. A coalgebra for a cut convolution has a different tensor, whose coefficient comparison must be specified before applying this obstruction.

33.6 Rational cooperation with exchange

There is a positive construction after localizing the spectral factors and changing the middle interchange. Fix j and retain R. Take ordered blocks X, Y of sizes m, n. Localize $R[X, Y]$ at differences of distinct variables and their translates by $\pm h_1, \pm h_2, \pm h_3$. Let $\mathcal{B}_{m,n}$ be its separately symmetric submodule.

The recipient is the algebraic direct sum over (m, n). For three or more blocks use the same localization, with invariance under permutations within each block. These localizations invert no parameter by definition. For $F \in \mathcal{A}^{(j)}_{m+n}$, put

$$Q(X, Y) = \prod_{x \in X,\, y \in Y} K_j(x - y), \qquad \Delta^{\mathrm{rat}}_{m,n} F = \frac{F(X, Y)}{Q(X, Y)}.$$

The numerator is the same symmetric function evaluated on the concatenated variables. The denominator is invertible in the specified recipient. Each charge has only finitely many splits. Further splits use this substitution and division formula.

For $F \in \mathcal{B}_{m,n}$ and $G \in \mathcal{B}_{p,q}$, define $F \circledast G$ by summing over $X = X_1 \sqcup X_2$, $Y = Y_1 \sqcup Y_2$, with $|X_1| = m$, $|Y_1| = n$, and summand

$$\begin{aligned} F(X_1, Y_1) G(X_2, Y_2) & \prod_{x \in X_1, x' \in X_2} K_j(x - x') \\ & \cdot \prod_{y \in Y_1, y' \in Y_2} K_j(y - y') \prod_{y \in Y_1, x' \in X_2} r(x' - y). \end{aligned} \tag{33.11}$$

The last product is the exchange of an earlier right input with a later left input. Its orientation is fixed by $r(u) = K_j(-u)/K_j(u)$.

Proposition 33.3 (Rational cooperation). *The product $\circledast$ is associative, with unit 1 in bidegree (0, 0). The maps Δ^{rat} preserve shuffle multiplication into this recipient, are coassociative, and have the two charge-zero counits. Coassociativity is an identity in the stated three-block localization.*

Proof. A summand in $\Delta^{\mathrm{rat}}(F *_j G)$ partitions the first input into X_1, Y_1 and the second into X_2, Y_2. Its cross factors before division are

$$K_j(X_1 - X_2)K_j(X_1 - Y_2)K_j(Y_1 - X_2)K_j(Y_1 - Y_2),$$

where each expression denotes the product over all pairs in the indicated blocks. Factor the total denominator as

$$Q(X, Y) = Q(X_1, Y_1)Q(X_1, Y_2)Q(X_2, Y_1)Q(X_2, Y_2).$$

The factor $K_j(X_1 - Y_2)$ cancels. The remaining middle ratio is $K_j(Y_1 - X_2)/K_j(X_2 - Y_1) = r(X_2 - Y_1)$. The two internal denominators give the corresponding summands of $\Delta^{\mathrm{rat}}F$ and $\Delta^{\mathrm{rat}}G$. This proves multiplicativity with (33.11).

For three inputs in the recipient, either parenthesization sums over the same partitions of X and of Y. Each earlier-later pair contributes its internal kernel, and each earlier-right, later-left pair contributes its exchange ratio exactly once. Thus the products are associative. Splitting into three blocks X, Y, Z divides by $Q(X, Y)Q(X, Z)Q(Y, Z)$ in either order, proving coassociativity. Empty blocks give the unit and both counits. All these are identities of rational functions with finite charge support. □

For example,

$$\Delta^{\mathrm{rat}}_{1,1}(1_2) = K_j(u)^{-1}.$$

This function is generally not polynomial: the construction has removed the shifted zero loci of K_j. After extending coefficients to $\mathrm{Frac}(R)$, an iterated Laurent field, such as $\mathrm{Frac}(R)((z_1))\cdots((z_n))$, gives a specified chamber expansion of these rational identities. That expansion retains the exchanged multiplication in the recipient. It does not restore the removed collision loci.

33.7 Geometric realization and the double

A geometric Hall comparison must identify the coefficient objects and the pull–push maps along extension flags with the product above. The preprojective Hall setting of [93] fixes such geometric data separately from the rational kernel. For the critical theory of the tripled Jordan quiver with potential $\mathrm{Tr}\, X[Y, Z]$, a comparison must also specify its vanishing-cycle coefficients, orientation, parameter ring and completion. An identification of generic positive algebras alone does not determine these integral coefficient objects.

The supported problem is to realize the shuffle multiplication, its charge-zero unit and its iterated flag comparisons. The realization must retain the true and shifted collision loci and their derived intersections. Any cooperation must come with a coefficient tensor whose interchange respects (33.10). The rational construction proves that an exchanged recipient can satisfy this relation. Extending it across the removed loci is a further construction.

A completed Drinfeld double requires negative and neutral currents and Hopf operations in the chosen tensor recipient. It also requires a continuous Hopf pairing between the appropriate halves. To obtain a quasitriangular structure, the canonical tensor of that pairing must exist in the declared completion. These data must be constructed and compared with the geometric operations. The positive shuffle product supplies their starting algebra.

Chapter 34

The ladder in one diagram

The nonlocal structures form the chain

$$\text{ordered coefficient} \longrightarrow \mathcal{Z}_R \longrightarrow \mathcal{T}_R \longrightarrow (V, Y, S),$$

$$(V, Y, S) \longrightarrow \mathrm{Sh}^{+} \longrightarrow D\big(\mathrm{Sh}^{+}\big).$$

The arrows retain different data:

Object	Defining operation	Coherence
$\mathcal{Z}_R$	exchange of charged fields	Yang–Baxter equation
$\mathcal{T}_R$	matrix-current product and coproduct	RTT and coassociativity
$\mathcal{K}_R$	boundary exchange	reflection equation
(V, Y, S)	formal field product	S-locality and weak associativity
Sh^{+}	extension shuffle	pairwise-kernel associativity
$D(\mathrm{Sh}^{+})$	positive–negative pairing	quasitriangularity

This ladder gives a concrete generator-and-relation theory of doubly noncommutative chiral observables. The first noncommutativity is ordered composition. The second is exchange between distinct order chambers. Chiral singularity remains the diagonal coefficient throughout.

Book VI

Field theory and master equations

Chapter 35

A finite cotangent BF model

The classical cotangent action exists for every finite dg Lie algebra. Its quantization depends on a further datum: the divergence of the Chevalley vector field. The obstruction is a cohomology class, and its vanishing depends on the algebra of functions in which a correction is allowed. We first prove this assertion for a finite homological vector field, then specialize it to Lie algebras.

35.1 Coordinates and the cotangent action

Let k be a field of characteristic zero and V a finite-dimensional graded vector space. Choose homogeneous linear coordinates q_i on V, of degrees d_i. On $E = T^*[-1]V$ their conjugate coordinates p_i have degrees $-1 - d_i$. Functions are graded commutative: odd coordinates anticommute and have square zero. We use either the polynomial pair

$$A = \mathrm{Sym}(V^*), \qquad B = \mathcal{O}(E),$$

or their completions in the ideals generated by all base coordinates and all cotangent coordinates, respectively. A statement about one pair does not assert the same result for the other.

All derivatives below are left derivatives. Fix the constant Berezinian, the graded analogue of a translation-invariant volume element, and set

$$\Delta = \sum_i (-1)^{d_i} \partial_{q_i} \partial_{p_i}, \qquad \{f, g\} = (-1)^{|f|}\big(\Delta(fg) - (\Delta f)g - (-1)^{|f|} f \Delta g\big). \tag{35.1}$$

The bracket and Δ have degree one. These conventions give $\{q_i, p_j\} = \delta_{ij}$ and $\{p_i, q_j\} = -\delta_{ij}$. They fix the signs in the geometric BV formalism of [82, 94].

Exactly one coordinate in each pair (q_i, p_i) is odd. Thus each summand of Δ squares to zero, and distinct summands anticommute. It follows that $\Delta^2 = 0$.

Expanding (35.1) shows that the bracket is a biderivation. Its Jacobi expression is a triderivation which vanishes on coordinate generators, hence vanishes on B. The same calculation gives

$$\Delta\{f, g\} = \{\Delta f, g\} + (-1)^{|f|+1}\{f, \Delta g\}. \tag{35.2}$$

In the completed case all these identities extend by continuity. Δ lowers the field-ideal filtration by at most two.

Let $Q : A \to A$ be a degree-one derivation with $Q^2 = 0$, and assume $Q(q_i)$ has no constant term. Such a derivation is a *homological vector field*. In the completed case it is continuous. Define its cotangent Hamiltonian, with the indicated order of factors, by

$$S_Q = -\sum_i Q(q_i)\, p_i \in B^0, \qquad D_Q = \{S_Q, -\}. \tag{35.3}$$

The action has momentum degree one and field order at least two.

Theorem 35.1 (Cotangent master equation). *The derivation D_Q restricts to Q on A, and*

$$\frac{1}{2}\{S_Q, S_Q\} = 0, \qquad D_Q^2 = 0.$$

Proof. For any even Hamiltonian S of degree zero, direct differentiation gives

$$\{S, q_i\} = (-1)^{d_i}\partial_{p_i} S, \qquad \{S, p_i\} = (-1)^{d_i}\partial_{q_i} S.$$

In (35.3), differentiating past $Q(q_i)$ contributes $(-1)^{d_i+1}$. Therefore $\{S_Q, q_i\} = Q(q_i)$.

The bracket $\{S_Q, S_Q\}/2$ is linear in the momenta. By Jacobi its Hamiltonian vector field restricts to $Q^2 = 0$ on the base. Its derivatives in every p_i consequently vanish. A function linear in the momenta with this property is zero. This proves the master equation without leaving an undetermined scalar term. Jacobi now gives $D_Q^2 = \{\{S_Q, S_Q\}/2, -\} = 0$ on all of B. □

35.2 The divergence class

For the chosen base Berezinian ρ, the divergence of an odd vector field is

$$\chi_Q = \operatorname{div}_\rho Q = \sum_i \partial_{q_i} Q(q_i) \in A^1.$$

The order and signs in (35.3) give

$$\Delta S_Q = \chi_Q, \qquad Q\chi_Q = 0. \tag{35.4}$$

Indeed, the two factors $(-1)^{d_i}$ in ΔS_Q cancel. Applying (35.2) to the classical master equation gives $\{S_Q, \Delta S_Q\} = 0$, which is the second assertion. The class $[\chi_Q] \in H^1(A, Q)$ is the *modular class* for this function algebra and density.

The base and cotangent divergences have different normalizations. If μ is the constant Berezinian on E, the coordinate formula for D_Q gives

$$\operatorname{div}_\mu D_Q = 2\Delta S_Q = 2\chi_Q. \tag{35.5}$$

The q and p derivatives each contribute χ_Q. Thus ΔS_Q is the full divergence of the base vector field and half the divergence of its Hamiltonian lift.

Give $\hbar$ degree zero. A *quantum correction* in the chosen function algebra is a series $S_\hbar = S_Q + \hbar S_1 + \hbar^2 S_2 + \cdots$ in $B^0[[\hbar]]$ satisfying the quantum master equation

$$\frac{1}{2}\{S_\hbar, S_\hbar\} + \hbar\Delta S_\hbar = 0. \tag{35.6}$$

The fields, coefficient algebra and Laplacian are fixed in this definition. No negative powers of $\hbar$ are allowed.

Theorem 35.2 (Cotangent quantization criterion). *A quantum correction exists if and only if $[\chi_Q] = 0$ in $H^1(A, Q)$. More precisely, if $F \in A^0$ satisfies $QF = \chi_Q$, then*

$$S_\hbar = S_Q - \hbar F \tag{35.7}$$

solves the quantum master equation to every order.

Proof. Set all momenta equal to zero to obtain $\pi : B \to A$. The q component of D_Q is Q, and its p component is linear in the momenta. Hence $\pi D_Q = Q\pi$. The coefficient of $\hbar$ in (35.6) is $D_Q S_1 + \chi_Q = 0$. Applying π gives $Q(\pi S_1) = -\chi_Q$, so a correction requires the stated vanishing. This argument permits arbitrary momentum dependence in S_1.

Conversely, a base function F satisfies $\Delta F = 0$ and $\{F, F\} = 0$. Substituting (35.7) into the left side of (35.6) therefore gives

$$\frac{1}{2}\{S_Q, S_Q\} + \hbar(\Delta S_Q - \{S_Q, F\}) + \frac{\hbar^2}{2}\{F, F\} - \hbar^2 \Delta F = \hbar(\chi_Q - QF) = 0.$$

There are no higher terms. □

For any degree-zero $S_\hbar$, put $\delta_{S_\hbar} = \{S_\hbar, -\} + \hbar\Delta$. Jacobi and (35.2) give the exact identity

$$\delta_{S_\hbar}^2 = \left\{\frac{1}{2}\{S_\hbar, S_\hbar\} + \hbar\Delta S_\hbar, -\right\}. \tag{35.8}$$

A quantum master action thus gives a differential on $B[[\hbar]]$. Its compatibility with multiplication is

$$\delta_{S_\hbar}(fg) = (\delta_{S_\hbar} f)g + (-1)^{|f|} f\,\delta_{S_\hbar} g + \hbar(-1)^{|f|}\{f, g\}. \tag{35.9}$$

Consequently the ordinary commutative product is generally not a product of complexes. Applying a construction that requires a derivation demands further algebraic data.

The exponential form of the master equation requires another completion. Let $\mathfrak{m}$ be the ideal of all field coordinates in B, and define

$$\mathcal{W} = \varprojlim_N (B/\mathfrak{m}^N)((\hbar)).$$

Here the lower bound on powers of $\hbar$ may depend on N.

Since $S_Q \in \mathfrak{m}^2$, its exponential $e^{S_Q/\hbar}$ exists in $\mathcal{W}$. In (35.7) one may subtract the constant of F and assume $F(0) = 0$. Then $e^{S_\hbar/\hbar}$ also exists and is invertible. Differentiation is defined by taking two extra orders of field precision before applying Δ.

The product identity for Δ gives

$$\hbar^2 e^{-S_\hbar/\hbar}\Delta(e^{S_\hbar/\hbar}) = \frac{1}{2}\{S_\hbar, S_\hbar\} + \hbar\Delta S_\hbar \tag{35.10}$$

in $\mathcal{W}$. It follows first for powers of an even action by induction and then for the exponential at each field precision. Thus the quantum master equation is equivalent there to $\hbar\Delta(e^{S_\hbar/\hbar}) = 0$. This formal identity asserts no convergence of a numerical integral.

In the formal coefficient algebra, the same primitive has a density interpretation. Set

$$\rho' = e^{-F}\rho, \qquad \mu' = e^{-2F}\mu.$$

The divergence rule gives $\operatorname{div}_{\rho'} Q = \chi_Q - QF = 0$. For the cotangent density the corresponding Laplacian is $\Delta' = \Delta - \{F, -\}$, so $\Delta' S_Q = 0$ as well.

The factor two in μ' agrees with (35.5). The exponentials use the formal completion. A polynomial primitive need not have a polynomial exponential.

35.3 Finite dg Lie algebras

Let L be a finite-dimensional dg Lie algebra and take $V = L[1]$. Then

$$E = L[1] \oplus L^*[-2].$$

For a homogeneous basis e_i of L the coordinates have degrees $|q_i| = 1 - |e_i|$ and $|p_i| = |e_i| - 2$.

Let Q be the Chevalley–Eilenberg derivation on $A = \mathrm{Sym}(L^*[-1])$, or its stated completion. It is the signed suspension of d_L and the bracket. The equations $d_L^2 = 0$, the derivation identity for d_L and Jacobi are respectively the linear, quadratic and cubic parts of $Q^2 = 0$. Equation (35.3) defines the cotangent BF action with these signs.

For a degree-zero endomorphism T of L, write $\mathrm{str}_L(T) = \sum_r (-1)^r \operatorname{Tr}(T|_{L^r})$. Define the degree-zero linear functional on L

$$\chi_L(x) = \mathrm{str}_L(\mathrm{ad}_x) \quad (x \in L^0), \qquad \chi_L|_{L^r} = 0 \quad (r \neq 0).$$

Its suspension is a degree-one linear function on $L[1]$. We call L *unimodular* if $\chi_L = 0$.

Proposition 35.3 (The modular term)**.** *The divergence χ_Q is the suspended functional χ_L. In particular, a unimodular finite dg Lie algebra has $\Delta S_Q = 0$, and its uncorrected cotangent action satisfies the quantum master equation.*

Proof. The linear part of Q has degree one, so it has no diagonal coefficient in the homogeneous coordinates. It contributes zero to $\sum_i \partial_{q_i} Q(q_i)$.

For $x \in L^0$, the terms involving its odd coordinate q^x in the quadratic part act on the other coordinates as $-q^x(\mathrm{ad}_x)^*$. Left differentiation in q_i contributes $-(-1)^{|q_i|}(\mathrm{ad}_x)_{ii}q^x = (-1)^{|e_i|}(\mathrm{ad}_x)_{ii}q^x$. Summing gives the stated supertrace. Only coordinates of degree one can occur in the resulting linear function, so these coefficients determine it.

Its closedness also follows from the vanishing of supertraces of graded commutators, including $\mathrm{ad}_{d_L y} = [d_L, \mathrm{ad}_y]$. The last assertion follows from Theorem 35.1. □

For an ordinary Lie algebra $\mathfrak{g}$ with $[e_a, e_b] = f_{ab}^c e_c$, write c^a for its degree-one coordinates and b_a for the degree -2 conjugates. Our convention becomes

$$Qc^a = -\frac{1}{2} f_{ij}^a c^i c^j, \qquad S_Q = \frac{1}{2} b_a f_{ij}^a c^i c^j, \qquad \Delta = -\sum_a \partial_{c^a}\partial_{b_a}. \tag{35.11}$$

Differentiation gives

$$\Delta S_Q = -\sum_{a,j} f_{aj}^a c^j = \sum_j \operatorname{Tr}(\mathrm{ad}_{e_j}) c^j.$$

Here $A^0 = k$ and $Q(A^0) = 0$. Thus an ordinary finite-dimensional $\mathfrak{g}$ admits a quantum correction on this cotangent space if and only if it is unimodular. Allowing momenta in the correction does not remove the obstruction. For a general dg Lie algebra, exactness of χ_Q is weaker than its vanishing as a function.

For the affine Lie algebra $[H, E] = E$, one has

$$S_Q = b_E c^H c^E, \qquad \Delta S_Q = c^H, \qquad \delta_{S_Q}^2(b_H) = \hbar.$$

The nonzero character prevents a quantum correction. For the Heisenberg algebra $[X, Y] = Z$, instead, $S_Q = b_Z c^X c^Y$ and $\Delta S_Q = 0$. More generally, every ordinary finite-dimensional nilpotent Lie algebra is unimodular because each adjoint endomorphism is nilpotent.

Nilpotence is sufficient but unnecessary. In $\mathfrak{sl}_2$, with $[H,E]=2E$, $[H,F]=-2F$ and $[E,F]=H$, the action is

$$S_Q = 2b_E c^H c^E - 2b_F c^H c^F + b_H c^E c^F .$$

The first two terms have opposite divergences, and the third has zero divergence. Jacobi supplies the classical master equation. The vanishing divergence supplies the quantum one.

35.4 A correction that requires completion

Take $A_{\mathrm{pol}} = k[t]\otimes\Lambda(c)$ with $|t|=0$, $|c|=1$, and define

$$Qt = c(1+t), \qquad Qc = 0.$$

The derivation is homological and has $\chi_Q = c$. The cotangent coordinates have degrees $|p_t|=-1$, $|p_c|=-2$, and

$$S_Q = -c(1+t)\,p_t, \qquad \Delta = \partial_t\partial_{p_t} - \partial_c\partial_{p_c}.$$

Polynomial differentiation is surjective on $k[t]$, so

$$H^1(A_{\mathrm{pol}},Q) = c\,k[t]\,\big/\,c(1+t)k[t] \cong k\,c.$$

Evaluation at $t=-1$ detects the nonzero class of c. Theorem 35.2 therefore excludes a polynomial quantum correction, including corrections depending on the momenta.

In $A_{\mathrm{for}} = k[[t]]\otimes\Lambda(c)$, however,

$$F = \log(1+t) = \sum_{m\geq 1}\frac{(-1)^{m+1}}{m}t^m, \qquad QF = c.$$

The derivative is again surjective and $1+t$ is now a unit, so $H^1(A_{\mathrm{for}},Q)=0$. The completed cotangent action

$$S_\hbar = -c(1+t)\,p_t - \hbar\log(1+t)$$

satisfies the quantum master equation exactly.

For the truncation $F_N = \sum_{m=1}^{N}(-1)^{m+1}t^m/m$, its residual is

$$\frac{1}{2}\{S_Q-\hbar F_N, S_Q-\hbar F_N\} + \hbar\Delta(S_Q-\hbar F_N) = (-1)^N\hbar c\,t^N.$$

It vanishes through total base-field degree N. The infinite identity follows from $Q\log(1+t)=c$. The displayed finite identities specify its output precision.

35.5 Local fields and interval gluing

Let X be a complex curve and $\mathfrak{g}$ an ordinary finite-dimensional nilpotent Lie algebra. The local dg Lie algebra

$$\mathcal{L}_X = \Omega^{0,*}(X)\widehat{\otimes}\Omega^*(\mathbb{R})\otimes\mathfrak{g}$$

has differential $\bar\partial + d_{\mathbb{R}}$ and the bracket obtained from wedge product and the bracket of $\mathfrak{g}$. The completed tensor product here is that of smooth form spaces. Its density dual pairs with compactly supported fields by integration. These data define the classical cotangent action: its master equation uses the differential and Jacobi identities, before a Laplacian at coincident points is introduced.

A finite cochain deformation retract alone does not identify the transferred interaction or its divergence. A finite model with an actual transferred homological vector field and a specified Berezinian falls under Theorem 35.2. The vector field may include higher Taylor terms. To compare its quantization with the local theory requires a compatible BV pushforward or another construction preserving the quantum master equation. An infinite-dimensional realization further requires gauge fixing, renormalized local interactions, the renormalized master equation and compatible operations on opens. The finite trace calculation supplies none of these constructions by itself.

There is nevertheless a definite algebraic gluing model. Give $U = U(\mathfrak{g})$ its scalar augmentation $\epsilon : U \to k$ and let $\overline{U} = \ker \epsilon$. The two boundary modules in this model are the right and left augmentation modules k. The normalized bar complex has $\mathrm{B}_q(U) = \overline{U}^{\otimes q}$ and differential

$$b[a_1|\cdots|a_q] = \sum_{i=1}^{q-1}(-1)^{i-1}[a_1|\cdots|a_i a_{i+1}|\cdots|a_q].$$

On $C_q(\mathfrak{g}) = \Lambda^q\mathfrak{g}$ use

$$\partial(x_1 \wedge \cdots \wedge x_q) = \sum_{i<j}(-1)^{i+j+1}[x_i, x_j] \wedge x_1 \wedge \cdots \widehat{x_i} \cdots \widehat{x_j} \cdots \wedge x_q.$$

The remaining entries keep their order. In particular, $\partial(x \wedge y) = [x, y]$.

Proposition 35.4 (Augmentation gluing)**.** *The antisymmetrizer*

$$\mathrm{Alt}(x_1 \wedge \cdots \wedge x_q) = \sum_{\sigma \in S_q} \mathrm{sgn}(\sigma)[x_{\sigma(1)}|\cdots|x_{\sigma(q)}]$$

gives a quasi-isomorphism $C_*(\mathfrak{g}) \to \mathrm{B}(U)$. Consequently $k \otimes^{\mathbb{L}}_U k \simeq C_*(\mathfrak{g})$ as complexes.

Proof. Adjacent permutations in b Alt pair to give $xy - yx = [x, y]$, with the displayed Chevalley sign. Thus Alt is a chain map.

The PBW filtration on U induces total word weight on its bar. Give $\Lambda^q\mathfrak{g}$ weight q. PBW identifies the associated graded algebra with $\mathrm{Sym}(\mathfrak{g})$.

Order a basis and rewrite yx as $xy + [y, x]$. The only overlaps involve words of length three, and their reductions differ by Jacobi. Disjoint reductions commute. Reduction lowers length or the lexicographic order at fixed length, so the ordered words form a basis.

The associated graded comparison is the antisymmetrizer from the exterior complex with zero differential to the polynomial bar complex. It is the Koszul comparison, a quasi-isomorphism: the polynomial Koszul resolution is exact by successive contraction in each basis variable. With negative first face in the left bar resolution the comparison has the degree sign $(-1)^q$. After augmentation the exterior differential is zero, so this sign does not change the quasi-isomorphism assertion.

The mapping cone of Alt has acyclic associated graded. Every cone cycle has finite largest weight. Subtracting a boundary in that weight lowers it and proves exactness by finite descent. Finally, the normalized left bar resolution is free over U, so its augmentation computes the stated derived tensor product. □

To obtain this gluing from the field theory requires an actual boundary quantization. Its algebra must be identified with $U(\mathfrak{g})$ and its two modules with the augmentation modules. Its composition must correspond to the derived tensor product. An interval polarization alone does not give these identifications. The scalar augmentation in this algebraic model also supplies no identification with a native transverse vacuum or a prepared state.

Chapter 36

BRST and ordered reduction

Reduction by a differential and the bar construction interact through the multiplication. If the differential is a derivation, it extends to tensor words and anticommutes with the operation that joins adjacent entries. This gives a bar complex before taking cohomology. Replacing its entries by cohomology classes requires a comparison of complexes, and replacing it by a chiral construction requires a comparison of operations.

36.1 The signed bar complex

Fix a field k of characteristic zero. Let (A, Q, ϵ) be a unital augmented differential graded associative k-algebra. Thus Q has cohomological degree one, $Q^2 = 0$, $Q(ab) = Q(a)b + (-1)^{|a|}aQ(b)$, and $\epsilon : A \to k$ is a unital algebra map satisfying $\epsilon Q = 0$. Write $I = \ker \epsilon$.

The suspension $I[1]$ places sa in degree $|a| - 1$. The *algebraic reduced bar* is

$$B(A) = \bigoplus_{r \geq 0} I[1]^{\otimes_k r}, \qquad [a_1|\cdots|a_r] = sa_1 \otimes \cdots \otimes sa_r.$$

Every element has finite support in word length. The empty word $[\,]$ spans the length-zero copy of k.

For homogeneous entries put $\sigma_i = \sum_{j<i}(|a_j| - 1)$ and define

$$Q_B[a_1|\cdots|a_r] = -\sum_i (-1)^{\sigma_i}[a_1|\cdots|Qa_i|\cdots|a_r],$$

$$b[a_1|\cdots|a_r] = \sum_{i<r} (-1)^{\sigma_i + |a_i|}[a_1|\cdots|a_i a_{i+1}|\cdots|a_r].$$

Both maps have total degree one and kill the empty word. Their sum is denoted by $D = Q_B + b$. Deconcatenation is the coproduct

$$\Delta[a_1|\cdots|a_r] = \sum_{i=0}^{r} [a_1|\cdots|a_i] \otimes [a_{i+1}|\cdots|a_r].$$

The inclusion of the empty word is its coaugmentation. Projection onto that word is its counit.

Proposition 36.1 (Bar differential and coproduct)**.** *The identities*

$$Q_B^2 = b^2 = 0, \qquad Q_B b + b Q_B = 0$$

hold on $B(A)$. Both Q_B and b are coderivations of Δ. Consequently $(B(A), D, \Delta)$ is a coaugmented differential graded coalgebra.

Proof. The unary differential on $I[1]$ is $sa \mapsto -sQa$. In its tensor extension, terms acting in distinct positions cancel in opposite orders, and terms acting twice in one position vanish by $Q^2 = 0$. For b^2, disjoint multiplications cancel by the same degree-one sign rule. On three adjacent entries the remaining expression is

$$(-1)^{|a_2|}[(a_1a_2)a_3 - a_1(a_2a_3)],$$

which vanishes by associativity.

For the mixed identity, operations in disjoint positions again cancel. On two adjacent entries the expression is

$$(-1)^{|a|}[Q(a)b] + [aQ(b)] - (-1)^{|a|}[Q(ab)] = 0.$$

This is precisely the derivation identity.

For the coproduct, a unary operation lies on one side of each cut. A multiplication lies on one side of every cut that survives the joining of its two entries. These are exactly the terms in $(D \otimes 1 + 1 \otimes D)\Delta$, with $(1 \otimes D)(u \otimes v) = (-1)^{|u|} u \otimes Dv$. The same argument applies separately to Q_B and b. □

The suspension signs can be seen without a general word. In $k[x] \otimes \Lambda(\theta)$, with $|x| = 0$, $|\theta| = 1$, $Qx = \theta$ and $Q\theta = 0$, they give

$$Q_B[x|x] = -[\theta|x] + [x|\theta], \qquad bQ_B[x|x] = 2[x\theta], \qquad Q_B b[x|x] = -2[x\theta].$$

An overall opposite sign for Q_B, with the same b, gives an isomorphic complex: conjugate by $[a_1|\cdots|a_r] \mapsto (-1)^{\sum_i |a_i|}[a_1|\cdots|a_r]$. The prefix signs within Q_B are still required.

36.2 Length and cohomology

Let $F_p B(A)$ be the subcomplex of words of length at most p, with $F_p = 0$ for $p < 0$. The filtration is increasing and exhaustive. The operator Q_B preserves length and b lowers it by one. Use an index q for the sum of the unsuspended entry degrees, so that a word in bidegree (p, q) has total degree $q - p$.

Proposition 36.2 (The length spectral sequence)**.** *The first page and the bidegrees of the differentials are*

$$E_1^{p,q} = \big(H(I, Q)^{\otimes_k p}\big)^q, \qquad d_r : E_r^{p,q} \longrightarrow E_r^{p-r,q-r+1}.$$

The differential d_1 is induced by b. Thus (E_1, d_1) is the bar complex of the augmented graded algebra $H(A, Q)$, and E_2 is its cohomology. For each filtration index, the eventual quotient is $\operatorname{gr}_F H(B(A), D)$. *This assertion does not require a common finite page on which all incoming differentials vanish.*

Proof. The associated graded differential is Q_B. Tensor products of complexes over a field obey the Künneth formula, including for unbounded complexes. This identifies the first page. The part of D lowering length by one is b, which gives d_1 and its stated signs. Total degree one gives the displayed bidegree of d_r.

For the limiting statement, a representative in F_p of a cycle on page r has differential in F_{p-r}. Once $r > p$, such representatives are actual cycles. The incoming boundaries form an increasing union. Every primitive in the algebraic bar has a finite largest word length, so this union contains exactly the boundaries in F_p. The resulting quotient is

$$\frac{\ker D \cap F_p}{(\ker D \cap F_{p-1}) + (\operatorname{im} D \cap F_p)} = \operatorname{gr}_p^F H(B(A), D).$$

The filtration on cohomology is exhaustive because every cycle has finite word support. The finite support argument supplies no uniform bound on the lengths of primitives for different classes. □

A completion needs its own convergence argument. Suppose, in addition, that $A = \bigoplus_{N \geq 0} A^{(N)}$ has an integral weight grading, $A^{(0)} = k$, $I = \bigoplus_{N>0} A^{(N)}$, Q preserves weight and multiplication adds weights. Let $B(A)^{(N)}$ denote total weight N. Every word in it has length at most N.

Define the *weight-completed bar* and its tensor recipient by

$$\widehat{B}_{\mathrm{wt}}(A) = \prod_{N \geq 0} B(A)^{(N)},$$

$$\widehat{B}_{\mathrm{wt}}(A) \widehat{\otimes} \widehat{B}_{\mathrm{wt}}(A) = \prod_{u,v \geq 0} B(A)^{(u)} \otimes_k B(A)^{(v)}.$$

All products here are taken degree by degree as products of complexes. The differential and coproduct extend componentwise.

Corollary 36.3 (Positive-weight convergence)**.** *At each weight N, the length spectral sequence has a finite filtration and computes $H(B(A)^{(N)}, D)$. Moreover*

$$H(\widehat{B}_{\mathrm{wt}}(A), D) = \prod_{N \geq 0} H(B(A)^{(N)}, D).$$

Proof. The bound $p \leq N$ bounds both outgoing and incoming differentials at fixed weight. Products of vector spaces are exact, so taking cohomology commutes with the displayed product of complexes. □

This completion is specified by weight. A product over arbitrary word lengths is a different complex, for which the finite support proof of proposition 36.2 does not apply.

36.3 Comparison before and after reduction

An augmented differential graded algebra map $f : A \to A'$ induces a coalgebra map $B(f)$ by applying f to every entry. It commutes with Q_B by the chain-map identity and with b by multiplicativity. This functorial map gives a comparison stronger than the identification of a spectral-sequence page.

Theorem 36.4 (Homotopy invariance of the algebraic bar)**.** *If $f : A \to A'$ is an augmented differential graded algebra quasi-isomorphism over k, then $B(f)$ is a quasi-isomorphism of differential graded coalgebras. No bound on the cohomological degrees is required. For positive-weight algebras and weight-preserving maps, the same holds for $\widehat{B}_{\mathrm{wt}}(f)$ when f is a quasi-isomorphism at every weight.*

Proof. The splittings $A = k \oplus I$ and $A' = k \oplus I'$ imply that $I \to I'$ is a quasi-isomorphism. Over a field an acyclic complex is contractible, so tensoring with any complex preserves quasi-isomorphisms. Every finite tensor power of $I \to I'$ is therefore a quasi-isomorphism.

Filter the mapping cone of $B(f)$ by word length. Its associated graded complexes are acyclic by this observation.

For a cone cycle of largest length p, solve for a primitive of its length-p component in the associated graded. Subtract its differential. The remaining cycle has smaller largest length. Repetition ends at length zero and expresses the original cycle as a boundary. This proves the algebraic assertion.

At fixed positive weight the same argument has at most $N + 1$ steps, and exactness of products proves the completed assertion. □

Suppose now that $H^i(A, Q) = 0$ for $i \neq 0$. The truncation $A_{\leq 0}$ is defined by

$$A^i_{\leq 0} = \begin{cases} A^i, & i < 0, \\ \ker(Q : A^0 \to A^1), & i = 0, \\ 0, & i > 0. \end{cases}$$

Its multiplication and augmentation are inherited from A. Products of degree-zero cycles are cycles, so it is an augmented differential graded subalgebra. Projection of its degree-zero cycles onto cohomology, and of its negative degrees onto zero, defines an algebra map $A_{\leq 0} \to H^0(A, Q)$.

Corollary 36.5 (Concentrated reduction)**.** *Under the concentration hypothesis there is a canonical zigzag of differential graded coalgebra quasi-isomorphisms*

$$B(A) \longleftarrow B(A_{\leq 0}) \longrightarrow B(H^0(A, Q)).$$

The positive-weight completed version holds with the preceding weight and tensor conventions.

Proof. The inclusion $A_{\leq 0} \to A$ is a quasi-isomorphism because the positive cohomology of A vanishes. The projection $A_{\leq 0} \to H^0(A, Q)$ is a quasi-isomorphism because its negative cohomology vanishes. Both preserve the augmentation. Apply theorem 36.4, weight by weight when the completion is present. □

The zigzag is canonical. A strict reverse map $B(H^0(A, Q)) \to B(A)$ needs an additional construction. Even a linear section of cycles need not preserve multiplication. Chapter 222 constructs a bar map with a nonzero quadratic correction and an example with nonzero d_2. Over a general coefficient ring, the tensor and Künneth arguments above require replacement hypotheses. In particular, they must hold for the intermediate truncation as well as for the original algebra.

36.4 The vertex-algebra application

For a finite-dimensional complex simple Lie algebra $\mathfrak{g}$, let $V^\kappa(\mathfrak{g})$ be its universal affine vertex algebra at level κ. The superscript distinguishes it from a simple quotient. For a principal nilpotent element, let $C^\kappa_{\mathrm{DS}}(\mathfrak{g})$ be the charged-ghost Drinfeld–Sokolov BRST complex defined in chapter 222. Its cohomology is concentrated in degree zero for every $\kappa \in \mathbb{C}$, by the universal-complex theorem of Kac–Wakimoto [64, Theorem 4.1]. Its degree-zero cohomology is the universal principal W-algebra. A noncritical level is needed for the Sugawara conformal vector, rather than for this concentration theorem.

This vertex-algebra theorem does not yet supply the augmented associative algebra in corollary 36.5. The normal product of vertex fields can fail associativity. In the principal $\mathfrak{sl}_2$ complex, the ordinary vacuum coefficient functional fails to commute with Q. Both failures are computed in chapter 222.

An application of the bar comparison therefore requires an associative realization of the coefficient operations with a compatible differential. It needs an augmentation, or specified relative boundary modules in place of that augmentation. It also requires the tensor product, completion and comparison with the full factorization operations. Neither word deconcatenation nor the empty word constructs those operations or identifies a physical state.

Chapter 37

Stokes and algebraic identities

The two bracketings of a product give two points of an interval. To turn that interval into an associativity homotopy, one needs an operation on its interior whose differential is the difference of the endpoint operations. The boundary of a pentagon then asks whether the chosen homotopy is compatible with a fourth input. Oriented boundaries determine these equations. Coefficient-valued operations must solve them.

We use cohomological grading over a field k. For cochain complexes E, F, the complex $\mathrm{Hom}^\bullet(E,F)$ consists of homogeneous linear maps, with differential

$$\delta f = d_F f - (-1)^{|f|} f d_E.$$

Tensor products use the Koszul rule $(f \otimes g)(a \otimes b) = (-1)^{|g||a|} f(a) \otimes g(b)$. In particular, if V is a cochain complex and $\mu : V \otimes V \to V$ is a degree-zero chain map, put

$$L_3 = \mu(\mu \otimes 1), \qquad R_3 = \mu(1 \otimes \mu).$$

An associativity homotopy is a degree--1 map $H : V^{\otimes 3} \to V$ satisfying $\delta H = L_3 - R_3$. Its existence is a condition on the product and the coefficient complex.

37.1 Cellular coefficients and the full square

Let K be a finite oriented regular CW complex. Write $\tau < \sigma$ when τ is a proper face of σ, and write $\epsilon_{\tau\sigma}$ for the cellular incidence number when $\dim \sigma - \dim \tau = 1$. Choose a cochain complex (E_σ, d_σ) for every cell. Give the orientation generator $[\sigma]$ degree $-\dim \sigma$ and set

$$C^m(K;E) = \bigoplus_\sigma [\sigma] \otimes E_\sigma^{m+\dim\sigma},$$

$$D_0([\sigma] \otimes e) = (-1)^{\dim\sigma} [\sigma] \otimes d_\sigma e.$$

Thus D_0 has degree one.

For each face pair of codimension r, choose a map

$$t_{\tau\sigma} : E_\sigma \longrightarrow E_\tau, \qquad |t_{\tau\sigma}| = 1 - r.$$

The chosen orientation coefficients are included in $t_{\tau\sigma}$. Define a degree-one endomorphism of the finite cellular direct sum by

$$D([\sigma] \otimes e) = D_0([\sigma] \otimes e) + \sum_{\tau<\sigma} [\tau] \otimes t_{\tau\sigma}(e).$$

Theorem 37.1 (Cellular curvature). *The component of D^2 from $[\sigma] \otimes E_\sigma$ to $[\tau] \otimes E_\tau$, for $\tau < \sigma$, is*

$$(-1)^{\dim \tau} \delta t_{\tau\sigma} + \sum_{\tau<\nu<\sigma} t_{\tau\nu} t_{\nu\sigma}. \tag{37.1}$$

Consequently D is a differential if and only if all these expressions vanish. For covers, the condition says that $t_{\tau\sigma}$ is a chain map.

Proof. The diagonal component is $d_\sigma^2 = 0$. For a face pair of codimension r, the two terms involving an internal differential are

$$\begin{aligned}(-1)^{\dim \tau} d_\tau t_{\tau\sigma} + (-1)^{\dim \sigma} t_{\tau\sigma} d_\sigma \\ = (-1)^{\dim \tau} \big(d_\tau t_{\tau\sigma} + (-1)^r t_{\tau\sigma} d_\sigma\big).\end{aligned}$$

Since $|t_{\tau\sigma}| = 1 - r$, the last parenthesis is $\delta t_{\tau\sigma}$. Every remaining term passes through exactly one intermediate cell ν, which gives the sum in (37.1). These are all components of the square. □

Suppose only cover maps occur, with $t_{\tau\sigma} = \epsilon_{\tau\sigma} F_{\tau\sigma}$ for degree-zero chain maps $F_{\tau\sigma}$. The theorem reduces to the codimension-two identities

$$\sum_{\tau<\nu<\sigma} \epsilon_{\tau\nu}\epsilon_{\nu\sigma} F_{\tau\nu} F_{\nu\sigma} = 0, \qquad \dim \sigma - \dim \tau = 2. \tag{37.2}$$

For constant coefficients and identity maps this is the cellular equation $\partial^2 = 0$. For general coefficients, opposite incidence signs cancel equal composites. Equality of the composites is a separate condition.

There is also a mixed term to retain when a face map is written as an identity plus a correction. For constant E, let d_K be the signed cellular boundary with identity coefficients, and write $D = D_0 + d_K + \rho$. Even when $D_0\rho + \rho D_0 = 0$, its square contains

$$d_K \rho + \rho d_K + \rho^2.$$

For example, orient the unit square Q by $dx \wedge dy$, and orient its bottom edge e_b by increasing x. With scalar coefficients in degree zero, set $\rho(Q) = e_b$ and let ρ vanish on every other cell. Writing v_{ab} for the vertex (a, b), one obtains

$$\rho^2 = 0, \qquad (d_K + \rho)^2(Q) = \partial e_b = v_{10} - v_{00} \neq 0.$$

The nonzero term is $d_K \rho(Q)$.

37.2 Choosing higher homotopies

Maps of codimension two have coefficient degree -1. They can therefore fill a failure of (37.2) when that failure is a boundary in the appropriate Hom complex. Subsequent codimensions impose further conditions on these choices.

Proposition 37.2 (Extension obstruction). *Fix the maps $t_{\tau\sigma}$ for all face pairs of codimension less than r, where $r \geq 2$, and suppose their curvature vanishes in those codimensions. For a face pair of codimension r, put*

$$O_{\tau\sigma} = \sum_{\tau<\nu<\sigma} t_{\tau\nu} t_{\nu\sigma}.$$

This is a closed element of $\mathrm{Hom}^{2-r}(E_\sigma, E_\tau)$. The chosen lower maps extend through codimension r if and only if every class $[O_{\tau\sigma}]$ vanishes in $H^{2-r}\,\mathrm{Hom}(E_\sigma, E_\tau)$. An extension is a choice of primitives satisfying

$$\delta t_{\tau\sigma} = (-1)^{\dim \tau + 1} O_{\tau\sigma}.$$

Proof. Set all as yet unchosen maps equal to zero and form the resulting D. Filter its matrix entries by the drop in cell dimension. The first possible nonzero drop in D^2 is r, and its component there is $O_{\tau\sigma}$. The identity $DD^2 - D^2D = 0$ in drop r involves only D_0 and O. Its (τ, σ) component is

$$(-1)^{\dim \tau}\big(d_\tau O_{\tau\sigma} - (-1)^r O_{\tau\sigma} d_\sigma\big) = 0.$$

This is $(-1)^{\dim \tau}\partial O_{\tau\sigma} = 0$, because $|O_{\tau\sigma}| = 2 - r$.

Adding maps of codimension r changes the curvature in that codimension only by $(-1)^{\dim \tau}\partial t_{\tau\sigma}$. Thus the displayed primitive equation is necessary and sufficient. The equations for different face pairs at this stage are independent. There are only finitely many stages for a finite-dimensional K. □

A cube shows why chosen face homotopies need not extend. Orient $Q^3 = [0,1]^3$ by $dx \wedge dy \wedge dz$, and orient each face by its increasing free coordinates. Take constant coefficients $E = k\alpha \oplus k\beta$, with $|\alpha| = 0$, $|\beta| = -1$ and $d_E = 0$. All cover maps are incidence numbers times the identity.

Let $F = \{z = 0\}$ and $v = (0, 0, 0)$. Among the codimension-two maps choose only

$$t_{vF} = h, \qquad h(\alpha) = \beta, \qquad h(\beta) = 0$$

to be nonzero. All equations through codimension two hold: the cover composites cancel, and $\partial h = 0$.

The incidence of F in Q^3 is -1, so

$$O_{v,Q^3} = t_{vF} t_{F,Q^3} = -h \neq 0.$$

The coefficient differential is zero and $\mathrm{Hom}^{-2}(E, E) = 0$. Hence this nonzero class has no primitive. The obstruction is to extending the specified choice of h. The identity coefficient system also admits the different choice in which every higher map is zero.

37.3 Integration constructs an operation homotopy

Now take $k = \mathbb{R}$ or $\mathbb{C}$ and a finite-dimensional cochain complex E. Suppose the closed cells of K are compact smooth oriented manifolds with corners, with compatible smooth face inclusions. A form on K means a family of forms on the closed cells whose restrictions agree on faces.

Let

$$\omega = \sum_{p \geq 0} \omega_p, \qquad \omega_p \in \Omega^p(K) \otimes E^{-p}, \qquad d_{\mathrm{tot}} = d_{\mathrm{form}} + (-1)^p d_E$$

on the summand of form degree p. Thus ω has total degree zero. For a cell σ of dimension r, define

$$\eta_r = (-1)^{r(r-1)/2}, \qquad I_\omega([\sigma]) = \eta_r \int_\sigma \omega_r \in E^{-r}. \tag{37.3}$$

The integral is taken coefficient by coefficient.

Proposition 37.3 (Signed Stokes map)**.** *If $d_{\mathrm{tot}}\omega = 0$, then I_ω is a chain map from cellular chains placed in degree minus cell dimension to E:*

$$d_E I_\omega([\sigma]) = I_\omega(\partial[\sigma]).$$

Proof. The component of $d_{\mathrm{tot}}\omega = 0$ in form degree r is $d_{\mathrm{form}}\omega_{r-1} + (-1)^r d_E \omega_r = 0$. For $r \geq 1$, Stokes' theorem gives

$$d_E I_\omega([\sigma]) = \eta_r (-1)^{r+1} \int_{\partial\sigma} \omega_{r-1} = \eta_{r-1} \int_{\partial\sigma} \omega_{r-1}.$$

The last equality uses $\eta_r(-1)^{r+1} = \eta_{r-1}$. The assertion at a vertex is the form-degree-zero equation $d_E\omega_0 = 0$. □

For operation coefficients, take $E = \mathrm{End}(V)(n) := \mathrm{Hom}(V^{\otimes n}, V)$, where V is a finite-dimensional cochain complex and E has differential δ. On an interval with coordinate t, a total-degree-zero form is

$$\omega = m_t + dt\, H_t, \qquad |m_t| = 0, \quad |H_t| = -1.$$

It is closed precisely when $\delta m_t = 0$ and $\delta H_t = \partial_t m_t$. Consequently

$$\delta\left(\int_0^1 H_t\, dt\right) = m_1 - m_0.$$

This formula constructs a homotopy from a closed coefficient-valued form.

The integration sign also respects products of cells. Let $\alpha \otimes f$ and $\beta \otimes g$ have total degree zero, with form degrees p, q and coefficient degrees $-p, -q$. For a composition of coefficient operations their product is

$$(\alpha \otimes f)(\beta \otimes g) = (-1)^{pq}(\alpha \wedge \beta) \otimes (f \circ g).$$

The factor $(-1)^{pq}$ moves f past β.

Since $\eta_{p+q}(-1)^{pq} = \eta_p \eta_q$, Fubini's theorem implies that (37.3) preserves this composition on product cells with product orientation. Suppose closed forms on a cellular operad restrict on every composition face to the prescribed products of forms. Their signed integrals then give an operation-valued chain map preserving substitution. Smooth extension to the closed cells and these face equations are hypotheses of this construction.

37.4 The associahedral equations

Write K_n for the associahedron of dimension $n-2$, for $n \geq 2$. Its cells are indexed by planar rooted trees with n inputs. The cell of a tree is a product of associahedra, one for each vertex. Grafting trees gives the nonsymmetric cellular operad. Its identification with the operad for associative algebras up to homotopy is proved in [76, §9.2, Proposition 9.2.8]. In that source the polytope is denoted K^{n-2}.

Place its cellular chains in negative degrees and change the orientation of every d-cell by $(-1)^d$ relative to that convention. This change respects product orientations and reverses each boundary coefficient. If θ_n is the oriented top cell, its boundary is

$$\partial\theta_n = -\sum_{r+s+t=n\,;\; s\geq 2\,;\; r+1+t\geq 2} (-1)^{r+st}\, \theta_{r+1+t} \circ_{r+1} \theta_s. \tag{37.4}$$

Here $\circ_i$ means substitution in the ith input. The unary operadic identity acts by 1_V. No nullary operation is assumed.

Proposition 37.4 (Operations on all associahedra)**.** *An operation-valued chain map from this cellular operad to* $\mathrm{End}(V)$ *preserving substitution is equivalent to maps*

$$m_n : V^{\otimes n} \longrightarrow V, \qquad |m_n| = 2 - n \quad (n \geq 2),$$

which, together with $m_1 = d_V$*, satisfy*

$$\sum_{r+s+t=n\,;\; s\geq 1} (-1)^{r+st}\, m_{r+1+t}(1^{\otimes r} \otimes m_s \otimes 1^{\otimes t}) = 0 \qquad (n \geq 1). \tag{37.5}$$

These are the cohomological A_∞ *equations. On homogeneous inputs the indicated tensor map has the additional Koszul sign* $(-1)^{(2-s)(|a_1|+\cdots+|a_r|)}$.

Proof. The graded cellular operad is freely generated by the corollas θ_n, $n \geq 2$: every other cell is the unique substitution specified by its planar tree. Hence a substitution-preserving map is determined by $m_n = I(\theta_n)$. It commutes with differentials if and only if $\delta m_n = I(\partial \theta_n)$ for each generator, since the cellular boundary is a derivation for substitution. Using (37.4) gives (37.5) after adding the terms involving m_1. Those terms are exactly $d_V m_n - (-1)^{2-n} m_n d_{V^{\otimes n}}$.

The arity-one equation is $d_V^2 = 0$. This also proves the converse on every tree cell. The resulting convention agrees with [66, §3.1]. □

37.5 A nonassociative product with a coherent extension

The interval operation and all subsequent equations can be made explicit. A strict A_∞ morphism will mean a chain map preserving each m_n. It is a quasi-isomorphism when it induces an isomorphism on cohomology. Over $\mathbb{Q}$, take

$$V^0 = \mathbb{Q}x \oplus \mathbb{Q}y, \qquad V^{-1} = \mathbb{Q}z, \qquad dz = y.$$

All other degrees and differentials vanish.

Define chain maps $\phi : V \to \mathbb{Q}$ and $i : \mathbb{Q} \to V$ by

$$\phi(x) = 1, \quad \phi(y) = \phi(z) = 0, \qquad i(1) = x - y,$$

and put $P = 1 - i\phi$. Let $h : V \to V$ have degree -1 with $h(x) = h(y) = z$ and $h(z) = 0$. Directly on this basis,

$$P^2 = P, \qquad dh + hd = P, \qquad h^2 = 0, \quad \phi h = 0, \quad hi = 0, \quad Ph = hP = h.$$

Set $R = 1 + P$ and define

$$\mu(a, b) = R(a)\phi(b), \qquad H(a, b, c) = 2h(a)\phi(b)\phi(c).$$

The product is a chain map, because R and ϕ are chain maps. Since $\phi R = \phi$ and $R^2 - R = 2P$, its associator is

$$L_3(a, b, c) - R_3(a, b, c) = 2P(a)\phi(b)\phi(c) = \delta H(a, b, c).$$

In particular, $\mu(x, x) = x + y$ and the associator at (x, x, x) is $2y$.

After extension to real coefficients, the form

$$R_3 + t(L_3 - R_3) + dt\, H$$

is closed and has signed integral H over the interval oriented from R_3 to L_3. The orientation in (37.4) is the opposite one, so the corresponding ternary operation is $m_3 = -H$.

Proposition 37.5. *The operations*

$$m_1 = d, \qquad m_2 = \mu, \qquad m_3 = -H, \qquad m_n = 0 \quad (n \geq 4)$$

satisfy (37.5) *in every arity. Moreover ϕ and i are strict A_∞ quasi-isomorphisms between this algebra and $\mathbb{Q}$, with $\phi i = 1$.*

Proof. The first three equations are $d^2 = 0$, the chain-map identity for μ, and $\delta m_3 + L_3 - R_3 = 0$, already proved. In arity four, the terms involving m_4 vanish. The term $\mu(1 \otimes m_3)$ vanishes because $\phi h = 0$. In the order

$$-\mu(m_3 \otimes 1), \quad m_3(\mu \otimes 1 \otimes 1),$$
$$-m_3(1 \otimes \mu \otimes 1), \quad m_3(1 \otimes 1 \otimes \mu),$$

the remaining terms are respectively $4, -4, 2, -2$ times $h(a)\phi(b)\phi(c)\phi(d)$. Here $Rh = hR = 2h$ and $\phi R = \phi$ give the coefficients. They sum to zero.

In arity five, each possible composite inserts m_3 into m_3. Insertion in the first input gives $h^2 = 0$. Insertion in either other input gives $\phi h = 0$. For $n > 5$, two nonzero operations cannot have combined arity n. This proves all equations, including their graded tensor signs.

For strictness, the identities are

$$\phi\mu(a, b) = \phi(a)\phi(b), \qquad \phi m_3 = 0,$$
$$\mu(i(1), i(1)) = i(1), \qquad m_3(i(1), i(1), i(1)) = 0.$$

All higher operations vanish, and the unary identities are the chain-map conditions for i and ϕ. Finally $dh + hd = 1 - i\phi$ and $\phi i = 1$ prove that both maps induce inverse isomorphisms on cohomology. □

The product μ has no strict two-sided unit. A right unit e would give $\phi(e) = 1$ on the input $x - y$ and $2\phi(e) = 1$ on the input y, which is impossible. The induced cohomology algebra is nevertheless $\mathbb{Q}$. The example gives an operation on every associahedral cell by the substitution construction of Proposition 37.4.

A fixed nonassociative product need not admit even the first homotopy. Take $W = \mathbb{Q}x \oplus \mathbb{Q}y$ in degree zero with zero differential, and set

$$\mu(x, x) = \mu(y, x) = y, \qquad \mu(x, y) = \mu(y, y) = 0.$$

Its associator at (x, x, x) is y. But $\mathrm{Hom}^{-1}(W^{\otimes 3}, W) = 0$, so no H can satisfy $\delta H = L_3 - R_3$. There is no A_∞ extension with this fixed binary operation.

For Fulton–MacPherson collision spaces or stable-graph parameter spaces, the corresponding construction requires an actual coefficient-valued map on the compactification. This map must respect every face restriction, composition and gluing. It also requires a tensor product and, where infinite sums occur, a completion in which the operations and integrals are defined. Singular forms require an extension or renormalization compatible with those equations before Stokes' theorem can be applied.

The associahedral construction proves what such data yield in its stated carrier. The chiral and modular coefficient maps remain to be constructed. Residue relations alone leave these operation homotopies and their higher compatibility equations to be supplied.

Chapter 38

Two edge colors and two loop operators

A loop is determined by its endpoints at the moment of contraction. Its color alone does not determine when it becomes a loop. Consider two genus-zero vertices joined by a transverse edge e and a chiral edge f. Give each vertex one labeled external leg, so that both vertices are stable.

Contracting e first leaves a chiral loop. Contracting f first leaves a transverse loop. Both orders produce the same genus-one corolla. This graph will determine which coefficients may be assigned to the two kinds of contraction.

38.1 Weights compatible with contraction

Let k be a field of characteristic zero and let R be a commutative k-algebra concentrated in degree zero. We use the stable graphs and coefficient maps of Proposition 20.3. Thus a graph G has a complex $C(G)$, and each edge e has an equivariant cochain map

$$c_e^G : C(G) \longrightarrow C(G/e).$$

The two composites for every pair of distinct edges agree after the specified identification of their targets. These equations include pairs of different colors and pairs containing loops.

Write r for the transverse color and b for the chiral color. Let $E_a(G)$ be the set of edges of color a. Choose scalars $u_r, u_b, v_r, v_b \in R$.

A contraction of color $a \in \{r, b\}$ has weight u_a when its endpoints are distinct, and weight v_a when it is a loop. The first condition allows the edge to lie on a cycle. Denote this weight by $w_a(e, G)$.

Put the edge orientation line $\mathrm{or}(G) = \det(kE(G))$ in degree $-|E(G)|$. Let d_C be the coefficient differential and let ι_e remove e from the oriented wedge with its exterior sign. For $x \in C^q(G)$ and $\omega \in \mathrm{or}(G)$, define

$$D_{u,v}(x \otimes \omega) = d_C x \otimes \omega + (-1)^q \sum_{e \in E(G)} w_{\mathrm{col}(e)}(e, G) c_e^G x \otimes \iota_e \omega. \tag{38.1}$$

The carrier is the direct sum over graph isomorphism classes, with automorphism coinvariants, as in that proposition. Each summand has finitely many outgoing terms.

Proposition 38.1 (The weighting equation)**.** *The identity $D_{u,v}^2 = 0$ holds for every such coefficient system on the full category of stable graphs if and only if*

$$u_r v_b = u_b v_r. \tag{38.2}$$

In particular, $v_a = \hbar u_a$ gives a differential over $k[u_r, u_b, \hbar]$. If $u_r = u_b = 1$, independent indeterminates v_r, v_b do not give a differential on this graph category.

Proof. The internal differential anticommutes with the edge terms because every c_e^G is a cochain map. For two distinct edges, the two contraction orders have opposite exterior signs and the same coefficient composite. Equal scalar products therefore cancel the pair.

Contraction can change another edge from a nonloop to a loop only when the two edges join the same pair of vertices. All other pairs retain their loop types, so their weights agree by commutativity of R. For parallel edges of the same color, both products are $u_a v_a$. For differently colored parallel edges, they are $u_r v_b$ and $u_b v_r$. Equation (38.2) consequently cancels every two-edge term.

For necessity, use the graph in the opening paragraph with $C(G) = k$ and all coefficient maps equal to the identity. Orient its edges by $e \wedge f$, with e transverse. The coefficient of the final corolla in $D^2_{u,v}$ is exactly

$$u_r v_b - u_b v_r.$$

The two labeled external legs and the different edge colors prevent an automorphism from reversing this orientation. Thus this coefficient must vanish. □

For a particular coefficient system, a nonzero scalar difference can annihilate the relevant composite. The proposition gives the condition valid for every coefficient system. It does not require either u_a to be invertible. For example, $u_r = u_b = 0$ permits arbitrary loop weights.

The common parameter $\hbar$ counts the total number of independent graph cycles. If a connected graph is contracted to a corolla, then exactly

$$b_1(G) = |E(G)| - |V(G)| + 1$$

of its contractions are loop contractions, in every ordering. The other contractions reduce the number of vertices by one. Consequently, when $v_a = \hbar u_a$, the scalar weight of the whole contraction is

$$u_r^{|E_r(G)|} u_b^{|E_b(G)|} \hbar^{b_1(G)}. \tag{38.3}$$

Both edge colors remain visible in this formula. The exponent of $\hbar$ is the ordinary graph cycle rank.

38.2 A domain with two invariant genera

There is also a contraction domain on which the two loop weights can remain independent. Let G_a be the spanning subgraph containing all vertices of G and just the edges of color a. Circuits include loops and pairs of parallel edges. Suppose every simple circuit of G is monochromatic. Equivalently,

$$H_1(G; k) = H_1(G_r; k) \oplus H_1(G_b; k),$$

where the summands have disjoint edge supports.

Indeed, simple circuits generate the cycle space. A circuit using both colors cannot split in this way, since neither of its proper color subgraphs contains a cycle.

Give each vertex two nonnegative genus labels $g_r(v)$, $g_b(v)$, whose sum is its ordinary genus. A nonloop contraction adds the two labels separately. A loop contraction of color a raises $g_a(v)$ by one. Define

$$\gamma_a(G) = \sum_{v \in V(G)} g_a(v) + b_1(G_a).$$

Proposition 38.2 (Monochromatic circuits)**.** *Graphs whose simple circuits are monochromatic are closed under contraction. On this category, each γ_a is invariant and the independently weighted operator $D_{u,v}$ squares to zero. For contraction of a connected graph to a corolla, the scalar weight is*

$$\prod_{a \in \{r,b\}} u_a^{|E_a(G)| - b_1(G_a)} v_a^{b_1(G_a)}.$$

Proof. A mixed circuit after contraction would lift to a mixed circuit before contraction, possibly by inserting the contracted edge. Thus the class is closed under contraction.

Contract a nonloop edge of color r. Its endpoints cannot be joined by a path consisting of blue edges, since that path together with the edge would be a mixed circuit. Identifying the endpoints therefore creates no blue cycle. The red cycle rank is unchanged by a red nonloop contraction. The same argument applies with the colors exchanged.

A loop contraction lowers the cycle rank of its color by one and raises the corresponding vertex genus by one. These observations prove invariance of both γ_a. They also show that a complete contraction has exactly $b_1(G_a)$ loop contractions of color a, which gives the displayed weight. There are no differently colored parallel edges in this class. Every two-edge square in the proof of Proposition 38.1 therefore cancels for arbitrary weights. □

This contraction category is not closed under arbitrary graph substitution. Start with a red loop on one vertex having two external legs. Replace that vertex by a blue edge joining two vertices, placing one flag of the red loop and one external leg at each vertex. Both graphs used in the substitution satisfy the circuit condition. The substituted graph is the mixed parallel-edge graph from the opening paragraph. An operadic sewing theory on the restricted domain must therefore specify which substitutions it admits.

38.3 The dual graph construction

The graph differential above contracts an existing edge. The Feynman transform uses the transpose operation: its differential expands a vertex by one edge. We describe the grading and the generator maps before formulating its master equation.

Use cohomological complexes throughout. For a finite edge set, let

$$\mathcal{K}(G) = \det(kE(G)) \quad \text{in degree } |E(G)|.$$

This is the degree-reversed version of the determinant cocycle in the homological convention of Getzler–Kapranov [43, §§4.5–5.2]. It is distinct from the negatively graded orientation line in (38.1).

A graph contraction partitions its edge set into quotient edges and edges of the contracted subgraphs. This partition identifies the determinant of all edges with the tensor product of the corresponding determinants. These identifications include their Koszul signs.

A cocycle $\mathcal{D}$ assigns graded lines $\mathcal{D}(G)$ with such coherent tensor identifications under contraction and substitution. Its value on a corolla is k. Put $\mathcal{D}^\vee = \mathcal{K} \otimes \mathcal{D}^{-1}$.

A modular $\mathcal{D}$-operad with two edge kinds is a stable collection $P(g,n)$, with symmetric-group actions and degree-zero cochain maps

$$\mu_G^P : \mathcal{D}(G) \otimes \bigotimes_{v \in V(G)} P(g(v), n(v)) \longrightarrow P(g(G), n(G)).$$

They are equivariant and compatible with every graph substitution. In particular, they obey all mixed contraction squares with the cocycle identifications. The corolla composition is the identity. The colors here distinguish edge operations on a common collection. They are not two independent genus labels.

Assume each $P(g,n)$ is a bounded complex of finite-dimensional vector spaces and vanishes unless $2g - 2 + n > 0$. Write $P(g,n)^\vee$ for its cochain dual. The graded Feynman transform is the free modular $\mathcal{D}^\vee$-operad generated by this dual collection:

$$(\mathsf{F}_{\mathcal{D}} P)(g,n) = \bigoplus_{[G]} \left(\mathcal{D}^\vee(G) \otimes \bigotimes_{v} P(g(v), n(v))^\vee \right)_{\mathrm{Aut}(G)}. \tag{38.4}$$

The sum is over stable graphs of type (g,n) with labeled external legs and either color on each edge. Graph substitution gives its free operadic composition.

For a one-edge graph H of type (g,n), transpose μ_H^P and tensor with its edge generator in $\mathcal{K}(H)$. This defines a degree-one expansion

$$\delta_H : P(g,n)^\vee \longrightarrow \mathcal{D}^\vee(H) \otimes \bigotimes_v P(g(v),n(v))^\vee.$$

In the free operad, apply this map at each vertex with the tensor-product differential sign. Weight a two-vertex expansion of color a by u_a and a one-vertex loop expansion by v_a. Together with the internal dual differential, these maps define a derivation $d_{\mathsf{F},u,v}$ of degree one.

If (38.2) holds, then

$$d_{\mathsf{F},u,v}^2 = 0.$$

To see this, expand a vertex by two edges. The two expansion orders are the transposes of the two contraction orders. Operadic compatibility identifies their coefficient maps, the determinant signs are opposite, and the scalar products agree by Proposition 38.1. The internal differential anticommutes with the expansion terms because the original compositions are cochain maps. This proves the assertion on generators and hence on the free operad. It is the two-kind, weighted version of the construction and square-zero argument in Getzler–Kapranov's Theorem 5.2.

38.4 Generator amplitudes and the master equation

Let Q be a modular $\mathcal{D}^\vee$-operad on the same graph category, whose components are also bounded complexes of finite-dimensional vector spaces. For example, if $\mathcal{D} = \mathcal{K}$, a bounded finite-dimensional complex V with a symmetric degree-zero perfect cochain pairing gives the untwisted target

$$Q(g,n) = V^{\otimes n}.$$

Its graph compositions contract the factors at the two flags of each edge with the pairing. Other pairing degrees require the corresponding complementary twist.

The generator carrier is

$$\mathcal{A} = \prod_{2g-2+n>0} \operatorname{Hom}_{S_n}\bigl(P(g,n)^\vee, Q(g,n)\bigr). \tag{38.5}$$

For homogeneous $\phi \in \mathcal{A}$, its internal differential is

$$(d\phi)_{g,n} = d_Q \phi_{g,n} - (-1)^{|\phi|} \phi_{g,n} d_{P^\vee}.$$

A degree-zero element Θ extends uniquely to a graded operad morphism

$$\widehat{\Theta} : \mathsf{F}_{\mathcal{D}} P \longrightarrow Q.$$

On a decorated graph this extension applies Θ at every vertex and then μ_G^Q. Thus the carrier in (38.5) records maps on the generators, rather than arbitrary linear maps on the whole free operad.

For a degree-zero generator map ϕ, define its loop term by

$$(\Delta_a \phi)_{g,n} = - \sum_{\substack{H \text{ one-vertex, one-edge} \\ \operatorname{col}(H)=a}} \mu_H^Q (\mathrm{id} \otimes \phi) \delta_H.$$

The internal-edge flags in this sum are unlabeled, and the external legs retain their labels. Equivalently, use the automorphism coinvariants in (38.4).

For the two-vertex graphs of color a, set

$$q_a(\phi)_{g,n} = -\sum_{\substack{H \text{ two-vertex, one-edge} \\ \mathrm{col}(H)=a}} \mu_H^Q(\mathrm{id}\otimes\phi\otimes\phi)\partial_H.$$

This sum uses unordered vertices. Let B_a be its polarization, so that $q_a(\phi) = \frac{1}{2}B_a(\phi,\phi)$.

For homogeneous inputs, introduce formal coefficients of opposite degrees to make the generator maps degree zero. Move those coefficients to the left by the Koszul rule in the preceding formulas. This defines the operations in every degree and gives

$$|\Delta_a| = |B_a| = 1, \qquad B_a(x,y) = (-1)^{|x||y|}B_a(y,x).$$

The factor one-half removes the double counting introduced by polarization.

The product in (38.5) has a concrete filtration. Give its (g,n) component the positive stability weight $w = 2g-2+n$ and let $F^p\mathcal{A}$ consist of components of weight at least p. There are finitely many stable pairs of each weight. A separating term adds its two input weights, and a loop term preserves the input weight. The one-edge sums at a fixed output have finitely many genus splittings and partitions of its external legs. Hence all the displayed operations are continuous on this completion.

Formal scalar parameters may be completed separately in their adic topology. Write $\mathcal{A}\widehat{\otimes}R$ for the product of the scalar-extended weight components. The stability condition excludes the unstable vacuum components from this carrier.

Theorem 38.3 (The two-kind master equation)**.** *Suppose* (38.2) *holds. Degree-zero amplitudes* $\Theta \in \mathcal{A}\widehat{\otimes}R$ *satisfying*

$$d\Theta + \frac{u_r}{2}B_r(\Theta,\Theta) + \frac{u_b}{2}B_b(\Theta,\Theta) + v_r\Delta_r\Theta + v_b\Delta_b\Theta = 0 \tag{38.6}$$

are in bijection with differential graded operad morphisms

$$(\mathsf{F}_{\mathcal{D}}P\otimes R, d_{\mathsf{F},u,v}) \longrightarrow Q\otimes R.$$

For adically complete R, interpret the target and morphism componentwise in that completion.

Proof. Freeness already gives the unique graded morphism $\widehat{\Theta}$. Both source and target differentials are operadic derivations. Their compatibility can therefore be checked on $P(g,n)^\vee$.

On a generator, the difference

$$d_Q\widehat{\Theta} - \widehat{\Theta}d_{\mathsf{F},u,v}$$

has one internal term, the two one-vertex loop terms, and the two two-vertex terms. Their definitions give exactly the left side of (38.6). Thus the equation is necessary and sufficient. The positive-weight filtration justifies the operations component by component. This is the generator-map argument of Barannikov, with the edge kinds and their weights retained [8, §§4–5]. □

The same construction specifies the Maurer–Cartan convention. On the graded space of generator maps, the difference of the two differentials defines an odd polynomial vector field $\boldsymbol{Q}$. Its linear term is $d + v_r\Delta_r + v_b\Delta_b$, and its quadratic term is $\frac{1}{2}(u_rB_r + u_bB_b)$. The identities $d_Q^2 = d_{\mathsf{F},u,v}^2 = 0$ imply $\boldsymbol{Q}^2 = 0$. Indeed, the difference is tangent to the space of graded operad maps, and applying it twice cancels

the two mixed compositions. Its linear, quadratic and cubic coefficients are respectively the differential, derivation and Jacobi identities.

To express these as an ordinary differential graded Lie algebra, put $L^i = \mathcal{A}^{i-1}$ and write $sx \in L^{|x|+1}$ for $x \in \mathcal{A}$. Set

$$d_L(sx) = s(d + v_r\Delta_r + v_b\Delta_b)x, \qquad [sx, sy]_a = (-1)^{|x|} sB_a(x, y).$$

Then $u_r[-,-]_r + u_b[-,-]_b$ is a degree-zero Lie bracket and d_L is its differential. For $\theta = s\Theta \in L^1$, equation (38.6) becomes

$$d_L\theta + \frac{1}{2}\big(u_r[\theta,\theta]_r + u_b[\theta,\theta]_b\big) = 0.$$

The amplitude has degree zero before this shift and degree one afterwards.

Taking all four weights to be one gives the ordinary two-kind Feynman transform. Taking $v_a = \hbar u_a$ gives a family with a common loop parameter.

One may instead mark a genus-g amplitude by $\hbar^g$. A loop composition raises the genus by one, while a separating composition adds the input genera. Thus the marking produces one factor of $\hbar$ in the loop term. It does not justify multiplying twice for the same genus increment.

38.5 The crossed derivation identity

There is an algebraic form of the parallel-edge obstruction. Use the operations on L just constructed, and denote the shifted loop operators again by Δ_a. For homogeneous $X, Y \in L$, define

$$T_{a,b}(X,Y) = \Delta_a[X,Y]_b - [\Delta_a X, Y]_b - (-1)^{|X|}[X, \Delta_a Y]_b.$$

The unweighted graph identities give

$$\Delta_a^2 = 0, \qquad \Delta_r\Delta_b + \Delta_b\Delta_r = 0,$$
$$T_{r,r} = T_{b,b} = 0, \qquad T_{r,b} + T_{b,r} = 0.$$

For example, regard $u_r, u_b, \hbar$ as independent indeterminates in the valid family $v_a = \hbar u_a$. Expand its identities and compare the coefficients of u_r^2, u_b^2 and $u_r u_b$. This gives the two individual derivation identities and their crossed sum. The separating brackets likewise satisfy the individual and mixed Jacobi identities.

Consequently, the failure of $v_r\Delta_r + v_b\Delta_b$ to be a derivation of $u_r[-,-]_r + u_b[-,-]_b$ is

$$(v_r u_b - v_b u_r)T_{r,b}. \tag{38.7}$$

With unweighted separating brackets and independent formal loop parameters, this vanishes identically precisely when $T_{r,b} = 0$. Then $T_{b,r} = 0$ as well. Thus independent loop parameters can occur on a particular amplitude carrier, but they require this additional compatibility. Anticommutation of the two loop operators supplies only the linear part of that requirement.

38.6 An explicit polynomial model

The crossed defect need not vanish even for constant-coefficient operators. Consider the graded commutative algebra

$$A = k[x,y] \otimes \Lambda(\xi,\eta,c), \qquad |x| = |y| = 0, \quad |\xi| = |\eta| = -1, \quad |c| = 1.$$

Give A the zero internal differential. Use left derivatives in the exterior variables and define

$$\Delta_r = \partial_x \partial_\xi, \qquad \Delta_b = \partial_y \partial_\eta.$$

Both operators have degree one, square to zero and annihilate 1. They anticommute because the two exterior derivatives anticommute.

For homogeneous $f, g \in A$, their second-order product defects define brackets

$$\{f, g\}_a = (-1)^{|f|}\big(\Delta_a(fg) - (\Delta_a f)g - (-1)^{|f|} f \Delta_a g\big). \tag{38.8}$$

These brackets have degree one and satisfy

$$\{f, g\}_a = -(-1)^{(|f|+1)(|g|+1)}\{g, f\}_a.$$

For example,

$$\{f, g\}_r = (\partial_x f)(\partial_\xi g) + (-1)^{|f|}(\partial_\xi f)(\partial_x g).$$

The blue formula replaces x, ξ by y, η.

These formulas prove the product derivation law

$$\{f, gh\}_a = \{f, g\}_a h + (-1)^{(|f|+1)|g|} g\{f, h\}_a.$$

Applying the product derivation law twice shows that the Jacobiator is a derivation in each of its three inputs. On the five algebra generators it vanishes, since every bracket of generators is constant. It therefore vanishes on A. This proof applies to every scalar linear combination of the two brackets. Thus they are compatible odd Poisson brackets.

Finally, expand $\Delta_a^2(fg) = 0$ using (38.8). The result is the derivation identity for Δ_a and its own bracket. Applying the same argument to $\Delta_r + \Delta_b$ gives the sum of the two crossed derivation identities.

For $f = xy\xi$ and $g = x\eta$, one obtains

$$\{f, g\}_b = x^2\xi, \qquad \Delta_r\{f, g\}_b = 2x, \qquad \{\Delta_r f, g\}_b = x, \qquad \Delta_r g = 0.$$

Hence the crossed derivation defect is x. The defect with the colors exchanged is $-x$. In the odd bracket convention the last term of a derivation identity has sign $(-1)^{|f|+1}$. It follows that $v_r\Delta_r + v_b\Delta_b$ fails to be a derivation of the unweighted bracket sum by $(v_r - v_b)x$ on these two inputs. This gives an actual coefficient realization of (38.7).

There are also nonzero amplitudes for which both colors contribute to a valid quantum master equation. Over $k[u_r, u_b, \hbar]$, put

$$\Theta = (xy)^2 + c(u_b x\xi - u_r y\eta).$$

It has degree zero.

The four color contributions are

$$\Delta_r\Theta = -u_b c, \qquad \Delta_b\Theta = u_r c,$$
$$\tfrac{1}{2}\{\Theta, \Theta\}_r = -2u_b(xy)^2 c, \qquad \tfrac{1}{2}\{\Theta, \Theta\}_b = 2u_r(xy)^2 c.$$

For the bracket calculation, the terms containing two factors of c vanish and the two cross terms agree. The displayed terms therefore prove

$$\frac{1}{2}\big(u_r\{\Theta, \Theta\}_r + u_b\{\Theta, \Theta\}_b\big) + \hbar(u_r\Delta_r + u_b\Delta_b)\Theta = 0. \tag{38.9}$$

Both colors are needed for these cancellations. Replacing the two loop coefficients by arbitrary v_r, v_b leaves the residual $(u_r v_b - u_b v_r)c$.

The factor one-half can also be seen without graph notation. Let Δ be either Laplacian or any scalar linear combination, and let $\{-,-\}$ be its bracket from (38.8). For an even element S, induction using the second-order product formula gives

$$\Delta(S^n) = nS^{n-1}\Delta S + \frac{n(n-1)}{2} S^{n-2}\{S,S\}.$$

With a degree-zero adic variable t and invertible $\hbar$, this implies the formal identity

$$\hbar^2 e^{-tS/\hbar}\Delta(e^{tS/\hbar}) = t\hbar\Delta S + \frac{t^2}{2}\{S,S\}.$$

Every coefficient is a finite algebraic expression. No analytic convergence is used.

The polynomial construction supplies a master equation on its specified graded algebra. A geometric realization for an ordered chiral algebra requires a common coefficient object and the relevant cyclic pairings. It also requires maps along both surface and curve clutchings. Those maps must preserve the tensor products, support conditions and completions, and must satisfy the mixed contraction identities or their coherent replacements.

If two geometric genera are retained, their behavior under all admitted sewings must also be defined. Theorem 38.3 applies once these data give its source and target operads and its generator amplitudes. It does not construct those data from an edge coloring or a scalar partition function.

Chapter 39

Typed anomalies

To cancel an anomaly one must specify the equation to be solved and the trivialization required. On $X = \mathbb{C}^\times$, consider the trivial holomorphic line with connection

$$\nabla = d + \frac{1}{2}\frac{dz}{z}.$$

Its curvature ∇^2 vanishes. A horizontal section satisfies $df + \frac{1}{2} f\, dz/z = 0$, so it is locally a multiple of $z^{-1/2}$. Continuation around a positively oriented circle changes its sign. Thus its holonomy, the parallel transport around that circle, is -1, and no nowhere-zero global horizontal section exists.

The same distinction between an equation and the data it retains occurs in deformation theory. A leading obstruction can be a cohomology class, while the curvature of an arbitrary partial construction need not belong to a complex at all. We first construct the obstruction class and then identify its different carriers. All algebraic sections use cohomological degrees over a field k of characteristic zero.

39.1 The differential at the classical solution

Let $(L, d, [-,-])$ be a differential graded Lie algebra, with bracket of degree zero. Its degree-one differential satisfies $d^2 = 0$ and is a graded derivation of the bracket. Fix a classical Maurer–Cartan element $a_0 \in L^1$, meaning

$$da_0 + \frac{1}{2}[a_0, a_0] = 0, \qquad \partial = d + [a_0, -].$$

The graded Jacobi identity gives $\partial^2 = 0$. This is the differential at the chosen classical solution.

For a degree-zero formal variable t, write $L[[t]] = \prod_{j \geq 0} t^j L$. The product denotes completion in powers of t. Each coefficient of a bracket is a finite sum. For $a \in tL^1[[t]]$, define its curvature and the corresponding operator by

$$\mathcal{R}(a) = \partial a + \frac{1}{2}[a, a], \qquad \partial_a = \partial + [a, -].$$

Direct expansion gives

$$\partial_a^2 = [\mathcal{R}(a), -], \qquad \partial_a \mathcal{R}(a) = 0. \tag{39.1}$$

Indeed, the square is the sum of $[\partial a, -]$ and $\frac{1}{2}[[a, a], -]$. For the second identity, $\partial \mathcal{R}(a) = -[a, \partial a]$, while $[a, [a, a]] = 0$ by graded Jacobi. The second equation is the Bianchi identity. The first shows why ∂_a need not define an ordinary cochain complex.

Proposition 39.1 (The leading obstruction). *Let $n \geq 2$ and let $a^{<n} = \sum_{j=1}^{n-1} t^j a_j$ satisfy $\mathcal{R}(a^{<n}) = 0$ modulo t^n. The coefficient*

$$\Omega_n = [t^n]\mathcal{R}(a^{<n}) \in L^2$$

is ∂-closed. An extension $a^{<n} + t^n b$ solves the Maurer–Cartan equation modulo t^{n+1} precisely when

$$\partial b = -\Omega_n.$$

Consequently the extension exists if and only if $[\Omega_n] = 0$ in $H^2(L, \partial)$.

Proof. In the Bianchi identity, the term $[a^{<n}, \mathcal{R}(a^{<n})]$ is divisible by t^{n+1}. Its coefficient of t^n therefore gives $\partial\Omega_n = 0$. Adding $t^n b$ changes the curvature, modulo t^{n+1}, by $t^n \partial b$. The bracket with $a^{<n}$ and the quadratic term in b have higher order. This proves both implications of the criterion. □

If $H^2(L, \partial) = 0$, every ∂-closed first coefficient a_1 extends recursively to a solution in $tL^1[[t]]$. At each step choose a primitive of $-\Omega_n$. The resulting coefficients define a formal series, so no additional convergence assertion is needed.

For a nontrivial extension, take $L^1 = kx \oplus ky$, $L^2 = kz$, with

$$dx = dz = 0, \qquad dy = -z, \qquad [x, x] = 2z,$$

and all other brackets zero. These formulas define a differential graded Lie algebra because z is central. At $a_0 = 0$, the first term tx has curvature $t^2 z$, and the exact solution is

$$a = tx + t^2 y.$$

Removing y instead gives a two-term graded Lie algebra with zero differential. For that algebra the same fixed first term tx cannot extend, since its second obstruction is the nonzero class of z.

Let $f : L \to M$ be a strict morphism of differential graded Lie algebras carrying a_0 to the chosen classical solution in M. It intertwines the classical differentials and satisfies

$$f\big(\mathcal{R}(a)\big) = \mathcal{R}\big(f(a)\big).$$

It therefore transports the leading obstruction classes. Both differential and bracket are used in this equation.

On $kx \oplus kz$, the identity map to the same complex with zero bracket is an isomorphism of complexes. It sends the source curvature $t^2 z$ to $t^2 z$, whereas the target curvature of tx is zero. An isomorphism of complexes alone does not compare these nonlinear equations.

39.2 Central extensions and nilpotency

For an ordinary Lie algebra $\mathfrak{g}$, use the Chevalley–Eilenberg complex

$$C^p(\mathfrak{g}; k) = \operatorname{Hom}_k(\Lambda^p \mathfrak{g}, k)$$

with trivial coefficient action. Our differential convention is

$$(\delta f)(x_0, \dots, x_p) = \sum_{i<j} (-1)^{i+j} f\big([x_i, x_j], x_0, \dots, \widehat{x_i}, \dots, \widehat{x_j}, \dots, x_p\big).$$

The remaining arguments retain their order. In δ^2, disjoint brackets cancel in pairs and the terms on three arguments cancel by Jacobi. Thus this is a cochain complex.

An alternating form $\omega \in C^2(\mathfrak{g}; k)$ defines a bracket on $\mathfrak{g} \oplus kK$ by

$$[(x, aK), (y, bK)] = ([x, y], \omega(x, y)K).$$

The element K is central. The central component of Jacobi vanishes exactly when

$$-\omega([x,y],z)+\omega([x,z],y)-\omega([y,z],x)=0,$$

which is $\delta\omega=0$.

Every central extension with fixed kernel kK has this form after choosing a linear section of its projection to $\mathfrak{g}$. Replacing the section σ by $\sigma+\beta K$ changes its form to

$$\omega'=\omega-\beta\circ[-,-]=\omega+\delta\beta.$$

Conversely, if E_ω denotes the displayed extension, this equation gives an isomorphism

$$E_{\omega'}\longrightarrow E_\omega,\qquad (x,aK)\longmapsto (x,(a+\beta(x))K).$$

It follows that $H^2(\mathfrak{g};k)$ classifies these extensions with identity maps on the quotient and kernel.

For instance, let $\mathfrak{g}=kh\oplus kx\oplus kz$ with $[h,x]=x$ and z central. The form with $\omega(h,z)=1$ and other basis values zero is closed and is not a coboundary, since every $\delta\beta$ vanishes on (h,z). The form with $\omega(x,z)=1$ instead has

$$(\delta\omega)(h,x,z)=-1$$

and does not define a central extension. A local central extension uses its specified local cochain complex. The choice of admissible cochains is part of that problem.

For BRST nilpotency, let (V,Q_0) be a cochain complex and set

$$L^j=\prod_i \operatorname{Hom}_k(V^i,V^{i+j}),\qquad [S,T]=ST-(-1)^{|S||T|}TS.$$

Composition is the multiplication, and the differential on L is $[Q_0,-]$. A formal degree-one operator $Q(t)=Q_0+a(t)$ satisfies

$$Q(t)^2=[Q_0,a(t)]+a(t)^2=[Q_0,a(t)]+\frac{1}{2}[a(t),a(t)].$$

Once the lower coefficients vanish, its leading failure of nilpotency defines the degree-two class of Proposition 39.1. If the classical operator is changed, the endomorphism differential must be changed with it.

Example 39.2. Let $V=ke_0\oplus ke_1\oplus ke_2$, with $|e_i|=i$, and let $Q_0=0$. Define $q(e_0)=e_1$, $q(e_1)=e_2$ and $q(e_2)=0$. The first term tq solves nilpotency modulo t^2, but

$$[t^2]Q(t)^2=q^2,\qquad q^2(e_0)=e_2.$$

The endomorphism differential is zero, so this nonzero obstruction cannot be removed by any coefficient of order t^2.

39.3 A master anomaly invisible to operator nilpotency

Let $\mathcal{A}$ be a unital graded-commutative BV algebra. Thus its degree-one operator Δ satisfies $\Delta 1=0$, $\Delta^2=0$, and is of order at most two. Use the derived bracket

$$\{f,g\}=(-1)^{|f|}\big(\Delta(fg)-(\Delta f)g-(-1)^{|f|}f\Delta g\big).\tag{39.2}$$

This bracket has degree one, is odd Poisson, and Δ is a graded derivation of it. These are the BV identities used below. Let Q be a degree-one derivation of the product and bracket, with $Q^2 = 0$ and $Q\Delta + \Delta Q = 0$.

Fix $S_0 \in A^0$ satisfying $QS_0 + \frac{1}{2}\{S_0, S_0\} = 0$. For a degree-zero formal variable $\hbar$ and $S \in S_0 + \hbar A^0[[\hbar]]$, put

$$C(S) = QS + \frac{1}{2}\{S, S\} + \hbar\Delta S, \qquad D_S = Q + \{S, -\} + \hbar\Delta.$$

The master equation is $C(S) = 0$, with curvature of degree one.

The BV identities give

$$D_S^2 = \{C(S), -\}, \qquad D_S C(S) = 0. \tag{39.3}$$

To verify the first formula, the mixed anticommutators with $\{S, -\}$ are $\{QS, -\}$ and $\hbar\{\Delta S, -\}$, while its square is $\frac{1}{2}\{\{S, S\}, -\}$. The second formula follows by expanding the curvature and using the same derivation identities and odd Jacobi. In particular, $D_S^2 = 0$ detects the master curvature only through its Hamiltonian action.

Proposition 39.3 (The quantum obstruction)**.** *Write $\delta_0 = Q + \{S_0, -\}$, so $\delta_0^2 = 0$. Suppose $S^{<n} = S_0 + \sum_{i=1}^{n-1} \hbar^i S_i$ solves the master equation modulo $\hbar^n$, for $n \geq 1$. Then*

$$\Omega_n = \Delta S_{n-1} + \frac{1}{2} \sum_{\substack{i+j=n \\ i,j\geq 1}} \{S_i, S_j\} \in A^1$$

is δ_0-closed. An extension $S^{<n} + \hbar^n b$ solves the equation modulo $\hbar^{n+1}$ if and only if $\delta_0 b = -\Omega_n$. Its obstruction therefore lies in $H^1(A, \delta_0)$.

Proof. The displayed expression is the coefficient of $\hbar^n$ in $C(S^{<n})$. Taking that coefficient in $D_{S^{<n}} C(S^{<n}) = 0$ gives $\delta_0 \Omega_n = 0$, since all lower curvature coefficients vanish. Adding $\hbar^n b$ changes this coefficient by $Qb + \{S_0, b\}$. Its Laplacian contributes only at order $n + 1$. This proves the extension criterion. □

The kernel of the Hamiltonian action can contain a nonzero obstruction class. Take

$$A = k[x] \otimes \Lambda(\xi, c), \qquad |x| = 0, \quad |\xi| = -1, \quad |c| = 1, \qquad Q = 0, \quad \Delta = \partial_x \partial_\xi,$$

where ∂_ξ is the left exterior derivative and monomials place ξ before c. The variable c has zero bracket with every element. For $S_0 = x\xi c$, direct calculation gives

$$\{S_0, S_0\} = 0, \qquad \Delta S_0 = c, \qquad \delta_0 x = xc, \quad \delta_0 \xi = \xi c, \quad \delta_0 c = 0.$$

Every degree-zero element is $f(x) + g(x)\xi c$, and

$$\delta_0\big(f(x) + g(x)\xi c\big) = xf'(x)c.$$

All degree-one elements $k[x]c$ are closed. Since $xk[x]c$ is the image from degree zero, there is an exact identification

$$H^1(A, \delta_0) = k \cdot [c], \qquad [c] \neq 0. \tag{39.4}$$

Thus no polynomial counterterm extends this fixed S_0 even to first quantum order. Nevertheless $C(S_0) = \hbar c$ and $D_{S_0}^2 = 0$ on all of $A[[\hbar]]$.

Changing the counterterm algebra can change the answer. In

$$\widetilde{A} = k[x, x^{-1}, u] \otimes \Lambda(\xi, c), \qquad |u| = 0, \quad \partial_x u = x^{-1},$$

extend ∂_x as a derivation and retain $\Delta = \partial_x \partial_\xi$. The symbol u is a formal logarithmic coordinate. Now $\{S_0, u\} = c$, $\Delta u = 0$ and $\{u, u\} = 0$, so

$$S = S_0 - \hbar u, \qquad C(S) = \hbar c - \hbar c = 0.$$

This is an exact solution in the enlarged algebra. The polynomial obstruction remains valid on its original carrier. For a local field theory, the admissible local functionals and renormalized operations must be specified before either obstruction calculation is applied.

39.4 Curvature as algebraic structure

A curved differential graded associative algebra consists of a graded unital algebra B, a degree-one derivation d_B, and an element $\kappa \in B^2$ such that

$$d_B^2(b) = [\kappa, b], \qquad d_B \kappa = 0.$$

The commutator is graded. The zero-input operation is the degree-two map $m_0 : k \to B$ with $m_0(1) = \kappa$. If $[\kappa, -] \neq 0$, the pair (B, d_B) is not a cochain complex, so κ is not a class in its ordinary cohomology.

For the graded vector space and operator q of Example 39.2, take

$$B = \mathrm{End}_k(V), \qquad d_B = [q, -], \qquad \kappa = q^2.$$

Associativity gives $d_B^2 = [q^2, -]$ and $d_B(q^2) = 0$. If p_0 is the projection onto ke_0, then

$$d_B^2(p_0)(e_0) = [q^2, p_0](e_0) = e_2.$$

The failure to be a complex is already visible on this one operator.

There is a precise way to change curved data. For $b \in B^1$, define

$$d_{B,b} = d_B + [b, -], \qquad \kappa_b = \kappa + d_B b + b^2. \tag{39.5}$$

Then $d_{B,b}^2 = [\kappa_b, -]$ and $d_{B,b}\kappa_b = 0$.

The square follows by expanding the two derivations. For the second identity, $d_B^2 b = [\kappa, b]$ cancels $[b, \kappa]$, $d_B(b^2) = (d_B b)b - b(d_B b)$ cancels $[b, d_B b]$, and $[b, b^2] = 0$. In the matrix example, $b = -q$ makes both the new differential and curvature zero. The existence of such a b is additional data in a general curved algebra.

If κ is central, $d_B^2 = 0$ and a class $[\kappa] \in H^2(B, d_B)$ is defined. Its meaning still depends on the construction. For example, the curvature in (24.1) records the scalar term of a Weyl relation in its quadratic dual. Identifying it with a quantum master obstruction requires a map between the stated carriers that preserves the differential, curvature and relevant operations.

39.5 The Atiyah extension, curvature and holonomy

Let L be a holomorphic line bundle on a complex manifold X. The sheaf $\mathrm{At}(L)$ consists of first-order $\mathbb{C}$-linear operators on its sections whose symbol is a vector field: an operator with symbol v obeys

$$P(fs) = fP(s) + v(f)s.$$

Its symbol sequence is

$$0 \longrightarrow \mathcal{O}_X \longrightarrow \mathrm{At}(L) \longrightarrow T_X \longrightarrow 0. \tag{39.6}$$

A holomorphic connection is an $\mathcal{O}_X$-linear splitting of this sequence. It preserves Lie brackets precisely when its curvature vanishes. The kernel consists of multiplication operators, with $[P, f] = v(f)$ for an operator of symbol v. These are the connection and extension constructions of Atiyah (1958), Sections 1–3 [4].

Choose local frames with $e_j = g_{ij}e_i$. Writing $\nabla e_i = A_i e_i$, the Leibniz rule gives

$$A_j = A_i + d\log g_{ij}, \qquad d\log g_{ij} = g_{ij}^{-1}\,dg_{ij}. \tag{39.7}$$

The forms $d\log g_{ij}$ are a Čech cocycle in Ω^1_X. Put $\alpha(L) = [\{d\log g_{ij}\}] \in H^1(X, \Omega^1_X)$. For the connection σ_i with zero one-form in the frame e_i, we have $\sigma_j - \sigma_i = -d\log g_{ij}$. Thus the extension class of (39.6), represented by those differences, is $-\alpha(L)$.

Proposition 39.4 (The holomorphic connection obstruction)**.** *The class $\alpha(L)$ vanishes if and only if L admits a holomorphic connection. When a connection exists, its curvature is a closed holomorphic two-form F_∇. Changing the connection by a global holomorphic one-form η changes the curvature by $d\eta$.*

Proof. Changing frames to $e'_i = h_i e_i$ replaces $d\log g_{ij}$ by

$$d\log g_{ij} + d\log h_j - d\log h_i,$$

so its class is unchanged. The class is zero precisely when, after refinement if necessary, there are holomorphic forms A_i satisfying (39.7). The corresponding local connections then glue, and every global connection supplies such forms. Their curvatures dA_i agree on overlaps because $d(d\log g_{ij}) = 0$. Thus $F_\nabla = dA_i$ is global and closed. Two connections differ by a global one-form, which proves the last assertion. □

On $\mathbb{P}^1$, take coordinates z and $w = z^{-1}$ and the line with transition $e_\infty = z^m e_0$. This is our frame convention for $\mathcal{O}(m)$. Its transition one-form is $m\,dz/z$.

A holomorphic form on the z-chart has only nonnegative powers of z in its Laurent expansion. A holomorphic form $h(w)\,dw$ on the other chart becomes $-h(z^{-1})z^{-2}dz$, with powers at most -2. Their difference has no $z^{-1}dz$ coefficient. Therefore (39.7) has a holomorphic solution exactly when $m = 0$. For $m = 0$, take both connection forms to be zero.

For a chosen connection, its curvature form, the de Rham class of that form, and the integral first Chern class $c_1(L) \in H^2(X, \mathbb{Z})$ are different data. For local logarithms ℓ_{ij} of the transition functions, the latter is represented by the integral cocycle

$$n_{ijk} = \frac{\ell_{ij} + \ell_{jk} - \ell_{ik}}{2\pi i}.$$

A flat connection additionally retains its holonomy. For $\lambda \in \mathbb{C}$, the trivial line on $\mathbb{C}^\times$ with $\nabla = d + \lambda\,dz/z$ has parallel transport around the positive generator of $\pi_1(\mathbb{C}^\times, 1)$ equal to

$$\exp(-2\pi i\lambda).$$

This follows from the local horizontal solution $z^{-\lambda}$. Both $\alpha(L)$ and $c_1(L)$ vanish because the underlying holomorphic line is trivial. Its curvature is zero for every λ, whereas a horizontal trivialization exists exactly when $\lambda \in \mathbb{Z}$.

The determinant gives an actual comparison of curvatures with a fixed coefficient. Let E be an ordinary rank-r complex vector bundle with connection $d + A$ in a local frame. The induced connection on $\det E$ has one-form $\operatorname{tr} A$, so

$$F_{\det E} = d \operatorname{tr} A = \operatorname{tr}(dA + A \wedge A) = \operatorname{tr} F_E. \tag{39.8}$$

Here $\operatorname{tr}(A \wedge A) = 0$, since its off-diagonal terms cancel in pairs and each diagonal one-form squares to zero. In particular, if $F_E = \omega \operatorname{id}_E$, then $F_{\det E} = r\omega$. The scalar coefficient is supplied by this determinant construction. Another family or normalization requires its own comparison.

39.6 Comparison maps

The preceding constructions locate their obstruction data as follows.

datum	carrier and condition
central extension	$H^2(\mathfrak{g}; k)$ in the declared cochain category.
leading BRST obstruction	$H^2(\operatorname{End}(V), [Q_0, -])$ after the lower nilpotency equations.
leading BV obstruction	$H^1(A, Q + \{S_0, -\})$ after fixing the classical solution.
holomorphic connection obstruction	$\alpha(L) \in H^1(X, \Omega^1_X)$.
connection curvature	$F_\nabla \in \Omega^2_X$, determined by a connection.
curved associative datum	$\kappa \in B^2$ with $d_B^2 = [\kappa, -]$, without an automatic cohomology class.

To compare cohomology classes of different degrees, specify a degree-r map $f : C^\bullet \to D^{\bullet+r}$ satisfying $d_D f = (-1)^r f d_C$ and prove the equality of the transported classes. To compare nonlinear deformation problems, also preserve the brackets and curvature equations, or supply the higher comparison maps and their identities. Numerical evaluations retain only the information detected by the specified cycle, trace, or background.

For a specified renormalized mixed bulk–boundary theory, start with its local BV deformation cochains. A chiral comparison must map these to the vacuum-normalized supported insertion complex, with a degree shift and an anomaly identity. The source, target differential, admissible counterterms, support, completion and compatibility with factorization and gluing must be constructed. The finite algebraic comparisons above do not supply that map.

Book VII

Calabi–Yau and automorphic interfaces

Chapter 40

Calabi–Yau categories

The trace pairing on matrices is a first example of the geometry in this chapter. On $\mathfrak{gl}_n(k)$, the form $(a, b) \mapsto \operatorname{tr}(ab)$ is symmetric and nondegenerate: the matrix units E_{ij} and E_{ji} are dual. The classifying stack BGL_n parametrizes rank-n vector bundles, and its tangent complex at a bundle is its endomorphism complex shifted by $[1]$. The shift changes the symmetric trace pairing into an alternating pairing with values in $k[2]$.

A Calabi–Yau structure extends this construction to categories whose endomorphism complexes have duality in dimension d. The resulting moduli form has degree $2 - d$. Its closedness requires the cyclic part of that structure, in addition to the nondegenerate pairing.

40.1 The two Calabi–Yau variances

Let k be a field of characteristic zero. Use cohomological grading, so $V[r]^i = V^{i+r}$ and the shift map $s : V \to V[1]$ has degree -1. Let C be a small stable idempotent-complete k-linear category, and put $\widehat{C} = \operatorname{Ind}(C)$. Thus C is the subcategory of compact objects of $\widehat{C}$. Call C *smooth* when its diagonal bimodule is perfect, and *proper* when every morphism complex between its objects is perfect over k. These conditions concern different duals.

Write I for the identity endofunctor of $\widehat{C}$. For smooth C, the left adjoint of categorical evaluation defines the inverse dualizing endofunctor $I^!$. For proper C, the right adjoint of evaluation defines the Serre endofunctor S. The trace identifications are

$$T = CC_*(C) \simeq \underline{\mathrm{RHom}}(I^!, I), \qquad T^\vee \simeq \underline{\mathrm{RHom}}(I, S),$$

under the respective hypotheses, with $T^\vee = \mathrm{RHom}_k(T, k)$. Here $CC_*(C)$ is the Hochschild chain complex, with homological degree j placed in cohomological degree $-j$. The Hom objects on the right are taken in the category of colimit-preserving k-linear endofunctors.

The canonical circle action on T is represented by its mixed-complex structure. Its operators b and B have cohomological degrees 1 and -1, with

$$b^2 = B^2 = bB + Bb = 0.$$

The negative cyclic complex is the completed product

$$CC_*^-(C) = (T[[u]], b + uB), \qquad |u| = 2. \tag{40.1}$$

It models the homotopy fixed points T^{hS^1}. A class $\theta : k[d] \to CC_*^-(C)$ has a representative $\theta_0 + u\theta_1 + u^2\theta_2 + \cdots$, where $|\theta_j| = -d - 2j$ and

$$b\theta_0 = 0, \qquad b\theta_{j+1} + B\theta_j = 0 \quad (j \geq 0). \tag{40.2}$$

The product in (40.1) retains all these components.

A *smooth Calabi–Yau structure of dimension d* is such a class whose underlying Hochschild class induces an equivalence

$$\alpha_\theta : I^!\,[d] \xrightarrow{\sim} I. \tag{40.3}$$

For proper C, a *proper Calabi–Yau structure of dimension d* is a map

$$\tau : T_{hS^1} \longrightarrow k[-d]$$

whose induced map $I \to S[-d]$ is an equivalence. The subscript hS^1 denotes homotopy orbits. The first structure is a class in a chain complex. The second is a cyclic trace functional. Both include coherent circle data. Neither is specified merely by choosing a nondegenerate bilinear form on a space of morphisms.

Put $R = CC^*(C) = \underline{\mathrm{RHom}}(I, I)$. Composition of endofunctors gives R its Hochschild E_2-structure. Under (40.3), precomposition gives the cap equivalence

$$D_\theta : R \xrightarrow{\sim} T[-d].$$

The proper structure instead gives $D_\tau : R \xrightarrow{\sim} T^\vee[-d]$ by postcomposition.

These are the orientation and coorientation of Theorem 19.3. The circle-equivariant Hochschild calculus and either cap equivalence give a framed E_2-structure on R, retaining its original E_2-operations [17, Theorems 4.1.7 and 4.3.4]. In this statement, *framed* means an action of $E_2 \rtimes SO(2)$, including the compatibility of rotation with every disk operation. The dimension d changes the cap shift while the operad remains E_2.

40.2 The moduli functor and its tangent complex

For a connective commutative differential graded k-algebra A, let $\mathrm{Perf}(A)$ denote the category of perfect A-complexes. Fix the covariant convention

$$\mathcal{M}_C(A) = \mathrm{Fun}_k^{\mathrm{ex}}(C, \mathrm{Perf}(A))^{\simeq}. \tag{40.4}$$

The superscript denotes the space of exact functors and their equivalences. These are pseudo-perfect functors: every object of C has perfect image over A. Using contravariant modules would replace the source by C^{op} and require the corresponding opposite Calabi–Yau class.

The functors in (40.4) assemble to a universal evaluation

$$F_C : C \longrightarrow \mathrm{Perf}(\mathcal{M}_C).$$

If C is smooth, an A-point F is corepresentable after scalar extension. Writing $C_A = \widehat{C} \otimes D(A)$, there is a compact object $E_A \in C_A$ with

$$F_A(-) \simeq \underline{\mathrm{RHom}}_{C_A}(E_A, -).$$

Here F_A is the A-linear extension of F, and its values on compact objects are perfect over A. This is the relative corepresentability theorem [16, Corollary 2.6 and Example 3.7]. The requirement on those values is the properness condition on the individual moduli object. The whole category need not be proper.

The tangent complex is

$$T_F\mathcal{M}_C \simeq \underline{\mathrm{RHom}}_{C_A}(E_A, E_A)[1]. \tag{40.5}$$

To see the shift, consider the loop space of $\mathcal{M}_C$ at F. It parametrizes automorphisms of F, or equivalently automorphisms of its corepresenter, using inversion for the opposite Yoneda variance.

Over a square-zero extension $A \oplus J$, an automorphism reducing to the identity has the form $1 + f$, with

$$f : E_A \longrightarrow E_A \otimes_A J.$$

Compactness and the properness condition identify this deformation functor with the one represented by the endomorphism complex of E_A. The tangent of the loop space is therefore that complex. Delooping gives (40.5), compatibly with scalar extension [16, Proposition 3.3].

For the symplectic assertion below, assume $\mathcal{M}_C$ is locally geometric and locally of finite presentation, with perfect cotangent complex $L_{\mathcal{M}_C}$. Thus it is covered by open derived Artin substacks on which the usual descent theory of differential forms applies. A sufficient hypothesis is that C be of finite type, meaning compact in the Morita category of small stable idempotent-complete k-linear categories [16, Remark 3.9]. More generally, the construction applies whenever these geometric and perfectness hypotheses hold.

40.3 From a cyclic class to a closed form

Closedness of a derived form includes homotopies. For an affine $U = \operatorname{Spec}(A)$, the derived de Rham complex is the graded mixed complex

$$\mathrm{DR}(A/k)(p) = \bigwedge_A^p L_{A/k}[p].$$

Its internal differential has degree 1. The de Rham differential raises the weight p by one and has degree -1 in this shifted complex.

For any such graded mixed complex V, put

$$\mathrm{NC}^w(V)(p) = \left(\prod_{j \geq 0} u^j V(p+j),\ d + u\epsilon \right), \qquad |u| = 2,$$

where ϵ is its mixed differential.

The space of closed p-forms of degree n is

$$\mathcal{A}^{p,\mathrm{cl}}(U, n) = \operatorname{Map}_{D(k)}\big(k, \mathrm{NC}^w(\mathrm{DR}(A/k))(p)[n-p]\big).$$

Projection to $j = 0$ gives the underlying p-form, a map $k \to \wedge_A^p L_{A/k}[n]$.

For a derived stack, take the limit of these spaces over the opposite of the category of its affine charts. Every limit over charts below uses this variance. These definitions retain the higher components that specify closedness [88, Definitions 1.7–1.12].

If Y has perfect cotangent complex, its tangent complex is $T_Y = L_Y^\vee$. A closed two-form ω of degree n is *n-shifted symplectic* when the adjoint of its underlying pairing is an equivalence

$$\omega^\flat : T_Y \xrightarrow{\sim} L_Y[n]. \tag{40.6}$$

Closedness and (40.6) are separate conditions.

Apply Hochschild chains to the universal functor F_C, restrict to each affine chart of $\mathcal{M}_C$, and use the circle-equivariant derived Hochschild–Kostant–Rosenberg map. Taking homotopy fixed points and projecting to weight two gives a map of complexes

$$\begin{aligned} \kappa_2 : CC_*^-(C) &\longrightarrow \lim_{U \to \mathcal{M}_C} CC_*^-(\mathrm{Perf}(U)) \\ &\longrightarrow \lim_{U \to \mathcal{M}_C} \mathrm{NC}^w(\mathrm{DR}(U/k))(2). \end{aligned} \tag{40.7}$$

All maps in the limit are pullbacks along affine charts. The circle-equivariance and the weight projection make this a map of the full negative cyclic complexes, rather than only a map on Hochschild homology [16, Theorem 4.6 and Proposition 5.2].

We use the trace normalization of the last arrow: the induced pairing on shifted endomorphisms is composition followed by trace, with the suspension sign below. Let $\mathrm{At}_E : E \to E \otimes L[1]$ be the Atiyah class. With the Chern-character convention $\mathrm{Ch}_2(E) = \frac{1}{2}\,\mathrm{Tr}(\mathrm{At}_E^2)$, this normalization uses twice the weight-two projection. The factor $\frac{1}{2}$ in the Chern-character convention is explicit in [88, proof of Theorem 2.12]. Scaling the whole closed-form map by 2 scales its homotopies as well as its underlying pairing.

Theorem 40.1 (Cyclic and shifted outputs). *Let C be smooth over the characteristic-zero field k, with a smooth Calabi–Yau structure θ of dimension d. Its Hochschild cochains carry a framed E_2-algebra structure refining their usual E_2-structure. If $\mathcal{M}_C$ satisfies the geometric and perfectness hypotheses above, the composite*

$$k[d] \xrightarrow{\theta} CC_*^-(C) \xrightarrow{\kappa_2} \lim_{U \to \mathcal{M}_C} \mathrm{NC}^w(\mathrm{DR}(U/k))(2)$$

defines a $(2-d)$-shifted symplectic form. At a field-valued point with corepresenter E, its pairing on homogeneous shifted endomorphisms is

$$\omega_\theta(sa, sb) = (-1)^{|a|}\,\mathrm{tr}_\theta(ab), \qquad \mathrm{tr}_\theta : \mathrm{RHom}(E,E) \to k[-d]. \tag{40.8}$$

Proof. The cap equivalence D_θ is induced by (40.3). The circle-equivariant Hochschild calculus makes $T[-d]$ a module for the circle factorization homology of R. The vector θ is circle-invariant, and its cap map identifies this module with R on the distinguished interval. The framed-algebra criterion used in Theorem 19.3 supplies the compatible disk and rotation operations. This proves the first assertion with its existing E_2-multiplication intact.

For the second, the composite in the statement is a cocycle of degree $-d$ in the weight-two negative cyclic complex. By the definition of $\mathcal{A}^{2,\mathrm{cl}}$, it is a closed two-form of degree $2-d$. Thus closedness is constructed before nondegeneracy is tested.

On a chart $\mathrm{Spec}(A) \to \mathcal{M}_C$, put $K_A = \underline{\mathrm{RHom}}_{C_A}(E_A, E_A)$. Relative Serre duality, applied to α_θ, gives an equivalence

$$K_A[d] \xrightarrow{\sim} K_A^\vee.$$

Its adjoint is the pairing

$$K_A \otimes_A K_A \xrightarrow{\circ} K_A \xrightarrow{\mathrm{tr}_\theta} A[-d].$$

The map on Hochschild chains induced by a corepresentable functor is computed by its unit, this duality map and evaluation. The weight-two Hochschild–Kostant–Rosenberg component symmetrizes the two endomorphism inputs. Cyclicity identifies the two terms, giving the displayed trace pairing with the chosen normalization. This is the functorial trace calculation of [16, Corollary 2.5 and Propositions 4.5 and 5.3]. It identifies the pairing with the underlying form of (40.7) on every affine chart, not only at individual points.

Shifting both inputs by [1] gives (40.8). Indeed, the interchange of a homogeneous endomorphism with the second suspension contributes $(-1)^{|a|}$. Cyclicity then yields

$$\omega_\theta(sa, sb) = -(-1)^{(|a|-1)(|b|-1)}\omega_\theta(sb, sa).$$

The shift differential is $d(sa) = -s(da)$. On a strict fiber model, the differential compatibility reduces to

$$(-1)^{|a|}\operatorname{tr}_\theta((da)b) + \operatorname{tr}_\theta(a\,db) = 0,$$

which is the closed-trace identity applied to $d(ab)$. The same suspension construction is a map of complexes over every chart.

Finally, $T_{\mathcal{M}_C}|_A \simeq K_A[1]$ and $L_{\mathcal{M}_C}|_A \simeq K_A^\vee[-1]$. The duality $K_A[d] \simeq K_A^\vee$ therefore identifies the adjoint of the form with

$$K_A[1] \xrightarrow{\sim} K_A^\vee[1-d] = L_{\mathcal{M}_C}|_A[2-d].$$

These equivalences are compatible with pullback. Descent proves (40.6) globally, completing the symplectic construction [16, Theorem 5.5(1)]. □

For $C = \mathrm{Perf}(k)$ and its unit orientation, the theorem recovers the degree-two form on the stack of perfect complexes. On BGL_n it is the trace pairing from the opening example.

For a graded complex V, the same calculation uses the supertrace $\operatorname{str}(a) = \sum_i (-1)^i \operatorname{tr}(a|_{V^i})$ and differential $[Q, -]$, where Q is the differential of V. For example, let $|e_0| = 0$, $|e_1| = 1$ and $Qe_0 = e_1$. Taking $a = E_{00}$ and $b = E_{01}$ in the shifted chain identity gives zero with (40.8). Omitting its suspension sign gives 2. The sign is required even though the underlying ungraded trace pairing is familiar.

40.4 A smooth category with nonproper total space

The category need not be proper for the moduli theorem. Let $A = k[t]$ and $C = \mathrm{Perf}(A)$. An exact functor from this category is specified by an object and one endomorphism. These finite data commute with filtered colimits of the recipient category, so C is of finite type.

The diagonal has the free bimodule resolution

$$0 \longrightarrow A^e \xrightarrow{t\otimes 1 - 1\otimes t} A^e \longrightarrow A \longrightarrow 0, \qquad A^e = k[t_1, t_2],$$

in degrees -1, 0 before augmentation. Multiplication identifies the cokernel with $k[t]$, and $t_1 - t_2$ is a non-zero-divisor. Thus C is smooth, with inverse dualizing bimodule $A^! \simeq A[-1]$. It is not proper, since $\mathrm{RHom}_A(A, A) = k[t]$ is not perfect over k.

The degree--1 generator of the diagonal resolution maps to $1[t]1$ in the normalized bimodule bar resolution, whose boundary is $t_1 - t_2$. After tensoring over A^e with A, its Hochschild class maps to dt under the Hochschild–Kostant–Rosenberg identification. This class has Hochschild degree one, is closed under the de Rham operator, and generates the dualizing line. It therefore gives a smooth one-dimensional Calabi–Yau structure.

For the covariant moduli convention, the corepresenters are the perfect complexes E for which $\mathrm{RHom}_A(E, A)$ is perfect over k. For $k[t]$, these are precisely the perfect complexes with finite-dimensional total cohomology, equivalently proper support on the affine line. Decompose their finitely generated cohomology modules over the principal ideal domain $k[t]$. Free summands persist under duality, while torsion summands have finite-dimensional dual cohomology.

At the residue module $k_a = A/(t-a)$, the two-term resolution

$$A \xrightarrow{t-a} A \longrightarrow k_a$$

gives

$$\operatorname{Ext}^*_A(k_a, k_a) = k \cdot 1 \oplus k \cdot \varepsilon, \qquad |\varepsilon| = 1, \quad \varepsilon^2 = 0.$$

The differential becomes zero after applying $\operatorname{Hom}_A(-, k_a)$, and there is no degree-two term.

Choose the generator so that $\operatorname{tr}_{dt}(\varepsilon) = 1$. The unshifted pairing has matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ in the basis $(1, \varepsilon)$. After shifting, it pairs the degree--1 automorphism direction with the degree-zero deformation direction and has values in $k[1]$. This is the $2 - d = 1$ shift of the general theorem. Discarding either direction destroys the duality.

40.5 Cyclicity and closedness

Nondegeneracy alone does not supply cyclicity. For $D = \operatorname{diag}(1, 2)$ on k^2, the functional $\tau_D(a) = \operatorname{tr}(Da)$ gives a nondegenerate pairing $\tau_D(ab)$ on $\operatorname{Mat}_2(k)$, since left multiplication by D is invertible. But

$$\tau_D(E_{12}E_{21}) = 1, \qquad \tau_D(E_{21}E_{12}) = 2.$$

It is not a cyclic trace and does not define the proper Calabi–Yau datum used above.

Nor does an everywhere nondegenerate geometric pairing have to be closed. On $U = \operatorname{Spec} k[x_1, x_2, x_3, x_4, (1 + x_1)^{-1}]$, put

$$\eta = dx_1 \wedge dx_2 + (1 + x_1)dx_3 \wedge dx_4.$$

Its determinant is $(1 + x_1)^2$, while

$$d_{\mathrm{dR}}\eta = dx_1 \wedge dx_3 \wedge dx_4 \neq 0.$$

Since U is an ordinary smooth affine scheme, a closed degree-zero two-form is an ordinary closed differential form. Thus η has no closing structure, despite being nondegenerate at every point.

On a derived complex, higher components can instead supply the required homotopy. For instance, let a mixed complex have basis v_0, a, v_1 in degrees $0, -1, -2$, with

$$Bv_0 = a, \qquad bv_1 = -a,$$

and all other values of b, B zero. Then $v_0 + uv_1$ is a negative cyclic cycle, although $Bv_0 \neq 0$. Its second component implements $bv_1 + Bv_0 = 0$ in (40.2). This example explains why a cyclic lift includes data beyond strict invariance of a chosen Hochschild representative.

40.6 The ordered chiral comparison

The categorical outputs above have a specified coefficient category and cyclic class. To obtain an ordered chiral algebra on a curve X, one needs geometric factorization data on $X^{\mathrm{an}} \times \mathbb{R}$.

Concretely, let $\mathcal{F}$ be a unital factorization theory whose interval coefficients $B_J(U) = \mathcal{F}(U \times J)$ lie in a symmetric monoidal category $(\mathcal{H}, \odot, \mathbf{1})$ of holomorphic factorization objects. Assume interval inclusions act by equivalences and that these coefficients have a compatible augmentation.

Supply a strong symmetric monoidal comparison

$$\Phi : (\mathcal{H}, \odot, \mathbf{1}) \longrightarrow (\mathcal{C}_D, \otimes^!, \omega_{\mathrm{Ran}}), \qquad \mathcal{C}_D \subseteq \operatorname{Fact}_D(X),$$

where C_D specifies the admitted algebraic coefficients and Φ preserves their coherent collision and separated factorization maps. Theorem 12.3 then constructs the ordered chiral algebra $\Phi(B_{J_o})$. If ϕ_2 is the tensorator of Φ, its multiplication is

$$\Phi(B_{J_o}) \otimes^{!} \Phi(B_{J_o}) \xrightarrow{\phi_2} \Phi(B_{J_o} \odot B_{J_o}) \xrightarrow{\Phi(\mu)} \Phi(B_{J_o}),$$

with unit and augmentation transported by the same monoidal comparison. Its locality and higher compatibilities follow from that comparison and the interval operations.

For a given Calabi–Yau category, constructing $\mathcal{F}$ and Φ, and identifying their cyclic pairing with θ, are additional geometric problems. The framed Hochschild operations do not themselves specify collision supports, algebraic coefficients or vacuum insertions on X. A further bulk-to-center equivalence must identify the actual local action and its factorization maps. The Calabi–Yau theorem supplies its categorical input and moduli form while leaving those comparisons to their stated hypotheses.

Chapter 41

Hall and BPS algebras

An extension product counts the ways to assemble an object from two others. For vector spaces over the finite field $\mathbb{F}_q$, let u_n denote the class of an n-dimensional space. An extension with quotient of dimension a and subobject of dimension b amounts to a b-dimensional subspace of $\mathbb{F}_q^{a+b}$. Thus the Hall product is

$$u_a * u_b = \begin{bmatrix} a+b \\ b \end{bmatrix}_q u_{a+b}, \qquad \begin{bmatrix} n \\ b \end{bmatrix}_q = \prod_{i=0}^{b-1} \frac{q^n - q^i}{q^b - q^i}. \tag{41.1}$$

The numerator counts ordered independent b-tuples, and the denominator counts ordered bases of a fixed subspace. In particular, $u_1 * u_1 = (q+1)u_2$ and u_0 is the unit.

Multiplying three classes counts flags. Choosing the smaller subspace first or the larger one first gives

$$\begin{bmatrix} a+b \\ b \end{bmatrix}_q \begin{bmatrix} a+b+c \\ c \end{bmatrix}_q = \begin{bmatrix} b+c \\ c \end{bmatrix}_q \begin{bmatrix} a+b+c \\ b+c \end{bmatrix}_q.$$

For $a = b = c = 1$ and $q = 2$, both sides count the 21 complete flags in $\mathbb{F}_2^3$. This identity is the elementary form of Hall associativity. Geometric Hall theories replace counting by operations on cohomology, retaining the same flag comparison. Their BPS generators and Yangian presentations require further geometry, which determines the coefficient objects and the resulting algebra.

41.1 The extension correspondence

Let $\mathcal{A}$ be an exact category with a moduli stack $\mathfrak{M} = \coprod_{\alpha \in \Gamma} \mathfrak{M}_\alpha$. For a stable infinity-category, use cofiber sequences and derived flag stacks in the same construction. Here Γ is an additive charge monoid, and the charge of an extension is the sum of the charges of its factors. Write $\mathfrak{E}_{\alpha,\beta}$ for the stack of exact sequences

$$0 \longrightarrow E_\beta \longrightarrow E \longrightarrow E_\alpha \longrightarrow 0.$$

The maps recording the two factors and the middle term give

$$\mathfrak{M}_\alpha \times \mathfrak{M}_\beta \xleftarrow{p} \mathfrak{E}_{\alpha,\beta} \xrightarrow{q} \mathfrak{M}_{\alpha+\beta}.$$

Fix a coefficient theory H, an external product $\boxtimes$, a pushforward q_* and a normalized pullback $p^\sharp$ for these maps. In a Borel–Moore theory, $p^\sharp$ is generally a refined or virtual Gysin map. Its shifts, orientation and coefficient class form part of the definition. Set

$$a * b = q_* p^\sharp (a \boxtimes b). \tag{41.2}$$

Proposition 41.1 (Associativity from flags)**.** *Assume that these operations are functorial and compatible with external products. Assume the base-change comparisons for both iterated extension diagrams, with the same shifts and orientations. Then* (41.2) *is associative. If extension by the zero object gives the identity correspondence with its identity coefficient map, its class is a unit.*

Proof. Let $\mathfrak{F}_{\alpha,\beta,\gamma}$ classify flags

$$0 \subset F_1 \subset F_2 \subset E, \qquad [E/F_2] = \alpha, \quad [F_2/F_1] = \beta, \quad [F_1] = \gamma.$$

There are canonical equivalences of stacks

$$\mathfrak{F}_{\alpha,\beta,\gamma} \simeq \mathfrak{E}_{\alpha,\beta} \underset{\mathfrak{M}_{\alpha+\beta}}{\times} \mathfrak{E}_{\alpha+\beta,\gamma},$$

$$\mathfrak{F}_{\alpha,\beta,\gamma} \simeq \mathfrak{E}_{\alpha,\beta+\gamma} \underset{\mathfrak{M}_{\beta+\gamma}}{\times} \mathfrak{E}_{\beta,\gamma}.$$

In the derived setting these are homotopy fiber products.

In the first, the inner extension is $0 \to F_2/F_1 \to E/F_1 \to E/F_2 \to 0$. Its inverse takes the preimage of the subobject in E/F_1. In the second, the inner extension is $0 \to F_1 \to F_2 \to F_2/F_1 \to 0$. These constructions respect isomorphisms and families.

Let r record the three successive factors and let s record E. Base change and functoriality identify each parenthesization with

$$s_* r^{\sharp}(a \boxtimes b \boxtimes c).$$

The equality uses the stipulated agreement of the two Gysin normalizations. The two zero-object correspondences prove the unit assertion. □

Normalization matters already for finite vector spaces. Multiply the coefficient in (41.1) by -1 only when $a = b = 1$. The unit is preserved, but at $q = 2$ the two products of charges $(1, 1, 2)$ have coefficients -105 and 105. The modified normalization fails the flag comparison.

The proposition concerns the product on $H(\mathfrak{M})$. A product on complexes with coherent higher associativity also requires the compatible operations for flags of every length. For a critical Hall theory, the vanishing-cycle coefficients and their orientation must participate in those comparisons.

One may form $\bigoplus_\alpha H(\mathfrak{M}_\alpha)$ when the product has finite charge support. If every charge has finitely many decompositions, the same formula extends to the product completion $\prod_\alpha H(\mathfrak{M}_\alpha)$. Each output then contains a finite convolution sum. The supported surface algebra below uses a different topology, determined by its moduli stack.

41.2 The BPS algebra and its Lie generators

We now work over $\mathbb{C}$ with rational mixed Hodge coefficients. Let $\mathcal{A}$ be an admissible finite-length Abelian subcategory of the homotopy category of a dg category C. Assume its objects form the classical truncation of an open 1-Artin substack of the derived moduli of C. The geometric hypotheses needed here are the following.

(i) The middle-term map q is proper and representable. The object stack has the resolution property, and the shifted relative RHom[1] complex on each pair of components has a global three-term vector-bundle presentation.

(ii) There is a good moduli map $\mathrm{JH} : \mathfrak{M} \to M$, where M is a separated scheme with finite-type connected components. The direct-sum map $\oplus : M \times M \to M$ is finite.

(iii) For every finite collection of simple objects, the full dg subcategory on those objects has a right 2-Calabi–Yau structure. The relative Hall construction is associative and unital with its specified virtual pullbacks.

These are Assumptions 1–5 and 7 of Davison–Hennecart–Schlegel Mejia [31, §§4,6]. They include more than the existence of a good moduli space. Choose a surjective charge map $\pi_0(M) \to \Gamma$ with finite

fibers, where Γ embeds in a lattice. Assume the Euler form of C descends to a symmetric form χ on that lattice.

For complexes of mixed Hodge modules on M, define convolution by

$$F \boxdot G = \oplus_*(F \boxtimes G).$$

The finiteness of $\oplus$ makes this tensor exact on mixed Hodge modules. The skyscraper sheaf $\mathbb{Q}$ at the zero object is its unit.

Let $\mathbb{L} = H_c^*(\mathbb{A}^1, \mathbb{Q})$ be the Tate complex, and use the normalization

$$\mathbb{Q}^{\mathrm{vir}}_{\mathfrak{M}_\alpha} = \mathbb{Q}_{\mathfrak{M}_\alpha} \otimes \mathbb{L}^{\chi(\alpha,\alpha)/2}, \qquad \mathscr{H} = \mathrm{JH}_* \, \mathbb{D}\mathbb{Q}^{\mathrm{vir}}_{\mathfrak{M}}.$$

Here $\mathbb{D}$ denotes Verdier duality. The displayed pushforward denotes the defined complex on M, constructed by the stack approximations in the relative Hall theory. It does not require a category of mixed Hodge modules on an arbitrary Artin stack. The Hall product makes $\mathscr{H}$ an algebra in $D^+(\mathrm{MHM}(M))$.

The 2-Calabi–Yau purity theorem gives $\mathcal{H}^i(\mathscr{H}) = 0$ for $i < 0$ and makes $\mathcal{H}^0(\mathscr{H})$ semisimple [31, Lemma 6.3]. Define the relative BPS algebra and its absolute algebra by

$$\mathscr{B}_{\mathrm{Alg}} = \mathcal{H}^0(\mathscr{H}), \qquad B_{\mathrm{Alg}} = \mathbb{H}^*(M, \mathscr{B}_{\mathrm{Alg}}).$$

The degree here is the mixed-Hodge-module degree, corresponding to perverse degree on the underlying sheaves. It is taken before global sections. By the decomposition theorem, $\mathscr{B}_{\mathrm{Alg}} \to \mathscr{H}$ has a retraction as a map of complexes. Consequently $B_{\mathrm{Alg}} \to \mathbb{H}^*(M, \mathscr{H})$ is an injective algebra map [31, Proposition 6.4].

The enveloping-algebra statement uses a sign twist. Choose a bilinear form $\psi : \Gamma \times \Gamma \to \mathbb{Z}/2$ such that

$$\psi(\alpha,\beta) + \psi(\beta,\alpha) = \chi(\alpha,\beta) \pmod 2, \qquad a *_\psi b = (-1)^{\psi(\alpha,\beta)} a * b.$$

Such a choice exists because the diagonal values of χ are even. Bilinearity gives the cocycle identity that preserves associativity. Superscript ψ will indicate this changed multiplication. The cohomological Koszul signs remain in force as well.

We describe the generators before stating the comparison. Write $p(\alpha) = 2 - \chi(\alpha,\alpha)$. Let Σ consist of the nonzero charges with $p(\alpha) \geq 0$ such that

$$p(\alpha) > \sum_{i=1}^{r} p(\alpha_i) \quad \text{for every decomposition} \quad \alpha = \sum_{i=1}^{r} \alpha_i, \quad r \geq 2, \quad \alpha_i \neq 0.$$

The simple-root set Φ^+ is the union of Σ with all positive multiples of its isotropic elements. Here isotropic means $\chi(\alpha,\alpha) = 0$.

For $\alpha \in \Sigma$, set $G_\alpha = \mathrm{IC}(M_\alpha)$. We normalize the intersection complex to have weight zero, restricting on a smooth component of dimension d to $\mathbb{Q}[d](d/2)$. For $\alpha = n\alpha_0$, with $\alpha_0 \in \Sigma$ isotropic and $n \geq 2$, set

$$G_\alpha = (\delta_n)_* \, \mathrm{IC}(M_{\alpha_0}), \qquad \delta_n(x) = x^{\oplus n}.$$

Thus a multiple isotropic root uses the small diagonal, rather than the intersection complex of the entire component. The definitions are taken componentwise before a finite-fiber charge map combines components.

Let $\mathfrak{n}^+_{M,G}$ be the Lie algebra object generated by $G = \bigoplus_{\alpha\in\Phi^+} G_\alpha$ under convolution, with the generalized Serre relations. For distinct simple-root generators x of charge α and y of charge β, these include

$$(\operatorname{ad} x)^{1-\chi(\alpha,\beta)}\, y = 0 \quad \text{if } \chi(\alpha,\alpha) = 2,$$
$$[x, y] = 0 \quad \text{if } \chi(\alpha,\beta) = 0.$$

The orthogonal relation also applies within an isotropic generator space. After global sections, brackets use the sign $(-1)^{|x||y|}$. The resulting graded Lie algebra is the positive half of the generalized Kac–Moody Lie superalgebra with this Euler form and these generator spaces.

Theorem 41.2 (Geometric BPS and PBW comparisons)**.** *Under the hypotheses above, the canonical generator map induces*

$$U_{\boxdot}(\mathfrak{n}^+_{M,G}) \xrightarrow{\sim} \mathscr{B}^{\psi}_{\mathrm{Alg}}, \qquad U(\mathfrak{n}^+_{\mathrm{BPS}}) \xrightarrow{\sim} B^{\psi}_{\mathrm{Alg}}, \qquad \mathfrak{n}^+_{\mathrm{BPS}} = \mathbb{H}^*(M, \mathfrak{n}^+_{M,G}).$$

These are algebra isomorphisms. Suppose additionally that $\mathfrak{M}$ has a positive determinant line bundle: scalar automorphisms of every nonzero object act on its fiber with positive weight. With $u = c_1(\mathcal{O}_{B\mathbb{G}_m}(1))$ of degree 2, there is a PBW isomorphism

$$\operatorname{Sym}\bigl(\mathfrak{n}^+_{\mathrm{BPS}} \otimes \mathbb{Q}[u]\bigr) \xrightarrow{\sim} \mathbb{H}^*(M, \mathscr{H}^{\psi}) \tag{41.3}$$

of charge-graded mixed Hodge structures with cohomological grading. The symmetric algebra uses the Koszul symmetry. The map is symmetrized Hall multiplication, not an asserted algebra isomorphism from a commutative algebra.

Proof. The canonical inclusions of the intersection-complex summands give $G \to \mathscr{B}^{\psi}_{\mathrm{Alg}}$. Their Hall products define a map from the free convolution algebra. Davison–Hennecart–Schlegel Mejia prove that it annihilates the Serre ideal and induces an isomorphism [31, Proposition 7.13 and Theorem 1.1]. This proves the first assertion.

Its proof uses the local Ext-quiver identification of the generator and BPS sheaves, including their products. The induction on dimension vector reduces the kernel and cokernel to the strictly seminilpotent quiver calculation [31, Proposition 7.7, Corollary 7.14, §7.7.2]. It therefore identifies the actual generator map, beyond its graded character. Taking global sections gives the second isomorphism by the same convolution comparison [31, Theorem 1.2].

For the PBW map, compose $\mathfrak{n}^+_{M,G} \to \mathscr{B}^{\psi}_{\mathrm{Alg}} \to \mathscr{H}^{\psi}$. The determinant line bundle lets $\mathbb{Q}[u] = H^*(B\mathbb{G}_m, \mathbb{Q})$ act, giving $\mathfrak{n}^+_{M,G} \otimes \mathbb{Q}[u] \to \mathscr{H}^{\psi}$. Apply iterated multiplication and average over permutations with their Koszul signs. This constructs the relative symmetric-tensor map.

The relative PBW theorem proves it is an isomorphism [31, Theorem 1.7]. Concretely, a nonzero cone would have a nonzero costalk at a closed point. The local Ext-quiver comparison identifies that costalk with the corresponding preprojective PBW comparison, whose cone vanishes. Global sections give (41.3), as in [31, Corollary 1.8]. □

The relative BPS Lie algebra is $\mathfrak{n}^+_{M,G}$. Its construction precedes any choice of a geometric coproduct on the full Hall algebra. The enveloping-algebra isomorphism transports a cocommutative Hopf structure to B^{ψ}_{Alg}. Identifying that structure with geometric cooperation, or extending it to the full CoHA, requires a separate comparison.

41.3 Point sheaves on a K3 surface

Let S be a projective K3 surface and let $\mathcal{A} = \mathrm{Coh}_0(S)$ be its category of zero-dimensional coherent sheaves. The charge is length, and $M_n = \mathrm{Sym}^n(S)$. The geometric hypotheses of Theorem 41.2 hold for

this surface category [31, §4]. The Euler form on point charges is zero, so $\Sigma = \{1\}$ and $\Phi^+ = \mathbb{Z}_{>0}$. Take $\psi = 0$.

Every simple root is isotropic. Therefore

$$G_n = (\delta_n)_* \operatorname{IC}(S), \qquad \mathfrak{n}^+_{\mathrm{BPS}} = \bigoplus_{n\geq 1} H^*(S, \operatorname{IC}(S))\, z^n$$

is abelian. Here z records length and $\operatorname{IC}(S) = \mathbb{Q}_S[2](1)$. The brackets vanish because every pair of charges is orthogonal.

The generator in each length has dimensions 1, 22, 1 in degrees -2, 0, 2. All these degrees are even. Forgetting the cohomological grading, the BPS dimension series is consequently

$$\sum_{n\geq 0} \dim(B_{\mathrm{Alg},n}) z^n = \prod_{r\geq 1} (1 - z^r)^{-24} = 1 + 24z + 324z^2 + 3200z^3 + \cdots . \tag{41.4}$$

The length-two coefficient consists of $\operatorname{Sym}^2(\mathbb{Q}^{24})$, of dimension 300, and the new isotropic generator space of dimension 24. Removing the latter would change the BPS algebra. In a category with odd generators, the corresponding symmetric factors are exterior factors, so this dimension formula cannot be reused without its parity hypothesis. For an odd isotropic generator θ, the relation $[\theta, \theta] = 0$ gives $2\theta^2 = 0$ in its enveloping algebra.

The determinant of the vector bundle of global sections is a line bundle on the object stack. A scalar acts with weight n in length n, so this determinant is positive. Writing t for cohomological degree, Theorem 41.2 gives

$$\operatorname{ch}_t B_{\mathrm{Alg},1} = t^{-2} + 22 + t^2, \qquad \operatorname{ch}_t \mathbb{H}^*(\mathcal{M}, \mathscr{H})_1 = \frac{t^{-2} + 22 + t^2}{1 - t^2}.$$

For instance, degree 4 contributes 24 classes to the full length-one CoHA and none to its BPS subalgebra. Thus the BPS enveloping theorem and the full Hall PBW theorem have different targets even in this basic symplectic-surface example.

41.4 Quiver and surface Yangian comparisons

Let Q be a finite quiver, possibly with loops, and let Π_Q be its preprojective algebra. It is the path algebra of the doubled quiver modulo $\sum_{a\in Q_1}[a, a^*]$. Use the arrow-scaling torus $T = (\mathbb{G}_m)^{Q_1} \times \mathbb{G}_m$, where the last factor scales the symplectic form. Its coefficient ring is

$$R = H^*_T(\mathrm{pt}, \mathbb{Q}) = \mathbb{Q}[t_a, \hbar \mid a \in Q_1].$$

Let $\mathcal{H}_R(Q)$ be the equivariant Borel–Moore CoHA of the stacks of finite-dimensional Π_Q-modules, with the virtual grading and Hecke-convolution normalization of Schiffmann–Vasserot [93, §2]. Let $\mathcal{H}^\vee_R(Q)$ use their Lusztig-nilpotent substacks instead. The superscript here distinguishes two coefficient carriers.

Their Yangian $\mathbf{Y}_R(Q)$ is defined by stable-envelope operators on Nakajima quiver varieties. The nilpotent version $\mathbf{Y}^\vee_R(Q)$ acts on the corresponding central fibers. Schiffmann–Vasserot construct algebra isomorphisms

$$\mathcal{H}_R(Q) \xrightarrow{\sim} \mathbf{Y}^-_R(Q), \qquad \mathcal{H}^\vee_R(Q) \xrightarrow{\sim} \mathbf{Y}^{-,\vee}_R(Q) \tag{41.5}$$

intertwining every such geometric action [93, Theorem 4.13]. The superscript minus is their convention for the half corresponding to positive dimension vectors in the Hall algebra. The comparison holds over R. No completion or passage to the fraction field is required in its statement.

The map first compares the Hecke classes with stable-envelope operators over $\mathrm{Frac}(R)$. Generation and the faithful family of vacuum evaluations make this comparison injective. The integral action lattices and the triangular decomposition then identify its image over R [93, Proposition 4.1, Theorem 4.8 and Proposition 4.10]. These operations distinguish (41.5) from a character identity. The central polynomial algebras of framing-tautological classes used in that proof are additional extensions of the halves.

For the surface comparison, let $\Gamma_0 \subset \mathrm{SL}_2(\mathbb{C})$ be a nontrivial finite subgroup and let $X \to \mathbb{C}^2/\Gamma_0$ be the minimal resolution. Let Z be its exceptional divisor and let T_0 be a diagonal torus commuting with Γ_0. The source is the stack of coherent sheaves with set-theoretic support in Z. It includes zero-dimensional sheaves and sheaves on infinitesimal thickenings of Z. Write $\mathcal{H}^{T_0}_{X,Z}$ for its nilpotent equivariant CoHA with rational coefficients. Its underlying topological space is the inverse limit of equivariant Borel–Moore homology over an admissible quasi-compact open exhaustion, taken in each grading [34, §III.7.2].

Fix an orientation of the finite ADE diagram. In the Grothendieck group, write α_i for its simple-root classes and δ for the point class. The comparison twists the Hall multiplication by $(-1)^{\Theta(d,e)}$, where

$$\Theta(\alpha_i, \alpha_j) = \langle \alpha_i, \alpha_j \rangle_Q \pmod 2, \qquad \Theta(\delta, d) = \Theta(d, \delta) = 0.$$

The brackets on the right denote the Euler form of the oriented finite quiver. We include this twist in the notation $\mathcal{H}^{T_0}_{X,Z}$.

The target half has a specific topology. Let $\mathbf{Y}^-_Q$ be the standard negative half of the affine ADE Yangian, specialized to $H^*_{T_0}(\mathrm{pt}, \mathbb{Q})$. Choose a coweight ϑ with positive coefficients at the finite vertices and $\vartheta(\delta) = 0$. For a nonzero dimension vector d, define its slope by $\mu_\vartheta(d) = \vartheta(d)/\sum_i d_i$. Let ϑ_f be its finite coweight and let T_λ denote the affine braid operator attached to a lattice translation λ.

Define the left ideal and translated quotients

$$J_{\vartheta,0} = \sum_{\mu_\vartheta(d)>0} \mathbf{Y}^-_Q (\mathbf{Y}^-_Q)_{-d}, \qquad V_{2\ell} = \frac{T_{4\ell\vartheta_f}(\mathbf{Y}^-_Q)}{T_{4\ell\vartheta_f}(J_{\vartheta,0})}, \qquad \ell \in \mathbb{Z}.$$

These are the quotients for the slope interval $(-\infty, 0]$ and its translates in [34, (III.3.6), §III.7.4].

For the transition from level 0 to level 2, first restrict the slope interval to $(-1/(4h), 0]$, where $h = \sum_i \delta_i$. Equivalently, quotient further by $\sum_{\mu_\vartheta(d) \le -1/(4h)} (\mathbf{Y}^-_Q)_{-d} \mathbf{Y}^-_Q$. The braid and PBW comparison identifies this quotient with V_2. Translating this map gives the restriction maps $V_{2\ell-2} \to V_{2\ell}$ of [34, (III.7.4)], and hence

$$\mathbf{Y}^+_\infty = \varprojlim_{\ell \to +\infty} V_{-2\ell}. \tag{41.6}$$

The quotients are vector-space quotients by left ideals, so their products cannot be assumed. For a fixed output level, multiply lifts at a sufficiently deeper level in the translated Yangian and project. The stabilization theorem makes the result independent of those lifts and defines the continuous multiplication on (41.6).

Theorem 41.3 (The supported surface half)**.** *With this coefficient ring, sign twist and topology, there is an isomorphism of graded topological algebras*

$$\Theta_{X,Z} : \mathcal{H}^{T_0}_{X,Z} \xrightarrow{\sim} \mathbf{Y}^+_\infty.$$

It intertwines tensoring sheaves by line bundles with the corresponding affine braid operators.

Proof. The derived McKay equivalence identifies perverse coherent sheaves supported in Z with nilpotent preprojective modules. Tensoring by increasingly dominant negative line bundles gives a sequence of hearts whose limiting heart is the category of coherent sheaves supported in Z. The continuity theorem for their flag stacks identifies the surface CoHA with the limit of the corresponding slope quotients [34, Theorem III.7.5].

The quiver Yangian comparison identifies each quotient with $V_{2\ell}$. Compatibility of derived reflection functors with braid operators identifies the transition maps and the multiplications of sufficiently deep lifts. Taking inverse limits therefore gives mutually inverse continuous maps preserving multiplication [34, Theorem III.7.8]. The same compatibility identifies the line-bundle action [34, Proposition III.7.14]. □

For the trivial torus, the target is a completed enveloping algebra. If $\mathfrak{g}_f$ is the finite ADE Lie algebra, with positive nilpotent part $\mathfrak{n}_f^+$ and Cartan $\mathfrak{h}_f$, its Lie algebra is

$$\mathfrak{n}_{\mathrm{ell}}^+ = \mathfrak{n}_f^+[s^{\pm 1}, t] \oplus s^{-1}\mathfrak{h}_f[s^{-1}, t] \oplus K^-.$$

Here K^- is the negative-s-degree part of the center in the universal central extension of $\mathfrak{g}_f[s^{\pm 1}, t]$. The completion of $U(\mathfrak{n}_{\mathrm{ell}}^+)$ is the slope completion corresponding to (41.6) [34, Theorem III.7.25]. In particular, the nonstandard half is not specified merely by choosing the usual positive roots of the affine Lie algebra.

41.5 Cooperation and geometric reconstruction

Even the vector-space Hall algebra illustrates the tensor data needed for a coproduct. Keep the prime power $q > 1$ fixed and work over $\mathbb{Q}$. In charge-graded vector spaces, use the braiding

$$v_b \otimes w_c \longmapsto q^{-bc} w_c \otimes v_b.$$

The induced product on two copies of the Hall algebra is

$$(u_a \otimes u_b)(u_c \otimes u_d) = q^{-bc}(u_a * u_c) \otimes (u_b * u_d).$$

Proposition 41.4 (The vector-space Hall coproduct)**.** *For this braided tensor product,*

$$\Delta u_n = \sum_{a+b=n} q^{-ab} u_a \otimes u_b, \qquad \epsilon(u_n) = \delta_{n,0}$$

defines a graded bialgebra. Its antipode is $S(u_n) = (-1)^n q^{-n(n-1)/2} u_n$. The same formulas are continuous in the positive charge completion.

Proof. Put $[n]_q! = \prod_{j=1}^{n}(1 + q + \cdots + q^{j-1})$ and $x = u_1$. Equation (41.1) gives $x^n = [n]_q! u_n$. In the braided tensor algebra, $Y = 1 \otimes x$ and $X = x \otimes 1$ satisfy $YX = q^{-1}XY$. The q^{-1}-binomial identity therefore gives

$$(X + Y)^n = \sum_{a+b=n} \begin{bmatrix} n \\ a \end{bmatrix}_{q^{-1}} X^a Y^b, \qquad \begin{bmatrix} n \\ a \end{bmatrix}_{q^{-1}} = q^{-ab} \begin{bmatrix} n \\ a \end{bmatrix}_q.$$

Dividing by $[n]_q!$ gives the displayed coproduct. It is multiplicative because it was defined by $x \mapsto X + Y$ on the polynomial algebra generated by x.

The coefficient of $u_a \otimes u_b \otimes u_c$ in either iterated coproduct is $q^{-ab-ac-bc}$. This proves coassociativity, and the endpoint terms prove the counit identities. The connected graded bialgebra has the recursively determined antipode. Braided reversal of n copies of x contributes $q^{-n(n-1)/2}$, giving its stated formula. Every fixed-charge sum is finite, so all operations extend continuously. □

If one deletes the crossing factor while keeping this coproduct, multiplicativity fails. The coefficient of $u_1 \otimes u_1$ in $\Delta(u_1 * u_1)$ is $1 + q^{-1}$, whereas the ordinary tensor square of Δu_1 has coefficient 2. This calculation fixes both the carrier and the failed operation. The shuffle obstruction (33.10) makes the corresponding tensor distinction on its own collision carrier. Its rational exchanged recipient remains a positive construction with the stated removed divisors.

To obtain an ordered chiral Hall object, one must construct the coefficient objects over configurations, their separated factorization maps and their collision operations. The extension-flag comparisons must commute with those maps and with the chosen charge-zero unit. For a critical Hall theory, this includes descent of the vanishing-cycle coefficients and their orientation. Supply the interval family, compatible augmentation and strong monoidal coefficient functor required by Theorem 12.3. These data transport the Hall multiplication to an ordered chiral multiplication. The Hall algebra on global cohomology alone does not give these objects or their descent.

A geometric coproduct must further specify its supported or localized recipient and the interchange governing that recipient. A completed double requires compatible negative and neutral halves, Hopf operations and a continuous Hopf pairing. A universal R-tensor additionally requires convergence of the pairing's canonical tensor. The BPS theorem and the Yangian-half comparisons preserve their full algebraic content while leaving those geometric constructions separate. For the supported surface algebra, even the geometric realization of its double is a further problem [34, Introduction, "Representations"].

Chapter 42

The Igusa denominator as a bar character

Let $\mathfrak{g}_\Delta$ be the Gritsenko–Nikulin Borcherds–Kac–Moody superalgebra and let $\mathfrak{n}_+$ be its positive nilpotent part. Root spaces are finite-dimensional and the positive cone is locally finite. Fix a proper integral height h that is positive on the effective root cone, and set

$$F^{>H}\mathfrak{n}_+ = \bigoplus_{h(\alpha)>H} \mathfrak{g}_{\Delta,\alpha}, \qquad \mathfrak{n}_+^{(H)} = \mathfrak{n}_+/F^{>H}\mathfrak{n}_+.$$

Height additivity makes $F^{>H}\mathfrak{n}_+$ an ideal; properness makes $\mathfrak{n}_+^{(H)}$ finite-dimensional and nilpotent. Every bar, Chevalley, enveloping, and character formula in this chapter is interpreted coefficientwise in the separated height completion.

Theorem 42.1 (Bar–Chevalley identification)**.** *For each height H there is a natural quasi-isomorphism of conilpotent coalgebras*

$$\mathrm{Bar}\big(U(\mathfrak{n}_+^{(H)})\big) \simeq C_*(\mathfrak{n}_+^{(H)}).$$

The quotient maps are compatible with these quasi-isomorphisms, and therefore induce a quasi-isomorphism in the separated height-complete category

$$\mathrm{Bar}_h^\wedge\big(U(\mathfrak{n}_+)\big) := \varprojlim_H \mathrm{Bar}\big(U(\mathfrak{n}_+^{(H)})\big) \simeq C_{*,h}^\wedge(\mathfrak{n}_+) := \varprojlim_H C_*(\mathfrak{n}_+^{(H)}).$$

The coefficientwise Euler supercharacter of the completed Chevalley complex is

$$\chi\, C_{*,h}^\wedge(\mathfrak{n}_+) = \prod_{\alpha>0}(1-e^{-\alpha})^{\mathrm{sdim}\,\mathfrak{g}_{\Delta,\alpha}}.$$

Proof. For every H, the PBW filtration on $U(\mathfrak{n}_+^{(H)})$ identifies the associated graded with $\mathrm{Sym}(\mathfrak{n}_+^{(H)})$. The bar complex of the enveloping algebra and the Chevalley complex have the same associated-graded Koszul resolution. The standard antisymmetrization map is a filtered quasi-isomorphism. Naturality for the quotient maps gives the inverse-limit comparison; proper height makes every homogeneous coefficient stabilize. For an even root vector the exterior Chevalley factor contributes $1-e^{-\alpha}$; for an odd root vector it contributes its inverse. Multiplication over the root spaces gives the product. □

With the positive-root convention $e^{-\rho} = q^{1/2}r^{1/2}s^{1/2}$, the denominator theorem reads

$$e^{-\rho}\,\chi\, C_{*,h}^\wedge(\mathfrak{n}_+) = D_5(2\Omega) = 2^{-6}\Delta_5(2\Omega)$$

[50, 52]. This identity is the exact meeting point of ordered bar theory and automorphic products.

Chapter 43

Finite-height currents and an exact physical model

Retain the finite-height quotients $\mathfrak{n}_+^{(H)}$ fixed in the preceding chapter.

The current Lie conformal superalgebra is

$$\operatorname{Cur}(\mathfrak{n}_+^{(H)}) = \Bbbk[\partial] \otimes \mathfrak{n}_+^{(H)}, \qquad [a_\lambda b] = [a, b].$$

Its universal enveloping vertex superalgebra is V_H. The current generators carry the cocommutative vertex-bialgebra coproduct

$$\Delta\big(a(-1)\mathbf{1}\big) = a(-1)\mathbf{1} \otimes \mathbf{1} + \mathbf{1} \otimes a(-1)\mathbf{1}, \qquad a \in \mathfrak{n}_+^{(H)},$$

extended uniquely as a vertex-algebra morphism.

Theorem 43.1 (Height-complete current realization). *Let $\mathfrak{n} = \bigoplus_{\alpha>0} \mathfrak{n}_\alpha$ be a root-graded Lie superalgebra with finite-dimensional root spaces and a proper additive height function. Put*

$$\mathfrak{n}^{(H)} = \mathfrak{n} \Big/ \bigoplus_{h(\alpha)>H} \mathfrak{n}_\alpha.$$

Then every $\mathfrak{n}^{(H)}$ is finite-dimensional and nilpotent, the quotient maps form an inverse system, and

$$\widehat{U}_h(\mathfrak{n}) = \varprojlim_H U(\mathfrak{n}^{(H)}), \quad \widehat{C}_{*,h}(\mathfrak{n}) = \varprojlim_H C_*(\mathfrak{n}^{(H)}), \quad \widehat{V}_h(\mathfrak{n}) = \varprojlim_H U^{\mathrm{ch}}(\operatorname{Cur}_X \mathfrak{n}^{(H)})$$

are separated complete algebras, coalgebras, and current algebras. Their associated graded objects are the corresponding completed symmetric objects, and the bar–Chevalley comparison is compatible with the inverse system.

Proof. Properness gives finitely many roots below a fixed height. Additivity implies that the roots of height $> H$ form a Lie ideal and that the quotient is nilpotent. PBW and Chevalley antisymmetrization are natural under quotient maps. Taking inverse limits in each finite total height therefore gives the displayed complete objects and preserves the coefficientwise quasi-isomorphism. □

Apply the finite BF model of Chapter 19 to every $\mathfrak{n}_+^{(H)}$. The inverse system gives a height-complete holomorphic–topological classical field theory with completed boundary algebra and interval complex

$$U_h^\wedge(\mathfrak{n}_+) := \varprojlim_H U(\mathfrak{n}_+^{(H)}), \qquad C_{*,h}^\wedge(\mathfrak{n}_+) := \varprojlim_H C_*(\mathfrak{n}_+^{(H)}).$$

This gives an exact algebraic realization, a local classical holomorphic–topological realization, and an exact quantum realization of the finite-dimensional zero-mode sectors of the positive enveloping algebra and its denominator character. The compact $K3 \times E$ identification is the geometric comparison studied in the companion volumes.

Chapter 44

Character, bracket, parity, and orientation

A character is an Euler invariant. Faithful reconstruction combines it with the bracket, parity decomposition, and orientation line carried by the intended object.

Theorem 44.1 (Three reconstruction fibres)**.** *The following fibres contain distinct structured objects with the same scalar image.*

(i) *The PBW-character fibre of a graded vector space with basis x, y, z in degrees a, b, $a + b$ contains both the abelian Lie algebra and the Heisenberg Lie algebra $[x, y] = z$.*

(ii) *The superdimension-zero fibre contains both the zero super vector space and $\Bbbk^{1|1}$.*

(iii) *The squaring fibre of a line-valued section contains s and $s \otimes \chi$ for every sign local system χ.*

Consequently a faithful reconstruction datum consists respectively of a bracket, a parity-refined dimension, and an oriented square-root line.

Proof. PBW identifies the graded vector-space character of an enveloping algebra with the symmetric-superalgebra character of its underlying graded Lie space, so the two displayed brackets have the same PBW character. The two super vector spaces in the second item both have superdimension zero and have parity dimensions $(0, 0)$ and $(1, 1)$. Finally $\chi^{\otimes 2} \cong \mathbf{1}$, hence

$$(s \otimes \chi)^{\otimes 2} \cong s^{\otimes 2}.$$

The extra structures listed in the conclusion distinguish each pair. □

Recognition Theorem 44.2 (Minimal PBW recognition)**.** *Let $\phi : \mathfrak{g} \twoheadrightarrow \mathfrak{h}$ be a surjective morphism of positively graded Lie superalgebras with finite-dimensional homogeneous pieces. If the parity-refined Hilbert series of $U(\mathfrak{g})$ and $U(\mathfrak{h})$ agree, then ϕ is an isomorphism. Equality of signed PBW characters suffices when the parity of every primitive degree is prescribed in advance.*

Proof. If the kernel is nonzero, choose its first positive degree. In that degree all decomposable PBW contributions agree by minimality, while the primitive even or odd dimension in $U(\mathfrak{h})$ is strictly smaller. This contradicts equality of parity-refined Hilbert series. With prescribed parity, the formal logarithm of the PBW product recovers the signed primitive dimension inductively in height, and prescribed parity recovers the ordinary dimension. □

Chapter 45

Exact dependency theorem

Theorem 45.1 (Ordered-chiral closure)**.** *Fix:*

(i) *a six-functor Ran category and its factorization subcategory.*

(ii) *a transversely ordered algebra, an associative chiral algebra, or a chiral Lie algebra, as appropriate.*

(iii) *every mixed distributive, completion, duality, trace, or analytic datum named by the intended construction.*

(iv) *for a formal-disc field realization, the state space, vacuum and field map, together with the operation comparisons of Proposition 11.9.*

The three bars, their complete cobars and their module/comodule Koszul transforms are defined functorially under their stated hypotheses. The same holds for the Hochschild center, ribbon theory and finite-height current realization under the hypotheses of those constructions. They satisfy the identities proved in this volume. The supplied formal-disc realization satisfies the corresponding conditional field-recognition statement. Every passage between the pointwise and chiral-convolution monoidal products uses the displayed mixed distributive datum.

Proof. Each output is one of the constructions in the preceding chapters, in its specified source and target categories. The comparison statements use the supplied distributive, PBW, completion, duality, trace, or realization datum. Composition of the constructed functors gives the asserted functorial system. The formal-disc assertion follows from Proposition 11.9. The reverse geometric functor remains the construction sought in Problem 11.10. □

Chapter 46

Synthesis

The invariant language has three algebraic axes. Transverse order is an E_1 structure for $\otimes^!$. Associative chirality is an E_1 structure for $\star^{\mathrm{ch}}$. Chiral Lie theory is a Lie structure for $\star^{\mathrm{ch}}$. Their bars are $\mathrm{Bar}^{\perp}$, $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}$, and $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}$. The aligned projector supplies the exact sector on which a mixed distributive law is defined. Every comparison in the theorem system factors through one of these typed maps.

The constructed operations form the two functorial rows

$$\text{HT boundary theory} \xrightarrow{\mathrm{Obs}_{\partial}} \mathrm{Alg}_{\mathsf{E}_1}(\mathrm{Fact}_D(X)) \xrightarrow{\mathrm{Bar}^{\mathrm{ord}}} \mathrm{Coalg}_{\mathrm{conil,cpl}}(\mathrm{Fact}_D(X)),$$

$$\mathrm{Alg}_{\mathsf{E}_1}(\mathrm{Fact}_D(X)) \xrightarrow{Z_{\mathrm{ord}}} \mathrm{Alg}_{\mathsf{E}_2}(\mathrm{Fact}_D(X)).$$

A primitive Lie algebra enters when a separate Chevalley or Hall comparison identifies the bar coalgebra with the Chevalley chains of that Lie algebra. Hall geometry supplies this comparison through extension correspondences. Quantum groups require an additional coproduct and exchange structure. Automorphic forms arise from graded Euler characters after a grading and completion have been fixed. Each arrow therefore preserves the geometric operation named at its source.

Book **VIII**

Technical appendices to the preceding book

Chapter 47

Sign calculation for the length-three bar

For homogeneous $a, b, c \in \bar{A}$, write $[a|b|c] = sa \otimes sb \otimes sc$. With the standard suspension convention,

$$b_m[a|b|c] = (-1)^{|a|}[ab|c] + (-1)^{|a|+|b|+1}[a|bc].$$

Applying b_m again gives

$$b_m^2[a|b|c] = (-1)^{|b|}\big[(ab)c - a(bc)\big],$$

which vanishes by associativity.

Chapter 48

Rational R-matrix verification

Let P_{ij} permute tensor factors i and j. Expanding

$$R_{12}(u)\,R_{13}(u+v)\,R_{23}(v)$$

and its reversed product shows equality coefficientwise. The cubic coefficient uses

$$P_{12}P_{13}P_{23} = P_{23}P_{13}P_{12},$$

the quadratic coefficients use the symmetric-group relations, and the linear coefficients agree termwise.

Chapter 49

Carriers and comparison maps

The five-carrier geometry, transverse bar construction, mixed-forest signs, and noncommutative examples use four kinds of categorical data:

(i) a factorization carrier is paired explicitly with its operation;

(ii) the Ran category is fixed before its tensor products are used;

(iii) the block-ordered operad is introduced after a representing chiral operad is chosen;

(iv) ribbon and stable-curve sewing are derived from nondegenerate Calabi–Yau and modular data.

A comparison between these constructions therefore specifies its source and target categories and the operations preserved by its map.

Chapter 50

Enveloping bars and planar path sequences

The dimension of a bar chain space and the dimension of its homology are different invariants. For an enveloping algebra, the Chevalley comparison gives a Lie-theoretic model for the homology. This makes it possible to compare Motzkin and Riordan path counts with the ordered bar on a specified carrier.

Bar length and Chevalley degree are homological degrees: H_n denotes degree-n homology. The cohomology of the dual Chevalley complex is written H^n. For finite-dimensional weight pieces their dimensions agree.

50.1 Chevalley comparison

Theorem 50.1 (Bar–Chevalley comparison)**.** *Let k be a field of characteristic zero and let $\mathfrak{g}$ be an ordinary Lie algebra over k. Give $U(\mathfrak{g})$ its augmentation killing $\mathfrak{g}$. There is a natural quasi-isomorphism of ordinary conilpotent differential graded coalgebras*

$$\kappa : C_*(\mathfrak{g}) \longrightarrow \mathrm{Bar}^{\perp}\big(U(\mathfrak{g})\big),$$

where both constructions use direct sums and ordinary tensors over k. It preserves the PBW filtration. If $\mathfrak{g} = \bigoplus_{w \geq 1} \mathfrak{g}_w$ and $[\mathfrak{g}_u, \mathfrak{g}_v] \subseteq \mathfrak{g}_{u+v}$, the map is a quasi-isomorphism in each total weight. It extends to the degreewise product over weights, using completed tensor products by total weight. The completed coalgebras are conilpotent in each weight; no global conilpotence is asserted for infinite sums.

Proof. Put $C_n(\mathfrak{g}) = \Lambda^n \mathfrak{g}$ in homological degree n, with differential

$$d(x_1 \wedge \cdots \wedge x_n) = \sum_{i<j} (-1)^{i+j+1} [x_i, x_j] \wedge x_1 \wedge \cdots \widehat{x_i} \cdots \widehat{x_j} \cdots \wedge x_n.$$

Its coproduct splits the exterior factors between two ordered sublists, with the sign of the corresponding permutation. The ordinary normalized bar differential joins positions $i, i+1$ with sign $(-1)^{i-1}$. Define

$$\kappa(x_1 \wedge \cdots \wedge x_n) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) [x_{\sigma(1)} | \cdots | x_{\sigma(n)}].$$

At degree zero this map preserves the unit. At degree two its differential is $[x_1 x_2 - x_2 x_1] = [[x_1, x_2]]$. In every degree, pair permutations whose merged adjacent entries occur in opposite orders. Their difference is the bracket of those entries, with the displayed exterior sign. This proves the chain-map equation. Cutting an antisymmetrized word amounts to choosing an ordered subset and antisymmetrizing its two subwords. The two procedures give the same permutations and signs, proving the coalgebra identity.

Filter $U(\mathfrak{g})$ by PBW degree, the bar by total PBW degree, and $\Lambda^n \mathfrak{g}$ by degree n. PBW identifies the associated graded algebra with $S(\mathfrak{g})$. The associated graded comparison is the antisymmetrization $\Lambda(\mathfrak{g}) \to B(S(\mathfrak{g}))$, with zero differential on its source. To verify that it is a quasi-isomorphism, consider the free Koszul resolution $S(V) \otimes \Lambda(V)$ for $V = \mathfrak{g}$. For a basis (x_i) of V, its differential is $\partial = \sum_i x_i \iota_i$, where ι_i contracts the exterior factor x_i. On polynomial degree p and exterior degree q, with $p + q > 0$, the map

$$h = \frac{1}{p+q} \sum_i (x_i \wedge -) \, \frac{\partial}{\partial x_i}$$

satisfies $\partial h + h\partial = 1$. Only finitely many summands act on each element, even when V is infinite-dimensional. The identity follows by cancellation of cross terms and the Euler identity in the two degrees. Thus this free resolution has augmentation homology k only. The free normalized bar resolution has the same homology by its extra-degeneracy contraction. Antisymmetrization defines an augmentation-preserving comparison between these free resolutions. The simplicial bar sign convention is related to the displayed bar convention by multiplication by $(-1)^n$ in degree n. Free bases permit a reverse comparison and homotopies to be lifted inductively, since every positive-degree cycle in a resolution is a boundary. Tensoring these homotopies with k proves the asserted quasi-isomorphism on associated graded complexes.

Every cycle in the cone of κ has finite PBW filtration degree. A primitive for its highest associated graded part lowers that degree; iteration terminates because the filtration is bounded below. The cone is acyclic, proving the ordinary assertion. For a positive weight grading, PBW degree is at most total weight. The same argument is therefore finite within each weight. Finally, kernels and images of a degreewise product differential are products of kernels and images, so the componentwise quasi-isomorphisms extend to the stated completion. □

The ordinary comparison does not itself construct a morphism in $\mathrm{FactD}(X)$. For geometric coefficient systems, one must give compatible enveloping and Chevalley constructions on every chart and coherent factorization equivalences for the whole Chevalley complex. On each disjointness locus, with restriction denoted by $j^!$, the required equation is

$$f_{B,I}\, j^! \kappa_I = (\kappa_{I_1} \boxtimes \kappa_{I_2})\, f_{C,I}, \qquad I = I_1 \sqcup I_2.$$

Here $f_{C,I}$ and $f_{B,I}$ are the specified Chevalley and bar factorization maps. The bar map may use the signed shuffle comparison under its geometric functor hypotheses. The Chevalley factorization maps, collision compatibility, refinement coherence, and continuity of the completed maps remain additional geometric data to construct or verify. Internal tensor formulas and PBW alone do not provide them.

Calculation 50.2 (Finite-dimensional simple case)**.** For a simple Lie algebra of rank r with exponents $m_1, \dots, m_r$, the Chevalley cohomology is the exterior algebra on primitive classes of degrees $2m_i + 1$. The corresponding homology Poincaré polynomials are

$$\mathrm{Bar}^{\perp} U(\mathfrak{sl}_2) : 1 + t^3, \qquad \mathrm{Bar}^{\perp} U(\mathfrak{sl}_3) : 1 + t^3 + t^5 + t^8, \qquad \mathrm{Bar}^{\perp} U(\mathfrak{g}_2) : 1 + t^3 + t^{11} + t^{14}.$$

The corresponding Betti sequences for $\mathfrak{sl}_2$ and $\mathfrak{sl}_3$ are $1, 0, 0, 1$ and $1, 0, 0, 1, 0, 1, 0, 0, 1$. For a finite-dimensional Lie algebra, the Chevalley chain dimensions are $\binom{\dim \mathfrak{g}}{n}$.

For an ordinary augmented algebra A over $\mathbb{C}$, let $I = \ker(A \to \mathbb{C})$. Its normalized two-sided bar with trivial coefficients has homological chain spaces

$$B_0(A) = \mathbb{C}, \qquad B_n(A) = I^{\otimes n} \quad (n \geq 1).$$

Here n is bar length, independently of any internal weight or PBW degree. If I has finite dimension d, then $\dim B_n(A) = d^n$; in particular, an augmentation ideal of dimension three gives the chain counts 3^n. In a cohomological convention that places B_n in degree $-n$, its homology is denoted H^{-n} instead of H_n.

For $A = U(\mathfrak{sl}_2)$, PBW gives the basis $e^a h^b f^c$, with $a, b, c \geq 0$, where $[h, e] = 2e$, $[h, f] = -2f$, and $[e, f] = h$. The augmentation ideal has all these basis elements except 1. Consequently $B_n(A)$ is countably infinite-dimensional for every $n \geq 1$. Its total PBW filtration has finite-dimensional associated graded pieces, with generating function

$$\sum_{d \geq 0} \dim \mathrm{gr}^{\mathrm{PBW}}_d B_n(A)\, z^d = \big((1 - z)^{-3} - 1\big)^n.$$

In particular, 3^n is the dimension in the least possible total PBW degree $d = n$. These least-degree dimensions are 3, 9, 27, 81, 243 for $n = 1, 2, 3, 4, 5$; the full bar chain spaces are infinite-dimensional. The quotient of indecomposables is zero: the perfectness of $\mathfrak{sl}_2$ implies $I/I^2 = 0$.

The Chevalley chain model is different: its degree-n space is $\Lambda^n \mathfrak{sl}_2$, of dimensions 1, 3, 3, 1 in degrees 0, 1, 2, 3. With the convention $d(x \wedge y) = [x, y]$, the map $d : \Lambda^2 \mathfrak{sl}_2 \to \mathfrak{sl}_2$ sends

$$e \wedge h \longmapsto -2e, \qquad e \wedge f \longmapsto h, \qquad h \wedge f \longmapsto -2f,$$

and is an isomorphism. The next differential vanishes, since

$$d(e \wedge h \wedge f) = -2e \wedge f - h \wedge h - 2f \wedge e = 0.$$

Together with $d : \mathfrak{sl}_2 \to \mathbb{C}$ equal to zero, this gives Chevalley homology dimensions 1, 0, 0, 1. The ordinary bar–Chevalley comparison identifies these homology groups with those of $U(\mathfrak{sl}_2)$; it does not identify the two chain spaces.

50.2 The Virasoro row

The transverse presentation of the Virasoro family is the enveloping algebra of the positive Witt algebra

$$L_1 = \bigoplus_{n \geq 1} \mathbb{C}\, e_n, \qquad [e_i, e_j] = (j - i) e_{i+j},$$

to which Theorem 50.1 applies: L_1 is positively graded with one-dimensional weight spaces, so every weight has finitely many decompositions.

Theorem 50.3 (Pentagonal rigidity). *The bar homology of $U(L_1)$ satisfies*

$$\dim H_n\big(\mathrm{Bar}^{\perp} U(L_1)\big) = 2 \qquad \text{for every } n \geq 1,$$

one dimension in each of the weights $(3n^2 - n)/2$ and $(3n^2 + n)/2$. Its graded Euler character is

$$\chi_q\big(\mathrm{Bar}^{\perp} U(L_1)\big) = \prod_{n \geq 1} (1 - q^n), \tag{50.1}$$

and (50.1) is accounted for degree by degree with no residual freedom.

Proof. By Theorem 50.1 the bar of $U(L_1)$ is quasi-isomorphic to $C_*(L_1)$. Each weight subcomplex is finite-dimensional, so its homology has the same dimensions as the weightwise dual cohomology. Goncharova's calculation gives two dimensions in each degree $n \geq 1$, in the weights $(3n^2 \mp n)/2$. Identity (50.1) is the denominator product applied to L_1, where every root space is one dimensional and even, so $\mathrm{smult}(L_1)_n = 1$. Euler's pentagonal number theorem expands the product as $\sum_{m \in \mathbb{Z}} (-1)^m q^{m(3m-1)/2}$, whose support is exactly $\{(3n^2 \pm n)/2\}$ with coefficient $(-1)^n$ at each of $(3n^2 - n)/2$ and $(3n^2 + n)/2$. Comparing with $\sum_n (-1)^n \dim H_{n,w}$, each weight of the Euler character is saturated by a single class in the degree Goncharova predicts, so no further classes and no cancellation are available. $\square$

For $n = 1, 2, 3, 4$, the nonzero homology groups occupy the weight pairs $(1, 2)$, $(5, 7)$, $(12, 15)$, $(22, 26)$, one dimension in each. The corresponding nonzero coefficients of $\prod_{n \geq 1}(1 - q^n)$ occur at 0, 1, 2, 5, 7, 12, 15, 22, 26 with signs $+, -, -, +, +, -, -, +, +$.

50.3 Planar path sequences

A Motzkin path is a lattice path from $(0,0)$ to $(n,0)$, staying at nonnegative height, with steps $(1,1)$, $(1,0)$, and $(1,-1)$. Let $M(n)$ count all such paths of length n, and let $R(n)$ count those with no horizontal step at height zero. Both counts include the empty path: $M(0) = R(0) = 1$, while $M(1) = 1$ and $R(1) = 0$. For $n \geq 1$ consider the two sequences

$$a_n = M(n+1) - M(n) = 1, 2, 5, 12, 30, 76, \ldots, \qquad b_n = R(n+3) = 3, 6, 15, 36, 91, \ldots.$$

Proposition 50.4 (Motzkin and Riordan identities)**.** *For every $n \geq 0$,*

$$M(n) = R(n) + R(n+1), \qquad M(n+1) - M(n) = R(n+2) - R(n).$$

The ordinary generating functions of the two path counts, and hence those of a_n and b_n, belong to $\mathbb{Q}(z, \sqrt{1-2z-3z^2})$.

Proof. Work in the formal power-series ring $\mathbb{Q}[[z]]$, and write $M(z) = \sum_{n\geq 0} M(n)z^n$ and $R(z) = \sum_{n\geq 0} R(n)z^n$. A nonempty Motzkin path either starts with a horizontal step followed by an arbitrary Motzkin path, or has a unique first-return decomposition $UPDQ$, where U and D are the up and down steps, P and Q are arbitrary Motzkin paths, and P is drawn one unit above the horizontal axis. Thus

$$M = 1 + zM + z^2M^2.$$

A nonempty path counted by R has no initial horizontal step. Its first return similarly has the form $UPDQ$, with P arbitrary and Q counted by R. Consequently

$$R = 1 + z^2MR, \qquad R = (1 - z^2M)^{-1}.$$

The inverses here exist because their denominators have constant term one. Using $z^2M^2 = M - 1 - zM$, compute

$$(1 + zM)(1 - z^2M) = 1 + zM - z^2M - z^3M^2 = 1 + z.$$

It follows that $(1+z)R = 1 + zM$. Comparing coefficients of z^{n+1} gives $R(n+1) + R(n) = M(n)$ for all $n \geq 0$. Subtracting consecutive identities gives the difference formula.

Solving the quadratic equation for the formal series with constant term one gives

$$M(z) = \frac{1 - z - \sqrt{1-2z-3z^2}}{2z^2}, \qquad R(z) = \frac{1 + zM(z)}{1+z},$$

where the square root has constant term one. The apparent pole in the formula for M is removable as a formal series. Finite index shifts and differences preserve the indicated quadratic function field, proving the last assertion. These identities are consequences solely of the path decompositions. □

Remark 50.5 (Low-degree homology). Theorem 50.3 gives homology dimensions 2, 2, 2 in degrees 1, 2, 3, whereas a_1, a_2, a_3 are 1, 2, 5. These triples agree only in degree two. The degree-one discrepancy already excludes equality of the full homology dimension sequences.

Theorem 50.6 (Enveloping bars and planar path counts)**.** *Let $B_\bullet(A)$ be the ordinary normalized bar over $\mathbb{C}$, with trivial coefficients and homological degree given by bar length. With $a_n = M(n+1) - M(n)$ and $b_n = R(n+3)$ indexed by $n \geq 1$ as above,*

$$\dim H_1B_\bullet(U(\mathfrak{sl}_2)) = 0 \neq b_1 = 3, \qquad \dim H_1B_\bullet(U(L_1)) = 2 \neq a_1 = 1,$$

where $L_1 = \bigoplus_{m\geq 1} \mathbb{C}e_m$ and $[e_i, e_j] = (j-i)e_{i+j}$. Hence neither sequence is the bar homology dimension sequence of the corresponding enveloping algebra. Neither is its full bar chain dimension sequence: in both cases B_n is countably infinite-dimensional for every $n \geq 1$.

Proof. For any augmented algebra A with augmentation ideal I, the bar differential out of $B_1 = I$ is zero and the image from $B_2 = I \otimes I$ is I^2. Therefore

$$H_1 B_\bullet(A) = I/I^2.$$

For an ordinary Lie algebra $\mathfrak{g}$, the inclusion of generators induces an isomorphism

$$\mathfrak{g}/[\mathfrak{g},\mathfrak{g}] \xrightarrow{\sim} I/I^2, \qquad I = \ker(U(\mathfrak{g}) \to \mathbb{C}).$$

Indeed the generators span I/I^2, and $[x, y] = xy - yx$ vanishes there. For injectivity, equip $\mathbb{C} \oplus \mathfrak{g}/[\mathfrak{g},\mathfrak{g}]$ with the square-zero product on its second summand. The quotient map from $\mathfrak{g}$ to that summand is a Lie map into the commutator Lie algebra of this associative algebra. The universal property of $U(\mathfrak{g})$ extends it to an augmented algebra map, which kills I^2 and provides an inverse on the displayed quotient.

The brackets $[h, e] = 2e$, $[h, f] = -2f$, and $[e, f] = h$ show that $\mathfrak{sl}_2$ equals its commutator algebra. For L_1, every nonzero bracket has weight at least three. Conversely, for $m \geq 3$,

$$e_m = \frac{1}{m-2}[e_1, e_{m-1}].$$

Thus $[L_1, L_1] = \bigoplus_{m\geq 3} \mathbb{C}e_m$, and its abelianization has basis the classes of e_1, e_2, in weights one and two. This proves both degree-one assertions without using any higher-degree calculation.

PBW gives a countably infinite basis of the augmentation ideal of either enveloping algebra. In the L_1 case the basis consists of nonempty ordered monomials in the countable set $e_1, e_2, \ldots$, each monomial having finite length. Finite tensor powers of either augmentation ideal are therefore countably infinite-dimensional. The path counts are finite in every degree, so they cannot count these full chain spaces. □

In particular, a complex whose homology dimensions are b_n cannot be quasi-isomorphic to $B_\bullet(U(\mathfrak{sl}_2))$, and one whose homology dimensions are a_n cannot be quasi-isomorphic to $B_\bullet(U(L_1))$, with the stated degree convention. A discrepancy of chain dimensions alone would not imply this conclusion; the degree-one homology discrepancy does.

The weight grading gives a different, finite comparison. An ordered PBW monomial of weight w specifies a partition of w, so for $n \geq 1$,

$$\sum_{w\geq 0} \dim B_n(U(L_1))_w q^w = \left(\prod_{m\geq 1}(1-q^m)^{-1} - 1\right)^n.$$

Each coefficient is finite. Thus a weightwise dimension, a dimension after a specified weight cutoff, and an ungraded dimension are distinct invariants. In a weight completion, dimensions must likewise be interpreted weightwise; the countable algebraic dimensions above refer to the ordinary direct-sum enveloping algebras.

Remark 50.7 (Chiral coefficients and transverse multiplication). Independently of these scalar calculations, a Virasoro chiral coefficient does not determine the transverse bar: a fixed coefficient may be combined with free, square-zero, enveloping, or quantum-plane transverse algebras whose bars differ, while the coefficient and the central charge are unchanged. A chain-level comparison with an ordered bar therefore also requires a specified transverse multiplication and a quasi-isomorphism from the proposed complex to that bar.

Chapter 51

The obstruction constitution

Each statement below names what is *not* determined, and therefore what must be supplied. Without them the preceding theorems admit false corollaries.

Proposition 51.1 (Where a mixed distributive law can live)**.** *The sector on which a mixed distributive law between $\otimes^!$ and $\star^{\mathrm{ch}}$ is geometrically defined is the common-decomposition sector, the image of the aligned idempotent e_{al}. Off that sector the two products admit no interchange, only the aligned retraction, so no distributive law can be written.*

Proof. A distributive law requires a natural transformation interchanging the two products in a fixed direction, subject to the coherence conditions of each. The canonical comparison in the additive model is the aligned inclusion ι with retraction π, and $\pi\iota = \mathrm{id}$ while $\iota\pi = e_{\mathrm{al}} \neq \mathrm{id}$ in general. A natural transformation subject to both coherence conditions therefore exists only where e_{al} acts as the identity. □

This locates a mixed law without constructing one. No example in the present corpus exhibits one; Zamolodchikov–Faddeev algebras are the natural candidate, since their spectral variable belongs to the chiral direction while their word belongs to the transverse direction.

Theorem 51.2 (No quantum group from dimension loss)**.** *The dimension of the kernel of $V^{\otimes n} \to \mathrm{Sym}^n V$ determines no R-matrix, associator, or quantum group. That kernel carries no multiplication, coproduct, monodromy, or coherence equation.*

Proof. The kernel is a representation of Σ_n and nothing more. Two algebras with the same underlying graded vector space, hence the same kernel dimensions in every arity, may have different multiplications and inequivalent representation categories; a free algebra and a quadratic quotient with the same Hilbert series is an example. A quantum-group structure is a coproduct together with coherence data, none of which is a function of a dimension. □

Theorem 51.3 (No modular object from topology alone)**.** *The identity $\partial^2 = 0$ on $\overline{\mathcal{M}}_{g,n}$ assigns no coefficient operations to boundary strata. A modular master element exists only after cyclic contractions and clutching-compatible coefficient maps have been supplied.*

Theorem 51.4 (No bulk from a boundary centre)**.** *The derived centre is terminal among algebraic central actions on the boundary algebra. It does not detect a decoupled bulk sector, so a physical bulk is not determined by the boundary chiral algebra unless a bulk-to-centre comparison is supplied and proved to be an equivalence.*

Proof. Terminality is the universal property of the centre. It is a statement about the boundary category and is insensitive to any sector acting trivially on it, so a bulk with trivial boundary action is invisible. □

Theorem 51.5 (No classification by central charge)**.** *Central charge determines the Virasoro anomaly coefficient. It does not determine the chiral operation, the representation category, the braid monodromy, or the higher-genus conformal blocks. Distinct holomorphic theories at equal central charge exist.*

Theorem 51.6 (No identification of anomalies of different types)**.** *A current-algebra level, a Virasoro central charge, a determinant-line curvature, a BRST ghost-number anomaly, and a Borcherds-product weight are invariants in different theories, living in different deformation complexes. They may be related by a theorem in a particular family. They are not one universal scalar, and no identity among them follows from numerical agreement.*

Proof. Each is a class in a distinct complex: an Atiyah-algebra extension class, a Virasoro cocycle, the curvature of a determinant line, a class in H^1 or H^2 of a BRST convolution complex, and a weight in the character lattice of an automorphic form. A map between two of these complexes is additional data; in its absence, equality of numbers is a coincidence of values, not an identification of classes. □

Theorem 51.7 (No unnamed comparison of bars)**.** *An asserted equality* $\mathrm{Bar}^{\perp}(A) = \mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A)$ *has no meaning without a functor carrying the pointwise multiplication problem to the chiral-convolution multiplication problem. The two have different underlying gradings, counting interval subdivisions and collision trees respectively. Agreement of Euler characteristics is not a comparison map.*

Theorem 51.8 (A scalar sequence is not homology data)**.** *A sequence of integers becomes bar homology only after a chain complex, a differential, a grading convention, and a rank computation or quasi-isomorphism are specified. Matching initial values, discriminants, or growth rates constructs no such map.*

Proof. Chain-space dimensions and homology dimensions differ by the ranks of the differentials, which are not functions of the chain dimensions. Calculation 50.2 exhibits an algebra whose bar chain dimensions are 3, 9, 27, 81, 243 and whose bar homology dimensions are 1, 0, 0, 1; Proposition 50.4 exhibits two sequences with the same discriminant and growth rate that are transforms of one another and of neither homology. □

Theorem 51.6 together with the central-charge statement retracts any programme treating a tuple of scalars drawn from these sources as a single classifying invariant, and any trace identity asserted among them without a named comparison morphism.

Book entrance: Mixed Holomorphic–Topological Deligne Theory

Hamiltonian jets, shifted cotangent fields, universal branes, Duflo quantization, and the all-arity open–closed map.

Book IX

Finite Hamiltonian geometry

Chapter 52

The formal symplectic plane

52.1 Hamiltonians and the Poisson bracket

Let $V = \mathbb{C}^2$ with coordinates (z_1, z_2) and symplectic form

$$\omega = \mathrm{d}z_1 \wedge \mathrm{d}z_2.$$

Let $\mathfrak{m} = (z_1, z_2)$. The pointed formal Hamiltonian Lie algebra is

$$\mathfrak{h} = \mathfrak{m}^2 \subset \mathbb{C}[\![z_1, z_2]\!], \qquad \{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g.$$

It consists of Hamiltonians whose vector fields fix the chosen Darboux point. The shifted order filtration $F^p\mathfrak{h} = \mathfrak{m}^{p+2}$ is complete and obeys

$$\{F^p\mathfrak{h}, F^q\mathfrak{h}\} \subset F^{p+q}\mathfrak{h}.$$

The affine Hamiltonian algebra $\mathfrak{h}^{\mathrm{aff}} = \mathbb{C}[\![z_1, z_2]\!]/\mathbb{C}$ also contains translations. Its linear sector is adjoined separately when a moving brane or center-of-mass coordinate is required.

Write $R_r = \mathrm{Sym}^r(V^\vee)$ for the homogeneous polynomials of degree r. The bracket obeys

$$\{R_r, R_s\} \subset R_{r+s-2}.$$

For $d \geq 2$ define the finite jet quotient

$$\mathfrak{h}_{\leq d} = \bigoplus_{r=2}^{d} R_r,$$

where every bracket component of degree greater than d is set equal to zero.

Theorem 52.1 (Finite Hamiltonian jet algebra)**.** *The truncated bracket makes $\mathfrak{h}_{\leq d}$ a finite-dimensional Lie algebra. Its dimension is*

$$\dim \mathfrak{h}_{\leq d} = \frac{d(d+3)}{2} - 2.$$

The quadratic part is

$$R_2 \cong \mathfrak{sl}_2,$$

and each R_r is the irreducible $\mathfrak{sl}_2$-module $\mathrm{Sym}^r(\mathbb{C}^2)$.

Proof. The Jacobiator of the polynomial Poisson bracket vanishes before truncation. Projection to degrees at most d preserves this identity because the discarded terms form a Lie ideal. The dimension is

$$\sum_{r=2}^{d} (r+1) = \frac{d(d+3)}{2} - 2.$$

Quadratic Hamiltonians act linearly on V and preserve ω; this identifies R_2 with $\mathfrak{sp}(V) = \mathfrak{sl}_2$. The standard action on homogeneous polynomials gives the final statement. □

52.2 Unimodularity

Theorem 52.2 (Unimodularity of every finite jet)**.** *For every $d \geq 2$ and every $x \in \mathfrak{h}_{\leq d}$,*

$$\mathrm{Tr}_{\mathfrak{h}_{\leq d}}(\mathrm{ad}_x) = 0.$$

Proof. A homogeneous Hamiltonian of degree greater than two raises polynomial degree. Its adjoint action is strictly upper triangular in the decomposition $\bigoplus_{r=2}^{d} R_r$, hence has trace zero. A quadratic Hamiltonian belongs to $\mathfrak{sl}_2$. Its action on each finite-dimensional $\mathfrak{sl}_2$-module R_r has trace zero. Linearity gives the result. □

Corollary 52.3 (Canonical finite BV measure)**.** *The linear odd symplectic space $T^*[-1]\mathfrak{h}_{\leq d}[1]$ has a translation-invariant Berezinian for which the cotangent BF Hamiltonian satisfies the quantum master equation.*

Proof. Choose dual linear coordinates on the shifted cotangent space and use their constant Berezinian. The classical term vanishes by Jacobi, while the BV divergence is the adjoint trace. Unimodularity makes that trace zero. □

Chapter 53

The shifted cotangent theory

53.1 The Lie algebra and its functions

Put

$$\mathfrak{g}_d = T^*[-1]\mathfrak{h}_{\le d} = \mathfrak{h}_{\le d} \ltimes \mathfrak{h}_{\le d}^\vee[-1],$$

where $\mathfrak{h}_{\le d}$ acts on the dual by the coadjoint action and the shifted dual is abelian.

Proposition 53.1 (Chevalley–Poisson identification)**.** *There is a canonical isomorphism of commutative dg algebras*

$$C^*_{\mathrm{CE}}(\mathfrak{g}_d) \cong C^*\big(\mathfrak{h}_{\le d}, \mathrm{Sym}(\mathfrak{h}_{\le d})\big).$$

The differential on the right is the Chevalley differential for the adjoint action on $\mathrm{Sym}(\mathfrak{h}_{\le d})$*. Under the identification* $\mathrm{Sym}(\mathfrak{h}_{\le d}) = \mathcal{O}(\mathfrak{h}^*_{\le d})$*, it is the linear Poisson differential.*

Proof. The dual of $\mathfrak{h}^\vee_{\le d}[-1]$ shifted by $[-1]$ is $\mathfrak{h}_{\le d}$ in degree zero. Chevalley cochains therefore contain polynomial functions on $\mathfrak{h}^*_{\le d}$ and exterior ghosts for $\mathfrak{h}_{\le d}$. The semidirect bracket produces the coadjoint action, which is the Hamiltonian action of the linear Poisson tensor. □

53.2 The BV action

Choose dual bases e_a and e^a. Let c^a be the ghost coordinates and b_a the cotangent coordinates. The finite action is

$$S_d = \frac{1}{2} b_c f^c{}_{ab} c^a c^b.$$

The canonical BV bracket satisfies

$$\frac{1}{2}\{S_d, S_d\} = 0$$

by Jacobi.

Theorem 53.2 (Exact finite quantum master equation)**.** *With the translation-invariant BV Laplacian,*

$$\Delta S_d = \sum_b \mathrm{Tr}(\mathrm{ad}\, e_b) c^b = 0.$$

Consequently

$$\frac{1}{2}\{S_d, S_d\} + \hbar \Delta S_d = 0$$

for every d.

Proof. Contracting one field and one antifield in the cubic Hamiltonian gives the divergence of the adjoint vector field. The preceding unimodularity theorem makes every coefficient vanish. □

Chapter 54

The universal boundary brane

54.1 Polarization

Place the cotangent theory on a half-plane. The standard Lagrangian boundary condition fixes the cotangent field and leaves the $\mathfrak{h}_{\leq d}$ field free. Boundary observables form the Rees enveloping algebra

$$U_\hbar(\mathfrak{h}_{\leq d}) = T(\mathfrak{h}_{\leq d})[\![\hbar]\!]/\langle x\,y - y\,x - \hbar[x,y]\rangle.$$

The Poincaré–Birkhoff–Witt symmetrization gives a filtered vector-space isomorphism

$$\mathrm{sym}_\hbar : \mathrm{Sym}(\mathfrak{h}_{\leq d})[\![\hbar]\!] \xrightarrow{\sim} U_\hbar(\mathfrak{h}_{\leq d}).$$

Theorem 54.1 (Boundary quantization)**.** *The factorization algebra of boundary observables of the finite cotangent theory is locally constant and its value on an interval is $U_\hbar(\mathfrak{h}_{\leq d})$. Fusion of adjacent intervals is multiplication in the enveloping algebra.*

Proof. Canonical quantization turns the linear boundary phase space $\mathfrak{h}^*_{\leq d}$ into the Gutt star algebra. For a linear Poisson bracket the Gutt product is transported by PBW to $U_\hbar(\mathfrak{h}_{\leq d})$. Excision identifies ordered interval fusion with multiplication. □

Book X

Relative Hochschild comparison

Chapter 55

Hochschild cochains of the universal brane

55.1 The coefficient ring and the Chevalley differential

The Rees relation scales both the Lie bracket and its adjoint action. Its Chevalley differential must retain the same parameter. For example, take the quadratic jet $\mathfrak{l} = \mathfrak{sl}_2$ with $[h, e] = 2e$. Modulo $\hbar$, the enveloping algebra is commutative, so the Hochschild differential of e vanishes. The unscaled Chevalley differential instead satisfies $(d_{\mathrm{CE}} e)(h) = 2e$.

Let $\mathfrak{l}$ be a finite-dimensional complex Lie algebra. For $N \geq 0$, put

$$R_N = \mathbb{C}[\hbar]/(\hbar^{N+1}), \qquad \mathfrak{l}_N = R_N \otimes_{\mathbb{C}} \mathfrak{l}, \qquad [x, y]_N = \hbar[x, y],$$

and

$$A_N = U_{R_N}(\mathfrak{l}_N) = T_{R_N}(\mathfrak{l}_N)/(xy - yx - \hbar[x, y]).$$

Here N denotes the order in $\hbar$, independently of a Chan–Paton rank. Write $\bar{A}_N$ for the augmentation ideal and $S = \mathrm{Sym}_{\mathbb{C}} \mathfrak{l}$. PBW symmetrization is the R_N-module isomorphism

$$\mathrm{sym}_N : S \otimes R_N \longrightarrow A_N, \qquad x_1 \cdots x_r \longmapsto \frac{1}{r!} \sum_{\sigma \in S_r} x_{\sigma(1)} \cdots x_{\sigma(r)}.$$

The source has its ordinary symmetric product. Use normalized relative Hochschild cochains

$$C^n_{\mathrm{HH}}(A_N/R_N, A_N) = \mathrm{Hom}_{R_N}(\bar{A}_N^{\otimes_{R_N} n}, A_N)$$

with differential

$$\begin{aligned}(bF)(a_1, \dots, a_{n+1}) = {} & a_1 F(a_2, \dots, a_{n+1}) \\ & + \sum_{i=1}^{n} (-1)^i F(a_1, \dots, a_i a_{i+1}, \dots, a_{n+1}) \\ & + (-1)^{n+1} F(a_1, \dots, a_n) a_{n+1}.\end{aligned}$$

Let d_{CE} act on $C^n(\mathfrak{l}, S) = \mathrm{Hom}_{\mathbb{C}}(\Lambda^n \mathfrak{l}, S)$ by the adjoint Chevalley differential.

Theorem 55.1 (Relative Rees comparison)**.** *Antisymmetrization defines a quasi-isomorphism of R_N-complexes*

$$\mathrm{Alt}_N : C^{\bullet}_{\mathrm{HH}}(A_N/R_N, A_N) \longrightarrow \big(C^{\bullet}(\mathfrak{l}, S) \otimes R_N, \hbar d_{\mathrm{CE}}\big),$$

given by

$$(\mathrm{Alt}_N F)(x_1, \dots, x_n) = \mathrm{sym}_N^{-1} \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) F(x_{\sigma(1)}, \dots, x_{\sigma(n)}).$$

These maps commute with coefficient reduction in N.

Proof. Write $A = A_N$, $R = R_N$, and $A^e = A \otimes_R A^{\mathrm{op}}$. In homological degree n, set

$$K_n = A \otimes_R \Lambda_R^n \mathfrak{l}_N \otimes_R A.$$

Its augmentation is multiplication. For $X = x_1 \wedge \cdots \wedge x_n$, let $X_{\widehat{i}}$ and $X_{\widehat{ij}}$ omit the indicated entries without reordering. Define

$$\begin{aligned}\partial(a \otimes X \otimes b) = {} & \sum_i (-1)^{i+1}\big(a x_i \otimes X_{\widehat{i}} \otimes b - a \otimes X_{\widehat{i}} \otimes x_i b\big) \\ & + \hbar \sum_{i<j} (-1)^{i+j} a \otimes [x_i, x_j] \wedge X_{\widehat{ij}} \otimes b.\end{aligned}$$

The normalized relative bimodule bar has terms $\mathcal{B}_n = A \otimes_R \bar{A}^{\otimes_R n} \otimes_R A$. Define

$$\iota_n(a \otimes X \otimes b) = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma)\, a[x_{\sigma(1)}| \cdots |x_{\sigma(n)}]b.$$

The outer bar faces give the first sum in ∂. Pair adjacent transpositions in the internal faces and use $xy - yx = \hbar[x, y]$ to obtain the second sum. For two inputs this reads

$$\begin{aligned}b\big(1[x|y]1 - 1[y|x]1\big) = {} & x[y]1 - y[x]1 + 1[x]y - 1[y]x \\ & - \hbar\, 1[[x, y]]1.\end{aligned}$$

Thus $b\iota = \iota\partial$. PBW makes $\mathfrak{l}_N$ a free summand of $\bar{A}$. Since every factorial is invertible in R, ι is injective. The identity $b^2 = 0$ then gives $\partial^2 = 0$.

Give K_n filtration degree equal to its outer PBW degrees plus n. The associated graded is the diagonal Koszul complex of $\mathrm{Sym}_R(\mathfrak{l}_N)$. Its differential uses differences of left and right polynomial generators. A linear change of variables makes these independent polynomial variables. They form a regular sequence even when R has nilpotents. For an augmented cycle, lift a primitive of its highest-degree symbol and subtract its differential. The filtration degree decreases, so this procedure terminates. Hence $K_\bullet \to A$ is a free bimodule resolution. The relative bar is also free over A^e, with its usual unit-insertion contraction. The map ι lifts the identity of A. Projectivity constructs its chain homotopy inverse degree by degree. Applying $\mathrm{Hom}_{A^e}(-, A)$ gives a cochain homotopy equivalence.

The induced differential on $\mathrm{Hom}_R(\Lambda_R^n \mathfrak{l}_N, A)$ is

$$\begin{aligned}(\delta_N \phi)(x_1, \dots, x_{n+1}) = {} & \sum_i (-1)^{i+1} [x_i, \phi(x_1, \dots, \widehat{x_i}, \dots, x_{n+1})]_A \\ & + \hbar \sum_{i<j} (-1)^{i+j} \phi([x_i, x_j], x_1, \dots, \widehat{x_i}, \dots, \widehat{x_j}, \dots, x_{n+1}).\end{aligned}$$

Commuting x past the factors of a symmetrized monomial gives

$$[x, \mathrm{sym}_N(f)]_A = \hbar\, \mathrm{sym}_N(\mathrm{ad}_x f).$$

Both sums therefore become $\hbar$ times the ordinary Chevalley differential. Precomposition with ι gives Alt_N. All formulas commute with coefficient reduction. □

55.2 The formal coefficient limit

Put $R = \mathbb{C}[\![\hbar]\!]$ and $A = \varprojlim_N A_N$. PBW identifies its underlying module with $S[\![\hbar]\!]$. Each coefficient of $\hbar$ is a polynomial. Define

$$A^{\widehat{\otimes}_R n} = \varprojlim_N A_N^{\otimes_{R_N} n}, \qquad C^\bullet_{\mathrm{HH,cont}}(A/R, A) = \varprojlim_N C^\bullet_{\mathrm{HH}}(A_N/R_N, A_N).$$

Equivalently, these are continuous R-multilinear normalized cochains on the completed relative tensors.

Corollary 55.2 (Formal relative Rees comparison)**.** *The compatible antisymmetrizations induce a quasi-isomorphism*

$$C^{\bullet}_{\mathrm{HH},\mathrm{cont}}(A/R, A) \longrightarrow \big(C^{\bullet}(\mathfrak{l}, S)[\![\hbar]\!], \hbar d_{\mathrm{CE}}\big).$$

Proof. Coefficient reduction is surjective in each cochain degree. For Hochschild cochains, lift outputs on the PBW input basis. The Chevalley exterior input has finite rank, so its outputs also lift. The comparison cones are acyclic and their transition maps are degreewise surjective. The long exact sequence shows that every transition kernel is acyclic. Choose a primitive of a compatible cycle at level zero. Lift it one level and correct its differential by a primitive in the transition kernel. Iteration gives compatible primitives at every order. The inverse-limit cone is therefore acyclic. □

The coefficient ring is part of this result. Absolute $\mathbb{C}$-linear cochains also detect variations of the coefficient parameter. For $\mathfrak{l} = 0$, the continuous absolute derivation $\partial_\hbar$ on R has no relative counterpart. For nonabelian $\mathfrak{l}$, the associated graded for the $\hbar$ filtration has zero differential, whereas d_{CE} need not vanish.

Chapter 56

The Kontsevich–Duflo map

56.1 Linear Poisson formality

For $\mathfrak{l} = \mathfrak{h}_{\leq d}$, put $A_d = U_\hbar(\mathfrak{h}_{\leq d})$ and

$$\mathcal{P}_d = \big(C^*(\mathfrak{h}_{\leq d}, \operatorname{Sym} \mathfrak{h}_{\leq d})[\![\hbar]\!], \hbar d_{\mathrm{CE}}\big).$$

This is the polynomial polyvector complex on $\mathfrak{h}^*_{\leq d}$ with differential $[\hbar\pi_d, -]$ for the linear Poisson tensor π_d. The relative comparison gives $C^\bullet_{\mathrm{HH,cont}}(A_d/R, A_d) \to \mathcal{P}_d$ as a map of complexes. Higher operations require a formality map with specified operadic structure.

Fix Kontsevich's formality morphism $\mathcal{U}$ on the affine space $\mathfrak{h}^*_{\leq d}$. Its twist by $\hbar\pi_d$ defines a star product $\star_d$ and an L_∞ quasi-isomorphism on the degree-shifted complexes. Let $T_d : (\operatorname{Sym} \mathfrak{h}_{\leq d}[\![\hbar]\!], \star_d) \to A_d$ be the algebra identification fixing the linear generators. This identification includes the equivalence between the Kontsevich and PBW products. PBW symmetrization alone identifies the underlying modules. The first Taylor coefficient of the twisted comparison is

$$\Phi_d(v) = (T_d)_* \sum_{r \geq 0} \frac{1}{r!} \mathcal{U}_{r+1}(\underbrace{\hbar\pi_d, \ldots, \hbar\pi_d}_{r}, v).$$

It has source $\mathcal{P}_d$ and target $C^\bullet_{\mathrm{HH,cont}}(A_d/R, A_d)$. At each order in $\hbar$, polynomial inputs have polynomial outputs. Its reduction modulo $\hbar$ is the Hochschild–Kostant–Rosenberg map, with the normalized antisymmetrization convention. That map is a quasi-isomorphism for the polynomial algebra. The complete $\hbar$ filtration then proves that Φ_d is a quasi-isomorphism. The higher Taylor coefficients supply the L_∞ identities after shifting by $[1]$. These statements use the affine formality construction [70, Sections 6 and 8.3].

On invariant polynomials, the standard Duflo normalization gives

$$\mathrm{Duf}_\hbar = \mathrm{sym}_\hbar \circ j^{1/2}(\hbar\partial), \qquad j^{1/2}(x) = \operatorname*{det}^{1/2}_{\mathfrak{h}_{\leq d}}\left(\frac{\sinh(\mathrm{ad}_x/2)}{\mathrm{ad}_x/2}\right).$$

It is an algebra isomorphism from $(\operatorname{Sym} \mathfrak{h}_{\leq d})^{\mathfrak{h}_{\leq d}}[\![\hbar]\!]$ to $Z(A_d)$, by the Duflo theorem [70, Section 8.3.4]. The square root has constant term one. This assertion concerns invariant degree-zero cohomology. It does not identify antisymmetrization with a homotopy-E_2 map.

56.2 The boundary action

The brace operad acts on $(C^\bullet_{\mathrm{HH,cont}}(A_d/R, A_d), A_d)$. Its closed color is the relative derived centre. A homotopy-E_2 refinement means a morphism or zigzag

$$\widetilde{\Phi}_d : \mathcal{P}_d \simeq C^\bullet_{\mathrm{HH,cont}}(A_d/R, A_d)$$

in complete R-linear homotopy-E_2 algebras, with the chosen Gerstenhaber structure on $\mathcal{P}_d$. A compatible Swiss-cheese refinement also specifies the mixed operations on the pair and their coherence homotopies.

Corollary 56.1 (Open–closed structure with a refinement)**.** *Given these refinements, the pair* $(\mathcal{P}_d, A_d)$ *has the transported homotopy Swiss-cheese action. Its open product is the enveloping product. The specified closed and mixed Taylor coefficients determine its closed bracket and boundary action.*

Proof. Transport the two-colored operations through the supplied equivalence of complete operadic algebras. The coherence homotopies are part of this equivalence. □

Chapter 57

The formal Hamiltonian limit

57.1 Weight completion and comparison data

The Hamiltonian Lie algebra and its boundary algebra have quotient systems

$$\mathfrak{h} = \varprojlim_{d} \mathfrak{h}_{\le d}, \qquad \widehat{U}_\hbar(\mathfrak{h}) = \varprojlim_{d} U_\hbar(\mathfrak{h}_{\le d}).$$

A Lie quotient does not by itself define a covariant map between Chevalley cochains with adjoint coefficients. Nor does an algebra quotient define a map between all Hochschild cochains with values in the algebra itself. A cochain must descend through the input quotient and the output quotient. Consequently the coefficient limit of Corollary 55.2 is distinct from this jet limit.

A *compatible jet comparison* consists of the following data. Choose complete R-linear cochain models P_d, H_d with transition maps, levelwise comparisons $P_d \simeq \mathcal{P}_d$ and $H_d \simeq C^\bullet_{\mathrm{HH,cont}}(A_d/R, A_d)$, and compatible quasi-isomorphisms $P_d \to H_d$. Require degreewise surjective transitions, or an explicit derived inverse-limit construction with the same exactness property. Specify identifications

$$\varprojlim_{d} P_d \simeq \mathcal{P}_\infty, \qquad \varprojlim_{d} H_d \simeq C^\bullet_{\mathrm{HH,cont}}(\widehat{U}_\hbar(\mathfrak{h})/R, \widehat{U}_\hbar(\mathfrak{h})),$$

where $\mathcal{P}_\infty$ is the proposed continuous Hamiltonian polyvector complex with differential $[\hbar\pi, -]$. The topology on the right combines the stated Hamiltonian weight completion with the $\hbar$ completion. The identifications must use this same topology. For higher operations, also supply compatible homotopy-E_2 and Swiss-cheese refinements whose operations converge in that topology.

Theorem 57.1 (Continuous comparison from compatible jets)**.** *A compatible jet comparison induces a continuous quasi-isomorphism*

$$\Phi_\infty : \mathcal{P}_\infty \xrightarrow{\sim} C^\bullet_{\mathrm{HH,cont}}(\widehat{U}_\hbar(\mathfrak{h})/R, \widehat{U}_\hbar(\mathfrak{h})).$$

Compatible homotopy-E_2 and Swiss-cheese refinements give the corresponding complete open–closed operations.

Proof. For surjective transitions, the cones of the levelwise maps are acyclic inverse systems with surjective maps in each degree. The primitive-lifting argument of Corollary 55.2 proves exactness of their limit. For a derived inverse limit, use its invariance under levelwise equivalences. Apply the supplied identifications of the two limits. Compatibility and convergence of the operadic maps give the final assertion. □

The finite PBW comparison constructs the relative centre at each fixed jet. A continuous field-theory comparison additionally requires the jet datum and a map from the specified field observables to $\mathcal{P}_\infty$. Universal graph formulas alone do not provide the cochain transition maps.

Principle 57.2 (Universal brane principle). The boundary algebra $\widehat{U}_\hbar(\mathfrak{h})$ is the universal formal Hamiltonian brane. Finite matrix branes are representation-theoretic images of this algebra. Its complete relative derived centre is the algebraic target for bulk observables. A field-theory identification requires the continuous comparison data stated above.

Book XI

The finite-jet laboratory

Chapter 58

The first three Hamiltonian jets

58.1 Quadratic Hamiltonians

Set

$$e = \frac{1}{2}z_1^2, \qquad f = -\frac{1}{2}z_2^2, \qquad h = -z_1 z_2.$$

A direct differentiation gives

$$\{h, e\} = 2e, \qquad \{h, f\} = -2f, \qquad \{e, f\} = h.$$

Thus $\mathfrak{h}_{\le 2} = R_2$ is $\mathfrak{sl}_2$ with its standard Chevalley basis. The finite cotangent theory for $d = 2$ is the linear BV theory of $T^*[-1]\mathfrak{sl}_2$ and its boundary algebra is $U_\hbar(\mathfrak{sl}_2)$.

58.2 Cubic Hamiltonians

Let

$$v_j = z_1^{3-j} z_2^j, \qquad 0 \le j \le 3.$$

Then

$$\{h, v_j\} = (3 - 2j)v_j, \qquad \{e, v_j\} = j v_{j-1}, \qquad \{f, v_j\} = (3 - j)v_{j+1},$$

with the boundary terms understood as zero. Hence R_3 is the irreducible highest-weight-three module.

Proposition 58.1 (The cubic jet algebra)**.** *The degree-three jet algebra is the semidirect product*

$$\mathfrak{h}_{\le 3} \cong \mathfrak{sl}_2 \ltimes \mathrm{Sym}^3(\mathbb{C}^2),$$

where the second summand is abelian.

Proof. The bracket of two cubic polynomials has degree four and vanishes in the degree-three quotient. The bracket with a quadratic polynomial is the displayed $\mathfrak{sl}_2$ action. □

58.3 The first nonlinear layer

Write $w_k = z_1^{4-k} z_2^k$, $0 \le k \le 4$. In $\mathfrak{h}_{\le 4}$ the cubic modes satisfy

$$\{v_i, v_j\} = 3(j - i)w_{i+j-1}, \tag{58.1}$$

whenever the index lies between 0 and 4; the coefficient makes every remaining case vanish.

Calculation 58.2 (Two brackets)**.**

$$\{z_1^3, z_2^3\} = 9 z_1^2 z_2^2 = 9w_2, \qquad \{z_1^2 z_2, z_1 z_2^2\} = 3 z_1^2 z_2^2 = 3w_2.$$

The two independent cubic pairs reach the same quartic channel with amplitudes in the ratio $3 : 1$.

Proof. Differentiate each monomial in the two symplectic coordinates and substitute in the Poisson formula. The first pair gives $3z_1^2 \cdot 3z_2^2$. The second gives $2z_1z_2 \cdot 2z_1z_2 - z_1^2 \cdot z_2^2 = 3z_1^2z_2^2$. □

Theorem 58.3 (The quartic jet presentation)**.** *The Lie algebra $\mathfrak{h}_{\le 4}$ is generated by $e, f, h, v_0, v_1, v_2, v_3$. Its relations are the $\mathfrak{sl}_2$ relations, the irreducible action on the v_j, formula* (58.1)*, and the induced $\mathfrak{sl}_2$ action on R_4. The quartic subspace is abelian in the quotient.*

Proof. Quadratic and cubic generators produce every quartic monomial through (58.1). Brackets involving a quartic and a cubic have degree five, and brackets of two quartics have degree six, so both vanish after projection. Jacobi follows from the polynomial Poisson bracket before projection. □

Chapter 59

Moyal quantization in coordinates

59.1 The bidifferential formula

For polynomials on $\mathbb{C}^2$, the Moyal product is

$$f \star g = \sum_{n \geq 0} \frac{1}{n!} \left(\frac{\hbar}{2} \right)^n \sum_{r=0}^{n} (-1)^r \binom{n}{r} (\partial_1^{n-r} \partial_2^r f)(\partial_1^r \partial_2^{n-r} g). \tag{59.1}$$

Every polynomial pair gives a finite sum.

Theorem 59.1 (Quadratic exactness)**.** *For polynomials f, g of degree at most two,*

$$[f, g]_\star = \hbar \{f, g\}.$$

The quadratic generators therefore define a Lie representation of the Rees-scaled $\mathfrak{sl}_2$. Their associative algebra has an additional Casimir relation.

Proof. The commutator retains the odd powers of the Poisson bidifferential. Derivatives of order three vanish on quadratics, so the first-order term is the entire commutator. □

Proposition 59.2 (The oscillator quotient)**.** *Let $R_N = \mathbb{C}[\hbar]/(\hbar^{N+1})$ and let $W_N = R_N[q, p]$ have the Moyal product for $\{q, p\} = 1$. Let U_N have generators E, F, H with relations*

$$[H, E] = 2\hbar E, \qquad [H, F] = -2\hbar F, \qquad [E, F] = \hbar H.$$

Set $\Omega = EF + FE + H^2/2$. The assignments

$$E \longmapsto e = q^2/2, \qquad F \longmapsto f = -p^2/2, \qquad H \longmapsto h = -qp$$

induce an algebra isomorphism

$$U_N/(\Omega + 3\hbar^2/8) \xrightarrow{\sim} W_N^{(q,p) \mapsto (-q, -p)}.$$

These compatible isomorphisms also hold after $\hbar$-adic completion. If linear Hamiltonians are retained, the generated associative algebra is the full Weyl algebra.

Proof. Quadratic exactness proves the three commutator relations. The associative products are

$$\begin{aligned} e \star f &= -q^2 p^2/4 - \hbar qp/2 - \hbar^2/8, \\ f \star e &= -q^2 p^2/4 + \hbar qp/2 - \hbar^2/8, \\ h \star h &= q^2 p^2 - \hbar^2/4. \end{aligned}$$

Thus Ω maps to $-3\hbar^2/8$. Its commutator with each of E, F, H vanishes by the defining relations. Give these generators PBW degree one. Give an even monomial $q^a p^b$ degree $(a + b)/2$. The leading symbol of $\Omega + 3\hbar^2/8$ is

$C = 2EF + H^2/2$. It is a non-zero-divisor in $R_N[E, F, H]$, since $2C$ is monic in H. Centrality and this property give

$$\operatorname{gr}\big(U_N/(\Omega + 3\hbar^2/8)\big) = R_N[E, F, H]/(C).$$

Indeed, every element of the two-sided ideal is a multiple of the central generator. Its leading symbol is the product of the two leading symbols. The quotient has basis $E^a H^b F^c$ with $a, c \geq 0$ and $b = 0, 1$. Their leading symbols map to unit multiples of the distinct monomials $q^{2a+b} p^{2c+b}$, precisely the even monomials. Hence the associated-graded map is an isomorphism. Lifting highest-degree terms proves surjectivity and comparing them proves injectivity. Every element at fixed N has finite polynomial degree. The formulas commute with reduction in N, so their inverse limit gives the formal isomorphism. Finally q, p generate W_N and topologically generate its formal limit. □

The primitive coproduct on U_N does not descend to this quotient. Modulo $\hbar$, it sends the classical relation to

$$\Delta(C) = C \otimes 1 + 1 \otimes C + 2E \otimes F + 2F \otimes E + H \otimes H.$$

The cross-term is nonzero in the tensor square of $\mathbb{C}[E, F, H]/(C)$. For example, evaluation at $(E, F, H) = (1, 0, 0)$ in the first factor and $(0, 1, 0)$ in the second gives 2. Thus the oscillator quotient is an associative algebra, with no coproduct inherited from these primitive generators.

Calculation 59.3 (Cubic correction)**.** For $f = z_1^3$ and $g = z_2^3$,

$$f \star g - g \star f = 9\hbar z_1^2 z_2^2 + \frac{3}{2}\hbar^3.$$

The first term quantizes the quartic Hamiltonian channel. The constant is the first wheel-free ordering correction visible beyond the quadratic Lie algebra.

Proof. In (59.1), the odd orders are $n = 1$ and $n = 3$. The first contributes $9\hbar z_1^2 z_2^2$. The third contributes $2(\hbar/2)^3(3!) = 3\hbar^3/2$. □

59.2 Weyl generators

With the convention $\{q, p\} = 1$, the symbols satisfy $[q, p]_\star = \hbar$. Their Weyl-ordered differential operators are

$$\widehat{q} = q, \qquad \widehat{p} = -\hbar\partial_q, \qquad [\widehat{q}, \widehat{p}] = \hbar.$$

The abstract Weyl algebra is defined by this commutator relation. Weyl ordering transports its multiplication to the Moyal product. The differential-operator representation realizes these relations over $\mathbb{C}[\hbar]$ and its formal completion. The finite quotient argument above uses the abstract R_N-algebra, so it does not require a faithful differential-operator representation at nilpotent $\hbar$.

Chapter 60

Chevalley coordinates and the Poisson differential

Choose a basis e_a of $\mathfrak{h}_{\leq d}$ with structure constants $[e_a, e_b] = f^c{}_{ab} e_c$. Let c^a be the odd Chevalley coordinate and let x_a be the even coordinate on $\mathfrak{h}^*_{\leq d}$. Then

$$d_{\mathrm{CE}} c^c = -\frac{1}{2} f^c{}_{ab} c^a c^b, \qquad d_{\mathrm{CE}} x_a = f^c{}_{ba} c^b x_c. \tag{60.1}$$

The first equation encodes the Lie bracket. The second is the adjoint action on $\mathrm{Sym}(\mathfrak{h}_{\leq d})$, viewed as functions on the dual.

Proposition 60.1 (Hamiltonian interpretation)**.** *For a polynomial $F(x)$,*

$$d_{\mathrm{CE}} F = c^a \{x_a, F\}_{\mathrm{lin}}.$$

Consequently $d_{\mathrm{CE}}^2 = 0$ is the Jacobi identity for the linear Poisson tensor.

Proof. The linear Poisson bracket is $\{x_a, x_b\} = f^c{}_{ab} x_c$. The Leibniz rule gives (60.1) on every polynomial. The square is one half of the Schouten bracket of the Poisson tensor with itself. □

60.1 The $\mathfrak{sl}_2$ invariant

For the quadratic jet, identify the coordinate functions with e, f, h. The invariant

$$q = 2ef + \frac{1}{2} h^2$$

is d_{CE}-closed. Its formal Duflo image is $\Omega + \hbar^2/2$, where $\Omega = ef + fe + h^2/2$ in the Rees enveloping algebra. This is the degree-zero invariant comparison. Under the oscillator quotient of Proposition 59.2, its image is $\hbar^2/8$.

Chapter 61

Residue duality and the formal current algebra

61.1 The continuous dual

Let

$$\rho_{ab} = z_1^{-a-1} z_2^{-b-1}\, \mathrm{d}z_1 \mathrm{d}z_2, \qquad a, b \geq 0.$$

The Grothendieck residue pairing is

$$\mathrm{Res}_0\Big(z_1^i z_2^j \, \rho_{ab} \Big) = \delta_{ia}\delta_{jb}.$$

The continuous dual of $\mathbb{C}[\![z_1, z_2]\!]$ is the direct sum of the ρ_{ab}. The continuous dual of the pointed algebra $\mathfrak{h} = \mathfrak{m}^2$ is the span of the modes with $a + b \geq 2$. The two modes of total degree one pair with the translation sector of $\mathfrak{h}^{\mathrm{aff}}$.

Proposition 61.1 (Coadjoint current formula)**.** *For $f = z_1^i z_2^j$ and ρ_{ab},*

$$\mathrm{ad}^*_f(\rho_{ab})(g) = -\rho_{ab}(\{f, g\}).$$

In the residue basis this is a finite linear combination of modes of bidegree

$$(a - i + 1, b - j + 1).$$

Proof. The definition is the coadjoint action. The Poisson bracket lowers the total degree of $f\,g$ by two, so residue conservation fixes the displayed bidegree. The Poisson formula contains two derivative terms. □

The shifted cotangent current algebra

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[-1]$$

is therefore a complete topological Lie algebra with a continuous invariant pairing. Its finite jet quotients are precisely the $\mathfrak{g}_d$ used above.

Chapter 62

All-arity open–closed operations

62.1 Hochschild braces

For Hochschild cochains $F \in C^{p}(A, A)$ and $G_i \in C^{q_i}(A, A)$, the brace

$$F\{G_1, \dots, G_r\}$$

is the signed sum over ordered insertions of the G_i into the inputs of F. The brace identities state that a nested insertion is the sum over all ways of distributing the inner insertions among the outer slots.

Theorem 62.1 (Brace realization with a refinement)**.** *For $A = A_d$, supply the homotopy-E_2 and mixed refinements of Corollary 56.1. They identify the selected homotopy Gerstenhaber operations on $\mathcal{P}_d$ with the relative Hochschild centre operations. The chosen mixed refinement also identifies their action on the boundary algebra.*

Proof. The relative Hochschild complex has its brace operations. The supplied operadic equivalence preserves their coherent homotopies. An equivalence of complexes alone would not imply this conclusion. □

62.2 The first mixed identities

A relative Hochschild 1-cocycle D of an algebra concentrated in degree zero is an R-linear derivation:

$$D(ab) = D(a)b + aD(b).$$

For two such cocycles D, E, their Gerstenhaber bracket acts by

$$[D, E](a) = D(E(a)) - E(D(a)).$$

For a graded algebra and homogeneous derivations, these formulas become

$$D(ab) = D(a)b + (-1)^{|D||a|}aD(b),$$
$$[D, E](a) = D(E(a)) - (-1)^{|D||E|}E(D(a)).$$

Higher Hochschild cochains act through multilinear evaluations and braces. They need not act by derivations or carry a coproduct defining a diagonal action. A compatible mixed refinement transports these operations, including their homotopies.

Corollary 62.2 (Continuous open–closed structure with compatible jets)**.** *Under the jet comparison and mixed refinement hypotheses of Theorem 57.1, the pair $(\mathcal{P}_\infty, \widehat{U}_\hbar(\mathfrak{h}))$ has the corresponding complete homotopy Swiss-cheese action. Its closed color is equivalent to the continuous relative derived centre.*

Proof. The supplied finite operations commute with the transition maps and converge in the specified completion. Their limit acts on the two limit objects. The comparison theorem identifies the closed complex with continuous relative Hochschild cochains. □

Chapter 63

Finite Chan–Paton ranks

63.1 Rank one

For $N = 1$, the commuting equation is automatic. The boundary coordinate algebra is $\mathbb{C}[x, y]$, and

$$J_1(f) = f(x, y).$$

The necklace bracket reduces to the ordinary Poisson bracket. Restriction to $f \in \mathfrak{m}^2$ gives the defining pointed-Hamiltonian representation. The two linear Hamiltonians act by translations and recover the affine center-of-mass extension.

63.2 Rank two and Cayley–Hamilton reduction

For a 2×2 matrix X,

$$X^2 - (\operatorname{Tr} X)X + (\det X)I = 0.$$

Hence every trace containing two consecutive powers of X reduces to traces with X-degree at most one after adjoining $\operatorname{Tr} X$ and $\det X$. The same statement holds for Y. On the commuting locus, simultaneous diagonalization identifies the invariant algebra with the symmetric polynomials in two ordered pairs.

Proposition 63.1 (First finite-rank trace relation)**.** *For commuting 2×2 matrices,*

$$p_{20} = p_{10}^2 - 2 \det X, \qquad p_{02} = p_{01}^2 - 2 \det Y,$$

and

$$p_{11}^2 - p_{20} p_{02} = -\big(x_1 y_2 - x_2 y_1\big)^2.$$

The stable generators remain independent until a relation forced by the rank is reached.

Proof. Write the joint eigenvalues as (x_1, y_1) and (x_2, y_2) and expand both sides. □

63.3 Stable passage

The transition from rank N to rank $N + 1$ adds a zero eigenvalue. Each fixed weighted degree stabilizes once N exceeds that degree. The inverse limit is therefore the completed multisymmetric algebra. The stable trace is an exact quotient of the universal centre on the polynomial trace sector.

Book XII

Global holomorphic symplectic descent

Chapter 64

Fedosov descent

64.1 The Weyl algebroid

All Hochschild cochains and Morita centres in this chapter are continuous and relative to $R = \mathbb{C}[\![\hbar]\!]$. The symplectic globalization uses its own sheafwise formality and Fedosov resolution data, independently of the finite Hamiltonian comparison. Let (S, ω) be a holomorphic symplectic surface. The bundle of formal Weyl algebras has fibre the completed Weyl algebra of a formal symplectic plane. A holomorphic symplectic connection and a Fedosov one-form define a flat connection

$$D_F = \nabla + \frac{1}{\hbar}[\gamma, -]$$

whose flat sections form a deformation-quantization algebroid $\mathcal{W}_{S,\hbar}$.

Theorem 64.1 (Globalized formality on a holomorphic symplectic surface)**.** *Fix a stable formality morphism and a compatible homotopy-E_2 refinement. There is a sheafwise homotopy-E_2 quasi-isomorphism*

$$\Phi_S : (\mathrm{PV}_S[\![\hbar]\!], [\hbar\pi, -]) \xrightarrow{\sim} C^{\bullet}_{\mathrm{HH}}(\mathcal{W}_{S,\hbar}, \mathcal{W}_{S,\hbar}),$$

where $\pi = \omega^{-1}$. A change of holomorphic symplectic connection or Fedosov one-form gives a gauge-equivalent resolution and a homotopic quasi-isomorphism. Holomorphic symplectomorphisms act equivariantly on the resulting homotopy class.

Proof. Fedosov resolutions replace both sheaves by acyclic sheaves of formal fibrewise polyvectors and polydifferential operators. The fixed covariant formality morphism acts fibrewise and intertwines the Fedosov differentials. A change of connection or Fedosov one-form is a gauge transformation in the pronilpotent resolution and therefore gives a homotopy. □

Corollary 64.2 (Global mixed-HT centre)**.** *The deformation-quantization category* $\mathrm{Perf}(\mathcal{W}_{S,\hbar})$ *has Morita centre*

$$Z_{\mathrm{Mor}}\big(\mathrm{Perf}(\mathcal{W}_{S,\hbar})\big) \simeq \mathbb{R}\Gamma(S, \mathrm{PV}_S[\![\hbar]\!], [\hbar\pi, -]).$$

For a K3 surface this computes the symplectic Poisson centre in the stated sheafwise model. A Hamiltonian BF identification additionally requires a comparison from its Chevalley observables to this symplectic polyvector complex.

Proof. The Morita centre of perfect modules over an algebroid is the derived global section complex of its Hochschild cochains. Apply the globalized formality equivalence and then derived global sections. A K3 surface is holomorphic symplectic, so the same sheafwise calculation applies whenever the chosen globalization data are available. □

Chapter 65

Factorization and descent

65.1 The algebraic mixed-HT realization

The sheaf

$$\mathcal{P}_S = (\mathrm{PV}_S[\![\hbar]\!], [\hbar\pi, -])$$

is a sheaf of complete homotopy-E_2 algebras. Dunn additivity gives its locally constant factorization realization in two oriented topological directions. The holomorphic sheaf variable retains the dependence on S.

Theorem 65.1 (Algebraic factorization realization)**.** *The homotopy-E_2 sheaf $\mathcal{P}_S$ determines a complete mixed holomorphic–topological factorization algebra*

$$\mathcal{A}^{\mathrm{alg}}_{S,\mathrm{bulk}} \quad \textit{on} \quad \mathbb{R}^2_{\mathrm{top}} \times S.$$

Its universal boundary category is $\mathrm{Perf}(\mathcal{W}_{S,\hbar})$*, and the bulk–boundary morphism is a local equivalence*

$$\mathcal{A}^{\mathrm{alg}}_{S,\mathrm{bulk}} \xrightarrow{\sim} Z^{\mathrm{der}}_{\mathrm{Mor}}\big(\mathrm{Perf}(\mathcal{W}_{S,\hbar})\big).$$

On a formal Darboux chart this map is the symplectic polyvector-to-Weyl Hochschild comparison. Comparing it with the pointed Hamiltonian centre additionally requires a map between those two polyvector complexes.

Proof. The factorization envelope of a complete homotopy-E_2 algebra is locally constant on oriented two-disks. Sheafwise application to $\mathcal{P}_S$ gives the first assertion. The Fedosov algebroid quantizes the symplectic Poisson tensor on S. Globalized formality identifies its Hochschild cochains with $\mathcal{P}_S$, and Morita invariance identifies Hochschild cochains with the centre of the perfect-module category. The construction is local on S and therefore respects factorization descent. □

65.2 BV realization

The cyclic Lie sheaf of formal Hamiltonians has the shifted-cotangent action

$$S_{\mathrm{BF}}(A, B) = \int_{\mathbb{R}^2 \times S} \Big\langle B, (\mathrm{d} + \bar{\partial})A + \frac{1}{2}[A, A] \Big\rangle.$$

A comparison of its classical observables with $\mathcal{A}^{\mathrm{alg}}_{S,\mathrm{bulk}}$ must specify a map from Hamiltonian Chevalley cochains to the symplectic polyvector complex. The finite Lie algebra identification alone does not supply that map. A compatible gauge fixing, extension of graph weights to compactified configurations, and solution of the quantum master equation produce a perturbative factorization realization of the same algebraic bulk–boundary map. The finite jet zero modes carry the exact quantum measure constructed above.

Book XIII

Matrix branes and trace geometry

Chapter 66

The derived commuting stack

Let $X, Y \in \mathfrak{gl}_N$ and let $\psi \in \mathfrak{gl}_N[1]$. The Koszul differential

$$QX = QY = 0, \qquad Q\psi = [X, Y]$$

resolves the commuting equation. The simultaneous conjugation action gives the boundary cdga

$$A_N = C^*_{\mathrm{CE}}\big(\mathfrak{gl}_N, \mathrm{Sym}(\mathfrak{gl}_N \oplus \mathfrak{gl}_N \oplus \mathfrak{gl}_N[1])\big).$$

Proposition 66.1 (Representation of the universal brane)**.** *Substitution of matrices defines a Lie morphism*

$$\rho_N : \mathfrak{h} \longrightarrow \mathrm{Der}(A_N), \qquad f \longmapsto \delta_f,$$

which integrates to an algebra morphism

$$\widehat{U_\hbar}(\mathfrak{h}) \longrightarrow \mathrm{End}(A_N)[\![\hbar]\!], \qquad f \longmapsto \hbar\, \delta_f.$$

Thus the N-brane stack is a representation of the universal boundary algebra.

Proof. For a polynomial Hamiltonian f, cyclic differentiation gives the matrix vector field

$$\delta_f(X) = -\partial^{\mathrm{cyc}}_Y f(X, Y), \qquad \delta_f(Y) = \partial^{\mathrm{cyc}}_X f(X, Y).$$

The cyclic Euler identity implies $\delta_f([X, Y]) = 0$. Hence δ_f commutes with the Koszul differential $Q\psi = [X, Y]$. The same formulas are equivariant for simultaneous conjugation, so δ_f descends to a derivation of the Chevalley–Eilenberg quotient A_N. The trace-moment-map calculation gives

$$[\delta_f, \delta_g] = \delta_{\{f,g\}},$$

first for polynomial Hamiltonians and then continuously for the formal completion. This proves that ρ_N is a continuous Lie morphism. The assignment $f \mapsto \hbar\delta_f$ satisfies

$$[\hbar\delta_f, \hbar\delta_g] = \hbar^2 \delta_{\{f,g\}} = \hbar\,(\hbar\delta_{\{f,g\}}),$$

which is the Rees relation. The universal property of the completed Rees enveloping algebra gives the displayed morphism. It is the universal boundary action evaluated in the N-dimensional Chan–Paton representation. □

Chapter 67

Necklace Hamiltonians

Let

$$F = \mathbb{C}\langle x, y\rangle, \qquad F_{\mathrm{cyc}} = F/[F, F].$$

For a cyclic word f, define cyclic derivatives by cutting the word at every occurrence of the selected letter. Put

$$\delta_f(x) = -\partial_y^{\mathrm{cyc}} f, \qquad \delta_f(y) = \partial_x^{\mathrm{cyc}} f.$$

Theorem 67.1 (Necklace Lie algebra)**.** *The bracket*

$$\{f, g\}_{\mathrm{neck}} = \left[\partial_x^{\mathrm{cyc}} f\ \partial_y^{\mathrm{cyc}} g - \partial_y^{\mathrm{cyc}} f\ \partial_x^{\mathrm{cyc}} g\right]_{\mathrm{cyc}}$$

makes $F_{\mathrm{cyc}}/\mathbb{C}$ a Lie algebra. The assignment $f \mapsto \delta_f$ is a Lie morphism. It preserves the moment map:

$$\delta_f([x, y]) = 0.$$

Proof. The cyclic Euler identity is

$$[x, \partial_x^{\mathrm{cyc}} f] + [y, \partial_y^{\mathrm{cyc}} f] = 0.$$

It gives the moment-map statement. The canonical double Poisson bracket on the free algebra descends to the displayed necklace bracket; its Jacobi identity gives the first assertion. Evaluation on generators gives $[\delta_f, \delta_g] = \delta_{\{f,g\}_{\mathrm{neck}}}$. □

Theorem 67.2 (Trace moment map)**.** *For*

$$J_N(f) = \operatorname{Tr} f(X, Y),$$

one has

$$\{J_N(f), J_N(g)\}_{\mathcal{M}_N} = J_N\big(\{f, g\}_{\mathrm{neck}}\big),$$

where $\mathcal{M}_N = \mathfrak{gl}_N^2$ carries $\operatorname{Tr}(\mathrm{d}X \wedge \mathrm{d}Y)$.

Proof. The differential of a trace is the matrix evaluation of the cyclic derivative. Contracting the two differentials with the inverse trace symplectic form gives the necklace formula. □

Chapter 68

Stable traces

On the semisimple commuting locus write the joint eigenvalues as (x_i, y_i). Define

$$p_{ab}^{(N)} = \sum_{i=1}^{N} x_i^a \, y_i^b.$$

Theorem 68.1 (Stable multisymmetric Poisson algebra)**.** *The inverse stable trace algebra is the completed polynomial algebra*

$$\mathcal{P}_\infty = \mathbb{C}[\![\, p_{ab} \mid a, b \geq 0,\ a + b > 0 \,]\!].$$

Its Poisson bracket is

$$\{p_{ab}, p_{cd}\} = (ad - bc)\, p_{a+c-1,b+d-1}.$$

Proof. The stable fundamental theorem for multisymmetric functions gives polynomial generators p_{ab}. The bracket follows from $\{x_i, y_j\} = \delta_{ij}$ and the Leibniz rule. □

Remark 68.2. The stable trace algebra is a quotient of the universal closed algebra after representation and scalarization. The universal centre retains Hochschild classes, stabilizer directions, and higher central operations that vanish under the trace.

Chapter 69

The two-brane cone

Separate center-of-mass and relative coordinates. Put

$$u = x_1 - x_2, \qquad v = y_1 - y_2, \qquad E = \frac{1}{2}u^2, \quad F = -\frac{1}{2}v^2, \quad H = -uv.$$

Then

$$H^2 + 4EF = 0.$$

Proposition 69.1 (The A_1 Kirillov–Kostant cone)**.** *With $\{u, v\} = 2$,*

$$\{H, E\} = 4E, \qquad \{H, F\} = -4F, \qquad \{E, F\} = 2H.$$

After the evident rescaling this is the $\mathfrak{sl}_2$ Lie–Poisson bracket on the nilpotent cone.

Proof. The three identities follow by direct differentiation. The quadratic equation is the determinant-zero equation for the traceless matrix

$$\begin{pmatrix} H & 2E \\ 2F & -H \end{pmatrix}.$$

□

Book XIV

Wheel corrections and exact quantum structure

Chapter 70

The Duflo wheel

First set the coefficient parameter to 1 in the polynomial Rees algebra. For the resulting ordinary $\mathfrak{sl}_2$ enveloping algebra, take generators

$$[h, e] = 2e, \qquad [h, f] = -2f, \qquad [e, f] = h.$$

The invariant polynomial and the quadratic Casimir are

$$q = 2ef + \frac{1}{2}h^2, \qquad \Omega = ef + fe + \frac{1}{2}h^2.$$

Calculation 70.1 (The zero-point shift)**.** The Duflo map gives

$$\mathrm{Duf}(q) = \Omega + \frac{1}{2}.$$

On the irreducible highest-weight-n module,

$$\left(\Omega + \frac{1}{2}\right)\Big|_{V_n} = \frac{(n+1)^2}{2}\,\mathrm{id}\,.$$

Proof. Write $x = ue + vf + wh$. Direct adjoint matrices give $\mathrm{Tr}(\mathrm{ad}_x^2) = 8uv + 8w^2$. Thus the quadratic term of $j^{1/2}(\partial)$ is $(\partial_e\partial_f + \partial_h^2)/6$. Applied to $q = 2ef + h^2/2$, it gives $(2+1)/6 = 1/2$. Higher differential terms annihilate this quadratic polynomial. Symmetrization sends q to Ω. On a highest-weight vector, ef acts by n, fe by zero, and $h^2/2$ by $n^2/2$. Centrality gives the eigenvalue on the entire irreducible module. □

Restoring the Rees parameter gives $\mathrm{Duf}_\hbar(q) = \Omega + \hbar^2/2$. The polynomial calculation permits specialization at 1; arbitrary formal power series do not admit that specialization. Higher wheels assemble into $j^{1/2}$ on higher-degree invariants.

Chapter 71

The local centre with comparison data

Assume a compatible jet comparison and a complete homotopy-E_2 refinement as in Theorem 57.1. Put

$$\mathcal{A} = \widehat{U}_\hbar(\mathfrak{h}), \qquad \mathcal{A}^{\mathrm{alg}}_{\mathrm{bulk}} = \mathrm{Fact}_{\mathsf{E}_2}(\mathcal{P}_\infty).$$

The factorization envelope uses the two oriented topological directions. The formal holomorphic coefficient remains at the Darboux point. Let C_∂ be the category of perfect complete $\mathcal{A}$-modules in the chosen continuous R-linear Morita theory. Assume its centre is computed by $C^\bullet_{\mathrm{HH,cont}}(\mathcal{A}/R, \mathcal{A})$.

Theorem 71.1 (Local derived-centre comparison). *These data give an equivalence of complete homotopy-E_2 factorization algebras*

$$\mathcal{A}^{\mathrm{alg}}_{\mathrm{bulk}} \xrightarrow{\sim} Z^{\mathrm{der}}_{\mathrm{Mor},R}(C_\partial).$$

At the formal insertion point the map is the selected refinement of Φ_∞. Every continuous boundary module has the induced derived centre action.

Proof. The compatible jet theorem gives the complete homotopy-E_2 comparison. The factorization envelope preserves this equivalence. The assumed continuous Morita model identifies its target with the centre. A centre element is a derived natural endomorphism of the identity functor, so evaluation gives the action on each boundary module. □

Corollary 71.2 (Global K3 comparison with descent data). *Let S be a K3 surface with holomorphic symplectic form. Suppose the local comparisons extend to a sheafwise equivalence with the relative Hochschild centre of* $\mathrm{Perf}(\mathcal{W}_{S,\hbar})$. *Suppose further that the selected cyclic and mixed refinements extend and satisfy descent in the same completed category. Then derived global sections give the corresponding global centre, cyclic, and open–closed comparisons.*

Proof. Derived global sections preserve equivalences of sheaves. Apply them to the supplied sheafwise maps and their coherence homotopies. □

Chapter 72

Cyclic classes, stable traces, and Lie homology

A trace of the unit records the representation dimension. For $B = \mathbb{C}$, one has $\mathrm{Tr}_N(1) = N$, so ordinary traces cannot define a rank-compatible map on all of $HC_0(B)$. A scalar-reduced trace removes this dependence after choosing an augmentation.

Let B be a unital associative $\mathbb{C}$-algebra. The affine representation scheme $\mathrm{Rep}_N B$ represents unital algebra maps $B \to \mathrm{Mat}_N(D)$ for commutative $\mathbb{C}$-algebras D. Put

$$\mathcal{R}_N = \mathcal{O}(\mathrm{Rep}_N B)^{GL_N}, \qquad \mathfrak{c}_B = HC_0(B) = B/[B,B],$$

where $[B,B]$ denotes the linear span of commutators. The ordinary trace is the linear map

$$\mathrm{Tr}_N : \mathfrak{c}_B \longrightarrow \mathcal{R}_N, \qquad \mathrm{Tr}_N([b])(\rho) = \mathrm{Tr}(\rho(b)).$$

Cyclicity makes it independent of the representative, and conjugation invariance puts its image in $\mathcal{R}_N$.

For Lie and Poisson compatibility, specify a Lie bracket on $\mathfrak{c}_B$, a Poisson bracket on each $\mathcal{R}_N$, and the identities

$$\{\mathrm{Tr}_N\,\beta, \mathrm{Tr}_N\,\gamma\}_N = \mathrm{Tr}_N[\beta,\gamma]_{\mathrm{cyc}} \qquad (\beta,\gamma \in \mathfrak{c}_B). \tag{72.1}$$

A Hamiltonian comparison also specifies a Lie map $j : \mathfrak{h} \to \mathfrak{c}_B$ for this bracket. These are additional data on B. The preceding necklace bracket and trace moment map supply the bracket and trace data for $B = \mathbb{C}\langle x, y\rangle$ with its canonical symplectic bracket. The map j remains an additional comparison. An arbitrary unital algebra does not carry these structures by definition.

For rank stabilization, choose an augmentation $\epsilon : B \to \mathbb{C}$. It defines the rank-stabilization morphism

$$s_N : \mathrm{Rep}_N B \longrightarrow \mathrm{Rep}_{N+1} B, \qquad \rho \longmapsto \rho \oplus \epsilon.$$

Restriction gives $s_N^* : \mathcal{R}_{N+1} \to \mathcal{R}_N$. Define the reduced cyclic space and trace by

$$\bar{\mathfrak{c}}_B = B/(\mathbb{C}1 + [B,B]), \qquad t_N(\bar{b})(\rho) = \mathrm{Tr}(\rho(b)) - N\epsilon(b).$$

Let $\mathfrak{m}_N \subset \mathcal{R}_N$ be the ideal of functions vanishing at the representation $b \mapsto \epsilon(b) I_N$. The completed invariant rings and stable target are

$$\widehat{\mathcal{R}}_N = \varprojlim_{r \ge 1} \mathcal{R}_N/\mathfrak{m}_N^r, \qquad \mathcal{R}_\infty^\wedge = \varprojlim_N \widehat{\mathcal{R}}_N.$$

Use the inverse-limit topologies. Write J for the augmentation ideal in $\mathrm{Sym}_{\mathbb{C}} \bar{\mathfrak{c}}_B$ and put

$$\widehat{\mathrm{Sym}}\,\bar{\mathfrak{c}}_B = \varprojlim_{r \ge 1} \mathrm{Sym}_{\mathbb{C}} \bar{\mathfrak{c}}_B / J^r.$$

The stable matrix Lie algebra is $\mathfrak{gl}_\infty B = \varinjlim_N \mathfrak{gl}_N B$ under block-zero Lie embeddings. For its homology, use trivial coefficients $\mathbb{C}$. Let $V_B = \bigoplus_{n \ge 1} HC_{n-1}(B)$, with $HC_{n-1}(B)$ placed in homological degree n. The notation $\mathrm{Sym}_{\mathrm{gr}} V_B$ denotes the free graded-commutative algebra, with the Koszul sign rule.

Theorem 72.1 (Cyclic traces and stable Lie homology). *Assume the compatible jet comparison and its homotopy Gerstenhaber refinement. For the augmented algebra and completion above, there are maps*

$$\mathcal{P}_\infty \xrightarrow{\Phi_\infty} C^*_{\mathrm{HH},\mathrm{cont}}(\mathcal{A}/R,\mathcal{A}), \qquad \mathrm{Tr}_N : \mathfrak{c}_B \longrightarrow \mathcal{R}_N,$$

$$\widehat{\mathrm{Sym}}\,\bar{\mathfrak{c}}_B \longrightarrow \mathcal{R}^\wedge_\infty, \qquad \mathrm{Sym}_{\mathrm{gr}} V_B \xrightarrow{\sim} H^{\mathrm{CE}}_*(\mathfrak{gl}_\infty B;\mathbb{C}).$$

The first preserves the selected homotopy Gerstenhaber structure. The second is linear and, under the bracket data (72.1), is a Lie map. Under those data, its composite with j preserves the Hamiltonian bracket. The third is a continuous homomorphism of commutative algebras, and the fourth is an isomorphism of graded Hopf algebras. These maps are natural for morphisms preserving the stated data.

Proof. The first arrow is the chosen comparison of Theorem 57.1. Its operadic refinement gives the asserted Gerstenhaber compatibility. The trace formula above constructs the second arrow. Equation (72.1) supplies its Lie compatibility, and the Lie property of j gives the composite assertion.

The augmentation annihilates commutators. Both terms of t_N vanish on a commutator, and their values on a scalar multiple of the unit cancel. Thus t_N is well defined on $\bar{\mathfrak{c}}_B$. At the trivial representation its value is zero, so its image lies in $\mathfrak{m}_N$. Moreover,

$$\begin{aligned}(s_N^* t_{N+1})(\bar{b})(\rho) &= \mathrm{Tr}(\rho(b)) + \epsilon(b) - (N+1)\epsilon(b)\\ &= t_N(\bar{b})(\rho).\end{aligned}$$

The point $N\epsilon$ maps to $(N+1)\epsilon$, so $s_N^*(\mathfrak{m}^r_{N+1}) \subset \mathfrak{m}^r_N$. Hence restriction defines the transition maps between the completed rings. The symmetric-algebra extension of t_N sends J^r into $\mathfrak{m}^r_N$. It therefore induces a ring homomorphism on each corresponding quotient. Taking the inverse limit in r gives a continuous map to $\widehat{\mathcal{R}}_N$. The displayed stabilization identity makes these maps compatible in N. Their inverse limit is the third arrow.

The diagonal Lie map $\mathfrak{gl}_\infty B \to \mathfrak{gl}_\infty B \oplus \mathfrak{gl}_\infty B$ induces the homology coproduct. Block sum in the opposite direction induces the homology product, using the Künneth isomorphism over $\mathbb{C}$. The Loday–Quillen–Tsygan theorem identifies primitive homology in degree n with $HC_{n-1}(B)$.[1] It identifies the full stable homology with the free graded-commutative Hopf algebra on these primitives, giving the fourth arrow. The trace formulas, the augmentation-preserving rank maps, and stable Lie homology are natural. Naturality of the first arrow requires preservation of the selected jet comparison. □

Scalar reduction need not preserve the supplied Poisson bracket. For example, for $B = \mathbb{C}[x, y]$ with $\{x, y\} = 1$ and $\epsilon(x) = \epsilon(y) = 0$, traces on N points satisfy

$$\{\mathrm{Tr}_N\, x, \mathrm{Tr}_N\, y\} = N, \qquad t_N(\bar{1}) = 0.$$

A reduced Poisson comparison consequently requires additional compatibility, such as vanishing of ϵ on the selected cyclic brackets, or a restriction to a Hamiltonian subalgebra with that property. Only multiplicativity and continuity are asserted for the third arrow. For $B = \mathbb{C}$, the reduced cyclic space is zero and that arrow is the identity on $\mathbb{C}$.

[1] J.-L. Loday and D. Quillen, *Cyclic homology and the Lie algebra homology of matrices*, Commentarii Mathematici Helvetici **59** (1984), 565–591, Theorem 6.2, pp. 584–586.

Chapter 73

Synthesis

At each fixed Hamiltonian jet, the Rees enveloping algebra has relative Hochschild complex equivalent to the Chevalley complex with differential $\hbar d_{\mathrm{CE}}$. PBW antisymmetrization proves this over every R_N and in the formal coefficient limit. The Duflo theorem gives the separate multiplicative comparison on invariant functions. Quadratic Moyal Hamiltonians generate the oscillator quotient of the enveloping algebra. Their associative Casimir relation prevents the primitive coproduct from descending.

The Hamiltonian jet limit requires the compatible cochain systems of Theorem 57.1. With its operadic refinement and continuous Morita model, the centre acts on every supplied continuous boundary module M through

$$\mathcal{P}_\infty \xrightarrow{\Phi_\infty} C^\bullet_{\mathrm{HH,cont}}(\mathcal{A}/R, \mathcal{A}) \longrightarrow \mathrm{RHom}_{\mathcal{A}}(M, M).$$

The last arrow evaluates a derived natural endomorphism at M. A matrix-brane realization must construct such a module and compare its centre action with the selected trace functions. A commuting square with scalar representation invariants requires these additional comparison maps. Likewise, Fedosov descent concerns symplectic polyvectors and a Weyl algebroid. Its relation to Hamiltonian Lie cochains requires the specified map between their carriers.

Book XV

Technical appendices to the preceding book

Chapter 74

Signs in the shifted cotangent complex

Let e_a have degree zero and let $e^a[-1]$ have degree -1. The Chevalley generators dual to them have degrees 1 and 0. Thus

$$C^*_{\mathrm{CE}}(T^*[-1]\mathfrak{l}) = \mathrm{Sym}\big(\mathfrak{l}^\vee[-1] \oplus \mathfrak{l}\big).$$

The degree-zero polynomial factor is $\mathrm{Sym}\,\mathfrak{l} = \mathcal{O}(\mathfrak{l}^*)$. The differential on a polynomial p is

$$(d_{\mathrm{CE}}p)(x) = \mathrm{ad}_x\, p,$$

which is the linear Poisson differential.

Chapter 75

A direct Jacobi check for Hamiltonian jets

For monomials

$$f = z_1^a z_2^b, \qquad g = z_1^c z_2^d,$$

one has

$$\{f, g\} = (ad - bc) z_1^{a+c-1} z_2^{b+d-1}.$$

Substitution of three monomials gives a scalar multiple of

$$\det((a,b),(c,d))\ \det((a+c-1, b+d-1),(e,f)) + \text{cyclic}.$$

Writing the three coefficients in terms of the 2×2 determinants of the exponent vectors shows pairwise cancellation. The coordinate-free proof is $[\pi, \pi]_{\mathrm{Sch}} = 0$ for $\pi = \partial_{z_1} \wedge \partial_{z_2}$.

Chapter 76

Bibliographic note

The finite affine formality map and the separate symplectic globalization use Kontsevich formality and its Fedosov resolution [21, 35, 70]. The relative Hochschild model is the Cartan–Eilenberg comparison, with both Chevalley terms scaled by $\hbar$. Its finite-base resolution proof is Theorem 55.1. Jet limits and operadic refinements require the additional data stated above. The all-arity centre action is the Deligne–Tamarkin brace action [71, 81]. The global deformation-quantization category is the holomorphic Weyl algebroid. The matrix trace and necklace formulas are the noncommutative symplectic moment-map construction.

Book entrance: The Igusa Square Root

The Jacobi seed, Borcherds denominator, root superalgebra, local–wrapped currents, quantum group, Hall model, and Fredholm realization.

Book XVI

The automorphic determinant

Chapter 77

The protected Jacobi input

77.1 The K3 elliptic genus

Let S be a K3 surface. The right-moving supersymmetric index of a unitary $\mathcal{N} = (4, 4)$ sigma model is the elliptic genus

$$Z_{K3}(\tau, z) = 2\phi_{0,1}(\tau, z).$$

The normalized weak Jacobi form has weight zero, index one, and Fourier expansion

$$\phi_{0,1}(\tau, z) = \sum_{n \geq 0} \sum_{\ell \in \mathbb{Z}} f(n, \ell) q^n r^\ell.$$

In Eichler–Zagier normalization,

$$\phi_{0,1}(\tau, z) = 4 \sum_{a=2}^{4} \frac{\vartheta_a(\tau, z)^2}{\vartheta_a(\tau, 0)^2} \in J^{\mathrm{w}}_{0,1}$$

[38]. The coefficient depends on the discriminant $4n - \ell^2$ and on ℓ modulo 2.

Theorem 77.1 (Exact low Fourier coefficients)**.** *The first four q-rows are*

$$\begin{aligned} \phi_{0,1} = {} & r^{-1} + 10 + r \\ & + q\left(10r^{-2} - 64r^{-1} + 108 - 64r + 10r^2\right) \\ & + q^2\left(r^{-3} + 108r^{-2} - 513r^{-1} + 808 - 513r + 108r^2 + r^3\right) \\ & + q^3\left(-64r^{-3} + 808r^{-2} - 2752r^{-1} + 4016 \right. \\ & \qquad \left. - 2752r + 808r^2 - 64r^3\right) + O(q^4). \end{aligned}$$

In particular,

$$f(0, 0) = 10, \qquad \frac{f(0, 0)}{2} = 5.$$

Proof. Insert the theta-series expansions into the preceding quotient formula. Work with $t = q^{1/8}$ and $x = r^{1/2}$; each quotient is an exact formal power series in t with Laurent-polynomial coefficients in x. The fractional powers cancel in the sum. Division through t^{24} gives the displayed rows. The arithmetic calculation is reproduced in Appendix 108. □

77.2 The virtual half-index

For every (n, ℓ) choose a finite-dimensional super vector space $U_{n,\ell}$ satisfying

$$\operatorname{sdim} U_{n,\ell} = f(n, \ell).$$

The integer $f(n, \ell)$ is its superdimension. In the ordinary Grothendieck group of finite-dimensional super vector spaces, the even and odd dimensions remain separate. The signed integer is their image under superdimension. The determinant depends only on that image.

Lemma 77.2 (Fock character and Berezinian)**.** *Let $V = \bigoplus_\gamma V_\gamma$ be a locally finite graded super vector space. Then*

$$\operatorname{sch}\big(\operatorname{Sym}(V_{\bar 0}) \otimes \bigwedge(V_{\bar 1})\big) = \prod_\gamma (1 - x^\gamma)^{-\operatorname{sdim} V_\gamma},$$

and

$$\operatorname{sdet}_V(1 - x) = \prod_\gamma (1 - x^\gamma)^{\operatorname{sdim} V_\gamma}.$$

Proof. An even one-dimensional summand contributes the geometric series $(1 - x^\gamma)^{-1}$. An odd one-dimensional summand contributes $1 - x^\gamma$. Multiplication over a homogeneous basis proves both identities. □

Chapter 78

The Borcherds product

78.1 The effective chamber

Define

$$\Gamma_{\mathrm{eff}} = \{(n,\ell,m) \in \mathbb{Z}^3 : m > 0,\ n \geq 0\} \cup \{(n,\ell,0) : n > 0\} \cup \{(0,\ell,0) : \ell < 0\}.$$

The ordering is lexicographic in $(m, n, -\ell)$ and is precisely the ordering selected by the cusp chamber. For $\gamma = (n,\ell,m)$ put

$$V_\gamma = U_{nm,\ell}, \qquad \operatorname{sdim} V_\gamma = f(nm,\ell).$$

Definition 78.1 (Monic virtual determinant)**.** The monic determinant of the half-index is

$$D_5(Z) = q^{1/2} r^{1/2} s^{1/2} \operatorname{sdet}_V(1 - x) = q^{1/2} r^{1/2} s^{1/2} \prod_{\Gamma_{\mathrm{eff}}} (1 - q^n r^\ell s^m)^{f(nm,\ell)}.$$

Theorem 78.2 (Borcherds–Gritsenko–Nikulin product)**.** *The preceding product converges normally on the chosen cusp chamber and extends to a holomorphic genus-two modular form of weight five with a character of order two. It is the monic Igusa form D_5. The theta-normalized form is*

$$\Delta_5^\theta = 2^6 D_5.$$

Proof. The singular theta lift of the vector-valued form corresponding to $\phi_{0,1}$ produces the product, divisor, weight, and automorphic character. The weight is $f(0,0)/2 = 5$. The genus-two identification with the Igusa form and the product formula are the Gritsenko–Nikulin theorems [14, 15, 50, 52]. The product of the ten even theta constants has coefficient 2^6 at $q^{1/2} r^{1/2} s^{1/2}$, fixing the scalar. □

78.2 The theta constant normalization

Let $\mathfrak{E}$ be the ten even genus-two characteristics. Define

$$\Delta_5^\theta(Z) = \prod_{\mathfrak{m} \in \mathfrak{E}} \vartheta[\mathfrak{m}](Z, 0).$$

The cusp leading terms of the ten factors contribute six factors of 2 and total exponent $(1/2, 1/2, 1/2)$. Thus

$$[q^{1/2} r^{1/2} s^{1/2}] \Delta_5^\theta = 64.$$

This coefficient selects a canonical scalar normalization of the automorphic section.

Proposition 78.3 (Ten-factor leading-term calculation). *Group the ten even characteristics by the first half-characteristic $a \in (\mathbb{Z}/2)^2$. Their leading contributions in the chosen cusp chamber are*

a	#	*scalar*	*monomial*
$(0,0)$	4	1	1
$(1,0)$	2	2	$q^{1/8}$
$(0,1)$	2	2	$s^{1/8}$
$(1,1)$	2	2	$q^{1/8}r^{1/4}s^{1/8}$.

Multiplication gives

$$1^4\,2^2\,2^2\,2^2\,q^{1/2}r^{1/2}s^{1/2} = 64q^{1/2}r^{1/2}s^{1/2}.$$

Proof. Insert each characteristic into the genus-two theta series and retain the unique terms of least order in the lexicographic cusp valuation $(m, n, -\ell)$. For $a = (0,0)$ the zero lattice vector contributes one. For each nonzero a, the two shortest lattice representatives are exchanged by sign and contribute the scalar two. Their quadratic exponents are the entries in the final column. Summing the exponents and multiplying the six factors of two gives the formula. □

Proposition 78.4 (Boundary factor of the Borcherds product). *The $m = 0$ part of the monic product is*

$$\prod_{n>0,\ell\in\mathbb{Z}} (1-q^n r^\ell)^{f(0,\ell)} \prod_{\ell<0} (1-r^\ell)^{f(0,\ell)} = (1-r^{-1}) \prod_{n\geq 1} (1-q^n r^{-1})(1-q^n)^{10}(1-q^n r).$$

Consequently the first Fourier–Jacobi coefficient is

$$q^{1/2}r^{1/2}(1-r^{-1}) \prod_{n\geq 1} (1-q^n r^{-1})(1-q^n)^{10}(1-q^n r) = \eta(\tau)^9 \vartheta_{11}(\tau, z),$$

where the odd theta function is fixed by the exact convention

$$\vartheta_{11}(\tau, z) = -q^{1/8}r^{-1/2} \prod_{n\geq 1} (1-q^{n-1}r)(1-q^n r^{-1})(1-q^n).$$

Proof. The q^0 row of $\phi_{0,1}$ has exactly three nonzero coefficients: $f(0,-1) = 1$, $f(0,0) = 10$, and $f(0,1) = 1$. Substitution into the $m = 0$ factors gives the first formula. The Jacobi triple product and $\eta(\tau) = q^{1/24} \prod_{n\geq 1}(1-q^n)$ give the second, including its leading sign and monomial. □

Corollary 78.5 (The scalar square). *The monic weight-ten form is*

$$\chi_{10} = D_5{}^2 = 2^{-12}\Delta_5^{\theta 2}.$$

Its automorphic character is trivial.

Proof. The identity $\Delta_5^\theta = 2^6 D_5$ gives the scalar factor. The character of D_5 has order two, so its square is trivial. □

Chapter 79

Divisor, character, and the meaning of the square root

The diagonal Humbert divisor is

$$H_1 = \{Z \in \mathbb{H}_2 : z = 0\}/\mathrm{Sp}_4(\mathbb{Z}).$$

Theorem 79.1 (Diagonal divisor)**.** *The divisor of D_5 is the $\mathrm{Sp}_4(\mathbb{Z})$-orbit of H_1, with multiplicity one. The form transforms by the Maass character ν_5 of order two, and $\chi_{10} = D_5^2$ is scalar-valued.*

Proof. This is the defining diagonal-divisor property of the Igusa square root and the first row of the Cléry–Gritsenko classification [24, 60, 61]. □

Real roots of the denominator algebra define Weyl reflections. Imaginary roots contribute root spaces and product exponents. This division of labor is intrinsic to a Borcherds algebra: the Weyl group is generated by real simple roots, while the imaginary simple roots enter the correction term of the denominator identity.

79.1 Automorphic section, orientation line, and Pfaffian operator

An automorphic square root is a section of a line bundle with character. An orientation is a square root of a determinant line. A Pfaffian operator is a skew family whose regularized Pfaffian is a section of that orientation line. These form a sequence of successive refinements:

$$\text{automorphic section} \longrightarrow \text{determinant-line identification}$$
$$\longrightarrow \text{oriented square-root line} \longrightarrow \text{Fredholm family and Pfaffian section.}$$

Proposition 79.2 (Logical separation of the four levels)**.** *The product theorem constructs the first object in the sequence. A determinant-line comparison constructs the second. A coherent square root of that line constructs the third. A first-order operator requires a symbol, domain, ellipticity or Fredholm condition, and regularization, and thereby produces the fourth refinement.*

Proof. Each stage adds a specified geometric structure. A determinant realization adds an isomorphism from a geometric determinant line to the automorphic line. An oriented realization adds a coherent square-root line and its parity refinement. An operator realization adds a principal symbol, a Fredholm domain, a spectrum, and a regularization whose Pfaffian is the resulting section. □

The title names the monic automorphic section D_5, characterized by the exact identity $D_5^2 = \chi_{10}$. Part IV formulates an analytic Pfaffian realization as a further geometric refinement.

Book **XVII**

The denominator algebra

Chapter 80

The Lorentzian root lattice

Let

$$L = \mathbb{Z} f_2 \oplus \mathbb{Z} f_3 \oplus \mathbb{Z} f_{-2}$$

with bilinear form

$$(f_2, f_{-2}) = -1, \qquad (f_3, f_3) = 2,$$

and all remaining basis pairings zero. Its signature is $(2, 1)$. Define the type-II root lattice

$$L_{II} = 2\mathbb{Z} f_2 \oplus \mathbb{Z} f_3 \oplus 2\mathbb{Z} f_{-2} \cong U(4) \oplus \langle 2 \rangle.$$

The Gram-index map is

$$\alpha : \Gamma_{\mathrm{ind}} = \mathbb{Z}^3 \longrightarrow L_{II}, \qquad \alpha(n, \ell, m) = 2n f_2 - \ell f_3 + 2m f_{-2}.$$

Then

$$(\alpha(n, \ell, m), \alpha(n', \ell', m')) = -2\big(2nm' + 2n'm - \ell\ell'\big),$$

and

$$(\alpha(n, \ell, m), \alpha(n, \ell, m)) = -2(4nm - \ell^2).$$

80.1 Real simple roots and Weyl vector

Put

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3.$$

Proposition 80.1 (Type-II Cartan matrix)**.** *The Gram matrix of the real simple roots is*

$$\big((\delta_i, \delta_j)\big)_{i,j=1}^{3} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

The reflections

$$s_i(x) = x - (x, \delta_i)\delta_i$$

are integral involutions preserving L_{II} and its form.

Proof. Substitution into the basis pairing gives the matrix. Since $(\delta_i, \delta_i) = 2$, the reflection formula has integral coefficients. Direct expansion gives $s_i^2 = 1$ and $(s_i x, s_i y) = (x, y)$. □

The Weyl vector is

$$\rho = f_2 - \frac{1}{2} f_3 + f_{-2}, \qquad (\rho, \delta_i) = -1.$$

In the Gram-index variables, $e^{-\rho} = q^{1/2} r^{1/2} s^{1/2}$.

Chapter 81

The Gritsenko–Nikulin Lie superalgebra

The Gritsenko–Nikulin construction specifies a generalized Kac–Moody Lie superalgebra $\mathfrak{g}_{\Delta_5}$, with real simple roots δ_1, δ_2, δ_3 and the imaginary correction attached to $\phi_{0,1}$. Its generators and relations supply the bracket whose denominator is compared with the Borcherds product.

Theorem 81.1 (Root supermultiplicities and denominator)**.** *On the active positive support,*

$$\operatorname{smult} \mathfrak{g}_{\Delta_5\,\alpha(n,\ell,m)} = f(nm,\ell).$$

In the Lorentzian coordinate convention of Gritsenko–Nikulin,

$$\operatorname{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z) = 2^{-6}\Delta_5^{\theta}(2Z).$$

Proof. The real roots and Weyl chamber are the type-II case of the Gritsenko–Nikulin construction. The imaginary simple-root correction is chosen so that its Weyl–Kac–Borcherds product has exponent $f(nm,\ell)$. The resulting product is the doubled-variable monic Igusa form [50–52]. □

The theorem gives a Lie superalgebra with a bracket, Cartan, Weyl group, root parity, and denominator. The scalar product alone records the signed dimensions of the root spaces; the Gritsenko–Nikulin construction supplies the algebraic operations.

Chapter 82

The ordered bar as the denominator complex

Let $\mathfrak{n}_{\Delta_5,+}$ be the positive nilpotent subalgebra of $\mathfrak{g}_{\Delta_5}$. Fix a proper integral height h on the positive root cone and define

$$F^{>H}\mathfrak{n}_{\Delta_5,+} = \bigoplus_{h(\alpha)>H} \mathfrak{g}_{\Delta_5\,\alpha}, \qquad {\mathfrak{n}_{\Delta_5,+}}^{(H)} = \mathfrak{n}_{\Delta_5,+}/F^{>H}\mathfrak{n}_{\Delta_5,+}.$$

Height additivity makes $F^{>H}\mathfrak{n}_{\Delta_5,+}$ an ideal. Properness and local finiteness make ${\mathfrak{n}_{\Delta_5,+}}^{(H)}$ finite-dimensional and nilpotent.

Theorem 82.1 (Height-complete Bar–Chevalley comparison)**.** *For every H, antisymmetrization gives a natural quasi-isomorphism of conilpotent dg coalgebras*

$$\mathrm{Bar}\big(U({\mathfrak{n}_{\Delta_5,+}}^{(H)})\big) \simeq C_*({\mathfrak{n}_{\Delta_5,+}}^{(H)}).$$

The quotient maps preserve these comparisons and induce a quasi-isomorphism in the separated height-complete category

$$\mathrm{Bar}_h^{\wedge}\big(U(\mathfrak{n}_{\Delta_5,+})\big) := \varprojlim_H \mathrm{Bar}\big(U({\mathfrak{n}_{\Delta_5,+}}^{(H)})\big) \simeq C_{*,h}^{\wedge}(\mathfrak{n}_{\Delta_5,+}) := \varprojlim_H C_*({\mathfrak{n}_{\Delta_5,+}}^{(H)}).$$

Its root-graded Euler supercharacter is

$$\chi_{\mathrm{root}}\, C_{*,h}^{\wedge}(\mathfrak{n}_{\Delta_5,+}) = \prod_{\alpha>0}(1-e^{-\alpha})^{\mathrm{sdim}\,\mathfrak{g}_{\Delta_5\,\alpha}}.$$

Proof. For each H, the antisymmetrization map sends $\mathrm{Sym}({\mathfrak{n}_{\Delta_5,+}}^{(H)}[1])$ to the normalized bar of $U({\mathfrak{n}_{\Delta_5,+}}^{(H)})$. The PBW filtrations have the same associated-graded Koszul resolution of $\mathrm{Sym}({\mathfrak{n}_{\Delta_5,+}}^{(H)})$, so antisymmetrization is a filtered quasi-isomorphism. Naturality gives a morphism of inverse systems. Every fixed root degree is visible in all sufficiently large height quotients and is unchanged thereafter; hence the inverse-limit map is a quasi-isomorphism coefficientwise. An even root vector becomes odd after the Chevalley shift and contributes $1-e^{-\alpha}$; an odd root vector becomes even and contributes its reciprocal. Multiplication over the finite-dimensional root spaces gives the product. □

Let $\mathbb{T}_{L_{II}}$ be the formal root torus and write Z_{root} for its logarithmic coordinate, normalized by

$$e^{-\alpha(n,\ell,m)} = Q^n R^\ell S^m, \qquad e^{-\rho} = Q^{1/2}R^{1/2}S^{1/2}.$$

Corollary 82.2 (Bar denominator identity)**.** *On the formal root torus,*

$$e^{-\rho}\chi_{\mathrm{root}} C_{*,h}^{\wedge}(\mathfrak{n}_{\Delta_5,+}) = D_5(Z_{\mathrm{root}}).$$

The Gritsenko–Nikulin embedding of the root torus into the Siegel period domain sends Z_{root} to $2Z$ [50, 52]*. Hence*

$$e^{-\rho}\chi_{\mathrm{root}} C_{*,h}^{\wedge}(\mathfrak{n}_{\Delta_5,+}) = \mathrm{den}(\mathfrak{g}_{\Delta_5}) = D_5(2Z)$$

in Siegel period coordinates.

Proof. Multiply the Euler product in the preceding theorem by the Weyl monomial $e^{-\rho}$. The denominator theorem identifies the result on the root torus, and the stated embedding gives the Siegel-coordinate formula. □

The denominator is therefore the Weyl-shifted Euler supercharacter of the height-complete interval bar of the positive enveloping algebra.

Chapter 83

Finite-height currents

Retain the finite-height quotients ${\mathfrak{n}_{\Delta_{\mathrm{s}},+}}^{(H)}$ of the preceding chapter. Define the current Lie conformal superalgebra

$$\mathrm{Cur}({\mathfrak{n}_{\Delta_{\mathrm{s}},+}}^{(H)}) = \Bbbk[\partial] \otimes {\mathfrak{n}_{\Delta_{\mathrm{s}},+}}^{(H)}, \qquad [a_\lambda b] = [a, b],$$

and its universal chiral enveloping algebra

$$V_H = U^{\mathrm{ch}}\Big(\mathrm{Cur}_E({\mathfrak{n}_{\Delta_{\mathrm{s}},+}}^{(H)})\Big).$$

Theorem 83.1 (Height-complete current envelope)**.** *The quotient maps induce a coefficientwise inverse limit*

$$\widehat{V}_{\Delta_{\mathrm{s}}} = \varprojlim_{H} V_H.$$

Every V_H is PBW-admissible as an associative chiral algebra, the current generators are primitive for the standard current coproduct, and

$$\mathrm{sch}\, \widehat{V}_{\Delta_{\mathrm{s}}} = \prod_{j \geq 1} \prod_{\alpha > 0} (1 - t^j e^{-\alpha})^{-\operatorname{sdim} \mathfrak{g}_{\Delta_{\mathrm{s}}\, \alpha}}.$$

Moreover,

$$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(V_H) \simeq \mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}\Big(\mathrm{Cur}_E({\mathfrak{n}_{\Delta_{\mathrm{s}},+}}^{(H)})\Big),$$

and these equivalences pass to the separated height completion.

Proof. PBW orders the negative modes $a(-j)$, $j \geq 1$. Even modes contribute symmetric powers and odd modes contribute exterior powers, giving the product. Proper height makes every coefficient stationary. The current coproduct sends $a(z)$ to $a(z) \otimes 1 + 1 \otimes a(z)$. Chiral antisymmetrization identifies the associative-chiral bar of the chiral enveloping algebra with the chiral Chevalley coalgebra; the complete separated PBW filtration is coefficientwise finite. □

83.1 Classical local theory and exact quantum zero modes

For every H, the finite-dimensional nilpotent Lie superalgebra ${\mathfrak{n}_{\Delta_{\mathrm{s}},+}}^{(H)}$ defines a cotangent BF theory on $E \times \mathbb{R}$. Its local dg Lie algebra is

$$\mathcal{L}_H = \Omega^{0,*}(E) \mathbin{\widehat{\otimes}} \Omega^*(\mathbb{R}) \otimes {\mathfrak{n}_{\Delta_{\mathrm{s}},+}}^{(H)}$$

and the classical action is

$$S_H(A, B) = \int_{E \times \mathbb{R}} \Omega_E \wedge \Big\langle B, (\bar{\partial} + \mathrm{d}_{\mathbb{R}})A + \frac{1}{2}[A, A] \Big\rangle.$$

Theorem 83.2 (Finite-height BF realization). *The action S_H satisfies the classical master equation and defines a classical holomorphic–topological factorization algebra. Harmonic reduction along a compact interval produces the finite-dimensional cotangent BV theory of ${\mathfrak{n}_{\Delta_5,+}}^{(H)}$. The positive-root height grading makes every even adjoint endomorphism strictly height-increasing and every odd adjoint endomorphism odd; hence the modular supercharacter vanishes and the reduced theory satisfies the quantum master equation.*

Choose the standard complementary Lagrangian boundary polarizations, fixing the ${\mathfrak{n}_{\Delta_5,+}}^{(H)}[1]$ coordinate at one endpoint and its cotangent coordinate at the other. The algebraic boundary quantization, the interval amplitude between the two augmentation states, and the holomorphic boundary current envelope are then

$$U({\mathfrak{n}_{\Delta_5,+}}^{(H)}), \qquad \mathbf{1} \otimes^{\mathbb{L}}_{U({\mathfrak{n}_{\Delta_5,+}}^{(H)})} \mathbf{1} \simeq C_*({\mathfrak{n}_{\Delta_5,+}}^{(H)}), \qquad V_H.$$

The height quotient maps preserve the polarizations and yield the inverse systems

$$\widehat{U}_h(\mathfrak{n}_{\Delta_5,+}), \qquad C^{\wedge}_{*,h}(\mathfrak{n}_{\Delta_5,+}), \qquad \widehat{V}_{\Delta_5}.$$

Proof. The classical master equation is the Jacobi identity paired with invariance of the cotangent pairing. Elliptic descent gives the classical factorization algebra. Harmonic reduction along the interval gives a finite-dimensional BV space. If x is even of positive root height, ad_x raises height and is nilpotent on the finite quotient, so its restrictions to the even and odd summands both have trace zero; if x is odd, ad_x is odd and has supertrace zero. Thus the modular character vanishes and the finite BV quantum master equation follows.

The chosen boundary polarization quantizes the linear Kirillov–Kostant boundary bracket to $U({\mathfrak{n}_{\Delta_5,+}}^{(H)})$. Gluing the two augmentation states gives the two-sided derived bar, and the PBW antisymmetrization identifies it with $C_*({\mathfrak{n}_{\Delta_5,+}}^{(H)})$. Holomorphic descendants generate $U^{\mathrm{ch}}(\mathrm{Cur}_E\, {\mathfrak{n}_{\Delta_5,+}}^{(H)}) = V_H$. All maps preserve root height, so they pass coefficientwise to the separated inverse limits. □

Book XVIII

The compact scalar character

Chapter 84

Reduced curve counting on $K3 \times E$

Let S be a nonsingular projective K3 surface and E an elliptic curve. For a primitive class $\beta_h \in H_2(S, \mathbb{Z})$ with $\beta_h^2 = 2h - 2$, let $N_{g,h,d}$ be the reduced disconnected Gromov–Witten invariant of class (β_h, d) with the standard incidence insertion. Define

$$\mathcal{Z}(u, q, s) = \sum_{g,h,d} N_{g,h,d} u^{2g-2} q^{h-1} s^{d-1}.$$

Theorem 84.1 (Primitive Igusa cusp-form theorem)**.** *Under $r = e^{iu}$, the partition function is the Laurent expansion*

$$\mathcal{Z}(u, q, s) = -\frac{1}{\chi_{10}(r, q, s)}.$$

Equivalently, in theta normalization,

$$\mathcal{Z} = -2^{12} \Delta_5^{\theta^{-2}}.$$

Proof. Oberdieck and Pixton identify the primitive reduced partition function with the Laurent expansion of $-\chi_{10}^{-1}$ [86]; their normalization is the monic Borcherds product χ_{10}. The second formula is the identity $\chi_{10} = 2^{-12} \Delta_5^{\theta^2}$. □

The theorem identifies the numerical generating series in its stated primitive sector. The Borcherds root algebra supplies a separate algebraic character to compare with that series.

Corollary 84.2 (Exact character comparison)**.** *Let $\jmath : \mathbb{H}_2 \to \mathbb{H}_2$ be the half-coordinate map $\jmath(Z) = Z/2$, and let*

$$\mathrm{Fock}(\mathfrak{n}_{\Delta_5,+}) = \mathrm{Sym}(\mathfrak{n}_{\Delta_5,+\bar{0}}) \otimes \bigwedge(\mathfrak{n}_{\Delta_5,+\bar{1}}).$$

Then

$$\mathcal{Z}^{\mathrm{red}}_{S \times E} = -\jmath^*\Big(e^{2\rho}\, \mathrm{sch}\big(\mathrm{Fock}(\mathfrak{n}_{\Delta_5,+})^{\otimes 2}\big)\Big) = -\chi_{10}^{-1}.$$

Proof. Put

$$P = \prod_{\alpha > 0} (1 - e^{-\alpha})^{\mathrm{sdim}\, \mathfrak{g}_{\Delta_5 \alpha}}.$$

The denominator theorem gives $\mathrm{den}(\mathfrak{g}_{\Delta_5}) = e^{-\rho} P = D_5(2Z)$, while PBW gives $\mathrm{sch}\,\mathrm{Fock}(\mathfrak{n}_{\Delta_5,+}) = P^{-1}$. Hence

$$e^{2\rho}\, \mathrm{sch}\big(\mathrm{Fock}(\mathfrak{n}_{\Delta_5,+})^{\otimes 2}\big) = \mathrm{den}(\mathfrak{g}_{\Delta_5})^{-2}.$$

Pullback by $\jmath$ sends $\mathrm{den}(\mathfrak{g}_{\Delta_5})$ to $D_5(Z)$, whose square is χ_{10}. The reduced incidence convention supplies the overall minus sign in the Oberdieck–Pixton theorem. □

The equality compares the completed root-Fock supercharacter with the numerical reduced generating series. Its proof uses the denominator identity and the half-coordinate pullback. A map from compact geometric states to this Fock space, preserving operators and their trace, is an additional comparison.

Chapter 85

Extensions over the elliptic classifying stack

Work on locally noetherian complex schemes with the fppf topology. Let BE be the stack of right E-torsors. For a left E-stack M, an object of $[M/E](T)$ is a right torsor $P \to T$ and an equivariant map $P \to M$. Forgetting the map gives $[M/E] \to BE$. All products of stacks below are two-fibre products.

Proposition 85.1 (The common translation torsor)**.** *There is a natural equivalence*

$$[(M \times N)/E_{\mathrm{diag}}] \simeq [M/E] \times_{BE} [N/E].$$

The equivalences for any finite number of factors commute with regrouping and insertion of the terminal E-stack.

Proof. An object on the right is (P, m), (Q, n) together with a torsor isomorphism $\alpha : P \to Q$. Send it to $(P, m, n\alpha)$. Conversely, use a common torsor twice and its identity isomorphism. A pair of torsor maps a, b on the right satisfies $b\alpha = \alpha' a$, so b is determined by a. The same construction identifies the maps and their descent isomorphisms. It commutes with base change. For several factors, every regrouping gives one torsor with the same list of equivariant maps, proving all stated coherences. □

For $M = N = E$ with translation, the diagonal quotient is E, with coordinate $y - x$. Both separate quotients are points. Their fibre product over BE retains the relative translation, while their ordinary product loses it.

Let $\mathfrak{M}$ be the stack of coherent sheaves on $X = S \times E$, with families flat over the parameter scheme, and put $\mathfrak{Q} = [\mathfrak{M}/E]$. An object of $\mathfrak{Q}(T)$ is a torsor $P \to T$ and a flat sheaf on $S \times P$. Fix a polarization and write $\mathfrak{Q}_\alpha$ for a Hilbert-polynomial component. Translation preserves these polynomials. Let $\mathfrak{H}_{\alpha,\beta}$ classify short exact sequences $0 \to A \to F \to B \to 0$ on a common $S \times P$, with all terms flat over T and polynomials α, $\alpha + \beta$, β. Its span over BE is

$$\mathfrak{Q}_\alpha \times_{BE} \mathfrak{Q}_\beta \xleftarrow{p} \mathfrak{H}_{\alpha,\beta} \xrightarrow{q} \mathfrak{Q}_{\alpha+\beta}.$$

The category of spans of fppf stacks over BE has composition by fibre product and tensor product $\times_{BE}$. Its unit object is the identity map of BE.

Theorem 85.2 (Geometric Hall correspondence)**.** *For fixed α, β, q is representable by algebraic spaces and proper. The spans form a coherently associative unital Hall object in the category of spans over BE. Its unit is the zero-sheaf span $BE \leftarrow BE \to \mathfrak{Q}$.*

Proof. Over a fixed (P, F), the fibre of q classifies quotients $F \twoheadrightarrow B$ of polynomial β. Its kernel is flat, because F and B are flat. An automorphism fixing the quotient and F is the identity. Trivializing P fppf-locally identifies this fibre with the fixed-polynomial Quot scheme, projective over the parameter scheme by Grothendieck's theorem [53, Theorem 3.2]. The quotient functor descends to an algebraic space, and properness descends with it.

Both composites of three Hall spans classify $0 \subset A \subset D \subset F$ on the same torsor. One bracketing concatenates inclusions. The other starts with $A \subset F$ and a subobject $B \subset F/A$, and sets $D = F \times_{F/A} B$. These constructions are inverse on objects and isomorphisms. They preserve flatness and commute with base change. For longer flags all regroupings identify the same subobjects and quotients. Their comparison maps agree, proving the pentagon and higher coherences. Finally $\mathfrak{Q}_0 = BE$. The sequences with zero subobject or zero quotient give the unit and its coherences. □

Properness here concerns the full coherent-sheaf stack on a fixed pair of Hilbert-polynomial components. A stability restriction can replace its fibre by an open part of a Quot scheme, which need not be proper. A bounded charge window also needs closure under the extensions and flags used in its operations. A reduced critical-coefficient realization must supply pullback, pushforward, and orientation comparisons on these same flags. Its unit has stabilizer E. A coefficient theory for this quotient must include that nonaffine stabilizer, or work relative to BE, on the unit and every flag.

Chapter 86

Additive physical charge and quadratic automorphic degree

Let

$$\Lambda = \widetilde{H}(S, \mathbb{Z})$$

be the Mukai lattice of signature $(4, 20)$. The electric–magnetic charge lattice is

$$\Gamma_{\mathrm{phys}} = \Lambda \oplus \Lambda, \qquad c = (Q, P).$$

Its addition is the physical superselection law. Define the Gram map

$$\Pi(Q, P) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2} \right) \in \Gamma_{\mathrm{ind}}.$$

Proposition 86.1 (Polarization identity). *For $c_i = (Q_i, P_i)$,*

$$\Pi(c_1 + c_2) = \Pi(c_1) + \Pi(c_2) + B(c_1, c_2),$$

where

$$B(c_1, c_2) = \big(Q_1 \cdot Q_2,\, Q_1 \cdot P_2 + Q_2 \cdot P_1,\, P_1 \cdot P_2 \big).$$

The cross-term is a symmetric vector-valued two-cocycle:

$$B(c_1, c_2) + B(c_1 + c_2, c_3) = B(c_2, c_3) + B(c_1, c_2 + c_3).$$

Proof. Expand the three quadratic expressions. The cocycle identity is bilinearity of B. □

86.1 The polarized central extension

Define the abelian central extension

$$0 \longrightarrow \Gamma_{\mathrm{ind}} \longrightarrow \widetilde{\Gamma}_\Pi \longrightarrow \Gamma_{\mathrm{phys}} \longrightarrow 0, \qquad \widetilde{\Gamma}_\Pi = \Gamma_{\mathrm{phys}} \times \Gamma_{\mathrm{ind}},$$

with multiplication

$$(c, \lambda) \cdot (d, \mu) = \big(c + d,\, \lambda + \mu + B(c, d) \big).$$

The cocycle identity makes this multiplication associative. The polarization identity makes the graph map

$$j_\Pi : \Gamma_{\mathrm{phys}} \longrightarrow \widetilde{\Gamma}_\Pi, \qquad c \longmapsto (c, \Pi(c))$$

a homomorphism.

Let $A = \bigoplus_{c \in \Gamma_{\mathrm{phys}}} A_c$ be a charge-graded associative algebra, and let $\Bbbk[\Gamma_{\mathrm{ind}}]$ have monomials u^λ. Define the polarized central-extension algebra

$$\mathcal{R}_\Pi(A) = A \otimes \Bbbk[\Gamma_{\mathrm{ind}}]$$

with product

$$(a_c u^\lambda) \star (b_d u^\mu) = a_c b_d \, u^{\lambda+\mu+B(c,d)}.$$

Give $a_c u^\lambda$ degree $(c, \lambda) \in \widetilde{\Gamma}_\Pi$.

Theorem 86.2 (Graph-multiplicative polarization)**.** *The product $\star$ is associative and $\widetilde{\Gamma}_\Pi$-graded. The map*

$$\iota_\Pi : A \longrightarrow \mathcal{R}_\Pi(A), \qquad a_c \longmapsto a_c u^{\Pi(c)}$$

is an algebra homomorphism whose homogeneous degree is the graph element $j_\Pi(c)$.

Proof. The two parenthesizations differ by

$$B(c,d) + B(c+d,e) - B(d,e) - B(c,d+e),$$

which vanishes by the cocycle identity. The polarization formula gives

$$\iota_\Pi(a_c) \star \iota_\Pi(b_d) = a_c b_d \, u^{\Pi(c+d)}.$$

Thus the graph of the quadratic Gram map is multiplicative inside the central extension. □

The same construction applies to Lie superalgebras and their enveloping algebras. It retains additive physical charge, Gram degree, and the interaction cross-term in one grading group.

Chapter 87

Point-local and wrapped sectors

Let $p_E : S \times E \to E$ be projection.

Lemma 87.1 (Positive elliptic degree is wrapped)**.** *Let $C \subset S \times E$ be a proper one-dimensional subscheme whose pushforward $(p_E)_*[C]$ has positive degree. Then $p_E(C) = E$.*

Proof. Properness makes $p_E(C)$ closed. Positive pushforward degree implies that at least one irreducible component of C dominates a one-dimensional closed subset of E. Every one-dimensional closed subset of the irreducible curve E is E itself, so the image of that component, and hence the image of C, is all of E. □

Projection-finite objects define point-supported sectors on $\mathrm{Ran}(E)$. Positive elliptic degree defines wrapped sectors supported over the entire elliptic curve.

Definition 87.2 (The local–wrapped factorization operad)**.** Let Ch_E denote the chosen chiral/factorization operad on E and let Ass denote the ordered associative operad. Define the colored operad Hyb_E with colors

$$\mathsf{l}, \qquad \mathsf{w}_m \quad (m \geq 1)$$

by the following operation objects.

(i) Operations with local output are

$$\mathsf{Hyb}_E(\mathsf{l}^{\otimes k}; \mathsf{l}) = \mathsf{Ch}_E(k).$$

(ii) For $r \geq 1$ and $m = m_1 + \cdots + m_r$, operations with wrapped output are

$$\mathsf{Hyb}_E(\mathsf{l}^{\otimes k}, \mathsf{w}_{m_1}, \ldots, \mathsf{w}_{m_r}; \mathsf{w}_m) = \mathsf{Ch}_E(k) \otimes \mathsf{Ass}(r).$$

All other operation objects are initial.

Operadic composition uses chiral insertion and disjoint union in the first factor and ordered substitution in the second. The equality

$$(\gamma_1 \circ \gamma_2) \otimes (a_1 \circ a_2) = (\gamma_1 \otimes a_1) \circ (\gamma_2 \otimes a_2)$$

is the mixed interchange law. The arities $(k, r) = (k, 0), (0, 2), (k, 1)$ recover, respectively, local factorization, wrapped multiplication, and the action of local insertions on a wrapped sector.

Definition 87.3 (Hybrid local–wrapped algebra)**.** A hybrid local–wrapped algebra over E is an algebra over Hyb_E. It consists of a factorization algebra $\mathbb{F}_{\mathrm{loc}}$, a height- or charge-complete family $W = \prod_{m>0} W_m$, ordered products

$$\mu_{m_1,\ldots,m_r} : W_{m_1} \otimes \cdots \otimes W_{m_r} \longrightarrow W_{m_1+\cdots+m_r},$$

and factorization-compatible actions of the local operation objects $\mathsf{Ch}_E(k)$ on every wrapped product. The operad axioms state associativity, local factorization, and compatibility of the two operations.

A compact Hall theory of $S \times E$ produces a correspondence-valued Hyb_E-algebra when disjoint-support direct sums, wrapped extensions, and the mixed extension correspondences satisfy the same base-change identities.

Book XIX

The Borcherds algebra and its quantum group

Chapter 88

The local–wrapped root algebra

The positive cone is graded by $\gamma = (n, \ell, m)$, with elliptic degree m. Set

$$\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}} = \bigoplus_{\substack{\alpha>0\\ m(\alpha)=0}} \mathfrak{g}_{\Delta_5\,\alpha}, \qquad \mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}} = \bigoplus_{\substack{\alpha>0\\ m(\alpha)>0}} \mathfrak{g}_{\Delta_5\,\alpha}.$$

Proposition 88.1 (Semidirect local–wrapped decomposition)**.** *The space $\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}$ is a Lie subsuperalgebra, $\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}$ is a Lie ideal, and*

$$\mathfrak{n}_{\Delta_5,+} = \mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}} \ltimes \mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}.$$

The proper height filtration preserves both summands.

Proof. The bracket adds root degrees. The sum of two roots of elliptic degree zero again has degree zero, and addition of a nonnegative degree to a positive degree remains positive. Height is additive on every nonzero bracket. □

Define the complete enveloping Hopf algebra and current envelope

$$\mathbb{U}_{\Delta_5} = \widehat{U}_h(\mathfrak{n}_{\Delta_5,+}), \qquad \mathbb{V}_{\Delta_5,E} := \widehat{V}_{\Delta_5} = \varprojlim_H U^{\mathrm{ch}}\Big(\mathrm{Cur}_E(\mathfrak{n}_{\Delta_5,+}{}^{(H)})\Big).$$

The adjoint action of $\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}$ on $\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}$ extends by derivations to $\widehat{U}_h(\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}})$. Let $\#$ denote the corresponding completed smash product.

Definition 88.2 (Current-generated local–wrapped operad)**.** The colored operad $\mathsf{Hyb}^{\mathrm{cur}}_{E,\Delta}$ is generated by the chiral current operations of $\mathrm{Cur}_E(\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}})$, the ordered multiplications of $\widehat{U}_h(\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}})$, and the adjoint derivation action of the former on the latter, modulo the chiral Jacobi, associative, derivation, and semidirect-product relations.

Theorem 88.3 (Local–wrapped Borcherds decomposition)**.** *Let $\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}$ be the sum of positive root spaces of elliptic degree $m = 0$ and let $\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}$ be the sum of those with $m > 0$. Then*

$$\mathfrak{n}_{\Delta_5,+} = \mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}} \ltimes \mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}},$$

where $\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}$ is a Lie subsuperalgebra and $\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}$ is a Lie ideal. For the proper-height completion,

$$\widehat{U}_h(\mathfrak{n}_{\Delta_5,+}) \cong \widehat{U}_h(\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}}) \# \widehat{U}_h(\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}),$$

and

$$\widehat{C}_{*,h}(\mathfrak{n}_{\Delta_5,+}) \simeq \mathrm{Tot}\, C_*(\mathfrak{n}_{\Delta_5,+_{\mathrm{loc}}}; \widehat{C}_{*,h}(\mathfrak{n}_{\Delta_5,+_{\mathrm{wr}}})).$$

The primitive Lie superalgebra of the completed enveloping algebra is the height completion $\widehat{\mathfrak{n}_{\Delta_5,+}}_h = \varprojlim_H \mathfrak{n}_{\Delta_5,+}{}^{(H)}$, and the Weyl-shifted Euler supercharacter of the completed Chevalley complex is $D_5(2Z)$.

Proof. Root addition adds elliptic degree. Thus the $m = 0$ summand is closed under the bracket, and every bracket with an $m > 0$ root again has $m > 0$. PBW orders wrapped generators before local generators and gives the completed smash product. The Chevalley complex of a semidirect product is the Hochschild–Serre total complex. Primitive elements of a completed enveloping algebra are recovered height by height. The denominator theorem supplies the last character identity. □

Remark 88.4. The theorem constructs a Borcherds root algebra with local and wrapped colors. The word Hall is reserved for an algebra obtained from geometric extension correspondences.

Chapter 89

The formal Igusa quantum group

Let $\mathfrak{b}_+$ be the positive Borcherds Borel half. Form the restricted graded dual $\mathfrak{b}_- := \mathfrak{b}_+^{\vee,\mathrm{gr}}$ with bracket and cobracket transported by the invariant pairing, identify the two Cartan copies along their common radical quotient, and complete by a proper root height. The completed Drinfeld double of this paired Borel datum is denoted ${\mathfrak{g}_{\Delta_5}}^{\mathrm{pair}}$. Its positive and negative halves, diagonal Cartan, invariant form, and canonical tensor form a height-admissible Manin triple and hence a quasitriangular Lie superbialgebra.

Definition 89.1 (Height-admissible Lie superbialgebra)**.** A root-graded Lie superbialgebra is height-admissible when every root space is finite-dimensional, every bounded-height region contains finitely many roots, and the bracket, cobracket, invariant tensor, and all iterated operations are coefficientwise finite in the height completion.

Proposition 89.2 (Height admissibility)**.** *The paired Gritsenko–Nikulin Borcherds Lie superbialgebra* ${\mathfrak{g}_{\Delta_5}}^{\mathrm{pair}}$ *is height-admissible.*

Proof. The Borcherds root spaces are finite-dimensional and the chosen height is proper. The bracket and cobracket preserve total root degree. In a bounded total height, finitely many decompositions of a root occur. The canonical tensor is therefore a coefficientwise finite sum in every height quotient. □

Theorem 89.3 (Quantized Borcherds superalgebra)**.** *Choose the symmetrizable Borcherds–Cartan superdatum and multiplicity index set that present* $\mathfrak{g}_{\Delta_5}$*. Hong's construction gives a quantized enveloping Hopf superalgebra*

$$U_q(\mathfrak{g}_{\Delta_5})$$

over $\mathbb{Q}(q)$*, generated by its Cartan elements and the real and imaginary simple-root generators, with triangular decomposition*

$$U_q(\mathfrak{g}_{\Delta_5}) \cong U_q^-(\mathfrak{g}_{\Delta_5}) \otimes U_q^0(\mathfrak{g}_{\Delta_5}) \otimes U_q^+(\mathfrak{g}_{\Delta_5}).$$

Its PBW root spaces have the same graded ranks as the classical Borcherds superalgebra. The canonical pairing of the two Borel halves yields a universal R-matrix in the root-height completion of

$$U_q^+(\mathfrak{g}_{\Delta_5}) \mathbin{\widehat{\otimes}} U_q^-(\mathfrak{g}_{\Delta_5}).$$

Consequently the quantization preserves the root-multiplicity data that enter $D_5(2Z)$*.*

Proof. The Gritsenko–Nikulin algebra is a symmetrizable Borcherds–Kac–Moody Lie superalgebra. Apply the generators-and-relations construction of the quantized Borcherds superalgebra. The nondegenerate bilinear form, triangular decomposition, centre, and universal R-matrix are established by Hong [58]. The PBW theorem identifies the associated graded root spaces with their classical counterparts. Since the denominator exponents are the classical signed root multiplicities, they are unchanged by the flat root-graded deformation. □

Remark 89.4. The quantum group is attached to the Borcherds generalized Cartan superdatum. A geometric Hall quantum group is identified with it through a Hall coproduct, pairing, completion, and primitive comparison.

Chapter 90

A holomorphic–topological BF realization

For every height bound H, let $\mathfrak{n}_{\Delta_5,+}{}^{(H)}$ be the finite-dimensional nilpotent quotient. On $E \times \mathbb{R}$ form the shifted cotangent theory of

$$\mathcal{L}_H = \Omega^{0,*}(E) \mathbin{\widehat{\otimes}} \Omega^*(\mathbb{R}) \otimes \mathfrak{n}_{\Delta_5,+}{}^{(H)}$$

with action

$$S_H(A, B) = \int_{E \times \mathbb{R}} \Omega_E \wedge \left\langle B, (\bar{\partial} + \mathrm{d}_{\mathbb{R}})A + \frac{1}{2}[A, A] \right\rangle.$$

Theorem 90.1 (Classical local and exact zero-mode realization)**.** *For every H:*

(i) *S_H satisfies the classical master equation and defines a classical holomorphic–topological factorization algebra;*

(ii) *harmonic reduction along a compact interval gives the finite-dimensional cotangent BV theory of $\mathfrak{n}_{\Delta_5,+}{}^{(H)}$, whose positive-root grading makes its modular supercharacter zero and hence gives the quantum master equation;*

(iii) *after choosing the complementary Lagrangian boundary polarizations, the boundary algebra is $U(\mathfrak{n}_{\Delta_5,+}{}^{(H)})$, the interval amplitude is $\mathbf{1} \otimes^{\mathbb{L}}_{U(\mathfrak{n}_{\Delta_5,+}{}^{(H)})} \mathbf{1} \simeq C_*(\mathfrak{n}_{\Delta_5,+}{}^{(H)})$, and the holomorphic boundary current envelope is V_H.*

The transition maps yield a pro-object with boundary algebra $\widehat{U}_h(\mathfrak{n}_{\Delta_5,+})$, interval complex $C^{\wedge}_{*,h}(\mathfrak{n}_{\Delta_5,+})$, and current envelope $\widehat{V}_{\Delta_5}$.

Proof. The classical master equation is the Jacobi identity and invariance of the cotangent pairing. The classical observables satisfy factorization descent. Harmonic reduction is finite-dimensional. An even adjoint operator raises positive-root height and is nilpotent on the finite quotient; an odd adjoint operator is odd. Both have supertrace zero, so the modular term in the finite BV Laplacian vanishes. The chosen polarization quantizes the boundary Lie–Poisson bracket to the enveloping algebra; the two augmentation states give the two-sided bar, identified with Chevalley chains by PBW. Holomorphic descendants give the current algebra. All structure maps preserve height. □

The theorem gives the precise field-theoretic realization used in this volume: a classical local theory together with an exact quantum protected zero-mode sector.

Chapter 91

The determinant and Fredholm square-root operators

Let

$$\mathcal{V}_H = \bigoplus_{\substack{\alpha>0\\ h(\alpha)\le H}} \mathfrak{g}_{\Delta_5\alpha}$$

and define

$$\mathscr{D}_H|_{\mathfrak{g}_{\Delta_5\alpha}} = (1 - e^{-\alpha})\,\mathrm{id}\,.$$

Theorem 91.1 (Universal determinant section)**.** *The finite-height Berezinian is*

$$\mathrm{Ber}(\mathscr{D}_H) = \prod_{\substack{\alpha>0\\ h(\alpha)\le H}} (1 - e^{-\alpha})^{\mathrm{sdim}\,\mathfrak{g}_{\Delta_5\alpha}}.$$

The coefficientwise limit satisfies

$$e^{-\rho}\,\mathrm{Ber}(\mathscr{D}_\infty) = D_5(2Z).$$

After pullback by $\jmath(Z) = Z/2$,

$$\jmath^*(e^{-\rho}\,\mathrm{Ber}(\mathscr{D}_\infty)) = D_5(Z), \qquad D_5(Z)^2 = \chi_{10}(Z).$$

Proof. The Berezinian of a diagonal endomorphism is the product of its eigenvalues with exponent equal to the superdimension of the eigenspace. Proper height makes every coefficient stationary. The denominator theorem supplies the Weyl line and the root-to-Siegel coordinate map. □

91.1 Trace-class realization on the root Hilbert superspace

Choose homogeneous orthonormal bases of the finite-dimensional root spaces and set

$$\mathcal{H}_\Delta = \widehat{\bigoplus}_{\alpha>0} \mathfrak{g}_{\Delta_5\alpha}$$

with its induced $\mathbb{Z}/2$ grading. Let $\mathfrak{C}_{\mathrm{abs}}$ be the subchamber on which

$$\sum_{\alpha>0} \dim \mathfrak{g}_{\Delta_5\alpha} \left|e^{-\alpha(Z)}\right| < \infty.$$

For $Z \in \mathfrak{C}_{\mathrm{abs}}$, define the trace-class diagonal operator

$$K_Z|_{\mathfrak{g}_{\Delta_5\alpha}} = e^{-\alpha(Z)}\,\mathrm{id}, \qquad D_Z = 1 - K_Z.$$

On each parity summand form the skew doubling

$$\mathbb{A}_{Z,\bar\epsilon} = \begin{pmatrix} 0 & D_{Z,\bar\epsilon} \\ -D_{Z,\bar\epsilon}^{\mathsf{t}} & 0 \end{pmatrix}.$$

Orient every finite-height truncation by the fixed root order and normalize its Pfaffian to 1 at $K_Z = 0$. The oriented Fredholm super-Pfaffian is the stabilized ratio

$$\mathrm{sPf}_F(\mathbb{A}_Z) = \frac{\mathrm{Pf}_F(\mathbb{A}_{Z,\bar{0}})}{\mathrm{Pf}_F(\mathbb{A}_{Z,\bar{1}})}.$$

Theorem 91.2 (Polarized Fredholm realization)**.** *Choose the protected parity refinement and the root-space Hilbert norms above. For every Z in the absolute-convergence chamber, the diagonal operator $K_Z|_{\mathfrak{g}_{\Delta_5\alpha}} = e^{-\alpha(Z)}\,\mathrm{id}$ is trace class on the even and odd root Hilbert spaces. For $D_Z = 1 - K_Z$,*

$$\mathrm{sdet}_F(D_Z) = \frac{\det_F(D_Z|_{\bar{0}})}{\det_F(D_Z|_{\bar{1}})} = \prod_{\alpha>0}(1 - e^{-\alpha(Z)})^{\mathrm{sdim}\,\mathfrak{g}_{\Delta_5\alpha}}.$$

After choosing the standard polarization of $\mathcal{H}_\Delta \oplus \mathcal{H}_\Delta^*$ and its orientation, the paritywise skew block operators $\mathbb{A}_{Z,\bar{0}}$ and $\mathbb{A}_{Z,\bar{1}}$ have an oriented Fredholm super-Pfaffian

$$\mathrm{sPf}_F(\mathbb{A}_Z) = \frac{\mathrm{Pf}_F(\mathbb{A}_{Z,\bar{0}})}{\mathrm{Pf}_F(\mathbb{A}_{Z,\bar{1}})} = \mathrm{sdet}_F(D_Z).$$

Hence

$$e^{-\rho}\,\mathrm{sdet}_F(D_Z) = D_5(2Z).$$

Proof. The defining summability condition for $\mathfrak{C}_{\mathrm{abs}}$ is precisely the trace norm of the diagonal operator, summed in both parities. The Fredholm determinant of $1 - K_Z$ is the convergent product over eigenvalues; taking the even/odd ratio gives the superdeterminant. On each finite root-height truncation the Pfaffian of the oriented skew doubling equals the determinant of D_Z. Trace-norm convergence of the truncations passes this identity to the Fredholm limit. The final equality is the Gritsenko–Nikulin denominator formula. □

Remark 91.3. The polarization and orientation in this theorem belong to the root Hilbert space. A compact moduli-space orientation is a separate line bundle and is compared with this one by a specified intertwiner.

This constructs the square-root operator on the root Hilbert superspace where its stated summability condition holds. The nonemptiness estimate below concerns the minimal virtual superspace, whose dimensions are $|\mathrm{sdim}\,\mathfrak{g}_{\Delta_5\alpha}|$. Applying that estimate to the full root superspace requires a bound on both parity dimensions. A compact geometric realization requires an operator intertwiner and a compatible map of determinant lines.

Chapter 92

What the automorphic character determines

Theorem 92.1 (Denominator data and reconstruction data)**.** *The Borcherds product determines:*

(i) *the signed root multiplicity* $\operatorname{sdim} \mathfrak{g}_{\Delta_s \alpha}$ *in every active root degree;*

(ii) *the Weyl vector and denominator character;*

(iii) *the divisor and automorphic multiplier of the section.*

A Lie bracket is supplied by the Borcherds–Kac–Moody presentation, parity dimensions by a parity-refined root datum, Hall multiplication by extension correspondences, and an oriented square root by a square-root line with monodromy. These reconstruction data refine the scalar product independently.

Proof. The first three items are read from the exponent, prefactor, and automorphy law. Independence is witnessed by explicit pairs: two brackets on the same graded vector space have the same PBW character; an even–odd pair has signed dimension zero; distinct Hall correspondences may have the same Euler product; and twisting a square-root line by a sign local system preserves its square. Hence each refined object is reconstructed from the additional datum named in the statement. □

Recognition Theorem 92.2 (Complete primitive recognition)**.** *Let* $\mathfrak{p}$ *be a positive graded Lie superalgebra and let* $\Psi : \mathfrak{n}_{\Delta_s,+} \twoheadrightarrow \mathfrak{p}$ *be a graded Lie map. The map is an isomorphism if either:*

(i) *the parity-refined Hilbert series of* $U(\mathfrak{n}_{\Delta_s,+})$ *and* $U(\mathfrak{p})$ *agree; or*

(ii) *each primitive degree has prescribed pure parity and the signed PBW characters agree.*

The induced height completions, Chevalley complexes, and current envelopes are then isomorphic.

Proof. Choose the first height containing a homogeneous kernel vector. All decomposable PBW terms in that height are determined by lower heights and agree. The primitive parity dimension therefore drops, contradicting the first hypothesis. In the second case the logarithm of the PBW product recovers signed primitive dimensions inductively, and prescribed parity recovers ordinary dimensions. Naturality of completion, Chevalley chains, and current envelopes gives the final assertion. □

Book XX

The universal Borcherds Hall and three-Calabi–Yau models

Chapter 93

Finite simple-root windows

Let I_Δ be the multiset of simple roots of the Gritsenko–Nikulin Borcherds superalgebra. A repeated imaginary simple root is written once for each copy. The parity map is

$$p : I_\Delta \longrightarrow \mathbb{Z}/2.$$

Choose a proper height function

$$h : I_\Delta \longrightarrow \mathbb{Z}_{>0}$$

whose sublevel sets are finite. Put

$$I_H = \{ i \in I_\Delta : h(i) \le H \}.$$

The restriction of the invariant form gives a finite symmetric Borcherds–Cartan matrix

$$\begin{aligned} &A_H = (a_{ij})_{i,j \in I_H}, \\ &a_{ii} = 2 \qquad\qquad \text{for real } i, \\ &a_{ii} \le 0 \qquad\qquad \text{for imaginary } i, \\ &a_{ij} \le 0 \qquad\qquad \text{for } i \ne j. \end{aligned}$$

The evenness of the root lattice makes every imaginary diagonal entry even.

Construction 93.1 (The Igusa Borcherds quiver)**.** Order I_H. For $i < j$ place $-a_{ij}$ arrows from i to j. At an imaginary vertex i place

$$g_i = 1 - \frac{a_{ii}}{2}$$

loops. A real vertex has loop number zero. Denote the resulting quiver by $Q_{\Delta,H}$.

Proposition 93.2 (Euler form)**.** *The symmetrized Euler form of $Q_{\Delta,H}$ is A_H:*

$$\langle e_i, e_j \rangle + \langle e_j, e_i \rangle = a_{ij}.$$

Proof. For $i \ne j$, the total number of arrows between the two vertices is $-a_{ij}$, so the symmetrized Euler entry is a_{ij}. At i the value is $2 - 2g_i = a_{ii}$. □

The inclusions $I_H \subset I_{H+1}$ identify $Q_{\Delta,H}$ with a full subquiver of $Q_{\Delta,H+1}$. Representations supported on the old vertices form an exact extension-closed subcategory. This gives compatible Hall embeddings.

Chapter 94

The real-root seed window

The three real simple roots are

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3,$$

with Gram matrix

$$A_{\mathrm{re}} = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

Choose the order $\delta_1 < \delta_2 < \delta_3$. The real-root quiver Q_{re} has two arrows from vertex i to vertex j for each $i < j$.

Proposition 94.1 (Real-root Hall seed)**.** *The symmetrized Euler form of Q_{re} is A_{re}. The Hall composition algebra generated by its three simple representations is the positive quantum Kac–Moody algebra with Cartan matrix A_{re}.*

Proof. The diagonal Euler entries are 2. The total number of arrows between distinct vertices is two, hence the symmetrized off-diagonal entries are -2. The Ringel–Green Hall theorem identifies the composition algebra with the positive quantum algebra defined by that symmetric Cartan matrix. □

The three vertices already contain the Weyl group and the reflective divisor. Imaginary vertices add the automorphic correction. The finite-height tower therefore begins with a completely explicit hyperbolic Hall algebra and enlarges it by one finite collection of imaginary simple roots at each stage.

94.1 The first Ginzburg differential

Let $a_{ij}^{(1)}, a_{ij}^{(2)} : i \to j$ be the two arrows for $i < j$. Add reverse arrows $(a_{ij}^{(r)})^* : j \to i$ of degree -1 and loops t_i of degree -2. The canonical three-Calabi–Yau completion has

$$d a_{ij}^{(r)} = 0, \qquad d(a_{ij}^{(r)})^* = 0,$$

and

$$dt_i = \sum_{a : s(a) = i} a a^* - \sum_{a : t(a) = i} a^* a. \tag{94.1}$$

The identity $d^2 = 0$ follows immediately. Formula (94.1) is the derived moment-map equation for the doubled real-root quiver.

Calculation 94.2 (Vertex one)**.** At vertex 1,

$$dt_1 = \sum_{r=1}^{2} \Big(a_{12}^{(r)} (a_{12}^{(r)})^* + a_{13}^{(r)} (a_{13}^{(r)})^* \Big).$$

At vertex 3,

$$dt_3 = -\sum_{r=1}^{2}\Big((a_{13}^{(r)})^* a_{13}^{(r)} + (a_{23}^{(r)})^* a_{23}^{(r)}\Big).$$

The sum over vertices is $\sum_a (aa^* - a^*a)$. It vanishes in the cyclic quotient and after matrix trace, but generally not in the path algebra.

Proof. The outgoing arrows at vertex 1 have targets 2 and 3. The incoming arrows at vertex 3 have sources 1 and 2. The displayed formulas follow from these incidence lists. Each arrow contributes one commutator to the sum over vertices. The trace of a commutator vanishes, giving the asserted trace identity. □

Chapter 95

Hall realization of the positive algebra

Let $\mathcal{A}_H = \mathrm{Rep}_{\mathbb{F}_q}(Q_{\Delta,H})$. Its twisted Hall product is

$$[M] * [N] = v^{\langle \dim M, \dim N\rangle} \sum_{[L]} |\{N' \subset L : N' \cong N,\ L/N' \cong M\}|[L], \qquad v^2 = q.$$

The class $[0]$ is the unit: the required subobject is unique on either side. Associativity follows by counting flags $N_1 \subset N_2 \subset L$ in two orders. The twist factors agree because the Euler form is bilinear:

$$\langle \alpha, \beta\rangle + \langle \alpha + \beta, \gamma\rangle = \langle \beta, \gamma\rangle + \langle \alpha, \beta + \gamma\rangle.$$

Thus this normalization defines the unital twisted Hall algebra on isomorphism classes.

Theorem 95.1 (Finite-window composition algebra)**.** *At each vertex i of $Q_{\Delta,H}$, choose the one-dimensional representation S_i supported at i, with every loop acting by zero. Use the generic composition algebra generated by these classes across finite fields, with the twisted subobject-count multiplication above. After extending coefficients from $\mathbb{Z}[v, v^{-1}]$ to $\mathbb{C}(v)$, this is the positive quantum generalized Kac–Moody algebra for the even Borcherds–Cartan matrix A_H. The parameter in Kang–Schiffmann is $v_{\mathrm{KS}} = v^{-1}$. Their specified semisimple perverse-sheaf construction on nilpotent representation varieties gives the corresponding canonical basis.*

Proof. This is the selected-simple composition construction of Kang–Schiffmann, Section 2.3 and Theorem 2.2, with one selected zero-loop simple at each vertex. Their geometric realization uses its split graded Grothendieck group over $\mathbb{Z}[v_{\mathrm{KS}}, v_{\mathrm{KS}}^{-1}]$ (Theorem 4.1), with the stated inversion of parameter [65]. The theorem does not use every simple object of the full representation category. A vertex with c loops has q^c one-dimensional simple representations over $\mathbb{F}_q$, whereas only the chosen zero-loop representation is the generator here. □

A semi-derived Hall realization of a full quantum group additionally specifies two-periodic complexes, the invertible acyclic classes and Cartan reduction [5, 78]. Its coproduct, pairing and completion are separate from the positive composition algebra used here.

95.1 Parity and the quantized superdatum

The Gritsenko–Nikulin denominator algebra carries a parity map on its simple-root multiset. It is additive on the free lattice of labelled simple generators. Projection to the Lorentzian degree can identify contributions of different parity. Both parity dimensions must therefore be retained in each projected root space. Hong's construction applied to this Borcherds superdatum gives the positive Borel

$$U_q^+(\mathfrak{g}_{\Delta_5}{}^{(H)})$$

and the full quantized Borcherds superalgebra $U_q(\mathfrak{g}_{\Delta_5}{}^{(H)})$.

Theorem 95.2 (Ordinary and super presentations)**.** *The ordinary quiver Hall presentation and Hong's quantum superalgebra use the same specified Cartan matrix and labelled simple generators. Their relation ideals and parity conventions are different inputs. Equality of their Cartan gradings alone does not identify their PBW characters.*

Proof. The two presentations are those of the preceding Hall construction and Hong's superalgebra construction. The difference already occurs for a single isotropic generator. The enveloping algebra of a one-dimensional even abelian Lie algebra is $\mathbb{C}[x]$. Declaring its generator odd only as a graded vector space leaves a nonzero vector x^2 in degree two. The enveloping algebra of a one-dimensional odd abelian Lie superalgebra is $\bigwedge(\xi)$, since its defining relation is $2\xi^2 = [\xi, \xi] = 0$. Its degree-two component is zero. Thus parity assignment to an ordinary PBW basis does not produce the super relation ideal or its character. A comparison must respect these relations degree by degree. □

Chapter 96

The three-Calabi–Yau completion

Let $kQ_{\Delta,H}$ be the path algebra. Its algebraic three-Calabi–Yau completion is

$$\Gamma_{\Delta,H} = \Pi_3(kQ_{\Delta,H}).$$

Write J for the ideal generated by all arrows in its graded quiver. Its completed version is the separate topological dg algebra $\widehat{\Gamma}_{\Delta,H} = \varprojlim_n \Gamma_{\Delta,H}/J^n$. The algebraic Calabi–Yau theorem is used for $\Gamma_{\Delta,H}$; a topological Calabi–Yau assertion for $\widehat{\Gamma}_{\Delta,H}$ requires its completed bimodule category. This is the Ginzburg model with zero potential: a degree-minus-one reverse arrow for every arrow and a degree-minus-two loop at each vertex. The loop differential is the signed sum of incident arrow–reverse-arrow products.

Theorem 96.1 (Borcherds–Ginzburg projections)**.** *For each finite quiver $Q_{\Delta,H}$, its undeformed three-Calabi–Yau completion $\Gamma_{\Delta,H}$ is the Ginzburg dg algebra with zero potential. If $H \leq H'$ and $Q_{\Delta,H}$ is the full subquiver on a subset of vertices, there is a surjective dg algebra map*

$$\pi_{H'H} : \Gamma_{\Delta,H'} \longrightarrow \Gamma_{\Delta,H}.$$

It kills the added vertices, every arrow incident to them, the corresponding reverse arrows, and the added degree-minus-two loops. These maps compose strictly and extend continuously to the completed algebras.

Proof. The finite Calabi–Yau assertion is the Ginzburg model of Keller's completion theorem [44, 68]. We concatenate paths from left to right, so aa^* is based at the source of a. Original and reverse arrows have zero differential. For the degree-minus-two loop t_i, the differential is

$$dt_i = \sum_{s(a)=i} aa^* - \sum_{t(a)=i} a^*a.$$

The projection deletes exactly the terms involving an added vertex, leaving the same sum for the full subquiver. It therefore commutes with the differential. It preserves path length and extends to the completed path algebra. Successive projections kill the union of their omitted vertices and arrows, proving strict composition. □

The evident inclusion of graded quivers does not give the reverse dg map. A single old vertex has loop differential zero. Adding an arrow $a : i \to j$ changes that differential to aa^* at i. The map fixing the old loop therefore fails to commute with the differential.

Write $\Gamma_\Delta = (\Gamma_{\Delta,H}, \pi_{H'H})_H$ for this pro-system of dg algebras. It is separate from the ordinary Hall embeddings obtained by extension by zero on representations of a full subquiver. A direct system of Calabi–Yau dg categories compatible with those Hall embeddings requires additional dg functors. It does not follow from the inclusion of the displayed Ginzburg generators.

The quantum group retains its separately specified root-height completion. Its transition maps must preserve the Cartan generators, coproduct, and pairing. Quotienting the positive nilpotent Lie algebra by height and truncating a full quantum group are different operations.

Chapter 97

Hall states and protected superstates

The representation varieties of $Q_{\Delta,H}$ carry the perverse-sheaf complexes used in the composition-algebra construction. The full equivariant derived category is a larger carrier. Already for one real vertex and dimension one, its quotient is $B\mathbb{G}_m$, with $H^*(B\mathbb{G}_m;\mathbb{Q}) = \mathbb{Q}[c_1]$, where $|c_1| = 2$. Its total equivariant cohomology is infinite-dimensional. Finite root multiplicities therefore refer to the specified composition algebra, not to total equivariant cohomology of this stack.

Definition 97.1 (Igusa Hall category)**.** For every dimension vector $\mathbf{d}$, let

$$\mathfrak{R}_{H,\mathbf{d}} = [\mathrm{Rep}_{\mathbf{d}}(Q_{\Delta,H})/G_{\mathbf{d}}].$$

The finite-height Hall category is the direct sum of the equivariant constructible derived categories on the $\mathfrak{R}_{H,\mathbf{d}}$, equipped with Hall convolution along the stack of short exact sequences. Denote it by $\mathcal{B}_{\Delta,H}$ and set

$$\mathcal{B}_{\Delta} = \varinjlim_{H} \mathcal{B}_{\Delta,H}.$$

Independently, the protected one-particle superspace is the root-graded superspace

$$\mathcal{H}_{\mathrm{ip},H} = \bigoplus_{\alpha>0} \mathfrak{g}_{\Delta_5,\alpha}^{(H)},$$

with the parity of the Borcherds superalgebra. Its multiparticle completion is the PBW/Fock superspace of $U(\mathfrak{n}_{\Delta_5,+}^{(H)})$.

Theorem 97.2 (Hall and protected-state comparison)**.** *The composition subalgebra in the finite-window Hall theorem is the ordinary positive quantum generalized Kac–Moody algebra. This statement does not identify the Grothendieck group of the full category $\mathcal{B}_{\Delta,H}$ with that composition algebra. The protected one-particle superspace has signed root multiplicities*

$$\operatorname{sdim} \mathcal{H}_{\mathrm{ip},H,\alpha} = \operatorname{sdim} \mathfrak{g}_{\Delta_5,\alpha}^{(H)}.$$

Its height-complete PBW character is

$$\operatorname{sch}\mathrm{Fock}(\mathfrak{n}_{\Delta_5,+}) = \prod_{\alpha>0} (1-e^{-\alpha})^{-\operatorname{sdim}\mathfrak{g}_{\Delta_5,\alpha}},$$

and its Weyl-shifted inverse is $D_5(2Z)$. The composition canonical basis and the protected root presentation use the specified simple-root indexing and Cartan grading; the parity refinement belongs to the Borcherds superdatum.

Proof. The composition-algebra assertion is the finite-window theorem. The protected assertion is PBW for the Borcherds Lie superalgebra. Proper height passes both constructions to the limit. The Gritsenko–Nikulin denominator theorem gives the final product identity. □

Chapter 98

Extension determinants and critical orientations

Let C be a three-Calabi–Yau dg category over $\mathbb{C}$, with its chosen trace. For families of objects A, B over a base T, let $C(A,B)$ denote the relative derived Hom complex on T. Suppose these complexes are perfect, and that the trace gives base-change-compatible duality

$$C(B,A) \simeq C(A,B)^{\vee}[-3].$$

Put $L(A,B) = \det C(A,B)$ and $D(A) = L(A,A)$. The determinant convention is $\det C = \bigotimes_i (\det C^i)^{(-1)^i}$, with symmetry sign specified by $\chi(C) \bmod 2$.

Theorem 98.1 (Extension determinant)**.** *For an exact triangle $A \to F \to B \to A[1]$, there is a base-change-compatible isomorphism*

$$D(F) \simeq D(A) \otimes D(B) \otimes L(A,B)^{\otimes 2}.$$

The determinant comparisons obtained by refining a flag agree, with the stated determinant symmetry signs.

Proof. Apply derived Hom to the triangle in each variable. Determinant additivity gives

$$D(F) \simeq L(B,A) \otimes D(A) \otimes D(B) \otimes L(A,B).$$

Dualization and the odd shift each invert the determinant line. Thus Calabi–Yau duality identifies $L(B,A)$ with $L(A,B)$. Reorder by determinant symmetry to obtain the formula. For a full flag, all iterated decompositions identify with the same ordered tensor product of $C(A_i, A_j)$, for its successive quotients A_i. Determinant additivity for refined filtrations proves coherence. Each operation commutes with base change. □

A square-root orientation consists of lines o_A and isomorphisms $o_A^{\otimes 2} \simeq D(A)$. Extension compatibility requires maps

$$o_A \otimes o_B \otimes L(A,B) \longrightarrow o_F$$

whose squares are the determinant comparisons. Choosing coordinates on a smooth critical chart trivializes its ambient canonical line. Hall convolution requires transition maps compatible with the cross line and the zero object. On three-step flags, the two orientation maps can differ by an element of μ_2, even when their squares agree. That sign must be trivialized coherently.

Example 98.2 (A nontrivial cross line)**.** Let X be a smooth projective Calabi–Yau threefold, with a chosen volume form, and let $x \in X$. On $X \times T$, where $T = \mathbb{P}^1$, take

$$A = \mathcal{O}_X \boxtimes \mathcal{O}_T(1), \qquad B = \mathcal{O}_x \boxtimes \mathcal{O}_T.$$

Then $C(A,B) = \mathcal{O}_T(-1)[0]$, and relative Serre duality gives $C(B,A) = \mathcal{O}_T(1)[-3]$. Thus $L(A,B) = L(B,A) = \mathcal{O}_T(-1)$. The self-Hom complexes of A and B have constant determinant lines, since the factors $\mathcal{O}_T(1)$ cancel in the first self-Hom. Nevertheless

$$D(A \oplus B) = D(A) \otimes D(B) \otimes \mathcal{O}_T(-2).$$

The orientation cross factor is $\mathcal{O}_T(-1)$. A sign change cannot replace this line of nonzero degree.

Corollary 98.3 (Oriented critical coefficients)**.** *Suppose the chosen critical charts have square-root orientation maps on overlaps and extension flags, with unit and flag coherences. Suppose their vanishing-cycle comparison maps preserve these data. Then the local vanishing-cycle complexes descend to the specified oriented coefficient object. Their Hall convolution is associative whenever its pullback and pushforward maps are defined and satisfy base change on the flag diagrams.*

Proof. The overlap cocycle identifies the local coefficient complexes by descent. On a three-step flag, orientation coherence identifies the two Thom–Sebastiani coefficient maps. Proper or compact-support base change, on the stipulated domain, identifies the two composites of convolution. The zero flag gives the unit equation. □

For the quiver charts, the remaining construction is an equivariant system of square roots and coefficient maps on the actual representation and extension stacks, compatible with changes of height. For compact $S \times E$, the reduced complexes and the common elliptic torsor must also be retained. A coordinate volume on an affine chart specifies neither descent system by itself.

Chapter 99

The colored elliptic Ran space

The elliptic degree is a global charge carried by the whole configuration. Define

$$\mathrm{Ran}^{\mathrm{wr}}(E) = \coprod_{m \geq 0} \mathrm{Ran}(E)_m$$

with monoidal operation

$$(I, m) \sqcup (J, n) = (I \sqcup J, m + n)$$

on disjoint supports. The component $m = 0$ is point-local. The components $m > 0$ are wrapped.

Let

$$\mathfrak{n}_{\Delta_s,+} = \bigoplus_{m \geq 0} \mathfrak{n}_{\Delta_s,+}[m]$$

be the elliptic-degree grading. The root bracket obeys

$$[\mathfrak{n}_{\Delta_s,+}[m], \mathfrak{n}_{\Delta_s,+}[n]] \subset \mathfrak{n}_{\Delta_s,+}[m + n].$$

Construction 99.1 (Hybrid factorization algebra)**.** Form the chiral Chevalley factorization coalgebra

$$\mathcal{F}^{\mathrm{hyb}}_{\Delta,E} = C^{\mathrm{ch}}_*\Big(\mathrm{Cur}_E(\mathfrak{n}_{\Delta_s,+})\Big).$$

Retain the elliptic-degree grading on every Chevalley word and place its degree-m summand on $\mathrm{Ran}(E)_m$. Disjoint union adds the degree labels.

Theorem 99.2 (Local–wrapped factorization)**.** *$\mathcal{F}^{\mathrm{hyb}}_{\Delta,E}$ is a graded factorization coalgebra on $\mathrm{Ran}^{\mathrm{wr}}(E)$. Its degree-zero color is the current theory of $\mathfrak{n}_{\Delta_s,+}[0]$. Every degree-$m$ color is a module for the degree-zero color. The factorization product of degrees m and n lands in degree $m + n$. Its chiral primitives are $\mathrm{Cur}_E(\mathfrak{n}_{\Delta_s,+})$, and constant modes recover $\mathfrak{n}_{\Delta_s,+}$.*

Proof. The chiral Chevalley construction is a factorization coalgebra for every chiral Lie algebra. The bracket preserves elliptic-degree addition, so its differential and coproduct preserve the graded Ran placement. The primitive theorem for the Chevalley coalgebra recovers the current Lie algebra. □

This construction retains the nonnegative root-degree label on a point-supported current theory. It does not replace a sheaf projecting onto the whole elliptic curve by a point-supported sheaf. The compact wrapped comparison must map its common-torsor extension correspondences to these graded current operations, preserving support and charge. The bracket on the present carrier is the Borcherds bracket.

Chapter 100

Physical charge and Gram degree

Let Γ_{phys} be an additive physical charge lattice and let

$$\Pi(c) = \left(\frac{Q^2}{2}, Q \cdot P, \frac{P^2}{2} \right)$$

be the Gram map. Its polarization is

$$B(c,d) = \Pi(c+d) - \Pi(c) - \Pi(d).$$

Define the central extension

$$\widetilde{\Gamma}_{\Pi} = \Gamma_{\mathrm{phys}} \times \Gamma_{\mathrm{ind}}$$

with multiplication

$$(c,\lambda)(d,\mu) = (c+d, \lambda+\mu+B(c,d)).$$

Theorem 100.1 (Graph homomorphism)**.** *The map*

$$j_{\Pi} : \Gamma_{\mathrm{phys}} \longrightarrow \widetilde{\Gamma}_{\Pi}, \qquad c \longmapsto (c, \Pi(c))$$

is a group homomorphism. For every Γ_{phys}-graded algebra A, the Rees lift

$$a_c \longmapsto a_c u^{\Pi(c)}$$

turns multiplication into an additive $\widetilde{\Gamma}_{\Pi}$-grading with cocycle $u^{-B(c,d)}$.

Proof. The defining multiplication gives

$$j_{\Pi}(c)\, j_{\Pi}(d) = (c+d, \Pi(c)+\Pi(d)+B(c,d)) = j_{\Pi}(c+d).$$

The algebra identity follows immediately. □

The universal Igusa category is graded directly by the Lorentzian root lattice. A geometric $K3 \times E$ theory enters through a charge functor to $\widetilde{\Gamma}_{\Pi}$.

Chapter 101

The universal protected field theory

For each H take cotangent BF theory with gauge Lie superalgebra ${\mathfrak{n}_{\Delta_5,+}}^{(H)}$ on $\mathbb{R}^2_{\mathrm{top}} \times E$. Its fields are

$$\Omega^\bullet(\mathbb{R}^2) \mathbin{\widehat{\otimes}} \Omega^{0,\bullet}(E) \mathbin{\widehat{\otimes}} \left({\mathfrak{n}_{\Delta_5,+}}^{(H)}[1] \oplus ({\mathfrak{n}_{\Delta_5,+}}^{(H)})^\vee[2] \right).$$

The action is

$$S_H = \int_{\mathbb{R}^2 \times E} \langle B, (\mathrm{d} + \bar{\partial}) A + \frac{1}{2}[A, A] \rangle.$$

Theorem 101.1 (Classical local theory and exact quantum harmonic sector)**.** *For every finite height H the shifted-cotangent action defines a classical mixed holomorphic–topological factorization algebra on $\mathbb{R}^2_{\mathrm{top}} \times E$. Harmonic reduction on a compact product cell gives the finite-dimensional cotangent BV theory of ${\mathfrak{n}_{\Delta_5,+}}^{(H)}$, and this reduced theory satisfies the quantum master equation. Complementary Lagrangian boundary polarizations give the boundary enveloping algebra $U({\mathfrak{n}_{\Delta_5,+}}^{(H)})$, the two-sided bar complex $C_*({\mathfrak{n}_{\Delta_5,+}}^{(H)})$, and the current envelope on E. The proper-height inverse system is coefficientwise finite and defines a complete classical local theory together with an exact quantum harmonic sector.*

Proof. The invariant cotangent pairing and the Jacobi identity give the classical master equation and factorization descent. Harmonic reduction leaves a finite positive nilpotent Lie superalgebra. Every even adjoint endomorphism raises proper height and is strictly upper triangular; every odd adjoint endomorphism has vanishing supertrace. The reduced BV divergence therefore vanishes. Canonical quantization in the selected polarization gives the enveloping algebra, and the augmentation boundary states give the two-sided bar, identified with Chevalley chains by PBW. Each map preserves height, so the inverse system is defined coefficientwise. □

Chapter 102

Absolute convergence of the virtual root operator

For each positive root α, let V_α be the minimal parity realization of the signed multiplicity: it is purely even of dimension $\operatorname{sdim} \mathfrak{g}_{\Delta,\alpha}$ when this integer is nonnegative and purely odd of dimension $-\operatorname{sdim} \mathfrak{g}_{\Delta,\alpha}$ when it is negative. Set

$$\mathcal{V}_\Delta = \widehat{\bigoplus}_{\alpha>0} V_\alpha .$$

This virtual root superspace is the canonical linear carrier of the denominator character; the Lie bracket remains carried by $\mathfrak{g}_{\Delta,1}$.

Let $h(\alpha)$ be the proper height. Jacobi-form coefficient estimates give constants $C_0, C_1 > 0$ such that

$$\dim V_\alpha = \left|\operatorname{sdim} \mathfrak{g}_{\Delta,\alpha}\right| \le C_0 \exp(C_1 h(\alpha)) \tag{102.1}$$

for every positive root. The number of Gram triples of height H is polynomial in H.

Theorem 102.1 (Nonempty trace-class chamber)**.** *Let Z_0 lie in the interior of the positive Weyl chamber. For sufficiently large $t > 0$,*

$$\sum_{\alpha>0} \dim V_\alpha \left| e^{-\alpha(tZ_0)} \right| < \infty .$$

Hence the absolute-convergence chamber contains a conical tail in every interior direction.

Proof. Interior positivity gives a constant $c(Z_0) > 0$ with $\Re\alpha(Z_0) \ge c(Z_0)h(\alpha)$. Combine this with (102.1). The contribution of roots of height H is bounded by a polynomial in H times

$$\exp\bigl((C_1 - tc(Z_0))H\bigr).$$

The series converges when $tc(Z_0) > C_1$. □

Corollary 102.2 (Trace-class root Hamiltonian)**.** *On the conical tail, the diagonal operator*

$$K_Z\big|_{V_\alpha} = e^{-\alpha(Z)}\,\mathrm{id}$$

is trace class on both parity summands. The Fredholm product is holomorphic and equals the absolutely convergent Borcherds product.

Proof. The trace norm is the summable series of the preceding theorem, separated by parity. Standard trace-class determinant theory gives holomorphy, and diagonalization gives the product over root eigenvalues. □

Chapter 103

The virtual root Fredholm operator

Let

$$\mathcal{H}_{\Delta}^{\mathrm{vir}} = \widehat{\bigoplus}_{\alpha>0} V_{\alpha}$$

with an orthonormal basis in each minimal parity realization. This is the virtual root Hilbert space used in the preceding absolute-convergence theorem. In the absolute-convergence chamber define

$$K_Z\big|_{V_\alpha} = e^{-\alpha(Z)}\,\mathrm{id}, \qquad D_Z = 1 - K_Z.$$

Theorem 103.1 (Oriented Fredholm square root)**.** *The operator K_Z is trace class on the even and odd summands of the virtual root Hilbert space. Its Fredholm Berezinian is*

$$\mathrm{sdet}_F(D_Z) = \prod_{\alpha>0} (1 - e^{-\alpha(Z)})^{\mathrm{sdim}\,\mathfrak{g}_{\Delta_5\alpha}}.$$

Choose an order of the positive roots, first by height and then lexicographically. This choice orients every finite-height root space. The skew doubling

$$\mathbb{A}_Z = \begin{pmatrix} 0 & D_Z \\ -D_Z^{\mathrm{t}} & 0 \end{pmatrix}$$

has an oriented Fredholm super-Pfaffian satisfying

$$\mathrm{sPf}_F(\mathbb{A}_Z) = \mathrm{sdet}_F(D_Z).$$

Consequently

$$e^{-\rho}\,\mathrm{sPf}_F(\mathbb{A}_Z) = D_5(2Z).$$

Proof. Absolute convergence of the Borcherds product is the trace-norm summability condition. The determinant formula follows from the spectrum. At finite height the Pfaffian of the oriented skew doubling is the determinant. Trace-norm convergence passes the identity to the limit. The denominator theorem supplies the Weyl shift. □

103.1 Automorphic orientation

The product defines a section of the automorphic line $\mathcal{L}^5 \otimes \nu_5$. On the absolute-convergence chamber, the oriented Fredholm Pfaffian is a trivialization of this line. Analytic continuation uses the automorphy factor of D_5 and therefore glues the chamberwise Fredholm models with transition character ν_5.

Corollary 103.2 (Orientation character)**.** *The automorphic orientation line of the root Dirac family is the sign local system ν_5. On every real simple reflection it has value -1.*

Proof. The Fredholm product equals D_5 on the convergence chamber, and D_5 transforms with automorphy factor ν_5. This transformation law is the gluing rule for the automorphic orientation line. A real simple root occurs with multiplicity one, so reflection reverses its oriented normal line. □

Chapter 104

The universal Igusa protected effective theory

Definition 104.1 (Universal Igusa completion)**.** The protected completion selected by the K3 half-index is the quintuple

$$\mathfrak{K}_{\Delta} = \left(\Gamma_{\Delta}, \mathcal{B}_{\Delta}, \mathcal{H}_{\mathrm{ip}}, \mathcal{F}^{\mathrm{hyb}}_{\Delta,E}, \mathbb{A}_{Z}\right).$$

Its components are the pro-system of finite three-Calabi–Yau completions, the ordinary quiver Hall category, the Borcherds one-particle Lie superalgebra, the local–wrapped current factorization algebra, and the virtual root Dirac family.

Theorem 104.2 (Igusa protected completion)**.** *The quintuple* $\mathfrak{K}_{\Delta}$ *satisfies:*

(i) *its one-particle Lie superalgebra is* $\mathfrak{g}_{\Delta_5}$*;*

(ii) *its ordinary Hall presentation and Hong's quantum supergroup retain the stated Cartan grading; comparison of their relation ideals and parity-refined PBW spaces requires an additional map;*

(iii) *its chiral primitive algebra on* E *is the local–wrapped current algebra of* $\mathfrak{g}_{\Delta_5}$*;*

(iv) *its raw multiparticle supercharacter is* P^{-1}*, where* $e^{-\rho}P = \mathrm{Den}_{\Delta}$*; equivalently* $e^{\rho}\,\mathrm{sch}\,\mathrm{Fock}(\mathfrak{n}_{\Delta_5,+}) = \mathrm{Den}_{\Delta}^{-1}$*;*

(v) *the half-coordinate pullback of the doubled Weyl-normalized character is* $-{\chi_{10}}^{-1}$ *in the Oberdieck–Pixton normalization;*

(vi) *its oriented first-order determinant is* Den_{Δ} *and its orientation monodromy is* ν_5*.*

Proof. The first three statements are the Borcherds root, quiver Hall, Hong quantum-supergroup, Ginzburg, and colored-Ran theorems. PBW gives the fourth. The identity $\chi_{10} = {D_5}^2$ and the established reduced scalar theorem give the fifth. The Fredholm theorem gives the sixth. □

The tuple retains algebraic constructions associated with the Igusa presentation and their scalar characters. A geometric type-IIA or topological-string realization must construct compatible functors from its oriented extension category to these targets. Primitive states and extension amplitudes are inputs to those functors. Their action on morphisms, convolution, orientations, and chiral operations must also be specified and proved compatible.

Chapter 105

The primitive comparison and its extensions

Let $\mathfrak{p}$ be a positive-root-graded Lie superalgebra with finite-dimensional homogeneous pieces and a proper positive height. A primitive realization of the Igusa presentation is a parity-preserving homogeneous map from its labelled simple-root spaces to $\mathfrak{p}$, satisfying the defining relations.

Theorem 105.1 (Universal presentation map)**.** *Let $\mathfrak{p}$ be a positive-root-graded Lie superalgebra over $\mathbb{C}$, with finite-dimensional homogeneous pieces. Suppose a homogeneous parity-preserving map from every simple-root generating space of $\mathfrak{n}_{\Delta_5,+}$ to $\mathfrak{p}$ satisfies the defining Borcherds–Serre relations. It extends uniquely to a graded Lie-superalgebra map*

$$f : \mathfrak{n}_{\Delta_5,+} \longrightarrow \mathfrak{p}.$$

It induces continuous maps on height completions, enveloping algebras, Chevalley chains, and level-zero currents. The map f is an isomorphism if and only if it is surjective and the root-graded PBW series agree in each parity.

Proof. Extend the chosen maps to the free Lie superalgebra on the simple-root spaces. The relation hypothesis makes the defining ideal lie in its kernel, giving f. The generators prove uniqueness. On enveloping algebras the map sends $x_1 \cdots x_j$ to $f(x_1) \cdots f(x_j)$. On Chevalley chains it is the graded symmetric map on the shifted generators. It commutes with the Chevalley differential because f preserves brackets. The current map is $\partial^j x \mapsto \partial^j f(x)$, preserving $[x_\lambda y] = [x, y]$. Root degree preservation gives compatible maps at each height and hence on the stated limits.

If f is surjective, $U(f)$ is surjective. Proper positive height makes each root-graded PBW component finite-dimensional. Equality of its even and odd dimensions makes $U(f)$ bijective in every component. PBW embeds each Lie superalgebra into its enveloping algebra, so f is injective. The converse follows by applying the constructions to its inverse. □

A geometric comparison also requires functors on the actual Hall and critical categories, a compatible coproduct and Hopf pairing, and maps of determinant lines and operator domains. These categories and lines are not the domain of the primitive Lie map. The additional maps must be constructed and checked against that map. They are not consequences of its universal property.

Chapter 106

The eight diagonal-divisor forms

Cléry and Gritsenko classify eight genus-two modular forms whose divisor is the diagonal with order one [24]. Gritsenko–Nikulin and the CHL denominator constructions give the sourced reflective presentations used for the corresponding rows [46–48, 50–52]. Every row has a minimal virtual product superspace obtained from its Fourier exponents, together with a Fock determinant and a diagonal Fredholm operator. A sourced denominator presentation refines this product model by its Borcherds–Cartan bracket, Weyl group, and quantum supergroup. The half-integral rows are formed on their multiplier covers.

form	group	weight	character order
Δ_5	$\mathrm{Sp}_4(\mathbb{Z})$	5	2
Δ_2	Γ_2	2	4
Δ_1	Γ_3	1	6
$\Delta_{1/2}$	Γ_4	1/2	8
∇_3	$\Gamma_0^{(2)}(2)$	3	2
∇_2	$\Gamma_0^{(2)}(3)$	2	2
$\nabla_{3/2}$	$\Gamma_0^{(2)}(4)$	3/2	4
Q_1	$\Gamma_2(2)$	1	4

Theorem 106.1 (Eight-row algebraic realization)**.** *Each diagonal-divisor row has a proper-height virtual root superspace, a super-Fock determinant, a colored abelian current theory, and a trace-class Fredholm denominator operator whose character is the corresponding automorphic product. For every row equipped with its reflective denominator presentation, the finite-window Borcherds quiver, ordinary Hall category, three-Calabi–Yau completion, and parity-refined quantum supergroup refine this product model and recover the same denominator character.*

Proof. Local finiteness of the product exponents gives the graded superspace and its proper-height completion. The Fock and Fredholm constructions depend only on these signed multiplicities and reproduce the product coefficientwise. A reflective denominator presentation supplies a symmetrizable Borcherds superdatum; the quiver, Hall, Ginzburg, current, and Hong constructions then apply to that datum and preserve its PBW denominator. □

Chapter 107

Algebraic and physical closure

Theorem 107.1 (Universal Igusa realization theorem)**.** *The constructions have the following inputs and outputs. An arrow labelled by a construction records its dependence on that input:*

$$\phi_{0,1} \xmapsto{\text{Borch}} D_5, \qquad \phi_{0,1} \xmapsto{\text{root datum}} \mathfrak{g}_{\Delta_5}, \qquad \mathcal{A}_\Delta \xmapsto{\text{quiver}} Q_{\Delta,H} \xmapsto{\text{Hall}} \mathcal{B}_\Delta,$$

$$kQ_{\Delta,H} \xmapsto{\Pi_3} \Gamma_{\Delta,H}, \qquad \mathfrak{g}_{\Delta_5} \xmapsto{\mathrm{Cur}_E} \mathcal{F}^{\mathrm{hyb}}_{\Delta,E}, \qquad \mathfrak{g}_{\Delta_5} \xmapsto{\text{Hong}} \widehat{U}_q(\mathfrak{g}_{\Delta_5}),$$

$$\mathfrak{g}_{\Delta_5} \xmapsto{\text{root spectrum}} \mathbb{A}_Z, \qquad e^{-\rho}\,\mathrm{sPf}_F(\mathbb{A}_Z) = \mathrm{Den}_\Delta(Z) = D_5(2Z),$$

$$\mathcal{Z}^{\mathrm{red}}_{K3\times E} = -(\jmath^*\,\mathrm{Den}_\Delta)^{-2} = -\chi_{10}{}^{-1}.$$

The system contains parity-refined one-particle spaces, an ordinary Hall extension product, the Borcherds root bracket, a local–wrapped collision law, a quantum supergroup, an automorphic orientation line, a first-order Fredholm determinant, and the reduced scalar square.

Proof. The Borcherds lift, Gritsenko–Nikulin presentation, and primitive reduced curve-counting theorem give the scalar identities. The enveloping and Chevalley maps are induced by the presentation. The quiver Hall algebra and finite Calabi–Yau completion have their respective quiver and path-algebra inputs. Their common grading specifies the degrees a comparison must preserve. A map between those constructions requires the additional functors, coefficient maps, and pairings described above. □

Book XXI

Technical appendices to the preceding book

Chapter 108

Exact theta-series calculation

Set $t = q^{1/8}$ and $x = r^{1/2}$. Then

$$\vartheta_2(\tau, z) = \sum_{n \in \mathbb{Z}} t^{4n^2+4n+1} x^{2n+1},$$
$$\vartheta_3(\tau, z) = \sum_{n \in \mathbb{Z}} t^{4n^2} x^{2n},$$
$$\vartheta_4(\tau, z) = \sum_{n \in \mathbb{Z}} (-1)^n t^{4n^2} x^{2n}.$$

Squaring numerator and denominator, dividing in $\mathbb{Q}[x^{\pm 1}][[t]]$, and summing the three quotients gives the coefficient table in Chapter 1. The calculation uses integer convolution and formal-series inversion; fractional powers cancel before conversion from (t, x) to (q, r).

Chapter 109

The root reflection matrices

In the basis (f_2, f_3, f_{-2}), the form is

$$G = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 0 \end{pmatrix}.$$

The three reflection matrices are

$$S_1 = \begin{pmatrix} 1 & 4 & 4 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{pmatrix}, \quad S_2 = \begin{pmatrix} 1 & 0 & 0 \\ -2 & -1 & 0 \\ 4 & 4 & 1 \end{pmatrix}, \quad S_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Direct multiplication gives

$$S_i^2 = I, \qquad S_i^{\mathsf{T}} G S_i = G.$$

Chapter IIO

Automorphic, homological, and Hall comparisons

The half-index determinant, theta normalization, Borcherds denominator, reduced scalar theorem, charge Gram map, and geometric Hall comparison datum belong to the following constructions:

(i) the automorphic square root is completed entirely within the Borcherds product theorem;

(ii) the BKM bracket comes from the Gritsenko–Nikulin algebra;

(iii) the bar–Chevalley comparison gives the denominator a homological source;

(iv) the current and BF constructions give an exact first-order algebraic model;

(v) orientation data refine determinant lines, while an operator Pfaffian requires an actual Fredholm family;

(vi) Weyl reflections are attached to real roots, and imaginary roots enter through multiplicities;

(vii) the compact Hall comparison uses additive Mukai charge, the polarized central extension, and hybrid local–wrapped sectors.

Book entrance: Calabi–Yau Quantum Groups

An extension of geometric objects defines a correspondence. Its action on cohomology depends on the coefficient theory and the allowed pushforward. Tensor products and exchange require a coproduct and compatible intertwiners. A trace on a specified representation then produces an amplitude. The reconstruction problem is to retain these operations through the comparisons between Calabi–Yau categories, boundary fields, Hall algebras, and quantum groups.

Book **XXII**

Categorical and field-theoretic foundations

Chapter III

Calabi–Yau categories

III.1 Smoothness, properness, and the Calabi–Yau class

Let k be a characteristic-zero field and C a small stable idempotent-complete k-linear category. Smoothness means that its diagonal bimodule is perfect. Properness means that every morphism complex between its objects is perfect over k. The distinction determines the variance of the Calabi–Yau datum.

For a smooth category, a dimension-d Calabi–Yau structure is a class

$$\theta : k[d] \longrightarrow CC_*^-(C)$$

whose underlying Hochschild class identifies $I^!\,[d]$ with the identity endofunctor I of $\mathrm{Ind}(C)$. Here $I^!$ is the inverse dualizing endofunctor and CC_*^- is the completed negative cyclic chain complex. For a proper category, the corresponding input is instead a cyclic trace $CC_*(C)_{hS^1} \to k[-d]$ identifying I with the shifted Serre functor. Chapter 40 constructs these two variances and their cap equivalences.

Fix the moduli convention

$$\mathcal{M}_C(A) = \mathrm{Fun}_k^{\mathrm{ex}}(C, \mathrm{Perf}(A))^{\simeq}.$$

For smooth C, a functor in this space has a relative corepresenter E_A with perfect endomorphism complex, and $T_F\mathcal{M}_C \simeq \mathrm{RHom}(E_A, E_A)[1]$. These pseudo-perfect objects impose a finiteness condition over the base, even when C itself is nonproper.

Theorem III.1 (The two universal categorical outputs)**.** *Let C be smooth with a smooth Calabi–Yau structure of dimension d. Its Hochschild cochains carry a framed E_2-algebra structure. If $\mathcal{M}_C$ is locally geometric and locally of finite presentation with perfect cotangent complex, it carries a $(2-d)$-shifted symplectic form. For a proper category equipped only with a proper Calabi–Yau structure, the framed Hochschild conclusion still holds.*

Proof. The smooth or proper cap equivalence supplies the circle-equivariant calculus required by Theorem 19.3. For the moduli assertion, first apply the universal functor to the smooth negative cyclic class. Then apply the circle-equivariant Hochschild–Kostant–Rosenberg map and the trace-normalized weight-two projection of (40.7). This constructs the closed form and all its homotopies. Relative Serre duality identifies its underlying pairing with

$$\mathrm{RHom}(E_A, E_A)[1]^{\otimes 2} \longrightarrow A[2-d],$$

where composition, trace and the suspension sign give the arrow. Its adjoint is the equivalence $T_{\mathcal{M}_C} \simeq L_{\mathcal{M}_C}[2-d]$ proved in Theorem 40.1. □

The categorical dimension controls the degree of the moduli form, while the Hochschild operation remains framed E_2. The nonproper example $\mathrm{Perf}(k[t])$ has dimension one and a degree-one moduli form on pseudo-perfect objects. To pass from either categorical output to an ordered chiral algebra, one must additionally supply the factorization theory and coefficient comparison of Theorem 12.3.

Chapter 112

Local BV theories and factorization observables

112.1 Cyclic elliptic local Lie algebras

Let X be a complex manifold. A classical local field theory is specified by a sheaf $\mathcal{L}$ of elliptic L_∞ algebras, a local invariant pairing, and an action functional S satisfying the classical master equation. For an open set $U \subset X$, the classical observables are

$$\mathrm{Obs}^{\mathrm{cl}}(U) = C^*_{\mathrm{CE,cont}}(\mathcal{L}(U)).$$

Here $\mathcal{L}(U)$ consists of unrestricted smooth fields with its nuclear smooth-section topology. Continuous cochains use the completed symmetric powers of its continuous dual, with the completed tensor products of the elliptic-observable construction [26, 27]. The dual consists of compactly supported distributions. For $U \subset V$, restriction of fields $\mathcal{L}(V) \to \mathcal{L}(U)$ induces the covariant map on observables. For pairwise disjoint $U_i \subset V$, restrict to $\bigoplus_i \mathcal{L}(U_i)$ and multiply the pulled-back cochains. These are the prefactorization maps.

Theorem 112.1 (Factorization observables)**.** *An elliptic classical BV theory produces a factorization algebra* $\mathrm{Obs}^{\mathrm{cl}}$*. A renormalized quantization satisfying the quantum master equation produces a factorization algebra* $\mathrm{Obs}^{\mathrm{q}}$ *over* $\Bbbk[[\hbar]]$.

Proof. Restriction is a morphism of local L_∞ algebras, so its contravariant continuous Chevalley map commutes with the differential. For disjoint opens, the completed cochains of the finite direct sum are the completed tensor product of the cochains. This constructs the displayed maps and their composition identities. The elliptic descent theorem in this completed coefficient category supplies Weiss descent [26, 27]. In the quantum theory, renormalized time-ordered products define the structure maps, and the quantum master equation makes their differential square-zero and compatible with nested ultraviolet contractions [26, 27]. □

112.2 Boundary and defect restriction

Let $Y = C \times \mathbb{R}_{\geq 0}$, with C a complex curve, and let a boundary condition be given at $C \times \{0\}$. Boundary observables are holomorphic along C and ordered along the inward normal line. A compact boundary object b in the boundary factorization category gives

$$A_b = \mathrm{RHom}(b, b).$$

Theorem 112.2 (Ordered boundary algebra)**.** *When boundary observables are holomorphic on C and locally constant along the oriented normal direction, A_b is an ordered chiral algebra*

$$A_b \in \mathrm{Alg}_{\mathsf{E}_1}\left(\mathrm{Fact}_D(C), \otimes^!\right).$$

If b is a local compact generator and the Morita equivalences obey factorization descent, the boundary category is equivalent to A_b-modules in factorization coefficients.

Proof. Composition of boundary endomorphisms gives the internal E_1 multiplication. Holomorphic collision gives the factorization coefficient on C. The componentwise locality identity makes multiplication compatible with every disjoint-support factorization map. Local compact-generator Morita theory descends over the Ran diagrams. These are the recognition and Morita theorems of Volume I. □

Chapter 113

Three curve-level algebraic outputs

Let C be a smooth curve and let $\mathcal{D}_C$ be the fixed Ran category. The same geometric source can produce three algebraic outputs, and their distinction is functorial.

Definition 113.1 (Curve-level faces)**.** A transverse face is

$$A^{\perp} \in \mathrm{Alg}_{\mathsf{E}_1}(\mathrm{Fact}_D(C), \otimes^!).$$

An associative chiral face is

$$A^{\mathrm{ch}}_{\mathsf{Ass}} \in \mathrm{Alg}_{\mathsf{E}_1}(\mathcal{D}_C, \star^{\mathrm{ch}}).$$

A chiral Lie face is

$$\mathfrak{g}^{\mathrm{ch}} \in \mathrm{Alg}_{\mathsf{Lie}}(\mathcal{D}_C, \star^{\mathrm{ch}}).$$

Their bars are $\mathrm{Bar}^{\perp}$, $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}$, and $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}$.

Theorem 113.2 (Source separation)**.** *A boundary normal direction determines the transverse face. A chiral-convolution multiplication determines the associative chiral face. A Lie bracket for chiral convolution determines the Lie face. The commutator functor sends the associative chiral face to the Lie face. A chiral enveloping algebra $U^{\mathrm{ch}}(\mathfrak{g})$ satisfies*

$$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(U^{\mathrm{ch}}(\mathfrak{g})) \simeq \mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}(\mathfrak{g})$$

on the nilcomplete PBW locus. A mixed source joins the transverse and associative chiral faces through an aligned distributive law.

Proof. Each statement is the corresponding operadic restriction or adjunction in the named monoidal category. The PBW equivalence is the enveloping lane of chiral Koszul duality. The mixed statement is the distributive-law theorem of Volume I. □

This theorem replaces every dimension-based dispatch of an E_n level. The geometry of the source selects the available directions; the operadic level records the operations actually constructed.

Chapter 114

Holomorphic Chern–Simons theory

Let X be a complex threefold with holomorphic volume form Ω_X, and let $\mathfrak{a}$ be a quadratic Lie algebra. The BV fields are

$$\mathcal{E}_X = \Omega^{0,*}(X, \mathfrak{a})[1],$$

with degree -1 pairing

$$\omega(A, B) = \int_X \Omega_X \wedge \langle A, B \rangle.$$

The action is

$$S_{\mathrm{hCS}}(A) = \int_X \Omega_X \wedge \left(\frac{1}{2} \langle A, \bar{\partial} A \rangle + \frac{1}{6} \langle A, [A, A] \rangle \right).$$

Theorem 114.1 (Classical hCS factorization algebra)**.** *The action S_{hCS} satisfies the classical master equation. Its classical observables form a holomorphic factorization algebra on X.*

Proof. The Hamiltonian vector field is the Maurer–Cartan differential $\bar{\partial} A + \frac{1}{2}[A, A]$. Its square is the Dolbeault square, the derivation identity, and the Jacobi identity. The elliptic Dolbeault complex then gives factorization observables by the preceding theorem. □

A perturbative quantization is controlled by the local BV deformation complex. In complex dimension three, its one-loop gauge anomaly occupies the degree-four invariant-polynomial summand; mixed and gravitational components enter when the background fields carry the corresponding couplings. A quantization is a trivialization of the complete local anomaly class, compatible with restriction and factorization [28].

Chapter 115

Pushforward to a curve

Let $p : X \to C$ be a holomorphic map to a complex curve. For a factorization algebra $\mathbb{F}$ on X, define

$$(p_*\mathbb{F})(U) = \mathbb{F}(p^{-1}U).$$

Theorem 115.1 (Factorization pushforward). *The assignment $p_*\mathbb{F}$ is a factorization algebra on C. Suppose in addition that p is proper and locally trivial in a topology compatible with the factorization theory, that $\mathbb{F}$ is locally constant in the fiber directions, and that product-chart excision and base change hold. Then on every trivializing open $U \subset C$ with fiber F_p there is a natural equivalence*

$$(p_*\mathbb{F})(U) \simeq \int_{F_p} \mathbb{F}|_{F_p \times U},$$

compatible on overlaps. If the resulting object is holomorphic and regular holonomic along C, Riemann–Hilbert algebraization gives a factorization D-module on C.

Proof. Disjoint opens in C have disjoint inverse images, so the factorization products push forward. Weiss descent is inherited from inverse images of Weiss covers. On a product chart, local constancy in the fiber and factorization excision identify the value with fiberwise factorization homology. The base-change hypothesis glues these local equivalences. Regular-holonomic Riemann–Hilbert transports the holomorphic factorization object to the algebraic D-module setting. □

Thus a curve shadow is attached to an actual map and an actual field theory. Different fibrations and boundary conditions produce different ordered chiral algebras from the same categorical source.

Chapter 116

The derived centre and the physical bulk

For an ordered boundary algebra A, define

$$Z_{\mathrm{ord}}(A) = \mathrm{RHom}_{A \otimes A^{\mathrm{op}}}(A, A).$$

It is a canonical E_2 algebra. A bulk theory acting on the boundary gives

$$\beta : \mathrm{Obs}_{\mathrm{bulk}}|_{\partial} \longrightarrow Z_{\mathrm{ord}}(A).$$

Definition 116.1 (Open–closed completeness)**.** A boundary condition is open–closed complete on a protected sector when β is an equivalence on that sector.

The definition turns the phrase ‘‘bulk equals centre’’ into a comparison theorem. The centre exists algebraically for every ordered boundary algebra; the physical bulk is identified with it by the open–closed map.

Book XXIII

Hall and BPS algebras

Chapter 117

Extension correspondences

Let $\mathfrak{M} = \coprod_{\alpha \in \Gamma} \mathfrak{M}_\alpha$ be a moduli stack in an exact category, with additive charge α. For a stable infinity-category, replace exact sequences by cofiber sequences and use derived flag stacks and homotopy fiber products. The extension stack $\mathfrak{E}_{\alpha,\beta}$ classifies

$$0 \longrightarrow E_\beta \longrightarrow E \longrightarrow E_\alpha \longrightarrow 0.$$

Its factor map p and middle-term map q form the correspondence

$$\mathfrak{M}_\alpha \times \mathfrak{M}_\beta \xleftarrow{p} \mathfrak{E}_{\alpha,\beta} \xrightarrow{q} \mathfrak{M}_{\alpha+\beta}.$$

Fix a coefficient theory with external products, pushforward q_* and normalized pullback $p^\sharp$. For Borel–Moore coefficients, the latter is the specified refined or virtual Gysin map.

Theorem 117.1 (Hall associativity)**.** *Assume functoriality, compatibility with external products and base change for both iterated extension diagrams, including their shifts and orientations. Then*

$$a * b = q_* p^\sharp (a \boxtimes b)$$

is associative. The zero object gives a unit when its extension correspondences have the identity coefficient maps.

Proof. Both iterated extension stacks identify with the stack of flags $0 \subset F_1 \subset F_2 \subset E$. The two products become $s_* r^\sharp$, where r records the three successive factors and s records E, as proved in Proposition 41.1. The zero-object correspondences prove the unit assertion. □

Critical CoHAs require compatible vanishing-cycle coefficients and orientations on all flag diagrams. An algebra on complexes requires the higher flag coherences as well. The completion must make each output operation continuous. Positive charge completion and the inverse limit over supported moduli opens in Theorem 41.3 are distinct choices.

Chapter 118

Orientation data and the threefold Hall product

For compact Calabi–Yau threefolds, natural orientation data on moduli of coherent sheaves and perfect complexes is constructed by Joyce and Upmeier [62]. A general cohomological Hall algebra for a 3-Calabi–Yau category uses strong orientation data, which carries the coherent multiplicativity required by the attractor and extension correspondences [69].

Theorem 118.1 (Threefold Hall algebra from strong orientation)**.** *Let C be a 3-Calabi–Yau category satisfying the geometric hypotheses of Kinjo–Park–Safronov, and assume strong orientation data. Then its cohomological Donaldson–Thomas complexes carry an associative cohomological Hall product. The product is the parabolic-induction correspondence supplied by the attractor Lagrangian correspondence.*

Proof. Kinjo–Park–Safronov prove the functoriality of the Donaldson–Thomas perverse sheaf under the attractor Lagrangian correspondence and use strong orientation to define the parabolic-induction morphism. Their base-change and associativity theorem gives the CoHA [69]. □

Definition 118.2 (Reduced $K3 \times E$ Hall datum)**.** A reduced $K3 \times E$ Hall datum is a strong-oriented threefold Hall datum together with descent along the elliptic translation action, a reduced deformation theory, and a separated completion in physical charge. A full quantum-group datum additionally includes a coproduct, Cartan extension, negative half, and nondegenerate continuous Hopf pairing.

Chapter 119

The geometric E-equivariant Hall source

Let S be a projective K3 surface, let E be an elliptic curve, and put

$$X = S \times E.$$

Write $\mathrm{Coh}_{\le 1}(X)$ for coherent sheaves whose support has dimension at most one. This is an extension-closed abelian category. Its numerical charge group is denoted Γ_X, and the degree of the projected one-cycle on E is the additive function

$$m : \Gamma_X \longrightarrow \mathbb{Z}_{\ge 0}.$$

For $\gamma \in \Gamma_X$, let $\mathcal{M}_\gamma$ be the moduli stack of objects of charge γ. Translation on the elliptic factor gives an action of E on every $\mathcal{M}_\gamma$ and on every extension stack.

Oberdieck and Shen construct the E-equivariant Grothendieck ring of stacks and its Hall convolution. In the present notation the completed algebra is

$$\mathcal{H}_X^{E,\mathrm{mot}} = \widehat{\bigoplus}_{\gamma \in \Gamma_X} K_0^E(\mathrm{St}/\mathcal{M}_\gamma),$$

where the completion is taken in effective curve class and Euler characteristic.

Theorem 119.1 (Geometric reduced Hall algebra)**.** *The extension correspondence in* $\mathrm{Coh}_{\le 1}(X)$ *defines an associative E-equivariant motivic Hall product on* $\mathcal{H}_X^{E,\mathrm{mot}}$*. The Oberdieck–Shen reduced integration morphism*

$$I_X^E : \mathcal{H}_X^{E,\mathrm{mot}} \longrightarrow \widehat{\mathbb{Q}[\Gamma_X]}[\epsilon]/(\epsilon^2)$$

is multiplicative on the regular subalgebra on which their integration theorem is defined. The coefficient of ϵ is the reduced contribution of the positive-dimensional E-orbits.

Proof. The stack of short exact sequences is E-equivariant because translation preserves exactness and numerical charge. The stack of two-step filtrations identifies the two composites of three Hall products, so equivariant base change proves associativity. The integration statement is the E-equivariant motivic Hall theorem of Oberdieck–Shen [87]. Their stabilizer stratification separates fixed contributions from moving elliptic orbits and packages the latter in the square-zero parameter ϵ. □

Chapter 120

Local and wrapped geometry inside the Hall algebra

Define the completed degree-zero and positive-degree pieces

$$\mathcal{H}_X^{\mathrm{loc}} = \widehat{\bigoplus}_{m(\gamma)=0} K_0^E(\mathrm{St}/\mathcal{M}_\gamma), \qquad \mathcal{H}_X^{\mathrm{wr}} = \widehat{\bigoplus}_{m(\gamma)>0} K_0^E(\mathrm{St}/\mathcal{M}_\gamma).$$

A pure one-dimensional support of degree zero projects to a finite subset of E. A support of positive degree projects onto E. The first sector is point-local on the elliptic curve; the second is wrapped.

Theorem 120.1 (Geometric local–wrapped decomposition)**.** *Hall multiplication respects elliptic degree:*

$$\mathcal{H}_{m_1} * \mathcal{H}_{m_2} \subseteq \mathcal{H}_{m_1+m_2}.$$

Consequently $\mathcal{H}_X^{\mathrm{loc}}$ is a subalgebra, $\mathcal{H}_X^{\mathrm{wr}}$ is a closed two-sided ideal, and

$$\mathcal{H}_X^{E,\mathrm{mot}} \cong \mathcal{H}_X^{\mathrm{loc}} \ltimes \mathcal{H}_X^{\mathrm{wr}}$$

as a complete vector space with multiplication determined by the Hall correspondences. The reduced integration morphism preserves this decomposition.

Proof. The cycle of the middle term of a short exact sequence is the sum of the cycles of its subobject and quotient. Projection to E therefore adds degrees. Degree zero is closed under addition, and positive degree absorbs addition by a nonnegative degree. The same grading is present on the target group algebra of I_X^E. □

This theorem supplies the geometric version of the local–wrapped splitting constructed from the Borcherds root algebra in the companion volume. The two constructions have the same additive elliptic degree and become comparable after a primitive root framing is chosen.

Chapter 121

Cohomological refinement and the Gram character

A cohomological refinement uses the Donaldson–Thomas sheaf on the derived moduli stack. The datum required for the convolution is the strong orientation coherence of Kinjo–Park–Safronov. A strong orientation enhancement of the natural Joyce–Upmeier orientation therefore produces

$$\mathcal{H}_X^{E,\mathrm{coh}} = \widehat{\bigoplus}_{\gamma\in\Gamma_X} H_*^{\mathrm{BM},E}(\mathcal{M}_\gamma, \Phi_\gamma),$$

with its cohomological Hall product.

Theorem 121.1 (Strong-oriented cohomological lift)**.** *For every strong orientation enhancement on the E-equivariant derived moduli problem, the attractor correspondence constructs an associative cohomological Hall algebra $\mathcal{H}_X^{E,\mathrm{coh}}$. Elliptic degree again gives a subalgebra in degree zero and a closed ideal in positive degree. Forgetting the cohomological coefficient system recovers the corresponding motivic Hall correspondences.*

Proof. The parabolic-induction theorem of Kinjo–Park–Safronov gives the cohomological product from the strong-oriented attractor Lagrangian correspondence [69]. Numerical charge and elliptic degree are constant on connected components of every correspondence, so the proof of the preceding decomposition applies verbatim. □

For the dyonic charge lattice

$$\Gamma_{\mathrm{phys}} = \widetilde{H}(S,\mathbb{Z}) \oplus \widetilde{H}(S,\mathbb{Z}),$$

write

$$\Pi(Q,P) = \left(\frac{Q^2}{2}, Q\cdot P, \frac{P^2}{2}\right).$$

To compare sheaf charges with this lattice, choose an additive map

$$c : \Gamma_X \longrightarrow \Gamma_{\mathrm{phys}}.$$

Put $B_\Pi(a,b) = \Pi(a+b) - \Pi(a) - \Pi(b)$ and give $\widetilde{\Gamma}_{\mathrm{phys}} = \Gamma_{\mathrm{phys}} \times \mathbb{Z}^3$ the group law

$$(a,u)(b,v) = (a+b, u+v+B_\Pi(a,b)).$$

Then $j_\Pi(a) = (a, \Pi(a))$ is a homomorphism. Assume $j_\Pi c$ preserves the selected charge completions. On the regular subalgebra $\mathcal{H}_{\mathrm{reg}}$ of the reduced integration theorem, define

$$I^E_{X,\Pi,c} = (j_\Pi c)_* I^E_X : \mathcal{H}_{\mathrm{reg}} \longrightarrow \widehat{\mathbb{Q}[\widetilde{\Gamma}_{\mathrm{phys}}]}[\epsilon]/(\epsilon^2).$$

Write $I^E_{X,\Pi}$ for this map when c has been fixed.

Proposition 121.2 (Polarized reduced character). *The map $I^{E}_{X,\Pi,c}$ is multiplicative. Its coefficients are the reduced invariants supplied by I^{E}_{X}, regraded by $j_{\Pi}c$. If $g \in 1 + F^{1}H$ is group-like in a complete filtered Hall bialgebra and the logarithm converges, then* $\log g$ *is primitive.*

Proof. Expanding $\Pi(a+b+d)$ gives

$$B_{\Pi}(a,b) + B_{\Pi}(a+b,d) = B_{\Pi}(b,d) + B_{\Pi}(a,b+d).$$

This proves associativity of the group law. The definition gives $j_{\Pi}(a+b) = j_{\Pi}(a)\, j_{\Pi}(b)$. Composing its group-algebra map with reduced integration proves multiplicativity on $\mathcal{H}_{\mathrm{reg}}$. For the last assertion, the two factors in $\Delta g = (g \widehat{\otimes} 1)(1 \widehat{\otimes} g)$ commute. Taking the convergent logarithm gives $\Delta \log g = \log g \widehat{\otimes} 1 + 1 \widehat{\otimes} \log g$. □

The primitive reduced Gromov–Witten theorem supplies the separate character $-\chi_{10}^{-1}$ in classes primitive in the K3 factor [86]. Comparing this character with the Hall character requires the same charge map, sector, and normalization. A primitive root framing must also preserve the extension bracket and parity of the chosen BPS states.

Chapter 122

BPS primitives

For a bialgebra H in graded vector spaces with Koszul symmetry, unit 1 and coproduct Δ, its primitive Lie algebra is

$$\operatorname{Prim}(H) = \{x : \Delta x = x \otimes 1 + 1 \otimes x\}.$$

Its bracket is the graded commutator. A geometric BPS Lie algebra can be constructed before a coproduct on the full Hall algebra. Let $\mathrm{JH} : \mathfrak{M} \to M$ be the good moduli map of a geometric 2-Calabi–Yau category satisfying the hypotheses of Theorem 41.2. Let $\mathscr{H} = \mathrm{JH}_* \, \mathbb{DQ}^{\mathrm{vir}}$ be its relative CoHA and let $\mathscr{B}_{\mathrm{Alg}} = \mathcal{H}^0(\mathscr{H})$ be its relative BPS algebra. This is degree zero in mixed Hodge modules, before taking global sections.

Theorem 122.1 (Two-Calabi–Yau BPS algebra)**.** *With the bilinear sign twist ψ of Theorem 41.2,*

$$\mathbb{H}^*(M, \mathscr{B}_{\mathrm{Alg}}^{\psi}) \simeq U(\mathfrak{n}_{\mathrm{BPS}}^+).$$

Here $\mathfrak{n}_{\mathrm{BPS}}^+$ is the positive generalized Kac–Moody Lie superalgebra whose simple-root spaces come from the specified intersection complexes and isotropic small diagonals. With a positive determinant line bundle, the full CoHA instead has the PBW isomorphism

$$\mathbb{H}^*(M, \mathscr{H}^{\psi}) \simeq \operatorname{Sym}(\mathfrak{n}_{\mathrm{BPS}}^+ \otimes \mathbb{Q}[u]), \qquad |u| = 2,$$

as graded mixed Hodge structures.

Proof. The canonical generator map identifies the relative BPS algebra with the convolution enveloping algebra by [31, Theorem 1.1]. Global sections give its absolute version [31, Theorem 1.2]. The determinant action followed by symmetrized Hall multiplication gives the separate PBW map [31, Theorem 1.7 and Corollary 1.8]. The construction and the hypotheses are given in Theorem 41.2. □

The K3 point-sheaf example (41.4) exhibits the isotropic generators and the extra u-classes explicitly. The enveloping algebra carries its usual primitive coproduct. Comparing that coproduct with geometric cooperation on the full CoHA remains additional data. Yangian structures use the separate equivariant correspondences of (41.5) and Theorem 41.3.

Chapter 123

The integral Hall–Borcherds amalgam

Let $\mathfrak{p}_X^+$ be a complete charge-graded BPS Lie superalgebra supplied by a geometric Hall source, and let $\mathfrak{n}_{\Delta_s,+}$ be the positive Igusa Borcherds Lie superalgebra. The polarized charge extension places both gradings over one abelian Cartan algebra $\mathfrak{h}_\Pi$. Form the two complete Borels

$$\mathfrak{b}_X = \mathfrak{h}_\Pi \ltimes \mathfrak{p}_X^+, \qquad \mathfrak{b}_\Delta = \mathfrak{h}_\Pi \ltimes \mathfrak{n}_{\Delta_s,+}.$$

The Cartan acts through the physical charge on the first Borel and through Gram degree on the second.

Definition 123.1 (Integral protected Lie algebra)**.** The integral Hall–Borcherds algebra is the completed amalgamated coproduct

$$\mathfrak{b}_{X\boxtimes\Delta} = \mathfrak{b}_X \widehat{\amalg}_{\mathfrak{h}_\Pi} \mathfrak{b}_\Delta$$

in complete charge-filtered dg Lie superalgebras.

Theorem 123.2 (Hall–Borcherds amalgamation)**.** *The completed amalgam exists and represents pairs of continuous Lie morphisms from $\mathfrak{b}_X$ and $\mathfrak{b}_\Delta$ that agree on $\mathfrak{h}_\Pi$. The Cartan projections of the two Borels induce canonical retractions*

$$r_X : \mathfrak{b}_{X\boxtimes\Delta} \longrightarrow \mathfrak{b}_X, \qquad r_\Delta : \mathfrak{b}_{X\boxtimes\Delta} \longrightarrow \mathfrak{b}_\Delta.$$

Its completed enveloping algebra is the amalgamated free product

$$\widehat{U}(\mathfrak{b}_{X\boxtimes\Delta}) \cong \widehat{U}(\mathfrak{b}_X) \widehat{*}_{\widehat{U}(\mathfrak{h}_\Pi)} \widehat{U}(\mathfrak{b}_\Delta).$$

The geometric Hall operations survive under r_X, while the Igusa bracket, PBW character, and automorphic determinant survive under r_Δ.

Proof. Construct the ordinary Lie coproduct from the free product of the two Borels and divide by the ideal identifying their two Cartan copies. Complete by total positive charge and proper root height. Every fixed filtered degree uses finitely many Lie words, so the completion is separated and has the stated mapping property. The projection $\mathfrak{b}_X \to \mathfrak{h}_\Pi \to \mathfrak{b}_\Delta$ together with the identity on $\mathfrak{b}_\Delta$ gives r_Δ by universality; the second retraction is symmetric. The enveloping functor preserves coproducts of Lie algebras, and PBW identifies the completed algebraic filtration with the displayed completed free product. □

Definition 123.3 (Comparison ideal)**.** A primitive comparison $\varphi : \mathfrak{n}_{\Delta_s,+} \to \mathfrak{p}_X^+$ preserving Cartan degree defines the closed ideal

$$\mathcal{J}_\varphi = \overline{\langle i_\Delta(x) - i_X(\varphi(x)) \mid x \in \mathfrak{n}_{\Delta_s,+} \rangle} \subset \mathfrak{b}_{X\boxtimes\Delta}.$$

The diagonal realization is

$$\mathfrak{b}_{X\simeq\Delta,\varphi} = \mathfrak{b}_{X\boxtimes\Delta} / \mathcal{J}_\varphi.$$

Proposition 123.4 (Graph quotient of a primitive comparison)**.** *Let $\varphi : \mathfrak{b}_\Delta \to \mathfrak{b}_X$ be a continuous Lie morphism restricting to the identity on $\mathfrak{h}_\Pi$. Use a completion in which the amalgam and closed quotient have their stated universal properties. Then*

$$\mathfrak{b}_{X\boxtimes\Delta}/\mathcal{J}_\varphi \cong \mathfrak{b}_X.$$

The two factor maps become id *and φ. If φ is an isomorphism, the quotient is also isomorphic to $\mathfrak{b}_\Delta$.*

Proof. The maps id and φ agree on $\mathfrak{h}_\Pi$. They give $q_\varphi : \mathfrak{b}_{X\boxtimes\Delta} \to \mathfrak{b}_X$, which kills the generators of $\mathcal{J}_\varphi$. The inclusion of $\mathfrak{b}_X$ gives a map s in the other direction. The composite $q_\varphi s$ is the identity. The other composite is the identity on the geometric factor. On the Borcherds factor, the defining relation identifies x with $s\varphi(x)$, so it is also the identity there. The two factor images topologically generate the quotient. Continuity proves that the maps are inverse on the completion. □

The graph quotient implements a specified primitive comparison. Constructing that comparison requires geometric simple states whose extension brackets satisfy the Borcherds relations. The quotient formula alone produces neither those states nor their orientation data.

Chapter 124

Coproducts, pairings, and doubles

A positive Hall algebra becomes a quantum group after additional structure is supplied. Let H^+ be a complete graded bialgebra, let H^0 be a Cartan algebra, and let H^- be the opposite or dual half. Let

$$\langle -, - \rangle : H^+ \mathbin{\widehat{\otimes}} H^- \longrightarrow \Bbbk$$

be a nondegenerate Hopf pairing.

Definition 124.1 (Completed Hall double)**.** The completed Drinfeld double is the topological algebra

$$\widehat{D}(H^+) = H^- \mathbin{\widehat{\otimes}} H^0 \mathbin{\widehat{\otimes}} H^+$$

with internal products in the three factors and cross-relations determined by the Hopf pairing. Its representation category is monoidal by the coproduct and braided by the universal R-matrix whenever the canonical tensor converges.

Theorem 124.2 (Double construction)**.** *Let H^+ and H^- be paired complete graded bialgebras, let H^0 be a Cartan bialgebra acting by the stated weight grading, and let*

$$\langle -, - \rangle : H^+ \mathbin{\widehat{\otimes}} H^- \longrightarrow \Bbbk$$

be a continuous nondegenerate Hopf pairing. Assume the triangular multiplication map from $H^- \mathbin{\widehat{\otimes}} H^0 \mathbin{\widehat{\otimes}} H^+$ is separated and complete. Then the pairing cross-relations determine a unique completed double with this triangular decomposition. On every continuous module category on which the canonical tensors converge, the completed canonical element gives the universal R-matrix.

Proof. The Hopf pairing defines the cross-relations between the paired halves and makes the three canonical embeddings bialgebra morphisms. The Hopf identities identify the two reductions of every triple product, proving associativity. Separated triangular decomposition gives uniqueness. Finite-degree canonical tensors form a compatible inverse system; convergence on the selected module category yields the quasitriangular element [37, 39]. □

Theorem 124.3 (Quantum Hall–Borcherds amalgam)**.** *Let H_X and H_Δ be complete Hopf algebras containing the same complete Cartan Hopf algebra H^0, with continuous Hopf retractions onto H^0. Their completed amalgamated free product*

$$H_{X \boxtimes \Delta} = H_X \mathbin{\widehat{*}_{H^0}} H_\Delta$$

has a unique complete Hopf structure extending the two coproducts, counits, and antipodes. It represents pairs of continuous Hopf morphisms agreeing on H^0 and has canonical Hopf retractions to both factors. A mixed universal R-matrix is equivalent to a continuous cross-pairing satisfying the two hexagon identities; the intrinsic R-matrices of the two retracts remain defined independently.

Proof. The algebraic amalgamated free product exists because complete filtered algebras are obtained degreewise from finite quotients. The two coproducts agree on H^{0}, so the universal property extends them to a coproduct on the free product. Coassociativity is checked on the two generating Hopf subalgebras. The same argument extends the counit and antipode. The Hopf retractions are induced by the factor retractions onto H^{0}. A quasitriangular structure extending both factors requires the additional commutation data between the two generating halves; writing this data as a cross-pairing turns the quasitriangular and Yang–Baxter equations into the two hexagon identities. □

The deconcatenation coproduct on a bar complex and the representation-theoretic coproduct on a quantum group occupy different structures. A geometric comparison between them is a theorem about a selected model.

Chapter 125

Ordered chiral bialgebras

A tensor product of representations requires compatible actions on threefold products and the unit. Retain the factorization coefficient category $(\mathrm{Fact}_D(C), \otimes^!)$ on the smooth complex curve C. Fix a category $\widehat{\mathcal{V}}_C$ of complete filtered objects with specified presentations $F \simeq \varprojlim_r F_r$, where $F_r \in \mathrm{Fact}_D(C)$. Use derived limits in the fixed Ran enhancement and define

$$F \mathbin{\widehat{\otimes}^!} G = \varprojlim_{r,s} (F_r \otimes^! G_s).$$

Assume these limits remain factorizing and give $\widehat{\mathcal{V}}_C$ a symmetric monoidal structure with unit $\mathbf{1}$. Its associativity, symmetry, and unit equivalences must preserve the presentations and the Ran structure. Morphisms are continuous maps compatible with these presentations. These are hypotheses on the chosen completion.

Definition 125.1 (Ordered chiral bialgebra)**.** An ordered chiral bialgebra is a counital E_1-coalgebra object in complete ordered chiral algebras:

$$A \in \mathrm{CoAlg}_{\mathsf{E}_1}\Big(\mathrm{Alg}_{\mathsf{E}_1}(\widehat{\mathcal{V}}_C, \mathbin{\widehat{\otimes}^!})\Big).$$

It has continuous multiplication m and unit η, and continuous morphisms of unital ordered chiral algebras

$$\Delta : A \longrightarrow A \mathbin{\widehat{\otimes}^!} A, \qquad \epsilon : A \longrightarrow \mathbf{1}.$$

The counit is the chosen augmentation. Write τ for the coefficient symmetry, including Koszul signs. Suppressing ambient associativity and unit equivalences, compatibility reads

$$\Delta m = (m \mathbin{\widehat{\otimes}^!} m)(\mathrm{id} \mathbin{\widehat{\otimes}^!} \tau \mathbin{\widehat{\otimes}^!} \mathrm{id})(\Delta \mathbin{\widehat{\otimes}^!} \Delta),$$
$$\Delta\eta = \eta \mathbin{\widehat{\otimes}^!} \eta, \qquad \epsilon m = \epsilon \mathbin{\widehat{\otimes}^!} \epsilon, \qquad \epsilon\eta = \mathrm{id}_{\mathbf{1}}.$$

Coassociativity and the counit identities read

$$(\Delta \mathbin{\widehat{\otimes}^!} \mathrm{id})\Delta = (\mathrm{id} \mathbin{\widehat{\otimes}^!} \Delta)\Delta,$$
$$(\epsilon \mathbin{\widehat{\otimes}^!} \mathrm{id})\Delta = \mathrm{id}_A = (\mathrm{id} \mathbin{\widehat{\otimes}^!} \epsilon)\Delta.$$

For a homotopy-coherent presentation, these identities include the higher coherences of the displayed coalgebra object. Every operation and coherence preserves diagonal transitions and disjoint-locus factorization maps. An antipode is a continuous convolution inverse $S : A \to A$ of id_A:

$$m(S \mathbin{\widehat{\otimes}^!} \mathrm{id})\Delta = \eta\epsilon = m(\mathrm{id} \mathbin{\widehat{\otimes}^!} S)\Delta.$$

A bialgebra with an antipode is a Hopf object in this coefficient category.

A continuous left A-module is an object $M \in \widehat{\mathcal{V}}_C$ with a unital E_1 action $\rho_M : A \mathbin{\widehat{\otimes}^!} M \to M$. Its action and coherences are morphisms in these factorization coefficients. Write $\mathrm{Mod}_A(\widehat{\mathcal{V}}_C)$ for their category.

Theorem 125.2 (Representation fusion)**.** *An ordered chiral bialgebra makes* $\mathrm{Mod}_A(\widehat{\mathcal{V}}_C)$ *monoidal. Its product is* $M \mathbin{\widehat{\otimes}^!} N$ *with action*

$$\begin{aligned} A \mathbin{\widehat{\otimes}^!} (M \mathbin{\widehat{\otimes}^!} N) &\xrightarrow{\Delta \mathbin{\widehat{\otimes}^!} \mathrm{id}} (A \mathbin{\widehat{\otimes}^!} A) \mathbin{\widehat{\otimes}^!} (M \mathbin{\widehat{\otimes}^!} N) \\ &\xrightarrow{\mathrm{id} \mathbin{\widehat{\otimes}^!} \tau_{A,M} \mathbin{\widehat{\otimes}^!} \mathrm{id}} (A \mathbin{\widehat{\otimes}^!} M) \mathbin{\widehat{\otimes}^!} (A \mathbin{\widehat{\otimes}^!} N) \\ &\xrightarrow{\rho_M \mathbin{\widehat{\otimes}^!} \rho_N} M \mathbin{\widehat{\otimes}^!} N. \end{aligned}$$

The unit is $\mathbf{1}$ *with action induced by* ϵ*. The ambient associator and unit equivalences are module equivalences.*

Suppose a continuous universal R*-matrix is also given. It is an invertible morphism* $R : \mathbf{1} \to A \mathbin{\widehat{\otimes}^!} A$ *for the tensor-product algebra, satisfying*

$$R\Delta(a) = \Delta^{\mathrm{op}}(a) R, \qquad \Delta^{\mathrm{op}} = \tau_{A,A} \Delta,$$
$$(\Delta \mathbin{\widehat{\otimes}^!} \mathrm{id})(R) = R_{13} R_{23}, \qquad (\mathrm{id} \mathbin{\widehat{\otimes}^!} \Delta)(R) = R_{13} R_{12},$$
$$(\epsilon \mathbin{\widehat{\otimes}^!} \mathrm{id})(R) = \eta = (\mathrm{id} \mathbin{\widehat{\otimes}^!} \epsilon)(R).$$

The first equation is an identity of morphisms from A *to its completed tensor square. Leg notation uses the unit and symmetry of* $\widehat{\mathcal{V}}_C$*. In a homotopy-coherent presentation, require compatible lifts of these identities. Then* $c_{M,N} = \tau_{M,N} \circ R_{M,N}$ *is a braiding, where* $R_{M,N}$ *acts through the external tensor-product module action.*

Proof. The symmetry combines the independent module actions into an $A \mathbin{\widehat{\otimes}^!} A$-action. Restriction along the unital algebra map Δ gives the displayed action. For three modules, the two actions use $(\Delta \mathbin{\widehat{\otimes}^!} \mathrm{id})\Delta$ and $(\mathrm{id} \mathbin{\widehat{\otimes}^!} \Delta)\Delta$. Coassociativity makes the ambient associator A-linear. The counit identities make the ambient unit equivalences A-linear. The coalgebra coherences give the pentagon and triangle coherences. The completion hypotheses keep these maps continuous and factorizing. Quasitriangularity makes $c_{M,N}$ an intertwiner and invertibility makes it an equivalence. The two coproduct identities for R give the hexagons. Its counit identities give compatibility with the unit. □

The tensor in this construction is the completed transverse tensor $\otimes^!$ on factorization coefficients. A chiral-convolution or Hall coproduct requires its own construction and a comparison in the specified category. A spectral version additionally requires a difference coordinate, a Laurent-series completion, and compatible domains for iterated expansions. Its identities must hold in those domains before defining module actions.

Example 125.3 (A counital algebra map without coassociativity)**.** In ordinary complex vector spaces, let $A = \mathbb{C}[x]$ and put

$$\Delta(x) = x \otimes 1 + 1 \otimes x + x \otimes x^2, \qquad \epsilon(x) = 0.$$

Polynomial substitution extends these assignments to unital algebra maps. Writing $F(u, v) = u + v + uv^2$, the equations $F(0, v) = v$ and $F(u, 0) = u$ prove both counit identities. But

$$\begin{aligned} F(F(u, v), w) - F(u, F(v, w)) &= -uv^2w^4 - uv^2w^2 - 2uvw^3 - 2uvw \\ &= -uvw(1 + w^2)(vw + 2) \neq 0. \end{aligned}$$

On three regular modules, the two actions differ already on $1 \otimes 1 \otimes 1$. Thus the canonical tensor associator is not A-linear. This ordinary example shows why multiplicativity and the counit identities must be supplemented by coassociativity. It makes no assertion about a geometric realization in factorization coefficients.

Book XXIV

Exact model cases

Chapter 126

The A_2 Hall algebra

For the quiver $1 \to 2$ over $\mathbb{F}_q$, the twisted Hall algebra has generators E_1, E_2 and $v^2 = q$. Counting subrepresentations gives

$$E_1^2 E_2 - (v + v^{-1}) E_1 E_2 E_1 + E_2 E_1^2 = 0.$$

For a representation of dimension vector $(2,1)$, the arrow has rank zero or one. The relevant flag counts are

rank	112	121	211
0	$q+1$	$q+1$	$q+1$
1	$q+1$	1	0.

After the Hall twists $v^{-1}, 1, v$, both rows give zero. This is the elementary geometric origin of the quantum Serre relation [49, 92].

Chapter 127

The tripled Jordan quiver and $\mathbb{C}^3$

Let Q have one vertex and loops x, y, z, with potential

$$W = \operatorname{Tr} x[y, z].$$

The Jacobi algebra is $\mathbb{C}[x, y, z]$. The critical CoHA uses the vanishing cycles of $\operatorname{Tr} W$ on the representation spaces. Its positive currents are generated by the one-vertex dimension sectors.

Theorem 127.1 (Affine-Yangian positive half)**.** *After the standard equivariant localization and completion, the critical CoHA of the tripled Jordan quiver is identified with the positive half of the affine Yangian of* $\mathfrak{gl}_1$*. The independently defined affine Yangian carries its standard Cartan and negative current halves; the positive-half comparison embeds the CoHA into that triangular presentation.*

Proof. The CoHA product becomes the shuffle product under localization. Dimensional reduction identifies the tripled-quiver critical theory with the equivariant surface CoHA, and Davison's affine-BPS theorem identifies its completed positive algebra with the strictly positive affine Yangian. The shuffle kernel satisfies the affine-Yangian current relations with equivariant parameters ϵ_1, ϵ_2, ϵ_3 and Calabi–Yau relation $\epsilon_1 + \epsilon_2 + \epsilon_3 = 0$. PBW and current comparison identify the positive algebra. The independently presented full affine Yangian is the triangular algebra generated by its negative currents, Cartan currents, and this positive half [30, 32, 93, 95]. □

The vacuum representation produces a $W_{1+\infty}$-type vertex algebra through an evaluation homomorphism. The CoHA, its double, and the vacuum algebra occupy successive levels: positive algebra, paired quantum group, and distinguished representation.

Chapter 128

Surface CoHAs and ADE Yangians

Let $X \to \mathbb{C}^2/\Gamma_o$ be the minimal resolution for a nontrivial finite subgroup $\Gamma_o \subset \mathrm{SL}_2(\mathbb{C})$, and let Z be the exceptional divisor. Choose a diagonal torus T_o commuting with Γ_o. Let $\mathcal{H}_{X,Z}^{T_o}$ be the rational equivariant CoHA of coherent sheaves set-theoretically supported in Z, with the sign twist and inverse-limit topology of Theorem 41.3. This source includes zero-dimensional sheaves and sheaves on thickenings of Z.

Theorem 128.1 (Curve-supported surface CoHA)**.** *There is a graded topological algebra isomorphism*

$$\mathcal{H}_{X,Z}^{T_o} \xrightarrow{\sim} \mathbf{Y}_{\infty}^{+}.$$

Here $\mathbf{Y}_{\infty}^{+}$ *is the limit* (41.6) *of the specified braid-translated slope quotients of the affine ADE Yangian. It intertwines tensor product by line bundles with the affine braid action.*

Proof. Derived McKay comparison and convergence of the translated hearts identify the source with the limit of nilpotent preprojective CoHA quotients. The compatible Yangian and braid comparisons identify that limit, including its multiplication, with $\mathbf{Y}_{\infty}^{+}$ [34, Theorems III.7.5, III.7.8 and Proposition III.7.14]. The maps and the topology are described in Theorem 41.3. □

A matching formal neighborhood of an ADE configuration on a K3 surface gives the corresponding supported CoHA comparison [34, Theorem III.7.2(ii)]. This identifies the specified local supported algebra. The global K3 category contains further support classes, and its inclusion and cooperation maps require their own geometric constructions.

Chapter 129

Coulomb branches and shifted Yangians

For a three-dimensional $\mathcal{N} = 4$ gauge theory, the BFN Coulomb branch is defined by equivariant Borel–Moore homology of an affine-Grassmannian correspondence. Its convolution algebra quantizes the Coulomb branch [18]. In quiver gauge theories, shifted Yangians and their truncations provide presentations of these quantizations. Boundary conditions select the shift and truncation data.

The BFN construction supplies a second geometric source of Yangians. Its grading and coproduct arise from loop rotation and convolution, while the surface CoHA source arises from sheaf extensions. A comparison between the two is a model-specific equivalence.

Book XXV

K3 symmetries

Chapter 130

The Mukai lattice

For a K3 surface S, the Mukai lattice is

$$\widetilde{H}(S,\mathbb{Z}) = H^0(S,\mathbb{Z}) \oplus H^2(S,\mathbb{Z}) \oplus H^4(S,\mathbb{Z})$$

with pairing

$$((r,c,s),(r',c',s')) = c\cdot c' - rs' - r's.$$

It is even unimodular of signature $(4,20)$ and rank 24.

Chapter 131

The Nakajima Heisenberg algebra

Let

$$\mathbb{H}_S = \bigoplus_{n \geq 0} H^*(S^{[n]}).$$

For $m \in \mathbb{Z} \setminus \{0\}$ and $\alpha \in H^*(S)$, the Nakajima correspondence gives an operator $p_m(\alpha)$.

Theorem 131.1 (K3 Heisenberg action)**.** *The operators satisfy*

$$[p_m(\alpha), p_n(\beta)] = m\, \delta_{m+n,0} \int_S \alpha \cup \beta \ \mathrm{id}\,.$$

They generate the rank-twenty-four Heisenberg algebra acting irreducibly on $\mathbb{H}_S$ with vacuum $1 \in H^*(S^{[0]})$.

Proof. The commutator is the excess intersection of the two incidence correspondences. Opposite creation and annihilation correspondences meet along the diagonal with multiplicity m and intersection pairing $\int_S \alpha\beta$. Correspondences of the same sign commute. The vacuum construction gives the Fock representation [85]. □

This action is global, compact, and abelian at the Lie level. It is an exact K3 symmetry and supplies the rank-twenty-four Fock sector of any broader K3 algebra.

Chapter 132

The orthogonal LLV algebra

For a compact hyperkähler manifold, every Lefschetz class gives an $\mathfrak{sl}_2$ triple on cohomology. The Lie algebra generated by all such triples is the Looijenga–Lunts–Verbitsky algebra.

Theorem 132.1 (Orthogonal LLV symmetries)**.** *For a K3 surface S, the LLV algebra acting on $H^*(S)$ is*

$$\mathfrak{g}_{\mathrm{LLV}}(S) \cong \mathfrak{so}\big(\widetilde{H}(S), (-,-)_{\mathrm{Muk}}\big) \cong \mathfrak{so}(4, 20).$$

For an irreducible hyperkähler manifold M of $K3^{[n]}$ type with $n \geq 2$, one has $b_2(M) = 23$ and

$$\mathfrak{g}_{\mathrm{LLV}}(M) \cong \mathfrak{so}(4, 21).$$

Moduli spaces of stable objects on S carry their own LLV actions; Mukai and monodromy correspondences relate these actions to the K3 lattice.

Proof. The Looijenga–Lunts–Verbitsky theorem identifies the Lie algebra generated by all Lefschetz triples with the orthogonal algebra of the Mukai extension $H^2(M) \oplus U$. For S, the extension has signature $(4, 20)$. For $K3^{[n]}$ type with $n \geq 2$, the Beauville–Bogomolov lattice has signature $(3, 20)$, so adjoining the hyperbolic plane gives signature $(4, 21)$ [77, 98]. □

The LLV algebra is a classical cohomological symmetry. A Yangian deformation of the full orthogonal action consists of two further structures: a Drinfeld current presentation and a compatible spectral coproduct.

Chapter 133

Local ADE Yangians

An ADE configuration of (-2)-curves on S gives a root sublattice of $H^2(S,\mathbb{Z})$. Sheaves supported on the configuration produce the surface CoHA of the preceding part and hence a completed positive affine ADE Yangian. Nakajima quiver varieties supply compatible Yangian actions on their equivariant cohomology.

Thus K3 geometry contains local nonabelian Yangian sectors indexed by ADE enhancements of the Picard lattice. Picard monodromy transports the ADE root sublattices over the corresponding Noether–Lefschetz loci. A family of sheaf moduli and Hecke correspondences compatible with that monodromy transports the Yangian actions themselves.

Chapter 134

The K3 symmetry trichotomy

The three exact structures are

Structure	Carrier	Operation
Mukai Heisenberg	$\bigoplus_n H^*(S^{[n]})$	creation and annihilation
LLV $\mathfrak{so}(4,20)$	$H^*(S)$; separately $\mathfrak{so}(4,21)$ on $K3^{[n]}$ type	Lefschetz Lie action
Local ADE Yangian	curve-supported equivariant moduli	Hecke and current correspondences

A global K3 quantum algebra is a system that contains these three actions with explicit comparison homomorphisms. The two-Calabi–Yau BPS theorem supplies a global generalized Kac–Moody positive algebra. A full Yangian or double is specified by a compatible coproduct, spectral parameter, negative half, Cartan part, pairing, and completion.

Chapter 135

K3 correspondences on fixed coefficients

Fix rational cohomology and a collection $\mathscr{M}_S$ of smooth projective moduli spaces on a projective K3 surface S. Include the Hilbert schemes $S^{[n]}$ and close the collection under finite products. For $M, N \in \mathscr{M}_S$, define the graded kernel space

$$K(M,N) = H^*(M \times N, \mathbb{Q})[2\dim_{\mathbb{C}} M].$$

A kernel of degree r acts from $H^j(M)$ to $H^{j+r}(N)$ by

$$T_\alpha(v) = (p_N)_*(p_M^* v \cup \alpha).$$

The shift in $K(M,N)$ accounts for integration over M. All pushforwards in this construction are proper.

Proposition 135.1 (Kernel composition)**.** *The product*

$$\beta \circ \alpha = (p_{13})_*(p_{12}^*\alpha \cup p_{23}^*\beta)$$

defines an associative graded category with diagonal identities. The kernel action identifies $K(M,N)$ *with* $\mathrm{Hom}^*(H^*(M), H^*(N))$.

Proof. The Künneth isomorphism and Poincaré duality identify $H^*(M \times N)[2\dim M]$ with $H^*(M)^\vee \otimes H^*(N)$. Under this identification, T_α is the associated linear map. Integration over the middle factor contracts its dual pair, so $T_{\beta\circ\alpha} = T_\beta T_\alpha$. Faithfulness then proves associativity and the diagonal identity. External products give tensor products with their Koszul signs. □

A geometric symmetry is specified by selected kernels in this category. Nakajima incidence kernels connect different Hilbert schemes. Lefschetz multiplication and its adjoint act on each individual cohomology space. Using every cohomology kernel would instead give all linear maps; the selected geometric kernels carry the representation-theoretic content.

Chapter 136

The algebra generated by geometric actions

Put $V_S = \bigoplus_{M \in \mathscr{M}_S} H^*(M, \mathbb{Q})$. Choose a set of homogeneous operators T_λ on V_S, each sending a vector with finite support to a vector with finite support. The Nakajima operators on the Hilbert-scheme summand satisfy this condition: each changes the point number by a fixed integer. Individual Lefschetz operators, extended by zero on other summands, also satisfy it. Let F be the free unital graded algebra on symbols t_λ.

Proposition 136.1 (Geometric action algebra)**.** *The map*

$$\rho : F \longrightarrow \mathrm{End}_{\mathbb{Q}}(V_S), \qquad t_\lambda \longmapsto T_\lambda$$

defines a faithful action of $A_S = F/\ker \rho$. For every algebra B, maps $A_S \to B$ are exactly maps $F \to B$ that kill $\ker \rho$*. Products and commutators of the selected kernels are their convolution products and commutators on the relevant moduli spaces.*

Proof. Finite words in the selected operators preserve finite support, so ρ is defined by composition. The quotient acts faithfully by definition of its kernel. The factorization property is the universal property of an algebra quotient. The convolution assertion follows from Proposition 135.1. □

An equivariant ADE construction uses a separate coefficient field, for example $\mathrm{Frac}\, H_T^*(\mathrm{pt}, \mathbb{Q})$, and its prescribed localization and proper pushforwards. Joining it to A_S requires maps between the actual state spaces that intertwine those operators and their coefficients. A completion of A_S requires a specified topology and continuity of every generator. The indefinite Mukai norm alone supplies neither condition. The desired common K3 quantum algebra must construct these local-to-global actions and their mixed convolution relations.

Chapter 137

Tensor reconstruction and geometric kernels

Let $\mathscr{C}$ be an essentially small rigid linear monoidal category and let $\omega : \mathscr{C} \to \mathrm{Vect}_{\mathrm{fd}}(\mathbb{Q})$ be a strong monoidal functor. A choice of geometric kernels specifies the morphisms of $\mathscr{C}$; it is part of the input. Define

$$H = \left(\bigoplus_{M \in \mathscr{C}} \omega(M)^* \otimes \omega(M) \right) \Big/ \left\langle [\lambda \circ \omega(f) \otimes v]_M - [\lambda \otimes \omega(f)v]_N \right\rangle,$$

where $f : M \to N$, $v \in \omega(M)$, and $\lambda \in \omega(N)^*$.

Proposition 137.1 (Matrix-coefficient coalgebra)**.** *For a basis (e_i) with dual basis (e^i), the formulas*

$$\Delta[\lambda \otimes v]_M = \sum_i [\lambda \otimes e_i]_M \otimes [e^i \otimes v]_M, \qquad \epsilon[\lambda \otimes v]_M = \lambda(v)$$

are basis-independent and make H a coalgebra. Tensor product gives its multiplication, and duals give its antipode. Each $\omega(M)$ is a comodule by

$$v \longmapsto \sum_i e_i \otimes [e^i \otimes v]_M.$$

Every selected kernel is a comodule morphism.

Proof. The identity tensor $\sum_i e_i \otimes e^i$ is basis-independent. Inserting it twice proves coassociativity; evaluation proves both counit identities. The defining relations express naturality under f and are preserved by these formulas. The strong monoidal constraints identify the coefficient of $M \otimes N$ with the product of coefficients of M and N. Their pentagon and unit identities give the bialgebra axioms. The evaluation and coevaluation identities for duals give the antipode identities. The same relations prove that $\omega(f)$ intertwines the coactions. □

The linear dual of H represents linear natural endomorphisms of ω. Tensor-compatible natural automorphisms are the corresponding multiplicative, unit-preserving families, not all elements of this dual. An equivalence with a comodule category requires the additional exactness and reconstruction hypotheses stated in the Tannaka theorem below. The action algebra A_S acts by selected kernels; the coend records symmetries commuting with those kernels.

Book XXVI

The exact $K3 \times E$ protected completion

Chapter 138

The protected input

The K3 elliptic genus is $2\phi_{0,1}$. The Borcherds half determines the weight-five denominator D_5 and its Borcherds Lie superalgebra $\mathfrak{g}_\Delta$. Fix its positive root spaces with their parity and proper height. The positive algebra and its root Fock space are defined from this Lie superalgebra, independently of a compact Hall realization.

For each finite zero-potential quiver window $Q_{\Delta,H}$, form the Ginzburg algebra $\Gamma_{\Delta,H}$. Restriction to full subquivers gives the projections of Proposition 144.1. Write Γ_Δ for this specified inverse system. A nondegenerate continuous Calabi–Yau structure on a completed limit is additional data. An ordinary Hall category $\mathcal{B}_\Delta$ must carry its chosen quiver coefficients and extension correspondences. Its identification with a root presentation requires the corresponding Hall theorem on that precise datum.

Definition 138.1 (Root-datum realizations)**.** A protected root-datum realization records

$$\mathfrak{P}(K3 \times E) = (\Gamma_\Delta, \mathcal{B}_\Delta, \mathcal{H}_{\mathrm{ip}}, \mathcal{F}^{\mathrm{hyb}}_{\Delta,E}, \widehat{U}_q(\mathfrak{g}_\Delta), \mathbb{A}_Z).$$

Here $\mathcal{H}_{\mathrm{ip}} = \bigoplus_{\alpha>0}(\mathfrak{g}_\Delta)_\alpha$ with its parity. The current-factorization, paired quantum-group, and Fredholm entries retain their own coefficient categories, completions, and defining hypotheses. Maps among these entries are specified realization maps, not consequences of using the same root labels.

The enveloping and Chevalley constructions are functorial under the primitive maps of Theorem 143.1. The root Fock trace is computed from the actual graded root spaces. A Fredholm realization adds a family of operators on a specified domain and the convergence and orientation data identifying its Pfaffian with the denominator. These are the analytic inputs to the square-root comparison below.

Definition 138.2 (Integral protected realization)**.** Let $\mathfrak{p}^+_X$ be the BPS primitive Lie algebra of a selected strong-oriented cohomological lift of the geometric E-equivariant Hall source. The integral protected realization is

$$\mathfrak{P}_{\mathrm{int}}(X, \Delta) = \left(\mathcal{H}^{E,\mathrm{mot}}_X, \mathfrak{P}(K3 \times E), \mathfrak{b}_{X\boxtimes\Delta}, r_X, r_\Delta, I^E_{X,\Pi}\right).$$

At the motivic level the first, second, and character components are defined. A strong orientation enhancement supplies $\mathfrak{p}^+_X$ and hence the amalgamated primitive component.

Proposition 138.3 (Primitive graph comparison)**.** *Suppose the two primitive Borels, their common Cartan action, and their completed amalgam are supplied as above. Their Cartan retractions induce retractions from the amalgam to each factor. For each specified primitive map φ preserving Cartan action, its graph quotient is canonically the geometric Borel.*

Proof. The identity on one Borel and the Cartan projection of the other agree on the common Cartan, so the amalgam gives the retraction. The quotient statement is Proposition 123.4. □

The motivic integration map and primitive comparison are different maps. Strong orientation gives the coefficient coherence of a cohomological Hall construction under its stated geometric hypotheses. Extracting a primitive Lie algebra additionally requires the appropriate BPS or coproduct construction.

Chapter 139

Cyclic open-string operations

Let A be a finite-dimensional, uncurved cyclic A_∞ algebra over $\mathbb{C}$, with a nondegenerate pairing of degree -3. Use the cohomological suspension $s : A \to A[1]$ of degree -1. The suspended operations $b_n : (A[1])^{\otimes n} \to A[1]$ have degree one. Our sign convention is the coderivation convention $b^2 = 0$ on the tensor coalgebra, with the Koszul rule for moving homogeneous inputs. Write ω for the resulting degree--1 symplectic form on $A[1]$. Fix the Hamiltonian convention $\iota_{X_F}\omega = dF$ and $[X_F, X_G] = X_{\{F,G\}}$.

For the universal degree-zero coordinate x on $A[1]$, put

$$S(x) = \sum_{n \geq 1} \frac{1}{n+1} \omega(x, b_n(x^{\otimes n})).$$

Cyclic invariance is read with the suspended Koszul signs. All functions are formal power series at the origin. An equivalent unsuspended convention writes this as $\frac{1}{2}\langle a, m_1 a\rangle + \sum_{n\geq 2} \frac{1}{n+1}\langle a, m_n(a^{\otimes n})\rangle$.

Proposition 139.1 (Cyclic master equation)**.** *The action satisfies* $\{S, S\} = 0$.

Proof. Cyclicity makes the $n+1$ derivatives of the arity-$n+1$ term equal, with the suspended Koszul signs. Thus $X_S(x) = \sum_{n\geq 1} b_n(x^{\otimes n})$. The coderivation identity implies $X_S^2 = 0$. The Hamiltonian identity gives $X_{\{S,S\}} = 2X_S^2 = 0$. Nondegeneracy of ω makes $\{S, S\}$ constant. Uncurvedness removes b_0, so S starts in order two and its bracket has no constant term. The constant is therefore zero. □

A finite-height Ginzburg realization requires such a cyclic model on the selected proper objects before this scalar action is applied. A Hall realization additionally requires a moduli problem with critical coefficient complexes. For an extension $A_0 \to F \to B_0$, put $L(A_0, B_0) = \det R\,\mathrm{Hom}(A_0, B_0)$. Three-Calabi–Yau duality gives

$$R\,\mathrm{Hom}(B_0, A_0) \simeq R\,\mathrm{Hom}(A_0, B_0)^\vee[-3].$$

Applying the determinant functor to the two derived Hom triangles gives

$$\det R\,\mathrm{Hom}(F, F) \cong \det R\,\mathrm{Hom}(A_0, A_0) \otimes \det R\,\mathrm{Hom}(B_0, B_0) \otimes L(A_0, B_0)^{\otimes 2}.$$

Indeed, the four subquotients are the two diagonal Hom complexes and the two cross complexes. Dualization and the odd shift cancel on their determinant lines, so the two cross determinants coincide. Orientation lines require square-root isomorphisms along this formula, compatible on triple flags and with any quotient descent. The cyclic master equation does not select these isomorphisms.

Chapter 140

An algebraic open–closed comparison

A finite-dimensional ordinary Lie algebra $\mathfrak{l}$ gives a precise coefficient model for an enveloping-algebra centre. Let $U = U(\mathfrak{l})$ over $\mathbb{C}$. The adjoint module U^{ad} has action $x \cdot u = xu - ux$. Its Chevalley cochains are

$$C^n(\mathfrak{l}, U^{\mathrm{ad}}) = \mathrm{Hom}_{\mathbb{C}}(\Lambda^n \mathfrak{l}, U),$$

with differential

$$\begin{aligned}(df)(x_1, \dots, x_{n+1}) = &\sum_i (-1)^{i+1} [x_i, f(x_1, \widehat{x_i}, \dots, x_{n+1})] \\ &+ \sum_{i<j} (-1)^{i+j} f([x_i, x_j], x_1, \widehat{x_i}, \widehat{x_j}, \dots, x_{n+1}).\end{aligned}$$

The second line places the bracket first and retains the order of the remaining arguments.

Theorem 140.1 (Hochschild–Chevalley comparison)**.** *Restriction followed by antisymmetrization defines a quasi-isomorphism*

$$C^{\bullet}_{\mathrm{HH}}(U, U) \longrightarrow C^{\bullet}(\mathfrak{l}, U^{\mathrm{ad}}), \qquad \phi \longmapsto \left[(x_1, \dots, x_n) \mapsto \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) \phi(x_{\sigma(1)}, \dots, x_{\sigma(n)}) \right].$$

This is a comparison of ordinary cochain complexes.

Proof. The Chevalley bimodule resolution has terms $K_n = U \otimes \Lambda^n \mathfrak{l} \otimes U$. Its differential has the left-minus-right multiplication terms and the bracket terms with the signs dual to the displayed differential. Filter K_n by total PBW degree, counting each exterior generator as degree one. The multiplication terms preserve this degree and bracket terms lower it. The associated graded is the Koszul resolution of the diagonal in $\mathrm{Sym}(\mathfrak{l}) \otimes \mathrm{Sym}(\mathfrak{l})$, with regular sequence $x_i \otimes 1 - 1 \otimes x_i$. That Koszul complex is exact away from degree zero. The filtration is bounded below and exhaustive, and each element has finite filtration degree, so lifting a leading term proves exactness of $K_\bullet$ by induction. Its augmentation has image U. Thus $K_\bullet$ is a free bimodule resolution.

Antisymmetrization gives a map from $K_\bullet$ to the ordinary bar resolution, lifting the identity of U. Adjacent products in its bar differential pair to commutators; $xy - yx = [x, y]$ gives exactly the bracket terms of $K_\bullet$. The outer products give its left-minus-right terms. A comparison between projective resolutions lifting the identity is a homotopy equivalence. Applying $\mathrm{Hom}_{U \otimes U^{\mathrm{op}}}(-, U)$ gives the stated formula and quasi-isomorphism. □

This is the classical Cartan–Eilenberg comparison. For a finite Lie superalgebra, the same construction uses the super-exterior powers and permutation Koszul signs. Odd generators produce symmetric exterior directions, so the resolution can be unbounded; its algebraic direct sums and cochain products remain part of the convention. The PBW and super-Koszul proof applies with those conventions.

A physical mixed holomorphic–topological realization requires a map from its independently defined bulk observables to this cochain object. The theorem specifies the adjoint coefficients and the actual algebraic comparison. An E_2 refinement, a current-factorization realization, and compatibility with root-height transitions require maps preserving their additional operations. A quotient of Lie algebras alone does not provide a covariant map of Hochschild centres.

Chapter 141

The exact scalar and first-order observables

Set

$$\mathrm{Den}_{\Delta}(Z) = D_5(2Z), \qquad \jmath(Z) = Z/2, \qquad \jmath^{*}\,\mathrm{Den}_{\Delta} = D_5.$$

The one-particle root spaces give

$$\operatorname{sch}\mathrm{Fock}(\mathfrak{g}_{\Delta_5+}) = \prod_{\alpha>0}(1-e^{-\alpha})^{-\operatorname{sdim}\mathfrak{g}_{\Delta_5\alpha}}.$$

The oriented root operator gives

$$e^{-\rho}\,\mathrm{sPf}_{F}(\mathbb{A}_Z) = D_5(2Z).$$

The primitive reduced curve-counting theorem gives

$$\mathcal{Z}^{\mathrm{red}}_{K_3\times E} = -(\jmath^{*}\,\mathrm{Den}_{\Delta})^{-2} = -\chi_{10}^{-1}.$$

Theorem 141.1 (Protected square-root theorem)**.** *Assume the stated root-space Fredholm comparison with its orientation, Weyl shift, and convergence domain. Within that realization, the denominator section* Den_{Δ} *is simultaneously:*

(i) *the Weyl denominator of* $\mathfrak{g}_{\Delta_5}$*;*

(ii) *the Euler character of the bar–Chevalley complex;*

(iii) *the oriented Fredholm Pfaffian of* $\mathbb{A}_Z$*;*

(iv) *the PBW determinant of the parity-refined root superspace* $\mathcal{H}_{\mathrm{rp}}$*;*

(v) *the automorphic denominator whose half-coordinate pullback squares to the reduced scalar denominator.*

All five identifications preserve root degree and Weyl transport.

Proof. The first two are the denominator and bar–Chevalley theorems. The third is the Fredholm theorem. PBW for the Borcherds Lie superalgebra identifies the root superspace determinant, giving the fourth. The half-coordinate identity $\jmath^{*}\,\mathrm{Den}_{\Delta} = D_5$, the theta normalization, and the reduced curve-counting theorem give the fifth. □

Chapter 142

Physical meaning

The root-datum realizations retain several distinct constructions. The finite Ginzburg algebras and their projections form an inverse system. A completed three-Calabi–Yau sector requires a continuous nondegenerate Calabi–Yau structure and compatible cyclic models on the selected proper objects. The quiver Hall category retains its chosen coefficients and extension correspondences. A Hall theorem on that datum must identify its root presentation. The Borcherds root superspace defines the root-graded Fock space and its supercharacter. Its identification with physical one-particle states requires a parity-preserving graded comparison intertwining the primitive actions.

A realization on the elliptic curve must construct the current object with its local and wrapped supports, collision operations, and comparison maps. The quantum Borcherds algebra, with its specified pairing and completion, acts as a line-operator symmetry only through a representation preserving fusion and exchange. The root operator gives the Fredholm product on its specified convergence domain. A physical determinant interpretation requires an operator comparison and an orientation-preserving identification of the determinant lines and sections. The Fock character gives the scalar square. An operator trace additionally requires its state space and trace domain.

Simple-root states satisfying the defining relations give a primitive Lie map. The explicit enveloping and Chevalley extensions preserve products and brackets. A compact string realization must additionally construct its brane functor, Hall orientation maps, paired quantum coproduct, and trace-line comparison.

Chapter 143

Primitive comparisons and their algebraic extensions

Let $\mathfrak{n}_\Delta$ be the positive Borcherds Lie superalgebra. Choose homogeneous parity-preserving images of its simple generators in a charge-graded Lie superalgebra $\mathfrak{p}$. Assume these images satisfy every defining relation.

Theorem 143.1 (Extension of a primitive comparison)**.** *The chosen images determine a unique Lie-superalgebra map $f : \mathfrak{n}_\Delta \to \mathfrak{p}$. It induces*

$$U(f)(x_1 \cdots x_r) = f(x_1) \cdots f(x_r), \qquad C_*(f) = \operatorname{Sym}^c(f[1]).$$

These are respectively an enveloping-algebra map and a Chevalley chain-coalgebra map. Filtration-preserving maps extend to the specified inverse-limit completions. If f is surjective and both algebras have finite charge components and proper positive height, equality of their parity-refined PBW characters implies that f and $U(f)$ are isomorphisms.

Proof. The defining presentation gives f. Its tensor-algebra extension carries $xy - (-1)^{|x||y|}yx - [x, y]$ to the corresponding relation. It therefore descends to $U(f)$. The shifted symmetric-coalgebra map commutes with the Chevalley differential because f preserves brackets. Compatible maps on filtration quotients give maps on their inverse limits. Surjectivity of f gives surjectivity of $U(f)$. Character equality makes each finite charge component an isomorphism, separately in both parities. The PBW embeddings then imply injectivity of f. □

At level zero the current Lie conformal algebra is $\mathbb{C}[\partial] \otimes \mathfrak{n}_\Delta$ with $[x_\lambda y] = [x, y]$ on constant generators. The map $\partial^j x \mapsto \partial^j f(x)$ preserves this bracket and sesquilinearity. A nonzero central level additionally requires preservation of the specified invariant pairing. Hall convolution, Calabi–Yau structures, quantum coproducts, and determinant lines require their own comparison data.

Chapter 144

Finite quivers and Calabi–Yau completions

For a finite quiver Q with zero potential, let $\Gamma(Q, 0)$ be its path-length-completed Ginzburg algebra. Path products are read from left to right: if $a : i \to j$ and $b : j \to k$, then $ab : i \to k$. For each arrow $a : i \to j$, add $a^* : j \to i$ of degree -1. At each vertex add a loop t_i of degree -2. Original arrows have degree zero, and

$$da = da^* = 0, \qquad dt_i = e_i\Big(\sum_a [a, a^*]\Big)e_i.$$

Proposition 144.1 (Restriction to a full subquiver)**.** *If Q_0 is a full subquiver, retaining its generators and killing all other vertices, arrows, reverse arrows, and loops defines a continuous dg-algebra projection*

$$\Gamma(Q, 0) \longrightarrow \Gamma(Q_0, 0).$$

The inclusion of retained graded generators need not commute with the differential.

Proof. The projection respects source and target idempotents and extends multiplicatively. At a retained vertex, every term involving a discarded arrow vanishes. The other terms are precisely dt_i in the smaller algebra. All arrow differentials vanish, so the projection is a chain map. It preserves the path-length filtration and is continuous. For the final assertion, start with one vertex i and no arrows, where $dt_i = 0$. Add a vertex j and an arrow $a : i \to j$. In the larger algebra $dt_i = aa^*$, a nonzero length-two path. Thus the graded inclusion is not a dg map. □

Ginzburg completion supplies a three-Calabi–Yau algebra for each finite quiver in its stated setting. The displayed projections specify an inverse system for compatible zero-potential windows. A Calabi–Yau structure on a completed limit requires its own nondegeneracy and continuity argument. Comparisons with Hall categories and quantized Borcherds algebras must also preserve orientations, root relations, and the paired Hopf data.

Book XXVII

Quantum groups, exchange, and representation theory

Chapter 145

From positive half to full quantum group

Suppose the compact or local positive algebra H^+ is paired continuously with a negative half H^-, carries a completed coproduct, and admits Cartan data and a separated triangular completion. The completed double

$$\mathcal{U} = \widehat{D}(H^+)$$

has triangular decomposition $H^- \mathbin{\widehat{\otimes}} H^0 \mathbin{\widehat{\otimes}} H^+$. Its universal R-matrix acts on continuous tensor products of modules. The classical limit is a Lie bialgebra whose cobracket is the first-order term of the coproduct asymmetry.

For $\mathbb{C}^3$, H^+ is the positive affine Yangian. For local ADE surfaces, it is the completed positive affine ADE Yangian. For the Igusa algebra, the paired Borel construction of the preceding part supplies a height-complete formal quantum group; a compact Hall realization maps to it through primitive, coproduct, and pairing comparisons.

Chapter 146

Tannaka reconstruction

Let $\mathcal{L}$ be an essentially small rigid $\Bbbk$-linear abelian monoidal category with finite-dimensional morphism spaces, finite-length objects, and exact tensor product. Let

$$\omega : \mathcal{L} \to \mathrm{Vect}_{\mathrm{fd}}$$

be a faithful exact strong monoidal functor. Assume the coend

$$H = \int^{M \in \mathcal{L}} \omega(M)^{*} \otimes \omega(M)$$

exists and that the canonical comparison functor to finite-dimensional H-comodules is comonadic. Tensor product makes H a bialgebra, and rigidity supplies its antipode. A braiding gives a coquasitriangular form; a meromorphic braiding gives its spectral version.

Theorem 146.1 (Rigid Tannaka reconstruction)**.** *Under the preceding finite and comonadic hypotheses, the comparison functor is a monoidal equivalence*

$$\mathcal{L} \simeq \mathrm{Comod}_{\mathrm{fd}}(H).$$

Braiding transports to the universal coquasitriangular form on H.

Proof. Matrix coefficients define the universal H-coaction on every $\omega(M)$. Tensor product and duality give the bialgebra and antipode structures. The Barr–Beck comonadic theorem identifies $\mathcal{L}$ with coalgebras for the induced comonad, and the coend identifies that comonad with $H \otimes -$. The hexagon identities transport to the coquasitriangular identities. □

This reconstruction explains why a line-operator category, a fiber functor, and exchange together produce a quantum group. The ordered boundary algebra provides ordered composition. A compatible bialgebra coproduct supplies tensor products of its modules, and exchange supplies additional braiding data.

Chapter 147

KZ and rational exchange

For a simple Lie algebra $\mathfrak{g}$, the Knizhnik–Zamolodchikov connection is

$$\nabla = d - \frac{\hbar}{2\pi i} \sum_{i<j} \Omega_{ij}\, d\log(z_i - z_j).$$

Flatness follows from the infinitesimal braid relations. Drinfeld–Kohno identifies its monodromy tensor category with the corresponding Drinfeld–Jimbo category after a choice of associator.

Rational difference exchange instead gives the Yangian R-matrix

$$R(u) = 1 - \frac{\hbar P}{u}$$

in the fundamental representation. The Yang–Baxter equation is the confluence identity for three exchanges. The KZ and Yangian constructions share the geometry of configuration spaces while using different flat connections.

Chapter 148

The centre and the double

The Hochschild centre $Z_{\mathrm{ord}}(A)$ controls central operations on the ordered algebra in its specified coefficient category. To form a Drinfeld centre of modules, first specify a monoidal module category. For an ordered chiral bialgebra satisfying Definition 125.1, use

$$\mathcal{M}_A = \mathrm{Mod}_A(\widehat{\mathcal{V}}_C)$$

with the monoidal product of Theorem 125.2. An object of $Z(\mathcal{M}_A)$ is a module M together with natural equivalences

$$c_{M,N} : M \widehat{\otimes}^{!} N \xrightarrow{\sim} N \widehat{\otimes}^{!} M \qquad (N \in \mathcal{M}_A)$$

satisfying the half-braiding hexagon and unit identities, with their coherences. This categorical centre is braided. A comparison with the Hochschild centre must specify a realization functor, the source and target of the comparison, and the operations it preserves. Dualizability alone does not supply such a comparison. Derived loop-space descriptions apply in the geometric categories and under the hypotheses of their respective centre theorems [11].

The centre and the double answer complementary questions. The centre classifies central actions in the chosen category. The double reconstructs a quantum group from paired positive and negative halves. A field-theory comparison requires explicit realizations of the relevant line and bulk operations and a map compatible with both constructions.

Book XXVIII

A compact six-dimensional protected source

Chapter 149

Holomorphic cotangent BF theory on $K3 \times E$

Let S be a K3 surface with holomorphic symplectic form σ_S, let E be an elliptic curve with holomorphic form $\mathrm{d}z$, and put

$$X = S \times E, \qquad \Omega_X = \sigma_S \wedge \mathrm{d}z.$$

For a finite proper-height quotient $\mathfrak{n}_{\Delta_s,+}{}^{(H)}$ of the positive Igusa Borcherds algebra, take fields

$$A \in \Omega^{0,\bullet}(X, \mathfrak{n}_{\Delta_s,+}{}^{(H)})[1], \qquad B \in \Omega^{0,\bullet}(X, (\mathfrak{n}_{\Delta_s,+}{}^{(H)})^\vee)[1].$$

The invariant evaluation pairing and Ω_X define the BV action

$$S_H(A,B) = \int_X \Omega_X \wedge \langle B, \bar\partial A + \frac{1}{2}[A,A]\rangle. \tag{149.1}$$

This is the holomorphic cotangent theory of the dg Lie algebra of $\mathfrak{n}_{\Delta_s,+}{}^{(H)}$-valued Dolbeault forms.

Theorem 149.1 (Classical compact-background source)**.** *The action* (149.1) *satisfies the classical master equation and defines a holomorphic factorization algebra on the compact Calabi–Yau threefold X.*

Proof. The shifted cotangent pairing is invariant. The classical master equation is the Jacobi identity of $\mathfrak{n}_{\Delta_s,+}{}^{(H)}$ together with $\bar\partial^2 = 0$ and the derivation property of $\bar\partial$. Local observables of an elliptic cyclic L_∞ algebra form a factorization algebra by the standard BV construction. □

Chapter 150

Tree exactness and the quantum master equation

Every interaction vertex in (149.1) has one B-leg and two A-legs. The propagator pairs an A-leg with a B-leg.

Lemma 150.1 (Loop bound)**.** *A connected Feynman graph of the cotangent theory has loop number at most one.*

Proof. A graph with V interaction vertices has exactly V internal or external B-half-edges. Every internal edge consumes one of them, hence $E \leq V$. The loop number is $E - V + 1 \leq 1$. □

The positive algebra is graded by strictly positive root height. For homogeneous $x \in \mathfrak{n}_{\Delta_s,+}{}^{(H)}$, the operator ad_x raises height. Its coadjoint operator lowers dual height. A product of adjoint operators labelled by positive roots has total nonzero height shift and therefore has zero trace.

Proposition 150.2 (Vanishing of one-loop Lie weights)**.** *For every finite positive-height quotient, the Lie weight of each one-loop graph in the cotangent theory vanishes.*

Proof. A one-loop graph has $E = V$. Every B-leg is consumed by an internal edge, so its external legs are A-legs. Its Lie weight is a cyclic trace of adjoint operators, or the dual trace with the conventionally equivalent signs. Each external label has positive root height. The product therefore strictly raises height on the finite-dimensional adjoint representation and has zero trace. The same argument applies to the dual representation. □

This vanishing removes the one-loop Lie coefficients in a perturbative construction with the stated propagator and vertex types. A renormalized quantum factorization algebra still requires admissible analytic kernels, local extensions, the quantum master equation, and compatible height transitions on the chosen observable spaces. These are the remaining quantum realization data.

Chapter 151

Classical reduction along K3

Choose a Kähler metric on S and normalize a closed harmonic $(0,2)$-form η by $\int_S \sigma_S \wedge \eta = 1$. Let $K = \mathbb{C}[\eta]/(\eta^2)$ with $|\eta| = 2$ and zero differential. There is a unital dg-algebra map

$$i : K \longrightarrow \Omega^{0,*}(S), \qquad \eta \longmapsto \eta.$$

Its image is closed under multiplication because $(0,4)$-forms vanish. The Dolbeault cohomology of a K3 surface is one-dimensional in degrees zero and two and zero in degree one, so i is a quasi-isomorphism. It preserves the trace defined by integration against σ_S.

Let $\mathfrak{l}$ be a finite-dimensional Lie algebra, and write $\mathfrak{d} = \mathfrak{l} \ltimes \mathfrak{l}^*$ for its cotangent Lie algebra, with invariant evaluation pairing. For an open $U \subset E$, tensor with $\Omega^{0,*}(U)$, using the completed smooth tensor product in the S direction. The field comparison is

$$K \otimes \Omega^{0,*}(U) \otimes \mathfrak{d} \xrightarrow{i \otimes \mathrm{id}} \Omega^{0,*}(S \times U) \otimes \mathfrak{d}.$$

The bracket is multiplication of forms followed by the Lie bracket, with Koszul signs.

Proposition 151.1 (Classical field comparison)**.** *The displayed map is a trace-preserving dg Lie quasi-isomorphism, compatible with restriction in U. Its induced minimal operations in the K3 factor have no higher products.*

Proof. Multiplicativity of i proves bracket compatibility. Hodge projection p and $h = \bar\partial^* G$ on the compact surface satisfy $pi = \mathrm{id}$ and $\bar\partial h + h\bar\partial = \mathrm{id} - i\,p$. Tensoring these continuous operators with forms on U gives the same contraction, with the tensor-product signs. Thus $i \otimes \mathrm{id}$ is a quasi-isomorphism. Restriction acts only on the U factor and commutes with it. The normalized trace on K is the restriction of the Dolbeault trace. Choose the contraction with $hi = 0$. In every higher transferred tree, the first product of inputs from $i(K)$ remains in $i(K)$ and the next internal homotopy kills it. Thus every higher transferred product vanishes. □

The coefficient algebra K retains both the degree-zero and degree-two K3 modes. The proposition compares classical field complexes and their brackets. Constructing an equivalence of quantum observable factorization algebras requires compatible quantization, support conventions, and the induced maps on observable completions. The ordinary Hochschild comparison of Theorem 140.1 is a separate algebraic calculation, not that quantum field map.

Chapter 152

Protected states and the Igusa trace

Let

$$\mathcal{H}_{\mathrm{ip}} = \widehat{\bigoplus}_{\alpha>0} \mathfrak{g}_{\Delta,\alpha}$$

with its Borcherds parity. The multiparticle state space is the completed super-Fock space

$$\mathcal{H}_{\mathrm{Fock}} = \widehat{\mathrm{Sym}}(\mathcal{H}_{\mathrm{ip},\bar{0}}) \otimes \widehat{\Lambda}(\mathcal{H}_{\mathrm{ip},\bar{1}}).$$

For Z in the absolute-convergence chamber, let e^{-H_Z} act by $e^{-\alpha(Z)}$ on the root space of degree α.

Theorem 152.1 (Protected Hamiltonian trace)**.** *The supertrace is*

$$\mathrm{str}_{\mathcal{H}_{\mathrm{Fock}}}(e^{-H_Z}) = \prod_{\alpha>0} (1 - e^{-\alpha(Z)})^{-\operatorname{sdim} \mathfrak{g}_{\Delta,\alpha}}.$$

After multiplication by the inverse Weyl line, its reciprocal is $\mathrm{Den}_\Delta(Z) = D_5(2Z)$*. After the half-coordinate pullback and doubling, the disconnected protected scalar is* $-(\jmath^* \mathrm{Den}_\Delta)^{-2} = -\chi_{10}^{-1}$ *in the Oberdieck–Pixton normalization.*

Proof. An even oscillator contributes $(1 - e^{-\alpha})^{-1}$ and an odd oscillator contributes $1 - e^{-\alpha}$. Multiplication over the root basis gives the product. The denominator identity and the reduced curve-counting theorem supply the final two equalities. □

Chapter 153

The compact-source comparison

The compact threefold provides the classical cotangent field theory and the dg Lie comparison of Proposition 151.1. The root datum separately gives the Borcherds algebra, its enveloping algebra, and the root-graded Fock space. The latter has the trace computed above. Proposition 150.2 supplies the finite one-loop Lie cancellation, while Theorem 140.1 computes the ordinary algebraic centre model with adjoint coefficients.

Problem 153.1 (Geometric compact comparison)**.** Construct objects E_i representing the simple branes and maps on their derived morphism complexes that preserve units, composition, and higher composition homotopies. Realize these maps on extension stacks with the reduced critical coefficients and orientation isomorphisms on triple flags. Construct compatible tensor and unit constraints on the resulting boundary functor. Compare the chosen quantum bulk action with the independently defined boundary centre, including support, collisions, and completion. Finally construct a trace and a map of its value line to the automorphic line in the same charge normalization.

A primitive map satisfying the root relations gives the enveloping and Chevalley maps of Theorem 143.1. A strong monoidal equivalence transports categorical centres. Neither construction determines the brane functor or its cyclic trace. These are the additional maps required for compact geometric reconstruction.

Book XXIX

Physics of the comparison maps

Chapter 154

Holomorphic–topological boundary theories

A three-dimensional holomorphic–topological theory on $C \times \mathbb{R}$ has local observables that are holomorphic along C and locally constant along $\mathbb{R}$. A boundary produces an ordered chiral algebra. A second topological direction permits exchange and gives an E_2 enhancement or a meromorphic braided line category. The bulk acts through the derived centre.

The finite cotangent BF model gives the exact local calculation:

$$\mathrm{Bar}(U(\mathfrak{l})) \simeq C_*(\mathfrak{l}).$$

For $\mathfrak{l} = \mathfrak{n}_{\Delta_5,+}{}^{(H)}$, this calculation realizes every finite-height Igusa denominator sector.

Chapter 155

Six-dimensional hCS on $K3 \times E$

Choose a holomorphic volume form $\Omega_S \wedge dz_E$. The hCS theory has classical fields

$$\Omega^{0,*}(S \times E, \mathfrak{a})[1]$$

and action S_{hCS}. Its factorization observables are native to the threefold. Pushforward along p_E gives a factorization algebra on E whose coefficient complex contains the Dolbeault modes of S.

A protected reduction to a chiral algebra consists of a deformation retract of the K3 Dolbeault complex, compatibility of the transferred L_∞ operations with the trace, a boundary or defect selecting ordered fusion, and a quantization trivializing the local anomaly. These data produce the ordered chiral algebra to which the Hall/BPS comparison may be applied.

The Hodge equality $\chi(\mathcal{O}_{S\times E}) = 0$ controls one compact supertrace. The gauge and gravitational anomalies are classes in the local BV deformation complex and are evaluated by their own characteristic forms.

Chapter 156

Omega backgrounds

Let a torus T act holomorphically on the local geometry and preserve the selected supercharge. For a generator V of the action, the equivariant differential is

$$Q_\Omega = Q + \iota_V, \qquad Q_\Omega^2 = \mathcal{L}_V.$$

After localization, fixed loci produce deformation parameters and often a spectral R-matrix. Toric threefolds and ADE surface resolutions carry such actions. A deformation family equipped with monodromy-compatible moduli correspondences and wall-crossing equivalences transports the localized algebra from algebraic or equivariant K3 fibers to the selected generic fiber.

Chapter 157

Anomalies as typed classes

The principal anomaly classes are

construction	deformation complex
hCS quantization	local BV cochains
critical CoHA orientation	Picard/gerbe theory of the critical stack
Hall coproduct	correspondence and localization complex
projective conformal blocks	Atiyah algebra of the determinant line
automorphic multiplier	character group of the modular cover.

A comparison theorem is a morphism between the corresponding complexes together with equality of the transported classes. This formulation turns numerical coincidences into geometric identities precisely when an intertwining map exists.

Book XXX

Automorphic characters and scalar shadows

Chapter 158

Borcherds products as characters

Let $\mathfrak{n}_+$ be a locally finite positive Lie superalgebra. The Chevalley Euler character is

$$\chi C_*(\mathfrak{n}_+) = \prod_{\alpha>0} (1 - e^{-\alpha})^{\operatorname{sdim} \mathfrak{g}_\alpha}.$$

A Borcherds product becomes the Weyl-shifted character of $C_*(\mathfrak{n}_+)$. This construction produces a character of an existing algebra. Hall geometry supplies a second algebra, and a primitive comparison theorem identifies the two.

Chapter 159

The universal Borcherds weight formula

For a weight-zero weak Jacobi form

$$\varphi(\tau, z) = \sum c(n, \ell) q^n r^\ell,$$

the Borcherds lift has weight

$$\mathrm{wt}(\mathrm{Borch}(\varphi)) = \frac{c(0,0)}{2}.$$

For the K3 half-index, $c(0,0) = 10$ and the weight is 5. This formula lives entirely on the automorphic layer. Hodge Euler characteristics, Heisenberg ranks, central charges, and BV anomalies are evaluated by their own traces.

Chapter 160

The Gritsenko–Cléry atlas

The eight diagonal-divisor forms are

$$\Delta_{5},\ \Delta_{2},\ \Delta_{1},\ \Delta_{1/2},\ \nabla_{3},\ \nabla_{2},\ \nabla_{3/2},\ Q_{1}$$

with weights

$$5,\ 2,\ 1,\ \frac{1}{2},\ 3,\ 2,\ \frac{3}{2},\ 1.$$

Each form is an automorphic product on its own paramodular or Hecke congruence group with its own character or multiplier [24]. A geometric realization is specified row by row by a charge lattice, Hall problem, trace, and primitive algebra comparison.

Chapter 161

Scalarization hierarchy

The constructions descend through four levels:

$$\text{factorization or Hall object} \longrightarrow \text{primitive Lie algebra} \longrightarrow \text{PBW character} \longrightarrow \text{scalar partition function}.$$

Each arrow retains the quotient of structure displayed at its target. The first object contains supports, extension maps, and brackets. The second retains the infinitesimal algebra. The third retains signed graded dimensions. The fourth adds a trace and normalization. Upward reconstruction is supplied by the corresponding geometric operations and comparison morphisms.

Book XXXI

The unified theorem system

Chapter 162

The foundational theorem

Definition 162.1 (Complete Calabi–Yau quantum-group datum)**.** A complete comparison datum specifies independent categorical, field, boundary, Hall, Hopf, and trace objects together with:

(i) PTVV-representable moduli for the smooth proper Calabi–Yau category;

(ii) a renormalized BV quantization satisfying the quantum master equation and a local compact factorization generator b;

(iii) a boundary reduction or proper locally trivial map to a curve satisfying factorization pushforward, together with the supplied curve-level faces and aligned distributive law;

(iv) a strong-oriented Hall coefficient theory whose pull–push maps satisfy base change and whose positive algebra is separated and complete;

(v) paired negative and Cartan halves, a completed coproduct, a continuous nondegenerate Hopf pairing, and a separated triangular completion on which the canonical tensors converge;

(vi) compatible open–closed and Hall–boundary morphisms;

(vii) a parity-refined primitive comparison for every automorphic identification.

Theorem 162.2 (Calabi–Yau quantum-group assembly)**.** *Let a complete datum consist of:*

(i) *a Calabi–Yau category and a representable moduli-of-objects problem;*

(ii) *a quantized cyclic elliptic BV theory and a boundary or defect generator;*

(iii) *a Hall extension correspondence with coefficient functor, base-change theorem, strong orientation data, and separated charge completion;*

(iv) *paired positive and negative bialgebras, Cartan data, a nondegenerate continuous Hopf pairing, and a completed double;*

(v) *every curve-level distributive, PBW, and analytic comparison used below.*

Then the datum determines functorially:

(i) *framed* E_2 *Hochschild cochains and the shifted-symplectic moduli structure;*

(ii) *the three curve-level algebraic faces and their three bars;*

(iii) *the positive Hall/BPS algebra and its primitive Lie algebra;*

(iv) *the completed quantum double and braided representation category;*

(v) *every supplied comparison among categorical, field-theoretic, Hall, current, quantum, and automorphic realizations;*

(vi) *the Weyl-shifted PBW/Chevalley character of any parity-refined primitive Lie-superalgebra identification.*

Proof. Cyclic Deligne and shifted symplectic geometry give the first output [16, 17, 88]. Boundary factorization gives the curve faces, and Volume I gives their bars and centres. The Hall correspondence and its base-change theorem give the associative product; strong orientation supplies the coefficient coherence required in the 3-Calabi–Yau setting [69]. The paired bialgebra data give the double and universal R-matrix. The comparison maps commute by hypothesis. PBW and Chevalley antisymmetrization give the final character. Each construction is natural under a morphism preserving the displayed data. □

The local quiver, equivariant surface, K3 correspondence, and Igusa Borcherds constructions have the carriers stated above. A geometric $K3 \times E$ comparison requires the primitive map and determinant-line map of Problem 153.1.

Chapter 163

Landscape of established realizations

Geometry	Positive algebra	Full or centred object	Character
A_2 quiver	$U_v^+(\mathfrak{sl}_3)$	$U_v(\mathfrak{sl}_3)$	PBW root product
$\mathbb{C}^3$	$Y^+(\widehat{\mathfrak{gl}}_1)$	$Y(\widehat{\mathfrak{gl}}_1)$	plane-partition and Fock traces
$\widetilde{\mathbb{C}^2/\Gamma}$	$Y^+(\widehat{\mathfrak{g}}_\Gamma)$	paired completion when the negative half and coproduct are supplied	surface Fock traces
$K3$	$U(\mathfrak{g}^+_{\mathrm{BPS}})$	Heisenberg and LLV actions	K3 Hall characters
$K3 \times E$	Borcherds root envelope; geometric reduced character	$U_q(\mathfrak{g}_{\Delta_5})$ and the root-current model and classical BF theory	$-\chi_{10}{}^{-1}$

The A_2 and $\mathbb{C}^3$ rows provide complete standard quantum-group models. The ADE row provides an established completed positive half together with its geometric current actions; its paired full double is formed from the negative half, Cartan data, coproduct, continuous Hopf pairing, and triangular completion stated in the table. The K3 row provides global two-Calabi–Yau BPS and cohomological symmetries. The final row records independently constructed root, character, and classical field objects. Their geometric Hall realization requires the comparison in Problem 153.1.

Chapter 164

Constructed comparisons and their geometric extensions

The Hall flag argument constructs associativity once its coefficient and orientation maps are defined. The tensor-action formula constructs representation fusion from a coassociative counital coproduct. The matrix-coefficient formulas construct the coend of a supplied fibre functor. The primitive presentation gives enveloping and Chevalley maps. The classical cyclic and K3 field calculations retain their actual differentials, traces, and brackets.

The quantum field realization requires admissible quantization and observable completions. The compact Hall realization requires the brane and orientation maps of Problem 153.1. The Fredholm and automorphic comparisons require their analytic domains and value-line isomorphisms. These constructions preserve the full geometric reconstruction problem beyond its finite algebraic and classical calculations.

Chapter 165

Actions and comparison maps

For a specified boundary algebra A_b, a coherent bulk action is a map

$$\beta : \mathrm{Obs}_{\mathrm{bulk}}|_{\partial} \longrightarrow Z_{\mathrm{ord}}(A_b).$$

The target is the derived endomorphism object of the diagonal bimodule in the chosen factorization coefficient category. A Hall construction instead starts with

$$\mathcal{M}_{\alpha} \times \mathcal{M}_{\beta} \xleftarrow{p} \mathcal{E}_{\alpha,\beta} \xrightarrow{q} \mathcal{M}_{\alpha+\beta}.$$

Critical coefficients, orientations, and admissible pull–push maps give its product. Framed moduli supply a representation after their action correspondences and triple-flag identities have been constructed. A paired Hopf structure gives $D(H^{+}, H^{\circ}, H^{-})$. Its coproduct defines tensor products of continuous modules, and a convergent universal R-matrix gives exchange.

A strong monoidal equivalence between this module category and a boundary category transports categorical centres. It does not identify that categorical centre with the object $Z_{\mathrm{ord}}(A_b)$ without a realization theorem. A trace on a specified representation then gives an amplitude. Its comparison with an automorphic section requires a map of value lines with the same normalization.

Chapter 166

Comparison data for Calabi–Yau constructions

Comparisons among Calabi–Yau categories, holomorphic field theory, chiral algebras, Hall algebras, quantum groups, and automorphic products require the following structures:

(i) for a smooth proper d-Calabi–Yau category with its nondegenerate negative-cyclic class, the categorical outputs are framed E_2 Hochschild cochains and, when the moduli stack of objects is representable in the PTVV setting, $(2-d)$-shifted symplectic moduli; the shift d does not change the Hochschild operadic level to E_n;

(ii) a geometric BV theory and boundary category produce factorization observables and transverse ordered chiral algebras; associative chiral convolution is recorded as a separate supplied structure;

(iii) pushforward to a curve uses an actual map $p : X \to C$; formal cycle-specialization notation alone specifies no pushforward;

(iv) extension stacks, coefficient sheaves, orientations, and completions define the Hall algebra;

(v) coproduct, Cartan, Hopf pairing, and topology define the double;

(vi) K3 Heisenberg, LLV orthogonal symmetry, local ADE Yangians, and global BPS generalized Kac–Moody algebras occupy distinct strata joined by explicit comparison maps where constructed;

(vii) the graph of the $K3 \times E$ Gram map becomes multiplicative in the polarized central extension;

(viii) point-local and wrapped sectors form a hybrid factorization problem;

(ix) the Igusa form is the Weyl-shifted PBW/Chevalley character of $\mathfrak{g}_{\Delta_5}$, and compact Hall geometry maps to it through the primitive comparison;

(x) anomaly identities are transported through explicit maps between deformation complexes.

The categorical, geometric, Hall, and automorphic operations thus have separate carriers and comparison maps.

Book **XXXII**

Technical appendices to the preceding book

Chapter 167

Typed data table

Object	Minimal data	Primary operation
CY category	nondegenerate negative-cyclic trace	Hochschild and shifted-symplectic structures
Factorization algebra	local observables and Weiss descent	disjoint-open product
Transversely ordered chiral algebra	factorization coefficient with an internal E_1 structure for $\otimes^{!}$	ordered fusion
Associative chiral algebra	an E_1 structure for chiral convolution $\star^{\mathrm{ch}}$	nonlocal chiral composition
CoHA	extension correspondence and coefficient theory	pull–push product
Bialgebra	compatible product and coassociative counital coproduct	fusion of representations
Double	positive and negative halves, Cartan data, Hopf pairing	universal R-matrix
Borcherds product	Jacobi or vector-valued modular input	graded character

Chapter 168

The three K3 brackets

The Heisenberg bracket is the excess intersection of two Nakajima incidence kernels. The LLV bracket is the commutator of diagonal Lefschetz kernels and their adjoints. The local Yangian bracket is the convolution commutator of ADE Hecke kernels with spectral equivariance. In $\mathrm{Corr}_{K3}(S)$ all three are compositions of geometric kernels. Their mixed commutators are the corresponding triple-intersection classes. The action algebra and Tannaka coend retain these classes while preserving the three geometric constructions.

Chapter 169

Exact arithmetic checks

The companion verification calculation establishes:

(i) the low Fourier rows of $\phi_{0,1}$ through q^3 by exact theta-series division;

(ii) the vector-valued cocycle identity for the Gram polarization;

(iii) the rational Yang–Baxter equation on $(\mathbb{C}^2)^{\otimes 3}$;

(iv) the A_2 Hall flag counts at ranks zero and one;

(v) the type-II Gram matrix and reflection identities.

Each calculation uses integer or rational arithmetic and appears as a worked formula in the relevant chapter.

Coherent boundary actions and BV fields

Boundary operations determine a deformation problem once their interface and coherence maps are supplied. A local realization gives a shifted cotangent BV theory. Comparing it with a physical bulk requires a map of fields, actions, and coherent boundary operations.

Book XXXIII

The full mixed boundary datum

Chapter 170

Three products and one interface

Holomorphic collisions and transverse composition act on different coefficient objects. A common boundary theory must specify their interaction on an interface. This interaction is part of the input to the following constructions.

Let X be a smooth complex curve and let $\mathcal{D}_X = D\text{-}\operatorname{Mod}(\operatorname{Ran} X)$ in a fixed six-functor enhancement. Write $\otimes^!$ for the pointwise product and $\star^{\mathrm{ch}}$ for chiral convolution.

Definition 170.1 (Complete mixed boundary datum)**.** A complete mixed boundary datum on X is a tuple

$$\mathbb{A} = (L, A^{\mathrm{ch}}, A^{\perp}, K, \iota, \lambda)$$

with the following components.

(i) L is a complete chiral Lie algebra.

(ii) A^{ch} is a complete associative algebra for $\star^{\mathrm{ch}}$.

(iii) $A^{\perp}$ is a complete E_1 algebra in $(\operatorname{Fact}_D(X), \otimes^!)$.

(iv) K is an $(A^{\mathrm{ch}}, A^{\perp})$ interface bimodule.

(v) $\iota : L \to (A^{\mathrm{ch}})_{\mathsf{Lie}}$ is a chiral Lie morphism.

(vi) λ is the aligned mixed distributive law relating the two module actions on K.

Every structure map is continuous for the chosen weight filtrations. The tuple includes the module actions, the mixed coherence homotopies, and their identities in the chosen enhancement. When a reduced bar is used, the relevant augmentations are supplied as additional data. Existence of this tuple for a physical boundary theory is a separate construction.

The three native bars are

$$\operatorname{Bar}^{\mathrm{ch}}_{\mathsf{Lie}}(L), \qquad \operatorname{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A^{\mathrm{ch}}), \qquad \operatorname{Bar}^{\perp}(A^{\perp}).$$

The interface has a fourth bar. Its degree-r terms are alternating words in the two associative colours with one marked interface letter. Its differential contracts adjacent compatible letters.

Construction 170.2 (Interface bar)**.** Let $\overline{A}^{\mathrm{ch}}$ and $\overline{A}^{\perp}$ be the augmentation ideals. Set

$$\operatorname{Bar}^{\mathrm{mix}}(\mathbb{A}; K) = \bigoplus_{p,q \geq 0} (s\overline{A}^{\mathrm{ch}})^{\otimes p} \otimes sK \otimes (s\overline{A}^{\perp})^{\otimes q}.$$

The differential is the sum of the internal differential, the two bar differentials, the left action on K, the right action on K, and the mixed terms supplied by λ.

Theorem 170.3 (Four-colour differential)**.** *The interface differential squares to zero. Its codimension-two terms are exactly the associative relations in the two colours, the two module relations, and the mixed distributive relation.*

Proof. Filter by total word length. Every component of the square contracts two adjacent edges. Disjoint contractions cancel by the determinant-line sign. Nested contractions in one colour cancel by associativity. A contraction meeting the interface cancels by the corresponding module axiom. The remaining square has one edge of each colour and is the aligned distributive square. The datum λ identifies its two composites. □

Chapter 171

Compatible modules and the mixed centre

An operation on a boundary theory must act on boundary-changing morphisms as well as on objects. Natural transformations retain this compatibility, and derived natural transformations retain its higher homotopies.

A compatible module is a triple $(M^{\mathrm{ch}}, M^{\perp}, \kappa)$ with an A^{ch} action, an $A^{\perp}$ action, and an interface comparison over K. Let $\mathrm{Mod}^{\mathrm{mix}}_{\mathbb{A}}$ be their stable category.

Definition 171.1 (Mixed derived centre)**.** The mixed derived centre is

$$Z_{\mathrm{mix}}(\mathbb{A}) = \mathrm{End}\Big(\mathrm{id}_{\mathrm{Mod}^{\mathrm{mix}}_{\mathbb{A}}}\Big).$$

It is computed by the endomorphisms of the diagonal compatible bimodule.

Theorem 171.2 (Universal interior action)**.** *$Z_{\mathrm{mix}}(\mathbb{A})$ carries an internal E_2 algebra structure. Every action already defined as a coherent natural endomorphism of the identity determines a point of this centre. The space of its factorizations through the representing endomorphism object is contractible. Morita-equivalent complete mixed data have equivalent centres.*

Proof. Endomorphisms of the identity functor form the higher centre of a module category. Composition and insertion of natural transformations give the brace operations and hence the E_2 structure. A supplied coherent natural action defines an endomorphism of the identity. A physical bulk insertion determines such an action only after its action on modules, morphisms, and higher compositions has been constructed. The factorization is the universal property of the endomorphism object. Morita equivalence identifies the module categories and their identity functors. □

Chapter 172

Relative induction from a chiral algebra

Let V be a chiral algebra with conformal vacuum. Its algebraic collar $\mathrm{Coll}(V)$ is the free transverse E_1 extension generated by one boundary copy of V, modulo the chiral relations already present in V. Weight completion gives $\widehat{\mathrm{Coll}}(V)$.

Theorem 172.1 (Universal collar)**.** *For every complete transversely ordered chiral algebra B and every chiral morphism $V \to B$, there is a unique continuous E_1 morphism*

$$\widehat{\mathrm{Coll}}(V) \longrightarrow B$$

extending the chiral map. The construction is left adjoint to restriction from transverse ordered algebras to chiral coefficients.

Proof. The algebraic collar is the relative operadic induction from the chiral operad to the mixed operad. Relative induction is left adjoint to restriction. Completion preserves the adjunction on positive complete objects because every map is determined weight by weight. □

When relative induction supplies a regular interface and a continuous aligned distributive law, these data define a mixed envelope with $L = V_{\mathsf{Lie}}$, $A^{\mathrm{ch}} = U^{\mathrm{ch}}(L)$, and $A^{\perp} = \widehat{\mathrm{Coll}}(V)$. Constructing that interface law in the selected geometric coefficient category is part of the envelope construction.

Book XXXIV

The deformation algebra and its BV envelope

Chapter 173

The centre-extended deformation algebra

Varying a boundary product changes its compatibility with the interface and the central action. The deformation complex must therefore include all these operations. A local field realization is further data needed to turn their deformation equations into a BV action.

Fix a cofibrant coloured operad $\mathcal{P}_{\mathrm{mix}}$ presenting the structures of a complete mixed datum and its central action. Let $\mathcal{P}_{\mathrm{mix}}^{!}$ be its bar cooperad. The convolution complex

$$\mathrm{Def}_{\mathbb{A}}^{\mathrm{full}} = \mathrm{Hom}_{\mathbb{S}}\left(\mathcal{P}_{\mathrm{mix}}^{!}, \mathrm{End}_{\mathbb{A}}\right)$$

contains the chiral operation, both associative products, the interface, and the centre action.

Theorem 173.1 (Complete deformation L_∞ algebra)**.** *The convolution complex has a canonical complete L_∞ structure. Its Maurer–Cartan points are complete mixed boundary structures together with a coherent central action. Twisting by the point defined by $\mathbb{A}$ gives the tangent algebra of deformations of the full datum.*

Proof. The infinitesimal decomposition maps of the bar cooperad insert collections of operations into one another. Composition in the endomorphism operad turns these decompositions into the convolution brackets. The cooperad identities imply the L_∞ identities. A Maurer–Cartan element is precisely an operad morphism from a cofibrant resolution of $\mathcal{P}_{\mathrm{mix}}$ to $\mathrm{End}_{\mathbb{A}}$. Twisting linearizes this mapping space at the selected structure. □

173.1 Field equations and Bianchi identities

Let α be a degree-one field in the localized deformation algebra. Its curvature is

$$\mathcal{F}(\alpha) = \sum_{n \geq 1} \frac{1}{n!} \ell_n(\alpha^{\otimes n}).$$

The linearization at α is

$$d_\alpha x = \sum_{n \geq 0} \frac{1}{n!} \ell_{n+1}(\alpha^{\otimes n}, x).$$

Theorem 173.2 (Universal Bianchi identity)**.** *For every α,*

$$d_\alpha \mathcal{F}(\alpha) = 0.$$

At a Maurer–Cartan point, $d_\alpha^2 = 0$.

Proof. Both identities are the exponential form of the L_∞ Jacobi relations. Expand the coderivation square on the cofree cocommutative coalgebra and evaluate it on e^{α} and $e^{\alpha} x$. □

Chapter 174

Shifted cotangent transgression

Let $\mathfrak{l}$ be a complete elliptic L_∞ algebra on a three-manifold M. Its shifted cotangent theory has fields

$$\mathcal{E} = T^*[-1]\mathfrak{l}[1] = \mathfrak{l}[1] \oplus \mathfrak{l}^\vee[-2].$$

The canonical action is

$$S(\alpha, \beta) = \langle \beta, \mathcal{F}(\alpha) \rangle.$$

Theorem 174.1 (Universal BV envelope)**.** *The action satisfies the classical master equation. Its Euler–Lagrange equations are*

$$\mathcal{F}(\alpha) = 0, \qquad d_\alpha^\vee \beta = 0.$$

The tangent complex at a solution is the cotangent complex of the Maurer–Cartan moduli problem.

Proof. The Hamiltonian vector field of S is the cotangent lift of the homological vector field defining the L_∞ structure. The square of that vector field is zero, hence $\{S, S\} = 0$. Variation in β gives the Maurer–Cartan equation. Variation in α gives the dual linearized equation. □

Definition 174.2 (Local realization and chiral BV envelope)**.** A local realization of the complete mixed datum consists of an elliptic local L_∞ algebra $\mathfrak{l}_{\mathbb{A}}^{\mathrm{full}}$ on $M = X^{\mathrm{an}} \times \mathbb{R}$, a specified continuous filtered L_∞ quasi-isomorphism

$$\Gamma(M, \mathfrak{l}_{\mathbb{A}}^{\mathrm{full}}) \longrightarrow \mathrm{Def}_{\mathbb{A}}^{\mathrm{full}},$$

and the local pairing needed for cotangent transgression. Here Γ denotes the chosen complex of global sections with its declared topology. The comparison includes compatibility with the specified boundary operations. The dual fields use the density-valued continuous dual, with compact support or boundary conditions that make the integrated pairing and integration by parts defined.

For a supplied local realization, define

$$\mathbf{BVEnv}_3^{\mathrm{full}}(\mathbb{A}) = T^*[-1]\,\mathrm{MC}(\mathfrak{l}_{\mathbb{A}}^{\mathrm{full}}).$$

Its action is the cotangent action displayed above. Constructing a local realization from an abstract mixed datum remains an additional existence problem.

Chapter 175

Intrinsic algebraic quantization

Let $\mathfrak{X}$ be a finite perfect derived formal moduli problem with a homological orientation μ: the corresponding Berezinian is invariant under the homological vector field. Functions on its shifted cotangent are completed polyvectors:

$$O(T^*[-1]\mathfrak{X}) = \widehat{\mathrm{PV}}(\mathfrak{X}) = \prod_{p \geq 0} \Gamma(\mathfrak{X}, \wedge^p T_{\mathfrak{X}})[p].$$

The homological orientation defines the divergence operator

$$\Delta_\mu : \widehat{\mathrm{PV}}(\mathfrak{X}) \longrightarrow \widehat{\mathrm{PV}}(\mathfrak{X})[1].$$

Theorem 175.1 (Oriented BV complex)**.** *The operator Δ_μ has order two and satisfies*

$$\Delta_\mu^2 = 0, \qquad [d_{\mathfrak{X}}, \Delta_\mu] = 0,$$

$$\Delta_\mu(fg) = \Delta_\mu(f)g + (-1)^{|f|} f \Delta_\mu(g) + (-1)^{|f|}\{f, g\}.$$

Consequently

$$\mathbf{QHdg}(\mathfrak{X}, \mu) = \left(\widehat{\mathrm{PV}}(\mathfrak{X})[\![\hbar]\!], d_{\mathfrak{X}} + \hbar\Delta_\mu\right)$$

is a square-zero quantum BV complex.

Proof. In oriented Darboux coordinates, Δ_μ is the sum of paired second derivatives. Their graded commutation gives $\Delta_\mu^2 = 0$. Invariance of the Berezinian under the homological vector field gives $[d_{\mathfrak{X}}, \Delta_\mu] = 0$. Expanding the operator on a product gives the bracket as the cross term. □

The orientation identifies polyvectors with differential forms by contraction. Under this identification,

$$\iota_\mu \Delta_\mu = d_{\mathrm{dR}}\, \iota_\mu.$$

Hence the BV complex is the Hodge-completed derived de Rham complex with differential

$$d_{\mathfrak{X}} + \hbar d_{\mathrm{dR}}.$$

The Rees parameter records Hodge degree.

Definition 175.2 (Intrinsic quantum complex of a mixed datum)**.** For a complete mixed datum $\mathbb{A}$ equipped with compatible homological orientations on its finite perfect charts, define $\mathbf{QHdg}_3^{\mathrm{full}}(\mathbb{A})$ by applying the preceding BV complex to finite perfect charts of

$$\mathrm{MC}(\mathbf{l}_{\mathbb{A}}^{\mathrm{full}})$$

and taking the separated inverse limit over the chart and weight filtrations.

Theorem 175.3 (Quantum master equation)**.** *A degree-zero action $S = S_0 + \hbar S_1 + \cdots$ defines a quantum state $e^{S/\hbar}\mu$ precisely when*

$$dS + \frac{1}{2}\{S, S\} + \hbar\Delta_\mu S = 0.$$

The equation is equivalent to

$$(d + \hbar\Delta_\mu)e^{S/\hbar} = 0.$$

Proof. Apply the second-order product formula to the exponential and collect powers of $\hbar$. □

Chapter 176

Perturbative factorization realization

The algebraic quantum differential does not specify a propagator or a space of singular kernels. Fix a local realization, elliptic gauge-fixing data, support and boundary conditions, and a renormalization prescription. A heat-kernel propagator is defined away from the diagonal. Extending its graph integrals to collisions requires counterterms and compatibility with the boundary conditions.

Definition 176.1 (Factorization realization data)**.** A factorization realization consists of renormalized local observable complexes, continuous disjoint-insertion maps, and coherent homotopies for their compositions. It includes extensions of graph weights to the required collision strata, their Stokes identities, and a quantum master equation in the permitted local functional complex. Weiss descent is a further required property of these actual complexes. A boundary realization also includes a comparison with the supplied mixed datum, preserving its two products, Lie operation, interface, and mixed law.

Under these hypotheses the disjoint-insertion maps and composition homotopies define the stated factorization object. The boundary comparison identifies its boundary operations with those of $\mathbb{A}$. The construction problem is to produce these data from the physical fields and action. A graphwise identity supplies the corresponding composition equation, whereas descent and boundary identification require their own maps and proofs.

Book XXXV

Boundary duality, exchange, and reduction

Chapter 177

The Koszul–Morita square

A change from modules to comodules must transport the interface as well as each algebra. The mixed bar supplies the comparison object. Exchange then asks for motion in an additional topological direction, while BRST reduction asks whether the differential respects every transported operation.

Let P be a dualizable boundary state for $A^{\perp}$. Its boundary dual is

$$A^{!}_{P} = \mathrm{RHom}_{A^{\perp}}(P, P).$$

The ordinary bar construction gives explicit twisted tensor functors. Their extension to the mixed datum must also transport the interface and its geometric operations.

Let k be a field and let $A_{\mathrm{alg}} = \bigoplus_{w \geq 0} A_w$ be an augmented associative algebra in cohomological degree zero. Assume $A_0 = k$, augmentation projects onto A_0, and multiplication adds weights. Set $A = \prod_{w \geq 0} A_w$ and $C = \prod_{w \geq 0} B(A_{\mathrm{alg}})_w$, with degreewise products and the weight topology. The bar suspension lowers cohomological degree by one, with $b_2[a|b] = [ab]$. The degree-one twisting map $\tau : C \to A$ sends $[a]$ to a and vanishes on the empty word and longer words.

A module or comodule has the form $\prod_{w \geq w_0} M_w$, with differential preserving weight, for some integer w_0. Actions and coactions preserve weight. Comodule coactions are counital and coassociative, and use the tensor product

$$(C \widehat{\otimes} M)_w = \bigoplus_{u+v=w} B(A_{\mathrm{alg}})_u \otimes M_v.$$

The same convention defines $A \widehat{\otimes} M$. These sums are finite in each weight. Repeated reduced coactions vanish in weight w after more than $w - w_0$ steps. Invert quasi-isomorphisms in each weight to obtain $D_{\mathrm{wt}}(A)$ and $D_{\mathrm{wt}}(C)$.

Proposition 177.1 (Ordinary twisted tensor equivalence)**.** *The functors*

$$F(M) = C \widehat{\otimes}_{\tau} M, \qquad G(N) = A \widehat{\otimes}_{\tau} N$$

induce inverse equivalences $D_{\mathrm{wt}}(A) \simeq D_{\mathrm{wt}}(C)$. Writing $\Delta c = \sum c_{(1)} \otimes c_{(2)}$ and $\rho n = \sum n_{(-1)} \otimes n_{(0)}$, their differentials are

$$d_F(c \otimes m) = d_C c \otimes m + (-1)^{|c|} c \otimes d_M m + \sum (-1)^{|c_{(1)}|} c_{(1)} \otimes \tau(c_{(2)}) m,$$
$$d_G(a \otimes n) = a \otimes d_N n - \sum a\tau(n_{(-1)}) \otimes n_{(0)}.$$

The counit and unit are

$$\epsilon_M(a \otimes c \otimes m) = a\epsilon_C(c)m, \qquad \eta_N(n) = \sum n_{(-1)} \otimes 1 \otimes n_{(0)}.$$

Proof. The twisting equation makes both differentials square to zero. The action, coaction, and counit identities show that the displayed maps are chain maps and satisfy the triangle identities. For $GF(M)$, the extra degeneracy

of the two-sided bar inserts $1[\bar{a}_0|a_1|\cdots|a_r]m$ into $a_0[a_1|\cdots|a_r]m$, where $\bar{a}_0 = a_0 - \epsilon_A(a_0)1$. With simplicial boundary $\sum_i (-1)^i d_i$, it contracts the augmented bar. Multiplying length-r words by $(-1)^r$ gives the displayed differential convention. This contraction preserves weight, so the counit is a weightwise quasi-isomorphism. For $FG(N)$, filter each weight by the weight of the final N-factor. The reduced coaction lowers that weight. The associated graded unit is the contracted right-sided bar $k \otimes N_v \to B(k, A_{\mathrm{alg}}, A_{\mathrm{alg}}) \otimes N_v$. The filtration is finite in each total weight, so the unit is also a quasi-isomorphism. The same finite filtrations show that F, G preserve weightwise quasi-isomorphisms, since tensoring complexes over k preserves them. Thus the unit and counit give the asserted inverse equivalences. □

For a right module R and a left module L with the same lower-weight bounds, the two-sided bar has underlying weight-completed complex

$$\prod_n \bigoplus_{u+v_1+\cdots+v_r+w=n} R_u \otimes s\bar{A}_{v_1} \otimes \cdots \otimes s\bar{A}_{v_r} \otimes L_w, \qquad v_i > 0,$$

where the sum also runs over $r \geq 0$ and $\bar{A} = \ker(A_{\mathrm{alg}} \to k)$. Internal differentials and the two endpoint actions and adjacent products form its bar differential. Its realization computes $R \otimes^{\mathbb{L}}_A L$ in these weight-graded categories. Comparing it with a derived cotensor product requires a two-sided twisting comparison on the chosen comodule category and completion. Weight completion here is not identified with derived augmentation-adic completion.

Proposition 177.2 (Centre transport under a mixed equivalence)**.** *Suppose an equivalence between the compatible mixed module categories is supplied, with coherent compatibility with all four bar objects and the interface operations. It identifies the derived endomorphism objects of their identity functors.*

Proof. Conjugate a natural endomorphism by the equivalence and a quasi-inverse. The unit and counit homotopies give inverse maps on derived endomorphism objects and preserve composition and its coherent operations. □

Constructing such a mixed equivalence requires the restriction, collision, interface, and distributive-law comparisons for the actual geometric coefficients. Its unit and counit must respect those maps and the chosen inverse limits. A diagonal bimodule comparison must produce an actual compatible bicomodule before tensor products can be identified with geometric cotensor products. Transport of mixed Hochschild chains and traces further requires their cyclic comparison and the trace domains used in their definition. These are additional mixed Koszul–Morita constructions. The ordinary weight-graded equivalence supplies its stated one-colour functors, without identifying the other mixed carriers or their physical traces.

Chapter 178

Planar exchange and Tannaka reconstruction

A second topological direction gives paths exchanging parallel defects. Their homotopy classes form braid groups. A rigid meromorphic line category $\mathscr{L}$ with a faithful fibre functor ω has coend

$$\mathcal{O}(\mathbb{G}) = \int^{M \in \mathscr{L}} \omega(M)^{*} \otimes \omega(M).$$

Theorem 178.1 (Quantum symmetry from exchange)**.** *The coend is a coquasitriangular bialgebra. Its continuous dual acts on every line object. The meromorphic braiding gives the universal R-matrix on completed matrix coefficients. A spectral rational braiding yields an RTT presentation.*

Proof. Tensor product in $\mathscr{L}$ gives the coproduct on matrix coefficients. Evaluation gives the counit. Rigidity gives the antipode on the Hopf envelope. The braiding defines the coquasitriangular pairing, and the hexagon identities give the Yang–Baxter equation. Writing this naturality on a generating fibre produces the RTT relation. □

Chapter 179

BRST reduction

Let Q be a square-zero derivation of a mixed datum. Extend Q factorwise to every bar word. Since Q is a derivation for each structure map, it anticommutes with every bar differential.

Theorem 179.1 (BRST–bar bicomplex)**.** *The total differential*

$$D = Q + b_{\mathrm{ch}} + b_{\perp} + b_{\mathrm{mix}}$$

satisfies $D^{2} = 0$. *A bounded-below complete filtration gives a convergent spectral sequence*

$$E_{1} = H_{Q}\big(\mathrm{Bar}^{\mathrm{mix}}(\mathbb{A}; K)\big) \Longrightarrow H_{D}\big(\mathrm{Bar}^{\mathrm{mix}}(\mathbb{A}; K)\big).$$

When Q-cohomology is concentrated and tensor-exact, the target is the mixed bar of the reduced datum.

Proof. The square vanishes by the derivation identity and the four-colour bar theorem. The filtration by BRST degree gives the spectral sequence. Concentration and tensor exactness identify the E_{1} differential with the bar differential of the reduced operations. □

Book XXXVI

Conformal covariance and Einstein completion

Chapter 180

Projective connections and the Virasoro tangent complex

A conformal boundary algebra transforms under changes of coordinates. Projective connections record its central term. Three-dimensional gravity supplies independently defined connection and coframe fields whose boundary transformation law can be compared with this algebraic structure.

Let $\mathcal{P}_c(X)$ be the formal moduli problem of projective connections with central parameter c. At a projective connection p, its tangent complex is locally

$$\Omega^{0,\bullet}(X, T_X) \xrightarrow{\mathcal{D}_{p,c}} \Omega^{0,\bullet}(X, K_X^{\otimes 2}),$$

where

$$\mathcal{D}_{p,c}(\epsilon) = \epsilon\, \partial p + 2(\partial \epsilon)\, p + \frac{c}{12} \partial^3 \epsilon.$$

Theorem 180.1 (Projective covariance)**.** *A conformal map* $\mathrm{Vir}_c \to A$ *defines a formal family of mixed structures over* $\mathcal{P}_c(X)$*. Its tangent algebra is the semidirect product of the Virasoro projective complex with the deformation algebra relative to a fixed projective connection.*

Proof. The conformal vector acts on fields by infinitesimal coordinate change. The central term is the affine cocycle in the transformation law of a projective connection. The action therefore gives a map from the projective tangent algebra to derivations of the mixed datum. The tangent algebra of the associated family is the semidirect product. □

Chapter 181

Three-dimensional gravity as a flat Cartan theory

Let M be an oriented three-manifold, let $\Lambda = -\ell^{-2}$, and let $\mathfrak{g}_\Lambda$ have basis J_a, P_a with

$$[J_a, J_b] = \epsilon_{ab}{}^{c} J_c, \qquad [J_a, P_b] = \epsilon_{ab}{}^{c} P_c, \qquad [P_a, P_b] = \ell^{-2} \epsilon_{ab}{}^{c} J_c.$$

The invariant pairing is $\langle J_a, P_b \rangle = \eta_{ab}$ and vanishes on the two diagonal summands. A Cartan connection is

$$\mathcal{A} = \omega^a J_a + e^a P_a.$$

Its curvature is

$$F_{\mathcal{A}} = \left(R^a + \frac{1}{2\ell^2} \epsilon^a{}_{bc} e^b \wedge e^c \right) J_a + T^a P_a.$$

Theorem 181.1 (Cartan–Einstein equivalence)**.** *On the locus where e is invertible, $F_{\mathcal{A}} = 0$ is equivalent to*

$$T^a = 0, \qquad G_{\mu\nu} + \Lambda g_{\mu\nu} = 0, \qquad g_{\mu\nu} = \eta_{ab} e^a_\mu e^b_\nu.$$

The Chern–Simons action for $\mathcal{A}$ equals the Einstein–Hilbert action with cosmological constant, up to its boundary term and normalization.

Proof. The P component of the flatness equation is the torsion equation. Invertibility of e gives the Levi–Civita spin connection. The J component states that the curvature two-form has constant sectional curvature $-\ell^{-2}$. In three dimensions the Riemann tensor is determined by the Ricci tensor, so this is the Einstein equation. Expanding the Chern–Simons three-form with the off-diagonal invariant pairing gives $e \wedge R + \ell^{-2} e^3$ plus an exact form [1, 99]. □

Chapter 182

Four equivalent linear complexes

Fix a constant-curvature solution (e, ω). There are four deformation complexes.

(i) The flat adjoint complex $\Omega^\bullet(M, \mathfrak{g}_\Lambda)$ with differential $d_{\mathcal{A}}$.

(ii) The Einstein–Cartan complex on $(\delta e, \delta\omega)$ with linearized torsion and curvature.

(iii) The metric Einstein complex on $h_{\mu\nu}$ with linearized Einstein operator.

(iv) The Calabi complex resolving local Killing fields by the Killing, linearized curvature, and Bianchi operators.

Theorem 182.1 (Chain equivalence of gravitational presentations)**.** *An invertible torsion-free dreibein gives chain maps among the four complexes. These maps are quasi-isomorphisms on every geodesically convex open set. Their cohomology describes infinitesimal isometries, physical linearized fields, and Bianchi identities in the four languages.*

Proof. The dreibein identifies internal translations with tangent vectors and internal Lorentz rotations with skew endomorphisms. Splitting $d_{\mathcal{A}}$ in this basis gives the linearized torsion and curvature operators. Eliminating $\delta\omega$ by the linearized torsion equation gives the metric Einstein operator. The symbol sequence is the Calabi symbol sequence and is exact on constant-curvature space. The Poincaré lemma for the flat adjoint connection and exactness of the symbol homotopies give local quasi-isomorphisms. □

Chapter 183

Brown–Henneaux and the projective boundary

Write the AdS connection as

$$A^{\pm} = \omega \pm \ell^{-1} e.$$

The Einstein action is the difference of two $\mathfrak{sl}_2$ Chern–Simons actions with level

$$k = \frac{\ell}{4G}.$$

Theorem 183.1 (Brown–Henneaux central charge)**.** *The classical asymptotic Poisson algebra of each chiral Chern–Simons sector has Virasoro central charge*

$$c = 6k = \frac{3\ell}{2G}.$$

The projective connection carried by the boundary stress tensor is the projective variable of the preceding conformal family.

Proof. Hamiltonian reduction of the boundary $\mathfrak{sl}_2$ current Poisson algebra gives the classical Virasoro bracket. Its central term is $6k$. Substitution of the gravitational level gives the second equality. The coadjoint transformation of the boundary stress tensor is the projective-connection transformation law [20]. □

Chapter 184

Einstein completion of a conformal pair

Let A_L and A_R be conformal mixed boundary data with equal Brown–Henneaux central charge. Their projective families map to the same $\mathcal{P}_c(X)$. Define their relative field algebra by the homotopy fibre product over the projective tangent algebra.

Definition 184.1 (Candidate Einstein BV completion)**.** Suppose the relative algebra and a continuous L_∞ action of the Cartan algebra have been supplied, together with their local realization. The associated cotangent candidate is

$$\mathbf{EinBV}_3(A_L, A_R) = T^*[-1]\,\mathrm{MC}\big(\mathfrak{g}_{\mathrm{grav}} \ltimes_{\mathrm{Ham}} \mathfrak{m}_{A_L,A_R}\big).$$

Here $\mathfrak{g}_{\mathrm{grav}}$ is the Cartan deformation algebra and $\mathfrak{m}_{A_L,A_R}$ is the relative mixed deformation algebra after projective matching.

Theorem 184.2 (Cotangent action for a supplied coupled algebra)**.** *Suppose projective matching and a continuous L_∞ action have constructed the displayed complete semidirect product. Suppose also that its local realization and integrated dual pairing are defined. Its shifted cotangent has action*

$$S(\alpha, \beta) = \langle \beta, \mathcal{F}(\alpha) \rangle$$

and satisfies the classical master equation. Its Bianchi identity is the L_∞ identity of the coupled algebra.

Proof. The supplied semidirect product has a square-zero homological vector field. Its cotangent lift is Hamiltonian for the displayed action and remains square-zero. This proves the classical master equation and the Bianchi identity by the preceding cotangent construction. □

The gravitational summand of this cotangent action is the cotangent theory of Cartan deformations. It introduces independent dual fields. To identify it with Einstein BV theory requires a comparison preserving the BV pairing and the action, including boundary terms, on the invertible-coframe sector. The flatness equations alone do not provide that comparison. A physical coupled completion must also construct the matter action and its equivariant moment map. The intended Einstein reconstruction therefore retains these field and action comparisons as existence problems.

Chapter 185

Higher-spin completion

Replace $\mathfrak{sl}_2$ by $\mathfrak{sl}_N$ and choose its principal $\mathfrak{sl}_2$ embedding. The adjoint representation decomposes as

$$\mathfrak{sl}_N \cong \bigoplus_{s=2}^{N} V_{2s-2}, \qquad \sum_{s=2}^{N} (2s-1) = N^2 - 1.$$

Theorem 185.1 (Principal higher-spin spectrum)**.** *The principal embedding gives one massless gauge field of each integer spin* $2, 3, \dots, N$*. Classical Drinfeld–Sokolov reduction of the boundary affine Poisson algebra gives the principal* W_N *algebra. A pair of principal* W_N *boundary data therefore has an* $\mathfrak{sl}_N \oplus \mathfrak{sl}_N$ *Einstein–Cartan completion with the same spin spectrum.*

Proof. The irreducible $\mathfrak{sl}_2$ module V_{2s-2} corresponds to a frame-like field of spin s. The dimension identity exhausts the adjoint. Principal Drinfeld–Sokolov reduction produces one generating current in each conformal weight $2, \dots, N$. The boundary currents and bulk gauge fields therefore match representation by representation [22, 54]. □

Book XXXVII

Exact gravitational complexes and quantum tests

Chapter 186

The four complexes in coordinates

The Cartan flatness equations and the metric Einstein equations describe the same classical solutions on the invertible-coframe locus. Their infinitesimal complexes make the comparison explicit. Quantum determinants additionally depend on gauge fixing, boundary conditions, and determinant-line data.

186.1 Flat adjoint complex

At a flat Cartan connection $\mathcal{A}$, the first complex is

$$\Omega^0(M,\mathfrak{g}_\Lambda) \xrightarrow{d_{\mathcal{A}}} \Omega^1(M,\mathfrak{g}_\Lambda) \xrightarrow{d_{\mathcal{A}}} \Omega^2(M,\mathfrak{g}_\Lambda) \xrightarrow{d_{\mathcal{A}}} \Omega^3(M,\mathfrak{g}_\Lambda).$$

Degree zero is the gauge parameter, degree one is the fluctuation, degree two is the linearized curvature, and degree three is the Bianchi identity.

186.2 Einstein–Cartan complex

Write a gauge parameter as (ξ^a,λ^a) and a fluctuation as $(u^a,v^a)=(\delta e^a,\delta\omega^a)$. The first differential is

$$\begin{aligned} u^a &= D_\omega \xi^a + \epsilon^a{}_{bc} e^b \lambda^c, \\ v^a &= D_\omega \lambda^a + \ell^{-2}\epsilon^a{}_{bc} e^b \xi^c. \end{aligned}$$

The second differential is

$$\begin{aligned} \delta T^a &= D_\omega u^a + \epsilon^a{}_{bc} v^b \wedge e^c, \\ \delta F^a &= D_\omega v^a + \ell^{-2}\epsilon^a{}_{bc} e^b \wedge u^c. \end{aligned}$$

The third differential is the linearized pair of Cartan Bianchi identities.

Proposition 186.1 (Chain identity)**.** *The composite of the first two Einstein–Cartan differentials vanishes on a constant-curvature torsion-free background.*

Proof. Apply D_ω^2 to ξ and λ. Replace the background curvature by $-\ell^{-2}e\wedge e$ and use $D_\omega e=0$. The curvature terms cancel the explicit ℓ^{-2} terms. The remaining expressions are Jacobi identities for $\epsilon_{ab}{}^c$. □

186.3 Metric complex

The dreibein identifies the translation parameter with a vector field. Modulo Lorentz rotations,

$$h_{\mu\nu} = 2\eta_{ab} e^a_{(\mu} u^b_{\nu)}.$$

The metric differential begins with the Killing operator

$$K(\zeta)_{\mu\nu} = \nabla_\mu \zeta_\nu + \nabla_\nu \zeta_\mu.$$

The linearized cosmological Einstein operator is

$$(E_\Lambda h)_{\mu\nu} = \frac{1}{2}\big(- \nabla^2 h_{\mu\nu} - \nabla_\mu\nabla_\nu h + \nabla_\mu\nabla^\rho h_{\rho\nu} + \nabla_\nu\nabla^\rho h_{\rho\mu} \\ - g_{\mu\nu}(\nabla^\rho\nabla^\sigma h_{\rho\sigma} - \nabla^2 h)\big) - \Lambda(h_{\mu\nu} - g_{\mu\nu}h).$$

It satisfies $E_\Lambda K = 0$ and $\nabla^\mu (E_\Lambda h)_{\mu\nu} = 0$.

Theorem 186.2 (Explicit metric reduction)**.** *Solving the linearized torsion equation for $v = \delta\omega$ and substituting into the linearized curvature equation gives $E_\Lambda h = 0$. The kernel of the map from (u, v) to h is the infinitesimal Lorentz action.*

Proof. The torsion equation is algebraic in the antisymmetric part of the covariant derivative of u and determines v. The symmetric part of u is $h/2$. Substitution into $\delta F = 0$ yields the standard variation of the Ricci tensor and the displayed Einstein operator. An antisymmetric $u_{\mu\nu}$ is exactly a Lorentz rotation of the dreibein. □

186.4 Calabi complex

On a constant-curvature manifold, the Calabi complex begins

$$\Gamma(TM) \xrightarrow{K} \Gamma(S^2 T^*M) \xrightarrow{R'} \Gamma(\mathcal{R}) \xrightarrow{B'} \Gamma(\mathcal{B}),$$

where R' is the linearized Riemann tensor and B' is the linearized algebraic and differential Bianchi operator. In three dimensions the Riemann tensor is equivalent to the Einstein tensor, so the metric and Calabi complexes are locally quasi-isomorphic.

Chapter 187

Boundary BFV theory and the Virasoro moment map

The variation of Chern–Simons theory on a manifold with boundary gives the boundary symplectic form

$$\Omega_\partial = \frac{k}{4\pi}\int_{\partial M} \langle \delta A \wedge \delta A\rangle.$$

A chiral polarization selects $A_{\bar z}$ as coordinate. The residual moment map is the flatness constraint.

Theorem 187.1 (Boundary current algebra)**.** *Canonical quantization of the boundary symplectic form gives the affine current algebra at level k. Drinfeld–Sokolov reduction of the principal $\mathfrak{sl}_2$ constraint gives the Virasoro moment map*

$$T(z) = \frac{1}{2(k+h^\vee)} :J^a J_a:(z) + \text{improvement},$$

with the classical Brown–Henneaux limit $c = 6k$.

Proof. The inverse of Ω_∂ is the current Poisson bracket. Quantization gives the affine central extension. The BRST complex of the principal constraint eliminates the gauge directions and retains the improved quadratic current. Its classical central term is $6k$ in the gravitational normalization. □

Chapter 188

Matter stress tensor and the Einstein moment map

Let $\mathfrak{m}$ be the relative field algebra of a conformal pair. Projective covariance gives a Hamiltonian action of vector fields and therefore a moment map

$$\mu_{\mathrm{matt}} : \mathfrak{m} \longrightarrow \Omega^{0,\bullet}(X, K_X^{\otimes 2})[-1].$$

Its classical component is the stress tensor.

Theorem 188.1 (Conservation from the master equation)**.** *The coupled BV action*

$$S_{\mathrm{tot}} = S_{\mathrm{grav}} + S_{\mathrm{matt}} + \langle h, \mu_{\mathrm{matt}} \rangle$$

satisfies the classical master equation precisely when the matter moment map is equivariant. The component of that equation linear in the diffeomorphism ghost is

$$\nabla^{\mu} T_{\mu\nu} = 0.$$

Proof. The master bracket of the coupling with itself is the defect of equivariance. The bracket with the gravitational action is the infinitesimal diffeomorphism of the moment map. Their cancellation is the Hamiltonian equivariance identity. Pairing the identity with the diffeomorphism ghost and integrating by parts gives stress-tensor conservation. □

Chapter 189

One-loop gravity and analytic torsion

Expand Chern–Simons theory around an acyclic flat connection $\mathcal{A}$. Gauge fixing gives the Laplacians of the flat adjoint de Rham complex.

Theorem 189.1 (One-loop determinant)**.** *The absolute value of the one-loop partition function is the square root of the Ray–Singer torsion of the flat adjoint bundle:*

$$|Z_{\text{1-loop}}(\mathcal{A})| = \tau_{\mathrm{RS}}(M, \operatorname{ad}\mathcal{A})^{1/2}.$$

The phase is the eta-invariant contribution fixed by the framing convention.

Proof. Bosonic and ghost Gaussian integrals give alternating determinants of the Laplacians in the flat de Rham complex. Their product is the analytic torsion. The odd signature operator supplies the determinant phase. This is the standard stationary-phase calculation for Chern–Simons theory [91, 99]. □

The same determinant complex is obtained from the Einstein–Cartan, metric, and Calabi presentations through their chain equivalences. One-loop equality is therefore a consequence of quasi-isomorphism together with compatible determinant-line orientations.

Chapter 190

BTZ, Cardy, and the area law

For a unitary modular boundary theory with left and right central charges c_L, c_R, the asymptotic entropy is

$$S_{\mathrm{Cardy}} = 2\pi\sqrt{\frac{c_L}{6}\left(L_0 - \frac{c_L}{24}\right)} + 2\pi\sqrt{\frac{c_R}{6}\left(\bar{L}_0 - \frac{c_R}{24}\right)}.$$

For the BTZ black hole,

$$L_0 - \frac{c}{24} = \frac{(r_+ + r_-)^2}{16G\ell}, \qquad \bar{L}_0 - \frac{c}{24} = \frac{(r_+ - r_-)^2}{16G\ell}, \qquad c = \frac{3\ell}{2G}.$$

Theorem 190.1 (Cardy–Bekenstein equality)**.** *Substitution gives*

$$S_{\mathrm{Cardy}} = \frac{2\pi r_+}{4G}.$$

This is the Bekenstein–Hawking entropy of the BTZ horizon.

Proof. Each square root becomes $\pi(r_+ \pm r_-)/(4G)$. Their sum is $2\pi r_+/(4G)$. The modular asymptotic is Cardy's theorem, and the geometric charges are the Chern–Simons holonomies of the BTZ solution [7, 23]. □

Chapter 191

Higher-spin holonomy and boundary charges

In $\mathfrak{sl}_N$ Chern–Simons gravity, a smooth Euclidean saddle is selected by the conjugacy class of the thermal holonomy. The invariant polynomials

$$\operatorname{Tr}(\mathrm{Hol}^{s}), \qquad 2 \le s \le N,$$

fix the spin-s charges. Drinfeld–Sokolov reduction identifies the same charges with zero modes of the W_N currents.

Theorem 191.1 (Bulk–boundary charge equality)**.** *For a principal $\mathfrak{sl}_N$ connection in Drinfeld–Sokolov gauge, the invariant polynomials of the spatial holonomy equal the classical W_N zero-mode Hamiltonians. Their Poisson brackets are the classical W_N brackets.*

Proof. Drinfeld–Sokolov gauge writes the connection as the principal raising operator plus lowest-weight components. The coefficients of the characteristic polynomial are the invariant polynomials and are polynomial functions of those components. Hamiltonian reduction gives the W_N Poisson structure on the same functions. □

Book XXXVIII

Anomalies, Hall sources, and automorphic completion

Chapter 192

Four native anomaly classes

A local anomaly, a determinant-line curvature, and an arity-zero bar term belong to different complexes. A cancellation in one complex becomes a geometric cancellation only through a comparison preserving the relevant class and its primitive. Hall and automorphic constructions require their coefficient and orientation maps for the same reason.

The construction carries four curvature classes.

(i) A local BV anomaly lies in the degree-one local deformation complex.

(ii) A projective anomaly is the curvature of the determinant line over parameter space.

(iii) A framing anomaly is the transgression of a characteristic class to a three-manifold.

(iv) A curved Koszul term is the arity-zero component of a curved coalgebra.

Theorem 192.1 (Curvature-killing extension)**.** *Let κ be a closed central degree-two class in the deformation complex of a theory. Adjoining a degree-one generator u with $du = \kappa$ produces a central extension in which the lifted curvature vanishes. Tensoring with the inverse invertible field theory gives the geometric realization of the same cancellation.*

Proof. The extended differential squares to zero because $d\kappa = 0$ and κ is central. The new Maurer–Cartan equation contains the compensating field u. An invertible theory with curvature $-\kappa$ adds the opposite class under tensor product. □

Chapter 193

Hall and Calabi–Yau sources

Let C be a Calabi–Yau category with a Hall moduli problem. An oriented extension correspondence gives a Hall algebra H^+. A factorization pushforward to the boundary curve gives a chiral-convolution algebra. Relative induction supplies the remaining colours.

Proposition 193.1 (Transport under a supplied mixed Morita equivalence)**.** *Suppose a Hall construction supplies a complete mixed boundary datum and a Morita equivalence of its compatible module category with that of $\mathbb{A}$. The equivalence identifies their derived centres. A comparison of local deformation algebras, BV pairings, and actions is additional data for a BV comparison. Quantum line and amplitude comparisons additionally require the monoidal and cyclic structures used to define them.*

Proof. Conjugating natural transformations by the equivalence and its inverse identifies the endomorphisms of the two identity functors. The unit and counit homotopies provide the inverse comparison and preserve composition. This proves the centre comparison. It does not construct the additional local, symplectic, or cyclic maps. □

An algebra map alone does not induce the required forward centre map. For example, the homomorphism $\mathbb{C}[x] \to M_2(\mathbb{C})$ with $x \mapsto \operatorname{diag}(0,1)$ sends the central element x to a noncentral matrix. Its commutator with the matrix unit E_{12} is $-E_{12}$. Thus even the degree-zero condition fails for a centre map extending the given algebra map. Hall multiplication maps must be supplemented by an actual coherent module comparison before centre transport can be applied.

For $X = K3 \times E$, suppose a geometric Hall source supplies the complete charge-graded BPS Lie superalgebra and Cartan action required in Volume IV. Its semidirect product with the common Cartan defines $\mathfrak{b}_X$. Volume IV then constructs the completed Lie amalgam with the Igusa Borcherds Borel $\mathfrak{b}_\Delta$. Applying the completed enveloping functor gives the cocommutative Hopf algebra

$$H_{X\boxtimes\Delta} = \widehat{U}(\mathfrak{b}_X)\widehat{*}_{\widehat{U}(\mathfrak{h}_\Pi)}\widehat{U}(\mathfrak{b}_\Delta).$$

The two Hopf retractions preserve the geometric and automorphic sectors separately. A q-deformed mixed Hopf algebra is supplied by the quantum amalgam theorem when a geometric Hall coproduct and pairing are present.

193.1 Mixed realization of the amalgam

The Hopf retractions above are maps of their stated algebraic objects. A mixed realization must construct chiral and transverse operations, an interface, and a compatible distributive law. To realize the retractions, it must then supply maps preserving those operations and the chosen completions.

A BV retraction requires further maps between the local deformation algebras, together with compatible pairings and actions. The counterexample to forward centre functoriality prevents deducing these maps from the Hopf retractions alone. The intended geometric–automorphic theory asks for this mixed realization and for a primitive comparison quotient on its actual boundary operators. Its interaction

ideal is the kernel of that constructed quotient, rather than an ideal determined by equality of scalar characters.

Chapter 194

Igusa root states and the mixed-boundary comparison

The Gritsenko–Nikulin presentation defines the root Lie superalgebra $\mathfrak{g}_{\Delta_5}$ and its positive subalgebra $\mathfrak{n}_+$. Let $d_\alpha = \operatorname{sdim}(\mathfrak{g}_{\Delta_5})_\alpha$, and fix the positive cone, its proper height, and Weyl vector ρ. Put

$$P = \prod_{\alpha>0}(1-e^{-\alpha})^{d_\alpha}, \qquad \mathrm{Den}_\Delta = e^{-\rho}P = D_5(2Z), \qquad \jmath(Z) = Z/2.$$

The products are first interpreted in the positive-height completion.

Theorem 194.1 (Root character and reduced scalar). *The root Fock space* $\mathrm{Sym}(\mathfrak{n}_+)$*, formed with super symmetry, has supercharacter* P^{-1}*. Consequently*

$$-\jmath^*\big(e^{2\rho}\operatorname{sch}(\mathrm{Sym}(\mathfrak{n}_+)^{\otimes 2})\big) = -(\jmath^*\mathrm{Den}_\Delta)^{-2} = -\chi_{10}^{-1}.$$

For a nonsingular projective K3 surface S and an elliptic curve E, this scalar also has an enumerative interpretation. Let $N_{g,h,d}$ be the reduced disconnected Gromov–Witten invariant in class (β_h, d), where β_h is nonzero and primitive with square $2h-2$. Use the incidence insertion $\beta_h^\vee \otimes [\mathrm{point}]$, with $\langle \beta_h, \beta_h^\vee\rangle = 1$. Then the series

$$\mathcal{Z}(u,q,s) = \sum_{g,h,d\ge 0} N_{g,h,d}\, u^{2g-2} q^{h-1} s^{d-1}$$

is the Laurent expansion of $-\chi_{10}(r,q,s)^{-1}$ under $r = e^{iu}$, by [86, Theorem 1].

Proof. An even root vector contributes $(1-e^{-\alpha})^{-1}$ to the supercharacter of its symmetric algebra. An odd root vector contributes $1-e^{-\alpha}$, because its symmetric algebra in super vector spaces is an exterior algebra. Multiplying the two contributions gives P^{-1}. Proper height makes every coefficient calculation finite. The Weyl factor gives $e^{2\rho}P^{-2} = \mathrm{Den}_\Delta^{-2}$. Pullback by $\jmath$ gives $D_5(Z)^{-2} = \chi_{10}(Z)^{-1}$. Oberdieck–Pixton, Theorem 1, identifies the separately defined reduced numerical generating series with its negative [86]. This last theorem is an enumerative input; the character calculation supplies the root side. □

A primitive realization starts with parity-preserving root classes in a charge-graded Lie superalgebra $\mathfrak{p}$ satisfying the defining relations. Require the classes to preserve charge. The free Lie presentation gives an even charge-preserving map $f : \mathfrak{n}_+ \to \mathfrak{p}$.

Applying f to tensor words constructs $U(f)$. With trivial coefficients, use Chevalley chains $C_*^{\mathrm{CE}}(\mathfrak{a}) = \mathrm{Sym}(\mathfrak{a}[1])$, where the homological suspension raises degree by one and the differential contracts pairs by the bracket. Applying the suspension of f to every factor gives a chain map because f preserves brackets.

For the level-zero Lie conformal current, define $\mathrm{Cur}(\mathfrak{a}) = \mathbb{C}[\partial]\otimes\mathfrak{a}$, with ∂ even. Its generator bracket is $[x_\lambda y] = [x,y]$. Extend it by $[\partial x_\lambda y] = -\lambda[x_\lambda y]$ and $[x_\lambda \partial y] = (\partial+\lambda)[x_\lambda y]$. The formula $\partial^j x \mapsto \partial^j f(x)$ defines $\mathrm{Cur}(f)$. Charge preservation makes these maps compatible with the specified proper-height quotients and their inverse limits.

A complete mixed boundary realization requires further geometric maps. The compact source consists of sheaves and extension flags on a common elliptic torsor. Its critical realization must supply reduced coefficient complexes, orientation cross-line maps, unit and flag coherences, and the admissible pull–push operations. A chiral comparison must preserve actual support and moving-insertion operations. A quantum-double comparison must preserve coproducts, pairings and the domain of exchange operators. The finite Calabi–Yau quiver completions require their specified dg maps; ordinary full-subquiver inclusions do not in general give those maps.

The mixed datum to which a BV envelope is applied must therefore include these operations and compatible maps between them. Identifying the primitive algebra of that envelope requires an actual map of its deformation complex to the root complex. A determinant comparison requires a skew perfect family, a square-root line and section, and an isomorphism to the automorphic line preserving transport and normalization. An operator-trace interpretation further requires its state space and trace domain. Constructing this datum and these comparisons is the Igusa mixed-boundary problem. The root character identity specifies the scalar that their final evaluation must recover.

Chapter 195

The carrier lattice

The complete theory uses the carrier set

$$\mathfrak{C} = \{\mathrm{diag}, \mathrm{ch}, \bot, \mathrm{int}, \mathrm{ex}, \mathrm{cyc}, \mathrm{mod}, \mathrm{BV}, \mathrm{Ein}\}.$$

For a subset $S \subset \mathfrak{C}$, let $\mathrm{Str}_S(\mathbb{A})$ be the homotopy limit of all mapping spaces attached to incidence cells supported in S.

Theorem 195.1 (Carrier descent)**.** *When S and T form a relation-excisive pair, the square*

$$\begin{array}{ccc} \mathrm{Str}_{S\cup T}(\mathbb{A}) & \longrightarrow & \mathrm{Str}_S(\mathbb{A}) \\ \downarrow & & \downarrow \\ \mathrm{Str}_T(\mathbb{A}) & \longrightarrow & \mathrm{Str}_{S\cap T}(\mathbb{A}) \end{array}$$

is a homotopy pullback. Thus a global structure is assembled from its carrier restrictions together with their common interface.

Proof. Relation-excision identifies the incidence category of $S \cup T$ with the pushout of the incidence categories for S and T over their intersection. Homotopy limits send this pushout to the displayed pullback. □

Definition 195.2 (Relation-complete carrier cover)**.** Let $U \subseteq \mathfrak{C}$ and let $\mathcal{U} = \{S_i\}_{i\in I}$ be a finite family with $U = \bigcup_i S_i$. The family is *relation-complete* when the canonical functor from the Cech homotopy colimit

$$\operatorname*{hocolim}_{\varnothing \neq J \subseteq I} \mathcal{I}_{\bigcap_{j\in J} S_j} \longrightarrow \mathcal{I}_U$$

is an equivalence. Equivalently, every incidence cell and every relation among its faces is contained in one member of the cover, with all overlaps recorded by the intersections.

Theorem 195.3 (Power-set coherence)**.** *The assignment*

$$S \longmapsto \mathrm{Str}_S(\mathbb{A})$$

is a contravariant functor on the full power set of $\mathfrak{C}$. For every relation-complete carrier cover $\mathcal{U} = \{S_i\}_{i\in I}$ of U, the restriction map is an equivalence

$$\mathrm{Str}_U(\mathbb{A}) \xrightarrow{\sim} \operatorname*{holim}_{\varnothing \neq J \subseteq I} \mathrm{Str}_{\bigcap_{j\in J} S_j}(\mathbb{A}).$$

Thus every subtheory, every intersection of subtheories, and the complete theory are values of one descent object.

Proof. An inclusion $S \subseteq T$ induces the full inclusion $\mathcal{I}_S \subseteq \mathcal{I}_T$ and therefore the restriction map on homotopy limits. Composition of inclusions gives functoriality. A relation-complete cover presents $\mathcal{I}_U$ as the displayed homotopy colimit of its intersection diagram. Mapping that colimit into the diagram of structure spaces converts it into the stated homotopy limit. The resulting equivalence is natural under refinement of the carrier cover. □

Definition 195.4 (Exact-support deformation complex)**.** For $S \subseteq \mathfrak{C}$, let $\mathrm{Def}_{\leq S}$ be the subcomplex generated by incidence cells supported in S, and set

$$\mathrm{Def}_{=S} = \mathrm{cofib}\Big(\operatorname*{hocolim}_{T \subsetneq S} \mathrm{Def}_{\leq T} \longrightarrow \mathrm{Def}_{\leq S}\Big).$$

Thus $\mathrm{Def}_{=S}$ retains the deformation cells whose minimal carrier support is exactly S.

Theorem 195.5 (Carrier-support spectral sequence)**.** *Let* $\mathrm{Def}_{\mathbb{A}}^{\mathrm{full}}$ *be the cellular convolution complex of the complete incidence diagram. Filter it by the cardinality of minimal carrier support. The finite filtration gives a convergent spectral sequence*

$$E_1^{p,q} = \bigoplus_{\substack{S \subseteq \mathfrak{C} \\ |S| = p}} H^{p+q}(\mathrm{Def}_{=S}) \Longrightarrow H^{p+q}\Big(\mathrm{Def}_{\mathbb{A}}^{\mathrm{full}}\Big).$$

A class on the S-summand is a deformation or curvature class whose first complete geometric carrier is S. The first differential is the signed incidence map between exact-support quotients.

Proof. The support-cardinality filtration is finite because $\mathfrak{C}$ is finite. Its pth associated-graded quotient is the direct sum of the exact-support complexes $\mathrm{Def}_{=S}$ with $|S| = p$. The spectral sequence of a finite filtered complex gives the displayed first page and converges strongly to the cohomology of the total convolution complex. The cellular determinant orientation gives the incidence signs. □

Chapter 196

Sixteen structural laws

The theory is governed by the following laws.

(i) A geometric carrier determines the type of its operation.

(ii) The diagonal stores the complete principal-part filtration.

(iii) The interval stores ordered fusion.

(iv) The plane stores exchange.

(v) The circle stores trace.

(vi) The stable curve stores modular sewing.

(vii) The transverse, associative-chiral, and chiral-Lie bars are distinct.

(viii) Chiral PBW compares the last two on enveloping objects.

(ix) The interface bar records the mixed distributive law.

(x) The derived centre is the universal interior action.

(xi) The deformation algebra is the tangent algebra of the centre-extended datum.

(xii) Maurer–Cartan curvature is the field equation.

(xiii) The L_∞ identity is the Bianchi identity.

(xiv) Shifted cotangent is the universal classical BV envelope.

(xv) Hodge-completed derived de Rham theory is the intrinsic algebraic quantum complex of the oriented formal moduli problem.

(xvi) A gravitational realization requires projective matching, a coupled local field algebra, and a comparison preserving the physical BV action and pairing.

Chapter 197

Coherent actions and gravitational reconstruction

A supplied mixed boundary datum defines compatible modules and their derived natural endomorphisms. A specified coloured operad organizes deformations of that datum together with its central action. These are constructions relative to the full operations and their coherence maps. The centre alone does not recover the datum from which the deformation algebra is formed.

A supplied local realization of the deformation algebra admits shifted cotangent transgression. With compatible invariant orientations, the finite perfect charts also carry the algebraic BV differential. The physical bulk is defined independently by its fields, action, gauge symmetry, boundary conditions, and quantization. Reconstruction requires a coherent action map from its observables to an independently defined geometric centre, followed by an equivalence theorem on the stated sector.

For gravity, the invertible coframe gives the classical Cartan–Einstein correspondence. The cotangent construction supplies a formal comparison theory with extra dual fields. Its identification with gravitational BV fields requires an action-preserving and pairing-preserving map. Quantum boundary currents, a continuum measure, global holonomy sectors, allowed fillings, and compatible states remain necessary for the quantum and global comparison.

Hall and Calabi–Yau geometry can supply boundary operations when their correspondences, coefficients, orientations, and comparison maps are constructed. Exchange additionally uses a monoidal category and a fibre functor. Cyclic pairings and sewing determine amplitudes only on their stated domains. These comparisons are the mathematical steps by which local boundary operations could reconstruct physical and geometric theories.

Book XXXIX

Technical appendices to the preceding book

Chapter 198

The Chern–Simons normalization

The gravitational charge calculation uses the action normalization. The mixed bar calculation uses the determinant orientation of its edges. The following formulas specify these conventions.

With invariant pairing $\langle J_a, P_b\rangle = \eta_{ab}$, the Chern–Simons action is

$$S_{\mathrm{CS}}[\mathcal{A}] = \frac{k}{4\pi}\int_M \langle \mathcal{A}\wedge \mathrm{d}\mathcal{A} + \tfrac{2}{3}\mathcal{A}\wedge\mathcal{A}\wedge\mathcal{A}\rangle.$$

For $A^{\pm} = \omega \pm \ell^{-1}e$ one finds

$$S_{\mathrm{CS}}[A^{+}] - S_{\mathrm{CS}}[A^{-}] = \frac{k}{2\pi\ell}\int_M \epsilon_{abc}e^{a}\wedge\left(R^{bc} + \frac{1}{3\ell^{2}}e^{b}\wedge e^{c}\right) + \int_{\partial M}(\cdots).$$

The equality with the Einstein–Hilbert normalization gives $k = \ell/(4G)$.

Chapter 199

The principal embedding count

The exponents of $\mathfrak{sl}_N$ are $1, 2, \dots, N-1$. Under the principal $\mathfrak{sl}_2$ embedding,

$$\mathfrak{sl}_N = \bigoplus_{m=1}^{N-1} V_{2m}.$$

The dimensions satisfy

$$\sum_{m=1}^{N-1} (2m+1) = (N-1)(N+1) = N^2 - 1.$$

Writing $s = m+1$ gives one field at every spin $s = 2, \dots, N$.

Chapter 200

Low-arity mixed signs

Let $x|k|y$ be a suspended interface word. The two length-lowering contractions are

$$b_L[x|k|y] = (-1)^{|x|}(xk)|y, \qquad b_R[x|k|y] = (-1)^{|x|+|k|}x|(ky).$$

Applying the remaining contraction gives the two composites $(xk)y$ and $x(ky)$ with opposite determinant-line signs. Their sum vanishes by the interface associativity relation. A transverse and a chiral contraction acquire the product orientation

$$\det(E_{\mathrm{ch}} \oplus E_{\perp}) = \det(E_{\mathrm{ch}}) \otimes \det(E_{\perp}),$$

so reversing their order contributes the Koszul sign and yields anticommutation.

Book XL

Detailed treatment of Volume I

Chapter 201

Five carriers, three noncommutativities, one type discipline

201.1 The local experiment

Let X be a smooth complex algebraic curve. Place observables in a product chart $U \times I \subset X^{\mathrm{an}} \times \mathbb{R}$, with U a small disc and I an oriented interval. Two independent short-distance motions are visible before any algebra is chosen.

In the first motion, the points in U collide while their interval positions remain fixed. The result is not merely a residue. It is a distribution supported on a partial diagonal, with all normal jets and their coordinate-change law. In the second motion, two subintervals become adjacent while the points of U remain separated. The result is an ordered associative composition. Thickening I to a plane permits one interval to pass around another and creates braid transport. Closing the interval creates a circle trace. Allowing the complex curve to degenerate creates clutching and sewing.

Thus the five carriers are

$$\text{diagonal}, \qquad \text{interval}, \qquad \text{plane}, \qquad \text{circle}, \qquad \text{stable curve}.$$

Their native algebraic structures are, respectively, a chiral distribution, an E_1-product, braid transport, Hochschild/cyclic trace, and a modular action. They are related by geometry, but none is a synonym for another.

The passage from a parameter space to an algebraic identity uses a map with values in the coefficient complex. The following form of Stokes' theorem gives such a map explicitly.

Theorem 201.1 (Stokes-to-master principle). *Let C be a finite regular cell complex with smooth face inclusions. Assume its closed cells are compact smooth oriented manifolds with corners. Let (E, d_E) be a finite-dimensional cochain complex over $\mathbb{R}$ or $\mathbb{C}$. Let $\omega = \sum_p \omega_p$, with $\omega_p \in \Omega^p(C) \otimes E^{-p}$, be smooth forms compatible under restriction to every face. Assume ω is closed for the total differential $d_{\mathrm{form}} + (-1)^p d_E$ on form degree p.*

For an oriented r-cell σ, put

$$\mu_\sigma = (-1)^{r(r-1)/2} \int_\sigma \omega_r.$$

Then $d_E \mu_\sigma = \mu_{\partial\sigma}$, where the right side is the incidence-weighted sum over boundary cells. Thus $[\sigma] \mapsto \mu_\sigma$ is a chain map from cellular chains placed in degree minus cell dimension to E.

Proof. Closure gives $d_E \omega_r = (-1)^{r+1} d_{\mathrm{form}} \omega_{r-1}$. For $r \geq 1$, ordinary Stokes' theorem and

$$(-1)^{r(r-1)/2 + r + 1} = (-1)^{(r-1)(r-2)/2}$$

give the stated boundary formula. For $r = 0$ the equation is $d_E \omega_0 = 0$. All integrals are coefficientwise on the compact closed cells, so no limiting operation is used. □

For a finite-dimensional cochain complex V, operation coefficients lie in $\mathrm{Hom}(V^{\otimes n}, V)$, with differential $\delta f = d_V f - (-1)^{|f|} f\, d_{V^{\otimes n}}$. An interval form $m_t + dt\, H_t$ of total degree zero is closed when $\delta m_t = 0$ and $\delta H_t = \partial_t m_t$. The theorem then gives the actual operation homotopy

$$\delta\left(\int_0^1 H_t\, dt\right) = m_1 - m_0.$$

Suppose the forms on composition faces factor as products of coefficient operations. Then the signed integral preserves composition. On a product of a p-cell and a q-cell, the integration signs cancel the Koszul sign $(-1)^{pq}$. Chapter 37 proves this compatibility, gives the complete cellular curvature and extension equations, and constructs an explicit nonassociative product with operations on all associahedra.

For collision and stable-graph spaces, one must supply the corresponding coefficient-valued forms or chain maps on the compactification. These must extend to its faces and respect composition and gluing. The tensor product, completion and any renormalized integration must support these maps. Associative, chiral, braid, BV and modular identities then have to be read in their respective coefficient complexes. An oriented boundary relation or a residue cancellation by itself does not supply those operations or their higher homotopies.

201.2 The Ran coefficient and its two products

Let $\mathrm{Fin}_{\neq\varnothing}^{\mathrm{surj}}$ denote nonempty finite sets and surjections. A surjection $\alpha : I \twoheadrightarrow J$ determines a partial diagonal

$$\Delta_\alpha : X^J \longrightarrow X^I, \qquad (x_j)_{j\in J} \longmapsto (x_{\alpha(i)})_{i\in I}.$$

A right D-module on Ran X is represented by a coherent family $F_I \in \mathrm{DMod}(X^I)$ with extraordinary pullback identifications along these diagonals. The unital completion adjoins the empty set.

For a partition $\pi = \{B_1, \dots, B_r\}$ of I, let $U_\pi \subset X^I$ be the locus where points in distinct blocks are disjoint, and write $j_\pi : U_\pi \hookrightarrow X^I$.

Definition 201.2 (Factorisable coefficient)**.** A factorisable right D-module is a Ran family F equipped with coherent equivalences

$$j_\pi^! F_I \simeq j_\pi^!\Big(\boxtimes_{B\in\pi} F_B\Big)$$

for every partition, compatible with refinement, permutation, and diagonal transition maps. The chosen stable presentable enhancement is denoted $\mathrm{FactD}(X)$.

There are two monoidal products.

The pointwise coefficient product is

$$(F \otimes_! G)_I = F_I \otimes^! G_I.$$

It does not add support cardinalities. It is the product in which endomorphisms of a fixed boundary condition compose.

Chiral convolution is the Day convolution for disjoint union:

$$(F \star_{\mathrm{ch}} G)_I = \bigoplus_{I = I_1 \sqcup I_2} (j_{I_1,I_2})_*\, j_{I_1,I_2}^!(F_{I_1} \boxtimes G_{I_2}).$$

It adds support cardinalities and is the ambient product for the Ran-space chiral Chevalley theory.

Proposition 201.3 (Cardinality pro-nilpotence)**.** *Let $\mathcal{R}_{\geq r}$ be Ran objects whose components vanish below cardinality r. Then*

$$\mathcal{R}_{\geq r} \star_{\mathrm{ch}} \mathcal{R}_{\geq s} \subseteq \mathcal{R}_{\geq r+s}.$$

Consequently, reduced chiral convolution is nilpotent modulo cardinality $> N$, and its inverse limit is pro-nilpotent.

Proof. Every summand is indexed by a disjoint union of the supporting finite sets, so cardinalities add. An $(N+1)$-fold product of reduced objects has support cardinality at least $N+1$ and vanishes in the N-th quotient. □

The same statement is false for $\otimes_!$, whose cardinality degree is zero. This elementary distinction is the first reason that the two bar constructions must not be conflated.

201.3 Three forms of noncommutativity

Definition 201.4 (Transversely ordered chiral algebra)**.** A transversely ordered chiral algebra is an augmented algebra

$$A \in \mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}(\mathrm{FactD}(X), \otimes_!).$$

The factorisable coefficient carries collision on X. The internal E_1-multiplication $m_\perp : A \otimes_! A \to A$ carries order along a transverse line.

Definition 201.5 (Associative chiral algebra)**.** An associative chiral algebra is an augmented E_1-algebra for chiral convolution,

$$C \in \mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}(\mathcal{R}_X, \star_{\mathrm{ch}}),$$

or, in a local geometric presentation, an algebra over the non-symmetric associative chiral operad $\mathsf{Ass}^{\mathrm{ch}}_X$. Its multiplication composes ordered chiral insertions and may be nonlocal in the vertex-algebraic sense.

Definition 201.6 (Braided or exchange enhancement)**.** An exchange enhancement is an E_2-lift, a flat meromorphic transport on labelled configuration spaces, or equivalent braided categorical data. It supplies paths that exchange two ordered factors and is not part of a bare E_1-algebra.

These are three separate noncommutativities:

(i) noncommutativity of an ordered word in the transverse direction;

(ii) noncommutativity or nonlocality of chiral fields under formal-disc composition;

(iii) nontrivial braid exchange after a second transverse direction or meromorphic connection is supplied.

A theory may carry any one, any compatible pair, or all three.

Definition 201.7 (Doubly associative chiral algebra)**.** A doubly associative chiral algebra is a Ran/factorisation object A equipped with

$$m_\perp : A \otimes_! A \to A, \qquad m_X : A \star_{\mathrm{ch}} A \to A,$$

coherent E_1-structures for each multiplication, and a mixed locality datum on every common disjoint decomposition. In a cofibrant model, this is an algebra over the two-coloured operadic tensor product

$$W\mathsf{Ass}_\perp \otimes W\mathsf{Ass}^{\mathrm{ch}}_X$$

with the mixed cells retained rather than collapsed.

The phrase "the mixed locality datum" is deliberate. The ambient products do not admit a global interchange isomorphism in general. The exact common-decomposition retract is constructed in chapter 202.

No-go. *There is no functor, natural on all factorisable coefficients, that reconstructs m_X from $m_\perp$, or $m_\perp$ from m_X. Indeed, fix a coefficient F. Distinct associative algebras B give distinct transverse products on $B \otimes F$ with the same chiral coefficient. Conversely, distinct nonlocal chiral products can be placed on a fixed underlying coefficient while the transverse product remains the same. A comparison requires extra geometric data, not a change of notation.*

201.4 Boundary and defect origin

Let C be a FactD(X)-linear factorisation category and $b \in C$ a factorisation-compatible boundary condition. The internal endomorphism object

$$A_b = \mathrm{RHom}_C(b, b)$$

composes in an order and varies factorisably in X.

Theorem 201.8 (Factorisation Morita chart)**.** *Assume that b restricts to a compact dualisable generator on a factorising basis, that these local generator equivalences commute with restriction, and that both sides satisfy Weiss descent. Then A_b is a transversely ordered chiral algebra and*

$$\mathrm{RHom}_C(b, -) : C \xrightarrow{\sim} \mathrm{Mod}_{A_b}(\mathrm{FactD}(X))$$

is an equivalence of factorisation categories.

Proof. Composition gives the internal E_1-structure. Factorisation of b on disjoint discs makes composition a morphism of coefficient systems. The ordinary compact-generator Morita theorem applies on each basis element. Compatibility on overlaps and Weiss descent glue the local equivalences. □

The algebra A_b is a chart, not the Morita-invariant totality. Changing the generator changes the algebra while preserving the factorisation category. This distinction becomes essential when comparing a boundary algebra with a physical bulk.

201.5 The complete diagonal coefficient

Let $D = \Delta(X) \subset X^2$. The meromorphic extension $\mathcal{O}_{X^2}(*D)$ has pole filtration

$$P_m \mathcal{O}_{X^2}(*D) = \mathcal{O}_{X^2}((m+1)D), \qquad m \geq 0,$$

and

$$\mathrm{gr}^P_m \mathcal{O}_{X^2}(*D) \simeq \mathcal{O}_D((m+1)D) \simeq N_{D/X^2}^{\otimes(m+1)}.$$

Because $N_{\Delta/X^2} \simeq T_X$, the coefficient of $(z-w)^{-m-1}$ is the local splitting of the $(m+1)$-st normal symbol. Coordinate changes act upper triangularly on these splittings. The filtered distribution is invariant; individual modes are local coordinates on it.

Proposition 201.9 (Complete-coefficient principle)**.** *A chiral operation is the entire diagonal extension and structure morphism. Residues, jets, delta derivatives, propagator coefficients, and formal modes are functorial projections or splittings of this object. No finite list of residues reconstructs the entire operation without a stated bounded-pole hypothesis.*

No-go. *There is no coordinate-free linear map from logarithmic one-forms to arbitrary finite principal parts that is inverse to inclusion. A logarithmic residue detects the simple normal coefficient. Higher principal parts transform through higher jets of a coordinate change and therefore require the full filtered normal object. Any formula that extracts all modes from* $d\log(z-w)$ *has silently chosen additional coordinates or a connection.*

Chapter 202

Aligned decompositions and the exact locality law

202.1 Linear species

Let $\mathcal{V}$ be an additive stable symmetric monoidal category with finite direct sums. A linear species is a functor $F : \mathrm{Fin}^{\simeq} \to \mathcal{V}$. It carries the Hadamard product

$$(F \boxtimes_H G)(I) = F(I) \otimes G(I)$$

and the Cauchy product

$$(F \boxtimes_C G)(I) = \bigoplus_{I = I_1 \sqcup I_2} F(I_1) \otimes G(I_2),$$

where the sum is over ordered decompositions.

The component of

$$(F_1 \boxtimes_C F_2) \boxtimes_H (G_1 \boxtimes_C G_2)$$

is indexed by a pair (d, e) of ordered decompositions of I, while the component of

$$(F_1 \boxtimes_H G_1) \boxtimes_C (F_2 \boxtimes_H G_2)$$

is indexed by one decomposition d. The latter is exactly the diagonal sector (d, d) of the former.

Theorem 202.1 (Aligned-decomposition splitting)**.** *There are canonical natural transformations*

$$\iota : (F_1 \boxtimes_H G_1) \boxtimes_C (F_2 \boxtimes_H G_2) \longrightarrow (F_1 \boxtimes_C F_2) \boxtimes_H (G_1 \boxtimes_C G_2),$$
$$\pi : (F_1 \boxtimes_C F_2) \boxtimes_H (G_1 \boxtimes_C G_2) \longrightarrow (F_1 \boxtimes_H G_1) \boxtimes_C (F_2 \boxtimes_H G_2),$$

with

$$\pi\iota = \mathrm{id}, \qquad e := \iota\pi, \qquad e^2 = e.$$

The image of e is the common-decomposition summand. These maps are compatible with associativity, units, and the symmetry of $\boxtimes_H$.

Proof. At arity I, include the summand indexed by d into the summand indexed by (d, d); this is ι. Project the double sum onto those pairs with equal decompositions; this is π. The identities are immediate componentwise. For three Cauchy factors, both coherence composites include, respectively project onto, triples with one common ordered decomposition. Hence every coherence diagram reduces to equality of the same direct-summand map. □

No-go. *Unless all independent-decomposition summands vanish, ι is not surjective and π is not injective. Therefore Hadamard and Cauchy products are not related by a universal interchange isomorphism. Any theorem using a strong duoidal law must either restrict to an aligned subcategory or retain the idempotent e.*

Definition 202.2 (Aligned window)**.** A full replete subcategory of species is aligned if it is closed under both products and if the projector e acts as the identity on every iterated mixed product of its objects. On such a window, ι and π are inverse and the two products form a strong duoidal pair.

Finite support in a fixed decomposition type gives an elementary aligned window. The full Ran theory is not one.

202.2 Sheaf-level locality

Fix an ordered decomposition $d = (I_1, I_2)$ and the corresponding disjointness inclusion $j_d : U_d \hookrightarrow X^I$. A factorisation structure gives

$$\delta_d : j_d^! A_I \xrightarrow{\sim} j_d^!(A_{I_1} \boxtimes A_{I_2}).$$

An ambient pointwise multiplication $m : A \otimes_! A \to A$ belongs to $\mathrm{FactD}(X)$ exactly when the square

$$\begin{array}{ccc} j_d^!(A_I \otimes_! A_I) & \xrightarrow{j_d^! m} & j_d^! A_I \\ \downarrow{\scriptstyle \delta_d \otimes_! \delta_d} & & \downarrow{\scriptstyle \delta_d} \\ j_d^!\big((A_{I_1} \otimes_! A_{I_1}) \boxtimes (A_{I_2} \otimes_! A_{I_2})\big) & \xrightarrow[m \boxtimes m]{} & j_d^!(A_{I_1} \boxtimes A_{I_2}) \end{array}$$

commutes for every d.

Theorem 202.3 (Exact locality criterion)**.** *A pointwise E_1-multiplication on the underlying Ran object defines a transversely ordered chiral algebra precisely when all of the displayed disjoint-locus squares and their higher E_1 coherences commute. Under finite-set indexing, these equations lie in the image of the aligned projector of theorem 202.1.*

Proof. A morphism in the category of factorisation objects is by definition a Ran morphism intertwining all factorisation equivalences. Applying this definition to every operation of the E_1-algebra gives the stated squares. The same decomposition d occurs before and after multiplication, which is exactly the common-decomposition sector. □

This criterion is stronger and more honest than saying that A is a "bimonoid" in a globally duoidal category. The bialgebra equation is a family of locality squares, not an unsupported isomorphism of mixed products.

202.3 The two coproducts on the transverse bar

Once $m_\perp$ is a morphism of factorisation objects, every face and degeneracy in the simplicial bar is a morphism of factorisation objects. Geometric realization therefore retains the chiral factorisation coproduct. Independently, the normalized word complex has the deconcatenation coproduct

$$\Delta_{\mathrm{dec}}[a_1|\cdots|a_r] = \sum_{p=0}^{r} [a_1|\cdots|a_p] \otimes [a_{p+1}|\cdots|a_r].$$

The first coproduct separates support on the curve; the second cuts the transverse word. They commute on the aligned locality sector because they operate on independent indices. They should not be called the same coproduct.

202.4 Aligned projection and mixed locality

The aligned projector has four consequences.

(i) It prevents an independent pair of decompositions from being mistaken for one laboratory decomposition.

(ii) It explains why locality squares are exact even though the ambient products are not strongly duoidal.

(iii) It makes the bar inherit factorisation without pretending that all mixed species summands participate.

(iv) It identifies the precise extra hypothesis needed to iterate transverse and chiral associative bars.

The idempotent records algebraically that the same decomposition is used in both directions.

The fourth item is a theorem rather than an observation.

Proposition 202.4 (Where a mixed distributive law can live)**.** *The sector on which a mixed distributive law between $\otimes_!$ and $\star_{\mathrm{ch}}$ is geometrically defined is the common-decomposition sector, that is the image of the idempotent e of theorem 202.1. Off that sector the two products admit no interchange, only the retraction of theorem 202.1, so no distributive law can be written.*

Proof. A distributive law requires a natural transformation interchanging the two products in a fixed direction, subject to the coherence conditions of each. By theorem 202.1 the canonical comparison in the additive model is ι, with retraction π, and $\pi\iota = \mathrm{id}$ while $\iota\pi = e \neq \mathrm{id}$ in general. A natural transformation subject to both coherence conditions therefore exists only where e acts as the identity, which is $\mathrm{im}\, e$ by definition 202.2. □

Remark 202.5 (The law is located, not constructed). Proposition 202.4 does not exhibit a λ. No example in the present corpus does. Zamolodchikov–Faddeev algebras of chapter 216 are the natural candidate, since their spectral variable belongs to the chiral direction while their word belongs to the transverse direction; producing λ for one of them, and verifying that it lands in $\mathrm{im}\, e$, is the outstanding construction on which the doubly noncommutative object depends for content.

Chapter 203

Rectangular recognition and its exact hypotheses

203.1 Product-disk factorisation

Let Disk_X be a factorising basis of sufficiently small analytic discs in X^{an}, and let $\mathrm{Disk}^{\mathrm{ord}}_{\mathbb{I}}$ be the little oriented intervals operad. The product basis consists of rectangles $U \times I$. A multimorphism is a family of pairwise disjoint rectangular embeddings, ordered in the interval coordinate.

Definition 203.1 (Rectangular holomorphic–topological factorisation algebra)**.** A rectangular factorisation algebra $\mathcal{O}$ on $X^{\mathrm{an}} \times \mathbb{R}$ satisfies:

(i) Weiss descent on the product-disk basis;

(ii) local constancy under isotopy in the interval variable;

(iii) a holomorphic algebraisation in the X-variable by a factorisable right D-module;

(iv) compatibility of that algebraisation with the finite colimits and tensor products used to form the structure maps.

Recognition Theorem 203.2 (Rectangular recognition)**.** *Under the preceding algebraisation and descent hypotheses, restriction to a standard interval induces an equivalence*

$$\mathrm{Fact}^{\mathrm{rect}}_{\mathrm{hol,lc}}(X \times \mathbb{R}) \simeq \mathrm{Alg}_{\mathsf{E}_1}(\mathrm{FactD}(X), \otimes_!).$$

The internal multiplication is fusion of ordered subintervals; every operation is a morphism of chiral coefficient systems.

Proof. A product-disk algebra is an algebra over the Boardman–Vogt tensor product $\mathrm{Disk}_X \otimes \mathrm{Disk}^{\mathrm{ord}}_{\mathbb{I}}$. The operadic exponential law identifies its algebras with E_1-algebras internal to Disk_X-algebras. Local constancy identifies all sufficiently small intervals through a contractible space of choices, and ordered embeddings produce the E_1-operations. Conversely, an internal E_1-algebra evaluates a family of rectangles by external factorisation in X, followed by ordered multiplication. The tensor-product relation is exactly the locality square. Weiss descent and the algebraisation hypothesis extend the equivalence from the basis to the asserted categories. □

Warning 203.3. The theorem is not an equivalence between arbitrary analytic factorisation algebras and algebraic D-modules. Irregular singularities, infinite products, non-algebraic monodromy, and failure of continuity under completion are genuine obstructions. These are hypotheses, not notation.

203.2 Block-Ran recognition

An ordered block profile is a finite ordered list $\mathbf{I} = (I_1, \dots, I_r)$ of finite chiral label sets. Its carrier is $X^{I_1} \times \cdots \times X^{I_r}$. Chiral refinements occur inside a block; transverse fusion joins adjacent blocks. Let BRan_X be the Grothendieck construction of the simplicial associative indexing category acting on the Ran diagram.

Theorem 203.4 (Block-Ran recognition)**.** *Transversely ordered chiral algebras are equivalent to coCartesian sections over* BRan_X *that are Segal in the block direction and factorising inside each chiral block. Marking a left endpoint, right endpoint, or both endpoints gives left modules, right modules, and bimodules. Gluing marked intervals computes relative tensor products.*

Proof. The simplicial nerve of an internal E_1-algebra has r-simplices $A^{\otimes_1 r}$ and satisfies the Segal condition. The active map $[2] \to [1]$ recovers multiplication, $[0] \to [1]$ recovers the unit, and the simplicial identities recover all coherence. Factorisation inside each block says these maps live in $\mathrm{FactD}(X)$. Straightening and unstraightening provide inverse constructions. The marked statements are the corresponding coloured operads. □

203.3 Opposites and three reversals

Reflection of the interval, $t \mapsto -t$, gives $A^{\mathrm{op},\perp}$ and reverses only the transverse multiplication. Reversal of an exchange path acts on a braid or R-matrix. Verdier duality acts on the D-module coefficient. These operations can agree in a rigid unitary model, but in general they have different domains and cannot be merged into a single "opposite" operation.

No-go. *The ordered chamber* $\mathrm{Conf}_n^{<}(\mathbb{R})$ *is contractible and contains no path exchanging two labelled points. Hence an* E_1*-product determines associativity but no braid generator and no R-matrix. Exchange requires a second transverse direction, a meromorphic connection, or braided categorical input.*

Chapter 204

Associative chiral algebras and nonlocal vertex fields

204.1 Why the second E_1 is necessary

The transverse theory keeps a conventional chiral coefficient and adds a line-order product. This is the correct language for boundary operators in a holomorphic–topological theory. It does not, by itself, construct an associative chiral multiplication whose formal fields are nonlocal. The original motivation for an E_1-chiral Koszul duality therefore requires an additional operad.

Let $\mathsf{Ass}^{\mathrm{ch}}_X(I)$ denote the complex of ordered chiral operations with inputs labelled by I. Geometrically, an operation is defined on the complement of collision diagonals, carries an ordering of the input flags, and extends as an allowed distribution across a specified diagonal. Operadic substitution is composition along nested partial diagonals, with the order of flags retained. Forgetting the order and antisymmetrising maps to the usual chiral Lie operad; imposing braided locality gives quantum-vertex-type quotients.

Definition 204.1 (Associative chiral multiplication)**.** An associative chiral multiplication on a coefficient C is an algebra structure over $\mathsf{Ass}^{\mathrm{ch}}_X$. In arity two it is a distributional map

$$\mu_X : j_* j^*(C \boxtimes C) \longrightarrow \Delta_! C$$

with ordered inputs. In arity three, the two nested collision composites are related by the specified associative homotopy rather than by antisymmetrised Jacobi alone.

The ordinary Beilinson–Drinfeld chiral algebra is Lie-like: its binary operation is skew after the appropriate transposition and satisfies Jacobi. An associative chiral algebra contains more data. Taking the commutator of ordered chiral multiplication gives a chiral Lie operation when the transposition and singular extensions are defined.

204.2 Formal-disc realization

Let V be a vector space with vacuum $\mathbf{1}$ and a field map

$$Y(-,z) : V \longrightarrow \mathrm{Hom}(V, V((z))), \qquad a \longmapsto Y(a,z) = \sum_{n \in \mathbb{Z}} a_{(n)} z^{-n-1}.$$

Thus $Y(a,z)b$ has a finite negative part for each pair a, b. There need not be one pole bound valid for every b. Write ι_{z_0,z_2} for expansion in nonnegative powers of z_2 relative to z_0.

Definition 204.2 (Nonlocal vertex algebra)**.** A nonlocal vertex algebra consists of $(V, Y, \mathbf{1})$ such that

(i) $Y(\mathbf{1}, z) = \mathrm{id}_V$.

(ii) $Y(a,z)\mathbf{1} \in V[[z]]$ and its constant term is a.

(iii) for every $a, b, c \in V$, there exists $N \geq 0$ with

$$(z_0 + z_2)^N \iota_{z_0,z_2} Y(a, z_0 + z_2)Y(b, z_2)c = (z_0 + z_2)^N Y(Y(a, z_0)b, z_2)c.$$

The exponent makes the substitution and equality of formal distributions defined. No commutativity or locality axiom is imposed.

For a geometric realization, the comparison of iterated operations must produce this precise weak associativity identity. The field identity alone does not construct the coefficients on collision configurations or their associative chiral operations.

Definition 204.3 (Quantum vertex enhancement)**.** Set $D(a) = \partial_z Y(a, z)\mathbf{1}|_{z=0}$. A quantum vertex enhancement adds to a nonlocal vertex algebra an invertible rational quantum Yang–Baxter operator

$$\mathcal{S}(z) : V \otimes V \longrightarrow V \otimes V \otimes \Bbbk((z)).$$

Write $\mathcal{S}_{21}$ for conjugation by tensor interchange, and $\mathcal{S}_{ij}$ for the operator on factors i, j. The required equations are

$$\begin{aligned}
&\mathcal{S}_{21}(-z)\mathcal{S}(z) = \mathrm{id},\\
&\mathcal{S}_{12}(z_1)\mathcal{S}_{13}(z_1 + z_2)\mathcal{S}_{23}(z_2) = \mathcal{S}_{23}(z_2)\mathcal{S}_{13}(z_1 + z_2)\mathcal{S}_{12}(z_1),\\
&[D \otimes 1, \mathcal{S}(z)] = -\frac{d}{dz}\mathcal{S}(z).
\end{aligned}$$

It also includes specified braided locality and hexagon compatibility relating $\mathcal{S}$ to Y.

The exchange operator is additional structure. Comparison with another quantum vertex convention requires maps preserving its locality, hexagon and completion conditions. In particular, a formal $\hbar$-adic convention requires topological freeness and the normalization $\mathcal{S} = 1 + O(\hbar)$. An arbitrary rational exchange operator does not supply that deformation.

Recognition Theorem 204.4 (Conditional formal-disc recognition)**.** *Suppose a translation-equivariant associative chiral algebra on the formal disc has a specified state space V, vacuum $\mathbf{1}$ and field map Y. Assume this realization satisfies vacuum and creation and has finite negative part on each vector. Assume its comparison of iterated geometric operations proves the weak associativity identity above. Then $(V, Y, \mathbf{1})$ is a nonlocal vertex algebra. If a specified exchange operator also satisfies the stated quantum vertex enhancement conditions, the field system has that enhancement.*

Proof. The realization hypotheses provide each defining nonlocal vertex axiom. The constant term of $Y(a, z)\mathbf{1}$ recovers a, so the state–field map is injective. The additional exchange conditions give the stated enhancement on this field system. Neither assertion constructs a geometric object from fields. □

Problem 204.5 (Geometric reconstruction from fields)**.** Given a complete nonlocal vertex algebra with covariance and continuity, construct an associative chiral algebra on the formal disc realizing it. Specify the coefficient category, supports and completions. Construct the coefficient objects on all finite configurations, their collision transitions, relabeling maps, factorization comparisons and associative operations with all coherences. Recover the given vacuum and fields through an explicit comparison. For a quantum vertex enhancement, also realize the exchange operator and its compatibility with these operations.

Determine the hypotheses under which both constructions are functorial and their composites are naturally equivalent to the identity. The full field class remains a target until those hypotheses and inverse comparisons are proved. Completeness and weak associativity alone do not specify these geometric maps.

Remark 204.6 (Field data and transverse multiplication). An object of $\mathrm{Alg}_{\mathsf{E}_1}(\mathrm{FactD}(X), \otimes_!)$ specifies a transverse multiplication. A formal-disc field realization must also specify Y and the comparison of its iterates with associative chiral operations. An exchange operator $\mathcal{S}$ requires further compatible data. These structures may come from the coefficient object or an additional construction. They are not part of the transverse multiplication datum itself.

204.3 The genuinely doubly noncommutative object

The local formal version of definition 201.7 consists of:

(i) a nonlocal vertex operation $Y_X(a,z)b$, possibly with $\mathcal{S}(z)$;

(ii) a transverse associative product $a \cdot_\perp b$;

(iii) mixed identities saying transverse multiplication is a morphism of nonlocal vertex algebras, or equivalently
$$Y_X(a \cdot_\perp b, z)(c \cdot_\perp d) = \sum \big(Y_X(a,z)c\big) \cdot_\perp \big(Y_X(b,z)d\big)$$
in the aligned decomposition sector, with the appropriate braiding when factors are exchanged;

(iv) all higher homotopies resolving the failure of strict equality in a derived model.

This is at least doubly noncommutative: the field product need not be local, and the transverse word need not commute. If $\mathcal{S} \neq 1$, exchange supplies a third noncommutative layer.

204.4 Elementary free construction

Let M be a complete translation module. The free nonlocal vertex algebra $T_{\mathrm{nv}}(M)$ is generated by fields corresponding to M, with only vacuum, translation, and weak associativity imposed. Let B be an augmented associative algebra. The tensor product
$$A = B \otimes T_{\mathrm{nv}}(M)$$
has transverse multiplication on B and nonlocal chiral multiplication on the second factor. Both are explicit, and the mixed law is strict because the operations act on separate tensor factors. Quotienting by an exchange relation generated by an R-matrix produces the Zamolodchikov–Faddeev models of chapter 216.

Chapter 205

Homotopy-coherent generators and relations

205.1 Why ordinary quotients are insufficient

A presentation $T(V)/(R)$ remembers generators and strict equations in an ordinary category. In a derived coefficient category, a relation can hold by a chosen null-homotopy, two null-homotopies can differ by a second homotopy, and these choices affect bar and deformation complexes. The correct quotient is therefore a homotopy pushout of free algebras.

Fix a combinatorial monoidal model category $\mathcal{M}$ presenting the desired stable symmetric monoidal ∞-category. Assume the relevant coloured operads are admissible, well pointed, and Σ-cofibrant after replacement. Let

$$\mathcal{P} = W\mathsf{Ass}_{\perp} \otimes W\mathsf{Ass}_{X}^{\mathrm{ch}}$$

be a cofibrant two-direction resolution.

Definition 205.1 (Derived presentation)**.** Let V be a collection of generators, R a collection of relation cells, and $\partial_0, \partial_1 : \mathrm{Free}_{\mathcal{P}}(R) \rightrightarrows \mathrm{Free}_{\mathcal{P}}(V)$ the two sides of each relation. The derived algebra presented by (V, R) is the homotopy coequalizer

$$A \simeq \mathrm{hcoeq}\big(\mathrm{Free}_{\mathcal{P}}(R) \rightrightarrows \mathrm{Free}_{\mathcal{P}}(V)\big),$$

equivalently the corresponding homotopy pushout after adjoining relation disks.

A cofibrant replacement of this homotopy coequalizer is the generators-and-relations object needed for derived mapping spaces. A map $A \to B$ is a choice of images of generators, null-homotopies of relations, and all higher compatibility data encoded by $\mathcal{P}$.

205.2 Two-coloured levelled forests

A cell of $\mathcal{P}$ is represented by a levelled forest whose internal edges are coloured $\perp$ or X. A transverse edge resolves associative composition along the ordered line. A chiral edge resolves nested collision. For a forest T, set

$$\mathrm{or}(T) = \det \Bbbk E_{\perp}(T) \otimes \det \Bbbk E_{X}(T).$$

The cellular complex is

$$\mathsf{Word}_{\mathcal{P}}(I) = \bigoplus_{T} \mathrm{Dec}(T) \otimes \mathrm{or}(T)$$

with

$$d = d_{\mathrm{dec}} + d_{\perp} + d_{X}.$$

Theorem 205.2 (Mixed boundary identity)**.** *With the product orientation,*

$$d_{\perp}^2 = d_{X}^2 = 0, \qquad d_{\perp} d_{X} + d_{X} d_{\perp} = 0,$$

and d_{dec} supercommutes with both edge differentials. Hence $d^2 = 0$.

Proof. A codimension-two cell has two contracted edges. If the colours agree, the two orders give the opposite boundary orientations of the same cellular operad. If the colours differ, both orders give the same decorated forest but exchange the two one-dimensional determinant factors, producing a minus sign. Naturality of the decoration differential gives the remaining anticommutators. □

When vertices are decorated by operations, $d_{\perp}^2 = 0$ is the Stasheff system, $d_X^2 = 0$ is associative chiral coherence, and the mixed anticommutator is the entire hierarchy of locality/interchange homotopies.

205.3 Rectification and the ∞-category

Theorem 205.3 (Cofibrant comparison)**.** *Assume:*

(i) $W\mathsf{Ass}_{\perp}$, $W\mathsf{Ass}_X^{\mathrm{ch}}$, *and their tensor product are admissible;*

(ii) *the tensor product is cofibrant;*

(iii) *the augmentations to* $\mathsf{Ass}_{\perp}$ *and* $\mathsf{Ass}_X^{\mathrm{ch}}$ *tensor to a weak equivalence;*

(iv) *change of operad along that weak equivalence is a Quillen equivalence.*

Then strict algebras over the target operad and homotopy-coherent algebras over $\mathcal{P}$ present equivalent ∞-categories. Derived presentations computed in either model represent the same mapping functor.

Proof. The hypotheses make $\mathcal{P} \to \mathsf{Ass}_{\perp} \otimes \mathsf{Ass}_X^{\mathrm{ch}}$ a cofibrant resolution and identify the localized algebra categories by change of operad. Homotopy pushouts between cofibrant diagrams compute colimits in the localized ∞-category. □

This theorem is the proper relation between model-category tooling and ∞-category semantics. A model is used to calculate; the localized mapping space is the invariant answer.

205.4 Low arity

At arity two there are two primitive binary operations $m_{\perp}$ and m_X. At arity three, the codimension-one faces produce

$$m_{\perp}(m_{\perp}(a,b),c) - m_{\perp}(a,m_{\perp}(b,c)),$$
$$m_X(m_X(a,b),c) - m_X(a,m_X(b,c)),$$

and the mixed square compares

$$m_X(m_{\perp}(a,b),m_{\perp}(c,d)) \quad \text{with} \quad m_{\perp}(m_X(a,c),m_X(b,d))$$

on a common decomposition. In a braided model, the middle factors are transported by the exchange operator before the second expression is evaluated. At arity four the pentagons, chiral nested-collision cells, and mixed prisms appear automatically. No independent "coherence conjecture" is needed: it is the cellular boundary of the cofibrant operad.

205.5 Homotopy transfer

Given a deformation retract

$$(H, d_H) \underset{p}{\overset{i}{\rightleftarrows}} (V, d_V), \qquad ip - \mathrm{id}_V = d_V h + h d_V,$$

evaluate a decorated forest by placing original operations at vertices and h on internal edges. The mixed boundary identity proves the Maurer–Cartan equation for the transferred two-direction structure. The same formula is the algebraic skeleton of perturbative Feynman rules: interaction vertices decorate vertices, propagators decorate edges, and boundary restrictions of compactified graph configurations implement edge contraction.

No-go. *The nerve of the incidence poset of collision strata can be contractible even when the geometric screen has nonzero cohomology. For three points on the affine complex line, the projectivized screen is* $\mathbb{P}^1 \setminus \{0, 1, \infty\}$*, whose first cohomology has rank two, whereas the incidence fibre with terminal corolla is contractible. Therefore a poset resolution cannot replace screen chains unless a comparison theorem proves that the omitted rows are acyclic.*

Chapter 206

The transverse interval bar

206.1 The simplicial object

Let

$$A \in \mathrm{Alg}^{\mathrm{aug}}_{\mathsf{E}_1}(\mathrm{FactD}(X), \otimes_!), \qquad \varepsilon : A \to \mathbf{1}, \qquad \bar{A} = \mathrm{fib}(\varepsilon).$$

The fibre $\bar{A}$ is taken in the ambient stable coefficient category. The two-sided simplicial bar is

$$B^{\perp}_r(A) = \mathbf{1} \otimes_! A^{\otimes_! r} \otimes_! \mathbf{1}, \qquad r \geq 0.$$

Interior faces multiply adjacent factors, outer faces use the augmentation, and degeneracies insert the unit. Its geometric realization is

$$\mathrm{Bar}^{\perp}_X(A) = \left| B^{\perp}_\bullet(A) \right| \simeq \mathbf{1} \otimes^{\mathbb{L}}_A \mathbf{1}.$$

Every simplicial map belongs to $\mathrm{FactD}(X)$ by the locality criterion. Its realization remains factorizing when the restriction and collision functors preserve realization and external tensors preserve realization separately, by the sifted-colimit construction.

206.2 Normalized words and signs

Choose a symmetric monoidal dg model in which the displayed tensors compute derived tensors and direct-sum normalization computes geometric realization. Write $s\bar{A}$ for cohomological suspension. Normalization gives the tensor coalgebra

$$T^c(s\bar{A}) = \bigoplus_{r \geq 0} (s\bar{A})^{\otimes_! r}.$$

For homogeneous $a_i \in \bar{A}$, abbreviate $[a_1 | \cdots | a_r] = sa_1 \otimes \cdots \otimes sa_r$. We fix

$$b_2(sa \otimes sb) = (-1)^{|a|} s(ab).$$

The multiplication coderivation is therefore

$$b_m[a_1 | \cdots | a_r] = \sum_{i=1}^{r-1} (-1)^{\sum_{j<i}(|a_j|+1)+|a_i|} [a_1 | \cdots | a_i a_{i+1} | \cdots | a_r].$$

The internal differential is the tensor differential on suspended letters. For an A_∞-algebra, add the coderivations induced by every m_k.

Proposition 206.1 (Bar identity). *The total bar differential $d_B = d_{\mathrm{int}} + \sum_{k \geq 2} b_{m_k}$ squares to zero exactly when the A_∞-relations hold. In the strict ungraded case, the square on a three-letter word is the suspended associator*

$$b^2_m[a|b|c] = \pm[(ab)c - a(bc)].$$

Proof. A coderivation of the cofree tensor coalgebra is determined by its projection to cogenerators. The projection of d_B^2 in arity n is precisely the signed sum of all substitutions $m_r(\mathrm{I}^{\otimes i} \otimes m_s \otimes \mathrm{I}^{\otimes j})$ with $r + s = n + 1$, which is the n-th Stasheff identity. □

The deconcatenation coproduct

$$\Delta_{\mathrm{dec}}[a_1|\cdots|a_r] = \sum_{p=0}^{r} [a_1|\cdots|a_p] \otimes [a_{p+1}|\cdots|a_r]$$

makes the underlying direct-sum normalized bar a conilpotent coassociative coalgebra in the ambient dg coefficient category. The bar differential is a coderivation because every multiplication contracts a consecutive subword. For separated factors P, Q, the comparison $BP \otimes BQ \to B(P \otimes Q)$ shuffles letters $p_i \otimes 1$ and $1 \otimes q_j$ with their suspended Koszul signs. Cross-factor merges cancel in pairs. Cutting a shuffle equals shuffling the two prefixes and the two suffixes, which proves the coalgebra identity. The comparison is a quasi-isomorphism: its bisimplicial two-sided bar has iterated realization $BP \otimes BQ$ and diagonal realization $B(P \otimes Q)$, identified by the sifted simplicial diagonal. The normalized realization comparison is the signed shuffle, obtained by the monotone-path triangulation of a product of simplices. Iterated shuffles enumerate the same multishuffles, proving refinement coherence. If the chosen restriction and collision functors preserve these comparisons, their derived inverses give the factorization maps compatible with deconcatenation. The shifted summands and the augmentation ideal are formed in the ambient category, rather than assumed to be factorization objects.

206.3 Interval factorisation homology

Theorem 206.2 (Bar as interval amplitude). *Let the endpoints of* $[0,1]$ *carry the augmentation module. Then*

$$\mathrm{Bar}^{\perp}_{X}(A) \simeq \mathbf{1} \otimes^{\mathbb{L}}_{A} \mathbf{1} \simeq \int_{([0,1],\partial[0,1])} A.$$

More generally, for a right module R *and left module* L*,*

$$\mathrm{Bar}(R, A, L) \simeq R \otimes^{\mathbb{L}}_{A} L \simeq \int_{([0,1],\{0,1\})} (A; R, L).$$

Proof. Cover the interval by an ordered chain of small subintervals and endpoint collars. The factorisation Cech diagram is the two-sided simplicial bar. Excision identifies its colimit with factorisation homology and with the derived relative tensor product. Refinement of the cover is cofinal. □

This is the exact physical meaning of a bar word: an ordered sequence of intermediate boundary operations. The differential fuses adjacent operations. The chiral coefficient accompanies each letter and each multiplication map, but the bar degree is transverse.

No-go. *The transverse bar differential has source* $(s\overline{A})^{\otimes_! r}$ *and target* $(s\overline{A})^{\otimes_!(r-1)}$*, and it is induced by* $m_\perp$*. A collision residue changes the normal degree of a distribution on* X^I *and is induced by restriction to a diagonal. Their source and target gradings differ. They may appear as anticommuting differentials in a bicomplex, but one cannot do the work of the other.*

206.4 Associahedra and the monadic bar

The simplicial bar presents the derived tensor product. The Stasheff associahedron resolves the multiplication used to evaluate it. If A is an algebra over a cofibrant A_∞-operad, the operadic bar has a cellular filtration with terms

$$\bigoplus_{r \geq 0} C_*(K_r) \otimes (s\bar{A})^{\otimes_! r}.$$

Rectification gives a zigzag of quasi-isomorphisms to the simplicial bar of a strict model. The simplex and associahedron therefore have different universal properties and must both be retained when the input multiplication is homotopy coherent.

206.5 Completion

The direct-sum bar is conilpotent. In formal or filtered field theory one often needs

$$\widehat{\mathrm{Bar}}^{\perp}(A) = \prod_{r \geq 0} (s\bar{A})^{\otimes_! r}.$$

Let $F^N A$ be a complete decreasing filtration such that every m_k raises total filtration by at least $k - 1$.

Proposition 206.3 (Completed-bar convergence)**.** *Assume that each quotient $A/F^N A$ has finite bar length in a fixed total weight and that the cohomology tower satisfies Mittag–Leffler. Then*

$$H^*(\widehat{\mathrm{Bar}}^{\perp}(A)) \cong \varprojlim_N H^*(\mathrm{Bar}^{\perp}(A/F^N A)).$$

The bar differential is continuous and the completion is separated.

Proof. The completed bar is the inverse limit of the finite quotient bars. The Mittag–Leffler hypothesis kills $\varprojlim^1$, and the filtration-growth assumption makes every coderivation continuous. □

Chapter 207

The associative chiral bar and the theorem of two bars

207.1 Operadic chiral bar

Let C be an augmented algebra over $\mathsf{Ass}^{\mathrm{ch}}_X$. The operadic bar is a conilpotent coalgebra over the bar cooperad $B\mathsf{Ass}^{\mathrm{ch}}_X$:

$$\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(C) = \bigoplus_T B\mathsf{Ass}^{\mathrm{ch}}_X(T) \otimes_{\mathrm{Aut}\, T} (s\overline{C})^{\otimes \mathrm{in}(T)}.$$

Here T runs over ordered chiral composition trees. The differential has three sources:

(i) the internal differential of C;

(ii) the cooperadic differential of the chiral resolution, including screen chains when present;

(iii) contraction of an internal chiral edge by the associative chiral multiplication.

The entire principal-part distribution remains attached to each vertex and edge. A residue is one component of this geometric coefficient, not the universal differential.

On the formal disc, this bar resolves iterated nonlocal fields. Its length is the number of chiral composition vertices, and its relations are weak associativity together with all higher homotopies. When an exchange operator is present, the chiral trees are braided and the differential uses the corresponding transport.

207.2 Chiral Chevalley chains are a third construction

A chiral Lie algebra $\mathfrak{g}$ has Chevalley chains

$$\mathrm{CE}^{\mathrm{ch}}_*(\mathfrak{g}) = \left(\mathrm{Cofree}^{\mathrm{ch}}_{\mathsf{Com}^!}(s\mathfrak{g}), d_{\mathrm{int}} + d_{[-,-]_{\mathrm{ch}}}\right).$$

This is the construction appearing in Francis–Gaitsgory chiral Koszul duality. It is not the associative chiral bar and not the transverse interval bar. The three constructions are related only after one supplies an enveloping map, a commutator map, or another explicit monoidal comparison.

Theorem 207.1 (The theorem of two associative bars)**.** *Let A be a doubly associative chiral algebra in the sense of definition 201.7. Then:*

(i) $\mathrm{Bar}^{\perp}(A)$ *is a conilpotent coalgebra for transverse deconcatenation and remains an associative chiral coefficient whenever the mixed locality maps are compatible with the bar faces;*

(ii) $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A)$ *is a conilpotent chiral cooperadic object and remains a transverse* E_1*-algebra whenever the same mixed data are compatible with chiral edge contraction;*

(iii) *the iterated objects* $\mathrm{Bar}^{\perp}\,\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A)$ *and* $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}\,\mathrm{Bar}^{\perp}(A)$ *are linked by a canonical comparison on the aligned-decomposition sector;*

(iv) *if the mixed operad is a genuine distributive law and both realizations preserve the relevant homotopy colimits, the comparison is an equivalence and the total differential is*

$$d_{\mathrm{tot}} = d_{\mathrm{int}} + d_{\perp} + d_X, \qquad d_{\perp} d_X + d_X d_{\perp} = 0.$$

Proof. The first two statements apply each bar functor internally to the algebra category for the other operation. This is possible precisely because the mixed cells say that every face map is a morphism for the remaining structure. The aligned projector supplies the comparison between the two orders of common decomposition. Under a distributive law, the two-sided bar construction is the geometric realization of one bisimplicial/cooperadic object; Fubini for homotopy colimits gives the equivalence. The anticommutator is the mixed edge identity of theorem 205.2. □

No-go. *An equality* $\mathrm{Bar}^{\perp}(A) = \mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(A)$ *has no meaning without a functor carrying the pointwise multiplication problem to the chiral-convolution multiplication problem. Even their underlying gradings count different geometric events. A similarity of Euler characteristics is not a comparison map.*

207.3 Universal enveloping comparison

The ordinary enveloping comparison uses the associative product of $U(\mathfrak{g})$. For this comparison, write C_n and B_n in homological degree n, equivalently cohomological degree $-n$.

Theorem 207.2 (Ordinary enveloping comparison)**.** *Let k be a field of characteristic zero and let $\mathfrak{g}$ be an ordinary Lie algebra over k. Give $U(\mathfrak{g})$ its augmentation killing $\mathfrak{g}$. There is a natural quasi-isomorphism of ordinary conilpotent differential graded coalgebras*

$$\kappa : C_*(\mathfrak{g}) \longrightarrow \mathrm{Bar}^{\perp}\big(U(\mathfrak{g})\big),$$

where both constructions use direct sums and ordinary tensors over k. It preserves the PBW filtration. If $\mathfrak{g} = \bigoplus_{w \geq 1} \mathfrak{g}_w$ and $[\mathfrak{g}_u, \mathfrak{g}_v] \subseteq \mathfrak{g}_{u+v}$, the map is a quasi-isomorphism in each total weight. It extends to the degreewise product over weights, using completed tensor products by total weight. The completed coalgebras are conilpotent in each weight; no global conilpotence is asserted for infinite sums.

Proof. Put $C_n(\mathfrak{g}) = \Lambda^n \mathfrak{g}$ in homological degree n, with differential

$$d(x_1 \wedge \cdots \wedge x_n) = \sum_{i<j} (-1)^{i+j+1} [x_i, x_j] \wedge x_1 \wedge \cdots \widehat{x_i} \cdots \widehat{x_j} \cdots \wedge x_n.$$

Its coproduct splits the exterior factors between two ordered sublists, with the sign of the corresponding permutation. The ordinary normalized bar differential joins positions $i, i+1$ with sign $(-1)^{i-1}$. Define

$$\kappa(x_1 \wedge \cdots \wedge x_n) = \sum_{\sigma \in S_n} \mathrm{sgn}(\sigma) [x_{\sigma(1)} | \cdots | x_{\sigma(n)}].$$

At degree zero this map preserves the unit. At degree two its differential is $[x_1 x_2 - x_2 x_1] = [[x_1, x_2]]$. In every degree, pair permutations whose merged adjacent entries occur in opposite orders. Their difference is the bracket of those entries, with the displayed exterior sign. This proves the chain-map equation. Cutting an antisymmetrized word amounts to choosing an ordered subset and antisymmetrizing its two subwords. The two procedures give the same permutations and signs, proving the coalgebra identity.

Filter $U(\mathfrak{g})$ by PBW degree, the bar by total PBW degree, and $\Lambda^n \mathfrak{g}$ by degree n. PBW identifies the associated graded algebra with $S(\mathfrak{g})$. The associated graded comparison is the antisymmetrization $\Lambda(\mathfrak{g}) \to B(S(\mathfrak{g}))$, with zero differential on its source. To verify that it is a quasi-isomorphism, consider the free Koszul resolution $S(V) \otimes$

$\Lambda(V)$ for $V = \mathfrak{g}$. For a basis (x_i) of V, its differential is $\partial = \sum_i x_i \iota_i$, where ι_i contracts the exterior factor x_i. On polynomial degree p and exterior degree q, with $p + q > 0$, the map

$$h = \frac{1}{p+q} \sum_i (x_i \wedge -) \frac{\partial}{\partial x_i}$$

satisfies $\partial h + h\partial = 1$. Only finitely many summands act on each element, even when V is infinite-dimensional. The identity follows by cancellation of cross terms and the Euler identity in the two degrees. Thus this free resolution has augmentation homology k only. The free normalized bar resolution has the same homology by its extra-degeneracy contraction. Antisymmetrization defines an augmentation-preserving comparison between these free resolutions. The simplicial bar sign convention is related to the displayed bar convention by multiplication by $(-1)^n$ in degree n. Free bases permit a reverse comparison and homotopies to be lifted inductively, since every positive-degree cycle in a resolution is a boundary. Tensoring these homotopies with k proves the asserted quasi-isomorphism on associated graded complexes.

Every cycle in the cone of κ has finite PBW filtration degree. A primitive for its highest associated graded part lowers that degree; iteration terminates because the filtration is bounded below. The cone is acyclic, proving the ordinary assertion. For a positive weight grading, PBW degree is at most total weight. The same argument is therefore finite within each weight. Finally, kernels and images of a degreewise product differential are products of kernels and images, so the componentwise quasi-isomorphisms extend to the stated completion. □

The ordinary comparison does not itself construct a morphism in $\mathrm{FactD}(X)$. For geometric coefficient systems, one must give compatible enveloping and Chevalley constructions on every chart and coherent factorization equivalences for the whole Chevalley complex. On each disjointness locus, with restriction denoted by $j^!$, the required equation is

$$f_{B,I}\, j^! \kappa_I = (\kappa_{I_1} \boxtimes \kappa_{I_2})\, f_{C,I}, \qquad I = I_1 \sqcup I_2.$$

Here $f_{C,I}$ and $f_{B,I}$ are the specified Chevalley and bar factorization maps. The bar map may use the signed shuffle comparison under its geometric functor hypotheses. The Chevalley factorization maps, collision compatibility, refinement coherence, and continuity of the completed maps remain additional geometric data to construct or verify. Internal tensor formulas and PBW alone do not provide them.

The associative chiral comparison uses the chiral tensor throughout. Let $\mathfrak{g}$ be a Lie object in positive Ran D-module coefficients, and let $K = (\mathbb{C}, 0)$ be the formally adjoined empty-configuration unit. Write $U^{\mathsf{Ass}}_{\mathrm{ch}}(\mathfrak{g})$ for its derived associative envelope for chiral convolution. There is an equivalence of counital coassociative coalgebras

$$K \oplus \mathrm{CE}^{\mathrm{ch}}_*(\mathfrak{g}) \longrightarrow \mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}\big(U^{\mathsf{Ass}}_{\mathrm{ch}}(\mathfrak{g})\big).$$

The map is the preceding permutation sum interpreted with chiral tensors and the Koszul signs of suspended inputs. It has no factorial denominator: cutting a permutation gives precisely the unshuffle coproduct. The envelope relation proves its chain-map identity.

Operadic PBW identifies the associated graded envelope with the chiral symmetric algebra. The free Koszul resolution and its Euler contraction then give the associated graded comparison, with the same multilinear formulas as above. A term with r positive original inputs vanishes on X^I for $r > |I|$. The filtration argument is therefore finite on each configuration and proves the equivalence in this Ran recipient.

For a different geometric or field-coefficient recipient, one must specify the envelope, the twisting map, and their collision and separated-factorization compatibilities. A product completion further requires continuous maps and convergence of the associated graded comparison. These conditions also remain necessary for the transverse geometric comparison described above. The associative envelope used here is distinct from the chiral Lie envelope associated with a change from the star tensor to the chiral tensor.

Chapter 208

Bar–cobar reconstruction and completion

208.1 Ordinary tensor cobar

Let k be a commutative ring. In this section, complexes are cohomologically graded k-modules and tensor products are over k. For a coaugmented conilpotent differential graded coalgebra C, put $\bar{C} = \ker(C \to k)$. The ordinary cobar is the direct-sum tensor algebra on $\bar{C}[-1]$. Here an element of degree j in $\bar{C}$ gives a generator of degree $j+1$. Its differential combines the differential and reduced coproduct of C. For an augmented differential graded algebra A, there is an adjunction

$$\mathrm{Hom}_{\mathrm{dgAlg}^{\mathrm{aug}}}(\Omega C, A) \cong \mathrm{Tw}(C, A) \cong \mathrm{Hom}_{\mathrm{dgCoalg}^{\mathrm{coaug}}}(C, BA).$$

The middle set consists of degree-one maps $C \to A$ vanishing on the coaugmentation and with zero composite with the augmentation of A, satisfying $d_A\tau + \tau d_C + \mu(\tau \otimes \tau)\bar{\Delta} = 0$, with the graded tensor sign convention. Thus Ω is left adjoint to B. The counit has direction $\Omega BA \to A$. A cobar defined by a completed totalization is a different construction. Its carrier and comparison direction must be specified separately.

208.2 A finite-word contraction

Let $A = k1 \oplus I$ be an augmented associative algebra in degree zero. The ordinary algebra $D = \Omega BA$ has multilinear generators $c(a_1, \dots, a_n)$ of degree $1 - n$, with $a_i \in I$. Every element has finite word support. Set

$$dc(a_1, \dots, a_n) = \sum_{i=1}^{n-1} (-1)^{i-1} \big(c(a_1, \dots, a_i)c(a_{i+1}, \dots, a_n) - c(a_1, \dots, a_i a_{i+1}, \dots, a_n) \big),$$

and extend by the graded Leibniz rule. Disjoint operations cancel in d^2. Two overlapping multiplications cancel by associativity. Splitting twice cancels by the two orders of inserting cuts. Mixed splitting and multiplication terms cancel with opposite signs. Thus $d^2 = 0$.

Theorem 208.1 (Ordinary associative reconstruction)**.** *The algebra map $\epsilon_D : D \to A$, given by $\epsilon_D(c(a)) = a$ and zero on longer generators, is a quasi-isomorphism. No flatness assumption on I is required.*

Proof. The two-entry differential maps to $ab - ab = 0$. All longer differential terms map to zero, so ϵ_D is a cochain map. Filter D by the total number N of entries in its block words. Only splitting survives in the associated graded. Its length-N complex is

$$I^{\otimes N} \otimes E^{\otimes (N-1)}, \qquad E^{-1} = ke_-, \quad E^0 = ke_+, \quad de_- = e_+ \qquad (N \geq 1).$$

An uncut gap contributes e_- and an outer cut contributes e_+. The tensor differential sign counts preceding uncut gaps. For a split after i letters in block r, with P preceding letters, the displayed derivation gives $(-1)^{r-1-P+i-1}$.

The uncut-gap count is $P + i - r$, with the same parity. For $N \geq 2$, the first-gap homotopy $e_+ \mapsto e_-$, $e_- \mapsto 0$ contracts this tensor complex, including its $I^{\otimes N}$ factor. It is an identity of maps, so tensor exactness is unnecessary. A negative-degree cycle has a finite largest length. Subtract a lifted primitive of its top-length part and repeat. Length strictly decreases, proving negative acyclicity. In degree zero the boundaries impose exactly $c(a)c(b) = c(ab)$. Every product reduces to a scalar plus one letter, and its image in $k1 \oplus I$ makes this expression unique. This proves the theorem. □

In particular, for $A = k[x]$ with augmentation $x \mapsto 0$, the ordinary counit recovers $k[x]$. The augmentation-adic completion is instead $\varprojlim_N k[x]/(x^N) = k[[x]]$. For a nonzero k, the latter contains $\sum_{n\geq 0} x^n$, which is not in the image of the canonical polynomial inclusion. Ordinary bar–cobar reconstruction therefore does not compute this completion.

208.3 Positive-weight completion

Let $A_{\mathrm{alg}} = \bigoplus_{w\geq 0} A_w$ be an augmented algebra in degree zero, with $A_0 = k$ and $A_u A_v \subseteq A_{u+v}$. The augmentation ideal has strictly positive weights. Form the ordinary $D = \Omega B(A_{\mathrm{alg}})$ and write D_w for its weight-w subcomplex. Define

$$\widehat{D} = \prod_{w\geq 0} D_w, \qquad \widehat{A} = \prod_{w\geq 0} A_w,$$

with products taken degreewise in complexes and with the weight topology. Multiplication is continuous: only finitely many pairs of nonnegative weights sum to a fixed weight.

Theorem 208.2 (Positive-weight reconstruction). *The componentwise counit $\widehat{D} \to \widehat{A}$ is a quasi-isomorphism of complete differential graded algebras.*

Proof. At weight w, a block word has at most w entries. The finite-word proof therefore applies within that weight and gives $H^j(D_w) = 0$ for $j < 0$ and $H^0(D_w) = A_w$. Kernels and images of the product differential are products of the corresponding kernels and images. Consequently cohomology commutes with this degreewise product, proving the claim. Equivalently, the finite weight quotients have surjective, split transition maps. This is a specified weight completion of the ordinary resolution. □

For a general filtration, the corresponding conclusion needs additional limit hypotheses. Suppose compatible maps $D_N \to A_N$ are quasi-isomorphisms of towers of complexes, both towers are degreewise surjective, and both cohomology towers are Mittag–Leffler in every degree. Then the inverse-limit map is a quasi-isomorphism: the inverse-limit cohomology exact sequence has zero $\varprojlim^1 H^{j-1}$ term, and the maps on $\varprojlim H^j$ are isomorphisms. The identification of these limits with a chosen completed algebra and its completed bar–cobar requires a separate continuity proof for that filtration.

The factorization category $\mathrm{FactD}(X)$ is not a stable coefficient category merely because its ambient Ran category is stable. Its bar realization uses the sifted-colimit statement above. An internal chiral duality theorem additionally requires specified coalgebra objects, completion functors, and geometric comparison maps with their continuity hypotheses. The ordinary tensor counit proved here and the completed double-centralizer map below have different definitions.

208.4 Deformation complexes

The convolution complex

$$\mathrm{Conv}(B\mathsf{E}_1, \mathrm{End}_A)$$

controls deformations of the associative multiplication. Its Maurer–Cartan elements are homotopy-coherent products, and its gauge equivalences are operadic homotopies. In the two-direction theory, the complete deformation complex is the total convolution algebra of the bicoloured cooperad. Its differential splits into internal, transverse, chiral, and mixed components. Curvature can be inserted only after its arity and degree are specified.

Chapter 209

Koszul–Morita theory, relative bases, and the centre

209.1 The augmentation boundary

For an augmented algebra A, the unit is a left and right module. The augmentation-boundary algebra is

$$A^{!} = \mathrm{RHom}_{A}(\mathbf{1}, \mathbf{1}).$$

When bar terms are dualisable, or continuous internal Hom is used,

$$A^{!} \simeq \mathrm{RHom}_{\mathcal{V}}(\mathrm{Bar}(A), \mathbf{1}).$$

This is the boundary Koszul dual. It is not automatically the quadratic dual, Verdier dual, or chiral dual.

Theorem 209.1 (Double centralizer)**.** *Assume A is connective, derived augmentation completion is computed by the cobar totalization of the Amitsur complex, and $\mathbf{1}$ is compact in the completed module category. Then*

$$A^{!!} := \mathrm{RHom}_{A^{!}}(\mathbf{1}, \mathbf{1}) \simeq A^{\wedge}_{\varepsilon}.$$

On the nilcomplete locus this is A.

Proof. The bar resolution identifies the thick subcategory generated by $\mathbf{1}$ with perfect $A^{!}$-modules. The endomorphisms of the corresponding Yoneda generator form the double centralizer. The universal algebra acting on the augmentation-generated subcategory is derived completion along $\ker \varepsilon$, computed by the assumed Amitsur totalization. □

209.2 One-sided and two-sided twisting

Put $C = \mathrm{Bar}(A)$ and let $\tau : C \to A$ be the universal twisting cochain. For an A-module M, define

$$K_{A}(M) = C \otimes_{!} M$$

with the sum of the internal differential and the twisting term obtained from Δ_{C}, τ, and the module action. The first factor makes it a C-comodule. The reverse functor is $L_{C}(N) = A \otimes_{!} N$ with the dual twisting differential.

Theorem 209.2 (Positive-complete module–comodule duality)**.** *For a connected positive-complete algebra with finite weight decompositions, the twisting functors are inverse equivalences between positive-complete modules and conilcomplete comodules. The two-sided version gives an equivalence between complete (A, B)-bimodules and conilcomplete $(\mathrm{Bar}\, A, \mathrm{Bar}\, B)$-bicomodules. It sends the regular diagonal bimodule to the regular diagonal bicomodule.*

Proof. Filter the composites by bar cuts and coradical length. In each fixed weight, the associated graded is the subdivision complex of an ordered word with an extra degeneracy, hence contractible onto the unsplit module or comodule. Finite decomposition gives convergence; completion passes from finite windows to the full object. □

For composable bimodules,

$$K^{\mathrm{bi}}_{A,D}(M \otimes^{\mathbb{L}}_{B} N) \simeq K^{\mathrm{bi}}_{A,B}(M) \square^{\mathbb{R}}_{\mathrm{Bar}\, B} K^{\mathrm{bi}}_{B,D}(N).$$

Thus tensor composition becomes cotensor composition. Mapping objects identify Hochschild cochains with coHochschild cochains, and Hochschild chains with coHochschild chains, in the complete categories.

209.3 Relative augmentation bases

Many important noncommutative algebras are not augmented over $\Bbbk$ but are augmented over a separable or semisimple base S.

Definition 209.3 (Relative bar)**.** For an S-algebra A equipped with a retraction $A \to S$, the relative reduced bar is

$$\mathrm{Bar}_S(A) = S \otimes^{\mathbb{L}}_{A} S \simeq T^c_S(s\bar{A}_S)$$

with tensors and multiplication taken over S.

For a quiver, $S = \bigoplus_i \Bbbk e_i$ is the vertex-idempotent algebra. Tensoring over S automatically retains only composable paths. This is the natural chain complex; choosing an arbitrary scalar augmentation destroys the object decomposition and can change the homology.

209.4 The internal derived center

Choose a presentable stable symmetric monoidal coefficient category $\mathcal{V}$ whose tensor $\otimes_!$ preserves small colimits separately. For a unital associative algebra A in that category, put

$$\mathbf{Z}(A) = \underline{\mathrm{RHom}}_{A^e}(A, A), \qquad A^e = A \otimes_! A^{\mathrm{op}}.$$

This is an internal mapping object, characterized by maps from v to $\mathbf{Z}(A)$ being bimodule maps $v \otimes_! A \to A$. Theorem 19.1 gives its E_2-structure and its universal coherent central action. The specified coefficient category and tensor are part of this assertion.

Here a linear Morita equivalence between A and A' means a $\mathcal{V}$-linear equivalence $\mathrm{LMod}_A(\mathcal{V}) \simeq \mathrm{LMod}_{A'}(\mathcal{V})$. The center is invariant under such an equivalence. Indeed, such an equivalence conjugates the categories of colimit-preserving linear endofunctors and carries the identity functor to the identity. It therefore identifies their internal endomorphisms of the unit, preserving both composition and the monoidal product. These endomorphism objects are the derived centers, with their E_2-structures.

A bulk algebra $\mathcal{B}$ realized as an E_2-algebra with a coherent central action on a boundary algebra A_b has the classifying map

$$\mathcal{B} \longrightarrow \mathbf{Z}(A_b).$$

To construct this map in a field theory, its operations must supply that action in the chosen coefficient category. For a specified protected-sector functor, the target comparison is (19.9). An open–closed theorem must prove that the resulting sector map is an equivalence.

No-go. *Example 19.2 constructs two oriented open–closed surface theories with open algebra k. Their closed algebras are k and $k \oplus k$. Their pairings, adjoint boundary maps and Cardy sewing identity are explicit. Both closed algebras have the same image acting on the boundary, but their circle state spaces have different dimensions. Thus the boundary algebra and this effective image do not recover the full bulk theory.*

The existence of a local bulk with specified stress tensor, global states and anomaly data is a further problem. In the mixed holomorphic–topological setting, its action must retain the supported collision maps and the chosen completion. Reconstruction through a chosen boundary object additionally requires the applicable generation and coherent-comparison hypotheses of the factorization Morita theorem.

Chapter 210

Quadratic, PBW, curved, and Verdier comparisons

210.1 Quadratic recognition

Let V be dualisable and

$$A = T(V)/(R), \qquad R \hookrightarrow V \otimes_! V.$$

The quadratic dual algebra is

$$A^{q!} = T(V^*[-1])/(R^\perp),$$

and the quadratic dual coalgebra is the largest subcoalgebra of $T^c(sV)$ whose quadratic core lies in s^2R.

Theorem 210.1 (Quadratic recognition)**.** *Assume every weight component is dualisable and the diagonal Koszul complex is exact in positive homological degree. Then the quadratic dual coalgebra includes quasi-isomorphically into* $\mathrm{Bar}(A)$*, and continuous duality gives*

$$A^! \simeq A^{q!}.$$

Proof. Filter the bar by tensor weight. Its linear strand is the quadratic Koszul complex. The exactness hypothesis collapses the spectral sequence onto the quadratic dual coalgebra. Continuous linear duality turns the coalgebra into the stated algebra. □

No-go. *The orthogonal-relations algebra $A^{q!}$ records only the quadratic strand. If the Koszul complex has higher homology, $A^{q!}$ is not quasi-isomorphic to* $\mathrm{RHom}_A(\mathbf{1},\mathbf{1})$*. Equality requires the Koszulness theorem, not merely a quadratic presentation.*

210.2 PBW deformations and curvature

For a quadratic algebra, lower-degree corrections preserve the leading relations only when their overlaps are compatible.

Work over a field k, and let V be finite-dimensional. Choose a relation subspace $R \subseteq V \otimes V$ and maps $\alpha : R \to V$, $\beta : R \to k$. The homogeneous and deformed algebras are

$$A_0 = T_k(V)/(R), \qquad U = T_k(V)/(r - \alpha(r) - \beta(r) : r \in R).$$

Filter U by word length. The leading relations give a canonical surjection $A_0 \to \mathrm{gr}\, U$. To decide when it is injective, consider the overlap space

$$W_3 = (R \otimes V) \cap (V \otimes R), \qquad \delta = (\alpha \otimes \mathrm{id} - \mathrm{id} \otimes \alpha)|_{W_3}.$$

Let $B = T_k(V^*)/(R^\perp)$ be the quadratic dual algebra, with generators in cohomological degree one. The pairing used to define $R^\perp$ is the ordinary tensor pairing, without a suspension sign. Thus $B^2 = R^*$ and $B^3 = W_3^*$. Set $d_1 = -\alpha^* : B^1 \to B^2$ and $c = -\beta^* \in B^2$.

Theorem 210.2 (Curved PBW duality). *Assume that A_0 is Koszul and that*

$$\delta(W_3) \subseteq R, \qquad \alpha\delta = -(\beta \otimes \mathrm{id} - \mathrm{id} \otimes \beta)|_{W_3}, \qquad \beta\delta = 0. \tag{210.1}$$

Then $A_0 \to \operatorname{gr} U$ is an isomorphism. The map d_1 extends to a degree-one derivation d on B, and

$$d^2 = [c, -], \qquad dc = 0.$$

Conversely, descent of this derivation and these curved identities imply the three overlap equations. That converse calculation does not require Koszulness. Koszulness is a hypothesis of the PBW implication asserted here.

Proof. The first equation makes the derivation of $T(V^*)$ defined by d_1 preserve the ideal $(R^\perp)$. Indeed, its value on a quadratic functional, restricted to W_3, is minus that functional composed with δ. It vanishes for every element of $R^\perp$ exactly when $\delta(W_3) \subseteq R$.

For $f \in V^*$ and $w \in W_3$, the second equation gives

$$\langle d^2 f, w\rangle = \langle f, \alpha\delta(w)\rangle = -\langle f, (\beta \otimes \mathrm{id} - \mathrm{id} \otimes \beta)(w)\rangle = \langle [c, f], w\rangle.$$

Both d^2 and $[c, -]$ are degree-two derivations, so equality on generators proves equality on B. Similarly, $\langle dc, w\rangle = \beta\delta(w) = 0$. Perfect pairings in degrees two and three prove the converse implications.

The PBW conclusion is Braverman–Gaitsgory's theorem for a Koszul homogeneous algebra [19, Theorem 4.1 and Lemma 3.3]. Its hypotheses are exactly (210.1), with the relation subspace the graph of $\alpha + \beta$. The Koszul hypothesis makes the higher deformation obstructions vanish. The degree-three identities alone do not supply that vanishing. The resulting curved algebra agrees with the nonhomogeneous quadratic construction of [89, Proposition 2.2] under $d_1 = -\alpha^*$ and $c = -\beta^*$. □

For Weyl and Clifford algebras, Theorem 24.1 gives direct basis proofs in every characteristic. Theorem 24.2 also constructs their curved cobar resolutions and fixes their suspension signs. These comparisons concern ordinary coefficient algebras. A Verdier or factorization comparison requires its own geometric map.

210.3 Verdier duality

Verdier duality acts on the factorisation D-module coefficient and reverses !- and $*$-extensions. Under holonomicity and finiteness hypotheses it can carry factorisation algebras to factorisation coalgebras. A comparison with a Koszul dual requires an explicit pairing between the bar resolution and the Verdier-dual coefficient. The operations agree in self-dual examples but are not definitionally identical.

No-go. *The interval bar $\mathbf{1} \otimes^{\mathbb{L}}_{A} \mathbf{1}$, boundary dual $\mathrm{RHom}_A(\mathbf{1}, \mathbf{1})$, quadratic dual $T(V^*[-1])/(R^\perp)$, and Verdier dual $\mathbb{D}A$ have different variances and hypotheses. They can be joined by comparison maps in a finite Koszul self-dual model. Replacing the maps by equality erases the theorem.*

Chapter 211

Free and square-zero ordered chiral algebras

211.1 Algebraic coefficients and factorization

The free and square-zero calculations use a specified tensor product and augmentation. Let k be a field. The ordinary calculations below take place in complexes of k-vector spaces. An exact strong monoidal dg functor to an ambient coefficient category transports the displayed bar formulas when it preserves their direct sums and normalization. This is a statement about that ambient tensor.

Tensoring a factorization object by a constant algebra requires further structure. For example, $k^2 \otimes \omega_{\mathrm{Ran}}$ has rank two on both a singleton and a separated pair. Its singleton tensor square has rank four, so the required factorization equivalence cannot exist. The generic formula $B \otimes F$ therefore does not by itself construct a factorizing ordered chiral algebra. The actual examples of Theorem 11.5 prove independence of the two products on a fixed factorizing family.

211.2 The free algebra

Let V be a vector space in degree zero, and give $T(V)$ its usual augmentation. Write $s = [1]$ for the cohomological suspension lowering degree by one. Equip $k \oplus sV$ with primitive coproduct on its reduced summand.

Theorem 211.1 (Bar of a free algebra)**.** *The inclusion of a generator as a one-input bar is a dg coalgebra quasi-isomorphism*

$$k \oplus sV \longrightarrow \mathrm{Bar}(T(V)).$$

Its underlying complexes admit a natural deformation retract. In homological bar degrees,

$$\mathrm{Tor}_0^{T(V)}(k,k) = k, \qquad \mathrm{Tor}_1^{T(V)}(k,k) = V, \qquad \mathrm{Tor}_r^{T(V)}(k,k) = 0 \quad (r \geq 2).$$

Proof. For a generator word of length n, index its bar decompositions by subsets of the $n-1$ possible cuts. The differential removes a cut, with alternating sign according to its position among the present cuts. When $n > 1$, insert the first cut if it is absent and send a word already containing that cut to zero. Removing this inserted cut gives the identity, and every other removal cancels with the opposite insertion order. Only weights zero and one remain.

This is the degree-zero instance of Theorem 21.1, whose proof also treats complexes with nonzero internal differential. The inclusion preserves the primitive coproduct. □

For $V = kx$, the chain projection need not preserve coproducts: on $[x|x]$ its reduced coproduct survives after projection to $sx \otimes sx$, while the projected word is zero. Thus the coalgebra comparison uses the inclusion specified in the theorem. For finite-dimensional V, dualizing that map gives the square-zero dg algebra model $k \oplus V^\vee[-1]$ for derived endomorphisms.

211.3 The square-zero algebra

Let M be a cochain complex and let $S(M) = k \oplus M$ have square-zero augmentation ideal.

Proposition 211.2 (Bar of a square-zero algebra)**.** *The multiplication part of the normalized bar differential vanishes, and*

$$\mathrm{Bar}(S(M)) = T^c(sM)$$

as dg coalgebras, with only the internal differential. If M is a d-dimensional vector space in degree zero, then

$$\dim_k \mathrm{Tor}_r^{S(M)}(k,k) = d^r, \qquad P_{S(M)}(z) := \sum_{r\geq 0} d^r z^r = \frac{1}{1-dz}.$$

Proof. Every adjacent multiplication in M is zero, so the multiplication terms vanish. The unary term remains $sm \mapsto -s(d_M m)$. In the stated degree-zero case it is zero as well, and the tensors of length r form the entire homology in bar degree r. This proves both formulas. □

For a bounded finite-dimensional complex M, the full linear dual is the completed tensor algebra on $M^\vee[-1]$. Its convolution multiplication models the derived endomorphism product by Theorem 21.3. The completion is a product over tensor lengths, and odd generators have no imposed exterior relation. The free and square-zero algebras therefore have different bars and dual products even when their generator complexes agree.

211.4 A geometric realization in both directions

A free geometric example in both associative directions requires more than a tensor product of its proposed state spaces. Given transverse generators $V_\perp$ and chiral generators M_X, construct the coefficient objects, their factorization maps and both associative operations with their mixed coherences. Construct the augmentation needed for each bar and specify the coefficient tensor used by it. If a field description is used, supply its geometric realization and the comparisons with iterated operations. Theorem 204.4 recognizes fields from such a realization and does not construct the reverse functor.

When the transverse bar is identified with the tensor-algebra coefficient model, Theorem 21.1 gives its primitive comparison. A square-zero replacement gives the tensor coalgebra of Proposition 21.2 under the corresponding realization hypotheses. Keeping the second associative bar unchanged requires an operation-preserving comparison of the remaining chiral structures. Constructing that comparison, and establishing the free geometric universal property, remain the two-direction realization problem.

Chapter 212

The quantum and Jordan planes

The quantum and Jordan planes have the same graded dimensions and augmentation Betti numbers. Their bar coproducts distinguish the relations, and their Yoneda products distinguish the resulting augmentation boundaries. We work over a field k, with zero differential and polynomial weight one on the algebra generators.

212.1 The quantum plane

Fix $q \in k^\times$ and set

$$A_q = k\langle x, y\rangle/(yx - qxy), \qquad \epsilon(x) = \epsilon(y) = 0.$$

The ordered monomials $x^a y^b$ form a basis, with product $(x^a y^b)(x^c y^d) = q^{bc} x^{a+c} y^{b+d}$, by Theorem 22.1. In particular, $H_{A_q}(z) = (1-z)^{-2}$. Write $A_q(-r)$ for a free graded left module whose generator has polynomial weight r.

Theorem 212.1 (Quantum-plane Koszul resolution)**.** *The augmented complex*

$$0 \longrightarrow A_q(-2) \xrightarrow{d_2} A_q(-1)^{\oplus 2} \xrightarrow{d_1} A_q \xrightarrow{\epsilon} k \longrightarrow 0,$$

with $d_1(u, v) = ux + vy$ and $d_2(c) = (cy, -qcx)$, is exact. Its augmentation Tor coalgebra is

$$k \oplus k\{sx, sy\} \oplus k\{\omega_q\}, \qquad \omega_q = [y|x] - q[x|y],$$

where sx, sy have homological degree one and ω_q has degree two. The degree-one classes are primitive, and

$$\overline{\Delta}\omega_q = sy \otimes sx - q\, sx \otimes sy.$$

The Poincare polynomial is $1 + 2t + t^2$.

Proof. Theorem 22.2 proves exactness by solving the coefficient equation $ux + vy = 0$ as $(u, v) = (cy, -qcx)$. Right multiplication by y is injective on the ordered basis, which proves injectivity at the left end. Tensoring with k kills both differentials. Proposition 22.3 identifies this calculation with the displayed subcoalgebra of the normalized bar. The bar cohomological degrees are the negatives of the homological degrees used here. □

For degree-one Yoneda generators ξ, η and a nonzero degree-two class ζ, Theorem 22.4 gives

$$\xi^2 = \eta^2 = 0, \qquad \xi\eta = q\zeta, \qquad \eta\xi = -\zeta.$$

Thus the quadratic dual is

$$Q_q = k\langle \xi, \eta\rangle/(\xi^2, \eta^2, \xi\eta + q\eta\xi).$$

The parameter survives in the product even though the Betti numbers agree with those of the commutative plane.

212.2 The Jordan plane

Set

$$A_J = k\langle x, y\rangle/(yx - xy - x^2), \qquad \epsilon(x) = \epsilon(y) = 0.$$

The rule $yx \mapsto xy + x^2$ decreases lexicographic order with $y > x$ among words of fixed length. It therefore expresses every word as a sum of ordered monomials.

Their independence has a concrete realization on $k[t, z]$:

$$X(f) = tf, \qquad Y(f) = zf + t^2\frac{\partial f}{\partial t}.$$

These operators satisfy $YX - XY = X^2$, and $X^aY^b(1) = t^a z^b$. The ordered monomials are therefore independent over any field. Consequently $H_{A_J}(z) = (1 - z)^{-2}$.

Theorem 212.2 (Jordan-plane Koszul resolution)**.** *The augmented complex*

$$0 \longrightarrow A_J(-2) \xrightarrow{d_2} A_J(-1)^{\oplus 2} \xrightarrow{d_1} A_J \xrightarrow{\epsilon} k \longrightarrow 0,$$

with

$$d_1(u, v) = ux + vy, \qquad d_2(c) = (c(y - x), -cx),$$

is exact. Its augmentation Tor groups have Poincare polynomial $1 + 2t + t^2$*. The degree-two bar class has representative*

$$\omega_J = [y|x] - [x|y] - [x|x].$$

Proof. The composite is $c((y - x)x - xy) = 0$. Filter the ordered basis by the exponent of y. The correction x^2 in the relation has smaller filtration degree, so the associated graded algebra is $k[x, y]$. Give the two middle free generators filtration degrees zero and one, respectively, and the left generator filtration degree one. The associated graded differential is then the quantum-plane resolution at $q = 1$. It is exact by Theorem 22.2.

At each fixed polynomial weight, this filtration is finite. For a cycle, lift a preimage of its leading filtered class and subtract its boundary. The filtration degree drops, so repeating the procedure terminates. This proves exactness. The same leading-class argument gives injectivity at the left end. The image at A_J is the augmentation ideal.

Tensoring with k kills both differentials. The left bar comparison sends the middle generators to $-[x], -[y]$ and the left generator to ω_J. Its chain-map equation is

$$d_P(1 \otimes \omega_J) = -y \otimes [x] + x \otimes [y] + x \otimes [x].$$

The free-resolution comparison used in Proposition 22.3 therefore identifies the indicated cycle with the nonzero Tor class. □

212.3 Products beyond the Betti numbers

Put the Jordan resolution in cohomological degrees $0, -1, -2$, with generators $1, e_x, e_y, e_r$. Let ξ, η, ζ be the dual augmentation cochains. Closed degree-one lifts of ξ and η are

	e_x	e_y	e_r
F_ξ	1	0	$e_x - e_y$
F_η	0	1	e_x

and vanish on degree zero. Their equations $dF + Fd = 0$ reduce on e_r to $(x - y) + (y - x) = 0$ and $x - x = 0$.

Composition, in the same Yoneda convention as above, gives

$$\xi^2 = \xi\eta = \zeta, \qquad \eta\xi = -\zeta, \qquad \eta^2 = 0.$$

These products generate the four-dimensional Ext space, so

$$\operatorname{Ext}^*_{A_J}(k,k) \cong k\langle \xi, \eta\rangle/(\eta^2, \xi\eta + \eta\xi, \xi^2 + \eta\xi).$$

Indeed, the relations reduce every word of length at least three to zero and leave at most four basis elements. They also annihilate the tensor $y \otimes x - x \otimes y - x \otimes x$, identifying this algebra with the quadratic dual.

In the Jordan dual, $(a\xi + b\eta)^2 = a^2\zeta$, so the degree-one elements with square zero form the single line $k\eta$. For the quantum dual, the square is $ab(q-1)\zeta$. Its zero set is the union of the two coordinate lines when $q \neq 1$, and the whole degree-one space when $q = 1$. Thus the graded Yoneda algebras are not isomorphic over any field.

In particular, no exact k-linear equivalence of the derived module categories can carry one augmentation module to the other. Such an equivalence would preserve this graded endomorphism algebra. This conclusion concerns that specified module, without asserting a classification of unrestricted module-category equivalences.

212.4 Finite bar dimensions and ranks

At polynomial weight $w > 0$, let $C_{w,r}$ be the normalized bar tensors with r nonunit monomial inputs. The bar differential is $b_r : C_{w,r} \to C_{w,r-1}$. Its matrix merges adjacent monomials with signs $(-1)^{j-1}$ at the jth join. The common Hilbert series gives

$$c_{w,r} := \dim C_{w,r} = [z^w]\big((1-z)^{-2} - 1\big)^r.$$

Write $\rho_{w,r} = \operatorname{rank} b_r$ and $h_{w,r} = \dim H_r(C_{w,\bullet})$. Then

$$h_{w,r} = c_{w,r} - \rho_{w,r} - \rho_{w,r+1}.$$

The exact resolutions determine $h_{1,1} = 2$, $h_{2,2} = 1$ and all other positive-weight homology dimensions as zero. Starting from $\rho_{w,1} = 0$, the displayed identity determines every rank.

Thus both algebras have the following table, for every allowed q and over every field:

w	r	$c_{w,r}$	$\rho_{w,r}$	$\rho_{w,r+1}$	$h_{w,r}$
1	1	2	0	0	2
2	1	3	0	3	0
2	2	4	3	0	1
3	1	4	0	4	0
3	2	12	4	8	0
3	3	8	8	0	0
4	1	5	0	5	0
4	2	25	5	20	0
4	3	36	20	16	0
4	4	16	16	0	0
5	1	6	0	6	0
5	2	44	6	38	0
5	3	102	38	64	0
5	4	96	64	32	0
5	5	32	32	0	0

The finite table illustrates the all-weight resolution theorems. It does not identify the coproducts of the two bar coalgebras or the products of their duals.

212.5 Internal chiral realization

To realize either plane geometrically, construct its coefficient objects, factorization equivalences, augmentation and transverse multiplication with the indicated relation. Then compare the entire geometric bar with the coefficient bar, including its coproduct and realization hypotheses. The formulas above give the algebraic side of that comparison.

Tensoring a constant plane algebra with a factorizing object is insufficient by the rank example of Chapter 21. The actual marked-point construction of Theorem 11.5 remains available. A second associative chiral operation requires its own construction and the mixed coherences with transverse multiplication. These maps are the additional content of a quantum- or Jordan-plane ordered chiral realization.

Chapter 213

Augmentation obstructions, Weyl curvature, and matrix sectors

A bar construction with scalar endpoints requires a character of the algebra. The Weyl relation forbids such a character, while a matrix algebra replaces the scalar endpoint by a projective module. An augmented algebra can nevertheless contain a full matrix sector. We compute an example in which the scalar bar and a cyclic trace retain different parts of that algebra.

213.1 The Weyl obstruction

Let k be a field and

$$W_1 = k\langle x, y\rangle/(yx - xy - 1).$$

Theorem 213.1 (No scalar augmentation for the Weyl algebra)**.** *There is no unital algebra map $W_1 \to k$. Consequently k cannot be made a left or right W_1-module by a scalar augmentation, and the derived tensor with those prescribed endpoints is undefined.*

Proof. Applying such a map to the relation gives $\epsilon(y)\epsilon(x) - \epsilon(x)\epsilon(y) = 1$. The left side is zero in k, a contradiction. □

The polynomial family

$$W_\hbar = k[\hbar]\langle x, y\rangle/(yx - xy - \hbar)$$

has a k-algebra character sending x, y, $\hbar$ to zero. Every k-algebra character sends $\hbar$ to zero, by the same commutator calculation. There is no $k[\hbar]$-algebra augmentation to $k[\hbar]$ fixing $\hbar$. Its fiber at zero is the polynomial plane. Its fiber at one is W_1. The genuine word-Rees algebra of W_1 has relation $YX - XY = z^2$, as proved in Chapter 24.

The curved dual of $W_\hbar$ is

$$E = \Lambda_{k[\hbar]}(\xi, \eta), \qquad d = 0, \qquad c = \hbar\xi\eta,$$

where ξ, η have cohomological degree one and are dual to $[x]$, $[y]$. With $\rho = [y|x] - [x|y]$, the cobar differential is

$$D(t\rho) = yx - xy - \hbar.$$

Theorem 24.2 proves that the resulting unital dg algebra resolves $W_\hbar$. Since c is central, $d^2 = [c, -] = 0$. Ordinary cohomology of the underlying complex exists and equals E. It forgets the curvature and hence the constant Weyl relation.

Modules also provide valid bar endpoints. For example, W_1 acts on $k[t]$ by $x = t$ and $y = \partial_t$. For a specified right module M and left module N, the two-sided bar computes $M \otimes^{\mathbb{L}}_{W_1} N$. This construction uses their actual module actions, without a scalar character.

213.2 The matrix obstruction

Theorem 213.2 (No scalar augmentation for a full matrix algebra)**.** *For $n > 1$, there is no unital algebra map $M_n(k) \to k$.*

Proof. Let E_{ij} be the matrix units. Under a hypothetical map ϕ, the orthogonal idempotents E_{ii} have values zero or one and sum to one. Exactly one, say E_{ii}, has value one. Choose $j \neq i$. The equations $E_{ij}E_{ji} = E_{ii}$ and $E_{ji}E_{ij} = E_{jj}$ would give the same scalar product both values one and zero. □

Put $A = M_n(k)$ and $e = E_{11}$. The right module eA and left module Ae are projective, and multiplication gives

$$eA \otimes^{\mathbb{L}}_{A} Ae = eA \otimes_A Ae \xrightarrow{\sim} eAe \cong k.$$

The inverse sends $a \in eAe$ to $e \otimes a$. Indeed, $er \otimes se = e \otimes rse = e \otimes erse$ by balancing. This is the scalar corner selected by these modules.

The ordinary trace is instead a cyclic linear functional. It does not supply their module actions or an algebra augmentation. A geometric realization by matrix-valued fields still requires the factorization coefficients and their compatible operations.

213.3 An augmented algebra with a matrix sector

Let

$$D = k[\epsilon]/(\epsilon^2), \qquad B_n = D \times M_n(k), \qquad \varepsilon_B(a + b\epsilon, M) = a.$$

This is a unital augmented algebra, noncommutative for $n > 1$. Its scalar augmentation is unrelated to the cyclic functional

$$\tau_B(a + b\epsilon, M) = b + \operatorname{Tr}(M).$$

The pairing $\tau_B(uv)$ is symmetric and nondegenerate. In D, the ordered basis $(1, \epsilon)$ has dual basis $(\epsilon, 1)$. In the matrix summand, E_{ij} is dual to E_{ji}. These dual bases prove nondegeneracy in every characteristic, including when the characteristic divides n.

For a symmetric Frobenius algebra with dual bases (u_a), (u^a), define its handle element to be $e = \sum_a u_a u^a$. This definition is independent of the dual bases, since their tensor is the inverse of the pairing.

Proposition 213.3 (Bar and handle element of B_n)**.** *Projection $\pi : B_n \to D$ induces a quasi-isomorphism of coaugmented dg coalgebras*

$$B\pi : B(B_n) \longrightarrow B(D) = T^c(sk\epsilon).$$

Consequently

$$k \otimes^{\mathbb{L}}_{B_n} k \simeq k \otimes^{\mathbb{L}}_{D} k, \qquad \dim_k \operatorname{Tor}^{B_n}_r(k,k) = 1 \quad (r \geq 0).$$

The handle element and its contractions are

$$e_{B_n} = (2\epsilon, nI_n), \qquad \tau_B(1) = n, \qquad \tau_B(e_{B_n}) = 2 + n^2, \qquad \tau_B(e^g_{B_n}) = n^{g+1} \quad (g \geq 2),$$

with the integer coefficients interpreted in k.

Proof. Let $\varepsilon_D : D \to k$ be its augmentation. The map

$$\sigma : D \longrightarrow B_n, \qquad \sigma(d) = (d, \varepsilon_D(d)I_n),$$

is unital, multiplicative and augmented, and $\pi\sigma = \mathrm{id}_D$. Thus $B\pi$ and $B\sigma$ are dg coalgebra maps whose composite is the identity on $B(D)$.

The periodic resolution of the augmentation module over D is

$$\cdots \xrightarrow{\epsilon} D \xrightarrow{\epsilon} D \xrightarrow{\epsilon} D \xrightarrow{\varepsilon_D} k \longrightarrow 0.$$

Its exactness follows from $\ker(\epsilon : D \to D) = k\epsilon = \mathrm{im}(\epsilon)$. Via π, each term is the projective B_n-module $B_n(1, 0)$. It therefore remains a projective resolution over B_n. Tensoring with the right augmentation module kills every differential and leaves one copy of k in every nonnegative homological degree.

The reduced bar of D has zero differential because $(k\epsilon)^2 = 0$. Both bar homologies thus have dimension one in every homological degree. Since $B\pi$ has the section $B\sigma$, it is an isomorphism on homology. This proves the asserted dg coalgebra comparison and the derived tensor calculation.

The displayed dual bases give

$$e_D = 1\epsilon + \epsilon 1 = 2\epsilon, \qquad e_{M_n} = \sum_{i,j} E_{ij}E_{ji} = nI_n.$$

Products and powers in B_n are componentwise. For $g \geq 2$, $(2\epsilon)^g = 0$, while $\mathrm{Tr}((nI_n)^g) = n^{g+1}$. The remaining trace values follow directly from the definition of τ_B. □

The scalar bar retains the dual-number sector. The Frobenius contractions also detect the matrix sector. These are algebraic contraction values. A closed oriented two-dimensional field theory requires a commutative closed state algebra and compatible sewing maps. Those data are not supplied by the noncommutative Frobenius algebra alone. Likewise, interpreting these values as one genus variable of a mixed theory requires its two families of operations and their interchange maps.

No-go. *A cyclic functional satisfies $\tau(ab) = \tau(ba)$, while an augmentation must satisfy $\varepsilon(ab) = \varepsilon(a)\varepsilon(b)$ and $\varepsilon(1) = 1$. The bar endpoint actions use the latter identities. Replacing them by cyclic pairings changes the construction and does not define the same derived tensor product.*

Chapter 214

Relative quiver bars and boundary conditions

The relative bar retains the objects at both ends of a composable sequence. For a chain of two arrows whose composite is zero, the relation gives a degree-two bar class and a nonzero Yoneda product. Keeping the vertex idempotents makes both statements visible.

214.1 The natural base

Let k be a field, Q a finite quiver and

$$S = \bigoplus_{i \in Q_0} k e_i, \qquad M = \bigoplus_{a : i \to j} k a, \qquad a = e_j a e_i.$$

The path algebra is $T_S(M)$, with paths multiplied from right to left. Give arrows weight one and vertex idempotents weight zero. Its augmentation fixes the vertex idempotents and kills the arrows. For a quotient A by homogeneous relations of positive weight, the same augmentation $\epsilon : A \to S$ remains defined. Write $\bar{A} = \ker \epsilon$.

Use the direct-sum relative bar

$$B_S A = \bigoplus_{r \geq 0} (s\bar{A})^{\otimes_S r}, \qquad s = [1], \qquad |sa| = |a| - 1,$$

with deconcatenation and the normalized differential of Theorem 15.1. For ordinary inputs, $b[a|a'] = [aa']$. The relative bar resolution identifies this complex with $S \otimes^{\mathbb{L}}_{A} S$, where the first input is a right A-module and the second a left A-module. In particular, $H^{-r}(B_S A) = \operatorname{Tor}^A_r(S, S)$.

To name individual endpoints, let $R_j = e_j S$ be the right simple and $L_i = S e_i$ the left simple, both regarded as A-modules through ϵ. Taking vertex summands of the bar resolution gives

$$e_j H^{-r}(B_S A) e_i \cong \operatorname{Tor}^A_r(R_j, L_i). \tag{214.1}$$

The balancing over S admits only composable inputs, and the two outer idempotents retain their target and source. When the quiver is realized by boundary conditions, these are the two boundary labels.

214.2 A relation and its coproduct

Consider

$$1 \xrightarrow{a} 2 \xrightarrow{b} 3, \qquad A = kQ/(ba).$$

This algebra has basis e_1, e_2, e_3, a, b. Its augmentation ideal is $ka \oplus kb$, and its only composable two-input tensor is $[b|a]$. There is no composable triple.

Theorem 214.1 (Relative bar of the relation quiver)**.** *The relative bar has zero differential and underlying S-bimodule*

$$B_S A = S \oplus k[a] \oplus k[b] \oplus k\omega, \qquad \omega = [b|a].$$

The cohomological degrees of $[a]$, $[b]$, ω *are* $-1, -1, -2$. *Its reduced coproduct vanishes on* $[a]$, $[b]$ *and satisfies*

$$\overline{\Delta}\omega = [b] \otimes_S [a]. \tag{214.2}$$

Consequently,

$$\operatorname{Tor}_0^A(S,S) = S, \qquad \operatorname{Tor}_1^A(S,S) = k[a] \oplus k[b], \qquad \operatorname{Tor}_2^A(S,S) = k\omega,$$

and all higher Tor groups vanish.

Proof. The basis and composability list exhaust the normalized relative bar. Its sole possible merger is $b[b|a] = [ba] = 0$, and there is no internal differential. Thus every listed element is a cycle and there are no boundaries. Deconcatenation at the unique internal cut gives (214.2). The endpoint terms in the full coproduct of ω are $e_3 \otimes \omega$ and $\omega \otimes e_1$. This proves the assertion as an S-coring as well as a homology calculation. □

The nonzero corners in positive homological degrees are therefore

$$\operatorname{Tor}_1^A(R_2, L_1) = k[a], \qquad \operatorname{Tor}_1^A(R_3, L_2) = k[b], \qquad \operatorname{Tor}_2^A(R_3, L_1) = k\omega.$$

The class ω records the relation and retains the two arrows in its coproduct.

For the free path algebra kQ, the augmentation ideal also contains $c = ba$. Its bar has the same two-input tensor, but now

$$b[b|a] = [c] \neq 0.$$

The pair $[b|a]$, $[c]$ is contractible. Only the two arrow classes survive in positive homological degree, both in degree one. Thus the relation changes the actual differential and creates the degree-two class.

214.3 Simple modules and the Yoneda product

Return to $A = kQ/(ba)$ and put $P_i = Ae_i$. The left projectives have bases

$$P_1 = \langle e_1, a \rangle, \qquad P_2 = \langle e_2, b \rangle, \qquad P_3 = ke_3.$$

The following augmented complexes resolve the left simples:

$$\begin{aligned} K_1 &: \quad 0 \longrightarrow P_3 \xrightarrow{u \mapsto ub} P_2 \xrightarrow{v \mapsto va} P_1 \longrightarrow L_1 \longrightarrow 0, \\ K_2 &: \quad 0 \longrightarrow P_3 \xrightarrow{u \mapsto ub} P_2 \longrightarrow L_2 \longrightarrow 0, \\ K_3 &: \quad P_3 \xrightarrow{\cong} L_3. \end{aligned}$$

In each complex, the projective immediately before the simple is in cohomological degree zero.

Right multiplication by a has image ka and kernel kb. Right multiplication by b identifies P_3 with that kernel, and the same map resolves the radical of P_2. This proves exactness of all three complexes. Assigning weight one to the degree-minus-one generators and weight two to the top generator of K_1 makes these resolutions linear, so A is Koszul.

Applying $\operatorname{Hom}_A(-, L_j)$ kills every differential, because arrows annihilate the simples. Besides the identity classes, the only Ext generators are

$$\alpha \in \operatorname{Ext}_A^1(L_1, L_2), \qquad \beta \in \operatorname{Ext}_A^1(L_2, L_3), \qquad \gamma \in \operatorname{Ext}_A^2(L_1, L_3).$$

Normalize each generator by the functional taking the corresponding projective generator to the vertex basis of its target simple.

To compute composition, lift α to a degree-one map $F_\alpha : K_1 \to K_2$. Its components are

$$F_\alpha^{-1} = \mathrm{id}_{P_2}, \qquad F_\alpha^{-2} = -\mathrm{id}_{P_3},$$

with every other component zero. Lift β to $F_\beta : K_2 \to K_3$ by $F_\beta^{-1} = \mathrm{id}_{P_3}$ and zero elsewhere.

The differential of a homogeneous Hom map is $\partial F = dF - (-1)^{|F|} F d$. For F_α, its only potentially nonzero component is

$$(dF_\alpha + F_\alpha d)(u) = -ub + ub = 0, \qquad u \in K_1^{-2} = P_3.$$

The other closure equations, including those for F_β, have zero source or zero target. Thus these are closed lifts of the normalized Ext classes. Their composite on K_1^{-2} is $-\mathrm{id}_{P_3}$, giving

$$\beta\alpha = -\gamma \neq 0. \tag{214.3}$$

This convention includes characteristic two, where the displayed minus sign equals a plus sign.

The sign also follows from the relative bar. The resolution comparison sends the arrow generators to $-[a]$ and $-[b]$, and the relation generator to $[b|a]$. Its compatibility with the top differential is the identity $d(1 \otimes [b|a]) = -b \otimes [a]$ in the left bar resolution. In signed evaluation on deconcatenation, the degree-one functional on $[a]$ crosses the degree-minus-one tensor $[b]$, contributing -1. The two minus signs from the arrow comparisons cancel, leaving (214.3).

Every other product of two positive-degree generators vanishes for lack of matching endpoints or a nonzero Ext group in its degree. For the free quiver, the degree-two group vanishes and hence $\beta\alpha = 0$. The two algebras therefore have different Yoneda multiplication, as well as different Tor groups.

214.4 Scalar boundary objects and geometric families

A graded scalar augmentation $A \to k$, with k in weight zero, sends every arrow to zero. On S, it sends orthogonal idempotents to orthogonal idempotents in a field. Their images are zero or one and sum to one, so exactly one vertex is selected. For that vertex i, the scalar derived tensor is

$$R_i \otimes^{\mathbb{L}}_A L_i \simeq e_i B_S A e_i.$$

In the relation example, each such diagonal corner is just ke_i. It omits all three positive-degree fields, whose endpoints are distinct.

No-go. *The relative bar $S \otimes^{\mathbb{L}}_A S$ retains all boundary labels. A graded scalar augmentation specifies one vertex and its diagonal derived corner. It cannot retain the off-diagonal boundary-changing fields of the full relative bar.*

To realize these algebras over X, first construct the vertex and arrow coefficient objects with their disjoint factorization maps. They must carry a compatible transverse product and augmentation to the vertex object. The relation maps must extend through collisions, and the full geometric bar must be compared with the relative coefficient bar above. The module interpretation then requires the marked block-Ran comparison for those actual coefficient objects.

A second associative chiral operation must act on composable coefficients and satisfy the mixed coherences of Chapter 205. Operations on the separate arrow objects do not by themselves supply these compositions. The relative calculation determines the vertex-labeled bar and Yoneda data that such a geometric realization must preserve.

Chapter 215

Enveloping algebras and planar-sequence obstructions

215.1 Cartan–Eilenberg comparison

Let $\mathfrak{g}$ be a dg Lie algebra object in the coefficient category and $U(\mathfrak{g})$ its augmented enveloping algebra.

Theorem 215.1 (Enveloping bar)**.** *The antisymmetrisation twisting map induces a natural quasi-isomorphism*

$$\mathrm{CE}_*(\mathfrak{g}) = \big(\mathrm{Sym}^c(s\mathfrak{g}), d_{\mathfrak{g}} + d_{[-,-]}\big) \xrightarrow{\sim} \mathrm{Bar}^{\perp}(U(\mathfrak{g})).$$

Proof. Filter both complexes by PBW degree. The associated graded of $U(\mathfrak{g})$ is $\mathrm{Sym}(\mathfrak{g})$. The comparison becomes the standard Koszul quasi-isomorphism from the exterior/symmetric coalgebra to the bar of a polynomial algebra. The complete, bounded-below filtration spectral sequence lifts the equivalence. $\square$

For a finite-dimensional semisimple Lie algebra over $\mathbb{C}$,

$$H^*(\mathfrak{g};\mathbb{C}) \cong \Lambda(\zeta_{2m_1+1}, \dots, \zeta_{2m_r+1}),$$

where m_i are the exponents. By duality the same Poincare polynomial describes homology.

Exact calculation. *Take $[h,e] = 2e$, $[h,f] = -2f$, $[e,f] = h$. In bases*

$$C_1 = (e, f, h), \qquad C_2 = (e \wedge f, e \wedge h, f \wedge h),$$

the Chevalley differential is

$$d_2 = \begin{pmatrix} 0 & -2 & 0 \\ 0 & 0 & 2 \\ 1 & 0 & 0 \end{pmatrix}, \qquad d_3 = 0.$$

Thus d_2 has rank three and

$$H_0 \cong \mathbb{C}, \quad H_1 = H_2 = 0, \quad H_3 \cong \mathbb{C}, \qquad P(t) = 1 + t^3.$$

The standard exact values are

$$\begin{aligned} U(\mathfrak{sl}_2) &: \ 1 + t^3, \\ U(\mathfrak{sl}_3) &: \ (1+t^3)(1+t^5) = 1 + t^3 + t^5 + t^8, \\ U(\mathfrak{g}_2) &: \ (1+t^3)(1+t^{11}) = 1 + t^3 + t^{11} + t^{14}. \end{aligned}$$

215.2 Three chain complexes

Three objects must be distinguished:

(i) the interval bar $\mathrm{Bar}^{\perp}(A)$, whose differential uses transverse multiplication;

(ii) chiral Chevalley chains, whose differential uses a diagonal-supported chiral Lie bracket;

(iii) a screen-enhanced bicomplex

$$B^{p,q}_{\mathrm{scr}}(A) = (s\overline{A})^{\otimes_! p} \otimes C^q_{\mathrm{scr}}$$

with total differential $b_A + d_A + d_{\mathrm{scr}} + \delta_{\mathrm{int}}$.

Only the third can contain both bar words and configuration-space cohomology, and only after the interaction differential is supplied and shown to square to zero.

Filtering by screen degree gives an E_1-page

$$E_1^{p,q} = H^q(C_{\mathrm{scr}}) \otimes B^{\perp}_p(A),$$

while filtering by bar length gives

$${}'E_1^{p,q} = \mathrm{Tor}^A_p(\mathbf{1},\mathbf{1}) \otimes C^q_{\mathrm{scr}}.$$

A product of dimensions on an E_1-page is not automatically the dimension of total homology.

215.3 The planar sequences

The Motzkin generating function is

$$M(z) = 1 + zM(z) + z^2M(z)^2 = \frac{1 - z - \sqrt{1 - 2z - 3z^2}}{2z^2}.$$

For $a_n = M_{n+1} - M_n$,

$$\sum_{n\geq 1} a_n z^n = \frac{4z}{\left(1 - z + \sqrt{1 - 2z - 3z^2}\right)^2},$$

with first values

$$1, 2, 5, 12, 30, 76, 196, 512, 1353, 3610.$$

The Riordan series is

$$R(z) = \frac{1 + z - \sqrt{1 - 2z - 3z^2}}{2z(1 + z)}.$$

Both have dominant singularity $z = 1/3$ and exponential growth rate three. These are correct combinatorial facts.

Theorem 215.2 (Non-identification of Motzkin and Riordan sequences with ordered bar homology)**.** *The displayed sequences are not the ordered bar homology of $U(\mathfrak{sl}_2)$, and a Virasoro chiral coefficient does not determine a transverse bar whose dimensions are Motzkin differences. The former is contradicted by theorem 215.1; the latter fails because the same Virasoro coefficient can be tensored with free, square-zero, enveloping, or quantum-plane transverse algebras having different bars.*

Proof. For $U(\mathfrak{sl}_2)$, the exact Chevalley calculation gives only degrees zero and three. For the Virasoro statement, choose a fixed Virasoro factorisation coefficient $F_{\mathrm{Vir},c}$ and form $B \otimes F_{\mathrm{Vir},c}$ for the four transverse algebras listed. The coefficient and central charge are unchanged while the interval bar varies with B. □

Theorem 215.2 refutes the $\mathfrak{sl}_2$ claim by computing the answer and the Virasoro claim by non-determination. The Virasoro row admits the stronger statement: in its transverse enveloping presentation the bar homology is known exactly, and it leaves no room for a Motzkin difference.

215.4 The Virasoro row computed

The transverse presentation of the Virasoro family is the enveloping algebra of the positive Witt algebra

$$L_1 = \bigoplus_{n\geq 1} \mathbb{k}\, e_n, \qquad [e_i, e_j] = (j-i)e_{i+j},$$

to which theorem 215.1 applies: L_1 is positively graded with one-dimensional weight spaces, so every weight has finitely many decompositions.

Theorem 215.3 (Pentagonal rigidity)**.** *The bar homology of $U(L_1)$ satisfies*

$$\dim H_n\big(\mathrm{Bar}^{\perp} U(L_1)\big) = 2 \qquad \text{for every } n \geq 1,$$

one dimension in each of the weights $(3n^2-n)/2$ and $(3n^2+n)/2$. Its graded Euler character is

$$\chi_q\big(\mathrm{Bar}^{\perp} U(L_1)\big) = \prod_{n\geq 1}(1-q^n), \tag{215.1}$$

and (215.1) is accounted for degree by degree with no residual freedom.

Proof. By theorem 215.1 the bar of $U(L_1)$ is quasi-isomorphic to $\mathrm{CE}_*(L_1)$, whose weight subcomplexes are finite-dimensional. Their homology dimensions agree with those of the weightwise dual cohomology, which Goncharova computes [45]: two dimensions in each degree $n \geq 1$, in the weights $(3n^2 \mp n)/2$. Identity (215.1) is theorem 228.1 applied to L_1, where every root space is one dimensional and even, so $\operatorname{sdim}(L_1)_n = 1$. Euler's pentagonal number theorem expands the product as $\sum_{m\in\mathbb{Z}}(-1)^m q^{m(3m-1)/2}$, whose support is exactly $\{(3n^2 \pm n)/2\}$ with coefficient $(-1)^n$ at each of $(3n^2-n)/2$ and $(3n^2+n)/2$. Comparing with $\sum_n (-1)^n \dim H_{n,w}$, each weight of the Euler character is saturated by a single class in the degree Goncharova predicts, so no further classes and no cancellation are available. □

Status. *Finite calculation. In homological degrees $n = 1, 2, 3, 4$, the Chevalley complex of L_1 has one-dimensional homology in each of the weight pairs $(1, 2)$, $(5, 7)$, $(12, 15)$, $(22, 26)$. Its differential squares to zero. The nonzero coefficients of $\prod_{n\geq 1}(1-q^n)$ in these weights and weight zero have signs $+, -, -, +, +, -, -, +, +$ at $0, 1, 2, 5, 7, 12, 15, 22, 26$. The all-degree assertion follows from Goncharova's theorem and Euler's identity.*

A Motzkin path is a lattice path from $(0, 0)$ to $(n, 0)$, staying at nonnegative height, with steps $(1, 1)$, $(1, 0)$, and $(1, -1)$. Let $M(n)$ count all such paths of length n, and let $R(n)$ count those with no horizontal step at height zero. Both counts include the empty path: $M(0) = R(0) = 1$, while $M(1) = 1$ and $R(1) = 0$. For $n \geq 1$ consider the two sequences

$$a_n = M(n+1) - M(n) = 1, 2, 5, 12, 30, 76, \ldots, \qquad b_n = R(n+3) = 3, 6, 15, 36, 91, \ldots.$$

Proposition 215.4 (Motzkin and Riordan identities)**.** *For every $n \geq 0$,*

$$M(n) = R(n) + R(n+1), \qquad M(n+1) - M(n) = R(n+2) - R(n).$$

The ordinary generating functions of the two path counts, and hence those of a_n and b_n, belong to $\mathbb{Q}(z, \sqrt{1-2z-3z^2})$.

Proof. Work in the formal power-series ring $\mathbb{Q}[[z]]$, and write $M(z) = \sum_{n\ge 0} M(n)z^n$ and $R(z) = \sum_{n\ge 0} R(n)z^n$. A nonempty Motzkin path either starts with a horizontal step followed by an arbitrary Motzkin path, or has a unique first-return decomposition $UPDQ$, where U and D are the up and down steps, P and Q are arbitrary Motzkin paths, and P is drawn one unit above the horizontal axis. Thus

$$M = 1 + zM + z^2M^2.$$

A nonempty path counted by R has no initial horizontal step. Its first return similarly has the form $UPDQ$, with P arbitrary and Q counted by R. Consequently

$$R = 1 + z^2MR, \qquad R = (1 - z^2M)^{-1}.$$

The inverses here exist because their denominators have constant term one. Using $z^2M^2 = M - 1 - zM$, compute

$$(1 + zM)(1 - z^2M) = 1 + zM - z^2M - z^3M^2 = 1 + z.$$

It follows that $(1 + z)R = 1 + zM$. Comparing coefficients of z^{n+1} gives $R(n + 1) + R(n) = M(n)$ for all $n \ge 0$. Subtracting consecutive identities gives the difference formula.

Solving the quadratic equation for the formal series with constant term one gives

$$M(z) = \frac{1 - z - \sqrt{1 - 2z - 3z^2}}{2z^2}, \qquad R(z) = \frac{1 + zM(z)}{1 + z},$$

where the square root has constant term one. The apparent pole in the formula for M is removable as a formal series. Finite index shifts and differences preserve the indicated quadratic function field, proving the last assertion. These identities are consequences solely of the path decompositions. □

Remark 215.5 (Low-degree homology). Theorem 215.3 gives homology dimensions 2, 2, 2 in degrees 1, 2, 3, whereas a_1, a_2, a_3 are 1, 2, 5. These triples agree only in degree two. The degree-one discrepancy already excludes equality of the full homology dimension sequences.

Theorem 215.6 (Enveloping bars and planar path counts). *Let $B_\bullet(A)$ be the ordinary normalized bar over $\mathbb{C}$, with trivial coefficients and homological degree given by bar length. With $a_n = M(n + 1) - M(n)$ and $b_n = R(n + 3)$ indexed by $n \ge 1$ as above,*

$$\dim H_1B_\bullet(U(\mathfrak{sl}_2)) = 0 \ne b_1 = 3, \qquad \dim H_1B_\bullet(U(L_1)) = 2 \ne a_1 = 1,$$

where $L_1 = \bigoplus_{m\ge 1} \mathbb{C}e_m$ and $[e_i, e_j] = (j - i)e_{i+j}$. Hence neither sequence is the bar homology dimension sequence of the corresponding enveloping algebra. Neither is its full bar chain dimension sequence: in both cases B_n is countably infinite-dimensional for every $n \ge 1$.

Proof. For any augmented algebra A with augmentation ideal I, the bar differential out of $B_1 = I$ is zero and the image from $B_2 = I \otimes I$ is I^2. Therefore

$$H_1B_\bullet(A) = I/I^2.$$

For an ordinary Lie algebra $\mathfrak{g}$, the inclusion of generators induces an isomorphism

$$\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}] \xrightarrow{\sim} I/I^2, \qquad I = \ker(U(\mathfrak{g}) \to \mathbb{C}).$$

Indeed the generators span I/I^2, and $[x, y] = xy - yx$ vanishes there. For injectivity, equip $\mathbb{C} \oplus \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$ with the square-zero product on its second summand. The quotient map from $\mathfrak{g}$ to that summand is a Lie map into the commutator Lie algebra of this associative algebra. The universal property of $U(\mathfrak{g})$ extends it to an augmented algebra map, which kills I^2 and provides an inverse on the displayed quotient.

The brackets $[h,e]=2e$, $[h,f]=-2f$, and $[e,f]=h$ show that $\mathfrak{sl}_2$ equals its commutator algebra. For L_1, every nonzero bracket has weight at least three. Conversely, for $m\geq 3$,

$$e_m=\frac{1}{m-2}[e_1,e_{m-1}].$$

Thus $[L_1,L_1]=\bigoplus_{m\geq 3}\mathbb{C}e_m$, and its abelianization has basis the classes of e_1, e_2, in weights one and two. This proves both degree-one assertions without using any higher-degree calculation.

PBW gives a countably infinite basis of the augmentation ideal of either enveloping algebra. In the L_1 case the basis consists of nonempty ordered monomials in the countable set $e_1, e_2, \ldots$, each monomial having finite length. Finite tensor powers of either augmentation ideal are therefore countably infinite-dimensional. The path counts are finite in every degree, so they cannot count these full chain spaces. □

In particular, a complex whose homology dimensions are b_n cannot be quasi-isomorphic to $B_\bullet(U(\mathfrak{sl}_2))$, and one whose homology dimensions are a_n cannot be quasi-isomorphic to $B_\bullet(U(L_1))$, with the stated degree convention. A discrepancy of chain dimensions alone would not imply this conclusion; the degree-one homology discrepancy does.

The weight grading gives a different, finite comparison. An ordered PBW monomial of weight w specifies a partition of w, so for $n\geq 1$,

$$\sum_{w\geq 0}\dim B_n(U(L_1))_w q^w=\left(\prod_{m\geq 1}(1-q^m)^{-1}-1\right)^n.$$

Each coefficient is finite. Thus a weightwise dimension, a dimension after a specified weight cutoff, and an ungraded dimension are distinct invariants. In a weight completion, dimensions must likewise be interpreted weightwise; the countable algebraic dimensions above refer to the ordinary direct-sum enveloping algebras.

No-go. *A sequence of integers becomes bar homology only after a chain complex, differential, grading convention, and quasi-isomorphism or rank computation are specified. Matching initial values, discriminants, or growth rates does not construct such a map.*

Chapter 216

Finite-label exchange and its bar coalgebra

216.1 The exchange algebra and its ordered basis

Fix finitely many spectral labels $t_1, \dots, t_m$. The notation $t_1 < \cdots < t_m$ denotes a chosen order of colors, independent of any ordering of the coefficient field. For copies V_a of a finite-dimensional vector space, put $c_{ab} = PR(t_a - t_b) : V_a \otimes V_b \to V_b \otimes V_a$ for $a > b$, where P interchanges the tensor factors. Work over a coefficient ring in which these matrices are defined. Impose only the descending exchange relations of (26.1), and impose no same-color relation.

Theorem 216.1 (Confluence is Yang–Baxter)**.** *The ordered-word map* (26.2) *is an isomorphism precisely when* (26.5) *holds on every three distinct chosen labels. This gives a basis for all words, including repeated labels. Reverse exchanges preserve this basis precisely when their maps are the inverses of the descending exchanges.*

Proof. A descending exchange lowers the color-inversion number. The only intersecting pair of possible exchanges has three strictly descending colors. Its two complete reductions are (26.3). Removing their common permutation gives Yang–Baxter. The induction and the inverse normal-form map of Theorem 26.1 apply to every word.

Reversing an ordered pair and sorting it gives $c_{ab}c_{ba}$, which must be the identity on the independent ordered quadratic tensors. These are square matrices over a commutative ring, so the prescribed maps are inverse in both directions. □

For $R(u) = 1 - \hbar P/u$, Proposition 26.2 proves Yang–Baxter and computes the reverse product $1 - \hbar^2/u^2$. Thus the raw matrix gives the oriented algebra. The matrix $\widehat{R}(u) = (u - \hbar P)/(u - \hbar)$ gives reversible exchanges over the specified localization. Their oriented algebras are related by the explicit color-count twist (26.9). These are statements about ordinary algebras with spectral coefficients. Collision extensions and state-field maps require further constructions.

216.2 The scalar algebra and its linear resolution

Let K be a field and fix units $q_{ab} \in K^\times$ for $1 \le a < b \le m$. The scalar exchange algebra is the augmented graded algebra

$$A = K\langle z_1, \dots, z_m\rangle/(z_b z_a - q_{ab} z_a z_b : a < b), \qquad \epsilon(z_a) = 0, \quad \deg z_a = 1. \tag{216.1}$$

Scalar exchanges satisfy the three-color equation because their coefficients commute. Hence the monomials $z^\alpha = z_1^{\alpha_1} \cdots z_m^{\alpha_m}$ form a K-basis, and

$$z^\alpha z^\beta = Q(\alpha, \beta) z^{\alpha+\beta}, \qquad Q(\alpha, \beta) = \prod_{a<b} q_{ab}^{\alpha_b \beta_a}. \tag{216.2}$$

Each factor counts crossings of a color b from the first monomial with a color a from the second.

For $I = \{i_1 < \cdots < i_n\} \subseteq \{1, \dots, m\}$, introduce a symbol e_I of homological degree and internal degree n. Set $e_\varnothing = 1$ and define free left A-modules and an A-linear map

$$\begin{aligned} L_n &= \bigoplus_{|I|=n} A e_I, \\ d(e_I) &= \sum_{r=1}^{n} (-1)^{r-1} \left(\prod_{s<r} q_{i_s i_r}^{-1} \right) z_{i_r} e_{I \setminus \{i_r\}}. \end{aligned} \tag{216.3}$$

For two labels, this reads $d(e_{\{a,b\}}) = z_a e_b - q_{ab}^{-1} z_b e_a$. Its second differential is zero by the defining relation.

Proposition 216.2. *The complex $L_\bullet \xrightarrow{\epsilon} K$ is a minimal graded free resolution of the trivial left A-module. In particular, A is Koszul: its trivial module has a resolution whose degree-n generators all have internal degree n.*

Proof. For any two removed indices, the two terms of d^2 have opposite signs. After removing the common unit factor, their coefficient is $z_a z_b - q_{ab}^{-1} z_b z_a$ for $a < b$. Thus $d^2 = 0$.

Give e_I the multidegree $\mathbf{1}_I$, the indicator vector of I. For a fixed multidegree $\nu \in \mathbb{N}^m$, the complex has basis

$$v_I = z^{\nu - \mathbf{1}_I} e_I, \qquad I \subseteq \operatorname{supp}(\nu).$$

Every coefficient $\lambda_{I,i}$ of $v_{I\setminus\{i\}}$ in $d(v_I)$ is a unit: explicitly, for $i = i_r \in I$ it is

$$\lambda_{I,i} = (-1)^{r-1} \prod_{s<r} q_{i_s i}^{-1} \prod_{b>i} q_{ib}^{\nu_b - \mathbf{1}_I(b)}.$$

If $\nu \neq 0$, fix $j \in \operatorname{supp}(\nu)$ and put

$$H(v_I) = \begin{cases} \lambda_{I\cup\{j\},j}^{-1} v_{I\cup\{j\}}, & j \notin I, \\ 0, & j \in I. \end{cases}$$

In $dH + Hd$, removing the added index j contributes v_I. The terms involving any other index cancel by the corresponding two-face identity in $d^2 = 0$. If $j \in I$, only the term removing and then reintroducing j survives. Thus $dH + Hd = 1$ in every positive multidegree.

Multidegree zero consists of $K e_\varnothing$, on which the augmentation is an isomorphism. This proves exactness. Finally, every differential coefficient lies in the augmentation ideal, so the resolution is minimal. □

216.3 The actual bar coalgebra

Use homological bar degree: $[a_1|\cdots|a_n] \in B_n A$ has degree n, where $B_n A = (\ker \epsilon)^{\otimes_K n}$ and $B_0 A = K$. The normalized bar differential and coproduct are

$$\begin{aligned} b[a_1|\cdots|a_n] &= \sum_{r=1}^{n-1} (-1)^{r-1} [a_1|\cdots|a_r a_{r+1}|\cdots|a_n], \\ \Delta[a_1|\cdots|a_n] &= \sum_{r=0}^{n} [a_1|\cdots|a_r] \otimes [a_{r+1}|\cdots|a_n]. \end{aligned}$$

An empty bar is the coaugmentation 1. For an ordered subset $I = \{i_1 < \cdots < i_n\}$ define

$$\omega_I = \sum_{\pi \in S_n} (-1)^{\ell(\pi)} \left(\prod_{r<s,\ \pi(r)>\pi(s)} q_{i_{\pi(s)} i_{\pi(r)}}^{-1} \right) [z_{i_{\pi(1)}}|\cdots|z_{i_{\pi(n)}}], \qquad \omega_\varnothing = 1, \tag{216.4}$$

where $\ell(\pi)$ is the number of inversions of the permutation π. For example, $\omega_{\{a,b\}} = [z_a|z_b] - q_{ab}^{-1}[z_b|z_a]$. Let C_n be the span of the ω_I with $|I| = n$.

Theorem 216.3. *The spaces C_n form a differential graded subcoalgebra $C \subseteq BA$ with zero differential. The inclusion is a quasi-isomorphism. Its coproduct is*

$$\Delta\omega_I = \sum_{J\sqcup H=I} \left(\prod_{j\in J,\ h\in H,\ j>h} (-q_{hj}^{-1}) \right) \omega_J \otimes \omega_H. \tag{216.5}$$

Consequently $\operatorname{Tor}_n^A(K,K)$ *has one basis element for each n-element subset of labels, and dimension* $\binom{m}{n}$.

Proof. Pair the terms of $b\omega_I$ obtained from permutations differing by one adjacent transposition. Their weights differ by $-q_{ab}^{-1}$, while their adjacent products differ by q_{ab}, so they cancel. For a fixed cut of a bar word, its inversions split into those internal to each side and those crossing the cut. The crossing inversions depend only on the two subsets of labels. Factoring their weights gives (216.5). The coefficient of the increasing bar word in ω_I is 1, so these elements are linearly independent.

For the quasi-isomorphism, use the normalized free left bar resolution $P_n = A \otimes_K B_n A$ with differential

$$d_P(a_0 \otimes [a_1|\cdots|a_n]) = -a_0a_1 \otimes [a_2|\cdots|a_n] + a_0 \otimes b[a_1|\cdots|a_n].$$

It is exact: writing $\bar{a}_0 = a_0 - \epsilon(a_0)$, the map $-1 \otimes [\bar{a}_0|a_1|\cdots|a_n]$, together with the unit in augmentation degree, is a contracting homotopy over K.

Grouping (216.4) by its first entry shows that

$$L_n \longrightarrow P_n, \qquad e_I \longmapsto (-1)^n(1 \otimes \omega_I)$$

is a chain map lifting the identity of K. The sign compensates for the negative first face of d_P.

Both complexes are free resolutions. A reverse comparison map and homotopies are constructed degree by degree, lifting cycles through the surjections onto boundaries. Hence this map is a chain homotopy equivalence. Tensoring with the right augmentation module K gives a quasi-isomorphism with image C in BA. The factors $(-1)^n$ are invertible in each degree, so the inclusion $C \hookrightarrow BA$ itself is a quasi-isomorphism. □

216.4 The Yoneda product and its variance

Applying $\operatorname{Hom}_A(-,K)$ to L kills the differential. Let $\xi_i \in \operatorname{Ext}_A^1(K,K)$ be dual to $e_{\{i\}}$. Products in Ext require lifts on the resolution. Define the A-linear map $F_i : L_n \to L_{n-1}$ by zero when $i \notin I$, and by

$$F_i(e_I) = (-1)^{r-1} \left(\prod_{j\in I,\ j>i} q_{ij}^{-1} \right) e_{I\setminus\{i\}}, \qquad i = i_r \in I.$$

For a removal of any $j \neq i$, the two orders of removal give opposite coefficients. Thus $dF_i + F_i d = 0$. Composing $F_i : L_1 \to L_0$ with the augmentation gives the cochain ξ_i.

With Yoneda multiplication represented by composition, the same coefficients give

$$F_i^2 = 0, \qquad F_bF_a = -q_{ab}^{-1}F_aF_b \quad (a < b).$$

For each subset I, the corresponding ordered product has a nonzero coefficient on e_I. It vanishes on all other subsets of the same size. These products therefore give all the dual basis classes. We obtain the algebra presentation

$$\operatorname{Ext}_A^*(K,K) = K\langle \xi_1, \dots, \xi_m \rangle / (\xi_i^2,\ \xi_b\xi_a + q_{ab}^{-1}\xi_a\xi_b : a < b), \qquad |\xi_i| = 1. \tag{216.6}$$

The bar model C is a coalgebra, whereas (216.6) is an algebra. Their comparison uses graded duality: bar homological degree n is cohomological degree $-n$, and its dual has degree n. Convolution uses

the Koszul tensor sign. For degree-one functionals f, g, $(f \otimes g)([z_a] \otimes [z_b]) = -f([z_a])g([z_b])$. Formula (216.5) then gives exactly the relations in (216.6). Under the displayed resolution comparison, ξ_i corresponds to minus the functional dual to $\omega_{\{i\}}$.

Thus the subset basis, its coproduct, and the dual Yoneda product are all specified. This computes ordinary associative Koszul duality for the scalar exchange algebra. Extending the calculation to associative chiral Koszul duality still requires geometric coefficient objects and maps preserving the collision and substitution operations.

Chapter 217

RTT Yangians and the bar filtration

The two Yangian filtrations answer different questions. Positive weight gives a polynomial associated graded and proves flatness. Loop degree gives a current enveloping algebra and controls the bar spectral sequence. Over the polynomial deformation ring, the current bracket retains a factor of $\hbar$. Its homology detects that factor before any higher differential is considered.

217.1 The augmented coefficient algebra

Fix $B = \mathbb{C}[\hbar]$ and $n \geq 1$. Let $A = Y_\hbar(\mathfrak{gl}_n)$ be the coefficient algebra of (27.1), with unscaled generators $t_{ij}^{(r)}$ and augmentation $\varepsilon(t_{ij}^{(r)}) = 0$ for $r \geq 1$. Theorem 27.2 gives its coproduct in the ordinary tensor product $A \otimes_B A$ and its polynomial antipode coefficients. Its ordered PBW monomials form a B-basis.

For $n = 2$ the central group-like series is

$$D(u) = t_{11}(u)t_{22}(u-\hbar) - t_{21}(u)t_{12}(u-\hbar).$$

The determinant-one quotient and its central tensor factor are those of Proposition 27.4. They require this column order.

Put $L = \mathfrak{gl}_n(\mathbb{C})[z]$ and $L_B = L \otimes_{\mathbb{C}} B$. Write $L_\hbar$ for this B-module with bracket $[x, y]_\hbar = \hbar[x, y]$. Give $t_{ij}^{(r)}$ loop degree $r-1$ and give B degree zero. The resulting increasing filtration F has

$$\operatorname{gr}_F A \cong U_B(L_\hbar), \qquad \overline{t}_{ij}^{(r)} \longleftrightarrow e_{ij} z^{r-1}, \tag{217.1}$$

by Theorem 27.3. The scalar counit on both sides makes this an isomorphism of augmented algebras.

217.2 The current bar differential

Let $\overline{A} = \ker \varepsilon$. Use the normalized relative bar complex over B, in homological degrees:

$$C_q = \mathrm{B}_B(A)_q = \overline{A}^{\otimes_B q}, \qquad C_0 = B,$$

$$b[a_1|\cdots|a_q] = \sum_{i=1}^{q-1} (-1)^{i-1} [a_1|\cdots|a_i a_{i+1}|\cdots|a_q]. \tag{217.2}$$

The empty word has degree zero. The internal algebra differential is zero. Since A and $\overline{A}$ are B-free, the normalized left bar resolution is a free A-resolution of the augmentation module B. Consequently $H_q(C) = \operatorname{Tor}_q^A(B, B)$.

The current analogue has a smaller complex. On $\Lambda_B^q L_B$ define the Chevalley differential by

$$\partial(x_1 \wedge \cdots \wedge x_q) = \sum_{i<j} (-1)^{i+j+1} [x_i, x_j] \wedge x_1 \wedge \cdots \widehat{x_i} \cdots \widehat{x_j} \cdots \wedge x_q. \tag{217.3}$$

The entries remaining after the bracket keep their original order. In particular, $\partial(x \wedge y) = [x, y]$. Jacobi gives $\partial^2 = 0$. The bracket of $L_\hbar$ therefore gives the complex $K_\hbar = (\Lambda_B^\bullet L_B, \hbar\partial)$.

Proposition 217.1 (Current bar comparison)**.** *The antisymmetrizer*

$$\mathrm{Alt}(x_1 \wedge \cdots \wedge x_q) = \sum_{\sigma \in S_q} \mathrm{sgn}(\sigma)[x_{\sigma(1)}|\cdots|x_{\sigma(q)}]$$

is a quasi-isomorphism from $K_\hbar$ *to* $\mathrm{B}_B U_B(L_\hbar)$.

Proof. Pair the permutations in the bar differential that differ by interchanging the two adjacent entries being multiplied. Their products differ by $xy - yx = \hbar[x, y]$. Moving this bracket to the first wedge position gives the sign in (217.3). Hence $b\,\mathrm{Alt} = \mathrm{Alt}\;\hbar\partial$.

Filter the enveloping algebra by PBW length and its bar by total PBW length. Filter $\Lambda_B^q L_B$ by assigning weight q to its elements. Both differentials preserve the increasing nonnegative filtrations, and $\hbar\partial$ lowers weight by one. On associated gradeds the map is the antisymmetrizer from the exterior module, with zero differential, to the bar of $\mathrm{Sym}_B(L_B)$.

For finitely many basis vectors this is the Koszul bar comparison: the polynomial Koszul resolution is exact by successive contraction in each variable. For the left bar resolution with negative first face, the comparison in degree q is $(-1)^q(1 \otimes \mathrm{Alt})$. After tensoring with B, the exterior differential is zero, so multiplication by this degree sign does not affect the quasi-isomorphism assertion for Alt. Every polynomial and bar word involves finitely many basis vectors. Taking their filtered union proves the same statement for L_B.

The mapping cone of Alt thus has acyclic associated graded. A cone cycle has a finite largest filtration degree. Exactness in that degree subtracts a boundary and lowers the degree of the cycle. Repeating finitely many times makes it zero. The cone is acyclic, which proves the proposition. □

Already in degree one,

$$H_1(K_\hbar) = L_B/\hbar[L_B, L_B]. \tag{217.4}$$

For $n = 2$, the decomposition into scalar and traceless matrices gives

$$H_1(K_\hbar) \cong (\mathbb{C}I[z] \otimes_{\mathbb{C}} B) \oplus (\mathfrak{sl}_2[z] \otimes_{\mathbb{C}} B/(\hbar)).$$

Indeed $[L, L] = \mathfrak{sl}_2[z]$: traces of commutators vanish, and brackets with constant matrix units generate each traceless current. Thus the traceless summand is genuine $\hbar$-torsion. For example,

$$b\big([e_{11}|e_{12}] - [e_{12}|e_{11}]\big) = \hbar[e_{12}].$$

Replacing $\hbar\partial$ by ∂ in the same comparison would leave the nonzero residual $(\hbar - 1)[e_{12}]$.

For $n = 2$, this torsion also occurs in the Yangian's actual bar homology. The first-mode inclusion $U_\hbar(\mathfrak{gl}_2) \longrightarrow A$ and ev_0 are augmentation-preserving maps whose composite is the identity. Applying the relative bar and then homology gives a split injection on Tor. The current-bar proof also applies to the constant Lie algebra. Its degree-one class of e_{12} is nonzero and killed by $\hbar$. Its image in $\mathrm{Tor}_1^A(B, B)$ has the same property.

217.3 The algebraic spectral sequence

Filter C_q by the sum of the loop degrees of its PBW entries. The differential preserves this filtration, $F_{-1}C = 0$, and every chain lies in some F_pC. The PBW basis also identifies the associated graded tensors without a flatness ambiguity.

Theorem 217.2 (Yangian bar spectral sequence)**.** *There is a spectral sequence of B-modules. Its first two pages are*

$$E^0_{p,q} = (\mathrm{B}_B U_B(L_\hbar))_{p+q,p}, \qquad E^1_{p,q} = H_{p+q}(K_\hbar)_p.$$

The last subscript on each right-hand side denotes total current degree. The differential has bidegree $(-r, r-1)$.

Its limiting terms satisfy

$$E^{\infty}_{p,q} \cong \operatorname{gr}^F_p \operatorname{Tor}^{A}_{p+q}(B, B)$$

for the exhaustive filtration induced by F *on homology. Here the limit uses eventual cycles modulo the union of page boundaries. No finite-page stabilization is asserted.*

Proof. Equation (217.1) identifies the associated graded bar with the current bar. Proposition 217.1, which preserves total current degree, identifies its homology and proves the first-page formula. The usual filtered-complex differential sends a representative in F_pC_{p+q} to its leading boundary in $F_{p-r}C_{p+q-1}$, giving the stated bidegree.

The limiting assertion can be checked directly. Inside $\operatorname{gr}^F_p C_n$, put

$$Z^p_r = \frac{(F_pC_n \cap b^{-1}F_{p-r}C_{n-1}) + F_{p-1}C_n}{F_{p-1}C_n}, \qquad B^p_r = \frac{(F_pC_n \cap b(F_{p+r-1}C_{n+1})) + F_{p-1}C_n}{F_{p-1}C_n}.$$

For $r \geq 1$ the page is Z^p_r / B^p_r.

When $r > p$, $F_{p-r} = 0$, so Z^p_r consists of the leading terms of actual cycles. A boundary in F_pC_n has a primitive in some finite F_mC_{n+1}, and hence lies in B^p_r for all sufficiently large r. Conversely every B^p_r is represented by such a boundary. The quotient by their union is exactly the associated graded of homology. Its filtration is exhaustive because every cycle is a finite chain. □

At $\hbar = 1$, the same construction has the ordinary current Chevalley homology as its first page. Over B, the first page is $H(K_\hbar)$, including the torsion in (217.4). These are separate coefficient statements. Specialization of homology need not commute with base change.

The displayed spectral sequence is one of modules. For example, the signed shuffle of the degree-one cycles $[e_{11}]$ and $[e_{12}]$ has the nonzero boundary computed above. That shuffle therefore does not define a product on this noncommutative bar complex.

Finite calculations can instead use the positive weight $\deg t^{(r)}_{ij} = r$. The bar subcomplex of total weight at most D is finite free over B: only finitely many PBW monomials occur. Bar length is at most D, and loop degree in length q is at most $D - q$. Its loop spectral sequence has a finite filtration. A bound on current indices alone leaves arbitrarily long products and bar words, and supplies no such finite filtration.

217.4 Completion and the inverse limit

Let $B_N = B/(\hbar^N)$, $A_N = A/(\hbar^N)$, and $C^{(N)} = \mathrm{B}_{B_N}(A_N)$. The continuous relative bar complex of $\widehat{A} = \varprojlim_N A_N$ is, by definition,

$$\widehat{C} = \varprojlim_N C^{(N)}.$$

Thus its tensors are completed by this specific inverse limit. This construction distinguishes continuous bar homology from the ordinary algebraic Tor of $\widehat{A}$. PBW makes the transition maps on every chain group surjective.

For the induced transition $f_N : C^{(N+1)} \to C^{(N)}$, define $\sigma((x_N)_N) = (f_N(x_{N+1}))_N$ on the product complex. There is a short exact sequence of complexes

$$0 \longrightarrow \widehat{C} \longrightarrow \prod_N C^{(N)} \xrightarrow{1-\sigma} \prod_N C^{(N)} \longrightarrow 0.$$

Surjectivity of $1-\sigma$ follows by solving its equations successively using the surjective chain transitions. Products of modules are exact, so product homology is the product of homologies. For an inverse system of homology modules, let $\varprojlim^1$ denote the cokernel of $1-\sigma$ on their product. The long exact sequence consequently gives

$$0 \longrightarrow \varprojlim{}^1_N H_{q+1}(C^{(N)}) \longrightarrow H_q(\widehat{C}) \longrightarrow \varprojlim_N H_q(C^{(N)}) \longrightarrow 0. \tag{217.5}$$

The sequence states the exact inverse-limit obstruction. It does not assume that this cokernel vanishes.

217.5 The abelian calculation and its geometric scope

For $n=1$, induction in (27.3) makes every commutator zero: its right side consists of commutators of smaller positive weight. PBW then gives

$$Y_\hbar(\mathfrak{gl}_1) = B[t^{(1)}, t^{(2)}, \dots], \qquad \operatorname{Tor}_*^{Y_\hbar(\mathfrak{gl}_1)}(B,B) = \Lambda_B(st^{(1)}, st^{(2)}, \dots).$$

The symbol $st^{(r)}$ has homological degree one. For the first M variables the polynomial Koszul resolution gives the exterior algebra on their M degree-one classes, represented in the bar by alternating tensors. Taking the filtered union over M proves the formula for all coefficients. If t records homological degree, the Poincaré polynomial for the first M variables is $(1+t)^M$.

The same Koszul argument works over every B_N. Its exterior homology modules have surjective transitions, so the $\varprojlim^1$ term in (217.5) vanishes. The continuous bar homology is the $\hbar$-adic completion of this exterior module: at each finite precision an element has finite exterior-monomial support.

These computations concern the augmented associative coefficient algebra. The rational intertwiner (27.6) also gives an actual exchange on tensor products of evaluation modules. An associative chiral realization requires geometric coefficient objects with collision and substitution maps. Its bar must then be compared with the coefficient bar by a map preserving these operations. Formal matrix series and a module intertwiner alone do not provide that reconstruction.

Chapter 218

The A_2 Hall algebra in complete detail

The order of the factors in a Hall product records the order of quotients in a flag. Reversing that order changes the structure constants. We make both length-three flag counts explicit, then describe the correspondence that gives associativity and the additional data required over a space of supports.

218.1 Untwisted products and the two flag tables

Let $k = \mathbb{F}_q$ and $Q : 1 \to 2$. Use the representations $S_1 = (k \to 0)$, $S_2 = (0 \to k)$ and $M = (k \xrightarrow{1} k)$ of Chapter 28. For this section, juxtaposition means the untwisted Hall product $*$: $[A] * [B]$ counts subobjects isomorphic to B with quotient A. Put $e_i = [S_i]$. Directly from the possible ranks,

$$e_1e_2 = [S_1 \oplus S_2] + [M], \qquad e_2e_1 = [S_1 \oplus S_2], \qquad e_i^2 = (q+1)[S_i^2].$$

The last coefficient counts lines in k^2.

Set $C_0 = S_1^2 \oplus S_2$ and $C_1 = S_1 \oplus M$. These are precisely the rank-zero and rank-one classes of dimension $(2,1)$. Reading successive quotients from the top of a flag gives

	112	121	211
C_0	$q+1$	$q+1$	$q+1$
C_1	$q+1$	1	0.

(218.1)

For the rank-one arrow, a flag of type 121 singles out its kernel line. A flag of type 211 would require a subrepresentation $(k^2 \to 0)$, which cannot occur. In type 112, every line in the source is allowed.

Thus

$$\begin{aligned} e_1^2e_2 &= (q+1)([C_0] + [C_1]), \\ e_1e_2e_1 &= (q+1)[C_0] + [C_1], \\ e_2e_1^2 &= (q+1)[C_0]. \end{aligned}$$

For dimension $(1,2)$ put $D_0 = S_1 \oplus S_2^2$ and $D_1 = M \oplus S_2$. The companion table is

	221	212	122
D_0	$q+1$	$q+1$	$q+1$
D_1	0	1	$q+1$.

(218.2)

For the rank-one arrow, type 221 would end in a subobject S_1 and would force the arrow to vanish. Type 212 forces the image line in the target. Type 122 allows every line in that target. The zero arrow imposes no condition in either table.

Theorem 218.1 (Untwisted Hall–Serre relations)**.** *The untwisted products satisfy*

$$e_1^2 e_2 - (q+1) e_1 e_2 e_1 + q e_2 e_1^2 = 0,$$
$$q e_2^2 e_1 - (q+1) e_2 e_1 e_2 + e_1 e_2^2 = 0.$$

Proof. On C_1 the first coefficient is $(q+1) - (q+1) = 0$. On C_0 it is $(q+1) - (q+1)^2 + q(q+1) = 0$.

On D_1 the second coefficient is $-(q+1) + (q+1) = 0$. On D_0 it is the same scalar cancellation as before. The rank classification exhausts both dimension vectors. □

For example, at $q = 2$ the three products in dimension $(2,1)$ have coordinate vectors $(3,3)$, $(3,1)$ and $(3,0)$ in the ordered basis $([C_0], [C_1])$. Their Serre combination is $(3,3) - 3(3,1) + 2(3,0) = (0,0)$.

218.2 The twist and divided powers

Put $v = \sqrt{q} > 0$ and work over $K_q = \mathbb{Q}(v)$. With the Euler form $\langle (a,b), (c,d) \rangle = ac + bd - ad$, use the twisted product

$$[A] \diamond [B] = v^{\langle \dim A, \dim B \rangle} [A] * [B].$$

The total Euler exponents for 112, 121, 211 are $-1, 0, 1$, while those for 221, 212, 122 are $1, 0, -1$. Multiplying the two untwisted relations by v^{-1} converts them into the two symmetric quantum Serre relations of Theorem 28.1. The middle coefficient changes because $v^{-1}(q+1) = v + v^{-1}$.

Now all powers use $\diamond$. Set $e_{12} = e_1 \diamond e_2 - v^{-1} e_2 \diamond e_1 = v^{-1}[M]$. For an element u define $u^{(m)} = u^m / [m]_v!$, with the nonzero symmetric factorial defined in Chapter 28. The full normalization is

$$e_2^{(a)} e_{12}^{(b)} e_1^{(c)} = v^{a(a-1)+b(b-1)+c(c-1)-b+ab+bc} [S_2^a \oplus M^b \oplus S_1^c]. \tag{218.3}$$

This follows from (28.5).

Each ordered monomial gives a single class. Together with the normal-form proof of Proposition 28.2, it identifies the whole twisted Hall algebra with the specialization of the positive quantum group.

218.3 Flags as correspondences

Let $\mathcal{R}$ be the groupoid of finite Q-representations and isomorphisms. Let $\mathcal{E}$ be the groupoid of inclusions $W \subset C$, with isomorphisms preserving the inclusion. There is a span

$$\mathcal{R} \times \mathcal{R} \xleftarrow{\pi} \mathcal{E} \xrightarrow{\rho} \mathcal{R}, \qquad \pi(W \subset C) = (C/W, W), \quad \rho(W \subset C) = C.$$

On finitely supported functions on isomorphism classes, its counting convolution is the actual formula

$$(f * g)(C) = \sum_{W \subset C} f(C/W) g(W). \tag{218.4}$$

The sum is over subobjects of the fixed representation C, each with weight one. This convention reproduces $g^C_{A,B}$ when f, g are the indicator functions of $[A]$, $[B]$.

Composing the span in either order gives the groupoid of flags $0 \subset W \subset U \subset C$. On subobjects, the comparison sends a subobject of C/W to its inverse image in C. Its inverse takes the quotient by W. The resulting bijection identifies the two expansions of (218.4).

For longer products, every parenthesization is a way of successively forgetting stages of one flag. The comparison maps agree under further forgetting. In particular, the five bracketings of four factors give the same map around the associativity pentagon. The zero representation supplies the unit. The Euler twist is compatible with every such comparison because its total exponent is $\sum_{i<j} \langle \alpha_i, \alpha_j \rangle$ for the ordered quotient dimensions $\alpha_1, \dots, \alpha_n$.

218.4 Coefficients over a space of supports

The flag construction suggests a geometric Hall algebra when representations carry support on a curve X. Such a construction starts with a coefficient category, a tensor operation, and objects on the spaces parametrizing finite collections of supports. Disjoint unions of supports must give specified factorization isomorphisms. An extension correspondence must then define a pull–push map in that category, with its shifts and orientation data fixed. Multiplication requires base-change identifications with the correspondences of longer flags. These must be coherent under every refinement and compatible with the zero-object unit.

If the coefficient category has an ordered chiral tensor, these identifications must also commute with its collision and substitution maps. Under these hypotheses, the flag comparisons give associativity and its higher coherences, while disjoint-support restriction gives the prescribed factorization. The resulting multiplication is an algebra object for the supplied ordered chiral tensor. Suppose the coefficient objects already carry a second associative product. The extension maps must preserve it and its interchange with chiral multiplication to produce a doubly associative chiral object.

Constructing these coefficient objects and compatible maps is the remaining geometric problem. Recovering the finite-field constants requires a trace or comparison map that carries the geometric pull–push operations to (218.4), with the Euler normalization (28.2). The flag counts determine the target of such a map. Its construction and compatibility with collisions require the geometric data.

Chapter 219

Necklace Hamiltonians and the mixed trace algebra

Matrix traces give Hamiltonian functions on matrix spaces and their commuting quotients. Their vector fields also act on the complex imposing the commuting equation. We retain the rank constant and distinguish algebraic conjugation invariants from an infinitesimal BRST complex.

219.1 The derived commuting equation

Fix $N \geq 1$ and use the polynomial coordinate ring $O_N = \mathbb{C}[X_{ij}, Y_{ij}]$ of Chapter 29. Put $\mu = [X, Y]$ and let $I_N \subset O_N$ be the ideal generated by its entries. The Koszul differential graded algebra of this equation is

$$K_N = O_N \otimes \Lambda(\psi_{ij}), \qquad |\psi_{ij}| = -1, \qquad dX = dY = 0, \qquad d\psi = \mu. \tag{219.1}$$

We use cohomological degrees, so d has degree one. All matrix entries supercommute with the signs dictated by their degrees. The equality $d^2 = 0$ follows from $d\mu = 0$.

This algebra is the derived zero fiber of the map $(X, Y) \mapsto [X, Y]$. It retains relations among the equations, without a regular-sequence assumption. For example, at $N = 1$ the equation vanishes identically and $K_1 = \mathbb{C}[X, Y] \otimes \Lambda(\psi)$ has zero differential.

The group $G = \mathrm{GL}_N$ acts on all three matrices by conjugation, and d is equivariant. The algebraic quotient has the complex K_N^G of equivariant global functions. Over $\mathbb{C}$, invariants under this reductive group are exact, so

$$H^0(K_N^G) = (O_N / I_N)^G. \tag{219.2}$$

Here the quotient means the algebraic group quotient of the derived zero fiber.

219.2 An infinitesimal BRST complex

Introduce a second odd matrix c of degree one and define $B_N = K_N \otimes \Lambda(c_{ij})$. Its degree-one derivation Q is specified by

$$QX = [c, X], \qquad QY = [c, Y], \qquad Q\psi = \mu + c\psi + \psi c, \qquad Qc = c^2. \tag{219.3}$$

The sum $c\psi + \psi c$ is the graded commutator of two odd matrices.

Proposition 219.1. *The derivation Q squares to zero. The inclusion $K_N^G \to B_N$ is a chain map, but need not be a quasi-isomorphism.*

Proof. Leibniz gives $Q\mu = [c, \mu]$. For any homogeneous matrix Z, the graded Jacobi identity gives $[c, [c, Z]] = \frac{1}{2}[[c, c], Z] = [c^2, Z]$. Consequently

$$Q^2 X = [Qc - c^2, X] = 0, \qquad Q^2 Y = [Qc - c^2, Y] = 0, \qquad Q^2 c = c^2 c - c c^2 = 0.$$

On ψ, the remaining terms are

$$Q^2\psi = [c,\mu] + [Qc,\psi] - [c,\mu] - [c,[c,\psi]] = 0.$$

The conjugation part of Q annihilates G-invariant elements of K_N, so its restriction there is d.

At $N = 1$, conjugation is trivial and both differentials vanish. Thus $K_1^G = \mathbb{C}[X,Y] \otimes \Lambda(\psi)$, whereas $B_1 = K_1^G \otimes \Lambda(c)$. The latter has a nonzero degree-one class c, and the former has no degree-one cohomology. The inclusion is therefore not a quasi-isomorphism. □

The complex B_N describes the infinitesimal conjugation action in these BRST conventions. Equation (219.2) refers to algebraic G-invariants. The rank-one example fixes the distinction between the two constructions.

219.3 The necklace action on the complexes

Use F_{cyc} and the cyclic derivatives from Chapter 29. Extend the derivation associated to $f \in F_{\mathrm{cyc}}$ by

$$\delta_f X = -D_y f(X,Y), \qquad \delta_f Y = D_x f(X,Y), \qquad \delta_f \psi = \delta_f c = 0.$$

Theorem 219.2 (Necklace Lie algebra and trace action)**.** *The necklace Lie algebra $F_{\mathrm{cyc}}/\mathbb{C}[1]$ acts by degree-zero differential graded derivations on K_N, K_N^G and B_N. Its action obeys*

$$[\delta_f, \delta_g] = \delta_{\{f,g\}_{\mathrm{neck}}}, \qquad \delta_f J_N(g) = J_N(\{f,g\}_{\mathrm{neck}}).$$

Every $J_N(g)$ is a degree-zero cycle. The trace map itself is defined on F_{cyc}, with $J_N([1]) = N$.

Proof. The Lie identity is (29.4). The cyclic Euler identity (29.5) gives $\delta_f \mu = 0$, so $[d, \delta_f] = 0$ on K_N. Each polynomial matrix $D_x f(X,Y)$ or $D_y f(X,Y)$ is conjugation equivariant. Thus the action preserves K_N^G and commutes with the conjugation part of Q, including its action on ψ and c.

The trace formula follows from Theorem 29.1, since δ_f evaluates to $\{J_N(f), -\}$. Moreover $QJ_N(g) = \mathrm{Tr}[c, g(X,Y)] = 0$, and $dJ_N(g) = 0$. Constants give zero derivations, while their traces retain the stated scalar. □

219.4 The ordinary commuting quotient

Give O_N its canonical matrix Poisson bracket and set $C_N = (O_N/I_N)^G$. For $a \in \mathfrak{gl}_N$, the function $\mathrm{Tr}(a\mu)$ generates conjugation, up to the choice of infinitesimal-action sign. Hence its bracket with an invariant function is zero. The ideal I_N is therefore stable under bracketing with invariants. Exactness of invariants lets us represent every element of C_N by an invariant function in O_N, and this defines a Poisson bracket on C_N.

Theorem 219.3 (Diagonal restriction)**.** *Restriction to diagonal matrices gives an isomorphism of Poisson algebras*

$$C_N \xrightarrow{\sim} R_N = \mathbb{C}[x_1, y_1, \dots, x_N, y_N]^{S_N}.$$

It sends $\mathrm{Tr}(X^a Y^b)$ *to* $p_{ab}^{(N)}$.

Proof. The algebra isomorphism is the characteristic-zero restriction theorem for the commuting scheme [97, Theorem 3]. Its hypotheses allow the scheme-theoretic ideal generated by all entries of $[X,Y]$. No reducedness assumption on that scheme is needed.

The orbit-sum argument below shows that the positive-degree power sums generate R_N. Modulo I_N, cyclic derivatives of $x^a y^b$ become the ordinary derivatives of the commutative monomial. Theorem 29.1 therefore identifies their trace brackets with (29.6). This proves Poisson compatibility on a generating set, and hence on the whole algebra. □

219.5 The stable commutative algebra and its rank-dependent bracket

For $N \geq 2$, setting $(x_N, y_N) = (0, 0)$ defines maps $R_N \to R_{N-1}$. Let $(R_N)_D$ denote the homogeneous component of total polynomial degree D. Define the bounded-degree stable algebra

$$\Lambda_2 = \bigoplus_{D \geq 0} \varprojlim_N (R_N)_D.$$

The compatible positive-degree power sums define elements $p_{ab} \in \Lambda_2$ for $a, b \geq 0$, $a + b > 0$.

Proposition 219.4 (Stable generators)**.** *The algebra Λ_2 is freely generated by these p_{ab}. For every N, their images generate R_N.*

Proof. For a list of nonzero exponent pairs $\alpha_1, \dots, \alpha_\ell$, put

$$M_{\boldsymbol{\alpha}}^{(N)} = \sum_{i_1,\dots,i_\ell \text{ distinct}} \prod_{j=1}^{\ell} x_{i_j}^{(\alpha_j)_1} y_{i_j}^{(\alpha_j)_2}.$$

Each orbit sum of monomials is one such expression divided by the factorials of repeated exponent-pair multiplicities. Thus these expressions span R_N over $\mathbb{C}$.

Let Π_ℓ be the set of partitions of $\{1, \dots, \ell\}$. Grouping particle indices by equality and inverting this grouping gives

$$M_{\boldsymbol{\alpha}}^{(N)} = \sum_{\pi \in \Pi_\ell} \prod_{B \in \pi} (-1)^{|B|-1} (|B| - 1)! \prod_{B \in \pi} p_{\sum_{j \in B} \alpha_j}^{(N)}. \tag{219.4}$$

To verify the coefficient, fix a partition specifying equal particle indices. The sum over its refinements factors over its blocks. For a block of size m, the signed sum counts permutations by cycles, giving zero for $m > 1$ and one for $m = 1$. This proves (219.4) and finite-rank generation.

In total degree D, every monomial uses at most D nonzero particle positions. Its orbit-sum basis therefore stabilizes for $N \geq D$. A product of power sums indexed by a multiset of exponent pairs has that orbit sum as its term of greatest support size. Its coefficient is the product of the multiplicity factorials, and the remaining terms use fewer positions. This triangular change of basis is invertible in each degree. It proves algebraic independence and the stated polynomial presentation.

This agrees with the characteristic-zero stable presentation in [96, Corollary 3.5]. □

Theorem 219.5 (Trace brackets with a rank parameter)**.** *Let $\mathcal{P} = \mathbb{C}[\nu, p_{ab} : a + b > 0]$, with nonnegative indices and $p_{00} = \nu$. The formulas*

$$\{\nu, p_{ab}\} = 0, \qquad \{p_{ab}, p_{cd}\} = (ad - bc)\, p_{a+c-1,b+d-1}$$

define a Poisson algebra. The maps π_N of (29.8) are surjective Poisson maps, so their finite-rank relation ideals are Poisson ideals.

Proof. This is the symmetric Poisson algebra of the Lie algebra $\mathbb{C}[x, y]$, as constructed in (29.7). The one-particle bracket gives the formula, and different particles Poisson commute. Proposition 219.4 supplies surjectivity. The kernel of a Poisson homomorphism is a Poisson ideal. □

The range $0 \leq a, b \leq 3$, $(a, b) \neq (0, 0)$ gives $15^3 = 3375$ ordered Jacobi expressions. Each vanishes, with the resulting constant label retained. The proof above establishes Jacobi for all labels.

The stable commutative algebra Λ_2 also admits the Poisson bracket obtained from $\mathcal{P}/(\nu)$. Its two linear generators commute. At rank N, their particle bracket is N. Thus the usual zero-particle restriction maps defining Λ_2 do not preserve the particle Poisson brackets. The family $\mathcal{P}$ specifies each rank separately.

219.6 Two particles and the nilpotent cone

For $N = 2$, set

$$u = \frac{x_1 + x_2}{2}, \quad w = \frac{y_1 + y_2}{2}, \qquad r = x_1 - x_2, \quad s = y_1 - y_2.$$

Then $\{u, w\} = 1/2$, $\{r, s\} = 2$, and every cross bracket vanishes. The transposition fixes u, w and negates r, s, so there is a Poisson splitting

$$R_2 = \mathbb{C}[u, w] \otimes \mathbb{C}[r, s]^{\{\pm 1\}}.$$

This splits off the center-of-mass plane as a tensor factor.

Put $A = r^2/2$, $B = rs/2$, $C = s^2/2$. Every invariant monomial has even total degree. If both exponents are even it is a monomial in A, C. If both are odd it is B times such a monomial. Hence

$$\mathbb{C}[r, s]^{\{\pm 1\}} = \mathbb{C}[A, B, C]/(B^2 - AC).$$

There are no further relations: reduction by $B^2 = AC$ leaves distinct monomials with B-exponent zero or one, whose images in r, s are independent.

Differentiating with $\{r, s\} = 2$ gives

$$\{A, B\} = 2A, \qquad \{B, C\} = 2C, \qquad \{A, C\} = 4B.$$

Thus $e = A/2$, $f = -C/2$, $h = -B$ satisfy the $\mathfrak{sl}_2$ brackets and $h^2 + 4ef = 0$ of Chapter 29. The gradient of $B^2 - AC$ vanishes only at the origin. Away from the origin at least one displayed bracket is nonzero, so the Poisson tensor has rank two on this smooth surface.

219.7 The formal closed comparison

The formal Hamiltonian Lie algebra is a different carrier. Define

$$R = \mathbb{C}[[z_1, z_2]], \qquad \{f, g\} = \partial_{z_1} f\, \partial_{z_2} g - \partial_{z_2} f\, \partial_{z_1} g, \qquad \mathfrak{h} = R/\mathbb{C}.$$

Use the (z_1, z_2)-adic topology. The continuous dual $\mathfrak{h}^\vee_{\mathrm{cont}}$ consists of finite-jet linear functionals. The bracket is continuous. For fixed f, bracketing lowers vanishing order by at most one, so its coadjoint action preserves the continuous dual.

The shifted cotangent Lie algebra for this datum is

$$\mathfrak{g} = \mathfrak{h} \ltimes \mathfrak{h}^\vee_{\mathrm{cont}}[1].$$

Here $\mathfrak{h}$ is in degree zero, the shifted dual is in degree -1, and the dual summand is abelian. The mixed bracket is the shifted coadjoint action, $(\mathrm{ad}^*_f \lambda)(g) = -\lambda(\{f, g\})$. This supplies the formal Lie datum used in Hamiltonian BF theory.

Abelianization gives a Lie map $F_{\mathrm{cyc}} \to \mathbb{C}[z_1, z_2]$, because cyclic derivatives become ordinary derivatives. Its kernel need not act trivially by the strict derivations of the unreduced matrix complex. For example, the cyclic class

$$f = [x^2 y^2 - xyxy]_{\mathrm{cyc}}$$

abelianizes to zero, whereas

$$\delta_f(x) = [\mu, x], \qquad \delta_f(y) = [\mu, y].$$

At $X = \mathrm{diag}(1, 0)$ and $Y = E_{12} + E_{21}$, the second expression is $\mathrm{diag}(2, -2)$. Thus this strict action does not factor through abelianization on that chain carrier. A homotopy-coherent action after reduction remains a separate possibility.

A full comparison with closed observables must specify their continuous cochain complex and the boundary derived center. It must construct a chain map preserving the required operations and prove that its comparison cone is acyclic. The necklace action supplies the explicit matrix Hamiltonians and their differential compatibility. The formal closed comparison, its completion and its higher operation coherences remain to be constructed.

Chapter 220

Modules, orbifolds, and higher E_n directions

220.1 Pointed bars and interfaces

For a right module M_R and left module M_L,

$$\mathrm{Bar}_A(M_R, M_L) = \left|[r] \mapsto M_R \otimes_! \overline{A}^{\otimes_! r} \otimes_! M_L\right| \simeq M_R \otimes^{\mathbb{L}}_A M_L.$$

A marked chiral insertion gives left and right coactions of $\mathrm{Bar}(A)$ by cutting the interval on either side. Interfaces are therefore bimodules and compose by relative tensor product; after Koszul transform, they compose by cotensor product.

220.2 Crossed products

Let a finite group G act by augmented automorphisms of A. Then

$$\mathrm{Mod}_{A \rtimes G} \simeq (\mathrm{Mod}_A)^G.$$

The crossed-product bar is the diagonal totalization of the group bar and the A-bar. In characteristic zero, averaging contracts positive group-homology rows. The fixed algebra A^G and crossed product $A \rtimes G$ remain different: the first describes invariant operators, the second equivariant modules and symmetry defects. A Galois or freeness theorem is needed for Morita equivalence.

220.3 Symmetric products and DMVV

For a graded vector space V with character $f(q)$,

$$\sum_{N \geq 0} p^N \operatorname{ch} \mathrm{Sym}^N(V) = \exp\left(\sum_{r \geq 1} \frac{p^r}{r} f(q^r)\right).$$

For a suitable seed elliptic genus $\sum c(m, \ell) q^m y^\ell$, the symmetric-product orbifold trace is

$$\sum_{N \geq 0} p^N \, \mathrm{Ell}_{\mathrm{orb}}(\mathrm{Sym}^N Y) = \prod_{n \geq 1, m \geq 0, \ell \in \mathbb{Z}} (1 - p^n q^m y^\ell)^{-c(nm, \ell)}.$$

The proof is cycle decomposition in symmetric groups. This is a trace identity. The orbifold operator algebra additionally requires twisted sectors, defect composition, and OPE maps. It must not be identified with a bar complex from the product formula alone.

220.4 Higher transverse dimensions

Dunn additivity gives

$$\mathrm{Alg}_{\mathsf{E}_{n+1}}(\mathcal{V}) \simeq \mathrm{Alg}_{\mathsf{E}_1}(\mathrm{Alg}_{\mathsf{E}_n}(\mathcal{V})).$$

An E_n-ordered chiral algebra has n locally constant transverse directions while retaining its chiral coefficient. Its n-fold bar is factorisation homology of the pointed n-disk and produces an E_n-coalgebra under suitable positivity conditions. The derived centre is an E_{n+1}-object.

Independently, one can form E_n-associative chiral algebras for the chiral product. The same theorem-of-two-bars warning persists: "higher E_n" must specify which monoidal direction is being iterated.

No-go. *A list of graded dimensions, even if it matches a root multiplicity or partition function, contains no coproduct, antipode, associator, or R-matrix. Tannaka reconstruction requires a tensor category and a fibre functor; Hall reconstruction requires extension correspondences and orientations. Scalar agreement is a shadow, not the structured object.*

Chapter 221

Perturbative BV realization of the ordered chiral object

221.1 Classical data

A perturbative classical field theory on a manifold M is described locally by an elliptic L_∞-algebra of fields $\mathcal{E}[-1]$, an invariant pairing of degree -1, and an action functional S solving the classical master equation

$$dS + \frac{1}{2}\{S, S\} = 0.$$

Classical observables on an open set are Chevalley cochains of compactly supported fields, with differential induced by the L_∞-structure. Disjoint support gives factorisation multiplication.

On $X^{\mathrm{an}} \times \mathbb{R}$, assume the kinetic complex is Dolbeault in X and de Rham or locally constant in the real direction. A boundary condition on $X \times \mathbb{R}_{\geq 0}$ restricts fields to a factorisation algebra on $X \times \mathbb{R}$. The ordered interval direction then supplies a transverse E_1-product.

221.2 Renormalised quantum observables

A gauge fixing produces a propagator P_ϵ^L between ultraviolet and infrared scales. Effective interactions $I[L]$ solve the homotopy renormalisation-group equation and the quantum master equation

$$(Q + \hbar\Delta_L)e^{I[L]/\hbar} = 0.$$

The quantum observable differential is

$$Q_L = Q + \{I[L], -\}_L + \hbar\Delta_L.$$

The QME is exactly $Q_L^2 = 0$.

Feynman weights are differential forms on configuration spaces of vertices. Renormalisation extends them to compactified collision spaces after local counterterms are added. Restriction to a boundary face factorises through contraction of the corresponding subgraph.

Theorem 221.1 (Feynman-weight morphism)**.** *Suppose:*

(i) *renormalised graph weights extend to the chosen two-direction compactifications;*

(ii) *restriction to every face agrees with graph contraction and the relevant local counterterm;*

(iii) *the renormalised QME holds;*

(iv) *boundary observables are holomorphically algebraizable on X and locally constant in the ordered direction.*

Then the graph weights define a morphism from the geometric two-coloured forest resolution to the endomorphism operad of boundary observables. Hence the boundary observables form a homotopy-coherent transversely ordered chiral algebra. If ordered chiral collision vertices are also present, they form the corresponding doubly associative object.

Proof. Face factorisation is operadic composition. Stokes' theorem for the extended weights identifies the de Rham boundary with the sum of graph contractions. The QME cancels the full codimension-one boundary, including BV and counterterm faces, so the assignment is a chain map. Algebraisation and local constancy identify the resulting operations with the coefficient and transverse structures. □

No-go. *A classical factorisation algebra can fail to admit a quantum lift because the one-loop obstruction class is nonzero. Gauge fixing can fail to preserve the chosen boundary condition, and the analytic observables can fail to algebraize as regular holonomic D-modules. The quantum ordered chiral object exists only after these obstructions vanish or are cancelled by an invertible anomaly theory.*

221.3 Free holomorphic–topological BF theory

Let V be finite dimensional and take fields

$$A \in \Omega^{0,*}(X)\widehat{\otimes}\Omega^*(\mathbb{R}) \otimes V[1], \qquad B \in \Omega^{1,*}(X)\widehat{\otimes}\Omega^*(\mathbb{R}) \otimes V^*[-1]$$

with action

$$S_{\mathrm{BF}} = \int_{X\times\mathbb{R}} \langle B, (\bar\partial + d_t)A\rangle.$$

The theory is Gaussian. The de Rham contraction on an interval proves local constancy in t; the Dolbeault complex gives the holomorphic coefficient. Complementary linear boundary conditions select a polarization. Quantization of paired boundary fields produces chiral Weyl or Clifford systems according to parity. The central constant in the Weyl relation is handled by the curved duality of chapter 213, not by a nonexistent scalar augmentation.

Chapter 222

BRST descent and Drinfeld–Sokolov reduction

The algebraic comparison between BRST reduction and the bar construction uses actual coalgebra maps. These preserve the differential and deconcatenation. Two small algebras show why these maps matter. One needs a quadratic correction to a section of cohomology. The other has a nonzero higher differential. The affine example then separates this associative calculation from the additional data needed for vertex-algebra reduction.

222.1 The comparison and its hypotheses

Retain the conventions of proposition 36.1. Thus A is an ordinary unital augmented differential graded associative algebra over a characteristic-zero field k, with differential Q and augmentation $\epsilon Q = 0$. Its algebraic bar $B(A) = \bigoplus_{r\geq 0}(\ker \epsilon)[1]^{\otimes r}$ has differential $D = Q_B + b$ and deconcatenation coproduct. If an augmentation weight is specified, it is positive on $\ker \epsilon$, is preserved by Q, and is added by multiplication. The weight completion and its completed tensor product are those of corollary 36.3.

Theorem 222.1 (Bar–BRST comparison). *If $H^i(A, Q) = 0$ for $i \neq 0$, then the truncation algebra $A_{\leq 0}$ gives a canonical zigzag*

$$B(A) \longleftarrow B(A_{\leq 0}) \longrightarrow B(H^0(A, Q))$$

of coaugmented differential graded coalgebra quasi-isomorphisms. The statement also holds for the positive-weight completions.

Proof. The two augmented algebra maps are the inclusion of the truncation and the projection onto degree-zero cohomology. They are quasi-isomorphisms, as proved in corollary 36.5. Homotopy invariance of the bar gives the displayed maps. The completed assertion follows weight by weight. □

This theorem does not select a strict reverse map from a collapsed spectral sequence. A choice of cycle representatives can require higher corrections, even when all cohomology lies in degree zero.

222.2 A comparison map from tilings

Consider

$$A = k[x] \otimes \Lambda(t), \qquad |x| = 0, \quad |t| = -1, \qquad Qx = 0, \quad Qt = x^2.$$

The augmentation kills x and t. Since multiplication by x^2 is injective on $k[x]$, the only cohomology is $H = H^0(A) = k[\bar{x}]/(\bar{x}^2)$. The quotient map $p : A \to H$ sends x to $\bar{x}$ and t to zero. It is an augmented algebra quasi-isomorphism. There is no algebra section $H \to A^0$: the image of $\bar{x}$ would be a nonzero nilpotent in $k[x]$.

The bar differential of H is zero. Its length-n space has basis $[\bar{x}|\cdots|\bar{x}]$. For each tiling of n consecutive positions by blocks of sizes one and two, replace a singleton by x and a pair by t. Let F_n be the sum of the resulting bar words, with coefficient one for every tiling, and set $F_0 = [\,]$. The first terms are

$$F_1 = [x], \qquad F_2 = [x|x] + [t], \qquad F_3 = [x|x|x] + [t|x] + [x|t].$$

The map $F : B(H) \to B(A)$ sends the length-n basis word to F_n. It has total degree zero: a singleton contributes -1 and a pair contributes -2 in the suspended grading.

Proposition 222.2 (Tiling comparison)**.** *The map F is a coaugmented differential graded coalgebra quasi-isomorphism, and $B(p)F = 1$. For weights $\mathrm{wt}(x) = 1$ and $\mathrm{wt}(t) = 2$, it extends to the weight completions.*

Proof. Replacing a t by $Qt = x^2$ in one tiling cancels the multiplication of two adjacent x entries in the tiling obtained by splitting that pair. The prefixes have the same suspended degree, while Q_B and the multiplication of two x entries have opposite signs.

The terms joining x to t cancel those joining t to x. The two tilings occupy the same three consecutive positions and have the same exterior prefix. Their local signs are $+1$ and -1. The products agree because x is even. Terms joining two t entries vanish because $t^2 = 0$. These exhaust DF_n, so $DF_n = 0$ for every n.

Cutting a bar word produced by a tiling cuts between its blocks. Equivalently, choose a position i and independent tilings of the first i and last $n - i$ positions. Thus

$$\Delta F_n = \sum_{i=0}^{n} F_i \otimes F_{n-i},$$

which proves the coalgebra identity.

Under $B(p)$ every tiling containing t vanishes. The singleton tiling gives the original word, so $B(p)F = 1$. Since $B(p)$ is a quasi-isomorphism by theorem 36.4, so is F. Every term of F_n has weight n, which proves the completed assertion componentwise. □

Already F_2 shows the necessity of the correction: $b[x|x] = [x^2]$ is cancelled by $Q_B[t] = -[x^2]$. The all-length map uses this correction coherently with every cut.

222.3 A nonzero higher differential

Let

$$A = \Lambda(x, y, z), \qquad |x| = |y| = |z| = 1, \qquad Qx = Qy = 0, \quad Qz = xy,$$

with augmentation zero on the generators. The weights $\mathrm{wt}(x) = \mathrm{wt}(y) = 1$ and $\mathrm{wt}(z) = 2$ are positive and preserved by Q. Its first cohomology has basis x, y, and all products of these classes vanish. Its second cohomology has basis xz, yz. These assertions follow directly: $xy = Qz$, whereas xz and yz are cycles and are not multiples of the only degree-two boundary xy.

In the length spectral sequence, the word $[x|x|y]$ gives a nonzero class in $E_2^{3,3}$. Its d_1 is zero because the products in $H^1(A)$ vanish. There is no incoming d_1: four augmentation entries have unsuspended degree at least four, whereas the source would require degree three. The class $[xz]$ is nonzero in $E_2^{1,2}$. It is a nonzero cohomology class of A, and an incoming d_1 could only be a product of two classes in $H^1(A)$.

The signed differentials are

$$D[x|x|y] = -[x|xy],$$
$$D[x|z] = -[x|xy] - [xz].$$

Consequently

$$D\big([x|x|y] - [x|z]\big) = [xz], \qquad d_2([x|x|y]) = [xz] \neq 0.$$

The correction has length two and the residual has length one, which is exactly the bidegree of d_2. Positive weight makes this a calculation in a finite length filtration. It gives no general collapse at E_2.

Concentration in degree zero is also essential to theorem 222.1. Take $A = k \oplus k\eta$, with $|\eta| = 1$, $\eta^2 = 0$ and $Q = 0$. Then $H^0(A) = k$, but every word $[\eta|\cdots|\eta]$ is a nonzero degree-zero bar class. All bar differentials vanish, so

$$H^0(B(A)) = \bigoplus_{n\geq 0} k[\eta|\cdots|\eta] \neq B(k) = k.$$

For weight one on η, the completed example has the product of these spaces. It is a separately specified completion of the same nonconcentrated algebra.

222.4 The universal affine reduction

Let $\mathfrak{g}$ be a finite-dimensional complex simple Lie algebra with invariant form normalized so that long roots have squared length two. The universal affine vertex algebra $V^\kappa(\mathfrak{g})$ has currents J_u with singular operator product

$$J_u(z)J_v(w) \sim \frac{\kappa(u,v)}{(z-w)^2} + \frac{J_{[u,v]}(w)}{z-w}.$$

Here $\sim$ records the singular terms. Choose a positive nilpotent subalgebra $\mathfrak{n}_+$, a basis u_α with $[u_\alpha, u_\beta] = f_{\alpha\beta}^{\ \gamma} u_\gamma$, and a principal character $\chi : \mathfrak{n}_+ \to \mathbb{C}$. It is nonzero on each simple-root direction and vanishes on $[\mathfrak{n}_+, \mathfrak{n}_+]$.

The charged-ghost vertex superalgebra has odd fields b_α, c^α, of cohomological degrees -1, $+1$, with $b_\alpha(z)c^\beta(w) \sim \delta_\alpha^\beta/(z-w)$. Its other generating self-products are regular. Colons denote the normal ordered product of fields. On the tensor product with $V^\kappa(\mathfrak{g})$ define

$$j = \sum_\alpha :(J_{u_\alpha} - \chi(u_\alpha))c^\alpha: - \frac{1}{2}\sum_{\alpha,\beta,\gamma} f_{\alpha\beta}^{\ \gamma} : b_\gamma c^\alpha c^\beta :, \qquad Q = \operatorname{Res}_z j(z).$$

The residue means the coefficient of z^{-1}. The character identity and Jacobi identity imply $Q^2 = 0$, and the residue acts as a derivation of the vertex operations [63, Section 2]. This defines $C^\kappa_{\mathrm{DS}}(\mathfrak{g})$.

The principal grading is integral, so this construction needs no neutral ghosts. General good half-integral gradings have additional neutral fields and the corresponding correction to j.

For this universal complex, Kac–Wakimoto's concentration and Poincaré–Birkhoff–Witt theorems give

$$H^i(C^\kappa_{\mathrm{DS}}(\mathfrak{g}), Q) = 0 \quad (i \neq 0),$$
$$\mathcal{W}^\kappa(\mathfrak{g}) = H^0(C^\kappa_{\mathrm{DS}}(\mathfrak{g}), Q)$$

for every $\kappa \in \mathbb{C}$ [64, Theorem 4.1]. The degree-zero algebra is the universal principal W-algebra. Statements for simple quotients or other affine modules require the corresponding cohomology theorem for those objects. Their behavior does not change the universal assertion above.

222.5 The $\mathfrak{sl}_2$ stress tensor

Fix $[H,E] = 2E$, $[H,F] = -2F$, $[E,F] = H$ and $(H,H) = 2$, $(E,F) = 1$. Write e, h, f for the currents, take $\chi(E) = 1$, and use one charged pair b, c. The current and differential become

$$j = (e-1)c, \qquad Q = \operatorname{Res} j.$$

The e–e and c–c operator products are regular, and the affine and ghost sectors have no mutual contractions. Thus the self-product of j is regular and $Q^2 = 0$. The simple poles of its products with the generators give

$$Qb = e - 1, \quad Qe = Qc = 0, \quad Qh = -2 : ec :, \quad Qf =: hc : +\kappa \partial c.$$

For $\kappa \neq -2$, define the Sugawara and ghost stress tensors by

$$T_{\mathrm{Sug}} = \frac{\frac{1}{2} : hh : + : ef : + : fe :}{2(\kappa + 2)}, \qquad T_{\mathrm{gh}} =: (\partial b)c : .$$

The latter assigns conformal weights zero and one to b and c. The improved field is

$$T = T_{\mathrm{Sug}} + \tfrac{1}{2}\partial h + T_{\mathrm{gh}}.$$

It is Q-closed. In fact,

$$\begin{aligned} QT_{\mathrm{Sug}} &=: e\partial c :, \\ Q(\tfrac{1}{2}\partial h) &= - : (\partial e)c : - : e\partial c :, \\ QT_{\mathrm{gh}} &=: (\partial e)c : . \end{aligned}$$

The three terms cancel exactly.

For a stress tensor with central charge c_T, the fourth-order pole of its self-product is $c_T/2$. The Sugawara contribution here is $3\kappa/(\kappa + 2)$. Since $h(z)h(w) \sim 2\kappa/(z - w)^2$, differentiation gives

$$\partial h(z)\partial h(w) \sim -\frac{12\kappa}{(z - w)^4}.$$

The coefficient $1/2$ in the improvement therefore shifts the central charge by -6κ.

For the ghosts, the two contractions in the fourth-order pole are $\partial b(z)c(w) \sim -(z - w)^{-2}$ and $c(z)\partial b(w) \sim (z - w)^{-2}$. Their nested fermionic pairing has positive sign, giving central charge -2. There is no scalar fourth-order pole in the mixed products. Hence

$$\begin{aligned} c_{\mathrm{DS}}(\kappa) &= \frac{3\kappa}{\kappa + 2} - 6\kappa - 2 \\ &= 13 - 6(\kappa + 2) - \frac{6}{\kappa + 2} = 1 - \frac{6(\kappa + 1)^2}{\kappa + 2}. \end{aligned}$$

The cohomology class of T generates the universal Virasoro vertex algebra at this central charge. This identification uses the generator and independence parts of the reduction theorem. For principal $\mathfrak{sl}_2$, the centralizer has one even generator, and the reduced algebra has the corresponding ordered monomial basis in its derivatives. The class of T has a nonzero leading coefficient on this generator when $\kappa + 2 \neq 0$.

It therefore generates, and the same monomial basis proves injectivity of the map from the universal Virasoro algebra [64, Theorems 4.1 and 5.1(b)]. Matching the central charge alone would not prove injectivity. At $\kappa = -2$, concentration of the universal BRST complex still holds, while the displayed Sugawara formula is undefined.

222.6 Associative coefficients and the vacuum

Two features of the vertex complex prevent an immediate use of theorem 222.1 with its normal product and vacuum coefficient. First, the normal product need not be associative. For an even Heisenberg current $a(z) = \sum_{n\in\mathbb{Z}} a_n z^{-n-1}$ at level one, fix

$$[a_m, a_n] = m\delta_{m+n,0}, \qquad a_n\mathbf{1} = 0 \quad (n \geq 0), \qquad a = a_{-1}\mathbf{1}.$$

Write $m(u, v) = u_{(-1)}v$ for the normal product of states. The normal ordered mode formula gives

$$(:aa:)_{(-1)} = \sum_{i<0} a_i a_{-2-i} + \sum_{i\geq 0} a_{-2-i}a_i.$$

On $a_{-1}\mathbf{1}$, the term $i = -1$ gives $a_{-1}^2\mathbf{1}$. The terms $i = -3$ and $i = 1$ each give $a_{-3}\mathbf{1}$ by $[a_1, a_{-1}] = 1$. Every other term vanishes.

Thus, if T denotes translation,

$$m(m(a, a), a) - m(a, m(a, a)) = 2a_{-3}\mathbf{1} = \mathsf{T}^2 a \neq 0.$$

This calculation fixes the current algebra and its vacuum throughout.

Second, in the affine BRST complex the usual vacuum coefficient functional ϵ_0 has $\epsilon_0(\mathbf{1}) = 1$ and $\epsilon_0(e) = 0$. The identity $Qb = e - 1$ gives

$$\epsilon_0(Qb) = -1.$$

It is therefore not a chain augmentation. The scalar 1 in the constraint is not an augmentation value of the current.

An associative or derived coefficient realization must specify a compatible differential, augmentation or relative boundary modules, tensor product and completion. An operation-preserving comparison is then required before the bar theorem applies. These examples obstruct the stated normal-product and vacuum choices, without excluding other such realizations. Likewise, the quantum BV operator of the finite cotangent model has a second-order product defect. Its square-zero identity does not make it the associative derivation required here.

Chapter 223

From order to exchange: KZ, Tannaka, and four-dimensional Chern–Simons

223.1 Infinitesimal braid relations

Let $\mathfrak{g}$ be a quadratic Lie algebra and $\Omega \in \mathfrak{g} \otimes \mathfrak{g}$ the inverse invariant form. On $V_1 \otimes \cdots \otimes V_n$, write Ω_{ij} for its action in factors i, j. Invariance gives

$$[\Omega_{ij}, \Omega_{kl}] = 0 \quad \text{for disjoint pairs,} \qquad [\Omega_{ij}, \Omega_{ik} + \Omega_{jk}] = 0.$$

These are the infinitesimal braid relations.

The KZ connection on $\mathrm{Conf}_n(\mathbb{C})$ is

$$\nabla_{\mathrm{KZ}} = d - \kappa \sum_{i<j} \Omega_{ij}\, d\log(z_i - z_j).$$

Proposition 223.1 (KZ flatness)**.** *The KZ connection is flat.*

Proof. The logarithmic one-forms are closed, so curvature is the wedge-square of the connection form. Terms on four distinct labels vanish by disjoint commutativity. On three labels, the coefficient is an infinitesimal braid commutator multiplied by Arnold's relation

$$\omega_{12} \wedge \omega_{23} + \omega_{23} \wedge \omega_{31} + \omega_{31} \wedge \omega_{12} = 0.$$

Both factors vanish in the required combination. □

Monodromy gives a braid-group representation. The Drinfeld–Kohno theorem identifies the formal monodromy category with the braided representation category of the corresponding quantum group after an associator and parameter identification are chosen. The associator is part of the comparison; it is not produced by the ordered line.

223.2 Meromorphic Tannaka reconstruction

Let C be a rigid meromorphic tensor category of line operators and let $\omega : C \to \mathrm{Vect}_{\Bbbk((u))}$ be a faithful exact strong monoidal fibre functor. The coend

$$H = \int^{V \in C} \omega(V)^* \otimes \omega(V)$$

reconstructs a bialgebra. Rigidity gives an antipode. Braided exchange gives a coquasitriangular structure. A topological completion may be required when spectral expansions are infinite.

No-go. *Tannaka reconstruction outputs an ordinary or topological Hopf algebra only after a suitable fibre functor is specified. Without it, the intrinsic output is the tensor category itself, or a Hopf object in a different base category. "The quantum group" is not a scalar extracted from braid matrices.*

223.3 Four-dimensional Chern–Simons theory

Let Σ be an oriented real surface, C a complex curve with meromorphic one-form ω, and consider the partial connection of four-dimensional Chern–Simons theory on $\Sigma \times C$:

$$S_{4d}(A) = \frac{1}{2\pi\hbar} \int_{\Sigma \times C} \omega \wedge \operatorname{Tr}\left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right).$$

The theory is topological on Σ and holomorphic on C. A Wilson line in Σ at spectral point $z \in C$ is a line object. Parallel fusion is ordered; exchange in Σ produces an R-matrix. In the rational case, the tree-level exchange is

$$R(z_1 - z_2) = 1 + \hbar \frac{\Omega}{z_1 - z_2} + O(\hbar^2).$$

Gauge invariance at the first nontrivial order imposes the classical Yang–Baxter equation. Quantisation produces the rational Yangian-type exchange algebra under the boundary, anomaly, and representation hypotheses of the model.

The geometry explains why ZF/RTT relations and Yang–Baxter occur. The complete identification with a specific Yangian remains a theorem about the line category and its generators, not a consequence of dimensional analysis.

Chapter 224

Cyclic traces, modular sewing, and the two-edge master equation

224.1 Closing the transverse interval

The surface construction needs a cyclic pairing and its inverse on a specified coefficient model. Let k be a field of characteristic zero, and let A be a finite Calabi–Yau ordered algebra as in Definition 20.1. Its pairing has the cohomological convention

$$B : A \otimes_k A \longrightarrow k[-d], \qquad A \cong A^{\vee}[-d].$$

For a proper derived algebra, a coherent cyclic trace has source $CC_*(A)_{hS^1}$ and the same shifted target. Passing from that datum to a finite cyclic model is a further hypothesis here. For an algebra in a chiral coefficient category, a realization must also preserve the chosen tensor, pairing and sewing operations.

Theorem 224.1 (Ribbon-graph action). *The algebra A determines an open topological conformal field theory and its universal open–closed extension. Both use Costello's admissible surface category with determinant coefficients $\mathcal{D}^{\otimes d}$. The extension has circle value quasi-isomorphic to $CC_*(A)$. Its closed restriction supplies surface operations on these chains with their coherent compositions. Passing to homology gives an action of the homology of the corresponding moduli spaces on $HH_*(A)$.*

Proof. Apply Theorem 20.2. The cyclic higher products label disk vertices and the inverse pairing contracts internal edges. The cellular disk relations are the cyclic A_∞ identities, while annuli compute the Hochschild circle object. Derived extension transports the cellular functor to the surface category and preserves its sewing compositions. The determinant line accounts for the grading of the pairing and its inverse. □

This construction gives the genus $g_\perp$ of the bordered surface. Its admissibility condition requires an incoming closed or a free boundary on every component. It is not an assertion about a finite cyclic extension of an arbitrary native state algebra.

224.2 Chiral sewing over stable curves

The chiral genus g_X is the arithmetic genus of a stable algebraic curve. A modular chiral coefficient must supply maps along separating and nonseparating clutching,

$$\overline{\mathcal{M}}_{g_1,n_1+1} \times \overline{\mathcal{M}}_{g_2,n_2+1} \longrightarrow \overline{\mathcal{M}}_{g_1+g_2,n_1+n_2}, \qquad \overline{\mathcal{M}}_{g-1,n+2} \longrightarrow \overline{\mathcal{M}}_{g,n}.$$

The bars denote the compactified moduli stacks, since the glued curves have nodes. The coefficient maps include their pullback, pushforward, support and completion conventions. A perfect pairing contracts the two flags at a node, but the modular composition identities and their compatibility with the coefficient maps must also hold. These are additional data beyond the transverse surface theory.

224.3 Two kinds of contraction

Suppose a common collection of decorated graphs admits both transverse and chiral contractions, with coherent identifications of every iterated contraction target. Assign a complex $C(G)$ to each graph and equivariant degree-zero cochain maps $c_e^G : C(G) \to C(G/e)$. Assume the single-kind composition laws, and require the mixed square

$$c_f^{G/e} c_e^G = c_e^{G/f} c_f^G \qquad (e \text{ transverse}, \ f \text{ chiral}).$$

Recognition Theorem 224.2 (Two-edge sewing theorem)**.** *Under these assumptions, the maps extend uniquely to the category of finite graph contractions. The determinant-oriented direct-sum complex of Proposition 20.3 has a square-zero differential. Its transverse and chiral edge differentials anticommute.*

Proof. Factor a contraction into single-edge contractions. Adjacent transpositions relate any two edge orders, and the single-kind laws and mixed square identify their coefficient composites. The coherent target identifications give an order-independent map and prove its compatibility with further contraction. In the graph differential the two edge orders have opposite exterior signs. Proposition 20.3 then gives the square-zero identity, including loops. □

If the mixed square is given only up to homotopy, higher compatible homotopies are required. A square-zero contraction differential is one part of a master-equation construction. Theorem 38.3 supplies the generator equation when the common sewing maps define a twisted modular operad and its complementary target. Its carrier consists of maps on the dual generators, completed by positive stability weight. An amplitude Θ has degree zero there and degree one after the stated Lie-algebra shift.

Write $[-,-]_\perp$ and $[-,-]_X$ for the polarized separating operations of that theorem, and $\Delta_\perp, \Delta_X$ for its loop operations. All four operations have degree one on the unshifted amplitude carrier. With separating weights one and a common degree-zero loop parameter $\hbar$, the equation is

$$d\Theta + \frac{1}{2}[\Theta,\Theta]_X + \frac{1}{2}[\Theta,\Theta]_\perp + \hbar(\Delta_X + \Delta_\perp)\Theta = 0. \tag{224.1}$$

The equation says that the internal boundary and the separating and nonseparating contributions cancel on every generator. It is equivalent to compatibility of the extended operad morphism with the differential. The common parameter records the ordinary total genus shift.

For a differential graded Lie structure with independent loop parameters $\hbar_X, \hbar_\perp$ and these separating brackets, the crossed derivation defects must vanish separately, as in (38.7). The anticommutator $\Delta_X\Delta_\perp + \Delta_\perp\Delta_X = 0$ does not imply that condition. On the full colored graph category, the universal contraction weights instead satisfy (38.2). Proposition 38.2 gives a restricted contraction domain with two invariant genera, together with its limitation under substitution. Realizing two geometric sewings therefore requires the common coefficient maps, their coherent compositions and their compatibility with the chosen completion. The equation cannot be inferred from an orientation or from scalar amplitudes alone.

Remark 224.3 (The two geometric genera). The surface genus $g_\perp$ and the curve genus g_X have different geometric definitions. A graph with two vertices and two differently colored parallel edges demonstrates a further issue. The color of the edge that becomes a loop depends on the contraction order. Assigning its genus increment to that color does not define two invariant genera on the ordinary colored graph category.

Thus a theory using both genera must specify a more structured common sewing object, or restrict its allowed contractions. It must prove the desired genus comparison there. The existence of such a comparison for an ordered chiral algebra remains a separate construction.

Chapter 225

Prime forms, plumbing, and the analytic chiral genus

225.1 The normalized bidifferential

Let Σ_g be a compact Riemann surface with symplectic homology basis and period matrix Ω. The prime form $E(p,q)$ is a $(-1/2,-1/2)$-form with a simple zero on the diagonal. The normalized fundamental bidifferential has local behaviour

$$B(p,q) = \left(\frac{1}{(z(p)-z(q))^2} + O(1) \right) dz(p)\,dz(q)$$

and vanishing A-periods. It may be expressed through logarithmic derivatives of the prime form with the normalization correction fixed by periods.

For a rank-one Heisenberg current,

$$\langle J(p)J(q)\rangle_{\Sigma_g} = kB(p,q).$$

Higher current correlators are sums over pairings. This is an exact Gaussian realization of the chiral coefficient at every genus once the determinant line and zero modes are included.

225.2 Plumbing a node

For local coordinates z, w at two marked points, plumbing identifies

$$zw = q, \qquad |q| \ll 1.$$

A separating node joins two components; a nonseparating node joins two points on one component. Formal sewing inserts a complete set of states:

$$\mathcal{F}_{\mathrm{sew}}(q) = \sum_{\alpha} q^{L_0(\alpha)-c/24}\mathcal{F}_1(\cdots, v_\alpha)\mathcal{F}_2(v^\alpha,\cdots).$$

The expression is formal in q. Analytic sewing additionally requires convergence, control of the completion, and compatibility of the chosen local coordinates.

No-go. *An identity in $V[[q]]$ can hold while the corresponding series has zero radius of convergence in a chosen topology. Trace-class estimates, energy bounds, or a rational/CFT finiteness theorem are separate analytic inputs. Formal modular-operad structure and analytic conformal blocks must be stated separately.*

225.3 Genus one

For a positive-energy module M, the torus character is

$$\chi_M(\tau) = \operatorname{Tr}_M q^{L_0 - c/24}, \qquad q = e^{2\pi i \tau}.$$

For a rank-one Heisenberg algebra,

$$\chi_{\mathrm{Heis}}(\tau) = \eta(\tau)^{-1}.$$

For an even lattice L,

$$\chi_{V_L}(\tau) = \frac{\Theta_L(\tau)}{\eta(\tau)^{\operatorname{rk} L}}.$$

The theta numerator records zero modes; the eta denominator is the oscillator determinant. Their modular transformations are sections of the anomaly line rather than ordinary invariant functions until the determinant factor is included.

225.4 Higher genus and Siegel theta series

The lattice zero-mode sum becomes

$$\Theta_L^{(g)}(\Omega) = \sum_{\lambda_1,\dots,\lambda_g \in L} \exp\left(\pi i \sum_{a,b} (\lambda_a, \lambda_b) \Omega_{ab} \right).$$

The oscillator factor is the corresponding chiral determinant. Separating degeneration factorises the theta series into the products of lower-genus theta series; nonseparating degeneration is a Fourier–Jacobi expansion. These are genuine analytic data that survive the correction from one genus to two independent genera: they belong to g_X, not to the ribbon loop count.

Chapter 226

A concrete two-genus model

226.1 The transverse Frobenius algebra

Take the augmented noncommutative Frobenius algebra $B_n = D \times M_n(\Bbbk)$ of proposition 213.3. Its cyclic trace produces a ribbon theory. Let

$$R_h(n) = \tau_B(e_{B_n}^h) = \begin{cases} n, & h = 0, \\ 2 + n^2, & h = 1, \\ n^{h+1}, & h \geq 2. \end{cases}$$

This is an exact closed-surface amplitude in the associated two-dimensional topological field theory convention.

226.2 A rank-one Deligne–Mumford theory

Let $C_\lambda = \Bbbk \mathbf{1}$ be the one-dimensional commutative Frobenius algebra with

$$\mathbf{1} \cdot \mathbf{1} = \mathbf{1}, \qquad \eta(\mathbf{1}, \mathbf{1}) = \lambda \in \Bbbk^\times.$$

The inverse metric is λ^{-1}, and the handle element is $\lambda^{-1}\mathbf{1}$. The associated rank-one cohomological field theory assigns the constant class

$$\Omega_{g,r} = \lambda^{1-g} \in H^0(\overline{\mathcal{M}}_{g,r}; \Bbbk)$$

to every stable pointed curve.

Proposition 226.1 (Clutching). *The classes $\Omega_{g,r}$ satisfy both separating and nonseparating CohFT gluing equations.*

Proof. A separating gluing contracts two classes with the inverse metric and gives

$$\lambda^{1-g_1}\lambda^{1-g_2}\lambda^{-1} = \lambda^{1-(g_1+g_2)}.$$

A nonseparating gluing gives

$$\lambda^{1-(g-1)}\lambda^{-1} = \lambda^{1-g}.$$

The unit axiom is immediate. □

226.3 Strict mixed interchange

Form the tensor-product state object

$$\mathcal{H} = B_n \otimes C_\lambda.$$

Ribbon contractions act on the first factor; Deligne–Mumford clutching acts on the second. Common corollas are tensor products of the two multiplications. Operators on different tensor factors commute before the determinant-line sign is inserted.

Theorem 226.2 (Realized two-edge recognition). *The state object $\mathcal{H}$ satisfies all five hypotheses of recognition Theorem 224.2. Its bivariate closed amplitude is*

$$\boxed{Z_{g_X,g_\perp}(n,\lambda) = \lambda^{1-g_X} R_{g_\perp}(n).}$$

Thus

$$Z_{g_X,0} = n\lambda^{1-g_X}, \qquad Z_{g_X,1} = (2+n^2)\lambda^{1-g_X}, \qquad Z_{g_X,h} = n^{h+1}\lambda^{1-g_X}\ (h \geq 2).$$

The two genus variables are independent and the mixed interchange square is strict.

Proof. Cyclicity and perfection for B_n give the transverse sewing. proposition 226.1 gives the chiral modular sewing. Tensor-product corollas satisfy each one-colour associativity law. A contraction on B_n commutes with a contraction on C_λ, and exchanging their odd edge orientations supplies the required sign. The recognition theorem applies. The amplitude factorises into the two exact one-colour amplitudes. □

This is the missing realized invocation of the recognition theorem. It is deliberately elementary enough that every hypothesis and scalar can be checked. It proves the formal two-genus architecture unconditionally. Tensoring the rank-one CohFT with a rational vertex-algebra modular functor gives richer analytic examples once convergence and anomaly-line hypotheses are supplied.

226.4 What the model proves and what it does not

The example proves existence of a noncommutative transverse sector, a genuine augmentation and bar, two independent modular contractions, and a mixed graph action. It does not claim that every holomorphic–topological gauge theory factorizes as a tensor product, or that its two loop parameters are physically unrelated. Interacting theories can have mixed vertices; then the mixed square is a substantive calculation rather than strict tensor-factor commutation.

Chapter 227

Typed anomalies and invertible theories

227.1 Classes, curved data and descent

A leading anomaly class is defined only after the lower equations give a square-zero differential on the relevant coefficient. Chapter 39 constructs the degree-two central-extension and BRST classes, the degree-one BV obstruction, and the separate holomorphic connection class. A curved associative element κ instead obeys $d^2 = [\kappa, -]$ and need not define an ordinary cohomology class. Likewise a chosen determinant curvature form, its characteristic class and its flat holonomy retain different information. The flat line with holonomy -1 and the central BV class in (39.4) show two ways that passing to a smaller invariant loses an obstruction.

Framing dependence is a descent problem for the geometric bordism category. Let π be the functor that forgets the framing. Descent of a framed theory T requires a theory T_0 on the unframed category and a monoidal identification $T \simeq \pi^* T_0$ compatible with all bordisms. It is not specified by a scalar curvature coefficient alone. A family index map, transgression, boundary map or open–closed comparison must give the actual transport between its declared anomaly data.

227.2 Cancellation requires a relative theory or a trivialization

Fix a bordism setting in which tensor products and monoidal transformations of theories are defined. We use the relative-theory convention of Freed–Teleman (2014), Definition 2.1 [41]. Let α be an invertible theory in dimension $d+1$. Put $A = \tau_{\leq d}\alpha$, its truncation to the dimensions used by the relative theory.

Write $\mathbf{1}$ for the trivial theory. In one variance a d-dimensional relative theory is a transformation

$$T : \mathbf{1} \longrightarrow A.$$

The opposite variance has a map $A \to \mathbf{1}$ and must be treated with that source and target. Choose an inverse A^{-1} and a monoidal evaluation isomorphism $e : A \otimes A^{-1} \xrightarrow{\sim} \mathbf{1}$.

Theorem 227.1 (Cancellation with a compensating theory)**.** *If a relative theory $U : \mathbf{1} \to A^{-1}$ is supplied, then*

$$\mathbf{1} \simeq \mathbf{1} \otimes \mathbf{1} \xrightarrow{T \otimes U} A \otimes A^{-1} \xrightarrow{e} \mathbf{1}$$

is an absolute theory in the same bordism setting. Alternatively, a monoidal trivialization $\varepsilon : A \xrightarrow{\sim} \mathbf{1}$ makes $\varepsilon \circ T$ absolute. For the reversed variance, supplied maps $\widetilde{T} : A \to \mathbf{1}$ and $\widetilde{U} : A^{-1} \to \mathbf{1}$ give $(\widetilde{T} \otimes \widetilde{U}) \circ e^{-1} : \mathbf{1} \to \mathbf{1}$.

Proof. Tensoring the transformations preserves their naturality and monoidal composition equations. Composing with the monoidal isomorphism e preserves those equations again, and the resulting source and target are both $\mathbf{1}$. In the relative-theory convention, a transformation relative to $\mathbf{1}$ is an absolute theory. On a closed d-manifold M, it evaluates the covector $U(M)(1)$ on the vector $T(M)(1)$. On a bordism, the same evaluation commutes with the structure maps by naturality, so it respects gluing. The construction with ε is composition of monoidal transformations. □

The inverse object alone gives a different map:

$$e \circ (T \otimes \mathrm{id}_{A^{-1}}) : A^{-1} \longrightarrow \mathbf{1}.$$

Its source is still the inverse anomaly theory. The isomorphism $A \otimes A^{-1} \simeq \mathbf{1}$ supplies neither the transformation U nor a trivialization of A. Existence of the required compensating boundary theory, with its locality and gluing, remains an additional construction.

227.3 A projective representation and its actual compensator

Let $G = (\mathbb{Z}/2)^2$, write its elements as (a, b), and set

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \rho(a, b) = X^a Z^b.$$

Since $ZX = -XZ$,

$$\rho(g)\rho(h) = \alpha(g, h)\rho(g + h), \qquad \alpha\big((a, b), (a', b')\big) = (-1)^{ba'}.$$

The exponent is bilinear modulo two, so $\alpha(g, h)\alpha(g + h, k) = \alpha(h, k)\alpha(g, h + k)$. Thus ρ is a projective representation with multiplier α.

A one-dimensional compensator would be nonzero scalars $b(g)$ satisfying

$$\frac{b(g)b(h)}{b(g + h)} = \alpha(g, h)^{-1}.$$

The left side is symmetric in g, h because G is abelian. The right side takes the values 1 and -1 on $((1, 0), (0, 1))$ and the reversed pair. Consequently no such one-dimensional compensator exists, although the inverse cocycle α^{-1} is defined.

An actual opposite projective system is the dual vector space with

$$\rho^{\vee}(g) = \big(\rho(g)^{-1}\big)^{\mathsf{T}}.$$

Taking inverses and transposes of the multiplier equation gives

$$\rho^{\vee}(g)\rho^{\vee}(h) = \alpha(g, h)^{-1}\rho^{\vee}(g + h).$$

Therefore $\rho \otimes \rho^{\vee}$ is an ordinary representation on the four-dimensional space $V \otimes V^{*}$. Its evaluation pairing is invariant because $\rho^{\vee}(g)$ is contragredient to $\rho(g)$. This supplies a compensating system with its maps. Its dimension is two, and the one-dimensional obstruction remains.

227.4 Scalar evaluations and central charge

An evaluation of an anomaly class is a map to scalars, with a chosen cycle, trace or background. Equality after that evaluation does not identify the original coefficient objects or their trivializations. The preceding projective multiplier and the flat holonomy example retain information not supplied by a curvature scalar.

Central charge also fails to classify general conformal vertex algebras. For a finite-dimensional commutative unital algebra B, put every state in degree zero and define $Y(a, z)b = ab$, translation $T = 0$,

and conformal vector $\omega = 0$. Associativity and commutativity verify the constant-field vertex identities, and the unit verifies the vacuum identities. All Virasoro operators vanish, with central charge zero, and $L_{-1} = T$ and $L_0 = 0$ agree with the chosen grading.

Taking $B = \mathbb{C}$ and $B = \mathbb{C}[\epsilon]/(\epsilon^2)$ gives nonisomorphic examples of dimensions one and two. The second example is outside the additional CFT-type condition $B_0 = \mathbb{C}$, which is not imposed here. The central charge does not recover their states or multiplication, much less the extra coefficient and sewing operations of an ordered chiral theory.

Chapter 228

Bar Euler products and Borcherds denominators

228.1 Root-graded Lie superalgebras

Let

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \neq 0} \mathfrak{g}_\alpha, \qquad \mathfrak{n}_+ = \bigoplus_{\alpha \in \Gamma_+} \mathfrak{g}_\alpha,$$

where every root space is finite dimensional and each degree has finitely many decompositions in the chosen completion. PBW and theorem 215.1 give

$$\operatorname{Bar}^\perp(U(\mathfrak{n}_+)) \simeq \operatorname{CE}_*(\mathfrak{n}_+) = \operatorname{Sym}(\mathfrak{n}_+[1]).$$

Theorem 228.1 (Bar denominator product)**.** *The completed graded Euler supercharacter is*

$$\chi_\Gamma \operatorname{Bar}^\perp(U(\mathfrak{n}_+)) = \prod_{\alpha \in \Gamma_+} (1 - e^{-\alpha})^{\operatorname{sdim} \mathfrak{g}_\alpha}.$$

If $\mathfrak{g}$ is a Borcherds–Kac–Moody superalgebra with Weyl vector ρ, multiplying by $e^{-\rho}$ gives the product side of its denominator identity.

Proof. An even root vector becomes odd after suspension and contributes $1 - e^{-\alpha}$ to the Euler supercharacter. An odd root vector becomes even and contributes its inverse. Multiplication over a homogeneous basis gives the superdimension exponent. Finite decomposition makes each coefficient of the completed product finite. □

The theorem is an object-to-character map: the Lie bracket controls the Chevalley differential, while the Euler character remembers only signed root multiplicities.

228.2 The Igusa product

Let

$$\phi_{0,1}(\tau, z) = \sum_{n \geq 0, \ell \in \mathbb{Z}} f(n, \ell) q^n y^\ell$$

be the normalized weak Jacobi form of weight zero and index one. In the standard chamber, the monic Borcherds product is

$$D_5(Z) = q^{1/2} y^{1/2} p^{1/2} \prod_{(n,\ell,m) > 0} (1 - q^n y^\ell p^m)^{f(nm,\ell)}.$$

For the Gritsenko–Nikulin Borcherds algebra, the signed root multiplicities are $f(nm, \ell)$. With the theta-constant normalization $\Delta_5 = 64 D_5$, the denominator convention gives

$$e^{-\rho} \chi_\Gamma \operatorname{Bar}^\perp(U(\mathfrak{n}_{\Delta_5,+})) = D_5(2Z) = 64^{-1} \Delta_5(2Z).$$

This equality is a typed Euler-character identity. It does not identify the entire bar complex with an automorphic function.

The scalar 64 is not a convention. Writing $\mathfrak{E}$ for the ten even genus-two characteristics and $\Delta_5^{\theta}(Z) = \prod_{\mathfrak{m}\in\mathfrak{E}} \vartheta[\mathfrak{m}](Z,0)$, the cusp leading terms of the ten factors group by first half-characteristic: four contribute the scalar 1, and the remaining six each contribute a factor 2 from a pair of shortest lattice representatives exchanged by sign. The exponents sum to $(1/2,1/2,1/2)$, so

$$[q^{1/2}r^{1/2}s^{1/2}]\,\Delta_5^{\theta} = 2^6 = 64,$$

which fixes $\Delta_5^{\theta} = 2^6\Delta_5^{\mathrm{mon}}$ and with it the normalization used above. The weight is $f(0,0)/2 = 5$, consistent with the normalization $f(0,0) = 10$ fixed above.

228.3 Finite positive windows

Let $h : \Gamma_+ \to \mathbb{N}_{>0}$ be proper and additive. The quotient

$$\mathfrak{n}_+(H) = \mathfrak{n}_+ / \bigoplus_{h(\alpha)>H} \mathfrak{g}_\alpha$$

is finite-dimensional nilpotent. Its level-zero current vertex algebra has PBW character, and the inverse limit over H is separated and complete. Coefficientwise finiteness follows because a fixed root and oscillator degree sees only finitely many positive roots and modes.

228.4 Polarized Gram transform

Let $\Pi : \Gamma_{\mathrm{phys}} \to \Lambda$ be quadratic with polarization

$$B(\gamma,\delta) = \Pi(\gamma+\delta) - \Pi(\gamma) - \Pi(\delta).$$

For a charge-graded algebra $A = \bigoplus_\gamma A_\gamma$, define

$$(a_\gamma u^\lambda) \star_B (b_\delta u^\mu) = a_\gamma b_\delta u^{\lambda+\mu+B(\gamma,\delta)}.$$

Biadditivity gives the cocycle identity and hence associativity. The Gram embedding

$$\iota_\Pi(a_\gamma) = a_\gamma u^{\Pi(\gamma)}$$

is an algebra homomorphism for $\star_B$. This is the exact device that turns quadratic Fourier indices into an additive grading without pretending that Π itself is additive.

Warning 228.2. A compact Hall realization of the Igusa root algebra is a further theorem. It must construct the moduli stack, orientation data, extension correspondence, PBW comparison, and map to the Borcherds algebra. The denominator product alone supplies none of these maps.

Chapter 229

Higher-dimensional sources and factorisation pushforward

229.1 Pushforward along an actual map

Let $f : M \to X$ be a map and $\mathcal{O}$ a factorisation algebra on M. Under properness/support and descent conditions, factorisation pushforward is

$$(f^{\mathrm{Fact}}_{*}\mathcal{O})(U) = \mathcal{O}(f^{-1}U).$$

Disjoint opens in X have disjoint inverse images, so the factorisation maps descend. Holomorphic or topological reduction to a curve is therefore a functor only after an actual map and support theory are specified.

229.2 Calabi–Yau categories and Hall products

A Calabi–Yau category of branes can produce:

(i) an E_1-algebra of endomorphisms of a chosen brane;

(ii) extension correspondences on moduli of objects, yielding Hall or cohomological Hall multiplication;

(iii) cyclic pairings and open-string traces;

(iv) closed-string maps into Hochschild invariants.

Shifted symplectic geometry can orient the correspondences and control quantization. Each arrow needs its own finiteness and orientation theorem. A slogan such as "Calabi–Yau source" is not yet an ordered chiral algebra.

229.3 Coulomb branches and shifted Yangians

In suitable supersymmetric gauge theories, convolution on affine Grassmannian-type spaces produces quantized Coulomb branches, while line defects organize modules over shifted Yangians. To place such an algebra in the present theory one must exhibit:

(i) the curve-valued factorisation coefficient;

(ii) the ordered line multiplication;

(iii) the augmentation or relative base used by the bar;

(iv) the exchange geometry or R-matrix;

(v) the comparison to the RTT/Drinfeld presentation.

Once these data are present, bar, centre, and BRST constructions are functorial. Without them, name matching is not a proof.

229.4 Source-to-chiral comparison

The correct general pattern is a diagram

$$\begin{array}{ccc} \text{higher-dimensional geometric category} & \xrightarrow{\text{pushforward/restriction}} & \text{factorisation category on } X \\ \Big\downarrow{\scriptstyle \text{Hall or convolution}} & & \Big\downarrow{\scriptstyle \mathrm{RHom}(b,b)} \\ \text{geometric algebra} & \xrightarrow[\text{comparison}]{} & \mathrm{Alg}_{\mathbb{E}_1}(\mathrm{FactD}(X)). \end{array}$$

Every square is a theorem about correspondences, supports, and descent. The diagram is the disciplined replacement for a universal "shadow tower."

Chapter 230

Obstructions to comparison

The theory must distinguish a genuine comparison map from a coincidence of shadows. Each prohibition below names the datum whose absence blocks the proposed identification and the construction that would make a comparison possible.

230.1 Carrier prohibitions

N1. Residue is not the transverse bar differential. Residue changes normal/collision degree; the bar differential contracts adjacent transverse letters. They can anticommute in a bicomplex.

N2. A logarithmic form does not extract all principal parts. Higher poles require the complete normal-jet filtration or a chosen connection.

N3. Incidence does not contract screen cohomology. A terminal corolla can make an incidence nerve contractible while the geometric screen has nontrivial cohomology.

N4. Order does not produce exchange. An ordered chamber on a line is contractible and has no braid generator.

N5. A trace is not an augmentation. Cyclic linear functionals do not define multiplicative bar faces.

N6. Graph topology does not determine coefficients. A stable graph specifies possible contractions; a pairing and state assignment specify their values.

230.2 Monoidal and operadic prohibitions

N7. Hadamard and Cauchy products are not globally strongly duoidal. The exact relation is a split inclusion/projection with aligned idempotent.

N8. Transverse E_1 does not imply associative chiral E_1. The two products have different carriers and operads.

N9. Associative order does not imply a quantum vertex S-operator. Unitarity, Yang–Baxter, shift, and braided locality are extra equations.

N10. Two pairwise exchange rules do not imply coherent three-field exchange. The critical-pair obstruction is Yang–Baxter.

N11. The interval bar, associative chiral bar, and chiral Chevalley complex are not one complex. A comparison requires an enveloping or distributive-law map.

N12. A quadratic dual is not automatically the derived Koszul dual. Exactness of the quadratic Koszul complex is the missing theorem.

N13. Ordinary tensor cobar and completed totalization have different carriers. The ordinary counit recovers a degree-zero augmented algebra without completion. For $k[x]$, it recovers $k[x]$, whereas augmentation completion is $k[[x]]$.

N14. Verdier, boundary, quadratic, and bar duals are not definitionally identical. Finite self-dual examples supply comparison maps.

230.3 Algebra and calculation prohibitions

N15. The Weyl algebra has no scalar augmentation. Use a Rees degeneration, a curved dual, or a relative module.

N16. A full matrix algebra has no scalar augmentation. Use Morita/relative bar data or embed it in an honestly augmented algebra.

N17. A multi-idempotent quiver is naturally relative over its vertex algebra. Scalar augmentation forgets the boundary objects.

N18. A chain-space count is not a homology dimension. The differential and its rank are part of the invariant.

N19. A matching integer sequence is not a quasi-isomorphism. Motzkin and Riordan arithmetic survives only in its actual planar combinatorial role.

N20. Central charge does not classify a chiral algebra. OPEs, modules, fusion, modular data, and transverse multiplication remain independent.

N21. A character does not reconstruct a Hopf algebra. Coproduct, antipode, associator, and exchange are structured maps.

N22. A denominator product does not construct a Hall realization. Moduli, orientations, extension correspondences, and PBW comparison are separate.

230.4 Physics and sewing prohibitions

N23. A boundary centre does not determine an arbitrary decoupled bulk. An open–closed equivalence is a theorem with generation and locality hypotheses.

N24. The two genera are not universally equal. Ribbon and Deligne–Mumford handles are different edge colours.

N25. Formal sewing does not imply analytic convergence. Energy bounds or trace-class estimates are additional inputs.

N26. Different anomaly classes are not one scalar. Curved, BV, BRST, determinant, and framing anomalies live in different complexes.

N27. Classical factorisation does not guarantee quantum factorisation. The QME obstruction and boundary anomaly must vanish.

N28. A physical analogy does not supply a functor. Every comparison between gauge theory, Hall geometry, vertex algebra, and automorphic form must specify source, target, and map.

230.5 Positive replacements

Each obstruction specifies an additional construction required for the proposed comparison:

False shortcut	Correct construction
residue as bar differential	total complex with typed collision and bar degrees
strong duoidal equivalence	aligned-decomposition retract and locality squares
‘‘nonlocal’’ from transverse order	independent associative chiral operad and formal-disc recognition
ordinary quotient by relations	homotopy pushout of free cofibrant algebras
bar of a Weyl/matrix algebra	curved, Rees, relative, or Morita bar
sequence recognition	explicit chain matrices and quasi-isomorphism
boundary centre equals bulk	bulk-to-centre map plus open–closed theorem
one genus	bicoloured stable graphs and mixed recognition
one anomaly number	comparison maps between typed deformation complexes
automorphic product as algebra	bar character theorem plus separate Hall realization

Chapter 231

Obstructions and finite calculations

231.1 Carrier-to-construction atlas

The mathematical claims separate according to their carriers.

Carrier	Available construction	Additional datum required
factorisable D-modules	aligned pointwise/chiral locality	a mixed distributive law off the aligned sector
augmented transverse algebra	interval bar and its word differential	a separate chiral multiplication
associative chiral algebra	collision-tree operadic bar	an enveloping or distributive-law comparison with the interval bar
relative quiver algebra over S	$S \otimes^{\mathbb{L}}_{A} S$	no scalar augmentation is inferred
exchange algebra	KZ, ZF, or RTT transport	flatness, unitarity, Yang–Baxter, and a fibre functor as applicable
cyclic or modular object	trace and sewing operations	pairing, convergence, and anomaly data
root-graded bar complex	Euler supercharacter denominator	a separate Hall correspondence and PBW comparison

231.2 Finite calculations and their boundaries

For the quantum plane at $q = 2$ and the Jordan plane, normalized bar words are graded by total polynomial weight $w \le 5$ and word length. The alternating adjacent-multiplication matrices are rational. Their nonzero homology in this window is

$$H_{1,1} \cong \Bbbk^2, \qquad H_{2,2} \cong \Bbbk,$$

with all other groups zero; Appendix 234 gives the bases, matrix ranks, and cycles. For $U(\mathfrak{sl}_2)$, the Chevalley–Eilenberg differential has $\det d_2 = -4$, $d_3 = 0$, and

$$(\dim H_0, \dim H_1, \dim H_2, \dim H_3) = (1, 0, 0, 1).$$

These calculations refute universal Motzkin or Riordan formulas for the corresponding ordered bar homology. A scalar sequence, chain-space count, or finite window does not establish a quasi-isomorphism outside its stated carrier and range.

231.3 Data in the worked examples

The examples distinguish the following algebraic data, with finite calculations restricted to their stated parameter ranges:

(i) the coefficient object and its diagonal extension;

(ii) the associative product and augmentation or relative base;

(iii) generators, relations, and a normal-form or PBW theorem;

(iv) the actual bar or Koszul differential;

(v) cycles, boundaries, and homology, not only chain dimensions;

(vi) any exchange, trace, modular, or physical realization as a named extra map;

(vii) an exact finite-window calculation, with its parameter range, when constants or signs are nontrivial.

These data appear in the quantum plane, Jordan plane, relation quiver, $\mathfrak{sl}_2$, ZF system, A_2 Hall algebra, stable trace bracket, and two-genus Frobenius/CohFT constructions.

Chapter 232

Structure theorem for the complete datum

232.1 The primitive datum

Let X be a smooth complex curve. The complete local object is the tuple

$$\mathbb{A} = \bigl(F, m_{\perp}, m_X, \lambda, \varepsilon;\ \text{completions}\bigr),$$

where:

(i) F is a factorisable D-module carrying every diagonal singularity and regular product;

(ii) $m_{\perp}$ is an augmented transverse E_1-multiplication for $\otimes_!$;

(iii) m_X is an augmented associative chiral E_1-multiplication for $\star_{\mathrm{ch}}$ or $\mathsf{Ass}^{\mathrm{ch}}_X$;

(iv) λ is the aligned mixed distributive law, including all higher homotopies;

(v) ε is a scalar or relative augmentation appropriate to the algebra;

(vi) the nilpotent, weight, pole, coradical, and analytic completions are specified separately.

Any subset of this tuple defines the corresponding partial theory. No missing component is inferred from the others.

Theorem 232.1 (Structure theorem)**.** *For a complete datum $\mathbb{A}$ as above, the following constructions are functorial and type preserving.*

(i) $\mathrm{Bar}^{\perp}(\mathbb{A})$ *is interval factorisation homology and a conilpotent transverse coalgebra retaining the associative chiral coefficient.*

(ii) $\mathrm{Bar}^{\mathrm{ch}}_{\mathsf{Ass}}(\mathbb{A})$ *is the operadic associative chiral bar retaining the transverse algebra.*

(iii) *The mixed distributive law gives a total two-direction bar with anticommuting edge differentials and an aligned Fubini comparison.*

(iv) *For ordinary augmented algebras in degree zero, the direct-sum tensor counit is a quasi-isomorphism. Its degreewise positive-weight completion reconstructs the specified weight-complete algebra. Geometric duality and the coalgebra-unit equivalence require their separately stated hypotheses.*

(v) *The augmentation boundary, relative module categories, two-sided twisting transform, and derived centre form the Koszul–Morita atlas.*

(vi) *An E_2-lift or flat meromorphic transport adds exchange; a fibre functor reconstructs the corresponding quantum symmetry.*

(vii) *A compatible BRST operator forms a bicomplex and descends all commuting structures to cohomology.*

(viii) *A cyclic transverse pairing and a modular chiral pairing satisfying the mixed square produce the two-edge master equation with independent genus variables.*

(ix) *A perturbative BV realization gives the preceding object when renormalised graph weights extend, the QME holds, and the observables algebraize.*

(x) *Root-graded Euler character, trace, period, index, and monodromy maps produce scalar shadows only after the structured object has been constructed.*

Conversely, every construction assembled from the displayed carriers factors through one or more of these functors. A proposed construction fails whenever it omits one of the required data or hypotheses.

Proof. Items 1–3 are theorems 206.2 and 207.1; item 4 is theorems 208.1 and 208.2; item 5 is theorems 209.1 and 209.2; item 6 follows from proposition 223.1 and theorem 216.1 and Tannaka reconstruction; item 7 is theorem 222.1; item 8 is recognition Theorem 224.2, realized by theorem 226.2; item 9 is theorem 221.1; item 10 is exemplified by theorems 219.5 and 228.1. Conversely, each operation factors by type as collision, ordered composition, exchange, trace, sewing, reduction, pushforward, or scalar evaluation, and each appears in the atlas. The obstructions above identify the additional input required for each proposed comparison. □

232.2 Two associative structures

The datum $\mathbb{A}$ distinguishes ordered fusion along a transverse interval from associative chiral composition on the curve. These are independent E_1-structures for $\otimes_!$ and $\star_{\mathrm{ch}}$, respectively. Their coherent combination uses the mixed distributive law λ and the two-coloured cofibrant operad. Each bar, centre, exchange, trace, sewing, or reduction construction uses its stated additional data and hypotheses. A partial datum does not supply the missing operations or comparison maps.

232.3 Mixed locality and derived presentations

The aligned projector selects the decompositions on which the mixed locality comparison is defined. It does not identify the two associative products. Generators and relations in the two-coloured theory are imposed by homotopy pushouts of free cofibrant algebras. The resulting differentials and homology groups provide tests of proposed comparisons; the finite calculations above apply only in their specified weights and parameter ranges.

232.4 Geometric sources of the operations

Diagonal collision describes singular composition, while an interval orders transverse multiplication and the formal disc carries associative chiral iteration. The exchange constructions here use an E_2-lift or the specified flat meromorphic transport; it is not obtained from interval order alone. Circle traces and stable-curve sewing use the cyclic and modular data, including the mixed compatibility square when the two sewing operations are combined. Analytic sewing also requires convergence estimates. Compactified-boundary identities use the chosen graph weights and their extensions to the boundary. Scalar evaluation records the value of a chosen functional on these structured objects. Equality of such values does not construct a comparison of their products, differentials, or sewing maps.

Chapter 233

Suspensions, orientations, and low-arity signs

233.1 Cohomological suspension

For homogeneous a, $|sa| = |a| + 1$. The Koszul rule for exchanging homogeneous factors u, v is

$$u \otimes v \longmapsto (-1)^{|u||v|} v \otimes u.$$

The binary bar coderivation is fixed by

$$b_2(sa \otimes sb) = (-1)^{|a|} s(ab).$$

On a word, moving this degree-one coderivation past the preceding suspended letters gives

$$b_2[a_1|\cdots|a_r] = \sum_{i=1}^{r-1} (-1)^{\sigma_i} [a_1|\cdots|a_i a_{i+1}|\cdots|a_r],$$

$$\sigma_i = \sum_{j<i} (|a_j| + 1) + |a_i|.$$

For degree-zero letters the signs alternate. Any convention differing by a global suspension conjugation produces an isomorphic complex; mixing conventions within one calculation does not.

233.2 Four-letter bar word

For ungraded letters and the alternating convention,

$$b[a|b|c|d] = [ab|c|d] - [a|bc|d] + [a|b|cd].$$

Applying b again gives

$$\begin{aligned} b^2[a|b|c|d] = {} & [(ab)c|d] - [ab|cd] \\ & - [a(bc)|d] + [a|b(cd)] \\ & + [ab|cd] - [a|(bc)d]. \end{aligned}$$

The middle terms cancel, and associativity cancels the remaining two pairs. This is the cellular boundary of the two-dimensional associahedron in its simplest word model.

233.3 Two edge colours

For a bicoloured graph G, choose

$$\operatorname{or}(G) = \det E_X(G) \otimes \det E_\perp(G).$$

Contracting an X-edge removes a basis vector from the first determinant; contracting a $\perp$-edge removes one from the second. Reversing the contraction order exchanges two odd normal directions and contributes -1. Hence

$$d_X d_\perp + d_\perp d_X = 0.$$

The same convention controls the two loop operators in the master equation.

233.4 Arnold relation and KZ curvature

For three points,

$$\omega_{ij} = d\log(z_i - z_j)$$

satisfy

$$\omega_{12}\wedge\omega_{23} + \omega_{23}\wedge\omega_{31} + \omega_{31}\wedge\omega_{12} = 0.$$

The arity-three KZ curvature is

$$F_\nabla = \kappa^2\big([\Omega_{12},\Omega_{13}]\omega_{12}\wedge\omega_{13} + [\Omega_{12},\Omega_{23}]\omega_{12}\wedge\omega_{23} + [\Omega_{13},\Omega_{23}]\omega_{13}\wedge\omega_{23}\big).$$

The infinitesimal braid relation rewrites the three commutators as one common commutator with the coefficients in Arnold's relation, so $F_\nabla = 0$.

233.5 Rational Yang–Baxter expansion

For $R_{ij}(u) = 1 - \hbar P_{ij}/u$, expand both sides of Yang–Baxter. The order-$\hbar$ terms agree trivially. The order-$\hbar^2$ identity reduces, after a common denominator, to the permutation identities

$$P_{12}P_{13} = P_{23}P_{12},\quad P_{13}P_{23} = P_{12}P_{13},\quad P_{12}P_{23} = P_{13}P_{12}.$$

The cubic term agrees because both sides represent the same longest permutation.

Chapter 234

Exact finite computations

234.1 Finite chain-matrix construction

For each total polynomial weight $w \le 5$ and word length r, let $C_{w,r}$ be the normalized bar space spanned over $\mathbb{Q}$ by words whose nonunit monomial letters have total weight w. The bar differential is the alternating sum of adjacent multiplications. In the quantum plane at the rational parameter $q = 2$, multiplication is

$$(x^a y^b)(x^c y^d) = q^{bc} x^{a+c} y^{b+d}.$$

In the Jordan plane it is determined by the exact reduction $yx \mapsto xy + x^2$. Thus every differential matrix in the stated window has rational entries; its kernel, image, and rank are taken over $\mathbb{Q}$.

234.2 Quantum-plane and Jordan-plane windows

The complete nonzero homology through weight five is:

$$H_{1,1} \cong \Bbbk^2, \qquad H_{2,2} \cong \Bbbk,$$

for both algebras, with every other $H_{w,r}$ zero in the window. The selected chain dimensions and ranks are:

algebra	w	r	$\dim C_r$	$\operatorname{rk} d_r$	$\operatorname{rk} d_{r+1}$	$\dim H_r$
$A_q, q=2$	1	1	2	0	0	2
$A_q, q=2$	2	1	3	0	3	0
$A_q, q=2$	2	2	4	3	0	1
$A_q, q=2$	3	1	4	0	4	0
$A_q, q=2$	3	2	12	4	8	0
$A_q, q=2$	3	3	8	8	0	0
$A_q, q=2$	4	1	5	0	5	0
$A_q, q=2$	4	2	25	5	20	0
$A_q, q=2$	4	3	36	20	16	0
$A_q, q=2$	4	4	16	16	0	0
$A_q, q=2$	5	1	6	0	6	0
$A_q, q=2$	5	2	44	6	38	0
$A_q, q=2$	5	3	102	38	64	0
$A_q, q=2$	5	4	96	64	32	0
$A_q, q=2$	5	5	32	32	0	0
A_J	1	1	2	0	0	2

algebra	w	r	$\dim C_r$	$\operatorname{rk} d_r$	$\operatorname{rk} d_{r+1}$	$\dim H_r$
A_J	2	1	3	0	3	0
A_J	2	2	4	3	0	1
A_J	3	1	4	0	4	0
A_J	3	2	12	4	8	0
A_J	3	3	8	8	0	0
A_J	4	1	5	0	5	0
A_J	4	2	25	5	20	0
A_J	4	3	36	20	16	0
A_J	4	4	16	16	0	0
A_J	5	1	6	0	6	0
A_J	5	2	44	6	38	0
A_J	5	3	102	38	64	0
A_J	5	4	96	64	32	0
A_J	5	5	32	32	0	0

The quantum cycle $[y|x] - 2[x|y]$ and Jordan cycle $[y|x] - [x|y] - [x|x]$ evaluate to the zero vector under the exact degree-two differential.

234.3 The $\mathfrak{sl}_2$ complex

The matrix in chapter 215 has determinant -4, so d_2 is an isomorphism. The top differential is zero by the Jacobi identity and direct calculation. The result is

$$(\dim H_0, \dim H_1, \dim H_2, \dim H_3) = (1, 0, 0, 1).$$

234.4 The A_2 Hall relation at $q = 2$

In the basis $([X], [Y])$,

$$e_1^2 e_2 = (3, 3), \qquad e_1 e_2 e_1 = (3, 1), \qquad e_2 e_1^2 = (3, 0).$$

Therefore

$$(3, 3) - 3(3, 1) + 2(3, 0) = (0, 0).$$

The symbolic proof in theorem 218.1 holds for arbitrary q; the finite check catches coefficient transpositions.

234.5 Finite stable-trace Jacobi calculation

For labels (a, b) with $0 \le a, b \le 3$ and $a + b > 0$, evaluate all ordered triples under

$$[(a, b), (c, d)] = (ad - bc)(a + c - 1, b + d - 1).$$

All $15^3 = 3375$ Jacobi sums vanish exactly in the free $\mathbb{Q}$-module on the resulting labels. This statement is confined to the displayed finite label set; the all-label Poisson identity is supplied by the separate symbolic proof.

Chapter 235

Generators-and-relations templates

235.1 Quadratic transverse presentation

Given coefficient generators $V \in \mathrm{FactD}(X)$ and relations $R \to V \otimes_! V$, the strict quadratic algebra is

$$A = T_{\otimes_!}(V)/(R).$$

The derived presentation is the homotopy pushout

$$\begin{array}{ccc} \mathrm{Free}_{\mathsf{E}_1}(R) & \overset{\partial_0}{\underset{\partial_1}{\rightrightarrows}} & \mathrm{Free}_{\mathsf{E}_1}(V) \\ \downarrow & & \downarrow \\ \mathbf{1} & \longrightarrow & A_{\mathrm{der}}. \end{array}$$

A cofibrant replacement adds syzygy cells whenever relations among relations occur.

235.2 Associative chiral presentation

For formal-disc generators M and ordered field relations R_X, replace $\mathrm{Free}_{\mathsf{E}_1}$ by $\mathrm{Free}_{\mathsf{Ass}^{\mathrm{ch}}_X}$. A relation can be a finite-pole identity

$$(z-w)^N \Big(Y(a,z)Y(b,w) - \sum_i f_i(z-w) Y(b_i,w)Y(a_i,z) \Big) = 0.$$

A quantum relation includes the S-matrix coefficients. The derived quotient remembers a null-homotopy of this field identity and its compatibility with nested substitutions.

235.3 Doubly associative presentation

Use the two-coloured operad $\mathcal{P}$. The data consist of

$$(V; R_\perp, R_X, R_{\mathrm{mix}}),$$

where R_{mix} contains the rectangular locality square and any braided permutation between middle factors. Resolving the presentation automatically adds:

- transverse overlap cells;
- chiral collision overlap cells;
- mixed prisms comparing the two reduction orders;
- higher cells filling the resulting spheres.

This is the promised model-category implementation of homotopically complete generators and relations.

235.4 Relative presentation

For a semisimple base S, use free S-bimodule algebras and relative tensor products throughout. The quiver template is

$$A = T_S(M)/(R), \qquad \mathrm{Bar}_S(A) = S \otimes^{\mathbb{L}}_A S.$$

A coloured chiral quiver replaces every $e_j M e_i$ by a coefficient object and imposes the factorisation compatibility on every source-target component.

Chapter 236

Logical status of comparison statements

A comparison is used only after its source, target, carrier category, and map have been specified. A finite chain-matrix calculation establishes its stated finite window only. A formal identity carries no analytic convergence claim, and a conditional comparison is invoked only with its named hypothesis. The obstructions in Chapter 230 and the exact calculations in Appendix 234 give the corresponding negative and finite criteria.

Chapter 237

Extended comparison table

Geometry	Object	Operation	Required theorem or datum
partial diagonal	factorisable D-module	singular OPE/principal part	diagonal extension, coordinate covariance, chiral identities
ordered interval	E_1-algebra for $\otimes_!$	transverse fusion	rectangular locality and augmentation/relative base
formal disc	$\mathsf{Ass}^{\mathrm{ch}}_X$-algebra	nonlocal field iteration	weak associativity and completion
transverse plane	E_2-lift or connection	exchange	braid paths, flatness, Yang–Baxter
circle	Hochschild chains	trace/closure	cyclic or Calabi–Yau structure
ribbon surface	cyclic A_∞-algebra	open TCFT	perfect pairing and ribbon graph action
stable curve	modular chiral object	conformal-block sewing	clutching maps, pairing, convergence/anomaly line
bicoloured graph	two-edge algebra	mixed master equation	common corollas and mixed interchange square
augmentation boundary	module $\mathbf{1}$	boundary Koszul dual	compactness and completion
annulus	Hochschild theory	centre/trace pairing	two-sided Morita transform
quiver	S-algebra	boundary-changing composition	relative base and path relations
ZF spectral line	exchange algebra	ordered normal form	unitarity and Yang–Baxter
RTT monodromy	Hopf algebra	coproduct and evaluation	RTT PBW theorem
Hall stack	correspondence algebra	extension product	proper pull–push, orientations, associativity
derived commuting scheme	necklace algebra	Hamiltonian trace action	cyclic derivatives and derived quotient
BV configuration	graph-weight complex	quantum observables	renormalisation and QME

Geometry	Object	Operation	Required theorem or datum
BRST complex	bicomplex	reduction	square-zero derivation, Kunneth, convergence
root lattice	graded bar	denominator character	PBW and finite-decomposition completion
scalar output	field/ring element	trace, period, index	specified functional on a structured object

Bibliography

[1] Achucarro Achucarro and Paul K. Townsend, *A chern–simons action for three-dimensional anti-de sitter supergravity theories*, Physics Letters B **180** (1986), no. 1–2, 89–92.

[2] Luigi Alfonsi, Hyungrok Kim, and Charles A. S. Young, *Raviolo vertex algebras, cochains and conformal blocks*, 2024.

[3] Josep Àlvarez Montaner, Ricardo García López, and Santiago Zarzuela Armengou, *Local cohomology, arrangements of subspaces and monomial ideals*, Advances in Mathematics **174** (2003), no. 1, 35–56.

[4] M. F. Atiyah, *Complex analytic connections in fibre bundles*, Symposium internacional de topología algebraica, 1958, pp. 77–82.

[5] Jonathan D. Axtell and Kyu-Hwan Lee, *Quantum generalized kac–moody algebras via hall algebras of complexes*, 2019.

[6] David Ayala, John Francis, and Hiro Lee Tanaka, *Factorization homology of stratified spaces*, Selecta Mathematica **23** (2017), 293–362, available at 1409.0848.

[7] Máximo Bañados, Claudio Teitelboim, and Jorge Zanelli, *The black hole in three-dimensional space-time*, Physical Review Letters **69** (1992), no. 13, 1849–1851.

[8] Serguei Barannikov, *Modular operads and Batalin–Vilkovisky geometry*, International Mathematics Research Notices **2007** (2007), rnm075, available at 1710.08442.

[9] Alexander Beilinson and Vladimir Drinfeld, *Chiral algebras*, American Mathematical Society Colloquium Publications, vol. 51, American Mathematical Society, 2004.

[10] Alexander Beilinson, Victor Ginzburg, and Wolfgang Soergel, *Koszul duality patterns in representation theory*, Journal of the American Mathematical Society **9** (1996), no. 2, 473–527.

[11] David Ben-Zvi, John Francis, and David Nadler, *Integral transforms and drinfeld centers in derived algebraic geometry*, Journal of the American Mathematical Society **23** (2010), no. 4, 909–966, available at 0805.0157.

[12] Clemens Berger and Ieke Moerdijk, *The Boardman–Vogt resolution of operads in monoidal model categories*, Topology **45** (2006), 807–849, available at math/0502155.

[13] ———, *Resolution of coloured operads and rectification of homotopy algebras*, Contemporary Mathematics **431** (2007), 31–58, available at math/0512576.

[14] Richard E. Borcherds, *Automorphic forms on $o_{s+2,2}(\mathbb{r})$ and infinite products*, Inventiones Mathematicae **120** (1995), 161–213.

[15] ———, *Automorphic forms with singularities on grassmannians*, Inventiones Mathematicae **132** (1998), 491–562, available at alg-geom/9609022.

[16] Christopher Brav and Tobias Dyckerhoff, *Relative calabi–yau structures ii: shifted lagrangians in the moduli of objects*, Selecta Mathematica (N.S.) **27** (2021), no. 4, 63, available at 1812.11913.

[17] Christopher Brav and Nick Rozenblyum, *The cyclic deligne conjecture and calabi–yau structures*, 2023.

[18] Alexander Braverman, Michael Finkelberg, and Hiraku Nakajima, *Towards a mathematical definition of coulomb branches of 3-dimensional $n = 4$ gauge theories, ii*, Advances in Theoretical and Mathematical Physics **22** (2018), no. 5, 1071–1147, available at 1601.03586.

[19] Alexander Braverman and Dennis Gaitsgory, *Poincaré–birkhoff–witt theorem for quadratic algebras of Koszul type*, 1994.

[20] J. David Brown and Marc Henneaux, *Central charges in the canonical realization of asymptotic symmetries: An example from three-dimensional gravity*, Communications in Mathematical Physics **104** (1986), no. 2, 207–226.

[21] Damien Calaque, Vasiliy Dolgushev, and Gilles Halbout, *Formality theorems for hochschild chains in the lie algebroid setting*, Journal für die reine und angewandte Mathematik **612** (2007), 81–127, available at math/0504372.

[22] Andrea Campoleoni, Stefan Fredenhagen, Stefan Pfenninger, and Stefan Theisen, *Asymptotic symmetries of three-dimensional gravity coupled to higher-spin fields*, Journal of High Energy Physics **2010** (2010), no. 11, 007, available at 1008.4744.

[23] John L. Cardy, *Operator content of two-dimensional conformally invariant theories*, Nuclear Physics B **270** (1986), 186–204.

[24] Fabien Cléry and Valery Gritsenko, *The siegel modular forms of genus 2 with the simplest divisor*, Proceedings of the London Mathematical Society **102** (2011), no. 6, 1024–1052, available at 0812.3962.

[25] Kevin Costello, *Topological conformal field theories and calabi–yau categories*, Advances in Mathematics **210** (2007), 165–214.

[26] Kevin Costello and Owen Gwilliam, *Factorization algebras in quantum field theory, volume 1*, New Mathematical Monographs, vol. 31, Cambridge University Press, 2017.

[27] ———, *Factorization algebras in quantum field theory, volume 2*, New Mathematical Monographs, vol. 41, Cambridge University Press, 2021.

[28] Kevin Costello and Si Li, *Anomaly cancellation in the topological string*, 2019.

[29] William Crawley-Boevey, Pavel Etingof, and Victor Ginzburg, *Noncommutative geometry and quiver algebras*, Advances in Mathematics **209** (2007), no. 1, 274–336, available at math/0502301.

[30] Ben Davison, *Affine bps algebras, w algebras, and the cohomological hall algebra of* $\mathbb{A}^2$, Homotopical methods in geometry and physics, 2026, pp. 163–199.

[31] Ben Davison, Lucien Hennecart, and Sebastian Schlegel Mejia, *Bps algebras and generalised kac–moody algebras from 2-calabi–yau categories*, 2025.

[32] Ben Davison and Sven Meinhardt, *Cohomological donaldson–thomas theory of a quiver with potential and quantum enveloping algebras*, Inventiones Mathematicae **221** (2020), 777–871, available at 1601.02479.

[33] Alberto De Sole, Matteo Gardini, and Victor G. Kac, *On the structure of quantum vertex algebras*, 2019.

[34] Duiliu-Emanuel Diaconescu, Mauro Porta, Francesco Sala, Olivier Schiffmann, and Eric Vasserot, *Cohomological hall algebras of one-dimensional sheaves on surfaces and yangians*, 2026.

[35] Vasiliy Dolgushev, *Covariant and equivariant formality theorems*, Advances in Mathematics **191** (2005), no. 1, 147–177, available at math/0307212.

[36] Vladimir Dotsenko and Pedro Tamaroff, *Endofunctors and Poincaré–Birkhoff–Witt theorems*, 2019.

[37] Vladimir Drinfeld, *Quantum groups*, Proceedings of the international congress of mathematicians, berkeley 1986, 1987, pp. 798–820.

[38] Martin Eichler and Don Zagier, *The theory of jacobi forms*, Progress in Mathematics, vol. 55, Birkhäuser, 1985.

[39] Pavel Etingof and Olivier Schiffmann, *Lectures on quantum groups*, International Press, 1998.

[40] John Francis and Dennis Gaitsgory, *Chiral koszul duality*, Selecta Mathematica **18** (2012), no. 1, 27–87, available at 1103.5803.

[41] Daniel S. Freed and Constantin Teleman, *Relative quantum field theory*, Communications in Mathematical Physics **326** (2014), no. 2, 459–476, available at 1212.1692.

[42] Ezra Getzler, *Lie theory for nilpotent* L_∞*-algebras*, Annals of Mathematics **170** (2009), no. 1, 271–301, available at math/0404003.

[43] Ezra Getzler and Mikhail Kapranov, *Modular operads*, Compositio Mathematica **110** (1998), 65–126.

[44] Victor Ginzburg, *Calabi–yau algebras*, arXiv preprint (2006), available at math/0612139.

[45] L. V. Goncharova, *Cohomology of lie algebras of formal vector fields on the line*, Funktsional. Anal. i Prilozhen. **7** (1973), no. 2, 6–14.

[46] Suresh Govindarajan, *Borcherds–kac–moody lie superalgebras from counting twisted CHL dyons*, Journal of High Energy Physics **2011** (2011), no. 5, 089, available at 1006.3472.

[47] Suresh Govindarajan and K. Gopala Krishna, *Generalized kac–moody algebras from CHL dyons*, Journal of High Energy Physics **2009** (2009), no. 4, 032, available at 0807.4451.

[48] Suresh Govindarajan and Sutapa Samanta, *Borcherds–kac–moody lie superalgebras from counting twisted CHL dyons--ii*, Nuclear Physics B **948** (2019), 114770, available at 1905.06083.

[49] James A. Green, *Hall algebras, hereditary algebras and quantum groups*, Inventiones Mathematicae **120** (1995), 361–377.

[50] Valery Gritsenko and Viacheslav Nikulin, *Siegel automorphic form corrections of some lorentzian kac–moody lie algebras*, American Journal of Mathematics **119** (1997), no. 1, 181–224, available at alg-geom/9504006.

[51] ———, *Automorphic forms and lorentzian kac–moody algebras. part i*, International J. Math. **9** (1998), no. 2, 153–199, available at alg-geom/9610022.

[52] ———, *Automorphic forms and lorentzian kac–moody algebras. part ii*, International J. Math. **9** (1998), no. 2, 201–275, available at alg-geom/9611028.

[53] Alexander Grothendieck, *Techniques de construction et théorèmes d'existence en géométrie algébrique IV: les schémas de Hilbert*, Séminaire bourbaki, 1961, pp. 249–276.

[54] Marc Henneaux and Soo-Jong Rey, *Nonlinear w_∞ as asymptotic symmetry of three-dimensional higher spin anti-de sitter gravity*, Journal of High Energy Physics **2010** (2010), no. 12, 007, available at 1008.4579.

[55] Gijs Heuts, *Koszul duality and a conjecture of francis–gaitsgory*, 2024.

[56] Vladimir Hinich, *Homological algebra of homotopy algebras*, Communications in Algebra **25** (1997), no. 10, 3291–3323, available at q-alg/9702015.

[57] ———, *Dwyer–kan localization revisited*, 2016.

[58] Jin Hong, *Center and universal r-matrix for quantized borcherds superalgebras*, Journal of Mathematical Physics **40** (1999), no. 6, 3123–3145, available at math/9810115.

[59] Ryoshi Hotta, Kiyoshi Takeuchi, and Toshiyuki Tanisaki, *D-modules, perverse sheaves, and representation theory*, Progress in Mathematics, vol. 236, Birkhäuser Boston, 2008.

[60] Jun-ichi Igusa, *On siegel modular forms of genus two*, American Journal of Mathematics **84** (1962), no. 1, 175–200.

[61] ———, *On siegel modular forms of genus two (ii)*, American Journal of Mathematics **86** (1964), no. 2, 392–412.

[62] Dominic Joyce and Markus Upmeier, *Orientation data for moduli spaces of coherent sheaves over calabi–yau 3-folds*, Advances in Mathematics **381** (2021), 107627, available at 2001.00113.

[63] Victor G. Kac, Shi-Shyr Roan, and Minoru Wakimoto, *Quantum reduction for affine superalgebras*, Communications in Mathematical Physics **241** (2003), 307–342, available at math-ph/0302015.

[64] Victor G. Kac and Minoru Wakimoto, *Quantum reduction and representation theory of superconformal algebras*, Advances in Mathematics **185** (2004), 400–458, available at math-ph/0304011.

[65] Seok-Jin Kang and Olivier Schiffmann, *Canonical bases for quantum generalized kac–moody algebras*, Advances in Mathematics **200** (2006), no. 2, 455–478, available at math/0311089.

[66] Bernhard Keller, *Introduction to A-infinity algebras and modules*, Homology, Homotopy and Applications **3** (2001), no. 1, 1–35, available at math/9910179.

[67] ———, *On differential graded categories*, 2006.

[68] ———, *Deformed calabi–yau completions*, Journal für die reine und angewandte Mathematik **654** (2011), 125–180, available at 0908.3499.

[69] Tasuki Kinjo, Hyeonjun Park, and Pavel Safronov, *Cohomological hall algebras for 3-calabi–yau categories*, 2024.

[70] Maxim Kontsevich, *Deformation quantization of poisson manifolds*, Letters in Mathematical Physics **66** (2003), 157–216, available at q-alg/9709040.

[71] Maxim Kontsevich and Yan Soibelman, *Deformations of algebras over operads and the deligne conjecture*, 2000.

[72] P. P. Kulish and E. K. Sklyanin, *Algebraic structures related to reflection equations*, Journal of Physics A: Mathematical and General **25** (1992), no. 22, 5963–5976, available at hep-th/9209054.

[73] Aaron D. Lauda and Hendryk Pfeiffer, *Open-closed strings: Two-dimensional extended TQFTs and Frobenius algebras*, 2008.

[74] Haisheng Li, *Nonlocal vertex algebras generated by formal vertex operators*, 2005.

[75] ———, *Quantum vertex algebras and their ϕ-coordinated quasi modules*, 2009.

[76] Jean-Louis Loday and Bruno Vallette, *Algebraic operads*, Grundlehren der mathematischen Wissenschaften, vol. 346, Springer, 2012.

[77] Eduard Looijenga and Valery Lunts, *A lie algebra attached to a projective variety*, Inventiones Mathematicae **129** (1997), 361–412.

[78] Ming Lu, *Quantum borcherds–bozec algebras via semi-derived ringel–hall algebras*, 2021.

[79] Jacob Lurie, *Higher topos theory*, Annals of Mathematics Studies, vol. 170, Princeton University Press, 2009. Updated author version, April 2017.

[80] ———, *Higher algebra*, 2017. Available from the author's webpage.

[81] James E. McClure and Jeffrey H. Smith, *A solution of deligne's hochschild cohomology conjecture*, Contemporary Mathematics **293** (2002), 153–193, available at math/9910126.

[82] Pavel Mnev, *Batalin–Vilkovisky formalism and applications in topological quantum field theory*, 2016. Lecture notes, University of Notre Dame, December 4.

[83] Alexander Molev, *Yangians and their applications*, Handbook of algebra, 2003, pp. 907–959.

[84] Alexander Molev, Maxim Nazarov, and Grigori Olshanskii, *Yangians and classical Lie algebras*, Russian Mathematical Surveys **51** (1996), no. 2, 205–282, available at hep-th/9409025.

[85] Hiraku Nakajima, *Heisenberg algebra and hilbert schemes of points on projective surfaces*, Annals of Mathematics **145** (1997), no. 2, 379–388.

[86] Georg Oberdieck and Aaron Pixton, *Holomorphic anomaly equations and the igusa cusp form conjecture*, Inventiones Mathematicae **213** (2018), no. 2, 507–587, available at 1706.10100.

[87] Georg Oberdieck and Junliang Shen, *Reduced donaldson–thomas invariants and the ring of dual numbers*, Proceedings of the London Mathematical Society **118** (2019), no. 1, 191–220, available at 1612.03102.

[88] Tony Pantev, Bertrand Toën, Michel Vaquié, and Gabriele Vezzosi, *Shifted symplectic structures*, Publications Mathématiques de l'IHÉS **117** (2013), 271–328, available at 1111.3209.

[89] Leonid E. Positselski, *Nonhomogeneous quadratic duality and curvature*, 2014. Corrected English version.

[90] Stewart Priddy, *Koszul resolutions*, Transactions of the American Mathematical Society **152** (1970), 39–60.

[91] Daniel B. Ray and Isadore M. Singer, *r-torsion and the laplacian on riemannian manifolds*, Advances in Mathematics **7** (1971), no. 2, 145–210.

[92] Claus Michael Ringel, *Hall algebras and quantum groups*, Inventiones Mathematicae **101** (1990), 583–591.

[93] Olivier Schiffmann and Eric Vasserot, *Cohomological hall algebras of quivers and yangians*, 2024.

[94] Albert Schwarz, *Geometry of Batalin–Vilkovisky quantization*, Communications in Mathematical Physics **155** (1993), 249–260, available at hep-th/9205088.

[95] Alexander Tsymbaliuk, *The affine yangian of* $\mathfrak{gl}_1$ *revisited*, Advances in Mathematics **304** (2017), 583–645, available at 1404.5240.

[96] Francesco Vaccarino, *The ring of multisymmetric functions*, Annales de l'Institut Fourier **55** (2005), no. 3, 717–731.

[97] ———, *Linear representations, symmetric products and the commuting scheme*, Journal of Algebra **317** (2007), no. 2, 634–641, available at math/0602660.

[98] Mikhail Verbitsky, *Cohomology of compact hyper-kähler manifolds and its applications*, Geometric and Functional Analysis **6** (1996), 601–611.

[99] Edward Witten, (2 + 1)*-dimensional gravity as an exactly soluble system*, Nuclear Physics B **311** (1988), no. 1, 46–78.